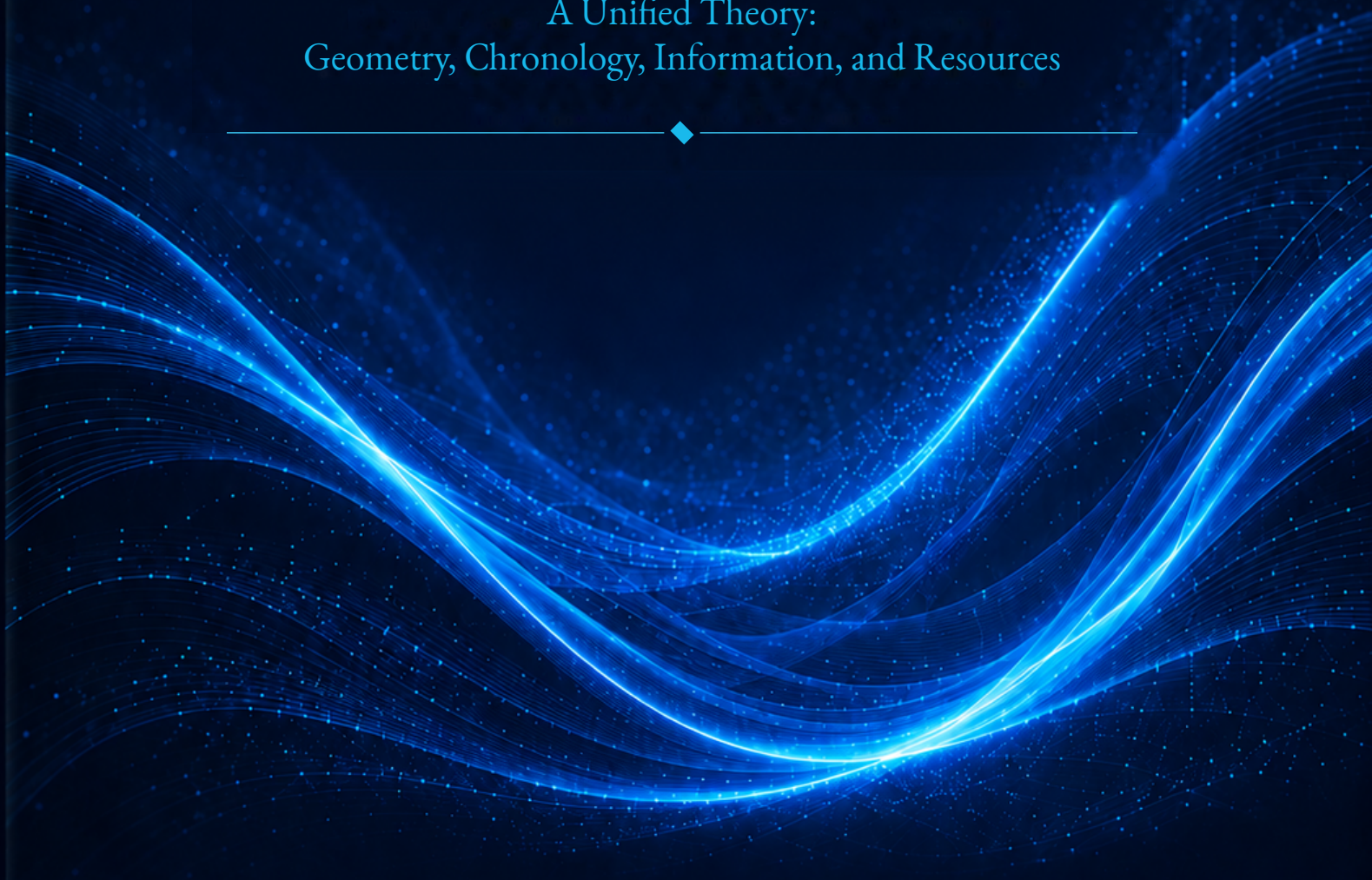
Semantic Normalization and Physical Realization

A Unified Theory:
Geometry, Chronology, Information, and Resources



ALP EREN BÜTÜN

THEORY · CONSTRUCTION · PROOFS · GENERALIZATION

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AUTOMATA THEORY · NORMALIZATION · COMPLETION · STRUCTURE

Research Monograph

Semantic Normalization and Physical Realization

A Unified Theory: Geometry, Chronology, Information, and Resources

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Scope note

This volume synthesizes a sequence of research studies developed between 2023 and 2026 as a unified mathematical theory. The exposition follows the logical development of its results while consolidating repeated arguments, separating representation-specific results from representation-independent ones, and recording computational certificates beside the symbolic results they support.

Proof and validation note

Unless a result is explicitly identified as computer-assisted, the mathematical claims in this volume rest on symbolic arguments. Computational audits are used as independent consistency checks; Appendix D records their scope, numerical outcomes, and verification status without making the symbolic theorems depend on implementation details.

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Preface

This book grew from a single deterministic pushdown automaton.

In 2023 I was studying the binary counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

The first object of interest was not a general formalism but a concrete machine: a center state, two side states, and a stack manipulated by cancellation and replacement rules. The design was initially understood operationally. What made it the starting point of a larger research program was the realization that its state and stack were carrying more information than acceptance alone. After a prefix had been read, the configuration encoded a precise residual quantity and, after completion normalization, a shortest way of repairing that residual.

That observation supplied the first transition from a machine diagram to a mathematical theory. The stack ceased to be merely storage. It became a representation of completion demand. The side states ceased to be auxiliary control. They became finite corrections to the meaning of the stack. The cancellation transitions became local steps in a directed shortest-completion geometry.

The chapters that follow preserve this order of discovery. The original completion-aware DPDA is first analyzed on its own terms. Its reachable stores and state potentials lead to an exact residual invariant. Residuals then give a quotient coordinate for prefixes and a directed geometry of shortest continuations. That geometry supports progressively more economical realizations and, after the representation class is fixed carefully, exact resource frontiers. A related resource question then leaves the original coordinate system and becomes the fixed-state stack-alphabet hierarchy problem left open after Masopust's realtime result [16]; the general ε-move case is resolved later in the monograph under its stated model conventions.

The research subsequently branches. Higher-dimensional counting systems turn shortest completions into cubical and local-action complexes. That geometry then admits a further abstraction: finite-phase periodic action systems support a finite cubical geodesic compiler for distance, optimal branching, shortest prefixes, and complete local optimal compatibility. The associated structure theory shows that higher-order compatibility is genuinely strict across the class: at every prescribed order new information can first appear, product constructions amplify the discrepancy Booleanly, and the resulting fiber viewpoint shows that this freedom is not merely a product artifact. On one fixed structural simplex, exact distance and all compatibility data through order $k-1$ can remain frozen while the entire order-k layer ranges over every k-uniform hypergraph. A later return to this same branch removes the distinguished-rank restriction: one fixed coefficient bank of exact Boolean-lattice-width size realizes every labeled simplicial complex, common-shell synchronization costs no extra row, and one periodic Bellman system realizes the full simplicial universe at one exact canonical distance. This later theorem is placed beside its geodesic prerequisites rather than being retroactively treated as an earlier motivation. A different branch follows one cancellation pattern into Fuss–Catalan

defect geometry and then into chronological and p-adic arithmetic. These branches are retained because they record genuine consequences of the completion viewpoint, but they are not allowed to obscure the central line of the book.

The central line returns to pushdown construction. For coprime positive p, q, a centered real-time literal-demand DPDA generalizes the original 2:1 architecture. Its finite-control rail and literal stack expose a WRITE–SEARCH–CANCEL normalization calculus. At that point a distinction that had been implicit from the beginning becomes explicit: the semantic process can be unique while its physical pushdown realization is not. This leads to the theory of exact semantic lifts, sections, representation debt, LIFO locality, normalization schedules, and causal obstruction. The six reduced 2:1 schedules are then upgraded to an arbitrary-coprime compiler and an exact causal-hypergraph classification, including the precise scope of downward closure and the category dependence of the named $A \to B$ persistence statement. Exact-lift minimality is sharpened before the General Theory by strict coordinate quotients, protected interfaces, bounded bi-recoding, and minimal protected cores. The same physical comparison layer then supports a separate confluence analysis: presentation schedules, literal completion fibers, finite-control sheets, and protected quotient classes are treated as four different rewrite objects rather than being conflated under one semantic normalizer.

The later mathematical branches are also pushed beyond their earlier endpoints. The optimal-fiber topology is no longer confined to rank two: a realization theorem identifies finite lattice-convex configurations as exactly the completion-total optimal fibers up to affine lattice equivalence, and product, apex, and prism operations give a rank-raising calculus with higher tori and unbounded higher homology at fixed active rank. The chronological-carry branch is extended from the CCR-3 jet to a typed fourth-order family and then to the CCR-k General Theory. The fourth-order stage remains the first point at which a new connected coefficient is not forced by the lower jet. At arbitrary order the resulting terminal-block hierarchy has one typed connected source per order, an exact higher-band Hessenberg realization, inverse normalized transport, polynomial fixed columns, and a universal raw Stirling shadow that forgets every p-divisible higher parameter modulo p. Finally, the representation-independent resource branch is sharpened from a state lower bound to the exact unrestricted state–stack antichain and then to an exact bounded-replacement law for arbitrary one-work-symbol deterministic recognizers. The former raw-only bounded staircase is thereby promoted only after a separate protected-phase/first-hit synchronization proof removes the representation assumption.

The final abstraction is deliberately broader than weighted counting languages. In the General Theory, a deterministic semantic system is given first, together with a physical policy domain and an exact projection from physical configurations to semantic states. Questions about canonical sections, local realizability, safe and fatal debt, schedule conflicts, synthesis, and verification can then be asked without assuming that the semantic state space is the integer residual line or that the physical machine is one of the centered p:q constructions. The weighted-balance family serves as a demanding test laboratory for the abstract theory, not as its definition.

After that abstraction was completed, the residual and completion viewpoints produce several post-theory branches. The first counts the physical CANCEL/WRITE history of centered rational normalization itself: primitive cancellation becomes a Schubert statistic, repeated normalization factors through Bizley and Raney structures, and phase normality compresses arbitrary scale to an exact one-counter process. A bridge then separates these operational statistics from the intrinsic path statistics of the next branch, where rational-Dyck return events become coarea barrier contacts and the two explicit enumeration questions posed by Dai–Fu–Qiu—run with return and ratio-run with return—are solved [23] before abstraction to a contact-preserving barrier-compression

framework. The shortest-completion map also generates the Farey-recursive arithmetic branch and its universal scalar spectra; that arithmetic line is then carried into the classical Cohn–Markoff monoid through an exact determinant-residual swap factorization and a completion-kernel/quadratic-return correspondence. Later branches quantify fixed-semantics LIFO chronology, extend rank-two optimal-fiber topology first to physical holonomy and sharp carrier cost and then to a displacement-marked higher-rank transfer calculus with exact prism/apex pure-sector laws, finite physical images, carrier girth, ancestry, connection-memory, and repair coordinates, prove unrestricted bounded-replacement state complexity, and connect universal higher-order semantic freedom to representation-independent source-stack exposure before synthesizing the exact protected-contract Pareto frontiers. A final algorithmic branch changes from resource optimization to one-counter decomposition and comparison. It first resolves Totzke's Prime-DOCA decidability question [99]: state-primality of deterministic one-counter presentations is shown undecidable by an elementary-abelian prime/composite switch coupled to deterministic two-counter transcripts. That result is then sharpened to exact two-factor decomposition by alternating-sentinel coupling: the hardness already occurs at $k = 2$, with half-size quotient factors, and therefore at every fixed factor budget $k \geq 2$. The same Part then turns back to fixed comparison and resolves both Automata Exchange one-counter questions treated here [109]. It first preserves the v11 solution of Problem 24.04(B): finite-word language equivalence between a deterministic one-counter net and a nondeterministic one-counter net is decidable, using exact forward envelopes, backward completion requirements, bounded-return compactness, a terminal last-occurrence ladder, and an exact LOW-state WQO tree. It then closes the zero-testing frontier in Problem 24.04(A): finite-word language equivalence between a deterministic zero-testing one-counter automaton and an arbitrary nondeterministic one-counter net is decidable. This second proof uses zero cuts, suffix-parametric positive first-return compactness, finite zero-state trees, and terminal-tail reduction. A further theorem shows that the A-side decidability survives standard deterministic epsilon moves on the OCA side and arbitrary epsilon moves on the OCN side, via finite threshold-affine weak macros and a guarded-macro transfer theorem. These post-theory branches are placed after the General Theory because they are consequences of the semantics already developed, not motivations invented after the fact; the one-counter algorithmic boundary is placed after the exact state/resource theory for the same reason.

The final structural enlargement comes after all source branches are available. Unified Semantic Normalization and Realization Theory does not collapse completion geometry, normalization fibers, chronology, protected information, and holonomy into one object; it supplies a typed host and explicit transfer/non-transfer bridges among them. This is followed by an external test in finite scalar fields. Extrema–Frame Foreground–Background theory develops observer-relative ownership, exact common-realization uncertainty semantics, shallow-tree ownership hardness, shallow spatial-frame hardness, cactus-graph tractability, all-orders obstruction families, finite-complex universality, parameterized ambiguity bounds, correlated polyhedral uncertainty, and finite layered semantics independently of the pushdown origin of the book. Its placement near the end is therefore intentional: it shows that the realization discipline extracted from the original machine can organize a mathematically different domain without making that domain a disguised automata problem.

Two editorial principles govern this monograph.

First, discovery history is preserved where it clarifies the mathematics. A later theorem is not retroactively presented as if it had motivated an earlier construction. In particular, the original 2023 machine remains the causal starting object, even when later chapters reveal cleaner coordinates or smaller realizations. The compact machines are

conclusions of analysis, not substitutes for the original motivation.

Second, scope is treated as part of the theorem. Several different notions of "optimality" occur in this program: raw residual-faithful resource optimality, optimality inside a deferred-SEARCH literal-demand model, and lower bounds for broader DPDA representation classes. They are not interchangeable. Likewise, normalization to a common terminal representative is not promoted to a general Church–Rosser statement, and a fixed-policy causal obstruction is not silently upgraded to an unrestricted impossibility theorem. The book repeatedly marks these boundaries because many of the strongest results are strong precisely because their model assumptions are explicit.

The same principle applies to computation. Exhaustive programs were useful in three roles: discovering finite patterns, auditing symbolic formulas over large bounded regions, and certifying explicitly finite reductions. The first two roles do not replace proof. Where a theorem contains a genuinely finite computer-assisted component, the certificate and its scope are stated as such. Appendix D records the computational validation and distinguishes independently rerun audits, reconstructed validation runs, source-reported numerical checks, and purely symbolic results.

The book therefore has two simultaneous purposes. It is a consolidation of a sequence of research studies, with repeated proofs merged and notation normalized. It is also a single mathematical narrative: a path from one completion-aware DPDA to a general theory of semantic normalization and physical realization, with geometry, chronology, information, and resources kept as distinct but interacting structural axes.

At book scale, the chapters are grouped into eleven macro-Parts. The macro-Parts follow the logical architecture: foundations and residual completion; higher-dimensional completion and geodesic compilation; cancellation/chronology and the return to the general $p{:}q$ machine; structural WSC/lift/causal theory; the abstract General Theory; enumerative and arithmetic consequences; chronology/topology/higher-rank structure; representation-independent physical complexity; state decomposition and its algorithmic boundary; unified realization theory with the EF–FG external scalar-field branch; and final synthesis. Distinct research branches remain visible as named chapter sequences inside these macro-Parts, with their theorem packages and scope preserved.

After the post-theory consequence and synthesis branches, the final chapters return to the original $C/L/R$ machine. This return is intentional. It allows the reader to see, in the same small automaton, the residual split between state and stack, literal completion demand, SEARCH debt, kernel materialization, schedule choice, operational event statistics, physical holonomy, and representation-sensitive resource tradeoffs. The aim of the abstraction is not to make the machine disappear. It is to explain why the machine was doing more than was initially visible.

The arithmetic, topological and recognition-theoretic boundaries are stated in their native mathematical categories: stationary integral sources are not conflated with Haar-varying local sources; a marked periodic quotient is not replaced by an unproved product decomposition; a protected paired service is not a uniform raw-input implementation; and deterministic tree-walking nonrecognizability is not a solution of the unambiguous–deterministic question.

Alp Eren Bütün
September 2026

Abstract

This monograph develops a theory of semantic normalization and physical realization beginning with a concrete completion-aware DPDA for

$$L_{2,1} = \{w : N_1(w) = 2N_0(w)\}$$

into a general framework separating semantic evolution from physical LIFO representation. The starting invariant is the signed residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w),$$

which determines the exact right quotient of a prefix and a canonical shortest completion demand. For coprime p, q, the primitive kernel (q, p) organizes the resulting directed completion geometry. These coordinates yield exact results on reachable stores, visibility, residual-faithful realizations, and resource frontiers, while keeping representation-specific and representation-independent lower bounds explicitly distinct.

The completion viewpoint is then lifted to higher rank and to positive-cost finite-phase periodic action systems. A finite cubical geodesic compiler recovers exact distance, optimal first moves, shortest-prefix membership, and the complete local optimal-compatibility complex from finite arithmetic data. Higher-order compatibility is strictly non-collapsing: at every prescribed order, lower-order data can remain fixed while the next layer varies freely. This culminates in full simplicial universality. For a structural simplex on s vertices, the exact number of coefficient rows required in advance to realize every labeled simplicial complex is

$$\binom{s}{\lfloor s/2 \rfloor},$$

and common-shell synchronization introduces no additional row cost. The same optimal coefficient system lifts to a periodic Bellman system realizing the full labeled simplicial universe at one canonical distance.

Returning to pushdown construction, the monograph gives a centered real-time literal-demand DPDA for every coprime ratio $p{:}q$ and isolates WRITE, SEARCH, and CANCEL as a normalization calculus. The semantic demand process is unique, whereas LIFO order, finite-control sheets, and materialization timing create representation debt and nonunique physical lifts. Exact causal-schedule classification, protected interfaces, bounded bi-recoding, minimal protected cores, and a typed confluence hierarchy lead to a General Theory of exact semantic lifts, bounded-local realizability, collapse capacity, debt, behavioral safety, synthesis, and effective verification.

A later synthesis layer places those theorem-bearing branches in a typed exact-realization architecture. Unified Semantic Normalization and Realization Theory (USNRT) distinguishes representation fibers, rooted path-outcome fibers, protected-obligation compatibility, marked transport, and resource evaluation; proves exact endpoint factorization and fixed-template section debt; develops protected-obligation products, asymptotic cover rate, transfer loss, and same-model saturation; and gives

source-faithful recovery of completion geometry, exact pushdown normalization, chronological arithmetic, and retained-information/exposure theory. The monograph then leaves pushdown automata entirely: Extrema–Frame Foreground–Background theory (EF–FG) gives a self-dual scalar-field semantics with an observer-record normal form, exact common-realization semantics, a polynomial local-witness algorithm on cactus graphs, NP-complete synchronized ownership already on radius-two trees, NP-complete complete spatial-frame realization on shallow bipartite graphs, universality of finite common-realization complexes, fixed-parameter tractable ambiguity regimes, correlated rational-polyhedral uncertainty, finite multichannel/spatiotemporal transfer, and common-realization obstructions of arbitrary order.

Several consequence branches reuse the same semantic objects. Operational CANCEL/WRITE histories admit exact enumerative descriptions and one-counter compression. For rational Dyck paths, residual rank turns returns into coarea barrier contacts and resolves both explicit joint-enumeration questions posed by Dai–Fu–Qiu: prescribed run with return and prescribed ratio-run with return [23]. The argument extends to a contact-preserving barrier-compression framework with fixed-composition, arbitrary-boundary, and parking-function applications. Shortest-completion minimizers also generate Farey-recursive spectra and an exact Cohn–Markoff completion geometry.

The semantic/physical separation is further sharpened by chronology, topology, and unrestricted complexity results. Chronological carry progresses from the CCR-3 refinement through the first independent fourth-order coordinate to an all-order CCR-k terminal-block theory with exact Hessenberg realization, inverse fixed-column transport, parameter identifiability, and universal raw Stirling shadows. The normalized CCR columns moreover admit exact all-odd-order Stirling source-obstruction lattices, fixed-integral finite-prime-support theorems, exceptional 419- and 73-adic folding laws, higher-column Galois specializations, and independent Haar-product and adelic resonance laws. These are theorems of the typed arithmetic source models, without an unproved identification with every physical chronology. Fixed-semantics orderings exhibit distinct LIFO scales; optimal completion fibers admit lattice-convex and higher-rank realization theorems; arbitrary unequal simplex-strand fibers have the same marked periodic-nerve homotopy product as their equal-width models; a source-native facet compiler exactly realizes shortest Parikh fibers, preserves them under scalar folding, and supports separate real-time one-stack recognition and clocked paired-session comparison. A displacement-marked group separates full and linear zero-displacement sectors, while finite physical holonomy, literal carrier locality, connection memory, and repair cost form distinct representation-dependent layers. For rational balance languages the monograph obtains exact unrestricted state–stack endpoints and an exact one-work-symbol bounded-replacement law. A representation-independent hierarchy branch answers the fixed-state general-DPDA stack-alphabet question left open after Masopust's realtime result [16], proving a strict binary hierarchy for $m \geq 2$ even against deterministic ε-move opponents. Filtered collapse capacity then converts universal retained semantic information into lower bounds on state/alphabet capacity, source-stack exposure, or representation debt.

A separate tree-walking recognition boundary is established by right-turn spectra on three-marker binary trees: regularity is equivalent to ultimate periodicity, deterministic zero-pebble recognition fails for every fixed threshold and certain periodic spectra, and all regular spectra have a deterministic one-visible-pebble recognizer. These results do not decide the UTWA–DTWA inclusion or the pure even-spectrum case.

Finally, the monograph crosses from resource optimization to decomposition. Totzke's Prime-DOCA question [99] is resolved negatively: presentation-relative state-primality for deterministic one-counter automata is undecidable, even for a strongly restricted

complete real-time one-work-symbol subclass. The boundary is then sharpened to the smallest nontrivial fixed factor budget: Factor-DOCA$_2$ is undecidable, with two explicit half-size factors in the composite branch; consequently every fixed $k \geq 2$ is undecidable under the same restrictions. The monograph then returns from existential decomposition to fixed comparison and resolves both Automata Exchange one-counter questions treated here [109]. Problem 24.04(B), comparing a deterministic one-counter net with a nondeterministic one-counter net, is decidable under finite-word final-state acceptance; exact forward envelopes, backward completion requirements, bounded-return compactness, a terminal last-occurrence ladder, and an exact LOW-state well-quasi-order tree yield the hard-direction small-witness bound under the deterministic-right inclusion promise. The adjacent zero-testing problem, Problem 24.04(A), is also decidable: a deterministic zero-testing one-counter automaton can be compared with an arbitrary nondeterministic one-counter net in the standard real-time model, and decidability survives the stated epsilon-transition extension. Its proof replaces the global net monotonicity argument by zero cuts, suffix-parametric positive first-return compactness, finite zero-state trees, and terminal-tail reduction; the epsilon extension uses finite threshold-affine weak macros and a guarded-macro transfer theorem, while the reverse inclusion reduces to one-zero-test VASS reachability. Thus two fixed comparison problems remain decidable even though unrestricted and fixed-budget existential decomposition of deterministic one-counter presentations is undecidable.

The resulting narrative is a single program of semantic normalization and physical realization: extract the semantic object represented by a system, identify the physical constraints that are genuinely forced, change coordinates when an obstruction is representational, and retain explicit debt when locality prevents canonical materialization.

New NP-Complete Problems, Resolved Open Problems, Impossibility Theorems, and Main Contributions

The theorem families include new NP-complete realization problems, solutions to problems explicitly posed in the literature, and independent impossibility and recognition-boundary results. Their geometric, arithmetic, topological, and operational consequences form one realization framework. The newly introduced NP-complete problems are listed first because they are independent complexity contributions of the EF–FG theory rather than answers to previously posed questions.

New NP-complete problems introduced by the theory

The two EF–FG stages introduce a $3 + 1$ complexity package: three native realization problems together with one abstract scheduling kernel that isolates the mechanism behind the ownership hardness.

1. **One-Polarity Common-Realization Observer Safety.** Deciding whether one scalar realization can make the one-polarity observer saliency safe is NP-complete even on interval-valued *radius-two trees*, with fixed $\theta = 1/2$ and $H = 0$.

2. **Possible Background Polarity-Core Ownership.** Deciding whether the self-dual observer state can be made background is NP-complete even on interval-valued *radius-two trees*, with fixed $\theta = 1/2$ and $\Delta = 0$.

3. **Complete Categorical Spatial-Frame Realization.** Deciding whether prescribed finite spatial-frame states can all be realized by one common scalar field is NP-complete even on *bipartite graphs of radius at most three*, with fixed seed threshold and uncertainty confined to identical terminal intervals. Its terminal-mask hardness core is already NP-complete on hub-radius-two bipartite graphs.

4. **Underlying kernel: Threshold-Band Owner-Scheduling Completion.** The threshold-band owner-scheduling kernel is NP-complete already for fixed $\theta = 1/2$ and $H = 0$. It isolates the source of the ownership hardness as global order selection under one-use owner resources, rather than evaluation of a fixed order.

Why these NP-complete problems are significant

These hardness results do not rely on deep trees, cycles, complicated ambient topology, or a growing collection of model parameters. NP-completeness is already present on radius-two trees and radius-two/three bipartite graphs. Thus the combinatorial difficulty is

intrinsic to global common-realization synchronization—coordinating order, ownership, terminal exposure, and spatial-frame consistency in one scalar realization—rather than being an artifact of graph-topological complexity.

Six externally posed problems resolved in the monograph

1. **Dai–Fu–Qiu: explicit run–return enumeration.** Dai, Fu, and Qiu introduced the rational run and ratio-run statistics and, in their concluding remarks, posed explicit enumeration with prescribed run values and returns [23]. The rational-path branch converts returns into coarea barrier contacts and gives a finite determinant formula without coefficient extraction for the joint distribution of ordinary run and return.

2. **Dai–Fu–Qiu: explicit ratio-run–return enumeration.** The same determinant framework treats the second statistic from [23]. With the sampling width specialized to the Dai–Fu–Qiu ratio-run, it gives an explicit finite formula for prescribed ratio-run and return. Thus the two joint enumeration questions are resolved separately, even though they are proved by one common barrier/contact mechanism.

3. **Masopust: the fixed-state stack-alphabet hierarchy beyond the realtime model.** Masopust established the binary stateless hierarchy, extended the result to m-state realtime deterministic pushdown automata, and left the corresponding m-state deterministic case open [16]. In the general deterministic model of the hierarchy branch, where deterministic ε-moves are allowed, the monograph proves for every $m \geq 2$ and $k \geq 1$ that

$$\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1},$$

   with binary witnesses. The proof in fact establishes the stronger representation-independent product lower bound $|Q|\,|\Gamma| \geq mk + 1$ for the witness family. The stateless convention is kept separate, exactly as required by the model comparison in the chapter.

4. **Totzke: Prime-DOCA decidability.** Totzke asked whether presentation-relative state-primality is decidable for deterministic one-counter automata [99]. The state-decomposition branch proves that Prime-DOCA is undecidable by an effective reduction from deterministic two-counter halting. The undecidability persists for complete real-time one-work-symbol DOCAs with a permanent bottom marker, all states final, and updates in $\{-1, 0, +1\}$. The same branch then gives a strict sharpening that is not needed to answer Totzke's question: undecidability already holds for the smallest nontrivial fixed budget $k = 2$, with two explicit factors each having exactly half as many control states as the target; hence every fixed $k \geq 2$ is undecidable under the same restrictions.

5. **Automata Exchange Problem 24.04(B): deterministic–nondeterministic one-counter-net language equivalence.** Problem 24.04(B) asks whether language equivalence is decidable between a deterministic one-counter net and a nondeterministic one-counter net [109]. The comparison branch proves decidability under finite-word acceptance by final control state, with arbitrary integer transition weights and initial counter value zero. The proof first decides the known deterministic-right inclusion using Hofman–Totzke [108]; under that reverse-inclusion promise it constructs a total computable small-witness bound for the hard inclusion by exact forward envelopes, backward completion requirements,

bounded-return compactness, a terminal last-occurrence ladder, and an exact LOW-state well-quasi-order tree.

6. **Automata Exchange Problem 24.04(A): deterministic zero-testing one-counter automata versus nondeterministic one-counter nets.** Problem 24.04(A) asks whether finite-word language equivalence is decidable for the asymmetric pair consisting of a deterministic zero-testing one-counter automaton and a nondeterministic one-counter net [109]. The second comparison branch proves decidability in the standard real-time unit-step model and, more strongly, under a deterministic-epsilon extension on the zero-testing side together with arbitrary epsilon moves on the net side. For the hard inclusion $L(A) \subseteq L(N)$ it develops exact forward envelopes and backward requirements, suffix-parametric positive first-return compactness, an effectively finite zero-state tree, and a terminal-tail reduction; the epsilon extension compiles weak visible net behavior into a finite threshold-affine macro calculus and transfers the compactness proof through guarded macros and stable/reset deterministic normalization. The reverse inclusion reduces to exact reachability in a vector addition system with one zero test. No useful complexity upper bound is claimed.

Impossibility theorems and precise recognition boundaries

For finite ordered full binary trees with exactly three marked leaves, let $\kappa(t)$ be the number of right-child edges between the ancestor of all three marks and the ancestor of the final two. The right-turn spectrum languages $\mathcal{L}_E = \{t : \kappa(t) \in E\}$ have an exact regularity criterion: $\mathcal{L}_E$ is regular if and only if $E \subseteq \mathbb{N}$ is ultimately periodic. The quantitative bottom-up construction has linear-order state complexity in the least preperiod and eventual period.

Uniform right-edge dilation and inverse finite-control simulation, starting from the established three-marker separation of Bojańczyk–Colcombet [156], exclude every fixed upper or lower threshold spectrum from deterministic tree-walking recognition; additional infinite ultimately periodic spectra admit the same obstruction. Every regular spectrum has a deterministic one-visible-pebble recognizer. These results distinguish regularity, deterministic nonrecognizability, and pebble recognition without asserting a separation of unambiguous tree-walking recognition from deterministic recognition. In particular, the first inclusion $\mathsf{DTWA} \subseteq \mathsf{UTWA}$ remains outside the established conclusion, and the ordinary deterministic recognizability of the pure even spectrum is not decided by the stated argument.

Main Contributions

The contributions of this monograph extend well beyond any short catalogue of individual results. It brings together a sequence of previously developed ideas and results, reorganizes them within a unified mathematical framework, and extends them through new definitions, structural principles, general theorems, exact classifications, and connections between previously separate parts of the theory. The resulting synthesis develops a broad theory of semantic normalization and physical realization, growing from deterministic pushdown normalization and organizing completion geometry, chronology, information, and resource complexity within one architecture. The contribution lies not only in the individual theorems but also in their generalization, interaction, and unification across the monograph.

Arithmetic, marked topology, and source-native realization

The full CCR-k hierarchy admits an arithmetic stratification of normalized columns, exceptional local folds at 419 and 73, all-odd-order integral-source obstruction lattices, finite-prime-support results, and independent local Haar and adelic resonance laws; stationary integer sources and independently varying p-adic coordinates remain explicitly different quantifier regimes. Marked periodic nerves of arbitrary unequal simplex-strand widths retain the homotopy product of a rank-s bouquet with a circle, extending the exact product geometry beyond literal Cartesian fibers. Source-native unit-cost compilers realize an entire prescribed optimal Parikh fiber, preserve it exactly under bounded radix folding, and admit explicitly separated real-time one-stack zero-return and protected paired-session comparison services. The paired service is not asserted to be a uniform unframed dual-service implementation.

Unified realization theory and external scalar-field generativity

After the source theorem families are developed, the monograph adds a typed synthesis rather than another coordinate-specific specialization. USNRT proves rooted endpoint factorization and exact fixed-template section debt, protected-obligation cover capacity, product and transfer laws, same-model saturation with separate pairwise and assembly defects, typed holonomy/resource transfer, and a four-axis recovery theorem showing that the source structures are hosted without being identified. EF–FG then provides an independent external specialization: observer records eliminate the full elder hierarchy from pointwise ownership, exact interval semantics require one common scalar realization, local possible exceedance is polynomial on cacti, synchronized global safety and possible-background ownership are NP-complete on radius-two trees, complete categorical spatial-frame realization is NP-complete on shallow bipartite graphs of radius at most three, and common-realization obstruction order is unbounded in both canonical and section-free forms.

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Notation and Conventions

This page records the notation used across several layers of the monograph. Symbols that are intentionally local to a chapter are defined again where they are used.

Words and weighted balance

$N_a(w)$ — Number of occurrences of the symbol a in the word w.

$L_{p,q}$ — For coprime positive integers p, q,

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

$\rho_{p,q}(w)$ — Signed residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

Thus reading 0 changes the residual by $-p$, while reading 1 changes it by $+q$.

m — The total weight $m = p + q$ when the $p{:}q$ family is under discussion.

$w_{\max}, w_{\min}$ — The larger and smaller balance weights, respectively,

$$w_{\max} = \max(p, q), \qquad w_{\min} = \min(p, q).$$

These symbols are used in the unrestricted resource theorems so that those global endpoint formulas do not overload the semantic-state symbol M of the General Theory. Earlier chapter-local uses of M remain local and are defined where they occur.

K — Primitive nonnegative kernel vector

$$K = (q, p),$$

which preserves the weighted completion equation.

$\mu(r)$ — The canonical minimum nonnegative completion-demand vector associated with residual r; when written (x, y), x counts appended zeros and y appended ones.

$\eta(x, y)$ — The canonical literal stack word associated with demand (x, y) in the relevant orientation, typically a fixed two-block representative such as $1^y 0^x$.

Original and centered machines

C, L, R — Center, left, and right states of the original 2:1 completion-aware DPDA. In the original-machine chapters their state potentials are $\Phi(C) = 0$, $\Phi(L) = 3$, and $\Phi(R) = -3$.

$D(S)$ — Original-machine logical-stack weight

$$D(S) = 2N_0(S) - N_1(S).$$

Together with the state potential it represents the prefix residual.

h_{ctr} — Centered-rail half-width

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor.$$

The subscript distinguishes this scalar from the WSC exchange vector below.

D_d — State at centered-rail coordinate d; $D_0 = C$. In the centered construction the state potential is proportional to dm.

WRITE–SEARCH–CANCEL coordinates

P — Physical count shadow of a literal stack or physical representation.

Θ — Canonical semantic demand/count target represented by a physical configuration. In the abstract General Theory the same letter denotes the semantic projection from physical configurations to semantic states.

Q — Quantity defect, typically

$$Q = \Theta - P.$$

Ω — Order defect: the part of physical noncanonicity that is invisible to Parikh counts but visible to LIFO order.

$\mathbf{h}$ — WSC exchange direction

$$\mathbf{h} = (1, -1).$$

It is typographically distinct from h_{ctr}.

M_t — Kernel-materialization count at time t in a physical WSC factorization.

General Theory

M — Semantic state space of a deterministic semantic system. This use of M is local to the General Theory and should not be confused with the WSC scalar M_t.

Σ — Input alphabet.

$U_a : M \rightharpoonup M$ — Semantic update induced by input symbol a on its defined semantic domain.

C — Physical policy/configuration domain in the abstract General Theory. This is a context-local reuse of the letter C; it is not the center state of the earlier machine chapters.

$\delta_a : C \rightharpoonup C$ — Physical settled-to-settled macro-update under input symbol a, on the policy domain where it is defined. A macro-update may include uniquely forced finite internal normalization before or after consuming a.

$\Theta : C \to M$ — Exact semantic projection satisfying domain reflection and

$$\Theta\delta_a = U_a\Theta$$

where the transitions are defined.

$\sigma : M \to C$ — Chosen canonical section, with $\Theta\sigma = \mathrm{id}_M$.

N — Canonical retraction

$$N = \sigma\Theta.$$

It satisfies $N^2 = N$ on its domain.

$\mathcal{D}_H$ — Immediate-disagreement or bad set associated with a fixed physical policy or schedule H, as defined in the relevant chapter.

$\mathcal{B}_H$ — Behavioral safe hull for policy H, equivalently the greatest forward-invariant subset that avoids eventual observable disagreement under the stated hypotheses.

$\mathrm{Adm}(\mathcal{E})$ — Admissible schedule family for a schedule system or environment $\mathcal{E}$.

Unified realization and EF–FG

$\mathrm{Ob}_h(P_h)$ — Protected-obligation compatibility complex at observation level h; its faces are source families that can share one observation cell while satisfying the declared confinement and protected-future contract.

$\mathrm{Sat}_{\mathsf{Cat}}(A, B)$ — Maximum joint protected-label rank simultaneously compilable inside one realization of the declared physical category, while preserving the two marginal protected families.

$\delta_{\mathrm{pair}}, \delta_{\mathrm{asm}}$ — Pairwise-availability and simultaneous-assembly defects in same-model transfer saturation.

$H_{c_0}(m)$ — Rooted path-outcome fiber: physical endpoints actually produced from the root by semantic words ending at m. It is generally smaller than the full representation fiber $\Theta^{-1}(m)$.

$\kappa_F(p, x)$ — EF–FG maximin attachment of a birth carrier p to observer x in scalar field F.

$\Sigma^{\pm}_{\theta}(x; F)$ — Bright/dark observer-core saliency in EF–FG; the dark quantity is the bright construction applied to $-F$.

$\mathsf{SF}_x, \mathsf{PO}_x$ — Spatial-frame and polarity-core projections. They are evaluated from the same scalar realization but neither determines the other in general.

τ — Local critical birth floor in the EF–FG threshold-band scheduling kernel; the symbol is distinct from the global $p{:}q$ balance ratio.

$\xi = \tau - D$ — EF–FG threshold-band deficit coordinate, paired with headroom $h = M - \tau$; the scalar-field observer coordinate remains x.

Rational-path enumeration

$\mathcal{D}_{m,n}$ — The set of $m \times n$ rational Dyck paths. In the rational-path chapters, m may denote a rectangle width; this local use is distinct from the earlier shorthand $m = p+q$ when that shorthand is explicitly in force.

$\mathcal{U}(P) = (u_1, \ldots, u_n)$ — The coarea sequence of a rational Dyck path, where u_i is the starting column of the ith north step.

b_i — The rational-Dyck upper coarea barrier

$$b_i = \left\lfloor \frac{m(i-1)}{n} \right\rfloor.$$

A return is an upper-barrier contact $u_i = b_i$.

run, $\widetilde{\text{run}}$, ret — Ordinary rational run, ratio-run, and return statistics. The sampled notation run_d unifies the two run specializations only where explicitly stated.

$\alpha \vDash n$ — Ordered vertical-run composition; RetComp and ReturnSet record its exact return components and positions.

Farey shortest-completion arithmetic

$\mu_{p,q}(r)$ — The unique shortest nonnegative completion vector

$$\mu_{p,q}(r) = \operatorname*{arg\,min}_{x,y \geq 0} \{x + y : px - qy = r\}.$$

This is the ratio-indexed form of the earlier canonical minimum-demand map $\mu(r)$.

$L_{p,q}(r)$ — The scalar completion length $|\mu_{p,q}(r)|_1$.

$((c,u),(v,d))$ — The oriented Farey parents of a proper coprime pair (p,q), normalized in the Farey-completion chapters by

$$(p,q) = (c,u) + (v,d), \qquad pu - qc = qv - pd = 1.$$

Scope conventions

"Optimal", "minimal", and "unique" are always read relative to the model specified in the theorem. In particular, the monograph distinguishes raw residual-faithful resource frontiers, deferred-SEARCH literal-demand width, and unrestricted DPDA recognition resources. A common terminal normal form is not, by itself, a claim of Church–Rosser confluence.

The symbol $\dashv$ denotes an explicit right endmarker. Completion callouts selected by $\dashv$ are not ordinary competing ε-moves.

Post-theory notation

$\mathcal{F}_{gq,gp}$	Balanced fixed-Parikh fiber used in the random-ordering experiment.
$\kappa_{p,q}$	Physical rematerialization constant of the block-boundary lift chain.
$\mathcal{O}_A(r)$	Optimal minimum-completion Parikh fiber for matrix system A and residual r.
$L_A(r)$	Active trade lattice generated by differences of optimal completions.
$I_A(r)$	Filled shortest-prefix quotient at residual r.
$\mathfrak{M}(P)$	Displacement-marked optimal-fiber group $\mathfrak{M}(P) = (\pi_1(Q(P)), L, \delta_P).$
$K_0(P), K_{\mathrm{lin}}(P)$	Full zero-displacement loop sector and its abelian shadow.
$\Pi_\rho(P)$	Realized pure holonomy image $\rho(K_0(P))$ for a chosen finite physical representation.
$g_k^{\mathrm{ord}}(A)$	Minimum zero-semantic literal carrier height whose Parikh vector has at least k distinct orderings.
$\chi(\mathcal{C})$	Deferred-connection terminal collision multiplicity controlling the transient path-state lower bound.
$Q_{\min}(L_{p,q})$	Minimum control-state count under the deterministic-pushdown convention stated in the relevant theorem.

Prologue — The Machine Before the Theory

This monograph begins with a machine, not with an abstract theory.

The starting problem is the binary counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

At the level of arithmetic, the condition is elementary. One can measure the imbalance between the numbers of zeros and ones by a signed linear expression. At the level of language recognition, this already suggests counter-like memory.

The research program developed in this monograph began somewhere more concrete. In 2023, a deterministic pushdown design was constructed for this language using a center state, two side states, and a stack whose contents were manipulated by cancellation and replacement rules. The machine was not originally designed as an instance of a general theory of semantic fibers, local realizability, normalization debt, or causal schedules. Those notions did not yet exist in the project.

What made the design mathematically productive was a more elementary observation: after a prefix had been processed, the machine appeared to remember not only whether the prefix was currently balanced, but also **what was still needed to make it balanced**.

That observation changed the direction of the work.

The first question was therefore not whether $L_{2,1}$ was deterministic context-free. The more useful questions were:

- Which stack contents can actually occur?
- Why do the side states matter arithmetically?
- What exactly is the cancellation operation doing?
- When the input ends away from balance, does the machine encode a continuation that repairs the deficit?
- Is that continuation shortest?
- Why does looking only a small distance into the stack reveal information about the entire store?
- Can the same structure be built for a general ratio $p{:}q$?
- If several machines realize the same counting semantics, which aspects of their design are forced and which are representational choices?

The answers do not arrive all at once. They are the subject of the book.

The development will repeatedly return to the original machine. First it will be analyzed as a concrete DPDA. Its state and stack behavior will then be reinterpreted arithmetically. The arithmetic will become a geometry of shortest completions. That geometry will lead to more efficient pushdown representations and exact resource frontiers. A resource question will then escape the original representation class and lead to

a stack-alphabet hierarchy theorem for general deterministic pushdown automata. The completion geometry will branch into higher-dimensional topology and then into finite geodesic compilation with a strict hierarchy of higher-order optimal compatibility, while one local cancellation transition will generate a separate Fuss–Catalan defect geometry and, later, an arithmetic p-adic offshoot.

After those branches, the book will return to the machine-construction problem. A uniform real-time p:q DPDA will explain the arithmetic hidden inside the original center/left/right architecture. WRITE, SEARCH, and CANCEL will then cease to be merely names for transition patterns and become components of a normalization calculus. Finally, the distinction between a unique semantic process and its nonunique physical pushdown realizations will lead to a general theory of locality, representation debt, causal schedules, synthesis, and verification.

The final chapters return once more to the 2023 design.

The aim is therefore not to replace the original machine by abstraction. It is to understand how much theory was already latent in it.

Part I

Foundations: From the 2023 Machine to Representation-Independent Questions

---

Part guide. **Question.** What semantic quantity is the original completion-aware DPDA really representing, and which resource constraints survive when its particular encoding is discarded? **Main result.** Signed residual and shortest-completion demand give the canonical quotient coordinates and exact raw frontiers, while a representation-independent hierarchy theorem resolves the fixed-state general-DPDA stack-alphabet case left open after Masopust's realtime result. **Why here.** This Part separates the historical machine from the invariant it realizes and then tests which resource information is visible in the language itself. **Dependency.** It requires only the original construction and supplies the semantic coordinates used throughout the book.

Chapter 1

The 2023 Completion-Aware DPDA

1.1 The motivating language

For a binary word w, let $N_0(w)$ and $N_1(w)$ denote the numbers of zeros and ones in w. The motivating language is

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Equivalently,

$$2N_0(w) - N_1(w) = 0.$$

It is useful to define already, without yet turning it into a general theory, the imbalance

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w).$$

Thus $\rho_{2,1}(w) = 0$ exactly when $w \in L_{2,1}$.

The original 2023 DPDA was built directly at the machine level. Its characteristic structure consists of three semantic core states:

$$C, \qquad L, \qquad R,$$

which we call the **center**, **left**, and **right** states.

The center is the balanced operational hub. The side states are not independent counters. They record a finite correction to the meaning of the physical stack. This interpretation will be proved rather than assumed.

Before doing so, we must distinguish two formal variants of the original design.

1.2 Two formal variants of the same core design

There are two variants of the 2023 machine that we will preserve throughout the monograph.

The first excludes the empty input and recognizes

$$L_{2,1} \setminus \{\varepsilon\}.$$

The second includes the empty input and recognizes the standard language

$$L_{2,1}.$$

Their center/left/right recognition logic is the same. Their difference lies in the initialization and acceptance wrapper.

This distinction is worth preserving for two reasons.

First, it records the actual development of the design rather than silently collapsing two formal choices into one picture. Second, it separates a semantic question—how the $C/L/R$ core represents imbalance—from an acceptance-convention question—whether ε is accepted by the wrapper.

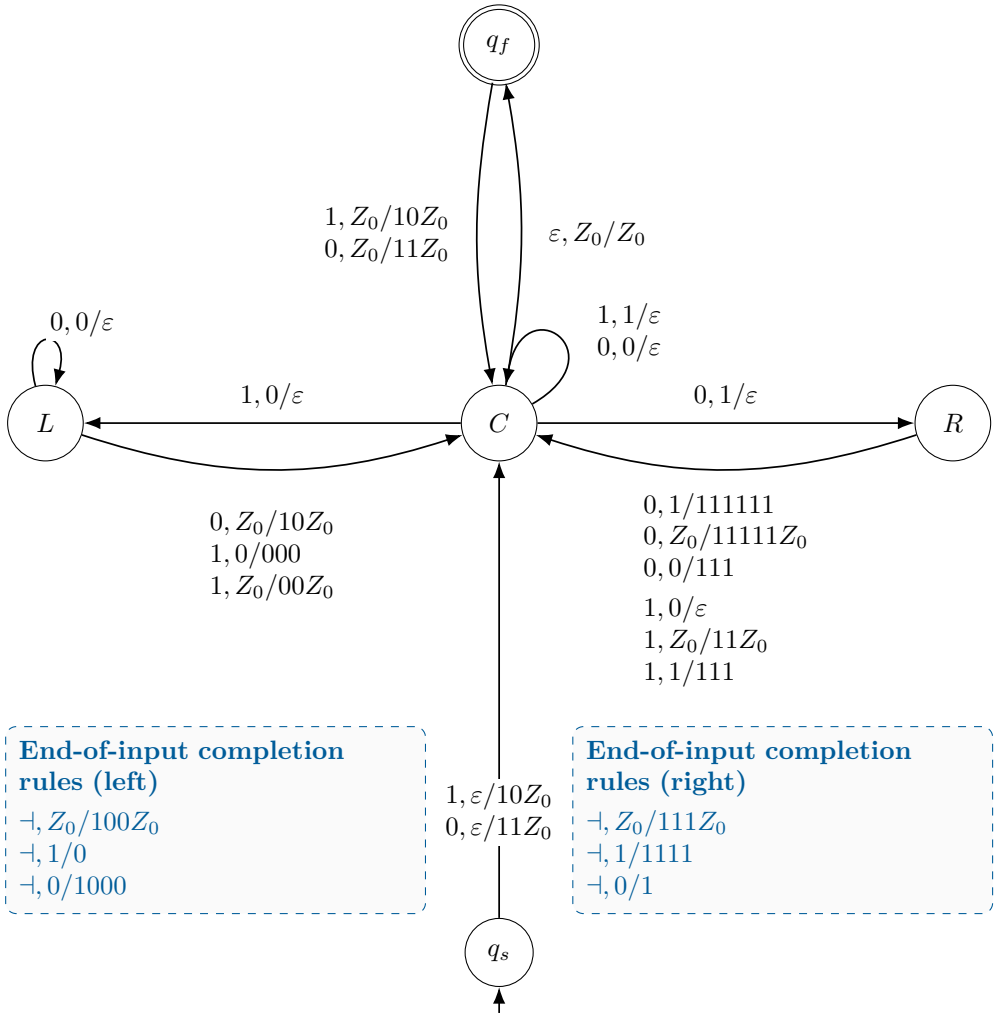


Figure 1.1: Formalized redrawing of the original 2:1 DPDA variant with the empty input excluded. Solid arrows are ordinary recognition transitions; dashed callout boxes contain only right-endmarker completion rules. The listed left-state completion rule with top 1 is retained for fidelity but has an unreachable source condition.

The logical core has states C, L, R. The wrapper contains the source state q_s and accepting state q_f.

The principal recognition transitions are the following.

From C:

$$1, 0/\varepsilon : C \to L,$$

$$0, 1/\varepsilon : C \to R,$$

and the center cancellation loops

$$1, 1/\varepsilon, \qquad 0, 0/\varepsilon.$$

The left state has the characteristic cancellation loop

$$\boxed{0, 0/\varepsilon}$$

and returns toward the center through the replacement rules

$$0, Z_0/10Z_0,$$

$$1, 0/000,$$

$$1, Z_0/00Z_0.$$

The right state returns toward the center through

$$0, 1/111111,$$

$$0, Z_0/11111Z_0,$$

$$0, 0/111,$$

$$1, 0/\varepsilon,$$

$$1, Z_0/11Z_0,$$

$$1, 1/111.$$

The wrapper moves between C and q_f at empty logical stack and initializes nonempty inputs appropriately.

This variant does not accept ε.

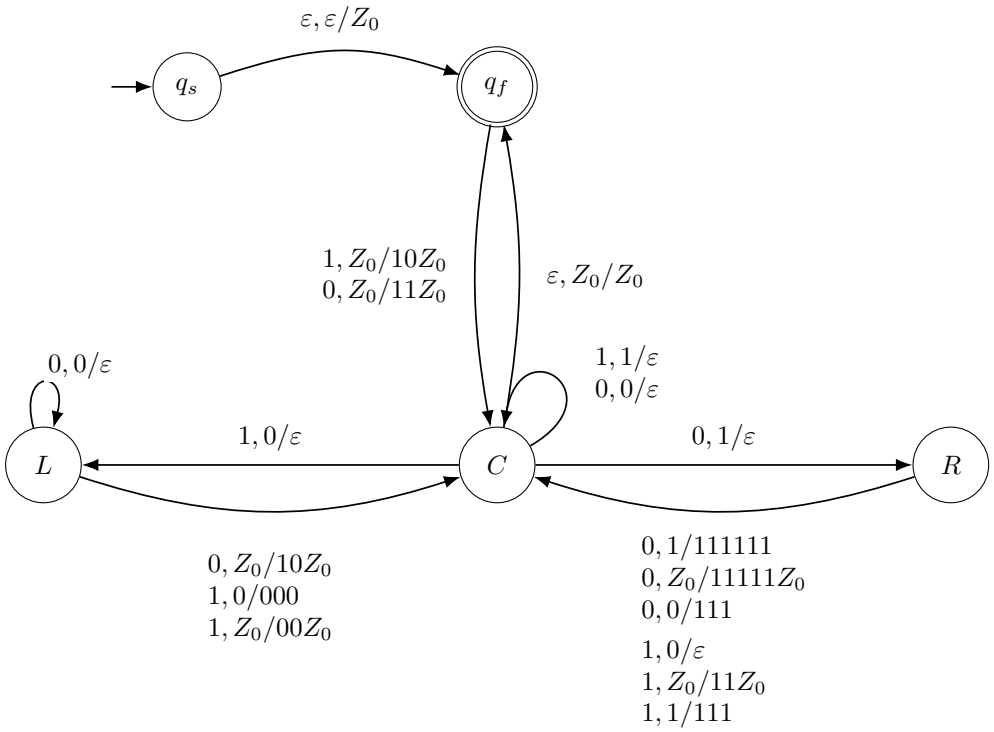


Figure 1.2: Formalized redrawing of the original 2:1 DPDA variant with the empty input included. The center/left/right recognition logic and end-of-input completion rules are identical to Figure 1.1; only the initialization/final-state wrapper changes so that ε is accepted.

The $C/L/R$ transition system is unchanged.

The difference is the initialization/final-state wrapper. In particular, an initial

$$\varepsilon, \varepsilon / Z_0$$

wrapper move allows the physical machine to represent the empty prefix at the accepting bottom-marker configuration.

Consequently this variant recognizes

$$L_{2,1},$$

including ε.

The two figures will remain separate in the final monograph, but their common recognition core will be analyzed only once.

1.3 Recognition and completion are different operations

The original design also contains rules whose purpose is not to consume another binary input symbol, but to expose the continuation represented by a nonbalanced final configuration.

These are **completion rules**.

In the formal monograph model, they are enabled only when a distinguished right endmarker

$$\dashv$$

is the next unread symbol.

For the left state, the completion list is

$$\dashv, Z_0 / 100 Z_0,$$

$$\dashv, 1/0,$$

$$\dashv, 0/1000.$$

For the right state, it is

$$\dashv, Z_0 / 111 Z_0,$$

$$\dashv, 1/1111,$$

$$\dashv, 0/1.$$

The rule with left-state top 1 is retained as part of the original completion list, but we will prove shortly that its source condition is unreachable.

The endmarker convention is essential. If these operations were interpreted as ordinary ε-moves, some state/top pairs would also have input-consuming moves, producing a conflict with standard DPDA determinism. With the right endmarker, recognition on 0 or 1 and completion on $\dashv$ are disjoint cases.

Lemma 1.1 — Endmarker determinism.

If every post-input completion rule is interpreted as a $\dashv$-labeled transition to a completion sink with no outgoing transitions, then the completion-extended machine is deterministic in the standard DPDA sense.

Proof.

Fix a state/top pair.

During binary recognition, the next input symbol is 0 or 1. A completion rule is unavailable.

After binary input has been exhausted, the next input symbol is $\dashv$. The binary recognition rules are unavailable, and at most one completion rule applies to the current state/top pair.

Thus every instantaneous configuration has at most one successor. The completion sink introduces no further conflict. □

This formalization will be used consistently in the rest of the monograph.

1.4 Removing the wrapper from the semantic analysis

The physical states q_s and q_f serve initialization and acceptance. They should not be confused with the semantic $C/L/R$ core.

Write a physical stack as

$$SZ_0, \qquad S \in \{0,1\}^*,$$

where S is the **logical stack**.

Define the core projection

$$\pi(C, SZ_0) = (C, S),$$

$$\pi(L, SZ_0) = (L, S),$$

$$\pi(R, SZ_0) = (R, S),$$

together with

$$\pi(q_s, \varepsilon) = (C, \varepsilon),$$

$$\pi(q_f, Z_0) = (C, \varepsilon).$$

Thus the empty prefix and every balanced prefix accepted at q_f receive the same zero-residual semantic core meaning.

From now on, unless the wrapper itself matters, "the configuration after a prefix" means its core-projected configuration.

1.5 The reachable stack is much more restricted than it looks

The transition table contains replacements of several different lengths. One might therefore expect the machine to generate a complicated set of stack words.

It does not.

Let

$$\mathcal{S}_C = 0^* \cup 1^* \cup \{10\},$$

$$\mathcal{S}_L = 0^*,$$

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

Theorem 1.2 — Reachable-store normal form.

For every binary prefix processed by either formal variant, if

$$(q, S)$$

is its core-projected configuration, then

$$S \in \mathcal{S}_q.$$

In particular:

- every reachable left-state stack is unary 0^k;
- every reachable right-state stack is either unary 1^k or the exceptional word 0;
- every reachable center stack is unary or the exceptional mixed word 10.

Proof.

We proceed by induction on the number of consumed input symbols.

The projected empty configuration is

$$(C, \varepsilon),$$

and

$$\varepsilon \in 0^* \cap 1^*.$$

Assume the current projected configuration belongs to the stated family.

The center loops remove matching top symbols. Hence they preserve unary forms.

A center mismatch on top 0 moves to L and removes that top. The resulting left-state stack is therefore still in 0^*.

A center mismatch on top 1 moves to R and removes that top. The resulting right-state stack lies in 1^*, with the special bottom-related construction accounting for the exceptional reachable 0 case.

The left-state loop

$$0, 0/\varepsilon$$

preserves 0^*. Its return transitions create only the allowed center forms 0^*, 1^*, or 10.

The right-state return transitions similarly produce only the allowed center families.

No recognition transition enters L with top 1. Therefore a left-state source condition with logical top 1 is unreachable.

All reachable configurations remain in the stated families. □

The theorem is stronger than a convenient regular-language description. It tells us that the apparently global question

> "Does a zero or a one occur somewhere in the stack?"

may have a bounded local answer.

Before proving that, we need to understand why the side states exist.

1.6 The state carries part of the arithmetic

Define the stack weight

$$D(S) = 2N_0(S) - N_1(S).$$

Define the state potential

$$\Phi(C) = 0,$$

$$\Phi(L) = 3,$$

$$\Phi(R) = -3.$$

Theorem 1.3 — Original-machine state-potential invariant.

After every consumed prefix w, let

$$(q, S)$$

be the core-projected configuration. Then

$$\boxed{\rho_{2,1}(w) = D(S) + \Phi(q).}$$

Proof.

For the empty prefix,

$$\rho_{2,1}(\varepsilon) = 0,$$

and the projected configuration is

$$(C, \varepsilon),$$

so

$$D(\varepsilon) + \Phi(C) = 0.$$

It remains to compare the change of the two sides under each recognition transition.

Reading 0 decreases $\rho_{2,1}$ by 2. Reading 1 increases it by 1.

Each stack replacement changes $D(S)$, and every state change changes Φ. The transition table is constructed so that

$$\Delta\big(D(S) + \Phi(q)\big) = \Delta\rho_{2,1}.$$

For the transition that will later generate the cancellation geometry,

$$(L, 0S) \xrightarrow{0} (L, S),$$

the stack weight decreases by 2, while the state potential remains 3. Thus the right-hand side changes by -2, exactly matching the change induced by consuming the relevant symbol under the selected residual sign convention.

The remaining transition types satisfy the same identity by direct finite verification. The complete transition-by-transition audit is recorded in Appendix A. □

Interpretation

The logical stack alone does **not** always equal the current arithmetic deficit.

The side state stores a finite correction:

$$+3 \quad \text{in } L,$$

$$-3 \quad \text{in } R.$$

This explains a configuration such as

$$(L, \varepsilon).$$

Its empty logical stack does not mean zero residual. The missing value is carried by the state.

This is the first place where the machine reveals a distinction that will become much more important later:

$$\boxed{\text{semantic information} = \text{stack information} + \text{finite-control information.}}$$

At this point, however, we keep the observation concrete.

1.7 End-of-input completion transfers the state correction into the stack

Suppose the binary input has been exhausted.

If the projected configuration is already in C, no side-state correction is required.

If it is in L or R, the unique endmarker completion rule converts the side-state potential into a normalized stack representation.

The result has an exceptionally simple form.

Theorem 1.4 — Original-machine shortest-completion extraction.

After end-of-input normalization, the logical stack is

$$\widehat{S}(w) = \begin{cases} 0^m, & \rho_{2,1}(w) = 2m, \quad m \geq 0, \\ 10^{m+1}, & \rho_{2,1}(w) = 2m+1, \quad m \geq 0, \\ 1^m, & \rho_{2,1}(w) = -m, \quad m \geq 1. \end{cases}$$

When interpreted as a continuation word, $\widehat{S}(w)$ is a shortest suffix that completes w to $L_{2,1}$.

Proof.

By Theorems 1.2 and 1.3, every side-state configuration belongs to a one-parameter stack family and differs from its stack weight by exactly the fixed state potential.

The endmarker completion rule removes that finite state correction and returns the representation to the zero-potential center convention.

Consequently, after normalization,

$$D(\widehat{S}(w)) = \rho_{2,1}(w).$$

The reachable normalized center family is

$$0^* \cup 1^* \cup \{10^+\},$$

so the displayed residual-to-stack map is injective.

Now let a completion contain x zeros and y ones. The completed word must satisfy

$$N_1(w) + y = 2\big(N_0(w) + x\big).$$

Since

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

this is equivalent to

$$\boxed{2x - y = \rho_{2,1}(w).}$$

The three displayed normalized forms give the unique minimum nonnegative Parikh pair for this equation. Hence the continuation selected by the normalized stack has minimum possible length. □

The important discovery is independent of notation:

the machine's end configuration can be decoded as a shortest repair word.

This is the observation from which the completion theory begins.

1.8 A worked run: the prefix 111

Consider the empty-input-excluded presentation and ignore Z_0 in the logical-stack notation.

Starting in the center:

$$(C, 111\ \dashv, \varepsilon) \vdash (C, 11\ \dashv, 10) \vdash (C, 1\ \dashv, 0) \vdash (L, \dashv, \varepsilon).$$

The consumed prefix has

$$\rho_{2,1}(111) = 3.$$

The physical logical stack is empty, but the machine is in L, whose potential carries the missing correction.

The endmarker completion rule

$$\dashv, Z_0/100Z_0$$

materializes the continuation

$$100.$$

Indeed,

$$111100$$

contains four ones and two zeros, so

$$111100 \in L_{2,1}.$$

No shorter continuation can repair the count imbalance.

Thus a side state that initially looks like an implementation detail has a precise role: it stores a finite amount of the completion demand outside the stack until the end-of-input normalizer materializes it.

1.9 The first structural surprise: two stack cells are enough

For a word v, let

$$\mathrm{alph}(v) = \{a \in \{0,1\} : a \text{ occurs in } v\}.$$

Let

$$\mathrm{pref}_d(v)$$

denote the prefix of v of length $\min(d, |v|)$.

Theorem 1.5 — Tight depth-two visibility.

For every completion-normalized stack S,

$$\boxed{\mathrm{alph}(S) = \mathrm{alph}(\mathrm{pref}_2(S)).}$$

Depth two is necessary.

Proof.

By Theorem 1.4, every normalized stack belongs to one of the families

$$0^*, \qquad 1^*, \qquad 10^+.$$

Every symbol that occurs anywhere in one of those words therefore occurs within the first two positions.

Depth one does not suffice. The normalized words

$$1$$

and

$$10$$

have the same top symbol but different global symbol supports.

Hence the exact visibility depth is two. □

This theorem formalizes an intuition present in the original design: a question that appears to require searching the entire stack can, on the reachable normalized family, be answered from only the first two cells.

The reason is not that arbitrary pushdown stacks are locally transparent. It is that the machine's reachable stores occupy a highly restricted language.

That distinction will matter repeatedly later.

1.10 The cancellation transition that will reappear much later

One transition deserves to be marked now even though its deeper significance will not be developed until the later cancellation-geometry branch:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

At the machine level it is simply a cancellation operation.

At the arithmetic level it participates in the state-potential invariant.

Much later, when prefixes are represented as nonnegative 2-Dyck paths, the same transition will be characterized by an exact family of landing heights. That observation

will generate correction signatures, an intrinsic defect, a Catalan boundary, and a binary cancellation skeleton.

We do not need any of that theory yet.

For now we record only the transition and its exact place in the original machine.

1.11 What has been learned without leaving the original machine

We can now answer several questions that were not visible from the transition diagram alone.

First, the reachable stack language is extremely small.

Second, the side states carry an exact finite arithmetic potential.

Third, the post-input completion rules can be formalized deterministically with a right endmarker.

Fourth, the final state/stack configuration can be normalized into a shortest continuation.

Fifth, global symbol support of the normalized store is exactly visible at depth two.

These facts suggest that the original DPDA is not merely storing an arbitrary history of the input. Its configuration is representing a much smaller mathematical object.

The next chapter identifies that object.

The progression will be:

$$\begin{array}{|c|}\hline \text{physical state and stack} \longrightarrow \text{residual} \\ \longrightarrow \text{minimum completion demand} \longrightarrow \text{directed completion geometry.} \\ \hline \end{array}$$

Only after that geometry is understood will we return to the engineering question:

> Can the representation itself be improved?

That question leads to the improved and scoped-optimal DPDA constructions, the exact resource frontiers, and the general $p{:}q$ architectures developed below.

Chapter 2

What the Machine Is Really Storing

Chapter 1 stayed deliberately close to the 2023 transition diagram. We saw that the reachable stores are highly restricted, that the side states carry a finite arithmetic potential, and that an end-of-input normalization converts a nonbalanced configuration into a shortest continuation.

Those observations suggest a simpler object behind the physical machine.

The purpose of this chapter is to identify it.

The central distinction will be:

> physical configuration $\longrightarrow$ arithmetic residual
> $\longrightarrow$ minimum completion demand.

The first object depends on a particular DPDA representation. The second and third do not.

That separation is the first decisive abstraction in the monograph.

2.1 From the 2:1 imbalance to the general p:q residual

The original language satisfies

$$N_1(w) = 2N_0(w).$$

The natural general family is

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\},$$

where

$$p, q \geq 1, \qquad \gcd(p,q) = 1.$$

The coprimality assumption removes a redundant common scale. Nothing about the language changes if p and q are divided by their greatest common divisor.

Define the residual

$$\boxed{\rho_{p,q}(w) = qN_1(w) - pN_0(w).}$$

Thus

$$w \in L_{p,q} \iff \rho_{p,q}(w) = 0.$$

Reading a symbol changes the residual by a fixed amount:

$$\rho_{p,q}(w0) = \rho_{p,q}(w) - p,$$

$$\rho_{p,q}(w1) = \rho_{p,q}(w) + q.$$

For the motivating ratio $(p, q) = (2, 1)$,

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

exactly the convention used in Chapter 1.

The residual is already a substantial simplification. Instead of retaining the whole Parikh pair

$$(N_0(w), N_1(w)),$$

we retain one integer.

But the stronger observation is that this one integer determines exactly what is needed on the right to reach the language.

2.2 The completion equation

Let w be a prefix with residual

$$r = \rho_{p,q}(w).$$

Suppose a continuation z contains

$$x = N_0(z)$$

zeros and

$$y = N_1(z)$$

ones.

Then

$$wz \in L_{p,q}$$

exactly when

$$q\big(N_1(w) + y\big) = p\big(N_0(w) + x\big).$$

Rearranging gives the basic completion equation.

Lemma 2.1 — Completion equation.

For every prefix w and continuation z with Parikh pair (x, y),

$$\boxed{z \in w^{-1}L_{p,q} \iff px - qy = \rho_{p,q}(w).}$$

Proof.

Expand the defining equality

$$qN_1(wz) = pN_0(wz).$$

Since

$$N_0(wz) = N_0(w) + x, \qquad N_1(wz) = N_1(w) + y,$$

we obtain

$$qN_1(w) + qy = pN_0(w) + px.$$

Therefore

$$px - qy = qN_1(w) - pN_0(w) = \rho_{p,q}(w).$$

The steps are reversible. □
This equation is the semantic heart of the original completion mechanism.
For $p = 2, q = 1$, it becomes

$$2x - y = r.$$

Thus the Chapter 1 example

$$w = 111, \qquad r = 3,$$

requires

$$2x - y = 3.$$

The minimum nonnegative solution is

$$(x, y) = (2, 1).$$

Hence every shortest completion has two zeros and one one.
There are three such words:

$$100, \qquad 010, \qquad 001.$$

The original DPDA selects 100.
This distinction will be important throughout the book:

minimum Parikh demand is unique; a shortest word need not be unique.

2.3 The primitive balanced cycle

The completion equation

$$px - qy = r$$

has an elementary but fundamental symmetry.
If (x, y) is a solution, then

$$(x + q,\ y + p)$$

is another solution, because

$$p(x + q) - q(y + p) = px - qy.$$

The vector

$$\boxed{K = (q, p)}$$

is therefore the primitive nonnegative kernel direction.
Its meaning is transparent:

- adding q zeros contributes pq;
- adding p ones contributes $-pq$;
- the two contributions cancel.

At the word level, any word containing exactly q zeros and p ones has net residual change zero.
Its length is

$$\boxed{p + q.}$$

Later this will become the fundamental directed cycle length of the residual graph.
At this stage we need only its Diophantine role.

2.4 Every completion lies on one affine ray

Because $\gcd(p, q) = 1$, the integer solutions of

$$px - qy = r$$

form a one-dimensional affine lattice.

Proposition 2.2 — Affine completion ray.
For every

$$r \in \mathbb{Z},$$

there is a unique nonnegative solution

$$\mu(r) = (x_r, y_r)$$

minimizing

$$x + y$$

subject to

$$px - qy = r.$$

Every nonnegative solution is

$$\boxed{(x_r + qk,\ y_r + pk), \qquad k \in \mathbb{N}_0.}$$

Consequently the possible completion lengths are exactly

$$\boxed{\Delta(r) + (p + q)k, \qquad k \in \mathbb{N}_0,}$$

where

$$\Delta(r) = x_r + y_r.$$

Proof.

Since p and q are coprime, Bézout's identity guarantees an integer solution

$$(x_0, y_0)$$

to

$$px - qy = r.$$

Every integer solution has the form

$$(x_0 + qk,\ y_0 + pk), \qquad k \in \mathbb{Z}.$$

For sufficiently large k, both coordinates are nonnegative.
Let k_* be the least integer for which this happens, and define

$$(x_r, y_r) = (x_0 + qk_*,\ y_0 + pk_*).$$

Every other nonnegative solution is obtained by replacing k_* with

$$k_* + j, \qquad j \in \mathbb{N}_0.$$

Therefore every other nonnegative solution adds

$$j(q, p).$$

Its length increases by

$$j(p + q).$$

Hence (x_r, y_r) is the unique minimum-length nonnegative Parikh pair. □
This proposition gives the first exact picture of the completion space:

$$\boxed{\mu(r), \quad \mu(r) + K, \quad \mu(r) + 2K, \quad \mu(r) + 3K, \ldots}$$

The minimum demand is the first lattice point of an affine ray inside the nonnegative quadrant.

The later geometry of the monograph begins here.

2.5 Minimum completion demand

We now give the central object a name.

Definition 2.3 — Minimum completion demand.

For residual

$$r \in \mathbb{Z},$$

let

$$\boxed{\mu_{p,q}(r) = (x_r, y_r)}$$

be the unique nonnegative solution of

$$px - qy = r$$

minimizing

$$x + y.$$

Define the minimum completion length

$$\boxed{\Delta_{p,q}(r) = x_r + y_r.}$$

When p, q are fixed, we write simply

$$\mu(r), \qquad \Delta(r).$$

The word **demand** is intentional.
The pair

$$(x_r, y_r)$$

does not describe the prefix already read. It describes what the prefix still requires:

- x_r additional zeros;
- y_r additional ones.

The physical DPDA need not store those two integers literally.
The semantic claim is only that a configuration representing residual r implicitly represents the unique pair $\mu(r)$.
For $L_{2,1}$,

$$2x - y = r.$$

The minimum demand has the closed form

$$\mu(2m) = (m, 0), \qquad m \geq 0,$$

$$\mu(2m + 1) = (m + 1, 1), \qquad m \geq 0,$$

and

$$\mu(-m) = (0, m), \qquad m \geq 1.$$

These are exactly the three families already materialized by the original 2023 DPDA:

$$0^m, \qquad 10^{m+1}, \qquad 1^m.$$

So the stack forms discovered operationally in Chapter 1 are not arbitrary normal forms.
They are physical encodings of $\mu(r)$.

2.6 A canonical shortest word

The pair

$$\mu(r) = (x_r, y_r)$$

is unique, but its letters can usually be ordered in several ways.
For semantic decoding it is convenient to choose one fixed representative.
The minimum pair has a simple modular description.

Theorem 2.4 — Canonical two-block shortest completion.

Let $r > 0$.
There is a unique

$$y_r \in \{0, \dots, p-1\}$$

such that

$$qy_r \equiv -r \pmod{p}.$$

Set

$$x_r = \frac{r + qy_r}{p}.$$

Then

$$\boxed{C(r) = 1^{y_r} 0^{x_r}}$$

is a shortest completion word.
For $r < 0$, let

$$x_r \in \{0, \dots, q-1\}$$

be the unique solution of

$$px_r \equiv r \pmod{q}.$$

Set

$$y_r = \frac{px_r - r}{q}.$$

Then

$$\boxed{C(r) = 0^{x_r} 1^{y_r}}$$

is a shortest completion.
Finally,

$$C(0) = \varepsilon.$$

Proof.

Suppose first that $r > 0$.
The congruence

$$qy \equiv -r \pmod{p}$$

has a unique solution modulo p because q is invertible modulo p.
Selecting the representative

$$0 \le y_r < p$$

makes

$$x_r = \frac{r + qy_r}{p}$$

a nonnegative integer.

Any other integer solution differs by the kernel vector

$$(q, p).$$

Subtracting one copy of K would make the bounded coordinate y_r negative. Therefore this is the first nonnegative point on the affine completion ray and hence the unique minimum Parikh pair.

The argument for $r < 0$ is symmetric, now bounding the zero coordinate modulo q.

The displayed block word is simply a fixed ordering of the unique minimum pair. □

The canonical word is a **decoder convention**, not a claim that other shortest orderings are invalid.

For example, for

$$p = 3, \qquad q = 2, \qquad r = 4,$$

we solve

$$2y \equiv -4 \pmod 3.$$

Thus

$$y = 1, \qquad x = \frac{4 + 2}{3} = 2.$$

Hence

$$\mu(4) = (2, 1)$$

and

$$C(4) = 100.$$

The other shortest words with the same Parikh pair are equally valid as completions. The canonical map simply chooses one of them reproducibly.

2.7 The original DPDA implements this decoder at 2:1

We can now restate the Chapter 1 completion theorem in semantic language.

Let

$$c = (q, S)$$

be any reachable core configuration of the original 2:1 DPDA.

Its residual is decoded by

$$\boxed{\Theta_{\text{orig}}(c) = D(S) + \Phi(q).}$$

Thus

$$\Theta_{\text{orig}}(c) = \rho_{2,1}(w)$$

for the prefix w that reached c.

The endmarker normalizer removes the side-state correction and produces

$$\widehat{S}(c).$$

Chapter 1 proved that

$$\widehat{S}(c) = \begin{cases} 0^m, & r = 2m, \\ 10^{m+1}, & r = 2m+1, \\ 1^m, & r = -m, \end{cases}$$

where

$$r = \Theta_{\text{orig}}(c).$$

But these are exactly the canonical two-block words

$$C(r)$$

for

$$p = 2, \qquad q = 1.$$

Therefore the original machine realizes the diagram

$$\boxed{c \overset{\Theta_{\text{orig}}}{\longmapsto} r \overset{\mu}{\longmapsto} (x_r, y_r) \overset{C}{\longmapsto} C(r).}$$

The machine is not being claimed to be an output transducer.

Rather, its configuration is a mathematical certificate from which the shortest canonical continuation can be decoded.

This distinction between **stored information** and **explicit output** will remain important later.

2.8 Physical information versus semantic information

We can now answer the question posed at the end of Chapter 1.

What is the machine really storing?

Not the entire prefix.

Not even the complete Parikh pair of the prefix.

At the semantic level, it stores only the residual

$$r.$$

From r, the minimum completion demand

$$\mu(r)$$

is uniquely determined.

From $\mu(r)$, the canonical shortest continuation

$$C(r)$$

is determined by convention.

This gives three layers.

Layer 1 — Physical representation

A concrete configuration such as

$$(L, 0^k),$$

$$(R, 1^k),$$

or

$$(C, 10)$$

belongs to the chosen 2023 machine.

Layer 2 — Semantic residual

The decoder

$$\Theta_{\text{orig}}$$

maps that configuration to one integer

$$r.$$

Layer 3 — Completion semantics

The residual determines

$$\mu(r), \qquad \Delta(r), \qquad C(r).$$

The crucial fact is that Layer 1 is not unique in principle.

Later chapters construct different physical DPDAs representing the same Layer 2 and Layer 3 objects.

That is the beginning of the distinction between semantics and representation.

The exact-lift, section, and semantic-fiber formalism is developed later in the book.

For now it is enough to observe that a complicated state/stack configuration may be merely a coordinate system for a much simpler arithmetic state.

2.9 The residual transition system

Because the residual changes by fixed amounts,

$$r \xrightarrow{0} r - p,$$

$$r \xrightarrow{1} r + q.$$

Thus every binary prefix traces a path on the integer line.

For the motivating language,

$$r \xrightarrow{0} r - 2,$$

$$r \xrightarrow{1} r + 1.$$

This graph is independent of the original DPDA's stack representation.

The 2023 machine is one physical realization of motion over this arithmetic system.
That observation immediately suggests a new question:

> If the semantic state is only one integer, why does the physical machine need the particular stack words and side states that it uses?

There are two very different possible answers.
One possibility is that the physical complexity is largely accidental and can be reduced by choosing better coordinates.
The other is that LIFO locality forces some of it.
Both possibilities will eventually occur.
Before studying resources, however, we need to understand the shortest paths in the residual graph.

2.10 Completion length as distance

The quantity

$$\Delta(r) = x_r + y_r$$

has an immediate graph interpretation.
It is the length of a shortest directed path from residual r to residual 0.
Indeed:

- appending a zero follows the edge
$$r \to r - p;$$
- appending a one follows
$$r \to r + q;$$
- a continuation reaches the language exactly when the resulting residual is zero.

Therefore

$$\boxed{\Delta(r) = \operatorname{dist}^{\to}(r, 0).}$$

We will prove and exploit this metric statement systematically in later chapters.
Here it already changes the interpretation of the minimum demand.
The pair

$$\mu(r) = (x_r, y_r)$$

does not merely count missing symbols.
It counts the two types of edges used by a shortest path to zero.
The canonical completion

$$C(r)$$

selects one distinguished shortest path among all orderings of those edges.
Thus the completion problem has become a geodesic problem.

2.11 How many shortest completions are there?

Because every shortest completion has the same Parikh pair

$$(x_r, y_r),$$

the number of shortest completion words is

$$\boxed{\binom{x_r + y_r}{x_r}.}$$

This simple count is the first indication that shortest completion has an internal geometry.

If both coordinates are positive, there are several ways to interleave the required zeros and ones.

For the prefix 111,

$$\mu(3) = (2, 1),$$

so there are

$$\binom{3}{1} = 3$$

shortest completions.

The original machine chooses the boundary ordering

$$100.$$

Later we will see that different shortest orderings are connected by elementary commuting moves, and the left branch of the original machine realizes a complete ladder of such shortest schedules at configuration level.

The multiplicity of shortest words is therefore not noise.

It is the first object that turns completion arithmetic into geometry.

2.12 Why the minimum pair is a better semantic object than the canonical word

It would be possible to define the semantics directly by the canonical word

$$C(r).$$

For the structural developments that follow, however, the pair

$$\mu(r) = (x_r, y_r)$$

is more fundamental.

There are several reasons.

First, the completion equation is linear in (x, y).

Second, the kernel direction

$$K = (q, p)$$

acts naturally on Parikh pairs.

Third, shortest-path descent becomes coordinate deletion:

- consuming a geodesic zero removes one unit of the zero demand;
- consuming a geodesic one removes one unit of the one demand.

Fourth, later physical representations will use different word orderings while preserving the same minimum pair.

Finally, in higher-dimensional counting languages the minimum demand will cease to be a single pair and become a fiber of optimal Parikh vectors. That generalization is visible only if we separate the Parikh object from its chosen word ordering now.

Thus our semantic hierarchy is

$$\boxed{r \quad \longrightarrow \quad \mu(r) \quad \longrightarrow \quad C(r),}$$

with the residual and minimum demand carrying the mathematics, and the canonical word supplying a convenient physical or decoding coordinate.

2.13 A first exact notion of redundancy

Suppose two different prefixes u and v have the same residual:

$$\rho_{p,q}(u) = \rho_{p,q}(v) = r.$$

Then both prefixes have exactly the same completion equation

$$px - qy = r.$$

Consequently they have:

- the same minimum demand;
- the same minimum completion length;
- the same set of feasible completion Parikh pairs;
- the same canonical shortest continuation.

Already, therefore, their past histories have become irrelevant to the completion problem.

This foreshadows the quotient theorem proved later:

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho_{p,q}(u) = \rho_{p,q}(v).$$

We postpone the full quotient interpretation to Chapter 4, where it will be connected to directed geometry.

For the present chapter, the important message is simpler:

the residual is a sufficient statistic for all right-completion arithmetic.

2.14 The original machine viewed through the new semantics

We can now revisit several Chapter 1 observations without introducing any new transition rule.

The side states

The state potential means that a side state stores part of the residual outside the logical stack.

For example,

$$(L, \varepsilon)$$

represents

$$r = 3.$$

This is why the stack may be physically empty even though the completion demand is nonzero.

Endmarker materialization

The completion rule does not create new semantic information.

It changes coordinates.

Before the endmarker, the residual may be split between:

$$\text{stack weight} + \text{state potential}.$$

After normalization, the state potential is zero and the entire minimum demand is visible in the canonical stack word.

Tight depth-two visibility

The normalized words

$$0^*, \qquad 1^*, \qquad 10^+$$

are exactly the 2:1 canonical completion words.

Their global support is visible in the first two positions because the canonical decoder has only these three structural forms.

Thus the depth-two theorem is not an isolated stack-language coincidence.

It is a property of the chosen canonical ordering of the minimum demand.

This distinction will later become crucial when we compare alternative physical representations.

2.15 The first major conceptual compression

The entire original configuration can now be summarized by the following commuting picture:

$$\begin{array}{ccc} \text{reachable DPDA configuration} & \xrightarrow{\Theta_{\mathrm{orig}}} & r \in \mathbb{Z} \\ & & \downarrow \mu \\ & & (x_r, y_r) \\ & & \downarrow C \\ & & \text{canonical shortest continuation} \end{array}$$

The physical DPDA is rich.

The semantic state is one-dimensional.

The completion demand is exact.

This is the first compression of the research program.

It also exposes the next structural question.

If

$$\Delta(r)$$

is the shortest distance to balance, what does a single input symbol do to that distance?

When does reading a zero move one step closer to acceptance?

When does reading a one do so?

Can both actions be shortest simultaneously?

And do the original DPDA's cancellation and completion moves physically realize those shortest paths?

Those questions lead directly to the **minimum-demand geodesic descent** and the **completion ladder**.

That is the subject of Chapter 3.

Chapter 3

Minimum Demand and the Completion Ladder

Research branch — Completion Becomes Geometry. *This branch develops the configuration ladder, minimum-demand geometry, and rectangular shortest-completion theory from the exact structural analysis of the original machine and its quotient/completion semantics. The use of shortest-path structure in pushdown graphs has prior literature; the claim here is the exact arithmetic specialization and configuration-level geometry of this family [5, 6].*

Chapter 2 replaced the original machine's physical configuration by a simpler semantic object:

$$r \longmapsto \mu(r) = (x_r, y_r).$$

For the motivating language,

$$L_{2,1} = \{w : N_1(w) = 2N_0(w)\},$$

the residual evolves by

$$r \xrightarrow{0} r - 2, \qquad r \xrightarrow{1} r + 1.$$

The minimum completion demand tells us how many zeros and ones are still required. We now ask what happens to that demand when another input symbol is read.

The answer reveals that the original DPDA is traversing shortest paths.

3.1 Geodesic input symbols

Fix

$$p = 2, \qquad q = 1.$$

For residual r, write

$$\mu(r) = (x_r, y_r),$$

and

$$\Delta(r) = x_r + y_r.$$

Recall:

$$\mu(2m) = (m, 0),$$

$$\mu(2m+1) = (m+1, 1),$$

and

$$\mu(-m) = (0, m).$$

Define the residual updates

$$T_0(r) = r - 2,$$

$$T_1(r) = r + 1.$$

A symbol $a \in \{0, 1\}$ will be called **geodesic at residual** r if reading it decreases the remaining shortest-completion distance by exactly one:

$$\boxed{\Delta(T_a(r)) = \Delta(r) - 1.}$$

This is the semantic form of the question:

> Does this next input symbol perform one unit of the completion work that was already required?

Lemma 3.1 — Minimum-demand geodesic descent.

Let

$$\mu(r) = (x_r, y_r).$$

If

$$x_r > 0,$$

then

$$\boxed{\mu(r-2) = (x_r - 1, y_r)}$$

and

$$\Delta(r-2) = \Delta(r) - 1.$$

If

$$y_r > 0,$$

then

$$\boxed{\mu(r+1) = (x_r, y_r - 1)}$$

and

$$\Delta(r+1) = \Delta(r) - 1.$$

Therefore the set of geodesic first symbols is

$$\boxed{\mathrm{GeoFirst}(r) = \{0 : x_r > 0\} \cup \{1 : y_r > 0\}.}$$

Proof.

Suppose

$$x_r > 0.$$

Since

$$2x_r - y_r = r,$$

we have

$$2(x_r - 1) - y_r = r - 2.$$

Thus

$$(x_r - 1, y_r)$$

is a completion demand for residual $r - 2$.

If $r - 2$ had a completion pair of length strictly less than

$$x_r + y_r - 1,$$

adding one zero to that pair would produce a completion of r shorter than

$$\mu(r),$$

contradicting minimality.

Hence

$$\mu(r - 2) = (x_r - 1, y_r).$$

The argument for $y_r > 0$ is symmetric:

$$2x_r - (y_r - 1) = r + 1.$$

Again, any shorter completion after reading 1 would yield a shorter completion before reading it.

Thus precisely those symbols whose coordinate in $\mu(r)$ is positive decrease the minimum distance by one. □

This lemma has a useful interpretation.

If the minimum demand is

$$(x, y),$$

then every geodesic input symbol simply deletes one required unit:

$$0 : (x, y) \mapsto (x - 1, y),$$

$$1 : (x, y) \mapsto (x, y - 1).$$

At the semantic level, shortest completion is therefore almost trivial.

The interesting question is whether the physical DPDA realizes the same simple coordinate deletion.

3.2 The left branch as an exact shortest-path process

Consider the reachable left configurations

$$c_k = (L, 0^k), \qquad k \geq 0.$$

From the state-potential invariant,

$$\rho(c_k) = 2k + 3.$$

Therefore

$$\boxed{\mu(c_k) = (k + 2, 1)}$$

and

$$\boxed{\Delta(c_k) = k + 3.}$$

So every such configuration requires:

- $k + 2$ zeros;
- one one.

The left cancellation transition

$$(L, 0^k) \xrightarrow{0} (L, 0^{k-1})$$

does exactly what Lemma 3.1 predicts:

$$(k + 2, 1) \mapsto (k + 1, 1).$$

It decreases the shortest-completion distance by exactly one.
This gives the original cancellation loop a new meaning.
It is not merely deleting matching physical stack symbols.
It is executing one edge of a shortest path in completion space.
The boundary sequence

$$(L, \varepsilon) \xrightarrow{0} (C, 10) \xrightarrow{0} (R, 0) \xrightarrow{1} (C, \varepsilon)$$

continues the same exact descent.
Thus the entire route from a left configuration to balance can be read geodesically.

3.3 The ladder sources are actually reachable

The configurations

$$(L, 0^k)$$

used in the next theorem are not hypothetical legal stack patterns.
They are reachable.
The prefix

111

reaches

$$(L, \varepsilon).$$

Two additional ones increase the left unary tail by one unit through the machine's center-side transition pattern.

Inductively,

$$\boxed{1^{2k+3} \text{ reaches } (L, 0^k).}$$

This reachability matters.

The geometry we are about to describe is therefore a geometry of actual machine computations, not merely of syntactically possible configurations.

3.4 Every position of the required one is shortest

From

$$(L, 0^k),$$

the minimum demand is

$$(k+2, 1).$$

Hence every shortest continuation contains exactly one 1 and $k+2$ zeros.

There are

$$k+3$$

possible positions for that unique one:

$$1\,0^{k+2},$$

$$0\,1\,0^{k+1},$$

$$0^2\,1\,0^k,$$

$$\ldots,$$

$$0^{k+2}1.$$

The remarkable fact is that the original DPDA realizes **every one of these shortest schedules**, and it realizes them in a rectangular ladder of configurations.

Theorem 3.2 — Exact geodesic ladder.

For every

$$k \geq 0,$$

define the lower row

$$\Psi_k(i,0) = \begin{cases} (L, 0^{k-i}), & 0 \leq i \leq k, \\ (C, 10), & i = k+1, \\ (R, 0), & i = k+2, \end{cases}$$

and the upper row

$$\Psi_k(i,1) = (C, 0^{k+2-i}), \qquad 0 \le i \le k+2.$$

Then:

1. every horizontal ladder edge is a solid input-0 transition;
2. every vertical ladder edge is a solid input-1 transition;
3. every monotone path from
$$(L, 0^k)$$
to
$$(C, \varepsilon)$$
consumes the minimum demand
$$(k+2, 1);$$
4. therefore every such path is shortest;
5. the shortest continuation words are exactly
$$\boxed{0^i 1\, 0^{k+2-i}, \qquad 0 \le i \le k+2.}$$

There are exactly

$$\boxed{k+3}$$

shortest continuations.

Proof.

Along the lower row, while a zero remains on the left stack, the horizontal edge is the cancellation loop

$$(L, 0^m) \xrightarrow{0} (L, 0^{m-1}).$$

At the lower boundary,

$$(L, \varepsilon) \xrightarrow{0} (C, 10),$$

followed by

$$(C, 10) \xrightarrow{0} (R, 0).$$

Along the upper row, each horizontal edge is a center cancellation

$$(C, 0^m) \xrightarrow{0} (C, 0^{m-1}).$$

The vertical edges consume the unique required 1.

For an interior left configuration,

$$(L, 0^m) \xrightarrow{1} (C, 0^{m+2}).$$

At the lower boundary,

$$(L, \varepsilon) \xrightarrow{1} (C, 00).$$

At

$$(C, 10),$$

the input 1 removes the visible 1 and reaches

$$(C, 0).$$

At

$$(R, 0),$$

the input 1 reaches

$$(C, \varepsilon).$$

Thus the entire two-row grid is physically realized by the DPDA.
Every source-to-target monotone path contains exactly:

$$k + 2$$

horizontal zero edges and one vertical one edge.
Hence every path consumes the unique minimum Parikh demand

$$(k + 2, 1).$$

By Chapter 2, no shorter continuation exists.
The position of the vertical edge determines the position of the unique 1 in the continuation, giving exactly

$$0^i 1\, 0^{k+2-i}, \qquad 0 \le i \le k + 2.$$

□

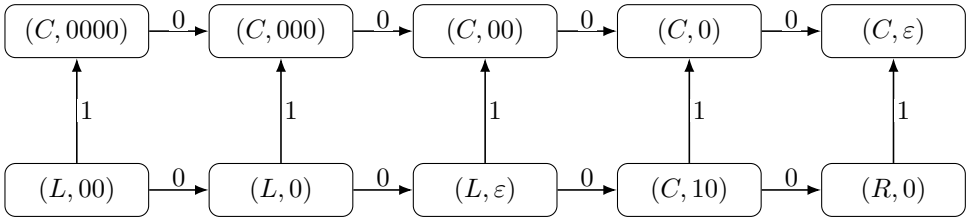


Figure 3.1: Exact configuration ladder for $k = 2$. The upper boundary is canonical completion; the lower boundary postpones the unique required 1 as long as possible.

The source is

$$(L, 00),$$

with residual

$$7$$

and minimum demand

$$(4, 1).$$

Its five shortest completions are

$$10000,$$

$$01000,$$

00100,

00010,

00001.

The upper route materializes the required 1 as early as possible.
The lower route postpones it as long as possible.
Both are shortest.
Everything between them is shortest as well.

3.5 Local commuting diamonds

A ladder square gives two ways to consume one zero and one one:

01

or

10.

The machine reaches the same configuration either way.

Lemma 3.3 — Local commuting diamonds.

At every square of the geodesic ladder,

$$\boxed{\delta^*(c, 01) = \delta^*(c, 10).}$$

Proof.

For an interior left configuration

$$(L, 0^m), \qquad m \geq 1,$$

one route is

$$(L, 0^m) \xrightarrow{0} (L, 0^{m-1}) \xrightarrow{1} (C, 0^{m+1}).$$

The other is

$$(L, 0^m) \xrightarrow{1} (C, 0^{m+2}) \xrightarrow{0} (C, 0^{m+1}).$$

Both end at the same configuration.

The two boundary squares are checked directly from the corresponding transition rules:

- at (L, ε), both 01 and 10 reach the same center configuration;
- at $(C, 10)$, both orders reach (C, ε).

These are exactly the ladder squares. □
The point is not that 0 and 1 commute as input symbols globally.
They do not.
The point is much more specific:

Within this shortest-completion interval, adjacent geodesic actions commute physically.

3.6 Parikh synchronization

Because adjacent 01 and 10 factors can be exchanged locally, any two shortest prefixes that have consumed the same numbers of zeros and ones reach the same ladder position.

Corollary 3.4 — Geodesic Parikh synchronization.

Let u and v be prefixes of shortest continuations from

$$(L, 0^k).$$

If

$$N_0(u) = N_0(v)$$

and

$$N_1(u) = N_1(v),$$

then

$$\boxed{\delta^*((L, 0^k), u) = \delta^*((L, 0^k), v).}$$

In particular, every shortest continuation reaches

$$(C, \varepsilon).$$

Proof.

Inside the ladder, repeatedly exchange adjacent factors

$$01 \leftrightarrow 10$$

using Lemma 3.3.

The consumed Parikh pair uniquely determines the grid coordinate. □

This is the first place where the physical machine's computation depends only on the consumed Parikh pair **inside a geodesic interval**, even though the machine is still a stack automaton processing an ordered word.

That phenomenon will later generalize.

3.7 Completion and cancellation are opposite boundaries of one object

The original machine suggested two seemingly different strategies.

One strategy is to expose or materialize the canonical completion as early as possible.

The other is to continue cancelling zeros on the left before finally consuming the required one.

The ladder shows that these are not different semantic processes.

They are boundary schedules of the same shortest-path rectangle.

Corollary 3.5 — Completion–cancellation boundary duality.

For the source

$$(L, 0^k),$$

the canonical completion

$$\boxed{10^{k+2}}$$

and the maximal-cancellation continuation

$$\boxed{0^{k+2}1}$$

are the two extremal boundary paths of the same geodesic ladder.
The word **duality** here is operational.
One boundary performs the unique required 1 first.
The other performs it last.
Both consume the same minimum demand.
This is a major reinterpretation of the original design:

completion and cancellation are scheduling choices inside one geodesic object.

3.8 A commutation potential

For residual r, define

$$\Xi(r) = x_r y_r$$

when

$$\mu(r) = (x_r, y_r).$$

For the left-ladder source,

$$\mu(2k+3) = (k+2, 1),$$

so

$$\Xi(2k+3) = k+2.$$

This number has several simultaneous meanings.
It is:

1. the number of elementary squares in the two-row ladder;
2. the maximum number of geodesic zeros that can be consumed before the unique one becomes unavoidable;
3. the number of adjacent $01 \leftrightarrow 10$ swaps needed to move the unique one from the first position to the last.

Corollary 3.6 — Cancellation–diamond identity.

For every reachable left configuration,

Ξ = diamond count = cancellation horizon = extremal adjacent-swap distance.

This identity is small, but it is conceptually useful.
A quantity defined purely from minimum-demand arithmetic,

$$x_r y_r,$$

simultaneously measures:

- a piece of configuration-space geometry;
- a cancellation horizon in the DPDA;
- a word-reordering distance.

The different descriptions are beginning to converge.

3.9 Why this is more than a counting argument

At the arithmetic level, it is easy to observe that a demand

$$(k+2, 1)$$

has

$$k+3$$

possible shortest word orderings.

The DPDA theorem says more.

It says that the machine realizes these orderings in a coherent configuration-level structure:

$$\boxed{\text{shortest words} \longleftrightarrow \text{monotone paths} \longleftrightarrow \text{reachable configurations.}}$$

Adjacent word commutations become literal commuting squares of machine transitions.

This is why the completion ladder matters.

It is the first exact bridge between:

- Diophantine completion arithmetic;
- shortest-path geometry;
- pushdown-store evolution.

3.10 The canonical completion is a boundary choice

The original endmarker normalizer selects

$$10^{k+2}$$

from the $k+3$ shortest completions of a left source.

Nothing about the minimum Parikh demand itself forces the 1 to occur first.

The ordering

$$10^{k+2}$$

is a canonical representation choice.

The geodesic ladder makes this distinction visible:

$$\boxed{\text{semantic minimum demand} \neq \text{unique physical schedule.}}$$

This sentence will acquire much deeper meanings later.

Later, the original and centered machines will differ by eager versus lazy normalization chronology.

The Structure Theory will separate semantic carry chronology from physical materialization chronology.

The Uniqueness/Lift Theory will exhibit several reduced physical lifts with the same semantics.

At this early stage, however, the ladder already contains the seed of that distinction.

3.11 Limits of the ladder description

The ladder gives a complete operational picture for the motivating left branch.

It does not yet answer three broader questions.

First, when we look at an arbitrary residual, how much of its shortest-path structure can be inferred from a bounded part of the physical stack?

Second, do equal residuals really represent identical future language behavior, not merely identical completion lengths?

Third, what replaces this two-row ladder for a general ratio

$$p{:}q,$$

where a shortest demand may contain several copies of both symbols?

These questions lead in two directions:

quotient and visibility

and

general $p{:}q$ rectangles.

The next chapter takes the first direction.

Chapter 4

Quotients, Geodesics, and What Can Be Seen Locally

Chapter 2 showed that the residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w)$$

determines the entire arithmetic of right completion. Chapter 3 then showed, in the original 2:1 machine, that shortest completions are realized by an actual configuration-level ladder of geodesic schedules.

We now strengthen both observations.

The residual does not merely determine the length or Parikh vector of a shortest repair. It determines the **entire future right language** of the prefix.

At the same time, the canonical shortest-completion word has a bounded visible frontier: one need not inspect its unbounded tail to know which types of geodesic action are available.

These two facts connect three levels:

prefix quotient $\longleftrightarrow$ residual vertex $\longleftrightarrow$ shortest-completion geometry.

4.1 The residual classifies all future behavior

For a language L and prefix u, the left quotient

$$u^{-1}L = \{z : uz \in L\}$$

is the set of all continuations that lead from u into L.

For $L_{p,q}$, Chapter 2 already gave the completion equation

$$z \in u^{-1}L_{p,q} \iff pN_0(z) - qN_1(z) = \rho_{p,q}(u).$$

The right-hand side depends on u only through its residual.

That observation is exact.

Lemma 4.1 — Every integer residual is reachable.

For every

$$r \in \mathbb{Z},$$

there exists a word

$$w \in \{0,1\}^*$$

such that

$$\rho_{p,q}(w) = r.$$

Proof.

Since

$$\gcd(p,q) = 1,$$

Bézout's identity gives integers x_0, y_0 satisfying

$$qy_0 - px_0 = r.$$

Every pair

$$(x_0 + qk,\ y_0 + pk)$$

has the same value of $qy - px$.

For sufficiently large k, both coordinates are nonnegative. Any binary word with those numbers of zeros and ones therefore has residual r. □

Thus the semantic residual graph will genuinely contain every integer, not merely a subset reached by special prefixes.

Theorem 4.2 — Prefix quotient classification.

For all

$$u, v \in \{0,1\}^*,$$

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho_{p,q}(u) = \rho_{p,q}(v).}$$

Proof.

If the residuals agree, the completion equation is identical for u and v. Hence every continuation completes one prefix exactly when it completes the other.

Conversely, suppose

$$\rho_{p,q}(u) = r, \qquad \rho_{p,q}(v) = s, \qquad r \neq s.$$

Choose any nonnegative solution

$$(x, y)$$

of

$$px - qy = r,$$

for example the minimum completion demand $\mu(r)$. Let z be any word with that Parikh pair.

Then

$$z \in u^{-1}L_{p,q}.$$

But the same pair cannot also satisfy

$$px - qy = s.$$

Therefore

$$z \notin v^{-1}L_{p,q}.$$

The quotients differ. □

This theorem gives the residual a stronger interpretation than the one used so far.

It is not merely a useful numeric invariant.

It is an exact coordinate for right-continuation behavior:

$$\boxed{\text{same residual} \iff \text{same future language.}}$$

Left quotients themselves are classical objects. The role of the theorem in this monograph is not to claim a new general quotient theory. Its importance is that the physical DPDA configurations developed from the original machine can later be shown to encode this quotient coordinate exactly.

4.2 The quotient graph

Define the directed labeled graph

$$\mathcal{G}_{p,q}$$

with vertex set

$$\mathbb{Z}$$

and edges

$$r \xrightarrow{0} r - p,$$

$$r \xrightarrow{1} r + q.$$

By Theorem 4.2 and Lemma 4.1, the vertices of this graph may equally well be regarded as the prefix left quotients of $L_{p,q}$.

Thus the arithmetic picture

$$\cdots, -2, -1, 0, 1, 2, \cdots$$

is simultaneously a quotient graph.

A word labels a path by applying its symbols from left to right.

The acceptance quotient is the vertex

$$0.$$

4.3 Completion length is exactly directed distance

Chapter 2 interpreted the minimum completion length

$$\Delta(r)$$

as a shortest path to zero.

The quotient graph allows a slightly stronger translation-invariant form.

Theorem 4.3 — Directed completion metric.

Let

$$d_{\mathcal{G}}(r, s)$$

denote directed shortest-path distance in $\mathcal{G}_{p,q}$. Then

$$\boxed{d_{\mathcal{G}}(r, s) = \Delta(r - s).}$$

Proof.

A path from r to s containing x zeros and y ones has displacement

$$s - r = qy - px.$$

Equivalently,

$$px - qy = r - s.$$

Thus the Parikh vectors of directed r-to-s paths are exactly the nonnegative solutions of the completion equation for residual $r - s$.

Minimizing path length

$$x + y$$

is therefore exactly the minimum-completion problem. □

The geometry is homogeneous:

$$d_{\mathcal{G}}(r, s)$$

depends only on the difference

$$r - s.$$

The language-specific shortest-completion problem is therefore one distinguished family of distances in a translation-invariant directed graph.

4.4 The fundamental balanced cycle

A directed closed walk has zero net residual displacement.

If it contains x zeros and y ones, then

$$qy - px = 0.$$

Coprimality forces a rigid form.

Theorem 4.4 — Fundamental directed cycle.

Every nonempty directed closed walk satisfies

$$N_0 = qk, \qquad N_1 = pk$$

for some integer

$$k \geq 1.$$

Consequently every directed cycle length is a positive multiple of

$$p + q,$$

and the shortest possible directed cycle length is

$$\boxed{g_{p,q} = p + q.}$$

Proof.

A closed walk has

$$qN_1 = pN_0.$$

Since p and q are coprime,

$$N_0 = qk, \qquad N_1 = pk$$

for some nonnegative integer k.

A nonempty walk has $k \geq 1$.

Conversely, any ordering of q zeros and p ones has net residual displacement zero and therefore gives a closed walk of length $p + q$. □

The primitive kernel vector

$$K = (q, p)$$

from Chapter 2 has now become a geometric object:

$$\boxed{K = \text{Parikh vector of a shortest directed balanced cycle.}}$$

This is the global analogue of cancellation.

4.5 Balanced-cycle reduction

Suppose a word z has Parikh pair

$$(x, y).$$

If

$$x \geq q \qquad \text{and} \qquad y \geq p,$$

then one full balanced kernel packet

$$(q, p)$$

can be removed without changing the displacement of the word.

Repeat this as often as possible.

Theorem 4.5 — Balanced-cycle reduction.

Let a word have Parikh pair

$$(x, y).$$

Define

$$k = \min\left(\left\lfloor \frac{x}{q} \right\rfloor, \left\lfloor \frac{y}{p} \right\rfloor\right).$$

Then

$$(x', y') = (x - kq,\ y - kp)$$

has the same residual displacement as (x, y) and is the unique minimum Parikh pair for that displacement.

Proof.

Subtracting

$$(q, p)$$

preserves

$$px - qy.$$

By maximality of k,

$$x' < q \qquad \text{or} \qquad y' < p.$$

Every integer solution with the same displacement differs by an integer multiple of

$$(q, p).$$

No further positive multiple can be removed while remaining nonnegative.

Therefore (x', y') is the first nonnegative point on the affine solution ray and hence the unique minimum pair. □

This theorem gives a global quotient-graph interpretation of the minimum demand:

minimum demand = Parikh vector after deleting every removable balanced cycle.

The original machine's cancellation behavior was local and representation-specific.

Balanced-cycle reduction is semantic and representation-independent.

4.6 Which words are geodesic?

A word is geodesic between its endpoints precisely when no balanced kernel packet can be deleted.

Corollary 4.6 — Exact geodesic domain.

A word z is shortest among all words with the same residual displacement if and only if

$$\boxed{N_0(z) < q \quad \text{or} \quad N_1(z) < p.}$$

Equivalently,

$$\mathrm{Geo}_{p,q} = \{z : N_0(z) < q\} \cup \{z : N_1(z) < p\}.$$

The geodesic-word language is therefore regular.

The criterion is strikingly simple.

A word ceases to be shortest exactly when it contains enough of **both** symbol types to hide one complete balanced cycle in its Parikh vector.

Consequently every binary word of length less than

$$p + q$$

is geodesic.

The first possible failure occurs exactly at the fundamental cycle length.

4.7 Exact quotient spheres

Let

$$S_n = \{r \in \mathbb{Z} : d_{\mathcal{G}}(r, 0) = n\}.$$

A length-n Parikh pair has the form

$$(x, n - x).$$

Its residual displacement is

$$(p + q)x - qn,$$

which is strictly increasing in x. Hence distinct values of x give distinct residual vertices.

Combining this with the geodesic criterion gives an exact growth law.

Proposition 4.7 — Exact quotient spheres.

For every

$$n \geq 0,$$

$$\boxed{|S_n| = \min(n + 1, p + q).}$$

Therefore

$$\boxed{\sum_{n \geq 0} |S_n| z^n = \frac{1 - z^{p+q}}{(1 - z)^2}.}$$

Proof.

If

$$n < p + q,$$

every length-n Parikh pair is geodesic, so all

$$n + 1$$

possible zero counts give distinct vertices at distance n.

If

$$n \geq p + q,$$

a length-n pair is geodesic precisely when

$$x < q$$

or

$$n - x < p.$$

These two regions are disjoint and contain respectively q and p pairs.

Hence exactly

$$p + q$$

residuals occur at distance n. □

The graph therefore has linear growth only until the first balanced cycle appears. After that, every directed sphere has constant size

$$p + q.$$

This small exact formula will later help distinguish semantic growth from the sometimes much larger physical state/stack resources used by a particular lift.

4.8 What the original two-cell observation really reveals

Chapter 1 proved that the original 2:1 normalized completion word exposes its global symbol support within two stack cells.

Chapter 3 gave a more operational interpretation.

For a reachable configuration c, define

$$\mathrm{GeoFirst}(c)$$

to be the set of symbols that begin some shortest continuation from the residual represented by c.

If

$$\mu(r) = (x_r, y_r),$$

then

$$\boxed{\mathrm{GeoFirst}(r) = \{0 : x_r > 0\} \cup \{1 : y_r > 0\}.}$$

For the original machine, the reachable qualitative roles are:

$$(C, \varepsilon) \mapsto \varnothing,$$

$$(C, 0^m) \mapsto \{0\},$$

$$(C, 1^m) \mapsto \{1\},$$

$$(C, 10) \mapsto \{0, 1\},$$

every reachable left configuration maps to

$$\{0, 1\},$$

and every reachable right configuration maps to

$$\{1\}.$$

Thus the old "bounded stack search" intuition can be stated exactly.

Theorem 4.8 — Exact depth-two geodesic locality of the original machine.

For the motivating 2:1 DPDA:

1. the complete geodesic first-action set is determined by the control state and the first two logical stack symbols;
2. depth one is insufficient;

3. after end-of-input normalization,

$$\boxed{\mathrm{alph}(\widehat{S}(c)) = \mathrm{GeoFirst}(c);}$$

4. no fixed finite observation depth determines the full quantitative demand vector on all reachable configurations.

Proof.

The reachable-store normal form from Chapter 1 leaves only the center forms

$$\varepsilon, \quad 0^m, \quad 1^m, \quad 10,$$

together with the fixed qualitative left and right roles.

At the center, two visible symbols distinguish exactly the four possible geodesic-action sets:

$$\varnothing, \quad \{0\}, \quad \{1\}, \quad \{0,1\}.$$

Depth one fails because

$$(C, 1)$$

and

$$(C, 10)$$

have the same state and top symbol but different geodesic first-action sets.

End-of-input normalization produces the canonical shortest continuation represented by the residual, and its symbol support is exactly the set of positive coordinates of the minimum demand.

Finally, arbitrarily large reachable unary center stacks

$$(C, 0^m)$$

share every fixed top-d observation once $m \geq d$, yet their demands

$$(m, 0)$$

have different magnitudes.

Thus qualitative geometry is exactly two-local, while quantitative demand is not fixed-depth local. □

This is the precise form of an important distinction:

$$\boxed{\text{bounded local observation can reveal the shape of the shortest-path choice}}$$

without revealing

$$\boxed{\text{the unbounded distance still remaining.}}$$

4.9 General canonical action visibility

For general p:q, recall the canonical shortest word

$$C(r) = \begin{cases} 1^{y_r}0^{x_r}, & r > 0, \\ 0^{x_r}1^{y_r}, & r < 0, \\ \varepsilon, & r = 0. \end{cases}$$

The minimum pair satisfies the bounded-coordinate property:

$$r > 0 \Longrightarrow 0 \le y_r < p,$$

$$r < 0 \Longrightarrow 0 \le x_r < q.$$

Therefore the first change of symbol, if it occurs, cannot be arbitrarily deep.

Definition 4.9 — Canonical visibility depth.

Let $d_{p,q}$ be the least integer d such that for every residual r,

$$\mathrm{alph}(C(r)) = \mathrm{alph}(\mathrm{pref}_d(C(r))).$$

Theorem 4.10 — Exact bounded visibility.

$$\boxed{d_{p,q} = \max(p, q).}$$

Proof.

Suppose first that

$$r > 0.$$

Then

$$C(r) = 1^{y_r}0^{x_r}$$

with

$$y_r < p.$$

If a zero occurs, the first zero appears no later than position

$$p.$$

Thus depth p suffices on the positive side.
For

$$r < 0,$$

the symmetric argument shows that depth q suffices.
Hence

$$d_{p,q} \le \max(p, q).$$

For tightness, suppose

$$p > q.$$

Take residual

$$r = q.$$

The minimum solution is

$$(x_r, y_r) = (q, p - 1),$$

so

$$C(q) = 1^{p-1}0^q.$$

Depth $p - 1$ sees only ones even though zeros occur later. Therefore

$$d_{p,q} \geq p.$$

If

$$q > p,$$

the symmetric residual

$$r = -p$$

gives

$$C(-p) = 0^{q-1}1^p,$$

forcing

$$d_{p,q} \geq q.$$

For

$$p = q = 1,$$

depth one is plainly necessary and sufficient.
Therefore

$$d_{p,q} = \max(p, q).$$

□
Since

$$\text{alph}(C(r)) = \text{GeoFirst}(r),$$

the same theorem can be read operationally:

The complete set of locally shortest first actions is visible within $\max(p, q)$ canonical symbols.

For the original machine,

$$(p, q) = (2, 1),$$

so

$$d_{2,1} = 2,$$

exactly reproducing the original tight depth-two theorem.

4.10 A second bounded quantity: the completion frontier

Visibility asks one question:

> How far into a canonical shortest completion must we inspect before every symbol type that occurs anywhere in it has appeared?

There is another bounded quantity that will become important when we redesign the physical DPDA.

Write positive and negative residuals in Euclidean form:

$$r = pm + s, \qquad 0 \le s < p,$$

and

$$r = -(qm + t), \qquad 0 \le t < q.$$

Define the finite residue-dependent bases

$$A_s = C(s),$$

$$B_t = C(-t),$$

with

$$A_0 = B_0 = \varepsilon.$$

Then

$$\boxed{C(pm + s) = A_s 0^m,}$$

and

$$\boxed{C(-(qm + t)) = B_t 1^m.}$$

Thus every canonical completion decomposes into:

$$\boxed{\text{bounded residue-dependent frontier} + \text{unbounded unary tail.}}$$

Definition 4.11 — Completion frontier.

Define

$$K_{p,q} = \max\left(\max_{0 \le s < p} |A_s|, \max_{0 \le t < q} |B_t|\right).$$

Theorem 4.12 — Exact almost-cycle frontier.

For every coprime pair (p, q),

$$\boxed{K_{p,q} = \begin{cases} 0, & (p,q) = (1,1), \\ p + q - 1, & \text{otherwise.} \end{cases}}$$

Proof.

For

$$(p, q) = (1, 1),$$

the only residue bases are

$$A_0 = B_0 = \varepsilon,$$

so

$$K_{1,1} = 0.$$

Now assume the coprime pair is nontrivial. Then

$$p \neq q.$$

The bounded-coordinate formulas for $C(r)$ imply

$$K_{p,q} \leq p + q - 1.$$

If

$$p > q,$$

the base residual

$$q$$

has

$$A_q = C(q) = 1^{p-1}0^q,$$

whose length is

$$p + q - 1.$$

If

$$q > p,$$

the symmetric base is

$$B_p = C(-p) = 0^{q-1}1^p,$$

again of length

$$p + q - 1.$$

Hence the upper bound is attained. □
Since the shortest balanced cycle length is

$$g_{p,q} = p + q,$$

we obtain, for every nontrivial coprime pair,

$$\boxed{K_{p,q} = g_{p,q} - 1.}$$

The largest finite frontier is literally one symbol short of a complete balanced cycle.

4.11 One extremal word, two different bounds

The same almost-cycle word makes both visibility and frontier tight.

If

$$p > q,$$

the witness is

$$C(q) = 1^{p-1}0^q.$$

If

$$q > p,$$

the witness is

$$C(-p) = 0^{q-1}1^p.$$

But the two parameters measure different things.

Visibility depth

$$\boxed{d_{p,q} = \max(p, q)}$$

asks:

> How many leading symbols must be inspected before the **qualitative support** of the canonical shortest completion is known?

Completion frontier

$$\boxed{K_{p,q} = p + q - 1}$$

for nontrivial pairs asks:

> How large can the complete **residue-dependent non-unary base** be before only an unbounded unary magnitude remains?

These should not be conflated.

In particular, their numerical values are generally different even though one extremal word witnesses both lower bounds.

This distinction is crucial for later resource arguments.

A finite observation depth is not the same thing as a finite amount of residue information that can be moved into control.

4.12 The next representation idea is now forced

We have reached a useful decomposition:

$$C(r) = \text{finite residue-dependent base} + \text{unbounded unary tail}.$$

The finite base has exact worst-case size

$$K_{p,q}.$$

The unary tail stores the unbounded magnitude.

This immediately suggests a new physical representation:

$$\boxed{\text{store the bounded frontier in finite control}}$$

and

$$\boxed{\text{leave only the unbounded unary magnitude on the pushdown.}}$$

Such a factorization can be lossless because the pair

$$\text{control residue role} + \text{unary tail length}$$

reconstructs the exact residual.

Once the residual is known, Theorem 4.2 gives the exact prefix quotient, Theorem 4.3 gives the exact distance to acceptance, and Chapter 2 gives a canonical shortest continuation.

This will eventually yield a compact physical realization.

But before we redesign the machine, one more piece of geometry is needed.

The 2:1 ladder of Chapter 3 was only two rows high because its minimum demand contained a single 1.

For a general residual,

$$\mu(r) = (x_r, y_r)$$

may have several copies of both symbols.

What is the full shortest-path interval then?

The answer is a rectangle.

That is the subject of Chapter 5.

Chapter 5

General p:q Completion Rectangles

The 2:1 ladder of Chapter 3 had two rows because every left-ladder source had minimum demand

$$(k+2, 1).$$

Only one 1 was required, so the position of that single action was the only scheduling choice.

For a general ratio p:q, the minimum demand

$$\mu(r) = (x_r, y_r)$$

may contain several copies of both symbols.

The ladder then becomes a rectangle.

This chapter identifies that rectangle exactly and shows that three objects coincide:

minimum-demand coordinates $\longleftrightarrow$ directed shortest-path interval $\longleftrightarrow$ commuting schedule geometry.

5.1 Remaining demand as a grid coordinate

Fix coprime positive integers

$$p, q$$

and residual

$$r.$$

Let

$$\mu(r) = (x_r, y_r).$$

Suppose a shortest completion has so far consumed

$$i$$

zeros and

$$j$$

ones.

Its current residual is

$$r - pi + qj.$$

If the prefix is still geodesic, the remaining demand should be

$$(x_r - i,\ y_r - j).$$

Thus the natural coordinate set is

$$[0, x_r] \times [0, y_r],$$

where the point

$$(i, j)$$

records how much of the unique minimum Parikh demand has already been consumed.

A horizontal step consumes a zero:

$$(i, j) \to (i + 1, j).$$

A vertical step consumes a one:

$$(i, j) \to (i, j + 1).$$

The source is

$$(0, 0),$$

and the balanced target is

$$(x_r, y_r).$$

The only possible problem is that two different grid coordinates might correspond to the same residual before the target is reached.

The minimum-demand property prevents exactly that.

5.2 The shortest interval is an actual rectangle

Theorem 5.1 — Rectangular shortest-completion geometry.

Let

$$\mu(r) = (x_r, y_r).$$

The map

$$\boxed{\varphi_r(i, j) = r - pi + qj}$$

is injective on

$$[0, x_r] \times [0, y_r].$$

Consequently the directed shortest-path interval from r to 0 in the residual graph is exactly the rectangular grid

$$\boxed{[0, x_r] \times [0, y_r].}$$

Every monotone source-to-target path is a shortest completion, and every shortest completion occurs in this way.

Proof.

Suppose two rectangle coordinates satisfy

$$r - pi + qj = r - pi' + qj'.$$

Then

$$p(i - i') = q(j - j').$$

Since

$$\gcd(p, q) = 1,$$

there is an integer t such that

$$i - i' = qt, \qquad j - j' = pt.$$

If

$$t \neq 0,$$

then the rectangle contains two points separated by at least one full kernel vector

$$(q, p).$$

In particular, the side lengths must satisfy

$$x_r \geq q, \qquad y_r \geq p.$$

But then

$$(x_r - q,\ y_r - p)$$

would be a shorter nonnegative solution of the same completion equation

$$px - qy = r,$$

contradicting the minimality of

$$\mu(r).$$

Hence

$$t = 0,$$

so

$$i = i', \qquad j = j'.$$

Thus φ_r is injective.

Every monotone path from

$$(0, 0)$$

to

$$(x_r, y_r)$$

consumes exactly x_r zeros and y_r ones, hence exactly the minimum demand.

Conversely, every shortest completion has that same unique minimum Parikh vector, so its sequence of consumed zero/one counts traces one monotone path through the rectangle. □

This theorem generalizes the Chapter 3 ladder without changing its meaning.

When

$$y_r = 1,$$

the rectangle has exactly two rows.

When

$$x_r = 0$$

or

$$y_r = 0,$$

it degenerates to a one-dimensional corridor.

When both coordinates are positive, the residual has a genuine scheduling fork.

5.3 Exact number of shortest completions

A monotone path in an

$$x_r \times y_r$$

rectangle consists of:

- x_r horizontal moves;
- y_r vertical moves.

Therefore the number of source-to-target paths is immediate.

Corollary 5.2 — Exact shortest-completion count.

For every residual r,

$$\boxed{|\operatorname{GeoComp}(r)| = \binom{x_r + y_r}{x_r} = \binom{x_r + y_r}{y_r}.}$$

Equivalently,

$$\operatorname{GeoComp}(r) = \{z : N_0(z) = x_r,\ N_1(z) = y_r\}.$$

The statement is stronger than "there exists a shortest completion."

It says that the unique minimum demand completely classifies **all** shortest words.

Their only freedom is order.

5.4 Exact number of commuting squares

Every elementary unit square of the rectangle corresponds to one local choice between

$$01$$

and

$$10.$$

There are

$$x_r y_r$$

such squares.
Define

$$\boxed{\Xi(r) = x_r y_r.}$$

Corollary 5.3 — Exact commutation potential.

The number of elementary commuting squares in the shortest-completion rectangle of residual r is

$$\boxed{\Xi(r) = x_r y_r.}$$

Thus the same arithmetic quantity introduced in the 2:1 ladder has a general meaning.

It measures the number of elementary places at which the two required symbol types can interchange their local chronological order while remaining inside the shortest interval.

For the Chapter 3 sources,

$$\mu(2k+3) = (k+2, 1),$$

so

$$\Xi(2k+3) = k+2,$$

exactly recovering the number of ladder diamonds.

5.5 A worked 3:2 rectangle

Take

$$p = 3, \qquad q = 2, \qquad r = 2.$$

The completion equation is

$$3x - 2y = 2.$$

The minimum nonnegative solution is

$$\boxed{\mu(2) = (2, 2).}$$

Thus

$$\Delta(2) = 4.$$

The shortest interval is

$$[0, 2] \times [0, 2].$$

It contains

$$(2 + 1)(2 + 1) = 9$$

residual vertices.
There are

$$\binom{4}{2} = 6$$

shortest completion words:

$$0011, \quad 0101, \quad 0110, \quad 1001, \quad 1010, \quad 1100.$$

The canonical two-block completion is

$$\boxed{C(2) = 1100.}$$

The rectangle contains

$$2 \cdot 2 = 4$$

elementary commuting squares.

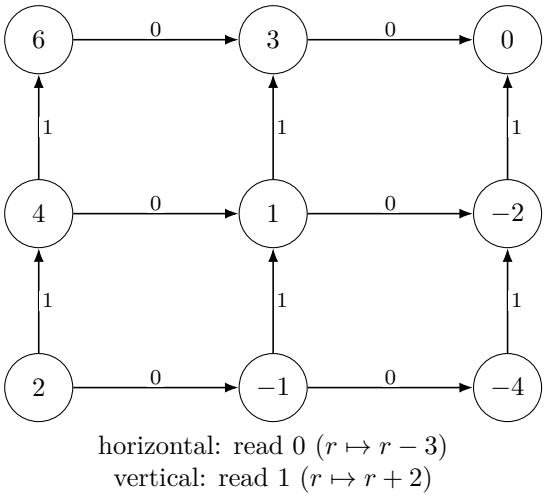


Figure 5.1: The 3:2, $r = 2$ shortest-completion rectangle. Horizontal edges read 0 and subtract 3 from the residual; vertical edges read 1 and add 2. The nine residual vertices form four commuting squares, and every monotone path from residual 2 to residual 0 is one of the six shortest completions with Parikh pair $(2, 2)$.

A useful feature of this example is that neither symbol type occurs only once. The two-row ladder has become a genuinely two-dimensional scheduling space.

5.6 Canonical order as a boundary normal form

For

$$r > 0,$$

our canonical representative is

$$C(r) = 1^{y_r} 0^{x_r}.$$

Among all shortest words with Parikh pair

$$(x_r, y_r),$$

this is one boundary ordering: every required 1 is performed before every required 0.
For

$$r < 0,$$

the canonical representative

$$C(r) = 0^{x_r} 1^{y_r}$$

uses the opposite boundary orientation.
The rest of the shortest words lie between these extremal orderings.
For positive residuals, repeatedly apply the adjacent rewrite

$$01 \longrightarrow 10.$$

Each rewrite preserves the Parikh pair and therefore remains inside the same shortest-completion rectangle.

If we count inversions of the form "a 0 occurring before a 1," every rewrite decreases that count by one.

The unique normal form is

$$1^{y_r} 0^{x_r} = C(r).$$

The maximum possible inversion count is

$$x_r y_r.$$

Thus:

$$\Xi(r) = \text{rectangle-square count}$$
$$= \text{maximum adjacent-swap normalization distance.}$$

For negative residuals the symmetric rewrite

$$10 \longrightarrow 01$$

normalizes toward

$$0^{x_r} 1^{y_r}.$$

The canonical shortest word is therefore not arbitrary.
It is a boundary normal form for a finite commuting geometry.

5.7 Exactly when does a geodesic fork occur?

A residual has a genuine first-action fork exactly when both minimum-demand coordinates are positive.

For positive residuals, the bounded coordinate satisfies

$$0 \le y_r < p.$$

The completion congruence gives

$$qy_r \equiv -r \pmod{p}.$$

Since q is invertible modulo p,

$$y_r = 0 \iff p \mid r.$$

The negative case is symmetric.

Corollary 5.4 — Exact geodesic-fork classification.

For

$$r > 0,$$

$$\boxed{\mathrm{GeoFirst}(r) = \begin{cases} \{0\}, & p \mid r, \\ \{0,1\}, & p \nmid r. \end{cases}}$$

For

$$r < 0,$$

$$\boxed{\mathrm{GeoFirst}(r) = \begin{cases} \{1\}, & q \mid r, \\ \{0,1\}, & q \nmid r. \end{cases}}$$

At

$$r = 0,$$

the set is empty.

This gives an exact arithmetic classification of local shortest-path shape.

There are only three nonterminal qualitative possibilities:

zero corridor,	one corridor,	commuting fork.

The original 2:1 machine has forks only on the positive side because

$$q = 1.$$

Every negative residual is divisible by q, so its shortest path is a one-way 1-corridor.

The one-sided appearance of the original transition diagram is therefore arithmetic, not merely graphical.

5.8 First obstruction at length $p + q$

Chapter 4 proved that a word is geodesic precisely when

$$N_0 < q \qquad \text{or} \qquad N_1 < p.$$

This immediately gives the first scale at which a word can fail to be shortest.
If

$$n < p + q,$$

a length-n word cannot simultaneously contain at least q zeros and at least p ones.
Therefore:

$$\boxed{\text{every binary word of length } n < p + q \text{ is geodesic.}}$$

At length

$$p + q,$$

the first failures appear.
They are exactly the words with Parikh vector

$$(q, p),$$

that is, the orderings of one primitive balanced cycle.
There are

$$\boxed{\binom{p+q}{q}}$$

such first non-geodesic words.
This is another appearance of the same threshold:

$$\boxed{\begin{aligned} p + q &= \text{kernel packet length} = \text{fundamental cycle length} \\ &= \text{first possible non-geodesic length.} \end{aligned}}$$

5.9 Exact geodesic-word growth

The quotient-sphere count of Chapter 4 counted **distinct residual endpoints** at a given directed distance.

We now count something different:

> How many length-n binary words themselves are geodesic between their endpoints?

Let

$$G_{p,q}(n)$$

denote that number.

Theorem 5.5 — Exact geodesic-word growth.

For

$$0 \le n < p+q,$$

$$\boxed{G_{p,q}(n) = 2^n.}$$

For

$$n \ge p+q,$$

$$\boxed{G_{p,q}(n) = \sum_{i=0}^{q-1} \binom{n}{i} + \sum_{j=0}^{p-1} \binom{n}{j}.}$$

The first non-geodesic words occur at length

$$p+q$$

and are exactly the

$$\binom{p+q}{q}$$

orderings of the primitive balanced Parikh vector

$$(q,p).$$

If

$$(p,q) \neq (1,1)$$

and

$$M = \max(p,q),$$

then

$$\boxed{G_{p,q}(n) \sim \frac{n^{M-1}}{(M-1)!}.}$$

For

$$(p,q) = (1,1),$$

$$G_{1,1}(0) = 1, \qquad G_{1,1}(n) = 2 \quad (n \ge 1).$$

Proof.

A length-n word with i zeros is geodesic exactly when

$$i < q$$

or

$$n - i < p.$$

For

$$n < p + q,$$

these inequalities cannot both fail, so every word is geodesic.
For

$$n \geq p + q,$$

the two geodesic regions are disjoint:

- words with fewer than q zeros;
- words with fewer than p ones.

Counting them gives the two fixed binomial tails.
For a nontrivial coprime pair,

$$p \neq q.$$

The larger fixed binomial tail has highest index

$$M - 1.$$

Hence the dominant term is

$$\binom{n}{M-1} \sim \frac{n^{M-1}}{(M-1)!}.$$

The balanced case $(1, 1)$ reduces to the two unary words at every positive length. □
This theorem shows a sharp change of growth.
Before the first balanced cycle, geodesic words grow exponentially:

$$2^n.$$

After the threshold, only two bounded-width binomial tails survive, and growth becomes polynomial.

Again, the primitive balanced cycle is the obstruction.

5.10 Distinct paths versus distinct quotient vertices

Two counting functions have now appeared:

$$|S_n| = \min(n + 1, p + q)$$

for quotient vertices at directed distance n, and

$$G_{p,q}(n)$$

for geodesic words of length n.
They measure different kinds of multiplicity.
The first forgets word order and remembers only the endpoint residual.
The second counts all geodesic orderings.
Inside a fixed shortest-completion rectangle, the same distinction appears as:

grid vertices	versus	monotone paths.

The semantic quotient collapses many word histories into one residual coordinate.

The rectangle restores exactly the scheduling information relevant to a shortest interval.

This interplay between quotienting and lifting will become a recurring theme later in the monograph.

5.11 The 2:1 ladder is now completely explained

Return to the Chapter 3 source

$$(L, 0^k).$$

Its residual is

$$2k + 3,$$

and its minimum demand is

$$(k + 2, 1).$$

The general rectangle is therefore

$$[0, k + 2] \times [0, 1].$$

That is precisely a two-row ladder.

Its path count is

$$\binom{k+3}{1} = k + 3.$$

Its square count is

$$(k + 2) \cdot 1 = k + 2.$$

The positive residual is odd, hence not divisible by $p = 2$, so it has a fork:

$$\mathrm{GeoFirst}(2k + 3) = \{0, 1\}.$$

Every feature of the original operational ladder is therefore a specialization of the general minimum-demand rectangle.

This is an important methodological point.

We did not begin by defining a rectangle and then force the original machine into it.

The rectangle was discovered by first analyzing a concrete cancellation/completion ladder in the machine and then identifying the arithmetic that generalized it.

That discovery order will be preserved in the monograph.

5.12 From geometry back to representation

We now have two complementary pieces of structure.

Chapter 4 gave the canonical completion decomposition

$$\boxed{C(r) = \text{finite residue frontier} + \text{unbounded unary tail}.}$$

This suggests moving the finite frontier into control.

Chapter 5 gives the full shortest-path rectangle

$$\boxed{[0, x_r] \times [0, y_r].}$$

This gives an exact semantic graph that a physical DPDA representation ought to realize.

These facts create a new engineering objective.

Can we construct a pushdown representation in which:

1. the exact residual is decodable;
2. the exact prefix quotient is therefore known;
3. the unbounded magnitude remains stack-like;
4. the finite residue information is moved into control;
5. ordinary input transitions reproduce the residual graph as faithfully as possible?

The first attempt will work semantically but reveal a boundary defect.

That defect will turn out to be useful.

It will teach us that the problem is not the residual semantics.

The problem is the coordinate system.

The next chapter begins that redesign.

Chapter 6

The First General Physical Lift and Its Defects

Research branch — Better Coordinates and Exact Resource Frontiers. *This branch consolidates the coordinate constructions and exact residual-faithful resource frontiers developed in the preceding structural stages of the monograph. State/stack-symbol tradeoffs and limited pushdown alphabets have a substantial descriptional-complexity literature [10, 9, 16, 20]; the frontier statements below are therefore kept explicitly scoped to their representation classes.*

The previous chapters separated two objects that were entangled in the original 2023 machine:

$$\text{semantic residual} \qquad \text{and} \qquad \text{physical pushdown representation.}$$

The residual graph is now completely explicit:

$$r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q.$$

Its shortest intervals are rectangles, its quotient coordinate is one integer, and its canonical completion has a bounded finite frontier followed by an unbounded unary tail.

This makes a new design problem unavoidable.

> Can the residual graph itself be lifted into a compact DPDA configuration graph?

The first answer is yes—but only after a deterministic normalization step at a finite boundary.

That apparent imperfection will be useful. It will identify exactly what the first coordinates got wrong.

6.1 Finite phase plus unary magnitude

Let

$$m = \max(p, q).$$

The idea is to move the bounded residue information into finite control while leaving only an unbounded magnitude on the stack.

Use stable control states

$$R_0, \ldots, R_{m-1}$$

and stack alphabet

$$\Gamma_4 = \{0, 1, X, Z_0\}.$$

The positive and negative ends are represented by different unary symbols:

- 0 for the positive end;
- 1 for the negative end.

The additional symbol X is a negative-side sentinel.
For

$$0 \le s < p,$$

define

$$a_s = \begin{cases} 0, & s = 0, \\ s - p, & 1 \le s < p. \end{cases}$$

For

$$0 \le t < q,$$

define

$$b_t = \begin{cases} 0, & t = 0, \\ q - t, & 1 \le t < q. \end{cases}$$

These are not arbitrary constants. They choose one finite base residual for each positive and negative residue class.

6.2 The stable four-symbol encoding

Define

$$\eta(0) = (R_0, Z_0).$$

If

$$r > 0,$$

let

$$s = r \bmod p$$

and write uniquely

$$r = pk + a_s, \qquad k \ge 1.$$

Set

$$\boxed{\eta(r) = (R_s, 0^k Z_0).}$$

If

$$r < 0,$$

let

$$t = (-r) \bmod q$$

and write uniquely

$$r = -qk + b_t, \qquad k \geq 1.$$

Set

$$\boxed{\eta(r) = (R_t, 1^k X).}$$

The representation is injective.

The control state records the finite phase. The unary tail records the unbounded quotient magnitude. The sign is visible physically from the top/sentinel convention.

This is the first systematic realization of the Chapter 4 factorization

$$\text{finite boundary role} + \text{unbounded unary magnitude}.$$

6.3 Stable rules away from the boundary

On the positive side, input 0 should subtract p. Deep inside the unary tail, this is simply a pop:

$$\delta(R_s, 0, 0) = (R_s, \varepsilon).$$

Input 1 adds q. Let

$$s' = (s + q) \bmod p$$

and define

$$c_s = \frac{a_s + q - a_{s'}}{p}.$$

Then

$$pk + a_s + q = p(k + c_s) + a_{s'}.$$

Thus the correct stable rule is

$$\delta(R_s, 1, 0) = (R_{s'}, 0^{c_s+1}).$$

The negative side is symmetric.

Let

$$t' = (t + p) \bmod q$$

and

$$d_t = \frac{p - b_t + b_{t'}}{q}.$$

Then

$$\delta(R_t, 1, 1) = (R_t, \varepsilon)$$

and

$$\delta(R_t, 0, 1) = (R_{t'}, 1^{d_t+1}).$$

Away from the finite boundary, the design is already raw: one input symbol produces the exact next stable residual encoding.

The only difficulty appears when a unary tail is completely consumed.

6.4 Forced boundary normalization

Suppose a positive configuration has only one unary 0 left.

Consuming that 0 exposes

$$Z_0.$$

For residue class $s = 0$, this is already the correct zero configuration.

For

$$1 \le s < p,$$

the configuration

$$(R_s, Z_0)$$

is only transient. A unique forced normalization moves it to the stable encoding of

$$a_s = s - p < 0.$$

Likewise, on the negative side, consuming the last 1 exposes

$$X,$$

and a unique forced normalization moves to the stable encoding of the corresponding nonnegative base residual b_t.

This motivates a semantic macro-step.

Definition 6.1 — Normalized input macro-step.

Write

$$c \overset{a}{\Rightarrow} c'$$

when the machine:

1. consumes the binary symbol a;
2. performs the unique forced finite normalization chain, if one is enabled;
3. stops at the next stable encoded configuration.

The normalization consumes no additional binary input and introduces no choice.

6.5 Exact normalized residual conjugacy

Theorem 6.2 — Normalized residual–configuration conjugacy.

For every

$$r \in \mathbb{Z},$$

$$\boxed{\eta(r) \overset{0}{\Rightarrow} \eta(r-p), \qquad \eta(r) \overset{1}{\Rightarrow} \eta(r+q).}$$

Consequently, for every binary word u,

$$\boxed{\widehat{\delta}^*(\eta(r), u) = \eta(r - pN_0(u) + qN_1(u))\,.}$$

Thus η is a label-preserving graph isomorphism between the residual graph and the stable configuration graph **modulo deterministic normalization**.

Proof.

Let

$$r = pk + a_s > 0.$$

On input 0, if $k > 1$, one unary 0 is popped and the result is immediately

$$\eta(r-p).$$

If $k = 1$, the pop reaches the finite boundary. The uniquely forced normalization sends the exposed boundary pair to the stable encoding of $r - p$.

On input 1,

$$pk + a_s + q = p(k + c_s) + a_{s'},$$

so the positive stable rule lands directly in

$$\eta(r+q).$$

The negative case is symmetric. Input 1 pops one unary 1, with the $k = 1$ case followed by its unique sentinel normalization; input 0 uses the identity defining d_t.

At residual zero the two binary moves are defined directly toward the encodings of $-p$ and q.

The one-step identities imply the word formula by induction. □

This theorem is already a major improvement in understanding.

The original $C/L/R$ machine encoded the residual through a more specialized stack/state interaction.

The new construction makes the quotient coordinate explicit:

finite residual phase in control + unbounded quotient magnitude on a unary tail.

6.6 The abstract rectangles lift to physical configurations

Let

$$\mu(r) = (x_r, y_r).$$

Define

$$\widehat{\Psi}_r(i,j) = \eta(r - pi + qj).$$

Corollary 6.3 — Exact normalized configuration rectangle.

The map

$$\widehat{\Psi}_r$$

is a label-preserving isomorphism from

$$[0, x_r] \times [0, y_r]$$

onto the normalized shortest-path interval from

$$\eta(r)$$

to

$$\eta(0).$$

Every elementary rectangle square is a genuine normalized commuting diamond:

$$\boxed{\widehat{\delta}^*\big(\widehat{\Psi}_r(i,j), 01\big) = \widehat{\Psi}_r(i+1, j+1) = \widehat{\delta}^*\big(\widehat{\Psi}_r(i,j), 10\big).}$$

The shortest-completion rectangle is therefore not merely an arithmetic diagram.
It is realized by actual DPDA configurations.
The qualification is important:

one rectangle edge may be	one input move + forced normalization.

The original 2:1 ladder was stronger: its relevant edges were ordinary binary moves.
The question is now whether the general construction can be made equally literal.

6.7 A concrete 3:2 boundary crossing

Take

$$p = 3, \qquad q = 2, \qquad r = 4.$$

Then

$$\mu(4) = (2, 1).$$

The normalized rectangle contains

$$\eta(4) = (R_1, 00Z_0),$$

$$\eta(1) = (R_1, 0Z_0),$$

$$\eta(-2) = (R_0, 1X),$$

$$\eta(6) = (R_0, 00Z_0),$$

$$\eta(3) = (R_0, 0Z_0),$$

$$\eta(0) = (R_0, Z_0).$$

The lower horizontal edge

$$1 \xrightarrow{0} -2$$

is semantically correct, but in these coordinates the binary pop first exposes a transient boundary pair before the forced normalization reaches

$$\eta(-2).$$

The rectangle remains exact.

The edge is simply not yet **raw**.

6.8 Raw residual conjugacy

We now strengthen the representation requirement.

Definition 6.4 — Raw residual–configuration conjugacy.

An injective encoding

$$\zeta : \mathbb{Z} \to \text{DPDA configurations}$$

is a raw residual conjugacy if

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)}$$

for every residual r, using exactly one ordinary binary-input transition and no normalization step.

A raw defect of an encoding is therefore an edge on which this stronger identity fails.

The four-symbol construction is normalized-exact.

We now ask exactly where it fails to be raw.

6.9 The defect set is finite and exact

Proposition 6.5 — Exact four-symbol raw-defect set.

For the four-symbol encoding η,

$$\boxed{\mathcal{D}_{p,q} = \{(s, 0) : 1 \le s < p\} \cup \{(-j, 1) : 1 \le j \le q\}.}$$

Therefore

$$\boxed{|\mathcal{D}_{p,q}| = p + q - 1.}$$

Every residual edge outside this set is already one raw binary transition between stable configurations.

Proof.

On the positive side, input 1 always uses a stable rule.

Input 0 is also raw whenever at least two unary zeros are present. The only problem occurs when the source residual is one of

$$1, 2, \ldots, p-1.$$

The last zero is popped, leaving the transient state/bottom pair

$$(R_s, Z_0)$$

instead of the required negative encoding.

These are exactly the $p-1$ positive defects.

On the negative side, input 0 is always raw.

Input 1 is raw while at least two unary ones remain. At the final unary one, the pop exposes X, and the unique normalization is still required.

The affected residuals are

$$-q, -(q-1), \ldots, -1,$$

giving exactly q negative defects.

The two families are disjoint. □

The count

$$p + q - 1$$

is numerically the same as the completion-frontier scale of Chapter 4 for nontrivial coprime pairs.

The meanings are different.

One counts the largest bounded canonical completion base.

The other counts the raw boundary defects of this particular four-symbol encoding.

The equality is evidence that the same finite arithmetic boundary is being seen from two physical viewpoints—not that the notions are identical.

6.10 Why the defects cannot be patched locally in the same coordinates

It is tempting to keep the encoding η and merely change the boundary transition rules.

That does not work.

Proposition 6.6 — Boundary obstruction for the fixed encoding.

Keeping the four-symbol physical representations $\eta(r)$, no standard top-rewrite PDA transition function can make all edges in

$$\mathcal{D}_{p,q}$$

raw while preserving the already correct behavior on the adjacent deeper unary tails.

Hence every defect of Proposition 6.5 is unavoidable **within these coordinates**.

Proof.

Fix

$$1 \le s < p.$$

Compare

$$\eta(p + s) = (R_s, 00Z_0)$$

with

$$\eta(s) = (R_s, 0Z_0).$$

The two sources have the same state and visible top symbol.

Correct realization of

$$p + s \xrightarrow{0} s$$

forces the transition on the local pair

$$(R_s, 0)$$

to pop one zero and remain in state R_s.

But the same deterministic rule applied to

$$(R_s, 0Z_0)$$

must produce

$$(R_s, Z_0),$$

whereas the correct target is the negative encoding

$$\eta(s - p).$$

The transition cannot distinguish the two hidden suffixes.

Thus the positive boundary edge cannot be raw without breaking the already correct deeper edge.

The negative case is analogous. Compare

$$(R_t, 11X)$$

and

$$(R_t, 1X).$$

Correct deep behavior forces the visible-1 rule to pop. Applied at the boundary, the same pop leaves

$$(R_t, X),$$

not the required nonnegative target.

The obstruction depends only on the fixed state/top encoding and one-top locality. □

This is the real lesson of the defect theorem.

The problem is not that we failed to invent a clever enough boundary rule.

The local rule is already determined by the deeper tail.

6.11 Which shortest rectangles are already raw?

Corollary 6.7 — Raw rectangle criterion in the four-symbol orientation.

The entire shortest-completion rectangle from residual r is raw under η exactly when

$$\boxed{r = 0 \quad \text{or} \quad (r > 0 \text{ and } p \mid r).}$$

Interpretation. If

$$r > 0$$

and

$$p \mid r,$$

then

$$\mu(r) = (r/p, 0),$$

so the rectangle degenerates to a pure zero corridor and never crosses a defective boundary.

Every positive residual with a genuine two-symbol fork eventually exposes one of the positive boundary defects.

Every negative nonzero rectangle contains the final edge

$$-q \xrightarrow{1} 0,$$

which is defective in this orientation.

The asymmetry is representational.

Zero shares the positive sentinel Z_0, while the negative side uses X.

That observation points directly toward the cure.

6.12 The defect is coordinate-relative

The fixed-coordinate obstruction proves:

$$\boxed{\text{these coordinates cannot be made globally raw.}}$$

It does **not** prove:

$$\boxed{\text{four stack symbols are necessary.}}$$

Those are very different statements.

The first is a theorem.

The second would be an unjustified extrapolation.

The right design question is therefore not:

> What extra symbol should we add?

It is:

> Can we align the finite residual bases differently so that the sign-changing boundary is itself an ordinary top rewrite?

The answer is yes.

Even better, the new coordinates remove the extra sentinel X.

The next chapter performs that coordinate change.

Chapter 7

From Boundary Defects to a Raw Three-Symbol DPDA

Chapter 6 ended with a precise failure.

The four-symbol representation

$$\eta$$

was semantically exact, but at

$$p + q - 1$$

boundary edges an ordinary binary transition could not land directly in the next stable residual encoding.

The obstruction was not global.

It was local:

> the same state/top pair had to serve both
> a deep unary tail and a sign boundary.

The first coordinates placed the finite residual bases on the wrong side of that boundary.

We now shift those bases.

The result is stronger in three ways:

1. every residual edge becomes one raw input transition;
2. the extra sentinel X disappears;
3. the construction still uses only

$$\max(p, q)$$

control states.

This is the first major example in the monograph where understanding the mathematics of a DPDA leads directly to a better DPDA.

7.1 Before using two work symbols: the unary baseline

It is useful to know what can be done with only one work-stack symbol.

Let

$$\Gamma_2 = \{A, Z_0\},$$

where Z_0 is a permanent bottom marker.

Every encoded stack then has the form

$$A^h Z_0.$$

The state must therefore distinguish every finite role that cannot be carried by stack-symbol type.

Theorem 7.1 — Exact one-work-symbol raw-conjugacy complexity.

For every coprime

$$p, q \geq 1,$$

any ε-free permanent-bottom raw residual conjugacy using only one work symbol requires

$$\boxed{|Q| \geq p + q.}$$

There is a right-endmarker construction attaining

$$\boxed{|Q| = p + q, \qquad |\Gamma| = 2.}$$

Hence $p + q$ is exact in this model.

Proof idea

Input 0 partitions the residual line into p bi-infinite orbits modulo p.

Far out on the positive end, injectivity forces the stack height to escape every bounded region. Since a single top rewrite can decrease work height by at most one, the eventual input-0 dynamics must contain enough negative-drift state phases to distinguish all p positive residue classes.

Thus at least p state roles are required.

Symmetrically, input 1 forces at least q state roles on the negative end.

With only one work symbol, those two tail families cannot safely share the same control roles: otherwise repeated negative drift would send infinitely many distinct residuals into finitely many bottom configurations.

Therefore the roles are disjoint:

$$|Q| \geq p + q.$$

A matching construction uses state families

$$P_0, \ldots, P_{p-1}$$

and

$$N_0, \ldots, N_{q-1},$$

with the single work symbol storing only unary magnitude.

The full transition table is recorded in Appendix B.

The theorem establishes an exact baseline:

$$\boxed{\text{one work symbol} \Longrightarrow p + q \text{ states.}}$$

The next question is whether a second work symbol can let the positive and negative phase roles share control states.

It can.

7.2 Boundary-compatible coordinates

Assume first

$$p \geq q.$$

Use only

$$\boxed{\Gamma_3 = \{0, 1, Z_0\}.}$$

Use control states

$$\boxed{Q_3 = \{R_0, \ldots, R_{p-1}\}.}$$

There are exactly

$$p = \max(p, q)$$

states.
For nonnegative residuals, use ordinary Euclidean division:

$$r = pk + s, \qquad k \geq 0, \qquad 0 \leq s < p.$$

Define

$$\boxed{\zeta(r) = (R_s, 0^k Z_0).}$$

For negative residuals, write uniquely

$$r = u - qh, \qquad h \geq 1, \qquad 0 \leq u < q.$$

Define

$$\boxed{\zeta(r) = (R_u, 1^h Z_0).}$$

This is the decisive shift.
The negative side no longer terminates in a separate sentinel X.
Its last unary 1 sits directly above the ordinary bottom marker.
The states

$$R_0, \ldots, R_{q-1}$$

are deliberately shared between:

- nonnegative boundary roles;
- negative residue phases.

The sign remains visible because:

- a negative configuration has top 1;
- a positive nonboundary configuration has top 0;
- a nonnegative boundary configuration has top Z_0.

Thus state sharing does not destroy injectivity.

7.3 Positive-tail arithmetic

For

$$0 \le s < p,$$

define

$$s^+ = (s+q) \bmod p$$

and

$$c_s = \frac{s+q-s^+}{p}.$$

Since

$$0 < q \le p,$$

we have

$$c_s \in \{0,1\}.$$

On a positive unary tail:

$$\boxed{\delta(R_s,0,0) = (R_s,\varepsilon),}$$

and

$$\boxed{\delta(R_s,1,0) = (R_{s^+},0^{c_s+1}).}$$

The arithmetic identity is

$$pk+s+q = p(k+c_s)+s^+.$$

So the visible rewrite is exactly the residual update.

7.4 The repaired sign boundary

For the nonnegative boundary, define

$$u_s = (s-p) \bmod q$$

and

$$h_s = \frac{u_s+p-s}{q}.$$

Then

$$s-p = u_s - qh_s.$$

Therefore input 0 at the bottom marker can cross directly into the negative encoding:

$$\boxed{\delta(R_s,0,Z_0) = (R_{u_s},1^{h_s}Z_0).}$$

No transient pair is created.
No normalization follows.

The boundary itself is now raw.

Input 1 at a nonnegative boundary uses

$$\boxed{\delta(R_s, 1, Z_0) = (R_{s^+}, 0^{c_s} Z_0).}$$

Again the target is already the exact stable encoding.

7.5 Negative-tail arithmetic and the second repaired boundary

For

$$0 \le u < q,$$

define

$$u^- = (u - p) \bmod q$$

and

$$d_u = \frac{u^- - u + p}{q}.$$

On the negative tail:

$$\boxed{\delta(R_u, 1, 1) = (R_u, \varepsilon),}$$

and

$$\boxed{\delta(R_u, 0, 1) = (R_{u^-}, 1^{d_u+1}).}$$

Now observe what happens when the negative stack has only one 1:

$$(R_u, 1Z_0) \xrightarrow{1} (R_u, Z_0).$$

But by definition

$$(R_u, Z_0) = \zeta(u).$$

The final negative 1-pop therefore crosses the sign boundary automatically.

The very rule that was forced by the deep negative tail is now also the correct boundary rule.

This is exactly what the four-symbol coordinates failed to achieve.

7.6 Global raw conjugacy

Theorem 7.2 — Three-symbol raw residual–configuration conjugacy.

For every coprime

$$p, q \ge 1,$$

there is an ε-free right-endmarker DPDA with

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3,}$$

whose stack alphabet is

$$\{0, 1, Z_0\}$$

and whose injective encoding

$$\zeta : \mathbb{Z} \to Q \times \Gamma^*$$

satisfies

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)}$$

for every residual r.
Hence, for every word w,

$$\boxed{\delta^*(\zeta(r), w) = \zeta(r - pN_0(w) + qN_1(w))\,.}$$

The right endmarker is accepted exactly at

$$\zeta(0).$$

Proof.
Assume

$$p \geq q.$$

The nonnegative and negative encodings are injective within their own Euclidean decompositions.

They are mutually disjoint physically because the negative family has top 1, whereas the nonnegative family has top 0 or Z_0.

If

$$r = pk + s$$

with

$$k \geq 1,$$

input 0 pops one zero:

$$pk + s - p = p(k - 1) + s.$$

Input 1 uses

$$pk + s + q = p(k + c_s) + s^+.$$

If

$$k = 0,$$

the same input-1 identity works at Z_0, while the boundary identity

$$s - p = u_s - qh_s$$

makes input 0 land directly in the negative encoding.
Now let

$$r = u - qh < 0.$$

Input 1 pops one unary 1.
If

$$h > 1,$$

the result is still negative.
If

$$h = 1,$$

the pop leaves

$$(R_u, Z_0) = \zeta(u),$$

which is exactly the nonnegative target.
Finally,

$$u - p = u^- - qd_u$$

gives

$$r - p = u^- - q(h + d_u),$$

so the input-0 rewrite produces exactly the required deeper negative encoding.
No case uses an ε-move.
For

$$q > p,$$

exchange the roles of 0 and 1, exchange p and q, and reverse the residual sign. □
This theorem removes both weaknesses of Chapter 6 at once:

$$\boxed{(\max(p, q), 4, \text{normalized macro-edges})}$$

becomes

$$\boxed{(\max(p, q), 3, \text{raw input edges}).}$$

7.7 The 3:2 boundary is now literal

Return to

$$p = 3, \qquad q = 2, \qquad r = 4.$$

The new encodings include

$$\zeta(4) = (R_1, 0Z_0),$$

$$\zeta(1) = (R_1, Z_0),$$

$$\zeta(-2) = (R_0, 1Z_0),$$

$$\zeta(0) = (R_0, Z_0).$$

The old defective edge

$$1 \xrightarrow{0} -2$$

is now exactly

$$\boxed{\delta(R_1, 0, Z_0) = (R_0, 1Z_0).}$$

The old negative normalization edge

$$-2 \xrightarrow{1} 0$$

is now the ordinary pop

$$\boxed{\delta(R_0, 1, 1) = (R_0, \varepsilon),}$$

which exposes Z_0.
The same semantic rectangle now exists literally in the raw configuration graph.

7.8 Raw rectangles

Corollary 7.3 — Three-symbol raw configuration rectangles.

Let

$$\mu(r) = (x_r, y_r).$$

Define

$$\Psi_r^{\mathrm{raw}}(i, j) = \zeta(r - pi + qj).$$

Then

$$\Psi_r^{\mathrm{raw}}$$

is a label-preserving isomorphism from

$$[0, x_r] \times [0, y_r]$$

onto the ordinary DPDA shortest-path interval from

$$\zeta(r)$$

to

$$\zeta(0).$$

Every elementary square is a raw commuting diamond:

$$\boxed{\delta^*\big(\Psi_r^{\mathrm{raw}}(i, j), 01\big) = \Psi_r^{\mathrm{raw}}(i + 1, j + 1) = \delta^*\big(\Psi_r^{\mathrm{raw}}(i, j), 10\big).}$$

The Chapter 5 rectangle has now been fully lifted into one-symbol DPDA dynamics. No quotient by normalization is needed.

7.9 What has actually been optimized?

The construction uses

$$\max(p, q)$$

states and two work symbols $0, 1$ above the bottom marker.

At this point it is tempting to call it simply "the optimal DPDA."

That wording is too broad.

We must specify the representation category.

The Chapter 7 machine is optimal under several increasingly strong statements, each tied to an explicit representation category. Unrestricted recognition complexity is treated separately in the representation-independent theory.

Theorem 7.4 — Exact homogeneous-tail frontier.

In the permanent-bottom, ε-free raw-conjugacy model with asymptotically homogeneous positive and negative tails:

$$\boxed{S_{\mathrm{hom}}(k) = \begin{cases} p + q, & k = 1, \\ \max(p, q), & k \geq 2, \end{cases}}$$

where k is the number of work symbols.

Thus a second work symbol gives the entire possible state reduction in this model.

Additional homogeneous-tail work symbols cannot reduce the state count below

$$\max(p, q).$$

The proof uses preserved deep suffixes and the cyclic action of:

- addition by q on positive classes modulo p;
- subtraction of p on negative classes modulo q.

Coprimality forces one common positive tail symbol and one common negative tail symbol. With one work symbol the two ends must remain state-disjoint; with two symbols they may share control phases.

This is already an exact state/stack tradeoff.

But it still assumes a homogeneous-tail representation shape.

7.10 The motivating 2:1 ratio survives arbitrary three-symbol encodings

The original research program began at

$$(p, q) = (2, 1).$$

Here the stronger lower bound can be proved without assuming homogeneous tails or sign separation.

Theorem 7.5 — Exact three-symbol state complexity for 2:1.

Consider the full permanent-bottom stack alphabet

$$\Gamma_3 = \{A, B, Z_0\}$$

with:

- no ε-moves;
- injective raw residual conjugacy;
- no homogeneous-tail assumption;
- no prescribed sign-separated encoding.

Then

$$\boxed{|Q| \geq 2.}$$

The construction of Theorem 7.2 uses exactly two states, so

$$\boxed{|Q|_{\min} = 2}$$

for three-symbol raw conjugacy of the motivating 2:1 residual graph in this model.

Proof strategy

Assume one state.

Let

$$G : S_r \mapsto S_{r+1}$$

be input 1 and

$$F : S_r \mapsto S_{r-2}$$

be input 0.

On the encoded orbit,

$$\boxed{FG^2 = G^2F = \mathrm{id}.}$$

Choose an encoded residual of minimum work-stack height.

The inverse identity bounds the lengths of all possible top replacements, reducing the problem to finitely many local rewrite patterns.

If the minimum height is positive, a common inert suffix can be stripped. The remaining possibilities for the first few G-iterates are finite, and each is eliminated by one of:

- preserved-suffix impossibility;
- forced rule reuse;
- violation of minimum height;
- collision of two residuals.

If the minimum height is zero, the bottom write is likewise bounded and only finitely many cases remain. Every case again contradicts injectivity or the inverse relation.

The scope and finite-certificate framework for these lower bounds are recorded in Appendix C.

Thus one state is impossible.

This theorem is particularly important for the monograph because it ties the improved machine back to the original ratio:

> The analysis of the 2023 2:1 DPDA leads to an exact two-state, three-symbol raw residual-conjugate realization.

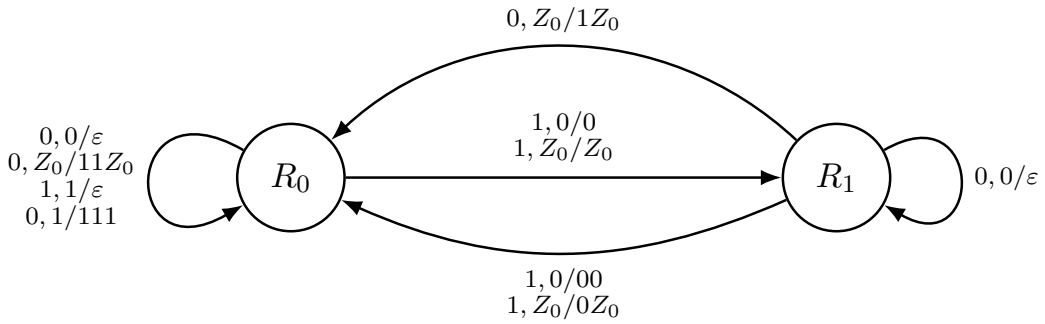


Figure 7.1: The improved two-state, three-symbol raw residual-conjugate DPDA for the motivating 2:1 residual graph. Labels have the form input, top/replacement; only transition cells occurring on the encoded residual orbit are shown. The right endmarker is accepted exactly at (R_0, Z_0).

Figure 7.1 is exact in the permanent-bottom, ε-free raw residual-conjugacy model. It is not a claim of unrestricted global optimality among all DPDAs recognizing $L_{2,1}$.

The scope words remain essential.

7.11 Further exact special ratios

The same unrestricted permanent-bottom three-symbol question can be solved at two further small ratios.

Theorem 7.6 — Exact 3:1 **three-symbol state complexity.**

For

$$(p, q) = (3, 1),$$

every ε-free permanent-bottom three-symbol raw conjugacy requires

$$\boxed{|Q| \geq 3.}$$

The construction of Theorem 7.2 attains three states.

Hence

$$\boxed{|Q|_{\min} = 3.}$$

The symmetric result holds for

$$(1, 3).$$

The proof again reduces an assumed smaller-state conjugacy to finitely many bounded top-rewrite possibilities, but the case structure is substantially larger than at 2:1. Its role and scope within the small-ratio finite certificates are recorded in Appendix C.

Theorem 7.7 — Computer-assisted exact 3:2 **result.**

For

$$(p, q) = (3, 2),$$

a finite mathematically bounded search eliminates every two-state permanent-bottom three-symbol raw conjugacy.

Therefore

$$\boxed{|Q| \geq 3,}$$

and the three-state construction is exact:

$$\boxed{|Q|_{\min} = 3.}$$

The symmetric result holds for

$$(2, 3).$$

This theorem is explicitly **computer-assisted**.

The computer is used only after the proof reduces the unrestricted-looking problem to a finite bounded search space. The survivor vectors and branch counts belong in the computational-validation appendix; the theorem should never be presented as a purely symbolic proof.

The small-ratio finite eliminations were an intermediate stage only. Chapter 8 closes the arbitrary-encoding theorem for every coprime pair by a representation-shape-free argument, so no ratio-by-ratio extension is required.

7.12 The right endmarker saves one control state in the scoped models

The constructions above use a distinguished right endmarker and accept exactly at the zero residual configuration.

If ordinary final-state acceptance on the binary alphabet is required instead, the zero configuration must use a final state that does not also occur on a nonzero residual.

In the scoped raw-conjugacy models this costs exactly one state.

Corollary 7.8 — Exact no-endmarker cost in the early raw models.

For the one-work-symbol model:

$$\boxed{|Q| = p + q + 1, \qquad |\Gamma| = 2.}$$

For the sign-separated homogeneous-tail three-symbol model:

$$\boxed{|Q| = \max(p, q) + 1, \qquad |\Gamma| = 3.}$$

Both bounds are tight.

This acceptance cost is separate from the stack-alphabet tradeoff.

It changes only how residual zero is represented.

7.13 The coordinate-shift improvement

The conclusion of Chapters 6 and 7 can now be stated cleanly.

Corollary 7.9 — Coordinate-shift improvement.

The $p + q - 1$ raw defects of the first four-symbol encoding are not a lower bound on the number of stack symbols.

They are properties of those coordinates.

A boundary-compatible shift gives

$$\boxed{(\max(p,q),\ 4,\ \text{normalized macro-steps}) \longrightarrow (\max(p,q),\ 3,\ \text{raw binary steps}).}$$

The missing ingredient was not an additional endpoint bit.
The finite bases had been aligned incorrectly with the sign-changing unary pops.
After moving the bases to adjacent residual classes:

- the same state can serve the post-pop boundary target;
- the ordinary bottom marker is sufficient;
- the extra sentinel X disappears;
- every residual edge becomes literal.

This is the first major design lesson of the monograph:

> a locality defect may be a coordinate defect
> rather than a semantic necessity.

7.14 Three different meanings of "better"

The evolution from the original machine to the raw three-symbol construction improves the representation in several non-equivalent ways.

Semantic transparency

The residual is decoded by a simple Euclidean state/tail coordinate.

Transition literalness

Every binary residual edge is exactly one DPDA input transition.

Stack alphabet

The general raw construction needs only

$$\{0, 1, Z_0\}.$$

Control width

It uses

$$\max(p, q)$$

states.

But the exactness claims established so far belong to stated raw-conjugacy categories.

Unrestricted recognition complexity is representation-independent and is treated separately in the later physical-complexity macro-Part.

That distinction is one of the central methodological themes of the book.

7.15 Removing the representation shape

The current lower bounds still come from several different levels of assumption:

- one work symbol;
- homogeneous positive/negative tails;
- special-ratio arbitrary three-symbol encodings.

The obvious next question is:

> Are the resource points
>
> $$p + q \quad \text{and} \quad \max(p, q)$$
>
> accidents of these structured coordinates, or are they forced by raw residual conjugacy itself?

To answer that, we must stop assuming homogeneous tails.
We will keep only the semantic requirement:

Every integer residual has one injective stable physical encoding, and every residual edge is one raw input transition.

From that weak assumption alone, the next chapter derives the complete exact frontier.

That result will turn the improved construction discovered here into a mature resource theorem.

Chapter 8

The Complete Raw Residual-Faithful Frontier

Chapter 7 produced a sharp construction:

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3,}$$

with one raw binary transition for every residual edge.
It also produced exact lower bounds in several structured settings:

- one work symbol;
- homogeneous positive/negative tails;
- unrestricted three-symbol encodings for several small ratios.

The remaining question is whether the same resource points are forced **without assuming a particular stack shape**.
This chapter removes that assumption.
The result is the mature form of the early resource story:

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}}$$

The proof is deliberately representation-shape-free.
It does not assume:

- homogeneous tails;
- sign separation;
- a canonical coordinate formula;
- fixed stack depth;
- bounded replacement length.

Only raw residual conjugacy remains fixed.

8.1 Model R

Let

$$\Gamma = \Delta \cup \{Z_0\},$$

where Z_0 is a permanent bottom marker and

$$g = |\Delta|$$

is the number of work symbols.

Definition 8.1 — Model R.

A raw residual conjugacy is an injective map

$$\eta : \mathbb{Z} \to Q \times \Delta^* Z_0$$

such that, for every residual r,

$$\boxed{\delta(\eta(r), 0) = \eta(r - p), \qquad \delta(\eta(r), 1) = \eta(r + q),}$$

with no ε-normalization on encoded residual configurations.

Let

$$S_{\text{raw}}(g)$$

be the minimum number of control states among such realizations, with value ∞ if no finite-state realization exists.

The semantic restriction is exact but narrow:

$$\boxed{\text{one residual} \leftrightarrow \text{one stable physical configuration.}}$$

The physical shape of that configuration is otherwise unrestricted.

This is the right setting for deciding whether the Chapter 7 optimum was merely a feature of its chosen coordinates.

8.2 Why bounded physical height is impossible

Fix a positive residue class

$$s \bmod p.$$

Consider the encoded chain

$$\eta(s), \quad \eta(s + p), \quad \eta(s + 2p), \quad \dots$$

or, if a finite initial segment crosses zero, begin sufficiently far into the positive tail.

Let

$$h(c)$$

denote the number of work symbols above Z_0.

Lemma 8.2 — Tail escape.

For every positive residue class $s \bmod p$,

$$\boxed{h(\eta(s + pk)) \to \infty \qquad (k \to \infty).}$$

The analogous statement holds on every negative residue class modulo q.

Proof.

Suppose infinitely many residuals on the chain were encoded at height at most H.

There are only finitely many physical configurations of bounded height because:

- Q is finite;
- Δ is finite;
- only finitely many stack words have length at most H.

Injectivity of η would then fail.

Therefore the height must escape every finite bound. □

This elementary argument replaces every earlier homogeneous-tail assumption.

No stack-shape hypothesis is available or needed here. The argument forces only unbounded growth along the residual chain.

8.3 The last occurrence of a height creates a stable suffix

Write

$$c_k = \eta(s + pk)$$

and

$$h_k = h(c_k).$$

Because

$$c_{k+1} \xrightarrow{0} c_k,$$

one ordinary PDA transition can remove at most the visible top work symbol.

Hence

$$\boxed{h_{k+1} \leq h_k + 1.}$$

This one-sided height bound, together with tail escape, creates a canonical cross-section.

Lemma 8.3 — Last-height crossing.

For every sufficiently large integer H, the sequence (h_k) has a last occurrence of height H.

If this last occurrence is at index k, then:

1. $$h_{k+1} = H + 1;$$
2. the transition
$$c_{k+1} \xrightarrow{0} c_k$$
is a pure one-symbol pop;
3. every later configuration on that residue-class tail has height $> H$;
4. all later configurations share the same bottom-H work-stack suffix.

Proof.

Tail escape implies that sufficiently large heights are eventually exceeded forever.

Because

$$h_{k+1} \leq h_k + 1,$$

the forward sequence cannot jump upward over a height H.

Thus every sufficiently large H is attained.

Choose its last occurrence.

The next height cannot be at most H, otherwise either H occurs again immediately or a later climb to values above H must pass through H once more.

Therefore

$$h_{k+1} = H + 1.$$

A one-step PDA transition that lowers height from $H + 1$ to H can only delete the visible work symbol. The lower H work symbols are untouched.

After the last height-H occurrence, every later configuration stays above H. Since subsequent transitions rewrite only the visible top region, that bottom-H suffix can never again be exposed or altered.

Hence it stabilizes forever. □

This is the key physical lemma.

No regular tail shape was assumed.

A stable suffix emerged from injectivity and one-top locality alone.

8.4 Positive residue classes synchronize

For each positive residue class

$$s \in \mathbb{Z}_p,$$

choose the last-height-H cross-section.

Let

$$\beta_s \in \Delta^H$$

be the resulting work-stack word above Z_0.

Lemma 8.4 — Positive synchronization.

For sufficiently large H,

$$\boxed{\beta_s = \beta_{s+q}}$$

for every residue class $s \bmod p$.

Consequently all p positive residue classes have the same last-height-H stack word, and their control states must be pairwise distinct.

Hence

$$\boxed{|Q| \geq p}$$

on the positive side.

Proof.

Take sufficiently deep positive configurations so that both source and target lie beyond their last-height-H crossings.

Input 1 maps residual class

$$s$$

to

$$s + q \bmod p.$$

A single top rewrite cannot modify the already stabilized bottom-H suffix.

Therefore

$$\beta_s = \beta_{s+q}.$$

Because

$$\gcd(p, q) = 1,$$

addition by q is one cycle on

$$\mathbb{Z}_p.$$

Thus all β_s are equal.

At the selected height-H cross-sections, the complete stack word is now identical across all p classes.

Since the residuals are distinct and η is injective, their control states must be distinct. □

The negative side is symmetric.

Lemma 8.5 — Negative synchronization.

At sufficiently large synchronized cross-sections on the negative tail,

$$\boxed{|Q| \geq q.}$$

Therefore every nonempty work alphabet already forces

$$\boxed{|Q| \geq \max(p, q).}$$

This is the first shape-free lower bound matching the Chapter 7 construction.

8.5 Why one work symbol is worse

Suppose now

$$g = 1.$$

Write

$$\Delta = \{A\}.$$

At a fixed height H, every work stack is literally

$$A^H Z_0.$$

The positive and negative synchronized cross-sections can therefore be chosen at the same height with exactly the same physical stack.

Lemma 8.6 — Cross-sign unary separation.

If

$$g = 1,$$

the p synchronized positive states and the q synchronized negative states must all be distinct.

Thus

$$\boxed{|Q| \geq p + q.}$$

Proof.

Choose the synchronized positive and negative cross-sections sufficiently far from zero.

All selected configurations have identical stack

$$A^H Z_0.$$

They represent

$$p + q$$

distinct residuals.

Injectivity can distinguish them only through control state.

Hence all $p + q$ states are distinct. □

The reason the second work symbol is powerful is now exact.

With one work symbol, sign cannot be carried by the tail symbol, so positive and negative residue roles cannot share states.

With two work symbols, the stack itself can separate the ends, allowing state reuse.

8.6 The complete raw frontier

Theorem 8.7 — Exact raw residual-faithful frontier.

For every coprime

$$p, q \geq 1,$$

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}}$$

Proof.

If

$$g = 0,$$

the only stack is the permanent bottom marker. With finitely many control states, only finitely many physical configurations exist, so no injective encoding of all

$$r \in \mathbb{Z}$$

is possible.

For

$$g \geq 1,$$

positive and negative synchronization give

$$|Q| \geq \max(p, q).$$

For

$$g = 1,$$

cross-sign unary separation strengthens the bound to

$$p + q.$$

The matching upper constructions are exactly the Chapter 7 machines:

- one work symbol with $p + q$ states;
- two work symbols with $\max(p, q)$ states.

Extra work symbols may simply remain unused. □

This theorem changes the status of several earlier results.

The Chapter 7 homogeneous-tail lower bound was not merely an artifact of homogeneous tails.

The exact same resource points survive after all such physical-shape assumptions are removed.

Thus the improved three-symbol machine discovered from the original DPDA lies on the complete raw residual-faithful frontier.

8.7 Scope of the theorem

The theorem is strong, but its category remains explicit.

It proves exact resource complexity for **raw residual conjugacy**.

It does not prove unrestricted recognition complexity for every DPDA recognizing

$$L_{p,q}.$$

A completely different recognizer may:

- fail to give every residual one stable configuration;
- encode several residuals indirectly through longer physical histories;
- use normalization;
- use a different semantic quotient;
- leave Model R entirely.

Therefore the correct statement is:

$$\boxed{\text{the raw residual-faithful frontier is exact.}}$$

Not:

$$\boxed{\text{the unrestricted DPDA state/stack frontier is solved.}}$$

This distinction becomes even more important in Chapter 9, where the analysis moves beyond residual-faithful representation to a language-level hierarchy problem.

8.8 Acceptance without a right endmarker

In the raw right-endmarker constructions, residual zero is accepted only when the endmarker arrives.

Without a right endmarker, ordinary final-state acceptance requires a state that is final exactly at residual zero.

The synchronized lower-bound states all occur on nonzero residuals, so none may be final.

Corollary 8.8 — Exact no-endmarker raw cost.

Under standard final-state acceptance without a right endmarker,

$$\boxed{S_{\mathrm{raw}}^{\mathrm{noend}}(g) = \begin{cases} \infty, & g = 0, \\ p + q + 1, & g = 1, \\ \max(p, q) + 1, & g \geq 2. \end{cases}}$$

The extra state is an acceptance-wrapper cost.

It does not change the nonzero residual geometry.

This is another place where the two original Chapter 1 variants become conceptually useful: acceptance conventions and semantic core complexity should not be conflated.

8.9 Can the optimum survive bounded pushes?

The raw three-symbol construction may use fixed replacement words longer than two symbols.

Suppose we impose the standard bounded-push convention

$$\max |\alpha| \leq 2.$$

A naive compilation could add many states.

The remarkable fact is that no new state is necessary.

Assume

$$p \geq q.$$

The positive-side writes are already of length at most two because

$$c_s \in \{0, 1\}.$$

Only some negative unary writes are long.

For each negative target phase v, define a family of suffix roles

$$H_{v,j},$$

meaning:

> append exactly j additional unary 1's, then enter stable state R_v.

The total number of such transient roles is exactly

$$\boxed{p - q.}$$

Now inspect the stable three-symbol construction.

Negative configurations use top 1 only in states

$$R_0, \ldots, R_{q-1}.$$

Therefore the state/top cells

$$(R_i, 1), \qquad q \leq i < p,$$

are unused by stable negative semantics.

There are exactly

$$p - q$$

such cells.

The transient suffix roles fit into them perfectly.

Theorem 8.9 — State-preserving bounded-push compiler.

For every coprime p, q, there is a deterministic right-endmarker DPDA whose stable configurations realize the residual graph and satisfy

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3, \qquad \max|\alpha| \le 2.}$$

The only ε-moves are uniquely forced terminating suffix-chain moves.

Proof idea. Split every long unary write into:

1. one input-consuming push of length at most two;
2. a strictly decreasing suffix-completion chain.

The total transient role count is

$$|p - q|.$$

Exactly that many stable state/top cells are unused.

Overlay the transient roles on those cells.

Since the base machine has no consuming transition on the overlaid cells, the forced ε-moves introduce no input/ε conflict.

Every chain terminates and reconstructs exactly the macro-result of the original raw transition.

The case

$$q > p$$

is symmetric. □

This theorem is a useful warning against counting "temporary states" too early.

A transient normalization role need not require a new control state if an unused state/top cell already exists.

8.10 A stronger semantic-visibility budget

Model R asks only for exact residual faithfulness.

We now intentionally demand more.

Fix

$$B \ge \max(p, q)$$

and write every residual as

$$r = s + Bz, \qquad 0 \le s < B.$$

Define the positive and negative phase periods

$$h_+(B) = \frac{p}{\gcd(B, p)},$$

$$h_-(B) = \frac{q}{\gcd(B, q)},$$

and

$$\boxed{H(B) = h_+(B) + h_-(B).}$$

In the **phase-local architecture**, each remainder s must expose its currently active signed quotient phase through:

- the control state;
- one visible work symbol.

Let

$$C$$

be the number of available work symbols.
For one fixed remainder, one state can provide at most C distinct state/top cells.
Therefore $H(B)$ phase roles require at least

$$\left\lceil \frac{H(B)}{C} \right\rceil$$

states.

Lemma 8.10 — Phase capacity.

For each remainder class s,

$$\boxed{|Q_s| \geq \left\lceil \frac{H(B)}{C} \right\rceil .}$$

The B remainder classes are semantically distinct in this architecture.
Summing gives the exact global frontier.

Theorem 8.11 — Exact normalized phase-local frontier.

For

$$B \geq \max(p, q),$$

$$\boxed{Q_{\min}^{\text{phase}}(B, C) = B \left\lceil \frac{\frac{p}{\gcd(B,p)} + \frac{q}{\gcd(B,q)}}{C} \right\rceil .}$$

For

$$C \geq 2,$$

the matching construction can be ε-free and raw.
For

$$C = 1,$$

the same state bound is attained with one forced bottom-cell normalization at the negative-to-zero crossing.
This theorem should not be confused with Theorem 8.7.
Theorem 8.7 asks for exact residual faithfulness only.
Theorem 8.11 additionally requires a chosen signed phase to be locally visible.
A stronger representation contract can require more states.
This is not a contradiction.
It is the beginning of a general principle:

> Resource complexity depends on what physical information the representation is required to expose locally.

8.11 A superlinear exact family

Take

$$p = k^2, \qquad q = k^2 - 1, \qquad B = k(k+1).$$

Then

$$H(B) = 2k - 1.$$

Hence

$$\boxed{Q_{\min}^{\text{phase}}(B, C) = k(k+1) \left\lceil \frac{2k-1}{C} \right\rceil .}$$

For fixed C,

$$Q_{\min}^{\text{phase}} = \Theta(k^3).$$

Since

$$\max(p, q) = \Theta(k^2),$$

this is

$$\boxed{\Theta\Big(\max(p, q)^{3/2}\Big) .}$$

The raw residual-faithful machine still needs only

$$\Theta(\max(p, q))$$

states.

The superlinear growth comes entirely from the extra phase-local visibility requirement.

This example makes category-relative resource claims impossible to ignore.

8.12 Direct lossless factorization: counting semantic cells

A still more structured model keeps the residual split into explicit semantic roles.

The stable roles are:

1. $p + q - 1$ boundary/bottom roles;
2. p deep positive roles with visible top 0;
3. q deep negative roles with visible top 1.

These role classes are disjoint.

Theorem 8.12 — Exact semantic-cell count.

Every exact implementation of this direct lossless unary-tail factorization requires

$$\boxed{2p + 2q - 1}$$

reachable stable state/top cells.

The canonical factorization attains this bound exactly.

The theorem counts semantic cells, not merely control states.

This is the correct unit when the representation contract explicitly requires different local roles to remain distinguishable.

8.13 Exact normalization-rule cost under push-at-most-two

Assume

$$p > q.$$

For each deep negative remainder t, the bounded-push normalization required by input 0 has a net stack-height increase

$$D_t = \left\lfloor \frac{p+t}{q} \right\rfloor - 1.$$

A terminating deterministic ε-computation that gains D persistent stack levels needs at least D distinct persistent-push rules.

Why?

If the same state/top rule created two different persistent levels, the intervening positive-drift ε-macro could repeat forever, contradicting termination.

Different source remainders also cannot share the same persistent rule, because after the common rule the lower context remains hidden and deterministic continuation would be identical, while the semantic target remainder is different.

The arithmetic sum is exact:

$$\sum_{t=0}^{q-1} D_t = p - q.$$

Theorem 8.13 — Exact ε-transition count.

In the direct lossless unary-tail factorization with:

- push-at-most-two;
- right endmarker;

the minimum number of forced normalization rules is

$$\boxed{|\delta_\varepsilon|_{\min} = |p - q|.}$$

Theorem 8.14 — Exact total transition count.

Under the same model,

$$\boxed{|\delta|_{\min} = 4(p+q) - 1 + |p - q|.}$$

Reason. The stable semantic graph forces:

$$4(p+q)-2$$

binary input rules.
The endmarker contributes one accepting rule.
The bounded-push normalization requires exactly

$$|p-q|$$

forced ε-rules.
All three lower bounds are attained simultaneously.
This is the most explicit descriptional-complexity layer so far.
We are no longer counting only states or stack symbols.
We are counting:

- semantic state/top cells;
- binary rules;
- forced normalization rules;
- total transition description size.

8.14 A top-window implementation lemma

For completeness, the following finite-control compilation records how a bounded semantic visibility window can be represented physically when only the current abstract top symbol is rewritten.

Proposition 8.14A — Top-λ annotation compiler.

Suppose an abstract pushdown system inspects the top λ work symbols but rewrites only the current top abstract symbol. If the abstract work alphabet has size g, the same one-step dynamics can be implemented with the same control states by annotating physical stack symbols, using at most

$$\boxed{\sum_{d=1}^{\lambda} g^d}$$

work annotations, plus the bottom marker.

Proof. Annotate each physical symbol by the abstract symbol at that level together with the next at most $\lambda-1$ abstract symbols below it. The visible physical top then contains the entire abstract window needed to choose the transition. Because only the current abstract top is rewritten, the new annotation can be computed locally from the old annotation and the replacement word. The number of nonempty words of lengths 1 through λ over a g-symbol alphabet gives the stated bound. □

Scope. The proposition does not cover an abstract operation that rewrites or pops an entire visible λ-window at once; that stronger operation requires a different compiler.

8.15 Four resource questions, four different answers

We can now place the resource results in a hierarchy.

Level 1 — Raw residual conjugacy

Only the residual semantics is fixed.

$$\boxed{S_{\mathrm{raw}}(g) = \infty,\ p+q,\ \max(p,q).}$$

Level 2 — Bounded-push normalized realization

The optimal three-symbol/state point survives:

$$\boxed{|Q| = \max(p,q), \quad |\Gamma| = 3, \quad \max|\alpha| \le 2.}$$

The price is uniquely forced normalization.

Level 3 — Phase-local architecture

A selected signed phase must be visible from one state/top cell.

$$\boxed{Q_{\min}^{\mathrm{phase}}(B,C) = B\left\lceil \frac{H(B)}{C} \right\rceil.}$$

Level 4 — Direct lossless unary-tail factorization

The semantic roles themselves are fixed and counted.

$$\boxed{C_{\mathrm{sem,min}} = 2p + 2q - 1,}$$

$$\boxed{|\delta_\varepsilon|_{\min} = |p-q|,}$$

$$\boxed{|\delta|_{\min} = 4(p+q) - 1 + |p-q|.}$$

These theorems do not compete.
They answer increasingly structured physical questions.

8.16 What is now genuinely settled

The early DPDA analysis has produced a complete statement:

> Within raw residual conjugacy, the improved three-symbol machine is not merely good—it is on the exact frontier.

The discovery sequence is now complete:

original machine

↓

completion semantics

↓

finite-frontier coordinates

↓

four-symbol normalized lift

↓

exact boundary defect

↓

coordinate shift

↓

three-symbol raw DPDA

↓

complete raw residual-faithful frontier.

At this point the natural resource question becomes more radical.

The entire analysis still assumes that a competing machine faithfully represents our residual semantics.

But a DPDA recognizing the same language has no obligation to use our residual coordinate at all.

It may use a completely different stack representation.

It may use deterministic ε-moves.

It may hide the arithmetic in a way our residual-faithful lower bounds do not see.

So we now ask:

> Can a stack-alphabet lower bound survive against **arbitrary equivalent deterministic pushdown representations**?

That question takes the monograph outside the original counting family and into a previously stated hierarchy problem.

That is the subject of the next research branch.

Chapter 9

From Residual Frontiers to a Representation-Independent Hierarchy Theorem

Research branch — When the Resource Question Leaves Our Representation. *This branch develops the representation-independent hierarchy question that emerges when the raw residual-faithful representation is no longer imposed; its scope is deliberately kept separate from the frontiers of the preceding resource-frontier branch. Related descriptional-complexity work on pushdown alphabets and determinization provides the surrounding context [16, 15, 9].*

The resource story of the previous chapters has reached a natural stopping point.

Within raw residual conjugacy, the frontier is exact:

$$S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}$$

The result is strong because the lower bound no longer assumes homogeneous tails, sign-separated coordinates, or a particular stack formula.

But it still assumes something fundamental:

the competing machine is required to realize the residual graph itself.

A DPDA recognizing the same language need not do that.

It may encode the future language in a completely different way. It may normalize internally. It may use deterministic ε-moves between input symbols. It may never expose anything resembling our residual coordinate.

The next step in the research program was therefore not another attempt to optimize the same representation.

It was to remove the representation from the lower bound.

9.1 The methodological jump

The first two resource stages asked:

> Given the semantic object we have discovered, how efficiently can it be represented?

The new question is different:

> Can the **continuation behavior of the language itself** force a state–stack lower bound, even if an equivalent DPDA chooses an unrelated internal representation?

This distinction is the reason Chapter 8 and the present research branch must not be conflated.

Chapter 8 is about **representation-faithful optimality**.

This research branch is about **arbitrary equivalent recognizers**.

The lower-bound invariant can therefore no longer refer to:

- our residual stack shape;
- our canonical completion word;
- our sign symbol;
- our chosen state phases.

It must be externally visible through the accepted continuation language.

Completion lengths provide exactly such an observable.

9.2 The external hierarchy question

Let

$$\mathcal{D}_{m,k}$$

denote the class of languages recognized by deterministic pushdown automata with at most:

$$m$$

control states and at most

$$k$$

pushdown symbols, in the general deterministic model where deterministic ε-moves are allowed.

The inclusion

$$\mathcal{D}_{m,k} \subseteq \mathcal{D}_{m,k+1}$$

is trivial: an extra stack symbol may simply remain unused.

The question is whether the inclusion is strict.

Masopust established a binary stack-alphabet hierarchy for stateless DPDAs and extended the separation to m-state **realtime** DPDAs for every $m \geq 1$ [16].

The remaining general-model case allows deterministic ε-moves. The theorem proved in this branch establishes the strict hierarchy

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1}}$$

over a fixed binary input alphabet for

$$m \geq 2, \qquad k \geq 1,$$

under the general single-initial-symbol convention used in this branch.

The stateless case $m = 1$ is kept separate because the earlier stateless formulation uses a slightly different initial-store convention.

That model distinction must remain explicit.

9.3 Why realtime separation does not automatically survive ε-moves

A realtime DPDA processes one input symbol per transition.

Its visible state/top pair is therefore synchronized with the input position.

A general DPDA can insert a deterministic ε-computation between two input symbols.

Such an internal computation can:

- rewrite the stack;
- move information between stack and control;
- expose a deeper symbol;
- normalize a temporary encoding;
- alter the local mode before the next symbol is consumed.

Therefore a lower bound based only on what a realtime machine must expose at input boundaries may collapse when the adversary is allowed internal stack work.

This is precisely why the lower bound must reason about **raw computation positions**, including positions inside forced ε-blocks.

The proof developed in this branch will do that.

9.4 Lineage from the original DPDA

The hierarchy theorem is formally self-contained.

It does not depend on the original 2:1 transition table.

Nevertheless, its mathematical route is directly connected to the earlier program.

The progression was:

> original completion-aware DPDA $\rightarrow$ shortest continuation semantics $\rightarrow$ exact state/stack tradeoffs $\rightarrow$ question of arbitrary alternative representations.

The key methodological lesson from the earlier machines was:

> Stack contents can be probed indirectly by the continuations they require.

The hierarchy witness turns that observation into an adversarial lower-bound tool.

Instead of assuming that a competitor stores a residual, we construct many prefix families whose required unary completion lengths grow at pairwise different rates.

Those rates are visible in the language itself.

The competitor may encode them however it wants.

9.5 The improved 2:1 raw DPDA as the immediate precursor

For the motivating ratio $(2, 1)$, Chapter 7 obtained a raw residual-conjugate construction with:

$$|Q| = 2, \qquad \Gamma = \{0, 1, Z_0\},$$

and no ε-normalization on binary residual edges.

This representation is more compact and more literal than the original center/left/right realization for the residual-conjugacy objective.

But that result still does not imply an unrestricted recognition lower bound.

An adversarial DPDA recognizing the same language is not required to use:

- the same two states;
- the same three stack symbols;
- the same residual coordinate;
- any stable residual encoding at all.

This is the conceptual bridge into the hierarchy problem.

9.6 Why the witness development did not end with a direct encoding trick

The route to the final binary hierarchy witness passed through two intermediate prototypes.

Four-letter prototype

The first general prototype used semantic modes

$$(i, t) \in \mathbb{Z}_m \times \mathbb{Z}_k$$

with distinct weights

$$\omega(i, t) = tm + i + 1 \in \{1, \ldots, mk\}.$$

Its purpose was to isolate the product phenomenon:

many externally different completion rates
$\Longrightarrow$ many recurrent state/top roles.

Three-letter prototype

A mixed-radix control symbol merged two semantic navigation coordinates.

This reduced the input alphabet but still retained a separate delimiter/cleanup phase.

Binary witness

A direct binary code of the three-letter system would have required decoder memory.

Under an exact m-state upper budget, that decoder memory is itself a resource and can ruin the comparison.

The final construction therefore changes the invariant instead of coding the alphabet.

Rather than forcing

$$mk \text{ slopes} + 1 \text{ cleanup role},$$

it forces directly

$$\boxed{mk + 1 \text{ distinct completion slopes.}}$$

The cleanup role disappears as a separate semantic mode.
The witness becomes binary without needing a phase decoder.
This is an important design lesson in its own right:

> When an encoding adds resource overhead, change the observable rather than merely recoding symbols.

9.7 The general DPDA model used in the lower bound

A DPDA is

$$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F),$$

with partial transition function

$$\delta : Q \times \Gamma \times (\Sigma \cup \{\varepsilon\}) \to Q \times \Gamma^*.$$

We write the stack top on the right.
Thus if

$$\delta(q, X, a) = (p, \beta),$$

then

$$(q, \gamma X, av) \vdash (p, \gamma\beta, v).$$

No transition is available from an empty stack.
Determinism requires that for each state/top pair:

- there is at most one transition for each input symbol;
- if an ε-transition is enabled, then no input-consuming transition is enabled from that same state/top pair.

Acceptance is by:

$$\boxed{\text{all input consumed} + \text{final state} + \text{empty stack.}}$$

A realtime DPDA is the special case with no ε-moves.
The upper witness will be realtime.
The lower-bound opponent is allowed to be fully general.

9.8 Modes

A nonempty control-store configuration has the form

$$(q, \eta X),$$

where X is the visible top symbol.
Its **mode** is the pair

$$\boxed{[q, X].}$$

A DPDA has at most

$$\boxed{|Q||\Gamma|}$$

nonempty modes.

The product lower bound will ultimately be a mode-counting theorem.

But merely counting modes is not enough.

We must first prove that different externally visible completion slopes cannot share one recurrent mode.

That is the hard part.

9.9 The lower-bound target

For every

$$m \geq 2, \qquad k \geq 1,$$

we will construct a binary language

$$B_{m,k}$$

such that:

1. $B_{m,k}$ is recognized by a realtime DPDA with exactly

   $$m$$

   states and

   $$k+1$$

   stack symbols;

2. every general DPDA recognizing $B_{m,k}$, even with arbitrary deterministic ε-moves, satisfies

   $$\boxed{|Q||\Gamma| \geq mk+1.}$$

The hierarchy corollary will then be immediate.

If a machine used at most m states and at most k stack symbols, then

$$|Q||\Gamma| \leq mk,$$

contradicting the product theorem.

The real work is therefore to build a binary witness exposing exactly

$$mk+1$$

pairwise incompatible continuation slopes.

That construction is the subject of Chapter 10.

Chapter 10

Completion Slopes and ε-Robust Pumping

Fix

$$m \geq 2, \qquad k \geq 1,$$

and write

$$\boxed{\ell = mk + 1.}$$

The witness will have ℓ distinguished live prefix families.

Each family will admit exactly one all-zero completion length after an accumulation block of 1's.

Those lengths will be affine functions with slopes

$$1, 2, \ldots, \ell.$$

The upper machine uses exactly:

$$m \text{ states}, \qquad k + 1 \text{ stack symbols}.$$

The lower bound then asks whether an arbitrary equivalent DPDA can hide those ℓ externally different rates inside fewer than ℓ state/top modes.

It cannot.

10.1 The binary realtime witness

Let

$$Q = \{q_0, \ldots, q_{m-1}\}$$

and

$$\Gamma = \{A, M_0, \ldots, M_{k-1}\}.$$

Hence

$$|Q| = m, \qquad |\Gamma| = k + 1.$$

Define the permutation

$$\pi(0) = 1, \qquad \pi(1) = 0,$$

and

$$\pi(r) = r \qquad (2 \le r < m).$$

Choose ℓ pairwise distinct slope modes

$$x_s = (r_s, X_s), \qquad 0 \le s < \ell,$$

by

$$x_0 = (q_1, A),$$

and, for

$$0 \le r < m, \qquad 0 \le t < k,$$

$$x_{1+rk+t} = (q_{\pi(r)}, M_t).$$

Thus

$$\{x_1, \ldots, x_{\ell-1}\} = Q \times \{M_0, \ldots, M_{k-1}\}.$$

The initial pair

$$(q_0, A)$$

is deliberately not a slope mode.
Define the fixed table word

$$\boxed{T = X_{\ell-1} X_{\ell-2} \cdots X_1 X_0,}$$

with top on the right, so X_0 is initially visible.

10.2 Four rule families

The witness DPDA $U_{m,k}$ is realtime.

Header

$$\boxed{\delta(q_0, A, 1) = (q_1, T).}$$

The first 1 installs the entire finite mode table.

Generic zero-pop on counter symbol

For every state q_i,

$$\boxed{\delta(q_i, A, 0) = (q_i, \varepsilon).}$$

Selector transitions

For

$$0 \le s < \ell - 1,$$

$$\boxed{\delta(r_s, X_s, 0) = (r_{s+1}, \varepsilon).}$$

At the last table mode,

$$\boxed{\delta(r_{\ell-1}, X_{\ell-1}, 0) = (q_0, \varepsilon).}$$

Thus a zero can walk down the installed table.

Slope transitions

For every

$$0 \le s < \ell,$$

$$\boxed{\delta(r_s, X_s, 1) = (r_s, A^{s+1}X_s).}$$

One accumulation 1 inserts exactly $s + 1$ copies of A immediately below the visible mode symbol.

The same mode remains visible.

This is the entire slope mechanism.

10.3 Upper-bound theorem

Proposition 10.1 — Realtime upper construction.

$U_{m,k}$ is a deterministic realtime DPDA with exactly

$$\boxed{m \text{ states}}$$

and

$$\boxed{k + 1 \text{ stack symbols.}}$$

Proof.

There are no ε-moves.

The slope modes are pairwise distinct, so their 1-rules and selector 0-rules do not conflict.

The only apparent overlap is at

$$x_0 = (q_1, A),$$

where the selector rule agrees exactly with the generic A-pop rule.

The header uses

$$(q_0, A, 1),$$

which is not one of the slope modes.

Thus the transition function is deterministic. □

Define

$$\boxed{B_{m,k} = L(U_{m,k}).}$$

The lower bound will not require a closed combinatorial description of the entire language.

One exact slice is enough.

10.4 The exact completion-slope slice

For

$$0 \le s < \ell,$$

define

$$\boxed{N_s(n) = \ell - s + (s+1)n.}$$

Let

$$p_s = 10^s.$$

Lemma 10.2 — Exact pure-zero slice.

For every

$$0 \le s < \ell$$

and every

$$n, r \ge 0,$$

$$\boxed{10^s 1^n 0^r \in B_{m,k} \iff r = N_s(n).}$$

Proof.

After the initial 1, the machine is in mode x_0 with table T.

Each of the next s zeros removes the current table symbol and exposes the next designated slope mode.

Thus after

$$10^s$$

the visible mode is

$$x_s$$

and the remaining table contains exactly

$$\ell - s$$

symbols.

Each subsequent 1 inserts

$$s + 1$$

copies of A below the visible mode symbol.

After n such ones, the total stack height is therefore

$$\ell - s + (s+1)n.$$

Now read zeros.

The first zero removes the visible mode symbol.

Inserted A's are then popped one by one, and whenever the next table symbol is exposed the machine is already in its designated state.

Thus every remaining zero removes exactly one stack symbol.

Because acceptance requires empty stack after the input has been exhausted, the all-zero suffix is accepted exactly when its length equals the current stack height. □

Hence the ℓ prefix families

$$p_s 1^n$$

have unique pure-zero completion slopes

$$\boxed{w_s = s + 1.}$$

So the language exposes:

$$\boxed{1, 2, \ldots, mk + 1}$$

as pairwise distinct affine continuation rates.

10.5 The smallest new case

Take

$$m = 2, \qquad k = 1.$$

Then

$$\ell = 3.$$

The slope modes are

$$x_0 = (q_1, A),$$

$$x_1 = (q_1, M_0),$$

$$x_2 = (q_0, M_0).$$

The exact slice becomes:

$$1\,1^n 0^{3+n},$$

$$10\,1^n 0^{2+2n},$$

$$100\,1^n 0^{1+3n}.$$

The three slopes are:

$$1, \qquad 2, \qquad 3.$$

The theorem of the next chapter will imply that no DPDA with:

$$2 \text{ states}$$

and

$$1 \text{ stack symbol}$$

can recognize the same language—even if that competitor uses deterministic ε-moves.

10.6 The lower-bound opponent is arbitrary

Let

$$M = (Q, \{0,1\}, \Gamma, \delta, q_0, Z_0, F)$$

be any general DPDA with

$$L(M) = B_{m,k}.$$

Fix one slope index s and consider the unique computation on

$$p_s 1^\omega.$$

Every finite prefix

$$p_s 1^n$$

is live because it has the accepted continuation

$$0^{N_s(n)}.$$

Therefore the computation cannot become permanently trapped in an ε-divergence before the next input symbol is consumed.

We call **raw positions** all individual computation boundaries, including positions inside forced ε-blocks.

This is essential.

The proof must not look only at input-boundary configurations.

10.7 Raw continuation principle

At any raw configuration

$$(q, \gamma),$$

determinism fixes the future computation once the unread input is fixed.

If an ε-transition is enabled, it is forced and independent of whether the next unread binary symbol is 0 or 1.

Therefore an equal raw control-store configuration can be reused with a changed future suffix.

This trivial-looking fact is the bridge from equality of physical configurations to equality of continuation behavior.

10.8 No raw configuration can repeat on a live slope ray

Lemma 10.3 — No raw configuration repetition.

Along the computation on

$$p_s 1^\omega,$$

the same full control-store configuration cannot occur twice at distinct raw positions.

Proof.

Suppose first that the two occurrences have consumed different numbers

$$n < n'$$

of accumulation ones.

The later prefix has accepted continuation

$$0^{N_s(n')}.$$

If the earlier and later raw control-store configurations are identical, determinism gives the same future computation from both.

Thus M would also accept

$$p_s 1^n 0^{N_s(n')}.$$

But the exact slice says that after

$$p_s 1^n$$

the unique all-zero completion length is

$$N_s(n),$$

and

$$N_s(n') \neq N_s(n).$$

Contradiction.

If the two repeated raw configurations occur after the same amount of input has been consumed, then the segment between them contains only ε-moves.

Determinism would repeat the same ε-cycle forever, preventing the next live input symbol from being read.

Again contradiction. □

This lemma absorbs arbitrary deterministic internal normalization into the argument.

Repeated physical configurations are impossible because the language exposes a unique continuation length.

10.9 Raw stack height must escape

Lemma 10.4 — Raw-height escape.

Along every slope ray,

$$\boxed{\text{raw stack height tends to infinity.}}$$

More explicitly, for every H, after some raw position the stack height is always greater than H.

Proof.

There are only finitely many control-store configurations of stack height at most H.

By Lemma 10.3, none can occur twice.

Therefore only finitely many raw positions have height at most H. □

The adversary may use any representation it likes.

But the infinitely many distinguishable live prefixes force its physical store to escape every bounded region.

10.10 Suffix-minimum positions

A raw position t is a **step** if its stack height is a minimum over the entire future suffix of the computation:

$$\boxed{h(t) \le h(t') \quad \text{for every } t' \ge t.}$$

Lemma 10.5 — Infinitely many steps.

Every slope ray contains infinitely many step positions.

Proof.

The height sequence tends to infinity.

For every starting position, the future tail has a minimum.

Choose a position where that minimum occurs.

As the starting point moves arbitrarily far to the right, these minima cannot all lie in a fixed finite set. □

Step positions are the correct pumping locations because the computation never descends below their initial stack height.

Therefore the lower stack context remains hidden.

10.11 The ε-robust step pump

Lemma 10.6 — Step-pump lemma.

On every slope ray there exist:

- a mode
$$[q, X];$$
- an integer
$$c \ge 1;$$
- a nonempty word
$$\rho \in \Gamma^{+};$$

such that for every lower context

$$\eta \in \Gamma^{*},$$

$$\boxed{(q, \eta X) \xRightarrow{1^c} (q, \eta\rho X),}$$

where the macro-computation:

1. consumes exactly c input symbols 1;
2. may contain forced ε-moves;

3. never goes below the initial height

$$|\eta X|.$$

The mode $[q, X]$ is called an **up-mode**.

Proof.

There are infinitely many step positions but only

$$|Q||\Gamma|$$

possible modes.

Choose two step positions

$$t_1 < t_2$$

with the same mode

$$[q, X].$$

Write the earlier stack as

$$\eta X.$$

Because t_1 is a suffix minimum, the computation between t_1 and t_2 never goes below height

$$|\eta X|.$$

Therefore the lower prefix η is never exposed.

The later stack has form

$$\eta \rho X.$$

If

$$\rho = \varepsilon,$$

the two positions would have the same full configuration, contradicting Lemma 10.3.

Hence

$$\rho \in \Gamma^+.$$

Let c be the number of input 1's consumed between the two steps.

Because the lower context is never inspected, the same deterministic transition sequence works for every replacement lower context.

Thus the pump is context-independent.

Finally,

$$c \neq 0.$$

If it consumed no input, it would be a height-increasing ε-macro from the same mode back to itself. Determinism would repeat it forever and prevent the next live symbol from being consumed.

Therefore

$$c \geq 1.$$

□

The pump iterates:

$$\boxed{(q, \eta X) \xRightarrow{1^{jc}} (q, \eta \rho^j X) \qquad (j \geq 0).}$$

This is the technical point at which the lower bound becomes genuinely ε-robust.
The proof did not forbid internal normalization.
It found a recurrent growth mechanism inside it.

10.12 The remaining proof obligation

Every slope ray now has an up-mode.

To obtain the product lower bound, we need one final structural fact:

$$\boxed{\text{one up-mode cannot serve two different completion slopes.}}$$

If that holds, then the $\ell = mk + 1$ slopes force ℓ different modes.
The proof requires one more determinism lemma about unary completion.
That is where the next chapter begins.

Chapter 11

One Mode, One Slope

The upper witness exposes

$$\ell = mk + 1$$

different affine completion slopes.

Chapter 10 showed that every arbitrary general DPDA recognizing the witness must, along each live slope ray, develop a context-independent height-increasing up-mode.

We now prove the separation theorem:

$$\boxed{\text{one up-mode} \Longrightarrow \text{at most one completion slope.}}$$

Once this is established, the product bound and the hierarchy follow immediately.

11.1 Unary completion uniqueness

The witness slice uses all-zero continuations.

Determinism makes those continuations rigid.

Lemma 11.1 — Unary completion uniqueness.

Fix any raw control-store configuration

$$(p, \gamma)$$

of a deterministic pushdown automaton in the present model.

There is at most one integer

$$r \geq 0$$

such that input

$$0^r$$

leads from (p, γ) to acceptance.

Proof.

Suppose

$$r < r'$$

and both

$$0^r$$

and

$$0^{r'}$$

were accepted.

The two deterministic computations are identical until the shorter input has consumed all r zeros, including every forced ε-move enabled independently of the presence of further input.

The accepting run on 0^r eventually reaches:

- empty unread input;
- a final state;
- empty stack.

The run on $0^{r'}$ reaches the same empty-stack configuration while

$$r' - r > 0$$

zeros remain unread.

No transition is available from the empty stack.

Thus the longer word cannot be accepted.

Contradiction. □

This is a very small lemma, but it is exactly what converts a recurrent physical mode into a constraint on completion length.

11.2 Slope separation

Lemma 11.2 — Slope separation.

Let

$$s \neq t.$$

The slope rays

$$p_s 1^\omega \qquad \text{and} \qquad p_t 1^\omega$$

cannot have the same up-mode in any DPDA recognizing

$$B_{m,k}.$$

Proof.

Assume both rays share an up-mode

$$[q, X].$$

Take the context-independent pump

$$(q, \eta X) \overset{1^c}{\Longrightarrow} (q, \eta\rho X),$$

with

$$c \geq 1, \qquad \rho \in \Gamma^+.$$

Suppose the mode occurs on the s-ray after n_s accumulation ones with lower context γ_s, and on the t-ray after n_t accumulation ones with lower context γ_t.

After j pump iterations the two rays reach:

$$(q, \gamma_s \rho^j X)$$

and

$$(q, \gamma_t \rho^j X)$$

after respectively

$$n_s + jc$$

and

$$n_t + jc$$

accumulation ones.
The exact witness slice provides accepted zero completions of lengths

$$N_s(n_s + jc)$$

and

$$N_t(n_t + jc).$$

Now run those two zero computations in parallel.
As long as the common upper segment

$$\rho^j X$$

has not been completely removed, the two computations have:

- the same state;
- the same visible active stack suffix;
- the same deterministic local behavior.

The lower contexts γ_s and γ_t remain hidden.
Thus the common upper segment is removed after the same number

$$e_j$$

of consumed zeros.
Immediately after the final active symbol of that segment disappears, the two raw configurations have the form

$$(p_j, \gamma_s) \qquad \text{and} \qquad (p_j, \gamma_t)$$

for the same state p_j.
Because Q is finite, there is an infinite set of indices J on which

$$p_j = p$$

is constant.
For $j \in J$, the remaining accepted zero lengths are

$$R_s(j) = N_s(n_s + jc) - e_j$$

from the fixed raw configuration

$$(p, \gamma_s),$$

and

$$R_t(j) = N_t(n_t + jc) - e_j$$

from

$$(p, \gamma_t).$$

Unary completion uniqueness forces each of these two quantities to be constant on J.
Therefore

$$N_s(n_s + jc) - N_t(n_t + jc)$$

must be constant on J.
But

$$N_s(n) = \ell - s + (s + 1)n,$$

so

$$\begin{aligned} &N_s(n_s + jc) - N_t(n_t + jc) \\ &= (s - t)cj + O(1). \end{aligned}$$

Because

$$s \neq t$$

and

$$c \geq 1,$$

this expression is unbounded along the infinite set J.
Contradiction. □
This lemma is the structural center of the hierarchy proof.
It says that a recurrent state/top mode has an externally measurable asymptotic identity:

$$\boxed{\text{its continuation slope.}}$$

Two different slopes cannot be hidden in the same up-mode.

11.3 The product lower bound

There are

$$\ell = mk + 1$$

slope rays.
Every slope ray has an up-mode.
Distinct slopes require distinct up-modes.
A DPDA with state set Q and stack alphabet Γ has at most

$$|Q||\Gamma|$$

nonempty modes.
Everything is now ready.

Theorem 11.3 — Product lower bound.

Let

$$m \geq 2, \qquad k \geq 1.$$

If a general deterministic pushdown automaton M, with arbitrary deterministic ε-moves allowed, recognizes

$$B_{m,k},$$

then

$$\boxed{|Q_M||\Gamma_M| \geq mk + 1.}$$

Proof.
There are

$$mk + 1$$

slope rays.
By the step-pump lemma, each has an up-mode.
By slope separation, no two slope rays share an up-mode.
Therefore M has at least

$$mk + 1$$

distinct nonempty state/top-symbol modes.
But there are at most

$$|Q_M||\Gamma_M|$$

such modes.
Hence

$$|Q_M||\Gamma_M| \geq mk + 1.$$

□
This is stronger than the adjacent hierarchy statement.
It excludes an entire hyperbolic resource region:

$$\boxed{|Q||\Gamma| \leq mk.}$$

11.4 The hierarchy theorem

Corollary 11.4 — Binary adjacent stack-alphabet hierarchy.

For every

$$m \geq 2, \qquad k \geq 1,$$

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1}.}$$

The strictness is witnessed over the binary alphabet

$$\{0,1\}.$$

Proof.

The realtime witness $U_{m,k}$ uses exactly:

$$m$$

states and

$$k+1$$

stack symbols.

Therefore

$$B_{m,k} \in \mathcal{D}_{m,k+1}.$$

If

$$B_{m,k} \in \mathcal{D}_{m,k},$$

some recognizing DPDA would satisfy

$$|Q| \leq m, \qquad |\Gamma| \leq k.$$

Hence

$$|Q||\Gamma| \leq mk,$$

contradicting Theorem 11.3.

The non-strict inclusion follows by allowing an unused extra stack symbol. □

Thus deterministic ε-moves do not collapse the fixed-state stack-alphabet hierarchy in the stated general model for

$$m \geq 2.$$

11.5 What exactly has been resolved

The bibliographic claim must remain narrow.

The theorem closes the fixed-state stack-alphabet hierarchy case left open after Masopust's realtime result [16], under:

- the general deterministic model of this branch;
- the single-initial-symbol convention;
- fixed state budget
$$m \geq 2;$$
- binary witness alphabet.

The stateless case is not silently folded into the same theorem because its earlier formulation uses a different initial-store convention.

Thus the monograph should say:

$$\boxed{\text{the general-model fixed-state hierarchy is resolved here for } m \geq 2}$$

and separately acknowledge the known stateless result.

11.6 Why this is not a realtime lower bound in disguise

The upper witness is realtime.

The lower-bound opponent is not.

The arbitrary competitor may use deterministic ε-moves throughout.

The proof survives because it deliberately reasons at raw positions.

The chain is:

$$\boxed{\text{exact continuation slice}}$$

$$\Downarrow$$

$$\boxed{\text{no raw configuration repetition}}$$

$$\Downarrow$$

$$\boxed{\text{raw-height escape}}$$

$$\Downarrow$$

$$\boxed{\text{suffix-minimum step pump}}$$

$$\Downarrow$$

$$\boxed{\text{slope separation}}$$

$$\Downarrow$$

$$\boxed{|Q||\Gamma| \geq mk + 1.}$$

The first box is a property of the explicit binary witness.

Every later box concerns an arbitrary equivalent DPDA.

That is why the theorem is representation-independent in a way the earlier residual-frontier results were not.

11.7 Why the product, rather than states or symbols separately, appears

The proof never forces a slope into a unique state alone.

It forces it into a unique **mode**:

$$[q, X].$$

Therefore the natural capacity is:

$$\boxed{|Q||\Gamma|.}$$

This is exactly the resource unit the proof can count.

A machine may trade states against stack symbols.

The product theorem allows that trade.

What it forbids is packing more than one distinct completion slope into one recurrent up-mode.

Thus the proof yields a quantitative state–stack tradeoff rather than only a yes/no alphabet separation.

11.8 Scope and excluded claims

The theorem is deliberately scoped and does not assert the following stronger statements.

Not a full Pareto frontier

The bound

$$|Q||\Gamma| \geq mk + 1$$

does not prove that every point above the hyperbola is attainable.

It excludes one region.

It does not determine the complete state/stack Pareto frontier of the witness.

Not a bounded-push hierarchy theorem

The upper witness installs a fixed table word of length

$$\ell = mk + 1$$

in one header transition.

The general PDA model permits replacement by an arbitrary fixed stack word.

If one imposed a push-length bound, compiling the header would change the resource accounting.

No bounded-push hierarchy claim is made in this theorem; that model is outside the present scope.

Not an unrestricted lower bound for $L_{p,q}$

The witness was engineered so that its continuation slopes are maximally separable.

The theorem does not automatically transfer to the original rational counting languages.

The representation-independent state question for the natural languages $L_{p,q}$ is resolved later by protected phase pumping, which yields the exact unrestricted state bound.

Not a claim that mode arguments are themselves new

State/top modes and state–stack-symbol tradeoffs have a pre-existing literature.

The contribution here is the particular binary witness, the ε-robust slope argument, the product theorem, and the resulting resolution in the stated model.

11.9 The methodological meaning for the monograph

This branch began with a problem created by our own DPDA analysis.

The original machine and its successors showed that the same semantic quantity can be redistributed between:

- control state;
- stack symbol;
- stack height;
- normalization.

That made representation-scoped lower bounds suspicious.
The hierarchy proof answers that suspicion by abandoning the representation.
It asks only what continuations the language forces.
This produces a new lower-bound pattern:

$$\boxed{\begin{array}{c}\text{externally visible continuation rates} \to \text{recurrent physical modes} \\ \to \text{representation-independent resource count.}\end{array}}$$

This method is conceptually distinct from the raw residual frontier.
Both belong in the monograph because they answer different questions.

11.10 Transition to higher-dimensional completion geometry

We have now completed the first major resource arc.
Starting from the 2023 2:1 machine, we obtained:

$$\begin{array}{c}\text{completion semantics} \\ \downarrow \\ \text{geodesic geometry} \\ \downarrow \\ \text{better coordinates} \\ \downarrow \\ \text{exact residual-faithful frontier} \\ \downarrow \\ \boxed{\text{representation-independent hierarchy theorem.}}\end{array}$$

The next research branch changes direction.
Instead of asking how many resources are needed to realize a one-dimensional residual geometry, we ask what happens when the counting constraint itself becomes higher-dimensional.
The binary rectangle will become an optimal Parikh fiber.
The absence of equal-length trade directions will become a kernel-rank condition.
And the first nontrivial equal-length trade will create topology.
We now leave resource complexity and return to completion geometry.

Part II

Completion Geometry, Geodesic Compilation, and Universal Compatibility

---

Part guide. **Question.** What remains of shortest completion after the scalar residual is replaced by a higher-dimensional demand fiber? **Main result.** Completion fibers become finite geodesic objects with exact local compatibility complexes, culminating in full simplicial universality, exact Boolean-lattice-width row complexity, and a common-distance Bellman lift. **Why here.** This is the geometric enlargement of the Part I invariant before the book returns to concrete pushdown normalization. **Dependency.** It uses shortest-completion semantics from Part I and is logically independent of the later WSC machinery.

Chapter 12

When Minimum Demand Becomes a Fiber

The first half of the monograph was built around one scalar residual:

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

That scalar residual had a unique minimum completion Parikh vector

$$\mu(r) = (x_r, y_r).$$

As a consequence, every shortest-completion interval was a rectangle.
The natural higher-dimensional question is:

> What survives if the counting constraint is no longer one scalar equation?

The answer preserves the completion viewpoint but changes the geometry fundamentally.
The minimum demand need no longer be one vector.
It can become a finite **fiber of equally short Parikh vectors**.
That is where higher-dimensional completion geometry begins.

12.1 Matrix counting languages

Let

$$\Sigma = \{a_1, \ldots, a_d\}$$

and let

$$\Psi : \Sigma^* \to \mathbb{N}^d$$

be the Parikh map.
Fix an integer matrix

$$A \in \mathbb{Z}^{k \times d}.$$

Define the vector residual

$$\boxed{\rho(w) = A\Psi(w)}$$

and the matrix counting language

$$\boxed{L_A = \{w : \rho(w) = 0\}.}$$

Residuals lie in the lattice

$$A\mathbb{Z}^d.$$

The binary rational counting family is a rank-one special case.
For

$$L_{p,q},$$

take

$$A = \begin{bmatrix} -p & q \end{bmatrix}.$$

Then

$$A\Psi(w) = qN_1(w) - pN_0(w).$$

Thus the higher-dimensional theory genuinely contains the earlier one.

12.2 Completion totality

A matrix counting language need not allow every residual to be completed by appending letters.

We therefore isolate a convenient hypothesis.

Definition 12.1 — Completion totality.

We call A **completion-total** if there exists

$$h \in \mathbb{N}^d_{>0}$$

such that

$$\boxed{Ah = 0.}$$

The vector h is a strictly positive balanced Parikh vector.
It plays the role of the primitive balanced packet

$$(q, p)$$

from the binary theory, except that it need not generate the entire integer kernel.

Lemma 12.2 — Every lattice residual is completable.

Assume completion totality.
For every

$$r \in A\mathbb{Z}^d,$$

there exists

$$x \in \mathbb{N}^d$$

such that

$$\boxed{Ax = -r.}$$

Proof.

Write

$$r = Az$$

with

$$z \in \mathbb{Z}^d.$$

Because every coordinate of h is positive, for sufficiently large t,

$$th - z \in \mathbb{N}^d.$$

Since

$$Ah = 0,$$

$$A(th - z) = -Az = -r.$$

Thus $th - z$ is a nonnegative completion vector. □

This hypothesis is deliberately strong enough to make completion semantics total on the entire residual lattice.

It lets us generalize the quotient theory cleanly.

12.3 Residuals still classify right quotients

Proposition 12.3 — Matrix residual quotient classification.

Under completion totality, for all prefixes

$$u, v \in \Sigma^*,$$

$$\boxed{u^{-1}L_A = v^{-1}L_A \iff \rho(u) = \rho(v).}$$

Proof.

A continuation z completes u exactly when

$$A\Psi(z) = -\rho(u).$$

Therefore equal residuals give equal quotients.

Conversely, suppose

$$\rho(u) \neq \rho(v).$$

By Lemma 12.2, choose

$$x \in \mathbb{N}^d$$

with

$$Ax = -\rho(u).$$

Choose a word z with

$$\Psi(z) = x.$$

Then

$$uz \in L_A.$$

But if

$$vz \in L_A,$$

we would have

$$Ax = -\rho(v)$$

as well, forcing

$$\rho(u) = \rho(v).$$

Contradiction. □
The semantic lesson from Chapter 4 survives unchanged:

$$\boxed{\text{the residual is an exact coordinate for future right behavior.}}$$

What changes is the geometry of shortest return to residual zero.

12.4 Minimum completion length

For residual

$$r \in A\mathbb{Z}^d,$$

define

$$\boxed{\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\}.}$$

This is the shortest number of appended letters required to cancel r.
In the binary case,

$$\ell_A(r) = \Delta_{p,q}(r).$$

So far the generalization looks formal.
The decisive change appears when we ask which Parikh vectors attain this minimum.

12.5 The optimal Parikh fiber

Definition 12.4 — Optimal Parikh fiber.

Define

$$\boxed{O_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.}$$

In the binary p:q theory,

$$O_A(r)$$

was always a singleton:

$$O_A(r) = \{\mu(r)\}.$$

In higher dimensions, several distinct nonnegative Parikh vectors can complete the same residual at the same minimum length.

Therefore the semantic object

$$\mu(r)$$

must be replaced by the entire set

$$\boxed{O_A(r).}$$

This is the first major qualitative change.

12.6 Equal-length trades

Introduce the augmented matrix

$$\boxed{C = \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix}.}$$

Define the integer equal-length kernel

$$\boxed{K_A = \ker_{\mathbb{Z}} C = \left\{ g \in \mathbb{Z}^d : Ag = 0,\ \mathbf{1}^\top g = 0 \right\}.}$$

A vector

$$g \in K_A$$

changes one completion Parikh vector into another while preserving:

1. residual contribution;
2. total completion length.

Thus K_A is the lattice of **equal-length trades**.

This is the higher-dimensional replacement for the binary kernel direction.

But there is a crucial difference.

For the binary minimum demand, subtracting one nonzero kernel packet always shortened the completion, so two distinct minimum Parikh vectors could not coexist.

In higher dimensions, a nonzero vector can have positive and negative coordinates while still satisfying

$$\mathbf{1}^\top g = 0.$$

Such a trade changes the distribution of letters without changing total length.

That is how an optimal fiber acquires multiple points.

12.7 Affine-semigroup viewpoint

The columns

$$C_i = \begin{bmatrix} A_i \\ 1 \end{bmatrix}$$

generate an affine semigroup.

For residual r, define the augmented optimum element

$$\boxed{b_r = (-r, \ell_A(r)).}$$

Then

$$O_A(r)$$

is exactly the factorization fiber of b_r into the generators C_i.

Every such factorization has the same last coordinate:

$$\ell_A(r).$$

This places the theory next to established affine-semigroup and factorization geometry.

The monograph uses that literature as background.

The new object of interest remains specifically the geometry of **minimum-completion schedules**.

12.8 Continuous optimum geometry is not the same as integer multiplicity

Define the continuous optimum polytope

$$P_r^A = \{x \in \mathbb{R}^d_{\geq 0} : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.$$

Let

$$S(r) = \{i : \exists x \in P_r^A \text{ with } x_i > 0\}$$

be its active support.

Proposition 12.5 — Dimension of the continuous optimum polytope.

If

$$P_r^A \neq \varnothing,$$

then

$$\boxed{\dim P_r^A = |S(r)| - \operatorname{rank}_{\mathbb{R}} \begin{bmatrix} A_{S(r)} \\ \mathbf{1}^\top \end{bmatrix}.}$$

Proof.

For every active coordinate choose a feasible point positive in that coordinate.

The average of these points is positive in every active coordinate and lies in the relative interior of the nonnegative orthant restricted to $S(r)$.

Thus no additional coordinate inequality is active in the relative interior.

The dimension is therefore the number of active variables minus the rank of the affine equality system on those variables. □

The proposition is useful mainly because it prevents a common confusion.

A positive-dimensional real optimum polytope does not automatically imply many integer optimum points.

Conversely, discrete equal-length trade structure can be significant even when a simple continuous dimension count does not capture its combinatorics.

The integer fiber

$$O_A(r)$$

is the primary shortest-completion object.

12.9 The shortest-prefix down-set

In the binary rectangle, a shortest-completion prefix was described by how many required zeros and ones had already been consumed.

For a higher-dimensional optimum vector

$$x \in O_A(r),$$

every coordinatewise subvector

$$0 \le u \le x$$

is the Parikh vector of some prefix of an ordering of that optimum completion.

Define

$$B_x = \{u \in \mathbb{N}^d : 0 \le u \le x\}.$$

Now take the union over all optimum vectors.

Definition 12.6 — Shortest-prefix down-set.

$$\boxed{\mathcal{C}_A(r) = \downarrow O_A(r) = \bigcup_{x \in O_A(r)} B_x.}$$

This is the higher-dimensional replacement for one rectangle.

If the optimum is unique,

$$O_A(r) = \{x\},$$

then

$$\mathcal{C}_A(r) = B_x,$$

a literal coordinate box.

If the optimum fiber contains several vectors, the shortest-prefix region becomes a union of overlapping boxes.

The next theorem says this down-set is not merely a convenient overapproximation.

It is exact.

12.10 Exact geodesic-prefix criterion

Theorem 12.7 — Exact geodesic-prefix criterion.

For every

$$u \in \mathbb{N}^d,$$

$$\boxed{u \in \mathcal{C}_A(r) \iff \ell_A(r + Au) = \ell_A(r) - |u|_1.}$$

Equivalently,

$$\mathcal{C}_A(r)$$

is exactly the set of Parikh vectors that occur as prefixes of shortest completions of residual r.

Proof.

Suppose

$$u \in \mathcal{C}_A(r).$$

Then

$$u \leq x$$

for some

$$x \in O_A(r).$$

The vector

$$x - u$$

completes residual

$$r + Au$$

and has length

$$\ell_A(r) - |u|_1.$$

If a shorter completion existed, appending it after u would produce a completion of r shorter than $\ell_A(r)$.

Therefore

$$\ell_A(r + Au) = \ell_A(r) - |u|_1.$$

Conversely, assume the displayed equality.

Let y be an optimal completion of

$$r + Au.$$

Then

$$A(u + y) = -r,$$

and

$$|u + y|_1 = |u|_1 + \ell_A(r + Au) = \ell_A(r).$$

Thus

$$u + y \in O_A(r),$$

so

$$u \leq u + y.$$

Hence

$$u \in \mathcal{C}_A(r).$$

□

This theorem is the higher-dimensional form of minimum-demand descent.

The difference is that there may be several optimal endpoints x, so the geodesic-prefix set is a union of coordinate boxes rather than one box.

12.11 First actions and compatible action sets

Let e_i denote the i-th unit vector.

Corollary 12.8 — First actions.

Letter type i can begin a shortest completion exactly when

$$\boxed{\ell_A(r + Ae_i) = \ell_A(r) - 1.}$$

More generally, for

$$J \subseteq [d],$$

there exists an optimal completion using every letter type in J at least once if and only if

$$\boxed{\ell_A(r + A\mathbf{1}_J) = \ell_A(r) - |J|.}$$

Thus shortest-action compatibility can be tested intrinsically from the distance function.

This is the higher-dimensional analogue of the binary rule:

$$x_r > 0 \Longleftrightarrow 0 \text{ is geodesic},$$

$$y_r > 0 \Longleftrightarrow 1 \text{ is geodesic}.$$

12.12 The conceptual transition

The completion object has now changed from

$$\boxed{\text{one minimum vector}}$$

to

$$\boxed{\text{a finite optimum fiber.}}$$

Accordingly, the shortest-prefix geometry has changed from

$$\boxed{\text{one rectangle}}$$

to

$$\boxed{\text{a union of boxes.}}$$

But the residual quotient still identifies prefixes that reach the same semantic state.

If two different shortest-prefix Parikh vectors reach the same residual at the same geodesic depth, they must differ by an equal-length trade in

$$K_A.$$

That observation leads to the next object:

$$\boxed{|\mathcal{C}_A(r)|/K_A.}$$

The next chapter will show that this quotient is exactly the filled shortest-path interval.

Chapter 13

The Filled Shortest-Prefix Quotient

Chapter 12 produced two objects:

$$\mathcal{C}_A(r) =\downarrow O_A(r),$$

the exact Parikh down-set of shortest prefixes, and

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix},$$

the lattice of equal-length residual-preserving trades.

We now combine them.

The result is the higher-dimensional completion object that replaces the binary shortest rectangle.

13.1 Why vertices alone are not enough

A shortest prefix is not only a point.

If several distinct next letters can be taken in any order while remaining geodesic, the geometry contains a commuting square, cube, or higher-dimensional cell.

The binary rectangle already had this property.

A pair of actions 0 and 1 produced an elementary commuting square.

In d letters, a compatible set

$$J \subseteq [d]$$

should produce a $|J|$-dimensional cube.

Thus we should fill the shortest-prefix down-set cubically.

13.2 Cubical realization of a down-set

Let

$$D \subseteq \mathbb{N}^d$$

be a finite coordinatewise down-set.

Define

$$|D|$$

to be the cubical complex whose vertices are the elements of D.

For

$$J \subseteq [d],$$

an elementary J-oriented cube based at u is present exactly when

$$\boxed{u + \mathbf{1}_J \in D.}$$

Because D is a down-set, all intermediate vertices of that cube are automatically present.

Applied to

$$D = \mathcal{C}_A(r),$$

every cube records a set of distinct letter actions that can be executed in any order while remaining on shortest paths.

This is the filled schedule geometry before residual identifications.

13.3 Equal-length identifications

Two shortest-prefix Parikh vectors can be different while representing the same reached residual.

Let

$$u, v \in \mathcal{C}_A(r).$$

They reach the same residual exactly when

$$A(u - v) = 0.$$

But because both are shortest prefixes reaching the same residual, they also have the same geodesic depth.

Thus

$$\mathbf{1}^\top (u - v) = 0.$$

Therefore

$$u - v \in K_A.$$

This gives the correct equivalence relation.

Two equally oriented cells are identified when their base vertices differ by an element of

$$K_A.$$

Definition 13.1 — Filled shortest-geodesic quotient.

Define

$$\boxed{\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A.}$$

This quotient keeps:

- every shortest prefix;
- every compatible commuting cell;

but identifies prefixes that reach the same residual at the same shortest depth.

13.4 The quotient theorem

Theorem 13.2 — Shortest-prefix quotient theorem.

The 0-skeleton of

$$\mathfrak{I}_A(r)$$

is label-preserving isomorphic to the complete directed shortest-path interval from residual r to residual 0.

Under this identification, elementary cubes are exactly compatible commuting sets of geodesic one-letter actions.

Proof.

Define

$$\pi_r : \mathcal{C}_A(r) \to A\mathbb{Z}^d$$

by

$$\boxed{\pi_r(u) = r + Au.}$$

By Theorem 12.7, every $u \in \mathcal{C}_A(r)$ is a geodesic prefix, so every image vertex lies on a shortest path from r to 0.

Conversely, every shortest path prefix has some Parikh vector u, and Theorem 12.7 places that vector in $\mathcal{C}_A(r)$.

Now suppose

$$\pi_r(u) = \pi_r(v).$$

Then

$$A(u - v) = 0.$$

Because u and v reach the same residual and both are geodesic,

$$\ell_A(r) - |u|_1 = \ell_A(\pi_r(u)) = \ell_A(\pi_r(v)) = \ell_A(r) - |v|_1.$$

Hence

$$|u|_1 = |v|_1,$$

so

$$\mathbf{1}^\top(u - v) = 0.$$

Therefore

$$u - v \in K_A.$$

Conversely, if

$$u - v \in K_A,$$

then

$$A(u - v) = 0,$$

so the two vertices have the same residual label.

Thus the vertex fibers of π_r are exactly the K_A-classes.

Finally, a J-cube based at u exists exactly when

$$u + \mathbf{1}_J \in \mathcal{C}_A(r).$$

By the geodesic-prefix criterion, every subcollection of the actions in J remains shortest, and coordinate addition commutes.

Passing to the equal-length kernel quotient therefore gives exactly the filled commuting shortest-path structure. □

This theorem is the central bridge from integer completion arithmetic to topology.

13.5 Recovery of the binary rectangle

For the binary scalar system

$$A = \begin{bmatrix} -p & q \end{bmatrix},$$

the augmented matrix is

$$\begin{bmatrix} -p & q \\ 1 & 1 \end{bmatrix}.$$

Its determinant is

$$-(p+q) \neq 0.$$

Therefore

$$\boxed{K_A = \{0\}.}$$

There are no nontrivial equal-length trades.

The optimum fiber is a singleton:

$$O_A(r) = \{\mu(r)\}.$$

Hence

$$\mathcal{C}_A(r) = [0, x_r] \times [0, y_r]$$

and no quotient identifications occur.

Thus

$$\boxed{\mathfrak{I}_A(r) = [0, x_r] \times [0, y_r].}$$

The binary rectangle is therefore the **rank-zero member** of the general theory.

This is an important reinterpretation.

The rectangle did not disappear in higher dimensions.

It became the simplest possible fiber-and-quotient case.

13.6 When is the quotient literally a box?

Theorem 13.3 — Literal-box criterion.

$$\boxed{|O_A(r)| = 1 \iff \mathfrak{I}_A(r) \text{ is a literal coordinate box.}}$$

Here "literal coordinate box" means that the quotient contains no nontrivial kernel identifications among the vertices and cells of the prequotient box.

Proof.

Suppose

$$O_A(r) = \{x\}.$$

Then

$$\mathcal{C}_A(r) = B_x.$$

Assume distinct

$$u, v \leq x$$

were identified.

Then

$$u - v \in K_A.$$

Consider

$$x - u + v.$$

It is nonnegative.

Because

$$A(u - v) = 0,$$

it has the same residual contribution as x.

Because

$$\mathbf{1}^\top (u - v) = 0,$$

it has the same length as x.

It is different from x.

Therefore it would be a second element of

$$O_A(r),$$

contradicting uniqueness.

Hence no nontrivial kernel collision occurs inside the box.

Conversely, suppose

$$x \neq y$$

both belong to

$$O_A(r).$$

Then x and y are distinct maximal vertices of $\mathcal{C}_A(r)$, both reach residual zero, and

$$x - y \in K_A.$$

Thus the quotient identifies distinct shortest schedules.

It cannot be the original literal coordinate box with all coordinate vertices distinct.

□

This theorem identifies the first source of global complexity:

nonunique optimum $\Longleftrightarrow$ nontrivial equal-length identification.

13.7 Unique optimum versus unique shortest word

The phrase "unique optimum" refers to the Parikh vector.

It does not mean there is only one shortest word.

Even when

$$O_A(r) = \{x\},$$

the letters represented by x can be ordered in many ways.

The literal box already records those schedules.

What uniqueness eliminates is a stronger ambiguity:

two different minimum Parikh endpoints.

Thus there are two levels of shortest-path multiplicity.

Ordering multiplicity

One Parikh vector, many permutations.

This produces a box.

Factorization multiplicity

Several distinct minimum Parikh vectors.

This creates kernel identifications among different boxes.

The second phenomenon is new beyond the binary line.

13.8 Local compatibility and global identification are different

The cubical complex

$$|\mathcal{C}_A(r)|$$

records which letter actions can coexist in one optimum and commute as schedule choices.

The quotient

$$\mathfrak{I}_A(r)$$

adds a different phenomenon:

globally different shortest prefixes can reach the same residual.

A later chapter will show that these two notions have different topology.
In rank one:

- every connected component of the local action-compatibility complex is contractible;
- the full global quotient can nevertheless be a circle.

Thus local coexistence and global schedule identification must not be conflated.
We are now ready to study that distinction.

13.9 The rank of equal-length trade

The integer lattice

$$K_A$$

measures the number of independent directions in which one optimal Parikh vector can trade letters for others without changing:

- residual;
- completion length.

Three regimes will become especially important.

Rank zero

$$K_A = \{0\}.$$

No equal-length trade exists.
Every optimum is unique.
The shortest interval is a literal box.
This is the binary p:q regime.

Rank one

One primitive equal-length trade direction exists.
The optimum fiber becomes an arithmetic interval.
The global quotient may acquire its first cycle.

Rank at least two

Several independent trade directions exist.
Even the local action complex can develop holes.
The next chapters will make this hierarchy exact.

Chapter 14

Local and Global Topology

Chapter 13 produced the filled shortest-prefix quotient

$$\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A.$$

Two different kinds of structure are present inside this object.
The first is **local**:

> Which distinct letter types can coexist inside one shortest completion?

The second is **global**:

> After all shortest prefixes are filled cubically and equal-length residual-preserving schedules are identified, what is the topology of the whole shortest interval?

These questions need not have the same answer.
The first genuinely new phenomenon occurs already when the equal-length kernel has rank one.
Locally, nothing topologically complicated happens.
Globally, a circle can appear.
That separation is one of the principal structural results of the higher-dimensional branch.

14.1 The local action complex

Recall the optimal Parikh fiber

$$O_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ |x|_1 = \ell_A(r)\}.$$

Define the simplicial complex

$$\boxed{\mathcal{H}_A(r) = \{J \subseteq [d] : \exists x \in O_A(r) \text{ with } x_j > 0 \text{ for every } j \in J\}.}$$

A face

$$J \in \mathcal{H}_A(r)$$

means that some shortest completion contains at least one copy of every action in J.
Equivalently, by Chapter 12,

$$J \in \mathcal{H}_A(r) \iff \ell_A(r + A\mathbf{1}_J) = \ell_A(r) - |J|.$$

Thus $\mathcal{H}_A(r)$ is the local compatibility complex of geodesic letter types.
It deliberately forgets:

- how many times an action occurs;
- the order in which actions occur;
- which shortest prefixes are globally identified.

It records only coexistence.

14.2 Squarefree-divisor interpretation

Let

$$C_i = \begin{bmatrix} A_i \\ 1 \end{bmatrix}$$

and let

$$S_C = \langle C_1, \dots, C_d \rangle$$

be the affine semigroup generated by the augmented columns.
For residual r, recall

$$b_r = (-r, \ell_A(r)).$$

Proposition 14.1 — Squarefree divisor identification.

$$\boxed{\mathcal{H}_A(r) = \left\{ J \subseteq [d] : b_r - \sum_{j \in J} C_j \in S_C \right\}.}$$

Proof.
The semigroup condition means that there exists

$$y \in \mathbb{N}^d$$

with

$$Cy = b_r - \sum_{j \in J} C_j.$$

Then

$$x = y + \mathbf{1}_J$$

satisfies

$$Cx = b_r.$$

Hence

$$x \in O_A(r)$$

and every coordinate indexed by J is positive.
The converse is obtained by subtracting $\mathbf{1}_J$ from any optimum vector supporting J.
□

Squarefree divisor complexes are established objects in affine-semigroup topology [3].
Their role here is specific:

they are exactly the local shortest-action complexes
of the completion problem.

The topology of $\mathcal{H}_A(r)$ therefore measures local compatibility among geodesic actions.

14.3 Rank zero: one simplex

If

$$K_A = \{0\},$$

the optimum fiber is a singleton

$$O_A(r) = \{x\}.$$

Then

$$\mathcal{H}_A(r)$$

is simply the simplex on

$$\operatorname{supp}(x).$$

Thus rank zero has no nontrivial local topology.
This includes every binary p:q completion problem.
The local action set may have one or two symbols, but it never develops a hole.

14.4 Rank one: the optimum fiber is an arithmetic interval

Assume now

$$K_A = \mathbb{Z}g$$

for a primitive nonzero vector

$$g \in \mathbb{Z}^d.$$

Because the optimum fiber is finite, it has the form

$$\boxed{O_A(r) = \{x_t = x_0 + tg : 0 \le t \le N\}.}$$

For each letter type i, define

$$T_i = \{t \in [0, N] \cap \mathbb{Z} : x_t(i) > 0\}.$$

Since

$$x_t(i) = x_0(i) + tg_i$$

is affine in t, the set T_i is an integer interval, possibly empty.
A set of actions

$$J$$

is locally compatible exactly when

$$\bigcap_{j\in J} T_j \neq \varnothing.$$

Thus the local action complex is the nerve of a finite family of intervals.

Theorem 14.2 — Rank-one local action topology.

If

$$\operatorname{rank}_{\mathbb{Z}} K_A \leq 1,$$

then every connected component of

$$\mathcal{H}_A(r)$$

is contractible.

In particular,

$$\boxed{\widetilde{H}_j(\mathcal{H}_A(r);\mathbb{Z}) = 0 \qquad (j \geq 1).}$$

Proof.

The rank-zero case was just described.

In rank one, $\mathcal{H}_A(r)$ is the nerve of the interval family

$$\{T_i\}.$$

Every connected component of the union of finitely many intervals is itself an interval.

By the nerve theorem [14], each connected component of the nerve is homotopy equivalent to the corresponding connected component of the union.

Hence every connected component is contractible. □

This theorem says that one equal-length trade direction is not enough to create a local homology hole.

Yet it is enough to create a global cycle.

Before proving that, we show that rank two changes the local picture immediately.

14.5 Rank two already permits a local hole

Consider the scalar weight vector

$$\boxed{\omega = (-11, -2, 1, 10)}$$

and residual

$$r = 7.$$

The minimum completion length is 5, and the complete optimum fiber is

$$\boxed{(0,4,1,0), \qquad (1,0,4,0), \qquad (1,3,0,1).}$$

The maximal local action faces are

$$\{2,3\}, \qquad \{1,3\}, \qquad \{1,2,4\}.$$

Therefore

$$f_0 = 4, \qquad f_1 = 5, \qquad f_2 = 1.$$

There is one filled triangle, but one independent graph cycle remains unfilled.
Hence

$$\boxed{H_1(\mathcal{H}_\omega(7);\mathbb{Z}) \cong \mathbb{Z}.}$$

The augmented equal-length kernel has rank two.
This example proves that the rank-one local theorem is sharp.
The first nontrivial local H_1 can appear as soon as two independent equal-length trades are available.

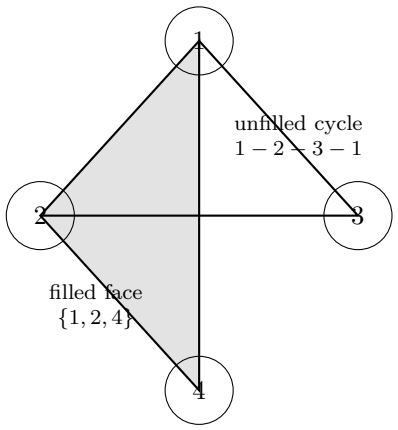


Figure 14.1: The local action complex for $\omega = (-11, -2, 1, 10)$ at residual 7. Its maximal faces are $\{2, 3\}$, $\{1, 3\}$, and the filled triangle $\{1, 2, 4\}$. The cycle $1 - 2 - 3 - 1$ remains unfilled, giving $H_1 \cong \mathbb{Z}$.

The figure should make the distinction visual:

- one triangular face is filled;
- another cycle is not.

This is not a global schedule-identification phenomenon.
The hole is already present in the local coexistence complex.

14.6 Rank-one global geometry

We now return to

$$K_A = \mathbb{Z}g.$$

Write the primitive generator as

$$g = g^+ - g^-,$$

where

$$g^+, g^- \in \mathbb{N}^d$$

have disjoint supports.
Let

$$O_A(r) = \{x_t = x_0 + tg : 0 \le t \le N\}.$$

Define the prequotient shortest-prefix complex

$$X = |\mathcal{C}_A(r)| = \left| \bigcup_{t=0}^{N} B_{x_t} \right|.$$

Because $\mathcal{C}_A(r)$ is a finite coordinate down-set,

$$\boxed{X \simeq *.}$$

So the topology of the final quotient cannot come from a hole already present in the prequotient region.

It must come from the kernel identification.

14.7 Overlap complexes

For

$$m \geq 1,$$

define

$$P_m = \{\sigma \subseteq X : \sigma + mg \subseteq X\},$$

where translation preserves cube orientation.

Lemma 14.3 — Exact m-step overlap.

For

$$1 \leq m \leq N,$$

translation by

$$-mg^-$$

identifies P_m with the anchored cubical down-set

$$\boxed{\bigcup_{t=0}^{N-m} B_{x_t - mg^-}.}$$

Hence P_m is nonempty and contractible.

For

$$m > N,$$

$$\boxed{P_m = \varnothing.}$$

Interpretation. The complete optimum fiber has parameter span

$$0, \ldots, N.$$

Two shortest-prefix cells can overlap after an mg translation exactly while that translation stays inside the span.

Thus the global quotient glues only finitely many neighboring translates.

The overlap itself remains topologically simple.

14.8 Periodic saturation

Define the periodic saturation

$$\boxed{\widetilde{X} = \bigcup_{k\in\mathbb{Z}} (X + kg).}$$

Let

$$X_k = X + kg.$$

Translation by g acts freely on $\widetilde{X}$.
Moreover,

$$\boxed{\widetilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r).}$$

The quotient by the integer translation action performs exactly the rank-one equal-length kernel identification.
To understand $\widetilde{X}$, we study the cover

$$\{X_k\}_{k\in\mathbb{Z}}.$$

Lemma 14.4 — Good-cover intersections.
Let

$$k_0 < \cdots < k_s.$$

Then

$$\boxed{X_{k_0} \cap \cdots \cap X_{k_s} \neq \varnothing \iff k_s - k_0 \leq N.}$$

Every nonempty finite intersection is contractible.

Proof idea. For any fixed cell, the set of translate indices k for which the cell lies in X_k is an integer interval.
Thus membership in the two extreme translates implies membership in all intermediate translates.
After translating by $-k_0 g$, the intersection reduces to the $(k_s - k_0)$-step overlap from Lemma 14.3.
That overlap is contractible exactly when the index gap is at most N. □
The nerve of this cover has vertices $\mathbb{Z}$.
A finite set of vertices spans a simplex exactly when its diameter is at most N.
For

$$N \geq 1,$$

this is the nerve of the interval cover

$$\{[k, k+N]\}_{k\in\mathbb{Z}}$$

of the real line.
Hence the nerve is contractible.

Theorem 14.5 — Contractible rank-one saturation.
If

$$N \geq 1,$$

then

$$\boxed{\widetilde{X} \simeq *.}$$

The periodic saturation is therefore a contractible universal-cover candidate.

14.9 The global circle

Theorem 14.6 — Rank-one global homotopy dichotomy.

If

$$\operatorname{rank}_{\mathbb{Z}} K_A = 1,$$

then

$$\boxed{\mathfrak{I}_A(r) \simeq \begin{cases} *, & |O_A(r)| = 1, \\ S^1, & |O_A(r)| > 1. \end{cases}}$$

Proof.

If the optimum is unique, Chapter 13 gives a literal coordinate box, hence a contractible complex.

Now suppose

$$|O_A(r)| > 1.$$

Then

$$N \geq 1.$$

By Theorem 14.5,

$$\widetilde{X}$$

is contractible.

The action of

$$\mathbb{Z}$$

by translation through g is free, properly discontinuous, and cellular.

Therefore

$$\widetilde{X} \to \widetilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r)$$

is a universal covering with deck group

$$\mathbb{Z}.$$

Hence

$$\pi_1(\mathfrak{I}_A(r)) \cong \mathbb{Z}$$

and all higher homotopy groups vanish.

Thus

$$\mathfrak{I}_A(r)$$

is a $K(\mathbb{Z}, 1)$, hence homotopy equivalent to

$$S^1.$$

□

This gives the first exact topological transition in the completion theory:

rank one + nonunique optimum $\Longrightarrow$ global circle.

14.10 Local triviality, global nontriviality

Combining Theorems 14.2 and 14.6 gives the sharp distinction:

rank-one local action complex: every component contractible
rank-one global shortest-prefix quotient: possibly S^1.

The circle is not caused by a local incompatibility hole.

It is caused by globally identifying different shortest schedules through an equal-length trade.

This difference will matter later when the monograph returns to normalization schedules.

Local feasibility and global identification are fundamentally different layers of structure.

14.11 What the topology means operationally

A point of

$$\mathcal{C}_A(r)$$

records a shortest-prefix Parikh vector.

A path through the cubical complex records gradual execution of a shortest schedule.

Translation by g changes the Parikh bookkeeping without changing:

- residual;
- geodesic depth.

The quotient therefore says that two apparently different shortest schedule histories have reached the same semantic point.

When one primitive trade direction exists, those identifications wrap the contractible periodic saturation around one generator.

That is the geometric origin of the circle.

The result is not merely "some complex has homology."

It is a statement about the topology created by schedule equivalence.

The next chapter turns kernel rank into an exact alphabet-size hierarchy for scalar counting languages.

Chapter 15

Scalar Rank Hierarchy and Exact Three-Letter Geometry

The previous chapter expressed the new topology in terms of

$$\operatorname{rank}_{\mathbb{Z}} K_A.$$

For a general matrix A, this rank is an independent structural parameter.
For a scalar counting language, it has a remarkably simple form.
This produces a sharp hierarchy by alphabet size.

15.1 Scalar counting systems

Let

$$\omega = (\omega_1, \ldots, \omega_d) \in \mathbb{Z}^d$$

have at least two distinct entries.
Treat

$$A = \omega$$

as a $1 \times d$ matrix.
The equal-length kernel is

$$K_\omega = \ker_{\mathbb{Z}} \begin{bmatrix} \omega \\ \mathbf{1}^\top \end{bmatrix}.$$

The two rows are linearly independent over $\mathbb{R}$.
Therefore:

$$\boxed{\operatorname{rank}_{\mathbb{Z}} K_\omega = d - 2.}$$

This single formula turns the topology of Chapter 14 into an alphabet-size threshold.

15.2 The $d = 2, 3, 4+$ hierarchy

Two letters

If

$$d = 2,$$

then

$$\operatorname{rank} K_\omega = 0.$$

Therefore:

- every optimum Parikh vector is unique;
- every shortest interval is a literal coordinate box;
- no equal-length schedule identification occurs.

This is exactly the binary theory of Chapters 2–5.

Three letters

If

$$d = 3,$$

then

$$\operatorname{rank} K_\omega = 1.$$

Therefore:

- the optimum fiber is an arithmetic interval;
- every connected component of the local action complex is contractible;
- the global quotient is either contractible or homotopy equivalent to

$$S^1.$$

Thus three letters are the first scalar alphabet size capable of creating a global shortest-schedule cycle.

Four or more letters

If

$$d \geq 4,$$

then

$$\operatorname{rank} K_\omega \geq 2.$$

Higher-rank equal-length trade geometry becomes possible.
The four-letter example of Chapter 14 already gives

$$H_1(\mathcal{H}; \mathbb{Z}) \cong \mathbb{Z}.$$

Thus four letters are the first scalar alphabet size at which a local shortest-action hole can occur.

The hierarchy is therefore:

d	shortest-completion geometry
2	rank 0: literal boxes
3	rank 1: first global S^1
≥ 4	higher rank: local holes possible

This gives a structural interpretation of alphabet size that is invisible in the original binary problem.

15.3 Three-letter normal form

Assume

$$\omega_1 < \omega_2 < \omega_3.$$

Define the adjacent gaps

$$a = \omega_3 - \omega_2, \qquad b = \omega_2 - \omega_1.$$

Let

$$h = \gcd(a, b),$$

and set

$$\alpha = \frac{a}{h}, \qquad \beta = \frac{b}{h}, \qquad m = \alpha + \beta.$$

A primitive equal-length kernel generator is

$$\boxed{g = (\alpha, -m, \beta).}$$

Indeed,

$$\alpha - m + \beta = 0,$$

and

$$\alpha\omega_1 - m\omega_2 + \beta\omega_3 = 0.$$

Thus one equal-length trade replaces m copies of the middle letter by:

- α copies of the first;
- β copies of the third.

Or vice versa.
This is the exact three-letter analogue of a primitive commuting schedule trade.

15.4 Parameterizing the optimum fiber

Write the complete optimum fiber as

$$\boxed{O_\omega(r) = \{x(t) = y + tg : 0 \le t \le N\},}$$

where

$$y = (y_1, y_2, y_3).$$

Thus

$$x(t) = (y_1 + \alpha t,\ y_2 - mt,\ y_3 + \beta t).$$

Let

$$z = y_2 - mN, \qquad 0 \le z < m.$$

Define

$$P_t = (y_1 + \alpha t + 1)(y_3 + \beta t + 1).$$

Slicing the prequotient down-set by the middle coordinate gives an exact vertex count before quotienting:

$$\boxed{|\mathcal{C}_\omega(r)| = m \sum_{t=0}^{N-1} P_t + (z+1)P_N.}$$

The kernel quotient then collapses contiguous segments along the g-direction.

Because every rank-one orbit intersects the down-set in an integer interval, the quotient can be counted by:

$$\text{vertices} - \text{adjacent orbit pairs}.$$

15.5 Exact three-letter quotient vertex count

Proposition 15.1 — Exact rank-one quotient vertex count.

Let

$$\Omega(y, N)$$

be the number of vertex orbits in

$$\mathcal{C}_\omega(r)/\mathbb{Z}g.$$

If

$$N = 0,$$

then

$$\boxed{\Omega(y, 0) = (y_1 + 1)(y_2 + 1)(y_3 + 1).}$$

If

$$N \geq 1,$$

then

$$\boxed{\Omega(y, N) = mP_{N-1} + (z+1)(P_N - P_{N-1}).}$$

Proof idea. Every kernel orbit meets the down-set in a finite integer interval.

An orbit containing s vertices contributes exactly $s - 1$ adjacent pairs

$$u,\ u + g.$$

Therefore:

$$\#\text{orbits} = \#\text{vertices} - \#\text{adjacent pairs}.$$

The same middle-coordinate slicing gives the adjacent-pair count, and subtraction yields the formula. □

The result is an exact finite formula for the number of semantic shortest-prefix states after equal-length schedule identification.

15.6 Exact cubical f-vector

The same reduction works separately for every cube orientation.

For

$$J \subseteq \{1,2,3\},$$

define

$$T_J = \{t : x_i(t) \geq 1 \text{ for every } i \in J\}.$$

Because each coordinate is affine in t,

$$T_J$$

is an integer interval, possibly empty.

If nonempty, write

$$T_J = [L_J, U_J] \cap \mathbb{Z},$$

set

$$N_J = U_J - L_J,$$

and define

$$y^{(J)} = x(L_J) - \mathbf{1}_J.$$

Proposition 15.2 — Exact rank-one cubical f-vector.

The number of quotient cubes oriented by J is

$$\boxed{f_J(r) = \begin{cases} 0, & T_J = \varnothing, \\ \Omega(y^{(J)}, N_J), & T_J \neq \varnothing. \end{cases}}$$

Hence

$$\boxed{f_j(r) = \sum_{|J|=j} f_J(r), \qquad 0 \leq j \leq 3.}$$

Thus the entire finite cubical shortest-interval geometry is computable from one rank-one orbit formula.

15.7 Worked example

Take

$$\boxed{\omega = (-5,-3,1), \qquad r = 9.}$$

Then:

$$a = 4, \qquad b = 2, \qquad h = 2,$$

so

$$\alpha = 2, \qquad \beta = 1, \qquad m = 3.$$

The primitive equal-length trade is

$$\boxed{g = (2, -3, 1).}$$

The complete optimum fiber is

$$\boxed{O_\omega(9) = \{(0, 3, 0), (2, 0, 1)\}.}$$

Thus the minimum completion has two distinct Parikh realizations of the same length:

$$(0, 3, 0)$$

and

$$(2, 0, 1).$$

The quotient cubical f-vector is

$$\boxed{(f_0, f_1, f_2, f_3) = (8, 10, 2, 0).}$$

Therefore the filled quotient contains:

- 8 vertices;
- 10 labeled edges;
- 2 elementary commuting squares;
- no 3-cube.

Yet by Theorem 14.6,

$$\boxed{\mathfrak{I}_\omega(9) \simeq S^1.}$$

This is an ideal example for the monograph because the topology is nontrivial while the finite complex remains small enough to draw completely.

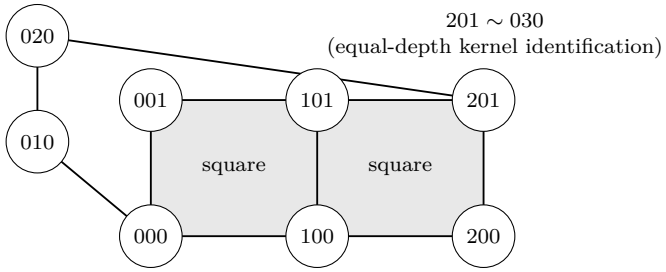


Figure 15.1: The complete rank-one quotient for $\omega = (-5, -3, 1)$ and $r = 9$. The down-set below $(2, 0, 1)$ contributes two filled squares; the path below $(0, 3, 0)$ shares the origin, and its far endpoint is identified with $(2, 0, 1)$ by the equal-length kernel translation. The quotient has $(f_0, f_1, f_2, f_3) = (8, 10, 2, 0)$ and is homotopy equivalent to S^1.

15.8 What changed relative to the binary rectangle?

In two letters, the minimum completion is one Parikh vector.

Different shortest words differ only by ordering.

In three letters, there can be two different minimum Parikh vectors:

$$x_0 \qquad \text{and} \qquad x_0 + g.$$

The prequotient geometry contains boxes beneath both.

The quotient then identifies equal-depth prefixes differing by g.

That is what creates the global circle.

Thus the transition

$$d = 2 \to d = 3$$

does not merely add another coordinate.

It adds the first equal-length trade direction.

And the transition

$$d = 3 \to d = 4$$

does not merely add one more trade.

It permits two independent trades, enough for local action topology itself to develop a hole.

15.9 Why this hierarchy belongs in the same monograph

The scalar alphabet-size hierarchy may look distant from the original 2:1 DPDA.

Its origin is nevertheless direct.

The original machine led to:

$$\text{shortest completion} \to \text{minimum Parikh demand} \to \text{rectangle}.$$

The higher-dimensional question was:

> Was the rectangle an accident of the binary alphabet?

The answer is no.

It was the rank-zero case of a general equal-length trade geometry.

The binary theory therefore sits at the first level of a topological hierarchy.

The next chapter studies a different direction: not how geometry changes with alphabet size, but how it behaves far out along a residual ray.

Chapter 16

Eventual Geodesic Arithmetic

The previous chapters studied the exact shortest-completion geometry of one fixed residual.

We now vary the residual systematically.

Fix two lattice residuals

$$r, b \in A\mathbb{Z}^d$$

and consider the ray

$$\boxed{r_n = r + nb, \qquad n \in \mathbb{N}.}$$

The minimum completion length

$$\ell_A(r_n)$$

need not be exactly linear in n.

Integer constraints can create periodic corrections.

The main result of this chapter is that those corrections eventually stabilize into a finite defect profile.

The same finite mechanism controls both:

- numerical shortest-completion distance;
- local shortest-action geometry.

16.1 The continuous directional slope

Consider the linear program

$$\boxed{\lambda_b = \min\{\mathbf{1}^\top x : Ax = -b,\ x \in \mathbb{R}^d_{\geq 0}\}.}$$

Completion totality guarantees feasibility.

Because the data are rational,

$$\lambda_b \in \mathbb{Q}_{\geq 0}.$$

The number

$$\lambda_b$$

is the continuous asymptotic completion cost per unit step in direction b.

If integer effects disappeared completely, one would expect

$$\ell_A(r + nb) \approx n\lambda_b + \text{constant}.$$

The eventual defect theory measures the exact failure of this naive linear law.

16.2 Dual certificate and reduced costs

Choose a rational dual optimum

$$y^*$$

for

$$\max\{-b^\top y : A^\top y \le \mathbf{1}\}.$$

Thus

$$-b^\top y^* = \lambda_b.$$

Choose an integer

$$L > 0$$

clearing all denominators needed below.
Define the reduced costs

$$\boxed{\delta_i = L(1 - A_i^\top y^*).}$$

Dual feasibility gives

$$\delta_i \in \mathbb{N}.$$

If

$$x_n$$

is any optimal integer completion of r_n, then

$$Ax_n = -r_n.$$

Therefore

$$\begin{aligned} L\big(\ell_A(r_n) + (y^*)^\top r - n\lambda_b\big) &= L\left(\mathbf{1}^\top x_n - (y^*)^\top A x_n\right) \\ &= \sum_i \delta_i x_n(i). \end{aligned}$$

This motivates the exact integer defect.

Definition 16.1 — Scaled integrality defect.

$$\boxed{\Delta_r(n) = L\big(\ell_A(r_n) + (y^*)^\top r - n\lambda_b\big).}$$

Then

$$\boxed{\Delta_r(n) \in \mathbb{N}.}$$

The defect is not an asymptotic error term.
It is an exact nonnegative integer reduced-cost total.
That arithmetic discreteness is what makes eventual stabilization possible.

16.3 Zero-defect shifts

Define

$$\boxed{\mathcal{S}_b = \left\{ t \geq 1 : \exists z \in \mathbb{N}^d,\ Az = -tb,\ |z|_1 = t\lambda_b \right\}.}$$

An element

$$t \in \mathcal{S}_b$$

is a direction length for which the continuous optimum is attained integrally with zero defect.

Lemma 16.2 — Zero-defect semigroup.

$$\mathcal{S}_b$$

is nonempty and closed under addition.

Proof sketch. A rational primal optimum for direction b becomes integral after clearing denominators, giving at least one zero-defect multiple.

If

$$t_1, t_2 \in \mathcal{S}_b$$

with witnesses z_1, z_2, then

$$z_1 + z_2$$

satisfies

$$A(z_1 + z_2) = -(t_1 + t_2)b$$

and

$$|z_1 + z_2|_1 = (t_1 + t_2)\lambda_b.$$

Thus

$$t_1 + t_2 \in \mathcal{S}_b.$$

□

Define the canonical arithmetic period

$$\boxed{\tau_b = \gcd \mathcal{S}_b.}$$

A zero-defect shift by any $t \in \mathcal{S}_b$ can be appended to an existing completion without increasing the defect.

Thus $\mathcal{S}_b$ transports good integer behavior forward along the ray.

16.4 Stabilized defect profile

For each residue class

$$s \in \mathbb{Z}/\tau_b\mathbb{Z},$$

define

$$\boxed{M_s = \min\{\Delta_r(n) : n \geq 0,\ n \equiv s \pmod{\tau_b}\}.}$$

Theorem 16.3 — Stabilized defect profile.
For every residue class s, there exists

$$N_s$$

such that

$$\boxed{\Delta_r(n) = M_s}$$

whenever

$$n \geq N_s$$

and

$$n \equiv s \pmod{\tau_b}.$$

Consequently, eventually,

$$\boxed{\ell_A(r_n) = -(y^*)^\top r + n\lambda_b + \frac{M_{n \bmod \tau_b}}{L}.}$$

Proof.
Choose finitely many elements

$$t_1, \ldots, t_m \in \mathcal{S}_b$$

whose gcd is

$$\tau_b.$$

The additive numerical semigroup they generate contains every sufficiently large multiple of τ_b.
For each residue class s, choose an index n_s attaining the minimum defect M_s.
If

$$n \equiv s \pmod{\tau_b}$$

is sufficiently large, then

$$n - n_s$$

is a nonnegative combination of the t_i.
Append the corresponding zero-defect completions to an optimum at n_s.
This gives a feasible completion at n with the same defect M_s.
By definition of M_s, the defect cannot be smaller.
Hence it is exactly M_s. □

The formula has a useful interpretation:

$$\boxed{\text{eventual distance} = \text{rational linear drift} + \text{finite periodic integer defect.}}$$

The eventual arithmetic is therefore controlled by a finite cyclic table.

16.5 Why τ_b already contains the slope denominator

Write

$$\lambda_b = \frac{a}{q}$$

in lowest terms.
For every

$$t \in \mathcal{S}_b,$$

the zero-defect completion has integer length

$$t\lambda_b.$$

Hence

$$q \mid t.$$

Therefore

$$\boxed{q \mid \tau_b.}$$

The canonical zero-defect period is thus automatically compatible with the denominator of the asymptotic slope.
But the **minimal eventual affine period** can be smaller than τ_b.
The remaining reduction comes from symmetry of the stabilized defect table.

16.6 Fundamental defect period

View

$$M = (M_0, \dots, M_{\tau_b - 1})$$

as a cyclic word.
Define its fundamental circular period

$$\boxed{d_M = \min\{p \geq 1 : M_{s+p} = M_s \text{ for every } s\}.}$$

Then

$$d_M \mid \tau_b.$$

We can now determine the exact eventual distance period.

Theorem 16.4 — Exact minimal eventual distance period.

Let

$$\lambda_b = \frac{a}{q}$$

in lowest terms.
The smallest positive integer p for which there exists an integer D_p satisfying

$$\ell_A(r_{n+p}) = \ell_A(r_n) + D_p$$

for all sufficiently large n is

$$\boxed{p_{\min}(r, b) = \operatorname{lcm}(q, d_M).}$$

Moreover,

$$\boxed{p_{\min}(r, b) \mid \tau_b,}$$

and

$$\boxed{D_p = p\lambda_b.}$$

Proof.
Suppose p is an eventual affine period.
Comparing the linear terms in Theorem 16.3 forces

$$D_p = p\lambda_b.$$

Since D_p is integral,

$$q \mid p.$$

The periodic correction term must also be invariant under shift by p, so

$$d_M \mid p.$$

Therefore

$$\operatorname{lcm}(q, d_M) \mid p.$$

Conversely, let

$$p_0 = \operatorname{lcm}(q, d_M).$$

Then

$$p_0\lambda_b$$

is integral and the stabilized defect table is invariant under shift by p_0.
The formula of Theorem 16.3 gives

$$\ell_A(r_{n+p_0}) = \ell_A(r_n) + p_0\lambda_b$$

for all sufficiently large n.
Thus p_0 is the minimal eventual affine period. □
This theorem separates two arithmetic sources of periodicity:

1. the denominator of the continuous slope;
2. the intrinsic symmetry period of the integer defect profile.

The actual distance period is their least common multiple.

16.7 Eventual local action geometry

The same mechanism controls local shortest-action compatibility.

For

$$J \subseteq [d],$$

let

$$c_J = A\mathbf{1}_J.$$

Recall:

$$J \in \mathcal{H}_A(r_n) \iff \ell_A(r_n + c_J) = \ell_A(r_n) - |J|.$$

Apply Theorem 16.3 not only to the ray

$$r_n$$

but also to every finitely many shifted rays

$$r_n + c_J.$$

The same:

- dual optimum;
- zero-defect semigroup;
- canonical period τ_b;

can be used.
Only the stabilized defect table changes with J.
Therefore the truth value of

$$J \in \mathcal{H}_A(r_n)$$

is eventually determined by

$$n \bmod \tau_b.$$

Let

$$\Theta_s$$

be the stabilized local action complex on residue class s.
Define

$$\boxed{d_{\mathcal{H}} = \min\{p \geq 1 : \Theta_{s+p} = \Theta_s \text{ for every } s\}.}$$

Corollary 16.5 — Exact minimal local-geometry period.

The smallest eventual period of the local action complex is

$$\boxed{p_{\min}^{\mathcal{H}}(r, b) = d_{\mathcal{H}}.}$$

Moreover,

$$\boxed{d_{\mathcal{H}} \mid \tau_b.}$$

The first-action set is eventually periodic for the same reason.

Thus one finite defect mechanism governs both:

$$\boxed{\text{numerical shortest distance}}$$

and

$$\boxed{\text{local shortest-action geometry.}}$$

16.8 What is and is not being claimed

Eventual periodicity in semilinear and parametric integer settings is not new in itself.

The point here is more specific.

The completion program produces one exact object:

$$\Delta_r(n).$$

From it we derive:

- eventual affine distance;
- exact minimal distance period;
- eventual local action complex;
- exact minimal local-geometry period.

Thus the numerical and geometric observables are not proved periodic by unrelated arguments.

They descend from the same minimum-completion defect profile.

This is the structural contribution relevant to the monograph.

16.9 The higher-dimensional branch in one picture

The path from the original binary machine to the present theorem is now complete:

$$\boxed{\begin{array}{c}\text{original 2:1 shortest completion}\\ \downarrow\\ \text{unique minimum Parikh demand}\\ \downarrow\\ \text{binary rectangle}\\ \downarrow\\ \text{matrix residual}\\ \downarrow\\ \text{optimal Parikh fiber}\\ \downarrow\\ \text{shortest-prefix down-set}\\ \downarrow\\ \text{equal-length kernel quotient}\\ \downarrow\\ \text{rank-sensitive local/global topology}\\ \downarrow\\ \text{eventual finite defect profile.}\end{array}}$$

The binary rectangle was therefore not a small isolated trick.
It was the rank-zero member of a much larger completion geometry.

16.10 From completion geometry to finite compilation

The higher-dimensional branch has now identified optimal fibers, shortest-prefix down-sets, local action complexes, quotient topology, and eventual arithmetic. The next step is not another special counting matrix. It is to ask which part of this shortest-path structure can be compiled from finite periodic data and how much information is carried by higher-order compatibility among commuting optimal actions.

This leads to a general finite-phase periodic model. The metric backend is classical periodic shortest-path arithmetic; the new layer developed in the next chapters compiles exact optimal compatibility of every structural order and then proves that no fixed lower-order truncation universally reconstructs the full local compatibility complex.

Chapter 17

From Completion Geometry to Finite Geodesic Compilers

Research branch — Geodesic Compilers and Higher-Order Optimal Structure. *This research branch abstracts the shortest-completion geometry of the preceding chapters to positive-cost finite-phase periodic action systems. Periodic minimum-cost paths, semilinear and Presburger descriptions, shortest-path preprocessing, cubical state complexes, and squarefree divisor complexes are established background; the contribution is an exact finite compiler for all-order optimal compatibility together with prescribed-order separation, Boolean amplification, and universal k-fiber realization.*

The preceding completion-geometry chapters started from a fixed linear counting system and studied the geometry of its shortest completions. We now isolate the mechanism that survived those examples. The resulting model is a finite-phase periodic shortest-path system: a finite phase graph carries integral translations and positive costs, while the lattice coordinate remains unbounded. This abstraction retains the arithmetic structure needed for exact completion queries but no longer assumes a particular pushdown implementation or a particular counting matrix.

The background/contribution boundary matters here. Minimum-cost paths in periodic graphs, semilinear and Presburger descriptions, and shortest-path preprocessing are established theories [70, 68, 69, 73, 76]. The contribution of this branch is their organization into an exact finite compiler for geodesic branching and all-order optimal compatibility, followed by a structure theory showing that the compiled compatibility hierarchy is universally non-collapsing across the class.

In the dependency chain of this monograph, the abstraction is reached from the earlier chapters as

$$\begin{array}{c}\text{completion-aware 2:1 DPDA} \longrightarrow \rho_{p,q} = qN_1 - pN_0 \\ \longrightarrow \text{ minimum completion } \longrightarrow \text{ geodesic rectangles and optimal fibers} \\ \longrightarrow \text{ finite-phase periodic geodesic compilation.}\end{array}$$

The original machine is therefore not a hypothesis of the results below; it records the concrete route by which the compiler abstraction was found.

17.1 Finite-phase periodic graphs

Let

$$\mathcal{B} = (\Phi, E)$$

be a finite directed graph. Every edge

$$e : \phi \to \psi$$

carries an integral displacement

$$g_e \in \mathbb{Z}^n$$

and a positive rational cost

$$c_e \in \mathbb{Q}_{>0}.$$

Its periodic lift has vertex set

$$\mathbb{Z}^n \times \Phi$$

and edges

$$(z, \phi) \xrightarrow{e} (z + g_e, \psi).$$

Fix a target phase $\phi_\star \in \Phi$ and target $t = (0, \phi_\star)$. Multiplying all costs by a common denominator preserves all shortest paths and scales all distances uniformly. Hence we assume

$$c_e \in \mathbb{N}_{>0}.$$

A state $v = (z, \phi)$ is *target-reachable* if there exists a path from v to t. For target-reachable states, let

$$d(v) = d_\phi(z)$$

be the minimum path cost to t. When $v = (z, \phi)$, we write $\mathrm{Reach}(v)$ as shorthand for $\mathrm{Reach}_\phi(z)$. The same convention is used for translated successor states.

Minimum-cost path problems in periodic graphs with integral gains and rational costs are classical; see Höfting and Wanke [70]. The role of the present section is to extract a finite arithmetic representation used by the compatibility compiler.

17.2 Gain-cost relations

Let

$$L_{\phi,\phi_\star} \subseteq E^*$$

be the regular language of base-graph walks from ϕ to $\phi_\star$. For $w = e_1 \cdots e_q$, define

$$G(w) = \sum_{j=1}^{q} g_{e_j}, \qquad C(w) = \sum_{j=1}^{q} c_{e_j}.$$

Set

$$R_\phi = \{(G(w), C(w)) : w \in L_{\phi,\phi_\star}\}.$$

Proposition 17.1 (Semilinear path relation)**.** *For every phase ϕ, the relation*

$$R_\phi \subseteq \mathbb{Z}^n \times \mathbb{N}$$

is effectively Presburger-definable.

Proof. The Parikh image of a regular language is effectively semilinear [68, 77]. Total gain and total cost are integral linear functions of the edge-count vector. Semilinear sets are closed under integral linear images and coincide with Presburger-definable sets [69].

We use the standard extension of semilinearity to subsets of $\mathbb{Z}^n \times \mathbb{N}$. Equivalently, signed coordinates may be represented by nonnegative coordinate pairs and projected by the integral map $x = x^+ - x^-$; this does not change Presburger definability. □

Define

$$\mathrm{Reach}_\phi(z) \iff \exists N \in \mathbb{N}\ R_\phi(-z, N).$$

Proposition 17.2 (Arithmetic distance predicate)**.** *There is an effectively constructible Presburger predicate*

$$\mathrm{Dist}_\phi(z, N)$$

such that

$$\mathrm{Dist}_\phi(z, N) \iff N = d_\phi(z)$$

for every target-reachable (z, ϕ)*.*

Proof. Define $\mathrm{Dist}_\phi(z, N)$ by

$$R_\phi(-z, N) \wedge \neg\exists N' \left(0 \le N' < N \wedge R_\phi(-z, N')\right).$$

For a target-reachable state, the feasible costs form a nonempty subset of $\mathbb{N}$, hence possess a least element. □

The finite family

$$\mathfrak{C}_{\mathrm{dist}} = \{\mathrm{Reach}_\phi, \mathrm{Dist}_\phi : \phi \in \Phi\}$$

will be called the *arithmetic distance compiler*. "Finite" refers to the description; the coordinate $z \in \mathbb{Z}^n$ remains unbounded.

17.3 Bellman guards

For a lifted edge $v \xrightarrow{e} T_e v$, define $e \in \mathrm{First}(v)$ when e is the first edge of some shortest path from v to t.

Lemma 17.3 (Optimal-edge criterion)**.** *For target-reachable* v*,*

$$\boxed{e \in \mathrm{First}(v) \iff \mathrm{Reach}(T_e v) \text{ and } c_e + d(T_e v) = d(v).}$$

Proof. If a shortest path begins with e, its remaining suffix must be shortest from $T_e v$. Conversely, if the equality holds, append any shortest suffix from $T_e v$. □

Lemma 17.4 (Prefix criterion)**.** *Let* $w = e_1 \cdots e_q$ *be executable from a target-reachable state* v*. Then* w *is a prefix of some shortest path from* v *if and only if*

$$\boxed{\mathrm{Reach}(T_w v) \quad \text{and} \quad C(w) + d(T_w v) = d(v).}$$

Proof. If w occurs on a shortest path, replacing its suffix by a shorter one would shorten the full path, so the suffix is shortest. Conversely, append a shortest suffix from $T_w v$. The displayed equality makes the resulting total cost equal to $d(v)$. □

17.4 Optimal unfolding

For a target-reachable state v, let $\mathrm{OptUnf}(v)$ be the directed subgraph reachable from v using only edges satisfying the Bellman guard of Theorem 17.3.

Theorem 17.5 (Finite geodesic compiler)**.** *For every fixed positive-cost finite-phase periodic graph, one can effectively construct a finite Presburger-guarded translation program such that, from every target-reachable state* v*,*

(i) it recovers $d(v)$*;*

(ii) it recovers every optimal first edge;

(iii) it decides shortest-prefix membership for every finite executable word; and

(iv) its optimal unfolding is exactly the union of all shortest v-to-t paths.

Proof. Items (i)–(iii) follow from Theorems 17.2 to 17.4. Suppose

$$v = v_0 \to v_1 \to \cdots \to v_q$$

is a path in the optimal unfolding. Each edge satisfies

$$d(v_{j-1}) = c_j + d(v_j),$$

hence

$$d(v_q) = d(v) - \sum_{j=1}^{q} c_j.$$

Every guarded successor is target-reachable, so a shortest suffix from v_q exists. Appending it yields a shortest path containing the given prefix. Conversely, every edge of a shortest path satisfies the Bellman equality by optimal substructure. □

Corollary 17.6 (Canonical and complete generation)**.** *Fix a total order on outgoing base edges. At every non-target target-reachable state, at least one optimal first edge exists. Choosing the least one gives a deterministic canonical shortest route.*

Moreover, because every cost is positive,

$$v \to v' \text{ optimal} \quad \Longrightarrow \quad d(v') < d(v).$$

Hence d is a ranking function on the optimal unfolding. Recursive branching over all optimal first edges generates all and only shortest paths and terminates on every branch.

Chapter 18

Cubical Geodesic Compilation

The local action complexes developed earlier in the higher-dimensional completion branch are recovered here as special cases. In a one-phase translation system whose structural action family is the previously fixed completion action set, the complex defined below is exactly the earlier shortest-completion local action complex. The new notation separates two layers that were intertwined in the special counting model: $\mathcal{S}_\phi$ records structural commutation, whereas $\mathcal{H}(v)$ records the faces selected by shortest-path optimality.

18.1 Periodic action systems

Let

$$I = \{1, \dots, s\}$$

be a finite action alphabet. For every $i \in I$, let

$$\tau_i : \Phi \rightharpoonup \Phi$$

be a partial phase transition. Whenever $\tau_i(\phi)$ is defined, associate

$$g_i(\phi) \in \mathbb{Z}^n, \qquad c_i(\phi) \in \mathbb{Q}_{>0}.$$

The lifted action is

$$T_i(z, \phi) = \big(z + g_i(\phi), \tau_i(\phi)\big).$$

For an executable action word $w = i_1 \cdots i_q$, T_w denotes the corresponding composition of lifted partial actions. If J is a structural cube, $T_J v$ denotes the common endpoint obtained from any permutation of the actions in J.

18.2 Structural cubes

Definition 18.1 (Structural cube)**.** Let $J \subseteq I$. We call J a *structural cube at phase* ϕ if, for every $K \subseteq J$, every permutation of the actions in K

(i) is executable from ϕ;

(ii) reaches the same final phase;

(iii) produces the same total displacement;

(iv) has the same total cost;

and, additionally,

$$K \neq L \quad \Longrightarrow \quad (\tau_K(\phi), g_K(\phi)) \neq (\tau_L(\phi), g_L(\phi)).$$

Write $\tau_K(\phi)$, $g_K(\phi)$, and $c_K(\phi)$ for the common data.

The final injectivity condition makes the $2^{|J|}$ subset vertices distinct, so the terminology "cube" is nondegenerate. Let

$$\mathcal{S}_\phi = \{J \subseteq I : J \text{ is a structural cube at } \phi\}.$$

Lemma 18.2 (Structural simpliciality)**.** *For every ϕ, $\mathcal{S}_\phi$ is a simplicial complex.*

Proof. Every defining condition is inherited by subsets, including injectivity of the subset-endpoint map. □

Because Φ and I are finite, each $\mathcal{S}_\phi$ is finite and can be computed offline from the finite action table. Cubical state complexes generated by independent local moves are classical in reconfiguration theory; see Ghrist and Peterson [72]. Our additional object is the subcomplex selected by shortest-path optimality.

18.3 Optimal compatibility

Definition 18.3 (Optimal-compatibility complex)**.** For a target-reachable state $v = (z, \phi)$ and $J \in \mathcal{S}_\phi$, define

$$J \in \mathcal{H}(v)$$

if

$$\mathrm{Reach}(T_J v)$$

and

$$\boxed{c_J(\phi) + d(T_J v) = d(v).}$$

For $h \geq 0$, define

$$\mathcal{H}_{\leq h}(v) = \{J \in \mathcal{H}(v) : |J| \leq h\}, \qquad \mathcal{H}_{=h}(v) = \{J \in \mathcal{H}(v) : |J| = h\}.$$

For later comparison across structural orders, write

$$\Gamma_h(v) = \big(d(v), \mathcal{H}_{\leq h}(v)\big).$$

The empty set belongs to both $\mathcal{S}_\phi$ and $\mathcal{H}(v)$, with $T_\varnothing v = v$ and $c_\varnothing = 0$. Hence $\mathcal{H}_{\leq 0}(v) = \{\varnothing\}$.

Proposition 18.4 (Cubical Bellman equivalence)**.** *Let $J \in \mathcal{S}_\phi$. For target-reachable $v = (z, \phi)$, the following are equivalent:*

(i) $J \in \mathcal{H}(v)$;

(ii) one permutation of the actions in J is a shortest prefix from v;

(iii) every permutation of the actions in J is a shortest prefix from v.

Proof. Every permutation of a structural cube has the same cost and endpoint. Hence the prefix criterion of Theorem 17.4 either holds for all permutations or for none. □

Proposition 18.5 (Optimal simpliciality)**.** *For every target-reachable v, $\mathcal{H}(v)$ is a simplicial subcomplex of $\mathcal{S}_\phi$.*

Proof. Let $J \in \mathcal{H}(v)$ and $K \subseteq J$. Order the actions of J so that the actions of K occur first. By Theorem 18.4 this permutation is a shortest prefix, and hence its initial K-segment is a shortest prefix. Therefore $K \in \mathcal{H}(v)$. □

18.4 Arithmetic face guards

For each defined action i at phase ϕ, let $\mathrm{Opt}_{\phi,i}(z)$ denote the Presburger Bellman guard supplied by Theorem 17.3. For $J \in \mathcal{S}_\phi$, define $\mathrm{Face}_{\phi,J}(z)$ by

$$\exists N, N' \left[\mathrm{Dist}_\phi(z, N) \wedge \mathrm{Dist}_{\tau_J(\phi)}(z + g_J(\phi), N') \wedge N = c_J(\phi) + N'\right].$$

Thus

$$J \in \mathcal{H}((z, \phi)) \iff \mathrm{Face}_{\phi,J}(z).$$

Definition 18.6 (Cubical geodesic compiler)**.** Let

$$\mathfrak{C}_{\mathrm{cube}} = \big(\{\mathrm{Reach}_\phi, \mathrm{Dist}_\phi\}, \{\mathrm{Opt}_{\phi,i}\}, \{\mathrm{Face}_{\phi,J}\}\big)$$

range over all phases, defined actions, and structural faces.

Theorem 18.7 (Finite cubical geodesic compilation)**.** *For every fixed positive-cost finite-phase periodic action system, the finite arithmetic compiler $\mathfrak{C}_{\mathrm{cube}}$ is effectively constructible and determines, at every target-reachable state,*

- *exact shortest distance;*
- *all optimal first actions;*
- *shortest-prefix membership for arbitrary finite executable words;*
- *the complete local optimal-compatibility complex;*
- *a canonical shortest route under fixed tie breaking; and*
- *an implicit exact representation of all shortest paths.*

Proof. The metric, first-action, prefix, canonical, and all-geodesic claims follow from Theorems 17.5 and 17.6. For each phase, the structural complex is finite. Membership of every structural face in $\mathcal{H}(v)$ is exactly one arithmetic face guard. Evaluating all such guards therefore recovers the complete local complex. □

18.5 Signature regions

For a subcomplex $K \subseteq \mathcal{S}_\phi$, define

$$\Omega_{\phi,K} = \{z : \mathrm{Reach}_\phi(z) \text{ and } \mathcal{H}((z, \phi)) = K\}.$$

Every such set is Presburger-definable, and

$$\boxed{\{z : \mathrm{Reach}_\phi(z)\} = \bigsqcup_K \Omega_{\phi,K}.}$$

Thus the target-reachable region admits a finite arithmetic partition by local optimal-compatibility type.

18.6 Affine-semigroup interpretation

In the one-phase unit-cost translation case, write $\mathcal{S}$ for the unique phase structural complex. The compatibility complex admits an exact divisor-complex interpretation. Let the distinct action translations be

$$a_1, \dots, a_s \in \mathbb{Z}^n$$

and let

$$D = d(r).$$

Define

$$b_i = (-a_i, 1) \in \mathbb{Z}^{n+1}, \qquad \beta_r = (r, D),$$

and the semigroup

$$\mathsf{M} = \langle b_1, \dots, b_s \rangle_{\mathbb{N}}.$$

Define the labeled squarefree divisor complex

$$\Delta_{\beta_r} = \left\{ J \subseteq [s] : \beta_r - \sum_{i \in J} b_i \in \mathsf{M} \right\}.$$

Proposition 18.8 (Homogenized semigroup correspondence)**.** *For every $J \subseteq [s]$,*

$$J \in \Delta_{\beta_r}$$

if and only if

$$|J| + d\left(r + \sum_{i \in J} a_i \right) = d(r).$$

Consequently,

$$\boxed{\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}.}$$

If every action subset is structurally nondegenerate, then $\mathcal{H}(r) = \Delta_{\beta_r}$.

Proof. Indeed,

$$\beta_r - \sum_{i \in J} b_i = \left(r + \sum_{i \in J} a_i, D - |J| \right).$$

Membership in M is therefore equivalent to the existence of a suffix consisting of exactly $D - |J|$ unit-cost actions that completes the state

$$r + \sum_{i \in J} a_i$$

to the target.

If the same state admitted a suffix of length strictly smaller than $D - |J|$, then preceding it by the J-prefix would give a completion of r of total length strictly smaller than

$$|J| + (D - |J|) = D = d(r),$$

a contradiction. Hence

$$d\left(r + \sum_{i \in J} a_i \right) = D - |J|,$$

which is equivalent to

$$|J| + d\left(r + \sum_{i \in J} a_i \right) = d(r).$$

Intersecting with the structural complex gives $\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}$. □

Squarefree divisor complexes are classical in affine-semigroup theory [71]; parametric Presburger methods can distinguish full divisor-complex types in affine-semigroup families [75]. We therefore do not claim novelty for divisor complexes or arithmetic definability in isolation.

18.7 Complexity of the Compiled Representation

The compiler has two distinct computational stages:

offline arithmetic compilation + online querying.

No polynomial-time claim is made for the general offline stage.

18.7.1 Query parameters

For $v = (z, \phi)$, let

$$\lambda(v) = 1 + \sum_{j=1}^{n} \lceil \log_2(1 + |z_j|) \rceil$$

be the coordinate encoding length.

Let $Q_{\mathrm{reach}}(\lambda)$ denote the cost of evaluating target reachability and let $Q_{\mathrm{dist}}(\lambda)$ denote the additional cost of recovering the exact distance once reachability has been established. We write

$$Q_{\mathrm{met}}(\lambda) = Q_{\mathrm{reach}}(\lambda) + Q_{\mathrm{dist}}(\lambda)$$

for a complete metric query. This convention is necessary because $d(v)$ is defined only at target-reachable states. In particular, successor states appearing in optimal-action, prefix, and face queries are tested for reachability before their distance values are evaluated.

Let $Q_{\mathrm{opt}}(\lambda)$ and $Q_{\mathrm{face}}(\lambda)$ denote the costs of evaluating one compiled optimal-action guard and one face guard, respectively. Let $Q_{\mathrm{first}}(\lambda)$ denote the cost of recovering the complete set of optimal first actions at one target-reachable state. Let $\deg^+_{\max}$ be the maximum outgoing base degree.

Proposition 18.9 (Distance and first-action queries)**.** *Exact distance at a target-reachable state is recovered in $Q_{\mathrm{dist}}(\lambda(v))$. All optimal first actions are recovered in*

$$O\big(\deg^+_{\max} Q_{\mathrm{opt}}(\lambda_*)\big)\,.$$

If optimal guards are evaluated through metric queries instead, then

$$O\big(\deg^+_{\max} Q_{\mathrm{met}}(\lambda_*) + Q_{\mathrm{dist}}(\lambda(v))\big)$$

suffices.

18.7.2 Prefix queries

Let w be an explicit executable word of length q. A single scan computes its endpoint phase, accumulated gain, and accumulated cost. For a fixed system,

$$\lambda(T_w v) = O(\lambda(v) + n \log(1 + q))\,.$$

Proposition 18.10 (Prefix-query complexity)**.** *Shortest-prefix membership is decidable in*

$$O(q\ \mathrm{poly}(\lambda_w) + Q_{\mathrm{met}}(\lambda_w) + Q_{\mathrm{dist}}(\lambda(v)))\,.$$

The linear dependence on q is unavoidable up to arithmetic costs because the explicit word must be read.

18.7.3 Face queries

A fixed structural face has precomputed phase, displacement, and cost data.

Proposition 18.11 (Face-query complexity)**.** *A specified structural face can be tested in* $Q_{\mathrm{face}}(\lambda_*)$*. Using only reachability and distance recovery gives*

$$O(Q_{\mathrm{dist}}(\lambda(v)) + Q_{\mathrm{met}}(\lambda(T_J v)))\,.$$

Let

$$F_h(\phi) = |\{J \in \mathcal{S}_\phi : |J| \le h\}|.$$

Then

$$F_h(\phi) \le \sum_{j=0}^{h} \binom{s}{j}.$$

Thus the complete explicit truncation $\mathcal{H}_{\le h}(v)$ *can be reported in*

$$O(F_h(\phi) Q_{\mathrm{face}}(\lambda_*) + |\mathcal{H}_{\le h}(v)|)\,.$$

Explicit recovery of the full complex can require exponential output, since even a full simplex has 2^s faces.

18.7.4 Canonical routes

Let

$$c_{\min} = \min_e c_e > 0.$$

Every optimal edge decreases distance by at least $c_{\min}$. Therefore any shortest route has at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor$$

edges. If $\ell_{\mathrm{can}}(v)$ is the canonical route length and Q_{canon} is the cost of selecting the canonical enabled optimal action, then route generation takes

$$O(\ell_{\mathrm{can}}(v) Q_{\mathrm{canon}}(\lambda_{\mathrm{path}}))\,.$$

This is path-length-sensitive. With binary-encoded coordinates, an explicitly printed shortest path may itself be exponentially long in the input bit length.

18.7.5 Enumeration of all shortest paths

Let $\mathrm{Geo}(v)$ be the finite set of shortest paths from v, and define

$$\mathrm{OUT}(v) = 1 + \sum_{\gamma \in \mathrm{Geo}(v)} (|\gamma| + 1).$$

The shortest-prefix tree has at most $O(\mathrm{OUT}(v))$ nodes.

Theorem 18.12 (Output-sensitive geodesic enumeration)**.** *All shortest paths from* v *can be enumerated in*

$$\boxed{O\big(\mathrm{OUT}(v)\big[Q_{\mathrm{first}}(\lambda_{\mathrm{path}}) + 1\big]\big)\,.}$$

The recursion depth is at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor.$$

We describe this as output-sensitive and path-length-sensitive rather than polynomial-delay enumeration, because a single shortest path may already be exponentially long in the binary coordinate encoding.

18.7.6 Normalized arithmetic compilers

A stronger online statement is available under an explicit normalization assumption. Suppose offline preprocessing produces

(i) a finite decision list of quantifier-free Presburger regions;

(ii) an explicit affine distance output on each region; and

(iii) quantifier-free optimal-action and face guards.

Let $\|\mathfrak{C}\|$ be the encoded compiler size and $L_{\mathfrak{C}}$ the maximum coefficient bit length.

Corollary 18.13 (Online bit complexity after normalization)**.** *Under this explicit normalization assumption,* DIST*,* FIRST*, and* FACE *are computable in*

$$\mathrm{poly}\big(\|\mathfrak{C}\|, L_{\mathfrak{C}}, \lambda(v)\big).$$

For a fixed compiled system, these queries are therefore polynomial in the queried coordinate bit length.

This is a conditional normalized-compiler result; the monograph does not claim that the required normalization is polynomial-time constructible in general. General Presburger elimination can incur substantial blow-up. Recent work obtains singly exponential elimination for an existential block, but this does not imply polynomial complexity for the complete compilation pipeline [78].

Chapter 19

Prescribed-Order Separation, Boolean Amplification, and Universal k-Fibers

We now show that the higher-order compatibility compiled in Chapter 18 cannot universally be reconstructed from any fixed lower-order truncation, then sharpen that nonreconstruction into a universal realization theorem. The first construction forces one discrepancy to appear at a prescribed order; independent products amplify such discrepancies Booleanly. The final construction asks whether that freedom is intrinsic to one lower-order signature fiber and proves that, on a fixed structural simplex, the entire next-order layer can vary arbitrarily.

Fix $k \geq 1$ and define

$$M = 2^{k+1}, \qquad \delta = 2^k - 1, \qquad N = 2k.$$

Consider the one-phase unit-cost translation actions

$$a_0 = -M,$$

$$a_i = -M + 2^{i-1}, \qquad 1 \leq i \leq k,$$

and

$$a_+ = 1.$$

Let

$$A_- = \{a_0, a_1, \ldots, a_k\}.$$

All action maps are translations, so they commute globally.

19.1 Completion and nondegeneracy

Lemma 19.1 (Completion-totality)**.** *Every integer state admits a finite path to* 0*.*

Proof. If $x \leq 0$, repeatedly apply $a_+ = 1$. If $x > 0$, apply $a_0 = -M$ until the state becomes nonpositive and then apply a_+. □

Lemma 19.2 (Nondegeneracy of the negative family)**.** *Distinct subsets of A_- have distinct total translations.*

Proof. For $K \subseteq A_-$, define

$$B(K) = \sum_{\substack{1 \le i \le k \\ a_i \in K}} 2^{i-1}.$$

Then

$$\sum_{a \in K} a = -|K|M + B(K).$$

If two subsets K, L have the same total translation, then

$$(|K| - |L|)M = B(K) - B(L).$$

The right side has absolute value $< M$, hence $|K| = |L|$. Then $B(K) = B(L)$, and uniqueness of binary expansion identifies all special actions. Cardinality then determines the presence of a_0. Thus $K = L$. □

In particular,

$$J_\star = \{a_1, \dots, a_k\}$$

is a genuine nondegenerate structural k-cube.

19.2 Binary-deficit distance

Lemma 19.3 (Binary-deficit distance lemma)**.** *If*

$$0 \le D < 2^k$$

and

$$\ell \ge k,$$

then

$$\boxed{d(\ell M - D) = \ell.}$$

Proof. Write

$$D = \sum_{i \in S} 2^{i-1}.$$

Apply a_i once for every $i \in S$ and apply a_0 another $\ell - |S|$ times. The total translation is $-\ell M + D$, so $d(\ell M - D) \le \ell$.

On the other hand, $D < 2^k < M$, so

$$\ell M - D > (\ell - 1)M.$$

One action decreases the state by at most M, so $\ell - 1$ actions cannot suffice. □

Action labels used in a cubical prefix may be reused later in the completion suffix; this is ordinary path semantics.

19.3 Adjacent equal-distance states

Define

$$r = NM - \delta$$

and

$$r' = r + 1 = NM - (\delta - 1).$$

Then

$$r \xrightarrow{a_+} r'.$$

Proposition 19.4 (Equal distance)**.**

$$\boxed{d(r) = d(r') = N.}$$

Proof. Apply Theorem 19.3 with deficits δ and $\delta - 1$. □

For a negative action set J, define

$$B(J) = \sum_{\substack{1 \le i \le k \\ a_i \in J}} 2^{i-1}.$$

Proposition 19.5 (Exact common lower-order truncation)**.** *For every action set J satisfying $|J| < k$,*

$$\boxed{J \in \mathcal{H}(r) \iff J \in \mathcal{H}(r') \iff J \subseteq A_-.}$$

Consequently,

$$\boxed{\mathcal{H}_{\le k-1}(r) = \mathcal{H}_{\le k-1}(r').}$$

Proof. Suppose first that $J \subseteq A_-$ and let $q = |J| < k$. After the prefix,

$$r_J = (N - q)M - (\delta - B(J)),$$

and

$$r'_J = (N - q)M - (\delta - 1 - B(J)).$$

Since $q < k$, not all k positive binary deficits occur, so $B(J) \le \delta - 1$. Thus both remaining deficits belong to $[0, 2^k)$, while $N - q \ge k + 1$. Theorem 19.3 gives remaining distance $N - q$ at both states. Hence J is optimal at both.

Now suppose $a_+ \in J$. Let q be the number of negative actions in J. Then $|J| = q + 1 < k$. After the prefix from r, the residual exceeds $(N - q - 1)M$, because

$$M - \delta + B(J) + 1 > 0.$$

Yet an optimal suffix would have only $N - q - 1$ unit-cost steps available, each decreasing by at most M. This is impossible. The same argument at r' has an extra 1 of residual and is again impossible. If such a set is structurally degenerate, it is excluded from $\mathcal{H}$ already by definition; in either case it is not an optimal face. □

19.4 First discrepancy at order k

Proposition 19.6 (Distinguished separating face)**.** *For*

$$J_\star = \{a_1, \dots, a_k\},$$

$$\boxed{J_\star \in \mathcal{H}(r), \qquad J_\star \notin \mathcal{H}(r').}$$

Proof. Because

$$\sum_{i=1}^{k} 2^{i-1} = \delta,$$

after applying $J_\star$ at r,

$$r + \sum_{i=1}^{k} a_i = (N - k)M = kM.$$

Theorem 19.3 with $D = 0$ gives $d(kM) = k = N - k$. Hence $J_\star$ is optimal.

At r', the same prefix leaves $kM + 1$. The required suffix would have k actions, which can decrease the residual by at most kM. Hence no such optimal suffix exists. □

Proposition 19.7 (Single-face difference through order k)**.**

$$\boxed{\mathcal{H}_{\le k}(r) = \mathcal{H}_{\le k}(r') \cup \{J_\star\}.}$$

Proof. The complexes agree below order k by Theorem 19.5. Any k-set containing a_+ fails the same maximum-descent test at both states.

A negative k-set other than $J_\star$ contains a_0 and omits exactly one special action a_i. Its deficit is $\delta - 2^{i-1}$. After the prefix, the two residuals are

$$kM - 2^{i-1}$$

and

$$kM - (2^{i-1} - 1),$$

respectively. Both deficits lie in $[0, 2^k)$, so Theorem 19.3 gives remaining distance k in both cases. □

Theorem 19.8 (Prescribed-order adjacent-state non-factorization)**.** *For every prescribed $k \ge 1$, there exists a completion-total one-phase unit-cost system of globally commuting translation actions and adjacent states $r \to r'$ such that*

$$\boxed{d(r) = d(r'),}$$

$$\boxed{\mathcal{H}_{\le k-1}(r) = \mathcal{H}_{\le k-1}(r'),}$$

and

$$\boxed{\mathcal{H}_{\le k}(r) = \mathcal{H}_{\le k}(r') \cup \{J_\star\}.}$$

Equivalently,

$$\Gamma_{k-1}(r) = \Gamma_{k-1}(r')$$

but

$$\Gamma_k(r) \neq \Gamma_k(r').$$

Thus the first compatibility discrepancy occurs exactly at the prescribed order k.

Corollary 19.9 (No fixed lower-order reconstruction)**.** *For every fixed $h \ge 0$, the data*

$$\boxed{\big(d(v), \mathcal{H}_{\le h}(v)\big)}$$

do not universally determine $\mathcal{H}(v)$.

Proof. Apply Theorem 19.8 with $k = h + 1$. □

The claim concerns this exact lower-order truncation. It does not assert insufficiency of an arbitrary auxiliary state description that might retain additional information such as the raw coordinates.

19.5 Exact full complexes

Proposition 19.10 (Full-complex description)**.**

$$\boxed{\mathcal{H}(r) = 2^{A_-}}$$

and

$$\boxed{\mathcal{H}(r') = 2^{A_-} \setminus \{J_\star, A_-\}.}$$

Proof. Every proper negative subset of size at most k is handled by Theorems 19.3, 19.5 and 19.7. For the full negative set,

$$r + \sum_{a \in A_-} a = (k-1)M.$$

If $k = 1$, this is already the target. If $k \geq 2$, its distance is $k-1$ by maximal descent and the obvious a_0-completion. Hence $A_- \in \mathcal{H}(r)$.

At r', the same prefix leaves $(k-1)M+1$, which cannot be completed in the required $k-1$ steps. No face containing a_+ is optimal. If its prefix cost already exceeds N, this is immediate; otherwise the same maximum-descent argument used above applies. □

Accordingly,

$$\boxed{\text{through order } k \text{ the complexes differ by one } k\text{-face,}}$$

whereas

$$\boxed{\text{the full complexes differ by two faces.}}$$

This distinction is retained throughout this branch.

19.6 Large optimization-induced minimal nonfaces

For $k \geq 2$, on the distinguished vertex set $J_\star$,

$$\mathcal{H}(r)|_{J_\star} = \Delta^{k-1},$$

while

$$\mathcal{H}(r')|_{J_\star} = \partial\Delta^{k-1}.$$

For $k = 1$, the distinguished vertex is present at r and absent at r'. Thus shortest-path optimality creates minimal nonfaces of arbitrarily large cardinality inside a globally commuting translation system. For $k \geq 3$, the resulting restricted complex is nonflag.

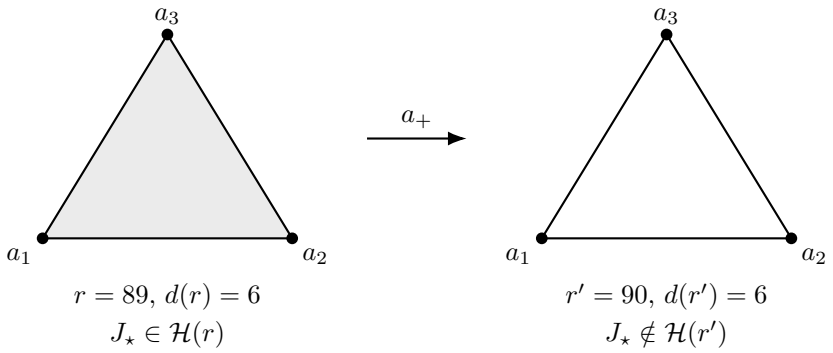


Figure 19.1: Prescribed-order separation for $k = 3$. The adjacent states $r = 89$ and $r' = 90$ have equal shortest distance and identical optimal-compatibility data through order 2. On the distinguished action set $J_\star = \{a_1, a_2, a_3\}$, the restriction of the optimal-compatibility complex changes from a filled triangle to its boundary. Thus the first discrepancy occurs exactly at order 3.

The novelty claim is not the existence of nonflag complexes or simplex boundaries in isolation; such phenomena are classical in semigroup topology [71]. The relevant constraint package is adjacency, equal shortest distance, one fixed action system, agreement of all lower compatibility orders, and first discrepancy at a prescribed order.

Remark 19.11 (Computational verification). As an independent regression check, the binary-deficit family was exhaustively verified for $1 \le k \le 10$. The distance formula, the exact lower-order agreement, and the full-complex identities of Theorem 19.10 matched the symbolic formulas in every case. Independent product-graph checks were also performed for representative small (k, m) pairs and recovered exactly the Boolean signatures of Theorem 19.16. These checks are not premises of any theorem.

19.7 Boolean Amplification

We now take independent products of the separating block.

19.7.1 Product formulas

Consider m periodic action systems with disjoint action alphabets.

Lemma 19.12 (Product distance)**.** *For*

$$V = (v_1, \dots, v_m),$$

$$\boxed{d(V) = \sum_{t=1}^{m} d_t(v_t).}$$

Proof. Every product path projects to a path in each factor, giving the lower bound. Interleaving shortest factor paths gives equality. □

For $v_t = (z_t, \phi_t)$, write $\mathcal{S}_t(\phi_t)$ for the structural complex in factor t, and write $\mathcal{S}(V)$ for the product structural complex at V. For an action set

$$J = J_1 \sqcup \cdots \sqcup J_m,$$

structural compatibility factors:

$$J \in \mathcal{S}(V) \iff J_t \in \mathcal{S}_t(\phi_t) \quad \forall t.$$

Distinct factor-subset tuples give distinct product endpoints, so nondegeneracy also factors.

Lemma 19.13 (Product-join lemma)**.**

$$\boxed{\mathcal{H}(V) = \mathcal{H}_1(v_1) * \cdots * \mathcal{H}_m(v_m).}$$

Proof. First, $T_J V$ is target-reachable exactly when every $T_{J_t} v_t$ is target-reachable. In that case define

$$\sigma_t = c_{J_t} + d_t(T_{J_t} v_t) - d_t(v_t) \ge 0.$$

Then

$$c_J + d(T_J V) - d(V) = \sum_{t=1}^{m} \sigma_t.$$

The sum vanishes exactly when every σ_t vanishes. □

19.7.2 Product family

Take m independent copies of the Chapter 19 block. Write

$$r^{(0)} = r, \qquad r^{(1)} = r',$$

and for $\varepsilon \in \{0,1\}^m$ define

$$R^{\varepsilon} = \big(r^{(\varepsilon_1)}, \dots, r^{(\varepsilon_m)}\big).$$

Let $J_\star^{(t)}$ be the distinguished k-face in factor t.

Proposition 19.14 (Constant lower-order signature)**.** *For every $\varepsilon, \eta \in \{0,1\}^m$,*

$$d(R^{\varepsilon}) = d(R^{\eta}) = mN$$

and

$$\boxed{\Gamma_{k-1}(R^{\varepsilon}) = \Gamma_{k-1}(R^{\eta}).}$$

Proof. Distance additivity gives the first assertion. Every factor component of an action set of total size at most $k-1$ has size at most $k-1$, where the two base states have identical compatibility. Apply Theorem 19.13. □

Proposition 19.15 (Exact order-k variation)**.** *There exists a fixed complex $\mathcal{K}_{k,m}$ such that*

$$\boxed{\mathcal{H}_{\le k}(R^{\varepsilon}) = \mathcal{K}_{k,m} \cup \{J_\star^{(t)} : \varepsilon_t = 0\}.}$$

Proof. If an order-$\le k$ face uses more than one factor, then every factor component has size $< k$, so its membership is determined entirely by the common lower-order base data. If all k actions lie in one factor, Theorem 19.7 says the only variable face is that factor's distinguished $J_\star$. □

Theorem 19.16 (Boolean fiber amplification)**.** *For every $k, m \ge 1$, there exists a completion-total one-phase unit-cost translation system containing 2^m states with one common order-$(k-1)$ signature and 2^m pairwise distinct order-k signatures.*

The variable part of the signature is precisely

$$\{J_\star^{(t)} : \varepsilon_t = 0\},$$

so the constructed order-k signatures form a Boolean lattice of rank m.

The states themselves form a directed m-cube in the underlying state graph, because changing $\varepsilon_t : 0 \to 1$ is exactly the factor action $a_+^{(t)}$. Throughout this state cube,

$$d(R^{\varepsilon}) = mN$$

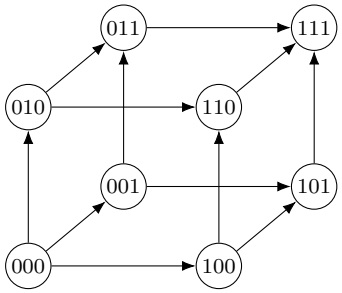
and $\Gamma_{k-1}(R^{\varepsilon})$ remain constant, whereas Γ_k distinguishes every vertex.

19.7.3 Exponential fibers

Each factor has $k+2$ action labels. Thus the product has

$$s = m(k+2)$$

actions.



$d(R^{\varepsilon}) = 3N$ for all ε

$\Gamma_{k-1}(R^{\varepsilon})$ is constant

Γ_k distinguishes all 8 vertices

coordinate t toggles $J_{\star}^{(t)}$

Figure 19.2: Boolean amplification for three independent factors. The eight product states form a directed state cube. Exact distance and the complete order-$(k-1)$ signature are constant throughout the cube. Moving in coordinate t toggles exactly the distinguished order-k face $J_{\star}^{(t)}$, producing eight distinct order-k signatures.

Corollary 19.17 (Exponential signature fibers)**.** *For every fixed $k \geq 1$ and every $m \geq 1$, there is a system with*

$$s = m(k+2)$$

actions for which one order-$(k-1)$ signature supports at least

$$\boxed{2^m = 2^{s/(k+2)}}$$

distinct order-k signatures. Thus exponentially large fibers occur along the infinite sequence of action dimensions divisible by $k+2$.

19.7.4 Conditional discrimination

Suppose an auxiliary summary $C(v)$ is required to support an exact decoder

$$\Gamma_k(v) = F\big(\Gamma_{k-1}(v), C(v)\big)$$

on the constructed family, with no raw state coordinate supplied separately.

Corollary 19.18 (Conditional discrimination bound)**.**

$$\boxed{|\operatorname{im} C| \geq 2^m}$$

and therefore

$$\boxed{\log_2 |\operatorname{im} C| \geq m = \frac{s}{k+2}.}$$

Proof. All R^{ε} have the same lower-order signature but pairwise distinct order-k signatures. Hence two distinct states cannot receive the same summary value. □

This is a discrimination bound for exact summaries under the stated decoder model. It is not a general RAM-space lower bound.

19.8 From Boolean amplification to k-fibers

The product theorem proves that a common order-$(k-1)$ signature can support many different order-k extensions, but its independent bits live in disjoint factors. This leaves a sharper structural question: after exact distance and the *entire* lower-order compatibility complex have been frozen, how much freedom can remain among overlapping k-faces on one structural simplex?

Definition 19.19 (k-fiber)**.** Fix a phase ϕ, let $k \geq 1$, and let γ be a realizable value of Γ_{k-1} at phase ϕ. The *k-fiber above γ* is

$$\mathcal{F}_k(\phi, \gamma) = \{\mathcal{H}_{=k}(v) : v \text{ is in phase } \phi,\ \Gamma_{k-1}(v) = \gamma\}\,.$$

The word *fiber* here refers to a fiber of the truncation map on optimal-compatibility signatures. It is distinct from the optimal Parikh fibers of Chapter 12, which collect minimum completion vectors rather than higher-order compatibility layers.

For $A \subseteq \{J \in \mathcal{S}_\phi : |J| = k\}$ define the exact realization region

$$\Omega^{(k)}_{\phi,\gamma,A} = \{z : \Gamma_{k-1}((z, \phi)) = \gamma,\ \mathcal{H}_{=k}((z, \phi)) = A\}.$$

Theorem 19.20 (Finite k-fiber compilation)**.** *For every fixed positive-cost finite-phase periodic action system, every phase ϕ, every $k \geq 1$, and every realizable order-$(k-1)$ signature γ:*

(i) every region $\Omega^{(k)}_{\phi,\gamma,A}$ is effectively Presburger-definable;

(ii) nonemptiness of every such region is decidable;

(iii) the complete finite fiber $\mathcal{F}_k(\phi, \gamma)$ is effectively computable.

Proof. The structural complex $\mathcal{S}_\phi$ is finite. Exact equality $\Gamma_{k-1}((z, \phi)) = \gamma$ is a finite Boolean combination of the distance predicate and the compiled face guards for structural faces of size at most $k-1$. Exact equality $\mathcal{H}_{=k}((z, \phi)) = A$ is a finite conjunction of positive guards for $J \in A$ and negative guards for structural k-faces not in A. Hence every exact realization region is Presburger-definable. Presburger decidability determines which of the finitely many candidate patterns occur, and those patterns are exactly the k-fiber. □

This theorem is an effective classification statement rather than a polynomial-time claim. It converts the statewise compiler of Chapter 18 into a finite classifier for all possible next-order extensions of a fixed lower-order signature.

19.9 The universal two-phase construction

Fix integers

$$s \geq k \geq 2, \qquad [s] = \{1, \dots, s\}, \qquad \mathcal{E} = \binom{[s]}{k}, \qquad m = |\mathcal{E}| = \binom{s}{k},$$

and put

$$R = \binom{s-1}{k-1}.$$

We construct one fixed two-phase periodic action system $\mathcal{U}_{s,k}$.

Let the phases be G and C, with target $(0, C)$. The coordinate set is

$$Q = \{b\} \sqcup \{p_1, \dots, p_s\} \sqcup \{h_E : E \in \mathcal{E}\},$$

so the lattice dimension is $1+s+\binom{s}{k}$. The coordinate b balances distances across witnesses, the private coordinates p_i force subset-endpoint injectivity, and the incidence coordinates h_E carry the next-order compatibility bits.

For each $i \in [s]$, introduce a distinguished action

$$a_i : G \to G$$

with translation

$$u_i = e_{p_i} + \sum_{\substack{E \in \mathcal{E} \\ i \in E}} e_{h_E}$$

and cost

$$c(a_i) = R + 1 = \|u_i\|_1.$$

Write

$$I_\star = \{a_1, \dots, a_s\}.$$

Completion is isolated in the second phase. Introduce a switch

$$\sigma : G \to C$$

with zero translation and cost 1. For every $q \in Q$, introduce unit-cost completion actions

$$\xi_q^+, \xi_q^- : C \to C$$

with translations $+e_q$ and $-e_q$, respectively. No completion action is defined in phase G, and no distinguished action is defined in phase C.

Lemma 19.21 (Completion-totality)**.** *Every state of $\mathcal{U}_{s,k}$ is target-reachable.*

Proof. From phase C, move coordinatewise to the origin. From phase G, apply σ and then complete coordinatewise in phase C. □

19.10 Exact metric and structural isolation

Lemma 19.22 (Completion-phase metric)**.** *For every $x \in \mathbb{Z}^Q$,*

$$d(x, C) = \|x\|_1.$$

Proof. Coordinate unit moves attain $\|x\|_1$. Conversely, the total cost of any completion path is the sum of the ℓ_1-norms of its unit translations and is therefore at least the norm of the total displacement $-x$. □

Lemma 19.23 (Geodesic-phase metric)**.** *For every $x \in \mathbb{Z}^Q$,*

$$d(x, G) = 1 + \|x\|_1.$$

Proof. Switching immediately and completing coordinatewise gives the upper bound. Conversely, suppose a path first performs a distinguished prefix with total displacement U and cost C_U, then switches to phase C. Every distinguished displacement is coordinatewise nonnegative and has cost equal to its ℓ_1-norm, so $C_U = \|U\|_1$. By Lemma 19.22, the remaining cost is at least $\|x + U\|_1$. Hence the total cost is at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1.$$

□

Lemma 19.24 (Distinguished structural simplex)**.** *The action set $I_\star$ spans a structural simplex in phase G.*

Proof. The distinguished actions are translations that preserve phase G, so every ordering of every subset is executable with the same displacement and cost. The p_i-coordinate of the total displacement of a subset is 1 exactly when a_i is present; therefore distinct subsets have distinct endpoints. □

Lemma 19.25 (Auxiliary structural isolation)**.** *The structural complex in phase G is*

$$\mathcal{S}_G = 2^{I_\star} \cup \{\{\sigma\}\}.$$

In particular, no structural face contains both σ and a distinguished action.

Proof. The singleton $\{\sigma\}$ is structural. If a set contains σ and some a_i, the ordering that executes σ first cannot execute a_i afterward because a_i is undefined in phase C. No completion action is defined in phase G. □

19.11 Hypergraph-coded witness states

For each

$$A \subseteq \mathcal{E},$$

let $L = m + 1$ and define $v_A \in \mathbb{Z}^Q$ by

$$\begin{aligned}(v_A)_b &= -(L - |A|),\\ (v_A)_{p_i} &= -1 \qquad (i \in [s]),\\ (v_A)_{h_E} &= -\big(k - 1 + \mathbf{1}_{E\in A}\big) \qquad (E \in \mathcal{E}).\end{aligned}$$

Thus $(v_A)_{h_E} = -k$ when $E \in A$ and $-(k-1)$ otherwise. We regard (v_A, G) as the witness state encoding the k-uniform hypergraph $([s], A)$.

Lemma 19.26 (Equal witness distance)**.** *All witness states have the same exact shortest distance*

$$D_{s,k} = s + 2 + k\binom{s}{k}.$$

Proof. Since $L = m + 1$,

$$\|v_A\|_1 = (L - |A|) + s + m(k - 1) + |A| = s + 1 + mk.$$

Apply Lemma 19.23. □

For $J \subseteq [s]$, write

$$U_J = \sum_{i\in J} u_i.$$

The incidence identity driving the construction is

$$(U_J)_{h_E} = |J \cap E| \qquad (E \in \mathcal{E}). \tag{19.1}$$

Also $(U_J)_{p_i} = \mathbf{1}_{i\in J}$, $(U_J)_b = 0$, and

$$\|U_J\|_1 = |J|(R + 1).$$

Lemma 19.27 (Coordinatewise geodesic criterion)**.** *For every $A \subseteq \mathcal{E}$ and every $J \subseteq [s]$,*

$$\{a_i : i \in J\} \in \mathcal{H}(v_A, G)$$

if and only if

$$U_J \le -v_A$$

coordinatewise.

Proof. By Lemma 19.23, the Bellman equality for the distinguished prefix is

$$\|U_J\|_1 + 1 + \|v_A + U_J\|_1 = 1 + \|v_A\|_1.$$

Every coordinate of v_A is nonpositive and every coordinate of U_J is nonnegative. Equality therefore holds exactly when no coordinate overshoots zero, equivalently $U_J \le -v_A$ coordinatewise. □

19.12 Universal k-fiber realization

Lemma 19.28 (Complete lower-order invariance)**.** *For every $A, B \subseteq \mathcal{E}$,*

$$\mathcal{H}_{\le k-1}(v_A, G) = \mathcal{H}_{\le k-1}(v_B, G).$$

More precisely, every distinguished subset of cardinality at most $k-1$ is optimal at every witness state, the switch singleton $\{\sigma\}$ is optimal at every witness state, and these are all structural faces of phase G through order $k-1$.

Proof. Let $J \subseteq [s]$ with $|J| \le k-1$. At a private coordinate,

$$(U_J)_{p_i} \le 1 = -(v_A)_{p_i}.$$

At an incidence coordinate, (19.1) gives

$$(U_J)_{h_E} = |J \cap E| \le |J| \le k-1 \le -(v_A)_{h_E}.$$

The balance coordinate receives zero distinguished displacement, so Lemma 19.27 makes every such J optimal for every A. The switch singleton is optimal because

$$1 + d(v_A, C) = 1 + \|v_A\|_1 = d(v_A, G).$$

By Lemma 19.25, there are no other structural faces in phase G. □

Let

$$K_{s,k} = \mathcal{H}_{\le k-1}(v_A, G), \qquad \gamma_{s,k} = (D_{s,k}, K_{s,k});$$

these are independent of A.

Lemma 19.29 (Exact k-face decoder)**.** *For every $A \subseteq \mathcal{E}$ and every $J \in \mathcal{E}$,*

$$\{a_i : i \in J\} \in \mathcal{H}(v_A, G) \quad \Longleftrightarrow \quad J \in A.$$

Proof. At the coordinate h_J, (19.1) gives $(U_J)_{h_J} = k$. If $J \in A$, the threshold is $-(v_A)_{h_J} = k$ and is saturated exactly. If $J \notin A$, the threshold is only $k-1$, so the prefix overshoots and cannot be geodesic. For $E \neq J$ with $|E| = |J| = k$,

$$|J \cap E| \le k-1 \le -(v_A)_{h_E},$$

while private coordinates are saturated at most once and the balance coordinate is unchanged. Apply Lemma 19.27. □

Theorem 19.30 (Universal k-Fiber Theorem)**.** *For every $s \ge k \ge 2$, there exists a fixed completion-total positive-cost two-phase periodic action system $\mathcal{U}_{s,k}$, a phase G, a distinguished structural s-simplex $I_\star$, and a realizable order-$(k-1)$ signature $\gamma_{s,k}$ whose complete k-fiber is the full power set of the distinguished k-faces:*

$$\boxed{\mathcal{F}_k(G, \gamma_{s,k}) = 2^{\binom{I_\star}{k}}.}$$

Indeed, the witness family

$$\{(v_A, G) : A \subseteq \binom{[s]}{k}\}$$

contains $2^{\binom{s}{k}}$ equal-distance states with one common complete lower-order compatibility truncation, and

$$\mathcal{H}_{=k}(v_A, G) = \big\{\{a_i : i \in J\} : J \in A\big\}.$$

Proof. Equal distance and the common order-$(k-1)$ signature follow from Lemmas 19.26 and 19.28. Since $k \geq 2$, Lemma 19.25 implies that every structural k-face in phase G lies entirely in the distinguished simplex. Hence every member of the fiber is a subset of $\binom{I_\star}{k}$. The exact decoder of Lemma 19.29 realizes every such subset as A ranges over $2^{\binom{[s]}{k}}$. □

Corollary 19.31 (Uniform-hypergraph realization)**.** *For every $s \geq k \geq 2$, every k-uniform hypergraph on $[s]$ occurs as the complete order-k optimal-compatibility layer on one fixed distinguished structural simplex, while exact shortest distance and the complete optimal-compatibility structure through order $k-1$ remain unchanged.*

Corollary 19.32 (Graphs inside one 2-fiber)**.** *For every $s \geq 2$, one fixed periodic shortest-path system has a common order-1 signature whose order-2 fiber realizes every simple graph on s distinguished vertices.*

Corollary 19.33 (Overlapping-face independence)**.** *Let $\mathcal{A} \subseteq \binom{[s]}{k}$ have arbitrary intersections among its members. Inside one common order-$(k-1)$ fiber, the membership bits of all faces in $\mathcal{A}$ can be prescribed independently.*

19.13 A worked universal fiber: $s = 3$, $k = 2$

Take

$$s = 3, \qquad k = 2, \qquad \mathcal{E} = \{12, 13, 23\}.$$

Then $m = 3$, $R = 2$, and $L = 4$. In the coordinate order

$$(b, p_1, p_2, p_3, h_{12}, h_{13}, h_{23}),$$

the distinguished translations are

$$\begin{aligned} u_1 &= e_{p_1} + e_{h_{12}} + e_{h_{13}}, \\ u_2 &= e_{p_2} + e_{h_{12}} + e_{h_{23}}, \\ u_3 &= e_{p_3} + e_{h_{13}} + e_{h_{23}}, \end{aligned}$$

all of cost 3.

For

$$A = \{12, 23\},$$

the witness vector is

$$v_A = (-2, -1, -1, -1, -2, -1, -2).$$

Its ℓ_1-norm is 10, so $d(v_A, G) = 11$; the same distance holds for all eight choices of $A \subseteq \{12, 13, 23\}$. Every distinguished singleton and the switch singleton are optimal at every witness, so

$$\mathcal{H}_{\leq 1} = \{\varnothing, \{\sigma\}, \{a_1\}, \{a_2\}, \{a_3\}\}$$

is fixed. At order two, the pairs 12 and 23 saturate their threshold-2 coordinates, while 13 overshoots the threshold-1 coordinate h_{13}. Hence

$$\mathcal{H}_{=2}(v_A, G) = \big\{\{a_1, a_2\}, \{a_2, a_3\}\big\}.$$

As A ranges over all subsets of $\{12, 13, 23\}$, the common order-1 signature remains fixed while the order-2 layer runs through all eight simple graphs on the three distinguished vertices.

19.14 Exact next-order capacity

Definition 19.34 (Distinguished k-fiber capacity)**.** For $s \geq k \geq 2$, define $\mathrm{Cap}_k(s)$ to be the supremum, over all positive-cost finite-phase periodic action systems, phases ϕ, realizable order-$(k-1)$ signatures γ, and distinguished action sets $I_\star$ with $|I_\star| = s$ and $2^{I_\star} \subseteq \mathcal{S}_\phi$, of

$$\left|\left\{\mathcal{H}_{=k}(v) \cap \binom{I_\star}{k} : \Gamma_{k-1}(v) = \gamma\right\}\right|.$$

Theorem 19.35 (Exact distinguished-layer capacity)**.** *For every $s \geq k \geq 2$,*

$$\boxed{\mathrm{Cap}_k(s) = 2^{\binom{s}{k}}.}$$

Proof. An s-vertex simplex has exactly $\binom{s}{k}$ possible k-faces, so no fiber can induce more than $2^{\binom{s}{k}}$ patterns on them. Theorem 19.30 realizes every such pattern inside one common fiber. □

Thus the universal theorem does not merely give an exponential lower bound: relative to a fixed distinguished structural simplex, it attains the maximum combinatorially possible next-order freedom.

19.15 From separation to universality

The progression inside this branch is now exact. Prescribed-order threshold separation arranges that every prefix of size at most $k-1$ lies below two nearby thresholds while one selected k-prefix distinguishes them. Boolean amplification places independent copies in disjoint factors. Universal k-fiber realization removes the product mechanism: all variable k-faces live on one fixed set of s distinguished actions and may overlap arbitrarily.

The arithmetic replacement for factor independence is the incidence identity

$$(U_J)_{h_E} = |J \cap E|.$$

Every proper lower-order prefix sees at most $k-1$ in every incidence coordinate, while a k-set reaches the value k only in its own coordinate. The architecture separates four jobs: h_E coordinates carry the variable next-order bits, p_i coordinates enforce structural injectivity, the balance coordinate b fixes the common distance, and the completion phase C supplies completion-totality without creating mixed structural faces.

Accordingly, the progression is

$$\boxed{\begin{array}{c}\text{prescribed-order threshold separation}\\ \Downarrow\\ \text{Boolean amplification in disjoint factors}\\ \Downarrow\\ \text{universal overlapping } k\text{-fiber realization.}\end{array}}$$

19.16 Relation to realization theories and scope

The surrounding ingredients remain classical: periodic shortest paths, semilinear and Presburger descriptions, cubical state complexes, and squarefree divisor complexes were already separated from the contribution in Chapters 17–18. Shortest-path computation on a *given* metric simplicial complex is another established setting [74]; here the simplicial

object is itself selected by joint shortest-prefix optimality in an infinite periodic graph. Broad realization behavior is also known for squarefree divisor complexes [71, 3], and parametric Presburger arithmetic controls complex types in affine-semigroup families [75].

No novelty is claimed for simplicial or hypergraph realization in isolation, for commuting actions forming cubes, or for Presburger definability. The structural constraint package is the point: one fixed periodic shortest-path system, one fixed phase, one fixed exact distance, one fixed complete compatibility truncation through order $k-1$, one fixed structural simplex, and every possible order-k pattern on that simplex. The capacity theorem is deliberately stated relative to the distinguished structural simplex; it makes no claim in terms of the total number of auxiliary action labels in the completion-total system.

19.17 Synthesis: finite compilation and universal next-order freedom

The compiler and structure theorems now give three levels of conclusion. First, every fixed positive-cost finite-phase periodic action system admits a finite arithmetic description of exact metric and local optimal-compatibility data. Second, the hierarchy does not collapse: for every prescribed k, a first discrepancy can occur exactly at order k, and independent products amplify such discrepancies Booleanly. Third, the ambiguity left after fixing all lower-order data can be maximal rather than sparse: one common lower-order signature can support every order-k pattern on a distinguished s-simplex.

Thus

shortest-path geometry is finite in description

while

optimal compatibility is unbounded in informational order and can be universally free at the next order.

Equivalently, fixing exact distance and the entire compatibility hierarchy through order $k-1$ imposes no universal restriction on the next-order layer. The $\binom{s}{k}$ possible distinguished k-faces can carry exactly $\binom{s}{k}$ independent binary bits, yielding the full pattern count $2^{\binom{s}{k}}$.

The symbolic proofs above are primary. Appendix D records independent small-parameter regression checks of common distance, complete lower-order invariance, and exhaustive realization of all prescribed k-face patterns.

Research branch — Full simplicial universality. *The universal k-fiber theorem varies one distinguished compatibility layer while freezing the lower layers. The next four chapters remove that distinguished-rank restriction: one fixed coefficient bank and one fixed Bellman system are required to realize every labeled simplicial complex. Discovery chronology is recorded in the Preface; here the branch is placed directly beside its geodesic prerequisites.*

Chapter 20

Universal Coefficient Banks and Simultaneous Simplicial Representation

The preceding k-fiber theorem isolates one rank at a time. The present chapter changes the quantifier order: the coefficient system is fixed before the target complex is known, and only the quota vector is allowed to vary. This produces a simultaneous representation problem rather than a target-dependent threshold-dimension problem.

20.1 Definitions and Main Results

Let $[s] = \{1, \dots, s\}$, and let $\mathrm{Simp}([s])$ denote the nonvoid downward-closed families of subsets of $[s]$. Ghost vertices are allowed unless stated otherwise. Throughout this branch, the rank of a face J means its cardinality $|J|$; thus a rank-k face has simplicial dimension $k-1$.

For

$$A \in \mathbb{R}^{d\times s}_{\geq 0}, \qquad b \in \mathbb{R}^{d}_{\geq 0},$$

define

$$\mathcal{K}(A,b) = \{J \subseteq [s] : A\mathbf{1}_J \leq b\}.$$

Every such family is simplicial because $A \geq 0$.

Definition 20.1. The *simultaneous universal row complexity* is

$$u(s) = \min\left\{ d : \exists A \in \mathbb{R}^{d\times s}_{\geq 0}\ \forall K \in \mathrm{Simp}([s])\ \exists b_K \geq 0 :\ K = \mathcal{K}(A, b_K) \right\}.$$

The *common-shell universal row complexity* is

$$u_\Sigma(s) = \min\Big\{ d : \begin{array}{l} \exists A \in \mathbb{R}^{d\times s}_{\geq 0},\ \exists R \geq 0, \\ \forall K \in \mathrm{Simp}([s])\ \exists b_K \geq 0 :\ K = \mathcal{K}(A, b_K),\ \|b_K\|_1 = R \end{array} \Big\}.$$

Put

$$w_s = \binom{s}{\lfloor s/2 \rfloor}, \qquad W_s = 2^s - 1, \qquad R_s = w_s W_s.$$

Theorem 20.2 (Exact Universal Row Complexity and Zero Shell Defect)**.** *For every $s \geq 2$,*

$$u(s) = u_\Sigma(s) = w_s.$$

Moreover, there exists

$$A_s \in \mathbb{Z}_{>0}^{w_s \times s}$$

such that for every $K \in \mathrm{Simp}([s])$ *there is* $b_K \in \mathbb{Z}_{\geq 0}^{w_s}$ *with*

$$K = \mathcal{K}(A_s, b_K), \qquad \|b_K\|_1 = R_s.$$

No nonnegative real coefficient matrix with fewer than w_s *rows is universal. Consequently* $\delta_\Sigma(s) = 0$.

This result is proved as theorem 21.5.

The notation $H(v, G)|_{I_\star}$, defined formally in Section 22, denotes the family of distinguished action subsets satisfying the corresponding Bellman optimality equality at v.

Theorem 20.3 (Universal Bellman Compatibility)**.** *For every* $s \geq 2$*, there exists one positive-cost translation-periodic shortest-path system* $\mathcal{U}_s$ *with distinguished action set* $I_\star = \{a_1, \ldots, a_s\}$ *such that for every* $K \in \mathrm{Simp}([s])$ *there exists a state* v_K *satisfying*

$$H(v_K, G)|_{I_\star} = K.$$

The witness may be chosen so that

$$d((v_K, \mathsf{P}), G) = 1 + R_s$$

for every K.

This result is proved as theorem 22.5.

Theorem 20.4 (Universal Extension-Fiber Completeness)**.** *Fix* $s \geq 2$*,* $0 \leq r \leq s$*, and a downward-closed lower truncation* $L \subseteq 2^{[s]}_{\leq r}$*. Let* $P_{s,r}(L)$ *be the poset of higher faces compatible with* L*. Then*

$$\mathrm{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)),$$

and the same fixed Bellman system realizes the entire fiber:

$$\mathrm{ExtFib}_r^{\mathcal{U}_s}(L) = \mathrm{ExtFib}_r(L).$$

Every member admits a witness at the same canonical distance $1 + R_s$.

This result is proved as theorem 23.2.

Corollary 20.5 (Simultaneous All-Rank Saturation)**.** *For every* $2 \leq k \leq s$*, the same system* $\mathcal{U}_s$ *realizes every subset of the rank-*k *layer while the complete lower truncation through rank* $k-1$ *is fixed. Hence*

$$|\,\mathrm{RankFib}_k\,| = 2^{\binom{s}{k}}.$$

This result appears as theorem 23.3.

20.2 Universal Coefficient Construction

Fix $s \geq 2$. Choose a symmetric chain decomposition

$$2^{[s]} = C_1 \sqcup \cdots \sqcup C_{w_s},$$

whose classical existence goes back to de Bruijn, van Ebbenhorst Tengbergen, and Kruyswijk [92]. For each chain C_q, choose an ordering $i_{q,1}, \ldots, i_{q,s}$ of $[s]$ such that every member of C_q is an initial segment for the corresponding admissible cardinalities. Define the qth row of A_s by

$$a_{q,i_{q,j}} = 2^{s-j}.$$

Thus every row is a permutation of $2^{s-1}, \ldots, 1$, and every row has total sum W_s.

20.2.1 Exact one-row separation

Let $E \in C_q$ with $|E| = m$, so

$$E = \{i_{q,1}, \dots, i_{q,m}\}.$$

Define

$$\tau_q(E) = W_s - 2^{s-m}.$$

Then $A_q \mathbf{1}_E = \tau_q(E) + 1$.

Lemma 20.6 (Exact chain separator)**.** *For every $J \subseteq [s]$,*

$$A_q \mathbf{1}_J \le \tau_q(E) \quad \Longleftrightarrow \quad E \nsubseteq J.$$

Proof. If $E \subseteq J$, then $A_q \mathbf{1}_J \ge A_q \mathbf{1}_E = \tau_q(E) + 1$. Conversely, suppose $E \nsubseteq J$. At least one element of E is omitted. The smallest weight carried by an element of E is 2^{s-m}, whereas the total weight of all positions outside E is $2^{s-m} - 1$. Hence the largest score of a set omitting an element of E is at most

$$A_q \mathbf{1}_E - 2^{s-m} + (2^{s-m} - 1) = \tau_q(E).$$

□

20.2.2 Minimal nonfaces and canonical quotas

For $K \in \mathrm{Simp}([s])$, let $\mathrm{MinNF}(K)$ denote its set of minimal nonfaces. This is an antichain, so each chain C_q contains at most one minimal nonface. Define

$$b^0_{K,q} = \begin{cases} \tau_q(E), & C_q \cap \mathrm{MinNF}(K) = \{E\}, \\ W_s, & C_q \cap \mathrm{MinNF}(K) = \varnothing. \end{cases}$$

Proposition 20.7 (Universal realization)**.** *For every $K \in \mathrm{Simp}([s])$,*

$$K = \{J \subseteq [s] : A_s \mathbf{1}_J \le b^0_K\}.$$

Proof. If $J \in K$, no minimal nonface is contained in J, so every active row accepts J by theorem 20.6; inactive rows have quota W_s and are automatic. If $J \notin K$, then J contains some minimal nonface E, and the row corresponding to the chain of E rejects J by theorem 20.6. □

Thus $u(s) \le w_s$.

20.2.3 Canonical quota mass

The canonical quota total is

$$\|b^0_K\|_1 = w_s W_s - \sum_{E \in \mathrm{MinNF}(K)} 2^{s-|E|}.$$

Define the canonical shell deficit by

$$\mathrm{def}(K) := R_s - \|b^0_K\|_1.$$

Then

$$\mathrm{def}(K) = \sum_{E \in \mathrm{MinNF}(K)} 2^{s-|E|}.$$

If $|\,\mathrm{MinNF}(K)| < w_s$, some row is inactive. Increasing its quota from W_s to $W_s + \mathrm{def}(K)$ leaves the represented family unchanged and brings the total quota to R_s. Thus only the extremal case

$$|\,\mathrm{MinNF}(K)| = w_s$$

remains.

20.2.4 Injectivity of subset scores

For every row q, the map $J \mapsto A_q \mathbf{1}_J$ is injective because the row entries are distinct powers of two. Hence

$$J \longmapsto A_s \mathbf{1}_J$$

is injective on $2^{[s]}$.

Chapter 21

Exact Row Complexity and Common-Shell Compression

The previous chapter gives a universal construction. We now determine its exact row cost and show that synchronizing all target quota vectors to one common ℓ_1 shell does not increase that optimum.

21.1 Maximum antichains

Suppose $|\operatorname{MinNF}(K)| = w_s$. Since $\operatorname{MinNF}(K)$ is an antichain, equality holds in Sperner's theorem [91]. We use the LYM inequality in the form proved by Lubell [93].

Proposition 21.1 (Maximum antichains are middle levels)**.** *If $\mathcal{A} \subseteq 2^{[s]}$ is an antichain with $|\mathcal{A}| = w_s$, then*

$$\mathcal{A} = \binom{[s]}{r}$$

for some middle rank r.

Proof. The LYM inequality gives

$$\sum_{E \in \mathcal{A}} \frac{1}{\binom{s}{|E|}} \leq 1.$$

Since $|\mathcal{A}| = w_s$ and $\binom{s}{|E|} \leq w_s$, equality must hold throughout, so every member lies on a maximum level. If s is even this level is unique.

Let $s = 2m + 1$. The two maximum levels are $\mathcal{L}_m$ and $\mathcal{L}_{m+1}$. Consider the bipartite inclusion graph between them; it is regular and connected. Put $X = \mathcal{A} \cap \mathcal{L}_m$. Since $\mathcal{A}$ is an antichain,

$$\mathcal{A} \cap \mathcal{L}_{m+1} \subseteq \mathcal{L}_{m+1} \setminus N(X).$$

For nonempty proper X, regularity gives $|N(X)| \geq |X|$. Equality would force $X \cup N(X)$ to be a union of connected components, contradicting connectedness. Hence $|N(X)| > |X|$, which forces $|\mathcal{A}| < w_s$. Therefore a maximum antichain cannot mix the two middle levels. □

Consequently, in the extremal case

$$\operatorname{MinNF}(K) = \binom{[s]}{r}, \qquad K = 2^{[s]}_{\leq r-1}.$$

Every canonical row then has quota

$$\tau = W_s - 2^{s-r},$$

and

$$\operatorname{def}(K) = w_s 2^{s-r}.$$

21.2 Two-for-one middle-level compression

Choose a chain C_{q_1} and write its rank-r member as

$$E = \{i_{q_1,1}, \dots, i_{q_1,r}\}.$$

Let

$$H = \{i_{q_1,1}, \dots, i_{q_1,r-1}\}, \qquad \beta = A_{q_1}\mathbf{1}_H = 2^s - 2^{s-r+1}.$$

Lemma 21.2 (Compressed row)**.** *Every J with $|J| \le r-1$ satisfies $A_{q_1}\mathbf{1}_J \le \beta$, while every r-set F containing H violates this inequality.*

Proof. H contains the $r-1$ largest row weights, so it maximizes the score among subsets of cardinality at most $r-1$. If $F = H \cup \{x\}$, then positivity gives $A_{q_1}\mathbf{1}_F = \beta + a_{q_1,x} > \beta$. □

There are $s - r + 1 \ge 2$ distinct r-sets containing H. Choose one $F \ne E$, and let q_2 be its chain. Modify the quotas by

$$\widetilde{b}_{K,q_1} = \beta, \qquad \widetilde{b}_{K,q_2} = W_s,$$

and keep all other rows at the canonical value τ.

Proposition 21.3 (Extremal compression preserves the complex)**.**

$$K = \{J \subseteq [s] : A_s\mathbf{1}_J \le \widetilde{b}_K\}.$$

Proof. If $|J| \le r-1$, the compressed row accepts by theorem 21.2, and every other active row is a canonical separator of an r-set, so all such J are accepted.

Now let $|J| \ge r$, and choose an r-subset $G \subseteq J$. If $G \notin \{E, F\}$, the canonical separator row belonging to the chain of G is unchanged and rejects J. If $G = E$ or $G = F$, then $H \subseteq G \subseteq J$, so the compressed row q_1 rejects J. Hence every set of cardinality at least r is rejected. □

The compressed row changes from τ to β, so

$$\beta - \tau = 1 - 2^{s-r}.$$

Making row q_2 inactive adds 2^{s-r}, for a net gain of 1. Therefore

$$\|\widetilde{b}_K\|_1 = w_s W_s + 1 - w_s 2^{s-r}.$$

The remaining deficit is

$$\Delta = w_s 2^{s-r} - 1 \ge 0.$$

Since row q_2 is inactive, increasing its quota by Δ preserves the represented family and yields total quota R_s. Therefore

$$u_\Sigma(s) \le w_s.$$

21.3 Lower-orthant shattering

For $b \in \mathbb{R}^d$, write

$$O_b = \{x \in \mathbb{R}^d : x \le b\}.$$

Lemma 21.4 (Lower-orthant shattering bound)**.** *If $x_1, \dots, x_n \in \mathbb{R}^d$ are shattered by lower orthants, then $n \le d$.*

Proof. For each i, shattering provides a lower orthant containing every point except x_i. Hence in some coordinate c_i, the point x_i is the unique strict maximum among all x_j. Two distinct points cannot be the unique strict maximum in the same coordinate, so $i \mapsto c_i$ is injective and $n \le d$. □

Let $A \in \mathbb{R}^{d\times s}_{\ge 0}$ be any simultaneous universal matrix. Choose a maximum antichain $\mathcal{A}$ with $|\mathcal{A}| = w_s$, and define score points

$$x_E = A\mathbf{1}_E, \qquad E \in \mathcal{A}.$$

For any $S \subseteq \mathcal{A}$, set $K_S = {\downarrow}S$ if $S \neq \varnothing$, and $K_\varnothing = \{\varnothing\}$. Since $\mathcal{A}$ is an antichain,

$$K_S \cap \mathcal{A} = S.$$

Universality of A gives a lower orthant cutting out exactly S from the score points. Thus the score points are shattered. They are also pairwise distinct: if $x_E = x_F$ for distinct $E, F \in \mathcal{A}$, no lower orthant could contain one without the other. By theorem 21.4, $d \ge w_s$. Since every common-shell universal matrix is, after forgetting the shell constraint, an ordinary universal matrix, we also have

$$u_\Sigma(s) \ge u(s).$$

Combining these lower bounds with the constructions above gives equality in both models.

Theorem 21.5 (Exact Universal Row Complexity and Zero Shell Defect)**.** *For every $s \ge 2$,*

$$u(s) = u_\Sigma(s) = w_s = \binom{s}{\lfloor s/2 \rfloor}.$$

Moreover, one positive integer matrix $A_s \in \mathbb{Z}^{w_s\times s}_{>0}$ simultaneously realizes every labeled simplicial complex on $[s]$, and every target admits a quota vector with

$$\|b_K\|_1 = w_s(2^s - 1).$$

Thus $\delta_\Sigma(s) = 0$. The row count is exact even against arbitrary nonnegative real coefficient matrices.

21.4 Full-vertex and one-vertex boundary cases

If ghost vertices are forbidden, then for $s \ge 3$ the same maximum-antichain shattering argument works using a middle level of rank at least two, so

$$u^{\mathrm{fv}}(s) = u^{\mathrm{fv}}_\Sigma(s) = w_s.$$

For $s = 2$,

$$u^{\mathrm{fv}}(2) = 1, \qquad u^{\mathrm{fv}}_\Sigma(2) = 2.$$

Indeed, the only full-vertex complexes are $\{\varnothing, \{1\}, \{2\}\}$ and $2^{[2]}$. A single row $(1, 1)$ with thresholds 1 and 2 realizes the two targets. A one-row common-shell realization is impossible because equal ℓ_1-mass would force the scalar thresholds to coincide. Two identical rows $(1, 1)$ with quota vectors $(1, 3)$ and $(2, 2)$ realize the two targets at common total 4.

For completeness, if $s = 1$ and ghosts are allowed,

$$u(1) = 1, \qquad u_\Sigma(1) = 2.$$

A common-shell realization is given by

$$A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad b_{\{\varnothing\}} = (0, 2)^T, \qquad b_{2^{[1]}} = (1, 1)^T.$$

One common-shell row is impossible because the two scalar quota values would have to be equal.

Chapter 22

Periodic Shortest-Path Lift and Exact Bellman Universality

The coefficient theorem is now returned to the shortest-completion language of Part II. The same fixed additive data are lifted to one positive-cost periodic system in which every target complex appears literally as a distinguished Bellman-optimal compatibility complex.

Write

$$A_s = [u_1 \ \cdots \ u_s], \qquad u_i \in \mathbb{Z}^{w_s}_{>0}.$$

Define the state space

$$V_s = \mathbb{Z}^{w_s} \times \{\mathsf{P}, \mathsf{C}\},$$

with goal $G = (0, \mathsf{C})$. For each $i \in [s]$, introduce a distinguished action

$$a_i : (x, \mathsf{P}) \to (x + u_i, \mathsf{P}), \qquad c(a_i) = \|u_i\|_1.$$

Introduce a commit action

$$\gamma : (x, \mathsf{P}) \to (x, \mathsf{C}), \qquad c(\gamma) = 1,$$

and completion moves

$$(x, \mathsf{C}) \to (x \pm e_q, \mathsf{C}), \qquad q = 1, \dots, w_s,$$

each of cost 1. There is no return from C to P. Denote the system by $\mathcal{U}_s$.

Proposition 22.1 (Exact completion metric)**.** *For every $x \in \mathbb{Z}^{w_s}$,*

$$d((x, \mathsf{C}), G) = \|x\|_1.$$

Proof. Coordinatewise cancellation gives a path of length $\|x\|_1$, and every unit move changes the ℓ_1-distance to the origin by at most one. □

Proposition 22.2 (Exact prefix metric)**.** *For every $x \in \mathbb{Z}^{w_s}$,*

$$d((x, \mathsf{P}), G) = 1 + \|x\|_1.$$

Proof. Committing immediately gives the upper bound. For an arbitrary path, suppose distinguished actions are used with multiplicities $n_i \geq 0$ before commit, and set $U = \sum_i n_i u_i \geq 0$. Their total cost equals $\|U\|_1$. The remaining path costs at least $1 + \|x + U\|_1$, so the total is at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1$$

by the triangle inequality. □

Thus no one-use gadget is required: distinguished actions may repeat arbitrarily.
Let

$$I_\star = \{a_1, \dots, a_s\}.$$

For $J \subseteq [s]$, define

$$U_J = \sum_{i \in J} u_i = A_s \mathbf{1}_J, \qquad c(J) = \sum_{i \in J} \|u_i\|_1 = \|U_J\|_1.$$

We declare J Bellman-optimal at (v, P) when

$$c(J) + d((v + U_J, \mathsf{P}), G) = d((v, \mathsf{P}), G).$$

The family of all such J is denoted $H(v, G)|_{I_\star}$. By theorem 22.2, this is equivalent to

$$\|U_J\|_1 + \|v + U_J\|_1 = \|v\|_1.$$

Lemma 22.3 (Coordinate identity)**.** *For $u, b \in \mathbb{R}^{w_s}_{\geq 0}$,*

$$\|u\|_1 + \|u - b\|_1 = \|b\|_1 \quad \iff \quad u \leq b.$$

Proof. Coordinatewise, for $u_q, b_q \geq 0$,

$$u_q + |u_q - b_q| = b_q$$

if and only if $u_q \leq b_q$. Summing gives the claim. □

Proposition 22.4 (Exact Bellman encoding)**.** *For every $b \in \mathbb{R}^{w_s}_{\geq 0}$,*

$$H(-b, G)|_{I_\star} = \{J \subseteq [s] : A_s \mathbf{1}_J \leq b\}.$$

Proof. At $v = -b$, the Bellman equality becomes

$$\|U_J\|_1 + \|U_J - b\|_1 = \|b\|_1,$$

which is equivalent to $U_J \leq b$ by theorem 22.3. □

If J is Bellman-optimal at $-b$, then every $I \subseteq J$ is Bellman-optimal because $U_I \leq U_J \leq b$. Therefore every ordering of the actions in J appears as an initial segment of a shortest path.

Theorem 22.5 (Universal Optimal-Compatibility Theorem)**.** *For every $s \geq 2$, the fixed system $\mathcal{U}_s$ realizes every labeled simplicial complex on $[s]$: for every $K \in \mathrm{Simp}([s])$ there exists $v_K \in \mathbb{Z}^{w_s}$ with*

$$H(v_K, G)|_{I_\star} = K.$$

The canonical choice is $v_K = -b_K$, where b_K is the common-shell quota vector of theorem 21.5. Every canonical witness satisfies

$$d((v_K, \mathsf{P}), G) = 1 + w_s(2^s - 1).$$

Proof. By theorem 21.5, choose b_K with $K = \{J : A_s \mathbf{1}_J \leq b_K\}$ and $\|b_K\|_1 = R_s$. Then theorem 22.4 gives the exact equality of complexes, and theorem 22.2 gives the common distance. □

The complete equivalence chain is

$$\begin{aligned} J \in K &\iff A_s \mathbf{1}_J \leq b_K \\ &\iff \|U_J\|_1 + \|U_J - b_K\|_1 = \|b_K\|_1 \\ &\iff J \in H(v_K, G)|_{I_\star}. \end{aligned}$$

Chapter 23

Extension Fibers, Full Metric-Shell Capacity, and Topological Universality

The final step removes any remaining rankwise viewpoint. Lower truncations are classified by their complete extension fibers, all ranks saturate simultaneously, and the common distance shell realizes the full labeled simplicial universe.

23.1 Extension fibers and simultaneous all-rank saturation

For $0 \le r \le s$, write

$$2^{[s]}_{\le r} = \{J \subseteq [s] : |J| \le r\}.$$

A lower truncation through rank r is a downward-closed family $L \subseteq 2^{[s]}_{\le r}$ containing $\varnothing$.

Define the compatible higher-face poset

$$P_{s,r}(L) = \{E \subseteq [s] : |E| > r,\ \forall F \subseteq E\ (|F| \le r \Rightarrow F \in L)\},$$

ordered by inclusion, and the extension fiber

$$\mathrm{ExtFib}_r(L) = \{K \in \mathrm{Simp}([s]) : K \cap 2^{[s]}_{\le r} = L\}.$$

Theorem 23.1 (Extension-Ideal Classification)**.** *For every lower truncation L,*

$$\mathrm{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)).$$

The bijection is $K \mapsto K \cap P_{s,r}(L)$, with inverse $I \mapsto L \cup I$.

Proof. If $K \in \mathrm{ExtFib}_r(L)$, then every higher face of K lies in $P_{s,r}(L)$, and the higher part is downward closed inside that poset. Conversely, if I is an order ideal of $P_{s,r}(L)$, then $L \cup I$ is simplicial: a subface of size at most r lies in L by compatibility, while a higher subface lies in $P_{s,r}(L)$ and hence in I by ideality. The two maps are inverse. □

The fiber is therefore a finite distributive lattice with meet and join given by intersection and union.

Define the system extension fiber

$$\mathrm{ExtFib}^{\mathcal{U}_s}_r(L) = \{H(v,G)|_{I_\star} : H(v,G)|_{I_\star} \cap 2^{[s]}_{\le r} = L\}.$$

Theorem 23.2 (Universal Extension-Fiber Completeness)**.** *For every $s \geq 2$, every $0 \leq r \leq s$, and every lower truncation L,*

$$\mathrm{ExtFib}_r^{\mathcal{U}_s}(L) = \mathrm{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)).$$

Every member admits a canonical witness at distance

$$D_s = 1 + w_s(2^s - 1).$$

Proof. Every system compatibility family is simplicial. Conversely, every $K \in \mathrm{ExtFib}_r(L)$ is realized by theorem 22.5, with canonical witness at distance D_s. □

Thus Bellman optimality imposes no additional higher-order restriction beyond simplicial downward closure in the constructed system.

For $2 \leq k \leq s$, let

$$L_{k-1} = 2^{[s]}_{\leq k-1},$$

and define

$$\mathrm{RankFib}_k = \left\{ K \cap \binom{[s]}{k} : K \in \mathrm{ExtFib}_{k-1}(L_{k-1}) \right\}.$$

Every $A \subseteq \binom{[s]}{k}$ is realized by

$$K_A = L_{k-1} \cup A,$$

with no faces above rank k.

Corollary 23.3 (Simultaneous All-Rank Saturation)**.** *For every $2 \leq k \leq s$,*

$$\mathrm{RankFib}_k = 2^{\binom{[s]}{k}}, \qquad |\,\mathrm{RankFib}_k\,| = 2^{\binom{s}{k}}.$$

The same system $\mathcal{U}_s$ works for every k.

If k_s is a middle rank, then

$$\log_2 |\,\mathrm{RankFib}_{k_s}\,| = w_s.$$

23.2 Full Metric-Shell Capacity and Entropy

Let

$$R_s = w_s(2^s - 1), \qquad D_s = 1 + R_s,$$

and define the full prefix-phase metric shell

$$\mathbb{S}_s = \{v \in \mathbb{Z}^{w_s} : d((v, \mathsf{P}), G) = D_s\} = \{v : \|v\|_1 = R_s\}.$$

Proposition 23.4 (Arbitrary-state classification)**.** *If $v \leq 0$ coordinatewise, then*

$$H(v, G)|_{I_\star} = \{J \subseteq [s] : A_s \mathbf{1}_J \leq -v\}.$$

If v has at least one positive coordinate, then

$$H(v, G)|_{I_\star} = \{\varnothing\}.$$

In particular, every prefix-phase state produces a simplicial compatibility complex.

Proof. The Bellman equality is coordinatewise. If $v \leq 0$, it holds exactly when $U_J \leq -v$. If some coordinate of v is positive, then every nonempty U_J is strictly positive in that coordinate, so equality fails there; only $J = \varnothing$ remains. □

Theorem 23.5 (Full-Shell Universality)**.** *For every $s \geq 2$,*

$$\{H(v, G)|_{I_\star} : v \in \mathbb{S}_s\} = \mathrm{Simp}([s]).$$

Proof. By theorem 23.4, every shell state produces a simplicial complex. Conversely, every $K \in \mathrm{Simp}([s])$ has a canonical witness $v_K \in \mathbb{S}_s$ by theorem 22.5. □

Let $M(s)$ denote the Dedekind number, equivalently the number of antichains in $2^{[s]}$. Since we count nonvoid simplicial complexes, their number is $M(s) - 1$. Define the semantic shell capacity by

$$\mathrm{Cap}(\mathbb{S}_s) = \left|\{H(v, G)|_{I_\star} : v \in \mathbb{S}_s\}\right|.$$

Then

$$\mathrm{Cap}(\mathbb{S}_s) = M(s) - 1.$$

This is semantic capacity, not geometric shell cardinality.

We use the phrase *semantic entropy* only as shorthand for the base-two logarithm of the number of distinct compatibility complexes; no probabilistic or Shannon-theoretic structure is intended. Kleitman's classical asymptotic solution of Dedekind's problem gives [94]

$$\log_2 M(s) = (1 + o(1))w_s.$$

Hence

$$\log_2 \mathrm{Cap}(\mathbb{S}_s) = (1 + o(1))w_s$$

and therefore

$$u(s) \sim \log_2 \mathrm{Cap}(\mathbb{S}_s).$$

The same middle-layer width controls both quantities, but no generic dimension–entropy law is claimed.

Moreover, for a middle rank k_s,

$$\frac{\log_2 |\,\mathrm{RankFib}_{k_s}\,|}{\log_2 \mathrm{Cap}(\mathbb{S}_s)} \to 1.$$

Thus complete all-order universality strengthens exact structure without changing the leading first-order information scale already visible at the middle layer.

Let C_s be the number of labeled simplicial complexes on $[s]$ containing every singleton. Inclusion–exclusion over ghost vertices gives

$$C_s = \sum_{j=0}^{s} (-1)^j \binom{s}{j} (M(s - j) - 1).$$

The number with at least one ghost vertex is at most $s(M(s - 1) - 1)$. Since $w_{s-1} = (1/2 + o(1))w_s$ and $\log_2 M(t) = (1 + o(1))w_t$, we have $M(s - 1)/M(s) \to 0$, hence

$$C_s = (1 - o(1))M(s), \qquad \log_2 C_s = (1 + o(1))w_s.$$

23.3 Topological Universality and Effective Consequences

Every finite labeled simplicial complex occurs literally as a distinguished compatibility complex. Consequently, across the uniform family $\{\mathcal{U}_s\}_{s\geq 2}$, every value of simplicial dimension, f-vector, Euler characteristic, homology or cohomology group, Betti-number pattern, torsion pattern, and finite simplicial homotopy type that is attained by a finite simplicial complex is also attained by a compatibility complex. In particular, every finitely presented group occurs as a fundamental group, as recorded below.

Corollary 23.6 (Distance does not determine topology)**.** *Let $\mathcal{I}$ be any invariant non-constant on* Simp($[s]$)*. Then there exist $v_1, v_2 \in \mathbb{S}_s$ with equal distance D_s from the goal but*

$$\mathcal{I}(H(v_1, G)|_{I_\star}) \neq \mathcal{I}(H(v_2, G)|_{I_\star}).$$

Every finitely presented group occurs as the fundamental group of a finite two-dimensional simplicial complex; see, for example, standard presentation-complex treatments such as [95].

Corollary 23.7 (Fundamental-group universality)**.** *For every finitely presented group Γ, there exist $s \geq 2$ and $v \in \mathbb{Z}^{w_s}$ such that*

$$\pi_1\big(|H(v, G)|_{I_\star}|\big) \cong \Gamma.$$

The realizing compatibility complex may be chosen two-dimensional, and v may be chosen as a canonical common-shell witness.

For algorithmic consequences, fix for each s a standard recursive symmetric-chain decomposition. Then $s \mapsto A_s$ is computable, and from a finite labeled simplicial complex K one can effectively enumerate minimal nonfaces, assign canonical quotas, perform ordinary shell padding or the explicit extremal compression, compute b_K, and output $v_K = -b_K$.

Proposition 23.8 (Effective universality)**.** *There is an effective transformation*

$$K \longmapsto (s, v_K)$$

such that

$$H(v_K, G)|_{I_\star} = K, \qquad d((v_K, \mathsf{P}), G) = D_s.$$

For fixed s, the semantic universe is finite, with $M(s) - 1$ objects, so no fixed-s undecidability claim is made. Undecidability can arise only when s varies over the uniform family.

Theorem 23.9 (Effective transfer principle)**.** *Let $\mathcal{P}$ be a property of finite simplicial complexes. If deciding $\mathcal{P}$ for finite simplicial complexes is undecidable, then deciding whether $H(v, G)|_{I_\star}$ has property $\mathcal{P}$ is undecidable across the uniform family $\{\mathcal{U}_s\}_{s \geq 2}$.*

Proof. Given K, compute (s, v_K) using theorem 23.8. A decision procedure for the Bellman version would decide $\mathcal{P}$ for K. □

Triviality of the fundamental group is undecidable for finite simplicial complexes, and this remains true in dimension two [96, 97]. Hence simple connectivity is undecidable across the uniform Bellman family. The reduction already uses canonical witnesses on their prescribed common-distance shells.

These topological and algorithmic statements are consequences of exact effective universality, not the structural center of this branch.

Bridge back to the main pushdown spine. The four chapters above complete the geodesic-compiler branch at the full simplicial level. They are logically independent of the cancellation and pushdown constructions that follow, so the monograph now returns to the original DPDA and resumes the pushdown-normalization spine.

23.4 Returning to the original machine from the opposite direction

The compiler and full-simplicial universality Part completes the outward abstraction of shortest-completion geometry. We now zoom back in.

In Chapter 1 we deliberately marked one local transition of the original 2:1 DPDA:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

So far that transition has been interpreted as:

- a stack cancellation;
- one unit of residual correction;
- one edge of a shortest-completion ladder.

But it contains another structure that the completion and compiler theories do not use. When the nonnegative 2:1 computations are drawn as lattice paths, the cancellation occurs at a precise arithmetic family of landing heights. That observation leads to an independent branch:

pushdown cancellation → correction signatures → intrinsic defect → Catalan geometry.

We now return to the 2023 machine and follow that transition.

Part III

Cancellation Geometry, Chronology, and the General Real-Time p:q Machine

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Part guide. **Question.** How does the original cancellation mechanism generalize from the historical 2:1 machine to every coprime positive ratio, and what arithmetic structure is exposed on the way? **Main result.** Defect and chronological-carry analysis leads to the centered real-time literal-demand p:q normalizer with exact WRITE, SEARCH, and CANCEL roles. **Why here.** It closes the constructional loop between scalar completion semantics and an explicit general machine. **Dependency.** It uses the residual/completion coordinates of Part I; the cancellation-enumeration offshoot is a consequence rather than a prerequisite.

Chapter 24

Zooming Back into the Original DPDA

The preceding geodesic-compiler branch generalized shortest completion outward.

We now move in the opposite direction.

Return to the original 2:1 DPDA of Chapter 1 and isolate one local transition:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

At the machine level, this is a cancellation.

At the residual level, it decreases

$$\rho = N_1 - 2N_0$$

by 2.

At the geodesic level, Chapter 3 showed that it consumes one required zero along a shortest-completion path.

There is, however, another structure hidden in the same event.

If the input word is drawn as a nonnegative 2-Dyck path, the cancellation occurs at a precise arithmetic set of landing heights.

That observation starts a second geometry.

24.1 The machine facts we reuse, not re-prove

The cancellation analysis needs only two facts already established in Chapter 1:

Reachable-store normal form

$$\mathcal{S}_C = 0^* \cup 1^* \cup \{10\},$$

$$\mathcal{S}_L = 0^*,$$

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

State-potential invariant

For every consumed prefix,

$$\boxed{\rho(w) = D(S) + \Phi(q),}$$

with

$$D(S) = 2N_0(S) - N_1(S),$$

$$\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.$$

We do not duplicate those proofs here.

Their consequence for positive odd residuals is immediate.

Corollary 24.1 — Positive odd residuals.

Every reachable core configuration with positive odd residual at least 3 is uniquely of the form

$$\boxed{(L, 0^k)}$$

for some

$$k \geq 0,$$

and its residual is

$$\boxed{2k + 3.}$$

The left state therefore has a very precise path-height meaning.

24.2 The nonnegative path sector

Encode input symbols as steps:

$$1 \longleftrightarrow U \quad \text{of height } +1,$$

$$0 \longleftrightarrow D \quad \text{of height } -2.$$

For a prefix w, the path height is

$$\boxed{h(w) = N_1(w) - 2N_0(w) = \rho(w).}$$

A word whose prefix heights are all nonnegative and whose final height is 0 is a 2-Dyck excursion.

The cancellation statistic will first be discovered inside this nonnegative sector.

This restriction matters.

The later combinatorial theory will be defined directly on generalized Dyck paths and will no longer depend on the remaining DPDA transition table.

24.3 Exact landing-height correspondence

Immediately before a distinguished cancellation,

$$(L, 0^k) \xrightarrow{0} (L, 0^{k-1}),$$

the logical top 0 must exist, so

$$k \geq 1.$$

The pre-step residual is

$$2k + 3.$$

Reading 0 decreases the height by 2, so the down-step lands at

$$2k + 1.$$

Therefore every cancellation lands at one of:

$$3, 5, 7, \ldots$$

The converse is equally rigid.

Proposition 24.2 — Motivating $d = 2$ correspondence.

Within the nonnegative 2-Dyck sector of the original computation, the transition

$$\boxed{0, 0/\varepsilon}$$

occurs if and only if the corresponding down-step lands at height

$$\boxed{3, 5, 7, \ldots}$$

Proof.

The forward direction was just computed.

Conversely, suppose a down-step lands at

$$2k + 1 \geq 3.$$

Its pre-step height is

$$2k + 3 \geq 5.$$

This is a positive odd residual.

Corollary 24.1 therefore forces the pre-step configuration to be

$$(L, 0^k)$$

with

$$k \geq 1.$$

The enabled input-0 transition is exactly the distinguished left cancellation. □

This theorem is the only point at which the full original machine is needed to motivate the correction statistic.

Everything that follows will be stated intrinsically for paths.

24.4 From one machine event to a d-Dyck statistic

For

$$d \geq 2,$$

a d-Dyck excursion has:

- up-step U of height $+1$;
- down-step D of height $-d$;
- nonnegative prefix heights;
- final height 0.

A size-n excursion has:

$$n$$

down-steps and

$$dn$$

up-steps.
The original $d = 2$ cancellation landing heights were

$$3, 5, 7, \ldots$$

or equivalently

$$h_{\text{end}} \equiv 1 \pmod 2, \qquad h_{\text{end}} \geq 3.$$

This suggests the intrinsic generalization.

Definition 24.3 — Correction.

A down-step is a **correction** if

$$\boxed{h_{\text{end}} \equiv 1 \pmod d, \qquad h_{\text{end}} \geq d + 1.}$$

Equivalently,

$$h_{\text{end}} = d + 1 + dr$$

for a unique

$$r \in \mathbb{N}.$$

The parameter d now belongs to the combinatorial path system.
For $d > 2$, we are **not** claiming that the original DPDA itself implements this generalized rule.
The correct causal statement is:

$$\boxed{\begin{array}{c}\text{original } d = 2 \text{ cancellation} \to \text{intrinsic landing-height condition} \\ \to \text{combinatorial } d\text{-generalization.}\end{array}}$$

24.5 Chronological correction signatures

Suppose a path contains $m \geq 1$ corrections, in chronological order, landing at heights

$$c_1, \ldots, c_m.$$

Write

$$c_i = d + 1 + dr_i.$$

Definition 24.4 — Normalized correction signature.

The ordered tuple

$$\boxed{r(P) = (r_1, \ldots, r_m)}$$

is the normalized correction signature.

The order is essential.

Two paths can have the same number of corrections and even the same multiset of correction heights while having different ordered signatures.

This branch therefore remembers more than the scalar count statistic.

24.6 Path defect

If a size-n path has $m \geq 1$ corrections, define

$$\boxed{\delta(P) = n - m - 2.}$$

The form may look artificial before the realization theorem.

It is chosen so that:

- maximal-correction paths have defect 0;
- every proposed ordered signature has an exact minimum defect;
- the excess above that minimum can be increased freely by zero-to-zero padding.

The theorem in Chapter 25 will prove in particular that

$$\delta(P) \geq 0.$$

24.7 The new inverse problem

The original question was forward:

> Given a path, where do cancellations/corrections occur?

The signature theory asks the inverse question:

> Given an ordered sequence
>
> $$r = (r_1, \ldots, r_m),$$
>
> can it occur as a correction history?

And, if so:

What is the smallest path defect at which it can occur?

This is much stronger than counting corrections.
The answer is one explicit functional

$$\Delta(r).$$

To derive it, we first need the exact local price of moving from one correction height to the next.

Chapter 25

Correction Signatures and Minimum Defect

Let

$$r = (r_1, \dots, r_m) \in \mathbb{N}^m, \qquad m \geq 1.$$

The goal is to determine exactly when r is realizable as a normalized correction signature.

The answer is built from two local costs:

1. the price of moving between consecutive corrections;
2. the price of terminating after the final correction.

The first cost contains the key nonlinearity.

25.1 The threshold function

Define

$$\boxed{\psi(a) = \begin{cases} 0, & a \leq 1, \\ a, & a \geq 2. \end{cases}}$$

Why is there a jump from zero to a?

Suppose consecutive normalized correction heights are

$$r \quad \text{and} \quad r',$$

and set

$$a = r - r'.$$

Let b be the number of non-correction down-steps strictly between the two corrections, and let u be the number of intervening up-steps.

Height balance gives

$$\boxed{u = d(b + 1 - a).}$$

This one equation explains the threshold.

25.2 Local Gap Cost

Lemma 25.1 — Local Gap Cost.

For two consecutive normalized correction heights r, r', with

$$a = r - r',$$

the number b of intervening non-correction down-steps satisfies

$$\boxed{b \geq \psi(a).}$$

The bound is sharp.

Proof.

If

$$a \leq 1,$$

choose

$$b = 0.$$

Then

$$u = d(1 - a) \geq 0.$$

The block

$$\boxed{U^{d(1-a)}D}$$

moves directly to the next correction without an intervening down-step.

Now assume

$$a \geq 2.$$

The balance equation gives

$$b \geq a - 1.$$

If

$$b = a - 1,$$

then

$$u = 0.$$

But the first intervening down-step would land again in the correction residue class, creating an unintended correction between the two prescribed ones.

Therefore

$$b \geq a.$$

Equality is attainable by the detour block

$$\boxed{UD^aU^{d-1}D.}$$

The intermediate down-steps lie one residue class away from the correction class, so they are not corrections. The final D returns exactly to the prescribed correction height. □

This theorem is the geometric origin of ψ.

A downward change of size at most 1 is free.

A drop of size at least 2 cannot be taken directly because the naive descent would create an unwanted correction.

The path must leave the correction residue class and return.

That detour costs exactly the full drop a.

25.3 Terminal Cost

Lemma 25.2 — Terminal Cost.

If the final correction has normalized height r, then at least

$$\boxed{r+2}$$

non-correction down-steps are required after it.

The bound is sharp.

Proof.

Immediately after the final correction, the height is

$$d + 1 + dr.$$

If b down-steps and u up-steps remain, then

$$d + 1 + dr + u - db = 0.$$

Hence

$$u = d(b - r - 1) - 1.$$

Since

$$u \geq 0,$$

we must have

$$b \geq r + 2.$$

Equality is attained by

$$\boxed{U^{d-1}D^{r+2}.}$$

All its down-steps land at multiples of d, so none is a correction. □

The final signature entry therefore contributes directly to the minimum defect.

25.4 Intrinsic signature defect

Define

$$\boxed{\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}).}$$

This quantity contains no d.
That universality is already striking.
The physical correction heights scale as

$$d + 1 + dr_i,$$

but the normalized realizability cost is independent of the Dyck parameter.

25.5 Canonical blocks

To prove sufficiency constructively, define:

Initial block

$$\boxed{I_r = U^{2d+1+dr} D.}$$

Its unique down-step lands at correction height

$$d + 1 + dr.$$

Gap block

For $a \in \mathbb{Z}$,

$$\boxed{G_a = \begin{cases} U^{d(1-a)} D, & a \le 1, \\ U D^a U^{d-1} D, & a \ge 2. \end{cases}}$$

Terminal block

$$\boxed{T_r = U^{d-1} D^{r+2}.}$$

Padding block

$$\boxed{P = U^d D.}$$

The padding block starts and ends at height zero and creates no correction.

Proposition 25.3 — Block Purity.

Each canonical block creates exactly the advertised correction behavior and no unintended correction.

Moreover, when started at its prescribed height, every intermediate height remains nonnegative.

This proposition is what makes the global realization theorem compositional.

25.6 Defect–Signature Characterization

Theorem 25.4 — Defect–Signature Characterization.

For every

$$d \geq 2$$

and every normalized signature

$$r = (r_1, \ldots, r_m),$$

the following are equivalent:

1. r is realized by a d-Dyck excursion of defect δ;
2. $$\boxed{\Delta(r) \leq \delta.}$$

Equivalently,

$$\boxed{\delta_{\min}(r) = \Delta(r).}$$

If the signature has length m, then its minimum path size is

$$\boxed{n_{\min}(r) = m + 2 + \Delta(r).}$$

Proof — lower bound. A path realizing r contains:

$$n - m$$

non-correction down-steps.
Between correction i and correction $i + 1$, Lemma 25.1 forces at least

$$\psi(r_i - r_{i+1})$$

such steps.
After the last correction, Lemma 25.2 forces at least

$$r_m + 2.$$

Any non-correction down-steps before the first correction only increase the count. Therefore

$$n - m \geq 2 + r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}) = 2 + \Delta(r).$$

Hence

$$\delta(P) = n - m - 2 \geq \Delta(r).$$

Proof — construction. Let

$$a_i = r_i - r_{i+1}.$$

For

$$s \geq 0,$$

define

$$\boxed{W_d(r;s) = I_{r_1} G_{a_1} \cdots G_{a_{m-1}} T_{r_m} P^s.}$$

Block Purity shows that this is a nonnegative d-Dyck excursion whose normalized correction signature is exactly r.

Its number of down-steps is

$$m + 2 + \Delta(r) + s.$$

Thus

$$\delta(W_d(r;s)) = \Delta(r) + s.$$

Taking $s = 0$ gives a minimum realization.

Taking

$$s = \delta - \Delta(r)$$

realizes every larger defect. □

This is the structural center of the cancellation branch.

It solves both directions:

$$\boxed{\text{path} \to \text{signature}}$$

and

$$\boxed{\text{signature} \to \text{minimum realizing path.}}$$

25.7 Canonical does not mean unique

Corollary 25.5 — Canonical minimum realization.

Every signature r has the explicit minimum witness

$$\boxed{W_d^{\min}(r) = W_d(r;0).}$$

The word **canonical** means:

> the distinguished block construction selected by the theory.

It does **not** mean:

> every minimum realizing path is identical to this word.

This scope matters.

The theorem determines the minimum defect uniquely, not the complete set of minimum path orderings.

25.8 Signature persistence

Corollary 25.6 — Signature Persistence.

A signature r is realizable at exactly the defect levels

$$\boxed{\Delta(r), \Delta(r)+1, \Delta(r)+2, \ldots}$$

because each copy of

$$P = U^d D$$

adds one defect unit without changing the correction signature.
Therefore the intrinsic defect is a **birth level**.
A signature appears for the first time at $\Delta(r)$, then persists forever.

25.9 Maximum number of corrections

Since

$$\Delta(r) \geq 0,$$

every path with $m \geq 1$ corrections satisfies

$$\boxed{m \leq n - 2.}$$

Thus defect zero is exactly the maximal-correction regime.
That boundary turns out to be Catalan.

25.10 Zero defect

Corollary 25.7 — Zero-defect signatures.

$$\boxed{\Delta(r) = 0}$$

if and only if

$$\boxed{r_m = 0}$$

and

$$\boxed{r_i - r_{i+1} \leq 1 \qquad (1 \leq i < m).}$$

Reverse the sequence:

$$a_i = r_{m+1-i}.$$

Then

$$a_1 = 0$$

and

$$\boxed{a_{i+1} \leq a_i + 1.}$$

These are exactly the usual Dyck area-sequence inequalities.

Therefore the number of zero-defect signatures of length m is

$$\boxed{C_m.}$$

The Catalan family is not the whole correction theory.
It is its exact zero-cost boundary.

25.11 Defect as penalized overshoot

After reversal,

$$a_i = r_{m+1-i},$$

the defect becomes

$$\boxed{\Delta^{\leftarrow}(a) = a_1 + \sum_{\substack{1 \le i < m \\ a_{i+1} - a_i \ge 2}} (a_{i+1} - a_i).}$$

Thus ordinary Dyck area sequences are precisely the nonnegative sequences with zero cost.

Positive defect allows overshoots above the Catalan nearest-neighbor condition, but every overshoot of size at least 2 is charged by its full magnitude.

This makes the geometry intuitive:

$$\boxed{\text{Catalan behavior} = \text{zero-defect regime;}}$$

$$\boxed{\text{non-Catalan jumps} = \text{quantified correction debt.}}$$

The next chapter shows that the Catalan boundary is not only enumerative.
Its maximal-correction paths themselves contain a full binary tree.

Chapter 26

The Catalan Boundary and the Binary Cancellation Skeleton

At defect zero,

$$m = n - 2.$$

Thus a size-n path contains the maximum possible number of corrections.
The signature count is Catalan:

$$C_{n-2}.$$

But the path count contains an additional factor d:

$$dC_{n-2}.$$

This chapter explains both factors structurally.
The intrinsic correction structure is a full binary tree.
The factor d records only how the path enters that binary regime.

26.1 Two unavoidable unmarked nodes

Use the standard first-return decomposition of a nonempty d-Dyck excursion:

$$\boxed{P = UP_1UP_2 \cdots UP_dDP_{d+1}.}$$

Each internal node corresponds to one down-step.
A node is marked exactly when its down-step is a correction.
A maximal-correction size-n path has:

$$n - 2$$

marked internal nodes and therefore exactly two unmarked internal nodes.
The root is necessarily unmarked because its baseline is zero.
Before a correction baseline

$$d + 1$$

can be reached, at least one further unmarked internal node is necessary.
In the maximal-correction regime, these are the only two unmarked nodes.
That rigid entrance forces the rest of the tree into a binary skeleton.

26.2 The d entrance choices

Choose the unique nonempty pre-down-step child of the root.

If it is slot

$$j \in [d],$$

its baseline is j.

Every other root child must be empty, otherwise another unmarked internal node would be created.

At this second unmarked node, the unique child that reaches correction baseline $d+1$ without introducing an additional unmarked node is slot

$$d + 1 - j.$$

Thus there are exactly

$$\boxed{d}$$

possible entrance choices.

After those two forced unmarked nodes, the remaining structure is entirely inside the correction residue class.

26.3 Why correction nodes become binary

Suppose a correction node has baseline

$$h \equiv 1 \pmod{d}, \qquad h \geq d + 1.$$

Its pre-down-step child baselines are

$$h + 1, \ldots, h + d,$$

and its trailing child has baseline h.

Among these, the only baselines that remain in the correction class are:

$$h + d$$

and

$$h.$$

Therefore, once all unmarked nodes have been used up, a correction node has exactly two recursive correction slots:

- slot d;
- slot $d + 1$.

All other child slots must be empty.

The correction structure is therefore binary.

26.4 Binary Cancellation Skeleton

Let

$$\mathcal{B}_m$$

be the set of plane full binary trees with m internal nodes.
Let

$$\mathcal{M}_{d,n}$$

be the class of size-n maximally corrected d-Dyck excursions.

Theorem 26.1 — Binary Cancellation Skeleton.

For every

$$n \geq 3,$$

$$\boxed{\mathcal{M}_{d,n} \cong [d] \times \mathcal{B}_{n-2}.}$$

Consequently,

$$\boxed{Q_d(n, n-2) = dC_{n-2}.}$$

Proof.

From a maximal-correction path:

1. record the root entrance slot
$$j \in [d];$$
2. delete the two forced unmarked entrance nodes;
3. retain the remaining marked subtree;
4. reinterpret its recursive slots d and $d+1$ as the two ordered binary children.

This gives an element of

$$[d] \times \mathcal{B}_{n-2}.$$

Conversely, given:

- an entrance choice j;
- a plane full binary tree;

reinsert the two forced unmarked nodes and place the binary tree recursively in the only two correction-preserving child slots.

The constructions are inverse. □

The Catalan factor is therefore not a generating-function accident.

It is the count of an actual binary cancellation skeleton.

26.5 Tree coordinates

Assign child weights

$$\omega(j) = j \qquad (1 \le j \le d),$$

and

$$\omega(d+1) = 0.$$

Lemma 26.2 — Tree-coordinate lemma.
If an internal node has root-to-node child address

$$i_1 \cdots i_s,$$

then the baseline height of its associated down-step is

$$\boxed{h(v) = \sum_{\nu=1}^{s} \omega(i_\nu).}$$

For a correction node, the two recursive binary directions are:

- **high**: slot d, increment d;
- **low**: slot $d+1$, increment 0.

The two entrance weights always sum to

$$d + 1.$$

Therefore the binary root always begins at correction height

$$d + 1,$$

independently of the entrance choice.

26.6 Correction height from the binary address

Let

$$L(v)$$

be the number of high edges on the path from the binary root to a correction node v.
Then:

$$\boxed{c(v) = d + 1 + dL(v),}$$

and hence its normalized correction height is

$$\boxed{r(v) = L(v).}$$

Thus the normalized signature has a purely binary-tree interpretation.
The parameter d only rescales the physical path heights.
The normalized correction values themselves count high edges.
This is another manifestation of the universality first seen in Chapter 25.

26.7 Inorder chronology

In the d-ary path traversal:

- the slot-d correction subtree is visited before the node's closing down-step;
- the slot-$(d+1)$ correction subtree is visited after it.

Therefore correction nodes appear in ordinary binary inorder:

$$\boxed{\text{high subtree} \to \text{node} \to \text{low subtree.}}$$

Let

$$v_1, \dots, v_m$$

be the internal binary nodes in inorder.
Then the normalized signature is

$$\boxed{r_i = L(v_i).}$$

So the chronological correction signature is exactly an inorder depth statistic on the binary skeleton.

26.8 The signature is lossless

A Catalan-counted statistic need not normally determine the full tree.
Here it does.

Theorem 26.3 — Lossless Signature Theorem.

In the zero-defect regime, the normalized correction signature uniquely determines the binary cancellation skeleton.

Proof.

Let

$$r = (r_1, \dots, r_m)$$

satisfy the zero-defect conditions.
Let j be the first index such that

$$r_j = 0.$$

In binary inorder, the first zero is the root.
Entries before j lie in the high subtree and each carries one additional high edge.
Therefore the normalized code of the high subtree is

$$\boxed{(r_1 - 1, \dots, r_{j-1} - 1).}$$

Entries after the root lie in the low subtree, whose code is

$$\boxed{(r_{j+1}, \dots, r_m).}$$

Both subsequences satisfy the same zero-defect conditions.
Recursively apply the same decomposition.
This reconstructs the tree uniquely. □
Hence:

binary cancellation skeleton $\longleftrightarrow$ zero-defect signature
$\longleftrightarrow$ Dyck area sequence.

The factor d is external to this intrinsic Catalan structure.
It remembers only the entrance choice.

26.9 Area statistic

If the zero-defect signature has length m, then

$$c_i = d + 1 + dr_i.$$

Therefore

$$\boxed{\sum_{i=1}^{m} c_i = m(d+1) + d\sum_{i=1}^{m} r_i.}$$

After reversal,

$$\sum_i r_i$$

is the ordinary area statistic of the corresponding Dyck path.
If

$$\mathcal{A}_m(v) = \sum_D v^{\mathrm{area}(D)}$$

is the Dyck area polynomial, then the maximal-correction paths satisfy

$$\boxed{\sum_{P \in \mathcal{M}_{d,m+2}} v^{\sum_i c_i} = d\, v^{m(d+1)} \mathcal{A}_m(v^d).}$$

Thus the Catalan boundary is structural at three levels:

1. number of signatures;
2. binary cancellation skeleton;
3. correction-height distribution.

26.10 What the original transition has become

We began with:

$$(L, 0S) \xrightarrow{0} (L, S).$$

At defect zero, the ordered history of occurrences of that transition becomes:

a full binary tree encoded losslessly by its normalized correction signature.

This is a much stronger statement than "Catalan numbers appear."
The original transition has generated a complete combinatorial coordinate system.
The next chapter moves away from individual signatures and counts:

- paths by number of corrections;
- signatures by exact intrinsic defect.

Those are different arrays and must remain separate.

Chapter 27

Exact Correction and Defect Enumeration

The structural theory has produced three different kinds of objects.

They must not be conflated.

$$\boxed{K_d \quad \text{auxiliary kernel coefficients,}}$$

$$\boxed{Q_d(n,k) \quad \text{paths with exactly } k \text{ corrections,}}$$

and

$$\boxed{N_{m,\ell} \quad \text{normalized signatures born at exact defect } \ell.}$$

The first appears naturally inside the scalar generating-function grammar.

The second is the actual path correction statistic.

The third belongs to the intrinsic signature geometry and is independent of d.

27.1 Scalar correction generating function

Let

$$G_h(z,t)$$

count d-Dyck excursions that begin and end at baseline h, never go below h, with:

- z marking down-steps;
- t marking corrections.

A down-step landing at baseline h has weight

$$\tau_h = \begin{cases} t, & h \in \{d+1, 2d+1, \dots\}, \\ 1, & \text{otherwise.} \end{cases}$$

The standard first-return decomposition gives

$$\boxed{G_h = 1 + z\tau_h G_{h+1} \cdots G_{h+d} G_h.}$$

Above the exceptional low boundary, the marking pattern is periodic modulo d.

This collapses the infinite family to three formal series:

- F: baseline zero;
- A: unmarked tail residue;
- B: marked correction residue.

They satisfy

$$\boxed{F = 1 + zA^d F,}$$

$$\boxed{A = 1 + zA^d B,}$$

$$\boxed{B = 1 + ztA^{d-1}B^2.}$$

Define

$$\boxed{Q_d(n,k) = [z^n t^k]F(z,t).}$$

Thus $Q_d(n,k)$ is the actual correction triangle.

27.2 The auxiliary kernel and A371395

Set

$$Y = zA^{d-1}B.$$

Then

$$A = \frac{1}{1-Y}, \qquad B = \frac{1}{1-tY},$$

and

$$\boxed{Y(1-Y)^{d-1}(1-tY) = z.}$$

Define

$$\boxed{K_d = \frac{Y}{z}.}$$

Lagrange inversion gives

$$\boxed{[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}\binom{dn+d-k-2}{n-k}.}$$

For

$$d = 2,$$

this becomes

$$\boxed{[z^n t^k]K_2 = \frac{1}{n+1}\binom{n+k}{k}\binom{2n-k}{n-k}.}$$

This is OEIS A371395 [19].
But K_2 is not the correction distribution.
The actual correction generating function is

$$\boxed{F_d(z,t) = \frac{1-Y}{1-2Y+tY^2}.}$$

Proposition 27.1 — Kernel–correction separation.

For $d = 2$, A371395 is the coefficient triangle of the auxiliary kernel K_2, not the path correction triangle Q_2.

This distinction is important because the two arrays diverge immediately.

A371395 is a useful algebraic waypoint.

It is not the final statistic generated by the DPDA-derived correction event.

27.3 Exact correction triangle

Eliminating B from the three-class system and writing

$$U = A - 1$$

gives

$$\boxed{U\big(1 + (1-t)U\big) = z(1+U)^{d+1}.}$$

Lagrange–Bürmann inversion yields the exact path count.

Theorem 27.2 — Exact correction triangle.

For

$$n \geq 1, \qquad k \geq 0,$$

$$\boxed{Q_d(n,k) = \mathbf{1}_{\{k=0\}} + \sum_{j=k+1}^{n-1} \frac{d(n-j)}{j} \binom{j+k-1}{k} \binom{dn-k-1}{j-k-1}.}$$

In particular,

$$\boxed{Q_d(n,k) = 0 \qquad (k > n-2).}$$

The complete Lagrange–Bürmann coefficient derivation is given in Appendix E.

The theorem statement and its structural consequences remain in the main text.

27.4 Row sums and boundary columns

Setting

$$t = 1$$

forgets the correction marking.

The row sum is the ordinary Fuss–Catalan number:

$$\boxed{\sum_{k \geq 0} Q_d(n,k) = \frac{1}{dn+1} \binom{(d+1)n}{n}.}$$

At the zero-correction boundary,

$$\boxed{Q_d(n,0) = \frac{1}{n}\sum_{j=0}^{n-1}(n-j)\binom{dn}{j}.}$$

For

$$d = 2,$$

this simplifies to

$$\boxed{Q_2(n,0) = \binom{2n-1}{n-1}.}$$

At the maximal-correction boundary,

$$\boxed{Q_d(n,n-2) = dC_{n-2}.}$$

Chapter 26 already explained the last identity bijectively through the binary cancellation skeleton.

This is a useful example of two proof styles meeting:

- algebraic coefficient extraction;
- structural combinatorial decomposition.

27.5 Intrinsic signature enumeration

Now forget the paths and enumerate normalized signatures directly.

Define

$$\boxed{N_{m,\ell} = \#\{r \in \mathbb{N}^m : \Delta(r) = \ell\}.}$$

Let

$$\boxed{S(x,q) = \sum_{m\geq 1}\sum_{\ell\geq 0} N_{m,\ell}x^m q^\ell.}$$

Here:

- x marks signature length;
- q marks exact intrinsic minimum defect.

This enumeration is fundamentally different from $Q_d(n,k)$.

A signature of defect ℓ can be realized at every larger path defect.

Thus $N_{m,\ell}$ records the level at which a signature is **born**.

27.6 Translation and first-zero decomposition

Let $Z(x,q)$ count nonempty signatures whose minimum entry is zero.

Every nonempty signature has a unique decomposition

$$r = \bar{r} + s\mathbf{1},$$

where

$$s = \min_i r_i$$

and

$$\min_i \bar{r}_i = 0.$$

Adding 1 to every entry leaves all adjacent differences unchanged and raises the terminal term by 1.

Therefore

$$\boxed{\Delta(r + \mathbf{1}) = \Delta(r) + 1.}$$

This gives

$$\boxed{Z = (1 - q)S.}$$

Now cut a minimum-zero signature at its first zero.

A second auxiliary series E counts nonempty signatures ending in zero.

The first-zero decomposition yields a closed algebraic system, which simplifies to the master equation.

27.7 Defect–Signature Master Equation

Theorem 27.3 — Defect–Signature Master Equation.

The formal power series $S(x, q)$ with zero constant term is uniquely determined by

$$\boxed{(1 - q)S(1 + qS) = x(1 + S)^2(1 + q^2S).}$$

Equivalently,

$$\boxed{S = x\frac{(1 + S)^2(1 + q^2S)}{(1 - q)(1 + qS)}.}$$

The equation contains no d.

Thus the universality of the Defect–Signature theorem persists at the enumerative level.

The path parameter d changes physical height scaling and path multiplicity.

It does not change the intrinsic normalized signature birth counts.

27.8 Exact defect coefficients

Lagrange inversion gives a finite coefficient formula for every

$$N_{m,\ell}.$$

The first four defect layers simplify to:

$$\boxed{N_{m,0} = C_m,}$$

$$\boxed{N_{m,1} = \frac{3m}{m + 2}C_m = C_{m+1} - C_m,}$$

$$\boxed{N_{m,2} = \frac{m(m^2+10m+1)}{(m+2)(m+3)} C_m,}$$

and

$$\boxed{N_{m,3} = \frac{5m(m^3+7m^2+2m+2)}{(m+2)(m+3)(m+4)} C_m.}$$

The zero layer is exactly Catalan.

The positive layers quantify how the signature class expands as non-Catalan overshoots are admitted.

27.9 Cumulative realizability

Define

$$\boxed{\widehat{N}_{m,\delta} = \#\{r \in \mathbb{N}^m : r \text{ is realizable at path defect } \delta\}.}$$

By signature persistence,

$$\boxed{\widehat{N}_{m,\delta} = \sum_{\ell=0}^{\delta} N_{m,\ell}.}$$

Thus:

- $N_{m,\ell}$ counts signatures born at exact intrinsic defect ℓ;
- $\widehat{N}_{m,\delta}$ counts all signatures visible by defect δ.

At defect one,

$$\boxed{\widehat{N}_{m,1} = C_{m+1}.}$$

The first cumulative layer is still exactly Catalan, one size higher.

Defect two is the first level at which a genuinely non-Catalan local overshoot such as

$$(2, 0)$$

can appear.

27.10 Three arrays, three questions

The distinction can now be stated cleanly.

K_d

An auxiliary kernel series produced by the scalar grammar.

Question:

> What coefficient array appears in the algebraic series-reversion core?

$Q_d(n, k)$

Path correction distribution.

Question:

> How many size-n d-Dyck excursions contain exactly k correction events?

$N_{m,\ell}$

Intrinsic ordered signature distribution.

Question:

> How many length-m correction histories are born at exact minimum defect ℓ?

The DPDA-derived project passes through all three, but they are not interchangeable. The next chapter studies the asymptotic hierarchy of the third array.

Chapter 28

The Asymptotic Defect Hierarchy

The master equation

$$(1-q)S(1+qS) = x(1+S)^2(1+q^2S)$$

provides exact coefficients.

It also reveals a surprisingly rigid asymptotic structure.

For fixed defect ℓ, the number of length-m signatures grows on the Catalan exponential scale:

$$4^m.$$

The defect level changes only the polynomial factor.

More precisely, defect levels occur in pairs:

$$(0,1), \quad (2,3), \quad (4,5), \ldots$$

and every two defect units increase the polynomial exponent by one.

28.1 Fixed-defect generating functions

Define

$$\boxed{S_\ell(x) = [q^\ell]S(x,q) = \sum_{m\geq 1} N_{m,\ell}x^m.}$$

Set

$$\boxed{t = \sqrt{1-4x}.}$$

At defect zero, the master equation gives

$$S_0(x) = C(x) - 1,$$

hence

$$\boxed{S_0(x) = \frac{1-t}{1+t}.}$$

Thus the dominant singularity is the Catalan point

$$x = \frac{1}{4}.$$

The same singular location controls every fixed defect layer.

28.2 Algebraicity and singularity location

Proposition 28.1 — Fixed-defect algebraicity.

For every fixed

$$\ell \geq 0,$$

$$\boxed{S_\ell(x) \in \mathbb{Q}(t), \qquad t = \sqrt{1-4x}.}$$

More precisely,

$$\boxed{S_\ell(x) \in \mathbb{Q}[t, t^{-1}, (1+t)^{-1}].}$$

Therefore, on the branch $t(0) = 1$, the only possible dominant singularity is

$$\boxed{x = \frac{1}{4}.}$$

Proof idea. Let

$$\mathcal{F}(S, q, x) = (1-q)S(1+qS) - x(1+S)^2(1+q^2S).$$

At

$$q = 0,$$

the coefficient of the unknown S_ℓ in the extracted q^ℓ equation is

$$\frac{\partial \mathcal{F}}{\partial S} = 1 - 2x(1+S_0) = t.$$

Thus each new fixed-defect layer is obtained rationally from the earlier ones by dividing by powers of t.

Induction gives the claimed field. □

This proposition justifies using singularity analysis uniformly across all fixed defect levels.

28.3 Critical scaling

To extract the leading singular behavior of all layers simultaneously, set

$$\boxed{q = ty.}$$

Seek an expansion

$$\boxed{S(x, ty) = 1 + tf(y) + t^2 g(y) + O(t^3).}$$

The branch compatible with

$$S(x, 0) = \frac{1-t}{1+t}$$

must satisfy

$$f(0) = -2.$$

Substitution into the master equation gives

$$f^2 - 4yf + 8y^2 - 4 = 0.$$

The correct branch is

$$\boxed{f(y) = 2y - 2\sqrt{1 - y^2}.}$$

At the next order,

$$\boxed{g(y) = 2\frac{4y^3 - 3y + \sqrt{1 - y^2}}{\sqrt{1 - y^2}}.}$$

The even coefficients of f and odd coefficients of g determine the dominant singular terms of successive defect layers.

This is the analytic origin of the pair structure.

28.4 Pairwise leading singularities

For

$$j \geq 1,$$

one obtains

$$S_{2j}(x) = A_j t^{1-2j} + O(t^{2-2j}),$$

and

$$S_{2j+1}(x) = B_j t^{1-2j} + O(t^{2-2j}),$$

with

$$A_j = \frac{C_{j-1}}{4^{j-1}},$$

and

$$B_j = (2j + 3)A_j.$$

The key point is that defects $2j$ and $2j + 1$ have the **same singular exponent**. Their leading constants differ, but their polynomial growth degree is identical.

28.5 Asymptotic Defect Hierarchy

Theorem 28.2 — Asymptotic Defect Hierarchy.

For every fixed

$$j \geq 0,$$

$$\boxed{N_{m,2j} \sim \frac{m^j}{j!} C_m,}$$

and

$$\boxed{N_{m,2j+1} \sim \frac{2j+3}{j!} m^j C_m.}$$

Equivalently,

$$\boxed{N_{m,2j} \sim \frac{4^m}{j!\sqrt{\pi}} m^{j-\frac{3}{2}},}$$

and

$$\boxed{N_{m,2j+1} \sim \frac{2j+3}{j!\sqrt{\pi}} 4^m m^{j-\frac{3}{2}}.}$$

Proof.

Apply the standard transfer theorem to the leading singular terms in powers of

$$t = \sqrt{1-4x}.$$

For $j \geq 1$, the singular exponent is

$$t^{1-2j} = (1-4x)^{-(j-\frac{1}{2})}.$$

The transfer coefficient, together with the explicit constants A_j, B_j, gives the displayed formulas.

The boundary case

$$j = 0$$

uses the exact identities

$$N_{m,0} = C_m$$

and

$$N_{m,1} = C_{m+1} - C_m \sim 3C_m.$$

□

Thus:

$$\boxed{\frac{N_{m,\ell}}{C_m} = \Theta\left(m^{\lfloor \ell/2 \rfloor}\right)}$$

for fixed ℓ.

28.6 The defect pairs

The first leading ratios to the Catalan scale are:

$$1, \quad 3, \quad m, \quad 5m, \quad \frac{1}{2}m^2, \quad \frac{7}{2}m^2, \quad \frac{1}{6}m^3, \quad \frac{3}{2}m^3, \ldots$$

The hierarchy is therefore:

$$\boxed{(0,1)}$$

on the Catalan polynomial scale,

then

$$\boxed{(2,3)}$$

one power of m higher,
then

$$\boxed{(4,5)}$$

two powers higher,
and so on.
Every two defect units add one asymptotically free polynomial degree.

28.7 Geometric interpretation

The theorem is analytic.

The following interpretation comes afterward.

The zero-defect Catalan condition permits adjacent reversed-signature rises of at most 1.

A true violation requires a rise of at least 2.

Such an overshoot costs at least two defect units.

Therefore the pair structure

$$(2j, 2j+1)$$

is consistent with j asymptotically free non-Catalan defect locations among m signature positions.

The factor

$$\frac{m^j}{j!}$$

has exactly the scale expected from choosing j widely separated defect locations.

This is an interpretation of the proven singularity result, not a replacement for its proof.

The monograph should keep that distinction explicit.

28.8 Cumulative asymptotics

Since

$$\widehat{N}_{m,\delta} = \sum_{\ell \le \delta} N_{m,\ell},$$

the top defect layer dominates within each fixed cumulative cutoff.

Thus:

$$\boxed{\widehat{N}_{m,2j} \sim \frac{m^j}{j!} C_m,}$$

and

$$\boxed{\widehat{N}_{m,2j+1} \sim \frac{2j+4}{j!} m^j C_m.}$$

At

$$j = 0,$$

this agrees with the exact identity

$$\widehat{N}_{m,1} = C_{m+1}.$$

28.9 What the cancellation branch has established

We can now summarize the complete branch generated by one transition of the original DPDA:

$$\boxed{0, 0/\varepsilon}$$

$$\Downarrow$$

$$\boxed{\text{odd landing-height event in the nonnegative 2-Dyck sector}}$$

$$\Downarrow$$

$$\boxed{d\text{-Dyck correction statistic}}$$

$$\Downarrow$$

$$\boxed{\text{ordered normalized signatures}}$$

$$\Downarrow$$

$$\boxed{\delta_{\min}(r) = \Delta(r)}$$

$$\Downarrow$$

$$\boxed{\text{Catalan zero-defect boundary}}$$

$$\Downarrow$$

$$\boxed{\text{lossless binary cancellation skeleton}}$$

$$\Downarrow$$

$$\boxed{\text{exact path and signature enumeration}}$$

$$\Downarrow$$

paired asymptotic defect hierarchy.

The DPDA was the discovery mechanism.

The final normalized signature theory is intrinsic and no longer depends on the machine's remaining states or stack alphabet.

28.10 Chronological carry as an arithmetic offshoot

The ordered correction signature remembers chronology.

Pushing that chronology one step further attaches an explicit third-order arithmetic interaction to:

- singleton correction positions;
- adjacent pairs;
- connected triples.

That construction produces the CCR-3 recurrence.

It then develops a modular Stirling shadow, a simple Hensel branch, singular extinction, and ramified p-adic recovery. The subsequent fourth-order chapter isolates the first independent higher connected coordinate, and the CCR-k chapter closes the same terminal-block mechanism at arbitrary finite order.

This arithmetic refinement is a genuine continuation of the correction branch, but it is not a prerequisite for the later general real-time p:q DPDA theory.

Accordingly, the following research branch is presented as a specialized offshoot.

Readers interested only in the main normalization spine may skip it and rejoin when the pushdown-machine line resumes.

Reader's guide.
Chapters 29–34 are a specialized arithmetic offshoot. They continue the ordered-correction branch of Chapters 24–28, but they are not prerequisites for the later real-time p:q pushdown-normalization theory. Readers following only the main DPDA spine may skip directly to Chapter 35.

Chapter 29

From Ordered Corrections to CCR-3

Research branch — Chronological Carry and the Arithmetic Offshoot. *This branch develops the chronological-carry refinement and p-adic root analysis that emerge from the cancellation branch. The surrounding arithmetic literature includes carries and the p-adic behavior of Stirling numbers [7, 1, 12, 18].*

The preceding cancellation branch ended with an intrinsic correction-signature theory.
The original DPDA supplied a distinguished cancellation event.
The path interpretation ordered those events chronologically.
The Defect–Signature theorem then showed that the realization cost is local along that chronology:

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}).$$

Thus the correction history carries a natural path graph

$$1 - 2 - \cdots - n.$$

This is exactly the amount of pushdown-derived structure needed for the next step.
It is also the boundary of what the DPDA itself forces.

29.1 Scientific boundary

The arithmetic refinement introduced in this branch must be stated carefully.
What is inherited from the earlier DPDA/correction branch is:

1. a distinguished cancellation/correction event;
2. chronological ordering of those events;
3. nearest-neighbor locality along the ordered positions.

What is **introduced here** is a new arithmetic polarization:

1. a numerical weight for a singleton chronological slot;
2. a weight for an adjacent pair interaction;
3. a weight for the first connected triple collision.

Therefore the correct implication is

DPDA-derived chronological locality + specified arithmetic polarization
$\Longrightarrow$ CCR-3.

The stronger statement

$$\text{original DPDA} \Longrightarrow \text{unique CCR-3 coefficients}$$

is not claimed.

This distinction is not editorial caution; it marks the exact scientific scope of the arithmetic branch.

29.2 Connected supports through order three

A connected subset of a path graph is an interval.

Therefore, up to size three, there are only three local connected support types.

Lemma 29.1 — Third-order connected support.

Every connected chronological support of cardinality at most three is one of

$$\boxed{\{j\}, \qquad \{j, j+1\}, \qquad \{j, j+1, j+2\}.}$$

These are, respectively:

- a singleton slot;
- an adjacent pair;
- a consecutive triple.

The lemma is elementary, but its role is structural: no other connected third-order motif exists on a one-dimensional chronology.

29.3 The CCR-3 recurrence

Fix a prime

$$p \geq 5$$

and define

$$\boxed{\ell_n = (p+1)n - p.}$$

Work in the truncated ring

$$\mathbb{Z}[u]/(u^4).$$

Set

$$P^{[3]}_{p,0}(u) = 1,$$

and

$$P^{[3]}_{p,-1}(u) = P^{[3]}_{p,-2}(u) = 0.$$

Definition 29.2 — CCR-3.

For

$$n \geq 1,$$

define

$$\boxed{\begin{aligned} P^{[3]}_{p,n}(u) &= (1+\ell_n u)P^{[3]}_{p,n-1}(u) \\ &\quad + p(n-1)\ell_{n-1}u^2 P^{[3]}_{p,n-2}(u) \\ &\quad - p(n-2)(n-1)u^3 P^{[3]}_{p,n-3}(u) \pmod{u^4}. \end{aligned}}$$

The three nontrivial sources represent:

$$\boxed{\text{singleton } + \text{ adjacent pair carry } + \text{ connected triple collision.}}$$

The first rows are

$$P^{[3]}_{p,0} = 1,$$

$$P^{[3]}_{p,1} = 1 + u,$$

$$P^{[3]}_{p,2} = 1 + (p+3)u + 2(p+1)u^2,$$

and

$$P^{[3]}_{p,3} = 1 + (3p+6)u + (4p^2+15p+11)u^2 + 6(p+1)^2u^3.$$

The superscript [3] is important.

The recurrence is intentionally third-order.

It should not be extrapolated blindly into an all-column theory: at order four a genuinely new connected motif may enter.

29.4 Hessenberg realization

Let

$$H_{p,n}$$

be the $n \times n$ leading principal truncation of the infinite lower-Hessenberg matrix defined by

$$(H_p)_{i,i} = \ell_i,$$

$$(H_p)_{i,i+1} = 1,$$

$$(H_p)_{i+1,i} = -pi\ell_i,$$

$$(H_p)_{i+2,i} = -pi(i+1),$$

with all entries below the second subdiagonal equal to zero.

Define

$$D_{p,n}(u) = \det(I_n + uH_{p,n}).$$

Proposition 29.3 — Hessenberg realization.

$$\boxed{\begin{aligned} D_{p,n} &= (1+\ell_n u)D_{p,n-1} \\ &\quad + p(n-1)\ell_{n-1}u^2 D_{p,n-2} \\ &\quad - p(n-2)(n-1)u^3 D_{p,n-3}. \end{aligned}}$$

Hence

$$\boxed{P^{[3]}_{p,n}(u) \equiv D_{p,n}(u) \pmod{u^4}.}$$

Proof.

Expand the determinant by the last row.

The diagonal contribution is

$$(1+\ell_n u)D_{p,n-1}.$$

The first subdiagonal closes with one superdiagonal entry and contributes

$$+p(n-1)\ell_{n-1}u^2 D_{p,n-2}.$$

The second subdiagonal closes with two consecutive superdiagonal entries and contributes

$$-p(n-2)(n-1)u^3 D_{p,n-3}.$$

No deeper band is present. □

Thus the three chronological interaction orders are encoded by a three-band Hessenberg structure.

29.5 Operator form

Use the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \qquad i \geq 1.$$

Let

$$D = \partial_x, \qquad Xf = xf, \qquad N = XD,$$

and define

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1},$$

$$Xe_i = ie_{i+1},$$

$$X^2 e_i = i(i+1)e_{i+2},$$

and

$$L_p e_i = \ell_i e_i.$$

Corollary 29.4 — Operator form.

The Hessenberg matrix H_p is the matrix, in the basis (e_i), of

$$\boxed{\mathcal{H}_p = D + L_p - pXL_p - pX^2.}$$

Equivalently,

$$\boxed{\mathcal{H}_p = \big[1 + (p+1)x - p(p+1)x^2\big]\partial_x + 1 - px - px^2.}$$

The determinant recurrence and differential-operator form are therefore two representations of the same CCR-3 dynamics.

29.6 What this chapter has and has not established

We have not derived CCR-3 from the original DPDA transition table.

We have established the narrower chain

ordered correction chronology → path-local connected supports
→ specified third-order polarization → Hessenberg/operator recurrence.

The next question is arithmetic.

What do the first three fixed columns of this recurrence look like?

The answer is unexpectedly rigid: they are polynomials, and modulo p the entire third jet collapses to a universal first-kind Stirling shadow.

Chapter 30

Fixed Columns and the Universal Stirling Shadow

The CCR-3 recurrence is nonlinear in its dependence on the chronological index but finite in interaction order.

To expose its arithmetic, we now freeze the power of u and let the row index vary.

This produces a transport system for the first three fixed columns.

30.1 Column coefficients

Define

$$\boxed{C^{(p)}_{n,k} = [u^k]P^{[3]}_{p,n}(u), \qquad 0 \le k \le 3.}$$

For

$$m \ge k + 1,$$

normalize by the falling factorial:

$$\boxed{\Pi_{p,k}(m) = \frac{C^{(p)}_{m-1,k}}{(m-1)_{\underline{k}}}.}$$

The normalization removes the obvious chronological placement multiplicity and reveals polynomial structure.

30.2 Transport equations

Proposition 30.1 — Transport system.

For the indicated ranges,

$$\boxed{m\Pi_{p,1}(m+1) = (m-1)\Pi_{p,1}(m) + \ell_m,}$$

$$\boxed{m\Pi_{p,2}(m+1) = (m-2)\Pi_{p,2}(m) + \ell_m\Pi_{p,1}(m) + p\ell_{m-1},}$$

and

$$\boxed{\begin{aligned} m\Pi_{p,3}(m+1) = (m-3)\Pi_{p,3}(m) + \ell_m \Pi_{p,2}(m) \\ + p\ell_{m-1}\Pi_{p,1}(m-1) - p. \end{aligned}}$$

Proof.

Extract the coefficients of

$$u, \quad u^2, \quad u^3$$

from CCR-3.

Then substitute

$$n = m$$

and divide by the corresponding falling factorial.

The chronological factors cancel exactly, leaving the displayed first-order transport equations. □

The third source can be separated into a connected form.

Since

$$\Pi_{p,1}(m-1) = \frac{\ell_{m-1} - p}{2},$$

the genuinely new third-order source is

$$\boxed{\mathfrak{T}_{p,3}(m) = p\left[\frac{1}{2}\ell_{m-1}(\ell_{m-1} - p) - 1\right].}$$

This isolates the connected triple correction from propagated lower-order contributions.

30.3 First three fixed columns

Theorem 30.2 — First three fixed columns.

For

$$p \geq 5,$$

$$\boxed{\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2},}$$

$$\boxed{\Pi_{p,2}(m) = \frac{p+1}{24}\left[3(p+1)m^2 - (5p+1)m - 12p\right],}$$

and

$$\boxed{\Pi_{p,3}(m) = \frac{p+1}{48}F_p(m),}$$

where

$$\boxed{\begin{aligned} F_p(m) = (p+1)^2 m^3 + (p^2-1)m^2 \\ - (34p^2 + 18p)m + 56p^2 - 8p. \end{aligned}}$$

The boundary values are

$$\Pi_{p,1}(2) = 1,$$

$$\Pi_{p,2}(3) = p + 1,$$

$$\Pi_{p,3}(4) = (p + 1)^2.$$

Proof.

The first transport equation is solved directly by the displayed linear polynomial.

Substituting that result into the second equation gives a first-order recurrence with polynomial source; the displayed quadratic satisfies both the recurrence and its boundary value.

Substituting the first two solutions into the third transport equation gives a cubic source. The displayed cubic satisfies the recurrence and boundary value.

Uniqueness follows at each stage from the first-order recurrence. □

The key methodological point is that these formulas arise from transport.

They should not be presented as unexplained computer-algebra output.

30.4 Reduction modulo p

Modulo p,

$$\ell_n = (p + 1)n - p \equiv n \pmod{p}.$$

Every pair and triple interaction source carries an explicit factor of p.

Therefore those interaction terms vanish modulo p.

CCR-3 degenerates to the free recurrence

$$P^{[3]}_{p,n}(u) \equiv (1 + nu)P^{[3]}_{p,n-1}(u) \pmod{(p, u^4)}.$$

Iteration gives the universal product.

Theorem 30.3 — Stirling shadow.

For every

$$n \geq 0,$$

$$\boxed{P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1 + ju) \pmod{(p, u^4)}.}$$

Consequently, for $k \leq 3$,

$$\boxed{C^{(p)}_{m-1,k} \equiv e_k(1, \ldots, m - 1) \equiv \begin{bmatrix} m \\ m - k \end{bmatrix} \pmod{p},}$$

where the bracket denotes the unsigned Stirling number of the first kind.

Thus the arithmetic interaction deformation has a universal free shadow.

30.5 Normalized shadow polynomials

The normalized fixed columns reduce to:

$$\boxed{S_1(m) = \frac{m}{2},}$$

$$\boxed{S_2(m) = \frac{m(3m-1)}{24},}$$

$$\boxed{S_3(m) = \frac{m^2(m-1)}{48}.}$$

So, as polynomial congruences,

$$\boxed{\Pi_{p,k}(m) \equiv S_k(m) \pmod{p} \qquad (k = 1, 2, 3).}$$

Important notation point

One must not derive these congruences by naively cancelling the falling factorial

$$(m-1)_{\underline{k}}$$

inside $\mathbb{F}_p$.
That denominator may vanish at exceptional residues.
The safe argument is:

1. derive the exact polynomial formulas over $\mathbb{Q}[m]$;
2. reduce those polynomial formulas modulo p.

Because

$$2, \quad 24, \quad 48$$

are units for $p \geq 5$, the reduction is legitimate.
This avoids illegal pointwise modular division.

30.6 Why the third column is the first singular one

The third normalized shadow is

$$S_3(m) = \frac{m^2(m-1)}{48}.$$

Therefore modulo p, the third-column zero set contains:

- a simple root at
$$m \equiv 1 \pmod{p};$$
- a double root at
$$m \equiv 0 \pmod{p}.$$

The exact cubic deformation $F_p(m)$ must decide what happens to these two shadow branches.
The next chapter shows that they behave completely differently.
The simple root survives integrally.
The double root becomes extinct in $\mathbb{Z}_p$.

Chapter 31

The Third-Column p-Adic Split

The third normalized column is

$$\Pi_{p,3}(m) = \frac{p+1}{48} F_p(m),$$

where

$$F_p(m) = (p+1)^2 m^3 + (p^2-1)m^2 - (34p^2+18p)m + 56p^2 - 8p.$$

Modulo p,

$$\boxed{F_p(m) \equiv m^2(m-1) \pmod{p}.}$$

Thus the shadow has two different types of roots:

$$m \equiv 1 \pmod{p}$$

is simple, while

$$m \equiv 0 \pmod{p}$$

is double.
The exact deformation separates them sharply.

31.1 The simple branch

At

$$m = 1,$$

$$F_p(1) \equiv 0 \pmod{p},$$

and

$$\boxed{F_p'(1) \equiv 1 \pmod{p}.}$$

The derivative is a unit.
Hensel's lemma therefore applies in its simplest form.

Theorem 31.1 — Simple Hensel tower.

There exists a unique

$$\boxed{\alpha_p \in \mathbb{Z}_p}$$

such that

$$\alpha_p \equiv 1 \pmod{p}$$

and

$$F_p(\alpha_p) = 0.$$

Its expansion begins

$$\boxed{\begin{aligned}\alpha_p = 1 + 24p - 888p^2 + 58008p^3 \\ - 4715256p^4 + 428583192p^5 + O(p^6).\end{aligned}}$$

Moreover, throughout the residue class

$$m \equiv 1 \pmod{p},$$

$$\boxed{\nu_p(F_p(m)) = \nu_p(m - \alpha_p).}$$

Proof.

Hensel gives existence and uniqueness.
Factor in $\mathbb{Z}_p[m]$:

$$F_p(m) = (m - \alpha_p)G_p(m).$$

Since

$$G_p(\alpha_p) = F_p'(\alpha_p)$$

is a unit, $G_p(m)$ remains a unit throughout the residue class

$$m \equiv \alpha_p \equiv 1 \pmod{p}.$$

Therefore the valuation is exactly the distance to the Hensel root. □
Thus the simple shadow branch survives exactly as expected.

31.2 The double shadow branch

Now set

$$m = pu.$$

Direct substitution gives

$$\boxed{\frac{F_p(pu)}{p} \equiv -8 \pmod{p}.}$$

Since $p \geq 5$,

$$-8$$

is nonzero modulo p.
Therefore:

Theorem 31.2 — Singular extinction.

For every

$$u \in \mathbb{Z}_p,$$

$$\boxed{\nu_p(F_p(pu)) = 1.}$$

Consequently, although

$$m \equiv 0 \pmod{p}$$

is a double root of the Stirling shadow, it has **no lift to a root modulo** p^2.

We call this phenomenon

$$\boxed{\textbf{singular extinction}.}$$

The term means exactly:

$$\boxed{\text{multiple root mod } p \quad \text{but no integral branch survives mod } p^2.}$$

It does not mean that the corresponding algebraic roots disappear from the algebraic closure.

Chapter 32 will show where they go.

31.3 Generic residue classes

If

$$m \not\equiv 0, 1 \pmod{p},$$

then

$$F_p(m) \not\equiv 0 \pmod{p}.$$

Hence

$$\boxed{\nu_p(F_p(m)) = 0.}$$

Thus the residue classes split into three exact arithmetic regimes:

1. generic;
2. simple Hensel;
3. singularly extinct.

31.4 The scaled third column

Define

$$\boxed{Q_{m,3}^{\mathrm{CCR}} = p^{m-4}\Pi_{p,3}(m), \qquad m \geq 4.}$$

Because

$$\frac{p+1}{48}$$

is a p-adic unit for $p \geq 5$,

$$\nu_p(\Pi_{p,3}(m)) = \nu_p(F_p(m)).$$

The previous three cases therefore give the complete valuation law.

Theorem 31.3 — Complete third-column valuation law.

For

$$m \geq 4,$$

$$\boxed{\nu_p(Q_{m,3}^{\mathrm{CCR}}) = \begin{cases} m-4, & m \not\equiv 0,1 \pmod{p}, \\ m-3, & m \equiv 0 \pmod{p}, \\ m-4+\nu_p(m-\alpha_p), & m \equiv 1 \pmod{p}. \end{cases}}$$

This theorem is best read after the branch analysis.
The three cases are not an arbitrary valuation table.
They are exactly:

$$\boxed{\text{generic unit}}$$

$$\boxed{\text{one forced } p\text{-factor from singular extinction}}$$

$$\boxed{\text{Hensel distance to the unique integral root.}}$$

31.5 Independent consistency checks

The monograph audit independently checked the arithmetic at several levels:

- the fixed-column formulas were regenerated symbolically through several rows;
- the displayed formulas matched 15 tested symbolic column/boundary instances;
- substituting the displayed Hensel expansion into F_p cancels every coefficient through p^5;
- digitwise Hensel lifting for

  $$p = 5, 7, 11, 13, 17, 19$$

  agrees through modulus p^6;
- hundreds of sample valuation checks agree with both the simple-branch and extinction laws.

These computations support the proof chain.
They do not replace it.

31.6 Ramified location of the remaining algebraic roots

The double shadow root has now disappeared from the integral p-adic locus:

$$0 \bmod p \longrightarrow \varnothing \bmod p^2.$$

But F_p is cubic.
One root has become the integral Hensel root α_p.
Two algebraic roots remain.
The discriminant will show that they have not disappeared.
They have moved into a ramified quadratic extension, with half-integral valuation.

Chapter 32

Ramified Recovery

The simple shadow root has survived inside

$$\mathbb{Z}_p.$$

The double shadow root has undergone singular extinction.

The two remaining algebraic roots are recovered in a ramified quadratic extension. Their location is encoded by the cubic discriminant.

32.1 Exact discriminant factorization

Theorem 32.1 — Discriminant and ramified field.

The discriminant of F_p factors as

$$\boxed{\mathrm{Disc}(F_p) = 4p(p+1)^2 R_p,}$$

where

$$\boxed{R_p = 9801p^5 + 22672p^4 + 32502p^3 + 14360p^2 - 927p - 8.}$$

Modulo p,

$$\boxed{R_p \equiv -8 \pmod{p}.}$$

Therefore

$$\boxed{\nu_p(\mathrm{Disc}(F_p)) = 1.}$$

The discriminant has odd valuation, so the residual quadratic factor after removing the simple Hensel root is ramified.

32.2 Removing the integral root

Factor

$$F_p(m) = (m - \alpha_p)Q_p(m)$$

over $\mathbb{Z}_p$.

The root α_p is simple and integral.

The remaining quadratic Q_p has discriminant with the same square class as the cubic discriminant, because the resultant contribution from the linear/quadratic splitting is a square.

Now

$$R_p/(-8) \equiv 1 \pmod{p}.$$

By Hensel's lemma,

$$R_p/(-8)$$

is a square in

$$\mathbb{Z}_p^\times.$$

The factors

$$4 \qquad \text{and} \qquad (p+1)^2$$

are also squares.

Hence the discriminant square class is

$$\boxed{[-2p].}$$

The quadratic splitting field is therefore

$$\boxed{\mathbb{Q}_p(\sqrt{-2p}).}$$

32.3 Half-integral roots

The quadratic factor is Eisenstein after the integral branch is removed.

Its two roots

$$\beta_{p,+}, \qquad \beta_{p,-}$$

therefore satisfy

$$\boxed{\nu_p(\beta_{p,\pm}) = \frac{1}{2}.}$$

Thus the double shadow root has not vanished algebraically.

Instead:

$$\boxed{\text{double root mod } p}$$

splits in the exact deformation into

$$\boxed{\text{two ramified roots of valuation } 1/2.}$$

This is the conceptual resolution of singular extinction.

Integral lifting fails because the two branches leave the integral valuation lattice.

32.4 Ramified square class by $p \bmod 8$

Corollary 32.2 — Ramified square class.

The ramified field is exactly

$$\boxed{\mathbb{Q}_p(\sqrt{-2p}).}$$

Its standard square-class description depends on

$$p \bmod 8.$$

For the residue classes in which -2 is a square unit, this is the standard p-ramified class.

For the remaining residue classes, it is the other ramified quadratic square class represented by a nonsquare unit times p.

This corollary is classical square-class arithmetic applied to the exact discriminant already proved.

32.5 The complete third-order arithmetic picture

The results of Chapters 29–32 can now be summarized without adding a new proof.

Corollary 32.3 — CCR-3 arithmetic geometry.

The third-order chronological carry refinement has the following structure.

1. Chronological locality allows only singleton, adjacent-pair, and consecutive-triple connected supports through order three.
2. The chosen arithmetic polarization produces the CCR-3 recurrence and its Hessenberg/operator realization.
3. The first three normalized columns are explicit polynomials.
4. Modulo p, all interaction terms disappear and
$$P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p,u^4)}.$$
5. The normalized columns reduce to near-diagonal first-kind Stirling polynomials.
6. In the third column:
   - the simple shadow root
   $$1 \pmod{p}$$
   lifts uniquely to
   $$\alpha_p \in \mathbb{Z}_p;$$
   - the double shadow root
   $$0 \pmod{p}$$
   undergoes singular extinction;

- the two missing algebraic branches survive with valuation

$$1/2$$

over

$$\mathbb{Q}_p(\sqrt{-2p}).$$

This is the complete third-order arithmetic offshoot.

32.6 Novelty boundary

Nothing in this chapter requires claiming novelty for the ingredients individually.

The following are classical or established areas:

- carry processes;
- Holte-type carry matrices;
- Stirling numbers;
- p-adic valuations of Stirling numbers;
- Hensel lifting;
- Eisenstein irreducibility;
- discriminants;
- Newton polygons;
- ramified quadratic extensions;
- lower-Hessenberg production matrices.

The family-specific contribution is the chain

> chronological local interactions → fixed-column polynomialization → Stirling shadow → simple lifting / singular extinction → exact ramified recovery.

The bibliographic claim is restricted to the proved family-specific chain stated here; no broader priority claim is made.

Chapter 33

CCR-4: Fourth-Order Chronological Carry, Connected Quadruples, and the Stirling Shadow

The preceding arithmetic Part established CCR-3 and its third-jet Stirling shadow. The fourth-order problem requires a scope correction: the complete third jet and chronological path-locality determine the new connected support type, but not its numerical weight. The new datum is therefore introduced as an explicit fourth-order polarization and then identified from the normalized fourth column.

33.1 The CCR-3 baseline

Fix a prime $p \geq 5$ and define

$$\ell_n = (p+1)n - p.$$

The third-order arithmetic polarization is taken as the baseline object of this chapter.

Definition 33.1 (CCR-3 baseline)**.** Work in $\mathbb{Z}[u]/(u^4)$ and set

$$P^{[3]}_{p,0}(u) = 1, \qquad P^{[3]}_{p,-1}(u) = P^{[3]}_{p,-2}(u) = 0.$$

For $n \geq 1$, define

$$\begin{aligned} P^{[3]}_{p,n}(u) = (1+\ell_n u)P^{[3]}_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2 P^{[3]}_{p,n-2}(u) \\ - p(n-2)(n-1)u^3 P^{[3]}_{p,n-3}(u) \pmod{u^4}. \end{aligned} \tag{33.1}$$

The three sources are called the singleton slot, adjacent-pair interaction, and connected triple interaction.

For $0 \leq k \leq 3$ put

$$C^{(p)}_{n,k} = [u^k]P^{[3]}_{p,n}(u), \qquad \Pi_{p,k}(m) = \frac{C^{(p)}_{m-1,k}}{(m-1)_{\underline{k}}} \quad (m \geq k+1).$$

Coefficient extraction from (33.1) gives the transport equations

$$\begin{aligned} m\Pi_{p,1}(m+1) &= (m-1)\Pi_{p,1}(m) + \ell_m, \\ m\Pi_{p,2}(m+1) &= (m-2)\Pi_{p,2}(m) + \ell_m\Pi_{p,1}(m) + p\ell_{m-1}, \\ m\Pi_{p,3}(m+1) &= (m-3)\Pi_{p,3}(m) + \ell_m\Pi_{p,2}(m) \\ &\qquad + p\ell_{m-1}\Pi_{p,1}(m-1) - p. \end{aligned} \tag{33.2}$$

Proposition 33.2 (First three normalized columns)**.** *For the indicated ranges,*

$$\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2},$$

$$\Pi_{p,2}(m) = \frac{p+1}{24}\left[3(p+1)m^2 - (5p+1)m - 12p\right],$$

and

$$\Pi_{p,3}(m) = \frac{p+1}{48}F_p(m),$$

where

$$F_p(m) = (p+1)^2m^3 + (p^2-1)m^2 - (34p^2+18p)m + 56p^2 - 8p.$$

The boundary values are

$$\Pi_{p,1}(2) = 1, \qquad \Pi_{p,2}(3) = p+1, \qquad \Pi_{p,3}(4) = (p+1)^2.$$

Proof. Substitution into (33.2) verifies the three displayed polynomials. Each transport equation is first order in its current column, so the corresponding boundary value gives uniqueness. No computer-algebra interpolation is used as a proof premise. □

Modulo p, every non-singleton source in (33.1) disappears and $\ell_n \equiv n$. Hence

$$P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p, u^4)}.$$

The purpose of this chapter is to determine exactly what survives, and what becomes new, when the truncation is raised to u^5.

33.2 What is genuinely new at order four

Definition 33.3 (Connected chronological support)**.** A set of correction positions is *connected* if it induces a connected subgraph of the chronological path $1 - 2 - \cdots - n$.

Lemma 33.4 (Unique connected quadruple)**.** *Every connected chronological support of cardinality four is an interval of the form*

$$\{j, j+1, j+2, j+3\}.$$

Thus there is exactly one new connected support type at order four.

Proof. Every connected subset of a path graph is an interval. An interval with four vertices has the displayed form. □

The structural issue is not the shape of the new support but the arithmetic weight assigned to it. The recurrence itself gives a clean way to separate propagated lower-order terms from the genuinely new fourth-order source.

Definition 33.5 (Terminal block form through order four)**.** A path-local fourth-order extension of CCR-3 is a recurrence in $\mathbb{Z}[u]/(u^5)$ of the form

$$\begin{aligned} P_{p,n}(u) = (1+\ell_n u)P_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2P_{p,n-2}(u) \\ - p(n-2)(n-1)u^3P_{p,n-3}(u) + q_{p,n}u^4P_{p,n-4}(u) \pmod{u^5}, \end{aligned} \tag{33.3}$$

with $P_{p,0} = 1$ and $P_{p,n} = 0$ for negative n. The first three terminal block weights are fixed by CCR-3; $q_{p,n}$ is the connected quadruple source.

Proposition 33.6 (Lower-order block separation)**.** *In* (33.3)*, every contribution to* $[u^4]P_{p,n}$ *obtained by appending a terminal connected block of size one, two, or three is already determined by CCR-3. The only term in the fourth coefficient not generated by those lower bands is* $q_{p,n}$ *itself.*

Proof. Extracting u^4 from (33.3) gives

$$\begin{aligned}[u^4]P_{p,n} = {} & [u^4]P_{p,n-1} + \ell_n[u^3]P_{p,n-1} \\ & + p(n-1)\ell_{n-1}[u^2]P_{p,n-2} \\ & - p(n-2)(n-1)[u]P_{p,n-3} + q_{p,n}.\end{aligned} \tag{33.4}$$

The middle three lines attach a terminal block of size one, two, or three to a previously generated lower-order history. Iterating those terms produces exactly the degree-four histories decomposable into lower connected block sizes, including the patterns 1+1+1+1, 2+1+1, 2+2, and 3+1 in chronological order. The final scalar $q_{p,n}$ is the contribution of a single terminal block of size four. By theorem 33.4, that block is the unique consecutive quadruple support. □

Remark 33.7 (Relation with cumulant language)*.* Classical cumulants isolate connected information by Möbius inversion on the lattice of set partitions [65, 64]; mixed cumulants vanish when the variables split into independent families. That theory motivates the word "connected," but theorem 33.6 does not identify the CCR coefficients with probabilistic cumulants. It is an internal recurrence statement about terminal interval blocks on a path.

Theorem 33.8 (Fourth-order non-identifiability from CCR-3)**.** *CCR-3 and chronological path-locality do not determine a unique fourth-order numerical weight. More precisely, for any integer sequence* $(q_{p,n})_{n\geq 4}$*, recurrence* (33.3) *has exactly the same coefficients through* u^3 *as CCR-3.*

Proof. The new term is divisible by u^4, so it cannot affect any coefficient of degree at most three. Induction on n therefore shows that every choice of $q_{p,n}$ has the same third jet as (33.1). The fourth coefficient is the first place where the choices can differ. □

This theorem is the central scope correction at order four: a numerical CCR-4 coefficient must come from an additional arithmetic polarization, not from third-order data alone.

33.2.1 Exact reconstruction of a general p-divisible fourth source

Suppose the connected source is p-divisible after the natural size-four placement factor:

$$q_{p,m} = p(m-1)_{\underline{3}}\eta_m, \qquad \eta_m \in \mathbb{Z}, \qquad m \geq 4.$$

For the corresponding solution of (33.3), write

$$C^{(q)}_{n,k} = [u^k]P_{p,n}(u), \qquad \Pi^{(q)}_{p,4}(m) = \frac{C^{(q)}_{m-1,4}}{(m-1)_{\underline{4}}}, \qquad m \geq 5.$$

Let $\Pi^{(0)}_{p,4}$ denote the zero-fourth-source solution and put

$$\Delta_m = \Pi^{(q)}_{p,4}(m) - \Pi^{(0)}_{p,4}(m).$$

Theorem 33.9 (Fourth-source reconstruction)**.** *For every $m \geq 5$,*

$$\boxed{m\Delta_{m+1} = (m-4)\Delta_m + p\eta_m.} \tag{33.5}$$

At the boundary,

$$\boxed{\Delta_5 = \frac{p\eta_4}{4}.} \tag{33.6}$$

Consequently the connected source is exactly identifiable from the fourth normalized column:

$$\boxed{\eta_m = \frac{m\Delta_{m+1} - (m-4)\Delta_m}{p} \qquad (m \geq 5),} \tag{33.7}$$

with $\eta_4 = 4\Delta_5/p$.

Proof. Extract u^4 from (33.3), put $n = m$, and divide by $(m)_{\underline{4}} = m(m-1)(m-2)(m-3)$. The lower three source bands are identical for the given source and for the zero-source recurrence because all columns of degree at most three agree by theorem 33.8. Their difference therefore cancels. The remaining fourth source contributes

$$\frac{q_{p,m}}{(m)_{\underline{4}}} = \frac{p(m-1)_{\underline{3}}\eta_m}{(m)_{\underline{4}}} = \frac{p\eta_m}{m},$$

while the preceding fourth coefficient contributes $(m-4)\Delta_m/m$. Multiplying by m gives (33.5). At $m = 4$, the only difference in $C_{4,4}$ is $q_{p,4} = 6p\eta_4$; division by $(4)_{\underline{4}} = 24$ gives (33.6). Solving (33.5) for η_m yields (33.7). □

Thus fourth-order identifiability is not restricted to the stationary model: the entire p-divisible connected chronology can be read back from the fourth column. Stationarity becomes a testable property of that column rather than merely an imposed coefficient ansatz.

33.3 The stationary p-divisible fourth-order family

We now specify one natural class in which the new fourth-order datum can be classified exactly.

Definition 33.10 (Stationary p-divisible fourth polarization)**.** A recurrence of the form (33.3) is *stationary p-divisible* if, for every $n \geq 4$,

$$p(n-1)_{\underline{3}} \;\big|\; q_{p,n},$$

and the normalized integer quotient

$$\eta_n = \frac{q_{p,n}}{p(n-1)_{\underline{3}}}$$

is independent of n. Thus stationarity is imposed only after removing the inherited factor p and the falling-factorial normalization of a size-four terminal block.

Theorem 33.11 (Stationarity and one-parameter classification)**.** *For every prime $p \geq 5$ and every p-divisible fourth-order extension, the following are equivalent:*

(i) *the normalized connected source η_n is independent of n;*

(ii) *the fourth-column difference $\Delta_m = \Pi^{(q)}_{p,4}(m) - \Pi^{(0)}_{p,4}(m)$ is independent of $m \geq 5$;*

(iii) *for one integer λ, the recurrence is exactly*

$$\boxed{\begin{aligned} P_{p,n}^{[4,\lambda]}(u) &= (1+\ell_n u)P_{p,n-1}^{[4,\lambda]}(u) + p(n-1)\ell_{n-1}u^2 P_{p,n-2}^{[4,\lambda]}(u) \\ &\quad - p(n-2)(n-1)u^3 P_{p,n-3}^{[4,\lambda]}(u) \\ &\quad + \lambda p(n-3)(n-2)(n-1)u^4 P_{p,n-4}^{[4,\lambda]}(u) \end{aligned}} \pmod{u^5}, \tag{33.8}$$

with $\lambda \in \mathbb{Z}$. In that case

$$\Delta_m = \frac{\lambda p}{4} \qquad (m \geq 5).$$

The member $\lambda = 0$ has no new connected size-four source, whereas every $\lambda \neq 0$ has a genuine connected quadruple contribution.

Proof. The equivalence of (i) and (iii) is the definition of the normalized source: $\eta_n \equiv \lambda$ is exactly $q_{p,n} = \lambda p(n-1)_{\underline{3}}$. If (i) holds, theorem 33.9 has boundary value $\Delta_5 = \lambda p/4$, and the constant sequence $\Delta_m = \lambda p/4$ satisfies (33.5); first-order uniqueness gives (ii) and the displayed constant. Conversely, if $\Delta_m \equiv c$, then (33.7) gives $\eta_m = 4c/p$ for every $m \geq 5$, while (33.6) gives the same value for $m = 4$. Hence (ii) implies (i), completing the equivalence. □

Definition 33.12 (Primitive same-sign CCR-4)**.** The choice

$$\lambda = 1$$

is called the *primitive same-sign fourth polarization*. It is the smallest positive integral extension compatible with the sign convention for the lower Hessenberg interaction bands introduced in section 33.4.

Remark 33.13 (What $\lambda = 1$ means). Definition 33.12 is a normalization choice. It gives one concrete CCR-4 recurrence, but it is not asserted that the original DPDA uniquely forces $\lambda = 1$. The theorem-level object is the full λ-family and the exact way in which λ is visible.

33.4 Four-band Hessenberg and operator realization

The extension parameter has a direct linear-algebraic realization.

Definition 33.14 (Fourth-band Hessenberg matrix)**.** Let $H_{p,\lambda}$ be the infinite lower-Hessenberg matrix with

$$(H_{p,\lambda})_{i,i} = \ell_i, \qquad (H_{p,\lambda})_{i,i+1} = 1,$$

$$(H_{p,\lambda})_{i+1,i} = -pi\ell_i, \qquad (H_{p,\lambda})_{i+2,i} = -pi(i+1),$$

$$(H_{p,\lambda})_{i+3,i} = -\lambda pi(i+1)(i+2),$$

and all entries below the third subdiagonal equal to zero. Write $H_{p,\lambda,n}$ for its $n \times n$ leading principal truncation and

$$D_{p,n}^{(\lambda)}(u) = \det(I_n + uH_{p,\lambda,n}).$$

Theorem 33.15 (Four-band Hessenberg realization)**.** *The determinants $D_{p,n}^{(\lambda)}$ satisfy recurrence* (33.8) *exactly. Consequently*

$$P_{p,n}^{[4,\lambda]}(u) \equiv D_{p,n}^{(\lambda)}(u) \pmod{u^5}.$$

Proof. Expand $\det(I_n + uH_{p,\lambda,n})$ by the last row. The diagonal term gives

$$(1 + \ell_n u)D^{(\lambda)}_{p,n-1}.$$

The first subdiagonal entry closes with one superdiagonal entry and gives

$$+p(n-1)\ell_{n-1}u^2 D^{(\lambda)}_{p,n-2}.$$

The second subdiagonal closes with two consecutive superdiagonal entries and gives

$$-p(n-2)(n-1)u^3 D^{(\lambda)}_{p,n-3}.$$

The third subdiagonal closes with three consecutive superdiagonal entries. Its cofactor sign reverses the negative matrix entry, producing

$$+\lambda p(n-3)(n-2)(n-1)u^4 D^{(\lambda)}_{p,n-4}.$$

No deeper band is present. This is exactly (33.8). □

Use the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \qquad i \geq 1.$$

Let $D = \partial_x$, let $Xf = xf$, set $N = XD$, and define

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1}, \qquad Xe_i = ie_{i+1}, \qquad X^k e_i = i(i+1)\cdots(i+k-1)e_{i+k},$$

and $L_p e_i = \ell_i e_i$.

Corollary 33.16 (Operator form)**.** *The matrix $H_{p,\lambda}$ is the matrix, in the basis (e_i), of*

$$\boxed{\mathcal{H}_{p,\lambda} = D + L_p - pXL_p - pX^2 - \lambda pX^3.} \tag{33.9}$$

Equivalently,

$$\boxed{\mathcal{H}_{p,\lambda} = \left[1 + (p+1)x - p(p+1)x^2\right]\partial_x + 1 - px - px^2 - \lambda px^3.} \tag{33.10}$$

Proof. The actions of D, L_p, XL_p, X^2, and X^3 on e_i reproduce respectively the superdiagonal, diagonal, first subdiagonal, second subdiagonal, and third subdiagonal of $H_{p,\lambda}$. Expanding $L_p = I + (p+1)XD$ gives (33.10). □

Banded Hessenberg determinants and their finite-term recurrences are classical; see, for example, [66]. The point here is the exact identification of the fourth chronological parameter with the third subdiagonal and the X^3 term.

33.5 Fourth-column transport and exact identifiability

For the fourth-order family define

$$C_{n,k}^{(p,\lambda)} = [u^k]P_{p,n}^{[4,\lambda]}(u), \qquad 0 \le k \le 4,$$

and, for $m \ge k+1$,

$$\Pi_{p,k}^{(\lambda)}(m) = \frac{C_{m-1,k}^{(p,\lambda)}}{(m-1)_{\underline{k}}}.$$

Because the λ-term starts in degree four, theorem 33.8 implies

$$\Pi_{p,k}^{(\lambda)} = \Pi_{p,k} \qquad (k = 1, 2, 3).$$

Theorem 33.17 (Fourth-column transport)**.** *For $m \ge 5$,*

$$\boxed{\begin{aligned} m\Pi_{p,4}^{(\lambda)}(m+1) = (m-4)\Pi_{p,4}^{(\lambda)}(m) + \ell_m\Pi_{p,3}(m) \\ + p\ell_{m-1}\Pi_{p,2}(m-1) - p\Pi_{p,1}(m-2) + \lambda p. \end{aligned}} \tag{33.11}$$

The last term is exactly the normalized connected quadruple source.

Proof. Extract u^4 from (33.8) and put $n = m$. Divide by

$$(m)_{\underline{4}} = m(m-1)(m-2)(m-3).$$

The four source terms simplify using

$$\frac{(m-1)_{\underline{3}}}{(m)_{\underline{4}}} = \frac{1}{m}, \qquad \frac{(m-1)(m-2)_{\underline{2}}}{(m)_{\underline{4}}} = \frac{1}{m},$$

$$\frac{(m-2)(m-1)(m-3)_{\underline{1}}}{(m)_{\underline{4}}} = \frac{1}{m}, \qquad \frac{(m-3)(m-2)(m-1)}{(m)_{\underline{4}}} = \frac{1}{m}.$$

Multiplying by m gives (33.11). □

The key payoff is that the new connected source does not distort the shape of the fourth normalized column; it translates it.

Theorem 33.18 (Connected-quadruple translation law)**.** *For every $\lambda, \mu \in \mathbb{Z}$ and every $m \ge 5$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m) = \frac{p(\lambda-\mu)}{4}.} \tag{33.12}$$

Equivalently,

$$\boxed{C_{m-1,4}^{(p,\lambda)} - C_{m-1,4}^{(p,\mu)} = \frac{p(\lambda-\mu)}{4}(m-1)_{\underline{4}}.} \tag{33.13}$$

Proof. Subtract the two instances of (33.11). With

$$\Delta_m = \Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m),$$

one obtains

$$m\Delta_{m+1} = (m-4)\Delta_m + p(\lambda-\mu).$$

The constant sequence

$$\Delta_m = \frac{p(\lambda-\mu)}{4}$$

satisfies this recurrence. At the first admissible row $m = 5$, direct extraction from $P_{p,4}^{[4,\lambda]}$ gives the same difference. Uniqueness of the first-order recurrence proves (33.12); multiplying by $(m-1)_{\underline{4}}$ gives (33.13). □

Corollary 33.19 (One-row identifiability)**.** *Once the lower-order polarization is fixed, any single normalized fourth-column value at an admissible row identifies λ exactly:*

$$\boxed{\lambda = \frac{4}{p}\left(\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(0)}(m)\right).}$$

At the first raw row $n = 4$, changing λ by one changes $[u^4]P_{p,4}$ by exactly $6p$.

Thus order four is simultaneously non-identifiable from the entire third jet and exactly identifiable from one fourth-column observation.

Theorem 33.20 (Explicit normalized fourth column)**.** *For $m \geq 5$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) = \frac{G_{p,\lambda}(m)}{5760},} \tag{33.14}$$

where

$$\begin{aligned} G_{p,\lambda}(m) = {} & 15(p+1)^4 m^4 + 30(p+1)^3(3p-1)m^3 \\ & - 5(p+1)^2(287p^2 + 158p - 1)m^2 \\ & + 2(p+1)^2(1645p^2 - 50p + 1)m \\ & + 240p(6\lambda - 5p^3 - 5p^2 + 11p + 5). \end{aligned} \tag{33.15}$$

In particular,

$$\Pi_{p,4}^{(\lambda)}(5) = \frac{\lambda p + 5p^3 + 15p^2 + 13p + 4}{4}. \tag{33.16}$$

Proof. Substitute the three explicit polynomials of theorem 33.2 into the source of (33.11). The resulting source is polynomial in m of degree four. Direct expansion verifies that (33.14) satisfies (33.11); evaluation at $m = 5$ gives (33.16). The first-order transport equation then gives uniqueness. The λ-dependent part of (33.15) is $1440\lambda p$, which is exactly the translation $\lambda p/4$ of theorem 33.18. □

33.6 The fourth-jet Stirling shadow

The new connected parameter is exact information, but it disappears modulo p.

Theorem 33.21 (Fourth-jet Stirling shadow)**.** *For every prime $p \geq 5$, every $\lambda \in \mathbb{Z}$, and every $n \geq 0$,*

$$\boxed{P_{p,n}^{[4,\lambda]}(u) \equiv \prod_{j=1}^{n}(1 + ju) \pmod{(p, u^5)}.} \tag{33.17}$$

Consequently,

$$\boxed{C_{m-1,4}^{(p,\lambda)} \equiv e_4(1, \ldots, m-1) \equiv \begin{bmatrix} m \\ m-4 \end{bmatrix} \pmod{p},} \tag{33.18}$$

where the bracket is the unsigned Stirling number of the first kind.

Proof. Modulo p, one has $\ell_n \equiv n$. Every pair, triple, and quadruple interaction source in (33.8) contains an explicit factor of p, so the recurrence reduces to

$$P_{p,n}^{[4,\lambda]}(u) \equiv (1 + nu)P_{p,n-1}^{[4,\lambda]}(u) \pmod{(p, u^5)}.$$

Iterating gives (33.17). The coefficient of u^4 in the product is the fourth elementary symmetric function of $1, \ldots, m-1$, which is the near-diagonal unsigned Stirling number shown in (33.18); see, for example, [67]. □

Over $\mathbb{Q}[m]$ the normalized fourth shadow is

$$\boxed{S_4(m) = \frac{m(15m^3 - 30m^2 + 5m + 2)}{5760},} \tag{33.19}$$

because

$$e_4(1, \dots, m-1) = (m-1)_{\underline{4}} S_4(m).$$

Corollary 33.22 (Normalized shadow away from $p = 5$)**.** *For every prime $p \geq 7$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) \equiv S_4(m) \pmod{p}}$$

as a polynomial congruence. The congruence is independent of λ.

Proof. Subtract (33.19) from the exact quartic (33.14). The resulting rational polynomial has an explicit factor of p. For $p \geq 7$, the denominator

$$5760 = 2^7 \cdot 3^2 \cdot 5$$

is invertible modulo p, so reduction is legitimate. □

Remark 33.23 ($p = 5$ denominator resonance). The raw coefficient congruence (33.18) remains valid at $p = 5$ because it is obtained before any division. However, 5760 is divisible by 5, so one must not obtain a normalized polynomial congruence by cancelling $(m-1)_{\underline{4}}$ or reducing (33.19) coefficientwise modulo 5. This is a normalization-denominator issue, not a failure of the fourth-jet Stirling shadow.

Corollary 33.24 (Shadow blindness versus exact identifiability)**.** *All stationary p-divisible fourth polarizations have the same mod-p fourth jet, but distinct parameters have distinct exact fourth columns. Thus the Stirling shadow forgets λ, while theorem 33.19 recovers it from one exact row.*

Chapter 34

CCR-k General Theory: Higher-Order Chronological Carry, Hessenberg Hierarchies, and Universal Stirling Shadows

Research branch — The all-order chronological-carry closure. *The preceding arithmetic branch established the third-order CCR theory and then isolated the genuinely new fourth-order coordinate. This chapter keeps those low-order chapters intact and passes to the all-order terminal-block hierarchy. The geometric input remains chronological path locality; the numerical connected sources are explicitly typed arithmetic data.*

Chapters 29–33 identify the first two nontrivial stages of chronological carry. The point of the present chapter is not to replace those stages, but to explain the general mechanism they reveal. At every order there is one connected interval support type, one new typed source, one corresponding Hessenberg band, and one new source term in normalized fixed-column transport. The exact source remains visible integrally and reconstructible from the normalized column, while the entire p-divisible hierarchy collapses to a common Stirling shadow modulo p.

Throughout this chapter fix a prime $p \geq 5$ unless stated otherwise and write

$$\ell_n = (p+1)n - p.$$

The determinant and transport identities are polynomial identities and make sense more generally; the prime hypothesis is needed for the mod-p arithmetic statements.

CCR-3 $\longrightarrow$ CCR-4 $\longrightarrow$ CCR-k:
low-order discovery $\longrightarrow$ first independent higher coordinate $\longrightarrow$ all-order source–band–transport–shadow theory.

34.1 Chronological support geometry and scope

The correction chronology is represented by the path graph on consecutive correction positions. The following elementary fact is the geometric reason that exactly one connected support *type* appears at each order.

Lemma 34.1 (Connected interval support)**.** *Every connected subset of a path having cardinality r is a consecutive interval*

$$\{j, j+1, \dots, j+r-1\}.$$

Hence there is exactly one connected support type of size r.

Proof. If $a < c$ lie in a connected subset S, then the unique path from a to c contains every integer between them. Connectivity forces each of those intermediate vertices into S. Thus S is an interval, and its cardinality determines its length. □

Remark 34.2 (Geometry does not determine arithmetic)*.* The lemma determines the support of a new connected interaction, not its numerical weight. The fourth-order chapter already exhibits this distinction: the complete third jet and chronological locality determine the consecutive-quadruple support but leave a genuinely new arithmetic coordinate. Numerical connected weights are therefore treated below as explicitly typed arithmetic data.

The structural dictionary is

$$\boxed{\begin{array}{c}\text{connected } r\text{-interval} \iff \text{source } a_{r,n} \iff \text{subdiagonal } r-1 \\ \iff \text{column-}r \text{ transport source}\end{array}}$$

The first arrow is a modeling interface: path geometry fixes the support type, while arithmetic polarization fixes its value.

34.2 General terminal-block CCR-k

Fix $k \geq 3$ and work in the truncated ring

$$R_k = \mathbb{Z}[u]/(u^{k+1}).$$

For $r \geq 3$, let $a_{r,n} \in \mathbb{Z}$ be the normalized connected source at order r, after removal of the chronological placement factor

$$n - 1_{\underline{r-1}} = (n-1)(n-2)\cdots(n-r+1).$$

The third-order baseline fixes

$$a_{3,n} = -p. \tag{34.1}$$

Definition 34.3 (Typed terminal-block CCR-k)**.** A typed terminal-block CCR-k family is a sequence $P^{[k]}_{p,n}(u) \in R_k$ satisfying

$$\boxed{\begin{aligned} P^{[k]}_{p,n}(u) = {} & (1+\ell_n u)P^{[k]}_{p,n-1}(u) \\ & + p(n-1)\ell_{n-1}u^2 P^{[k]}_{p,n-2}(u) \\ & + \sum_{r=3}^{k} n - 1_{\underline{r-1}} a_{r,n} u^r P^{[k]}_{p,n-r}(u) \pmod{u^{k+1}}, \end{aligned}} \tag{34.2}$$

with $P_{p,0} = 1$, $P_{p,n} = 0$ for $n < 0$, and (34.1).

The definition is intentionally explicit: it is the class of chronological recurrences that factor by the size of the terminal connected interval block. It does not assert that every conceivable path-local model must have this factorization.

Write

$$C_{n,j} = [u^j]P^{[k]}_{p,n}(u).$$

Theorem 34.4 (All-order terminal-block separation)**.** *For every $r \le k$,*

$$\boxed{\begin{aligned} C_{n,r} = C_{n-1,r} + \ell_n C_{n-1,r-1} + p(n-1)\ell_{n-1} C_{n-2,r-2} \\ + \sum_{s=3}^{r-1} n - 1_{\underline{s-1}} a_{s,n} C_{n-s,r-s} + n - 1_{\underline{r-1}} a_{r,n}. \end{aligned}} \tag{34.3}$$

After all sources of orders below r have been fixed, the only new recurrence datum entering degree r is $a_{r,n}$.

Proof. Extract u^r from (34.2). A terminal source of size s contributes

$$n - 1_{\underline{s-1}} a_{s,n} [u^{r-s}] P^{[k]}_{p,n-s}(u).$$

When $s < r$, this depends only on a coefficient of degree $r - s < r$, hence only on lower-order data. For the $s = r$ term there are two boundary regimes. If $1 \le n < r$, then the placement factor $n - 1_{\underline{r-1}}$ contains a zero factor, so the entire term vanishes. If $n \ge r$, then $P_{p,n-r}$ has constant term one (inductively from $P_{p,0} = 1$ and the recurrence), so $[u^0]P_{p,n-r} = 1$ and the final term in (34.3) follows. Sources of order greater than r are divisible by a power of u too large to contribute. □

Corollary 34.5 (All-order non-identifiability)**.** *Two typed terminal-block families that agree through order $r-1$ but differ in $a_{r,n}$ have identical $(r-1)$-jets. Thus lower-order data do not identify the next connected arithmetic source.*

Proof. The first term involving $a_{r,n}$ has degree r, so all lower coefficients are independent of it. □

34.3 Higher-band lower-Hessenberg realization

For $i \ge 1$ and $m \ge 0$, write

$$i^{\overline{m}} = i(i+1)\cdots(i+m-1), \qquad i^{\overline{0}} = 1.$$

Definition 34.6 (General Hessenberg matrix)**.** Let $H^{[k]}$ be the infinite lower-Hessenberg matrix with

$$H_{i,i} = \ell_i, \qquad H_{i,i+1} = 1, \tag{34.4}$$

$$H_{i+1,i} = -pi\ell_i, \tag{34.5}$$

and, for $3 \le r \le k$,

$$\boxed{H_{i+r-1,i} = (-1)^{r-1} i^{\overline{r-1}}\, a_{r,i+r-1}.} \tag{34.6}$$

All entries below the $(k-1)$-st subdiagonal are zero. Let $H^{[k]}_n$ denote the leading $n \times n$ principal truncation.

Theorem 34.7 (Exact Hessenberg realization)**.** *The determinant polynomials*

$$D^{[k]}_n(u) = \det(I_n + uH^{[k]}_n)$$

satisfy (34.2) *exactly. Therefore*

$$\boxed{D^{[k]}_n(u) = P^{[k]}_{p,n}(u) \quad \text{in } R_k.}$$

Proof. Expand $\det(I_n + uH_n^{[k]})$ along the last row. The diagonal term contributes

$$(1 + \ell_n u)D_{n-1}^{[k]}.$$

The first subdiagonal entry is $-p(n-1)\ell_{n-1}$. Its cofactor sign is negative, and the remaining forced superdiagonal closure contributes one factor of u, so this term is

$$+p(n-1)\ell_{n-1}u^2 D_{n-2}^{[k]}.$$

Now choose the entry in column $n-r+1$, where $r \geq 3$. By (34.6),

$$H_{n,n-r+1} = (-1)^{r-1}n - r + 1^{\overline{r-1}}a_{r,n}.$$

The cofactor sign is $(-1)^{r-1}$. Lower-Hessenberg structure forces the last $r-1$ columns of the cofactor to close through the consecutive unit superdiagonal entries, contributing u^{r-1}; the selected last-row entry contributes the remaining factor u. The two signs cancel, and

$$n - r + 1^{\overline{r-1}} = n - 1_{\underline{r-1}}.$$

The untouched leading block is $I_{n-r} + uH_{n-r}^{[k]}$. Hence the contribution is

$$n - 1_{\underline{r-1}}a_{r,n}u^r D_{n-r}^{[k]}.$$

Summing the available bands gives (34.2). The initial conditions agree, so the two families coincide. □

The theorem turns the chronological order filtration into an exact band filtration. The next result shows that this filtration is also visible coefficientwise.

34.4 The subdiagonal light cone

Theorem 34.8 (Subdiagonal light cone)**.** *Let $M = (m_{ij})$ be any lower-Hessenberg matrix, so $m_{ij} = 0$ for $j > i+1$, and let*

$$Q_n(u) = \det(I_n + uM_n).$$

Then $[u^j]Q_n(u)$ depends only on the diagonal, superdiagonal, and first $j-1$ subdiagonals. Equivalently, an entry on the s-th subdiagonal can first affect degree $s+1$.

Proof. The coefficient $[u^j]Q_n(u)$ is the sum of the principal $j \times j$ minors of M_n. Suppose a permutation term in such a minor uses an entry on the s-th subdiagonal. Relative to the diagonal, that move has displacement $-s$. Because M is lower-Hessenberg, every positive displacement in the same permutation is at most $+1$. Since the total permutation displacement is zero, compensating $-s$ requires at least s positive unit displacements. The permutation therefore moves at least $s+1$ indices and cannot occur in a minor smaller than $(s+1) \times (s+1)$. Thus the s-th subdiagonal is invisible below degree $s+1$. □

Corollary 34.9 (Exact order filtration)**.** *In the typed CCR hierarchy, the order-r source affects no fixed column below r, while every band deeper than $r-1$ is invisible to the r-th fixed column.*

This provides a matrix-theoretic proof that higher interactions cannot retroactively change lower jets.

34.5 Normalized fixed-column transport and inverse reconstruction

For $j \geq 0$, set

$$C_{n,j} = [u^j] P_{p,n}^{[k]}(u).$$

For $m \geq j+1$, define the falling-factorial normalized column

$$\boxed{\Pi_{p,j}(m) = \frac{C_{m-1,j}}{m-1_{\underline{j}}}.} \tag{34.7}$$

Also set $\Pi_{p,0}(m) = 1$.

The first admissible row $m = j$ must be treated as a boundary equation; this avoids the meaningless expression $0 \cdot \Pi_{p,j}(j)$.

Theorem 34.10 (Boundary and transport equations)**.** *For $1 \leq j \leq k$, the first normalized row is determined by*

$$\boxed{\begin{aligned} j\Pi_{p,j}(j+1) = \ell_j \Pi_{p,j-1}(j) + p\ell_{j-1}\Pi_{p,j-2}(j-1) \\ + \sum_{r=3}^{j} a_{r,j}\Pi_{p,j-r}(j-r+1), \end{aligned}} \tag{34.8}$$

where terms with negative column index are omitted. For every $m \geq j+1$,

$$\boxed{\begin{aligned} m\Pi_{p,j}(m+1) = (m-j)\Pi_{p,j}(m) + \ell_m \Pi_{p,j-1}(m) \\ + p\ell_{m-1}\Pi_{p,j-2}(m-1) \\ + \sum_{r=3}^{j} a_{r,m}\Pi_{p,j-r}(m-r+1). \end{aligned}} \tag{34.9}$$

Proof. Extract u^j from (34.2) with $n = m$:

$$\begin{aligned} C_{m,j} = C_{m-1,j} + \ell_m C_{m-1,j-1} + p(m-1)\ell_{m-1} C_{m-2,j-2} \\ + \sum_{r=3}^{j} m-1_{\underline{r-1}} a_{r,m} C_{m-r,j-r}. \end{aligned}$$

Use

$$C_{m,j} = m_{\underline{j}}\Pi_{p,j}(m+1) = mm-1_{\underline{j-1}}\Pi_{p,j}(m+1),$$

$$C_{m-1,j} = m-1_{\underline{j}}\Pi_{p,j}(m) = (m-j)m-1_{\underline{j-1}}\Pi_{p,j}(m),$$

and

$$m-1_{\underline{r-1}}m-r_{\underline{j-r}} = m-1_{\underline{j-1}}.$$

For $m \geq j+1$, division by $m-1_{\underline{j-1}}$ gives (34.9). At $m = j$, the term $C_{j-1,j}$ is zero because its degree exceeds its row size, so the same coefficient extraction yields (34.8) without ever defining $\Pi_{p,j}(j)$. □

The transport equation is triangular in the interaction order and therefore invertible.

Theorem 34.11 (Exact source reconstruction)**.** *Fix $r \geq 3$ and suppose all sources of orders below r are known. The first source value is recovered from the boundary row by*

$$\boxed{\begin{aligned} a_{r,r} = {} & r\Pi_{p,r}(r+1) - \ell_r \Pi_{p,r-1}(r) - p\ell_{r-1}\Pi_{p,r-2}(r-1) \\ & - \sum_{s=3}^{r-1} a_{s,r}\Pi_{p,r-s}(r-s+1). \end{aligned}} \tag{34.10}$$

For every $m \geq r+1$,

$$\boxed{\begin{aligned} a_{r,m} = {} & m\Pi_{p,r}(m+1) - (m-r)\Pi_{p,r}(m) - \ell_m \Pi_{p,r-1}(m) \\ & - p\ell_{m-1}\Pi_{p,r-2}(m-1) \\ & - \sum_{s=3}^{r-1} a_{s,m}\Pi_{p,r-s}(m-s+1). \end{aligned}} \tag{34.11}$$

Thus the complete normalized source sequence is identifiable from the r-th column together with the already known lower-order data.

Proof. Set $j = r$ in theorem 34.10. The term with source order r is multiplied by $\Pi_{p,0} = 1$. Solving the boundary and transport equations for that source gives (34.10) and (34.11). □

34.6 Stationary p-divisible extensions

The preceding theory allows arbitrary row-dependent source sequences. A particularly rigid and useful class is obtained by removing that row dependence after the chronological placement factor.

Definition 34.12 (Stationary p-divisible class)**.** For $r \geq 4$, a source is stationary p-divisible if

$$a_{r,n} = (-1)^r p\lambda_r$$

for an integer λ_r independent of n. We set $\lambda_3 = 1$, so that $a_{3,n} = -p$ agrees with CCR-3.

Theorem 34.13 (Stationary extension classification)**.** *A stationary p-divisible extension through order k is uniquely determined by the parameter vector*

$$\boxed{(\lambda_4, \lambda_5, \ldots, \lambda_k) \in \mathbb{Z}^{k-3}.}$$

Its recurrence is

$$\boxed{\begin{aligned} P_{p,n}^{[k,\boldsymbol{\lambda}]}(u) = {} & (1+\ell_n u)P_{p,n-1} + p(n-1)\ell_{n-1}u^2 P_{p,n-2} \\ & + p\sum_{r=3}^{k}(-1)^r \lambda_r n - 1_{\underline{r-1}} u^r P_{p,n-r} \pmod{u^{k+1}}, \end{aligned}} \tag{34.12}$$

with $\lambda_3 = 1$.

Proof. By Lemma 34.1, there is one connected support type at each order. By theorem 34.4, a typed terminal-block model introduces one new normalized source sequence at that order. Stationarity makes the normalized source independent of n, and p-divisibility expresses it uniquely as p times an integer. The factor $(-1)^r$ is a sign convention chosen so that positive λ_r produces a common negative sign on the corresponding Hessenberg subdiagonal; see equation (34.14) below. □

Definition 34.14 (Primitive same-sign hierarchy)**.** The primitive same-sign normalization is the stationary point

$$\boxed{\lambda_r = 1 \qquad (3 \le r \le k).}$$

Thus

$$\boxed{\begin{aligned} P_{p,n}^{[k],\mathrm{prim}}(u) &= (1+\ell_n u)P_{p,n-1} + p(n-1)\ell_{n-1}u^2 P_{p,n-2} \\ &\quad + p\sum_{r=3}^{k}(-1)^r n - 1_{\underline{r-1}} u^r P_{p,n-r} \pmod{u^{k+1}}. \end{aligned}} \tag{34.13}$$

The word *primitive* here names a selected arithmetic normalization. It is not a claim that the underlying pushdown transition table forces $\lambda_r = 1$.

34.7 Operator realization

Under Definition 34.12, the deeper Hessenberg bands simplify to

$$\boxed{H_{i+r-1,i} = -\lambda_r p\, i^{\overline{r-1}}, \qquad 3 \le r \le k.} \tag{34.14}$$

All higher connected bands therefore have the same matrix sign when $\lambda_r > 0$.

In the divided-power basis $e_i = x^{i-1}/(i-1)!$, one has

$$X^s e_i = i^{\overline{s}} e_{i+s}.$$

Theorem 34.15 (General differential-operator realization)**.** *The stationary* CCR-k *Hessenberg matrix is the matrix of*

$$\boxed{\mathcal{H}_{p,\boldsymbol{\lambda}}^{[k]} = D + L_p - pXL_p - p\sum_{r=3}^{k}\lambda_r X^{r-1}.} \tag{34.15}$$

For the primitive same-sign hierarchy,

$$\boxed{\mathcal{H}_{p,\mathrm{prim}}^{[k]} = D + L_p - pXL_p - p(X^2 + X^3 + \cdots + X^{k-1}).} \tag{34.16}$$

Proof. The terms D, L_p, and $-pXL_p$ reproduce the unit superdiagonal, diagonal ℓ_i, and first lower band $-pi\ell_i$. For $r \ge 3$,

$$-p\lambda_r X^{r-1} e_i = -p\lambda_r i^{\overline{r-1}} e_{i+r-1},$$

which is exactly (34.14). □

34.8 Polynomial fixed columns

The normalized transport system has a useful consequence not visible from the determinant alone: stationary fixed columns are polynomial at every order.

For $j \ge 1$, define the linear operator on $\mathbb{Q}[m]$

$$(\mathcal{T}_j f)(m) = mf(m+1) - (m-j)f(m). \tag{34.17}$$

If $f(m) = c_d m^d + \cdots$, then

$$\mathcal{T}_j f = (d+j)c_d m^d + \text{lower-degree terms}. \tag{34.18}$$

Thus $\mathcal{T}_j$ preserves degree and is an automorphism of $\mathbb{Q}[m]$.

Theorem 34.16 (Arbitrary-order fixed-column polynomiality)**.** *For every stationary p-divisible* CCR-k *family and every* $0 \le j \le k$*, the normalized column* $\Pi_{p,j}(m)$ *is the restriction, for* $m \ge j+1$*, of a unique polynomial in*

$$\mathbb{Q}[p, \lambda_4, \dots, \lambda_j][m].$$

For $j \ge 1$*, its degree is exactly* j*, and its leading coefficient is*

$$\boxed{[m^j]\Pi_{p,j}(m) = \frac{(p+1)^j}{2^j j!}.} \tag{34.19}$$

Proof. Proceed by induction on j. The case $j = 0$ is $\Pi_{p,0} = 1$. Suppose the assertion holds below j. Under stationarity, the right-hand side of (34.9) excluding the term $(m-j)\Pi_{p,j}(m)$ is a polynomial $G_j(m)$. Its only degree-j contribution comes from

$$\ell_m \Pi_{p,j-1}(m),$$

because the pair term has degree at most $j-1$, while the source term of order r multiplies a column of degree $j-r$. Hence G_j has degree j.

Equation (34.9) is precisely

$$\mathcal{T}_j \Pi_{p,j} = G_j.$$

By (34.18), $\mathcal{T}_j$ is triangular with nonzero diagonal $d+j$ on the monomial basis, so it is invertible on $\mathbb{Q}[m]$. There is therefore a unique polynomial solution of degree j. At $m = j$, the polynomial identity reduces to the boundary equation (34.8), so the polynomial solution has the same first admissible value as the normalized column. The first-order recurrence then identifies the two for all $m \ge j+1$.

If c_{j-1} is the leading coefficient of $\Pi_{p,j-1}$, the leading coefficient of G_j is $(p+1)c_{j-1}$. Since $\mathcal{T}_j$ multiplies the leading coefficient of a degree-j polynomial by $2j$,

$$c_j = \frac{p+1}{2j} c_{j-1}.$$

Starting from $c_0 = 1$ gives (34.19). □

The leading term is independent of every higher connected parameter. Those parameters change lower-degree structure while leaving the universal top-degree growth fixed.

34.9 Exact parameter translation and identifiability

Suppose two stationary families agree at every order below r but have order-r parameters λ_r and μ_r. Let

$$\Delta_m = \Pi_{p,r}^{(\lambda_r)}(m) - \Pi_{p,r}^{(\mu_r)}(m).$$

Theorem 34.17 (Stationary translation law)**.** *The first admissible row satisfies*

$$\boxed{\Delta_{r+1} = \frac{(-1)^r p(\lambda_r - \mu_r)}{r}.} \tag{34.20}$$

For $m \ge r+1$*,*

$$\boxed{m\Delta_{m+1} = (m-r)\Delta_m + (-1)^r p(\lambda_r - \mu_r).} \tag{34.21}$$

The unique solution is the constant translation

$$\boxed{\Delta_m = \frac{(-1)^r p(\lambda_r - \mu_r)}{r} \qquad (m \ge r+1).} \tag{34.22}$$

Proof. Subtract the two order-r instances of theorem 34.10. Every lower-order term cancels, leaving only the difference

$$(-1)^r p(\lambda_r - \mu_r).$$

The boundary equation gives (34.20); the ordinary transport rows give (34.21). The constant in (34.22) satisfies both equations, and the first-order recurrence has a unique solution after its first value is fixed. □

Corollary 34.18 (One-row identifiability)**.** *Once lower parameters are known, a single exact normalized observation in column r identifies the stationary parameter λ_r.*

At $r = 4$, equation (34.22) becomes the recovered fourth-order translation

$$\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m) = \frac{p(\lambda - \mu)}{4}.$$

At $r = 5$, it predicts the alternating next case

$$\Pi_{p,5}^{(\lambda_5)}(m) - \Pi_{p,5}^{(\mu_5)}(m) = -\frac{p(\lambda_5 - \mu_5)}{5}.$$

34.10 Universal Stirling shadows

Assume now that every connected source is divisible by p; in particular this holds throughout the stationary class. Since

$$\ell_n = n + p(n-1) \equiv n \pmod{p},$$

and every interaction term beyond the singleton term has an explicit factor of p, the full recurrence collapses modulo p.

Theorem 34.19 (Arbitrary-order raw Stirling shadow)**.** *For every $k \geq 3$,*

$$\boxed{P_{p,n}^{[k]}(u) \equiv \prod_{j=1}^{n}(1 + ju) \pmod{(p, u^{k+1})}.} \tag{34.23}$$

Consequently, for $0 \leq r \leq k$,

$$\boxed{[u^r]P_{p,n}^{[k]}(u) \equiv e_r(1, 2, \ldots, n) = \begin{bmatrix} n+1 \\ n+1-r \end{bmatrix} \pmod{p},} \tag{34.24}$$

where the bracket denotes an unsigned Stirling number of the first kind.

Proof. Modulo p, recurrence (34.2) becomes

$$P_{p,n}^{[k]}(u) \equiv (1 + nu)P_{p,n-1}^{[k]}(u) \pmod{(p, u^{k+1})}.$$

Iterating from $P_{p,0} = 1$ gives (34.23). Coefficient extraction gives the elementary symmetric function in $1, \ldots, n$, which is the displayed near-diagonal unsigned Stirling number. □

The raw form (34.24) is the safest universal theorem because it requires no modular division. The normalized shadow admits a precise rational-polynomial formulation.

Proposition 34.20 (Normalized shadow and denominator resonances)**.** *For fixed r, define*

$$S_r(m) = \frac{e_r(1, 2, \dots, m-1)}{m - 1_{\underline{r}}}. \tag{34.25}$$

Then $S_r(m) \in \mathbb{Q}[m]$ has degree r. In the stationary p-divisible hierarchy,

$$\Pi_{p,r}(m) - S_r(m) \in p\,\mathbb{Q}[p, \lambda_4, \dots, \lambda_r, m]. \tag{34.26}$$

Hence there exists an integer $D_r > 0$, depending only on r, such that for every prime $p \nmid D_r$ the normalized polynomial congruence

$$\Pi_{p,r}(m) \equiv S_r(m) \pmod{p} \tag{34.27}$$

is legitimate coefficientwise after denominators are cleared. For primes dividing the chosen denominator, the raw congruence (34.24) *remains valid and no cancellation should be performed.*

Proof. For fixed r, the quantity $e_r(1, \dots, m-1)$ is a polynomial in m of degree $2r$. One direct justification uses Newton's identities

$$r e_r = \sum_{j=1}^{r} (-1)^{j-1} e_{r-j} \sum_{t=1}^{m-1} t^j,$$

together with Faulhaber's polynomial formulas for the power sums; induction on r gives polynomiality, and the leading term is $m^{2r}/(2^r r!)$. This polynomial vanishes at $m = 1, 2, \dots, r$, because fewer than r nonzero slots are present. Hence it is divisible in $\mathbb{Q}[m]$ by $m - 1_{\underline{r}}$, proving that S_r is a polynomial of degree r.

By theorem 34.16, $\Pi_{p,r}$ is polynomial over $\mathbb{Q}[p, \boldsymbol{\lambda}]$. Specializing the recurrence at the formal value $p = 0$ removes every interaction term and sends ℓ_n to n, so the family becomes exactly the free product $\prod_{j=1}^{n}(1+ju)$. Therefore $\Pi_{0,r} = S_r$. Polynomial division in the indeterminate p gives (34.26). Clearing rational denominators yields some integer D_r, and coefficientwise reduction modulo primes not dividing D_r gives (34.27). If $p \mid D_r$, only the pre-division raw theorem is automatic. □

Remark 34.21 (Exact information versus shadow information)*.* The same parameter that is recovered exactly from one normalized row by Corollary 34.18 disappears from the entire raw mod-p jet by theorem 34.19. Thus the hierarchy exhibits a systematic exact-versus-shadow information loss at every order.

34.11 The all-order closure theorem

We can now package the preceding results into one statement.

Theorem 34.22 (General higher-order chronological-carry theorem)**.** *Fix $k \geq 3$. Within the typed terminal-block class of Definition 34.3:*

(i) *at every order $r \leq k$, the chronological path has exactly one connected support type, namely the consecutive r-block;*

(ii) *after all lower sources are fixed, the sole new degree-r recurrence datum is the source sequence $a_{r,n}$;*

(iii) *the recurrence has the exact lower-Hessenberg realization* (34.4)–(34.6)*;*

(iv) *the r-th fixed column is insensitive to every subdiagonal deeper than $r-1$;*

(v) *normalized columns satisfy the boundary/transport system* (34.8)–(34.9)*;*

(vi) *every connected source sequence is reconstructed exactly by* (34.10)–(34.11)*;*

(vii) *in the stationary p-divisible subclass, the extension space through order k is $\mathbb{Z}^{k-3}$, with operator* (34.15)*;*

(viii) *stationary normalized columns are polynomials of exact degree equal to their column index;*

(ix) *every p-divisible member has the universal raw Stirling shadow* (34.23)*.*

No additional connected support type or deeper band datum is required to determine the k-jet.

Proof. Part (i) is Lemma 34.1. Parts (ii) and the absence of an additional order-r source are theorem 34.4. Part (iii) is theorem 34.7. Part (iv) is theorem 34.8. Parts (v) and (vi) are theorems 34.10 and 34.11. Part (vii) is theorems 34.13 and 34.15. Part (viii) is theorem 34.16. Part (ix) is theorem 34.19. Finally, a new connected support type cannot appear because connected subsets of the chronological path are already classified by Lemma 34.1, and a deeper subdiagonal cannot alter the k-jet by theorem 34.8. This closes the order induction. □

34.12 The first genuinely new checkpoint beyond CCR-4: order five

The all-order theorem contains the preceding CCR-3 and CCR-4 chapters as its first two nontrivial levels. The first new checkpoint not already developed there is order five, where a second independent stationary parameter appears. Its recurrence is

$$\begin{aligned} P^{[5]}_{p,n}(u) = (1+\ell_n u)P_{p,n-1} + p(n-1)\ell_{n-1}u^2P_{p,n-2} - pn - 1_{\underline{2}}u^3P_{p,n-3} \\ + \lambda_4 pn - 1_{\underline{3}}u^4P_{p,n-4} - \lambda_5 pn - 1_{\underline{4}}u^5P_{p,n-5} \pmod{u^6}. \end{aligned} \tag{34.28}$$

The new fifth-order source occupies the fourth subdiagonal

$$H_{i+4,i} = -\lambda_5 p\, i(i+1)(i+2)(i+3)$$

and contributes the operator term $-\lambda_5 pX^4$. Changing only λ_5 translates the fifth normalized column by

$$-\frac{p(\lambda_5-\mu_5)}{5}.$$

The sign alternation in the recurrence is therefore converted into a common negative sign in the stationary Hessenberg bands.

34.13 Synthesis

The CCR-k General Theory has a simple structural message. Chronological path geometry is rigid enough to determine where a new connected interaction can live, but not strong enough to determine its numerical weight. Once that weight is admitted as typed arithmetic data, the entire hierarchy becomes exact and recursively observable.

Table 34.1: The first orders of the stationary hierarchy. The numerical source is additional arithmetic data; only its support type is fixed by the path geometry.

order r	recurrence source a_r	Hessenberg band	normalized translation per unit λ_r
3	$-p$	$-p\, i^{\overline{2}}$	fixed baseline
4	$+p\lambda_4$	$-p\lambda_4\, i^{\overline{3}}$	$+p/4$
5	$-p\lambda_5$	$-p\lambda_5\, i^{\overline{4}}$	$-p/5$
6	$+p\lambda_6$	$-p\lambda_6\, i^{\overline{5}}$	$+p/6$

At every order r, four descriptions of the same new datum coexist:

$$\boxed{\begin{array}{c}\text{connected interval source } a_{r,n}\\ \Updownarrow\\ \text{terminal recurrence block } n-1_{\underline{r-1}}a_{r,n}u^rP_{n-r}\\ \Updownarrow\\ \text{Hessenberg subdiagonal } (-1)^{r-1}i^{\overline{r-1}}a_{r,i+r-1}\\ \Updownarrow\\ \text{source term in normalized-column transport.}\end{array}}$$

In the stationary p-divisible class, this reduces to one integer λ_r per order. The exact normalized column sees λ_r immediately through the constant translation $(-1)^r p\lambda_r/r$, while reduction modulo p sees none of it. The free Stirling shadow is therefore universal precisely because it is an information-losing projection.

The resulting all-order spine is

$$\boxed{\begin{array}{c}\text{chronological locality + typed connected arithmetic data}\\ \Longrightarrow \text{ higher-band Hessenberg transport}\\ \Longrightarrow \text{ exact identifiability + universal Stirling shadow.}\end{array}}$$

34.14 Proof-dependency ledger

For audit purposes, the logical dependencies of the main results are summarized in table 34.2.

34.15 Scope safeguards and nonclaims

The following statements are part of the mathematical scope of this chapter.

1. The original pushdown correction process supplies a motivating chronology and path-local support geometry. The numerical CCR sources are an added arithmetic polarization.

2. The typed terminal-block factorization (34.2) is an explicit model class. We do not assert that every possible path-local deformation must factor in this form.

3. The stationary p-divisible classification is a classification *inside that stated class.*

4. The primitive same-sign point $\lambda_r = 1$ is a normalization convention, not a theorem of machine-forced uniqueness.

Table 34.2: Dependency ledger for the all-order theorem package.

Result	Direct inputs	Role
Connected interval support	path graph	one support type per order
Terminal-block separation	Definition 34.3	isolates the new order-r source
Hessenberg realization	source normalization + last-row determinant expansion	recurrence/band equivalence
Light cone	lower-Hessenberg displacement bound	excludes deeper-band contamination
Normalized transport	recurrence coefficient extraction + falling-factorial cancellation	row evolution of fixed columns
Source reconstruction	normalized transport	inverse identifiability
Stationary classification	one source type/order + stationarity + p-divisibility	parameter lattice $\mathbb{Z}^{k-3}$
Operator realization	divided-power basis identities	compact all-band representation
Polynomiality	stationary transport + invertibility of $\mathcal{T}_j$	exact degree and leading coefficient
Translation law	subtraction of two stationary transports	one-row parameter identifiability
Stirling shadow	p-divisibility + $\ell_n \equiv n \pmod p$	universal information-losing projection
Main theorem	all preceding rows	all-order closure

5. The universal raw mod-p shadow is stronger and safer than any normalized congruence. Normalized polynomial reduction requires denominator control as in Proposition 34.20.

6. Classical results on cumulants, Möbius inversion, carry matrices, Stirling numbers, continued fractions, and lower-Hessenberg recurrences are used only as context or standard tools.

34.16 Arithmetic stratification of the normalized CCR columns

The terminal-block source hierarchy and the universal Stirling shadow are established in Chapter 34. Here the normalized fixed-column polynomial is studied over local and global arithmetic fields. The fourth *column* is the index $r = 4$ in the full CCR-k hierarchy, not a fourth-order replacement for that hierarchy. Algebraic sources and physical chronological realizations remain distinct.

34.16.1 Fourth-column normalization and the integral numerator

For a prime $p \geq 5$ and a stationary integer parameter λ, retain the integral quartic numerator $G_{p,\lambda}$ from (33.15). Its normalized fourth column satisfies

$$G_{p,\lambda}(m) = 5760\Pi_{p,4}^{(\lambda)}(m). \tag{34.29}$$

The roots of the normalized column in $\mathbb{Q}_p$ are those of $G_{p,\lambda}$. Crucially, at $p = 5$ one works with the *integral numerator* G, not a coefficientwise reduction of Π. Write

$$H(m) = 15m^3 - 30m^2 + 5m + 2, \qquad G_{p,\lambda}(m) \equiv mH(m) \pmod{p}. \tag{34.30}$$

The parameter λ disappears in this reduction. Direct computation gives

$$\operatorname{disc}(H) = 125700 = 10^2 \cdot 3 \cdot 419, \qquad \operatorname{disc}(mH) = 502800 = 2^4 \cdot 3 \cdot 5^2 \cdot 419. \tag{34.31}$$

34.17 The complete local factorization classification

Theorem 34.23 (All-prime root and factor stratification)**.** *For each integer λ and prime $p \geq 5$:*

1. *If $p \geq 7$ and $p \neq 419$, the roots of $G_{p,\lambda}$ in $\mathbb{Q}_p$ are precisely the unique lifts of the roots of $mH(m)$ in $\mathbb{F}_p$. Their number is 1, 2, or 4, independent of λ, and all residual irreducible factors lift unramified.*

2. *If $p = 5$, there is exactly one root in $\mathbb{Q}_5$. For $\lambda \neq 115$ its valuation is $2+v_5(\lambda-115)$, and at $\lambda = 115$ the root is 0. The remaining degree-three factor is irreducible and totally ramified over $\mathbb{Q}_5$.*

3. *If $p = 419$, the simple residues $m \equiv 0, 20 \pmod{419}$ always lift to two distinct roots. A further quadratic pair in $m \equiv -9 \pmod{419}$ is controlled by a unique 419-adic critical parameter Λ. Its exact splitting and ramification rule is stated in Theorem 34.25 below for all integer λ.*

Proof. For $p \geq 7$, $p \neq 419$, the coefficient of m^4 in G is a unit; the reduction mH is squarefree by (34.31) (and $H(0) = 2$). Ordinary Hensel lifting gives a bijection of linear residue factors with $\mathbb{Q}_p$-roots. Hensel factorization also lifts each separable irreducible factor of the reduction, yielding unramified extensions.

At $p = 5$, direct specialization is

$$G_{5,\lambda}(m) = 19440m^4 + 90720m^3 - 1433520m^2 + 2943072m + 7200(\lambda - 115).$$

The coefficients of m, m^2, m^3, m^4 have valuations $0, 1, 1, 1$, respectively. For $\lambda \neq 115$ the constant term has valuation $2 + d$, where $d = v_5(\lambda - 115)$. Its Newton polygon has segments $(0, 2 + d) \to (1, 0) \to (4, 1)$: the first slope gives one integral root of valuation $2+d$; the second gives three conjugate roots of valuation $-1/3$. A degree-three factor whose root valuation has denominator 3 is totally ramified and irreducible. When $\lambda = 115$, factor out m and apply the same final segment to the cubic.

At $p = 419$, one checks the residue factorization

$$G_{419,\lambda}(m) \equiv 15m(m - 20)(m + 9)^2 \pmod{419}.$$

The two simple residues lift. The double residue is analyzed below with the stationary derivative and a p-adic quadratic normal form. □

34.18 The 419-adic critical coordinate

Let $f_\lambda(m) = G_{419,\lambda}(m)$. Its derivative is independent of λ and satisfies

$$f'_\lambda(-9) \equiv 0, \qquad \frac{1}{2} f''_\lambda(-9) \equiv 144 \pmod{419}.$$

Hensel's lemma yields a unique $c \in \mathbb{Z}_{419}$ such that $c \equiv -9 \pmod{419}$ and $f'_\lambda(c) = 0$. Since the parameter occurs as $1440 \cdot 419\lambda$, there is a unique $\Lambda \in \mathbb{Z}_{419}$ satisfying $f_\Lambda(c) = 0$.

Lemma 34.24 (Exact quadratic normal form)**.** *There is an analytic invertible change of variable near c, defined over $\mathbb{Z}_{419}$, under which the quadratic branch of $f_\lambda(m) = 0$ becomes*

$$z^2 = -10 \cdot 419\,(\lambda - \Lambda). \tag{34.32}$$

For $\lambda \neq \Lambda$, put $h = v_{419}(\lambda - \Lambda)$. The quadratic is totally ramified if h is even; if h is odd, it splits over $\mathbb{Q}_{419}$ exactly when the leading unit of $-10(\lambda - \Lambda)/419^h$ is a square modulo 419*, and otherwise it is an unramified irreducible quadratic.*

Proof. For $m = c + y$, stationary Taylor expansion yields

$$f_\lambda(c + y) = 1440 \cdot 419(\lambda - \Lambda) + y^2 U(y), \quad U(0) \equiv 144 \pmod{419}.$$

For $y \in 419\mathbb{Z}_{419}$, the unit $U(y)/144$ is congruent to 1 modulo 419. Its unique square root congruent to 1 defines the analytic coordinate $z = y\sqrt{U(y)/144}$; moreover it is an isometry of the entire residue disk, not merely a local map with unit derivative. Indeed, for $s(y) = \sqrt{U(y)/144}$, the integral convergent power series on $419\mathbb{Z}_{419}$ satisfies $s(y) \equiv 1 \pmod{419}$ and $s(y) - s(y') \in (y - y')\mathbb{Z}_{419}$. Thus

$$z(y) - z(y') = (y - y') \left(s(y) + y' \frac{s(y) - s(y')}{y - y'} \right),$$

and the factor in parentheses is congruent to 1 modulo 419. Hensel lifting from $y = z = 0 \pmod{419}$ gives surjectivity and a unique inverse modulo every 419^n, hence a bi-analytic disk isometry. It gives (34.32) exactly on that disk, and therefore all integral-root and finite-congruence assertions. In the nonresonant case $v_{419}(\lambda - \Lambda) = 0$, however, the two algebraic zeros have $v_{419}(y) = 1/2$ and *lie outside* the integral residue disk: their irreducibility and total ramification follow separately from the translated polynomial's Newton segment joining $(0, 1)$ to $(2, 0)$, using $f'(c) = 0$ and $U(0)$ a unit. The same Newton-polygon argument handles all odd critical-value valuations; even critical-value valuations give the split/unramified cases by the local square-unit criterion. This separation of the integral disk from extension-valued roots is essential, and is consistent with the established p-adic Morse/inverse-function framework [175]. □

34.19 An exact integer discriminant certificate

The preceding description refers to the non-rational critical coordinate Λ. Our next result eliminates it from every integer-parameter decision.

$$\boxed{\begin{aligned} F(T) = {} & 127983969T^3 - 773766474332943627T^2 \\ & + 19464840642527447214601687T \\ & + 25531064848926331469188878661677 79. \end{aligned}} \tag{34.33}$$

Theorem 34.25 (Integer discriminant certificate and effective resonance bound)**.** *The polynomial $F(T) \in \mathbb{Z}[T]$ is irreducible over $\mathbb{Q}$, and its unique zero in $\mathbb{Z}_{419}$ is Λ. Consequently, $\Lambda \notin \mathbb{Q}$; in particular, $\lambda \neq \Lambda$ for every integer λ.*

For any $\lambda \in \mathbb{Z}$, calculate the nonzero integer *$F(\lambda)$ and set*

$$k = v_{419}(F(\lambda)), \qquad u = 419^{-k} F(\lambda) \pmod{419} \in \mathbb{F}_{419}^{\times}.$$

The full local factorization of $G_{419,\lambda}$ is given by the following finite certificate:

Criterion	*Additional quadratic factor*	$\mathbb{Q}_{419}$*-roots*
k *even*	*irreducible, totally ramified*	2
k *odd,* $\left(\frac{374u}{419}\right) = +1$	*splits into linear factors*	4
k *odd,* $\left(\frac{374u}{419}\right) = -1$	*irreducible, unramified*	2

In particular $G_{419,\lambda}$ *is separable for every integer* λ *and the exceptional repeated-root branch of Lemma 34.24 is* not attained by integer sources. *Moreover*

$$k \le \log_{419} |F(\lambda)| \le \log_{419} C + 3 \log_{419}(1 + |\lambda|), \quad C = \sum_{j=0}^{3} |[T^j]F|. \tag{34.34}$$

This is an explicit input-height bound on resonance depth and a terminating integral decision algorithm.

Proof. Independently calculate the discriminant of the quartic (34.29) with $p = 419$ as a polynomial in the parameter:

$$\begin{aligned} \operatorname{disc}_m G_{419,T} &= C_0 F(T), \\ C_0 &= 44676505705166237495036215296 \cdot 10^{18} \neq 0. \end{aligned} \tag{34.35}$$

Modulo 23, we have

$$F(T) \equiv 9(T^3 + 5T^2 + 2T - 6) \pmod{23}.$$

The displayed cubic has no root in $\mathbb{F}_{23}$; as its degree is three, it is irreducible. Thus F is irreducible in $\mathbb{Q}[T]$. Modulo 419,

$$F(T) \equiv 233(T - 132) \pmod{419},$$

and 233 is a unit. Hensel gives a unique zero $\Lambda_0 \in \mathbb{Z}_{419}$. But $G_{419,\Lambda}$ has a multiple root c by construction, so (34.35) implies $F(\Lambda) = 0$. Therefore $\Lambda_0 = \Lambda$.

The exact difference identity

$$F(T) - F(\Lambda) = (T - \Lambda)W(T, \Lambda), \qquad W(T, \Lambda) \equiv 233 \pmod{419}$$

for all $T \in \mathbb{Z}_{419}$ yields

$$v_{419}(F(\lambda)) = v_{419}(\lambda - \Lambda), \qquad \frac{\lambda - \Lambda}{419^k} \equiv 233^{-1} u \pmod{419}.$$

Substitute these equalities into Lemma 34.24. The quadratic has valuation $1 + k$ on the right of (34.32). For even k, it is totally ramified. For odd k, it has even valuation and splits precisely when

$$\left(\frac{-10 \cdot 233^{-1} u}{419}\right) = \left(\frac{374u}{419}\right) = +1;$$

otherwise it is irreducible unramified. Since F is irreducible cubic, it has no rational zero: its critical repeated-root equality is impossible for integer λ. Finally $419^k \mid F(\lambda) \neq 0$, and the triangle inequality gives $|F(\lambda)| \le C(1 + |\lambda|)^3$, proving (34.34). □

Remark 34.26. For example, $\lambda = 132$ gives $k = 1$ and an unramified quadratic, whereas $\lambda = 551 = 132 + 419$ gives $k = 1$ and a split quadratic. The deeper class $\lambda = 38261$ gives $k = 2$ and a totally ramified quadratic. The theorem needs no precomputed digits of Λ.

34.20 Prime-axis density: S_3 Chebotarev law

Theorem 34.27 (Uniform prime-density law)**.** *Fix any integer λ. Among primes $p \geq 7$, excluding the single prime 419, the natural densities of those for which $G_{p,\lambda}$ has exactly 1, 2, or 4 roots in $\mathbb{Q}_p$ are, respectively,*

$$\boxed{\delta_1 = \frac{1}{3}, \qquad \delta_2 = \frac{1}{2}, \qquad \delta_4 = \frac{1}{6}.} \tag{34.36}$$

The classification at each individual generic prime, not merely the densities, is independent of λ. In particular the average number of $\mathbb{Q}_p$-roots across primes tends to 2.

Proof. The reduction of H modulo 7 is $m^3 - 2m^2 - 2m + 2$ and has no root in $\mathbb{F}_7$, so the cubic H is irreducible over $\mathbb{Q}$. Its nonsquare discriminant $125700 = 100 \cdot 3 \cdot 419$ shows that its splitting-field Galois group is S_3. Frobenius cycle types in S_3 produce 0, 1, or 3 distinct roots of H modulo almost every prime with natural densities 2/6, 3/6, or 1/6, respectively, by the Chebotarev theorem [178]. Equation (34.30) adds exactly the simple root $m = 0$ and Theorem 34.23 lifts the roots. The weighted mean is $(1/3) + 2(1/2) + 4(1/6) = 2$. □

34.21 Parameter-axis density and precision law

Endow $\mathbb{Z}_{419}$ with normalized additive Haar measure μ; we may extend the quartic formally to all 419-adic parameters. The following distributions also hold as natural densities of integer parameters $0 \leq \lambda < X$ when $X \to \infty$.

Theorem 34.28 (Exact parameter density)**.** *For the quadratic pair at the exceptional prime 419,*

$$\boxed{\begin{aligned} \mu(\textit{totally ramified}) &= \frac{419}{420}, \\ \mu(\textit{unramified irreducible}) &= \frac{1}{840}, \\ \mu(\textit{split into two linear factors}) &= \frac{1}{840}. \end{aligned}} \tag{34.37}$$

The critical equality $\lambda = \Lambda$ has measure zero and never occurs for integer λ. The mean number of distinct $\mathbb{Q}_{419}$-roots under Haar-random parameter is 841/420.

Proof. Since translation preserves Haar measure, $\lambda - \Lambda$ is Haar-uniform. For every $j \geq 0$, $\mu(v_{419}(\lambda - \Lambda) = j) = 418/419^{j+1}$. The normal form (34.32) is totally ramified for j even, yielding the geometric sum

$$\sum_{a \geq 0} \frac{418}{419^{2a+1}} = \frac{419}{420}.$$

For j odd the valuation is even and the leading unit is uniformly random; precisely half are squares. Hence each of the split and unramified cases has measure

$$\frac{1}{2} \sum_{a \geq 0} \frac{418}{419^{2a+2}} = \frac{1}{840}.$$

At finite precision all decided branches are unions of congruence classes, while the single undecided congruence class has density tending to zero. This establishes the same natural densities for integer λ. The mean is $2 + 2(1/840) = 841/420$. □

Proposition 34.29 (Exact finite-precision residue counts)**.** *For each $N \geq 1$ there are exactly the following numbers of parameter classes modulo 419^N whose quadratic factorization is decided for* every *lift to $\mathbb{Z}_{419}$:*

$$S_N = U_N = 209 \sum_{j=0}^{\lfloor N/2 \rfloor - 1} 419^{N-2j-2},$$
$$R_N = 418 \sum_{j=0}^{\lceil N/2 \rceil - 1} 419^{N-2j-1}.$$

Here S_N, U_N, R_N denote split, unramified and totally ramified classes, respectively. Exactly one class, $\lambda \equiv \Lambda \pmod{419^N}$, remains undecided, and $S_N + U_N + R_N + 1 = 419^N$. For $N = 2$, the counts are $(209, 209, 175142, 1)$.

Proof. Among classes with exact $j = v_{419}(\lambda - \Lambda) < N$ there are $418 \cdot 419^{N-j-1}$ choices. If j is even all are ramified; if j is odd half split and half are unramified, by equidistribution of their leading units. Sum over the allowed parity classes. The only class with valuation $\geq N$ is $\lambda \equiv \Lambda \pmod{419^N}$. □

34.22 Prime-power congruence fibers and a resonance moment law

The number of distinct $\mathbb{Q}_{419}$-roots does not determine the number of roots modulo 419^n: singular residue classes can persist for several precision levels before disappearing. To capture this finer arithmetic, define

$$N_n(\lambda) = \#\{m \bmod 419^n : G_{419,\lambda}(m) \equiv 0 \pmod{419^n}\}, \qquad n \geq 1. \tag{34.38}$$

Write $P = 419$ for brevity, and retain $k = v_P(F(\lambda))$, $u = P^{-k}F(\lambda) \bmod P$ and $a = k + 1$ from Theorem 34.25.

Theorem 34.30 (Exact root counts at every prime-power precision)**.** *For every integer λ and $n \geq 1$, let $S_n(\lambda) = N_n(\lambda) - 2$. Then*

$$\boxed{S_n(\lambda) = \begin{cases} P^{\lfloor n/2 \rfloor}, & 1 \leq n \leq a, \\ 2P^{a/2}, & n > a,\ a \text{ even},\ \left(\frac{374u}{P}\right) = 1, \\ 0, & \text{otherwise.} \end{cases}} \tag{34.39}$$

In particular, $N_n(\lambda)$ is determined by finite integer arithmetic applied to $F(\lambda)$, for arbitrary n. Writing $N_0 = 1$, the unnormalized root-count series is rational and explicit:

$$\sum_{n \geq 0} N_n(\lambda) T^n = 1 + \frac{2T}{1-T} + \sum_{n=1}^{a} P^{\lfloor n/2 \rfloor} T^n + \mathbf{1}_{\text{split}} \frac{2P^{a/2} T^{a+1}}{1-T}, \tag{34.40}$$

where $\mathbf{1}_{\text{split}}$ is one exactly in the second case of (34.39)*.*

Proof. The reduction of $G_{419,\lambda}$ has two simple roots at $0, 20$ and one double root at -9. The first two contribute exactly two roots modulo each P^n by Hensel's lemma. In the disk $m \equiv c \equiv -9 \pmod{P}$, the coordinate $z = (m - c)\sqrt{U(m-c)/144}$ from Lemma 34.24 is an integral analytic isometry: its derivative is a unit congruent to one modulo P, and it induces a bijection on residue classes modulo P^n. Thus the singular contribution

S_n equals the number of $z \bmod P^n$ with $z \equiv 0 \pmod P$ and $z^2 \equiv B \pmod{P^n}$, where $B = -10P(\lambda - \Lambda)$ and $v_P(B) = a$.

If $n \le a$, this is $z^2 \equiv 0 \pmod{P^n}$ and has $P^{\lfloor n/2 \rfloor}$ solutions. If $n > a$, a solution requires $a = 2h$ even and $z = P^h w$. The reduced unit equation $w^2 \equiv B/P^{2h} \pmod{P^{n-2h}}$ has two Hensel roots exactly when its unit is a square modulo P; each gives P^h choices of z modulo P^n. The square criterion is precisely $\left(\frac{374u}{P}\right) = 1$ by Theorem 34.25. This proves the formula, and summing its eventually constant terms proves (34.40). □

Theorem 34.31 (Exact parameter moments and rare-resonance limit)**.** *Let λ be Haar-uniform in $\mathbb{Z}_P$. For every $n \ge 1$,*

$$\boxed{\mathbb{E}_\lambda N_n = 3, \qquad \mathrm{Var}_\lambda(N_n) = (P-1)(n-1) = 418(n-1).} \tag{34.41}$$

More precisely, with $S_n = N_n - 2$,

$$\Pr(S_n = P^{\lfloor n/2 \rfloor}) = P^{-(n-1)},$$
$$\Pr(S_n = 2P^h) = \frac{P-1}{2P^{2h}}, \quad 1 \le h \le \left\lfloor \frac{n-1}{2} \right\rfloor,$$

and the remaining probability is at $S_n = 0$. Moreover $N_n(\lambda)$ stabilizes almost surely as $n \to \infty$ to a finite N_∞, with

$$\Pr(N_\infty = 2 + 2P^h) = \frac{P-1}{2P^{2h}} \quad (h \ge 1), \qquad \Pr(N_\infty = 2) = \frac{839}{840}. \tag{34.42}$$

The limit has $\mathbb{E}N_\infty = 3$ but $\mathbb{E}N_\infty^2 = +\infty$, whereas the finite-$n$ variances in (34.41) *grow exactly linearly.*

Proof. Translation by the critical parameter Λ preserves Haar measure, so $\Pr(v_P(\lambda - \Lambda) \ge n-1) = P^{-(n-1)}$. The first atom follows from (34.39). The second occurs when $v_P(\lambda - \Lambda) = 2h - 1$ and the leading unit has the required quadratic character; its probability is $(P-1)/(2P^{2h})$. Since P is odd, the displayed positive values of S_n are distinct.

One may sum these atoms to obtain the first moment. Alternatively, for each $m \bmod P^n$ reducing to one of the three zeros $0, 20, -9$ modulo P, the congruence $G_{P,\lambda}(m) \equiv 0 \pmod{P^n}$ has Haar probability $P^{-(n-1)}$: the coefficient of λ is $1440P$ with $1440 \in \mathbb{Z}_P^\times$. Exactly $3P^{n-1}$ choices of m are eligible, whence $\mathbb{E}N_n = 3$. The second moment of the singular contribution is

$$\mathbb{E}S_n^2 = P^{2\lfloor n/2 \rfloor - (n-1)} + 2(P-1)\left\lfloor \frac{n-1}{2} \right\rfloor = 1 + (P-1)(n-1).$$

Since $\mathbb{E}S_n = 1$, the variance identity follows. Except on the singleton $\lambda = \Lambda$, the singular contribution eventually vanishes or stabilizes at $2P^h$, proving (34.42). Finally $\sum_{h \ge 1} 2P^h(P-1)/(2P^{2h}) = 1$, but $\sum_{h \ge 1}(2P^h)^2(P-1)/(2P^{2h}) = +\infty$. □

Remark 34.32 (Distinguishing local roots from congruence fibers). The average number of *distinct* $\mathbb{Q}_{419}$-roots is $841/420$ by Theorem 34.28, not 3. An actual 419-adic root with derivative divisible by 419^h can account for 419^h residue solutions at every sufficiently high precision. The general theory and rationality of univariate local zeta/Poincaré functions are prior work; the present formulas are a CCR-specialized calculation, not a claim of inventing the generic method [176].

Corollary 34.33 (The exceptional prime 5 over parameters)**.** *For Haar-random $\lambda \in \mathbb{Z}_5$ the unique $\mathbb{Q}_5$-root α satisfies, for all $d \geq 0$,*

$$\Pr(v_5(\alpha) = 2 + d) = \frac{4}{5^{d+1}}, \quad \mathbb{E}\, v_5(\alpha) = \frac{9}{4}.$$

The other cubic is totally ramified for all parameters.

Proof. Use Theorem 34.23 and the geometric distribution of $v_5(\lambda - 115)$; the singleton parameter 115 has measure zero. □

34.23 Universal odd-column fold and p-adic source resonance

This section derives an extension to *all odd normalized CCR column orders* $r \geq 5$. Throughout, $\Pi_{p,r}(m)$ denotes the stationary normalized column supplied by the established higher-order CCR transport, not the d-orthogonal characteristic polynomial Q_r. The source parameters below r are fixed integers and the top source λ_r varies. The series identity used below belongs to classical Nörlund theory [172]; its role in the CCR deformation and the ensuing p-adic counts are the application here.

34.23.1 The all-order Stirling generating function

Let $S_r(m) = \Pi_{0,r}(m)$ be the formal $p = 0$ specialization and $S_0 = 1$. The original normalized polynomial transport yields

$$T_r S_r(m) = m S_{r-1}(m), \qquad (T_r f)(m) = m f(m+1) - (m-r) f(m). \tag{34.43}$$

Lemma 34.34 (Nörlund identity and universal double root)**.** *As an identity in $\mathbb{Q}[m][[t]]$,*

$$\sum_{r \geq 0} S_r(m) t^r = \left(\frac{t}{1 - e^{-t}}\right)^m = e^{mt/2} \left(\frac{t}{2\sinh(t/2)}\right)^m. \tag{34.44}$$

For every $s \geq 1$, the polynomial S_{2s+1} has an exact double *root at zero:*

$$S_{2s+1}(m) = c_s m^2 + O(m^3), \qquad c_s = -\frac{B_{2s}}{4s(2s)!} \neq 0. \tag{34.45}$$

Here B_{2s} is the classical Bernoulli number, and $O(m^3)$ is a polynomial divisible by m^3.

Proof. Let $A(t) = t/(1 - e^{-t})$. The generating form of (34.43) is $m(F(m+1,t) - F(m,t)) + t\partial_t F = mtF$. The proposed $F = A(t)^m$ satisfies this because $A - 1 + tA'/A = t$. Uniqueness follows since $T_r(m^d)$ is triangular with diagonal coefficient $r + d \neq 0$. The logarithm identity

$$\log A(t) = \frac{t}{2} - \sum_{j \geq 1} \frac{B_{2j}}{2j(2j)!} t^{2j}$$

shows that the coefficient of mt^{2s+1} vanishes and the coefficient of $m^2 t^{2s+1}$ is half the coefficient of t^{2s} in $\log A(t)$, giving (34.45). Classical Bernoulli numbers B_{2s} are nonzero. □

34.23.2 An all-odd-order arithmetic fold theorem

By stationary polynomiality, for each fixed r there exists an integer $D_r \geq 1$ which clears all rational coefficients of S_r and $(\Pi_{p,r} - S_r)/p$ in their independent formal variables. Define a *good prime at order* $r = 2s + 1$ as an odd prime p not dividing rD_r and for which c_s in (34.45) is a p-adic unit. Only finitely many primes are excluded for each fixed r.

Theorem 34.35 (Universal odd-column CCR p-adic fold)**.** *Fix any odd $r = 2s + 1 \geq 5$, any good prime p, and integer lower source parameters $\lambda_4, \ldots, \lambda_{r-1}$. The dependence of the stationary normalized polynomial on its top source is* exactly

$$\Pi_{p,r}^{(\lambda_r)}(m) = \Pi_{p,r}^{(0)}(m) - \frac{p}{r}\lambda_r. \tag{34.46}$$

There is a unique critical center $c \in p\mathbb{Z}_p$, independent of λ_r, and a unique critical parameter $\Lambda_r \in \mathbb{Z}_p$ for which $\Pi_{p,r}^{(\Lambda_r)}(c) = 0$. An integral analytic isometric coordinate z on $m \equiv 0 \pmod{p}$ converts the local root equation exactly into

$$\boxed{z^2 = K_r p(\lambda_r - \Lambda_r), \qquad K_r = \frac{1}{rc_s} = -\frac{4s(2s)!}{rB_{2s}} \in \mathbb{Z}_p^{\times}.} \tag{34.47}$$

If $k = v_p(\lambda_r - \Lambda_r) < \infty$, the local degree-two factor is totally ramified and irreducible when k is even. When k is odd, it splits into two $\mathbb{Q}_p$-linear factors precisely when $K_r(\lambda_r - \Lambda_r)/p^k$ is a square unit modulo p, and otherwise is an unramified irreducible quadratic. At $\lambda_r = \Lambda_r$, the local factor has a repeated root. No nonattainment claim for integer parameters is made in this generality.

Proof. In the order-r transport, the source λ_r first appears as the constant $(-1)^r p\lambda_r = -p\lambda_r$. Since T_r maps constants to r times themselves, subtracting the two transport equations gives (34.46) independently of the lower parameters. The good-prime condition ensures $\Pi_{p,r} \equiv S_r \pmod{p}$ coefficientwise, and Lemma 34.34 gives a nondegenerate double root at zero. Thus Hensel's lemma applied to the derivative of $\Pi_{p,r}^{(0)}$ yields the unique $c \in p\mathbb{Z}_p$. Because $\Pi_{p,r}^{(0)}(c) \equiv 0 \pmod{p}$ and p/r is a p times a unit, there is exactly one $\Lambda_r \in \mathbb{Z}_p$ canceling its critical value. Taylor expansion at c is

$$\Pi_{p,r}^{(\lambda_r)}(c + y) = -\frac{p}{r}(\lambda_r - \Lambda_r) + y^2 U(y), \qquad U(0) \equiv c_s \pmod{p}.$$

For $y \in p\mathbb{Z}_p$, the unit $U(y)/c_s$ has a unique analytic square root congruent to one modulo p, and $z = y\sqrt{U(y)/c_s}$ induces an isometric bijection of the disk. This proves (34.47). Odd-valuation critical values have quadratic Newton slope with denominator two, giving an Eisenstein/total-ramification conclusion; at even valuation the local equation lies within the integral disk and the p-adic square-unit criterion distinguishes split from unramified quadratic cases. The nonsingular complementary factor is separated by Hensel factorization. □

34.23.3 Universal local prime-power fiber and moment laws

For $n \geq 1$, count *only* the zero residue disk, rather than assuming all other residues are simple:

$$M_{p,r,n}(\lambda_r) = \#\{m \bmod p^n : m \equiv 0 \pmod{p}, \quad \Pi_{p,r}^{(\lambda_r)}(m) \equiv 0 \pmod{p^n}\}.$$

Theorem 34.36 (All-odd-order rare-resonance statistics)**.** *Under Theorem 34.35, write* $a = 1 + v_p(\lambda_r - \Lambda_r)$ *for a noncritical parameter. If* a *is even let* ε *be the Legendre symbol of the unit* $K_r(\lambda_r - \Lambda_r)/p^{a-1}$*. Then*

$$M_{p,r,n}(\lambda_r) = \begin{cases} p^{\lfloor n/2 \rfloor}, & n \le a, \\ 2p^{a/2}, & n > a,\ a \text{ even},\ \varepsilon = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{34.48}$$

At the critical singleton, $M_{p,r,n} = p^{\lfloor n/2 \rfloor}$ *for every* n*. For Haar-uniform* $\lambda_r \in \mathbb{Z}_p$*, with all lower parameters fixed, the exact local moments are*

$$\boxed{\mathbb{E}M_{p,r,n} = 1, \qquad \operatorname{Var}(M_{p,r,n}) = (p-1)(n-1).} \tag{34.49}$$

The almost-sure limiting count M_∞ *has mass* $(p-1)/(2p^{2h})$ *at* $2p^h$ *for every* $h \ge 1$*, and remaining mass* $1 - 1/[2(p+1)]$ *at zero; it has mean one and infinite second moment.*

Proof. The local coordinate of Theorem 34.35 bijects every finite residue disk, so the count equals the solutions of $z^2 \equiv K_r p(\lambda_r - \Lambda_r) \pmod{p^n}$ with $z \equiv 0 \pmod p$. The same valuation and Hensel count as in Theorem 34.30 gives (34.48). Haar translation makes $\lambda_r - \Lambda_r$ uniform. The positive atoms have probabilities $p^{-(n-1)}$ at $p^{\lfloor n/2 \rfloor}$ and $(p-1)/(2p^{2h})$ at $2p^h$, for $1 \le h \le \lfloor (n-1)/2 \rfloor$. Their first moment is one; their second moment is $1 + (p-1)(n-1)$. The limit follows by summing the stable split atoms; its second moment diverges by constant contributions over infinitely many values of h. □

34.23.4 The entire fifth column away from $p = 73$

For $r = 5$, the sharp denominator $D_5 = 11520$ is an earlier fixed-order CCR result and $c_2 = 1/5760$. Thus *every prime* $p \ge 7$ *is good for the local fifth-column fold*, with $K_5 = 1152$. The inherited Stirling polynomial is

$$S_5(m) = \frac{m^2(m-1)(3m^2 - 7m - 2)}{11520}. \tag{34.50}$$

Moreover, writing $N_5 = 11520\Pi_{p,5}$, direct use of the original transport yields

$$\begin{aligned}
[m^5]N_5 &= 3(p+1)^5, \\
[m^4]N_5 &= 10(p+1)^4(4p-1), \\
[m^3]N_5 &= -5(p+1)^3(107p^2 + 74p - 1), \\
[m^2]N_5 &= 2(p+1)^2(280p^3 + 705p^2 + 106p + 1), \\
[m]N_5 &= 4p(p+1)(360\lambda_4 + 1183p^3 + 1157p^2 + 629p + 295), \\
[1]N_5 &= -96p(66\lambda_4 p + 36\lambda_4 + 24\lambda_5 + 85p^4 + 138p^3 + 76p^2 + 30p + 13).
\end{aligned}$$

Consequently

$$\Lambda_5(p, \lambda_4) \equiv -\frac{36\lambda_4 + 13}{24} \pmod p. \tag{34.51}$$

Corollary 34.37 (Global fifth-column congruence moments)**.** *Fix* $p \ge 7$*,* $p \ne 73$*, and integer* λ_4*. Let* $N_{p,n}(\lambda_5)$ *count every root of* $\Pi_{p,5}$ *modulo* p^n*. With* $\chi_p = \left(\frac{73}{p}\right) \in \{-1, 1\}$*, the complete count is*

$$N_{p,n}(\lambda_5) = \underbrace{(2 + \chi_p)}_{\text{simple residual roots}} + M_{p,5,n}(\lambda_5). \tag{34.52}$$

For Haar-uniform $\lambda_5 \in \mathbb{Z}_p$*,*

$$\boxed{\mathbb{E}N_{p,n} = 3 + \chi_p, \qquad \operatorname{Var}(N_{p,n}) = (p-1)(n-1).} \tag{34.53}$$

Proof. After excluding $p = 73$, the reduction (34.50) has exactly one double zero at 0 and all other zeros are simple. The simple zero at $m = 1$ always lifts. The quadratic $3m^2-7m-2$ has discriminant 73, hence has exactly $1+\chi_p \in \{0, 2\}$ simple $\mathbb{F}_p$-roots. These yield $2 + \chi_p$ lifts at all precisions. Add the singular contribution of Theorem 34.36. □

Remark 34.38 (Source and model scope). The exceptional prime 73 has a second repeated residue; its complete global classification is proved separately in Section 34.24. For $r \geq 7$ the all-odd-order theorem is intentionally local to the zero residue disk: other singular residues are not assumed away. The Nörlund identity [172], the standard analytic square coordinate and the moment-summation method are classical; the qualified research advance is this exact CCR-specific source deformation for every odd order.

34.24 The exceptional fifth-column prime 73: two coupled folds

The discriminant of $3m^2 - 7m - 2$ is 73. Thus the prime excluded from Corollary 34.37 is not merely a bad denominator: it creates a *second* nondegenerate critical residue. Throughout this section put $p = 73$, $L = \lambda_4$, $T = \lambda_5$, and denote the integral fifth-column numerator by $N_{L,T}(m) = 11520\Pi_{73,5}^{(L,T)}(m)$. The coefficient formula above shows that the dependence on T is exactly $-2304 \cdot 73T$ and

$$N_{L,T}(m) \equiv 9m^2(m-1)(m-62)^2 \pmod{73}. \tag{34.54}$$

The residue $m = 1$ is simple; the two repeated residues 0 and 62 have nonzero quadratic Taylor coefficients 2 and 24, respectively, modulo 73.

Theorem 34.39 (Two-fold normal form and critical-value collision)**.** *For every $L \in \mathbb{Z}_{73}$ there are unique critical centers $c_0(L) \equiv 0$ and $c_{62}(L) \equiv 62 \pmod{73}$, and two critical source values $\Lambda_0(L), \Lambda_{62}(L) \in \mathbb{Z}_{73}$. The two residue disks admit integral analytic isometric coordinates in which their equations are, respectively,*

$$z_0^2 = 1152 \cdot 73\,[T - \Lambda_0(L)], \qquad z_{62}^2 = 96 \cdot 73\,[T - \Lambda_{62}(L)]. \tag{34.55}$$

Their critical parameters have first digits

$$\Lambda_0(L) \equiv 39 + 35L, \qquad \Lambda_{62}(L) \equiv 37 + 19L \pmod{73}. \tag{34.56}$$

In particular, their reductions coincide precisely when $L \equiv 9 \pmod{73}$. There is exactly one $L_* \in \mathbb{Z}_{73}$ for which $\Lambda_0(L_*) = \Lambda_{62}(L_*)$; its first four base-73 digits are

$$L_* \equiv 9 + 11 \cdot 73 + 30 \cdot 73^2 + 35 \cdot 73^3 \pmod{73^4}. \tag{34.57}$$

At the pair $\big(L_*, \Lambda_0(L_*)\big)$, the quintic has two distinct double roots and one simple root in $\mathbb{Q}_{73}$.

Proof. Apply Hensel's lemma to $\partial_m N_{L,T}$ in the two disks: its reduction vanishes simply at both residue centers by the nonzero second derivatives. The derivative does not depend on T, hence has unique 73-adic centers $c_i(L)$ independent of T. Put $\Lambda_i(L) = N_{L,0}(c_i)/(2304 \cdot 73)$; integrality follows from (34.54). In each disk the exact Taylor expansion is

$$N_{L,T}(c_i + y) = -2304 \cdot 73(T - \Lambda_i) + y^2 U_i(y),$$

where $U_0(y) \equiv 2$ and $U_{62}(y) \equiv 24 \pmod{73}$. Therefore $z_i = y\sqrt{U_i(y)/u_i}$, $u_0 = 2$, $u_{62} = 24$, is integral analytic and induces an isometric bijection on every congruence

quotient of its residue disk. To see the global assertion, write $s_i(y) = \sqrt{U_i(y)/u_i}$. On $73\mathbb{Z}_{73}$, $s_i(y) \equiv 1 \pmod{73}$ and $s_i(y) - s_i(y') \in (y-y')\mathbb{Z}_{73}$. Consequently $z_i(y) - z_i(y') = (y-y')\big(s_i(y) + y'\,(s_i(y) - s_i(y'))/(y-y')\big)$, whose second factor is a 73-adic unit. The inverse is obtained by successive unique Hensel lifts. The equations become (34.55) because $2304/2 = 1152$ and $2304/24 = 96$.

Since $c_i \equiv i \pmod{73}$ and $N'_{L,0}(c_i) = 0$, the first digit of Λ_i is obtained by evaluating $N_{L,0}(i)/(2304 \cdot 73)$ modulo 73. The displayed exact coefficient formula yields (34.56). Differentiating through the critical center,

$$\Lambda_i'(L) = \frac{\partial_L N_{L,0}(c_i(L))}{2304 \cdot 73},$$

so $\Lambda_0' - \Lambda_{62}' \equiv 35 - 19 = 16 \pmod{73}$ is a unit for all L. Analytic Hensel lifting of $\Lambda_0(L) - \Lambda_{62}(L)$ gives a unique zero in the residue class $L \equiv 9 \pmod{73}$. Successive exact derivative-center lifts give (34.57). The residue disks remain distinct, so at the simultaneous critical parameter there are two double roots, plus the simple lift of 1. □

For $i \in \{0, 62\}$ and $n \geq 1$, let $M_{i,n}(T)$ count roots modulo 73^n in residue class i of $N_{L,T}$. These are the separate quadratic-fiber counts from Theorem 34.36, with constants $K_0 = 1152$, $K_{62} = 96$ and critical sources Λ_i. The ratio $K_0/K_{62} = 12$ is a square in $\mathbb{Z}_{73}^{\times}$: indeed $31^2 \equiv 12 \pmod{73}$.

Theorem 34.40 (Exact 73-adic global count and covariance law)**.** *For fixed $L \in \mathbb{Z}_{73}$ and any $T \in \mathbb{Z}_{73}$, the entire quintic root count at every precision $n \geq 1$ is*

$$N_n(L,T) = 1 + M_{0,n}(T) + M_{62,n}(T). \tag{34.58}$$

For Haar-uniform T and fixed L, $\mathbb{E}_T N_n = 3$ for every $n \geq 1$. Put $d = v_{73}(\Lambda_{62}(L) - \Lambda_0(L)) \in \{0, 1, 2, \ldots, \infty\}$, with $v_{73}(0) = \infty$. At $n = 1$, $N_1 = 3$ deterministically. For $n \geq 2$, the exact mixed moment of the two singular contributions is

$$\mathbb{E}_T[M_{0,n}M_{62,n}] = \begin{cases} 0, & d = 0, \\ 1 + 72(n-1), & d \geq 1, \quad n \leq d+1, \\ 72d, & 1 \leq d < \infty, \quad n \geq d+2. \end{cases} \tag{34.59}$$

Consequently, with $\mathrm{Var}(N_1) = 0$,

$$\boxed{\mathrm{Var}_T(N_n) = \begin{cases} 144(n-1) - 2, & d = 0, \\ 288(n-1), & d \geq 1, \quad n \leq d+1, \\ 144(n-1) + 144d - 2, & 1 \leq d < \infty, \quad n \geq d+2. \end{cases}} \tag{34.60}$$

For $d = \infty$ the second line applies at all $n \geq 2$, reflecting perfectly correlated fold counts.

Proof. The factorization (34.54) and Hensel's lemma give one root in the simple disk at every 73^n, establishing (34.58). Each quadratic disk has mean one and variance $72(n-1)$ by Theorem 34.36.

For $n \geq 2$ write the local coordinate $z_i = 73a_i$ with $a_i \pmod{73^{n-1}}$. Equation (34.55) becomes

$$T \equiv \Lambda_i + \frac{73}{K_i} a_i^2 \pmod{73^{n-1}}.$$

Let $\delta = \Lambda_{62} - \Lambda_0$. Expanding the product of the finite solution counts and averaging over uniform $T \pmod{73^{n-1}}$ gives

$$\mathbb{E}[M_{0,n}M_{62,n}] = 73^{-(n-1)} \#\big\{(a,b) \pmod{73^{n-1}} : 73\Big(\frac{a^2}{1152} - \frac{b^2}{96}\Big) \equiv \delta \pmod{73^{n-1}}\big\}.$$

If $d = 0$, this set is empty. For $d \geq 1$, put $t = n - 2$ and divide by 73. Each variable has one additional free base-73 digit, so the last display equals

$$73^{1-t}\#\{(a,b) \pmod{73^t} : a^2 - 12b^2 \equiv 1152\delta/73 \pmod{73^t}\}.$$

Choose $s \in \mathbb{Z}_{73}^{\times}$ with $s^2 = 12$. The invertible linear transformation $(a,b) \mapsto (u,v) = (a - sb, a + sb)$ reduces the count to $uv \equiv 1152\delta/73 \pmod{73^t}$. If $0 \leq d - 1 < t$, each valuation $v_{73}(u) = 0, \ldots, d-1$ supplies $72 \cdot 73^{t-1}$ pairs, giving $72d$ after normalization. If $d - 1 \geq t$ (including $d = \infty$), the zero-product congruence has $73^t + t \cdot 72 \cdot 73^{t-1}$ pairs, giving $1 + 72(n-1)$. The case $t = 0$ is the single trivial residue class, with the same count by direct interpretation. This proves (34.59). Since each local mean is one, the variance of the sum of the two folds is $2 \cdot 72(n-1) + 2(\mathbb{E}[M_{0,n}M_{62,n}] - 1)$, proving (34.60). □

Corollary 34.41 (Attainability of complete local splitting)**.** *For every fixed ordinary integer L, there exists an ordinary integer T for which $N_{L,T}$ has five distinct $\mathbb{Q}_{73}$ roots if and only if*

$$L \equiv 9 \pmod{73}. \tag{34.61}$$

In particular, $L = 9$, $T = 135$ supplies a witness with

$$(N_1, N_2, N_3, N_4) = (3, 147, 293, 293),$$

and both quadratic factors split over $\mathbb{Q}_{73}$.

Proof. If $L \not\equiv 9 \pmod{73}$, the critical parameter residues are distinct. Splitting of a fold requires T to lie in its critical residue class modulo 73, so both cannot split. If $L \equiv 9$, put $d = v_{73}(\Lambda_{62} - \Lambda_0) \geq 1$. For $d \geq 2$, take T with $v_{73}(T - \Lambda_0) = 1$ and with the square leading unit specified by K_0; then the second fold has the same valuation and compatible square character, because $K_0/K_{62} = 12$ is a 73-adic square. Such T is available as an ordinary integer by choosing its residue modulo 73^2. For $d = 1$, write $T = \Lambda_0 + 73u$ and $\delta = 73\Delta$ with Δ a unit. Let χ be the quadratic character modulo 73 extended by $\chi(0) = 0$. Both split if $\chi(K_0 u) = \chi(K_{62}(u - \Delta)) = 1$. Since $\chi(K_0) = \chi(K_{62})$ and $\chi(-1) = 1$, a standard quadratic-character correlation gives exactly

$$\frac{73 - 3 - 2\chi(K_0\Delta)}{4} \in \{17, 18\}$$

admissible residues $u \pmod{73}$, hence an integer T exists. Direct evaluation at $L = 9$ yields $\Lambda_0 \equiv 5026$, $\Lambda_{62} \equiv 1887 \pmod{73^2}$; $T = 135$ is one of the admissible residues. The displayed modular counts are independently verified by direct integer-polynomial Hensel lifting, and the split local normal forms prove the assertion about five $\mathbb{Q}_{73}$-roots. □

Remark 34.42 (Scope of the collision phenomenon). The parameter L_* is a 73-adic analytic critical-value coincidence, not a claim that L_* is an ordinary integer. The local analytic folds and character-sum/product-congruence counts are standard methods; the explicit two-fold coupling and the exact finite-precision covariance stratification belong to the concrete CCR fifth-column family. They do not establish global all-order physical DPDA realization or a new generic p-adic Morse theorem.

34.25 Joint source equidistribution and the exact five-root density at 73

The preceding section fixed the lower stationary source $L = \lambda_4$ and varied $T = \lambda_5$. We now vary both L, T independently according to normalized Haar measure on $\mathbb{Z}_{73}^2$. This change reveals a sharp distinction between *conditional* dependence and *unconditional* independence of the two singular residue disks.

Theorem 34.43 (Two-source critical-value isometry)**.** *Retain the two critical parameter functions $A(L) = \Lambda_0(L)$ and $B(L) = \Lambda_{62}(L)$ of Section 34.24. The difference $D(L) = B(L) - A(L)$ is an isometric bijection $\mathbb{Z}_{73} \to \mathbb{Z}_{73}$. In fact, for every $L \neq L'$ in $\mathbb{Z}_{73}$,*

$$\frac{D(L) - D(L')}{L - L'} \equiv 19 - 35 = -16 \pmod{73}. \tag{34.62}$$

Consequently the map

$$\Phi : (L, T) \longmapsto \big(T - A(L),\ T - B(L)\big)$$

is a measure-preserving bijection of $\mathbb{Z}_{73}^2$. Thus its two output coordinates are independent Haar-uniform 73-adic integers when (L, T) is Haar-uniform.

Proof. The established quintic numerator $N_{L,T}(m)$ is affine in L and T, with L- and T-coefficients divisible by 73; moreover the T-coefficient is the constant $-2304 \cdot 73$. Let $c_i(L)$ be the stationary derivative-zero center for $i = 0, 62$. Since $\partial_m^2 N_{L,0}(c_i(L))$ is a unit modulo 73, implicit Hensel lifting makes c_i an integral analytic function of L. As $\partial_L \partial_m N_{L,0}$ is divisible by 73,

$$c_i(L) - c_i(L') \in 73(L - L')\mathbb{Z}_{73}.$$

By definition $A(L) = N_{L,0}(c_0(L))/(2304 \cdot 73)$ and likewise $B(L) = N_{L,0}(c_{62}(L))/(2304 \cdot 73)$. The vanishing of the m-derivative at each center, combined with the preceding Lipschitz estimate, shows that the critical-value difference quotients reduce modulo 73 to the explicit L-coefficient at the corresponding residue center. The exact coefficient formula in the previous section gives

$$\frac{A(L) - A(L')}{L - L'} \equiv 35, \qquad \frac{B(L) - B(L')}{L - L'} \equiv 19 \pmod{73}.$$

This proves (34.62). It also proves that D is an isometry. Since $D(L) \equiv -2 - 16L \pmod{73}$ is a permutation, for each target $t \in \mathbb{Z}_{73}$ and every solution L_j of $D(L_j) \equiv t \pmod{73^j}$ there is exactly one next digit $h \pmod{73}$ such that $D(L_j + h73^j) \equiv t \pmod{73^{j+1}}$: the difference quotient above makes the digit coefficient -16, a unit. Induction proves that D is bijective modulo every 73^j and preserves Haar measure.

Now put $(U, V) = \Phi(L, T)$. The inverse is explicit:

$$L = D^{-1}(U - V), \qquad T = U + A(L).$$

Both the forward and inverse maps preserve congruence classes modulo every 73^j and give bijections of their finite residue grids; therefore Φ preserves product Haar measure. In particular U, V are independent Haar variables. □

Theorem 34.44 (Complete joint law for the fifth column)**.** *For Haar-independent $L, T \in \mathbb{Z}_{73}$ and every $n \geq 1$, the two singular-disk solution counts $M_{0,n}, M_{62,n}$ are independent and identically distributed. The complete fifth-column congruence count $N_n = 1 + M_{0,n} + M_{62,n}$ obeys*

$$\mathbb{E}_{L,T} N_n = 3, \qquad \mathrm{Var}_{L,T}(N_n) = 144(n - 1). \tag{34.63}$$

More precisely, write $h = \lfloor (n-1)/2 \rfloor$, $q_n = 73^{-(n-1)}$, $a_n = 73^{\lfloor n/2 \rfloor}$, and $w_j = 36/73^{2j}$. The exact probability generating polynomial is

$$P_n(X) = \left(1 - q_n - \sum_{j=1}^{h} w_j\right) + q_n X^{a_n} + \sum_{j=1}^{h} w_j X^{2 \cdot 73^j}, \tag{34.64}$$

$$\mathbb{E}_{L,T}[X^{N_n}] = X P_n(X)^2.$$

Let R be the number of distinct roots in $\mathbb{Q}_{73}$ of the fifth-column quintic. Then

$$\begin{aligned} \Pr(R=1) &= \frac{21609}{21904}, \qquad \Pr(R=3) = \frac{294}{21904}, \\ \Pr(R=5) &= \boxed{\frac{1}{21904}}, \qquad\qquad \mathbb{E}R = \frac{38}{37}. \end{aligned} \tag{34.65}$$

These probabilities are also the natural densities of ordinary integer source pairs (L,T) in expanding integer squares. The almost-surely stabilized congruence count has expectation 3 but infinite variance.

Proof. The local formulas of Section 34.23 depend on (L,T) only through the appropriate difference $U = T - A(L)$ or $V = T - B(L)$. Theorem 34.43 makes these independent Haar inputs. Each local count has mean 1, variance $72(n-1)$ and probability polynomial $P_n(X)$, as proved previously; their sum gives (34.63) and (34.64). Each quadratic splits over $\mathbb{Q}_{73}$ with probability $s = 1/(2\cdot 74) = 1/148$, independently. The simple residue root $m \equiv 1$ always survives; therefore $R = 1 + 2I_0 + 2I_{62}$, where I_0, I_{62} are independent Bernoulli variables with parameter s, outside a Haar-null repeated-root locus. This yields (34.65). The sets specifying splitting are unions of 73-adic cylinders up to null critical singletons. Approximating by finite cylinders proves equality of Haar probabilities with the natural densities among integer pairs. The limit-moment statements follow from the independent heavy-tailed local limits already proved for one fold. □

Proposition 34.45 (Finite two-digit certificate)**.** *Modulo 73^2, the critical gap map $L \mapsto D(L)$ permutes all 5329 residue classes. Among the $73^4 = 28\,398\,241$ source pairs (L,T) modulo 73^2, exactly*

$$\left(\frac{73-1}{2}\right)^2 = \boxed{1296}$$

classes certify that both folds split at their first possible resonance depth $v_{73}(T-\Lambda_i) = 1$. For the 73 values of $L \pmod{73^2}$ with $L \equiv 9 \pmod{73}$, precisely 36 admit 18 such T, 36 admit 17, and one admits 36. Moreover the full N_2 distribution on $(L,T) \pmod{73}$ is

$$\#\{N_2 = 1\} = 5184, \qquad \#\{N_2 = 74\} = 144, \qquad \#\{N_2 = 147\} = 1.$$

Proof. By Theorem 34.43, (U,V) ranges bijectively over the pair of residue classes modulo 73^2. A first-depth split in each coordinate requires a nonzero square leading unit, for which there are $(73-1)/2 = 36$ possibilities. This gives 1296. The finer $36, 36, 1$ census follows from the two explicit critical functions and a quadratic character sum (or direct enumeration of all 73 coincident lower-source residues). At precision $n = 2$, each disk contributes 73 solutions iff its critical source residue matches $T \pmod{73}$, otherwise none. The bijection $\Phi \pmod{73}$ therefore gives the stated binomial census. □

Remark 34.46 (Conditional and unconditional source correlations). The independent global law does not negate the sharply nonzero *conditional* covariance of Theorem 34.40: fixing L creates dependence, which cancels exactly on averaging through the measure-preserving lower-source coordinate. The isometry/Haar principle is standard in p-adic dynamics; the specific CCR two-critical-value map, explicit simultaneous-splitting density, and joint source distribution are the concrete arithmetic statements established here. The conclusions do not include a generic isometry theorem or a theorem on physical source realizability.

34.26 Fixed integer sources: finite resonance support across primes

The preceding root-density results distinguish two different sampling schemes: variation of a stationary source in $\mathbb{Z}_p$ with p fixed, versus variation of the prime while the sources are fixed *ordinary integers*. The latter exhibits a rigid arithmetic obstruction not apparent from the Haar probability of splitting at any one prime.

Theorem 34.47 (Fixed-source prime support and exact two-digit stability)**.** *Fix $L = \lambda_4, T = \lambda_5 \in \mathbb{Z}$ and put*

$$A(L,T) = 24T + 36L + 13 = 12(2T + 3L + 1) + 1 \neq 0.$$

For every rational prime $p \geq 7$ satisfying $p \neq 73$ and $p \nmid A(L,T)$, the integer fifth-column numerator $N_{p,L,T}(m) = 11520\Pi_{p,5}^{(L,T)}(m)$ has exactly

$$b_p = 2 + \left(\frac{73}{p}\right) \in \{1, 3\}$$

distinct roots in $\mathbb{Q}_p$. More precisely, letting $C_{p,n}$ count solutions of $N_{p,L,T}(m) \equiv 0 \pmod{p^n}$,

$$C_{p,1} = 3 + \left(\frac{73}{p}\right), \qquad C_{p,n} = 2 + \left(\frac{73}{p}\right) \quad (n \geq 2).$$

All surviving roots arise from simple residues, while the double residue $m \equiv 0 \pmod{p}$ yields a totally ramified irreducible quadratic over $\mathbb{Q}_p$ and disappears already modulo p^2.

Proof. The exact coefficient formula of the previous section gives

$$N_{p,L,T}(m) \equiv m^2(m-1)(3m^2 - 7m - 2) \pmod{p}.$$

For $p \geq 7$, $p \neq 73$, the nonzero roots of the right side are simple: $m = 1$, plus two roots of its quadratic factor exactly when $\left(\frac{73}{p}\right) = 1$. Hensel's lemma lifts each such root uniquely through all p^n and to $\mathbb{Q}_p$.

The exact constant and linear coefficients satisfy

$$N_{p,L,T}(0) \equiv -96p(24T + 36L + 13) \pmod{p^2}, \qquad [m]N_{p,L,T} \equiv 0 \pmod{p}.$$

Consequently, for every $z \in \mathbb{Z}_p$,

$$N_{p,L,T}(pz) \equiv -96pA(L,T) \pmod{p^2}.$$

The right side is nonzero modulo p^2 when $p \nmid A$, as $-96 \in \mathbb{Z}_p^\times$. Thus *no* zero-disk solution survives even to p^2. Its local quadratic factor has Newton slope $-1/2$ (nonzero constant valuation one, unit quadratic coefficient), hence is irreducible and totally ramified. This proves the exact congruence and field-root counts. Finally $A(L,T) \equiv 1 \pmod{12}$, so no integral source pair makes $A = 0$. □

Corollary 34.48 (Source-independent prime-axis densities)**.** *For every fixed integer pair (L,T) the fifth-column quintic has exactly one $\mathbb{Q}_p$-root at a natural density $1/2$ of rational primes, and exactly three at density $1/2$; primes exhibiting five roots are contained in the* finite *exceptional set*

$$\{73\} \cup \{p \geq 7 : \ p \mid (24T + 36L + 13)\}$$

(aside from the excluded small primes). The average local-root count over primes tends to 2. Moreover, for any $B \geq 0$, all integer sources $|L|, |T| \leq B$ satisfy Theorem 34.47 simultaneously for every $p > \max\{73, 60B + 13\}$.

Proof. Since $73 \equiv 1 \pmod 4$, quadratic reciprocity gives $\left(\frac{73}{p}\right) = \left(\frac{p}{73}\right)$. Exactly 36 of the 72 nonzero residues modulo 73 are squares; Dirichlet's theorem on primes in arithmetic progressions assigns density $1/72$ to each residue class, so the two root counts each have density $1/2$. Only finitely many primes divide the nonzero integer A. Finally $|A| \le 24|T| + 36|L| + 13 \le 60B + 13$, which yields the uniform threshold. □

Remark 34.49 (Comparison with source-Haar statistics). The density-1/2 prime dichotomy concerns *fixed integer* sources; it is not the source-Haar distribution of Theorem 34.36, and it does not override the exceptional prime 73 two-fold model. Standard Dirichlet and Hensel methods are used; the new CCR-specific content is the exact integer resonance obstruction $A(L, T)$ and its two-digit criterion.

34.27 All odd orders: a Stirling source obstruction and finite prime support

The fifth-column obstruction of the previous section is not isolated. In this section we derive a closed source-obstruction formula *for every odd column*, using the pre-existing normalized transport and the classical Nörlund generating function. The odd-index double-zero property itself has appeared in the Stirling-polynomial literature [167]; the CCR source obstruction and its prime-support application are the subject here.

Write $\left\{ {n \atop k} \right\}$ for a Stirling number of the second kind. Throughout, $r \ge 3$, $\lambda_3 = 1$, and all λ_a are independent stationary source parameters unless explicitly specialized to integers. Denote by $S_r(m) = \Pi_{0,r}(m)$ the Stirling specialization and set $E_r(p, m, \boldsymbol{\lambda}) = (\Pi_{p,r}(m) - S_r(m))/p$ in the rational coefficient ring supplied by the original polynomial transport.

Theorem 34.50 (Exact first-order source obstruction)**.** *For every $r \ge 3$ one has*

$$A_r(\boldsymbol{\lambda}) := \sum_{a=3}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a - 1, \qquad E_r(0, 0, \boldsymbol{\lambda}) = \frac{(-1)^{r-2}}{r!} A_r(\boldsymbol{\lambda}). \tag{34.66}$$

For integer stationary sources, A_r is always odd. For every odd $r \ge 5$, the stronger arithmetic rigidity holds:

$$A_r = 2^{r-1} - 3 + \sum_{a=4}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a \equiv 1 \pmod 6. \tag{34.67}$$

In particular $A_r(\boldsymbol{\lambda}) \neq 0$ for every *integer vector $\boldsymbol{\lambda}$.*

Proof. The original transport with $f_j = \Pi_{p,j}$ is

$$T_j f_j(m) = \ell_m f_{j-1}(m) + p\ell_{m-1} f_{j-2}(m-1) + \sum_{a=3}^{j} (-1)^a p \lambda_a f_{j-a}(m-a+1),$$

where $\ell_m = m + p(m-1)$ and $T_j f(m) = mf(m+1) - (m-j)f(m)$. Differentiate formally at $p = 0$, put $m = 0$, use $S_j(0) = 0$ for $j \ge 1$, and note $T_j f(0) = jf(0)$. This gives

$$jE_j(0, 0) = -S_{j-2}(-1) + \sum_{a=3}^{j} (-1)^a \lambda_a S_{j-a}(1-a). \tag{34.68}$$

The established Nörlund identity $\sum_{q \ge 0} S_q(m) t^q = (t/(1-e^{-t}))^m$ yields

$$S_{j-2}(-1) = \frac{(-1)^{j-2}}{(j-1)!}, \qquad S_{j-a}(1-a) = \frac{(-1)^{j+a-2}(a-1)!}{(j-1)!} \left\{ {j-1 \atop a-1} \right\}.$$

The second formula follows by expanding $(1-e^{-t})^{a-1}/t^{a-1}$ and using the standard finite-difference identity defining Stirling numbers. Substituting these coefficients into (34.68) proves (34.66). All $(a-1)!$ are even for $a \geq 3$, so the sum minus one is odd. Since $2!\left\{ {r-1 \atop 2} \right\} = 2(2^{r-2}-1)$ and $(a-1)!$ is divisible by 6 for $a \geq 4$, equation (34.67) follows when $r-1$ is even and at least four. □

Theorem 34.51 (Exact CCR source-obstruction lattice and coherent integral witness)**.** *For every $r \geq 4$ put $n = r-1$ and define the positive integer*

$$d_r = \gcd_{3 \leq j \leq n} \left(j! \left\{ {n \atop j} \right\} \right).$$

With $\lambda_3 = 1$ and all subsequent stationary sources ranging freely over $\mathbb{Z}$, the exact image *of the first-order source obstruction is*

$$\left\{ A_r(1, \lambda_4, \ldots, \lambda_r) : \lambda_a \in \mathbb{Z} \right\} = (-1)^{r-1} + d_r\mathbb{Z}. \tag{34.69}$$

There is a single coherent choice of integer sources for every order,

$$\lambda_a^* = (-1)^{a-1} \quad (a \geq 3), \qquad A_r(\boldsymbol{\lambda}^*) = (-1)^{r-1} \quad (r \geq 3). \tag{34.70}$$

For every odd $r \geq 5$, the image in (34.69) *is $1 + d_r\mathbb{Z}$, and $12 \mid d_r$. In the first three odd cases,*

$$d_5 = 12, \qquad d_7 = 60, \qquad d_9 = 252.$$

Thus (34.67) *may be strengthened to the exact lattice image, not just a necessary congruence.*

Proof. The established Stirling finite-difference identity [171, Eq. 26.8.36] is $\sum_{j=0}^{n} (-1)^j j! \left\{ {n \atop j} \right\} = (-1)^n$. At $n \geq 2$ the $j = 0$ term vanishes and the $j = 1$ term equals -1. Substituting $\lambda_{j+1}^* = (-1)^j$ into (34.66) therefore yields $A_r(\boldsymbol{\lambda}^*) = (-1)^n$ exactly. Subtracting the canonical-source equality from the general source gives

$$A_r(\boldsymbol{\lambda}) - (-1)^n = \sum_{j=3}^{n} j! \left\{ {n \atop j} \right\} \left(\lambda_{j+1} - (-1)^j \right).$$

Every right-hand term is divisible by d_r. Conversely, Bézout's identity shows that arbitrary integer values of the freely chosen source differences generate *every* element of $d_r\mathbb{Z}$. This proves the exact image rather than only a containment. For odd $r \geq 5$, $n \geq 4$ is even. The terms $j!$ for $j \geq 4$ are multiples of 24. The remaining generator satisfies

$$3! \left\{ {n \atop 3} \right\} = 3^n - 3 \cdot 2^n + 3 \equiv 0 \pmod{12},$$

since $3^n \equiv 9 \pmod{12}$ and $3 \cdot 2^n \equiv 0 \pmod{12}$. Hence $12 \mid d_r$. Exact Stirling arithmetic gives the three displayed small-order gcd values. □

Proposition 34.52 (Closed form of the odd-order source-lattice index)**.** *Let $r \geq 5$ be odd, and let d_r be the index in Theorem 34.51. Then*

$$\boxed{d_r = 4 \prod_{\substack{\ell \geq 3 \text{ prime} \\ \ell - 1 \mid r-3}} \ell^{1 + \mathbf{1}_{\ell \mid r-3}}.} \tag{34.71}$$

Equivalently, for every odd prime ℓ,

$$\ell \mid d_r \iff \ell - 1 \mid r-3, \qquad \ell^2 \mid d_r \iff \ell(\ell-1) \mid r-3, \qquad \ell^3 \nmid d_r,$$

and $v_2(d_r) = 2$. In particular, the exact source image $1 + d_r\mathbb{Z}$ is determined solely by the elementary prime factorizations of the divisors of $r-3$; its first values are $12, 60, 252, 60, 132, 16380$ at $r = 5, 7, 9, 11, 13, 15$.

Proof. Write $n = r-1 \geq 4$ (even), set $a = 2^{n-1}-1$, and introduce the integer polynomial

$$P_n(X) = X^n - X - aX(X-1).$$

By $X^n = \sum_{j=0}^{n} j! \left\{ {n \atop j} \right\} \binom{X}{j}$ and $2! \left\{ {n \atop 2} \right\} = 2^n - 2 = 2a$, its binomial-basis expansion is

$$P_n(X) = \sum_{j=3}^{n} j! \left\{ {n \atop j} \right\} \binom{X}{j}.$$

Integral forward differences and binomial inversion therefore identify d_r with the fixed divisor $\gcd\{P_n(x) : x \in \mathbb{Z}\}$: taking differences recovers every binomial coefficient, and conversely every integer polynomial value is an integral combination of them.

Let $\ell \geq 5$ be prime. The condition $\ell \mid d_r$ means that P_n vanishes as a function on $\mathbb{F}_\ell$. At nonzero x this requires

$$x^{n-1} = 1 + a(x-1) \quad (x \in \mathbb{F}_\ell^\times).$$

Reducing X^{n-1} modulo $X^{\ell-1}-1$, the left side is a single monomial of degree $0, \ldots, \ell-2$, while the right side is a linear polynomial. Since evaluation on $\mathbb{F}_\ell^\times$ is injective for polynomials of degree at most $\ell-2$, the exponent must reduce to 0 or 1 modulo $\ell-1$. The case 0 is impossible because $n-1$ is odd and $\ell-1$ is even. Thus $\ell - 1 \mid n-2$. Conversely that divisibility gives $x^{n-1} = x$ for every nonzero $x \in \mathbb{F}_\ell$ and $a \equiv 1 \pmod{\ell}$, so P_n vanishes. For $\ell = 3$ the same criterion holds directly: $2 \mid n-2$ and $a \equiv 1 \pmod 3$.

Suppose an odd ℓ divides d_r. Evaluating the polynomial at $1+\ell$ yields

$$P_n(1+\ell) \equiv \ell\,(n-2) \pmod{\ell^2},$$

so $\ell^2 \mid d_r$ forces $\ell \mid n-2$. Conversely, if both $\ell-1$ and ℓ divide $n-2$, Euler's theorem modulo ℓ^2 implies $x^{n-2} \equiv 1$ for every ℓ-adic unit, and also $a \equiv 1 \pmod{\ell^2}$. Consequently

$$P_n(x) = x^2(x^{n-2}-1) - (a-1)x(x-1) \equiv 0 \pmod{\ell^2}$$

for units, and separately for multiples of ℓ; hence $\ell^2 \mid d_r$. No odd ℓ^3 can divide d_r, because

$$P_n(2\ell) - 2P_n(\ell) \equiv -2a\ell^2 \not\equiv 0 \pmod{\ell^3},$$

where $a \equiv 1 \pmod{\ell}$. Finally $4 \mid P_n(x)$ for every integer x since n is even, $n \geq 4$, and $a \equiv -1 \pmod 4$; but $3! \left\{ {n \atop 3} \right\} = 3^n - 3 \cdot 2^n + 3 \equiv 4 \pmod 8$, so $v_2(d_r) = 2$. Combining these local valuations proves (34.71). □

Remark 34.53 (Classical Stirling divisibility versus the CCR application). The fixed-divisor object $\Delta_{n,3} = \gcd\{j! \left\{ {n \atop j} \right\} : 3 \leq j \leq n\}$ is *not* introduced here: Lundell studied prime-power divisibility of precisely $\Delta_{n,m}$ in 1978 [168]. Formula (34.71) is included with a self-contained elementary proof as a useful classical specialization, *not claimed as an independently original Stirling theorem.* The CCR-specific interpretation uses the previously established first-order source transform to identify the exact source image and its permitted-prime resonance consequences. In particular, $\ell \mid d_r$ occurs only when $\ell \leq r-2$; these potentially small primes need not satisfy the independent p-integrality hypotheses of Theorem 34.55, and no local splitting claim is inferred there merely from this divisibility statement.

Corollary 34.54 (All-odd-order fixed-source nonresonance)**.** *Choose the coherent integral stationary source family $\lambda_a = (-1)^{a-1}$ for every $a \geq 3$. At each fixed odd $r \geq 5$, and for* every *prime satisfying the order-r good-prime hypotheses of Theorem 34.55, the zero-residue fold has no roots modulo p^2 or over $\mathbb{Q}_p$ and is an irreducible totally ramified quadratic factor. There are no additional* source-dependent *exceptional primes for this family.*

Proof. Theorem 34.51 gives $A_r = 1$, which is a unit at every prime. The proof of Theorem 34.55 applies verbatim. □

The first new higher-order example beyond the fifth column is

$$A_5 = 13 + 36\lambda_4 + 24\lambda_5,$$
$$A_7 = 61 + 540\lambda_4 + 1560\lambda_5 + 1800\lambda_6 + 720\lambda_7. \tag{34.72}$$

At order nine one obtains $A_9 = 253+5796\lambda_4+40824\lambda_5+126000\lambda_6+191520\lambda_7+141120\lambda_8+40320\lambda_9$.

Theorem 34.55 (Universal fixed-source finite prime-support theorem)**.** *Let $r = 2s + 1 \geq 5$ be fixed and $\lambda_4, \ldots, \lambda_r \in \mathbb{Z}$. Choose a positive integer D_r clearing the rational coefficients of both S_r and E_r as polynomials in their independent variables, as guaranteed by stationary polynomiality. Put $c_s = -B_{2s}/(4s(2s)!) \neq 0$. At every prime p such that*

$$p > r, \qquad p \nmid D_r, \qquad v_p(c_s) = 0, \qquad p \nmid A_r(\boldsymbol{\lambda}),$$

the disk $m \equiv 0 \pmod{p}$ contributes exactly one double residual root modulo p*, but* no solutions modulo p^2*, no solutions modulo higher powers, and no $\mathbb{Q}_p$ roots. Its degree-two local factor is irreducible and totally ramified. Consequently the primes at which this CCR odd-column zero-disk fold can split form a* finite *set, contained in the union of the finite order-dependent bad-prime set and the prime divisors of the explicit nonzero integer $A_r(\boldsymbol{\lambda})$.*

Proof. At the stated primes, the established integral coefficientwise reduction gives $\Pi_{p,r} \equiv S_r \pmod{p}$. The classical Nörlund coefficient identity gives $S_r(m) = c_s m^2 + O(m^3)$, with c_s a p-unit, so the disk is a nondegenerate double residual root. For $m = pz$, $z \in \mathbb{Z}_p$, we have $S_r(pz) \equiv 0 \pmod{p^2}$, and polynomial integrality yields

$$\Pi_{p,r}(pz) \equiv pE_r(0, 0, \boldsymbol{\lambda}) = -\frac{pA_r(\boldsymbol{\lambda})}{r!} \not\equiv 0 \pmod{p^2}.$$

The stated conditions make $r!$ invertible and A_r a unit modulo p. Thus the whole disk vanishes at the next digit. The derivative-zero center in that disk exists uniquely by Hensel's lemma applied to the derivative; its critical value has p-valuation exactly one. Its quadratic Newton polygon therefore has slope $-1/2$, proving irreducibility and total ramification. The nonvanishing congruence (34.67) makes A_r a nonzero integer, hence it has finitely many prime divisors. □

Corollary 34.56 (Finite-prime simultaneous tuning versus fixed-source rigidity)**.** *Fix an odd order $r \geq 5$, fixed lower integer sources $\lambda_4, \ldots, \lambda_{r-1}$, and a finite set $\mathcal{P}$ of pairwise distinct order-r good primes. There exist* ordinary integer *top source values λ_r for which the zero-disk quadratic factor splits into two distinct $\mathbb{Q}_p$-linear factors* simultaneously for all $p \in \mathcal{P}$*. Precisely, $\prod_{p\in\mathcal{P}}(p-1)/2$ residue classes of λ_r modulo $\prod_{p\in\mathcal{P}} p^2$ realize splitting at the first resonance depth $v_p(\lambda_r - \Lambda_r) = 1$ at every such prime. Nevertheless for every fixed integral source vector the number of primes at which this quadratic factor can split is finite.*

Proof. At each good prime, the top-source translation $\Pi_{p,r}^{(\lambda_r)} = \Pi_{p,r}^{(0)} - p\lambda_r/r$ and the local normal form in Theorem 34.35 give a unique critical $\Lambda_r \in \mathbb{Z}_p$ and a unit K_r with $z^2 = K_r p(\lambda_r - \Lambda_r)$. Exactly one class modulo p satisfies $\lambda_r \equiv \Lambda_r \pmod{p}$. Of its p lifts modulo p^2, exactly $(p-1)/2$ satisfy $v_p(\lambda_r - \Lambda_r) = 1$ and $\left(\frac{K_r(\lambda_r-\Lambda_r)/p}{p}\right) = 1$. These lifts split as claimed. The Chinese remainder theorem combines independent prime-square conditions, giving the exact product count and integer representatives. For any one such integer representative, Theorem 34.55 limits the split primes to the fixed finite order-dependent bad-prime set and prime divisors of the nonzero integer A_r. □

Remark 34.57 (Exact finite example, order five). With $\lambda_4 = 0$ and $\lambda_5 = 774982$, both zero-disk roots split at $p = 7, 11, 13$. The corresponding local field-root counts are three at each prime, whereas the unnormalized polynomial's full congruence-root counts for $n = 1, 2, 3, 4$ are, respectively,

$$p = 7 : \ (2, 8, 15, 15), \quad p = 11 : \ (2, 12, 23, 23), \quad p = 13 : \ (2, 14, 27, 27).$$

The difference between the field-root number and stable congruence-fiber cardinality is essential: a split local root with nonunit derivative represents p distinct classes modulo each sufficiently high power. The exact integer witness is a finite consistency check, not a substitute for the CRT proof.

Corollary 34.58 (Uniform source-height threshold)**.** *If $|\lambda_a| \le B$ for all $4 \le a \le r$, then*

$$|A_r| \le C_r(B) := 2^{r-1} - 3 + B \sum_{a=4}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\}.$$

For every prime meeting the order-r good-prime conditions and satisfying $p > C_r(B)$, the zero-residue disk disappears at precision p^2 uniformly for every integral source vector of that height. *For $r = 7$, $C_7(B) = 61 + 4620B$; the old $r = 5$ bound is $13 + 60B$.*

Remark 34.59 (Precise global scope and mathematical antecedents). Theorem 34.55 concerns the CCR-specific zero-residue quadratic factor. It does *not* assert an all-prime factorization of the remaining degree-$r-2$ factor for every $r \ge 7$. If $S_r(m)/m^2$ is squarefree modulo a particular good prime p, then all nonzero residual roots are simple and the complete root count modulo every p^n, $n \ge 2$, equals the number of nonzero roots of $S_r \pmod p$; in characteristic zero the quotient is explicitly squarefree for $r = 5, 7, 9, 11$ by finite symbolic calculation. The alternating Stirling sum and Bézout argument in Theorem 34.51 are classical combinatorial facts, not an assertion of new Stirling-number identities. Their interpretation as an exact CCR source-obstruction lattice and simultaneous p-adic splitting criterion is the qualified specialized contribution. The Nörlund identity, and the odd-index double-zero factor itself, are established independently: see Bencherif–Garici [167]. In the CCR hierarchy, the source-deformation formula (34.66), together with parity and the fixed-source condition, yields the exact obstruction lattice and finite-prime-support conclusion.

34.28 Chronological integer-source observations and their limits

The preceding stationary theory concerns a single vector of source parameters. It is important to ask exactly which *terminal raw coefficient* constraints survive when only the top order-r source varies in time. This section gives finite, closed criteria; its assertions are *not* identities of entire normalized polynomial columns, and there is no claim that the prescribed source amplitudes arise from a physical DPDA transition system.

Fix an odd integer $r \ge 5$. Retain the stationary lower integer sources $\lambda_3 = 1, \lambda_4, \ldots, \lambda_{r-1}$ and let the top source at temporal row $t \ge r$ be an independent integer $\eta_{r,t}$. Write

$$w_t = \binom{t-1}{r-1}, \quad B_N = \binom{N}{r} = \sum_{t=r}^{N} w_t, \quad Z_N = \sum_{t=r}^{N} w_t \eta_{r,t}, \quad \lambda_{\text{eff}}(N) = \frac{Z_N}{B_N}. \tag{34.73}$$

The r-th connected source enters the raw CCR recurrence in degree r, so changes to its temporal sequence affect the degree-r coefficient linearly through the terminal-block

weights w_t; no lower-degree coefficient is affected. Thus $\lambda_{\mathrm{eff}}(N)$ is the *unique stationary top-source shadow* of that single raw terminal coefficient. From Theorem 34.50 the first-order obstruction of this shadow is

$$A_r^{\mathrm{shadow}}(N) = L_r + (r-1)!\lambda_{\mathrm{eff}}(N), \quad L_r = 2^{r-1} - 3 + \sum_{a=4}^{r-1}(a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a. \quad (34.74)$$

Because $L_r \equiv 1 \pmod 6$ and $(r-1)!$ is divisible by 6, the cancellation target $c_r = -L_r/(r-1)!$ is a noninteger rational.

Theorem 34.60 (Signed integer multisnapshot criterion)**.** *Let $r \le N_1 < \cdots < N_m$, put $N_0 = r-1$, and fix $c \in \mathbb{Q}$. With unrestricted signed integer sources $\eta_{r,t}$, one history satisfies $\lambda_{\mathrm{eff}}(N_i) = c$ for all i if and only if*

$$c\left(\binom{N_i}{r} - \binom{N_{i-1}}{r}\right) \in g_i\mathbb{Z}, \qquad g_i = \gcd\{w_t : N_{i-1} < t \le N_i\}, \quad 1 \le i \le m. \quad (34.75)$$

In particular no integer history can cancel A_r^{shadow} at two consecutive terminal rows, but there are single signed integer histories attaining the cancellation at infinitely many suitably separated rows.

Proof. The independent increment $Z_{N_i} - Z_{N_{i-1}}$ is an integer linear combination of the weights in that interval. Bézout's identity characterizes all such increments as $g_i\mathbb{Z}$; separate intervals do not share source cells, giving necessity and sufficiency. For consecutive snapshots subtraction in (34.73) forces $\eta_{r,N+1} = c_r$, impossible since $c_r \notin \mathbb{Z}$. Conversely, the $(r-1)$-st finite difference gives $\gcd(w_{N+1}, \ldots, w_{N+r}) = 1$. Writing $c_r = a/D$ in lowest terms, the sequence $N_j = j\,r!D$ has integral $c_r\binom{N_j}{r}$ and gaps at least r. Formula (34.75) then produces the claimed infinite history. □

Theorem 34.61 (Nonnegative integer sources and exact reachability)**.** *Keep the same lower stationary parameters, which may be signed, but impose $\eta_{r,t} \in \mathbb{Z}_{\ge 0}$. For the snapshot rows of Theorem 34.60, let $\mathcal{S}_i = \langle w_t : N_{i-1} < t \le N_i \rangle_{\mathbb{Z}_{\ge 0}}$. One nonnegative integer history realizes the rational target c at all snapshots if and only if*

$$c\left(\binom{N_i}{r} - \binom{N_{i-1}}{r}\right) \in \mathcal{S}_i \qquad (1 \le i \le m). \quad (34.76)$$

For $c < 0$ no snapshot is attainable; for $c = 0$ all are attainable using zero sources; for every $c > 0$ one history attains infinitely many sparse snapshots, without a uniform amplitude or gap bound.

Proof. Equation (34.76) says exactly that each independent nonnegative weighted increment is representable by source pulses in its time interval. Negative increments are impossible and zero increments are attained by the zero history. For $c = a/D > 0$, choose an initial large multiple N_1 of $r!D$ and place the integral amount cB_{N_1} at weight $w_r = 1$. Given an attained N_j, the first r weights past N_j have gcd one by finite differences. Their nonnegative numerical semigroup has a finite complement in $\mathbb{Z}_{\ge 0}$ [184]. Choose the next sufficiently large multiple N_{j+1} of $r!D$ so that the integral required increment $c(B_{N_{j+1}} - B_{N_j})$ lies beyond the conductor. Represent it using those first r weights and set all other intervening sources to zero. This proves induction. □

Theorem 34.62 (Sharp mixed-sign lower-band threshold)**.** *Let all $\eta_{r,t} \in \mathbb{Z}_{\ge 0}$ remain unrestricted in amplitude. Define positive Stirling weights and the signed lower-source*

budgets

$$K_{r,a} = (a-1)!\left\{ {r-1 \atop a-1} \right\} > 0,$$
$$P_r = \sum_{a=4}^{r-1} K_{r,a} \max(\lambda_a, 0),$$
$$D_r = \sum_{a=4}^{r-1} K_{r,a} \max(-\lambda_a, 0).$$

Then $A_r^{\mathrm{shadow}}(N) = 0$ *is attainable for some terminal row* N *if and only if*

$$\boxed{D_r \geq 2^{r-1} + 2 + P_r.} \tag{34.77}$$

For each odd $r \geq 5$*, the* separate *choice* $\lambda_4 = -1$ *and* $\lambda_5 = \cdots = \lambda_{r-1} = 0$ *satisfies this threshold. In the typed higher-band Hessenberg realization, writing* $a_{r,t} = (-1)^r p \eta_{r,t}$ *gives*

$$H_{i+r-1,i} = -p\, i(i+1)\cdots(i+r-2)\eta_{r,i+r-1};$$

thus a negative lower stationary λ_a *is equivalent to a* positive *corresponding lower Hessenberg entry. This equivalence is formal and does not assert DPDA realizability.*

Proof. By (34.74), $L_r = 2^{r-1} - 3 + P_r - D_r \equiv 1 \pmod 6$. Nonnegative top sources imply $\lambda_{\mathrm{eff}}(N) \geq 0$, so cancellation requires $L_r < 0$, equivalently (34.77); equality $L_r = 0$ is excluded by its congruence. Conversely if $L_r < 0$, set $c = -L_r/(r-1)! = a/D > 0$ in lowest terms and choose $N = r!D$. Since $D \mid \binom{N}{r}$, the single nonnegative source pulse $\eta_{r,r} = c\binom{N}{r}$ attains the required shadow. Finally $K_{r,4} = 3^{r-1} - 3 \cdot 2^{r-1} + 3$, so the stated lower choice gives $L_r = 4 \cdot 2^{r-1} - 3^{r-1} - 6 < 0$ for odd $r \geq 5$. The Hessenberg identity is the explicit signed source realization of the established typed CCR recurrence. □

Remark 34.63 (Interpretation and a checkable example). At $r = 5$, $\lambda_4 = -1$ gives $L_5 = -23$ and $c_5 = 23/24$. The source $\eta_{5,5} = 759$ yields the target at $N = 12$, as $759/\binom{12}{5} = 23/24$. It cannot simultaneously be attained at $N = 16$ with nonnegative pulses confined to the interval $(12, 16]$, although that interval has weights of gcd one; 3427 is outside the semigroup generated by $495, 715, 1001, 1365$. At $(12, 18]$ the pulses $\eta_{5,13} = 3$, $\eta_{5,14} = 3$, $\eta_{5,15} = 2$, and $\eta_{5,17} = 1$ give the required increment 7452. This separates signed Bézout reachability from nonnegative semigroup reachability. The comparison remains at *one terminal raw degree-r coefficient*, not at the level of full p-adic normalized columns.

34.29 Explicit Galois laws at the seventh and ninth columns

The odd-column result above removes the double zero residue disk, but it leaves the nonzero residual roots to be determined. Orders seven and nine provide complete global root distributions at almost every prime, as an arithmetic consequence of the *previously known* Stirling polynomials. Bencherif–Garici [167] explicitly list the underlying quartic and sextic (with the opposite overall sign). We neither claim their polynomial formulas nor general Chebotarev theory as new. The present statement connects their Frobenius cycle types to *fixed-source CCR* prime-power root counts by Theorem 34.55.

Write

$$q_7(m) = 9m^4 - 54m^3 + 51m^2 + 58m + 16, \tag{34.78}$$
$$q_9(m) = 15m^6 - 165m^5 + 465m^4 + 17m^3 - 648m^2 - 548m - 144. \tag{34.79}$$

The Nörlund identities already used in Section 34.23 give

$$S_7(m) = \frac{m^2(m-1)q_7(m)}{5806080}, \qquad S_9(m) = \frac{m^2(m-1)q_9(m)}{2786918400}. \tag{34.80}$$

The factors $m-1$ are simple, and the two quotients are squarefree over $\mathbb{Q}$: directly,

$$\operatorname{disc}(q_7) = -2^4 3^5 \cdot 83 \cdot 97 \cdot 271,$$
$$\operatorname{disc}(q_9) = -2^{16} 3^3 5^2 \cdot 7 \cdot 263 \cdot 67891 \cdot 63049079.$$

Furthermore $q_7(0) = 16$, $q_7(1) = 80$, $q_9(0) = -144$ and $q_9(1) = -1008$, so these extra roots do not collide with 0 or 1 for sufficiently large primes away from the displayed discriminants.

Theorem 34.64 (Exact small-order Galois groups)**.** *For the explicit polynomials* (34.78)–(34.79)*,*

$$\operatorname{Gal}(q_7/\mathbb{Q}) \cong S_4, \qquad \operatorname{Gal}(q_9/\mathbb{Q}) \cong S_6.$$

Proof. The squarefree modular factor-degree certificates for q_7 are (4) at $p = 13$ and $(1,3)$ at $p = 17$. Thus the Galois group is transitive and contains both a four-cycle and a three-cycle. The only transitive subgroup of S_4 with both types is S_4.

For q_9, the factor-degree certificates are (6) at $p = 29$, $(1,5)$ at $p = 17$, and $(1,1,1,1,2)$ at $p = 607$. None of these primes divides $\operatorname{disc}(q_9)$. Consequently the Galois group $G \le S_6$ is transitive and contains a five-cycle and a transposition. It is primitive: a nontrivial system of blocks in a transitive degree-six action has block size two or three, and a five-cycle cannot act nontrivially on either the blocks or any such block. Finally, the graph on six letters whose edges are the G-conjugates of the supplied transposition has a G-invariant component partition. Primitivity forces that graph to be connected; its edge transpositions generate S_6. Hence $G = S_6$. The stated factor degrees are determined by factoring the explicit integer polynomials (34.78)–(34.79) in the indicated finite fields. The discriminants follow by exact resultant evaluation, $\operatorname{disc}(q) = (-1)^{d(d-1)/2} \operatorname{Res}(q, q')/\operatorname{lc}(q)$ for $d = \deg q$. Thus the cycle-type witnesses and discriminant assertions are finite, reproducible algebraic certificates rather than additional hypotheses. □

For each odd r, let $A_r(\boldsymbol{\lambda})$ be the nonzero integer in (34.66). Define the two explicit finite bad-prime sets

$$\mathcal{B}_7(\boldsymbol{\lambda}) = \{p \text{ prime} : p \le 14 \text{ or } p \mid A_7(\boldsymbol{\lambda})\, 83 \cdot 97 \cdot 271\},$$
$$\mathcal{B}_9(\boldsymbol{\lambda}) = \{p \text{ prime} : p \le 18 \text{ or } p \mid A_9(\boldsymbol{\lambda})\, 263 \cdot 67891 \cdot 63049079\}.$$

Here the small-prime cutoffs also ensure that every denominator in the order-r normalized transport is p-invertible. Indeed the triangular operator T_j in Section 34.27 has diagonal entries $j+d$ for $0 \le d \le j$, all at most $2r$, and its right-hand sides have integral coefficients once the earlier columns are integral. Thus, by induction, the coefficients of S_r and its p-deformation are p-integral for all $p > 2r$. No claim is made at the explicitly excluded primes.

Theorem 34.65 (Fixed-source seventh/ninth prime-axis laws)**.** *Fix a stationary integer CCR source vector $\lambda_3 = 1, \lambda_4, \dots, \lambda_r \in \mathbb{Z}$ and $r \in \{7, 9\}$. For any rational prime $p \notin \mathcal{B}_r$, and every $n \ge 2$, the number of solutions of the normalized r-th CCR column modulo p^n equals*

$$1 + \#\{u \in \mathbb{F}_p : q_r(u) = 0\},$$

and is also the number of distinct roots in $\mathbb{Q}_p$. Their natural densities, among all rational primes, are

root count $(r = 7)$	1	2	3	5
density	9/24	8/24	6/24	1/24

root count $(r = 9)$	1	2	3	4	5	7
density	265/720	264/720	135/720	40/720	15/720	1/720.

In both cases the prime-axis mean number of distinct roots converges to 2*, independently of the fixed integral source vector.*

Proof. For $p > 2r$, the original normalized transport reduces coefficientwise to S_r modulo p. By (34.80) and the exclusion of $\mathcal{B}_r$, every nonzero residual root comes either from $m = 1$ or a simple root of q_r and lifts uniquely modulo all p^n by Hensel's lemma. The double root at zero contributes no solution modulo p^2, by Theorem 34.55, because $p \nmid A_r$. This proves the exact count for every such prime, not merely a statistical limit.

Apply Chebotarev to the groups in Theorem 34.64. The number of roots of q_r modulo an unramified prime is the number of fixed points of the corresponding permutation. For S_4, the permutation counts with $0, 1, 2, 4$ fixed points are $9, 8, 6, 1$ out of $4! = 24$. For S_6, the counts with $0, 1, 2, 3, 4, 6$ fixed points are $265, 264, 135, 40, 15, 1$ out of $6! = 720$ (the classical rencontres numbers $\binom{6}{k} D_{6-k}$). The unavoidable root $m = 1$ adds one to each. The finite excluded sets do not change the natural densities. In both symmetric groups the mean number of fixed points is one, so the total mean is two. □

Corollary 34.66 (Conditional all-odd orbit-count mean)**.** *Fix any odd $r \geq 5$. Over $\mathbb{Q}$, factor $S_r(m) = m^2(m-1)q_r(m)$ with $q_r \in \mathbb{Q}[m]$. Assume q_r is squarefree and $q_r(1) \neq 0$, and let t_r be its number of distinct irreducible factors over $\mathbb{Q}$. For any fixed integral stationary source vector, the mean number of $\mathbb{Q}_p$ roots of the normalized r-th column, averaged over rational primes $p \leq X$, tends to $1 + t_r$ as $X \to \infty$. The same mean holds for the number of roots modulo p^n for every fixed $n \geq 2$ outside finitely many source-dependent primes. In particular, if q_r is irreducible, the mean is exactly* 2*.*

Proof. The classical odd Nörlund factorization and its exact double zero at the origin are already cited in Section 34.23; the factor $m - 1$ follows from vanishing odd Bernoulli numbers at $m = 1$. For a fixed r, away from finitely many coefficient-denominator, discriminant, resultant and A_r prime divisors, all nonzero residual roots are simple and lift, whereas the zero disk disappears modulo p^2 by Theorem 34.55. The remaining roots are the single root at $m = 1$ and the fixed points of Frobenius acting on the roots of q_r. By the Chebotarev density theorem the limiting average number of such fixed points is the group-average number of fixed points. Burnside's lemma identifies it with the number of Galois orbits, namely t_r, the number of irreducible factors of the squarefree rational polynomial q_r. The finite excluded prime set has zero density. □

Remark 34.67 (Exactly what is new here). The explicit q_7, q_9 appeared in the 2017 Stirling-polynomial literature [167]. Their algebraic Galois computations and the Frobenius fixed-point technique are standard mathematical approaches. The *CCR-specific arithmetic result* of this section is the combination with the previously established integral source obstruction A_r, producing source-independent global p-adic root-count laws at all but finitely many primes. This is an arithmetic specialization, not a theorem about all odd r or actual DPDA source realizability; the underlying unperturbed polynomial identities are properly attributed to the Stirling literature.

34.30 Exact certificates, higher odd columns, and adelic laws

The following finite-depth congruence certificates are obtained by reducing the displayed integer coefficient formulas modulo prime powers, lifting derivative-critical residues by Hensel's lemma, and counting the resulting congruence classes. They supplement, but do not replace, the exact all-depth proofs in Lemma 34.24 and Theorem 34.40.

The 419-adic coordinate and its domain. A digitwise Hensel lift of the derivative-zero center followed by evaluation of the critical value gives the first six base-419 digits of the critical parameter as

$$\Lambda \equiv 132 + 91 \cdot 419 + 73 \cdot 419^2 + 157 \cdot 419^3 + 402 \cdot 419^4 + 44 \cdot 419^5 \pmod{419^6}.$$

For $\lambda = 0, 132, 551, 38261$, direct congruence lifting from the integer quartic, without using the closed solution-count formula, gives the following root counts:

λ	N_1	N_2	N_3
0	3	2	2
132	3	421	2
551	3	421	840
38261	3	421	421

The isometry claim concerns integral residue classes. As clarified in Lemma 34.24, half-integral-valuation algebraic roots are instead classified by the Newton polygon of the translated integral polynomial.

The 73-adic covariance. For the fifth-column quintic, reducing the original integer polynomial modulo 73, 73^2, and 73^3 and averaging over all $73^2 = 5329$ source residues $T \pmod{73^2}$ for fixed L gives

L	$d = v_{73}(\Lambda_{62} - \Lambda_0)$	$\mathbb{E}_T[M_{0,3}M_{62,3}]$	$\mathrm{Var}_T(N_3)$
0	0	0	286
9	1	72	430
812	2	145	576

Each individual fold has mean 1. These counts independently validate all three regimes of Theorem 34.40 at precision $n = 3$. For clarity, the *ordered* critical gap throughout this calculation is $\delta = \Lambda_{62} - \Lambda_0$, and its first residue is $\delta \equiv -2 - 16L \pmod{73}$. The opposite sign refers to the reverse ordered gap and must not be substituted into the product congruence without also reversing its sign. All formulas and independent counts below use this ordered gap convention; reversing the order also reverses the sign in the corresponding product congruence.

Methodological boundary. Analytic Morse coordinates, Hensel lifting, and such finite modular counts are established tools, not claimed as new [175]. These finite congruence identities apply at the stated precisions and under the given coordinate hypotheses; the general conclusions follow from the preceding symbolic proofs.

34.31 Haar-product independence across different odd CCR columns

The preceding fifth-column two-fold theorem couples two critical disks *inside one column* at the exceptional prime 73. We now consider one distinguished zero-residue fold in *each of several different odd columns*, all at the same prime. This is a separate assertion whose proof uses the original CCR-k triangular source structure [166], not the special 73-adic two-curve isometry.

Normalized p-integrality at large primes. The good-prime hypothesis below can be checked directly from the original stationary transport, rather than assumed implicitly. For $1 \le j \le R$ and $0 \le k \le j$, its triangular operator satisfies

$$T_j(m^k) = (j+k)m^k + \sum_{h=0}^{k-2} \binom{k}{h} m^{h+1},$$

where an empty sum means zero. On polynomials of degree at most j, T_j has integral triangular matrix and diagonal $j, j+1, \dots, 2j$. If $p > 2R$, all these diagonal entries are p-adic units, so T_j^{-1} preserves the module of p-integral polynomials of degree at most j. The original normalized CCR recurrence has p-integral polynomial right-hand side whenever the preceding normalized columns and the stationary sources are p-integral. Induction from $\Pi_{p,0} = 1$ therefore shows that every $\Pi_{p,j}$, $j \le R$, is coefficientwise p-integral for every source vector in $\mathbb{Z}_p$; specialization at $p = 0$ gives $\Pi_{p,j} \equiv S_j \pmod p$. The additional unit-curvature condition stated below ensures that the residual zero-disk double root is nondegenerate. No assertion is made for the finitely many smaller or curvature-bad primes.

Theorem 34.68 (Odd-column Haar product law)**.** *Fix a finite set $I \subseteq \{5, 7, 9, \dots, R\}$ of odd indices and put $d = |I|$. Let $p > 2R$ be an odd prime such that, for every $r = 2s+1 \in I$, the curvature*

$$c_s = -\frac{B_{2s}}{4s(2s)!} \in \mathbb{Z}_p^{\times}.$$

Work with the stationary normalized CCR polynomial extension, $\lambda_3 = 1$, and regard all source parameters with indices outside I as arbitrary fixed elements of $\mathbb{Z}_p$. Choose $\{\lambda_r : r \in I\}$ independently with normalized Haar measure on $\mathbb{Z}_p$. For $n \ge 1$, denote by $M_{r,n}$ the number of zeros of $\Pi_{p,r}(m)$ modulo p^n with $m \equiv 0 \pmod p$.

Then the variables $M_{r,n}$, $r \in I$, are independent and have the same single-fold law. More precisely, define

$$h = \left\lfloor \frac{n-1}{2} \right\rfloor, \quad q = p^{-(n-1)}, \quad a = p^{\lfloor n/2 \rfloor}, \quad w_j = \frac{p-1}{2p^{2j}}$$

and the probability polynomial

$$P_{p,n}(X) = \left(1 - q - \sum_{j=1}^{h} w_j\right) + qX^a + \sum_{j=1}^{h} w_j X^{2p^j}. \tag{34.81}$$

For $S_n = \sum_{r \in I} M_{r,n}$,

$$\mathbb{E}[X^{S_n}] = P_{p,n}(X)^d, \qquad \mathbb{E}[S_n] = d, \qquad \operatorname{Var}(S_n) = d(p-1)(n-1). \tag{34.82}$$

The number B of indices in I whose zero-residue quadratic factor has two distinct $\mathbb{Q}_p$*-roots satisfies*

$$B \sim \text{Binomial}\left(d, \frac{1}{2(p+1)}\right). \tag{34.83}$$

The possible repeated-root critical parameters form a Haar-null set.

Proof. For each $r \in I$, the established normalized shadow has $S_r(m) = c_s m^2 + O(m^3)$ and a unit quadratic coefficient modulo p. The p-integral transport admits a unique derivative-critical center $c_r(\lambda_{<r}) \in p\mathbb{Z}_p$. This center depends only on the lower-order

sources, because the already established exact stationary translation law for the top source is

$$\Pi_{p,r}^{(\lambda_r)}(m) = \Pi_{p,r}^{(0)}(m) - \frac{p}{r}\lambda_r$$

for odd r, with all lower sources fixed. Define $\Lambda_r(\lambda_{<r}) \in \mathbb{Z}_p$ by $\Pi_{p,r}^{(\Lambda_r)}(c_r) = 0$. Since the critical value of $\Pi^{(0)}$ is divisible by p and r is invertible, such a parameter exists uniquely. The elementary p-adic quadratic coordinate in the disk $m \in p\mathbb{Z}_p$ produces an isometric coordinate $z_r \in p\mathbb{Z}_p$ satisfying

$$z_r^2 = K_r p U_r, \qquad U_r = \lambda_r - \Lambda_r(\lambda_{<r}), \qquad K_r = \frac{1}{rc_s} \in \mathbb{Z}_p^{\times}. \tag{34.84}$$

Here the analytic square-root normalization is applied only on the *integral residue disk*; field-factor ramification outside this disk, if any, follows separately from its Newton polygon.

The source substitution

$$(\lambda_4, \dots, \lambda_R) \longmapsto (\widetilde{\lambda}_4, \dots, \widetilde{\lambda}_R), \quad \widetilde{\lambda}_r = \begin{cases} \lambda_r - \Lambda_r(\lambda_{<r}), & r \in I, \\ \lambda_r, & r \notin I \end{cases}$$

is triangular with unit diagonal, and is invertible recursively: recover $\lambda_4, \lambda_5, \dots$ in increasing order. All critical functions are p-integral and compatible with reduction modulo every p^a by Hensel's implicit-function theorem. Hence this map is a bijection on each finite residue vector space and preserves product Haar measure. With parameters outside I fixed, the same recursive argument gives independent Haar U_r for $r \in I$.

The zero-disk count $M_{r,n}$ is now a function only of U_r via (34.84). Multiplication by each fixed unit K_r does not change a Haar-uniform distribution. Elementary quadratic valuation counting gives probability q for the fully resonant count a, probability w_j for a split count $2p^j$ at each $1 \le j \le h$, and the remaining mass at zero. This is (34.81). In particular, $\mathbb{E}M_{r,n} = 1$ and $\mathrm{Var}(M_{r,n}) = (p-1)(n-1)$, the universal one-fold moment formulas already established in Section 34.23. Independence proves (34.82).

Finally, a quadratic branch splits with two distinct $\mathbb{Q}_p$ roots exactly when $v_p(U_r) = 2j-1$ for some $j \ge 1$, and the leading unit of $K_r U_r$ is a square. Its probability is

$$\sum_{j \ge 1} \frac{p-1}{2p^{2j}} = \frac{1}{2(p+1)}.$$

The branch events are independent, proving (34.83). □

Remark 34.69 (What is and is not independent). Even when the lower sources are correlated through the higher transport, the upper-source coordinates give exact independence of these *distinguished local folds*. Nothing here asserts independence of the *full* root counts from all other residue disks, or of arbitrary nonstationary/physically realized source data. Triangular Haar changes and one-dimensional quadratic congruence counting are classical methods; the precise CCR application follows from the pre-existing top-source translation theorem.

Finite-precision two-column illustration. Direct polynomial reduction using the full normalized CCR transport (rather than the local normal form (34.84)) fixes $\lambda_4 = \lambda_6 = 0$ and enumerates every $(\lambda_5, \lambda_7) \in \mathbb{F}_p^2$ for $p = 17, 19, 23$. For every $m \in p\mathbb{Z}/p^2\mathbb{Z}$ the resulting joint distribution shows that $(M_{5,2}, M_{7,2})$ takes values $(0,0), (p,0), (0,p), (p,p)$ with multiplicities $(p-1)^2, p-1, p-1, 1$, respectively. These three complete finite-precision distributions are consistent with the independent all-depth triangular/Hensel proof.

34.32 Adelic resonance dichotomy: fixed integer sources versus profinite Haar sources

The all-odd fixed-integer source obstruction and the multicolumn Haar-product theorem have complementary quantifiers. The former fixes an integer vector and varies the prime; the latter varies source coordinates in a single p-adic space. They admit an exact common adelic formulation, provided these two ambient spaces are not conflated.

Fix a nonempty finite set of odd column orders $I \subseteq \{5, 7, \dots, R\}$ and write $d = |I|$. Let $\mathcal{G}_R$ be the cofinite set of primes $p > 2R$ at which all selected normalized columns are coefficientwise p-integral and their quadratic curvatures $c_{(r-1)/2}$ are p-units. This exclusion set depends on R, not on the source parameters. Define the profinite source space

$$\widehat{\mathbb{Z}}^{R-3} = \prod_p \mathbb{Z}_p^{R-3},$$

with independent normalized Haar factors and $\lambda_3 = 1$. For $p \in \mathcal{G}_R$ and $r \in I$, let $E_{p,r}$ be the event that the distinguished $m \equiv 0 \pmod p$ fold of the r-th CCR column has two distinct $\mathbb{Q}_p$ roots. No condition is placed on other root residue disks.

Theorem 34.70 (Adelic CCR prime-by-column product law)**.** *On the above profinite Haar source space, the events $\{E_{p,r} : p \in \mathcal{G}_R,\ r \in I\}$ are mutually independent, with*

$$\mathbb{P}(E_{p,r}) = q_p := \frac{1}{2(p+1)}. \tag{34.85}$$

Writing $N_I(X) = \sum_{\substack{p \le X \\ p \in \mathcal{G}_R}} \sum_{r \in I} \mathbf{1}_{E_{p,r}}$, we have

$$\mu_I(X) := \mathbb{E}N_I(X) = \frac{d}{2} \log\log X + O_R(1), \tag{34.86}$$

$$\mathrm{Var}(N_I(X)) = \frac{d}{2} \log\log X + O_R(1). \tag{34.87}$$

Moreover, almost surely,

$$\frac{N_I(X)}{\frac{d}{2}\log\log X} \longrightarrow 1, \tag{34.88}$$

and the centered count satisfies the standard central limit theorem

$$\frac{N_I(X) - \mu_I(X)}{\sqrt{\mathrm{Var}(N_I(X))}} \implies \mathcal{N}(0,1).$$

For some positive constant $C_{I,R}$ (depending on the finite excluded prime set),

$$\mathbb{P}(N_I(X) = 0) = \prod_{\substack{p \le X \\ p \in \mathcal{G}_R}} (1 - q_p)^d \sim \frac{C_{I,R}}{(\log X)^{d/2}}. \tag{34.89}$$

In particular, almost every profinite source vector has infinitely many split zero-disk CCR folds in each *chosen odd column.*

Proof. At each fixed good prime, the multicolumn triangular coordinates of the construction in Section 34.31 express the critical source difference as $U_r = \lambda_r - \Lambda_r(\lambda_{<r})$. As the chosen source coordinates vary independently in $\mathbb{Z}_p$, the U_r are independent Haar-uniform. The local quadratic equation $z_r^2 = K_r p U_r$ has two distinct $\mathbb{Q}_p$ roots precisely

when $v_p(U_r)$ is positive odd and the corresponding leading unit satisfies the quadratic-character condition. Since every K_r is a unit, the exact probability is

$$\sum_{j\ge 1}\frac{p-1}{2p^{2j}}=\frac{1}{2(p+1)}.$$

The profinite Haar space is the product of the local source spaces *over primes*, so the already-proved within-prime independence extends to the entire prime-by-column family.

The reciprocal-prime Mertens estimate [177] and $q_p = \frac{1}{2p} + O(p^{-2})$ imply $\sum_{p\in\mathcal{G}_R,\ p\le X} q_p = \frac{1}{2}\log\log X + O_R(1)$. The sum of q_p^2 converges, proving the first two asymptotics. For completeness, enumerate the independent indicators as Y_j, put $\mu_n=\sum_{j\le n}\mathbb{E}Y_j\to\infty$, and observe

$$\sum_j \frac{\operatorname{Var}(Y_j)}{\mu_j^2}\le\sum_j\frac{\mathbb{E}Y_j}{\mu_j^2}<\infty,$$

where the initial finite terms are ignored and the final convergence follows by comparing increments of μ_j with $u^{-2}du$. Kolmogorov's convergence theorem and Kronecker's lemma then give $\sum_{j\le n}(Y_j-\mathbb{E}Y_j)/\mu_n\to 0$ almost surely. Because the centered indicators satisfy $|Y_j-\mathbb{E}Y_j|\le 1$ while their cumulative variance tends to infinity, Lindeberg's condition eventually has no nonzero summands, so the stated central limit theorem follows.

The constant in the no-splitting asymptotic has an explicit convergent Euler-product expression. Put $\mathcal{B}_R=\{p:\ p\notin\mathcal{G}_R\}$ and $a_p=(1-q_p)/(1-1/p)^{1/2}$ for $p\in\mathcal{G}_R$. Since $\log a_p=O(p^{-2})$, the product $\prod_{p\in\mathcal{G}_R}a_p$ converges to a strictly positive value. The prime-product form of Mertens' theorem gives

$$C_{I,R}=e^{-d\gamma/2}\prod_{p\in\mathcal{B}_R}(1-p^{-1})^{-d/2}\prod_{p\in\mathcal{G}_R}\left(\frac{1-1/(2(p+1))}{\sqrt{1-p^{-1}}}\right)^d>0,$$

where γ is Euler's constant. This supplies (34.89) without leaving the positive constant implicit. Applying the almost-sure conclusion to each of the finitely many individual columns proves the final assertion. □

Theorem 34.71 (Integral-source finiteness and the height-limit dichotomy)**.** *Fix all unselected source coordinates as ordinary integers, and let the selected coordinates λ_r range over ordinary integers. For* every fixed *integer source vector, only finitely many pairs (p,r) with $p\in\mathcal{G}_R$, $r\in I$ can satisfy $E_{p,r}$. In sharp contrast, let selected integer coordinates be chosen independently and uniformly in $[-H,H]\cap\mathbb{Z}$ and write $N_I(\infty)$ for their total number of split good-prime folds. Then, for every fixed nonnegative integer K,*

$$\lim_{H\to\infty}\mathbb{P}_H\big(N_I(\infty)\le K\big)=0. \tag{34.90}$$

Nevertheless $N_I(\infty)$ is finite for every *integer source vector, including all those sampled at height H.*

Proof. For any odd order $r\in I$, the CCR first-order source obstruction $A_r(\boldsymbol{\lambda})$ belongs to $1+d_r\mathbb{Z}$, so it cannot vanish for integral sources (in particular $A_r\equiv 1 \pmod 6$). The fixed-source theorem proves that a good-prime splitting implies $p\mid A_r(\boldsymbol{\lambda})$, giving finitely many such primes for each fixed integral source vector.

For a *fixed finite* collection of good primes and column orders, Chinese-remainder equidistribution of integer source vectors in large boxes gives the same joint limiting distribution as the product Haar source measure. Indeed, each local split event is a

union of p-adic residue cylinders, and its only possible boundary lies on the critical-value equality locus $U_r = 0$, of Haar measure zero. Thus its frequency among integer sources can be approximated from above and below by finite residue conditions. For fixed X, the law of $N_I(X)$ under the integer-box model therefore converges to the finite-prime law in Theorem 34.70. Since always $N_I(\infty) \geq N_I(X)$,

$$\limsup_{H\to\infty} \mathbb{P}_H(N_I(\infty) \leq K) \leq \mathbb{P}_{\mathrm{Haar}}(N_I(X) \leq K).$$

In fact this limit can be obtained without invoking an additional almost-sure theorem: whenever $\mu_I(X) > K$, Chebyshev's inequality and the exact mean/variance estimates yield

$$\mathbb{P}_{\mathrm{Haar}}(N_I(X) \leq K) \leq \frac{\mathrm{Var}(N_I(X))}{(\mu_I(X) - K)^2} = O_{I,R,K}\left(\frac{1}{\log\log X}\right).$$

Because X is arbitrary after taking the height limit, the preceding lim sup is zero, proving (34.90). This does not claim any uniform bound in H and X simultaneously. □

Remark 34.72 (Quantifiers and historical scope). The integer lattice is countable and Haar-null inside the profinite source space. It is therefore essential *not* to conclude from the adelic almost-sure statement that any fixed ordinary integer source has infinitely many resonance primes. The derived zero–one, strong-law, central-limit, and Mertens-product arguments use classical probability and analytic number theory; their proposed CCR contribution is solely the specific source-dependent resonance events to which the product laws apply. Neither a general full-root-count independence claim nor a theorem of physical DPDA source realization is intended.

Exact small-prime CRT certificate. For the fifth column with $\lambda_4 = 0$, reduce the integer numerator modulo p^3 and classify all $\lambda_5 \pmod{p^2}$ for $p = 7, 11, 13$ by the number of roots with $m \equiv 0 \pmod{p}$. The number of first-depth splitting source classes is $(p-1)/2 = 3, 5, 6$, respectively, with the unique critical class giving p roots and the nonsquare classes giving none. CRT therefore gives $3 \cdot 5 \cdot 6 = 90$ simultaneously splitting classes modulo $(7 \cdot 11 \cdot 13)^2 = 1002001$, an exact integer-source density $90/1002001$. In contrast, allowing *all* higher odd splitting valuations under profinite Haar gives the larger simultaneous probability $1/(16 \cdot 24 \cdot 28) = 1/10752$. The ordinary integer source $\lambda_5 = 774982$ is one explicit CRT witness. These finite certificates illustrate, but do not establish, the infinite-prime probability theorems.

34.33 Boundary of arithmetic specialization

The polynomial, stationary-source, and Haar-source conclusions do not identify arbitrary p-adic roots with physically realizable chronological machine histories. In particular, the results do not classify all nonstationary raw sources or unbounded physical realization constraints. Hessenberg principal-determinant polynomials Q_n and fixed normalized column polynomials $\Pi_{p,r}$ are different objects.

34.34 Re-entry to the main DPDA story

The arithmetic offshoot is now complete at all finite chronological orders within the typed terminal-block class.

We return to the main machine problem.

The monograph began with the original 2:1 center/left/right DPDA.

Chapters 2–8 extracted its shortest-demand semantics and produced compact residual-faithful machines.

But those compact machines changed the physical representation.

> The operational idea encoded in the original $C/L/R$ architecture admits a uniform generalization to every $p{:}q$.

The resulting real-time centered construction is developed next.

From Chapter 35 onward, the main line will remain uninterrupted:

> general real-time $p{:}q \to$ Structure Theory $\to$ Uniqueness/Lift Theory $\to$ General Theory.

Chapter 35

Why Static Two-Visibility Fails

Research branch — Returning to the Machine: General Real-Time p:q. *This branch develops the uniform centered real-time literal-demand construction for all coprime positive p, q.*

We now return to pushdown machines.
This is not another residual-counter construction.
The goal is more specific:

> Reconstruct, for every coprime p:q, the operational idea hidden in the original center/left/right machine.

The original 2:1 design had three striking features:

1. a literal completion demand was physically visible in the stack;
2. a mismatch could temporarily move information into finite control;
3. a later long replacement could rematerialize that information.

The general construction will preserve this style rather than replace it by the compact Euclidean residual encoding of Chapters 7–8.
That is why its resource count is different.

35.1 We do not re-prove minimum demand

The semantic foundation is already complete.
For

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\},$$

the residual is

$$\rho = qN_1 - pN_0.$$

For every residual r, the unique minimum completion demand is

$$\mu(r) = (x_r, y_r),$$

and the primitive zero-weight kernel is

$$\boxed{K = (q, p).}$$

Every nonnegative vector of the same weight differs from the minimum demand by a nonnegative multiple of K.

Those results are imported from the earlier chapters.

The new problem is physical ordering.

35.2 A two-visible canonical word

For a demand vector

$$(x, y),$$

define

$$\boxed{\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}}$$

This word has an attractive property.

If both symbols occur, the stack begins with

$$10.$$

More generally, every suffix that remains mixed begins with either

$$10$$

or

$$01.$$

Thus the presence of both demand types is always visible within two stack cells.

This is the general literal-demand analogue of the original 2:1 depth-two phenomenon.

But static visibility is not enough.

35.3 The 3:2 obstruction

Consider the canonical stack

$$\boxed{1010.}$$

Both demand symbols are visible immediately.

Suppose the next input is 0, but the required zero is the **second** visible cell.

A naive LIFO strategy could:

1. pop the top 1;
2. delete the 0;
3. immediately restore the displaced 1.

This produces

$$1010 \longrightarrow 110.$$

The surviving zero has moved to depth three.

Two-cell visibility has been destroyed.

The problem is not lack of static information.

It is chronology.

35.4 Static two-visible obstruction

Lemma 35.1 — Static two-visible obstruction.

Suppose a mixed binary stack contains at least two zeros and two ones and begins with

10.

No purely static strategy can support both deletion directions while:

1. deleting either visible demand type;
2. immediately restoring a displaced first-cell blocker;
3. guaranteeing that a surviving mixed suffix still exposes both symbol types in its first two cells.

Proof.

If deleting the top 1 must leave a mixed two-visible suffix, then the next two surviving symbols must differ. Therefore the third original symbol must be 1.

If instead the second-cell 0 is deleted and the first 1 is immediately restored, preserving two-visibility requires the next surviving symbol beneath it to be 0. Therefore the third original symbol must be 0.

The same third cell cannot satisfy both conditions. □

This lemma isolates the exact reason the original architecture needs side-state memory.

A LIFO blocker sometimes must leave the stack temporarily.

35.5 SEARCH

The repair is:

SEARCH = move one displaced blocker into finite control.

The blocker is **not** immediately restored.

Its semantic effect is carried as finite-control debt while the remaining physical stack continues to be popped.

A later WRITE will materialize that debt.

This is the general form of the role played by L and R in the original machine.

The crucial design question becomes:

> How much signed debt must finite control be able to carry before WRITE becomes safe?

That will produce the centered state rail.

Before building the rail, we need to formalize what a debt state means semantically.

Chapter 36

Debt, Virtual Demand, and Minimal WRITE

Let

$$m = p + q.$$

A one-cell mismatch changes the relationship between the physical stack and the semantic demand by one kernel-length unit m.

This motivates a centered integer debt coordinate.

36.1 Virtual demands

For

$$d \in \mathbb{Z},$$

define the virtual demand

$$\boxed{V_d = \mu(dm).}$$

Its residual weight is

$$W(V_d) = dm.$$

A debt state D_d therefore carries finite-control potential

$$\boxed{\Phi(D_d) = dm.}$$

Let a physical logical stack S have Parikh vector

$$P(S) = (N_0(S), N_1(S)).$$

The stack and state together may contain more material than the minimum demand. The semantic projection removes any complete kernel packets.

36.2 Virtual-demand projection

Define kernel reduction

$$\mathrm{red}_{p,q}(x, y)$$

to mean repeated subtraction of

$$K = (q, p)$$

while both coordinates remain nonnegative.
The result is exactly the unique minimum demand of the same residual weight.

Definition 36.1 — Rich-to-demand projection.
For configuration

$$(D_d, S),$$

define

$$\boxed{\Theta(D_d, S) = \mathrm{red}_{p,q}\big(P(S) + V_d\big).}$$

Theorem 36.2 — Virtual-demand projection.
If

$$\boxed{\rho = W(P(S)) + dm,}$$

then

$$\boxed{\Theta(D_d, S) = \mu(\rho).}$$

Proof.
The nonnegative vector

$$P(S) + V_d$$

has weight

$$W(P(S)) + W(V_d) = W(P(S)) + dm = \rho.$$

Kernel reduction of any nonnegative vector of weight ρ is the unique minimum demand $\mu(\rho)$. □
This theorem is one of the main semantic bridges in the monograph.
It says:

$$\boxed{\text{rich state+stack configuration} \longrightarrow \text{unique shortest-demand semantics.}}$$

The rich machine may retain normalization history.
Its semantic projection does not.

36.3 The WRITE equation

Suppose one transition:

- consumes input a;
- removes visible top symbol z;
- changes debt state from D_s to D_t;
- writes replacement word u.

Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Ignoring the unchanged stack suffix, residual preservation forces

$$\boxed{P(u) = e_z - e_a + V_s - V_t + \kappa K}$$

for some integer κ.

At the bottom marker, the term e_z is omitted.

Thus every invariant-preserving rewrite lies on one affine kernel ray.

36.4 Minimal kernel carry

Let

$$B = e_z - e_a + V_s - V_t.$$

The first kernel shift making both coordinates nonnegative is

$$\boxed{\kappa_* = \max\left(\left\lceil -\frac{B_0}{q} \right\rceil, \left\lceil -\frac{B_1}{p} \right\rceil\right).}$$

Theorem 36.3 — Minimal kernel-carry rewrite.

Among all nonnegative invariant-preserving replacement vectors, the unique shortest one is

$$\boxed{R_* = B + \kappa_* K.}$$

Every other solution has the form

$$\boxed{R_* + \ell K, \qquad \ell \geq 1,}$$

and is longer by exactly

$$\ell(p + q)$$

symbols.

Proof.

All integer solutions differ by a multiple of the primitive kernel K.

The displayed κ_* is the least shift that places both coordinates in the nonnegative quadrant.

Every further kernel packet adds

$$q + p = m$$

symbols. □

When a physical word is needed, we write the vector in canonical order using η.

36.5 Why the historical long writes existed

The original 2:1 table contained replacements such as:

$$10, \qquad 11, \qquad 000, \qquad 111111.$$

Before the general theory, these can look ad hoc.
The WRITE equation changes their interpretation.
A long replacement is determined by:

1. the consumed input symbol;
2. the visible stack symbol removed;
3. the source debt potential;
4. the target debt potential;
5. the least kernel carry needed to make the replacement vector nonnegative.

Thus the long words are not arbitrary repair strings.
They are shortest solutions of a constrained residual-preservation equation.
This is one of the main retrospective explanations of the original machine.

36.6 From arbitrary debt to a finite rail

The virtual-demand definition allows every

$$d \in \mathbb{Z}.$$

A finite-state DPDA cannot.
The construction therefore needs a bounded debt interval.
Set

$$\boxed{h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.}$$

The centered core will use exactly

$$\boxed{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}.}$$

with

$$D_0 = C.$$

Why is this width sufficient?
Because the canonical word created at every WRITE has bounded minority count.
Why is it exact in the deferred architecture?
Because one common canonical witness can exhaust the physical stack while leaving every debt value in this interval purely in finite control.
Those are the two halves of Chapters 37 and 38.

Chapter 37

The Centered Real-Time p:q DPDA

Set

$$m = p + q, \qquad h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

The centered state set is

$$\boxed{Q = \{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}\},}$$

with

$$D_0 = C.$$

The stack alphabet is

$$\boxed{\{0, 1, Z_0\}.}$$

The bottom marker Z_0 remains fixed throughout binary-input processing and is removed only by the final right-endmarker acceptance move.

The construction has four binary transition roles:

$$\boxed{\text{MATCH,} \qquad \text{SEARCH,} \qquad \text{outer WRITE,} \qquad \text{bottom WRITE.}}$$

Every binary move consumes exactly one input symbol.

There are no ε-moves during binary processing.

37.1 The centered rail

MATCH

For every

$$|d| \le h_{\mathrm{ctr}}$$

and $z \in \{0, 1\}$,

$$\boxed{(D_d, zT) \xrightarrow{z} (D_d, T).}$$

A matching input simply cancels one physically present demand cell.

Interior SEARCH

For

$$d < h_{\mathrm{ctr}},$$

$$\boxed{(D_d, 0T) \xrightarrow{1} (D_{d+1}, T).}$$

For

$$d > -h_{\mathrm{ctr}},$$

$$\boxed{(D_d, 1T) \xrightarrow{0} (D_{d-1}, T).}$$

A mismatch still pops the visible cell, but instead of restoring it the state debt moves one step.

Outer-boundary WRITE

Define

$$U_+ = \eta\big(\mu((h_{\mathrm{ctr}} + 1)m)\big),$$

$$U_- = \eta\big(\mu(-(h_{\mathrm{ctr}} + 1)m)\big).$$

Then

$$\boxed{(D_{h_{\mathrm{ctr}}}, 0T) \xrightarrow{1} (C, U_+T),}$$

and

$$\boxed{(D_{-h_{\mathrm{ctr}}}, 1T) \xrightarrow{0} (C, U_-T).}$$

At the outer debt boundary, another outward mismatch triggers physical rematerialization and resets debt to zero.

Bottom WRITE

For every

$$|d| \le h_{\mathrm{ctr}},$$

$$\boxed{(D_d, Z_0) \xrightarrow{1} (C, \eta(\mu(dm + q))Z_0)\,,}$$

and

$$\boxed{(D_d, Z_0) \xrightarrow{0} (C, \eta(\mu(dm - p))Z_0)\,.}$$

When no pre-existing demand remains, the next input directly creates the exact new minimum-demand word.

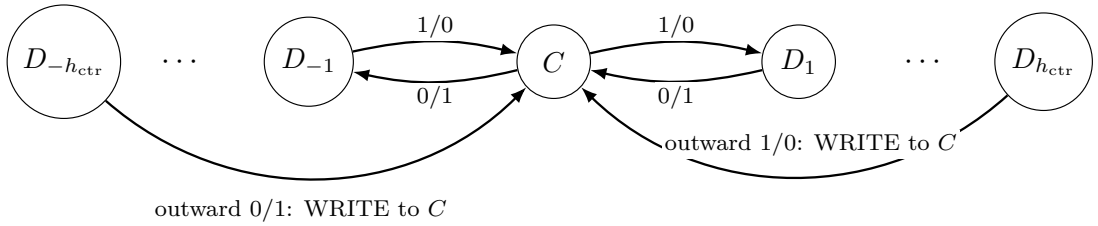
Acceptance

A mandatory right endmarker is allowed only at

$$\boxed{(C, Z_0).}$$

It removes Z_0 and accepts by empty stack.

No other reachable configuration has a right-endmarker acceptance move.



Adjacent mismatch rules repeat along the rail.
Matching 0/0 or 1/1 pops and keeps the same debt state.

Figure 37.1: Schematic centered debt rail. Labels 1/0 and 0/1 denote input/top mismatches. Interior mismatches pop the visible cell and move debt by one; an outward mismatch at $\pm h_{\mathrm{ctr}}$ triggers WRITE back to the center.

The figure should visually emphasize:

- adjacent mismatch SEARCH moves;
- center state C;
- outer WRITE arcs back to C;
- MATCH as state-preserving pops.

37.2 Residual invariant

Let the logical stack be S.

Theorem 37.1 — Residual invariant.

After every processed binary prefix, if the centered configuration is

$$(D_d, SZ_0),$$

then

$$\boxed{\rho = W(P(S)) + dm.}$$

Proof.

Initially,

$$(C, Z_0)$$

has residual zero.

A matching 0-pop decreases stack weight by p, exactly matching the input-0 residual update.

A matching 1-pop increases stack weight by q.

For mismatch 1/0, popping one physical zero changes stack weight by $-p$, while moving

$$d \to d + 1$$

adds state potential

$$m = p + q.$$

The net change is

$$-p + m = q,$$

exactly the input-1 residual update.
The 0/1 case is symmetric:

$$q - m = -p.$$

At an outer WRITE, the replacement word is defined to carry exactly the missing boundary potential.

At bottom WRITE, the target demand is defined directly from the new residual.

Therefore every binary transition preserves the invariant. □

Combined with Chapter 36:

$$\boxed{\Theta(D_d, S) = \mu(\rho)}$$

for every reachable configuration.

37.3 Why the debt rail can stop at h_{ctr}

The key is the **minority size**

$$\mathrm{minc}(x, y) = \min(x, y).$$

Lemma 37.2 — WRITE-frontier minority bound.

Let

$$h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

Then:

1. for every $|d| \le h_{\mathrm{ctr}}$, both bottom-WRITE demands

$$\mu(dm + q)$$

and

$$\mu(dm - p)$$

have minority size at most h_{ctr};

2. if

$$\mu((h_{\mathrm{ctr}}+1)m) = (x_+, y_+),$$

then

$$\boxed{y_+ \le h_{\mathrm{ctr}} < x_+;}$$

3. if

$$\mu(-(h_{\mathrm{ctr}}+1)m) = (x_-, y_-),$$

then

$$\boxed{x_- \le h_{\mathrm{ctr}} < y_-.}$$

Proof.

For the bottom-WRITE family, both residual forms can be written

$$r = jm + q$$

with

$$-h_{\mathrm{ctr}} - 1 \le j \le h_{\mathrm{ctr}}.$$

Let

$$\mu(r) = (x, y), \qquad \ell = x + y.$$

Modulo m, since

$$p \equiv -q \pmod{m},$$

the completion equation gives

$$-q\ell \equiv q \pmod{m}.$$

Because

$$\gcd(q, m) = 1,$$

$$\ell \equiv -1 \pmod{m}.$$

Hence

$$\ell = km - 1.$$

If both x and y were at least $h_{\mathrm{ctr}} + 1$, then $\ell > m - 1$, so $k \ge 2$.
But $\mu(r)$ is kernel-free, so either

$$x < q$$

or

$$y < p.$$

Each alternative, together with $x, y \geq h_{\mathrm{ctr}} + 1$, forces the quotient index j outside the allowed interval.

Contradiction.

For the outer positive boundary, write $D = h_{\mathrm{ctr}} + 1$. Integer solutions to weight Dm have form

$$x = D + qk, \qquad y = pk - D.$$

The minimum nonnegative solution uses the first feasible k, giving

$$y \leq h_{\mathrm{ctr}}$$

and

$$x \geq h_{\mathrm{ctr}} + 1.$$

The negative case is symmetric. □

This lemma explains the centered width geometrically.

Every freshly written canonical word contains at most h_{ctr} copies of its minority symbol.

The debt rail is exactly wide enough to search through that minority frontier.

37.4 Boundary guard

The centered construction also requires a boundary guarantee at $D_{h_{\mathrm{ctr}}}$. An outward mismatch must not occur while the surviving suffix is still mixed, because writing a new canonical prefix in front of such a suffix could destroy the normal form.

The minority bound provides exactly this guarantee.

Lemma 37.3 — Boundary guard.

Assume the current stack is a suffix of a canonical word

$$\eta(x, y)$$

with

$$\min(x, y) \leq h_{\mathrm{ctr}}.$$

Then:

1. if the state is $D_{h_{\mathrm{ctr}}}$ and the stack begins with 0, an outward mismatch 1/0 can occur only on a unary suffix

$$\boxed{0^{t+1};}$$

2. symmetrically, at $D_{-h_{\mathrm{ctr}}}$, an outward 0/1 mismatch can occur only on a unary suffix

$$\boxed{1^{t+1}.}$$

Proof.

Consider the positive case.

Reaching debt h_{ctr} requires at least h_{ctr} positive mismatch events.

Each such mismatch has popped one zero from the original canonical stack.

Suppose the surviving suffix begins with 0 but still contains a later 1.

Then the current zero lies inside the alternating part of $\eta(x, y)$, not in its final unary tail.

Every previously popped zero in that alternating region has a distinct preceding one that must also have been popped.

Together with:

- the current surviving zero;
- the later surviving one;

the original word would contain at least

$$h_{\mathrm{ctr}} + 1$$

zeros and at least

$$h_{\mathrm{ctr}} + 1$$

ones.

This contradicts

$$\min(x, y) \leq h_{\mathrm{ctr}}.$$

Therefore the suffix must be unary.

The negative case is symmetric. □

This is the LIFO safety theorem of the construction.

37.5 Canonical segments

Theorem 37.4 — Canonical-segment invariant.

Immediately after every WRITE, the machine is in the center state with stack

$$\boxed{\eta(\mu(r))Z_0}$$

for its current residual r.

The corresponding demand has minority size at most h_{ctr}.

Between two WRITEs, the logical stack is a suffix of that canonical word.

Proof.

Initially the logical stack is empty.

A bottom WRITE is canonical by definition, and Lemma 37.2 gives the minority bound.

Between WRITEs, MATCH and interior SEARCH only pop one pre-existing work symbol, so the stack remains a suffix of the last canonical WRITE word.

Now consider a positive outer WRITE.

By Lemma 37.3, its source suffix is

$$0^{t+1}.$$

Write

$$\mu((h_{\mathrm{ctr}} + 1)m) = (x, y).$$

Lemma 37.2 gives

$$y \le h_{\mathrm{ctr}} < x.$$

Thus the canonical word $\eta(x, y)$ ends in 0, and the WRITE produces

$$\eta(x, y)0^t = \eta(x + t, y).$$

The residual invariant shows that this vector has exactly the new residual weight.

Moreover, $y < p$, so no kernel packet can be subtracted; the vector is therefore already the minimum demand.

The negative boundary is symmetric. □

The entire reachable-store theorem has now been reduced to episodes:

$$\boxed{\text{WRITE creates a canonical word;}}$$

$$\boxed{\text{MATCH/SEARCH consume a suffix of it;}}$$

$$\boxed{\text{boundary WRITE can occur only after the mixed part is gone.}}$$

37.6 Two-cell qualitative visibility

Corollary 37.5 — Two-cell visibility.

For every reachable logical stack S,

$$\boxed{\mathrm{alph}(S) = \mathrm{alph}(\mathrm{pref}_2(S)).}$$

Equivalently, if both 0 and 1 occur anywhere in S, then both occur among the first two cells.

Proof.

Every canonical word $\eta(x, y)$, and every suffix of it, avoids

$$001 \qquad \text{and} \qquad 110.$$

Therefore a mixed suffix cannot begin with two equal symbols.

It begins with

$$10 \qquad \text{or} \qquad 01.$$

Apply Theorem 37.4. □

This is qualitative, not quantitative.

The first two cells reveal which demand types remain.

They do not reveal the unbounded magnitudes.

37.7 Exact WRITE normalization

Theorem 37.6 — Exact WRITE normalization.

After every bottom or outer-boundary WRITE,

$$\boxed{(D_d, S) = (C, \eta(\mu(\rho)))\,,}$$

where ρ is the residual after the consumed binary symbol.

WRITE therefore does more than preserve the invariant.

It returns the physical stack to the exact minimum-demand normal form.

This result is the normalization analogue of the original endmarker materialization theorem, but now it occurs online.

37.8 Zero residual

Theorem 37.7 — Zero-residual correctness.

At the end of every binary prefix,

$$\boxed{\rho = 0 \iff (D_d, S) = (C, \varepsilon).}$$

Proof.

The reverse implication follows from the invariant.

For the forward direction, consider a nonempty processed prefix.

Immediately after the most recent WRITE, the machine was at

$$(C, \eta(\mu(r_0)))$$

for some residual r_0.

Write

$$\mu(r_0) = (x, y).$$

Since then, each MATCH or SEARCH step has popped exactly one pre-existing demand cell.

Suppose the intervening input contains:

$$u$$

zeros and

$$v$$

ones, so

$$u + v \le x + y.$$

If the current residual is zero, this segment completes r_0.

By minimality of $\mu(r_0)$, every completion requires length at least

$$x + y.$$

Hence

$$u + v = x + y.$$

The entire canonical stack has been consumed.

So

$$S = \varepsilon.$$

The residual invariant then gives

$$dm = 0,$$

hence

$$d = 0.$$

Therefore the state is C. □

This proof is important because acceptance is obtained from shortest-demand minimality, not from a large case analysis of reachable stacks.

37.9 Main construction theorem

Theorem 37.8 — Real-time centered rich DPDA.

For every coprime

$$p, q \geq 1,$$

the centered schema defines a deterministic pushdown machine with:

$$\boxed{|Q| = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1}$$

semantic core states and stack alphabet

$$\boxed{\{0, 1, Z_0\}.}$$

Every binary transition consumes exactly one symbol.

No ε-transition is used during binary processing.

With the mandatory right endmarker, the machine recognizes

$$\boxed{L_{p,q}.}$$

Furthermore:

1. every WRITE returns to the exact minimum-demand normal form;
2. every reachable stack has two-cell qualitative visibility;
3. the residual is exactly decoded by the state-plus-stack invariant.

The state count is

$$\boxed{|Q| = \begin{cases} p+q, & p+q \text{ odd}, \\ p+q-1, & p+q \text{ even}. \end{cases}}$$

This is the exact size of the centered construction.

This is a construction-size statement rather than an unrestricted state lower bound. The representation-independent state theorem is established in the later physical-complexity macro-Part.

37.10 Shortest-demand quotient

The centered machine is physically richer than a residual counter.

Two configurations can store the same semantic demand while distributing it differently between:

- physical stack;
- finite-control debt.

Corollary 37.9 — Shortest-demand quotient.

For reachable configurations, define

$$c_1 \sim_\Theta c_2 \iff \Theta(c_1) = \Theta(c_2).$$

Then the residual/minimum-demand transition system is a semantic quotient of the rich configuration graph.

The quotient forgets normalization chronology.

The physical fiber may remember it.

That observation will later become a central theorem family in the Uniqueness/Lift Theory.

For now, it is already visible concretely.

Chapter 38

What "Optimal" Means in the Deferred-SEARCH Architecture

The centered construction uses

$$2h_{\mathrm{ctr}} + 1$$

debt states, where

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor .$$

Is that number necessary?

The answer is yes—but only for a precisely stated physical interface.

This theorem must not be confused with the raw residual-faithful frontier of Chapter 8.

The two results optimize different representation categories.

38.1 Canonical deferred-SEARCH model

Definition 38.1 — Canonical deferred-SEARCH realization.

A realization belongs to the canonical deferred-SEARCH model if:

1. the normalized minimum demand
$$\mu(r)$$
is physically represented by the canonical word
$$\eta(\mu(r));$$

2. during a SEARCH episode, each binary transition consuming a pre-existing demand cell removes exactly one such cell;

3. a mismatch is stored only in finite control—no auxiliary debt marker is inserted into the remaining pre-existing word;

4. in the witness episode used below, the transition consuming the final pre-existing demand cell performs no materializing push and leaves the same bottom marker Z_0;

5. debt materialization is deferred to a later transition.

The last condition is what makes the model **deferred**.

An eager implementation lies outside this model.

38.2 Residual distinguishability

Lemma 38.2 — Residual distinguishability.

If two prefixes have distinct residuals

$$r_1 \neq r_2,$$

then a common binary continuation followed by the right endmarker distinguishes them.

Proof.

Choose a word z whose Parikh vector is

$$\mu(r_1).$$

Appending z sends the first residual to zero.

The second residual becomes

$$r_2 - r_1 \neq 0.$$

The right endmarker is accepted after the first prefix and rejected after the second. □

This is a language property.

It is not assumed by the architecture.

38.3 One witness containing enough of both symbols

Define

$$\boxed{\nu_{h_{\mathrm{ctr}}} = \begin{cases} (h_{\mathrm{ctr}}, h_{\mathrm{ctr}}), & m = p + q \text{ odd}, \\ (h_{\mathrm{ctr}}, h_{\mathrm{ctr}} + 1), & m \text{ even}. \end{cases}}$$

Its canonical word is

$$(10)^{h_{\mathrm{ctr}}}$$

in the odd case and

$$(10)^{h_{\mathrm{ctr}}}1$$

in the even case.

Lemma 38.3 — Canonical witness.

$$\boxed{\nu_{h_{\mathrm{ctr}}} = \mu(W(\nu_{h_{\mathrm{ctr}}})).}$$

Thus $\nu_{h_{\mathrm{ctr}}}$ is kernel-free and contains at least h_{ctr} zeros and h_{ctr} ones.

Proof.

If m is odd and a kernel packet

$$(q, p)$$

could be subtracted from $(h_{\mathrm{ctr}}, h_{\mathrm{ctr}})$, then

$$p \leq h_{\mathrm{ctr}}, \qquad q \leq h_{\mathrm{ctr}},$$

so

$$m = p + q \le 2h_{\mathrm{ctr}} = m - 1,$$

impossible.

If m is even, subtraction from $(h_{\mathrm{ctr}}, h_{\mathrm{ctr}} + 1)$ would require

$$q \le h_{\mathrm{ctr}}, \qquad p \le h_{\mathrm{ctr}} + 1,$$

giving again

$$m \le 2h_{\mathrm{ctr}} + 1 = m - 1.$$

Contradiction. □

This single canonical block contains enough of both symbol types to generate every debt in the centered interval.

38.4 Exhaust the same stack, leave different debts

Lemma 38.4 — Debt distinguishability witness.

Starting from the same canonical witness word $\eta(\nu_{h_{\mathrm{ctr}}})$, a deferred SEARCH episode can consume the entire pre-existing work stack while leaving any debt

$$\boxed{d \in \{-h_{\mathrm{ctr}}, -h_{\mathrm{ctr}} + 1, \ldots, h_{\mathrm{ctr}}\}.}$$

The resulting residual is

$$\boxed{dm.}$$

Proof.

For

$$d \ge 0,$$

choose exactly d witness zero-cells and mismatch them by reading 1 instead of 0.

Match every other pre-existing cell.

Debt increases monotonically from 0 to d.

For

$$d < 0,$$

mismatch exactly $|d|$ witness one-cells by reading 0.

Because the witness contains at least h_{ctr} copies of both symbols, every debt in the interval is possible.

Let the witness contain x zeros and y ones.

If A zero-cells are mismatched by input 1 and B one-cells are mismatched by input 0, then

$$d = A - B.$$

The consumed input contains

$$x - d$$

zeros and

$$y + d$$

ones.

Starting from residual

$$px - qy,$$

the final residual is

$$px - qy + q(y + d) - p(x - d) = d(p + q) = dm.$$

□

By the deferred-model assumption, after the last witness cell is consumed, every case has the **same physical stack**:

$$\boxed{Z_0.}$$

Only finite control can distinguish the residuals.

38.5 Exact deferred width

Theorem 38.5 — Exact deferred-SEARCH control width.

Every canonical deferred-SEARCH realization requires at least

$$\boxed{2h_{\mathrm{ctr}} + 1}$$

finite-control states.

The centered construction uses exactly $2h_{\mathrm{ctr}} + 1$.

Therefore

$$\boxed{|Q|_{\min}^{\mathrm{deferred}} = 2\left\lfloor \frac{p + q - 1}{2} \right\rfloor + 1}$$

or equivalently

$$\boxed{|Q|_{\min}^{\mathrm{deferred}} = \begin{cases} p + q, & p + q \text{ odd}, \\ p + q - 1, & p + q \text{ even}. \end{cases}}$$

Proof.

Lemma 38.4 produces

$$2h_{\mathrm{ctr}} + 1$$

different residuals:

$$-hm, \ldots, -m, 0, m, \ldots, hm.$$

In every case the work stack has been exhausted and the remaining physical stack is the same bottom marker Z_0.

If two distinct residuals shared the same finite-control state, the two instantaneous configurations would be physically identical.

But Lemma 38.2 gives a common continuation that must distinguish the residuals.

Determinism from identical configurations would force identical behavior.

Contradiction.
Thus the $2h_{\mathrm{ctr}} + 1$ residual debts require $2h_{\mathrm{ctr}} + 1$ distinct states.
The centered rail attains the bound. □

38.6 Scope — place this beside the theorem

The theorem does **not** say:

$$\boxed{\text{every DPDA recognizing } L_{p,q} \text{ needs } 2h_{\mathrm{ctr}} + 1 \text{ states.}}$$

A smaller machine may:

- abandon literal-demand storage;
- use another stack encoding;
- materialize debt eagerly;
- insert auxiliary physical markers;
- use another normalization schedule;
- leave the deferred-SEARCH interface entirely.

This scope is part of the theorem.
It is not a technical footnote.

38.7 Why Chapter 8 gives a different optimum

Chapter 8 proved the raw residual-faithful frontier:

$$S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}$$

With two work symbols, the raw residual graph can therefore be realized using only

$$\max(p, q)$$

states.
The centered deferred machine can require

$$p + q \quad \text{or} \quad p + q - 1.$$

There is no contradiction.

Raw residual-faithful question

> Give every residual one stable physical configuration and make every residual edge one raw input transition.

Deferred literal-demand question

> Store the actual canonical minimum-demand word, search it by pure pops, and keep displaced mismatch information only in finite control until later materialization.

These are different representation contracts.
The correct conclusion is:

resource optimality is representation-category relative.

This observation is more important than either number individually.
It is one of the direct motivations for the later Structure and Uniqueness theories.

38.8 Why this chapter belongs before the 2:1 recovery

The width theorem tells us what the centered architecture itself costs.

Only after its category has been made explicit should we specialize back to the historical 2:1 machine.

Otherwise the reader could incorrectly interpret the three-state historical architecture as globally state-optimal.

The next chapter will instead say something more precise and more interesting:

The general centered theory reconstructs the unusual architecture of the original machine.

And the remaining transition differences will turn out to be schedule choices.

Chapter 39

Closing the First Loop: Recovering the Original 2:1 Machine

The strongest test of a proposed generalization is not that it accepts the right family.

It is that, when specialized back to the motivating case, it explains the machine from which the theory arose.

Set

$$\boxed{p = 2, \qquad q = 1.}$$

Then

$$m = p + q = 3,$$

and

$$h_{\text{ctr}} = \left\lfloor \frac{3-1}{2} \right\rfloor = 1.$$

The centered rail has exactly three states:

$$D_{-1}, \qquad D_0, \qquad D_1.$$

Identify them as

$$\boxed{D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L.}$$

The state potentials become

$$\boxed{-3, \qquad 0, \qquad +3.}$$

These are exactly the state potentials discovered directly in the original DPDA.

The coincidence is not imposed.

It falls out of the general debt formula

$$\Phi(D_d) = d(p + q).$$

39.1 Center mismatch directions

From $C = D_0$:

- a mismatch $1/0$ moves to $D_1 = L$;
- a mismatch $0/1$ moves to $D_{-1} = R$.

Thus the general SEARCH rail reproduces the original side-state directions.
The original left state is the positive debt state.
The original right state is the negative debt state.
This gives the old L/R asymmetry a uniform arithmetic interpretation.

39.2 Left cancellation rail

At $L = D_1$, matching $0/0$ is simply MATCH:

$$\boxed{(L, 0T) \xrightarrow{0} (L, T).}$$

This is exactly the original cancellation transition that generated the entire Fuss–Catalan branch.

Thus the transition that later produced correction geometry is also a literal specialization of the general MATCH rule inside positive debt.

39.3 Right locality-repair rail

At $R = D_{-1}$, the general rules reproduce the right-side locality repair behavior.

The state is not an unrelated special-purpose gadget.

It is the negative SEARCH-debt sheet required when a one-cell blocker has been displaced from the canonical demand stack.

This is the general explanation of why the original machine needed a right side at all.

39.4 Historical WRITE arithmetic

The primitive kernel is

$$\boxed{K = (1, 2).}$$

The minimal kernel-carry WRITE equation now reproduces the characteristic replacement Parikh vectors of the historical table.

Depending on:

- source debt;
- target debt;
- input;
- removed top symbol;

the first nonnegative kernel shift produces the historical families represented by words such as

10,

11,

00,

000,

11111,

111111,

together with the corresponding bottom-WRITE forms.

The precise ordering is the selected canonical literal-demand order.

Thus the long WRITE strings are recovered from the general arithmetic compiler rather than copied into the generalization by hand.

39.5 Specialization and recovery theorem

Theorem 39.1 — 2:1 **specialization and recovery.**

Specializing the centered real-time p:q construction to

$$(p, q) = (2, 1)$$

recovers:

1. the three-state
$$C/L/R$$
skeleton;
2. the state potentials
$$0, +3, -3;$$
3. the center mismatch directions;
4. the left cancellation rail;
5. the right locality-repair rail;
6. the arithmetic of the historical WRITE words.

The only reachable transition families not reproduced literally are two right-state matching families.

Those differences are purely normalization-schedule differences.

This theorem closes the first large loop of the monograph:

original machine → general theory → original architecture reconstructed.

39.6 Lazy versus eager discharge

Consider the reachable configuration

$$(R, 0).$$

The centered specialization performs the lazy matching pop

$$\boxed{(R, 0) \xrightarrow{0}_{\text{lazy}} (R, \varepsilon).}$$

The historical machine instead immediately materializes the represented demand and moves to the center:

$$\boxed{(R, 0) \xrightarrow{0}_{\text{eager}} (C, 111).}$$

Both configurations represent the same minimum demand.

Likewise, for a unary right stack

$$(R, 1^k),$$

the centered specialization uses

$$\boxed{(R, 1^k) \xrightarrow{1}_{\text{lazy}} (R, 1^{k-1}),}$$

whereas the historical machine uses

$$\boxed{(R, 1^k) \xrightarrow{1}_{\text{eager}} (C, 1^{k+2}).}$$

Again, the semantic demand agrees.

Only the time at which virtual debt is rematerialized differs.

39.7 Canonical materialization map

Define

$$\boxed{\mathcal{N}(c) = (C, \eta(\Theta(c)))\,.}$$

This map forgets the physical distribution of demand between state and stack and returns the canonical center representation of the same shortest demand.

The historical eager successor is exactly the canonical materialization of the centered lazy successor on the two differing right-state families.

39.8 Eager–lazy semantic equivalence

Theorem 39.2 — Eager–lazy semantic equivalence.

Let

$$c_{\text{hist}}(w)$$

and

$$c_{\text{cent}}(w)$$

be the core configurations reached after the same binary prefix w by:

- the historical 2:1 machine;
- the centered 2:1 specialization.

Then

$$\boxed{\Theta(c_{\mathrm{hist}}(w)) = \Theta(c_{\mathrm{cent}}(w))}$$

for every prefix w.
On every transition family where the two tables differ,

$$\boxed{\delta_{\mathrm{hist}}(c, a) = \mathcal{N}\big(\delta_{\mathrm{cent}}(c, a)\big).}$$

Thus:

$$\boxed{\text{historical 2:1} = \text{eager discharge},}$$

while

$$\boxed{\text{centered 2:1} = \text{lazy discharge}.}$$

The semantics is the same.
The normalization chronology is different.

39.9 Historical machine as an eager refinement

Corollary 39.3 — Historical eager refinement.

Start with the centered 2:1 specialization.

Retain every specialized transition except on the two reachable right-state matching families.

On exactly those families, replace the lazy successor by its canonical materialization $\mathcal{N}$.

The resulting core is the historical 2:1 machine.

Therefore the return from the general p:q construction is exact at the level of:

- control architecture;
- state potentials;
- WRITE arithmetic;
- shortest-demand semantics;

and becomes literal transition-table identity after choosing the historical eager normalization schedule.

Direct specialization alone is not claimed to be literal table identity.

The remaining difference is precisely the eager/lazy schedule choice.

39.10 Computational audit

The centered family was independently checked by enumerating its reachable transitions for every ordered coprime pair

$$1 \le p, q \le 20,$$

including

$$(1, 1).$$

The exact finite enumeration covers:

$$\boxed{255 \text{ ratios},}$$

$$\boxed{1{,}909{,}191 \text{ distinct reached configurations},}$$

and

$$\boxed{3{,}738{,}864 \text{ binary edges.}}$$

It checked:

1. residual invariant;
2. residual-collision consistency;
3. two-cell qualitative visibility;
4. zero-residual correctness.

Result:

$$\boxed{\text{PASS.}}$$

These computations are consistency checks.
The symbolic theorems remain the proof.

39.11 The structural consequence of the $p{:}q$ generalization

The general machine has done more than extend 2:1.

It has exposed a structural nonuniqueness.

The semantic evolution is rigid:

$$\boxed{\text{one residual} \to \text{one minimum demand.}}$$

But the physical normalization chronology is not rigid:

$$\boxed{\text{same semantics} \not\Rightarrow \text{unique physical schedule.}}$$

This is the hinge of the remainder of the monograph.

The Structure Theory developed in the following Part explains why WRITE, SEARCH, and CANCEL are the primitive operational roles and separates the components forced by semantic arithmetic from those forced by one-top LIFO locality.

That is the beginning of the **Structure Theory**.

From this point onward the main sequence is uninterrupted:

$$\boxed{\text{Structure Theory} \to \text{Uniqueness/Lift Theory} \to \text{General Theory.}}$$

Part IV

Structural Pushdown Normalization: WSC, Lift Freedom, Causality, and Confluence

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Part guide. **Question.** Which features of a correct normalizer are forced by semantics, which by one-top LIFO locality, and which are merely choices of physical schedule? **Main result.** WRITE–SEARCH–CANCEL factorization separates the unique semantic quotient from nonunique physical lifts, classifies causal schedules, protected interfaces, bounded recodings, and typed confluence. **Why here.** This Part converts one family of machines into structural objects suitable for abstraction. **Dependency.** It starts from the general $p{:}q$ machine of Part III and provides the concrete model for the General Theory of Part V.

Chapter 40

Semantic Arithmetic Before Representation

Chapter 35 ended with a fact that the original 2:1 machine could not reveal by inspection alone:

the same semantic demand can be normalized on different physical schedules.

The centered generalization is lazy on two historical right-state families.

The original machine is eager.

Both are correct.

The next task is therefore no longer to construct a machine.

It is to separate what is forced by arithmetic from what is forced by LIFO locality and what is merely a normalization choice.

This is the beginning of the Structure Theory.

40.1 Dominating kernel level

Let a prefix contain

$$a = N_0(w), \qquad b = N_1(w).$$

The primitive balanced kernel is

$$K = (q, p).$$

Define

$$\boxed{\kappa(w) = \max\left(\left\lceil \frac{a}{q} \right\rceil, \left\lceil \frac{b}{p} \right\rceil\right).}$$

Theorem 40.1 — Dominating kernel level.

The unique minimum completion demand is

$$\boxed{\mu(w) = \big(q\kappa(w) - a,\ p\kappa(w) - b\big).}$$

Equivalently,

$$\boxed{(a, b) + \mu(w) = \kappa(w)K.}$$

Proof.

If (x, y) completes w, then

$$q(b + y) = p(a + x).$$

Coprimality gives an integer k such that

$$a + x = qk, \qquad b + y = pk.$$

Thus

$$x = qk - a, \qquad y = pk - b.$$

Nonnegativity requires

$$k \geq \left\lceil \frac{a}{q} \right\rceil$$

and

$$k \geq \left\lceil \frac{b}{p} \right\rceil.$$

The least feasible k is therefore $\kappa(w)$.

The completion length is

$$(p + q)k - (a + b),$$

which is strictly increasing in k.

Hence the first dominating kernel level gives the unique minimum demand. □

Geometrically, κ is the first point

$$kK = (kq, kp)$$

on the kernel ray that dominates the current count point (a, b) coordinatewise.

The minimum demand is the slack vector from the current count point to that first dominating kernel point.

40.2 One input symbol: CANCEL plus optional carry

Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Write

$$\mu_t$$

and

$$\kappa_t$$

for the minimum demand and dominating kernel level after t input symbols.

Theorem 40.2 — Unique semantic CANCEL–carry factorization.

If the next input symbol is

$$a \in \{0, 1\},$$

then

$$\boxed{\mu_{t+1} = \mu_t - e_a + \chi_t K,}$$

where

$$\boxed{\chi_t = \kappa_{t+1} - \kappa_t \in \{0, 1\}.}$$

Moreover:

- on input 0,
$$\chi_t = 1 \iff \mu_t^{(0)} = 0;$$
- on input 1,
$$\chi_t = 1 \iff \mu_t^{(1)} = 0.$$

Proof.

Reading one symbol increments exactly one prefix-count coordinate.
Each ceiling in the formula for κ therefore changes by at most one.
Hence

$$\chi_t \in \{0, 1\}.$$

Before the input,

$$(N_0, N_1) + \mu_t = \kappa_t K.$$

After the input,

$$(N_0, N_1) + e_a + \mu_{t+1} = \kappa_{t+1} K.$$

Subtracting gives the factorization.
For input 0, the zero slack is

$$q\kappa_t - N_0.$$

The kernel level rises precisely when that coordinate is already saturated, i.e. when the zero-demand coordinate is 0.
The input-1 case is symmetric. □
Thus every semantic input step has exactly two roles:

$$\boxed{\text{CANCEL one unit of the incoming symbol}}$$

and, if the relevant demand coordinate would become negative,

$$\boxed{\text{add one primitive kernel packet.}}$$

This is realization-independent.
It is the semantic reason WRITE exists at all.

40.3 Semantic WRITE count

Every semantic carry increments κ by one.

Corollary 40.3 — Semantic WRITE count.

For every prefix,

$$\boxed{W_{\mathrm{sem}}(w) = \kappa(w).}$$

If

$$w \in L_{p,q}$$

and

$$N_0(w) = qn, \qquad N_1(w) = pn,$$

then

$$\boxed{W_{\mathrm{sem}}(w) = n.}$$

The number of semantic kernel arrivals is fixed by the input.
What can vary is when those arrivals are physically materialized.

40.4 Demand-length clock

Let

$$r = p + q.$$

Corollary 40.4 — Demand-length clock.

At prefix length t,

$$\boxed{|\mu_t| = r\kappa_t - t.}$$

Hence

$$\boxed{|\mu_t| \equiv -t \pmod{r}.}$$

The minimum-demand length carries a rigid modular clock.
This clock will later have a physical analogue.

40.5 Semantic chronology is unique

Corollary 40.5 — Semantic chronology uniqueness.

For a fixed input word, the entire sequences

$$(\rho_t), \qquad (\kappa_t), \qquad (\mu_t), \qquad (\chi_t)$$

are uniquely determined by the input.
They do not depend on which correct pushdown representation is used.
This is the semantic backbone of the Structure Theory.
It says:

the semantic chronology is unique

before we say anything about:

- stack order;
- side states;
- SEARCH debt;
- eager/lazy materialization.

To see which part of this chronology survives one-top execution, we next impose an exact literal-demand lift and isolate the suffix obstruction created by LIFO locality.

Chapter 41

What One-Top LIFO Locality Forbids

We now formalize a physical realization of the unique demand dynamics.

The abstraction appears here, not at the beginning of the monograph, because the reader has already seen why it is needed.

41.1 Exact literal-demand lift

Let

$$\mathcal{M}_{p,q} = \{\mu(\rho) : \rho \in \mathbb{Z}\}$$

be the semantic minimum-demand system.

Definition 41.1 — Exact real-time literal-demand lift.

An exact literal-demand lift consists of:

- a reachable physical configuration set
$$\mathcal{C};$$
- deterministic binary transitions
$$\delta_a;$$
- a semantic projection
$$\Theta : \mathcal{C} \to \mathcal{M}_{p,q};$$
- a canonical section
$$\sigma : \mathcal{M}_{p,q} \to \mathcal{C};$$

such that

$$\boxed{\Theta \circ \sigma = \mathrm{id}}$$

and

$$\boxed{\Theta(\delta_a(c)) = U_a(\Theta(c)),}$$

where U_a is the unique semantic update from Chapter 36.

The section chooses one preferred physical representative of each semantic demand.

The other physical representatives form the semantic fiber.

That vocabulary will become central later.

41.2 Fixed block orientation

For the present Structure Theory, choose the literal canonical encoding

$$\boxed{\eta(x, y) = 1^y 0^x Z_0,}$$

with stack top on the left.

A one-top real-time transition has the form

$$\boxed{XT \longrightarrow \gamma T.}$$

The suffix T is immutable during that transition.

This one fact gives an exact direct-realizability criterion.

41.3 Top-rewrite suffix criterion

Theorem 41.2 — Top-rewrite suffix criterion.

Suppose a one-top transition begins from

$$XT.$$

A desired target word S^* can be reached in that one transition if and only if

$$\boxed{T \text{ is a suffix of } S^*.}$$

Proof.

Every one-top successor has the form

$$\gamma T.$$

Therefore direct realization requires T to be a suffix of the target.

Conversely, if

$$S^* = \gamma T,$$

replace X by γ. □

This theorem is one of the simplest and most important results in the entire structural line.

It converts one-top LIFO locality into an exact string condition.

41.4 Complete obstruction classification

Apply the suffix criterion to the canonical word

$$1^y 0^x Z_0.$$

Theorem 41.3 — Canonical direct-realizability classification.

For the fixed orientation $1^y 0^x Z_0$:

1. input 1 is directly canonical-realizable for every canonical demand;
2. input 0 fails to be directly canonical-realizable exactly in two families:

Buried-CANCEL obstruction

$$\boxed{x > 0, \qquad y > 0.}$$

Carry-order obstruction

$$\boxed{x = 0, \qquad y \geq 2, \qquad q \geq 2.}$$

For the reverse block orientation, exchange 0 and 1.

Proof — input 1. If $y > 0$, pop the top 1.
If $y = 0$ and $x > 0$, the semantic target is

$$1^{p-1}0^{x+q}Z_0.$$

The untouched suffix after removing the top zero is

$$0^{x-1}Z_0,$$

which is a suffix of the target.
Thus a single top replacement suffices.
The bottom case is direct as well.

Proof — input 0. If $y = 0$ and $x > 0$, pop the top 0.
Now let $x > 0, y > 0$.
The semantic target is

$$1^y0^{x-1}Z_0.$$

But after removing the top 1, the untouched suffix is

$$1^{y-1}0^xZ_0.$$

That suffix is not a suffix of the target.
Direct canonicalization is impossible.
Finally let $x = 0$.
The carry target is

$$1^{y+p}0^{q-1}Z_0.$$

For $y = 0$, write at the bottom.
For $y = 1$, the untouched suffix is empty and direct replacement is possible.
For $y \geq 2$, the untouched suffix is

$$1^{y-1}Z_0.$$

It is a suffix of the target exactly when

$$q - 1 = 0,$$

that is,

$$q = 1.$$

So for $q \geq 2$ the second obstruction family is exact. □

41.5 Two fundamentally different failures

The classification reveals that "cannot stay canonical" has two different causes.

Buried cancellation

The symbol that semantic CANCEL wants to remove exists in the physical word, but a blocker lies above it.

This is a visibility/locality failure.

Carry-order obstruction

A semantic carry can create the correct **number** of zeros and ones in one transition, yet the immutable suffix forces those letters into the wrong order.

This is not a quantity failure.

It is an ordering failure.

Thus:

$$\boxed{\text{count-exact} \not\Rightarrow \text{canonical.}}$$

This failure of direct canonicalization is the source of representation debt; the following analysis separates that debt into quantity, order, and finite-control channels.

Chapter 42

Quantity, Order, and Control Debt

A configuration may leave the canonical section in more than one way.

Parikh counts see only one of them.

The Structure Theory therefore distinguishes:

1. quantity debt;
2. order debt;
3. control/sheet debt.

These channels will remain separate through the Uniqueness and General Theory Parts.

42.1 Quantity defect

Let $S(c)$ be the literal work-stack word of a rich configuration c.

Define its physical Parikh vector

$$P(c) = \mathrm{Par}(S(c)).$$

Definition 42.1 — Quantity defect.

$$\boxed{Q(c) = \Theta(c) - P(c).}$$

A canonical configuration has

$$Q = 0.$$

But $Q = 0$ does not imply canonicality because Parikh counts forget literal order.

42.2 Universal representation conservation

Proposition 42.2 — Universal representation conservation.

For every exact transition on input a,

$$\boxed{\Delta P + \Delta Q = -e_a + \chi K.}$$

Proof.

Since

$$Q = \Theta - P,$$

$$\Delta Q = \Delta\Theta - \Delta P.$$

The semantic update is

$$\Delta\Theta = -e_a + \chi K.$$

Rearrange. □
This equation holds before we impose the later WSC lattice restrictions.
It is the universal accounting identity.

42.3 Order defect

For literal word

$$S = S_1 \cdots S_m,$$

define

$$\boxed{\Omega(S) = \#\{(i,j) : i < j,\ S_i = 0,\ S_j = 1\}.}$$

Lemma 42.3 — Block-order test.

$$\boxed{\Omega(S) = 0 \iff S \in 1^*0^*.}$$

Thus Ω measures departure from the chosen canonical block order.

42.4 Three-channel canonicality

A third channel is finite control.
A configuration can have:

- correct counts;
- correct literal order;

but still lie on a noncanonical control sheet.

Proposition 42.4 — Three-channel canonicality.
A rich configuration is canonical exactly when:

1. quantity defect is zero;
2. order defect is zero;
3. the control state is the canonical section state.

Thus:

$$\boxed{\text{canonicality} = \text{quantity} + \text{order} + \text{sheet/control}.}$$

This is the first mature answer to the question:

Where can normalization debt live?

Proposition 42.4A — Configuration-level separation of the three defect channels.

At the level of exact semantic representations used in this theory, none of the three canonicality channels is implied by the other two.

Each channel has a witness for which the other two vanish.

- **Quantity-only defect.** In the 2:1 schedule interaction studied later, the kernel-excess configuration

  $$(C, 1110)$$

  is compared with canonical

  $$(C, 1).$$

  The control sheet and block order are canonical, but the quantity defect is one nonzero kernel packet.

- **Order-only defect.** The carry-order repair of Theorem 42.6 preserves the exact canonical Parikh vector while forcing positive inversion count

  $$\Omega > 0.$$

- **Control-only defect.** Take any exact literal lift and adjoin a semantically ignored two-sheet control decoration. A balanced input can return the base machine to a canonical stack while leaving the auxiliary sheet noncanonical. Quantity and order are then canonical while control defect remains nonzero.

Therefore

$$\boxed{Q = 0, \quad \Omega = 0, \quad \text{canonical control}}$$

are genuinely independent physical requirements.

This point becomes important in the Uniqueness Theory: a count-shadow invariant alone cannot reconstruct the full physical semantic fiber.

42.5 Kernel economy

Define a literal-demand representation to be **kernel-economical** if

$$\boxed{|P(c)| \le |\Theta(c)|}$$

for every reachable configuration.

This prohibits adding complete zero-residual kernel packets merely as redundant workspace.

Under economy, the buried-CANCEL obstruction forces an exact quantity debt.

42.6 Forced unit SEARCH

Theorem 42.5 — Forced unit SEARCH for buried cancellation.

Assume:

- one-top real-time locality;
- kernel economy;

- canonical mixed demand
$$\Theta(c) = (x, y)$$
with
$$x > 0, \quad y > 0;$$
- literal encoding
$$1^y 0^x Z_0.$$

On input 0, the one-step successor has

$$\boxed{Q' = (-1, 1) = -(1, -1).}$$

Proof.
The semantic target is

$$(x - 1, y),$$

whose length is

$$x + y - 1.$$

A one-top transition can replace only the blocking top 1 by some word γ.
The successor physical length is

$$x + y - 1 + |\gamma|.$$

Kernel economy requires this to be at most the semantic target length.
Hence

$$|\gamma| = 0.$$

The top 1 must be popped.
The physical successor counts are

$$(x, y - 1),$$

while the semantic target is

$$(x - 1, y).$$

Therefore

$$Q' = (-1, 1).$$

□
Thus SEARCH is not merely one convenient implementation trick.
Under the stated locality and economy assumptions, the buried cancellation produces exactly one forced exchange defect.

42.7 Carry obstruction with zero quantity debt

The second obstruction behaves differently.

Assume:

$$x = 0, \qquad y \geq 2, \qquad q \geq 2.$$

On input 0, the semantic target is

$$1^{y+p}0^{q-1}Z_0.$$

A one-top transition can replace the top 1 by

$$\gamma = 1^{p+1}0^{q-1}.$$

The resulting word is

$$1^{p+1}0^{q-1}1^{y-1}Z_0.$$

Theorem 42.6 — Count-exact carry-order repair.

The successor has exactly the target Parikh counts:

$$\boxed{Q' = 0.}$$

But it has nonzero order defect.

Thus the representation can be semantically count-exact and still noncanonical.

42.8 Minimum order debt

Theorem 42.7 — Minimum carry-order debt.

Under the same hypotheses, among all one-top successors with exact target counts,

$$\boxed{\Omega_{\min} = (q-1)(y-1).}$$

Proof.

The untouched suffix is

$$1^{y-1}Z_0.$$

Exact target counts force the replacement word to contain:

$$q - 1$$

zeros and

$$p + 1$$

ones.

Every one of those $q - 1$ zeros lies before all $y - 1$ untouched suffix ones.

Therefore at least

$$(q-1)(y-1)$$

inversions are forced.

The replacement

$$1^{p+1}0^{q-1}$$

introduces no additional inversions inside the replacement and attains the bound. □
The obstruction/debt principle is now clear:

LIFO obstruction forces departure from the canonical section,

but the debt may be stored in:

- quantity;
- order;
- finite control;
- or several channels at once.

With the three debt channels separated, we can now freeze a count-shadow policy and ask which part of the physical arithmetic becomes rigid.

Chapter 43

The WSC Count Shadow

The preceding debt theory was general within the literal-demand setting.

We now impose an explicit representation policy to obtain a sharper arithmetic structure.

This policy is not claimed to describe every correct DPDA.

The scope is part of the theorem.

Set

$$\boxed{\mathbf{h} = (1, -1), \qquad K = (q, p), \qquad r = p + q.}$$

43.1 Phase coherence

Definition 43.1 — Phase coherence.

A reachable configuration is phase-coherent if

$$\boxed{|P(c)| \equiv |\Theta(c)| \pmod{r}.}$$

A realization is phase-coherent if every reachable configuration satisfies this congruence.

Since

$$Q = \Theta - P,$$

phase coherence is equivalent to

$$Q_0 + Q_1 \equiv 0 \pmod{r}.$$

This congruence produces the WSC lattice.

43.2 WSC defect lattice

Define

$$\boxed{\mathcal{D}_{p,q} = \mathbb{Z}\mathbf{h} + \mathbb{Z}K.}$$

Theorem 43.2 — WSC defect lattice.

$$\boxed{\mathcal{D}_{p,q} = \{(u, v) \in \mathbb{Z}^2 : u + v \equiv 0 \pmod{r}\}.}$$

Consequently every phase-coherent quantity defect has a unique decomposition

$$\boxed{Q = s\mathbf{h} + wK, \qquad s, w \in \mathbb{Z}.}$$

Proof.

Both generators have coordinate sum divisible by r:

$$(\mathbf{h})_0 + (\mathbf{h})_1 = 0,$$

$$K_0 + K_1 = r.$$

Conversely, if

$$u + v \equiv 0 \pmod{r},$$

define

$$\boxed{w = \frac{u+v}{r},}$$

and

$$\boxed{s = \frac{pu - qv}{r}.}$$

The first is integral by assumption.
Since

$$p \equiv -q \pmod{r},$$

the numerator of s is also divisible by r.
Direct substitution gives

$$(u, v) = s\mathbf{h} + wK.$$

Uniqueness follows because

$$\det(\mathbf{h}, K) = p + q = r \neq 0.$$

□

43.3 Defect coordinates

Define

$$\Phi(Q) = pQ_0 - qQ_1,$$

$$\Lambda(Q) = Q_0 + Q_1.$$

Corollary 43.3 — Defect coordinates.

For

$$Q = s\mathbf{h} + wK,$$

$$\boxed{\Phi(Q) = rs,}$$

$$\boxed{\Lambda(Q) = rw.}$$

Hence

$$\boxed{s = \frac{\Phi(Q)}{r}, \qquad w = \frac{\Lambda(Q)}{r}.}$$

The coordinate s measures the SEARCH/exchange sheet in the quantity shadow.
The coordinate w measures deferred kernel quantity.

43.4 Defect quotient

Corollary 43.4 — Defect quotient.

$$\boxed{\mathbb{Z}^2/\mathcal{D}_{p,q} \cong \mathbb{Z}_r.}$$

The homomorphism is

$$(u, v) \mapsto u + v \pmod{r}.$$

This quotient is a useful arithmetic shadow.
It is not the physical semantic fiber.

43.5 WSC lattice is not the full LIFO fiber

The map

$$S \mapsto \mathrm{Par}(S)$$

forgets stack order.
The quantity defect also forgets finite-control sheets.
Therefore two physical configurations can have the same coordinates

$$(s, w)$$

while having:

- different literal stack words;
- different control states;
- different future locality properties.

Thus:

$$\boxed{\text{WSC defect lattice} \neq \text{full physical representation fiber.}}$$

This warning is essential.
Chapter 43 onward will develop the missing lift/fiber theory.

43.6 Kernel economy in lattice coordinates

Since

$$|\Theta| - |P| = Q_0 + Q_1 = rw,$$

we get:

Corollary 43.5 — Kernel economy.

Within the phase-coherent class,

$$\boxed{\text{kernel economy} \iff w \geq 0.}$$

Thus the deferred-kernel coordinate acquires a direct nonnegative backlog interpretation under economy.

43.7 Abstract count-shadow fiber region

Suppose the semantic demand is

$$\Theta(c) = \mu = (x, y)$$

and

$$Q = s\mathbf{h} + wK.$$

Then

$$P = \mu - Q = (x, y) - s(1, -1) - w(q, p).$$

So

$$P_0 = x - s - qw,$$

$$P_1 = y + s - pw.$$

Proposition 43.6 — Count-shadow fiber region.

Physical nonnegativity is equivalent to

$$\boxed{pw - y \leq s \leq x - qw,}$$

with

$$\boxed{0 \leq w \leq \left\lfloor \frac{x+y}{r} \right\rfloor.}$$

For fixed w, the number of integer s-positions is

$$\boxed{x + y - rw + 1.}$$

The count-shadow fiber therefore has a finite triangular/layered shape over each semantic demand.

Again, this is an abelian shadow of the full physical fiber, not the complete fiber itself.

These coordinates determine how much kernel mass exists, but not when it is physically written. Materialization timing is therefore the remaining variable.

Chapter 44

Physical Materialization, Backlog, and WSC Factorization

The semantic carry schedule

$$\chi_t$$

is unique.
The physical machine may materialize those kernel packets at different times.
We now make that chronology explicit.
Let

$$Q_t = s_t h + w_t K$$

along a phase-coherent run.

44.1 Materialization level

Definition 44.1 — Physical materialization level.
Define

$$\boxed{\lambda_t = \kappa_t - w_t.}$$

The semantic system has generated κ_t kernel packets.
The quantity shadow still defers w_t packets.
Therefore λ_t is the number of kernel packets already materialized physically at the count-shadow level.

44.2 Physical length formula

Theorem 44.2 — Physical length formula.
At prefix length t,

$$\boxed{|P_t| = r\lambda_t - t.}$$

Proof.

$$|P_t| = |\Theta_t| - (Q_{t,0} + Q_{t,1}).$$

Chapter 36 gives

$$|\Theta_t| = r\kappa_t - t.$$

Chapter 39 gives

$$Q_{t,0} + Q_{t,1} = rw_t.$$

Subtract and use

$$\lambda_t = \kappa_t - w_t.$$

□

The physical stack length therefore has its own modular clock:

$$\boxed{|P_t| \equiv -t \pmod{r}.}$$

44.3 Materialization amount

Define

$$\boxed{M_t = \lambda_{t+1} - \lambda_t.}$$

Theorem 44.3 — Materialization amount from stack length.

For every phase-coherent one-top transition,

$$\boxed{M_t = \frac{\Delta|P_t| + 1}{r}.}$$

In particular,

$$\boxed{M_t \in \mathbb{N}.}$$

Proof.

Subtract the physical length formula at t and $t+1$:

$$\Delta|P_t| = rM_t - 1.$$

Thus the formula follows.

Because a one-top rewrite can delete at most the visible literal cell,

$$\Delta|P_t| \geq -1.$$

Hence

$$M_t \geq 0.$$

□

The number of kernel packets physically materialized by a transition can therefore be reconstructed from the physical stack-length change alone.

44.4 Backlog law

Theorem 44.4 — Backlog law.

$$\boxed{w_{t+1} - w_t = \chi_t - M_t.}$$

Equivalently, if

$$A_t = \kappa_t$$

is cumulative semantic arrival and

$$B_t = \lambda_t$$

is cumulative physical materialization, then

$$\boxed{w_t = A_t - B_t.}$$

This is a queue-like law:

$$\boxed{\text{backlog change} = \text{semantic arrivals} - \text{physical service.}}$$

The analogy is structural, not stochastic.

44.5 Canonical carry dichotomy

Assume phase coherence and kernel economy, and begin a transition from canonical quantity state

$$w_t = 0.$$

Theorem 44.5 — Canonical carry dichotomy.

If

$$\chi_t = 0,$$

then

$$\boxed{M_t = 0.}$$

If

$$\chi_t = 1,$$

then

$$\boxed{M_t \in \{0, 1\}.}$$

Thus a semantic carry is either:

$$\boxed{\text{eagerly materialized}}$$

or

$$\boxed{\text{deferred as exactly one kernel packet of backlog.}}$$

This is the structural generalization of the eager/lazy distinction discovered concretely in Chapter 35.

44.6 Unique physical WSC role factorization

Let

$$\alpha_t = s_{t+1} - s_t.$$

Theorem 44.6 — Unique physical WSC count-shadow factorization.
Every phase-coherent one-top transition on input a satisfies

$$\boxed{\Delta P = -e_a - \alpha_t \mathbf{h} + M_t K.}$$

The coefficients

$$\alpha_t \in \mathbb{Z}$$

and

$$M_t \in \mathbb{N}$$

are unique.

Proof.
The defect change is

$$\Delta Q = \alpha_t \mathbf{h} + (w_{t+1} - w_t)K.$$

Use the backlog law:

$$\Delta Q = \alpha_t \mathbf{h} + (\chi_t - M_t)K.$$

Insert this into the universal conservation identity

$$\Delta P + \Delta Q = -e_a + \chi_t K.$$

The $\chi_t K$ terms cancel:

$$\Delta P = -e_a - \alpha_t \mathbf{h} + M_t K.$$

Uniqueness follows from linear independence of $\mathbf{h}$ and K. □

Meaning

This is a **role decomposition**, not a syntactic transition taxonomy.
The terms mean:

$$-e_a$$

= semantic CANCEL forced by the input;

$$-\alpha \mathbf{h}$$

= SEARCH/exchange effect in the abelian quantity shadow;

$$MK$$

= physically materialized kernel packets.
One concrete PDA transition may combine several of these roles.

44.7 Reconstructing the charges

Corollary 44.7 — Reconstruction.

$$\boxed{M = \frac{\Delta|P| + 1}{r},}$$

and

$$\boxed{\alpha = -\frac{p(\Delta P_0 + e_{a,0}) - q(\Delta P_1 + e_{a,1})}{r}.}$$

Thus the WSC charges can be recovered from the physical count effect.

Corollary 44.7A — General physical phase clock.

Every phase-coherent run satisfies

$$\boxed{|P_t| \equiv -t \pmod{r}.}$$

This follows immediately from

$$|P_t| = r\lambda_t - t.$$

The semantic demand clock of Chapter 36 and the physical stack clock therefore have the same modulus $r = p + q$, even though the physical materialization schedule may be eager, lazy, or mixed.

44.8 Global materialization budget

Let an accepted word have

$$N_0 = qn, \qquad N_1 = pn.$$

Theorem 44.8 — Global materialization budget.

For a phase-coherent kernel-economical run starting canonically,

$$\boxed{\sum_t M_t = n.}$$

This equals the total semantic carry count:

$$\boxed{\sum_t \chi_t = n.}$$

Proof.

The accepted input length is

$$rn.$$

At acceptance, semantic demand is zero.

Kernel economy forces the physical literal count to zero as well.

The physical length formula gives

$$0 = r\lambda_{rn} - rn.$$

Hence

$$\lambda_{rn} = n.$$

Since $\lambda_0 = 0$,

$$\sum_t M_t = n.$$

The semantic equality follows from Chapter 36. □
The schedule may move materialization earlier or later.
The total materialized kernel budget cannot change.

44.9 Deferral capacity

Corollary 44.9 — Deferral capacity bound.

At prefix length t,

$$\boxed{\left\lceil \frac{t}{r} \right\rceil \le \lambda_t \le \kappa_t,}$$

and

$$\boxed{0 \le w_t \le \kappa_t - \left\lceil \frac{t}{r} \right\rceil .}$$

Thus the physical representation has an exact finite capacity for delaying kernel materialization at a given semantic prefix.

Once materialization timing has been separated from semantic count, the natural question is uniqueness: which coordinates belong to the semantic quotient, and which belong only to a chosen physical presentation?

Chapter 45

Unique Semantic Quotient, Nonunique Physical Presentation

The Structure Theory has now separated:

- the unique semantic demand trajectory;
- the physical normalization trajectory.

We can state that distinction as a quotient.

45.1 Semantic fibers

For semantic demand

$$m \in \mathcal{M}_{p,q},$$

define

$$\boxed{F_m = \Theta^{-1}(m).}$$

Two rich configurations are semantically equivalent when

$$\boxed{c \sim c' \iff \Theta(c) = \Theta(c').}$$

A fiber can contain several physical configurations representing the same minimum demand.

45.2 Fiber congruence

Theorem 45.1 — Fiber congruence and semantic quotient.

In every exact literal-demand lift, the relation $\sim$ is preserved by binary input transitions.

Hence

$$\boxed{\mathcal{C}/\sim \cong \mathcal{M}_{p,q}.}$$

Proof.

If

$$\Theta(c) = \Theta(c'),$$

then exactness gives

$$\Theta(\delta_a(c)) = U_a(\Theta(c)) = U_a(\Theta(c')) = \Theta(\delta_a(c')).$$

Thus successors remain in the same semantic fiber.

The quotient state is therefore exactly the common semantic demand, and the quotient transitions are the canonical demand updates. □

This is a concrete precursor of the machine-independent semantic quotient theorem developed below.

The stronger language-level statement will come in the Uniqueness Theory.

45.3 Canonical retraction

Define

$$\boxed{N = \sigma \circ \Theta.}$$

This sends every physical configuration to the canonical section representative of its semantic state.

Proposition 45.2 — Canonical absorption.

$$\boxed{N^2 = N.}$$

Moreover, for every input word w,

$$\boxed{N \circ \delta_w = N \circ \delta_w \circ N.}$$

Proof.

Since

$$\Theta \circ \sigma = \mathrm{id},$$

$$N^2 = \sigma\Theta\sigma\Theta = \sigma\Theta = N.$$

For one symbol,

$$N\delta_a = \sigma\Theta\delta_a = \sigma U_a\Theta.$$

Also,

$$N\delta_a N = \sigma\Theta\delta_a\sigma\Theta = \sigma U_a\Theta\sigma\Theta = \sigma U_a\Theta.$$

Induct on $|w|$. □

Interpretation:

Once we normalize to the canonical section, earlier physical normalization history is erased.

This is a structural form of the eager/lazy semantic equivalence.

45.4 What is actually unique

The following are semantic and unique:

- residual;
- minimum demand;
- dominating kernel level;
- semantic carry indicator;
- canonical quotient path.

The following need not be unique:

- physical stack word;
- finite-control sheet;
- timing of materialization;
- normalization schedule;
- full pushdown presentation.

Thus the correct statement is:

unique semantic core; nonunique physical normalization lift.

45.5 Cheap nonuniqueness already exists

Proposition 45.3 — Unrestricted physical presentation nonuniqueness.

The canonical semantic quotient does not determine a unique deterministic pushdown presentation.

Proof.

Start with any exact deterministic lift with finite control Q.

For integer

$$m \geq 2,$$

adjoin an independent cyclic coordinate

$$j \in \mathbb{Z}_m.$$

On every consumed input symbol, update

$$j \mapsto j + 1 \pmod{m}.$$

Let:

- stack dynamics;
- semantic projection;
- acceptance

ignore the extra coordinate.

The result is still an exact deterministic lift.

Its control set is

$$Q \times \mathbb{Z}_m.$$

For infinitely many suitable pairwise distinct m, these presentations have different finite-control cardinalities and are therefore non-isomorphic, while their semantic quotient is unchanged. □

This proposition proves physical presentation nonuniqueness.

But it is deliberately cheap.

It uses redundant control decoration.

It does **not** by itself distinguish genuine reduced presentations from redundant decoration. Chapters 58–59 supply the protected quotient/recoding framework that removes this ambiguity and proves existence of minimal protected cores. The conclusion used here is existence and minimality under the protected quotient/recoding framework, not a Church–Rosser assertion for arbitrary terminal protected cores.

45.6 Scope of the confluence claim

Canonical absorption says that normalizing after a fixed input word forgets earlier physical history.

It establishes endpoint coherence for the chosen normalizer. A Church–Rosser theorem for arbitrary rewrite schedules would be a different claim because the present framework fixes a deterministic normalizer rather than an arbitrary-choice rewrite relation.

Therefore the safe conclusion is:

canonical endpoint coherence under the chosen normalizer

not

global confluence.

This scope distinction is used immediately in the lift-comparison chapters below and again in the General Theory of Part V.

Chapter 46

The Original 2:1 Machine as WSC

The Structure Theory now returns to the original machine for a second reinterpretation.

Chapter 35 explained the historical transition table as an eager schedule of the centered 2:1 construction.

The present chapter goes deeper.

It interprets:

- the historical side states as SEARCH sheets;
- the historical stack-length changes as materialization bursts;
- an old transition-count identity as a global kernel budget law.

Set

$$\boxed{p = 2, \qquad q = 1, \qquad r = 3, \qquad K = (1, 2).}$$

46.1 Historical states as SEARCH sheets

The original state potentials were

$$\Phi(C) = 0,$$

$$\Phi(L) = 3,$$

$$\Phi(R) = -3.$$

Since the WSC sheet coordinate s contributes residual quantity

$$rs = 3s,$$

we obtain:

Corollary 46.1 — Historical states as SEARCH sheets.

$$\boxed{s(C) = 0, \qquad s(L) = 1, \qquad s(R) = -1.}$$

Thus the original $C/L/R$ architecture is exactly a three-sheet realization of the WSC exchange coordinate.

The side states are not merely "left" and "right."

They are the ± 1 SEARCH sheets.

46.2 Historical stack-length changes

For $r = 3$, the physical materialization formula is

$$\Delta|P| = 3M - 1.$$

Therefore:

$$M = 0 \Longrightarrow \Delta|P| = -1,$$

$$M = 1 \Longrightarrow \Delta|P| = +2,$$

$$M = 2 \Longrightarrow \Delta|P| = +5.$$

Corollary 46.2 — Historical stack actions as materialization bursts.

The characteristic historical stack-length changes

$$\boxed{-1, \qquad +2, \qquad +5}$$

are exactly the WSC materialization levels

$$\boxed{M = 0, \qquad M = 1, \qquad M = 2.}$$

Therefore a $+5$ transition is not an anomalously long push.

It is a two-kernel-packet materialization burst.

This explains why some original transition replacements were much longer than others.

46.3 The global identity $b + 2c = n$

Suppose an accepted 2:1 run has:

- b transitions with stack-length change $+2$;
- c transitions with stack-length change $+5$.

Those transitions carry materialization amounts:

$$1 \qquad \text{and} \qquad 2.$$

The global materialization budget says that an accepted word with

$$N_0 = n, \qquad N_1 = 2n$$

must satisfy

$$\sum_t M_t = n.$$

Hence:

Corollary 46.3 — Structural derivation of the historical identity.

$$\boxed{b + 2c = n.}$$

The identity is no longer a transition-table counting curiosity.

It is the 2:1 specialization of the universal materialization budget.

46.4 What Structure Theory has accomplished

The generalization and Structure Theory now give three successive interpretations of the same original machine.

Chapter 1

$$\boxed{\text{state+stack residual invariant.}}$$

Chapter 35

$$\boxed{C/L/R = \text{centered } p{:}q \text{ debt rail at } (2,1),}$$

with historical behavior as eager discharge.

Chapter 42

$$\boxed{C/L/R = \text{WSC SEARCH sheets,}}$$

and the physical stack actions are kernel-materialization bursts.
This repeated return to the original machine is intentional.
Each return explains something that was previously only observed.

46.5 The Structure Theory endpoint

The strongest conclusion is:

$$\boxed{\text{CANCEL and semantic WRITE are forced by arithmetic;}}$$

$$\boxed{\begin{array}{c}\text{SEARCH and normalization debt arise when that semantic update meets} \\ \text{one-top LIFO locality.}\end{array}}$$

The Structure Theory has shown that physical presentations are nonunique, but the examples so far do not yet separate genuine lift freedom from redundant decoration.
The next branch therefore asks the sharper question:

> After fixing the semantic quotient, the canonical section, and an admissible LIFO morphism category, can two **reduced** physical lifts still be genuinely different?

The answer is yes. The next four chapters make that statement precise: they derive the semantic core from the language, define exact sectioned lifts and category-relative minimality, isolate forced fiber branching, and exhibit genuinely distinct reduced 2:1 lifts. Only after that reduction step does the schedule enumeration begin.
That is the Uniqueness/Lift Theory.

Chapter 47

The Semantic Core Belongs to the Language

Research branch — Semantic Uniqueness and Physical Lift Freedom. *This branch develops the language-level semantic quotient, exact sectioned lifts, reduced-lift nonuniqueness, and causal schedule analysis. The terminology is kept separate from classical VPL congruence/minimization and DPDA equivalence results [2, 8, 21].*

The Structure Theory established a canonical semantic quotient inside exact literal-demand lifts.

The Uniqueness Theory strengthens that result.

The semantic core will now be derived from the **language itself**, not from a chosen WSC construction.

47.1 Right residuals

For a prefix u, define the right-continuation language

$$u^{-1}L_{p,q} = \{z : uz \in L_{p,q}\}.$$

Chapter 4 already proved that this continuation language depends only on the arithmetic residual

$$\rho(u) = qN_1(u) - pN_0(u).$$

We now promote that result to the canonical semantic theorem.

Theorem 47.1 — Right-residual classification.

For all prefixes u, v,

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).}$$

Every integer residual occurs.

Thus the reachable right-residual classes are canonically indexed by

$$\mathbb{Z}.$$

The semantic transition system is therefore

$$\boxed{\mathcal{R}_{p,q} : \qquad r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q.}$$

This is the language-determined behavioral core.

Corollary 47.1A — Canonical behavioral core.

The reachable right-residual classes of $L_{p,q}$ are canonically identified with

$$\boxed{\mathbb{Z}},$$

and their induced labeled transition system is exactly

$$\boxed{\mathcal{R}_{p,q}: \quad r \xrightarrow{0} r-p, \qquad r \xrightarrow{1} r+q.}$$

Thus the integer residual line is not a coordinate convention inherited from a particular DPDA. It is the canonical continuation-behavior core of the language.

47.2 Universal semantic quotient

Consider any correct deterministic real-time recognizer of $L_{p,q}$.

Look only at reachable configurations at input boundaries.

Map every such physical configuration c, reached by prefix u, to

$$\boxed{\Theta_{\mathrm{res}}(c) = \rho(u).}$$

Determinism and correctness make this map well defined on behavioral equivalence classes.

Theorem 47.2 — Universal semantic quotient.

Every correct deterministic real-time recognizer of $L_{p,q}$ admits a unique surjective transition-preserving quotient onto

$$\boxed{\mathcal{R}_{p,q}.}$$

Thus the residual system is not merely the semantic quotient of our chosen constructions.

It is the canonical behavioral quotient forced by the language.

This strengthens Chapter 41.

47.3 Minimum-demand semantics is a coordinate change

The minimum-demand system

$$\mathcal{M}_{p,q}$$

has one state

$$\mu(r)$$

for every residual r.

The map

$$\boxed{r \longmapsto \mu(r)}$$

is bijective.

It also intertwines the input transitions.

Theorem 47.3 — WSC semantic core equals the residual core.

$$\boxed{\mathcal{R}_{p,q} \cong \mathcal{M}_{p,q}.}$$

Therefore:

$$\boxed{\text{language semantics} = \text{right-residual system},}$$

while

$$\boxed{\text{completion semantics} = \text{a useful isomorphic coordinate system.}}$$

This distinction is conceptually important.

The minimum-demand representation was discovered first because it grew naturally out of the original DPDA.

But the language-level semantic object is the residual quotient.

47.4 Three levels of description

We can now separate:

Language level

$$\mathcal{R}_{p,q}.$$

This is canonical.

Semantic coordinate level

$$\mathcal{M}_{p,q}.$$

This is isomorphic to the residual system.

Physical lift level

A DPDA realization mapping onto that semantic system.

This level is not unique.

The rest of the Uniqueness Theory asks how to compare such lifts without quotienting away the very pushdown structure we want to study.

Chapter 48

Exact Sectioned Lifts and Category-Relative Minimality

If arbitrary behavioral quotienting is allowed, every correct machine collapses to the same residual core.

That is too coarse for physical comparison.

We need a category that remembers:

- the chosen semantic projection;
- the canonical section;
- the LIFO coordinates.

48.1 Reachable versus ambient configurations

Let

$$C^{\mathrm{amb}}$$

be all syntactically possible physical configurations.

Let

$$\boxed{C^{\mathrm{reach}} \subseteq C^{\mathrm{amb}}}$$

be the configurations actually reachable from the initial condition.

All minimality, fiber, and schedule claims in this branch concern reachable configurations unless explicitly stated otherwise.

This prevents unreachable syntactic artifacts from distorting the theory.

48.2 Exact sectioned lift

Definition 48.1 — Exact sectioned lift.

An exact sectioned lift is

$$\boxed{\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma)}$$

such that:

1. Θ is a surjective semantic projection;
2. σ is a reachable section;

3.
$$\Theta \circ \sigma = \mathrm{id};$$

4. physical transitions commute exactly with semantic transitions.

The section is part of the data.

Two lifts with the same quotient but different preferred physical representatives are therefore not automatically identified.

48.3 Section-preserving morphisms

A morphism

$$F : \mathcal{L}_1 \to \mathcal{L}_2$$

is section-preserving when

$$\boxed{\Theta_2 F = \Theta_1,}$$

$$\boxed{F\delta_a^{(1)} = \delta_a^{(2)} F,}$$

and

$$\boxed{F\sigma_1 = \sigma_2.}$$

For concrete pushdown comparison, a **coordinate LIFO morphism** must also respect the finite-control and stack coordinates letterwise rather than replacing the pushdown by an arbitrary behavioral quotient.

This is the correct category for asking whether two physical lifts are genuinely the same representation.

48.4 Minimality is category-relative

Proposition 48.2 — Category-relative minimality.

There is no meaningful representation-independent notion of a "minimal physical lift" without specifying:

- the admissible physical policy;
- the allowed morphisms;
- whether the canonical section is preserved;
- which stack/control coordinates may be identified.

The same semantic system can have different minima in different categories.

This is not a weakness of the theory.

It is the same lesson already encountered in the resource frontiers:

physical optimality is meaningful only relative to a representation category.

With the comparison category fixed, we can now ask what one-top locality forces inside one literal policy rather than across arbitrary presentations.

Chapter 49

When LIFO Locality Forces Fiber Branching

The Structure Theory proved the top-rewrite suffix criterion.

We now use it at the level of an entire observation fiber.

49.1 Uniform local rule

A deterministic one-top transition cannot inspect the hidden stack suffix.

Therefore all reachable configurations sharing the same locally visible cell must be handled by the same rule.

Theorem 49.1 — Uniform one-top rewrite criterion.

Fix one deterministic local observation class.

A single one-top rule can send every source in that class to its canonical semantic target if and only if the untouched suffix of every source is compatible with that source's canonical target under the suffix criterion.

This theorem is the fiberwise form of Chapter 37.

The obstruction is no longer one bad stack.

It is incompatibility across a family of sources that the local rule cannot distinguish.

49.2 Canonical-section closure dichotomy

Use the fixed literal section

$$\boxed{\sigma(x, y) = (C, 1^y 0^x Z_0).}$$

Theorem 49.2 — Canonical-section closure dichotomy.

The canonical literal section is transition-closed under one-top real-time updates if and only if

$$\boxed{(p, q) = (1, 1).}$$

Meaning. For the balanced one-to-one language, the literal section can remain canonical after every input step.

For every nontrivial coprime ratio, some exact semantic successor cannot be realized while staying on the canonical section.

Thus the use of noncanonical physical representatives is not merely a feature of our centered construction.

Inside this literal one-top policy, it is unavoidable.

49.3 Essential physical fiber branching

Corollary 49.3 — Essential branching.

For every nontrivial ratio

$$(p, q) \neq (1, 1),$$

every exact one-top realization of the fixed literal policy must reach some noncanonical physical configuration in the same semantic fiber as a canonical one.

Therefore:

nontrivial semantic fibers are forced physically

inside the policy.

This is a much stronger statement than:

> "our machine happens to use side states."

It says the canonical section itself cannot be closed.

49.4 Quotient-protected branching

Could one simply identify the noncanonical representative with the canonical one?

Not if the quotient must preserve legal one-top transitions in the same policy.

Corollary 49.4 — Quotient-protected branching.

The forced branch cannot be quotiented away while preserving:

- the fixed literal section;
- the one-top LIFO policy;
- exact transition behavior.

Thus some physical branching is structurally protected inside the representation category.

But the claim remains policy-relative.

Chapter 53 will show that a broader residual re-coordination can remove this branching entirely.

Chapter 50

Two Reduced 2:1 Lifts

We now revisit the eager and lazy 2:1 machines.

Chapter 35 proved semantic equivalence.

The Structure Theory explained their chronology.

The Uniqueness Theory asks whether one is merely a redundant presentation of the other.

The answer is no.

50.1 Historical eager lift

The historical machine eagerly materializes two reachable right-state matching families.

It returns immediately to the canonical center representation.

50.2 Centered lazy lift

The centered specialization leaves those same semantic debts on the right sheet and postpones materialization.

Both lifts use:

- the same semantic residual core;
- the same three control labels C, L, R;
- the same two work symbols;
- the same canonical section.

Their difference is temporal.

50.3 Section incomparability

Theorem 50.1 — Eager–lazy section incomparability.

There is no section-preserving transition morphism

$$\mathcal{L}_{\text{eager}} \to \mathcal{L}_{\text{lazy}}$$

and no such morphism

$$\mathcal{L}_{\text{lazy}} \to \mathcal{L}_{\text{eager}}.$$

The two lifts are therefore genuinely incomparable in the chosen category.

Their semantic quotient is identical.

Their physical transition structures are not.

50.4 Coordinate reducedness

Reachable witnesses force all three control states to remain distinct.

Likewise, the two work symbols cannot be merged without collapsing physically distinguishable reachable behavior.

Proposition 50.2 — Coordinate reducedness.

Both eager and lazy lifts are reduced inside the stated:

$$\boxed{\text{three-state / two-work-symbol coordinate category.}}$$

Thus their nonisomorphism is not caused by obvious redundant states or symbols.

50.5 Morphism-relative nonuniqueness

Corollary 50.3 — Reduced lift nonuniqueness.

The same semantic system admits multiple:

- coordinate-reduced;
- section-preserving;
- non-isomorphic

physical lifts.

Therefore:

$$\boxed{\begin{array}{c}\text{same semantics + same coordinate budget + reducedness} \\ \not\Rightarrow \text{ unique physical lift.}\end{array}}$$

This strengthens Chapter 41's cheap cyclic-decoration nonuniqueness into a substantive structural theorem.

These two reduced lifts are not isolated examples. The next chapter enumerates the complete selected three-site schedule family in which they sit.

There are exactly six.

Chapter 51

The Six 2:1 Normalization Schedules

The historical and centered machines differ at specific reachable sites.

To understand the space of schedule choices, isolate three eager-discharge sites.

51.1 Sites

Site A

Left-state 0/0 matching family.

Site B

Right-state, top 0, input 0 family.

Site C

Right-state, top 1, input 1 family.

A schedule is a bit vector

$$\boxed{(a, b, c) \in \{0, 1\}^3,}$$

where 1 means that the corresponding eager rule is enabled.

There are eight possible schedules.

Exactly six are admissible.

51.2 Exact schedule classification

Theorem 51.1 — Exact three-site schedule classification.

$$\boxed{\mathrm{Adm}_{2,1} = \{(a, b, c) \in \{0, 1\}^3 : ab = 0\}.}$$

Thus the six admissible schedules are

$$\boxed{000,\ 001,\ 010,\ 011,\ 100,\ 101.}$$

The forbidden schedules are

$$\boxed{110,\ 111.}$$

The unique minimal conflict is

$$\boxed{\{A, B\}.}$$

Site C is compatible with either side of that conflict.

51.3 Schedule geometry

Order schedules by inclusion of eager sites.

Corollary 51.2 — Schedule poset.

The admissible schedule poset is not a lattice.
Its two maximal elements are

$$\boxed{\{A, C\}}$$

and

$$\boxed{\{B, C\}.}$$

Their union

$$\{A, B, C\}$$

is inadmissible.
So no join exists inside the admissible poset.

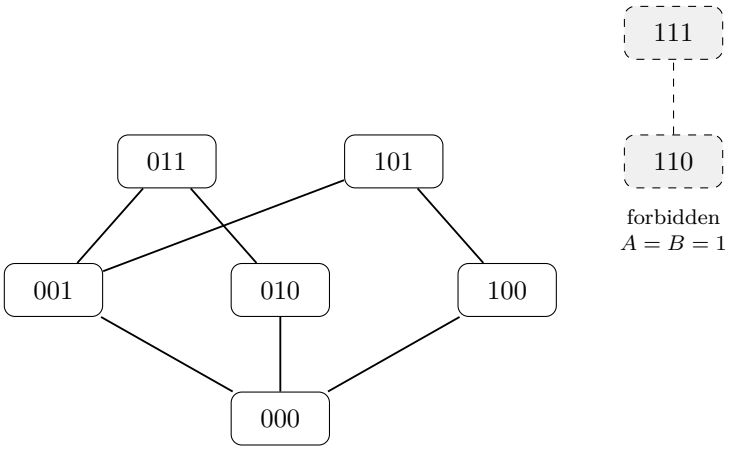


Figure 51.1: Hasse diagram of the six admissible 2:1 schedules. The schedule bits are ordered by inclusion of eager sites A, B, C. The two omitted cube vertices 110 and 111 are shown separately as forbidden because A and B cannot be simultaneously eager.

The missing 110 and 111 vertices should be shown separately as forbidden.

51.4 Six reduced lifts

Proposition 51.3 — Six coordinate-reduced schedule lifts.

All six admissible schedules define pairwise non-isomorphic coordinate-reduced lifts in the selected 2:1 category.

Thus schedule nonuniqueness is not merely eager versus lazy.

The physical realization space already contains a finite family of six reduced points.

51.5 Terminal normalization coherence

Let

$$\mathfrak{N}$$

be the stated terminal canonical normalizer.

Theorem 51.4 — Six-schedule terminal coherence.

For every admissible schedule and every common processed prefix, applying $\mathfrak{N}$ yields the same canonical semantic endpoint.

Thus all six systems have common normalized endpoint behavior.

Terminology boundary

This is **not** a Church–Rosser theorem.

The safe term is:

$$\boxed{\text{normalizer-relative terminal schedule coherence.}}$$

The six lifts are physically non-isomorphic.

They nevertheless agree after the specified terminal normalization.

Corollary 51.4A — Reduced non-isomorphism with common normalized behavior.

There exist coordinate-reduced, pairwise non-isomorphic physical lifts with identical terminal-normalization behavior on every processed prefix.

Thus even after obvious coordinate redundancy is removed,

$$\boxed{\text{common normalized semantics} \not\Rightarrow \text{unique reduced physical schedule.}}$$

Part V will abstract precisely this separation between a fixed semantic update and several physically admissible realizations.

Chapter 52

Why A Causes B

The schedule classification tells us that

$$A + B$$

is forbidden.

Its directionality is supplied by one of the clearest causal mechanisms in the corpus.

52.1 Local carry visibility

A fixed eager rule is chosen from a bounded local observation.

If two sources share that observation but require different semantic carry behavior, the same eager rule cannot be backlog-neutral on both.

Theorem 52.1 — Local carry visibility criterion.

A fixed eager local rule can remain backlog-neutral throughout an observation fiber only if the semantic carry indicator is constant on that fiber.

Thus carry aliasing is a local source of schedule failure.

52.2 Portability

For an eager rule r, define its portability set

$$\mathrm{Port}(r)$$

to be the physical source configurations on which applying that rule is safe relative to the selected canonical policy.

If F_o^α is the observation fiber exposed by schedule α, then:

Proposition 52.2 — Eager-rule portability criterion.

The eager rule is portable over that observation class exactly when

$$\boxed{F_o^\alpha \subseteq \mathrm{Port}(r).}$$

Schedule interactions therefore arise when enabling one rule changes which configurations another rule encounters.

52.3 A expands the B-fiber

Without A, site B sees only its baseline portable contexts.

Enabling A changes earlier normalization chronology.

That change exposes to B the infinite family

$$\boxed{(R, 0^m), \qquad m \geq 2.}$$

Lemma 52.3 — Schedule-induced fiber expansion.

Enabling A enlarges the first-hit observation fiber of B by the family $(R, 0^m)$, $m \geq 2$.

This is the causal mechanism.

A does not make B's local rule intrinsically different.

It changes which hidden suffixes reach the same local B-observation.

52.4 Direct canonicalization becomes impossible

For source

$$(R, 0^m)$$

on input 0, the canonical semantic target is

$$\boxed{10^{m-2}.}$$

A one-top rewrite must preserve the untouched suffix

$$\boxed{0^{m-1}.}$$

But 0^{m-1} is not a suffix of 10^{m-2}.

Theorem 52.4 — Schedule-induced direct-canonicality obstruction.

No one-top rule at site B can directly canonicalize every newly exposed source $(R, 0^m)$, $m \geq 2$.

This is the suffix-theorem diagnosis.

52.5 Forced kernel excess

Suppose the B-rule nevertheless insists on returning directly to the center while preserving residual.

Then it must create a nonminimal amount of literal demand.

Theorem 52.5 — Forced kernel-excess return.

Every such center-return repair introduces at least one extra kernel packet:

$$\boxed{(t+1)K}$$

relative to the minimum demand.

The historical replacement

$$\boxed{111}$$

realizes the minimum one-packet excess.

Thus the same conflict has three exact readings.

Corollary 52.6 — Three-way diagnosis.

Within this $A \to B$ family,

$$\boxed{\text{suffix failure}}$$

$$\Longleftrightarrow$$

$$\boxed{\text{carry aliasing}}$$

$$\Longleftrightarrow$$

$$\boxed{\text{forced kernel excess.}}$$

This triangle is important because the three notions are not globally identical in arbitrary representations.

They coincide here because of the specific one-top literal policy and source family.

The next chapter turns this concrete explanation into a directed causal relation.

Chapter 53

Directed Causal Scheduling

A minimal conflict such as

$$\{A, B\}$$

is set-theoretic.

It does not identify which rule caused the other to become unsafe.

The first-hit fiber expansion gives a directed notion.

53.1 First-hit exposure

For target site b and previously enabled schedule set H, let

$$\boxed{\mathcal{E}_b(H)}$$

be the first-hit exposure fiber: the configurations at which a computation first reaches site b under the schedule H.

A directed cause

$$H \rightsquigarrow b$$

occurs when H is minimal among schedule sets whose first-hit exposure contains a b-source outside the portability set of the b-rule.

Thus causality is about **newly exposed unsafe contexts**.

53.2 Exact 2:1 primary dependency

Theorem 53.1 — Exact primary dependency classification.

For the selected 2:1 three-site family,

$$\boxed{\mathcal{D}_{2,1}^{\mathrm{prim}} = \{\{A\} \rightsquigarrow B\}.}$$

So the directed causal graph is

$$\boxed{A \longrightarrow B \qquad \sqcup \qquad C.}$$

There is:

- no $B \to A$;
- no edge involving C.

Thus the forbidden pair $\{A, B\}$ has a unique primary direction.

Corollary 53.2 — Directed cause recovers the conflict.

Forgetting orientation gives the unique minimal schedule conflict:

$$\boxed{A \to B \quad \mapsto \quad \{A, B\}.}$$

53.3 Concrete causal certificate

One first-hit witness is

$$\boxed{1111100 \rightsquigarrow (R, 00).}$$

Applying the B-rule yields

$$\boxed{(C, 1110)}$$

instead of canonical

$$\boxed{(C, 1).}$$

The quantity excess is one kernel packet in the relevant sign convention:

$$\boxed{-K.}$$

Appending one more 1 reaches residual zero at the noncanonical configuration

$$\boxed{(C, 110).}$$

This gives a finite causal certificate:

1. enabling A creates a new first hit at B;
2. B's eager rule is not portable there;
3. the resulting debt is behaviorally exposed by a short continuation.

Proposition 53.2A — 2:1 causal certificate.

There exists a finite word, beginning with the witness prefix

$$\boxed{1111100,}$$

whose first use of the B-site after enabling A reaches

$$(R, 00)$$

and whose eager B-successor is the noncanonical kernel-excess configuration

$$(C, 1110).$$

A finite continuation then distinguishes this successor behaviorally from the canonical semantic successor.

Hence the directed dependency

$$\boxed{A \rightsquigarrow B}$$

admits a finite first-hit certificate.

This concrete witness is the prototype for the Fatal Frontier certificates of the General Theory.

53.4 Coordinate invariance

Suppose two lift presentations are related by a site-preserving coordinate LIFO isomorphism.

Then:

- first-hit exposure fibers correspond;
- portability sets correspond;
- directed causal hyperedges correspond.

Theorem 53.3 — Causal hypergraph invariance.

The directed causal schedule structure is invariant under site-preserving coordinate isomorphism.

Thus the $A \to B$ graph is not an artifact of renaming states or stack symbols inside the same coordinate category.

53.5 But semantics does not determine causality

A raw residual-conjugacy machine can realize the same language semantic quotient without any literal A, B, C normalization sites.

Theorem 53.4 — Semantic non-determination of causal structure.

The same semantic quotient can admit physical lifts with different causal schedule structures.

Therefore:

$$\boxed{\text{semantic quotient} \not\Rightarrow \text{causal schedule signature.}}$$

Corollary 53.5 — Coordinate/semantic separation principle.

A property can be invariant under coordinate isomorphism and still fail to be language-intrinsic.

Thus:

$$\boxed{\text{coordinate-intrinsic} \not\Rightarrow \text{semantic-intrinsic.}}$$

This distinction is central to the obstruction theorems that follow.

Chapter 54

Which Obstructions Are Really Forced?

The Uniqueness Theory has proved several kinds of necessity.

But necessity depends on the class of allowed representations.

This final chapter of the branch makes that dependence explicit.

54.1 Nested representation classes

Consider nested classes

$$\boxed{\mathfrak{C}_0 \subseteq \mathfrak{C}_1 \subseteq \mathfrak{C}_2.}$$

A typical interpretation is:

- $\mathfrak{C}_0$: a narrow coordinate category;
- $\mathfrak{C}_1$: a fixed literal-demand policy with more presentation freedom;
- $\mathfrak{C}_2$: a broader realization class allowing re-coordination.

For an obstruction $\mathcal{O}$, ask whether $\mathcal{O}$ is unavoidable in each class.

54.2 Obstruction monotonicity

Theorem 54.1 — Obstruction monotonicity.

If the realization class is enlarged, an essential obstruction can disappear but cannot newly become essential.

Thus:

$$\boxed{\mathrm{Ess}(\mathfrak{C}_2) \subseteq \mathrm{Ess}(\mathfrak{C}_1) \subseteq \mathrm{Ess}(\mathfrak{C}_0).}$$

For fully resolved three-level binary profiles, only

$$\boxed{111, \quad 110, \quad 100, \quad 000}$$

are possible.

54.3 Branching profile

For nontrivial semantic-fiber branching in the stated nested classes:

Theorem 54.2 — Policy-essential but broad-class removable branching.

$$\boxed{\Pi(\mathcal{O}_{\text{branch}}) = 110.}$$

Thus branching is:

- essential in the narrow coordinate class;
- essential in the fixed literal policy;
- removable in a broader class containing raw residual re-coordination.

This theorem connects the early raw DPDA constructions to the later lift theory.
The compact raw representation did not contradict forced branching.
It left the literal-demand policy in which branching was forced.

54.4 Category-relative status of $A \to B$

The specific causal obstruction has an exact category-relative profile once the schedule presentation is typed separately from the underlying lift.

Proposition 54.3 — Exact category bifurcation for the classical $A \to B$ edge.

In the site-flexible fixed-literal-policy presentation class, the optional menu may vary while the baseline literal-demand lift is held fixed. The named B site can then be omitted, so

$$\boxed{\Pi_{\text{flex}}(A \to B) = (1, 0, 0).}$$

If instead the designated A, B cells and the finite first-hit causal certificate are part of the preserved interface, the dependency persists:

$$\boxed{\Pi_{\text{cert}}(A \to B) = (1, 1, 0).}$$

Broad exact re-coordination can remove the named phenomenon in either case. Chapter 57 gives the full presentation typing and the arbitrary-coprime extension.

54.5 Representation–obstruction separation

Theorem 54.4 — Representation-obstruction separation.

Every theorem claiming that a physical obstruction is "forced" must specify the representation class in which the necessity is proved.

This is one of the main methodological conclusions of the project. It applies to state lower bounds, stack-symbol lower bounds, fiber branching, normalization debt, schedule conflicts, and causal edges.

54.6 Computational validation note

The associated semantic-uniqueness/lift validation study reports

$$\boxed{\text{458,776 assertion-level checks}},$$

including exhaustive words through length 14 for all eight schedules and the first common forbidden-schedule failure.

This historical count is recorded as a reported finite consistency check, not as a separately established theorem. The enumeration ranges and schedule conditions specify what was checked. The mathematical assertions above rest on their symbolic proofs and do not use the reported count as an assumption.

54.7 Transition to General Theory

The Uniqueness Theory has now produced a family of concepts that are no longer specific to weighted binary balance:

- semantic transition system;
- exact physical lift;
- canonical section;
- semantic fiber;
- bounded local observation;
- forced branching;
- portability;
- causal schedule structure;
- representation-relative obstruction.

Before the abstraction to General Theory, Part IV must close three scope questions created by the 2:1 analysis: whether the schedule geometry persists for arbitrary coprime ratios; which coordinate quotients and bounded recodings preserve the named physical interface; and what kind of confluence is actually justified at each normalization layer. The next structural branches answer these questions in turn.

Once those distinctions are fixed, Part V asks:

> Which of these notions survive for a general semantic system and a bounded-local physical normalization policy?

The abstract chain will then organize them into:

$$\text{locality} \to \text{debt} \to \text{behavioral safety} \to \text{causality} \to \text{synthesis} \to \text{verification}.$$

Chapter 55

From the Six 2:1 Schedules to the Arbitrary-Coprime Compiler

The preceding chapters isolated the six reduced schedules of the original 2:1 literal-demand lift and their directed causal relation. We now keep that same shortest-completion semantics but remove the special ratio. The result is a uniform compiler and a finite arithmetic site universe for every coprime p:q.

55.1 Centered literal-demand normalization

Put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad K = (q, p).$$

The centered literal-demand lift has finite-control debt states

$$D_{-h}, D_{-h+1}, \ldots, D_h$$

and a center state C. The semantic residual is represented by a canonical shortest nonnegative demand vector

$$\mu(r) = \arg\min\{x + y : x, y \geq 0,\ px - qy = r\},$$

and the chosen literal stack order is denoted by $\eta(x, y)$.

For debt state D_d define the virtual demand

$$V_d = \mu(dm).$$

A local work-top observation is a triple

$$o = (d, a, z), \qquad |d| \leq h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d, the next input letter is a, and the visible stack symbol is z.

Definition 55.1 (Schedule presentation)**.** A schedule presentation consists of a fixed mandatory centered baseline policy together with a finite optional-site menu and, for each site, a designated eager-return rule. A schedule is a subset of the menu; enabled optional rules may pre-empt the corresponding baseline transition.

Definition 55.2 (Admissibility and fatal supports)**.** A schedule H is *admissible* if the resulting deterministic machine recognizes exactly $L_{p,q}$. A support F is *fatal* if every schedule containing F is inadmissible, and is *minimal fatal* if no proper subset is fatal.

The distinction between a lift and a schedule presentation will be essential in Section 57.2: the physical transition system does not determine which optional sites are named or offered.

55.2 The uniform eager-return compiler

Let $e_0 = (1, 0)$ and $e_1 = (0, 1)$. For an observation $o = (d, a, z)$ define

$$B_o = e_z - e_a + V_d$$

and

$$\kappa(o) = \max\left\{\left\lceil -\frac{(B_o)_0}{q}\right\rceil, \left\lceil -\frac{(B_o)_1}{p}\right\rceil\right\}.$$

Set

$$R_o = B_o + \kappa(o)K.$$

The associated local eager rule is

$$E_o : (D_d, zT) \xrightarrow{a} (C, \eta(R_o)T).$$

Theorem 55.3 (Uniform centered eager-return compiler)**.** *For every local observation o, the vector R_o is the unique shortest nonnegative replacement vector that preserves the residual invariant and returns directly to the center. Hence E_o is the canonical direct eager-return candidate at o.*

Proof. Any direct return to the center must compensate the change in the visible stack cell and the consumed input, hence its replacement vector lies in the affine lattice

$$B_o + \mathbb{Z}K.$$

The only admissible replacements are nonnegative vectors. Adding one copy of $K = (q, p)$ increases both coordinates, so the smallest integer carry making both coordinates nonnegative is exactly $\kappa(o)$. Any smaller carry has a negative coordinate, while any larger carry increases total length by $p + q$. Therefore R_o is feasible and uniquely length-minimal. □

This formula recovers, for example, the six explicit $3 : 2$ eager words used in the finite stress test. The important point here is not those special values but the existence of one ratio-uniform compiler from which the whole site universe can be derived.

55.2.1 The mandatory reset envelope

Write

$$\mu((h+1)m) = (x_+, y_+), \qquad \mu(-(h+1)m) = (x_-, y_-).$$

Define

$$\begin{aligned}\mathcal{R}^0_{p,q} = &\{\mu(jm+q) : -h-1 \le j \le h\}\\ &\cup \{(x_+ + n, y_+) : n \ge 0\}\\ &\cup \{(x_-, y_- + n) : n \ge 0\}.\end{aligned}$$

Lemma 55.4 (Mandatory-WRITE reset envelope)**.** *Every mandatory WRITE in the centered $p : q$ baseline lands at a canonical center pair belonging to $\mathcal{R}^0_{p,q}$.*

Proof. A bottom WRITE from D_d on input 1 produces $\mu(dm + q)$, whereas input 0 produces

$$\mu(dm - p) = \mu((d-1)m + q),$$

which gives the finite seed range $-h - 1 \le j \le h$. At the positive outer boundary, the untouched suffix is a unary 0-tail, so the reset ray is $(x_+ + n, y_+)$. The negative side is symmetric. □

Definition 55.5 (Baseline-portable site)**.** A baseline episode starts at a center configuration $(C, \eta(x, y))$ with $(x, y) \in \mathcal{R}^0_{p,q}$, follows only mandatory MATCH/SEARCH transitions, and stops before the next mandatory bottom or outer WRITE. A compiler site is *baseline-portable* if its eager rule lands on the canonical center representative for every source exposed in every baseline episode.

Baseline portability is deliberately weaker than singleton admissibility: an eager reset can be safe on all baseline sources and later become unsafe on a source created by its own earlier activation. This self-exposure mechanism is responsible for the extra singleton-fatal vertices below.

55.3 The finite baseline-portable site universe

Assume first that $p \geq q \geq 2$. For $j \geq 0$, the bottom-seed formula is

$$\mu(jm + q) = (j + q, p - j - 1).$$

For $j = -t < 0$, write $t = kq + s$, $0 \leq s < q$; then

$$\mu(-tm + q) = \begin{cases} (0, km - 1), & s = 0, \\ (q - s, p + km + s - 1), & 1 \leq s < q. \end{cases}$$

For $1 \leq d \leq h$,

$$V_d = (d + q, p - d).$$

The three nontrivial positive-debt compiler count vectors are

$$M_d = (d + q, p - d), \qquad P_d = (d + q + 1, p - d - 1), \qquad N_d = (d + q - 1, p - d + 1),$$

with imbalances

$$\Delta(M_d) = 2d + q - p,$$
$$\Delta(P_d) = 2d + q - p + 2,$$
$$\Delta(N_d) = 2d + q - p - 2.$$

Negative-debt compiler words are one-heavy except for the trivial forced return that is already part of the baseline.

A central reset word excludes almost every interior observation. If m is odd, the diagnostic seed is (h, h) with word $(10)^h$; if m is even, it is $(h, h + 1)$ with word $(10)^h 1$. Tracking untouched suffixes shows that all negative-interior candidates fail portability and leaves at most two positive-interior survivors.

Theorem 55.6 (Eight-Site Theorem)**.** *Let $p \geq q \geq 2$ be coprime and let $h = \lfloor (p + q - 1)/2 \rfloor$.*

If $p + q$ is odd, the baseline-portable site universe consists exactly of the six core sites

$$(-h, 0, 0),\ (-h, 1, 0),\ (-h, 1, 1),\ (h - 1, 0, 0),\ (h - 1, 1, 0),\ (h, 0, 0),$$

together with

$$X = (h - q + 1, 0, 1) \quad \Longleftrightarrow \quad p > 5q - 5,$$

and

$$Y = (h - q, 1, 1) \quad \Longleftrightarrow \quad p > 5q - 1.$$

If $p + q$ is even, the universe consists exactly of the four core sites

$$(-h, 1, 1),\ (h, 0, 0),\ (h, 0, 1),\ (h, 1, 1),$$

together with

$$X = (h - q + 1, 0, 0) \quad \Longleftrightarrow \quad p > 5q - 5,$$

and

$$Y = (h - q, 1, 0) \quad \Longleftrightarrow \quad p > 5q.$$

In particular,

$$|\mathcal{S}^{\mathrm{bp}}_{p,q}| \leq 8.$$

For $q > p \geq 2$, the complete list is obtained by the physical duality

$$(p, q, d, a, z) \longleftrightarrow (q, p, -d, 1 - a, 1 - z).$$

Proof. The proof is an exact arithmetic elimination. The central diagnostic removes every negative-interior observation and all but two positive-interior observations; exact boundary fibers give the core; the two survivors are decided by primitive-kernel minimality. All formulas, including the negative-seed decomposition and the parity-specific threshold simplifications, are given in Section 56.4. □

Chapter 56

Exact Causal-Hypergraph Classification for All Coprime Ratios

The compiler reduces schedule admissibility to a finite obstruction problem. The classification separates the stable high-minimum regime from the two low-minimum lines where finite fork phenomena survive.

56.1 The high-minimum regime: every core schedule is safe

Let $\mathrm{Core}_{p,q}$ denote the parity-dependent core in Theorem 55.6. For a named site A, write $\mathcal{E}_A$ for the set of visible source stack words that can occur at that site during an episode under the chosen core schedule. These source envelopes are physical reachability sets, not new semantic states.

Theorem 56.1 (Core downward closure)**.** *If* $\min(p,q) \geq 3$*, then every subset* $H \subseteq \mathrm{Core}_{p,q}$ *is admissible. Equivalently,*

$$2^{\mathrm{Core}_{p,q}} \subseteq \mathrm{Adm}_{p,q}\,.$$

No minimal fatal support is contained entirely in the core.

Proof. By duality assume $p \geq q \geq 3$. Induct over reset episodes. Every enabled core activation returns to a canonical minimum-demand word. It therefore suffices to show that the source fibers of core sites stay within protected unary envelopes after any sequence of core activations.

If $m = 2h+1$, label the six core sites

$$A = (-h,0,0),\ B = (-h,1,0),\ C = (-h,1,1),$$

$$D = (h-1,0,0),\ E = (h-1,1,0),\ F = (h,0,0).$$

Uniformly over all core schedules,

$$\mathcal{E}_A = \mathcal{E}_B = \{0\}, \qquad \mathcal{E}_C \subseteq 1^+,$$

and

$$\mathcal{E}_D, \mathcal{E}_E, \mathcal{E}_F \subseteq 0^+.$$

Negative-core resets contain fewer than $q \leq h$ zeros, so they cannot create a new negative-boundary top-0 source. Positive-core outputs have one-count at most $h-1$, so their mixed

part is exhausted before the next positive-boundary source. The eager compiler has the same tail orientation as the protected source, and kernel minimality is preserved.

If $m = 2h + 2$, the four core sites satisfy

$$\mathcal{E}_A \subseteq 1^+, \qquad \mathcal{E}_B \subseteq 0^+, \qquad \mathcal{E}_C = \mathcal{E}_D = \{1\}.$$

The same minority-count argument prevents a negative reset from generating a new positive-boundary source and prevents a positive reset from generating an additional top-1 source at D_h. Hence every optional activation is canonical and kernel-reduced. The residual invariant and the baseline terminal argument then imply correctness. □

56.2 Extra-site self-exposure for $\min(p, q) \geq 3$

The extra vertices X, Y are baseline-portable precisely because their first activation stays below the primitive kernel threshold. Repeating the same activation increases a hidden alternating prefix until one full kernel packet becomes trapped.

For odd $m = 2h + 1$, define

$$S_k^X = (10)^{k(q-1)}1^{2q-2}, \qquad S_k^Y = (10)^{kq}1^{2q}.$$

For even $m = 2h + 2$, define

$$S_k^X = 0(10)^{k(q-1)}1^{2q-3}, \qquad S_k^Y = 0(10)^{kq}1^{2q-1}.$$

The first source S_0 is reached by an explicit monotone baseline prefix. Between consecutive X-activations one reads $1^{p-q+1}0$; between consecutive Y-activations one reads 1^{p-q}. Thus the alternating exponent grows by $q - 1$ or q respectively.

Lemma 56.2 (One-kernel trapping)**.** *Let*

$$k_X = \left\lceil \frac{p - (h + 2q - 2)}{q - 1} \right\rceil, \qquad k_Y = \left\lceil \frac{p - (h + 2q)}{q} \right\rceil.$$

At the first nonminimal repeated activation, the physical target differs from the canonical minimum demand by exactly one primitive kernel packet $K = (q, p)$.

Proof. At the fatal activation the one-count is $p + e$ with $0 \leq e \leq q - 1$ (and $e \leq q - 2$ in the X case), while the zero-count is at least q. Subtracting one K therefore remains nonnegative, while subtracting two cannot. Thus the target is exactly $\mu(r) + K$. □

Feeding the shortest semantic completion of r removes the minimal component and leaves exactly one balanced kernel packet physically present. The final semantic residual is zero but the machine is not in the balanced accepting configuration.

Theorem 56.3 (Complete schedule classification for $\min(p, q) \geq 3$)**.** *For every coprime p, q with $\min(p, q) \geq 3$,*

$$\boxed{\mathrm{Adm}_{p,q} = 2^{\mathrm{Core}_{p,q}}.}$$

Every present extra site is an upward-fatal singleton conflict, and these are the only minimal fatal supports.

Proof. Every core subset is admissible by Theorem 56.1. The Y self-exposure witness meets no other optional site, hence every schedule containing Y is bad. The X witness meets no core site and at most the neighboring Y observation; if Y is disabled it proves X fatal, while if Y is enabled the Y witness already proves failure. Thus every schedule containing either extra is inadmissible. □

56.3 Low-minimum regimes and arbitrary-coprime classification

56.3.1 The line $\min(p, q) = 2$

By duality it suffices to take $q = 2$, so $p \geq 3$ is odd. The six-site core has exactly two minimal fatal supports,

$$\{A, D\}, \qquad \{B, D\}.$$

The two possible extra sites are

$$X_1 = (h - 1, 0, 1) \quad (p \geq 7),$$
$$X_2 = (h - 2, 1, 1) \quad (p \geq 11).$$

and both are singleton-fatal.

Theorem 56.4 (Exact $\min(p, q) = 2$ classification)**.** *For $q = 2$,*

$$\mathrm{Adm}_{p,2} = \{H \subseteq \mathcal{S}^{\mathrm{bp}}_{p,2} : X_1, X_2 \notin H,\ \{A, D\} \not\subseteq H,\ \{B, D\} \not\subseteq H\},$$

with absent extras omitted. Exactly 40 *schedules are admissible.*

Proof. Section 56.5.1 gives a self-contained reset-closure proof of sufficiency and explicit robust witnesses for the two fork pairs $\{A, D\}$ and $\{B, D\}$; the six-site core therefore has exactly 40 safe schedules. Each present extra site is an upward-fatal singleton by the one-kernel trapping certificate, so every admissible expanded schedule excludes all extras. Conversely, after the extras are excluded the machine is exactly one of the safe core schedules. Thus the admissible count remains 40 independently of how many extras are present. □

56.3.2 The exceptional line $q = 1$

For $q = 1$, negative minimum demands are unary, so the negative corridor has no order freedom. The complete baseline-portable site universe is exceptional.

For $p = 1$ it is empty. For $p = 2$ the four sites are

$$X = (0, 1, 0), \quad A = (-1, 1, 1), \quad B = (-1, 0, 0), \quad G = (1, 0, 0).$$

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \qquad D = (h, 1, 1), \qquad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h - 1, 1, 0), \qquad C = (h, 0, 1).$$

For even $p = 2h \geq 4$, the sites

$$A = (-h, 1, 1),\ B = (-h, 0, 0),\ C = (-h, 1, 0),$$

$$E = (h - 1, 0, 0),\ F = (h - 1, 1, 0),\ G = (h, 0, 0)$$

are always present, and $D = (h - 1, 1, 1)$ appears for $p \geq 6$.

Theorem 56.5 (Exact $q = 1$ schedules)**.** *Up to physical duality:*

1. $(p, q) = (1, 1)$*: the site universe is empty and there is one admissible schedule.*

2. *$(2,1)$: the minimal fatal supports are $\{B,X\}$ and $\{B,G\}$; exactly* 10 *schedules are admissible.*

3. *odd $p \geq 3$:*
$$\mathrm{Adm}_{p,1} = \{\varnothing, \{A\}\}.$$

4. *even $p \geq 4$: the maximal safe sectors are*
$$\mathcal{N} = \{A,B,C\}, \qquad \mathcal{P} = \{A,F,G\},$$
while D and E are singleton-fatal when present and the cross-pairs
$$\{B,F\}, \{B,G\}, \{C,F\}, \{C,G\}$$
are fatal. Exactly 14 *schedules are admissible.*

Proof. The site-universe elimination and the balanced fatal certificates are recorded in Section 56.5. The safe cases are proved by episode-closed reset families: for odd p, only the unary negative ray generated by A is added; for even p, every safe schedule lies in one of the two closed sectors $2^{\mathcal{N}}$ or $2^{\mathcal{P}}$. The listed singleton and cross-pair witnesses trap a primitive kernel packet at residual zero. The small case $(2,1)$ is verified by its four-site transition geometry, with explicit witnesses 111001 and 111110001. □

Theorem 56.6 (Arbitrary-coprime causal-hypergraph classification)**.** *For every coprime positive pair (p,q), the canonical baseline-portable weighted-balance schedule family has an explicit finite set of minimal fatal supports given by Theorems 56.3, the* min $= 2$ *theorem, and Theorem 56.5, together with physical duality. Consequently*
$$H \in \mathrm{Adm}_{p,q}, \quad H' \subseteq H \quad \Longrightarrow \quad H' \in \mathrm{Adm}_{p,q}\,.$$
Equivalently, $\mathrm{Adm}_{p,q}$ *is the independence complex of the exact fatal-support hypergraph, in the standard simplicial-complex sense [48].*

56.4 Arithmetic details for the Eight-Site Theorem

This appendix records the elimination argument behind Theorem 55.6; it is included so that the arbitrary-coprime vertex classification can be checked without relying on a finite search.

Put $m = p+q$ and $h = \lfloor (m-1)/2 \rfloor$, and assume $p \geq q \geq 2$. For the bottom-WRITE seed family $B_j = \mu(jm+q)$ one has, for $j \geq 0$,
$$B_j = (j+q, p-j-1).$$
For $j = -t < 0$, write $t = kq+s$, $0 \leq s < q$. Then
$$B_{-t} = \begin{cases} (0, km-1), & s = 0, \\ (q-s, p+km+s-1), & 1 \leq s < q. \end{cases}$$
For $1 \leq d \leq h$,
$$V_d = \mu(dm) = (d+q, p-d).$$
Thus the positive-debt compiler count vectors are
$$M_d = (d+q, p-d), \quad P_d = (d+q+1, p-d-1), \quad N_d = (d+q-1, p-d+1),$$
with imbalances
$$2d+q-p, \qquad 2d+q-p+2, \qquad 2d+q-p-2.$$
Every nontrivial negative-debt compiler word is one-heavy; the only exception is the forced $d = -1$, input/top 1/0 return already contained in the baseline.

Lemma 56.7 (Central diagnostic exclusion)**.** *For $p \geq q \geq 2$, no negative-interior observation is baseline-portable. If m is odd, the only possible positive-interior survivors are*

$$(h - q + 1, 0, 1), \qquad (h - q, 1, 1),$$

and if m is even the only possible positive-interior survivors are

$$(h - q + 1, 0, 0), \qquad (h - q, 1, 0).$$

Proof. If $m = 2h + 1$, the seed $B_{h-q} = (h, h)$ has canonical word $(10)^h$. At an interior debt d, the untouched suffix is $(10)^{h-d-1}$ under a visible top 0 and $0(10)^{h-d-1}$ under a visible top 1. A top-0 eager concatenation is canonical only when its compiler word is balanced; all relevant interior imbalances are odd. A top-1 concatenation is canonical only when the compiler has exactly one excess 1, which gives precisely the two displayed equations for d.

If $m = 2h + 2$, the central seed is $(h, h + 1)$ with word $(10)^h1$. The corresponding suffixes are $(10)^{h-d-1}1$ and $0(10)^{h-d-1}1$. A top-0 candidate must have zero imbalance, while the parity of the positive-debt compiler imbalances excludes every top-1 candidate. Solving the zero-imbalance equations gives the two even-parity survivors. For negative debt, the compiler has a terminal 1-tail whereas the central diagnostic exposes an untouched 0 beyond the visible cell; the eager target is therefore not in canonical alternating-tail form. This excludes all negative-interior sites. □

Lemma 56.8 (Boundary fibers)**.** *The always-present boundary sites are exactly the parity-dependent cores in Theorem 55.6.*

Proof. For odd m, the negative boundary has a singleton top-0 source and unary top-1 sources; at $d = h - 1$ every top-0 source is unary, and at $d = h$ every source is unary top-0. A source 11 at $d = h - 1$ eliminates the remaining top-1 candidates. For even m, the negative boundary has unary top-1 sources but also the diagnostic source 00, while at $d = h$ the top-0 sources are unary and the only top-1 source is 1; the central sources 01 and 101 eliminate the remaining $d = h - 1$ candidates. Direct substitution into the compiler gives exactly the core lists stated in Theorem 55.6. □

For the two odd-parity survivors the common compiler word is $(10)^h1$. Their exposed source one-counts are bounded by $2q - 2$ and $2q$, with equality attained by baseline sources. Kernel reduction is therefore equivalent to

$$h + 2q - 2 < p, \qquad h + 2q < p,$$

which simplify to $p > 5q - 5$ and $p > 5q - 1$. For even m the common compiler is $(10)^{h+1}$, the extremal untouched one-counts are $2q - 3$ and $2q - 1$, and the same calculation gives $p > 5q - 5$ and, using coprimality of odd p, q, $p > 5q$. This completes the proof of Theorem 55.6.

56.5 Low-minimum certificates

The low-minimum cases are included explicitly because they are the only places where non-singleton fatal supports survive.

56.5.1 Self-contained odd-p:2 fork certificate

Fix $q = 2$ and write

$$p = 2h - 1, \qquad m = p + q = 2h + 1, \qquad h \geq 2.$$

For $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + pk_t, \qquad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The six core rules are, in literal stack notation,

$$\begin{array}{c|c} A : (D_{-h}, 0T) \xrightarrow{0} (C, \Gamma_h T) & B : (D_{-h}, 0T) \xrightarrow{1} (C, \Gamma_{h-1} T) \\ C : (D_{-h}, 1T) \xrightarrow{1} (C, \Gamma_h T) & D : (D_{h-1}, 0T) \xrightarrow{0} (C, (10)^h 0T) \\ E : (D_{h-1}, 0T) \xrightarrow{1} (C, (10)^{h-1}000T) & F : (D_h, 0T) \xrightarrow{0} (C, (10)^{h-1}000T). \end{array}$$

The arithmetic identities needed below are

$$\mu(-tm) = (\epsilon_t, \beta_t), \qquad \mu(jm + 2) = (j + 2, p - j - 1) \quad (0 \leq j \leq h),$$

$$\mu(-tm + 2) = (\epsilon_t, \beta_t - 1) \quad (1 \leq t \leq h + 1), \qquad \mu((h + 1)m) = (h + 3, h - 2).$$

They follow by direct substitution in $px - 2y = r$ and kernel reduction by $(2, p)$.

Define the canonical baseline reset family

$$\begin{aligned} \mathcal{R}_h^{\text{base}} = {} & \{\mu(jm + 2) : -h - 1 \leq j \leq h\} \\ & \cup \{(x, h - 2) : x \geq h + 3\} \\ & \cup \{(\epsilon_{h+1}, y) : y \geq \beta_{h+1}\}. \end{aligned}$$

Two episode-closed repertoires separate the safe schedule regimes:

$$\begin{aligned} \mathcal{R}_{\neg D,h} = {} & \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h - 1) : x \geq h + 2\}, \\ \mathcal{R}_{D,h} = {} & \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(x, h - 1) : x \geq h + 2\} \cup \{(x, h) : x \geq h + 1\}. \end{aligned}$$

Every pair in either repertoire has minority coordinate at most h. Consequently an episode beginning in one of these repertoires has the following exact boundary-source geometry: in $\mathcal{R}_{\neg D,h}$, a D_{-h} source with visible 0 has remaining stack exactly 0; in $\mathcal{R}_{D,h}$ it has remaining stack 0^n, $n \geq 1$, with $n \geq 2$ arising only from the D-ray. In either repertoire, D_{-h} with visible 1, D_{h-1} with visible 0, and D_h with visible 0 have unary remaining stacks. This follows because MATCH and interior SEARCH only delete the visible top cell, so every pre-reset stack is a suffix of a canonical reset word; an opposite symbol beyond the displayed unary suffix would force minority size at least $h + 1$.

For a unary source of length $n \geq 1$, the six eager targets have canonical count pairs

$$\begin{array}{c|c} A : 0 & (\epsilon_h, \beta_h) \\ B : 0 & (\epsilon_{h-1}, \beta_{h-1}) \\ C : 1^n & (\epsilon_h, \beta_h + n - 1) \\ D : 0^n & (n + h, h) \\ E : 0^n & (n + h + 1, h - 1) \\ F : 0^n & (n + h + 1, h - 1). \end{array}$$

Hence baseline resets together with any enabled subset of A, B, C, E, F preserve $\mathcal{R}_{\neg D,h}$ when D is disabled; when D is enabled but A, B are disabled, baseline resets together with C, D, E, F preserve $\mathcal{R}_{D,h}$. Therefore all $2^5 = 32$ schedules with $D = 0$ are safe, and with $D = 1$ precisely the $2^3 = 8$ schedules having $A = B = 0$ are safe.

For necessity, fix any $x \geq h + 1$. Then

$$(C, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the common A/B source fiber from the singleton $\{0\}$ to the entire unary ray $\{0^n : n \geq 1\}$. There is a uniform prefix u_h reaching $(D_{-h}, 000)$ and meeting no optional site except D. Put

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0\,1^{2h+(h-2)m/2}, & h \text{ even}, \end{cases} \qquad u_h = v_h 1^{2h-1} 0^{2h}.$$

Then u_h has the stated property: the increment block from an empty stack advances the centered debt by two, the middle block reaches $(D_{h-1}, 00)$, the next zero invokes D, and the final zeros expose $(D_{-h}, 000)$ without meeting A, B, C, E, F earlier.

Finally, Γ_t has a suffix 1^p for every $t \geq 1$; write $\Gamma_t = z_t 1^p$. At the common source $(D_{-h}, 000)$,

$$A: \ (D_{-h}, 000) \xrightarrow{0} (C, z_h 1^p 00), \qquad B: \ (D_{-h}, 000) \xrightarrow{1} (C, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} by MATCHes leaves $(C, 1^p 00)$. This word contains p ones and two zeros, so its residual is zero, but it is a nonempty primitive kernel packet and is therefore nonaccepting. The complete fatal witnesses

$$w_A^{(h)} = u_h 0 z_h, \qquad w_B^{(h)} = u_h 1 z_{h-1}$$

show that $\{A, D\}$ and $\{B, D\}$ are robust fatal supports. Thus the six-site core has exactly 40 admissible schedules and no other minimal fatal support.

The extra baseline-portable sites

$$X_1 = (h - 1, 0, 1) \quad (p \geq 7), \qquad X_2 = (h - 2, 1, 1) \quad (p \geq 11)$$

are individually upward-fatal by the one-kernel trapping mechanism proved above. Excluding them reduces the expanded problem to the proved six-site fork classification, so the admissible count remains exactly 40.

For $q = 1$ the full vertex sets are as follows. At $(2, 1)$,

$$\mathcal{S}_{2,1}^{\mathrm{bp}} = \{X = (0, 1, 0),\ A = (-1, 1, 1),\ B = (-1, 0, 0),\ G = (1, 0, 0)\}.$$

For odd $p = 2h+1 \geq 3$, $A = (-h, 1, 1), D = (h, 1, 1), E = (h, 0, 0)$ are always present and $B = (h - 1, 1, 0), C = (h, 0, 1)$ occur for $p \geq 5$. For even $p = 2h \geq 4$, the always-present sites are

$$A = (-h, 1, 1), \quad B = (-h, 0, 0), \quad C = (-h, 1, 0),$$
$$E = (h - 1, 0, 0), \quad F = (h - 1, 1, 0), \quad G = (h, 0, 0).$$

and $D = (h - 1, 1, 1)$ is present iff $p \geq 6$.

Lemma 56.9 (Explicit $q = 1$ fatal certificates)**.** *The fatal supports in Theorem 56.5 admit balanced witnesses that end at residual zero in a nonaccepting physical configuration.*

Proof. For odd $p = 2h + 1$, the positive sites are witnessed by

$$w_B = 1^{p(p-2)}0^{p-2},$$

$$w_C = 1^{2h(h+1)-1}(0\,1^{p-1})^{h-1}0^h,$$

$$w_D = 1^{2h(h+1)}0^p1^{2h(h+1)+1}, \qquad w_E = 1^{2h^2+4h+1}0^{p+1}1^{2h(h+1)+1}.$$

The B-witness meets no other optional site; with B disabled the C- and D-witnesses are unpreempted, and with B, D disabled the E-witness is unpreempted. Site A acts only on the unary negative corridor and preserves an episode-closed unary reset family. Hence only $\varnothing$ and $\{A\}$ are admissible.

For even $p = 2h$, the two safe sectors are $\mathcal{N} = \{A, B, C\}$ and $\mathcal{P} = \{A, F, G\}$. The individually fatal sites have witnesses

$$w_D = 1^{p(p-3)}0^{p-3} \quad (h \geq 3),$$

$$w_E = 1^{(h+1)(p-1)}(0\,1^{p-1})^{h-1}0^h.$$

The four cross-pair certificates are

$$w_{BF} = 1^{h(p+1)}0^p1^{h(p-1)}, \qquad w_{CF} = 1^{h(p+1)}0^{p-1}1^{h(p-3)},$$
$$w_{BG} = 1^{h(p+3)}0^{p+1}1^{h(p-1)}, \quad w_{CG} = 1^{h(p+3)}0^p1^{h(p-3)}.$$

They trap exactly one primitive kernel packet after the indicated mixed pair has been activated. Finally, for $(p, q) = (2, 1)$ the two balanced words 111001 and 111110001 witness $\{B, X\}$ and $\{B, G\}$ and both end at $(C, 110)$, a nonaccepting ordering of the primitive kernel packet. These certificates, together with the episode-closed safe sectors, prove the stated low-minimum classification. □

Chapter 57

Downward Closure and Category-Relative Causal Persistence

The exact weighted-balance family is downward closed, but that fact is not a theorem of exact lifts in general. The correct boundary statement is obtained by separating the lift from the schedule presentation and by typing the category in which a named causal edge is preserved.

57.1 Downward closure is not universal

Theorem 56.6 is a theorem about the canonical weighted-balance family, not a theorem about arbitrary exact lifts. The distinction is real.

Consider one semantic state m with two semantic letters x, y, both acting as the identity. Let the physical states be c, d, b, all projecting to m. Assign terminal values

$$\alpha(c) = \alpha(d) = 0, \qquad \alpha(b) = 1.$$

The baseline transitions are

$$c \xrightarrow{x} c, \quad c \xrightarrow{y} c,$$
$$d \xrightarrow{x} c, \quad d \xrightarrow{y} b,$$

with b absorbing. Add two optional sites:

$$A : c \xrightarrow{x} d, \qquad B : d \xrightarrow{y} c.$$

Theorem 57.1 (Two-site failure of universal downward closure)**.** *For this exact schedule environment,*

$$\mathrm{Adm} = \{\varnothing, \{B\}, \{A, B\}\}.$$

In particular $\{A, B\}$ is admissible but $\{A\}$ is not.

Proof. The empty schedule and $\{B\}$ never leave c. Under $\{A\}$, the word xy follows $c \xrightarrow{x} d \xrightarrow{y} b$ and fails. Under $\{A, B\}$, the same first step reaches d but the enabled B rule repairs the second step and returns to c. Thus adding an optional rule can repair a failure created by another optional rule. □

This two-site example is minimal among counterexamples with an admissible empty schedule: with only one optional site, the only possible failure of downward closure would require the empty schedule to be bad and the singleton schedule good.

57.2 Causal persistence depends on the presentation category

A named edge such as $A \to B$ refers to two designated optional sites and to a first-hit or pre-emption relation between them. This data is not contained in the underlying lift alone.

Proposition 57.2 (Typing of named causal edges)**.** *A fixed exact lift does not determine a named causal edge. The optional-site menu and the local alternative rules are part of the schedule presentation, so forgetting the presentation can change the causal graph without changing the baseline lift.*

For a named edge e, let $\Pi(e) = (\pi_0, \pi_1, \pi_2)$ record persistence under (i) site-preserving coordinate isomorphism, (ii) a specified fixed-literal-policy presentation class, and (iii) broad exact physical re-coordination. Because the middle coordinate depends on what presentation data are frozen, its class is written as a subscript.

For the classical $2:1$ literal one-top policy, consider two presentation categories. In the first, the baseline policy is fixed but the optional-site menu may vary. In the second, the designated A, B cells and the finite causal certificate are preserved.

Theorem 57.3 (Exact $A \to B$ category bifurcation)**.** *For the classical* $2:1$ *baseline literal one-top policy,*

$$\Pi_{\text{flex}}(A \to B) = (1, 0, 0)$$

in the site-flexible fixed-policy class, while

$$\Pi_{\text{cert}}(A \to B) = (1, 1, 0)$$

in the certificate/site-preserving class. Hence the middle persistence bit has no category-independent value.

Proof. The same all-lazy baseline lift admits two presentations. With menu $\{A, B, C\}$ the classical first-hit certificate yields $A \to B$; with menu $\{A, C\}$ the named B site is absent and the directed edge cannot be formed. Hence the middle bit is 0 in the site-flexible class. In the certificate-preserving class the cells

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

and the distinguishing continuation 1 are part of the morphism data; the same non-canonical first-hit fiber therefore survives every allowed change. Broad re-coordination can remove the named phenomenon. Section 57.3 gives the typed formulation. □

57.3 Persistence signature and category typing

A schedule presentation is a triple

$$\mathfrak{P} = (\mathcal{L}_0, E, \{\delta^e\}_{e \in E}),$$

where $\mathcal{L}_0$ is the fixed baseline lift, E is the named optional-site menu, and each e carries its alternative local rule. A named directed edge therefore belongs to $\mathfrak{P}$, not to $\mathcal{L}_0$ alone.

For a named edge e define its three-level persistence signature

$$\Pi(e) = (\pi_0, \pi_1, \pi_2) \in \{0, 1\}^3,$$

where π_0 records persistence under site-preserving coordinate isomorphism, π_1 records persistence in a specified fixed-literal-policy presentation class, and π_2 records persistence under broad exact physical re-coordination. The middle class must therefore be named as part of the datum.

For the classical 2:1 edge $A \to B$, the all-lazy baseline admits both the menu $\{A, B, C\}$, where the directed dependency is present, and the menu $\{A, C\}$, where it is absent. This proves the middle bit is 0 in the site-flexible class. If instead the designated A, B cells and the finite first-hit certificate

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

with distinguishing continuation 1 are preserved, the same unsafe first-hit fiber survives every allowed presentation change; the middle bit is then 1. Broad re-coordination can eliminate the named phenomenon. Thus

$$\Pi_{\text{flex}}(A \to B) = (1, 0, 0), \qquad \Pi_{\text{cert}}(A \to B) = (1, 1, 0).$$

57.4 Monograph synthesis

The 2:1 schedule geometry is therefore not an isolated finite accident. It sits inside an arbitrary-coprime finite causal-hypergraph classification. What is special to the canonical weighted-balance family is the downward-closed independence-complex structure; arbitrary exact-lift schedule families need not share it. Likewise, the edge $A \to B$ is coordinate-stable but not language-intrinsic: its middle persistence bit depends on whether the optional-site presentation is held fixed. These conclusions close the schedule questions generated in the preceding lift chapters and prepare the protected comparison theory that follows.

Chapter 58

Strict Coordinate Quotients and Protected Interfaces

Exact sectioned lifts admit both genuine reductions and vacuous semantic collapses. To compare physical presentations without erasing the LIFO information under study, we first restrict reduction to coordinate quotients that preserve the exact section and the permanent bottom marker, and then declare explicitly which finite physical interface is protected.

58.1 Exact sectioned lifts and used coordinates

Let M be a semantic transition system over input alphabet Σ. An exact sectioned pushdown lift is a tuple

$$\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma),$$

where C^{reach} is the reachable physical configuration set, δ is the one-top pushdown transition structure, $\Theta : C^{\mathrm{reach}} \to M$ is the semantic projection, and $\sigma : M \to C^{\mathrm{reach}}$ is a chosen section satisfying

$$\Theta\sigma = \mathrm{id}_M.$$

Exactness means that the physical transitions project to the semantic transitions.

For a finite-coordinate lift, define

$$Q_{\mathrm{use}} = \{q : \exists S\ (q, S) \in C^{\mathrm{reach}}\}$$

and let Γ_{use} be the stack symbols appearing in reachable stacks, including exactly one permanent bottom marker Z_0.

The basic resource vector is

$$\mathbf{R}(\mathcal{L}) = \big(|Q_{\mathrm{use}}|, |\Gamma_{\mathrm{use}}|\big).$$

Unreachable ambient states and symbols are ignored.

58.2 Strict coordinate quotients

Definition 58.1 (Strict coordinate quotient)**.** Let $\mathcal{L}_1, \mathcal{L}_2$ be exact sectioned one-top lifts of the same semantic system. A strict coordinate quotient (SCQ)

$$F : \mathcal{L}_1 \twoheadrightarrow_{\mathrm{SCQ}} \mathcal{L}_2$$

consists of surjections

$$f_Q : Q^{(1)}_{\mathrm{use}} \twoheadrightarrow Q^{(2)}_{\mathrm{use}}, \qquad f_\Gamma : \Gamma^{(1)}_{\mathrm{use}} \twoheadrightarrow \Gamma^{(2)}_{\mathrm{use}}$$

such that

$$f_\Gamma^{-1}(Z_0) = \{Z_0\},$$

and such that the induced stack map is letterwise and non-erasing:

$$F(q, X_1 \cdots X_k Z_0) = \big(f_Q(q), f_\Gamma(X_1) \cdots f_\Gamma(X_k) Z_0\big).$$

It must satisfy

$$F(C_1^{\mathrm{reach}}) = C_2^{\mathrm{reach}}, \quad \Theta_2 F = \Theta_1, \quad F\sigma_1 = \sigma_2,$$

and

$$F\delta_a^{(1)} = \delta_a^{(2)} F$$

on every reachable transition.

The bottom-marker condition is essential: a work symbol may not be merged with the unique permanent bottom cell.

Theorem 58.2 (SCQ reduction theorem)**.** *SCQ morphisms compose and define a preorder. Mutual SCQ reducibility is exactly SCQ isomorphism. If*

$$\mathcal{L}_1 \twoheadrightarrow_{\mathrm{SCQ}} \mathcal{L}_2,$$

then

$$\mathbf{R}(\mathcal{L}_2) \le \mathbf{R}(\mathcal{L}_1)$$

componentwise; if the quotient is not an isomorphism, at least one coordinate decreases strictly.

Proof. Composition is coordinatewise and preserves the transition, semantic, section, and bottom-marker conditions. If SCQ morphisms exist in both directions, finiteness and surjectivity force equality of the used state and symbol cardinalities; all coordinate maps are bijective and their inverses preserve the same structure. For monotonicity, each coordinate map is surjective. Equality in both resource coordinates forces both maps to be bijections, hence an isomorphism. □

Corollary 58.3 (Finite minimal-core existence)**.** *Every finite-coordinate exact sectioned lift has at least one SCQ-minimal quotient descendant.*

Proof. Each strict SCQ step strictly decreases a point of $\mathbb{N}^2$. No infinite strictly descending chain exists. □

Cheap finite-control decorations are automatically removed: if a redundant cyclic coordinate $j \in \mathbb{Z}_m$ is added to control but ignored by semantics and stack dynamics, projection $(q, j) \mapsto q$ is a strict SCQ quotient. On the other hand, an arbitrary behavioral quotient that erases the LIFO store is excluded by the letterwise non-erasing coordinate requirement.

58.3 Protected interfaces and maximal safe quotients

SCQ is intentionally rigid. To formulate which physical distinctions a quotient is allowed to erase, we attach a finite interface signature.

Definition 58.4 (Protected-interface lift)**.** A protected-interface lift is a pair

$$(\mathcal{L}, \lambda), \qquad \lambda : C^{\mathrm{reach}} \to \Lambda,$$

where Λ is finite and λ records physical information declared to be part of the comparison contract.

For a literal one-top policy, a useful interface is

$$\lambda_{\mathrm{lit}}(c) = \big(o_{\mathrm{lit}}(c), \chi_\sigma(c)\big),$$

where o_{lit} is the locally visible state/top observation and

$$\chi_\sigma(c) = \mathbf{1}_{\{c \in \sigma(M)\}}$$

records whether the configuration lies on the chosen canonical section.

Theorem 58.5 (Maximal interface-safe quotient criterion)**.** *Let $F : C_1^{\mathrm{reach}} \twoheadrightarrow C_2^{\mathrm{reach}}$ be surjective and let $\lambda_1 : C_1^{\mathrm{reach}} \to \Lambda$. The following are equivalent:*

1. *there is a unique $\lambda_2 : C_2^{\mathrm{reach}} \to \Lambda$ such that*

$$\lambda_1 = \lambda_2 \circ F;$$

2. *λ_1 is constant on every fiber of F.*

Hence λ-separated configurations can never be identified by an interface-safe quotient.

Proof. Necessity is immediate. Conversely define $\lambda_2(F(c)) = \lambda_1(c)$; fiber constancy makes the definition well posed, and surjectivity gives uniqueness. □

This criterion is maximal at the configuration level: it imposes exactly the identifications forbidden by the protected observable and no others.

Chapter 59

Bounded Bi-Recoding and Minimal Protected Cores

Coordinate quotients reduce resources, whereas bounded invertible recodings merely change the physical alphabet in a locally recoverable way. Treating these operations separately produces recoding-invariant minimal cores and finite protected Pareto envelopes.

59.1 Bounded bi-recoding and recoding-invariant resource spectra

A bounded invertible change of stack code should not be counted as a reduction.

Definition 59.1 (Bounded bi-recoding)**.** A bounded bi-recoding (BBR) between protected-interface lifts consists of a bijective control recoding and a non-erasing work-stack block code

$$\varphi : \Gamma_{1,\mathrm{use}} \setminus \{Z_0\} \to (\Gamma_{2,\mathrm{use}} \setminus \{Z_0\})^+$$

extended multiplicatively above the fixed bottom marker, such that:

1. $1 \le |\varphi(X)| \le B$ for a uniform finite B;
2. an inverse bounded non-erasing block code exists on the reachable stack languages;
3. the induced configuration bijection intertwines semantics, section, transitions, and the protected interface.

BBR is therefore an actual conjugacy of reachable dynamics. It preserves every obstruction expressible in semantic fibers, paths, section membership, and protected-interface values.

The identity code, the stipulated bounded inverse, and composition of bounded block codes show that BBR is an equivalence relation; see Section 59.2.

Define the protected recoding spectrum

$$\mathcal{R}_\lambda(\mathcal{L}) = \{(|Q_{\mathrm{use}}(\mathcal{L}')|, |\Gamma_{\mathrm{use}}(\mathcal{L}')|) : \mathcal{L}' \simeq_{\mathrm{BBR},\lambda} \mathcal{L}\}.$$

Its minimal antichain is

$$\mathcal{P}_\lambda(\mathcal{L}) = \operatorname{Min} \mathcal{R}_\lambda(\mathcal{L}).$$

Theorem 59.2 (Finite protected Pareto envelope)**.** *For every finite-coordinate protected lift, $\mathcal{P}_\lambda(\mathcal{L})$ is nonempty and finite and depends only on the BBR-equivalence class.*

Proof. The original presentation belongs to its spectrum, so the set is nonempty. Every nonempty subset of $\mathbb{N}^2$ has minimal elements, and Dickson's lemma [49] implies that the antichain of all componentwise-minimal elements is finite. BBR-equivalent lifts have the same spectrum by transitivity. □

A normalized protected core step first chooses a representative whose resource vector lies in $\mathcal{P}_\lambda$, then takes a strict λ-preserving SCQ quotient, and finally passes to the target BBR class. By Theorem 58.2, every such step strictly decreases the normalized resource pair. Hence terminal protected BBR-classes exist. Uniqueness is not asserted.

59.2 Typed protected contracts and BBR equivalence

For complete type correctness, a protected representation contract is written

$$\mathfrak{K} = \big(\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}\big),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned lifts of one semantic system and each $\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda$ is a protected interface. The family is required to be compatible with BBR conjugacies: if $F : \mathcal{L} \to \mathcal{L}'$ is a BBR equivalence, then $\lambda_{\mathcal{L}} = \lambda_{\mathcal{L}'} \circ F$. In the body we suppress the object subscript when no ambiguity is possible.

Proposition 59.3 (BBR is an equivalence relation)**.** *Bounded bi-recoding is reflexive, symmetric, and transitive on finite-coordinate protected lifts.*

Proof. The identity code gives reflexivity. The defining bounded non-erasing inverse on reachable stack languages gives symmetry. If block codes φ and ψ have maximum block lengths B_φ, B_ψ, then $\psi \circ \varphi$ has maximum block length at most $B_\varphi B_\psi$; the inverse is the reverse composition of the bounded inverses. The induced configuration bijections compose and continue to intertwine semantics, section, transitions, and the protected-interface family. Hence transitivity holds. □

Under contract refinement $\mathfrak{K}' \succeq \mathfrak{K}$, the representation class is included and the protected interfaces refine objectwise. Therefore the resource-set inclusion used in Theorem 99.4 is literal, rather than an informal comparison between unrelated coordinate systems.

59.3 Interface with General Theory

The outcome is a comparison category strong enough to exclude semantic trivialization and weak enough to identify harmless bounded recodings. Terminal protected cores exist in every finite-coordinate class, but no uniqueness or Church–Rosser statement is asserted here. The remaining confluence chapter completes Part IV by comparing exact normalization layers under this protected interface; Part V can then formulate General Theory over a cleaner physical comparison layer without conflating exact semantics, coordinate quotient, and recoding equivalence.

Chapter 60

Confluence Across Exact Pushdown Normalization Layers

Research branch — Confluence and Normalization Choice. *The preceding causal-schedule and protected-interface theory distinguishes semantic correctness from physical presentation. This branch makes the rewrite layer explicit: schedule activation, full completion-fiber normalization, finite-control sheets, and protected-core descent are treated as different confluence problems rather than conflated under one endpoint-coherence statement.*

The exact-lift, causal-schedule, SCQ, and BBR objects used here have already been developed in the preceding chapters. The purpose of this chapter is to add the missing rewrite-theoretic layer and to separate four questions: canonical endpoint coherence, schedule-presentation confluence, physical completion-fiber confluence, and confluence of protected quotient descent.

60.1 Exact lifts and four notions of normalization coherence

Definition 60.1 (Exact sectioned lift)**.** Let M be a deterministic semantic transition system over input alphabet Σ, with deterministic partial semantic transitions $U_a : M \rightharpoonup M$. An *exact sectioned pushdown lift* is a reachable physical transition system

$$\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma)$$

with semantic projection $\Theta : C^{\mathrm{reach}} \to M$, partial physical settled macro-transitions δ_a, and section $\sigma : M \to C^{\mathrm{reach}}$ satisfying

$$\Theta\sigma = \mathrm{id}_M, \qquad c \in \mathrm{Dom}(\delta_a) \iff \Theta(c) \in \mathrm{Dom}(U_a), \qquad \Theta\delta_a = U_a\Theta$$

on the common defined domain. The canonical retraction is $N = \sigma\Theta$.

Proposition 60.2 (Canonical retraction and absorption; baseline)**.** *The canonical retraction satisfies*

$$N^2 = N, \qquad \Theta N = \Theta,$$

and for every input word w for which the transition is defined,

$$N\delta_w = N\delta_w N = \sigma U_w \Theta.$$

Thus the canonical endpoint after reading w depends only on the initial semantic state. This is endpoint coherence, not a Church–Rosser theorem for an internal normalization rewrite.

Proof. Since $\Theta\sigma = \mathrm{id}_M$, one has $N^2 = \sigma\Theta\sigma\Theta = N$ and $\Theta N = \Theta$. Exactness gives $\Theta\delta_w = U_w\Theta$, hence

$$N\delta_w = \sigma U_w\Theta = \sigma U_w\Theta\sigma\Theta = N\delta_w N.$$

□

The following distinctions will be used throughout.

Definition 60.3 (Coherence notions)**.** Let $\Rightarrow$ be an internal normalization relation, distinct from the ordinary input transition δ.

1. *Endpoint coherence*: the designated normalizer sends every reachable configuration to the chosen section representative of its semantic value.
2. *Physical confluence*: every fork under $\Rightarrow$ joins to one physical configuration.
3. *Confluence modulo* $\approx$: every fork joins to configurations equivalent under a declared relation $\approx$.
4. *Section-complete normalization*: every configuration in a semantic fiber reduces to the chosen section point.

Proposition 60.4 (Bare semantic modulo-confluence is vacuous)**.** *Suppose*

$$c \Rightarrow d \quad \Longrightarrow \quad \Theta(c) = \Theta(d).$$

Then every fork $c \Rightarrow^* u$, $c \Rightarrow^* v$ already satisfies $\Theta(u) = \Theta(v)$. Thus confluence modulo the equivalence $u \equiv_\Theta v$ iff $\Theta(u) = \Theta(v)$ is automatic.

Proof. Semantic preservation is inherited by the reflexive-transitive closure. Both descendants have semantic value $\Theta(c)$. □

Consequently, a substantive modulo-confluence theorem must protect more physical information than Θ alone.

Definition 60.5 (Strictly canonicalizing physical rewrite)**.** A tuple $(C, \Rightarrow, \Theta, \sigma, D)$ is *strictly canonicalizing* if D takes values in a well-founded ordered set and:

1. $c \Rightarrow d$ implies $\Theta(c) = \Theta(d)$;
2. every rewrite strictly decreases D;
3. $D(c) = 0$ iff $c = \sigma(\Theta(c))$;
4. every noncanonical c admits a rewrite step.

Proposition 60.6 (Canonical-normal-form criterion)**.** *Every strictly canonicalizing physical rewrite is strongly normalizing and confluent. The unique normal form in the fiber $\Theta^{-1}(m)$ is $\sigma(m)$.*

Proof. Well-founded strict descent forbids infinite chains. Every maximal chain terminates, and the last two conditions imply that a terminal point is exactly the section point in its semantic fiber. Semantic preservation keeps every descendant of a fork in the same fiber, so both branches terminate at the same section representative. □

60.2 Finite schedule-activation rewriting

Let E be a finite set of optional eager-normalization sites. A schedule is a subset $H \subseteq E$, and $\mathrm{Adm}(E) \subseteq 2^E$ denotes the schedules preserving the intended recognition/normalization contract.

Definition 60.7 (Schedule activation)**.** Define

$$H \rightsquigarrow_E H \cup \{e\}$$

when $e \notin H$ and $H \cup \{e\} \in \mathrm{Adm}(E)$. A schedule is normal when no further one-site activation is admissible.

Proposition 60.8 (Termination)**.** *Schedule activation terminates for every finite E.*

Proof. The rank $|E| - |H|$ decreases by one at every step. □

Theorem 60.9 (Exact criterion for downward-closed schedule families)**.** *Assume* $\mathrm{Adm}(E)$ *is nonempty and downward closed. The following are equivalent:*

1. $(\mathrm{Adm}(E), \rightsquigarrow_E)$ *is confluent;*
2. *there is a unique normal form reachable from $\varnothing$;*
3. $\mathrm{Adm}(E)$ *has a unique inclusion-maximal member.*

Proof. In a downward-closed family, if $H \subseteq K$ are admissible then every intermediate set obtained by adding elements of $K \setminus H$ remains admissible. Hence rewrite normal forms are exactly the inclusion-maximal schedules and every admissible schedule reaches every maximal extension containing it. A unique maximal member therefore joins every branch. Two distinct maximal members are distinct terminal descendants of $\varnothing$ and cannot join. □

Suppose now that admissibility is the independence complex of a finite hypergraph $\mathcal{F}$ of minimal fatal supports:

$$\mathrm{Adm}(E) = \mathrm{Ind}(\mathcal{F}) = \{H \subseteq E : \text{ no } F \in \mathcal{F} \text{ is contained in } H\}.$$

Theorem 60.10 (Fatal-support criterion)**.** *Schedule activation is confluent iff every minimal fatal support is a singleton. If*

$$S_1 = \{e \in E : \{e\} \in \mathcal{F}\},$$

then, in the confluent case, the unique normal form is $E \setminus S_1$.

Proof. If every fatal support is singleton, admissibility is exactly $2^{E \setminus S_1}$. Conversely, let $F \in \mathcal{F}$ contain distinct x, y. Minimality makes $F \setminus \{x\}$ and $F \setminus \{y\}$ admissible. Extend each to a maximal admissible schedule. They cannot have the same maximal extension, since such an extension would contain their union F. Thus there are at least two normal forms, and theorem 60.9 gives nonconfluence. □

60.3 Rational schedule confluence

The canonical baseline-portable literal-demand presentation of rational $p{:}q$ normalization has a finite optional-site family whose minimal fatal supports are summarized in table 60.1. Appendix 60.7 records the schedule data and the low-width certificates needed for the present rewrite application. The important point here is not the site coordinates themselves but the size of the minimal supports.

Regime	minimal fatal-support structure	maximal-schedule count
$\min(p,q) \geq 3$	all present fatal extras are singleton; the canonical core is free	1
$\min(p,q) = 2$	two non-singleton fork supports $\{A,D\},\{B,D\}$, plus singleton extras	2
$(p,q) = (1,1)$	no optional site	1
$\min(p,q) = 1$, odd maximum ≥ 3	all positive optional sites singleton-fatal; one negative-boundary site is free	1
$(2,1)$ up to duality	$\{B,X\},\{B,G\}$	2
$\min(p,q) = 1$, even maximum ≥ 4	singleton fatal sites plus the four pairs between two negative- and two positive-sector sites	2

Table 60.1: Fatal-support geometry needed for the rational schedule-confluence theorem. Physical duality exchanges p and q.

Corollary 60.11 (The canonical 2:1 fork)**.** *For the complete four-site baseline-portable family*

$$E_{2,1} = \{X, A, B, G\},$$

one has

$$\mathrm{Adm}_{2,1} = \{H \subseteq E_{2,1} : \{B,X\} \nsubseteq H,\ \{B,G\} \nsubseteq H\}.$$

Schedule activation terminates but is not confluent. Its two normal forms are

$$\{A,B\}, \qquad \{A,X,G\}.$$

Proof. Appendix 60.7 gives the exact minimal fatal supports $\{B,X\}$ and $\{B,G\}$; the site A is free. Hence the maximal admissible schedules are exactly the two displayed sets, and theorem 60.9 gives nonconfluence. □

Theorem 60.12 (Exact arbitrary-coprime schedule-confluence dichotomy)**.** *For every coprime $p,q \geq 1$, activation in the canonical baseline-portable rational schedule family is confluent exactly when*

$$\boxed{\min(p,q) \geq 3 \quad \textit{or} \quad \min(p,q) = 1 \ \textit{and}\ \max(p,q)\ \textit{is odd.}}$$

It is nonconfluent exactly when

$$\boxed{\min(p,q) = 2 \quad \textit{or} \quad \min(p,q) = 1 \ \textit{and}\ \max(p,q)\ \textit{is even.}}$$

Proof. Read the minimal fatal supports from table 60.1 and apply theorem 60.10. In the high-minimum and odd boundary regimes all minimal supports are singleton. In the width-two, $(2,1)$, and even boundary regimes a non-singleton fatal pair is present. □

Corollary 60.13 (One-or-two normal-form law)**.** *Every canonical rational schedule-activation system has exactly one or two normal forms. It has one precisely in the confluent regimes of theorem 60.12 and two precisely in the nonconfluent regimes.*

Proof. The maximal independent sets are visible in table 60.1. For min = 2, either the fork vertex D is chosen and A, B are excluded, or D is excluded and A, B are chosen, with all free sites added. The $(2, 1)$ and even boundary cases similarly split into exactly two safe sectors. The remaining regimes have one maximal admissible schedule. □

This classification is a presentation-level result. It says that some valid ways of adding eager normalization sites cannot be reconciled by further admissible activations. It does not yet describe rewriting inside a fixed physical semantic fiber.

60.4 The full rational completion fiber

Fix coprime $p, q \geq 1$ and use the completion equation

$$pN_0(z) - qN_1(z) = r.$$

For each $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r) \in \mathbb{N}_0^2$$

be the unique minimum-length nonnegative solution. The elementary one-dimensional kernel structure gives

$$(x, y) = \mu(r) + k(q, p), \qquad k \in \mathbb{N}_0, \tag{60.1}$$

for every other nonnegative solution (x, y). Thus $K = (q, p)$ is the primitive redundant completion packet.

Define the canonical shortest completion

$$C(r) = \begin{cases} 1^{y_r}0^{x_r}, & r \geq 0, \\ 0^{x_r}1^{y_r}, & r < 0. \end{cases}$$

For $r = 0$ this is ε.

Definition 60.14 (Full completion-fiber rewrite)**.** Let

$$\mathcal{C}_{p,q} = \{(r, z) : r \in \mathbb{Z},\ z \in \{0, 1\}^*,\ pN_0(z) - qN_1(z) = r\}.$$

Within a fixed residual fiber, generate $\Rightarrow_{p,q}$ by

$$\begin{aligned} r \geq 0: &\quad 01 \Rightarrow 10, \quad 1^p0^q \Rightarrow \varepsilon, \\ r < 0: &\quad 10 \Rightarrow 01, \quad 0^q1^p \Rightarrow \varepsilon. \end{aligned}$$

The swap discharges boundary-order debt and the block deletion removes one primitive kernel packet.

Lemma 60.15 (Semantic preservation and exact kernel debt)**.** *Every rewrite in theorem 60.14 preserves r. If $\mathrm{Par}(z) = (x, y)$, then the integer k in* (60.1) *is unique and*

$$\mu(r) - \mathrm{Par}(z) = -k(q, p).$$

Thus k is the exact kernel-excess quantity-debt level.

Proof. Swaps preserve Parikh counts. A deletion removes q zeros and p ones, so the value $pN_0 - qN_1$ changes by $pq - qp = 0$. The solution lattice of the coprime equation is generated by (q, p), and minimality of $\mu(r)$ forces the coefficient to be nonnegative. □

Theorem 60.16 (Full completion-fiber confluence)**.** *For every coprime $p, q \geq 1$ and every residual r, the rewrite of theorem 60.14 is strongly normalizing and confluent on the entire completion fiber. Its unique normal form is $C(r)$.*

Proof. Define the sign-adapted inversion number

$$\Omega_r(z) = \begin{cases} \#\{i < j : z_i = 0, z_j = 1\}, & r \geq 0, \\ \#\{i < j : z_i = 1, z_j = 0\}, & r < 0, \end{cases}$$

and let $k(z)$ be the kernel level from theorem 60.15. Use lexicographic order on

$$D_r(z) = (k(z), \Omega_r(z)) \in \mathbb{N}_0^2.$$

A swap fixes k and decreases Ω_r by one. A kernel deletion decreases k by one. Hence every rewrite strictly decreases D_r, so the system is strongly normalizing.

Let z be irreducible. If $r \geq 0$, absence of 01 forces $z = 1^y 0^x$; if $r < 0$, absence of 10 forces $z = 0^x 1^y$. Write $(x, y) = \mu(r) + k(q, p)$. If $k \geq 1$, then the block junction contains $1^p 0^q$ in the nonnegative orientation, respectively $0^q 1^p$ in the negative orientation, contradicting irreducibility. Therefore $k = 0$ and $z = C(r)$. Conversely $C(r)$ has neither an order redex nor a removable kernel packet. Thus every fiber has one normal form, and termination implies confluence (equivalently, apply theorem 60.6). □

Corollary 60.17 (Shortest-completion stratum)**.** *On the stratum $k = 0$, kernel deletion is unavailable and theorem 60.16 reduces to the boundary-oriented adjacent-swap normalization of shortest completions. The theorem therefore extends geodesic normalization to every nonminimal kernel level.*

Corollary 60.18 (Two-channel physical debt)**.** *The pair*

$$(k(z), \Omega_r(z))$$

is an exact normalization debt on completion words: its first coordinate measures redundant primitive-kernel quantity and its second coordinate measures word-order debt.

The point of theorem 60.16 is physical, not merely semantic. The local relation has many possible reduction orders; the theorem proves that all of them reach the same literal word. This is strictly stronger than applying the retraction $N = \sigma\Theta$ after the fact.

60.5 Finite-control sheets and three-channel confluence

Completion words have no independent finite-control coordinate. To model the third debt channel, let W_r be the word fiber above and let $(S_r, \to_r)$ be a separately declared control-sheet rewrite system with distinguished sheet s_r^*.

Definition 60.19 (Sheet-separable normalization)**.** Set

$$\widehat{C}_r = W_r \times S_r, \qquad \widehat{\sigma}(r) = (C(r), s_r^*).$$

The combined rewrite is generated by

$$(z, s) \Longrightarrow_r (z', s) \quad \text{if } z \Rightarrow_{p,q} z',$$

and

$$(z, s) \Longrightarrow_r (z, s') \quad \text{if } s \to_r s'.$$

The two move types commute exactly because they act on disjoint coordinates.

Theorem 60.20 (Exact product reduction)**.** *For every residual r:*

1. *$\Longrightarrow_r$ terminates iff $\to_r$ terminates;*

2. $\Longrightarrow_r$ *is confluent iff* $\to_r$ *is confluent;*

3.

$$\mathrm{NF}(\widehat{C}_r) = \{C(r)\} \times \mathrm{NF}(S_r);$$

4. every point of $\widehat{C}_r$ *reduces to* $(C(r), s_r^*)$ *iff every sheet reduces to* s_r^*.

Proof. The word relation is terminating and confluent with one normal form by theorem 60.16. An infinite combined chain would contain infinitely many moves in one coordinate; this is possible exactly when the sheet relation has an infinite chain. For confluence, project a combined fork to the two coordinates. Word projections join by theorem 60.16, sheet projections join exactly when $\to_r$ is confluent, and exact cross-commutation combines the joins. Conversely, any sheet fork lifts to a fork from $(C(r), s)$, so combined confluence projects back to sheet confluence. Normal forms and section reachability are coordinatewise. □

Corollary 60.21 (Confluence does not imply section-complete normalization)**.** *Let* S_r *contain at least two sheets and let* $\to_r$ *be empty. Then the product rewrite is terminating and confluent but has one normal form* $(C(r), s)$ *for each sheet.*

Proof. The empty sheet relation is terminating and confluent. Apply theorem 60.20. □

Thus

branching coherence $\not\Rightarrow$ collapse of the whole semantic fiber to the section.

Theorem 60.22 (Rooted-sheet canonicalization)**.** *Suppose there is a well-founded rank* $\eta_r : S_r \to W_r^{\mathrm{sh}}$ *such that every sheet move strictly decreases* η_r*,* $\eta_r(s) = 0$ *iff* $s = s_r^*$*, and every noncanonical sheet admits a move. Then the combined rewrite is strictly canonicalizing with lexicographic debt*

$$\widehat{D}_r(z, s) = (k(z), \Omega_r(z), \eta_r(s)).$$

It is therefore strongly normalizing and confluent, with unique normal form $(C(r), s_r^*)$.

Proof. Kernel deletion decreases the first coordinate, sorting decreases the second while fixing the first, and sheet discharge decreases the third while fixing the first two. The debt is zero exactly at the section point. Apply theorem 60.6. □

The separability hypothesis is deliberate. We do not assert that an arbitrary DPDA configuration fiber decomposes as word × sheet. The theorem identifies a natural class in which the independent control channel can be handled without semantically erasing it.

60.6 Protected confluence and canonical Pareto anchoring

We finally allow harmless changes of physical code while retaining declared interface information. The minimum structure needed for the argument is stated abstractly so that the protected confluence theorem is self-contained.

Definition 60.23 (Protected exact-lift class)**.** Fix a class of finite-coordinate exact sectioned lifts of one semantic system. Each lift $\mathcal{L}$ carries a finite protected interface

$$\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda.$$

A *bounded bi-recoding* (BBR) is an invertible bounded physical recoding of finite control and reachable stack blocks that intertwines semantics, section, input transitions, and λ. We write $[\mathcal{L}]$ for its BBR class.

A *strict protected coordinate quotient* (SCQ) is a surjective coordinate quotient preserving semantics, transitions, section, the unique bottom marker, and λ, and inducing a componentwise nonincreasing map on the used resource coordinates, with at least one strict decrease:

$$\mathbf{R}(\mathcal{K}) < \mathbf{R}(\mathcal{L}) \quad \text{componentwise, with at least one strict coordinate,}$$

where $\mathbf{R}(\mathcal{L}) = (|Q_{\mathrm{use}}|, |\Gamma_{\mathrm{use}}|)$.

For a BBR class $X = [\mathcal{L}]$, define its protected recoding spectrum

$$\mathcal{R}_\lambda(X) = \{\mathbf{R}(\mathcal{L}') : \mathcal{L}' \simeq_{\mathrm{BBR},\lambda} \mathcal{L}\},$$

and its Pareto envelope

$$\mathcal{P}_\lambda(X) = \operatorname{Min} \mathcal{R}_\lambda(X) \subset \mathbb{N}^2.$$

Dickson's lemma implies that this antichain is finite. Every member is realized by an actual representative because it is a spectrum point.

Definition 60.24 (Protected modulo-equivalence)**.** Let X be the state set of a normalization rewrite and let

$$\tau : X \to \Lambda^\sharp$$

be a declared protected signature. An equivalence $\approx_\tau$ is *protected* when

$$x \approx_\tau y \quad \Longrightarrow \quad \tau(x) = \tau(y).$$

At configuration level one may take $\tau = \lambda$. When the rewrite states are BBR classes, τ may be any BBR-invariant signature induced by the protected interface, such as the isomorphism type of a specified finite interface-labelled observable. Passing from representatives to BBR classes already quotients out BBR equivalence itself.

Proposition 60.25 (Terminal protected fork)**.** *In a terminating rewrite presentation equipped with a protected signature τ, if a source has terminal descendants u, v with $\tau(u) \neq \tau(v)$, then the system is not confluent modulo any protected equivalence $\approx_\tau$.*

Proof. Terminality leaves no further reduction. A modulo-join would therefore force $u \approx_\tau v$, contradicting preservation of τ. □

A subtlety arises if each reduction step is allowed to choose an arbitrary incomparable Pareto point of the current BBR class: one no longer has a single resource pair that visibly decreases along every possible path. We remove this hidden source choice canonically.

Definition 60.26 (Lex-normalized protected-core rewrite)**.** Let

$$\nu_\lambda(X) = \min_{\mathrm{lex}} \mathcal{P}_\lambda(X).$$

A step

$$X \Longrightarrow_\lambda^{\mathrm{lex}} Y$$

chooses a BBR-equivalent representative of X attaining the actual resource pair $\nu_\lambda(X)$, takes a strict protected SCQ quotient to a lift $\mathcal{K}$, and sets $Y = [\mathcal{K}]$.

Theorem 60.27 (Strong normalization of canonical protected-core descent)**.** *Every lex-normalized protected-core step strictly decreases ν_λ in lexicographic order on $\mathbb{N}^2$. Hence $\Longrightarrow_\lambda^{\mathrm{lex}}$ is strongly normalizing.*

Proof. Put $r = \nu_\lambda(X)$. A strict protected SCQ quotient from a representative attaining r has raw resource vector $r' < r$ componentwise, with at least one strict coordinate. The quotient is a representative of the target BBR class Y. Hence some Pareto point $r^\sharp \in \mathcal{P}_\lambda(Y)$ satisfies $r^\sharp \le r'$. By lexicographic minimality,

$$\nu_\lambda(Y) \le_{\mathrm{lex}} r^\sharp \le_{\mathrm{lex}} r' <_{\mathrm{lex}} r.$$

Lexicographic order on $\mathbb{N}^2$ is well-founded. □

Theorem 60.28 (Protected-core confluence criterion)**.** *On the descendant cone of any starting BBR class under $\Longrightarrow_\lambda^{\mathrm{lex}}$, the following are equivalent:*

1. *confluence;*
2. *local confluence;*
3. *exactly one terminal BBR class is reachable from the starting class.*

Proof. Termination is theorem 60.27, so Newman's lemma gives equivalence of confluence and local confluence [62, 61]. Every descendant of a terminating system reaches a terminal class. Confluence forces all terminal descendants to coincide. Conversely, if one terminal class T is reachable from the source and is the only terminal descendant of that source, then every descendant also terminates at T, because any terminal descendant of a descendant is a terminal descendant of the source. Thus any two descendants join at T. □

Corollary 60.29 (Protected signature certificate)**.** *If two terminal lex-normalized descendants have different values of a BBR-invariant protected signature τ, then the protected-core rewrite is nonconfluent from that source (and cannot be repaired by passing to any equivalence that preserves τ).*

Proof. Apply theorem 60.25. □

The canonical anchor is essential to the theorem as stated. We do not claim that a larger unanchored relation allowing arbitrary re-selection of incomparable Pareto points is strongly normalizing or confluent.

60.7 Rational fatal-support data used in Section 60.3

This appendix makes the rational schedule application self-contained at the level needed by the confluence theorem: it specifies the optional-site contract, the complete site universe, and the exact minimal-fatal-support data. No theorem in the abstract rewrite sections depends on these coordinates.

Put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad K = (q, p),$$

and let

$$\mu(r) = \arg\min\{x + y : x, y \in \mathbb{N},\ px - qy = r\}$$

be the unique minimum nonnegative demand vector. Write $\eta(x, y)$ for the fixed canonical literal stack word representing the demand pair (x, y). The centered literal-demand lift

has debt states $D_{-h}, \dots, D_h$ and center state C; write $V_d = \mu(dm)$. A local work-top observation is a triple

$$o = (d, a, z), \qquad |d| \le h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d, the next input letter is a, and the visible stack symbol is z. With $e_0 = (1, 0)$, $e_1 = (0, 1)$, define

$$\begin{aligned} B_o &= e_z - e_a + V_d, \\ \kappa(o) &= \max\left\{ \left\lceil -\frac{(B_o)_0}{q} \right\rceil, \left\lceil -\frac{(B_o)_1}{p} \right\rceil \right\}, \\ R_o &= B_o + \kappa(o) K. \end{aligned}$$

The optional eager candidate at o consumes a, replaces the visible cell by the literal minimum-demand word representing R_o, and returns directly to C. This is the unique shortest nonnegative direct return preserving the residual invariant, because every other direct-center replacement belongs to the affine line $B_o + \mathbb{Z}K$.

A *baseline episode* begins at a canonical center demand reached by a mandatory WRITE, then follows only mandatory MATCH/SEARCH transitions, and ends before the next mandatory bottom or outer WRITE. A site is *baseline-portable* when its eager rule lands on the canonical center representative for every source exposed in every baseline episode. A schedule is a subset of the resulting finite menu; it is admissible when the deterministic machine still recognizes exactly $L_{p,q}$. A set F of sites is *minimal fatal* when every schedule containing F is bad but no proper subset has that property. Thus the terms used in table 60.1 refer to a completely specified finite presentation rather than to unnamed abstract vertices.

60.7.1 Complete site universe for $\min(p, q) \ge 2$

By physical duality it suffices to take $p \ge q \ge 2$. If m is odd, the complete baseline-portable universe consists of the six core sites

$$(-h, 0, 0),\ (-h, 1, 0),\ (-h, 1, 1),\ (h-1, 0, 0),\ (h-1, 1, 0),\ (h, 0, 0),$$

together with

$$\begin{aligned} X = (h - q + 1, 0, 1) \quad &\Longleftrightarrow \quad p > 5q - 5, \\ Y = (h - q, 1, 1) \quad &\Longleftrightarrow \quad p > 5q - 1. \end{aligned}$$

If m is even, the complete universe consists of the four core sites

$$(-h, 1, 1),\ (h, 0, 0),\ (h, 0, 1),\ (h, 1, 1),$$

together with

$$\begin{aligned} X = (h - q + 1, 0, 0) \quad &\Longleftrightarrow \quad p > 5q - 5, \\ Y = (h - q, 1, 0) \quad &\Longleftrightarrow \quad p > 5q. \end{aligned}$$

Hence the menu has at most eight sites. For $q > p$, apply

$$(p, q, d, a, z) \longleftrightarrow (q, p, -d, 1 - a, 1 - z).$$

For completeness, the exclusion mechanism is exact rather than computational. For the bottom-WRITE seed family

$$B_j = \mu(jm + q)$$

one has, for $j \geq 0$,

$$B_j = (j+q, p-j-1),$$

while for $j = -t < 0$, writing $t = kq + s$ with $0 \leq s < q$,

$$B_{-t} = \begin{cases} (0, km-1), & s = 0, \\ (q-s, p+km+s-1), & 1 \leq s < q. \end{cases}$$

For $1 \leq d \leq h$, $V_d = (d+q, p-d)$ and the three nontrivial positive-debt compiler count vectors have imbalances

$$2d+q-p, \qquad 2d+q-p+2, \qquad 2d+q-p-2.$$

If $m = 2h+1$, the central seed $B_{h-q} = (h,h)$ has word $(10)^h$. At an interior debt d, the untouched suffix is $(10)^{h-d-1}$ below a visible 0 and $0(10)^{h-d-1}$ below a visible 1. A top-0 eager concatenation can be canonical only when its compiler word is balanced, while a top-1 concatenation requires exactly one excess 1; solving these conditions leaves precisely the two odd-parity candidates displayed above. If $m = 2h+2$, the central seed is $(h, h+1)$ with word $(10)^h 1$; the corresponding parity calculation eliminates every top-1 interior candidate and leaves precisely the two displayed top-0 candidates. On the negative side, every nontrivial compiler has a terminal 1-tail whereas the diagnostic exposes an untouched 0, so no negative-interior observation is portable.

The boundary source fibers give exactly the stated cores. For the two odd-parity survivors the common compiler word is $(10)^h 1$ and the extremal exposed one-counts are $2q-2$ and $2q$, so kernel reduction is equivalent to

$$h + 2q - 2 < p, \qquad h + 2q < p,$$

which simplify to $p > 5q-5$ and $p > 5q-1$. In even parity the common compiler is $(10)^{h+1}$, the extremal counts are $2q-3$ and $2q-1$, and the same calculation gives $p > 5q-5$ and, using coprimality of odd p, q, $p > 5q$. Thus no unlisted optional site is being silently ignored.

60.7.2 High-minimum fatal supports

Assume $\min(p,q) \geq 3$. Every subset of the parity-dependent core above is admissible. Indeed, induction over reset episodes keeps every core source inside a protected unary envelope, and every enabled core activation returns to a canonical minimum-demand word. Minority counts stay below the opposite boundary threshold, so a core reset cannot create a new incompatible source on the other side.

Each present extra site X or Y is singleton-fatal. Repeated self-activation increases a hidden alternating prefix until the first nonminimal target is reached. At that first failure the target is exactly

$$\mu(r) + K,$$

not $\mu(r) + jK$ with $j \geq 2$. Feeding a shortest semantic completion for r removes the minimum component and leaves exactly one balanced primitive-kernel packet physically present at semantic residual zero. The witness for Y meets no other optional site; the witness for X meets no core site and at most the neighboring Y observation. Therefore the singleton fatality is robust under every larger schedule. Consequently the only minimal fatal supports in this regime are the present singleton extras.

60.7.3 Width two

By duality take $q = 2$ and odd $p \geq 3$, with the six odd-parity core sites labelled

$$A = (-h, 0, 0), \qquad B = (-h, 1, 0), \qquad C = (-h, 1, 1),$$
$$D = (h-1, 0, 0), \quad E = (h-1, 1, 0), \quad F = (h, 0, 0).$$

The core has exactly the two non-singleton minimal supports

$$\{A, D\}, \qquad \{B, D\}.$$

The possible extras are

$$X_1 = (h-1, 0, 1) \quad (p \geq 7), \qquad X_2 = (h-2, 1, 1) \quad (p \geq 11),$$

and each present extra is singleton-fatal. Hence

$$\mathrm{Adm}_{p,2} = \{H : X_1, X_2 \notin H,\ \{A, D\} \not\subseteq H,\ \{B, D\} \not\subseteq H\},$$

with absent extras omitted.

Here the sufficiency and necessity can be checked directly. Write $p = 2h - 1$, $m = 2h + 1$, and for $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + pk_t, \qquad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The mandatory reset family is explicitly

$$\begin{aligned} \mathcal{R}_h^{\mathrm{base}} = {} & \{\mu(jm + 2) : -h - 1 \leq j \leq h\} \\ & \cup \{(x, h-2) : x \geq h + 3\} \\ & \cup \{(\epsilon_{h+1}, y) : y \geq \beta_{h+1}\}. \end{aligned}$$

Two episode-closed repertoires are

$$\begin{aligned} \mathcal{R}_{\neg D,h} = {} & \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \geq h + 2\}, \\ \mathcal{R}_{D,h} = {} & \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(x, h-1) : x \geq h + 2\} \cup \{(x, h) : x \geq h + 1\}. \end{aligned}$$

Every pair in either repertoire has minority coordinate at most h. Before the first reset, MATCH and interior SEARCH only delete the visible top cell, so a boundary source is a suffix of its starting canonical word. This forces the exact source geometry: with D disabled, the common A/B source at D_{-h} is the one-cell stack 0; with D enabled it is an arbitrary 0^n, $n \geq 1$; all C sources are unary 1^n and all D, E, F sources are unary 0^n.

The eager targets are therefore exactly

site	canonical target pair
A	(ϵ_h, β_h)
B	$(\epsilon_{h-1}, \beta_{h-1})$
C	$(\epsilon_h, \beta_h + n - 1)$
D	$(n + h, h)$
E, F	$(n + h + 1, h - 1)$.

Thus $\mathcal{R}_{\neg D,h}$ is closed under every subset of A, B, C, E, F, giving $2^5 = 32$ safe schedules with $D = 0$, while $\mathcal{R}_{D,h}$ is closed when $D = 1$ and $A = B = 0$, giving another $2^3 = 8$. This proves the 40 safe schedules without enumeration.

For necessity, the D-ray gives, for every $x \geq h+1$,

$$(C, \eta(x,h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the A/B source fiber from $\{0\}$ to the whole unary ray. A uniform prefix reaches $(D_{-h}, 000)$ while meeting no optional site except D. At that common source

$$A : (D_{-h}, 000) \xrightarrow{0} (C, \Gamma_h 00), \qquad B : (D_{-h}, 000) \xrightarrow{1} (C, \Gamma_{h-1} 00).$$

Each Γ_t ends in at least p consecutive ones; writing $\Gamma_t = z_t 1^p$, subsequent MATCHes along z_t leave the nonempty stack $1^p 00$. Its residual is $2p - 2p = 0$, so it is one primitive kernel packet at semantic zero and is nonaccepting. Hence $\{A, D\}$ and $\{B, D\}$ are robust minimal fatal supports. The two maximal core schedules are $\{A, B, C, E, F\}$ and $\{C, D, E, F\}$; the extra sites are independently singleton-fatal.

60.7.4 The boundary line $q = 1$

For $(p, q) = (1, 1)$ the optional-site universe is empty. For $(2, 1)$ the complete four-site universe is

$$X = (0, 1, 0), \qquad A = (-1, 1, 1), \qquad B = (-1, 0, 0), \qquad G = (1, 0, 0),$$

and the minimal fatal supports are

$$\{B, X\}, \qquad \{B, G\}.$$

The balanced words 111001 and 111110001 are explicit fatal certificates; they end at semantic residual zero with a nonaccepting ordering of one primitive kernel packet. There are exactly ten admissible schedules and two maximal ones.

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \qquad D = (h, 1, 1), \qquad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h-1, 1, 0), \qquad C = (h, 0, 1).$$

Every positive optional site is singleton-fatal, while A alone generates an episode-closed unary negative reset corridor. For $h \geq 2$ one may use

$$w_B = 1^{p(p-2)} 0^{p-2}, \qquad w_C = 1^{2h(h+1)-1} (0\, 1^{p-1})^{h-1} 0^h,$$

and for every $h \geq 1$,

$$w_D = 1^{2h(h+1)} 0^p 1^{2h(h+1)+1}, \qquad w_E = 1^{2h^2+4h+1} 0^{p+1} 1^{2h(h+1)+1}.$$

The B-witness meets no other optional site; after B is excluded the C- and D-witnesses are unpreempted; after B, D are excluded the E-witness is unpreempted. The last two end at the noncanonical zero-residual kernel word $(C, 1^p 0)$, and the first two likewise end at explicit nonaccepting zero-residual configurations. Hence no positive-site failure can be repaired by enlarging the schedule, whereas the unary corridor generated by A stays reset-closed. Thus

$$\mathrm{Adm}_{p,1} = \{\varnothing, \{A\}\},$$

so there is one maximal schedule.

For even $p = 2h \geq 4$, the always-present sites are

$$A = (-h, 1, 1), \qquad B = (-h, 0, 0), \qquad C = (-h, 1, 0),$$
$$E = (h-1, 0, 0), \quad F = (h-1, 1, 0), \quad G = (h, 0, 0),$$

and $D = (h-1, 1, 1)$ is also present when $p \geq 6$. The vertices D and E are singleton-fatal when present, and the non-singleton minimal supports are exactly

$$\{B, F\}, \quad \{B, G\}, \quad \{C, F\}, \quad \{C, G\}.$$

After singleton-fatal sites are removed, every safe schedule lies in one of the two closed sectors

$$\mathcal{N} = \{A, B, C\}, \qquad \mathcal{P} = \{A, F, G\}.$$

For subsets of $\mathcal{N}$, the only new optional reset words are unary-one words and the source fibers satisfy $\mathcal{E}_A \subseteq 1^+$ and $\mathcal{E}_B = \mathcal{E}_C = \{0\}$; the mandatory reset family plus this unary ray is episode-closed. For subsets of $\mathcal{P}$, the only new positive reset ray is $(10)^h 0^n$, $n \geq 1$, together with the unary-one negative ray from A; all positive sources remain unary zero and all targets remain canonical and kernel-reduced. Thus every subset of either sector is safe.

Necessity has uniform balanced certificates. For $h \geq 3$ one may take

$$w_D = 1^{p(p-3)} 0^{p-3},$$

and for $h \geq 2$,

$$w_E = 1^{(h+1)(p-1)} (0\, 1^{p-1})^{h-1} 0^h.$$

The four cross-pair supports are certified by

$$w_{BF} = 1^{h(p+1)} 0^p 1^{h(p-1)}, \qquad w_{CF} = 1^{h(p+1)} 0^{p-1} 1^{h(p-3)},$$
$$w_{BG} = 1^{h(p+3)} 0^{p+1} 1^{h(p-1)}, \quad w_{CG} = 1^{h(p+3)} 0^p 1^{h(p-3)}.$$

In each cross-pair certificate the positive site first expands the negative-boundary top-0 source from 0 to 00, after which the indicated negative site traps exactly one primitive kernel packet. Consequently the only maximal schedules are $\mathcal{N}$ and $\mathcal{P}$, and the union of their subset families has $8 + 8 - 2 = 14$ schedules because the overlap is $2^{\{A\}}$. Physical duality gives the cases $p = 1$.

Combining the four regimes proves the exact fatal-support table used in table 60.1. In particular, its use in theorem 60.12 is no longer an appeal to an unnamed external classification: all vertices, the admissibility contract, the exhaustive universe, and the fatal-support mechanisms required by the confluence argument have been specified here.

60.8 Interface with the General Theory

The structural program of Part IV is now closed at the level needed for abstraction. Exact semantics has been separated from schedule choice; arbitrary-coprime causal geometry has been typed by presentation category; protected quotients and bounded recodings specify which physical information is retained; and confluence has been split into distinct endpoint, schedule, fiber, sheet, and protected-descent questions.

Part V can therefore remove the weighted-balance arithmetic without erasing the physical distinctions that made the preceding theorems meaningful. Its starting data are the reusable objects isolated here: a semantic transition system, an exact physical lift, a canonical section, bounded local observation, and an explicitly chosen comparison contract.

Part V

General Theory of Exact Pushdown Normalization

Part guide. **Question.** Which conclusions of the WSC analysis survive after weighted-balance arithmetic is removed? **Main result.** Rooted exact semantic lifts, canonical sections, bounded-local realizability, collapse capacity, representation debt, behavioral safety, conflict complexes, synthesis, re-coordination, and verification are developed as an abstract framework. **Why here.** This is the conceptual center of the monograph: it states the reusable theory rather than another special-purpose automaton. **Dependency.** Part IV supplies the motivating structures; subsequent Parts reuse the framework but do not define its scope.

Chapter 61

Semantic Fibers and Canonical Retraction

Parts I–IV progressively separated semantic evolution from physical normalization. Part IV completed that separation by distinguishing schedule causality, protected comparison, and confluence at different physical layers. The reusable conclusion is

> semantic evolution can be unique while physical normalization is not.

The General Theory now removes the weighted-balance specifics.
Its objects are:

- an abstract semantic transition system;
- an exact physical lift;
- a chosen section;
- bounded-local physical update rules;
- explicitly stated representation categories.

The word **General** is scoped accordingly.
This macro-Part does not claim:

- a theory of every pushdown automaton;
- existence of a global section for every correct DPDA;
- unrestricted DPDA minimization;
- unrestricted infinite-lift synthesis;
- universal schedule downward closure;
- universal confluence.

61.1 Rooted exact semantic lifts and defined domains

Let M be a semantic state set with a distinguished root m_0 and deterministic partial actions

$$U_a : M \rightharpoonup M, \qquad a \in \Sigma.$$

Let C be a physical policy domain with root c_0 and deterministic partial settled-to-settled macro-updates

$$\delta_a : C \rightharpoonup C.$$

A macro-update may stand for one input-consuming raw transition together with uniquely forced finite internal normalization before or after that input. Thus the abstract theory is compatible both with real-time lifts and with deterministic ε-pushdown realizations viewed at settled boundaries.

Write M^{reach} and C^{reach} for the states and physical configurations reachable from the two roots by words whose successive actions are defined. An *exact rooted lift* is a surjective projection

$$\boxed{\Theta : C^{\mathrm{reach}} \to M^{\mathrm{reach}}}$$

such that, for every reachable c and symbol a,

$$\boxed{c \in \mathrm{Dom}(\delta_a) \iff \Theta(c) \in \mathrm{Dom}(U_a)}$$

and, on that common defined domain,

$$\boxed{\Theta(\delta_a(c)) = U_a(\Theta(c)).}$$

This domain-reflection clause prevents a physical policy from silently disabling a semantic action or inventing one that the semantic system does not possess.

Choose a reachable section

$$\boxed{\sigma : M^{\mathrm{reach}} \to C^{\mathrm{reach}}}, \qquad \Theta\sigma = \mathrm{id}.$$

When zero representation debt from the root is discussed we additionally assume

$$\boxed{c_0 = \sigma(m_0).}$$

Define the canonical retraction

$$\boxed{N = \sigma\Theta.}$$

For readability, the superscript "reach" is suppressed in the remainder of the General Theory: M and C henceforth mean these rooted reachable domains unless an ambient domain is explicitly named.

61.2 Canonical fibers

Theorem 61.1 — Canonical fiber and retraction.

The map N satisfies

$$\boxed{N^2 = N,}$$

$$\boxed{\Theta N = \Theta,}$$

and

$$\boxed{\mathrm{Fix}(N) = \mathrm{im}\,\sigma.}$$

Moreover,

$$\boxed{\Theta(c) = \Theta(c') \iff N(c) = N(c').}$$

Thus the fibers of the canonical retraction are exactly the semantic fibers.

Proof.

Idempotence follows from

$$\Theta\sigma = \mathrm{id}.$$

If $N(c) = c$, then $c = \sigma(\Theta(c))$, so c lies in the section image.

Conversely every section point is fixed.

Finally, equal semantic values clearly give equal canonical representatives, and equal canonical representatives project to equal semantic states. □

This is the abstract form of the canonical-fiber idea first seen concretely in WSC.

61.3 Canonical absorption

Theorem 61.2 — Canonical absorption.

For every input word w for which the rooted transition is defined,

$$\boxed{N\delta_w = N\delta_w N = \sigma U_w \Theta.}$$

Thus the canonical endpoint after reading w depends only on the initial semantic state.

All prior physical normalization history is forgotten by the final retraction.

This theorem is the general form of:

- eager/lazy semantic equivalence;
- WSC canonical absorption;
- terminal normalization coherence.

It remains weaker than an unrestricted confluence theorem.

61.4 Zero debt and raw conjugacy

A reachable section is **transition-closed** when, for every $m \in M$ and every symbol a with $U_a(m)$ defined,

$$\boxed{\delta_a(\sigma(m)) = \sigma(U_a(m)).}$$

Domain reflection guarantees that the left-hand side is defined whenever the right-hand side is.

Theorem 61.3 — Rooted zero-debt/raw-conjugacy equivalence.

Assume $c_0 = \sigma(m_0)$ and that M, C denote the rooted reachable domains just defined. Then the following are equivalent:

1. the chosen section is transition-closed;
2. every reachable physical configuration lies on the section, equivalently

$$\boxed{C = \sigma(M).}$$

Proof. If the section is transition-closed, induction on the consumed word gives $\delta_w(c_0) = \sigma(U_w(m_0))$ whenever the word is defined, so every reachable physical configuration lies on the section. Conversely, let $m \in M$ and let a be defined at m. Because m is rooted-reachable, some word w satisfies $U_w(m_0) = m$. By the second hypothesis the corresponding physical endpoint lies on the section; exactness forces that endpoint to be the unique section point $\sigma(m)$. The next physical macro-update is reachable and again lies on the section, while exactness gives semantic value $U_a(m)$. Hence it must equal $\sigma(U_a(m))$. Thus the section is transition-closed. □

The rooted hypotheses are essential: bijectivity of Θ on some unrelated reachable representatives does not imply closure of a separately chosen section.

61.5 Three levels now separated permanently

From this point onward we distinguish:

$$\boxed{\text{semantic state}}$$

canonical section representative

arbitrary reachable physical representative in the same fiber.

The next chapter asks whether a bounded-local physical policy can keep every semantic transition on the chosen section.

Chapter 62

Bounded-Local Realizability, Collapse Capacity, and the Three-Stage Obstruction

The top-rewrite suffix criterion was discovered in the literal WSC setting. The General Theory now separates the part that depends on a pushdown stack from the more abstract information-capacity statement.

The fundamental question is:

> Can many semantically distinct sources that share one next semantic state be represented by canonical physical sources whose one-symbol macro-updates all hit one chosen canonical target without using more local information, exposing hidden source information, or retaining representation debt?

62.1 Collapse interfaces and retained information

Fix a symbol $a \in \Sigma$ and a reachable semantic target $m^* \in M$.

Collapse family. A nonempty finite set

$$\mathcal{S} \subseteq U_a^{-1}(m^*)$$

is an a-collapse family with target m^*. Put

$$N = |\mathcal{S}|, \qquad I_{\mathrm{ret}} = \log_2 N.$$

The quantity I_{ret} measures distinctions present immediately before the common semantic action; it is not the size of an earlier ambient fiber.

For $x \in \mathcal{S}$, exactness and domain reflection make the physical macro-update

$$\delta_a(\sigma(x)) \in \Theta^{-1}(m^*)$$

well defined. Call x a *canonical hit* when

$$\delta_a(\sigma(x)) = \sigma(m^*),$$

and a *debt source* otherwise. Write

$$D = \#\{x \in \mathcal{S} : x \text{ is a debt source}\}, \qquad K = N - D.$$

62.2 Observation filtrations and hidden information

Let

$$\omega_h : \sigma(\mathcal{S}) \longrightarrow \Omega_h, \qquad h = 0, 1, 2, \ldots,$$

be a nested family of finite source observations: ω_{h+1} refines ω_h. Put

$$V(h) = |\Omega_h|.$$

The index h is an abstract information depth. In a pushdown specialization it will be the number of source-stack cells made available; in another physical model it may describe a different nested observation window.

A canonical source computation is called *h-confined* when, before its first canonical hit, it never accesses source information beyond that represented by ω_h. Let

$$X_h = \#\{x \in \mathcal{S} : x \text{ is a canonical hit but not } h\text{-confined}\}.$$

Hidden-tail noncoalescence. The filtration has hidden-tail noncoalescence on this collapse interface if two distinct canonical sources with the same ω_h-observation cannot both remain h-confined and reach the same chosen physical target $\sigma(m^*)$. This is the precise hypothesis needed by the abstract counting argument. It is not assumed merely because the physical device is called a pushdown automaton; the later pushdown-exposure chapter proves it from untouched-source-cell provenance.

Theorem 62.1 — Filtered collapse-capacity theorem.

If hidden-tail noncoalescence holds, then for every $h \geq 0$,

$$\boxed{D + X_h + V(h) \geq N.}$$

Equivalently,

$$\boxed{X_h \geq (K - V(h))_+.}$$

Proof. Among the K canonical sources, exactly $K - X_h$ are h-confined. Hidden-tail noncoalescence makes ω_h injective on this subset, so

$$K - X_h \leq V(h).$$

Substituting $K = N - D$ gives the claim. □

Corollary 62.2 — Exposure-rank form.

Assume zero debt and that every canonical computation is h-confined for some finite h. Let $e(x)$ be the least such depth and sort the values as

$$e_{(1)} \leq \cdots \leq e_{(N)}.$$

With generalized inverse

$$V^{-1}(r) = \min\{h \geq 0 : V(h) \geq r\},$$

one has

$$\boxed{e_{(r)} \geq V^{-1}(r)} \qquad (1 \leq r \leq N).$$

Corollary 62.3 — Bit strong converse.

Let

$$H_{\text{phys}}(h) = \log_2 V(h).$$

Then

$$\boxed{\frac{D + X_h}{N} \geq \left(1 - 2^{H_{\mathrm{phys}}(h) - I_{\mathrm{ret}}}\right)_+.}$$

In particular, if $H_{\mathrm{phys}}(h) \leq I_{\mathrm{ret}} - \lambda$, then

$$\boxed{\frac{D + X_h}{N} \geq 1 - 2^{-\lambda}.}$$

62.3 Uniform one-top portability

For ordinary one-top rewriting, sources with the same local observation must share one rewrite rule. The concrete model-specific alignment test is already proved in Chapter 37: after the common local rewrite, every part of the source below the inspected top position is untouched, so direct canonical realization requires the corresponding untouched suffix to occur as the required suffix of the chosen canonical target. We use that established suffix criterion here as the one-top specialization of Stage III rather than promoting a second, abstractly worded portability lemma.

62.4 Finite observation capacity

Corollary 62.5 — Zero-debt finite-observation obstruction.

Assume hidden-tail noncoalescence and fix h. If all sources are canonical hits and h-confined, then

$$\boxed{N \leq V(h).}$$

Consequently, a finite observation regime cannot keep a transition-closed canonical section across a collapse family larger than its observation capacity. If one target has arbitrarily large finite collapse subfamilies while $V(h)$ is fixed, total h-confined transition closure is impossible.

62.5 Bounded-d-top pushdown capacity

For a top-access pushdown interface that can distinguish a control state and at most d source-stack symbols, one may take the source signature consisting of that state and the complete shorter stack or the top d symbols. Hence a coarse universal bound is

$$\boxed{V(d) \leq |Q| \sum_{r=0}^{d} |\Gamma|^r.}$$

The later source-exposure specialization refines this count by charging only states and work symbols that occur on the source interface.

Corollary 62.6 — Bounded-d-top capacity.

Under hidden-tail noncoalescence, a zero-debt d-confined collapse family must satisfy

$$\boxed{N \leq |Q| \sum_{r=0}^{d} |\Gamma|^r.}$$

Thus semantic collapse can force control/alphabet capacity even when no particular residual encoding is assumed.

62.6 The three stages

Synthesis Principle 62.7 — Three-stage bounded-local obstruction decomposition.

Failure of direct canonical bounded-local realizability can be localized to one of three ordered tests:

Stage I — semantic capacity overflow

The collapse load exceeds the observation capacity allowed by the physical interface.

Stage II — chosen-section observation collision

Capacity is sufficient in principle, but the chosen section assigns sources to one observation class even though their canonical updates require physically incompatible behavior.

Stage III — hidden-tail/action alignment failure

The sources are locally separated, but the available local action cannot transform the hidden remainder into the chosen canonical target. For one-top pushdown rewriting this becomes the suffix/tail obstruction.

The three stages are a diagnostic organization, not three new independent lower-bound theorems. The quantitative theorem is the filtered collapse-capacity inequality above; the later model-specific chapters supply the alignment tests.

62.7 One-top specialization

Corollary 62.8 — One-top specialization.

In the literal one-top setting, the Stage III test is exactly

$$\boxed{\text{untouched suffix is a suffix of the canonical target.}}$$

Thus the theorem of Chapter 37 is the one-top boundary case of the general locality decomposition.

62.8 Locality–debt–recoordination principle

Synthesis Principle 62.9 — Locality–debt–recoordination trilemma.

When direct canonical realization fails, an exact implementation cannot keep all three of the following fixed:

1. the local observation regime;
2. the chosen canonical section;
3. direct canonical successor behavior.

At least one response is necessary:

$$\boxed{\text{refine observation}} \quad \text{or} \quad \boxed{\text{allow representation debt}}$$

$$\text{or} \quad \boxed{\text{re-coordinate the section}}.$$

This is a synthesis principle extracted from the exact alternatives already proved; it is intentionally not presented at the same theorem level as the collapse-capacity inequality.

The next chapter asks an equally important question:

If debt is forced, is it necessarily dangerous?

The answer is no.

Chapter 63

Representation Debt and Behavioral Safety

A physical configuration can be noncanonical and still behave perfectly.

The General Theory therefore separates:

$$\boxed{\text{representation debt}}$$

from

$$\boxed{\text{behavioral incorrectness.}}$$

This distinction prevents a common mistake:

> "The machine left the canonical section, therefore it is already wrong."

That implication is false.

63.1 Terminal semantics

Let

$$f : M \to O$$

be the semantic terminal observable and

$$\alpha : C \to O$$

the physical terminal observable.

A physical configuration c is behaviorally correct for all continuations if

$$\boxed{\alpha(\delta_w(c)) = f(U_w(\Theta(c)))}$$

for every continuation w.

63.2 Immediate disagreement

Define the immediate disagreement set

$$\boxed{\mathcal{D} = \{c : \alpha(c) \neq f(\Theta(c))\}.}$$

A configuration can be outside $\mathcal{D}$ now but still have a future continuation reaching $\mathcal{D}$.

Thus immediate agreement is not enough.

Proposition 63.2A — Terminal factorization.

The identity

$$\boxed{\alpha = f\Theta}$$

holds on all physical configurations if and only if:

1. α is constant on every semantic fiber of Θ; and
2. $$\boxed{\alpha\sigma = f.}$$

Proof.

If $\alpha = f\Theta$, equal semantic projections have equal terminal values, and

$$\alpha\sigma = f\Theta\sigma = f.$$

Conversely, let c be any physical configuration. Fiber constancy gives

$$\alpha(c) = \alpha(\sigma(\Theta(c))).$$

Using $\alpha\sigma = f$,

$$\alpha(c) = f(\Theta(c)).$$

□

This proposition separates two requirements that are easy to conflate:

$$\boxed{\text{terminal correctness of the chosen section}}$$

and

$$\boxed{\text{terminal constancy across every physical representative of a semantic fiber.}}$$

63.3 Behavioral hull

Define

$$\boxed{\mathcal{B} = \{c : \alpha(\delta_w(c)) = f(U_w(\Theta(c))) \text{ for all } w\}.}$$

Theorem 63.1 — Behavioral hull characterization.

$$\boxed{\mathcal{B} = \nu X.\Phi(X) = C \setminus \mathrm{Pre}^*(\mathcal{D}).}$$

That is:

- $\mathcal{B}$ is the greatest forward-safe fixed point;
- a configuration is behaviorally safe exactly when no continuation reaches immediate disagreement.

This is the exact physical correctness object.

Corollary 63.2 — Root correctness.

A fixed policy is correct from root c_0 exactly when

$$\boxed{c_0 \in \mathcal{B}.}$$

63.4 Forward invariance

Theorem 63.3 — Forward invariance of behavioral safety.

If

$$c \in \mathcal{B},$$

then every reachable successor

$$\delta_w(c)$$

also lies in $\mathcal{B}$.

Thus safe debt remains safe under the fixed policy.

63.5 Debt birth dichotomy

Observation 63.4 — Debt birth dichotomy.

When a transition first leaves the canonical section, the resulting debt is born in exactly one of two regimes:

1. $$\boxed{c \in \mathcal{B}}$$

   — behaviorally harmless debt;

2. $$\boxed{c \notin \mathcal{B}}$$

   — debt that is already behaviorally unsafe.

A later continuation does not transform safe debt into unsafe debt.

If a continuation eventually reveals an error, the configuration was outside $\mathcal{B}$ from the moment the unsafe debt was created.

This corrects the tempting narrative:

"debt becomes fatal later."

The correct statement is:

$$\boxed{\text{a later suffix reveals pre-existing unsafety.}}$$

63.6 Two opposite examples

Weighted-balance schedule conflict

Some noncanonical debts produced by forbidden eager schedules lie outside the behavioral hull and are exposed by short continuations.

Multi-type Dyck collapse

At a rejecting collapse boundary, bounded-locality may force noncanonical physical representatives, yet every such state can lie in an absorbing rejecting semantic fiber.

The debt is forced but harmless.

Thus:

$$\boxed{\text{forced debt} \not\Rightarrow \text{fatal debt.}}$$

The next chapter develops semantic certificates for proving harmlessness without constructing the exact physical behavioral hull.

Chapter 64

Semantic Containment Cores

The behavioral hull

$$\mathcal{B}$$

is exact.

But it lives in the full physical configuration space.

A coarser proof certificate can sometimes be obtained entirely at the semantic level.

64.1 Semantic trap containment

Let

$$S \subseteq M$$

be forward invariant:

$$U_a(S) \subseteq S.$$

Suppose every physical representative in every fiber over S is immediately terminally correct.

Theorem 64.1 — Semantic trap containment.

Under those hypotheses,

$$\boxed{\Theta^{-1}(S) \subseteq \mathcal{B}.}$$

Proof idea. A run beginning over S remains over S semantically.

Every reached physical representative lies in a fiber whose entire ambient fiber is terminally correct.

Therefore no continuation can reach disagreement. □

This is conservative because it requires **every** representative in the full fiber to be safe.

64.2 Absorbing-fiber harmlessness

Corollary 64.2 — Absorbing-fiber harmlessness.

If a semantic state is absorbing and all physical representatives in its fiber give the correct terminal observation, then the entire fiber lies in the behavioral hull.

This is the clean explanation of multi-type Dyck collapse debt:

- the debt can be forced physically;
- the semantic state is already an absorbing reject;
- the entire fiber is harmless.

64.3 Maximal full-fiber containment core

Define the semantic good set of states whose full fibers are terminally correct.

Iteratively remove any semantic state whose successor leaves the current set.

Theorem 64.3 — Maximal fiberwise semantic containment core.

There exists a greatest forward-invariant semantic set

$$\boxed{\mathcal{T}}$$

such that

$$\boxed{\Theta^{-1}(\mathcal{T}) \subseteq \mathcal{B}.}$$

This is the maximal **ambient full-fiber** semantic containment certificate.

It is deliberately conservative.

An unreachable bad representative can remove an otherwise useful semantic state from $\mathcal{T}$.

64.4 Reachable-fiber refinement

Let

$$F_m^{\mathrm{reach}}$$

be only the representatives actually reachable under the policy.

Use those reachable fibers in the containment construction.

Theorem 64.4 — Reachable semantic containment.

There is a greatest reachable-fiber containment core

$$\boxed{\mathcal{T}^{\mathrm{Reach}}}$$

whose reachable physical representatives are behaviorally safe.

It satisfies the conceptual hierarchy

$$\boxed{\mathcal{T} \subseteq \mathcal{T}^{\mathrm{Reach}}.}$$

The reachable core removes ambient syntactic contamination.

64.5 Root correctness via containment

Theorem 64.5 — Root correctness via reachable containment.

If the root semantic state belongs to

$$\mathcal{T}^{\mathrm{Reach}},$$

then the physical root lies in

$$\mathcal{B}.$$

Thus the three correctness scales are:

$$\boxed{\mathcal{T} \longrightarrow \mathcal{T}^{\mathrm{Reach}} \longrightarrow \mathcal{B}.}$$

They answer increasingly exact questions:

- ambient semantic certificate;
- reachable semantic certificate;
- exact physical behavioral safety.

The next chapter returns to normalization schedules and uses the behavioral hull to distinguish merely structural unsafe targets from genuinely fatal ones.

Chapter 65

Safe Reset and the Fatal Frontier

The 2:1 causal theory introduced first-hit exposure and portability.

The General Theory now separates two notions:

1. a structural reset certificate;
2. exact behavioral safety.

This separation is essential because leaving a convenient reset domain need not be fatal.

65.1 Wordwise safe-reset domain

Let $\mathcal{R} \subseteq C$ be a set of physical configurations.

Informally, $\mathcal{R}$ is a safe-reset domain if every run that stays within the certified reset behavior can be normalized correctly word by word.

Proposition 65.1 — Safe-reset certificate.

If $\mathcal{R}$ is a wordwise safe-reset domain for a schedule family $\mathfrak{H}$, then every schedule whose eager activations remain within the certified reset behavior is correct.

The domain is a proof certificate.

It need not equal the exact behavioral hull.

65.2 Structural unsafe targets

For schedule H, let

$$U_{\mathcal{R}}(H)$$

be the set of actual eager activation targets that lie outside $\mathcal{R}$.

These are structurally uncertified targets.

Some may still lie in the behavioral hull.

Define that difference as behavioral slack.

The exact badness criterion therefore needs both layers.

65.3 Fatal Frontier

Theorem 65.2 — Fatal Frontier.

Assume $\mathcal{R}$ is a wordwise safe-reset domain for the full schedule family.

Then

$$\boxed{H \notin \mathrm{Adm} \iff U_{\mathcal{R}}(H) \setminus \mathcal{B}_H \neq \varnothing.}$$

Thus a schedule is bad exactly when it actually produces an eager target that is:

1. outside the reset certificate;
2. outside the exact behavioral hull.

This is the abstract form of what Chapter 53 saw concretely.

A structural deviation becomes a **fatal frontier event** only when no behavioral slack saves it.

65.4 Activation tightness

Corollary 65.3 — Activation tightness.

If every actual eager target outside $\mathcal{R}$ is behaviorally unsafe, then

$$\boxed{H \notin \mathrm{Adm} \iff U_{\mathcal{R}}(H) \neq \varnothing.}$$

In that case the structural unsafe frontier is exact.

Without activation tightness, structural detection is only sufficient, not necessary.

65.5 Site-free diagnostics

Sometimes one can certify fatality by a continuation that uses no additional optional eager site.

Theorem 65.4 — Site-free diagnostic frontier.

Under the stated safe-reset hypotheses, a finite site-free continuation can certify that an activation target is already outside the behavioral hull.

This gives a causal witness local to the activation event.

The 2:1 certificate

$$1111100 \rightsquigarrow (R, 00)$$

followed by the short exposing suffix is the motivating example.

65.6 Fatal causality

Proposition 65.5 — Fatal causality implies conflict.

If a minimal enabled site set H creates a fatal first-hit target at site b, then

$$H \cup \{b\}$$

is an inadmissible schedule set.

Thus directed fatal cause induces set-theoretic schedule conflict.

The next chapter develops the combinatorics of those conflicts and stress-tests the theory on 3:2 and an infinite odd-p:2 family.

Chapter 66

Conflict Complexes and Weighted-Balance Stress Tests

The Fatal Frontier tells us when one schedule is wrong.

We now ask how the whole family of schedules is organized.

66.1 Minimal conflicts

A conflict is a set of optional sites that cannot be enabled together.

A conflict is minimal if all of its proper subsets are admissible.

Lemma 66.1 — Minimal-conflict participation.

Every site belonging to a minimal conflict participates in some fatal mechanism for that conflict.

No member can be completely irrelevant.

Otherwise the conflict would not be minimal.

66.2 Robust fatal supports

Suppose a family of minimal site sets has fatal certificates that remain valid whenever additional sites outside the support are enabled.

These are robust fatal supports.

Theorem 66.2 — Robust fatal supports give an independence complex.

Under the robust-certificate hypotheses, the admissible schedule family is exactly the family of site subsets containing no minimal fatal support.

Therefore it is the independence complex of the hypergraph of minimal fatal supports.

The phrase **independence complex** is combinatorial packaging.

The project-specific content is the derivation of the fatal supports from physical normalization causality.

Corollary 66.3 — Finite diagnostic basis.

If the robust minimal fatal support family is finite, the entire finite schedule classification has a finite diagnostic basis.

66.3 Reset-ray fork

Proposition 66.4 — Reset-ray fork template.

Suppose one site D changes the reachable reset-ray geometry so that it exposes fatal first hits at two victim sites A and B, while schedules not containing D remain reset-safe and schedules containing D but neither victim remain safe.

Then the minimal fatal supports are

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The admissible schedules are the independence complex avoiding those two pairs.

This template abstracts the directed fork mechanism later verified uniformly.

66.4 Imported weighted-balance foundations

The General Theory does **not** re-prove:

- primitive-kernel minimum demand;
- centered residual exactness;
- centered canonical-episode terminal correctness.

Those results were already established in the p:q construction.

Here they are used as a stress-test environment for the new causal machinery.

66.5 Exact 3:2 case

A finite reset analysis produces an exact certificate for the centered 3:2 schedule family.

Lemma 66.5A — Exact 3:2 finite reset certificate.

For the centered 3:2 schedule family there is an explicit finite reset repertoire whose episode closure certifies every schedule avoiding the two fatal pairs

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The same finite analysis produces first-hit fatal witnesses when either forbidden pair is enabled.

This lemma is the finite reset engine behind the exact schedule count; the detailed reset catalogue is retained for the technical appendix rather than repeated in the narrative.

Theorem 66.5 — Exact 3:2 schedule classification.

There are

$$\boxed{\text{40 admissible schedules}}$$

and

$$\boxed{\text{24 inadmissible schedules.}}$$

The minimal fatal fork is

$$\boxed{D \to A, \qquad D \to B.}$$

Equivalently, the minimal fatal supports are

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The technical finite reset catalogue supporting this result belongs in the appendix; the classification remains in the main narrative.

66.6 Infinite odd-p:2 family

Now let

$$q = 2$$

and

$$p \geq 3$$

be odd.
The underlying calculation gives:

- a detailed arithmetic catalogue;
- source geometry;
- reset closure;
- uniform D-induced fiber expansion;
- reachability of a common length-three fatal frontier;
- kernel-suffix fatality.

The long arithmetic/source catalogues are appendix material.
The structural conclusion is main-text material.

Lemma 66.6A — Odd-p:2 reset closure.

Let

$$q = 2, \qquad p = 2h - 1 \geq 3.$$

There are two explicit safe reset repertoires:

- $\mathcal{R}_{\neg D}$, used when site D is absent;
- $\mathcal{R}_D$, used when D is present but A and B are absent.

They satisfy

$$\boxed{D \notin H \Longrightarrow H \in \mathrm{Adm}_{p,2},}$$

and

$$\boxed{D \in H, \quad A, B \notin H \Longrightarrow H \in \mathrm{Adm}_{p,2}\,.}$$

This accounts for

$$32 + 8 = 40$$

safe schedules before the fatal pairs are analyzed.

Lemma 66.6B — Uniform D-induced fiber expansion.

For every

$$x \geq h + 1,$$

the D-activated canonical ray satisfies

$$\boxed{(C, \eta(x,h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}).}$$

Therefore enabling D expands the common A/B observation fiber from the singleton unary source

$$\{0\}$$

to

$$\boxed{\{0, 00, 000, \ldots\}.}$$

This is the uniform directed mechanism

$$\boxed{D \rightsquigarrow A, \qquad D \rightsquigarrow B.}$$

Lemma 66.6C — Uniform reachability of the length-three frontier.

For every

$$h \geq 2,$$

there is an explicit prefix u_h, using no optional eager site except D, that reaches the common fatal-frontier source

$$\boxed{(D_{-h}, 000).}$$

One convenient construction is

$$\boxed{u_h = v_h\, 1^{\,2h-1} 0^{\,2h},}$$

where v_h is the parity-dependent prefix that first reaches

$$(D_{h-1}, \varepsilon).$$

Thus the same length-three unary source is not merely syntactically possible; it is uniformly reachable across the entire odd-p:2 family.

Lemma 66.6D — Uniform kernel-suffix fatality.

For

$$p{:}2,$$

the primitive kernel is

$$\boxed{K = (2, p).}$$

The zero-weight physical kernel word

$$\boxed{1^p 00}$$

appears as a suffix in the eager A- and B-targets generated from the common source

$$(D_{-h}, 000).$$

A site-free continuation removes the non-kernel prefix and exposes this residual-zero kernel excess as behavioral disagreement.

Hence both A and B are uniformly fatal after the D-induced fiber expansion.

Appendix migration note

Two long supporting results are intentionally not expanded here:

- the **odd-p:2 arithmetic catalogue**;
- the **odd-p:2 source geometry catalogue**.

They are retained as appendix-level proof infrastructure. Their purpose is to justify the closed forms and source restrictions used in Lemmas 61.6A–61.6D, not to create a second narrative inside the main theorem.

Theorem 66.6 — Odd-p:2 fatal-fork family.

For every odd

$$p \geq 3,$$

$$\boxed{\mathrm{Adm}_{p,2} = \mathrm{Ind}\{\{A, D\}, \{B, D\}\}.}$$

Thus every member of the infinite family has:

$$\boxed{40 \text{ admissible}, \qquad 24 \text{ inadmissible}}$$

schedules and the same directed fatal fork

$$\boxed{D \to A, \qquad D \to B.}$$

This is an important stress test.
The General Theory is not merely repackaging the 2:1 example.
Its causal mechanism persists uniformly across an infinite parameter family.
The next chapter changes direction from diagnosis to construction:

Can a transition-closed section be synthesized when one exists?

Chapter 67

Rooted Section Synthesis

The theory has so far explained why canonical sections fail.

It now becomes constructive.

Given:

- a semantic system;
- a physical transition template;
- a chosen physical representative of the root;

can we synthesize a transition-closed section?

The answer is an exact path-outcome coherence criterion.

67.1 Rooted path-outcome fibers

Fix semantic root

$$m_0$$

and physical representative

$$c_0$$

with

$$\Theta(c_0) = m_0.$$

For semantic state m, define

$$\boxed{\mathcal{H}_{c_0}(m)}$$

to be the set of all physical outcomes obtained by following physical transitions along semantic words that take m_0 to m.

Different semantic paths to the same state can produce different physical endpoints.

That is exactly the synthesis obstruction.

67.2 Path-outcome coherence

Theorem 67.1 — Rooted path-outcome coherence criterion.

A rooted transition-closed section through c_0 exists if and only if

$$\boxed{|\mathcal{H}_{c_0}(m)| = 1}$$

for every reachable semantic state m.

When it exists, the section is uniquely determined by the root representative.

Proof.

If a transition-closed section exists, every semantic path to m must end at the single section point $\sigma(m)$.

Conversely, suppose every path-outcome fiber is a singleton.

Define $\sigma(m)$ to be that unique physical outcome.

If $m \xrightarrow{a} U_a(m)$, append a to any path reaching m.

Path-outcome uniqueness gives

$$\delta_a(\sigma(m)) = \sigma(U_a(m)).$$

Thus the section is transition-closed. □

Corollary 67.2 — One root representative determines the section.

For a fixed physical root candidate, there is at most one rooted transition-closed section.

67.3 Finite synthesis algorithm

For finite semantic systems, the theorem yields an exact propagation procedure.

Start from a root candidate.

Propagate its physical image along semantic edges.

Whenever a semantic state is reached again:

- if the physical endpoint agrees, continue;
- if it differs, synthesis fails.

Theorem 67.3 — Finite rooted synthesis.

For a finite semantic transition graph, this propagation decides exactly whether a rooted transition-closed section exists and constructs it when it does.

The algorithm is graph-theoretic once the physical transition template is fixed.

67.4 Short obstruction certificate

Theorem 67.4 — Short parallel-path obstruction certificate.

If finite rooted synthesis fails, there exist two semantic paths from the root to the same semantic state whose physical endpoints differ.

A shortest such pair gives a finite bounded certificate of failure.

Thus synthesis failure is witnessed by path disagreement.

67.5 Counting sections

Corollary 67.5 — Counting synthesized sections.

If the physical root fiber is finite, the number of rooted transition-closed sections equals the number of root representatives that pass the path-outcome coherence test.

The theory therefore turns section existence into a concrete search over root candidates.

67.6 Infinite semantic cores require symbolic synthesis

The coherence theorem itself does not require finiteness.

But brute-force propagation over an infinite semantic graph is not an algorithm.

The next chapter shows one important infinite case where a finite symbolic rule skeleton can be synthesized exactly.

That construction also demonstrates again that re-coordination can remove debt.

Chapter 68

Symbolic Re-Coordination and Forced-Debt Frontiers

The locality trilemma gave three responses to a failed canonical section:

refine observation, allow debt, re-coordinate.

This chapter gives a constructive example of the third option, then quantifies what remains unavoidable when only a partial section can be maintained.

68.1 Modular quotient/remainder synthesis

Consider integer translation semantics generated by fixed signed weights.

Choose a modulus large enough to absorb every one-step weight change.

Represent each semantic integer by:

- a finite sign/remainder state;
- a unary quotient stack.

Theorem 68.1 — Modular quotient/remainder synthesis.

There exists an explicit one-top zero-debt lift in which:

- the finite control stores the signed remainder;
- the stack stores the unbounded quotient;
- every semantic translation is realized directly;
- the chosen section is transition-closed.

This is a general constructive example of:

re-coordination removes normalization debt.

Scope

The construction is:

- pseudo-polynomial in the numerical weight bound;
- not polynomial in the bit length of weights;
- not a state-minimality theorem;
- not a general algorithm for arbitrary infinite lifts.

68.2 Finite symbolic skeleton

Theorem 68.2 — Finite symbolic skeleton and global path-outcome coherence.

The modular representation is governed by a finite set of symbolic local rules.

Those rules satisfy global rooted path-outcome coherence on the infinite semantic core.

Thus an infinite transition-closed section can sometimes be certified by finite symbolic arithmetic.

Corollary 68.3 — Zero-weight loop coherence.

Every semantic zero-weight loop returns the synthesized physical representation to the same section point.

This is an immediate path-outcome consequence, not a separate confluence theorem.

68.3 Partial canonical sections

Total transition closure may be impossible.

Let

$$G \subseteq M$$

be a semantic region over which a chosen section is transition-closed.

Proposition 68.4 — Canonical interior.

As long as the semantic run remains inside G, the physical run remains exactly on the chosen canonical section.

Representation debt is born only on an edge leaving the canonical interior.

This turns debt creation into a boundary phenomenon.

68.4 Canonical-hit capacity

Suppose many semantic source states share one target under the same input.

Only finitely many source representatives can land directly on one canonical target if the local observation space is finite.

Theorem 68.5 — Canonical-hit capacity.

For a fixed bounded-local policy, the number of source edges that can hit one chosen canonical target directly is bounded by the available local observation capacity.

This theorem no longer assumes total transition closure.

It applies specifically at the boundary of a partial section.

68.5 Quantitative forced debt

Let $\mathcal{S}$ be a finite set of relevant same-target source states and let Ω_a be the input-a observation set.

Corollary 68.6 — Quantitative forced-debt frontier.

At least

$$\boxed{|\mathcal{S}| - |\Omega_a|}$$

source edges are forced to leave the chosen section.

If $\mathcal{S}$ is infinite and Ω_a is finite, infinitely many debt-producing edges are forced.

This is the quantitative form of the earlier fiber-branching theorem.

68.6 Multi-type Dyck frontier

For one bracket type,

$$\boxed{\mathrm{Dyck}_1 : \text{ total canonical synthesis.}}$$

For

$$k \geq 2,$$

the multi-type Dyck semantic system has an infinite collapse boundary that exceeds bounded local observation capacity.

Theorem 68.7 — Dyck forced-debt frontier.

For

$$k \geq 2,$$

there is:

$$\boxed{\text{zero-debt live interior} + \text{infinite forced-debt collapse boundary.}}$$

Yet that boundary lies in an absorbing rejecting semantic fiber.
Therefore the forced debt is behaviorally harmless.
This example is crucial because it separates:

$$\boxed{\text{forced physical noncanonicality}}$$

from

$$\boxed{\text{behavioral failure.}}$$

The theory can therefore quantify unavoidable debt without equating debt with error.

Chapter 69

Effective Verification Hierarchy

The General Theory ends with algorithms.

The objective is not to claim new classical automata algorithms.

The contribution is a hierarchy of reductions from normalization correctness to established decision problems.

There are three principal verification levels.

69.1 Level I — Structural safe-reset saturation

Suppose a regular set Bad_{str} contains every structurally unsafe eager target outside a certified safe-reset domain.

Theorem 69.1 — Regular structural safe-reset certification.

If pushdown reachability saturation proves that no configuration in Bad_{str} is reachable under the schedule, then the safe-reset certificate proves policy correctness.

This test is:

$$\boxed{\text{sufficient in general.}}$$

Under activation tightness, it becomes exact.

This is the cheapest level when a good structural reset certificate is available.

69.2 Level II — Exact immediate-disagreement saturation

Suppose the exact immediate disagreement set

$$\mathcal{D}_H$$

is regular and effectively represented.

Theorem 69.2 — Regular immediate-disagreement saturation.

Then

$$\boxed{\mathcal{B}_H = C \setminus \mathrm{Pre}_H^*(\mathcal{D}_H)}$$

is regular and effectively constructible by pushdown predecessor saturation.

Thus the exact behavioral hull can be computed.

Corollary 69.3 — Exact root classification.

The fixed-policy root is correct exactly when it lies outside the saturated predecessor set of $\mathcal{D}_H$.

Corollary 69.4 — Exact finite schedule classification.

For a finite optional-site family, if the regular disagreement construction is effective for every schedule, the full schedule family is exactly decidable by enumerating schedules and applying the saturation test.

This turns the Fatal Frontier theory into a complete finite classifier in the regular subclass.

69.3 Reachable-containment shortcut

Sometimes one only needs to prove correctness of one root, not construct the entire exact hull.

Corollary 69.5 — Reachable-containment root shortcut.

A root can be certified through the reachable semantic containment core without explicitly constructing the whole physical behavioral hull.

This can be substantially cheaper when semantic containment is small.

69.4 Finite semantic containment computation

If the semantic system M is finite and the initial semantically good set G_0 is known:

Theorem 69.6 — Finite containment-core computation.

The maximal forward-invariant semantic containment core is computed by iterative removal of states with outgoing edges leaving the current set.

Once G_0 is known, the edge-processing complexity is

$$\boxed{O(|M||\Sigma|).}$$

Corollary 69.7 — Finite containment-failure certificate.

If a semantic root is removed, there is a semantic witness path of length at most

$$\boxed{|M| - 1}$$

leading from the root to a containment failure.

Thus semantic containment failure has a short certificate.

69.5 Level III — DPDA-equivalence fallback

Regular immediate disagreement is powerful but not universal.

When no useful regular representation of $\mathcal{D}_H$ is available, compare continuation languages directly.

For physical configuration c, construct:

- the physical continuation language from c;
- the semantic continuation language from $\Theta(c)$.

If both have effective deterministic pushdown presentations, classical deterministic pushdown equivalence can decide equality [21, 13].

Theorem 69.8 — DPDA-equivalence behavioral fallback.

Behavioral safety of a fixed physical root reduces exactly to equivalence between its physical and semantic deterministic pushdown continuation languages.

Corollary 69.9 — Exact fixed-policy root verification.

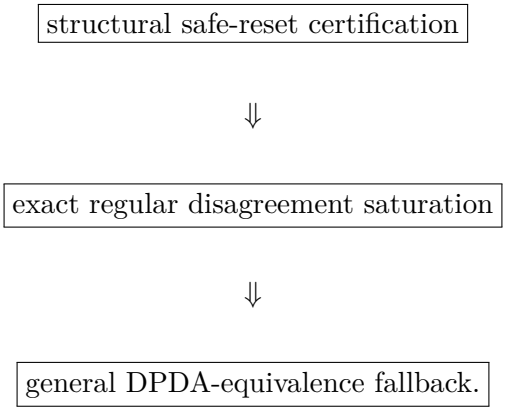
Whenever those two continuation languages are effectively represented as DPDAs, fixed-policy root correctness is decidable.

The imported equivalence algorithm is classical.

The contribution here is the reduction from normalization safety to that equivalence problem.

69.6 Verification hierarchy

The three levels are therefore:

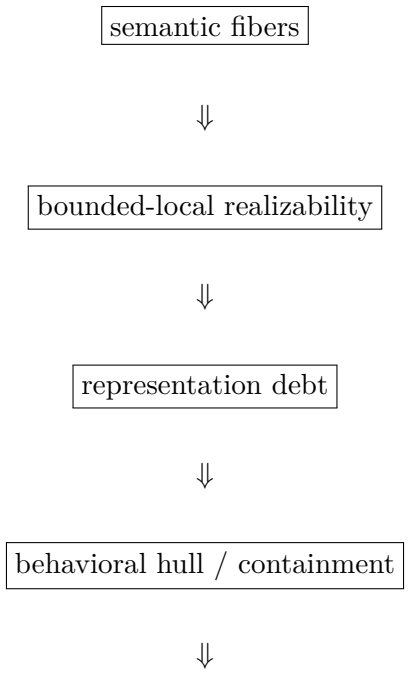


They trade:

- structural specificity;
- computational convenience;
- generality.

69.7 General Theory synthesis

The final theoretical chain is now complete:



safe reset / fatal causality

$\Downarrow$

conflict complexes

$\Downarrow$

section synthesis / forced debt

$\Downarrow$

effective verification.

But the monograph should not end on an abstract verification theorem. The post-theory branches below first record the enumerative, arithmetic, probabilistic, topological, and unrestricted-complexity consequences that emerged from the completed framework.

The final Part then returns one last time to the two 2023 DPDA figures.

The point of the General Theory is not to replace the original machine.

It is to explain it.

Part VI

Enumerative and Arithmetic Consequences of Completion Semantics

Part guide. **Question.** What exact combinatorial and arithmetic structures are encoded by the same residual and shortest-completion semantics? **Main result.** The Part derives operational event enumerations, solves the Dai–Fu–Qiu run–return and ratio-run–return problems, develops contact-preserving barrier compression, and obtains Farey-recursive and Cohn–Markoff completion structures. **Why here.** These are consequence branches sharing semantic coordinates, not additional axioms of the General Theory. **Dependency.** The branches reuse Parts I–V selectively and are intentionally separable from one another.

Chapter 70

Primitive Operational Statistics and Schubert Cancellation Geometry

The original 2:1 construction has already supplied the provenance for CANCEL, SEARCH, WRITE, and shortest-completion demand. We therefore begin directly with the centered p:q event process rather than repeating the automaton construction.

70.1 The centered p:q event process

Fix coprime positive integers p, q, put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor,$$

and write D_d for the centered debt state with $-h \le d \le h$. For an integer residual r, let

$$\mu(r) = (x, y) = \arg\min\{x + y : x, y \in \mathbb{Z}_{\ge 0},\ px - qy = r\},$$

which is unique after restricting to the standard nonnegative kernel strip. The canonical binary realization of a demand pair is

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \ge y, \\ (10)^x 1^{y-x}, & x < y. \end{cases}$$

The centered real-time process has four event classes. If the stack top matches the next input bit, the top cell is popped (MATCH/CANCEL). A mismatch is absorbed by changing d by one while $|d| < h$ (SEARCH$\pm$). At the outer boundary a further same-direction mismatch performs OUTER-WRITE and returns to debt 0. If the stack is empty, the next symbol performs BOTTOM-WRITE using the shortest-demand map μ.

The exact transition rule is the following. With input bit a and stack word S whose leftmost symbol is the top,

$$\begin{array}{ll} \text{top } z = a: & (d, zT) \mapsto (d, T), \\ (a, z) = (1, 0),\ d < h: & (d, 0T) \mapsto (d + 1, T), \\ (a, z) = (0, 1),\ d > -h: & (d, 1T) \mapsto (d - 1, T), \\ (a, z) = (1, 0),\ d = h: & (h, 0T) \mapsto (0, \eta(\mu((h + 1)m))T), \\ (a, z) = (0, 1),\ d = -h: & (-h, 1T) \mapsto (0, \eta(\mu(-(h + 1)m))T), \\ S = \varepsilon: & (d, \varepsilon) \mapsto (0, \eta(\mu(dm + w(a)))), \end{array}$$

where $w(1) = q$ and $w(0) = -p$. The language-recognition correctness of this centered normalization is background for the present event-enumeration problem; all statistics below refer to these displayed transitions.

A word is *rational-Dyck* when every prefix has

$$\rho = qN_1 - pN_0 \geq 0.$$

A scale-k balanced word has kp ones and kq zeros. Let $C(P)$ be its number of MATCH/CANCEL events and $O(P)$ its number of OUTER-WRITE events.

70.2 Primitive balanced words: fixed-stack Hamming geometry

A primitive balanced word has p ones and q zeros.

Lemma 70.1 (No later WRITE in the primitive sector)**.** *After the first input symbol of a primitive balanced word, the shortest-demand stack has length $m-1$. Exactly $m-1$ input symbols remain. Hence no later WRITE occurs: every remaining transition is SEARCH or CANCEL.*

Proof. The first BOTTOM-WRITE realizes the unique shortest completion of the first-symbol residual. Because the completed word has total composition (p, q), the demand length is $m - 1$. Every subsequent non-WRITE transition consumes one input symbol and one stack cell. A new WRITE before the end would create more stack cells than remaining input can remove, contradicting balanced acceptance. □

Thus $C(P)$ is exactly Hamming agreement between the suffix of P and its fixed canonical stack reference.

Theorem 70.2 (Primitive balanced CANCEL distribution)**.** *For every coprime $p, q > 0$ with $p \neq q$,*

$$\boxed{c_{\min}(p, q) = |p - q| - 1.}$$

The number of primitive balanced words attaining this minimum is

$$\boxed{\binom{\max(p,q)-1}{|p-q|-1} = \binom{\max(p,q)-1}{\min(p,q)}.}$$

In the unique coprime diagonal case $(p, q) = (1, 1)$, both primitive balanced words 10 *and* 01 *have exactly one CANCEL, so*

$$\boxed{c_{\min}(1, 1) = 1}$$

with multiplicity 2*. More generally, for $j \geq 0$,*

$$\boxed{\#\{P : C(P) = m - 1 - 2j\} = \binom{p-1}{j}\binom{q}{j} + \binom{p}{j}\binom{q-1}{j},}$$

with all other values zero.

Proof. Condition on the first bit. Lemma 70.1 reduces the suffix to comparison of two binary strings of equal length and fixed composition. If the first bit is 1, the input suffix contains $p - 1$ ones and q zeros; if it is 0, it contains p ones and $q - 1$ zeros. A Hamming disagreement swaps one selected 1-position with one selected 0-position, so disagreements

occur in pairs. Choosing j swapped positions in the two composition classes gives the two binomial products. For $p \neq q$, maximizing the number of paired disagreements gives $|p-q|-1$ agreements and the stated multiplicity. In the remaining coprime case $(1,1)$ the distribution formula itself gives two words at CANCEL value 1, yielding the separate diagonal statement. □

Corollary 70.3 (Cancellation-free primitive orbit)**.** *A cancellation-free primitive balanced word exists if and only if $|p-q|=1$. It is unique: for $p=q+1$ it is $(10)^q 1$, while for $q=p+1$ it is $0(01)^p$.*

This is the first operational appearance of the Christoffel/mechanical-word boundary: the extremal word is a cyclic representative of the corresponding balanced mechanical pattern, but no Christoffel theorem is needed in the proof.

70.3 Primitive rational-Dyck paths and the Schubert statistic

Assume first $p=q+d$ with $d \geq 1$; the opposite inequality follows by the exact bit/physical duality. Define

$$R_{q,d} = 1(10)^q 1^{d-1}, \qquad A_q = \{3,5,\ldots,2q+1\}.$$

For a primitive rational-Dyck path P, let $Z(P)$ be the set of zero positions.

Theorem 70.4 (Canonical-demand agreement theorem)**.** *For every primitive rational-Dyck P of type $(q+d):q$,*

$$\boxed{C(P) = d - 1 + 2|Z(P) \cap A_q|.}$$

Moreover, writing $Z(P)=\{z_1<\cdots<z_q\}$, rational-Dyck nonnegativity is equivalent to

$$\boxed{z_i \geq \left\lceil \frac{(2q+d)i}{q} \right\rceil, \qquad 1 \leq i \leq q.}$$

Hence the zero sets form the bases of a Schubert, equivalently lattice-path, matroid.

Proof. The first BOTTOM-WRITE fixes the entire remaining demand by Lemma 70.1. Direct comparison with that demand yields the reference word $R_{q,d}$ and therefore the displayed agreement formula. At the position z_i of the ith zero, the prefix has $z_i - i$ ones and i zeros; $q(z_i-i)-pi \geq 0$ is exactly the stated lower flag. Such coordinatewise lower flags define the standard Schubert/lattice-path basis family [55]. □

Define the normalized polynomial

$$\mathcal{R}_{q,d}(x) = \sum_B x^{|B \cap A_q|},$$

where B runs over these bases.

Corollary 70.5 (Ultra-log-concavity)**.** *If $\mathcal{R}_{q,d}(x) = \sum_j r_j x^j$, then*

$$\boxed{\left(\frac{r_j}{\binom{q}{j}}\right)^2 \geq \frac{r_{j-1}}{\binom{q}{j-1}} \frac{r_{j+1}}{\binom{q}{j+1}}.}$$

Thus the primitive normalized CANCEL coefficients are ultra-log-concave, and hence log-concave and unimodal with no internal zero.

Proof. The multivariate basis polynomial of a matroid is Lorentzian; nonnegative specialization to a marked subset preserves the required Lorentzian consequences. Apply the strongest Mason inequality of Brändén–Huh [56]. □

The support is exact:

$$j_{\max} = \left\lfloor \frac{q(2q+1)}{2q+d} \right\rfloor, \qquad j_{\min} = \begin{cases} 1, & d = 1, \\ 0, & d \geq 2. \end{cases}$$

After reversal to an upper flag, put

$$s_i = 1 + \left\lfloor \frac{(2q+d)(i-1)}{q} \right\rfloor, \qquad \bar{A}_q = \{d, d+2, \ldots, d+2q-2\}.$$

If $u_i = |\bar{A}_q \cap [1, s_i]|$, $v_i = s_i - u_i$, and

$$\sum_{r \geq 0} H_{i,r}(x) z^r = (1 - xz)^{-u_i} (1 - z)^{-v_i},$$

then the standard flagged Jacobi–Trudi/Lindström–Gessel–Viennot specialization [60, 24] gives

$$\boxed{\mathcal{R}_{q,d}(x) = \det[H_{i,1-i+j}(x)]_{1 \leq i,j \leq q}.}$$

Because the determinant is upper Hessenberg it also has the matrix-free recurrence

$$D_{q+1} = 1, \qquad D_i = \sum_{r=1}^{q-i+1} (-1)^{r-1} H_{i,r} D_{i+r}, \qquad \mathcal{R}_{q,d} = D_1.$$

70.3.1 Narayana windows

The near-diagonal cases recover classical distributions after the operational identification. If $d = 1$, deleting the initial 1 gives an ordinary Dyck path of semilength q, and

$$\#\{P : C(P) = 2j\} = \frac{1}{q} \binom{q}{j} \binom{q}{j-1}.$$

If $d = 2$ (so q is odd by coprimality), a last-level decomposition $P = UAUB$ gives

$$\boxed{\#\{P : C(P) = 2j-1\} = \frac{1}{j} \binom{q+1}{j-1} \binom{q-1}{j-1}.}$$

These are ordinary/rational Narayana distributions; the second equality is distributional, not pathwise equality with the classical peak statistic [51].

Chapter 71

Repeated Kernels, Carry Histories, and Algebraic Closure

Primitive normalization has a fixed demand stack. Repeated kernels introduce physical resets, so the event law becomes history-sensitive. The correct decomposition is a Bizley factorization refined by operational events, followed by a Raney carry skeleton and a stationary shifted-Schubert phase kernel.

71.1 Repeated kernels and operational Bizley factorization

For scale k, let $\mathcal{D}_k$ be the rational-Dyck paths with kp ones and kq zeros, and let $\mathcal{I}_k$ be those with no proper zero-residual contact. Put

$$D_k(x,y) = \sum_{P \in \mathcal{D}_k} x^{C(P)} y^{O(P)}, \qquad I_k(x,y) = \sum_{P \in \mathcal{I}_k} x^{C(P)} y^{O(P)}.$$

A zero-residual contact returns the centered machine to the canonical accepting configuration. Both statistics are additive under concatenation, hence

$$\boxed{D(z;x,y) = 1 + \sum_{k \geq 1} D_k(x,y) z^k = \frac{1}{1 - I(z;x,y)}.}$$

Define operational Bizley cumulants by

$$\boxed{\log D(z;x,y) = \sum_{k \geq 1} \frac{A_k(x,y)}{k} z^k.}$$

At $x = y = 1$ this reduces to the classical rational-Dyck/Bizley decomposition; in particular

$$\boxed{A_k(1,1) = \frac{1}{p+q} \binom{k(p+q)}{kp}.}$$

The unweighted factor-free structure is classical [54, 57, 53]; the refinement here is by actual normalization events.

71.2 Physical carry hierarchy and WRITE-weight conservation

Set

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \qquad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

A positive OUTER-WRITE is called a *carry*.

Theorem 71.1 (Carry-onset hierarchy)**.** *The first scale at which $r \geq 1$ carries are possible is*

$$\boxed{\kappa_r(p,q) = a + rc.}$$

Equivalently,

$$\boxed{O_{\max}(k) = \max\left\{0, \left\lfloor \frac{k-a}{c} \right\rfloor\right\}.}$$

At $k = \kappa_r$, every path realizing r carries is factor-free, so

$$\boxed{[y^r]A_{\kappa_r}(x,y) = \kappa_r[y^r]D_{\kappa_r}(x,y).}$$

Proof. Before the first positive outer carry, SEARCH+ must accumulate enough positive centered debt to cross the outer boundary. Among rational-Dyck prefixes with a fixed number of completed scale blocks, replacing a zero by a one cannot delay that positive boundary: a zero lowers the rational height and never supplies positive mismatch budget. Hence the all-ones trajectory is extremal for earliest carry onset. Writing $h = q(a-1)+s$ with $0 \leq s < q$ shows that $a = 1 + \lfloor h/q \rfloor$ scale units are necessary and sufficient before the first OUTER+ trigger.

After an OUTER+ reset the written frame has length $cm - 1$, where $c = 1 + \lfloor h/p \rfloor$. The WRITE-weight invariant of Theorem 71.2 assigns this reset weight c. Between two consecutive carries no event can create scale budget; SEARCH and CANCEL only consume existing cells. Thus each further carry costs at least an additional c units of total scale, giving the lower bound $k \geq a + rc$ for r carries. The same extremal positive-boundary trajectory, followed at each stage by its shortest balancing continuation, realizes equality, so $\kappa_r = a + rc$ and the displayed formula for $O_{\max}(k)$ follows.

At $k = \kappa_r$, suppose a path with r carries had a proper zero-residual factorization into positive scales $k_1 + \cdots + k_s = k$, $s \geq 2$, with carry counts $r_1 + \cdots + r_s = r$. Every factor with $r_i > 0$ satisfies $k_i \geq a + r_i c$, while a factor with $r_i = 0$ still has $k_i \geq 1$. If carries occur in two or more factors, summing the onset bounds contributes at least one extra copy of a; if all carries occur in one factor, every remaining factor contributes at least one extra unit of scale. In either case $k > a + rc = \kappa_r$, a contradiction. Hence every onset path with r carries is factor-free. The standard logarithmic relation between all paths and factor-free factors then gives $[y^r]A_{\kappa_r} = \kappa_r[y^r]D_{\kappa_r}$. □

The hierarchy is a consequence of a stronger pathwise invariant.

Theorem 71.2 (WRITE-weight conservation)**.** *Assign every WRITE event a positive integer scale weight ω: a written frame of length $\omega m - 1$ at an empty stack has weight ω, while OUTER+ has fixed weight c and OUTER− fixed weight a. For a balanced accepted scale-k computation,*

$$\boxed{\sum_{\text{WRITE } j} \omega_j = k.}$$

Proof. Every WRITE of weight ω changes physical stack height by $\omega m - 1$, while every SEARCH or CANCEL decreases height by one. If W is the number of WRITEs and N the number of non-WRITEs, empty-to-empty height balance gives

$$N = m \sum_j \omega_j - W.$$

Since every input symbol produces exactly one event, $W + N = km$, and the identity follows. □

For BOTTOM-WRITE at debt d the exact weight is

$$\omega_B(d,1) = \max\left\{1, \left\lceil -\frac{d}{q} \right\rceil, \left\lceil \frac{d+1}{p} \right\rceil \right\},$$

$$\omega_B(d,0) = \max\left\{1, \left\lceil \frac{1-d}{q} \right\rceil, \left\lceil \frac{d}{p} \right\rceil \right\}.$$

Thus scale is literally the total physical shortest-demand WRITE budget.

71.3 Minimal multi-carry paths: Raney signatures and Schubert edge weights

Define the first-carry reserve

$$\boxed{R_0 = aq - h}, \qquad Q = qc.$$

At minimal r-carry onset let z_j be the number of zeros read before the jth carry. The feasible signatures are exactly

$$0 \le z_1 \le \cdots \le z_r, \qquad z_j \le R_0 - 1 + (j-1)Q.$$

Introduce the zero-tail slack

$$\lambda_j = R_0 - 1 + (j-1)Q - z_j.$$

Then

$$\boxed{0 \le \lambda_1 \le R_0 - 1, \qquad 0 \le \lambda_j \le \lambda_{j-1} + Q.}$$

Theorem 71.3 (Raney signature law)**.** *The number of feasible r-carry signatures is*

$$\boxed{|\mathcal{Z}_r| = \frac{R_0}{(Q+1)r + R_0} \binom{(Q+1)r + R_0}{r}.}$$

Hence the signature skeleton is the Raney number $R_r^{(Q+1,R_0)}$.

Proof. Interpret z_j as the number of horizontal steps before the jth vertical step. The displayed bounds are exactly the strict ballot boundary $x < R_0 + Qy$. The classical Raney/cycle lemma gives the formula [59]. □

Before a carry with slack λ, the state is $(D_h, 0^{\lambda+1})$. After the carry, define

$$X = h + 1 + Q, \qquad Y = pc - h - 1.$$

The written frame is

$$W_+ = (10)^Y 0^{X-Y}.$$

For a transition $\lambda \to \mu$ put

$$\delta = Q + \lambda - \mu, \qquad n = pc - 1 + \delta.$$

The no-WRITE phase contains $pc - 1$ ones and δ zeros and is compared with

$$Q_{\lambda,\mu} = (10)^Y 0^{n-2Y}.$$

If $B = \{b_1 < \cdots < b_\delta\}$ is the input-zero set and

$$H_\lambda = (h+1)m + p\lambda,$$

then rational nonnegativity is the lower flag

$$\boxed{b_i \geq \ell_i(\lambda) = \max\left\{i, \left\lceil \frac{mi - H_\lambda}{q} \right\rceil \right\}.}$$

Let $A_{\lambda,\mu}$ be the zero-position set of $Q_{\lambda,\mu}$. The inter-carry polynomial is

$$\boxed{M_{\lambda,\mu}(x) = \sum_B x^{n-|A_{\lambda,\mu}|-\delta+2|A_{\lambda,\mu}\cap B|}.}$$

Thus each edge of the Raney signature skeleton is a shifted-Schubert Hamming enumerator.

After reversal, $M_{\lambda,\mu}$ has the standard flagged determinant and upper-Hessenberg recurrence. This gives an exact finite product/sum over signatures for every minimal-onset polynomial $G^{(r)}_{p,q}(x)$.

71.3.1 Stationary bulk kernel and finite wall

When the rational lower flag is inactive, $M_{\lambda,\mu}$ depends only on $\delta = Q + \lambda - \mu$:

$$\boxed{\widehat{M}_\delta(x) = \sum_{j=\max(0,\delta-Y)}^{\delta} \binom{h+\delta}{j}\binom{Y}{\delta-j} x^{Y-\delta+2j}.}$$

Its generating function is the rational symbol

$$\boxed{\widehat{K}(v;x) = \sum_{\delta\geq 0} \widehat{M}_\delta(x) v^\delta = \frac{(x + (1-x^2)v)^Y}{(1-xv)^{pc}}.}$$

For every $p > q$ the wall is absent. For steep $q > p$, let

$$\Theta = pQ - m(h+1), \qquad b = \left\lceil \frac{\Theta}{p} \right\rceil.$$

If $\Theta > 0$, then

$$\boxed{b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor, \qquad 1 \leq b \leq Q-1,}$$

and only target slacks $0, \ldots, b-1$ differ from the Toeplitz bulk. The exact operator is therefore Toeplitz-Hessenberg bulk plus a finite-rank rational-wall correction.

71.4 Algebraic closure of the minimal-onset family

The bulk kernel equation is

$$\boxed{v^Q = t\widehat{K}(v;x),}$$

or

$$\boxed{v^Q(1-xv)^{pc} = t(x + (1-x^2)v)^Y.}$$

Let $v_1, \dots, v_Q$ be the Q small Puiseux branches, taken in an algebraic Puiseux extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric Green-function expressions below descend to algebraic formal series over $\mathbb{Q}(x,t)$; see Section 71.6. The bulk excursion Green function is

$$\boxed{\mathcal{E}_Q(t;x) = \frac{(-1)^{Q-1}}{tx^Y} \prod_{i=1}^{Q} v_i,}$$

and from slack 0 to slack μ,

$$\boxed{\mathcal{R}_\mu(t;x) = \mathcal{E}_Q(t;x) h_\mu(v_1, \dots, v_Q),}$$

where h_μ is the complete homogeneous symmetric polynomial. These are the standard small-root half-line formulas specialized to the DPDA-derived symbol [52].

For a general start $s < Q$, the Green row has the finite form

$$\mathcal{R}_\mu^{(s)} = \sum_{i=1}^{Q} c_i^{(s)} v_i^\mu.$$

The coefficients are determined by a $Q \times Q$ boundary matrix. Writing

$$K_{<n}(v;x) = \sum_{d=0}^{n-1} \widehat{M}_d(x) v^d, \qquad \beta_n(v) = tv^{-n} K_{<n}(v;x),$$

define

$$\mathsf{C}_{\mu i} = \beta_{Q-\mu}(v_i), \qquad 0 \le \mu < Q.$$

Then

$$\boxed{\mathsf{C} c^{(s)} = e_s.}$$

The first-carry prefix vector has finite support $0 \le s < R_0 \le Q$, while the terminal slack generating series is rational. For $p > q$, this boundary system closes the entire minimal-onset family. For $q > p$, the finite wall is added by the matrix inversion lemma; hence:

Theorem 71.4 (Complete minimal-onset algebraicity)**.** *For every fixed coprime p, q, the generating series*

$$\mathcal{G}_{p,q}(t;x) = \sum_{r \ge 1} G_{p,q}^{(r)}(x) t^{r-1}$$

is algebraic over $\mathbb{Q}(x,t)$ and is given by a finite small-root/boundary system. No infinite-dimensional boundary problem remains.

For $q = 1$, $Q = 1$ and ordinary Lagrange inversion gives a scalar coefficient theorem. At 2:1 this simplifies further: if $W = 1 + xtW^3$, then

$$\boxed{\mathcal{G}_{2,1}(t;x) = x^4 W^3,}$$

so at the minimal r-carry scale $\kappa_r = r + 2$,

$$\boxed{G_{2,1}^{(r)}(x) = \frac{1}{r} \binom{3r}{r-1} x^{r+3}.}$$

71.5 Carry-onset lower bound and wall-rank lemma

Let

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \qquad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

Before the first positive outer carry, the maximal all-1 trajectory is the cheapest way to accumulate the required positive mismatch debt. It consumes exactly a units of scale before the first carry. After each OUTER+ reset, the written frame has length $cm - 1$, hence, by WRITE-weight conservation, each additional carry consumes at least one further block of scale weight c. Zeros cannot reduce either requirement because they decrease the rational height available to sustain the positive boundary trajectory. The all-1 extremal prefix followed by the unique shortest balancing continuation realizes equality at every level. Consequently the first scale supporting r carries is $\kappa_r = a + rc$.

At $k = \kappa_r$, a proper zero-residual factor would split the total scale into two positive balanced blocks. The carry-onset bound applied separately to the factors would then leave strictly less than the scale required for r carries in the factor containing them, contradicting minimality. Thus every r-carry path at first onset is factor-free.

For steep $q > p$, recall

$$\Theta = pQ - m(h+1) > 0, \qquad b = \left\lceil \frac{\Theta}{p} \right\rceil.$$

Then

$$b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor,$$

and because $m(h+1)/p > 1$ in every genuine wall case,

$$1 \le b \le Q - 1.$$

Hence every row needed by the finite Woodbury wall correction lies inside the already solved $Q \times Q$ boundary Green block. No additional start-state formula is required.

71.6 Formal domain of the small-root formulas

The Q small roots of

$$v^Q = tK_c(v; x)$$

are naturally Puiseux series in an algebraic extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric combinations used for the Green functions descend to algebraic formal series over $\mathbb{Q}(x, t)$; the underlying coefficient recurrence has polynomial coefficients in x. Formulas containing x^{-Y} are therefore interpreted first over $\mathbb{Q}(x)$ and specialized at singular values such as $x = 0$ only through the polynomial recurrence or by a removable-limit argument. Likewise, Vandermonde-denominator expressions are interpreted for distinct generic roots and extended to collisions by symmetric continuation.

Chapter 72

Phase Normality and Exact One-Counter Compression

Despite the history dependence created by WRITE resets, the centered rational chamber has a much smaller exact state description than the literal binary stack suggests. A phase-normal reconstruction theorem reduces the full reachable demand to one signed residual together with a finite phase bit, and a fixed shift makes the residual a nonnegative counter.

72.1 Centered phase normal form and arbitrary-scale compression

Minimal onset still uses a distinguished reset skeleton. At arbitrary scale, BOTTOM and OUTER frames can be nested. The decisive fact is that the apparent binary-stack complexity collapses on the rational chamber.

For a stack word S, write

$$\rho(S) = qN_1(S) - pN_0(S), \qquad R = -\rho(S) = pN_0(S) - qN_1(S).$$

The variable R is signed; on the rational chamber it is bounded below by $-hm$. The shifted counter $J = R + hm$ is therefore nonnegative. Let $A_{10}(S)$ and $A_{01}(S)$ denote the numbers of adjacent factors 10 and 01. A word is called *phase normal* if it consists of an alternating prefix followed by a monochromatic tail. The following theorem is proved by simultaneous induction on the physical transitions; the induction constructs two canonical reconstruction maps ν_0, ν_1 on the reachable shortest-demand pairs.

Theorem 72.1 (Centered phase-normal form)**.** *For every configuration reachable from a rational-Dyck prefix:*

1. *the stack Parikh vector is the shortest-demand pair $\mu(R)$;*
2. *there is a phase bit $b \in \{0, 1\}$ such that*

$$\boxed{S = \nu_b(\mu(R));}$$

   thus the complete bit stack is reconstructible from (R, b);

3. *the centered debt satisfies*

$$\boxed{A_{10}(S) - h \le d \le h - A_{01}(S).}$$

The three assertions are transition-closed only when proved simultaneously.

Proof. Define the two reconstruction maps explicitly by

$$\nu_0(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y, \end{cases} \qquad \nu_1(x,y) = \begin{cases} (01)^y 0^{x-y}, & x \geq y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

They have the same Parikh vector (x, y) and are exactly the two phase-normal orderings compatible with a shortest-demand pair. We prove simultaneously by induction over the displayed centered transitions that (i) the current Parikh vector is $\mu(R)$, (ii) the stack equals one of these two words, and (iii) the debt inequality holds.

For a BOTTOM-WRITE the exact shortest-demand formulas are

$$\begin{aligned} (x,y) &= (d + q\omega,\ p\omega - d - 1) \quad \text{on input 1,} \\ (x,y) &= (d + q\omega - 1,\ p\omega - d) \quad \text{on input 0.} \end{aligned}$$

with the minimal positive weight ω of Theorem 71.2. Minimality of ω makes the minority coordinate at most h, so the canonical frame lies in the centered alternating interval and satisfies the debt bounds.

For CANCEL, deleting the visible matching cell decreases exactly one Parikh coordinate. Because the pre-transition pair is kernel-reduced and the deleted letter is visible in a phase-normal representative, the remaining pair is again the unique shortest pair for the updated signed demand; deleting the first letter of ν_b leaves a phase-normal word and changes A_{10}, A_{01} by at most one in the direction compensated by the unchanged debt.

For an interior SEARCH, the visible mismatching cell is consumed while d moves by one. If a SEARCH+ sees a stack beginning 00, phase normality forces the stack to be monochromatic from that point onward; hence $A_{01} = 0$ and the upper debt inequality remains strict enough after $d \mapsto d+1$. The SEARCH− case beginning 11 is the physical dual. All alternating-prefix SEARCH cases simply shorten the alternating prefix and preserve both inequalities.

At OUTER+, the new frame has

$$X_+ = h + 1 + qc, \qquad Y_+ = pc - h - 1,$$

with $Y_+ \leq h$ and a terminal zero-run, so it is one of the displayed ν_b words and resets d to 0 inside the required interval. OUTER− is the dual transition at the level of the induction; Theorem 72.2 below then excludes it on the rational-Dyck chamber. These cases exhaust the transition table, proving the three assertions simultaneously. The explicit reconstruction and compressed successor table are collected in Section 72.5. □

Theorem 72.2 (No negative outer carry)**.** *OUTER− never occurs on a rational-Dyck prefix.*

Proof. At a hypothetical OUTER− trigger, $d = -h$ and the top symbol is 1. The lower centered inequality in Theorem 72.1 forces $A_{10}(S) = 0$. Phase normality therefore makes the entire nonempty stack 1^r. Hence the global rational height is

$$\rho_{\text{global}} = dm - \rho(S) = -hm - rq < 0,$$

contradicting rational-Dyck nonnegativity. □

72.1.1 Exact one-counter recurrence

Theorem 72.1 makes the physical stack a derived object. The exact compressed state is

$$\boxed{\sigma = (d, J, b), \qquad J = R + hm,}$$

where $d \in [-h, h]$, the signed demand residual $R = pN_0(S) - qN_1(S)$ satisfies $R \geq -hm$ on the rational chamber, hence $J \in \mathbb{Z}_{\geq 0}$ is a genuine one-counter coordinate, and b is the phase bit. When evaluating transitions we recover $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$; Section 72.5 gives the exact case table. If $\mu(R) = (x, y)$, popping a top 1 sends

$$\boxed{R \mapsto R + q,}$$

while popping a top 0 sends

$$\boxed{R \mapsto R - p.}$$

The next phase bit is uniquely determined by the reconstructed remaining shortest-demand word. OUTER+ sends

$$\boxed{R' = (R - p) + (h + 1)m, \qquad d' = 0,}$$

and an empty-stack BOTTOM-WRITE on bit a sends

$$\boxed{R' = dm + w(a), \qquad d' = 0.}$$

By Theorem 72.2, no OUTER− transition is present in the rational chamber.

For each positive-height compressed state $\sigma = (d, J, b)$, let $F_\sigma(z, x, y)$ be the generating series of factor-free accepting continuations, with x marking CANCEL, y marking OUTER+, and z marking WRITE scale weight. Section 72.5 gives, for each next bit a, an explicit deterministic successor $T_a(\sigma)$ and monomial $M(\sigma, a)$. Therefore

$$\boxed{F_\sigma = \sum_{a \in \{0,1\}} M(\sigma, a)\, \Xi(T_a(\sigma)),}$$

where Ξ returns 1 at the first legal empty-stack terminal state, 0 if the rational height becomes negative or a proper zero-residual contact occurs, and otherwise returns the corresponding successor series. The initial factor-free series $I(z; x, y)$ is obtained by the two possible first BOTTOM transitions, and

$$\boxed{I_{k,r}(x) = [z^k y^r] I(z; x, y).}$$

The only unbounded coordinate is the nonnegative integer J; (d, b) is finite control. This is thus an exact weighted one-counter recurrence, to which standard one-counter/PDA generating-function methods apply [58, 50].

Corollary 72.3 (Arbitrary-scale operational enumeration)**.** *For every coprime p, q and every scale k, the factor-free bivariate polynomial $I_k(x, y)$ is determined exactly by the compressed one-counter recurrence above. All rational-Dyck paths and operational Bizley cumulants follow from*

$$\boxed{D(z; x, y) = \frac{1}{1 - I(z; x, y)}, \qquad \log D(z; x, y) = \sum_{k \geq 1} \frac{A_k(x, y)}{k} z^k.}$$

The primitive and minimal-onset formulas are specializations of the same compressed dynamics.

72.2 The original 2:1 family at arbitrary scale

The route returns to the motivating ratio with a particularly rigid all-scale law. In a factor-free scale-k 2:1 path, each WRITE has scale weight 1, every nonterminal WRITE episode contains exactly one SEARCH+, and the terminal episode contains none. If W, S, C are the numbers of WRITEs, SEARCHes, and CANCELs, stack-height and input-length balance give

$$C + S = 2W, \qquad W + C + S = 3k.$$

Hence $W = k$, $S = k - 1$, and

$$\boxed{C = k + 1}$$

for every factor-free path.

If an arbitrary rational-Dyck path of scale k has $s(P)$ zero-residual factors, additivity therefore gives the pathwise identity

$$\boxed{C(P) = k + s(P).}$$

This is the operational contribution; the factor counts themselves belong to the classical Fuss–Catalan/factor-free theory.

The number of paths with s factors is

$$\boxed{\#\{P : C(P) = k + s\} = \frac{s}{k}\binom{3k - s - 1}{k - s}, \qquad 1 \le s \le k,}$$

so

$$\boxed{D_k(x, 1) = \sum_{s=1}^{k} \frac{s}{k}\binom{3k - s - 1}{k - s} x^{k+s}.}$$

The operational Bizley cumulants simplify to

$$\boxed{A_k(x, 1) = x^k \sum_{s=1}^{k} \binom{3k - s - 1}{k - s} x^s,}$$

and $A_k(1, 1) = \frac{1}{3}\binom{3k}{k}$ as required.

72.3 Computational validation and structural checks

The proofs above are symbolic. Exact computation was used independently as regression testing while the formulas were developed. The carry-onset hierarchy was checked on coprime $p, q \le 7$ through scales up to 10 (350 scale tests and 138 onset tests for the first four carry layers) with no discrepancy. The stationary inter-carry determinant/Hessenberg formula was compared with direct local-word enumeration on 481 small parameter/slack pairs, and the rational Toeplitz bulk formula on 660 tested transitions, again with zero discrepancies. The scalar p:1 formulas were compared polynomial-by-polynomial with direct centered-machine enumeration in representative 2:1, 3:1, and 4:1 cases. Before Theorem 72.2 was proved, exact state-DP searches over 34 ordered coprime ratios with $p, q \le 7$ and 176 ratio/scale instances found no accepted rational-Dyck example containing OUTER−; the theorem now explains that observation.

These computations are not used as proof. They serve to detect sign, parity, boundary, and reset errors in formulas whose symbolic proofs are independent.

72.4 Operational synthesis: the loss of primitive log-concavity

The event statistics exhibit a sharp structural transition. Primitive normalization has one fixed reference word; CANCEL is therefore a single Hamming/Schubert statistic and inherits the corresponding matroid convexity. A physical WRITE reset replaces that static geometry by sums and products of Schubert pieces. For example, the first-carry polynomial at 5:3 is

$$36x^{10} + 67x^{11} + 153x^{12} + 180x^{13} + 135x^{14} + 39x^{15} + 18x^{16} + 2x^{17},$$

whose coefficients are not log-concave. Thus the loss of primitive Lorentzian behavior is not an artifact of proof technique; it records genuine history sensitivity introduced by a reset.

The repeated-kernel theory can be summarized as

physical WRITE budget + Raney carry skeleton
+Schubert cancellation phases + Bizley zero-contact factorization.

The phase-normal theorem shows that this history-sensitive system is nevertheless much smaller than the literal binary stack suggests: on the rational chamber it is exactly a one-counter process with finite phase control.

72.5 Phase reconstruction and exact compressed transition table

For a shortest-demand pair (x, y) define the two phase-normal words

$$\nu_0(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y, \end{cases}$$

$$\nu_1(x,y) = \begin{cases} (01)^y 0^{x-y}, & x \geq y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

When one of the formulas coincides with the other, the phase bit is immaterial. The simultaneous induction in Theorem 72.1 proves that every stack reachable from a rational-Dyck prefix is one of these two words for its shortest-demand pair.

It is important that the signed stack-demand residual need not itself be nonnegative. Put

$$R = pN_0(S) - qN_1(S) = -\rho(S).$$

The rational-chamber condition is

$$dm + R \geq 0.$$

Since $-h \leq d \leq h$, it follows that $R \geq -hm$. Thus the genuine one-counter coordinate is the shifted variable

$$J := R + hm \in \mathbb{Z}_{\geq 0}.$$

This shift corrects the misleading shorthand $R \geq 0$: R is signed, whereas J is the nonnegative counter.

Given a compressed state (d, J, b), recover the signed value $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$ (with the empty-stack case treated separately by the BOTTOM rule). The exact transition and its weight monomial are:

condition	(d', R')	weight
$S = \zeta T,\ \zeta = a$	$(d, R + w_S(a))$	x
$S = 0T,\ a = 1,\ d < h$	$(d + 1, R - p)$	1
$S = 1T,\ a = 0,\ d > -h$	$(d - 1, R + q)$	1
$S = 0T,\ a = 1,\ d = h$	$(0, (R - p) + (h + 1)m)$	yz^c
$S = 1T,\ a = 0,\ d = -h$	excluded on the rational chamber	$--$
$S = \varepsilon$	$(0, dm + w(a))$	$z^{\omega_B(d,a)}$

where $w(1) = q$, $w(0) = -p$, $w_S(1) = q$, $w_S(0) = -p$ records the change in R under a pop, $c = 1 + \lfloor h/p \rfloor$, and ω_B is the exact bottom-WRITE weight from Theorem 71.2. In the first row, equivalently, popping a 1 sends $R \mapsto R + q$ and popping a 0 sends $R \mapsto R - p$. After computing the successor stack by the centered rule, b' is the unique phase for which $S' = \nu_{b'}(\mu(R'))$ (or is ignored when the stack is empty). This gives an executable deterministic recurrence on the finite control (d, b) and the single nonnegative counter $J = R + hm$; after each row one stores $J' = R' + hm$.

For a factor-free positive-height state, the corresponding generating series therefore satisfies

$$F_{d,J,b} = \sum_{a \in \{0,1\}} M(d, J, b; a)\, \Xi(T_a(d, J, b)),$$

where M is the monomial in the table and Ξ is 1 on the first legal empty-stack terminal state, 0 after a negative rational height or a proper zero-residual contact, and otherwise the successor F-series. This is the precise one-counter recurrence used in Corollary 72.3.

72.6 Operational-to-intrinsic bridge

This research branch has counted statistics of a chosen physical normalization process. The next enumerative branch changes the viewpoint: the residual becomes the intrinsic rank of a rational path, returns become barrier contacts, and the counting problem is expressed without reference to a physical execution history. The two branches share the same scalar residual but answer different questions—operational event history here, intrinsic path statistics there.

Chapter 73

From Residual Rank to Explicit Run–Return Enumeration

Research branch — From Residual Contacts to Refined Rational-Path Enumeration. *This branch first develops the explicit run–return solution and its determinant reduction, then extracts the contact-preserving barrier-compression framework, its four fixed-composition rational-Dyck enumerations, and its arbitrary-boundary and parking-function applications. The determinant technology itself is classical [25, 24, 26]; the contribution is scoped to the contact-sensitive reduction and the resulting refined enumerations.*

The preceding macro-Parts developed the residual

$$\rho_{p,q} = qN_1 - pN_0$$

from the completion semantics of the original 2:1 machine and then from the centered real-time p:q family. The same scalar invariant also applies directly to a rational Dyck rectangle. Write

$$m = gq, \qquad n = gp.$$

At a lattice point (x, y) the usual rational-path rank is

$$R(x, y) = my - nx = g(qy - px) = g\rho_{p,q}.$$

Thus the rank is exactly the pushdown residual up to the dilation factor g. At a non-terminal return position the rank condition $0 \le R < n$ is therefore equivalent to the low-residual band

$$0 \le \rho_{p,q} < p.$$

This identity is the semantic bridge; none of the determinant proofs below depends logically on an automaton construction.

Dai, Fu, and Qiu introduced the rational run and ratio-run statistics and proved composition-preserving symmetry results with the return statistic [23]. Their concluding remarks posed explicit enumeration of rational Dyck paths with prescribed ordinary run and returns, and with prescribed ratio-run and returns. This chapter identifies the return/contact bridge; Chapter 74 gives explicit finite determinant solutions to both questions.

The key step was not the determinant technology itself. Binomial barrier determinants and the Lindström–Gessel–Viennot method are classical [25, 24, 26]. The new step in the solution was to turn run constraints into avoidance of selected coarea values while

retaining return positions as contacts of the compressed upper barrier. The resulting structural chain is

residual → rational rank → return as contact → forbidden-value compression → weighted determinant

The specialized reduction below solves the two posed enumeration questions; the subsequent chapters then extract the reusable barrier-compression principle and its refinements.

73.1 Rational Dyck paths and sampled runs

Throughout, m, n are positive integers with $m \geq n$. A path consists of north steps $N = (0, 1)$ and east steps $E = (1, 0)$.

Definition 73.1 — Rational Dyck path.

An $m \times n$ *rational Dyck path* is a path from $(0, 0)$ to (m, n) that remains weakly above $y = nx/m$. Its set is denoted by $\mathcal{D}_{m,n}$.

For $P \in \mathcal{D}_{m,n}$, let

$$\mathcal{U}(P) = (u_1, \ldots, u_n)$$

be its coarea sequence, where u_i is the horizontal coordinate of the ith north step, equivalently the number of cells to its left. Define

$$b_i := \left\lfloor \frac{m(i-1)}{n} \right\rfloor, \qquad 1 \leq i \leq n. \tag{73.1}$$

Proposition 73.2 — Coarea characterization.

A sequence $u = (u_1, \ldots, u_n)$ is the coarea sequence of a path in $\mathcal{D}_{m,n}$ if and only if

$$0 = u_1 \leq u_2 \leq \cdots \leq u_n, \qquad u_i \leq b_i \quad (1 \leq i \leq n). \tag{73.2}$$

Proof. The weak inequalities express the fact that successive north steps occur in nondecreasing columns. Immediately before the ith north step the path is at $(u_i, i-1)$, which lies weakly above the diagonal precisely when

$$i - 1 \geq \frac{n}{m} u_i,$$

or equivalently $u_i \leq m(i-1)/n$. Since u_i is integral, this is $u_i \leq b_i$. Finally $b_1 = 0$, so $u_1 = 0$. □

Dai–Fu–Qiu define a return position as a nonterminal lattice point on the path having rank smaller than n, where the rank of (x, y) is $my - nx$ [23]. Their return statistic counts the corresponding north steps.

Definition 73.3 — Sampled run.

For $d \geq 1$ and $P \in \mathcal{D}_{m,n}$, define

$$\operatorname{run}_d(P) := \min\{a \in [n] : da \notin \{u_1, \ldots, u_n\}\}. \tag{73.3}$$

The minimum exists because the sequence contains only n entries, one of which is 0.

For $d = 1$, Definition 73.3 is the ordinary rational run of [23]. If

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor,$$

then $d = \kappa$ gives their ratio-run $\widetilde{\text{run}}$.

73.2 Returns are coarea contacts

The following observation is the conceptual bridge used throughout this branch.

Theorem 73.4 — Return–contact correspondence.

For every $P \in \mathcal{D}_{m,n}$ with coarea sequence $(u_1, \dots, u_n)$,

$$\boxed{\operatorname{ret}(P) = \#\{i \in [n] : u_i = b_i\}.} \tag{73.4}$$

Proof. Immediately before the ith north step the path is at $(u_i, i-1)$ and hence has rank

$$R_i = m(i-1) - nu_i.$$

Write

$$m(i-1) = nb_i + r_i, \qquad 0 \le r_i < n.$$

Then

$$R_i = r_i + n(b_i - u_i). \tag{73.5}$$

By Proposition 73.2, $b_i - u_i \ge 0$. Therefore

$$R_i < n \iff b_i - u_i = 0 \iff u_i = b_i.$$

Every nonterminal path point of rank $< n$ must be followed by a north step: an east step would decrease the rank by n and make it negative, contradicting the rational-Dyck condition. Hence the north steps whose initial points have rank $< n$ are precisely the returns of [23]; the terminal endpoint is not an initial point of a north step. Summing over i gives (73.4). □

Remark 73.R1 — Pushdown residual interpretation.

Let $g = \gcd(m, n)$ and write $m = gq$, $n = gp$ with $\gcd(p, q) = 1$. Under the step encoding $0 = E$, $1 = N$, a prefix containing y north steps and x east steps has the weighted-balance residual

$$\rho = qy - px,$$

which is the scalar residual obtained by generalizing the arithmetic carried by the original 2:1 pushdown machine. The rational-path rank is $R = g\rho$, so a return position satisfies

$$0 \le \rho < p.$$

Thus Theorem 73.4 may equivalently be read as the statement that the finite low-residual band singled out by the pushdown semantics becomes upper-barrier contact in coarea coordinates. This remark records the origin of the viewpoint; the theorem itself depends only on the rank calculation above.

73.3 Deleting forbidden coarea values

Let $F \subset \mathbb{Z}_{>0}$ be finite. We wish to enumerate paths whose coarea sequence avoids F, while retaining the return weight from Theorem 73.4.

Define

$$\phi_F(t) := t - \#(F \cap [1, t]), \qquad t \in \mathbb{Z}_{\ge 0}, \tag{73.6}$$

and

$$c_i(F) := \phi_F(b_i) = b_i - \#(F \cap [1, b_i]). \tag{73.7}$$

Also put

$$\epsilon_i(F) := \mathbf{1}_{\{b_i \notin F\}}. \tag{73.8}$$

Lemma 73.5 — Forbidden-value compression.

The map

$$(u_1, \dots, u_n) \longmapsto (v_1, \dots, v_n), \qquad v_i = \phi_F(u_i), \tag{73.9}$$

is a bijection between

$$\left\{ \begin{array}{c} 0 = u_1 \le \cdots \le u_n, \quad u_i \le b_i, \\ \{u_1, \dots, u_n\} \cap F = \varnothing \end{array} \right\} \tag{73.10}$$

and

$$\{0 = v_1 \le \cdots \le v_n : \ v_i \le c_i(F)\}. \tag{73.11}$$

Moreover, whenever $b_i \notin F$,

$$u_i = b_i \iff v_i = c_i(F). \tag{73.12}$$

If $b_i \in F$, the equality $u_i = b_i$ is impossible on the domain.

Proof. For each $B \ge 0$, deletion of the elements of $F \cap [1, B]$ makes ϕ_F an order-preserving bijection

$$\{0, 1, \dots, B\} \setminus F \longrightarrow \{0, 1, \dots, \phi_F(B)\}.$$

Applying this coordinatewise with $B = b_i$ proves that (73.9) is well-defined, preserves weak order, and has a unique inverse obtained by reinserting the deleted values. This proves the bijection. If $b_i \notin F$, the maximum allowed value b_i maps to the maximum compressed value $c_i(F)$, proving (73.12). The final assertion is immediate. □

Chapter 74

Weighted Determinants and Explicit Run–Return Enumeration

Chapter 73 converted returns into upper-barrier contacts and isolated the specialized forbidden-value compression that arose in the explicit-enumeration solution. We now retain the complete determinant and inclusion–exclusion argument, including the coefficient-free form and the independent exact checks. The sampled statistic run_d is used only as a unifying notation: $d = 1$ is ordinary run and $d = \lfloor m/n \rfloor$ is the Dai–Fu–Qiu ratio-run. No independent novelty claim is made for intermediate sampling widths.

74.1 A weighted binomial determinant

We use the convention

$$\binom{a}{b} = 0 \quad \text{if } b < 0 \text{ or } b > a,$$

for integers $a \geq 0$.

Definition 74.1 — Weighted forbidden-value determinant polynomial.

For a finite $F \subset \mathbb{Z}_{>0}$, define

$$\boxed{D_F(z) := \det_{1 \leq i,j \leq n} \left[\binom{c_i(F)+1}{j-i+1} + (z-1)\epsilon_i(F) \binom{c_i(F)}{j-i} \right].} \tag{74.1}$$

Theorem 74.2 — Weighted forbidden-value determinant.

For every finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_F(z) = \sum_{\substack{P \in \mathcal{D}_{m,n} \\ \mathcal{U}(P) \cap F = \varnothing}} z^{\mathrm{ret}(P)}.} \tag{74.2}$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$. For $1 \leq i, j \leq n$, set

$$S_i = (-c_i - 1, i), \qquad T_j = (0, j + 1).$$

A path from S_i to T_j has horizontal displacement c_i+1 and vertical displacement $j-i+1$, hence there are

$$\binom{c_i+1}{j-i+1}$$

such paths.

If $\epsilon_i = 1$, give weight z to the northeast edge leaving S_i and weight 1 to all other edges. If $\epsilon_i = 0$, all edges leaving S_i have weight 1. The number of $S_i \to T_j$ paths using that first northeast edge is

$$\binom{c_i}{j-i},$$

so the weighted single-path generating function is exactly the (i,j) entry of (74.1).

By the Lindström–Gessel–Viennot lemma [24], the determinant is the signed generating function of vertex-disjoint path families connecting the sources to the targets. The source–target configuration is nonpermutable. Indeed, since c_i is nondecreasing, for $i<k$ we have $x(S_i) \geq x(S_k)$. On the common horizontal range from $x(S_i)$ to 0, an S_i-path starts strictly below an S_k-path. If their assigned targets were inverted, say T_j and T_ℓ with $j > \ell$, their vertical order at $x = 0$ would be reversed. Because every east or northeast step advances the horizontal coordinate by one, each path has one vertex at every integer abscissa in this common range; the change of vertical order therefore forces a common vertex. Thus a vertex-disjoint family has no inversion, and only the identity matching $S_i \to T_i$ contributes.

An identity path $\pi_i : S_i \to T_i$ contains exactly one northeast step and c_i east steps. Let $e_i \in \{0,\dots,c_i\}$ be the number of east steps occurring before its unique northeast step, and set

$$v_i := c_i - e_i. \tag{74.3}$$

For adjacent paths, vertex-disjointness is equivalent to

$$-c_{i+1} - 1 + e_{i+1} < -c_i + e_i,$$

which, since all quantities are integral, is equivalent to

$$c_i - e_i \leq c_{i+1} - e_{i+1},$$

that is, $v_i \leq v_{i+1}$. Thus identity nonintersecting families are in bijection with

$$0 \leq v_1 \leq \cdots \leq v_n, \qquad v_i \leq c_i.$$

This is precisely the compressed sequence class in Lemma 73.5.

Finally, π_i uses its first northeast edge if and only if $e_i = 0$, equivalently $v_i = c_i$. Such an edge is weighted by z exactly when $\epsilon_i = 1$. By Lemma 73.5, these weighted events correspond exactly to the original contacts $u_i = b_i$, and Theorem 73.4 identifies their number with $\mathrm{ret}(P)$. Therefore each family has weight $z^{\mathrm{ret}(P)}$, proving (74.2). □

Corollary 74.3 — Return polynomial.

For $F = \varnothing$,

$$\sum_{P \in \mathcal{D}_{m,n}} z^{\mathrm{ret}(P)} = \det_{1\leq i,j\leq n}\left[\binom{b_i}{j-i+1} + z\binom{b_i}{j-i}\right]. \tag{74.4}$$

Proof. When $F = \varnothing$, $c_i = b_i$ and $\epsilon_i = 1$. Pascal's identity gives

$$\binom{b_i+1}{j-i+1} + (z-1)\binom{b_i}{j-i} = \binom{b_i}{j-i+1} + z\binom{b_i}{j-i}.$$

Apply Theorem 74.2. □

74.2 Explicit run–return enumeration

For $d \geq 1$ and $a \geq 1$, define the required sampled values

$$S_{d,a-1} := \{d, 2d, \ldots, (a-1)d\}, \tag{74.5}$$

with $S_{d,0} = \varnothing$.

Theorem 74.4 — Main enumeration theorem.

Let $m \geq n \geq 1$, $d \geq 1$, and $a, b \in [n]$. Then

$$\boxed{\begin{aligned} N_{m,n}^{(d)}(a,b) &:= \#\{P \in \mathcal{D}_{m,n} : \mathrm{run}_d(P) = a,\ \mathrm{ret}(P) = b\} \\ &= [z^b] \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z) \end{aligned}} \tag{74.6}$$

Proof. By Definition 73.3, $\mathrm{run}_d(P) = a$ if and only if

$$S_{d,a-1} \subseteq \mathcal{U}(P) \qquad \text{and} \qquad ad \notin \mathcal{U}(P). \tag{74.7}$$

First restrict to paths avoiding ad. To force every value of $S_{d,a-1}$ to occur, apply inclusion–exclusion to the events that a specified value of $S_{d,a-1}$ is missing. For $T \subseteq S_{d,a-1}$, the corresponding class avoids the forbidden set $T \cup \{ad\}$ and therefore has return polynomial $D_{T \cup \{ad\}}(z)$ by Theorem 74.2. Thus the return polynomial of paths satisfying (74.7) is

$$\sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z).$$

Extracting $[z^b]$ gives (74.6). □

Corollary 74.5 — Ordinary run–return.

For $a, b \in [n]$,

$$\#\{P \in \mathcal{D}_{m,n} : \mathrm{run}(P) = a,\ \mathrm{ret}(P) = b\} = [z^b] \sum_{T \subseteq \{1,\ldots,a-1\}} (-1)^{|T|} D_{T \cup \{a\}}(z). \tag{74.8}$$

Corollary 74.6 — Ratio-run–return.

Let $\kappa = \lfloor m/n \rfloor$. For $a, b \in [n]$,

$$\boxed{\begin{aligned} &\#\{P \in \mathcal{D}_{m,n} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{ret}(P) = b\} \\ &\quad = [z^b] \sum_{T \subseteq \{\kappa, 2\kappa, \ldots, (a-1)\kappa\}} (-1)^{|T|} D_{T \cup \{a\kappa\}}(z) \end{aligned}} \tag{74.9}$$

Corollaries 74.5 and 74.6 give explicit finite formulas for the two joint enumeration problems posed in [23].

74.2.1 A coefficient-free form

Formula (74.6) is already finite, but the coefficient extraction can be removed. For a finite forbidden set F, define $n \times n$ matrices A^F and B^F by

$$A_{ij}^F := \binom{c_i(F) + 1}{j - i + 1} - \epsilon_i(F) \binom{c_i(F)}{j - i}, \tag{74.10}$$

$$B_{ij}^F := \epsilon_i(F) \binom{c_i(F)}{j - i}. \tag{74.11}$$

Then

$$D_F(z) = \det(A^F + zB^F). \tag{74.12}$$

For $I \subseteq [n]$, let M_I^F be the matrix whose ith row is the ith row of B^F if $i \in I$, and the ith row of A^F otherwise.

Theorem 74.7 — Coefficient-free explicit formula.

For $a, b \in [n]$,

$$\boxed{N_{m,n}^{(d)}(a,b) = \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^{T \cup \{ad\}}.} \tag{74.13}$$

In particular, the joint count is expressed using only finite subset sums, binomial coefficients, and integer determinants.

Proof. The determinant in (74.12) is multilinear in its rows. Choosing the zB^F contribution in exactly the rows indexed by I and the A^F contribution in all remaining rows gives

$$[z^b]D_F(z) = \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^F.$$

Substitute this identity into Theorem 74.4. □

Remark 74.R1 — Coefficient-free scope.

The formula is effective without symbolic polynomial expansion. For fixed m, n, d, a, b, every determinant in (74.13) is an $n \times n$ integer determinant. The two subset sums have respectively 2^{a-1} and $\binom{n}{b}$ terms before cancellations or structural simplifications.

74.3 Examples

We display joint matrices with rows indexed by the run value and columns by the return value.

Example 74.E1 — Ordinary run, $(m, n) = (12, 4)$.

With $d = 1$, Theorem 74.4 gives

$$\left(N_{12,4}^{(1)}(a,b)\right)_{1 \le a,b \le 4} = \begin{pmatrix} 52 & 30 & 8 & 1 \\ 30 & 8 & 1 & 0 \\ 8 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{74.14}$$

This is the ordinary run–return matrix displayed in [23].

Example 74.E2 — Ordinary run, $(m, n) = (13, 5)$.

The formula gives

$$\begin{pmatrix} 106 & 106 & 39 & 9 & 1 \\ 106 & 39 & 9 & 1 & 0 \\ 39 & 9 & 1 & 0 & 0 \\ 9 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \tag{74.15}$$

again reproducing the published example [23].

Example 74.E3 — Ratio-run, $(m, n) = (11, 5)$.

Here $\kappa = \lfloor 11/5 \rfloor = 2$. Corollary 74.6 gives

$$\boxed{\begin{pmatrix} 85 & 43 & 13 & 2 & 0 \\ 43 & 32 & 11 & 2 & 0 \\ 13 & 11 & 7 & 2 & 0 \\ 2 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}}. \tag{74.16}$$

For instance, there are exactly 11 paths with $\widetilde{\mathrm{run}} = 2$ and $\mathrm{ret} = 3$. The sum of all entries is 273.

Example 74.E4 — Ratio-run, $(m, n) = (10, 4)$.

Here $\kappa = 2$ and

$$\begin{pmatrix} 30 & 11 & 2 & 0 \\ 11 & 9 & 4 & 1 \\ 2 & 4 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{74.17}$$

The total number of paths is 76.

The symmetry of the displayed matrices is guaranteed independently by the involutions of Dai–Fu–Qiu. In the present approach, that symmetry becomes a family of identities between the determinant sums in Theorem 74.4.

74.4 Computational verification

The proofs above are purely combinatorial and do not depend on computation. We nevertheless used exact computation as an audit of indexing, endpoint conventions, and the inclusion–exclusion signs. Two independent exact checks were performed:

1. direct enumeration of all weakly increasing coarea sequences satisfying (73.2), followed by direct computation of run_d and ret;

2. independent evaluation of Theorem 74.4 from the forbidden-value determinants.

The displayed matrices in Section 74.3 agree entry by entry under these calculations. A small exhaustive determinant audit covered 33 sampled-run cases with no mismatches. An independent audit of the coefficient-free formula covered 86 cases with no mismatches. These computations are consistency checks only and are not used in place of the symbolic proofs.

74.5 Interpretation and scope

The main formula separates the solution into three transparent operations:

$$\boxed{\text{return} = \text{barrier contact}} \longrightarrow \boxed{\text{delete forbidden values}}$$

$$\longrightarrow \boxed{\text{weighted determinant} + \text{inclusion–exclusion}}.$$

The determinant technology itself is classical: the binomial determinant belongs to a well-established family associated with nondecreasing sequences below barriers [25, 26, 24]. The new step in the present solution is the reduction of the run constraints to forbidden-value avoidance while retaining return positions as weighted contacts of the compressed upper barrier; inclusion–exclusion then converts that refinement into the requested joint enumeration.

The sampled statistic run_d is used here as a unifying notation: it packages the ordinary case $d = 1$ and the ratio case $d = \lfloor m/n \rfloor$ into one theorem. No additional structural claim about intermediate sampling widths is needed for the explicit enumeration established here.

The automata-theoretic origin and the combinatorial proof play different roles in this branch. The completion-aware DPDA suggested that a weighted residual, rather than a particular stack representation, is the stable semantic quantity. Under the rational-path encoding this residual becomes the usual rank up to scale, and the low-residual band becomes the contact statistic of Theorem 73.4. No automata construction is needed after this bridge: forbidden-value compression, the weighted LGV determinant, and inclusion–exclusion provide the complete enumeration.

Chapter 75

Contact-Preserving Barrier Compression

Chapters 73–74 resolve the two unrestricted joint enumeration questions. The next step is to separate what was problem-specific from what is a reusable reduction principle. The answer is *contact-preserving barrier compression*.

The novelty boundary is deliberately narrow. Lindström–Gessel–Viennot determinants, Kreweras-type barrier determinants, order-preserving relabeling, and inclusion–exclusion are classical. What is used here as one reduction is the coupling

value deletion + induced barrier compression
+exact contact marking + presence/absence constraints

This coupling permits arbitrary required and forbidden values, exact contact sets, and simultaneous first-missing statistics without losing the original boundary-contact information. The theory from this point is self-contained combinatorics; the residual-to-rank bridge explains its discovery but is not a proof hypothesis.

Throughout this section let

$$\mathbf{b} = (b_1, \dots, b_n), \qquad 0 \le b_1 \le \cdots \le b_n,$$

and define

$$\mathcal{A}_{\mathbf{b}} = \{u \in \mathbb{Z}^n_{\ge 0} : 0 \le u_1 \le \cdots \le u_n,\ u_i \le b_i\}.$$

For $u \in \mathcal{A}_{\mathbf{b}}$, set

$$\mathrm{Val}(u) = \{u_1, \dots, u_n\}, \qquad C_{\mathbf{b}}(u) = \{i \in [n] : u_i = b_i\}.$$

Presence and absence concern values of the sequence; contacts concern row positions. Keeping these two kinds of information separate is the central point of the method. We use the convention $\binom{a}{b} = 0$ whenever $b < 0$ or $b > a$.

75.0.1 Deleting forbidden values

Let $F \subset \mathbb{Z}_{>0}$ be finite and define

$$\phi_F(t) = t - \#(F \cap [1, t]), \qquad c_i(F) = \phi_F(b_i), \qquad \epsilon_i(F) = \mathbf{1}_{\{b_i \notin F\}}. \tag{75.1}$$

Lemma 75.1 — Contact-preserving value compression.

The coordinatewise map

$$\Phi_F(u_1, \dots, u_n) = (\phi_F(u_1), \dots, \phi_F(u_n))$$

is a bijection

$$\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing\} \longrightarrow \mathcal{A}_{\mathbf{c}(F)}.$$

Moreover, if $b_i \notin F$, then

$$u_i = b_i \iff \phi_F(u_i) = c_i(F),$$

whereas if $b_i \in F$ then no F-avoiding sequence can contact the original barrier in row i.

Proof. The restriction of ϕ_F to $\mathbb{Z}_{\geq 0} \setminus F$ is the unique order-preserving bijection from the allowed nonnegative integers onto $\mathbb{Z}_{\geq 0}$. Hence an F-avoiding weakly increasing sequence remains weakly increasing after applying ϕ_F, and $u_i \leq b_i$ implies $\phi_F(u_i) \leq \phi_F(b_i) = c_i(F)$. Conversely, applying the inverse order isomorphism coordinatewise to a sequence below $\mathbf{c}(F)$ gives an F-avoiding weakly increasing sequence below $\mathbf{b}$. If $b_i \notin F$, injectivity on the allowed alphabet gives $\phi_F(u_i) = \phi_F(b_i)$ if and only if $u_i = b_i$. If $b_i \in F$, that original contact is forbidden. □

75.0.2 Multivariate contacts

Define

$$D_{\mathbf{b},F}(z_1, \ldots, z_n) = \det_{1 \leq i,j \leq n} \left[\binom{c_i(F)+1}{j-i+1} + (z_i - 1)\epsilon_i(F) \binom{c_i(F)}{j-i} \right]. \tag{75.2}$$

Theorem 75.2 — Multivariate contact determinant.

For every finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_{\mathbf{b},F}(\mathbf{z}) = \sum_{\substack{u \in \mathcal{A}_{\mathbf{b}} \\ \mathrm{Val}(u) \cap F = \varnothing}} \prod_{i \in C_{\mathbf{b}}(u)} z_i.} \tag{75.3}$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$, sources

$$S_i = (-c_i - 1, i),$$

and targets

$$T_j = (0, j + 1).$$

A path from S_i to T_j has $c_i + 1$ horizontal steps and $j - i + 1$ northeast steps, so there are $\binom{c_i+1}{j-i+1}$ such paths. If $\epsilon_i = 1$, give weight z_i to the northeast edge leaving S_i and weight 1 to all other edges; if $\epsilon_i = 0$, give all initial edges weight 1. The number of S_i–T_j paths using the initial northeast edge is $\binom{c_i}{j-i}$, giving exactly the matrix entry in (75.2).

By the Lindström–Gessel–Viennot lemma [24], the determinant is the signed generating function of vertex-disjoint path families. Since c_i is nondecreasing, the source–target configuration is nonpermutable: an inversion of targets would reverse vertical order over a common horizontal interval and force an intersection. Thus only the identity matching contributes.

An identity path $\pi_i : S_i \to T_i$ has one northeast step and c_i east steps. Let e_i be the number of east steps before that northeast step and set $v_i = c_i - e_i$. Adjacent paths are vertex-disjoint exactly when $v_i \leq v_{i+1}$. Hence nonintersecting identity families are in bijection with

$$0 \leq v_1 \leq \cdots \leq v_n, \qquad v_i \leq c_i.$$

Moreover, π_i uses its initial northeast edge exactly when $v_i = c_i$. Lemma 75.1 identifies the weighted instances of these compressed contacts with the original contacts $u_i = b_i$. Therefore the family corresponding to u has weight $\prod_{i \in C_{\mathbf{b}}(u)} z_i$. □

For exact contacts define

$$B_{ij}^{F} = \epsilon_i(F)\binom{c_i(F)}{j-i}, \qquad A_{ij}^{F} = \binom{c_i(F)+1}{j-i+1} - B_{ij}^{F}. \tag{75.4}$$

For $J \subseteq [n]$, let $\mathsf{M}_J^{\mathbf{b},F}$ be the matrix whose ith row is B_i^F for $i \in J$ and A_i^F for $i \notin J$.

Corollary 75.3 — Exact contact-set determinant.

For every $J \subseteq [n]$,

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing,\ C_{\mathbf{b}}(u) = J\} = \det \mathsf{M}_J^{\mathbf{b},F}.}$$

Proof. The ith row in (75.2) is $A_i^F + z_i B_i^F$. Expand by row multilinearity and compare the coefficient of $\prod_{i\in J} z_i$ with Theorem 75.2. □

75.0.3 Presence, absence, and simultaneous first-missing statistics

Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. Set

$$G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{T \subseteq R} (-1)^{|T|} D_{\mathbf{b},F\cup T}(\mathbf{z}). \tag{75.5}$$

Theorem 75.4 — Presence–absence contact calculus.

For finite disjoint $R, F \subset \mathbb{Z}_{>0}$,

$$\boxed{G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{\substack{u\in\mathcal{A}_{\mathbf{b}} \\ R\subseteq \mathrm{Val}(u),\ \mathrm{Val}(u)\cap F=\varnothing}} \prod_{i\in C_{\mathbf{b}}(u)} z_i.}$$

Consequently, for every $J \subseteq [n]$,

$$\boxed{\begin{aligned}\#\{u \in \mathcal{A}_{\mathbf{b}} : R \subseteq \mathrm{Val}(u),\ \mathrm{Val}(u) \cap F = \varnothing,\\ C_{\mathbf{b}}(u) = J\} = \sum_{T\subseteq R} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b},F\cup T}\end{aligned}} \tag{75.6}$$

Proof. Among the objects avoiding F, require every $r \in R$ to occur. Inclusion–exclusion over the events $r \notin \mathrm{Val}(u)$ gives (75.5); Theorem 75.2 supplies the weighted determinant, and Corollary 75.3 gives (75.6). □

Let $s^{(1)}, \dots, s^{(q)}$ be injective sequences of positive integers and define

$$\mathrm{fm}_{s^{(h)}}(u) = \min\{a \geq 1 : s_a^{(h)} \notin \mathrm{Val}(u)\}.$$

For prescribed $a_1, \dots, a_q$, set

$$R(\mathbf{a}) = \bigcup_{h=1}^{q} \{s_1^{(h)}, \dots, s_{a_h-1}^{(h)}\}, \qquad F(\mathbf{a}) = \{s_{a_1}^{(1)}, \dots, s_{a_q}^{(q)}\}.$$

Corollary 75.5 — Simultaneous first-missing statistics.

If $R(\mathbf{a}) \cap F(\mathbf{a}) \neq \varnothing$, no sequence satisfies all prescribed first-missing conditions. Otherwise, for every $J \subseteq [n]$,

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{fm}_{s^{(h)}}(u) = a_h\ \forall h,\ C_{\mathbf{b}}(u) = J\} = \sum_{T\subseteq R(\mathbf{a})} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b},F(\mathbf{a})\cup T}.}$$

Proof. Each condition $\mathrm{fm}_{s^{(h)}}(u) = a_h$ requires $s_1^{(h)}, \dots, s_{a_h-1}^{(h)}$ to be present and $s_{a_h}^{(h)}$ to be absent. Combine these conditions and apply Theorem 75.4. □

Remark 75.R1 — Exact-contact specialization.

The ordinary probe $s_a = a$ and the sampled probe $s_a = \delta a$ are immediate special cases. No rational-slope, Dyck-path, or automata assumption occurs anywhere in this section.

75.1 Fixed-composition rational Dyck paths

Assume henceforth that $m > n \geq 1$ and let

$$\alpha = (\alpha_1, \dots, \alpha_d) \vDash n, \qquad A_0 = 0, \quad A_i = \alpha_1 + \cdots + \alpha_i.$$

Let $\mathcal{D}_{m,\alpha}$ be the set of $m \times n$ rational Dyck paths with ordered vertical-run composition α.

75.1.1 Component-column coordinates and returns

For $P \in \mathcal{D}_{m,\alpha}$, let x_i be the horizontal coordinate of the initial vertex of the ith vertical component and set

$$\beta_i = \left\lfloor \frac{mA_{i-1}}{n} \right\rfloor. \tag{75.7}$$

Lemma 75.6 — Fixed-composition component encoding.

The map $P \mapsto (x_1, \dots, x_d)$ is a bijection from $\mathcal{D}_{m,\alpha}$ onto

$$\mathcal{X}_{m,\alpha} = \{(x_1, \dots, x_d) : 0 = x_1 < x_2 < \cdots < x_d,\ x_i \leq \beta_i\}.$$

Proof. The ith component starts at (x_i, A_{i-1}). The Dyck condition gives $A_{i-1} \geq nx_i/m$, hence $x_i \leq \beta_i$. Distinct vertical components are separated by east steps, so their columns are strictly increasing. Conversely, from such a column sequence reconstruct

$$N^{\alpha_1} E^{x_2 - x_1} N^{\alpha_2} \cdots E^{x_d - x_{d-1}} N^{\alpha_d} E^{m - x_d}.$$

Each horizontal segment stays above the diagonal because its right endpoint satisfies the corresponding bound $x_{i+1} \leq \lfloor mA_i/n \rfloor$. □

Let $\mathrm{RetComp}(P)$ be the set of component indices whose first north step is a return.

Lemma 75.7 — Return components are barrier contacts.

For $P \leftrightarrow (x_1, \dots, x_d)$,

$$\boxed{i \in \mathrm{RetComp}(P) \iff x_i = \beta_i.}$$

Consequently $\mathrm{ret}(P) = |\,\mathrm{RetComp}(P)|$.

Proof. At the start of component i, the rank is

$$R_i = mA_{i-1} - nx_i.$$

Write $mA_{i-1} = n\beta_i + r_i$ with $0 \leq r_i < n$. Then

$$R_i = r_i + n(\beta_i - x_i).$$

Since $x_i \leq \beta_i$, one has $R_i < n$ exactly when $x_i = \beta_i$. This is precisely the return condition of Dai–Fu–Qiu [23]. A later north step in the same component has rank at least $R_i + m \geq n$, so a component contributes at most one return. □

Corollary 75.8 — Exact north-step return positions.

If $J = \mathrm{RetComp}(P)$, then

$$\boxed{\mathrm{ReturnSet}(P) = \{A_{i-1} + 1 : i \in J\}.}$$

75.1.2 Forbidden columns and the effective barrier

For finite $F \subset \mathbb{Z}_{>0}$, set

$$c_i(F) = \beta_i - \#(F \cap [1, \beta_i]), \qquad g_i(F) = c_i(F) - (i-1),$$

and define the suffix-minimum effective barrier

$$\boxed{h_i(F) = \min_{j \geq i} g_j(F).} \tag{75.8}$$

Then $h_1(F) \leq \cdots \leq h_d(F)$.

Lemma 75.9 — Effective-barrier regularization.

Let

$$\mathcal{X}_{m,\alpha}(F) = \{x \in \mathcal{X}_{m,\alpha} : \{x_1, \ldots, x_d\} \cap F = \varnothing\}.$$

If $h_1(F) < 0$, the class is empty. If $h_1(F) \geq 0$, the maps

$$y_i = \phi_F(x_i), \qquad v_i = y_i - (i-1)$$

give a bijection

$$\mathcal{X}_{m,\alpha}(F) \longleftrightarrow \{0 \leq v_1 \leq \cdots \leq v_d : v_i \leq h_i(F)\}.$$

Proof. Because ϕ_F is strictly increasing on the allowed alphabet, strict x-inequalities are equivalent to strict y-inequalities. After the shift $v_i = y_i - (i-1)$, strictness becomes weak monotonicity and the raw upper bound is $v_i \leq g_i(F)$. For a weakly increasing sequence, the full system $v_j \leq g_j(F)$ is equivalent to $v_i \leq \min_{j \geq i} g_j(F) = h_i(F)$ for every i. Since $\beta_1 = 0$, feasibility forces $h_1(F) = 0$; if $h_1(F) < 0$ no nonnegative sequence exists. □

Define the surviving-contact indicator

$$\eta_i(F) = \mathbf{1}_{\{\beta_i \notin F,\ h_i(F) = g_i(F)\}}. \tag{75.9}$$

Lemma 75.10 — Surviving return contacts.

Under Lemma 75.9,

$$\boxed{x_i = \beta_i \iff \eta_i(F) = 1 \text{ and } v_i = h_i(F).}$$

Proof. If $x_i = \beta_i$, then $\beta_i \notin F$ and $v_i = c_i(F) - (i-1) = g_i(F)$. Since $v_i \leq h_i(F) \leq g_i(F)$, equality is forced. Conversely, if $\eta_i(F) = 1$ and $v_i = h_i(F) = g_i(F)$, then $y_i = c_i(F) = \phi_F(\beta_i)$; injectivity on the allowed alphabet gives $x_i = \beta_i$. □

Define

$$D_{\alpha,F}(z_1, \ldots, z_d) = \det_{1 \leq i,j \leq d} \left[\binom{h_i(F)+1}{j-i+1} + (z_i - 1)\eta_i(F) \binom{h_i(F)}{j-i} \right] \tag{75.10}$$

when $h_1(F) \geq 0$, and set it equal to zero otherwise. Also define

$$B_{ij}^{\alpha,F} = \eta_i(F) \binom{h_i(F)}{j-i}, \qquad A_{ij}^{\alpha,F} = \binom{h_i(F)+1}{j-i+1} - B_{ij}^{\alpha,F},$$

and let $M_J^{\alpha,F}$ use row $B_i^{\alpha,F}$ when $i \in J$ and $A_i^{\alpha,F}$ otherwise.

Theorem 75.11 — Fixed-composition forbidden-column determinant.

For finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_{\alpha,F}(\mathbf{z}) = \sum_{\substack{P \in \mathcal{D}_{m,\alpha} \\ \{x_i(P)\} \cap F = \varnothing}} \prod_{i \in \mathrm{RetComp}(P)} z_i.}$$

Consequently,

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \{x_i(P)\} \cap F = \varnothing,\ \mathrm{RetComp}(P) = J\} = \det M_J^{\alpha,F}.} \tag{75.11}$$

Proof. Lemma 75.9 gives a weakly increasing sequence below $\mathbf{h}(F)$. Apply Theorem 75.2 to that effective barrier, but mark row i only when $\eta_i(F) = 1$. Lemma 75.10 identifies exactly those marked effective contacts with the original return components. The exact-set formula follows by row multilinearity. □

Remark 75.R2 — Effective-barrier interpretation.

For $F = \varnothing$, $g_i = \beta_i - (i-1)$ is already nondecreasing because $m > n$ and every part of α is positive. Thus $h_i = g_i$ and $\eta_i = 1$. The suffix-minimum regularization is needed only after forbidden values are deleted.

Chapter 76

Four Composition-Refined Enumeration Problems

With the general contact calculus and the fixed-composition effective barrier in place, the framework yields four distinct rational-Dyck enumeration results at fixed ordered vertical-run composition α:

1. ordinary run together with returns;
2. ratio-run together with the exact return components;
3. simultaneous ordinary run, ratio-run, and exact returns;
4. fixed ratio-signature, where every class on the associated skew diagonal has one common determinant cardinality.

The fourth result also yields a positive complete-homogeneous expansion of the fixed-signature polynomial. The fixed-signature discussion retains an important scope correction: a ratio-signature is *not* generally an initial component prefix; that prefix interpretation becomes valid only after passage to the minimal-return representative used in the proof.

76.1 Explicit fixed-composition run–return enumeration

Dai–Fu–Qiu define a statistic $\mathrm{rr}(\alpha, m)$ depending only on the composition and m; if $I_1(\alpha)$ is the length of the maximal initial string of parts equal to 1, then [23]

$$r_\alpha := \mathrm{rr}(\alpha, m) = \min\left\{ I_1(\alpha) + 1, \left\lceil \frac{n}{m-n} \right\rceil \right\}. \tag{76.1}$$

Every path of composition α has $\mathrm{run}(P), \mathrm{ret}(P) \geq r_\alpha$. Their fixed-composition bijections imply that for $K = k + \ell - r_\alpha$,

$$|\mathcal{D}_{m,\alpha}^{(k,\ell)}| = |\mathcal{D}_{m,\alpha}^{(K,r_\alpha)}|, \tag{76.2}$$

whenever the class is nonempty [23, Theorem 3.2 and Corollary 3.5].

For $K < d$, put $p = d - K$ and define

$$B_i^{(\alpha,K)} = \beta_{K+i} - (K+i) - 1 = \left\lfloor \frac{mA_{K+i-1}}{n} \right\rfloor - (K+i) - 1, \qquad 1 \leq i \leq p. \tag{76.3}$$

Lemma 76.1 — Minimal-return suffix barrier.

For a path in $\mathcal{D}_{m,\alpha}^{(K,r_\alpha)}$, the first K component columns are forced:

$$x_i = i - 1 \qquad (1 \le i \le K).$$

If $K < d$, then after the shift

$$v_i = x_{K+i} - (K + i),$$

the free suffix is in bijection with

$$0 \le v_1 \le \cdots \le v_p, \qquad v_i \le B_i^{(\alpha,K)}.$$

Moreover $B_1^{(\alpha,K)} \le \cdots \le B_p^{(\alpha,K)}$.

Proof. For the minimal-return representative, Dai–Fu–Qiu use $M_1 \cdots M_K ET$, where $M_i = N^{\alpha_i} E$ [23, Corollary 3.5]. Thus the first K component columns are $0, 1, \ldots, K-1$, while the forced separator gives $x_{K+1} \ge K + 1$. Since the suffix contributes no further returns, Lemma 75.7 gives $x_i \le \beta_i - 1$ for $i > K$. The stated shift yields the weak sequence and its barrier. Finally

$$B_{i+1}^{(\alpha,K)} - B_i^{(\alpha,K)} = \beta_{K+i+1} - \beta_{K+i} - 1 \ge 0,$$

because $m > n$ and $\alpha_{K+i} \ge 1$. □

Theorem 76.2 — Explicit composition-refined run–return enumeration.

Let $k, \ell \ge 1$ and set $K = k + \ell - r_\alpha$. If any of

$$k < r_\alpha, \qquad \ell < r_\alpha, \qquad K > d$$

holds, the class is empty. If $K < d$ and $B_1^{(\alpha,K)} < 0$, it is likewise empty. Otherwise,

$$\boxed{|\mathcal{D}_{m,\alpha}^{(k,\ell)}| = \det_{1 \le i,j \le d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].} \tag{76.4}$$

For $K = d$, the empty determinant is interpreted as 1.

Proof. Use the Dai–Fu–Qiu reduction (76.2), then apply the classical bounded weak-sequence determinant to the suffix barrier in Lemma 76.1. □

Set

$$C_{\alpha,K} = \det_{1 \le i,j \le d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].$$

Corollary 76.3 — Positive complete-homogeneous expansion.

With

$$F_{m,\alpha}(u, v) = \sum_{P \in \mathcal{D}_{m,\alpha}} u^{\mathrm{run}(P)} v^{\mathrm{ret}(P)},$$

one has

$$\boxed{F_{m,\alpha}(u, v) = (uv)^{r_\alpha} \sum_{K=r_\alpha}^{d} C_{\alpha,K} h_{K-r_\alpha}(u, v),} \tag{76.5}$$

where $h_j(u, v) = u^j + u^{j-1}v + \cdots + v^j$.

Proof. For fixed K, the feasible statistic pairs are $(r_\alpha + a, K - a)$ for $0 \le a \le K - r_\alpha$, and Theorem 76.2 gives the common cardinality $C_{\alpha,K}$. Summing their monomials gives $(uv)^{r_\alpha} h_{K-r_\alpha}(u, v)$. □

Example 76.E1 — Example.

Let $(m, n) = (11, 5)$ and $\alpha = (2, 1, 1, 1)$. Then $\beta = (0, 4, 6, 8)$ and $r_\alpha = 1$. For $(\mathrm{run}, \mathrm{ret}) = (1, 2)$, $K = 2$ and the suffix barrier is $(2, 3)$. Hence

$$|\mathcal{D}^{(1,2)}_{11,(2,1,1,1)}| = \det \begin{pmatrix} 3 & 3 \\ 1 & 4 \end{pmatrix} = 9.$$

Remark 76.R1 — Ordinary-run scope.

The determinant in Theorem 76.2 is classical. The new enumerative step is the reduction from a fixed ordered composition and prescribed $(\mathrm{run}, \mathrm{ret})$ to a single bounded monotone suffix. In particular, it makes the cardinality in the partial-path characterization of Dai–Fu–Qiu Corollary 3.5 explicit.

76.2 Composition-refined ratio-run and returns

Set

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor.$$

For $P \in \mathcal{D}_{m,\alpha}$, let $X(P) = \{x_1(P), \dots, x_d(P)\}$.

Lemma 76.4 — Coarea values and component columns.

The set of distinct north-step coarea values of P is $X(P)$. Consequently

$$\boxed{\widetilde{\mathrm{run}}(P) = \min\{a \ge 1 : a\kappa \notin X(P)\}.} \tag{76.6}$$

Proof. Every north step in one vertical component has the same horizontal coordinate, and every component contains at least one north step. Thus the distinct coarea values are exactly the component columns. The second statement is the definition of ratio-run [23]. □

For $a \ge 1$, define

$$R_{\kappa,a} = \{\kappa, 2\kappa, \dots, (a-1)\kappa\}, \qquad F_{\kappa,a} = \{a\kappa\}.$$

Then $\widetilde{\mathrm{run}}(P) = a$ exactly when $R_{\kappa,a} \subseteq X(P)$ and $X(P) \cap F_{\kappa,a} = \varnothing$.

Theorem 76.5 — Composition-refined ratio-run with exact returns.

For every $a \ge 1$ and $J \subseteq [d]$,

$$\boxed{\begin{aligned} &\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{RetComp}(P) = J\} \\ &\qquad = \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T} \end{aligned}} \tag{76.7}$$

Proof. The ratio-run condition is exactly the presence–absence condition above. Inclusion–exclusion over the required values, followed by Theorem 75.11, gives the formula. □

Corollary 76.6 — Composition-refined ratio-run–return enumeration.

For $a, b \geq 1$,

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{ret}(P) = b\} = \sum_{\substack{J \subseteq [d] \\ |J| = b}} \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T}.}$$

Example 76.E2 — Example.

For $(m, n) = (11, 5)$ and $\alpha = (2, 1, 1, 1)$, one has $\beta = (0, 4, 6, 8)$ and $\kappa = 2$. Prescribe $\widetilde{\mathrm{run}} = 2$ and $J = \{1, 4\}$. Then $R_{2,2} = \{2\}$ and $F_{2,2} = \{4\}$. The two inclusion–exclusion determinants are

$$\det M_{\{1,4\}}^{\alpha,\{4\}} = 6, \qquad \det M_{\{1,4\}}^{\alpha,\{2,4\}} = 3,$$

so the required number is $6 - 3 = 3$. The three component-column sequences are

$$(0, 1, 2, 8), \qquad (0, 2, 3, 8), \qquad (0, 2, 5, 8).$$

76.3 Simultaneous run, ratio-run, and exact returns

Lemma 76.7 — Ordinary run as a first-missing component column.

For every $P \in \mathcal{D}_{m,\alpha}$,

$$\boxed{\mathrm{run}(P) = \min\{a \geq 1 : a \notin X(P)\}.}$$

Proof. By Lemma 76.4, $X(P)$ is the set of distinct coarea values. The ordinary run statistic is the first missing positive coarea value [23]. □

For positive integers a, c, define

$$R_{a,c} = \{1, \ldots, a-1\} \cup \{\kappa, 2\kappa, \ldots, (c-1)\kappa\}, \qquad F_{a,c} = \{a, c\kappa\}. \tag{76.8}$$

Theorem 76.8 — Simultaneous run–ratio-run–return-set enumeration.

For $J \subseteq [d]$, if $R_{a,c} \cap F_{a,c} \neq \varnothing$, then

$$\#\{P \in \mathcal{D}_{m,\alpha} : \mathrm{run}(P) = a,\ \widetilde{\mathrm{run}}(P) = c,\ \mathrm{RetComp}(P) = J\} = 0.$$

Otherwise,

$$\boxed{\begin{aligned}&\#\{P \in \mathcal{D}_{m,\alpha} : \mathrm{run}(P) = a,\ \widetilde{\mathrm{run}}(P) = c,\ \mathrm{RetComp}(P) = J\} \\ &\quad = \sum_{T \subseteq R_{a,c}} (-1)^{|T|} \det M_J^{\alpha, F_{a,c} \cup T}\end{aligned}} \tag{76.9}$$

Proof. The two first-missing conditions together are precisely $R_{a,c} \subseteq X(P)$ and $X(P) \cap F_{a,c} = \varnothing$. If the sets intersect, the requirements are inconsistent. Otherwise apply inclusion–exclusion and the exact-return determinant (75.11). □

Corollary 76.9 — Exact north-step return set.

If $S \subseteq [n]$ is not of the form $S = \{A_{i-1} + 1 : i \in J\}$ for some $J \subseteq [d]$, then no path of composition α has $\mathrm{ReturnSet}(P) = S$. For such a set S and disjoint $R_{a,c}, F_{a,c}$, the number of paths with

$$\mathrm{run}(P) = a, \qquad \widetilde{\mathrm{run}}(P) = c, \qquad \mathrm{ReturnSet}(P) = S$$

is the right-hand side of (76.9).

Example 76.E3 — Simultaneous run and ratio-run.

Again let $(m, n) = (11, 5)$ and $\alpha = (2, 1, 1, 1)$, so $\beta = (0, 4, 6, 8)$ and $\kappa = 2$. Prescribe run $= 3$ and $\widetilde{\text{run}} = 2$. Then

$$R_{3,2} = \{1, 2\}, \qquad F_{3,2} = \{3, 4\}.$$

The admissible component-column sequences are

$$(0, 1, 2, 5),\ (0, 1, 2, 6),\ (0, 1, 2, 7),\ (0, 1, 2, 8).$$

The first three have RetComp $= \{1\}$, while the last has RetComp $= \{1, 4\}$. Hence the corresponding exact counts are 3 and 1.

Remark 76.R2 — Automatic compatibility detection.

The condition $R_{a,c} \cap F_{a,c} = \varnothing$ automatically detects incompatibilities. For example, if $\kappa = 2$, then run $= 2$ requires column 2 to be absent while $\widetilde{\text{run}} = 2$ requires it to be present, so the simultaneous class is empty.

76.4 Explicit enumeration at fixed ratio-signature

This section uses the first ratio-signature $\text{sign}_1(P)$ of Dai–Fu–Qiu [23]. Their auxiliary path P_1 is formed by scanning P and collecting each κ-east-step block contributing to ratio-run or return. Thus P_1 is generally a shuffle-selected subpath rather than an initial segment. Let

$$\mathcal{D}_{m,\alpha,s} = \{P \in \mathcal{D}_{m,\alpha} : \text{sign}_1(P) = s\},$$

and assume this class is nonempty. Write

$$r_{\alpha,s} = \widetilde{\text{rr}}(\alpha, m, s),$$

for the minimum value attained by either $\widetilde{\text{run}}$ or ret on this fixed-signature class in the notation of Dai–Fu–Qiu.

Remark 76.R3 — The signature is not generally a prefix.

No argument below identifies a general P_1 with a prefix. The prefix interpretation arises only after passing, via the fixed-signature bijections of Dai–Fu–Qiu Theorem 3.7, to the representative with minimal return.

Lemma 76.10 — Minimal-return prefix collapse.

Suppose $P \in \mathcal{D}_{m,\alpha,s}$ has $\widetilde{\text{run}}(P) = a$ and $\text{ret}(P) = b$. Set

$$h = a + b - r_{\alpha,s}.$$

The fixed-signature bijections of Dai–Fu–Qiu reduce the class to a representative Q with

$$\widetilde{\text{run}}(Q) = h, \qquad \text{ret}(Q) = r_{\alpha,s}, \qquad \text{sign}_1(Q) = s.$$

For this minimal-return representative, Q_1 is exactly the initial segment occupying the first $h\kappa$ east-step columns. Consequently

$$\boxed{|s| = h\kappa.} \tag{76.10}$$

Proof. Dai–Fu–Qiu Theorem 3.7 constructs fixed-signature bijections along the skew diagonal and states that the corresponding minimal value is $r_{\alpha,s}$ [23]. In the minimal-return representative, every return block already belongs to the initial family of blocks contributing simultaneously to ratio-run and return; there are no additional return-only blocks. The ratio-run value h contributes the first h sampled κ-blocks, so the union defining Q_1 is precisely the initial $h\kappa$ east-step segment. Since s is the ordinary east-step signature of Q_1, its length is $h\kappa$. □

Corollary 76.11 — Fixed-signature skew diagonal.

If $\mathcal{D}_{m,\alpha,s} \neq \varnothing$, then $\kappa \mid |s|$. With

$$h_s = \frac{|s|}{\kappa},$$

every nonempty fixed-signature statistic class satisfies

$$\boxed{\widetilde{\mathrm{run}}(P) + \mathrm{ret}(P) = h_s + r_{\alpha,s}.}$$

Let the positions of the 1's in s be

$$1 \le p_1 < \cdots < p_t \le |s|, \qquad t = |s|_1.$$

Lemma 76.12 — Prefix components determined by the signature.

For a minimal-return representative Q as above, the first t component columns are forced:

$$\boxed{x_i = p_i - 1 \qquad (1 \le i \le t).}$$

Proof. By Lemma 76.10, s is the ordinary east-step signature of the first $|s|$ east steps of Q. An east step has signature 1 exactly when it immediately follows a north step. The east step in signature position p_i therefore closes the ith vertical component, whose horizontal coordinate is $p_i - 1$. □

Corollary 76.13 — Reading the minimal return from the signature.

For an admissible signature,

$$\boxed{r_{\alpha,s} = \#\{i \in [t] : p_i - 1 = \beta_i\}.}$$

Proof. In the minimal-return representative, all returns occur among the forced prefix components. Apply Lemma 75.7 and Lemma 76.12. □

Lemma 76.14 — Forced separator and free suffix.

For a minimal-return representative Q,

$$x_{t+1} \ge h_s\kappa + 1 = |s| + 1$$

when $t < d$, and

$$x_{t+j} \le \beta_{t+j} - 1 \qquad (1 \le j \le d - t).$$

Proof. Because $\widetilde{\mathrm{run}}(Q) = h_s$, the sampled column $h_s\kappa$ is missing, so the next component after the prefix starts at least one column later. Since $\mathrm{ret}(Q) = r_{\alpha,s}$ and the prefix already accounts for all returns, no suffix component is a return; Lemma 75.7 gives the strict upper bound. □

Put $p = d - t$ and define

$$c_j^{(\alpha,s)} = \beta_{t+j} - |s| - j - 1, \qquad 1 \le j \le p. \tag{76.11}$$

Then $c_1^{(\alpha,s)} \le \cdots \le c_p^{(\alpha,s)}$ because $\beta_{i+1} - \beta_i \ge 1$.

Theorem 76.15 — Explicit fixed-ratio-signature cardinality.

Let s be admissible and use the notation above. Then

$$\boxed{\begin{aligned} C_{m,\alpha,s} &:= |\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\mathrm{run}}(P) = h_s,\ \mathrm{ret}(P) = r_{\alpha,s}\}| \\ &= \det_{1\le i,j\le d-t}\left[\binom{c_i^{(\alpha,s)}+1}{j-i+1}\right] \end{aligned}} \tag{76.12}$$

The empty determinant is 1.

Proof. The signature fixes the prefix columns by Lemma 76.12. For the free suffix set

$$v_j = x_{t+j} - (|s| + j).$$

Lemma 76.14 gives

$$0 \le v_1 \le \cdots \le v_p, \qquad v_j \le c_j^{(\alpha,s)}.$$

The barrier is nondecreasing, so the classical bounded-sequence determinant gives (76.12). □

Corollary 76.16 — Common cardinality on the fixed-signature skew diagonal.

For every $0 \le j \le h_s - r_{\alpha,s}$,

$$\boxed{|\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\mathrm{run}}(P) = r_{\alpha,s} + j,\ \mathrm{ret}(P) = h_s - j\}| = C_{m,\alpha,s}.} \tag{76.13}$$

All other fixed-signature statistic classes are empty.

Proof. Apply the fixed-signature skew-diagonal bijection of Dai–Fu–Qiu Theorem 3.7 to the minimal-return class, then use Corollary 76.11 to exhaust the possible pairs. □

Corollary 76.17 — Explicit fixed-signature polynomial.

$$\boxed{\sum_{P\in\mathcal{D}_{m,\alpha,s}} u^{\widetilde{\mathrm{run}}(P)} v^{\mathrm{ret}(P)} = C_{m,\alpha,s}(uv)^{r_{\alpha,s}} h_{h_s - r_{\alpha,s}}(u,v).} \tag{76.14}$$

Example 76.E4 — The fixed signature 11111110**.**

Let

$$(m,n) = (23,11), \qquad \alpha = (1,1,2,1,1,1,1,1,1,1), \qquad s = 11111110.$$

Then $\kappa = 2$, $|s| = 8$, $h_s = 4$, and $t = 7$. The component barrier is

$$\beta = (0,2,4,8,10,12,14,16,18,20),$$

and the signature forces $(x_1,\dots,x_7) = (0,1,2,3,4,5,6)$. Hence $r_{\alpha,s} = 1$. The free suffix barrier is $(6,7,8)$, so

$$C_{23,\alpha,s} = \det\begin{pmatrix} 7 & 21 & 35 \\ 1 & 8 & 28 \\ 0 & 1 & 9 \end{pmatrix} = 154.$$

Therefore each of the four classes

$$(\widetilde{\mathrm{run}}, \mathrm{ret}) = (1,4), (2,3), (3,2), (4,1)$$

has cardinality 154, and the fixed-signature polynomial is $154\,uv\,h_3(u,v)$.

Chapter 77

Arbitrary Boundaries and Labeled Applications

The framework is not tied to one rational slope. Two application directions of the barrier-compression development are retained here in full. First, the contact determinant works below an arbitrary northeast boundary and therefore specializes to partial Dyck paths. Second, fixed-composition path counts transfer to column-increasing rational parking-function labelings by the standard multinomial labeling factor. These applications are mathematically separate from the four core rational-Dyck specializations and are kept visible as such.

77.1 Paths below an arbitrary fixed boundary

Let B be a northeast lattice path whose north-step starting columns are

$$\mathbf{b} = (b_1, \dots, b_n), \qquad b_1 \le \cdots \le b_n.$$

A path P lying weakly northwest of B has a north-step column sequence $u(P) \in \mathcal{A}_{\mathbf{b}}$. Thus Section 75 applies verbatim.

Corollary 77.1 — Arbitrary-boundary presence and exact contacts.
Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. For $J \subseteq [n]$,

$$\boxed{\begin{aligned}&\#\{P \le B : R \subseteq \mathrm{Val}(P),\ \mathrm{Val}(P) \cap F = \varnothing,\ C_B(P) = J\} \\ &\qquad = \sum_{T \subseteq R} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b}, F \cup T}\end{aligned}}$$

77.1.1 Partial Dyck paths

Let $0 \le m \le n$ and let $\mathcal{B}_{m,n}$ be the paths from $(0,0)$ to (m,n) remaining weakly above $y = x$. Their north-step barrier is

$$\boxed{b_i = \min(i-1, m).}$$

For $i \le m+1$, contact $u_i = b_i = i-1$ is a diagonal contact; for $i > m+1$, contact $u_i = b_i = m$ is contact with the terminal vertical wall. The multivariate variables therefore separate two geometrically different kinds of boundary contact without changing the compression argument.

Define

$$\mathrm{Ret}_\Delta(P) = \{i \in [m+1] : u_i = i-1\}, \qquad \mathrm{ret}_\Delta(P) = |\mathrm{Ret}_\Delta(P)|.$$

Specializing $z_i = z$ for $i \le m+1$ and $z_i = 1$ afterward gives a polynomial $D^{\mathrm{par}}_{m,n;F}(z)$ satisfying

$$D^{\mathrm{par}}_{m,n;F}(z) = \sum_{\substack{P \in \mathcal{B}_{m,n} \\ \mathrm{Val}(P) \cap F = \varnothing}} z^{\mathrm{ret}_\Delta(P)}.$$

If

$$\mathrm{fm}(P) = \min\{a \ge 1 : a \notin \mathrm{Val}(P)\},$$

then $\mathrm{fm}(P) = a$ is the required/forbidden condition $R_a = \{1, \dots, a-1\}$ and $F_a = \{a\}$. Hence the framework gives both the return-number polynomial and exact diagonal-return-set determinant sums for partial Dyck paths.

77.2 Column-increasing rational parking-function labelings

Classical rational parking functions are commonly represented as rational Dyck paths whose north steps carry labels increasing within each column; see Armstrong, Loehr, and Warrington [22]. For the present application, we use the same column-increasing labeling convention for the $m \times n$ paths considered here, without requiring coprimality. The path statistics run, $\widetilde{\mathrm{run}}$, and ret are inherited from the underlying path.

Lemma 77.2 — Labelings of a fixed composition.

If $P \in \mathcal{D}_{m,\alpha}$, then the number of column-increasing labelings of its n north steps by $1, \dots, n$ is

$$\boxed{L(\alpha) = \frac{n!}{\alpha_1! \cdots \alpha_d!}.}$$

Proof. Choose which α_i labels occupy each vertical component. Once those sets are chosen, the increasing condition determines their within-column order uniquely. □

Define the fixed-composition labeled run–return polynomial

$$\mathcal{P}_{m,\alpha}(u,v) = \sum_{\pi:\, \mathrm{comp}(\underline{\pi}) = \alpha} u^{\mathrm{run}(\pi)} v^{\mathrm{ret}(\pi)},$$

where $\underline{\pi}$ is the underlying path.

Theorem 77.3 — Composition-refined labeled run–return expansion.

$$\boxed{\mathcal{P}_{m,\alpha}(u,v) = \frac{n!}{\alpha_1! \cdots \alpha_d!} (uv)^{r_\alpha} \sum_{K=r_\alpha}^{d} C_{\alpha,K} h_{K-r_\alpha}(u,v).} \tag{77.1}$$

Proof. Every underlying path of composition α has the same multiplicity $L(\alpha)$ by Lemma 77.2. Multiply Corollary 76.3 by this factor. □

Summing over all compositions gives an explicit labeled polynomial

$$\boxed{\mathcal{P}_{m,n}(u,v) = \sum_{\alpha \vDash n} \frac{n!}{\prod_i \alpha_i!} (uv)^{r_\alpha} \sum_{K=r_\alpha}^{\ell(\alpha)} C_{\alpha,K} h_{K-r_\alpha}(u,v).} \tag{77.2}$$

For ratio-run, let $\Sigma_{m,\alpha}$ be the admissible first ratio-signatures. Corollary 76.17 similarly gives

$$\boxed{\widetilde{\mathcal{P}}_{m,\alpha}(u,v) = \frac{n!}{\prod_i \alpha_i!} \sum_{s \in \Sigma_{m,\alpha}} C_{m,\alpha,s} (uv)^{r_{\alpha,s}} h_{h_s - r_{\alpha,s}}(u,v).} \tag{77.3}$$

Thus both labeled distributions have manifest positive complete-homogeneous expansions once the determinant coefficients are fixed.

Remark 77.R1 — Labeling scope.

No symmetric-function coefficient claim is needed here. The application uses only the standard column-increasing labeling rule and the fact that the statistics under consideration depend on the underlying path.

77.3 Computational validation

All results in this branch are established theoretically. As an independent check, we implemented exhaustive enumeration of the underlying bounded sequences and fixed-composition rational Dyck paths and compared the cardinalities with the determinant formulas.

For the general exact-contact theorem, every nondecreasing barrier of length at most 5 with entries in $\{0, 1, 2, 3, 4\}$ was tested, together with every forbidden subset of $\{1, 2, 3, 4\}$ and every exact contact set. This produced 88,032 determinant–enumeration comparisons.

For the rational-path specializations, the ordinary fixed-composition formula was tested on all $m > n$ with $n \leq 7$ and $m \leq 12$, every composition of n, and every tested run–return class: 755 parameter instances and 9,983 statistic queries. The ratio-run formula with exact return components was tested on 628 parameter instances and 54,664 exact-return queries. The simultaneous run–ratio-run formula was tested on 501 parameter instances and 246,392 simultaneous queries. No discrepancy was found.

The fixed-ratio-signature construction was tested independently by forming sign_1 directly from the selected κ-column blocks in the Dai–Fu–Qiu definition, rather than by using the determinant reduction. The test covered 2,357 fixed-signature classes over 12,081 underlying paths, with no mismatch.

As a larger stress test, for

$$(m, n) = (23, 11), \qquad \alpha = (1, 1, 2, 1, 1, 1, 1, 1, 1, 1),$$

all 63,920 fixed-composition paths were enumerated. Exactly 616 have first ratio-signature 11111110, split into the four classes

$$(1, 4),\ (2, 3),\ (3, 2),\ (4, 1)$$

with exactly 154 paths in each, agreeing with Example 76.E4. No mismatch occurred in any exhaustive test.

77.4 Conclusion

We developed a contact-preserving barrier-compression framework for refined lattice-path enumeration. The method separates value constraints, upper-bound geometry, and boundary contacts. Deleting forbidden values produces an order-preserving compression of the alphabet and the barrier, while a multivariate Lindström–Gessel–Viennot determinant retains the surviving contact positions. Inclusion–exclusion then incorporates arbitrary required values and simultaneous first-missing statistics.

The framework arose from a concrete arithmetic route. A completion-aware deterministic pushdown automaton for the 2:1 balance relation led to the scalar residual $\rho_{p,q} = qN_1 - pN_0$. For $m = gq$ and $n = gp$, the identity $R = g\rho_{p,q}$ converted the

low-residual condition into the rational-path return condition, which in coarea coordinates became an upper-barrier contact. This branch isolates the resulting compression mechanism from that discovery route and develops it as a self-contained enumerative method.

For rational Dyck paths with fixed ordered vertical-run composition, the method gives explicit formulas at several levels of refinement. Ordinary run–return classes reduce to one determinant. Ratio-run is handled by sampled presence–absence constraints and exact return-component determinants. Ordinary run and ratio-run can be prescribed simultaneously with exact return positions. At fixed ratio-signature, the minimal-return representative converts the signature data into a forced prefix and a bounded free suffix, giving a single determinant for the common cardinality on the associated skew diagonal. The same framework applies to arbitrary northeast boundaries and partial Dyck paths, while standard column-increasing labeling transfers the fixed-composition formulas to labeled rational parking-function enumerations.

All determinant identities are proved combinatorially, and independent exhaustive computation verified the general framework and the four rational-path specializations over broad finite parameter ranges. The method can therefore be summarized as

$$\boxed{\begin{array}{c}\text{pushdown residual} \longrightarrow \text{rank} \longrightarrow \text{boundary contact} \\ \longrightarrow \text{contact-preserving compression} \longrightarrow \text{explicit determinants}\end{array}}$$

Chapter 78

Canonical Completion Profiles and Farey Unit Lifts

Research branch — Farey Arithmetic of Shortest Completion. *This branch develops the arithmetic of residual-indexed shortest-completion minimizers, the cross-parameter Ordered Farey Recursion, and the universal completion spectra. Farey/Stern–Brocot, Beatty, Christoffel/mechanical, digital-line, and Dedekind structures are classical neighboring theories [27, 28, 32, 29, 33, 30, 31]; novelty is not claimed for those objects in isolation.*

The rational-Dyck branch used the residual as a bridge to enumeration. A distinct arithmetic consequence branch studies the shortest-completion map itself. For coprime positive integers p, q and residual r, define

$$\mu_{p,q}(r) = \arg \min_{x,y \in \mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\}.$$

This is the same minimum-demand semantics developed earlier in the book, now studied as an ordered arithmetic profile in r and across different rational slopes.

The operational provenance remains the original completion-aware DPDA. For example, the prefix 111 has $\rho_{2,1} = 3$, and the machine materializes the completion word 100; its Parikh vector $(2, 1)$ is exactly $\mu_{2,1}(3)$. The automaton is not redrawn here because Chapters 1–8 already preserve it in detail. From this point onward the Farey results are self-contained statements about the minimizer map.

Novelty boundary. The local arithmetic has strong classical shadows. We do not claim novelty for Diophantine-strip reduction, Bézout/Farey unit steps, modular representatives or inverses, mechanical/Christoffel coding, standard Christoffel factorization, digital-line Stern–Brocot updates, Beatty partitions, or Dedekind reciprocity in isolation. Indeed, after a unimodular change of coordinates the local completion walk is a classical rational mechanical/Christoffel path. The principal completion-specific statement appears in Chapter 79: the *cross-parameter Ordered Farey Recursion*, which compares the canonical fixed-residual ℓ^1 minimizer sections of a mediant and both Farey parents and identifies two interleaved child residual subsequences with an unshifted parent profile and a fixed translate of the other. This novelty boundary is a cautious literature assessment, not an absolute priority guarantee.

78.1 Shortest-completion profiles and canonical strips

Fix coprime positive integers p, q. Define

$$W_{p,q}(x, y) = px - qy.$$

The primitive positive kernel vector of $W_{p,q}$ is

$$K_{p,q} = (q, p).$$

Thus any two integer solutions of $px - qy = r$ differ by an integral multiple of (q, p).

Definition 78.1 — Shortest completion.

For $r \in \mathbb{Z}$, define

$$\mu_{p,q}(r) = (x_r, y_r) = \operatorname*{arg\,min}_{x,y \in \mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\},$$

and set

$$L_{p,q}(r) = x_r + y_r.$$

When p, q are fixed, we abbreviate these as $\mu(r)$ and $L(r)$.

Lemma 78.2 — Canonical-strip characterization.

For $r \geq 0$, $\mu(r)$ is the unique nonnegative solution of

$$px - qy = r$$

satisfying

$$0 \leq y < p.$$

For $r \leq 0$, $\mu(r)$ is the unique nonnegative solution satisfying

$$0 \leq x < q.$$

Proof. Suppose $r \geq 0$. Since $\gcd(p, q) = 1$, the congruence

$$qy \equiv -r \pmod{p}$$

has a unique solution $y_0 \in \{0, \ldots, p - 1\}$. Put

$$x_0 = \frac{r + qy_0}{p} \geq 0.$$

Every integer solution is

$$(x_0, y_0) + k(q, p), \qquad k \in \mathbb{Z}.$$

Because $0 \leq y_0 < p$, nonnegativity forces $k \geq 0$. Every other nonnegative solution therefore has length

$$x_0 + y_0 + k(p + q),$$

which is strictly larger for $k > 0$. Hence (x_0, y_0) is the unique minimizer. The case $r \leq 0$ is dual, using the unique representative $x_0 \in \{0, \ldots, q - 1\}$. □

Corollary 78.3 — Kernel reduction.

If (x, y) is any nonnegative solution of residual r, then

$$(x, y) = \mu(r) + k(q, p)$$

for a unique $k \geq 0$, and

$$x + y = L(r) + k(p + q).$$

Proposition 78.4 — Boundary families.

The cases $p = 1$ or $q = 1$ are explicit. If $p = 1$, then

$$\mu_{1,q}(0) = (0, 0), \qquad \mu_{1,q}(1) = (1, 0),$$

and for $1 \leq s \leq q$,

$$\mu_{1,q}(-s) = (q - s, 1).$$

If $q = 1$, then

$$\mu_{p,1}(0) = (0, 0), \qquad \mu_{p,1}(-1) = (0, 1),$$

and for $1 \leq r \leq p$,

$$\mu_{p,1}(r) = (1, p - r).$$

Proof. Each displayed pair solves the relevant equation and lies in the canonical strip of Lemma 78.2. □

78.2 Canonical Farey unit-lift dynamics

Assume throughout this section that $p, q > 1$. Let

$$(p, q) = (c, u) + (v, d) \tag{78.1}$$

be the oriented Farey-parent decomposition characterized by

$$pu - qc = 1, \qquad qv - pd = 1. \tag{78.2}$$

It follows that

$$uv - cd = 1. \tag{78.3}$$

The existence and uniqueness of this oriented parent pair are classical Farey/Stern–Brocot facts; see, for example, [27].

Theorem 78.5 — Canonical Farey Unit-Lift Theorem.

Define

$$a = (u, c), \qquad b = (-d, -v).$$

Then

$$W_{p,q}(a) = W_{p,q}(b) = 1$$

and

$$a - b = (q, p) = K_{p,q}.$$

If

$$\mu(r) = (x_r, y_r), \qquad 0 \leq r < p,$$

then

$$\mu(r + 1) = \begin{cases} \mu(r) + (u, c), & 0 \leq y_r < v, \\ \mu(r) + (-d, -v), & v \leq y_r < p. \end{cases} \tag{78.4}$$

Proof. By equation (78.2),

$$W_{p,q}(u, c) = pu - qc = 1$$

and

$$W_{p,q}(-d, -v) = -pd + qv = 1.$$

Moreover,

$$(u, c) - (-d, -v) = (u + d, c + v) = (q, p).$$

Thus the two vectors are kernel-equivalent lifts of residual increment $+1$.

If $y_r < v$, then

$$0 \le y_r + c < v + c = p,$$

so $\mu(r) + (u, c)$ has residual $r + 1$ and remains in the positive canonical strip. By Lemma 78.2, it is $\mu(r + 1)$.

If $y_r \ge v$, then adding (u, c) leaves the strip because $y_r + c \ge p$. Subtracting one kernel gives

$$(u, c) - (q, p) = (-d, -v),$$

and the new second coordinate is $y_r - v \in [0, p - 1]$. Hence $\mu(r) + (-d, -v)$ is the unique canonical representative of residual $r + 1$. □

Remark 78.R1 — Canonical unit lifting.

The mechanism in Theorem 78.5 is intrinsic to shortest completion:

$$\text{unit residual lift} \quad + \quad \text{canonical kernel reduction.}$$

The Farey parent vectors appear as the two kernel-equivalent lifts available to this reduction.

Theorem 78.6 — Negative unit-lift theorem.

Define

$$a^- = (d, v), \qquad b^- = (-u, -c).$$

Then

$$W_{p,q}(a^-) = W_{p,q}(b^-) = -1, \qquad a^- - b^- = (q, p).$$

If

$$\mu(-s) = (x_s, y_s), \qquad 0 \le s < q,$$

then

$$\mu(-(s + 1)) = \begin{cases} \mu(-s) + (d, v), & 0 \le x_s < u, \\ \mu(-s) + (-u, -c), & u \le x_s < q. \end{cases} \tag{78.5}$$

Proof. The two increments have residual -1 because

$$pd - qv = -1, \qquad -pu + qc = -1,$$

and their difference is (q, p). If $x_s < u$, adding (d, v) keeps the first coordinate below $u + d = q$. If $x_s \ge u$, this lift leaves the negative canonical strip, and subtracting the kernel replaces (d, v) by $(-u, -c)$. The conclusion follows from Lemma 78.2. □

Corollary 78.7 — Closed vector profiles.

For $0 \le r \le p$,

$$\mu_{p,q}(r) = r(u,c) - \left\lfloor \frac{cr}{p} \right\rfloor (q,p). \tag{78.6}$$

For $0 \le s \le q$,

$$\mu_{p,q}(-s) = s(d,v) - \left\lfloor \frac{ds}{q} \right\rfloor (q,p). \tag{78.7}$$

In particular,

$$\mu(p) = (1,0), \qquad \mu(-q) = (0,1).$$

Proof. In the positive walk, the kernel correction occurs exactly when the second coordinate wraps under addition of c modulo p. The number of wraps among the first r steps is $\lfloor cr/p \rfloor$, giving equation (78.6). The negative formula is identical with d modulo q. At $r = p$ and $s = q$, use $pu - qc = 1$ and $qv - pd = 1$. □

Example 78.E1 — The pair 7:4.

For $(p,q) = (7,4)$ the oriented parents are $(c,u) = (5,3)$ and $(v,d) = (2,1)$. Hence the two positive unit lifts are $(3,5)$ and $(-1,-2)$. The positive fundamental profile is

$$(0,0), (3,5), (2,3), (1,1), (4,6), (3,4), (2,2), (1,0),$$

whose successive increments are

$$(3,5), (-1,-2), (-1,-2), (3,5), (-1,-2), (-1,-2), (-1,-2).$$

The ordered recursion places the left-parent profile at indices

$$\left\lceil \frac{7r}{5} \right\rceil \in \{0,2,3,5,6,7\}$$

and the translated right-parent block at indices

$$\left\lfloor \frac{7s}{2} \right\rfloor + 1 \in \{1,4\}.$$

Thus the local two-jump law and the global floor/ceiling shuffle are visible in the same small example.

Chapter 79

Ordered Farey Recursion and the Completion Tree

The unit-lift laws identify the two possible local jumps. The central result of the Farey development is stronger and cross-parametric: it reconstructs the entire mediant profile from the two parent profiles at complementary interleaved residual positions. Parent recovery and the recursively decorated Stern–Brocot family are retained as mechanisms and consequences of that theorem rather than promoted to separate novelty claims.

79.1 Local Farey recovery and ordered parent recursion

79.1.1 Local encoding and parent recovery

Theorem 79.1 — Vector jump law.

The positive fundamental profile has exactly two local increments,

$$\mu(r+1) - \mu(r) \in \{(u, c), (-d, -v)\}, \qquad 0 \le r < p.$$

The jump (u, c) occurs v times and the jump $(-d, -v)$ occurs c times.

The negative fundamental profile has exactly the two increments

$$(d, v), \qquad (-u, -c),$$

with multiplicities u and d, respectively.

Proof. The two-value statements are Theorems 78.5 and 78.6. In the positive profile, the number of kernel corrections is

$$\sum_{r=0}^{p-1} \left(\left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \right) = c.$$

Since there are $p = c + v$ jumps in total, the other jump occurs v times. The negative case is identical. □

Corollary 79.2 — Parent recovery.

For $p, q > 1$, the ordered positive completion profile uniquely determines the oriented Farey parents. If its two local increments are

$$(a, b), \qquad (-e, -f),$$

then

$$(c, u) = (b, a), \qquad (v, d) = (f, e),$$

and consequently

$$p = b + f, \qquad q = a + e.$$

Proof. By Theorem 79.1, $(a, b) = (u, c)$ and $(-e, -f) = (-d, -v)$. □

79.1.2 Ordered parent-to-mediant recursion

Write

$$\mathcal{M}^{+}_{p,q} = (\mu_{p,q}(0), \mu_{p,q}(1), \ldots, \mu_{p,q}(p))$$

and

$$\mathcal{M}^{-}_{p,q} = (\mu_{p,q}(0), \mu_{p,q}(-1), \ldots, \mu_{p,q}(-q)).$$

Theorem 79.3 — Cross-parameter Ordered Farey Recursion.

Let $(p, q) = (c, u) + (v, d)$ be as in equations (78.1) and (78.2). For $0 \le r \le c$,

$$\mu_{p,q}\Big(\Big\lceil \frac{pr}{c} \Big\rceil\Big) = \mu_{c,u}(r). \tag{79.1}$$

For $0 \le s < v$,

$$\mu_{p,q}\Big(\Big\lfloor \frac{ps}{v} \Big\rfloor + 1\Big) = (u, c) + \mu_{v,d}(s). \tag{79.2}$$

Dually, for $0 \le r \le u$,

$$\mu_{p,q}\Big(-\Big\lfloor \frac{qr}{u} \Big\rfloor\Big) = \mu_{c,u}(-r), \tag{79.3}$$

and for $1 \le s \le d$,

$$\mu_{p,q}\Big(-\Big(\Big\lceil \frac{qs}{d} \Big\rceil - 1\Big)\Big) = (u, c) + \mu_{v,d}(-s). \tag{79.4}$$

Moreover, the positive index families

$$I^{+}_{L} = \Big\{\Big\lceil \frac{pr}{c} \Big\rceil : 0 \le r \le c\Big\}, \qquad I^{+}_{R} = \Big\{\Big\lfloor \frac{ps}{v} \Big\rfloor + 1 : 0 \le s < v\Big\}$$

form a disjoint partition of $\{0, 1, \ldots, p\}$. The corresponding negative index families in equations (79.3) and (79.4) form a disjoint partition of $\{0, 1, \ldots, q\}$.

Proof. For equation (79.1), let $\mu_{c,u}(r) = (x, y)$. Then

$$cx - uy = r, \qquad 0 \le y < c.$$

Set $R = px - qy$. Since $pu = qc + 1$,

$$\frac{pr}{c} = px - \frac{pu}{c}y = px - qy - \frac{y}{c} = R - \frac{y}{c}.$$

As $0 \le y/c < 1$, we get

$$R = \Big\lceil \frac{pr}{c} \Big\rceil.$$

Since $y < c < p$, the same vector lies in the child canonical strip, proving equation (79.1).

For equation (79.2), let $\mu_{v,d}(s) = (x, y)$, so

$$vx - dy = s, \qquad 0 \le y < v.$$

Since $pd = qv - 1$,

$$\frac{ps}{v} = px - \frac{pd}{v}y = px - qy + \frac{y}{v},$$

and therefore

$$px - qy = \left\lfloor \frac{ps}{v} \right\rfloor .$$

Adding (u, c) raises the child residual by one. Moreover $0 \le c + y < c + v = p$, so the translated vector is canonical. This proves equation (79.2).

For equation (79.3), let $\mu_{c,u}(-r) = (x, y)$, so $cx - uy = -r$ and $0 \le x < u$. Then

$$\frac{qr}{u} = qy - \frac{qc}{u}x = qy - px + \frac{x}{u} = -(px - qy) + \frac{x}{u}.$$

Hence $px - qy = -\lfloor qr/u \rfloor$, and the same vector lies in the child negative strip because $x < u < q$.

For equation (79.4), let $\mu_{v,d}(-s) = (x, y)$, so $vx - dy = -s$ and $0 \le x < d$. Using $qv = pd + 1$,

$$\frac{qs}{d} = qy - \frac{qv}{d}x = qy - px - \frac{x}{d} = -(px - qy) - \frac{x}{d}.$$

Thus $\lceil qs/d \rceil = -(px - qy)$. Adding (u, c) raises the residual by one and keeps the first coordinate below $u + d = q$, which gives equation (79.4).

It remains to verify the two index partitions. We give the positive one directly. Both index lists are strictly increasing. For $1 \le n < p$, the number of left indices at most n is

$$\left\lfloor \frac{cn}{p} \right\rfloor + 1,$$

while the number of right indices at most n is

$$\left\lceil \frac{vn}{p} \right\rceil = n - \left\lfloor \frac{cn}{p} \right\rfloor ,$$

because $v = p - c$ and cn/p is nonintegral. Their sum is $n + 1$, exactly the number of positions in $\{0, \dots, n\}$. The endpoint counts at $n = 0, p$ are immediate, so every position $0, \dots, p$ occurs exactly once. The negative partition is identical with $q = u + d$. These index sets are the familiar finite rational Beatty partition; the vector labels supplied by equations (79.1) to (79.4) are the completion-specific content. □

Remark 79.R1 — Vector-valued Beatty shuffle.

Theorem 79.3 says more than set inclusion. The entire residual order is reconstructed: the left parent populates the positions I_L^+ unchanged, while the translated right parent populates the complementary positions I_R^+. The same holds on the negative side. The endpoint $\mu_{v,d}(v) = (1, 0)$ is omitted from equation (79.2) because the child endpoint $\mu_{p,q}(p) = (1, 0)$ is already inherited through the left parent.

Corollary 79.4 — Local–global Farey correspondence.

For a proper pair $p, q > 1$, the oriented Farey parents determine both the local jump law and the global ordered recursion. Conversely, the local positive jump set recovers the oriented parents and hence determines the ordered parent recursion.

Proof. Forward determination is Theorems 78.5 and 79.3; reverse determination is Corollary 79.2. □

79.2 The global Farey completion family

The preceding results concern a single Farey mediant. Iteration over the Stern–Brocot tree produces one recursively coherent completion family.

Definition 79.5 — Farey completion tree.

Decorate every coprime node q/p of the Stern–Brocot tree by the pair of ordered profiles

$$\mathfrak{C}(p,q) = \big(\mathcal{M}^{+}_{p,q}, \mathcal{M}^{-}_{p,q}\big).$$

We call the resulting decorated tree the *Farey completion tree*.

Theorem 79.6 — Recursive coherence.

Whenever

$$(p,q) = (c,u) + (v,d)$$

is a Farey mediant, both ordered profiles in $\mathfrak{C}(p,q)$ are determined uniquely from the two parent profiles by Theorem 79.3. Consequently the family

$$\{\mathfrak{C}(p,q) : p,q \geq 1,\ \gcd(p,q) = 1\}$$

is recursively coherent under Farey mediants.

Proof. Apply Theorem 79.3 at every proper interior node and stop when $p = 1$ or $q = 1$, where Proposition 78.4 supplies the explicit boundary profiles. The two index families partition every proper child profile, so the recursion determines every child entry exactly once. □

Remark 79.R2 — Classical tree, new decoration.

The Stern–Brocot tree and its continued-fraction complexity are classical. The point of Definition 79.5 and Theorem 79.6 is not a new rational tree, but a new vector-valued decoration: the shortest-completion profiles form a single parent-generated system rather than an unrelated collection indexed by $p : q$.

Chapter 80

Universal Completion Spectra and Arithmetic Interfaces

The vector profile remembers the ratio $p{:}q$, but scalar projection reveals a universal phenomenon controlled only by $m = p + q$. This chapter retains the fundamental and shell spectra, their moment consequences, the one-sided reflection law, and the precise interfaces with Beatty, mechanical/Christoffel, digital-line, and Dedekind arithmetic. These classical interfaces are included because they explain what the completion theory projects to; they are not used to overstate the novelty of the local coding.

80.1 Scalar projection and universal spectra

For the Farey-coordinate formulas in the first part of this section assume $p, q > 1$; the boundary families are given by Proposition 78.4. Set

$$m = p + q, \qquad A = c + u, \qquad B = v + d.$$

Then $A + B = m$. From equation (78.2),

$$pA - mc = 1, \qquad qB - md = 1. \tag{80.1}$$

Thus $A \equiv p^{-1} \pmod{m}$ and $B \equiv q^{-1} \pmod{m}$.

Proposition 80.1 — Scalar Farey profiles.

For $0 \le r \le p$,

$$L_{p,q}(r) = Ar - m \left\lfloor \frac{cr}{p} \right\rfloor. \tag{80.2}$$

For $0 \le s \le q$,

$$L_{p,q}(-s) = Bs - m \left\lfloor \frac{ds}{q} \right\rfloor. \tag{80.3}$$

Proof. Take coordinate sums in equations (78.6) and (78.7). □

Corollary 80.2 — Scalar jump law and affine periodicity.

For $0 \le r < p$,

$$L(r+1) - L(r) \in \{A, -B\},$$

with multiplicities v and c. Dually,

$$L(-(s+1)) - L(-s) \in \{B, -A\},$$

with multiplicities u and d. Moreover,

$$L(r+p) = L(r) + 1 \qquad (r \geq 0) \tag{80.4}$$

and

$$L(-(s+q)) = L(-s) + 1 \qquad (s \geq 0). \tag{80.5}$$

Proof. The jump statements are the ℓ^1 projections of Theorem 79.1. Their net positive drift is

$$vA - cB = v(c+u) - c(v+d) = uv - cd = 1,$$

and the negative drift is $uB - dA = 1$. Alternatively, equations (80.4) and (80.5) follow directly from the canonical strips by adding $(1,0)$ or $(0,1)$. □

80.1.1 Fundamental spectra

Define

$$S_+(p,q) = \{L(r) : 0 \leq r < p\}, \qquad S_-(p,q) = \{L(-s) : 0 \leq s < q\}.$$

Proposition 80.3 — Completion–Beatty identification.

As sets,

$$S_+(p,q) = \left\{ \left\lceil \frac{mj}{p} \right\rceil : 0 \leq j < p \right\}, \tag{80.6}$$

and

$$S_-(p,q) = \left\{ \left\lceil \frac{mj}{q} \right\rceil : 0 \leq j < q \right\}. \tag{80.7}$$

Proof. For $0 \leq r < p$, the positive canonical coordinate y_r runs through every residue $j \in \{0, \ldots, p-1\}$ exactly once. For fixed j, the least nonnegative first coordinate satisfying $px - qj \geq 0$ is $\lceil qj/p \rceil$, and therefore the corresponding completion length is

$$j + \left\lceil \frac{qj}{p} \right\rceil = \left\lceil \frac{(p+q)j}{p} \right\rceil.$$

The negative case is dual. □

Theorem 80.4 — Universal Fundamental Spectrum.

For every coprime pair p, q with $m = p + q$,

$$S_+(p,q) \uplus S_-(p,q) = \{0, 0, 2, 3, \ldots, m-1\} \tag{80.8}$$

as multisets.

Proof. By Proposition 80.3, it is enough to show that the two nonzero ceiling sets partition $\{2, \ldots, m-1\}$. For $1 \leq \ell \leq m$, define

$$\chi_p(\ell) = \left\lfloor \frac{p\ell}{m} \right\rfloor - \left\lfloor \frac{p(\ell-1)}{m} \right\rfloor, \qquad \chi_q(\ell) = \left\lfloor \frac{q\ell}{m} \right\rfloor - \left\lfloor \frac{q(\ell-1)}{m} \right\rfloor.$$

Each quantity is 0 or 1. The condition $\chi_p(\ell) = 1$ is equivalent to $\ell = \lceil mj/p \rceil$ for some $1 \leq j \leq p-1$ when $2 \leq \ell \leq m-1$, and similarly for q. Since $q = m-p$ and $\gcd(p,m) = 1$, for every $2 \leq \ell \leq m-1$ neither $p\ell/m$ nor $p(\ell-1)/m$ is integral; using $\lfloor \ell - x \rfloor = \ell - \lceil x \rceil$ gives

$$\chi_p(\ell) + \chi_q(\ell) = 1.$$

Thus every integer $2, \ldots, m-1$ belongs to exactly one of the two nonzero spectra. At $\ell = 1$ neither occurs, while both spectra contain 0. This proves equation (80.8). The resulting partition is the finite rational Beatty complementarity seen in Proposition 80.3. □

80.1.2 Universal shell spectra

For $k \geq 0$, define

$$\Sigma_k^+ = \{L(kp + r) : 0 \leq r < p\}, \qquad \Sigma_k^- = \{L(-kq - s) : 0 \leq s < q\}.$$

Theorem 80.5 — Universal Shell Spectrum.
For every $k \geq 0$,

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k+2, k+3, \ldots, k+m-1\}. \tag{80.9}$$

Proof. By equations (80.4) and (80.5),

$$L(kp + r) = k + L(r), \qquad L(-kq - s) = k + L(-s).$$

Translate Theorem 80.4 by k. □

Corollary 80.6 — Shell generating polynomial.
If

$$C_k(z) = \sum_{r=0}^{p-1} z^{L(kp+r)} + \sum_{s=0}^{q-1} z^{L(-kq-s)},$$

then

$$C_k(z) = z^k\big(2 + z^2 + z^3 + \cdots + z^{m-1}\big),$$

and hence

$$\sum_{k\geq 0} C_k(z)t^k = \frac{2 + z^2 + z^3 + \cdots + z^{m-1}}{1 - tz}.$$

Corollary 80.7 — Universal moment hierarchy.
For every integer $j \geq 1$,

$$M_j(k) := \sum_{\ell \in \Sigma_k^+ \uplus \Sigma_k^-} \ell^j = 2k^j + \sum_{\ell=2}^{m-1} (k + \ell)^j.$$

In particular,

$$M_j(0) = \sum_{\ell=2}^{m-1} \ell^j.$$

Thus all combined shell power moments depend only on $m = p + q$.

80.1.3 Reflection symmetry

Write the nonzero positive spectrum in increasing order as

$$\ell_j^+ = \left\lceil \frac{mj}{p} \right\rceil, \qquad 1 \leq j < p,$$

and define ℓ_j^- analogously with q.

Proposition 80.8 — One-sided reflection.
For $1 \leq j < p$,

$$\ell_j^+ + \ell_{p-j}^+ = m + 1,$$

and for $1 \leq j < q$,

$$\ell_j^- + \ell_{q-j}^- = m + 1.$$

Consequently, with $h = (m + 1)/2$, every odd centered one-sided moment vanishes.

Proof. Because $\gcd(m,p)=1$, mj/p is nonintegral for $1 \le j < p$. Hence

$$\left\lceil \frac{m(p-j)}{p} \right\rceil = m - \left\lfloor \frac{mj}{p} \right\rfloor,$$

which sums with $\lceil mj/p \rceil$ to $m+1$. The negative identity is dual, and centered odd powers cancel in symmetric pairs. □

80.2 Classical arithmetic interfaces

The results through section 80.1 were obtained from shortest-completion minimization, canonical kernel reduction, and Farey unimodularity. We now record how classical arithmetic structures appear as projections. The purpose of this section is interpretive rather than foundational.

80.2.1 Mechanical and Christoffel coding

The positive wrap indicator is

$$\varepsilon_r = \left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \in \{0,1\}. \tag{80.10}$$

Then

$$\mu(r+1) - \mu(r) = (u,c) - \varepsilon_r(q,p).$$

Thus $\varepsilon_r = 0$ selects (u,c) and $\varepsilon_r = 1$ selects $(-d,-v)$. The finite binary sequence $(\varepsilon_r)_{r=0}^{p-1}$ is a rational lower mechanical word; under the standard finite conventions it is a Christoffel encoding, up to the usual choice of slope orientation and alphabet. Standard factorization, continued fractions, and the relation with the Stern–Brocot tree are classical; see [27].

The completion interpretation is that the mechanical/Christoffel coding records exactly when canonical kernel reduction occurs. This should not be read as a novelty claim for the coding itself.

80.2.2 Unimodular realization and the distinction from standard factorization

Let

$$k_r = \left\lfloor \frac{cr}{p} \right\rfloor \qquad (0 \le r \le p)$$

and define

$$J_+ = \begin{pmatrix} u & -d \\ c & -v \end{pmatrix}.$$

Since $uv - cd = 1$, one has $\det J_+ = -1$. Using $(q,p) = (u,c) - (-d,-v)$ in equation (78.6) gives

$$\mu_{p,q}(r) = (r-k_r)(u,c) + k_r(-d,-v) = J_+ \begin{pmatrix} r-k_r \\ k_r \end{pmatrix}. \tag{80.11}$$

Therefore

$$J_+^{-1}\mu_{p,q}(r) = \begin{pmatrix} r - \lfloor cr/p \rfloor \\ \lfloor cr/p \rfloor \end{pmatrix},$$

which is the vertex sequence of a rational mechanical lattice path with v steps of one type and c of the other. Thus the local completion walk is a unimodular image of classical Christoffel/mechanical geometry. This is consistent with digital-straight-line

theory, where arithmetic remainders, minimal $a + b$ parameters, Bézout vectors and Stern–Brocot slope updates already occur together [29], and with classical Christoffel duality and modular-inverse phenomena [28].

It is important, however, not to identify Theorem 79.3 with standard Christoffel factorization. Standard factorization decomposes a proper Christoffel word into a *contiguous concatenation* of two Christoffel factors associated with the Farey parents [27]. By contrast, Theorem 79.3 compares three different canonical minimizer maps,

$$\mu_{p,q}, \qquad \mu_{c,u}, \qquad \mu_{v,d},$$

and partitions the child residual positions into two *interleaved* sets

$$I_L^+ = \left\{ \left\lceil \frac{pr}{c} \right\rceil : 0 \le r \le c \right\}, \qquad I_R^+ = \left\{ \left\lfloor \frac{ps}{v} \right\rfloor + 1 : 0 \le s < v \right\}.$$

The corresponding child-vector subsequences are exactly the left-parent completion profile and a fixed translate of the right-parent completion profile. Hence standard factorization explains why Farey parents and Christoffel structure are nearby, but it is not the same residual-indexed cross-parameter identity.

The two scalar jump amplitudes are

$$A = c + u, \qquad B = v + d,$$

so, under the standard Christoffel factorization conventions, the corresponding factor lengths also appear as the ℓ^1 projections of the two completion unit lifts.

80.2.3 Dedekind structure of the one-sided second moment

The combined spectrum is universal, but its split into positive and negative parts depends on $p : q$. This ratio dependence is already visible in the centered second moment.

Let

$$((x)) = \begin{cases} \{x\} - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases}$$

and define the classical Dedekind sum

$$s(h,k) = \sum_{j=1}^{k-1} \left(\left(\frac{j}{k} \right) \right) \left(\left(\frac{hj}{k} \right) \right).$$

Put $h_0 = (m+1)/2$. For the positive sorted spectrum,

$$\ell_j^+ - h_0 = m \left(\left(\frac{j}{p} \right) \right) - \left(\left(\frac{qj}{p} \right) \right). \tag{80.12}$$

Since multiplication by q permutes the nonzero residue classes modulo p,

$$\sum_{j=1}^{p-1} \left(\left(\frac{qj}{p} \right) \right)^2 = \sum_{j=1}^{p-1} \left(\left(\frac{j}{p} \right) \right)^2 = \frac{(p-1)(p-2)}{12p}.$$

Squaring equation (80.12) therefore gives

$$D_2^+(p,q) := \sum_{r=1}^{p-1} \left(L(r) - \frac{m+1}{2} \right)^2 = \frac{(m^2+1)(p-1)(p-2)}{12p} - 2m\, s(q,p). \tag{80.13}$$

Similarly,

$$D_2^-(p,q) = \frac{(m^2+1)(q-1)(q-2)}{12q} - 2m\, s(p,q). \tag{80.14}$$

By Theorem 80.4,

$$D_2^+(p,q) + D_2^-(p,q) = \frac{(m-1)(m-2)(m-3)}{12}. \tag{80.15}$$

Substituting equations (80.13) and (80.14) into equation (80.15) recovers the classical Dedekind reciprocity law

$$s(q,p) + s(p,q) = -\frac{1}{4} + \frac{1}{12}\left(\frac{q}{p} + \frac{p}{q} + \frac{1}{pq}\right);$$

see [31]. Dedekind reciprocity is not a new result here. Its completion interpretation is that Dedekind sums quantify the ratio-dependent one-sided splitting of an otherwise universal completion spectrum.

Remark 80.R1 — Higher even moments.

By Proposition 80.8, all odd centered moments vanish. Higher even moments expand into products of periodic Bernoulli functions and hence into generalized Dedekind-type sums. Since such sums and their reciprocity laws have an extensive independent literature, including [34], we do not pursue new reciprocity statements here.

80.3 Computational discovery and validation

The results above are proved symbolically and do not depend on computation. Numerical experimentation served first as a discovery device and later as an independent audit of signs, endpoints, and floor/ceiling conventions.

An initial audit covered every coprime ordered pair $2 \le p, q \le 100$. A second reviewer-oriented audit extended the exhaustive range to

$$2 \le p, q \le 200,$$

covering 24,064 coprime ordered pairs, and added 30,000 randomized coprime pairs with $2 \le p, q \le 5000$. Shortest-completion vectors were constructed independently from Lemma 78.2. For every residual in the positive and negative fundamental ranges, these direct solutions were compared with the formulas of sections 78.2 and 79.1. The audits verified, without exception:

- the positive and negative canonical unit-lift laws;
- the closed vector formulas equations (78.6) and (78.7);
- all four ordered parent-to-mediant identities equations (79.1) to (79.4);
- the corresponding positive and negative index partitions.

No counterexample was found in either the exhaustive or randomized vector audits.

A separate scalar audit tested every coprime ordered pair

$$1 \le p, q \le 150,$$

comprising 13,715 ratios. For each pair, the fundamental spectrum and the translated shells $k = 0, 1, \ldots, 7$ were generated directly from shortest completions. In every case,

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k+2, k+3, \ldots, k+p+q-1\}.$$

These computations are not proof premises. The direct completion oracle used only the defining equation $px - qy = r$ and the canonical-strip bounds, rather than the closed Farey formulas being tested. The experiments therefore function only as independent consistency checks.

80.4 Conclusion

We have studied the ordered family of shortest nonnegative integer solutions of

$$px - qy = r,$$

with p and q coprime, motivated by shortest completion in rational balance languages.

For a proper Farey decomposition

$$(p, q) = (c, u) + (v, d), \qquad uv - cd = 1,$$

the positive shortest-completion profile advances through two kernel-equivalent unit-residual lifts,

$$(u, c) \qquad \text{and} \qquad (-d, -v).$$

Canonical reduction in the strip $0 \le y < p$ selects between them, and the negative profile has the corresponding dual dynamics. After a unimodular coordinate change this local walk is a classical rational mechanical path, so its Christoffel form is best viewed as the local arithmetic mechanism rather than as the principal novelty.

The principal completion-specific result is the cross-parameter Ordered Farey Recursion Theorem. It reconstructs the entire mediant minimizer section from the two parent minimizer sections through complementary floor/ceiling index maps: the two parent profiles appear as interleaved child subsequences, one after a fixed translation. The local jump vectors recover the oriented parents, and iteration yields a recursively coherent family over the Stern–Brocot tree.

Returning to the motivating automaton, the map $r \mapsto \mu_{2,1}(r)$ is exactly the shortest-demand semantics realized by its distributed state–stack representation; the present results show that this semantic object belongs to a much larger Farey-recursive arithmetic family.

Scalar projection produces a complementary universal phenomenon. Writing $m = p + q$, the combined positive and negative fundamental spectrum is always

$$\{0, 0, 2, 3, \ldots, m - 1\},$$

and every residual shell is an affine translate of this multiset. Hence all combined shell moments depend only on m. Classical Beatty, mechanical/Christoffel, and Dedekind structures arise as arithmetic shadows of this vector theory rather than as proof premises.

Taken together, these results organize the coprime rational balance family as a single Farey-recursive system of shortest-completion profiles, with locally recoverable parent data and universal scalar spectra.

The Farey chapters preceding this branch studied the shortest nonnegative solutions of $px - qy = r$ as the ratio (p, q) varies through the Stern–Brocot/Farey tree. The next question is external rather than internal: whether the same completion coordinates reveal exact structure in a classical unimodular word system. We therefore pass to the Cohn generators

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad Y = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix},$$

and study the word-state map they generate. No pushdown theorem is used as a hypothesis in the results below; the link to the earlier machine is one of discovery and coordinate choice.

The central identity of the branch is

$$\boxed{Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s)},$$

which refines the classical sign/equality picture by giving an exact magnitude factorization through a Farey determinant residual. The zero set is then rigid on the word orbit, and the homogeneous completion kernel is transported to the fundamental quadratic-return translations.

Chapter 81

Cohn Word States and Farey Completion Coordinates

Research branch — From Farey Completion Arithmetic to Cohn–Markoff Geometry. *The preceding arithmetic Part extracted Farey recursion, mechanical coding, and universal spectra from shortest completion. The present Part carries that completion viewpoint into the classical Cohn–Markoff matrix monoid. Cohn matrices, Christoffel words, Markoff triples, and flip methods are classical background; the contribution developed here is the exact completion-coordinate layer, especially the quadratic adjacent-swap magnitude factorization, the completion-kernel/return correspondence, and the centered reversal-defect identities.*

81.1 The Cohn word-state system

Define

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad Y = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Both matrices are symmetric, positive and unimodular. Let

$$\mu : \{0,1\}^* \longrightarrow \mathrm{SL}_2(\mathbb{Z})$$

be the morphism with $\mu(0) = X$, $\mu(1) = Y$. We write words on the right, so

$$\mu(s0) = \mu(s)X, \qquad \mu(s1) = \mu(s)Y.$$

Let

$$v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad q(s) = \mu(s)^T v.$$

Because X and Y are symmetric,

$$q(s0) = Xq(s), \qquad q(s1) = Yq(s). \tag{81.1}$$

Every $q(s)$ is primitive because $X, Y \in \mathrm{SL}_2(\mathbb{Z})$ and v is primitive.

Lemma 81.1 (Exact terminal decoder)**.** *Every word state has slope in the invariant interval*

$$I = \left(1, \frac{5}{2}\right).$$

For a nonempty word, the last letter is determined by the slope:

$$q(s0): \quad \frac{3}{2} < \frac{q_1}{q_2} < \frac{12}{7},$$

$$q(s1): \quad \frac{7}{3} < \frac{q_1}{q_2} < \frac{29}{12}.$$

These intervals are disjoint. The corresponding inverse maps are

$$T_0(a,b) = (a-b,\, 2b-a), \qquad T_1(a,b) = (a-2b,\, 5b-2a).$$

Consequently the map $s \mapsto q(s)$ is injective.

Proof. The initial vector $v = (2,1)^T$ has slope $2 \in I$. If a positive vector has slope $r \in I$, then multiplication by X and Y gives slopes

$$\phi_0(r) = \frac{2r+1}{r+1}, \qquad \phi_1(r) = \frac{5r+2}{2r+1}.$$

Both maps are increasing on I, and their endpoint images are

$$\phi_0(I) = \left(\frac{3}{2}, \frac{12}{7}\right), \qquad \phi_1(I) = \left(\frac{7}{3}, \frac{29}{12}\right).$$

Thus I is invariant and the last letter is read from the unique terminal interval. Since

$$X^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad Y^{-1} = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix},$$

the displayed inverse maps recover the predecessor state. Iteration reaches the unique empty state v, proving injectivity. □

We use the scalar

$$F(s) = v^T q(s). \tag{81.2}$$

If $q(s) = (a,b)^T$, then

$$F(s) = 2a + b.$$

Lemma 81.2 (Reversal transpose)**.** *For every word s,*

$$\mu(s^R) = \mu(s)^T.$$

In particular, if $p = p^R$ is a palindrome, then $P = \mu(p)$ is symmetric.

Proof. Transpose reverses the order of a product, and both generators are symmetric. □

For a palindromic base p, write

$$P = \mu(p) = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \qquad m = F(p) = v^T P v, \tag{81.3}$$

and

$$x = q_1(p) = (Pv)_1 = 2A + B. \tag{81.4}$$

Then

$$Pv = \begin{pmatrix} x \\ m - 2x \end{pmatrix}.$$

Lemma 81.3 (Characteristic divisibility)**.** *With the notation above,*

$$x^2 + 1 = Am.$$

In particular

$$x^2 \equiv -1 \pmod{m}.$$

Proof. Since $\det P = AD - B^2 = 1$,

$$\begin{aligned} x^2 + 1 &= (2A + B)^2 + 1 \\ &= 4A^2 + 4AB + B^2 + AD - B^2 \\ &= A(4A + 4B + D) \\ &= Am. \end{aligned}$$

□

The congruence in theorem 81.3 is classical in the Markoff setting. We will use it only as an arithmetic identity attached to a palindromic base.

81.2 Farey completion coordinates

Define a new coordinate vector by

$$\Psi(s) = \begin{pmatrix} q_1(s) \\ F(s) \end{pmatrix}. \tag{81.5}$$

If $q(s) = (a, b)^T$, then

$$\Psi(s) = \begin{pmatrix} a \\ 2a + b \end{pmatrix} = Cq(s), \qquad C = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

The matrix C is unimodular, so $\Psi(s)$ is primitive whenever $q(s)$ is primitive.

Theorem 81.4 (Word-to-Farey coordinate conjugacy)**.** *For every word s, writing $\Psi(s) = (a, f)^T$, one has*

$$\Psi(s0) = \begin{pmatrix} f \\ 3f - a \end{pmatrix}, \tag{81.6}$$

and

$$\Psi(s1) = \begin{pmatrix} a + 2f \\ 2a + 5f \end{pmatrix}. \tag{81.7}$$

Equivalently, for $\rho(s) = a/f$,

$$\rho(s0) = \frac{1}{3 - \rho(s)}, \qquad \rho(s1) = \frac{\rho(s) + 2}{2\rho(s) + 5}. \tag{81.8}$$

Proof. Conjugating the two append maps in equation (81.1) by C gives

$$CXC^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}, \qquad CYC^{-1} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}.$$

Applying these matrices to $(a, f)^T$ gives equations (81.6) and (81.7); division of the two coordinates gives equation (81.8). □

Remark 81.5 (Relation with the $a = 0$ Cohn normalization). The two conjugated generators

$$\begin{pmatrix} 0 & 1 \\ -1 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

are exactly Aigner's $a = 0$ Cohn generators. Veselov [84] uses this normalization to identify Cohn-matrix indices with Markov fractions. Thus theorem 81.4 is recorded as the coordinate bridge needed for the quadratic results below, not as a new Cohn–Farey correspondence.

Fix from now on a palindromic base p and write

$$\Psi(p) = \begin{pmatrix} x \\ m \end{pmatrix}.$$

Definition 81.6 (Farey residual)**.** For any word s with $\Psi(s) = (a, f)^T$, define

$$R_p(s) = ma - xf. \tag{81.9}$$

Thus

$$R_p(s) = -\det(\Psi(p), \Psi(s)). \tag{81.10}$$

The sign of $R_p(s)$ records on which side of x/m the fraction a/f lies. In the original q-coordinates,

$$R_p(s) = (m - 2x)q_1(s) - xq_2(s). \tag{81.11}$$

This residual is the natural linear coordinate for the Diophantine equation

$$mu - xd = r,$$

whose homogeneous kernel is generated by (x, m). The next section shows that the same completion lattice appears canonically from the quadratic form defined by P.

81.3 Relative completion coordinates

For $u \in \mathbb{Z}^2$, put

$$Q_P(u) = u^T P u.$$

For a vector $q \in \mathbb{Z}^2$, define

$$\ell(q) = v^T P q, \qquad \delta(q) = \det(v, q). \tag{81.12}$$

Since $v = (2, 1)^T$, if $q = (q_1, q_2)^T$ then

$$\delta(q) = 2q_2 - q_1.$$

Theorem 81.7 (Relative completion coordinates)**.** *For every $q \in \mathbb{Z}^2$,*

$$mQ_P(q) = \ell(q)^2 + \delta(q)^2. \tag{81.13}$$

Moreover,

$$\ell(q) + x\delta(q) = mq_2, \tag{81.14}$$

and

$$\ell(q) - (m - x)\delta(q) = m(q_1 - q_2). \tag{81.15}$$

Consequently

$$\ell(q) \equiv -x\delta(q) \pmod{m}. \tag{81.16}$$

Proof. The two-dimensional Gram determinant identity gives

$$(v^T Pv)(q^T Pq) - (v^T Pq)^2 = \det(P)\det(v,q)^2.$$

Since $\det P = 1$, this is exactly equation (81.13).

Using $Pv = (x, m-2x)^T$ and $\delta(q) = 2q_2 - q_1$,

$$\begin{aligned}\ell(q) + x\delta(q) &= xq_1 + (m-2x)q_2 + x(2q_2 - q_1)\\ &= mq_2,\end{aligned}$$

which proves equation (81.14). Similarly,

$$\begin{aligned}\ell(q) - (m-x)\delta(q) &= xq_1 + (m-2x)q_2 - (m-x)(2q_2 - q_1)\\ &= m(q_1 - q_2).\end{aligned}$$

Reducing equation (81.14) modulo m gives equation (81.16). □

Corollary 81.8 (Exact completion lattice)**.** *The image of the integral map $q \mapsto (\ell(q), \delta(q))$ is exactly*

$$\Lambda_{m,x} = \{(\ell, d) \in \mathbb{Z}^2 : \ell + xd \equiv 0 \pmod m\}.$$

In particular, this map identifies $\mathbb{Z}^2$ with an index-m sublattice of $\mathbb{Z}^2$.

Proof. In standard coordinates the map has matrix

$$L = \begin{pmatrix} x & m-2x \\ -1 & 2 \end{pmatrix}, \qquad \det L = m.$$

By equation (81.16), its image is contained in $\Lambda_{m,x}$. The latter has index m, since it is the kernel of the surjective homomorphism $(\ell, d) \mapsto \ell + xd \pmod m$. The image of L also has index $|\det L| = m$, so the two lattices coincide. □

Remark 81.9 (Diophantine completion interpretation)*.* For $d \geq 0$, equation (81.14) reads

$$mu - xd = \ell, \qquad u = q_2.$$

For each fixed value of ℓ, the corresponding slice of $\Lambda_{m,x}$ is the integer solution set of this two-variable completion equation, whose homogeneous kernel is generated by (x, m).

The sign of δ also remembers the terminal branch.

Lemma 81.10 (Terminal sign)**.** *For every word s,*

$$\delta(q(s0)) = q_2(s) > 0, \qquad \delta(q(s1)) = -q_1(s) < 0.$$

Proof. Use equation (81.1) and compute

$$\delta(X(a,b)^T) = b, \qquad \delta(Y(a,b)^T) = -a.$$

All word states have positive coordinates. □

Proposition 81.11 (Palindromic sandwich identity)**.** *For every word t and every palindromic base p,*

$$\boxed{Q_P(q(t)) = F(tpt^R).} \tag{81.17}$$

Moreover,

$$F(w) = \mu(0w1)_{12} \tag{81.18}$$

for every word w. Hence the quadratic observable is also an ordinary Cohn matrix-entry statistic of a palindromic sandwich.

Proof. Since $p = p^R$, one has

$$\mu(tpt^R) = \mu(t)P\mu(t)^T,$$

which is symmetric. Therefore

$$F(tpt^R) = v^T\mu(tpt^R)^T v = v^T\mu(t)P\mu(t)^T v = Q_P(q(t)).$$

For equation (81.18), write $\mu(w) = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. A direct multiplication gives

$$\mu(0w1)_{12} = 4A + 2B + 2C + D = v^T\mu(w)^T v = F(w).$$

□

Chapter 82

Quadratic Swap Curvature and Completion Returns

82.1 Quadratic chronology and Farey curvature

We now compare the two opposite two-letter chronologies 10 and 01. If $q(s) = q$, then

$$q(s10) = XYq, \qquad q(s01) = YXq.$$

Theorem 82.1 (Exact adjacent-swap factorization)**.** *Let p be a palindromic base, $P = \mu(p)$, and define m, x as above. Then for every word s,*

$$\boxed{Q_P(q(s10)) - Q_P(q(s01)) = 12F(s)R_p(s).} \tag{82.1}$$

Equivalently,

$$Q_P(q(s10)) - Q_P(q(s01)) = -12F(s)\det(\Psi(p), \Psi(s)). \tag{82.2}$$

Proof. Write

$$P = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \qquad q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix}.$$

A direct multiplication gives

$$(XY)^T P(XY) - (YX)^T P(YX) = \begin{pmatrix} 48B + 24D & -24A + 6D \\ -24A + 6D & -24A - 12B \end{pmatrix}.$$

The associated quadratic polynomial factors as

$$\begin{aligned} q^T\big((XY)^T P(XY) - (YX)^T P(YX)\big)q \\ = 12(2q_1 + q_2)\big((2B + D)q_1 - (2A + B)q_2\big). \end{aligned}$$

Now $2q_1 + q_2 = F(s)$, $x = 2A + B$, and

$$2B + D = m - 2x.$$

Therefore the second factor is

$$(m - 2x)q_1 - xq_2 = mq_1 - xF(s) = R_p(s),$$

which proves equation (82.1). Equation equation (82.2) follows from equation (81.10). □

Corollary 82.2 (Word-level swap factorization)**.** *For every word s,*

$$F(s10p01s^R) - F(s01p10s^R) = 12F(s)R_p(s).$$

Equivalently, after the standard $0(\cdot)1$ wrapping, the same identity is a difference of upper-right Cohn matrix entries.

Proof. Apply theorem 81.11 with $t = s10$ and $t = s01$, noting that $(s10)^R = 01s^R$ and $(s01)^R = 10s^R$. The wrapped statement follows from equation (81.18). □

Remark 82.3 (Relation to flip methods). Lagisquet et al. [82] obtain sign and equality rules for simple and composite flips of a related matrix-word m-value. The flip idea itself is therefore classical background. The point of theorems 82.1 and 82.2 is the closed base-dependent factorization of the palindromic-sandwich defect into $12F(s)$ times a single determinant residual.

Corollary 82.4 (Farey sign of the swap)**.** *The sign of the quadratic chronology defect is the sign of*

$$\frac{q_1(s)}{F(s)} - \frac{x}{m}.$$

More precisely,

$$Q_P(q(s10)) > Q_P(q(s01)) \iff \frac{q_1(s)}{F(s)} > \frac{x}{m},$$

and the inequality reverses on the other side.

Proof. Since $F(s), m > 0$,

$$R_p(s) = mF(s)\left(\frac{q_1(s)}{F(s)} - \frac{x}{m}\right).$$

Apply theorem 82.1. □

The zero set is rigid because both Farey coordinate vectors are primitive.

Theorem 82.5 (Zero-curvature return rigidity)**.** *For every word s,*

$$\boxed{Q_P(q(s10)) = Q_P(q(s01)) \quad \iff \quad s = p.} \tag{82.3}$$

No palindromicity hypothesis on s is required.

Proof. By theorem 82.1, equality is equivalent to $R_p(s) = 0$, since $F(s) > 0$. Thus $\Psi(s)$ is rationally parallel to $\Psi(p)$. Both are primitive integer vectors with positive coordinates. Hence the rational proportionality factor is positive, and primitivity forces it to be 1. Therefore

$$\Psi(s) = \Psi(p).$$

Because C is unimodular, this gives $q(s) = q(p)$. By the exact decoder in theorem 81.1, equality of word states implies equality of the underlying words. Hence $q(s) = q(p)$ implies $s = p$. The converse is immediate from $R_p(p) = 0$. □

Example 82.6 (The base $p = 0$)**.** Take $p = 0$. Then

$$P = X, \qquad (x, m) = (5, 13), \qquad x^2 + 1 = 2m,$$

and

$$R_p(s) = 13q_1(s) - 5F(s).$$

The first few values illustrate the exact sign and zero laws:

s	$F(s)$	$R_p(s)$	$Q_P(q(s10)) - Q_P(q(s01))$
ε	5	1	60
0	13	0	0
1	29	11	3828
00	34	-1	-408

The unique zero in these rows occurs at the base progenitor $s = p = 0$, as theorem 82.5 predicts.

Remark 82.7 (Local, not global, Markoff rigidity). Theorem 82.5 is a same-progenitor statement. It does not assert that two arbitrary distinct Christoffel words with the same Markoff number have a common progenitor to which the theorem can be applied. Thus it is independent of, and does not prove, the Frobenius uniqueness conjecture.

82.2 Kernel lifts and quadratic returns

The preceding two coordinate systems are related more tightly than the common residual notation suggests. We first express the relative quadratic coordinates after the two terminal chronologies directly in terms of $\Psi(s)$.

Theorem 82.8 (Terminal transport formula)**.** *Let*

$$\Psi(s) = \begin{pmatrix} a \\ f \end{pmatrix}, \qquad R = ma - xf.$$

Then

$$(\ell, \delta)(q(s10)) = (3mf + R,\ f), \tag{82.4}$$

and

$$(\ell, \delta)(q(s01)) = (3mf - R,\ -f). \tag{82.5}$$

Proof. Write $q(s) = (a, b)^T$, so $f = 2a + b$. Directly from XYq and YXq one obtains

$$\delta(XYq) = f, \qquad \delta(YXq) = -f.$$

For the first coordinates, substitute $P = \left(\begin{smallmatrix} A & B \\ B & D \end{smallmatrix}\right)$, $m = 4A + 4B + D$ and $x = 2A + B$ into $\ell = v^T P(\cdot)$; simplification gives

$$\ell(XYq) = 3mf + (ma - xf),$$

$$\ell(YXq) = 3mf - (ma - xf).$$

These are equations (82.4) and (82.5). □

Now consider the Diophantine completion equation

$$ma - xf = R. \tag{82.6}$$

Its integer solution set is invariant under the primitive kernel translation

$$(a, f) \longmapsto (a + x, f + m). \tag{82.7}$$

Theorem 82.9 (Kernel lift equals quadratic-return translation)**.** *Fix a residual R. Under the terminal* 10 *transport, the kernel lift equation* (82.7) *induces*

$$(\ell, \delta) \longmapsto (\ell, \delta) + (3m^2, m). \tag{82.8}$$

Under the terminal 01 *transport it induces*

$$(\ell, \delta) \longmapsto (\ell, \delta) + (3m^2, -m). \tag{82.9}$$

Proof. The kernel lift leaves R invariant:

$$m(a + x) - x(f + m) = ma - xf = R.$$

Apply theorem 82.8. On the 10 branch,

$$(3m(f + m) + R, f + m) - (3mf + R, f) = (3m^2, m).$$

On the 01 branch,

$$(3m(f + m) - R, -f - m) - (3mf - R, -f) = (3m^2, -m).$$

□

Remark 82.10 (Ambient lattice versus word orbit). The kernel translate $(a, f) \mapsto (a + x, f + m)$ is an operation on the ambient Diophantine completion lattice. A translated lattice point need not itself belong to the Cohn word orbit. The theorem is therefore an exact lattice-transport identity; theorem 82.11 records the distinguished return points that do arise from the base word.

The base state itself has $\Psi(p) = (x, m)$ and $R_p(p) = 0$; in particular, $\Psi(p)$ is the primitive homogeneous kernel vector. Applying the two terminal transports to this distinguished kernel point gives the fundamental blocks $p10$ and $p01$.

Corollary 82.11 (Fundamental quadratic returns)**.** *For the two blocks*

$$C_1 = p10, \qquad C_0 = p01,$$

one has

$$(\ell, \delta)(q(C_1)) = (3m^2, m), \qquad (\ell, \delta)(q(C_0)) = (3m^2, -m). \tag{82.10}$$

Consequently

$$m \mid \ell(q(C_i)), \qquad m \mid \delta(q(C_i)), \qquad i = 0, 1.$$

Proof. Set $(a, f) = (x, m)$ and $R = 0$ in theorem 82.8. □

This is the promised kernel/return correspondence: a homogeneous translation in the completion lattice becomes a fixed Euclidean translation in the relative quadratic plane.

Chapter 83

Characteristic Parents, Centered Fibers, and Reversal Defects

83.1 Characteristic roots and Farey parents

The characteristic congruence in theorem 81.3 determines the oriented Farey parents of the pair (m, x) explicitly.

Theorem 83.1 (Characteristic root determines the oriented Farey parents)**.** *Assume*

$$m > x > 0, \qquad \gcd(m, x) = 1, \qquad x^2 + 1 = km.$$

Then

$$\boxed{(c, u) = (x, k), \qquad (v, d) = (m - x, x - k)} \tag{83.1}$$

satisfy

$$(m, x) = (c, u) + (v, d),$$

and the oriented unit-determinant relations

$$mu - xc = 1, \qquad xv - md = 1.$$

Moreover, they are the unique such decomposition with $0 < c < m$ and $0 < v < m$. (The boundary case $(m, x) = (2, 1)$ has $d = 0$; otherwise the displayed parent coordinates are positive.) Hence the two unit-residual lifts are

$$(k, x), \qquad (-(x - k), -(m - x)). \tag{83.2}$$

Proof. The characteristic identity gives $x^2 - km = -1$. Therefore the displayed first pair satisfies

$$mk - x \cdot x = 1.$$

For the second pair,

$$x(m - x) - m(x - k) = mk - x^2 = 1,$$

and their sum is (m, x). Thus the displayed pairs have the required oriented unit-determinant relations.

For uniqueness, any normalized first parent satisfies $xc \equiv -1 \pmod{m}$. Since $x^2 \equiv -1 \pmod{m}$ and x is invertible modulo m, one gets $c \equiv x \pmod{m}$. The condition $0 < c < m$ forces $c = x$, and then $u = k$. Similarly $xv \equiv 1 \pmod{m}$ gives $v \equiv -x \equiv m - x \pmod{m}$; the condition $0 < v < m$ forces $v = m - x$, and then $d = x - k$. Finally, $k = (x^2+1)/m \leq x$, with equality only for $(m, x) = (2, 1)$, which proves the stated positivity qualification. □

Remark 83.2. The arithmetic congruence $x^2 \equiv -1 \pmod{m}$ and the classical Farey/Cohn correspondence are part of the established Markoff literature. The role of theorem 83.1 here is to identify the characteristic data with the same completion kernel used in theorem 82.9, so that the local Farey and quadratic-return descriptions share one coordinate system.

83.2 Centered fibers and reversal defects

The completion viewpoint also gives a compact reconstruction statement on a fixed F-fiber. This part does not require the base word to be palindromic.

Theorem 83.3 (Centered-fiber reconstruction)**.** *Let s be any word with $F(s) = m$, and write*

$$\mu(s) = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

$$r = q_1(s) = 2A + C, \qquad r' = q_1(s^R) = 2A + B.$$

Then

$$rr' + 1 = Am. \tag{83.3}$$

Thus, with

$$K = \frac{rr' + 1}{m},$$

the full matrix is reconstructed from (m, r, r') as

$$\boxed{\mu(s) = \begin{pmatrix} K & r' - 2K \\ r - 2K & m - 2r - 2r' + 4K \end{pmatrix}.} \tag{83.4}$$

Proof. Since $F(s) = v^T \mu(s)^T v$,

$$m = 4A + 2B + 2C + D.$$

Also $\det \mu(s) = AD - BC = 1$. Hence

$$\begin{aligned} rr' + 1 &= (2A + C)(2A + B) + 1 \\ &= 4A^2 + 2AB + 2AC + BC + AD - BC \\ &= A(4A + 2B + 2C + D) \\ &= Am. \end{aligned}$$

Thus $A = K$. The identities $r = 2A + C$ and $r' = 2A + B$ give $C = r - 2K$ and $B = r' - 2K$, and the formula for D follows from the equation for m. □

Theorem 83.4 (Wrapped trace defect equals reversal defect)**.** *Under the hypotheses of theorem 83.3, let*

$$H = \mu(0s1) = X\mu(s)Y.$$

Then

$$H_{12} = F(s) = m, \tag{83.5}$$

and

$$\boxed{\operatorname{Tr}(H) - 3H_{12} = r - r'.} \tag{83.6}$$

Consequently

$$\boxed{\operatorname{Tr}(\mu(0s1)) = 3\mu(0s1)_{12} \quad \Longleftrightarrow \quad s = s^R.} \tag{83.7}$$

Proof. A direct multiplication of $X\mu(s)Y$ gives $H_{12} = 4A + 2B + 2C + D = m$ and

$$\mathrm{Tr}(H) - 3H_{12} = C - B.$$

But

$$C - B = (r - 2A) - (r' - 2A) = r - r'.$$

Thus the trace identity is equivalent to $r = r'$. By theorem 83.3, $r = r'$ is equivalent to $B = C$, hence to symmetry of $\mu(s)$. By theorem 81.2,

$$\mu(s)^T = \mu(s^R).$$

Hence $\mu(s) = \mu(s^R)$, so in particular $q(s) = q(s^R)$. Injectivity of the word-state map from theorem 81.1 gives $s = s^R$. □

83.3 Relation to the classical Markoff setting

When p is a central palindrome, the word $0p1$ is the corresponding lower Christoffel word, and the matrices above lie in the standard Cohn–Markoff parametrization initiated in Cohn's work on Markoff forms and primitive words [80, 81]. Modern accounts identify Christoffel/Farey data with Markoff triples and study the resulting order and uniqueness questions [79, 83, 82]. The results of this chapter add a local completion layer to that picture.

There are three distinct structures:

(a) the classical tree coordinate, which locates a Christoffel/Cohn object by a Farey fraction;

(b) the completion residual $R_p(s)$, which measures the determinant displacement of an arbitrary word state from a fixed palindromic base;

(c) the quadratic chronology defect, which compares the two opposite terminal orders 10 and 01.

Theorem 82.1 identifies (b) and (c) exactly. In particular, the swap energy has no additional nonlinear zero set: on the word orbit, zero curvature is exactly the base state itself. Theorem 82.9 then identifies the completion kernel with the quadratic return lattice.

This local rigidity should not be confused with global Markoff injectivity. In a global collision, the two competing branches need not have the same progenitor. The result proved here is deliberately local: once a common progenitor is fixed, the opposite chronology is completely controlled by a single Farey determinant; no global core-identification theorem is claimed.

83.4 Return to the monograph's completion language

The Cohn–Markoff results close an arithmetic loop rather than introduce a new automata hypothesis. In the rational-balance chapters, a residual chose a canonical completion modulo a primitive kernel. Here a determinant residual again measures displacement from a base, relative coordinates again expose a completion lattice, and terminal chronology transports the homogeneous kernel into explicit opposite return vectors. The shared theme is therefore not the particular monoid but the separation of a canonical completion section from its unimodular translation freedom. With the arithmetic consequence branches closed, Part VII returns to fixed semantics and asks what chronology, topology, and physical transport can still vary inside a completion fiber.

Part VII

Chronology, Topology, and Higher-Rank Completion Structure

Part guide. **Question.** After the semantic quotient is fixed, what structure can still remain in ordering, optimal fibers, physical transport, and higher rank? **Main result.** Random chronology has distinct LIFO scales; optimal fibers admit lattice-geometric classification; physical holonomy separates topological information from carrier and repair cost; higher-rank realization recovers broad topological freedom. **Why here.** The Part measures structure invisible to the semantic quotient itself. **Dependency.** It builds on completion fibers and semantic/physical separation, while remaining distinct from the resource lower bounds of Part VIII.

Chapter 84

Fixed-Parikh Random Ordering and Direct Obstruction

The centered real-time $p{:}q$ normalizer was developed earlier as a deterministic physical lift of shortest-completion semantics. We now keep that semantic content fixed. For coprime p, q, write $m = p + q$ and sample uniformly from

$$\mathcal{F}_{gq,gp} = \{w \in \{0,1\}^{mg} : N_0(w) = gq,\ N_1(w) = gp\}.$$

Every word in this fiber has the same zero endpoint residual and the same endpoint completion demand. Only chronology varies. The resulting experiment therefore isolates costs created by LIFO order rather than by semantic variation. We restate below only the shortest-demand and transition data needed for the quantitative analysis; the original 2:1 provenance figure and the full construction have already appeared in earlier Parts.

84.1 Shortest completion demand

For $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r)$$

be the unique nonnegative solution of

$$px - qy = r$$

that minimizes $x + y$. All other nonnegative solutions are obtained by adding copies of the primitive zero-weight kernel

$$K = (q, p).$$

We use the canonical stack ordering

$$\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases} \tag{84.1}$$

The leftmost symbol is the visible stack top.

Let

$$h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad D_{-h}, \ldots, D_h$$

be the centered debt states. For a logical stack word S define

$$W(S) = pN_0(S) - qN_1(S).$$

The state–stack residual invariant is

$$\rho = W(S) + dm \qquad \text{at configuration } (D_d, S). \tag{84.2}$$

The semantic projection reduces the physical stack plus virtual debt by complete kernel packets and returns exactly $\mu(\rho)$.

84.2 Transition classes

Suppose the current input is $a \in \{0,1\}$. If $S \neq \varnothing$ and a equals the visible top, the machine performs a *MATCH*: it pops one cell and leaves d unchanged. If the visible top is 0 and $a = 1$, a mismatch increments debt while $d < h$; this is *SEARCH*. At $d = h$ the next such mismatch is an *outer WRITE*: the visible zero is removed, debt is reset to zero, and

$$U_+ = \eta(\mu((h+1)m))$$

is inserted before the surviving suffix. The negative rule is symmetric: input 0 over visible top 1 decrements debt while $d > -h$, and at $d = -h$ an outer WRITE inserts

$$U_- = \eta(\mu(-(h+1)m)).$$

If the logical stack is empty, the next symbol triggers a *bottom WRITE*. For every $|d| \le h$,

$$(D_d, \varnothing) \xrightarrow{1} \big(D_0, \eta(\mu(dm + q))\big), \tag{84.3}$$

$$(D_d, \varnothing) \xrightarrow{0} \big(D_0, \eta(\mu(dm - p))\big). \tag{84.4}$$

Every WRITE therefore returns the physical stack to the exact canonical shortest-demand section for the new residual.

The construction has the following properties, all consequences of shortest-demand minimality and the centered debt bound.

Lemma 84.1 (WRITE frontier bound)**.** *Every canonical word created by a bottom or outer WRITE has shortest-demand vector (x, y) satisfying*

$$\min(x, y) \le h.$$

For the outer-WRITE vectors $\mu(\pm(h+1)m)$, the minority coordinate is at most h while the majority coordinate is at least $h+1$; in the positive case the minority one-count is $< p$, and in the negative case the minority zero-count is $< q$.

Proof. For a bottom WRITE the target residual has the form

$$r = jm + q, \qquad -h - 1 \le j \le h,$$

because $dm - p = (d-1)m + q$. Write $\mu(r) = (x, y)$ and $\ell = x + y$. Modulo m, the relation $px - qy = r \equiv q$ and $p \equiv -q \pmod m$ give

$$-q\ell \equiv q \pmod m.$$

Since $\gcd(q, m) = 1$, $\ell \equiv -1 \pmod m$, so $\ell = km - 1$. If both $x, y \ge h + 1$, then $k \ge 2$. The minimum demand is kernel-free, hence either $x < q$ or $y < p$. In the first case $q \ge h + 2$ and

$$r = m(x - qk) + q$$

forces $j \le -q - 1 \le -h - 3$, a contradiction. In the second case $p \ge h + 2$ and

$$j = pk - y - 1 \ge p > h,$$

again a contradiction. Thus $\min(x, y) \le h$.

For the positive outer WRITE, put $c = h + 1$. The minimum solution is

$$\mu(cm) = (qk + c, pk - c), \qquad k = \left\lceil \frac{c}{p} \right\rceil .$$

Its second coordinate lies in $[0, h]$, is strictly less than p, and its first is at least $c = h + 1$. The negative case is symmetric, with the zero-coordinate strictly less than q. □

Lemma 84.2 (Boundary guard)**.** *Suppose the current stack is a suffix of a canonical word whose minority count is at most h.*

(a) If the current state is D_h and the stack begins with 0, then an outward $1/0$ mismatch can occur only when the surviving stack is unary zero.

(b) If the current state is D_{-h} and the stack begins with 1, then an outward $0/1$ mismatch can occur only when the surviving stack is unary one.

Proof. Consider the positive case. Since the most recent WRITE, debt started at zero. Reaching D_h requires at least h previous positive mismatches, each of which popped a zero from the pre-existing canonical word. If the current visible zero were still in the mixed alternating part and a one survived later in the stack, then the h previously popped zeros would each have a distinct preceding one in the alternating part. Together with the current zero and a later surviving one, the original canonical word would contain at least $h + 1$ zeros and at least $h + 1$ ones, contradicting Lemma 84.1. Hence the surviving suffix is unary zero. The negative case is symmetric. □

Lemma 84.3 (Canonical-segment property)**.** *Immediately after every WRITE the machine is in state D_0 with stack $\eta(\mu(\rho))$, whose minority count is at most h. Until the next WRITE, every transition removes exactly one cell from that pre-existing canonical word. Hence every reachable logical stack is a suffix of the canonical word produced by the most recent WRITE.*

Proof. A bottom WRITE has the displayed canonical form by definition and satisfies Lemma 84.1. Between WRITEs, MATCH and SEARCH only remove the visible pre-existing cell. At an outer WRITE, Lemma 84.2 shows that the surviving suffix is unary. The newly inserted canonical word and that unary suffix concatenate to the canonical ordering of the new shortest demand; kernel-freeness follows from the minority bound in Lemma 84.1. Induction over WRITE episodes proves the claim. □

Lemma 84.4 (Zero-residual correctness)**.** *At the end of every processed binary prefix,*

$$\rho = 0 \quad \Longleftrightarrow \quad (D_d, S) = (D_0, \varnothing).$$

Proof. The reverse implication follows from (84.2). For the forward implication, consider the most recent WRITE. Immediately after it the stack was $\eta(\mu(r_0))$ for some residual r_0. Until the current time no further WRITE has occurred, so every intervening symbol has popped exactly one of those cells. If the current residual is zero, that intervening input is a completion of r_0. By minimality of $\mu(r_0)$, no completion can use fewer symbols than the length of the written stack. Since no more than that many cells can have been popped, equality is forced: the entire stack has been consumed. Equation (84.2) then gives $dm = 0$, hence $d = 0$. □

Operational observables. For a word w, let $S(w), O(w), B(w)$ be the numbers of SEARCH, outer-WRITE, and bottom-WRITE transitions. The primary LIFO-obstruction count is

$$R(w) = S(w) + O(w).$$

A bottom WRITE is not counted as an obstruction: it occurs when the physical stack is already empty and represents rematerialization rather than LIFO search.

84.3 The fixed-Parikh model as a critical bridge

For arbitrary $a, b \geq 0$ define the fixed-Parikh fiber

$$\mathcal{F}_{a,b} = \{w \in \{0,1\}^{a+b} : N_0(w) = a,\ N_1(w) = b\}.$$

Our balanced model is

$$W_g \sim \mathrm{Unif}(\mathcal{F}_{gq,gp}), \qquad n = mg.$$

Introduce i.i.d. symbols X_i under the critical law

$$\mathbb{P}(X_i = 1) = \frac{p}{m}, \qquad \mathbb{P}(X_i = 0) = \frac{q}{m},$$

and increments

$$\xi_i = \begin{cases} q, & X_i = 1, \\ -p, & X_i = 0. \end{cases}$$

Then

$$\mathbb{E}\xi_i = 0, \qquad \mathrm{Var}(\xi_i) = pq.$$

Write $\rho_t = \sum_{i \leq t} \xi_i$.

Lemma 84.5 (Canonical critical-bridge representation)**.** *For $n = mg$,*

$$\mathcal{L}(X_1, \ldots, X_n \mid \rho_n = 0) = \mathrm{Unif}(\mathcal{F}_{gq,gp}).$$

Proof. The condition $\rho_n = 0$ gives

$$mN_1(n) - pn = 0,$$

so $N_1(n) = pg$ and $N_0(n) = qg$. Every word with these counts has the same i.i.d. probability

$$\left(\frac{p}{m}\right)^{pg} \left(\frac{q}{m}\right)^{qg}.$$

Conditioning therefore makes the fiber uniform. □

At block boundaries set

$$Z_j = \frac{\rho_{mj}}{m}.$$

If K_j is the number of ones in the next m symbols, then under the critical i.i.d. law

$$Z_{j+1} - Z_j = K_j - p, \qquad K_j \sim \mathrm{Bin}\Big(m, \frac{p}{m}\Big). \tag{84.5}$$

Hence

$$\mathbb{E}(Z_{j+1} - Z_j) = 0, \qquad \mathrm{Var}(Z_{j+1} - Z_j) = \frac{pq}{m}. \tag{84.6}$$

The block skeleton is therefore a centered, irreducible, aperiodic, finite-range random walk on $\mathbb{Z}$.

84.4 Direct realization and the first LIFO obstruction

Call a word *direct* if no SEARCH or outer-WRITE transition occurs, equivalently if $R(w) = 0$.

84.4.1 Primitive direct blocks

The shortest demands after the first symbol are

$$\mu(-p) = (q-1, p), \qquad \mu(q) = (q, p-1).$$

Define

$$B_0 = 0\,\eta(q-1,p), \qquad B_1 = 1\,\eta(q,p-1). \tag{84.7}$$

Both words have length m and Parikh vector (q, p).

Theorem 84.6 (Direct-realization factorization)**.** *A balanced word $w \in \mathcal{F}_{gq,gp}$ satisfies $R(w) = 0$ if and only if*

$$w = B_{\varepsilon_1} B_{\varepsilon_2} \cdots B_{\varepsilon_g}$$

for a unique $(\varepsilon_1, \ldots, \varepsilon_g) \in \{0,1\}^g$. Consequently

$$\#\{w \in \mathcal{F}_{gq,gp} : R(w) = 0\} = 2^g \tag{84.8}$$

and

$$\mathbb{P}(R(W_g) = 0) = \frac{2^g}{\binom{mg}{qg}}. \tag{84.9}$$

Proof. Start from empty stack and zero debt. The first input symbol triggers a bottom WRITE. If it is 0, the newly written stack is $\eta(\mu(-p))$; if it is 1, the stack is $\eta(\mu(q))$. If no LIFO obstruction occurs thereafter, every subsequent input until the stack empties must match the visible top. Thus the entire first episode is forced to be B_0 or B_1. Each block has length m, consumes exactly (q, p) symbols, and returns the machine to $(D_0, \varnothing)$. Iteration gives the factorization. Conversely, each displayed block is realized by one bottom WRITE followed only by MATCH transitions, so any concatenation is direct. Uniqueness follows from the first symbol of each block. □

By Stirling's formula,

$$\mathbb{P}(R(W_g) = 0) \sim \sqrt{\frac{2\pi pqg}{m}} \left(\frac{2p^p q^q}{m^m}\right)^g. \tag{84.10}$$

The base

$$\lambda_{p,q} = \frac{2p^p q^q}{m^m}$$

is strictly less than one.

84.4.2 Exact first-obstruction law

Let

$$\tau_g = \min\{t \geq 1 : \text{the transition at time } t \text{ is SEARCH or outer WRITE}\},$$

with $\tau_g = mg + 1$ if no obstruction occurs. For $\varepsilon \in \{0,1\}$ and $0 \leq s < m$, write

$$z_{\varepsilon,s} = N_0(\mathrm{pref}_s(B_\varepsilon)).$$

Theorem 84.7 (Exact first LIFO-obstruction survival law)**.** *For* $0 \le k \le g$,

$$\mathbb{P}(\tau_g > km) = \frac{2^k \binom{m(g-k)}{q(g-k)}}{\binom{mg}{qg}}. \tag{84.11}$$

For $0 \le k < g$ *and* $1 \le s < m$,

$$\mathbb{P}(\tau_g > km + s) = \frac{2^k \sum_{\varepsilon=0}^{1} \binom{m(g-k)-s}{q(g-k)-z_{\varepsilon,s}}}{\binom{mg}{qg}}. \tag{84.12}$$

Proof. The event $\{\tau_g > km\}$ is exactly the event that the first k primitive blocks are direct. By Theorem 84.6, there are 2^k possible direct prefixes, each consuming (kq, kp) symbols. The remaining suffix is an arbitrary balanced fixed-Parikh word of size $g-k$, giving (84.11). For a partial block, after the first k complete direct blocks the next s symbols must equal $\mathrm{pref}_s(B_0)$ or $\mathrm{pref}_s(B_1)$. These two events are disjoint because B_0 and B_1 begin with different symbols. After fixing such a prefix, the remaining number of zeros is $q(g-k) - z_{\varepsilon,s}$, which gives (84.12). □

Define

$$R_{p,q}(0) = 1,$$

and, for $1 \le s < m$,

$$R_{p,q}(s) = \sum_{\varepsilon=0}^{1} \left(\frac{q}{m}\right)^{z_{\varepsilon,s}} \left(\frac{p}{m}\right)^{s-z_{\varepsilon,s}}.$$

Taking $g \to \infty$ with k, s fixed in Theorem 84.7 gives

$$\mathbb{P}(\tau_g > km + s) \longrightarrow \lambda_{p,q}^k R_{p,q}(s). \tag{84.13}$$

In particular,

$$\boxed{\tau_g = O_{\mathbb{P}}(1).} \tag{84.14}$$

84.4.3 Primitive balanced SEARCH distribution

At $g = 1$ no outer WRITE is required before return, and SEARCH occurs in mismatch pairs.

Proposition 84.8 (Primitive SEARCH law)**.** *On* $\mathcal{F}_{q,p}$,

$$S(W_1) \in \{0, 2, 4, \ldots, 2\min(p, q)\},$$

and

$$\#\{w \in \mathcal{F}_{q,p} : S(w) = 2j\} = \binom{p}{j}\binom{q-1}{j} + \binom{q}{j}\binom{p-1}{j}. \tag{84.15}$$

Consequently

$$\mathbb{E}S(W_1) = \frac{2pq(m-2)}{m(m-1)}. \tag{84.16}$$

Proof. Condition on the first symbol. If it is 0, the bottom WRITE creates the deterministic target word $\eta(q-1, p)$, with $q-1$ zero-cells and p one-cells. The remaining input suffix has exactly the same Parikh vector. During this primitive episode no outer WRITE can occur: the debt at any prefix is the difference between the number of input ones and target ones seen so far, whose absolute value is at most $\min(p, q-1) \le h$. Thus

every step pops one target cell. If exactly j target one-cells are hit by input zero, then conservation of the two symbol counts forces exactly j target zero-cells to be hit by input one, giving $2j$ SEARCH mismatches. The number of such suffixes is

$$\binom{p}{j}\binom{q-1}{j}.$$

If the first symbol is 1, the same argument with target $\eta(q, p-1)$ gives

$$\binom{q}{j}\binom{p-1}{j}.$$

Adding proves (84.15). For the mean, use

$$\sum_{j} j\binom{a}{j}\binom{b}{j} = \frac{ab}{a+b}\binom{a+b}{a}$$

in the two terms and divide by $\binom{m}{p}$. □

Chapter 85

Physical-Lift Excursions and Rayleigh Rematerialization

The first obstruction is local, but rematerialization of the canonical shortest-demand section is diffusive. The block-boundary lift chain below records precisely that phenomenon.

85.1 The block-boundary physical-lift chain

We now pass from exact finite obstruction counts to the recurrent physical structure that governs rematerialization.

Lemma 85.1 (Empty stack only at block boundaries)**.** *If the logical stack is empty after t processed symbols, then $m \mid t$.*

Proof. If $S_t = \varnothing$, the residual invariant (84.2) gives $\rho_t = dm$ for some integer d. On the other hand,

$$\rho_t = mN_1(t) - pt.$$

Hence $m \mid pt$. Since $\gcd(p, m) = \gcd(p, q) = 1$, we obtain $m \mid t$. □

Let X_j be the full physical configuration after jm symbols under the critical i.i.d. input law, and let

$$\pi(X_j) = Z_j = \rho_{jm}/m.$$

Let

$$o = (D_0, \varnothing).$$

By Lemma 84.4, $X_j = o$ exactly when $Z_j = 0$. Since the scalar block walk Z_j is recurrent, so is the reachable class of the physical block chain.

Let

$$T_o^+ = \min\{j \geq 1 : X_j = o\}.$$

Definition 85.2 (Physical-lift invariant measure)**.** For a reachable block-boundary physical configuration x, define

$$\nu(x) = \mathbb{E}_o \left[\sum_{j=0}^{T_o^+ - 1} \mathbf{1}_{\{X_j = x\}} \right]. \tag{85.1}$$

The standard excursion construction for an irreducible recurrent Markov chain makes ν an invariant measure normalized by $\nu(o) = 1$; see, for example, [46, Ch. 1].

Theorem 85.3 (Unit semantic-shell mass)**.** *For every $z \in \mathbb{Z}$,*

$$\boxed{\sum_{x:\pi(x)=z} \nu(x) = 1.} \tag{85.2}$$

Proof. Push ν forward under the factor map π. The result is an invariant measure for the irreducible recurrent random walk Z_j. Such an invariant measure is unique up to a multiplicative constant. Because the block walk is translation-invariant, counting measure on $\mathbb{Z}$ is invariant. At shell zero, Lemma 84.4 shows that the only physical state is o, so the pushed-forward mass equals $\nu(o) = 1$. The normalization therefore fixes the pushed-forward measure to be counting measure, proving (85.2). □

This is the quantitative form of the semantic/physical distinction: the residual quotient assigns one unit of invariant mass to each shell, while physical chronology determines how that unit is split among its lifts.

Definition 85.4 (Physical rematerialization constant)**.** For $-h \le d \le h$, let

$$E_d = (D_d, \varnothing), \qquad \theta_d^{p,q} = \nu(E_d),$$

and define

$$\boxed{\kappa_{p,q} = \sum_{d=-h}^{h} \theta_d^{p,q}.} \tag{85.3}$$

Because E_d belongs to semantic shell d, Theorem 85.3 gives

$$0 \le \theta_d^{p,q} \le 1, \qquad \theta_0^{p,q} = 1, \qquad 1 \le \kappa_{p,q} \le 2h + 1. \tag{85.4}$$

Let $\mathcal{E} = \{E_d : -h \le d \le h\}$ and consider the trace chain obtained by observing X_j only when it enters $\mathcal{E}$. Restriction of an invariant measure to a recurrent trace set is invariant for the trace chain. Therefore its stationary distribution is

$$\pi_d^{\mathcal{E}} = \frac{\theta_d^{p,q}}{\kappa_{p,q}},$$

and in particular

$$\boxed{\pi_0^{\mathcal{E}} = \frac{1}{\kappa_{p,q}}.} \tag{85.5}$$

85.2 Rayleigh law for Bottom-WRITE rematerialization

The fixed-Parikh bridge can return to physical empty states only at block boundaries by Lemma 85.1. This allows an excursion decomposition at the block level.

A *primitive semantic excursion* is a block path beginning at o, avoiding o at positive intermediate block times, and ending at its first return to o. For such an excursion e, let

$$L(e) = \text{its length in blocks}$$

and let

$$R_B(e) = \#\{\text{empty-stack block states visited in } [0, L(e))\}.$$

Every such visit is followed, unless it is the terminal boundary of the full word, by exactly one bottom WRITE. Concatenating primitive semantic excursions therefore adds the rewards R_B exactly.

Lemma 85.5 (Excursion reward mean and exponential moments)**.** *Under the critical i.i.d. law,*

$$\mathbb{E}R_B(e) = \kappa_{p,q}. \tag{85.6}$$

Moreover there exists $s > 1$ such that

$$\mathbb{E}s^{R_B(e)} < \infty. \tag{85.7}$$

Proof. Equation (85.6) is the excursion formula (85.1) summed over the finite empty-state set $\mathcal{E}$. For (85.7), consider the finite trace chain on reachable empty states. From every such state there is a positive-probability finite input continuation that completes its residual and reaches E_0. Finiteness gives constants $L < \infty$ and $\varepsilon > 0$ such that from every trace state the probability of hitting E_0 within at most L trace transitions is at least ε. Hence the trace hitting time has a geometric tail, which implies an exponential moment for $R_B(e)$. □

Let

$$b_g = \mathbb{P}(Z_g = 0) = \binom{mg}{pg} \left(\frac{p}{m}\right)^{pg} \left(\frac{q}{m}\right)^{qg}.$$

Define the bridge return generating function

$$G(z,1) = \sum_{g \geq 0} b_g z^g.$$

Let $F(z,u)$ be the probability generating function of primitive semantic excursions, with z marking block length and u marking R_B. Unique decomposition into primitive excursions gives

$$G(z,u) = \frac{1}{1 - F(z,u)}. \tag{85.8}$$

At $u = 1$, Stirling's formula yields

$$b_g \sim \sqrt{\frac{m}{2\pi pq}}\, g^{-1/2}. \tag{85.9}$$

A standard Abelian/Tauberian transfer therefore gives

$$G(z,1) \sim \sqrt{\frac{m}{2pq}}(1-z)^{-1/2} \qquad (z \uparrow 1), \tag{85.10}$$

and hence

$$1 - F(z,1) \sim \sqrt{\frac{2pq}{m}}\sqrt{1-z}. \tag{85.11}$$

The same square-root singularity is a standard source of Rayleigh local-time laws for zero-drift lattice-path bridges; compare [38]. We use only the elementary regular-variation consequences of this square-root behavior below. Here the reward is not scalar zero-local-time itself but the physical empty-lift reward carried by each primitive semantic excursion.

Theorem 85.6 (Bottom-WRITE Rayleigh law)**.** *Let $B_g = B(W_g)$. Then*

$$\boxed{\frac{B_g}{\sqrt{mg}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right).} \tag{85.12}$$

Equivalently, for every fixed integer $r \geq 1$,

$$\mathbb{E}\left[\left(\frac{B_g}{\sqrt{mg}}\right)^r\right] \longrightarrow \left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right)^r 2^{r/2}\Gamma\Big(1 + \frac{r}{2}\Big). \tag{85.13}$$

Proof. Write

$$H_j(z) = \partial_u^j F(z,u)\big|_{u=1}.$$

By Lemma 85.5, $H_j(1) < \infty$ for every fixed j, and

$$H_1(1) = \kappa_{p,q}.$$

Differentiating (85.8) r times in u, the term with r factors of H_1 is

$$r! H_1(z)^r G(z,1)^{r+1}. \tag{85.14}$$

Every other Faà di Bruno term contains at most $r-1$ derivative factors of F and hence at most $G(z,1)^r$. More precisely, each such term is a product of some $H_{j_1}(z), \dots, H_{j_k}(z)$ with $k \le r-1$ and a factor $G(z,1)^{k+1}$. Since every H_j has summable nonnegative coefficients by Lemma 85.5, the convolution theorem for summable sequences against regularly varying coefficients shows that its coefficient is $O(g^{(k-1)/2}) = o(g^{(r-1)/2})$. Thus only (85.14) contributes at leading order.

Using (85.10), Karamata coefficient asymptotics, and the same standard convolution lemma (equivalently, the corresponding singularity-transfer calculation; cf. [40]),

$$[z^g]\partial_u^r G(z,1) \sim \frac{r!\kappa_{p,q}^r}{\Gamma((r+1)/2)} \left(\frac{m}{2pq}\right)^{(r+1)/2} g^{(r-1)/2}.$$

Dividing by b_g in (85.9) gives the conditional factorial moment

$$\mathbb{E}[(B_g)_r] \sim \kappa_{p,q}^r \left(\frac{m}{pq}\right)^{r/2} 2^{r/2}\Gamma\Big(1+\frac{r}{2}\Big)\, g^{r/2},$$

where the duplication formula for the Gamma function was used. Ordinary moments have the same leading term. These are exactly the moments of a Rayleigh variable with scale $\kappa_{p,q}\sqrt{m/(pq)}$ for $B_g/\sqrt{g}$, or equivalently scale $\kappa_{p,q}/\sqrt{pq}$ for $B_g/\sqrt{mg}$. The Rayleigh distribution is moment-determinate, so the method of moments proves (85.12). □

Corollary 85.7 (Mean and variance of rematerialization)**.** *With $n = mg$,*

$$\mathbb{E}B_g \sim \kappa_{p,q}\sqrt{\frac{\pi}{2pq}}\sqrt{n}, \tag{85.15}$$

$$\mathrm{Var}(B_g) \sim \kappa_{p,q}^2 \frac{4-\pi}{2pq}\, n. \tag{85.16}$$

Chapter 86

Macroscopic LIFO Cost, Duality, and Finite Kernels

We now pass from the diffusive rematerialization scale to the macroscopic SEARCH/outer-WRITE regime, then return to finite kernels that compute the physical rematerialization constant.

86.1 Two regenerative bulk regimes

We now analyze the n-scale LIFO cost. Put

$$c = h + 1, \qquad k_+ = \left\lceil \frac{c}{p} \right\rceil, \qquad k_- = \left\lceil \frac{c}{q} \right\rceil.$$

The canonical outer-WRITE demands are

$$\mu(cm) = (qk_+ + c,\ pk_+ - c), \tag{86.1}$$
$$\mu(-cm) = (qk_- - c,\ pk_- + c). \tag{86.2}$$

To define the bulk constants, use the translation-invariant positive-tail extension obtained by quotienting complete unary packets (equivalently, adjoin an unbounded unary zero tail and retain the same finite phase dynamics). A finite deep-positive interval in the actual normalizer follows this extension until it hits the central boundary; only its first and last regeneration cycles can be truncated, and those endpoint pieces are handled in Section 86.2. In the homogeneous positive extension, consider the interval from one positive outer WRITE to the next. The mixed head of the newly written word is finite, while the surviving tail is unary zero. The next positive outer WRITE occurs after exactly pk_+ input ones have been read. Under the critical i.i.d. law the cycle length is therefore

$$T_+ \sim \mathrm{NB}\Big(pk_+, \frac{p}{m}\Big), \tag{86.3}$$

where T_+ counts trials through the final success. Similarly,

$$T_- \sim \mathrm{NB}\Big(qk_-, \frac{q}{m}\Big). \tag{86.4}$$

Thus

$$\mathbb{E}T_+ = mk_+, \qquad \mathrm{Var}(T_+) = \frac{k_+ qm}{p}, \tag{86.5}$$
$$\mathbb{E}T_- = mk_-, \qquad \mathrm{Var}(T_-) = \frac{k_- pm}{q}. \tag{86.6}$$

In particular the outer-WRITE rates are

$$o_+ = \frac{1}{mk_+}, \qquad o_- = \frac{1}{mk_-}. \tag{86.7}$$

Theorem 86.1 (Exact bulk LIFO-obstruction rates)**.** *For $R = S + O$, the critical i.i.d. positive- and negative-bulk rates are*

$$\boxed{r_+ = \frac{2pq}{m^2} + \frac{(p-q)c}{m^2 k_+},} \tag{86.8}$$

$$\boxed{r_- = \frac{2pq}{m^2} - \frac{(p-q)c}{m^2 k_-}.} \tag{86.9}$$

Hence the SEARCH rates are

$$s_+ = r_+ - o_+, \qquad s_- = r_- - o_-. \tag{86.10}$$

Proof. We prove the positive case. Write

$$(x, y) = \mu(cm) = (qk_+ + c, pk_+ - c).$$

A positive cycle ends at the pk_+-th input one. All y one-cells in the canonical head are consumed before the outer boundary can be reached, by Lemma 84.2. Let M be the number of those one-cells that are consumed by input zero; each such event is a negative-direction SEARCH mismatch. Since these cells are reached before stopping and the input is i.i.d.,

$$\mathbb{E}M = y\frac{q}{m}.$$

Among the pk_+ input ones in the cycle, exactly $y - M$ match one-cells. Therefore the number of one-over-zero mismatches is

$$pk_+ - (y - M) = c + M.$$

The total number of LIFO obstructions in the cycle is consequently

$$c + 2M,$$

with mean

$$c + 2(pk_+ - c)\frac{q}{m}.$$

Dividing by the mean cycle length mk_+ gives (86.8). The negative calculation is obtained by interchanging zero and one, p and q, and reversing the debt sign; this gives (86.9). □

The difference

$$r_+ - r_- = \frac{(p-q)c}{m^2}\left(\frac{1}{k_+} + \frac{1}{k_-}\right) \tag{86.11}$$

is nonzero whenever $p \neq q$. Since coprime $p = q$ forces $p = q = 1$, the only symmetric ratio is 1:1.

86.2 Finite-to-macroscopic operational transfer

Let f_t be a bounded operational reward determined by the physical transition at time t; examples include indicators of SEARCH, outer WRITE, MATCH, or bounded functions of the debt state. Put

$$C_g(f) = \sum_{t=1}^{n} f_t, \qquad n = mg.$$

The positive-residual occupation count is

$$A_g = \#\{1 \le t \le n : \rho_t > 0\}.$$

Let $v_+(f)$ and $v_-(f)$ denote the corresponding critical i.i.d. positive- and negative-bulk reward rates. For R, these are r_+ and r_- from Theorem 86.1.

The proof uses a symbol-level finite-phase quotient in the two deep tails. We record the required periodicity explicitly because the rewards in this section are accumulated one input symbol at a time.

Lemma 86.2 (Symbol-level affine tail periodicity)**.** *For every $r \ge 0$,*

$$\mu(r + pm) = \mu(r) + (m, 0), \tag{86.12}$$

and for every $r \le 0$,

$$\mu(r - qm) = \mu(r) + (0, m). \tag{86.13}$$

Consequently, after quotienting complete unary m-packets, the reachable configurations in each sufficiently deep positive or negative residual tail have a finite symbol-level phase space.

Proof. For $r \ge 0$, the minimum solution $\mu(r) = (x, y)$ has $0 \le y < p$: among the p residue classes for y, exactly one makes $r + qy$ divisible by p, and the resulting x is nonnegative. Hence $(x + m, y)$ is a nonnegative solution of weight $r + pm$ with the same $y < p$, so no kernel packet (q, p) can be subtracted. By uniqueness of the minimum demand it is $\mu(r + pm)$. The negative identity is symmetric, using the representative with $0 \le x < q$.

By Lemma 84.3, between WRITEs the stack is a suffix of a canonical shortest-demand word and the mixed alternating part has uniformly bounded length. Equations (86.12)–(86.13) show that translating a deep residual tail changes only the number of complete unary m-packets. Thus a phase may be specified by the debt state, the bounded mixed suffix, the unary symbol and unary length modulo m, together with the residual class modulo pm or qm. Only finitely many phases occur. □

Choose once and for all

$$H \ge \max(p, q)$$

large enough that the finite symbol-level quotients of Lemma 86.2 are homogeneous whenever $\rho > H$ or $\rho < -H$. In the positive tail, feeding sufficiently many 1's first consumes the bounded mixed head and then accumulates positive mismatches on the unary zero tail until the positive outer-WRITE reset; for H large enough there are at least $h{+}1$ unary zero cells available, and the residual only increases along this forcing word. The negative statement is symmetric, using a run of 0's. Thus every sufficiently deep phase can reach the sign-specific reset phase with positive probability without leaving its tail. Consequently, after transient phases are discarded, the finite phase kernel has a unique recurrent class. Restrict to that class and let P_+ and P_- denote the critical i.i.d. one-symbol phase kernels. For a bounded transition reward f, let

$$\bar{f}_\sigma(x) = \frac{q}{m} f(x, 0) + \frac{p}{m} f(x, 1), \qquad \sigma \in \{+, -\},$$

where $f(x,a)$ denotes the reward of the deterministic transition from phase x on symbol a. If π_σ is the invariant probability of the recurrent phase class, define

$$v_\sigma(f) = \pi_\sigma \bar{f}_\sigma.$$

The finite Poisson equation

$$g_\sigma - P_\sigma g_\sigma = \bar{f}_\sigma - v_\sigma(f) \tag{86.14}$$

has a bounded solution, unique up to an additive constant.

86.2.1 Bridge perturbation

Under the fixed-Parikh bridge, if the current residual is ρ_t, then

$$\mathbb{P}(X_{t+1} = 1 \mid \mathcal{F}_t) = \frac{p}{m} - \frac{\rho_t}{m(n-t)}. \tag{86.15}$$

Thus the deviation from the critical i.i.d. symbol law is

$$\delta_t = -\frac{\rho_t}{m(n-t)}.$$

For a fixed-Parikh bridge,

$$\mathrm{Var}(\rho_t) = pq\frac{t(n-t)}{n-1}. \tag{86.16}$$

Therefore

Lemma 86.3 (Total bridge-kernel perturbation)**.**

$$\mathbb{E}\sum_{t=1}^{n-1} |\delta_t| \le \frac{\pi\sqrt{pq}}{2m}\sqrt{n} + O(1). \tag{86.17}$$

Proof. By Cauchy–Schwarz and (86.16),

$$\mathbb{E}|\delta_t| \le \frac{\sqrt{pq}}{m\sqrt{n-1}}\sqrt{\frac{t}{n-t}}.$$

Summing and comparing with

$$\int_0^1 \sqrt{\frac{x}{1-x}}\,dx = \frac{\pi}{2}$$

gives (86.17). □

86.2.2 The central band

Let

$$L_H = \#\{1 \le t \le n : |\rho_t| \le H\}.$$

Since $N_1(t)$ is hypergeometric with parameters (n, pg, t) and

$$\rho_t = mN_1(t) - pt,$$

a direct Stirling bound on the hypergeometric probability mass function gives, uniformly in admissible r,

$$\mathbb{P}(\rho_t = r) \le C_{p,q}\sqrt{\frac{n}{t(n-t)}}.$$

(The finitely many endpoint values of t are absorbed into the constant.) The band contains only finitely many residual values, hence

$$\boxed{\mathbb{E}L_H = O(\sqrt{n}).} \tag{86.18}$$

Because $H \ge \max(p,q)$, a path cannot jump directly from the deep positive tail $\{\rho > H\}$ to the deep negative tail $\{\rho < -H\}$ in one input step. Therefore the number of maximal deep-tail intervals is at most $L_H + 1$ up to an absolute endpoint constant.

86.2.3 Poisson decomposition under the bridge

At time t in a fixed deep sign, write Q_t for the bridge phase kernel. Since the next-symbol probabilities differ from the critical law only by δ_t,

$$\|(Q_t - P_\sigma)u\|_\infty \le 2\|u\|_\infty|\delta_t| \tag{86.19}$$

for every bounded phase function u, and likewise the bridge conditional reward mean differs from $\bar{f}_\sigma$ by at most $2\|f\|_\infty|\delta_t|$.

For an interior step of a deep-tail interval, combine (86.14) with Q_t to write

$$f_t - v_\sigma(f) = \big(g_\sigma(Y_t) - g_\sigma(Y_{t+1})\big) + \Delta M_t + \varepsilon_t, \tag{86.20}$$

where (ΔM_t) is a martingale-difference sequence with uniformly bounded increments and

$$|\varepsilon_t| \le C_{p,q,f}|\delta_t|. \tag{86.21}$$

Indeed, subtract first the bridge conditional mean of f_t, and then replace both the bridge reward mean and $Q_t g_\sigma$ by their critical counterparts; the two replacement errors are controlled by (86.19). The remaining centered terms form the martingale difference, while the g_σ terms telescope. The at most two transitions adjacent to each deep-tail interval endpoint are charged to the interval endpoint error.

Theorem 86.4 (Finite-to-macroscopic operational transfer)**.** *For every bounded operational reward f there is a constant $K_{p,q,f} < \infty$ such that*

$$\boxed{\mathbb{E}\,|C_g(f) - [nv_-(f) + (v_+(f) - v_-(f))A_g]| \le K_{p,q,f}\sqrt{n}.} \tag{86.22}$$

Proof. Compare f_t with $v_+(f)$ when $\rho_t > 0$ and with $v_-(f)$ when $\rho_t \le 0$. On the central band the absolute discrepancy is bounded by a constant times L_H, hence has expectation $O(\sqrt{n})$ by (86.18).

On every maximal deep positive or negative interval, sum (86.20). The g_σ terms telescope. Since g_σ is bounded, all interval endpoint and omitted boundary-step contributions have total L^1 size $O(\mathbb{E}(L_H + 1)) = O(\sqrt{n})$. The accumulated perturbation has expectation

$$O\left(\mathbb{E}\sum_{t<n}|\delta_t|\right) = O(\sqrt{n})$$

by Lemma 86.3. The sum of all martingale differences has bounded increments and predictable quadratic variation $O(n)$, so Cauchy–Schwarz gives L^1 norm $O(\sqrt{n})$.

Finally,

$$\sum_{t=1}^{n}\big[v_-(f) + (v_+(f) - v_-(f))\mathbf{1}_{\{\rho_t>0\}}\big] = nv_-(f) + (v_+(f) - v_-(f))A_g.$$

Combining the central, endpoint, perturbation, and martingale contributions proves (86.22). □

86.3 Macroscopic operational limit laws

The critical bridge invariance principle gives

$$\frac{\rho_{\lfloor nt\rfloor}}{\sqrt{pq\,n}} \Longrightarrow B^{\mathrm{br}}(t)$$

in $C[0,1]$, where B^{br} is a standard Brownian bridge. The zero set of a Brownian bridge has Lebesgue measure zero almost surely, so the positive-occupation functional is almost surely continuous at the limit. The positive occupation time of a Brownian bridge is uniform on $[0,1]$; uniform sojourn laws also admit exchangeable-increment and Lévy-bridge formulations [36, 37]. Hence

$$\frac{A_g}{n} \Longrightarrow U, \qquad U \sim \mathrm{Unif}(0,1). \tag{86.23}$$

Classical fixed-score ballot/election-return results provide a discrete counterpart to this occupation phenomenon [35].

Theorem 86.5 (Fixed-semantics operational limit)**.** *For every bounded operational reward f,*

$$\boxed{\frac{C_g(f)}{n} \Longrightarrow v_-(f) + U\big(v_+(f) - v_-(f)\big), \qquad U \sim \mathrm{Unif}(0,1).} \tag{86.24}$$

Proof. Divide (86.22) by n. The L^1 error tends to zero. Apply (86.23) and Slutsky's theorem. □

Corollary 86.6 (Uniform LIFO-obstruction law)**.** *If $p \neq q$, then*

$$\boxed{\frac{R_g}{n} \Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big).} \tag{86.25}$$

If $p = q = 1$, then

$$\frac{R_g}{2g} \longrightarrow \frac{1}{2}$$

in probability.

Corollary 86.7 (Rank-one macroscopic fluctuations)**.** *Let $f_1, \ldots, f_k$ be bounded operational rewards and*

$$\boldsymbol{C}_g = (C_g(f_1), \ldots, C_g(f_k)).$$

Write

$$V_\pm = (v_\pm(f_1), \ldots, v_\pm(f_k)).$$

Then

$$\frac{\boldsymbol{C}_g}{n} \Longrightarrow V_- + U(V_+ - V_-), \tag{86.26}$$

and

$$\boxed{\frac{\mathrm{Cov}(\boldsymbol{C}_g)}{n^2} \longrightarrow \frac{1}{12}(V_+ - V_-)(V_+ - V_-)^T.} \tag{86.27}$$

In particular the limiting covariance matrix has rank at most one.

Proof. Apply Theorem 86.5 componentwise; the same occupation variable U drives every coordinate. Since each normalized count is bounded, second moments converge. The covariance formula follows from $\mathrm{Var}(U) = 1/12$. □

Theorem 86.8 (Three-scale LIFO realization)**.** *For the balanced fixed-Parikh random-ordering model,*

$$\boxed{\begin{aligned} &\tau_g = O_{\mathbb{P}}(1), \\ &\frac{B_g}{\sqrt{n}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right), \\ &\frac{R_g}{n} \Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big). \end{aligned}} \tag{86.28}$$

for $p \neq q$, with the last law degenerating to $1/2$ for $p = q = 1$.

86.4 Physical-lift duality

For a word w, let $\bar{w}$ denote the bitwise complement. Interchanging p and q and complementing input reverses the residual:

$$\rho_t^{q,p}(\bar{w}) = -\rho_t^{p,q}(w). \tag{86.29}$$

Moreover shortest demands are coordinate-swapped:

$$\mu_{q,p}(-r) = \mathrm{swap}(\mu_{p,q}(r)). \tag{86.30}$$

The two canonical words $\eta(x,y)$ and $\eta(y,x)$ have the same alternating prefix and opposite unary tails. By the canonical-segment property, outer WRITE cannot occur until the mixed prefix has disappeared. During that common mixed phase, complemented input produces opposite debt. The unary phases are then mirror images.

Theorem 86.9 (Physical-lift duality)**.** *For every finite word w,*

$$B_{p,q}(w) = B_{q,p}(\bar{w}), \qquad O_{p,q}(w) = O_{q,p}(\bar{w}). \tag{86.31}$$

Consequently, for every g,

$$(B_g, O_g)_{p,q} \overset{d}{=} (B_g, O_g)_{q,p}$$

under the corresponding balanced fixed-Parikh laws, and

$$\boxed{\kappa_{p,q} = \kappa_{q,p}.} \tag{86.32}$$

Proof. Couple the p:q run on w with the q:p run on $\bar{w}$. At the start of every WRITE episode the two canonical stacks have the same length, the same alternating-prefix length $2\min(x,y)$, and complementary unary tails. The boundary guard prevents an outer WRITE while that mixed prefix is still present, so both runs necessarily pop the entire mixed prefix. If v ones occur in the original input during those $2\min(x,y)$ steps, then the original debt is $v - \min(x,y)$, whereas the complemented run has debt $2\min(x,y) - v - \min(x,y)$; the two debts are therefore opposite exactly when the unary tails begin. From that point onward the visible unary symbols are complementary and the coupled input symbols are complementary, so MATCH/mismatch status agrees and the debts remain opposite. Hence a positive outer WRITE in one run is paired with a negative outer WRITE at the same time in the other. If no outer WRITE occurs, the equal-length stacks empty simultaneously, so the next input is a bottom WRITE in both runs. Induction over WRITE episodes proves equality of bottom- and outer-WRITE indicators at every time, yielding (86.31). The complement map is a bijection between $\mathcal{F}_{gq,gp}$ and $\mathcal{F}_{gp,gq}$. Finally, (86.32) follows from the bottom-WRITE Rayleigh scale or directly from the mirrored invariant measures. □

86.5 Near-diagonal universality

Let

$$p = q + 1, \qquad m = 2q + 1, \qquad h = q.$$

The central physical-lift fibers simplify sharply.

Proposition 86.10 (Near-diagonal central fibers)**.** *At block boundaries, for $0 \le z \le q$ the central semantic shell z has only the empty lift*

$$E_z = (D_z, \varnothing).$$

For $z = -r$, $1 \le r \le q$, the only central lifts are

$$E_{-r} = (D_{-r}, \varnothing), \qquad F_{-r} = (D_{q-r}, 1^m). \tag{86.33}$$

Proof. The proof is by most-recent-WRITE analysis. A positive outer WRITE has

$$\mu((q+1)m) = (m, 0),$$

so it creates only a zero tail. Before the next WRITE, a nonempty block-boundary configuration on that tail has stack 0^{km} for some $k \geq 1$ and nonnegative debt, hence semantic shell at least $(q+1)k \geq q+1$. It cannot lie in the central range.

A negative outer WRITE has

$$\mu(-(q+1)m) = (q-1, 3q+3),$$

so its mixed prefix has length $2(q-1) < m$ and its final tail is unary one. Let a later block-boundary suffix have length km and contain a zeros. Debt was reset to zero at the WRITE, and positive debt can subsequently be created only by one-over-zero mismatches that pop zero cells. Since the written word contained only $q-1$ zeros, the current debt satisfies

$$d \leq (q-1) - a.$$

Hence its semantic shell obeys

$$z = a + d - qk \leq q - 1 - qk.$$

If $k \geq 2$, then $z \leq -q-1$, outside the central range. Thus a central nonempty suffix forces $k = 1$; since the mixed prefix is shorter than m, the surviving length-m suffix is exactly 1^m.

It remains to inspect bottom WRITEs. Both possible bottom-WRITE residuals have the form

$$r = jm + q, \qquad -q - 1 \leq j \leq q.$$

If $\mu(r) = (x, y)$ and $\ell = x + y$, then

$$mx - q\ell = jm + q,$$

so $\ell \equiv -1 \pmod{m}$. Write $\ell = km - 1$. Solving gives

$$x = j + qk, \qquad y = k(q+1) - 1 - j.$$

For $-q \leq j \leq q$, the minimum is $k = 1$, so the newly written word has length $m-1$ and is completely consumed by the remaining $m-1$ symbols of the block. The single exceptional value $j = -q-1$ requires $k = 2$; it occurs exactly for input zero from E_{-q}, and

$$\mu(r) = (q-1, 3q+2).$$

Its canonical word is $(10)^{q-1}1^{m+2}$, so after the remaining $m-1$ pops the next block boundary contains precisely 1^m. These cases give exactly (86.33). □

Let

$$D_q = \sum_{r=1}^{q} \nu(F_{-r}).$$

Since every central semantic shell has total invariant mass one,

$$\kappa_{q+1,q} = 2q + 1 - D_q. \tag{86.34}$$

Lemma 86.11 (Near-diagonal predecessor bound)**.** *If a block ends in one of the nonempty central lifts F_{-r}, then the preceding semantic shell satisfies*

$$Z_j \leq -q.$$

Proof. If the block contains a WRITE, consider its last WRITE. A positive outer WRITE cannot produce a one-tail. A negative outer WRITE inserts a word of length $2m$ before its surviving suffix, so with at most $m-1$ input symbols left in the block there are too few subsequent pops for the final stack to be exactly 1^m. A bottom WRITE can occur only on the first symbol of a block by Lemma 85.1, and the only such case that can end in a nonempty central lift is the transition from E_{-q}. It remains to consider a block with no WRITE. Then every one of the m input steps pops an old cell, so the starting stack has the form $T1^m$ with $|T| = m$. Let a be the number of zeros in this stack, and let x_0 be the zero-count of the canonical word produced by the most recent WRITE. Because at least m trailing ones survive and the demand is kernel-free, $x_0 \le q-1$. Positive debt can have increased only by popping zero cells through one-over-zero mismatches, so if the current debt is d then

$$d \le x_0 - a.$$

The residual shell of a length-$2m$ stack is

$$Z_j = a + d - 2q \le x_0 - 2q \le -q - 1.$$

This proves the claim. □

Let

$$Y_q = K - (q+1), \qquad K \sim \mathrm{Bin}\left(2q+1, \frac{q+1}{2q+1}\right).$$

Then

$$\mathbb{E}Y_q = 0, \qquad \mathrm{Var}(Y_q) = \frac{q(q+1)}{2q+1}.$$

By Lemma 86.11 and unit shell mass,

$$\begin{aligned} D_q &\le \sum_{z \le -q} \mathbb{P}\{z + Y_q \in [-q, -1]\} \\ &\le \mathbb{E}\big[\min(Y_q + 1, q)\mathbf{1}_{\{Y_q \ge 0\}}\big] \\ &\le 1 + \mathbb{E}Y_q^+ \\ &\le 1 + \frac{1}{2}\sqrt{\frac{q(q+1)}{2q+1}}. \end{aligned} \tag{86.35}$$

Theorem 86.12 (Near-diagonal universal Rayleigh scale)**.** *For $p = q+1$,*

$$\boxed{\frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \longrightarrow 2.} \tag{86.36}$$

More quantitatively,

$$2\sqrt{\frac{q}{q+1}} - \frac{1}{2\sqrt{2q+1}} \le \frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \le \frac{2q+1}{\sqrt{q(q+1)}}. \tag{86.37}$$

Consequently, after first taking $g \to \infty$ and then $q \to \infty$,

$$\frac{B_g}{\sqrt{(2q+1)g}} \Longrightarrow \mathrm{Rayleigh}(2).$$

Proof. Combine (86.34) with (86.35) and divide by $\sqrt{q(q+1)}$. Both sides of (86.37) tend to two. The final statement follows from Theorem 85.6. □

86.6 Finite physical-lift kernels and explicit computation of $\kappa_{p,q}$

The distributional results above require only the invariant-measure definition of $\kappa_{p,q}$. We now explain why the same constant is effectively computable from a finite matrix-kernel system.

86.6.1 Affine shortest-demand tails

At block boundaries the shortest demands have explicit forms. For $z \geq 0$,

$$\mu(mz) = \left(z + q \left\lceil \frac{z}{p} \right\rceil, p \left\lceil \frac{z}{p} \right\rceil - z \right). \tag{86.38}$$

For $n \geq 0$,

$$\mu(-mn) = \left(q \left\lceil \frac{n}{q} \right\rceil - n, n + p \left\lceil \frac{n}{q} \right\rceil \right). \tag{86.39}$$

Hence

$$\mu(m(z+p)) = \mu(mz) + (m, 0), \tag{86.40}$$

$$\mu(-m(n+q)) = \mu(-mn) + (0, m). \tag{86.41}$$

Thus every p positive semantic shells add one unary packet 0^m, while every q negative shells add one unary packet 1^m.

Proof of (86.38)–(86.41). For $mz > 0$, all integer solutions of $px - qy = mz$ are

$$x = z + qk, \qquad y = pk - z.$$

The least k making both coordinates nonnegative is $k = \lceil z/p \rceil$, giving (86.38). The negative formula is symmetric. The affine identities follow by replacing z by $z+p$ and n by $n+q$. □

By Lemma 84.3, a reachable stack is a suffix of a canonical shortest-demand word. The mixed alternating head has bounded length because every freshly written shortest demand has minority count at most h; the only unbounded part is a unary tail. Write the unary-tail length as

$$L = m\ell + r, \qquad 0 \leq r < m.$$

Modulo complete unary packets a^m, the remaining information consists of the debt state, the unary symbol, the bounded mixed head, and r. There are finitely many possibilities.

Theorem 86.13 (Finite physical-lift kernel)**.** *For every coprime p, q, there exist finite positive- and negative-tail phase sets and finite families of rational matrices $\{M_j^+\}$ and $\{M_j^-\}$ such that the invariant shell vectors $v_\ell^\pm$ satisfy, for all sufficiently deep levels,*

$$v_\ell^\pm = \sum_j v_{\ell-j}^\pm M_j^\pm. \tag{86.42}$$

The corresponding matrix Laurent polynomials

$$\mathcal{K}_\pm(\lambda) = I - \sum_j M_j^\pm \lambda^j \tag{86.43}$$

have rational coefficients. Together with finitely many central balance equations and the normalization $\nu(o) = 1$, the bounded nonnegative solutions of (86.42) *uniquely determine $\kappa_{p,q}$.*

Proof. The affine identities (86.40)–(86.41) show that translation in a deep semantic tail adds only complete unary m-packets. The canonical-segment property bounds the mixed head independently of level. Quotienting the unary length modulo m therefore leaves finitely many physical phases and one integer level. During one m-symbol input block, the level can change by only a bounded amount, and once the tail is sufficiently deep the transition probabilities depend only on the phase and the level increment, not on the absolute level. Under the critical law every block word has rational probability $(p/m)^k(q/m)^{m-k}$, so the matrices are rational. The invariant equation in the homogeneous tail is exactly (86.42). Irreducibility and the normalization $\nu(o) = 1$ make the recurrent invariant measure unique, hence the central empty-state coordinates and $\kappa_{p,q}$ are uniquely determined. □

This is a finite-phase, level-homogeneous matrix-analytic structure of the same general type that underlies quasi-birth-and-death and matrix-geometric methods [39]. The matrix-analytic machinery itself is classical; the point here is that the shortest-demand pushdown quotient canonically produces such a kernel whose central boundary values encode physical LIFO rematerialization.

86.6.2 Small explicit kernel examples

The two smallest asymmetric $q = 1$ cases reduce to scalar tail recurrences. We record their eliminations as compact reproducibility checks, and then give the first genuinely multi-phase example at 3:2.

The ratio 2:1**.** In negative shell $-n$ there are two deep phases,

$$A_n = (D_{-1}, 1^{3(n-1)}), \qquad B_n = (D_0, 1^{3n}).$$

Write $a_n = \nu(A_n)$, $b_n = \nu(B_n)$ and $u_n = a_n - b_n$. The unit-shell identity gives $a_n + b_n = 1$. Exact enumeration of the eight critical three-symbol blocks gives

$$u_{n-2} - 6u_{n-1} + 39u_n - 8u_{n+1} = 0 \tag{86.44}$$

in the homogeneous tail, while the finite central balance equations are

$$31u_1 - 8u_2 = 15, \qquad 4u_1 - 39u_2 + 8u_3 = 3. \tag{86.45}$$

The characteristic polynomial is

$$Q_{2,1}(R) = 8R^3 - 39R^2 + 6R - 1.$$

It has a unique real root $R > 1$. Boundedness of the invariant lift profile cancels that mode, which is equivalent to

$$u_1(31R^2 - 4R + 1) = 15R^2 - 3R.$$

With $\theta = (1 + u_1)/2 = a_1$, elimination of R between this relation and $Q_{2,1}(R) = 0$ gives

$$8\theta^3 - 3\theta - 1 = 0. \tag{86.46}$$

Thus

$$\kappa_{2,1} = 2 + \theta = 2.737843258897860\ldots. \tag{86.47}$$

The ratio 3:1**.** The same two negative phases occur with unary packet length four. Writing again $u_n = a_n - b_n$, exact sixteen-block enumeration gives

$$81u_{n+1} - 364u_n + 54u_{n-1} - 12u_{n-2} + u_{n-3} = 0, \tag{86.48}$$

and the finite central equations are

$$283u_1 - 81u_2 = 148, \tag{86.49}$$
$$-27u_1 + 364u_2 - 81u_3 = -40, \tag{86.50}$$
$$9u_1 - 54u_2 + 364u_3 - 81u_4 = 4. \tag{86.51}$$

The characteristic polynomial is

$$Q_{3,1}(R) = 81R^4 - 364R^3 + 54R^2 - 12R + 1.$$

Cancelling its unique unstable real mode gives

$$u_1(81R^3 - 81R^2 + 27R - 3) = 148R^2 - 40R + 4.$$

Eliminating R and setting $\theta = (1 + u_1)/2$ yields

$$54\theta^4 - 18\theta^2 - 8\theta - 1 = 0, \tag{86.52}$$

so

$$\kappa_{3,1} = 2 + \theta = 2.750694578264632\ldots. \tag{86.53}$$

The ratio 3:2**.** Here the deep negative tail has seven phases. Writing the negative semantic shell as $-n$, the four odd-shell phases for $n = 2j + 1$ are

$$\begin{aligned} O_0(j) &= (D_{-1}, 1^{5j}), \\ O_1(j) &= (D_0, 0\,1^{5j+4}), \\ O_2(j) &= (D_0, 10\,1^{5j+3}), \\ O_3(j) &= (D_1, 1^{5(j+1)}), \end{aligned}$$

and the three even-shell phases for $n = 2j$ are

$$\begin{aligned} E_0(j) &= (D_{-2}, 1^{5(j-1)}), \\ E_1(j) &= (D_{-1}, 0\,1^{5j-1}), \\ E_2(j) &= (D_0, 1^{5j}). \end{aligned}$$

Enumerating the 32 five-symbol blocks with their exact critical probabilities $3^k 2^{5-k}/5^5$ gives a rational seven-phase Laurent kernel whose determinant factors as

$$\det \mathcal{K}_{3,2}(\lambda) = \frac{(\lambda - 1)^2 C_3(\lambda) K_5(\lambda)}{5^{20}\lambda^6}, \tag{86.54}$$

where

$$\begin{aligned} C_3(\lambda) &= 59049\lambda^3 - 1531872\lambda^2 + 9472\lambda - 1024, \\ K_5(\lambda) &= 59049\lambda^5 - 1387530\lambda^4 + 16632265\lambda^3 \\ &\quad - 1448160\lambda^2 + 11520\lambda + 1024. \end{aligned}$$

The critical phase kernel has invariant vector

$$\left(\frac{1}{4}, \frac{3}{50}, \frac{1}{10}, \frac{9}{100}, \frac{1}{4}, \frac{1}{25}, \frac{21}{100}\right) \tag{86.55}$$

in the order $(O_0, O_1, O_2, O_3, E_0, E_1, E_2)$. Coupling this exact tail kernel to the finite central balance equations gives numerically

$$\kappa_{3,2} = 4.64089572244120\ldots. \tag{86.56}$$

An independent direct probability propagation, extrapolated from block horizons 100 and 200, gives $4.6408956068\ldots$, independently confirming the first six decimal places. The rematerialization constant need not be rational even though the underlying transition probabilities are rational: already the cubic (86.46) has no rational root, so $\kappa_{2,1}$ is irrational.

Remark 86.14 (Algebraicity and regular kernels). The finite kernels provide an exact rational boundary-value problem for $\kappa_{p,q}$. When the associated matrix Laurent pencil is regular, companion linearization shows that its characteristic modes are algebraic, and the finite central boundary equations then make the normalized bounded solution algebraic. This applies to the explicit kernels above. We do not require, and do not use, an unconditional algebraicity theorem for every coprime p, q in the probabilistic results of this Part.

Chapter 87

Optimal-Fiber Compression and Active Trade Geometry

Research branch — Optimal-Fiber Topology and Physical Holonomy. *Earlier chapters introduced higher-dimensional minimum-completion fibers and showed how equal-length kernel trades create nontrivial quotient topology. This branch sharpens that picture through the complete active-rank-two classification and then carries the topology into a category-relative physical transport theory. Kernel rank controls when higher topology can appear; the global homotopy type is governed by the translation geometry of the active optimal fiber, while the displacement kernel identifies the pure loops on which physical LIFO holonomy can live.*

For an integer counting matrix $A \in \mathbb{Z}^{k\times d}$ and completable residual r, write

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\},$$
$$\mathcal{O}_A(r) = \{x : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.$$

The earlier higher-dimensional completion theory supplied the filled shortest-prefix quotient. The purpose here is to compress its global topology to finite optimal-fiber translation data and then determine what active rank two can actually support.

87.1 Minimum-completion geometry

Let $A \in \mathbb{Z}^{k\times d}$. A residual is an element $r \in A\mathbb{Z}^d$ for which the completion equation

$$Ax = -r, \qquad x \in \mathbb{N}^d$$

is feasible. A convenient global hypothesis is the following.

Definition 87.1 (Completion totality)**.** The matrix A is *completion-total* if there exists $h \in \mathbb{N}^d$ with every coordinate positive and $Ah = 0$.

If A is completion-total, every lattice residual is completable: if $r = Az$ with $z \in \mathbb{Z}^d$, then $th - z \in \mathbb{N}^d$ for all sufficiently large t, and $A(th - z) = -r$.

For a completable residual r define

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\} \tag{87.1}$$

and the optimal Parikh fiber

$$\mathcal{O}_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}. \tag{87.2}$$

The set is finite because all of its points have the same coordinate sum.

The equal-length kernel is

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix} = \{g \in \mathbb{Z}^d : Ag = 0,\ \mathbf{1}^\top g = 0\}. \tag{87.3}$$

Differences of optimal completions lie in K_A.

87.1.1 The shortest-prefix down-set

For $x \in \mathbb{N}^d$ put

$$B_x = \{u \in \mathbb{N}^d : 0 \le u \le x\},$$

where inequalities are coordinatewise, and define

$$\mathcal{C}_A(r) =\downarrow \mathcal{O}_A(r) = \bigcup_{x \in \mathcal{O}_A(r)} B_x. \tag{87.4}$$

Proposition 87.2 (Exact geodesic-prefix criterion)**.** *For $u \in \mathbb{N}^d$,*

$$u \in \mathcal{C}_A(r) \quad \Longleftrightarrow \quad \ell_A(r + Au) = \ell_A(r) - \mathbf{1}^\top u.$$

Thus $\mathcal{C}_A(r)$ is exactly the set of Parikh vectors that occur as prefixes of shortest completions of r.

Proof. If $u \le x$ for some $x \in \mathcal{O}_A(r)$, then $x - u$ completes $r + Au$ and has length $\ell_A(r) - \mathbf{1}^\top u$. A strictly shorter completion would combine with u to contradict optimality of x.

Conversely, if the displayed equality holds and y is a shortest completion of $r + Au$, then $u + y$ completes r and has length $\ell_A(r)$. Hence $u + y \in \mathcal{O}_A(r)$ and $u \in \mathcal{C}_A(r)$. □

Give every finite down-set $D \subset \mathbb{N}^d$ its standard cubical realization $|D|$: a J-oriented elementary cube based at u is present when $u + \mathbf{1}_J \in D$. The down-set condition supplies all faces automatically.

Two equally oriented cells in $|\mathcal{C}_A(r)|$ are identified when their base vertices differ by an element of K_A. We write

$$I_A(r) = |\mathcal{C}_A(r)|/K_A \tag{87.5}$$

for the resulting filled shortest-prefix quotient. By theorem 87.2, its vertices are precisely shortest-path residual states, and its higher cubes record commuting one-letter shortest moves.

87.2 Active trades and optimal-fiber lattice geometry

The full kernel K_A can contain directions that never occur inside the optimal fiber for a fixed residual. The correct residual-dependent lattice is smaller.

Definition 87.3 (Active trade lattice)**.** Fix a completable residual r. Define

$$L_A(r) = \langle x - y : x, y \in \mathcal{O}_A(r) \rangle_{\mathbb{Z}} \subseteq K_A. \tag{87.6}$$

For $x_0 \in \mathcal{O}_A(r)$ define the normalized optimal fiber

$$P_A(r) = \mathcal{O}_A(r) - x_0 \subseteq L_A(r). \tag{87.7}$$

Its active rank is $\operatorname{rank}_{\mathbb{Z}} L_A(r)$.

Lemma 87.4 (Active collision lemma)**.** *If $u, v \in \mathcal{C}_A(r)$ and $u - v \in K_A$, then*

$$u - v \in \mathcal{O}_A(r) - \mathcal{O}_A(r) \subseteq L_A(r).$$

Proof. The condition $u - v \in K_A$ implies

$$Au = Av, \qquad \mathbf{1}^\top u = \mathbf{1}^\top v.$$

Hence u and v reach the same residual at the same geodesic depth. Choose a shortest completion y of that common residual. By theorem 87.2, both $u + y$ and $v + y$ lie in $\mathcal{O}_A(r)$. Their difference is $u - v$. □

Proposition 87.5 (Active-fiber saturation)**.** *For every $x_0 \in \mathcal{O}_A(r)$,*

$$\mathcal{O}_A(r) = (x_0 + L_A(r)) \cap \mathbb{N}^d. \tag{87.8}$$

Consequently

$$P_A(r) = \operatorname{conv}(P_A(r)) \cap L_A(r). \tag{87.9}$$

Proof. The inclusion $\mathcal{O}_A(r) \subseteq (x_0 + L_A(r)) \cap \mathbb{N}^d$ follows from the definition of $L_A(r)$. Conversely, if $x = x_0 + \lambda \in \mathbb{N}^d$ with $\lambda \in L_A(r) \subseteq K_A$, then

$$Ax = Ax_0 = -r, \qquad \mathbf{1}^\top x = \mathbf{1}^\top x_0 = \ell_A(r),$$

so $x \in \mathcal{O}_A(r)$.

For the second statement, put $P = P_A(r)$. Equation (87.8) says

$$P = L_A(r) \cap \{z \in \mathbb{R}^d : z_i \geq -(x_0)_i \ \forall i\}.$$

The right-hand real region is convex. Since $\operatorname{conv}(P)$ lies inside it, every lattice point of $\operatorname{conv}(P)$ belongs to P. □

Thus every normalized optimal fiber is a finite lattice-convex configuration in its active lattice. This constraint is decisive in active rank two.

87.3 Optimal-fiber nerve compression

Let

$$X = |\mathcal{C}_A(r)|, \qquad L = L_A(r).$$

Form the periodic saturation

$$\widetilde{X} = \bigcup_{\lambda \in L} (X + \lambda). \tag{87.10}$$

By theorem 87.4, the quotient $\widetilde{X}/L$ is canonically the same cell quotient as $I_A(r)$.

Lemma 87.6 (Contractible translate intersections)**.** *Every nonempty finite intersection*

$$\bigcap_{\lambda \in F} (X + \lambda), \qquad F \subset L \text{ finite},$$

is contractible.

Proof. Work first on vertices. Put $C = \mathcal{C}_A(r)$ and

$$D_F = \bigcap_{\lambda \in F} (C + \lambda).$$

Assume $D_F \neq \varnothing$ and set

$$m_i(F) = \max_{\lambda \in F} \lambda_i.$$

Every $u \in D_F$ satisfies $u \geq m(F)$ coordinatewise: for each coordinate choose a translate attaining the maximum. If $0 \leq z \leq u - m(F)$, then for every $\lambda \in F$,

$$0 \leq z + m(F) - \lambda \leq u - \lambda.$$

Since $u - \lambda \in C$ and C is a down-set, $z + m(F) - \lambda \in C$. Thus $D_F - m(F)$ is a finite coordinate down-set in $\mathbb{N}^d$. Its cubical realization is a union of coordinate boxes anchored at the origin and is therefore contractible. Translating back proves the claim. □

Choose $x_0 \in \mathcal{O}_A(r)$ and write $P = P_A(r)$. Define the periodic simplicial complex

$$\mathcal{N}(P) = \{F \subseteq L : \ F \text{ finite and } \exists c \in L \text{ with } c - F \subseteq P\}. \tag{87.11}$$

Equivalently, $\mathcal{N}(P)$ is the nerve of the family of finite convex-lattice sets

$$\{\lambda + P : \lambda \in L\}.$$

The lattice L acts by translation on $\mathcal{N}(P)$.

Theorem 87.7 (Optimal-Fiber Nerve Compression)**.** *For every completable residual r,*

$$I_A(r) \simeq \mathcal{N}(P_A(r))/L_A(r). \tag{87.12}$$

The homotopy type of the global shortest-prefix quotient is therefore determined by the finite optimal fiber and its translation lattice.

Proof. The translates $\{X + \lambda\}_{\lambda \in L}$ cover $\widetilde{X}$ and are permuted by the L-action. By theorem 87.6, every nonempty finite intersection is contractible. The equivariant simplicial nerve theorem [45, Theorem 2.8] gives an L-equivariant homotopy equivalence between $\widetilde{X}$ and the nerve of this cover.

It remains to identify that nerve with (87.11). A finite $F \subset L$ spans a simplex exactly when there exists a common vertex u with

$$u - \lambda \in \mathcal{C}_A(r) \qquad (\lambda \in F).$$

All points $u - \lambda$ have the same residual and the same geodesic depth because their pairwise differences lie in $L \subseteq K_A$. Choose a shortest suffix y from that common residual. Then

$$c = u + y$$

satisfies $c - \lambda \in \mathcal{O}_A(r)$ for every $\lambda \in F$. Conversely, if $c - F \subseteq \mathcal{O}_A(r)$ then c belongs to every translated shortest-prefix set. After translating by x_0, this is exactly the simplex criterion (87.11).

Finally, the L-action is free and cellular, so passage to quotients preserves the equivariant homotopy equivalence. Using theorem 87.4, $\widetilde{X}/L$ is $I_A(r)$. □

Remark 87.8 (Rank-one rigidity). If $\operatorname{rank} L_A(r) = 1$, lattice-convexity forces $P_A(r)$ to be an interval of consecutive points in a rank-one lattice. If the fiber is nontrivial, $\mathcal{N}(P_A(r))$ is contractible and its translation quotient is S^1. Thus the first non-rigid regime is active rank two.

87.4 Finite presentations and first homology

The compressed quotient has one vertex orbit. Its 2-skeleton can be read directly from differences in the optimal fiber.

Proposition 87.9 (Finite fundamental-group presentation)**.** *Assume* $0 \in P = P_A(r)$*. Then* $\pi_1(I_A(r))$ *is isomorphic to*

$$G(P) = \left\langle X_p \ (p \in P),\ X_0 = 1 \ \middle|\ X_p^{-1} X_q = X_s^{-1} X_t \text{ whenever } q - p = t - s \right\rangle. \tag{87.13}$$

Proof. In $\mathcal{N}(P)/L$, oriented edge orbits are indexed by differences $d = q - p \in P - P$. Write the corresponding edge generator as a_d, with $a_{-d} = a_d^{-1}$. Every triple $p, q, s \in P$ supplies the triangular relation

$$a_{q-p} a_{s-q} = a_{s-p}.$$

Set $X_p = a_p$ for $p \in P$ and $X_0 = 1$. The triangle on $0, p, q$ gives

$$a_{q-p} = X_p^{-1} X_q.$$

Two edges represent the same edge orbit exactly when their differences agree, which gives precisely the relations in (87.13). Higher simplices do not affect the fundamental group. □

There is a canonical displacement homomorphism

$$\delta : G(P) \longrightarrow L, \qquad \delta(X_p) = p. \tag{87.14}$$

It is surjective because $L = \langle P - P \rangle = \langle P \rangle$.

Corollary 87.10 (Parallelogram–commutator principle)**.** *If*

$$p, \quad p + u, \quad p + v, \quad p + u + v \in P,$$

then the corresponding difference generators satisfy

$$[a_u, a_v] = 1.$$

Proof. The two triangles give

$$a_u a_v = a_{u+v} = a_v a_u.$$

□

87.4.1 Freiman dimension and first homology

Let F be a field of characteristic 0. A normalized Freiman 2-homomorphism on P is a function $f : P \to F$ with $f(0) = 0$ such that

$$p + q = s + t \quad \Longrightarrow \quad f(p) + f(q) = f(s) + f(t).$$

Let $\dim_F(P)$ denote the dimension of this vector space. This agrees with the usual normalized Freiman-dimension convention.

Proposition 87.11 (First homology as the Freiman shadow)**.** *For every finite normalized fiber* P*,*

$$\beta_1(\mathcal{N}(P)/L) = \dim_F(P).$$

Proof. Abelianizing (87.13) replaces X_p by additive variables x_p and the defining relations by

$$x_q - x_p = x_t - x_s \quad \text{whenever} \quad q - p = t - s.$$

Equivalently,

$$x_p + x_t = x_q + x_s \quad \text{whenever} \quad p + t = q + s.$$

After $x_0 = 0$, this is exactly the vector space of normalized Freiman homomorphisms. Tensoring H_1 with F gives the stated equality of ranks. □

Theorem 87.12 (Rank-two first-Betti bound)**.** *If* $\operatorname{rank} L_A(r) = 2$*, then*

$$\beta_1(I_A(r)) \in \{2, 3\}.$$

Proof. Put $P = P_A(r)$, $m = |P|$, and $Q = \operatorname{conv}(P)$ inside $L \otimes \mathbb{R} \cong \mathbb{R}^2$. By theorem 87.5, $P = Q \cap L$. Let B be the number of lattice points on ∂Q. Pick's theorem and the Ehrhart polynomial give

$$|2Q \cap L| = 4m - B - 3 \leq 4m - 6. \tag{87.15}$$

Since $P + P \subseteq 2Q \cap L$,

$$|P + P| \leq 4m - 6.$$

If $d_F = \dim_F(P)$, Freiman's dimension lemma [47, Chapter 5] gives

$$|P + P| \geq (d_F + 1)m - \binom{d_F + 1}{2}.$$

For $d_F \geq 4$ and $m \geq 5$, the right side is at least $5m - 10 > 4m - 6$. If $m \leq 4$, then trivially $d_F \leq m - 1 \leq 3$. Hence $d_F \leq 3$.

On the other hand, the two coordinate linear functionals on the full-rank planar lattice L give two independent normalized Freiman homomorphisms, so $d_F \geq 2$. Apply theorem 87.11 and theorem 87.7. □

87.5 The optimal-fiber Taylor slice

The nerve model has a second description in commutative algebra. This description is classical on the lattice-ideal side; the point here is that the minimum-completion quotient selects a canonical principal slice.

Fix a field $\Bbbk$, put

$$S = \Bbbk[x_1, \ldots, x_d],$$

and let $L = L_A(r) \subseteq \mathbb{Z}^d$. Because $L \subseteq K_A$, every $\lambda \in L$ has coordinate sum 0, hence

$$L \cap \mathbb{N}^d = \{0\}.$$

Define the lattice ideal

$$I_L = \langle x^{u^+} - x^{u^-} : u \in L \rangle \subseteq S. \tag{87.16}$$

Give S the $\mathbb{Z}^d/L$ grading $\deg(x^a) = a + L$. For $\alpha \in \mathbb{N}^d/L$ write

$$\operatorname{fiber}(\alpha) = \{x^a : a \in \mathbb{N}^d,\ a + L = \alpha\}.$$

Choose $x_0 \in \mathcal{O}_A(r)$ and set

$$\alpha = x_0 + L.$$

By theorem 87.5,

$$\operatorname{fiber}(\alpha) \quad \text{has exponent set exactly } \mathcal{O}_A(r). \tag{87.17}$$

For each degree β, let $E_i(\beta)$ be the family of i-element subsets of $\mathrm{fiber}(\beta)$ with monomial gcd equal to 1. Bayer–Sturmfels identify these sets as the homogeneous basis of the lattice-ideal Taylor resolution and give the differential explicitly [44, Proposition 3.10].

Write $\beta \preceq \alpha$ when $\alpha - \beta$ is the degree of a monomial.

Theorem 87.13 (Optimal-Fiber Taylor Slice)**.** *Let*

$$Q(P) = \mathcal{N}(P_A(r))/L_A(r).$$

For every $j \geq 0$ there is a canonical basis correspondence

$$C_j(Q(P); \Bbbk) \cong \bigoplus_{\beta \preceq \alpha} \Bbbk\, E_{j+1}(\beta), \tag{87.18}$$

under which the cellular boundary is the Bayer–Sturmfels Taylor differential specialized by

$$\varepsilon : S \to \Bbbk, \qquad x_i \mapsto 1.$$

Equivalently, $C_\bullet(Q(P); \Bbbk)$ is the dehomogenized principal degree slice of the lattice-ideal Taylor complex.

Proof. Represent a j-simplex orbit of $Q(P)$ by a $(j+1)$-element subset

$$F = \{u_0, \dots, u_j\} \subseteq \mathcal{O}_A(r).$$

Let $g = \min(u_0, \dots, u_j)$ coordinatewise. Subtracting g gives a gcd-one set

$$\overline{F} = F - g$$

in the fiber of the degree

$$\beta = \alpha - \deg(x^g) \preceq \alpha.$$

If F is replaced by an L-translate, the gcd changes by the same translate and $\overline{F}$ is unchanged. Hence the construction depends only on the simplex orbit.

Conversely, let $I \in E_{j+1}(\beta)$ with $\beta \preceq \alpha$. Choose $g \in \mathbb{N}^d$ with $\alpha = \beta + \deg(x^g)$. Multiplying every monomial of I by x^g gives a subset of $\mathrm{fiber}(\alpha) = \mathcal{O}_A(r)$. A different choice g' differs from g by an element of L and therefore gives the same simplex orbit. The two constructions are inverse.

If one vertex is removed, the remaining set acquires a new common gcd h. Bayer–Sturmfels' differential is

$$\partial(I) = \sum_{m \in I} \mathrm{sign}(m, I)\ \gcd(I \setminus \{m\})\, [I \setminus \{m\}],$$

where the bracket denotes removal of that common factor. Under $x_i \mapsto 1$, every monomial coefficient becomes 1, leaving exactly the oriented simplicial boundary of the quotient. The shift $E_{j+1} \leftrightarrow C_j$ is the usual one: E_1 is homological degree 0 in the resolution of S/I_L. □

87.5.1 Principal truncation and character twists

Let $\mathbb{T}_\bullet$ be the full lattice Taylor resolution and $\mathbb{F}_\bullet$ a minimal $\mathbb{Z}^d/L$-graded free resolution of S/I_L.

Lemma 87.14 (Principal truncation lemma)**.** *The full graded chain-homotopy equivalence between $\mathbb{T}_\bullet$ and $\mathbb{F}_\bullet$ restricts to a chain-homotopy equivalence on the principal degree order ideal*

$$\{\beta : \beta \preceq \alpha\}.$$

Proof. Comparison maps and homotopies may be chosen homogeneous of degree 0. A homogeneous map from a basis summand of degree β to one of degree γ has a monomial coefficient of degree $\beta - \gamma$, so it can be nonzero only when $\gamma \preceq \beta$. Hence every map and homotopy preserves every degree down-set, in particular the principal down-set below α. □

Let $\chi : L \to \Bbbk^\times$ be a character. After a faithfully flat extension of the field if necessary, extend χ to a homomorphism

$$\widetilde{\chi} : \mathbb{Z}^d \to \overline{\Bbbk}^\times.$$

Let $\Bbbk_\chi$ be the corresponding rank-one local system on $Q(P)$.

Proposition 87.15 (Character-twisted Taylor slice)**.** *After a degree-dependent rescaling of basis vectors, the complex*

$$C_\bullet(Q(P); \Bbbk_\chi)$$

is isomorphic to the principal Taylor slice evaluated at

$$x_i = \widetilde{\chi}(e_i).$$

Proof. For every $\beta \preceq \alpha$, choose a lift $g_\beta \in \mathbb{N}^d$ with

$$\alpha = \beta + \deg(x^{g_\beta}).$$

Suppose a Taylor boundary term deletes one vertex, produces common gcd exponent h, and passes from degree β to γ. The corresponding face in the periodic nerve differs from the chosen representative in degree γ by the deck translation

$$\tau = h + g_\beta - g_\gamma \in L.$$

After rescaling the degree-β basis by $\widetilde{\chi}(g_\beta)$, the evaluated Taylor coefficient becomes

$$\widetilde{\chi}(h)\frac{\widetilde{\chi}(g_\beta)}{\widetilde{\chi}(g_\gamma)} = \widetilde{\chi}(\tau) = \chi(\tau),$$

which is exactly the local-system coefficient attached to that face. The signs are the common simplicial incidence signs. □

Chapter 88

Lattice-Ideal Slices and Rank-Two Structural Rigidity

The compression theorem admits an algebraic refinement: the cellular chains of the quotient can be read as a principal dehomogenized slice of the Taylor resolution of the active lattice ideal. This bridge controls homology and prepares the planar classification.

88.1 Rank-two homological dimension

Assume from now on that

$$\operatorname{rank}_{\mathbb{Z}} L_A(r) = 2.$$

Peeva–Sturmfels prove the general projective-dimension bound

$$\operatorname{pd}_S(S/I_L) \le 2^{\operatorname{rank} L} - 1,$$

so in rank two the minimal resolution has no term above homological degree 3 [42].

Lemma 88.1 (Torus-point injectivity)**.** *For every field $\Bbbk$ and every character $\chi : L \to \Bbbk^\times$, the third differential of the full minimal resolution, after the corresponding torus-point specialization, is injective.*

Proof. Extend scalars to an algebraic closure and represent χ by a point $\xi \in (\overline{\Bbbk}^\times)^d$.

If χ is nontrivial, then some lattice binomial evaluates nonzero at ξ, so ξ lies outside the support of S/I_L. The specialized full free resolution is therefore acyclic.

If χ is trivial, localize at the algebraic torus. For a lattice basis g_1, g_2 of L, the Laurent lattice ideal is generated by

$$x^{g_1} - 1, \qquad x^{g_2} - 1.$$

These two independent equations form a regular sequence in the regular Laurent ring. Hence the localized module is a codimension-two complete intersection and its Tor vanishes above degree 2. In both cases the third homology of the specialized full minimal resolution is zero. Since there is no fourth term, the third differential is injective. □

Theorem 88.2 (Rank-Two Homological-Dimension Theorem)**.** *If* $\operatorname{rank} L_A(r) = 2$*, then*

$$H_j(I_A(r); \mathbb{Z}) = 0 \qquad (j \ge 3), \tag{88.1}$$

and $H_2(I_A(r); \mathbb{Z})$ is free abelian.

Proof. Over any field $\Bbbk$, theorems 87.13 and 87.14 identify the chain complex of $Q(P)$ up to chain homotopy with the principal slice of a minimal free resolution of length at most 3. Thus homology vanishes above degree 3. By theorem 88.1, the third differential remains injective after restriction to the principal slice, so

$$H_j(Q(P);\Bbbk) = 0 \qquad (j \geq 3).$$

The same holds for every prime field and for $\mathbb{Q}$. Since $Q(P)$ is finite, the universal coefficient theorem implies (88.1). Applying the mod-p universal coefficient sequence in degree 3 gives

$$\operatorname{Tor}_1^{\mathbb{Z}}(H_2(Q(P);\mathbb{Z}), \mathbb{F}_p) = 0$$

for every prime p, so $H_2(Q(P);\mathbb{Z})$ has no torsion. Finally use theorem 87.7. □

88.2 Planar lattice geometry and the fundamental group

By theorem 87.5, a rank-two normalized fiber is a finite lattice-convex set in a rank-two lattice. We now show that a single geometric feature—a nondegenerate parallelogram—separates the abelian and free branches of the fundamental group.

88.2.1 The parallelogram branch

Lemma 88.3 (Minimal parallelogram lemma)**.** *Let $P \subset L \cong \mathbb{Z}^2$ be finite and lattice-convex. If P contains a nondegenerate lattice parallelogram, then it contains one of normalized area 1 or 2 whose edge directions are primitive.*

Proof. Choose a nondegenerate parallelogram

$$R = \{a, a+u, a+v, a+u+v\} \subseteq P$$

of minimum positive normalized area $D = |\det_L(u,v)|$. If u were not primitive, an interior lattice point on each of the two u-parallel edges would belong to P by lattice-convexity and would produce a smaller parallelogram. Thus u and v are primitive. Hence the boundary of $\operatorname{conv}(R)$ contains exactly four lattice points. Pick's theorem gives

$$I(R) = D - 1$$

interior lattice points.

Let c be the center of R. Reflection in c preserves the lattice and the parallelogram. If there is an interior lattice point $z \neq c$, its reflection $z' = 2c - z$ is another interior lattice point. One of the two parallelograms formed by the pair z, z' and one of the two opposite vertex pairs of R is nondegenerate; it lies strictly inside R and has smaller normalized area, contradicting minimality. Hence R has at most one interior lattice point, so $D - 1 \leq 1$. □

A finite lattice-convex set $X \subset L$ will be called *parity-complete* if it meets every coset of $2L$.

Lemma 88.4 (Observer chain)**.** *If $X \subsetneq P$ are finite lattice-convex subsets of L, there exists $z \in P \setminus X$ such that*

$$\operatorname{conv}(X \cup \{z\}) \cap L = X \cup \{z\}.$$

Iterating gives a chain from X to P by one-point lattice-convex extensions.

Proof. Choose $z \in P \setminus X$ minimizing the positive Euclidean distance to $\mathrm{conv}(X)$. If a further lattice point $y \notin X \cup \{z\}$ belonged to $\mathrm{conv}(X \cup \{z\})$, then $y \in P$ by lattice-convexity and, along a segment from z toward $\mathrm{conv}(X)$, would have strictly smaller distance to $\mathrm{conv}(X)$, a contradiction. □

Lemma 88.5 (Parity propagation)**.** *Suppose $X \subseteq P$ is lattice-convex and parity-complete, and suppose that in $G(P)$ all X_x with $x \in X$ lie in the subgroup generated by commuting elements U, V whose displacements form a basis of L. If z is an observer of X, then X_z lies in the same subgroup, with exponent vector equal to the lattice coordinates of z.*

Proof. Choose $x \in X$ with $x \equiv z \pmod{2L}$ and put $y = (x+z)/2 \in L$. The observer property forces $y \in X$. Since

$$z - y = y - x,$$

relation (87.13) gives

$$X_y^{-1} X_z = X_x^{-1} X_y, \qquad X_z = X_y X_x^{-1} X_y.$$

If $X_x = U^a V^b$ and $X_y = U^c V^d$, commutativity gives

$$X_z = U^{2c-a} V^{2d-b},$$

whose displacement is $2y - x = z$. □

Lemma 88.6 (Determinant-two parity completion)**.** *Let*

$$C_2 = \{(0,0), (1,0), (0,1), (-1,2), (0,2)\} \subset \mathbb{Z}^2.$$

If a lattice-convex set $P \supseteq C_2$ contains a point in the missing parity class $(1,1) + 2\mathbb{Z}^2$, then it contains at least one point of

$$M = \{(1,-1), (1,1), (-1,1), (-1,3)\}.$$

For every $m \in M$, the set $C_2 \cup \{m\}$ is lattice-convex and parity-complete, and the generator X_m is already a word in the generators from C_2.

Proof. Consider the linear map

$$\Phi(x, y) = (y, 2x + y).$$

It sends $\mathrm{conv}(C_2)$ to the square $[0,2]^2$. If z lies in the missing parity class, write

$$\Phi(z) = (s, t), \qquad A = s - 1, \quad B = t - 1.$$

Then A, B are even and $B - A \equiv 2 \pmod 4$, so $|A| \neq |B|$ and therefore

$$\big||A| - |B|\big| \geq 2.$$

One of the four inequalities

$$B \geq |A| + 2, \quad -B \geq |A| + 2, \quad A \geq |B| + 2, \quad -A \geq |B| + 2$$

holds. In the first case, the point $(1,3)$ lies in $\mathrm{conv}([0,2]^2 \cup \{\Phi(z)\})$: indeed, with $\theta = 1/(B-1)$, write it as a convex combination of $\Phi(z)$ and a point of the top side using $|A| \leq B - 2$. The other three cases are symmetric. The four resulting points are

precisely the images under Φ of the four elements of M. Since P is lattice-convex, the corresponding element of M lies in P.

A direct check of the four small polygons shows

$$\operatorname{conv}(C_2 \cup \{m\}) \cap \mathbb{Z}^2 = C_2 \cup \{m\}$$

for each $m \in M$, and the added point supplies the fourth parity class. Finally, in each case there are $p, a, b \in C_2$ with $m = p + b - a$. The relation

$$X_p^{-1} X_m = X_a^{-1} X_b$$

expresses X_m in the subgroup generated by C_2. □

Theorem 88.7 (Parallelogram–abelian fundamental group)**.** *Let $P = P_A(r)$ have active rank two. If P contains a nondegenerate lattice parallelogram, then*

$$\pi_1(I_A(r)) \cong L_A(r) \cong \mathbb{Z}^2.$$

Proof. By theorem 88.3, choose a minimal parallelogram of normalized area 1 or 2.

If the area is 1, translate one vertex to 0 and let u, v be its primitive edge directions. They form a basis of L, the four vertices represent all four classes in $L/2L$, and theorem 87.10 gives commuting basis generators X_u, X_v. The four-point core is parity-complete. Apply theorems 88.4 and 88.5 inductively to reach all of P.

If the area is 2, let the primitive edge directions be a, b. Modulo $2L$ they are equal, so the center

$$v = \frac{a + b}{2}$$

is a lattice point. With $u = a$, the pair u, v is a lattice basis and the five lattice points of the minimal parallelogram have coordinates

$$C_2 = \{0, u, v, 2v - u, 2v\}.$$

The relations

$$X_{2v} = X_v^2, \qquad X_{2v-u} = X_v X_u^{-1} X_v, \qquad X_u = X_{2v-u}^{-1} X_{2v}$$

force $[X_u, X_v] = 1$. This core represents three parity classes. If P contains only those three classes, theorems 88.4 and 88.5 again propagate the basis expressions to all of P. If the fourth class occurs, theorem 88.6 enlarges the core by one already-dependent generator to a parity-complete lattice-convex core, after which the same induction applies.

Thus $G(P)$ is generated by two commuting elements whose displacements are a basis of L. The displacement map (87.14) is therefore an isomorphism $G(P) \cong L$. Use theorems 87.7 and 87.9. □

88.2.2 The parallelogram-free branch

Call an affine lattice line ℓ *rich* if $|P \cap \ell| \geq 3$. Lattice-convexity implies that $P \cap \ell$ is a consecutive arithmetic progression in the primitive direction of ℓ.

Lemma 88.8 (Line-locality of repeated differences)**.** *If P contains no nondegenerate parallelogram and*

$$q - p = t - s$$

for two distinct ordered pairs of points of P, then the two segments lie on the same affine lattice line.

Proof. If the two segments lie on different parallel lines, the four endpoints form a nondegenerate parallelogram. If the lines are not parallel, equal displacement is impossible. □

A point lying on at least two rich lines will be called *shared*.

Lemma 88.9 (Transverse determinant lemma)**.** *Let P be lattice-convex and parallelogram-free. Suppose*

$$0, u, 2u, -v, v \in P$$

with primitive, nonparallel u, v. Then

$$|\det_L(u, v)| = 1.$$

Proof. Choose coordinates $u = (1, 0)$ and $v = (a, b)$ with $b > 0$ and $\gcd(a, b) = 1$. The triangle

$$T = \operatorname{conv}\{-v, v, 2u\}$$

lies in $\operatorname{conv}(P)$, hence every lattice point of T lies in P.

For $b \geq 3$, the intersection $T \cap (2u - T)$ has, in the u, v coordinate system, the form

$$\{xu + yv : |x - 1| + 2|y| \leq 1\}.$$

Take the ordinary lattice point $z = (m, 1)$ where m is nearest to $1 + a/b$. Then $y = 1/b$ and

$$|x - 1| = \operatorname{dist}(a/b, \mathbb{Z}) \leq \frac{\lfloor b/2 \rfloor}{b} \leq 1 - \frac{2}{b}.$$

Thus z and $2u - z$ both lie in $T \cap L \subseteq P$. They are not on the u-line, and

$$0 + 2u = z + (2u - z)$$

gives a nondegenerate parallelogram, a contradiction.

If $b = 2$, a lattice shear gives $v = (1, 2)$. Then the four points

$$(1, 2), \quad (0, -1), \quad (0, 0), \quad (1, 1)$$

lie in $T \cap L \subseteq P$ and satisfy

$$(1, 2) + (0, -1) = (0, 0) + (1, 1),$$

again a nondegenerate parallelogram. Hence $b = 1$. □

Lemma 88.10 (Endpoint-bridge exclusion)**.** *Let P be lattice-convex and parallelogram-free. A rich segment of lattice length at least 2 cannot have both endpoints interior to transverse rich lines.*

Proof. Write the rich segment as

$$0, u, 2u, \ldots, ku, \qquad k \geq 2,$$

and suppose the two endpoints are interior to transverse rich lines, so

$$-v, 0, v \in P, \qquad ku - w, ku, ku + w \in P.$$

By theorem 88.9,

$$|\det(u, v)| = |\det(u, w)| = 1.$$

Choose coordinates $u = (1, 0)$, $v = (0, 1)$. Then $w = (a, \varepsilon)$ with $\varepsilon = \pm 1$. Hence at the levels $y = 1$ and $y = -1$ the right transverse line contributes points with horizontal coordinates $k + a$ and $k - a$ (in some order). At least one of these is nonzero. On the same level the left transverse line contributes $(0, \pm 1)$. Convexity therefore forces a horizontal unit edge on that level. The main rich segment contains the horizontal unit edge from $(0, 0)$ to $(1, 0)$. The two equal unit edges form a nondegenerate parallelogram, a contradiction. □

Lemma 88.11 (Rich-line incidence)**.** *If P is lattice-convex and parallelogram-free, then every rich line contains at most two shared points; if it contains two, they are consecutive lattice points on that line.*

Proof. A point cannot be interior to two transverse rich lines: the four neighboring points have the same midpoint and form a nondegenerate parallelogram. Thus if a shared point p is interior to a rich line D, it is an endpoint of every transverse rich line through it. Let u be the primitive direction of D and let the transverse line contain $p, p+v, p+2v$. If D extended two steps from p in the same direction, then $p+2u, p+2v \in P$ and lattice-convexity would give

$$p+u+v = \frac{(p+2u)+(p+2v)}{2} \in P,$$

producing the parallelogram $p, p+u, p+v, p+u+v$. Hence a shared interior point can be at most one lattice step from each endpoint of D.

A point cannot be an endpoint of two transverse rich lines either: the two two-step extensions yield the same midpoint argument. Consequently, if a rich line had three shared points, its middle shared point would force the line to consist of exactly three consecutive points, while the two endpoints would be interior to transverse rich lines, contradicting theorem 88.10. If two shared points were nonconsecutive, neither could be an interior point of the rich line, so both would be endpoints and both would be interior to transverse rich lines, again contradicting theorem 88.10. □

Theorem 88.12 (Parallelogram-free homotopy theorem)**.** *Let $P = P_A(r)$ have active rank two and contain no nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq \bigvee^{\beta_1(I_A(r))} S^1. \tag{88.2}$$

In particular,

$$\pi_1(I_A(r)) \cong F_{\beta_1(I_A(r))}.$$

Proof. By theorem 88.8, every nontrivial positive-dimensional face identification in a fundamental simplex on P is supported inside the simplex on $P \cap \ell$ for some rich line ℓ. Let R be the union of these rich-line simplices together with the full 0-skeleton. Then the quotient $Q(P)$ is the pushout obtained from the full simplex $\Delta(P)$ by replacing R by its translation quotient $\overline{R}$. Since $\Delta(P)$ is contractible and $R \hookrightarrow \Delta(P)$ is a cofibration,

$$Q(P) \simeq \mathrm{Cofib}(R \to \overline{R}).$$

Distinct rich-line simplices meet only at shared vertices. By theorem 88.11, each rich line contains at most two such vertices, and if there are two they are consecutive. Hence R deformation retracts to a finite incidence graph. The quotient of a single rich-line simplex is the rank-one completion quotient, hence a circle; no positive-dimensional face is identified across distinct rich lines by theorem 88.8. Thus $\overline{R}$ retracts to a wedge of one circle for each rich line.

Choose a spanning forest in the incidence graph. In the cellular mapping cone of $R \to \overline{R}$, every non-tree incidence edge gives one 2-cell relation. Because the two shared points on the corresponding rich line are consecutive, its attaching word traverses that rich-line circle generator with coefficient ± 1. Choosing fundamental cycles with respect to the forest makes that circle generator unique to the corresponding non-tree relation. Successive elementary 2-cell/1-cell cancellations remove all 2-cells. The mapping cone therefore collapses to a finite graph. Its first Betti number is the first Betti number of $Q(P)$, so the graph is homotopy equivalent to the wedge in (88.2). Use theorem 87.7. □

Combining theorems 87.12, 88.7 and 88.12 gives the complete rank-two fundamental-group list

$$F_2, \qquad F_3, \qquad \mathbb{Z}^2.$$

88.3 The parallelogram branch and exact homotopy type

Assume $P = P_A(r)$ contains a nondegenerate parallelogram. By theorem 88.7,

$$\pi_1(Q(P)) \cong L \cong \mathbb{Z}^2.$$

Hence the periodic nerve

$$\widetilde{Q} := \mathcal{N}(P)$$

is the universal cover of $Q(P)$.

Let

$$R_L = \mathbb{Z}[L] \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}], \qquad M = H_2(\widetilde{Q}; \mathbb{Z}).$$

The next result is the module-theoretic step that turns the homology classification into a homotopy classification.

Lemma 88.13 (Universal-cover higher homology)**.** *For active rank two,*

$$H_j(\mathcal{N}(P); \mathbb{Z}) = 0 \qquad (j \geq 3).$$

Proof. The complex $\mathcal{N}(P)$ is the nerve of the family of translates $\{\lambda + P\}_{\lambda \in L}$. Since $P = \mathrm{conv}(P) \cap L$, these are convex lattice sets in $L \cong \mathbb{Z}^2$. Alon–Kalai–Matoušek–Meshulam prove that the nerve of every finite family of convex lattice sets in $\mathbb{Z}^d$ is $(2^d - 1)$-collapsible [41]; in dimension two it is therefore 3-Leray. Any finite cellular cycle in $\mathcal{N}(P)$ is supported on a finite induced subcomplex, so homology in degree at least 3 vanishes. □

Proposition 88.14 (Projectivity of the π_2 module)**.** *In the parallelogram branch, M is a finitely generated projective R_L-module.*

Proof. The universal-cover cellular chains are finitely generated free R_L-modules in each degree, and R_L is Noetherian, so M is finitely generated.

Let $\mathfrak{m}$ be a maximal ideal of R_L and let $k(\mathfrak{m})$ be its residue field. Since the group elements of L are units in R_L, their images define a character

$$\chi_{\mathfrak{m}} : L \to k(\mathfrak{m})^{\times}.$$

For the free R_L-chain complex $C_\bullet(\widetilde{Q})$, the hyperhomology spectral sequence gives

$$E^2_{p,q} = \mathrm{Tor}_p^{R_L}(k(\mathfrak{m}), H_q(\widetilde{Q}; \mathbb{Z})) \Longrightarrow H_{p+q}(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}).$$

Because $\widetilde{Q}$ is simply connected, $H_1(\widetilde{Q}) = 0$, and by theorem 88.13 only the rows $q = 0, 2$ can be nonzero. The $q = 0$ row is the Koszul homology of the two augmentation elements $s - 1, t - 1$ after evaluation at the residue character; it vanishes in degrees above 2, and for a nontrivial character it is acyclic.

On the other hand, theorems 87.15 and 88.1 give

$$H_3(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}) = 0.$$

No differential can kill or create the term $E^2_{1,2}$, so

$$\mathrm{Tor}_1^{R_L}(k(\mathfrak{m}), M) = 0$$

for every maximal ideal $\mathfrak{m}$. Localizing, the finite $R_{L,\mathfrak{m}}$-module $M_{\mathfrak{m}}$ has vanishing first Tor with its residue field. A minimal local presentation and Nakayama's lemma therefore show that $M_{\mathfrak{m}}$ is free. Thus M is locally free at every maximal ideal, hence finitely generated projective. □

Theorem 88.15 (Parallelogram-branch homotopy theorem)**.** *Let $P = P_A(r)$ have active rank two and contain a nondegenerate lattice parallelogram. Put*

$$b = \beta_2(I_A(r)).$$

Then $b \geq 1$ and

$$I_A(r) \simeq T^2 \vee \bigvee^{b-1} S^2. \tag{88.3}$$

Proof. By theorem 88.7, $\pi_1(Q(P)) \cong \mathbb{Z}^2$. The Hopf exact sequence therefore gives a surjection

$$H_2(Q(P);\mathbb{Z}) \twoheadrightarrow H_2(\mathbb{Z}^2;\mathbb{Z}) \cong \mathbb{Z},$$

so $b \geq 1$.

By theorem 88.14, M is finitely generated projective over

$$R_L \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}].$$

Swan's Laurent polynomial theorem [43] implies that every finitely generated projective module over this ring is free, so

$$M \cong R_L^q$$

for some $q \geq 0$.

To determine q, take any nontrivial character over a field. In the spectral sequence used above, the $q = 0$ row is acyclic and the only nonzero homology of the local-system complex can occur in degree 2. Hence

$$\dim H_2(Q(P);\Bbbk_\chi) = q.$$

The Euler characteristic of a rank-one local-system chain complex is the ordinary Euler characteristic. By theorem 88.2,

$$\chi(Q(P)) = 1 - 2 + b = b - 1.$$

Thus $q = b - 1$, and

$$\pi_2(Q(P)) \cong H_2(\widetilde{Q}) \cong R_L^{b-1}$$

by Hurewicz on the simply connected universal cover.

Let

$$Y = T^2 \vee \bigvee^{b-1} S^2.$$

Choose a map $T^2 \to Q(P)$ inducing the isomorphism on π_1; the torus 2-cell extends because the two basis loops commute. Choose maps of the 2-spheres representing an R_L-basis of $\pi_2(Q(P))$. Together they define

$$f : Y \to Q(P)$$

inducing isomorphisms on π_1 and on the R_L-module π_2. Lift to universal covers. Both universal covers are simply connected; both have zero homology above degree 2 by theorem 88.13, and the lifted map is an isomorphism in degree 2. Hence it is an integral homology equivalence. The simply connected homological Whitehead theorem [11] makes the lifted map a homotopy equivalence, and therefore so is f.

Finally use theorem 87.7. □

Chapter 89

Complete Rank-Two Classification by Lattice Geometry

The remaining distinction is geometric. In active rank two, the presence of a nondegenerate lattice parallelogram separates a toral branch from a free branch. The following results complete the classification and show that fixed rank does not bound second homology.

89.1 Complete active-rank-two classification

We can now combine the geometric and algebraic branches.

Theorem 89.1 (Complete rank-two homotopy classification)**.** *Let r be a completable residual with*

$$\operatorname{rank} L_A(r) = 2.$$

Then exactly one of the following holds.

(a) *The normalized optimal fiber $P_A(r)$ is nondegenerate-parallelogram-free. Then*

$$I_A(r) \simeq \bigvee^{\beta_1} S^1, \qquad \beta_1 \in \{2, 3\}.$$

(b) *The normalized optimal fiber contains a nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq T^2 \vee \bigvee^{\beta_2 - 1} S^2, \qquad \beta_2 \geq 1.$$

Consequently the possible active-rank-two homotopy types are precisely

$$S^1 \vee S^1, \qquad S^1 \vee S^1 \vee S^1, \qquad T^2 \vee \bigvee^{m} S^2 \quad (m \geq 0). \tag{89.1}$$

Proof. Apply theorems 87.12, 88.12 and 88.15. The two cases are complementary by definition. □

Corollary 89.2 (Integral homology profiles)**.** *In active rank two the integral homology is exactly one of*

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_j = 0 \ (j \geq 2),$$
$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^3, \quad H_j = 0 \ (j \geq 2),$$

or

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_2 \cong \mathbb{Z}^b, \quad H_j = 0 \ (j \geq 3), \qquad b \geq 1.$$

89.2 Explicit scalar realizations

The classification is not merely formal: all allowed types occur in four-letter scalar completion systems.

89.2.1 A fixed-system residual bifurcation

For an integer $d \geq 5$, define

$$\omega_d = (1-2d,\ 1-d,\ 1,\ 1+d). \tag{89.2}$$

This row vector is completion-total; for example

$$h_d = (1, 1, 2d-3, 1) > 0, \qquad \omega_d h_d^{\top} = 0.$$

The equal-length kernel is

$$K_{\omega_d} = \mathbb{Z}u \oplus \mathbb{Z}v, \qquad u = (1,-2,1,0), \quad v = (2,-3,0,1). \tag{89.3}$$

(We write the first basis vector as u below.)

Theorem 89.3 (Fixed-system rank-two bifurcation)**.** *For the single scalar system* (89.2)*, define*

$$r_W = 5d-4, \qquad r_T = 3d-4.$$

Then

$$\ell_{\omega_d}(r_W) = \ell_{\omega_d}(r_T) = 4,$$

and in both cases the active lattice is the full equal-length kernel $K_{\omega_d} \cong \mathbb{Z}^2$. Nevertheless,

$$I_{\omega_d}(r_W) \simeq S^1 \vee S^1, \qquad I_{\omega_d}(r_T) \simeq T^2.$$

Proof. For any $x \in \mathbb{N}^4$ of length n put

$$s = x_2 + 2x_3 + 3x_4.$$

Then

$$\omega_d \cdot x = (1-2d)n + ds.$$

For both residuals the completion equation implies

$$n \equiv 4 \pmod{d}.$$

Since $d \geq 5$ and length-4 solutions exist, both minimum lengths equal 4.

For r_W, setting $n = 4$ reduces the equation to $s = 3$. Its complete set of nonnegative solutions is

$$\mathcal{O}_W = \{(1,3,0,0), (2,1,1,0), (3,0,0,1)\}. \tag{89.4}$$

Relative to the first point, the two nonzero differences are

$$u = (1,-2,1,0), \qquad v = (2,-3,0,1),$$

which form the kernel basis. The normalized fiber has three affinely independent lattice points and no nondegenerate parallelogram, so theorem 89.1 gives $S^1 \vee S^1$.

For r_T, the length-4 equation is $s = 5$ and the complete optimal fiber is

$$\mathcal{O}_T = \{(0,3,1,0), (1,1,2,0), (1,2,0,1), (2,0,1,1)\}. \tag{89.5}$$

Relative to the first point, let

$$g = u, \qquad h = v - u = (1, -1, -1, 1).$$

Then the four normalized points are

$$\{0, g, h, g + h\},$$

a fundamental lattice parallelogram. Its quotient has Euler characteristic 0, so theorems 88.2 and 88.7 give $\beta_2 = 1$, and theorem 88.15 gives T^2. □

Thus even the matrix, the full equal-length kernel, the active lattice, its rank, and the minimum completion length can be held fixed while the global topology changes. The missing datum is the residual-dependent optimal-fiber translation geometry.

89.2.2 Unbounded second homology at fixed rank two

For $m \geq 2$ put

$$\alpha = 2m + 1$$

and define

$$\omega_m = \big(1 - m\alpha,\ 1 - (m-1)\alpha,\ 1,\ 1 + \alpha\big), \tag{89.6}$$

$$r_m = \alpha m(m-1) - 2m. \tag{89.7}$$

The system is completion-total; for example

$$h_m = \big(1, 1, (2m-2)\alpha - 3, 1\big) > 0$$

satisfies $\omega_m h_m^\top = 0$.

Theorem 89.4 (Unbounded rank-two sphere family)**.** *For every $m \geq 2$,*

$$\ell_{\omega_m}(r_m) = 2m.$$

The complete optimal fiber is

$$\mathcal{O}_m = \{A_0, \ldots, A_m, B, C\}, \tag{89.8}$$

where

$$A_t = (t, m-t, m-t, t), \qquad 0 \leq t \leq m,$$

$$B = (0, m+1, 0, m-1), \qquad C = (m-1, 0, m+1, 0).$$

Its active lattice equals the full equal-length kernel and has rank two. Moreover

$$I_{\omega_m}(r_m) \simeq T^2 \vee \bigvee^{m-2} S^2, \tag{89.9}$$

so

$$\beta_2(I_{\omega_m}(r_m)) = m - 1.$$

Proof. All four weights in (89.6) are congruent to 1 modulo α. Hence every completion of r_m with length n satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m+1}.$$

The vectors in (89.8) have length $2m$, so $\ell = 2m$.

At length $2m$, eliminating the common congruence term gives

$$x_2 + mx_3 + (m+1)x_4 = m(m+1).$$

Modulo m,

$$x_2 + x_4 \equiv 0 \pmod{m},$$

and $0 \le x_2 + x_4 \le 2m$. Thus $x_2 + x_4$ is 0, m, or $2m$. The three cases give respectively C, the family A_t, and B, proving (89.8).

Set

$$g = (1, -1, -1, 1), \qquad h = (0, 1, -m, m-1).$$

Every equal-length kernel vector is uniquely $ag + bh$: indeed the equations $\mathbf{1}^\top z = 0$ and $\omega_m \cdot z = 0$ imply

$$z_3 = -(m+1)z_1 - mz_2, \qquad z_4 = mz_1 + (m-1)z_2,$$

which corresponds to $a = z_1$, $b = z_1 + z_2$. Hence g, h form a basis of the full kernel. Relative to A_0,

$$A_t - A_0 = tg, \qquad B - A_0 = h, \qquad C - A_0 = (m-1)g - h.$$

Thus in active coordinates

$$P_m = \{(t, 0) : 0 \le t \le m\} \cup \{(0, 1), (m-1, -1)\}. \tag{89.10}$$

It contains a nondegenerate parallelogram, for example the four points corresponding to $(0, 1)$, $(m-1, -1)$, $(0, 0)$, and $(m-1, 0)$.

It remains to compute β_2. Count simplex orbits in the finite quotient using (89.10). The horizontal-only part is the rank-one quotient and has Euler characteristic 0. The total alternating contribution of faces containing exactly one of the two off-line points, excluding the already-counted singleton vertex, is -1 for each off-line point. Faces containing both off-line points have total alternating contribution 0. The only cross-identifications between the two one-off-line families occur for the m edge pairs

$$\{(0, 1), (t, 0)\} \sim \{(m-1, -1), (m-1-t, 0)\}, \qquad 0 \le t \le m-1.$$

No face of size at least 3 has such a cross-identification because the two off-line levels are unique. Each edge identification raises the Euler characteristic by 1. Therefore

$$\chi(I_{\omega_m}(r_m)) = m - 2.$$

The parallelogram branch has $\beta_1 = 2$, and theorem 88.2 gives no homology above degree 2. Hence

$$m - 2 = 1 - 2 + \beta_2, \qquad \beta_2 = m - 1.$$

Apply theorem 88.15 to obtain (89.9). □

Corollary 89.5 (Fixed-rank unbounded complexity)**.** *There exist completion-total four-letter scalar systems with*

$$\operatorname{rank} K = \operatorname{rank} L = 2, \qquad \pi_1 \cong \mathbb{Z}^2,$$

for which β_2 is arbitrarily large.

89.2.3 A free-rank-three example

Example 89.6 (The F_3 branch)**.** Take

$$\omega = (-11, -8, -5, 4), \qquad r = 25.$$

All weights are 1 modulo 3, so every completion length satisfies $n \equiv 2 \pmod 3$. Length 2 is impossible because the smallest sum of two weights is $-22 > -25$, while length 5 solutions exist. At length 5 the equation becomes

$$x_2 + 2x_3 + 5x_4 = 10,$$

whose complete nonnegative solution set is

$$\mathcal{O} = \{(0, 0, 5, 0), (0, 3, 1, 1), (1, 1, 2, 1), (3, 0, 0, 2)\}.$$

The active lattice has rank two, and the six unordered pair differences are distinct up to sign. Hence there is no nontrivial repeated-difference relation and no nondegenerate parallelogram. Presentation (87.13) is therefore free on the three nonzero normalized fiber points. By theorem 88.12,

$$I_\omega(25) \simeq S^1 \vee S^1 \vee S^1.$$

Together, theorems 89.3, 89.4 and 89.6 realize every homotopy type in (89.1).

89.3 Computational consistency checks

The theorems above are symbolic and do not depend on computation. Small exhaustive enumerations were used only as independent checks of the fiber formulas, cell-orbit conventions, and the rank-two dichotomy.

For the fixed-system family (89.2), direct enumeration for $5 \le d \le 20$ recovers exactly the two fibers in equations (89.4) and (89.5) and minimum length 4 for the stated residuals. For the unbounded family (89.6), direct enumeration for $2 \le m \le 12$ recovers exactly (89.8). The first cases are shown in table 89.1.

m	α	r_m	$\lvert\mathcal{O}_m\rvert$	predicted homotopy type
2	5	6	5	T^2
3	7	36	6	$T^2 \vee S^2$
4	9	100	7	$T^2 \vee 2S^2$
5	11	210	8	$T^2 \vee 3S^2$
6	13	378	9	$T^2 \vee 4S^2$

Table 89.1: Initial cases of the explicit unbounded family. The computations are consistency checks only; theorem 89.4 is proved symbolically.

A separate finite-grid audit of planar lattice-convex fibers was used during discovery to test the parallelogram/free-group dichotomy and the Betti patterns. These experiments motivated the geometric lemmas in section 88.2; they are not proof premises and no computational assumption enters theorem 89.1.

Chapter 90

From Displacement Kernel to Physical Completion Holonomy

The preceding rank-two classification determines the intrinsic homotopy type of the filled optimal-fiber quotient. We now ask what survives when loops are transported by an explicitly specified physical LIFO policy. The displacement map separates lattice motion from pure loops; only the latter can carry holonomy invisible to the semantic endpoint.

90.1 Optimal fibers, displacement, and pure loops

Let $\mathcal{O}(r) \subset \mathbb{N}^s$ be the set of minimum-length completion count vectors at residual r. Fix $x_0 \in \mathcal{O}(r)$ and put

$$P = \mathcal{O}(r) - x_0, \qquad L = \langle \mathcal{O}(r) - \mathcal{O}(r) \rangle_{\mathbb{Z}}.$$

Let $I = I(r)$ denote the filled shortest-prefix quotient associated with this optimal fiber. In active rank two, the completion-fiber construction gives a canonical surjection

$$\delta : \pi_1(I) \twoheadrightarrow L \cong \mathbb{Z}^2,$$

which records the lattice displacement of a loop.

Definition 90.1 (Pure-loop sector)**.** Define

$$K_0 = \ker \delta$$

and

$$K_{\mathrm{lin}} = \ker \big(H_1(I; \mathbb{Z}) \to L \big).$$

The first group is the full zero-displacement sector; the second records the part already visible after abelianization.

Theorem 90.2 (Rank-two displacement-kernel visibility trichotomy)**.** *Assume* $\operatorname{rank} L = 2$*. Then the active-rank-two alternatives have the following pure-loop structure.*

1. *If I is toral, $\pi_1(I) \cong \mathbb{Z}^2$ and δ is an isomorphism. Hence*

$$K_0 = 1, \qquad K_{\mathrm{lin}} = 0.$$

2. *If $I \simeq S^1 \vee S^1$, then $\pi_1(I) \cong F_2$ and*

$$K_0 = [F_2, F_2], \qquad K_{\mathrm{lin}} = 0.$$

Thus pure topology exists but is nonabelian-only.

3. *If* $I \simeq S^1 \vee S^1 \vee S^1$, *then* $\pi_1(I) \cong F_3$ *and*

$$K_{\mathrm{lin}} \cong \mathbb{Z}.$$

The full kernel $K_0 = \ker(F_3 \twoheadrightarrow \mathbb{Z}^2)$ *is free of countably infinite rank, but one pure direction is already visible in* H_1.

Proof. In the toral branch the displacement theorem identifies $\pi_1(I)$ with L. In the F_2 branch, abelianization gives a surjection $\mathbb{Z}^2 \to \mathbb{Z}^2$, hence an isomorphism, so the kernel of $F_2 \to \mathbb{Z}^2$ is exactly the commutator subgroup. In the F_3 branch, abelianization gives a surjection $\mathbb{Z}^3 \to \mathbb{Z}^2$ with primitive rank-one kernel. The assertions about the full kernels follow from Nielsen–Schreier and the standard structure of infinite-index normal subgroups of nonabelian free groups. □

Definition 90.3 (Displacement-compatible physical transport)**.** A flat physical transport is a functor from the fundamental groupoid $\Pi_1(I)$ to a groupoid of invertible physical comparison maps. At a basepoint it induces

$$\mathrm{Hol} : \pi_1(I) \to G_{\mathrm{phys}}.$$

It is displacement-compatible if there is a homomorphism $c : G_{\mathrm{phys}} \to L$ such that

$$c \circ \mathrm{Hol} = \delta.$$

Then $\mathrm{Hol}(K_0)$ lies in the count-invisible subgroup $\ker c$.

Proposition 90.4 (Gauge persistence of pure holonomy)**.** *If* $\mathrm{Hol}|_{K_0}$ *is nontrivial, no objectwise change of physical trivialization within the same flat invertible transport category can make every pure loop trivial. Such a change only conjugates the holonomy representation.*

This proposition is deliberately category-relative. It does not say that topology forces debt in an arbitrary DPDA.

90.2 A fixed completion system with a free/toral residual bifurcation

Fix $d \geq 5$ and the four weights

$$\omega_d = (1 - 2d,\ 1 - d,\ 1,\ 1 + d).$$

The equal-length kernel contains the basis

$$u = (1, -2, 1, 0), \qquad v = (2, -3, 0, 1).$$

Consider the two residuals

$$r_W = 5d - 4, \qquad r_T = 3d - 4.$$

Both have minimum completion length 4 and the same active rank two.

At r_W the optimal completions are

$$x_0 = (1, 3, 0, 0), \qquad x_u = (2, 1, 1, 0), \qquad x_v = (3, 0, 0, 1),$$

so the normalized fiber is $\{0, u, v\}$ and

$$I(r_W) \simeq S^1 \vee S^1.$$

At r_T the optimal completions are

$$(0,3,1,0), \quad (1,1,2,0), \quad (1,2,0,1), \quad (2,0,1,1),$$

which form a lattice parallelogram; hence

$$I(r_T) \simeq T^2.$$

By Theorem 90.2, the first residual has a nonabelian-only pure sector $[F_2, F_2]$, whereas the second has no pure sector at all.

90.3 Stable-LIFO comparison transport

For an equal-length count vector $x = (x_1, \dots, x_s)$ define the slot word

$$\eta_{\mathrm{slot}}(x) = 1^{x_1} 2^{x_2} \cdots s^{x_s}.$$

Temporarily label its slots from left to right. For two equal-length count vectors x, y, define the Stable-LIFO matching $\tau_{x,y}$ by preserving same-type occurrences stably from left to right and matching all remaining source positions to remaining target positions in reverse order. Then

$$\tau_{y,x} = \tau_{x,y}^{-1}.$$

For the free optimal endpoints, the canonical words are

$$w_0 = 1222, \qquad w_u = 1123, \qquad w_v = 1114.$$

The three comparison permutations are

$$\tau_{0u} = (234), \qquad \tau_{0v} = (24), \qquad \tau_{uv} = (34).$$

On moved slots $\{2,3,4\}$ identify

$$r = (012), \qquad s = (02).$$

Then

$$[r, s] = (021),$$

corresponding on the original slots to (243).

Theorem 90.5 (Pure nonabelian Stable-LIFO holonomy)**.** *The Stable-LIFO comparison transport on the free optimal fiber is flat and sends a zero-displacement commutator in $K_0 = [F_2, F_2]$ to the nontrivial 3-cycle*

$$(243).$$

The image is invisible to every abelian permutation observable.

Proof. The triangle relation is

$$\tau_{0u}\tau_{uv} = \tau_{0v}$$

under chronological right action, so the endpoint comparisons define a flat local system on the three-point free quotient. The two displacement generators map to r and s, and their commutator maps to $(021) \neq 1$. Since every homomorphism from S_4 to an abelian group kills commutators, the certificate is genuinely nonabelian. □

90.4 The full free shortest-completion episode

A prefix count vector y determines

$$n = \mathbf{1}^\top y, \qquad s = y_2 + 2y_3 + 3y_4.$$

For r_W,

$$r_W + \omega_d \cdot y = r_W + (1 - 2d)n + ds.$$

Since $0 \le n \le 4 < d$, residual equality on shortest prefixes is equivalent to equality of (n, s). The full quotient therefore has vertex set

$$\begin{aligned} V_W = {} & \{(0,0)\} \cup \{(1,s) : 0 \le s \le 3\} \\ & \cup \{(2,s) : 0 \le s \le 3\} \\ & \cup \{(3,s) : 0 \le s \le 3\} \cup \{(4,3)\}, \end{aligned}$$

so $|V_W| = 14$. Its cell vector is

$$(f_0, f_1, f_2, f_3) = (14, 28, 15, 2).$$

Theorem 90.6 (Full free-episode realization)**.** *The entire* 14*-vertex shortest-prefix quotient admits a flat S_3 connection whose endpoint products are*

$$w_0 \mapsto e, \qquad w_u \mapsto r, \qquad w_v \mapsto s.$$

All square relations and both 3*-cube boundaries are flat. The pure commutator of Theorem 90.5 therefore survives on the actual one-letter shortest-completion episode, not merely on a compressed comparison complex.*

Construction. Assign edge labels by a spanning-tree gauge and extend them over the remaining edges so that each of the 15 squares has trivial boundary product. The two 3-cubes then have flat boundary automatically. The explicit connection uses only the four local values e, r, r^{-1}, s and the transposition $a = (01)$ at the mixed vertices; direct multiplication gives the endpoint products above for all 20 shortest words. □

Chapter 91

Sharp LIFO Carrier Costs and Toral Correction

The free branch supports genuine pure nonabelian holonomy. The toral branch fills the commutator cell, so global descent instead imposes a correction problem. The resulting comparison gives exact carrier, ancestry, and correction costs inside the stated Stable-LIFO/literal-stack categories.

91.1 Sharp literal-stack carrier cost

Before measuring a carrier, we fix the implementation category. A *buffered ancestry refinement* of capacity b allows at most b temporary finite-control handles that remember distinguished stack-cell ancestries during a one-top macro; the handles are cleared at the endpoint. A macro is ancestry-bijective if every initial distinguished ancestry has exactly one final descendant. Section 91.7 proves the general bottom-move lower bound $b \geq n-1$ when the bottom of an n-cell distinguished prefix changes ancestry. We now impose a specific physical persistence category: no epsilon moves, no permanent finite-control sheet register, common endpoint control and zero potential, literal stack alphabet $\{1,2,3,4\}$ only, and identical endpoint Parikh counts across sheets. Persistent monodromy must therefore live in stack order.

For a zero-weight literal count vector $z \in \mathbb{N}^4$, define its order capacity by the multinomial number

$$M(z) = \frac{|z|_1!}{\prod_i z_i!}.$$

For $k \geq 2$, set

$$g_k^{\mathrm{ord}}(\omega_d) = \min\{|z|_1 : z \neq 0,\ \omega_d z = 0,\ M(z) \geq k\}.$$

Lemma 91.1 (Arithmetic carrier girth)**.** *Every nonempty zero-weight literal word has length at least d. Moreover*

$$g_3^{\mathrm{ord}}(\omega_d) = g_6^{\mathrm{ord}}(\omega_d) = d.$$

Proof. Each weight is congruent to 1 modulo d. Hence $\omega_d z = 0$ implies

$$|z|_1 \equiv 0 \pmod{d},$$

so every nonempty zero-weight word has length at least d. Equality is attained by

$$z^{(3)} = (0, 1, d-1, 0)$$

for three orderings and by

$$z^{(6)} = (0, 2, d-3, 1)$$

for six orderings. □

For the six-state carrier, let

$$T_d = 2\,3^{d-4}$$

and index the six permutations P_g of the three letters 234 by $g \in S_3$. Set

$$Z_g = P_g T_d.$$

All Z_g have the same Parikh vector $z^{(6)}$ and zero total weight.

Theorem 91.2 (Sharp free literal carrier)**.** *There is a real-time one-top realization of the full free connection such that*

$$Z_g \xrightarrow{w_0} Z_g, \qquad Z_g \xrightarrow{w_u} Z_{gr}, \qquad Z_g \xrightarrow{w_v} Z_{gs},$$

with no epsilon moves and no permanent sidecar. The minimum persistent literal order-carrier height in the stated category is exactly

$$\boxed{d}.$$

The chosen realization needs exactly two transient ancestry handles. In the deferred-carrier subclass, exactly three transient path states are necessary and sufficient.

Proof. The lower bound on persistent height is Lemma 91.1: every nonempty zero-weight literal carrier has length divisible by d, hence length at least d. For the upper bound use the six same-Parikh words $Z_g = P_g T_d$, $g \in S_3$. Only the distinguished three-cell prefix P_g is changed; the common suffix T_d remains untouched throughout.

For a shortest word w, let $\sigma(w)$ be the endpoint product supplied by the flat connection of Theorem 90.6. During the four input-synchronous steps, finite control records the current quotient vertex and two temporary ancestry handles. The first two distinguished carrier cells are exposed into those handles, the third remains visible, and on the terminal step the visible third cell together with the two handles determines the incoming prefix permutation g. The transition rewrites that visible cell by the three-cell prefix $P_{g\sigma(w)}$. Thus

$$Z_g \xrightarrow{w_0} Z_g, \qquad Z_g \xrightarrow{w_u} Z_{gr}, \qquad Z_g \xrightarrow{w_v} Z_{gs},$$

with one input symbol consumed per transition and no ε-move. The semantic shortest-episode coordinate in control supplies the unique potential needed to verify the residual identity at each step, so this is an exact lift. Section 91.6 gives the full edge certificate for all twenty shortest words.

Necessity of two ancestry handles follows from Lemma 91.6: the generator $s = (02)$ changes the ancestry occupying the bottom of the distinguished three-cell prefix, so any ancestry-bijective top-access macro needs at least $3 - 1 = 2$ handles. The construction above uses two.

Finally restrict to the deferred-carrier subclass, in which the distinguished carrier order is not changed before the terminal rewrite. The three histories 222, 231, and 411 reach the same preterminal semantic vertex with the same next input but with accumulated products s, a, e; their common last input must therefore produce three distinct endpoint multipliers e, r, s. Determinism forces three transient path states. The explicit connection table has at most three accumulated-product classes at every vertex, so three states suffice. Hence both locality bounds are sharp in the stated subclasses. □

91.2 Toral curvature and the difference between local flattening and torus descent

For the toral residual write the four optimal endpoints as $0, g, h, g+h$. Raw Stable-LIFO endpoint comparisons are

$$A = (1243), \qquad B = (24), \qquad C = (1234), \qquad D = (23).$$

The two chronological route products are

$$AB = (143), \qquad CD = (134) = (143)^{-1}.$$

Thus the raw parallelogram has nontrivial 3-cycle curvature.

On one isolated square, one edge change is enough to make the two route products equal. But a torus local system must descend globally. If normalized path products to the three nonzero lattice points are U, V, W, torus descent requires

$$[U, V] = e, \qquad W = UV = VU.$$

Hence the corrected endpoint tuple must have the form

$$(\widetilde{A}, \widetilde{B}, \widetilde{C}, \widetilde{D}) = (U, V, V, U), \qquad UV = VU.$$

Theorem 91.3 (Minimum number of globally altered endpoint comparisons)**.** *Any torus-descending correction of the raw tuple (A, B, C, D) changes at least three of the four endpoint comparisons. Three changes suffice.*

Proof. Because $A \neq D$, at least one of the g-pair must change; because $B \neq C$, at least one of the h-pair must change. If exactly two comparisons changed, the retained commuting pair would have to be one of $(A, B), (A, C), (D, B), (D, C)$. Each pair has nontrivial commutator (243), so two changes are impossible. Taking $U = A$ and $V = e$ gives (A, e, e, A) and changes exactly three entries. □

Let d_T be Cayley distance in S_4 with respect to transpositions and define

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V)$$

subject to $[U, V] = e$.

Theorem 91.4 (Exact global toral correction Pareto law)**.** *The minimum total correction distance is*

$$\boxed{F_{\min} = 4},$$

attained by

$$U = (243), \qquad V = (234) = U^{-1},$$

with four single-transposition corrections. Among corrections changing exactly three endpoint comparisons, the minimum total transposition length is 6 *and the minimum possible maximum support is* 4*. Hence the exact nondominated profiles*

$$(\#\ \textit{altered},\ \textit{total transposition length}, \max\ \textit{support})$$

are

$$\boxed{(3, 6, 4), \qquad (4, 4, 2).}$$

Proof. For any commuting U, V,

$$F(U,V) \geq d_T(A,D) + d_T(B,C) = 2 + 2 = 4$$

by the triangle inequality. Equality is realized by the displayed U, V, using

$$A(12) = U, \quad B(34) = V, \quad C(12) = V, \quad D(34) = U.$$

For the three-change branch, retain one raw edge and minimize over its centralizer. The four symmetric cases give minimum total length 6 and force one correction to move all four slots. These two profiles are incomparable and every other correction is dominated by one of them. □

91.3 The full toral episode and repair/debt tradeoff

For r_T the residual quotient again uses (n, s), now with vertex set

$$\begin{aligned} V_T = {} & \{(0,0)\} \cup \{(1,s) : 0 \leq s \leq 3\} \\ & \cup \{(2,s) : 0 \leq s \leq 5\} \\ & \cup \{(3,s) : 2 \leq s \leq 5\} \cup \{(4,5)\}. \end{aligned}$$

Thus

$$|V_T| = 16, \qquad (f_0, f_1, f_2, f_3) = (16, 38, 28, 6).$$

There are 40 shortest completion words.

For the minimum-distance correction of Theorem 91.4, $V = U^{-1}$ and the image is the cyclic group

$$\langle U \rangle \cong C_3.$$

Define on the active lattice

$$\rho_T(ag + bh) = U^{a-b}.$$

Because the toral displacement map is an isomorphism, this defines a flat local system on the entire quotient. The 40 shortest words split into endpoint products

$$16 \text{ words} \mapsto e, \qquad 12 \mapsto U, \qquad 12 \mapsto V.$$

Theorem 91.5 (Full toral realization and exact repair/debt tradeoff)**.** *The minimum-distance toral repair admits a full* 16*-vertex real-time realization with nontrivial displacement-visible C_3 transport. If this transport is stored persistently in literal stack order, the minimum carrier height is d; the three-slot carrier uses two ancestry handles and, in the deferred subclass, three transient path states.*

Complete erasure of transport is possible by setting $U = V = e$. Its total correction distance from the raw tuple is

$$\boxed{8}.$$

Thus toral topology forces no pure debt: one may retain the nearest nontrivial flat transport at distance 4*, or pay distance* 8 *and remove persistent order memory entirely.*

Proof. The corrected image C_3 acts only on three slots, so the free height-d carrier construction applies. The lower bound is again the zero-weight congruence. For complete erasure,

$$\ell_T(A) + \ell_T(B) + \ell_T(C) + \ell_T(D) = 3 + 1 + 3 + 1 = 8.$$

Since $K_0 = 1$ in the toral branch, none of this retained transport is pure zero-displacement holonomy; it factors through the displacement lattice. □

91.4 The free–toral topology-to-LIFO dichotomy

The fixed family gives a clean contrast with every algebraic parameter except the residual held fixed.

free: local flatness permits global pure nonabelian holonomy,
toral: the filled commutator cell forbids pure holonomy
and converts raw noncommutativity into correction cost.

The distinction is not simply "topology requires memory." In the free branch, the missing parallelogram leaves a nonabelian zero-displacement sector in which order memory can live. In the toral branch, the parallelogram fills the commutator relation, so a flat physical representation must commute on the lattice generators. Noncommutativity survives only as the cost of repairing a chosen raw comparison policy.

91.5 Scope and representation dependence

The following distinctions are essential.

- The group K_0, the existence of nontrivial pure holonomy, the free/toral split, and the literal zero-weight height lower bound are intrinsic within the stated completion-fiber and carrier categories.
- The concrete permutations r, s, A, B, C, D and the correction profiles of Theorem 91.4 are relative to the fixed Stable-LIFO endpoint-comparison policy, although they are invariant under simultaneous conjugation of slot labels.
- The two-handle and three-state bounds are exact for the chosen three-slot and deferred-carrier subclasses; they are not claimed as universal lower bounds over all possible pushdown policies.
- No theorem here says that nontrivial topology alone forces physical debt under unrestricted finite control.

91.6 Full free-episode connection certificate

The full shortest-prefix quotient at r_W has vertices

$$V_W = \{(0,0)\} \cup \{(1,s),(2,s),(3,s) : 0 \le s \le 3\} \cup \{(4,3)\}$$

and f-vector $(14, 28, 15, 2)$. A directed edge is written $(n,s) \xrightarrow{i} (n+1, s+i-1)$. With chronological right multiplication, put

$$r = (012), \qquad r^{-1} = (021), \qquad s = (02), \qquad a = (01).$$

The following table is an explicit flat S_3 connection; omitted edges do not exist.

source	i	σ
$(0,0)$	$1,2,3,4$	e
$(1,0)$	$1,2,3,4$	e
$(1,1)$	1	e
$(1,1)$	2	r^{-1}
$(1,1)$	3	a
$(1,2)$	1	e
$(1,2)$	2	a
$(1,3)$	1	e
$(2,0)$	1	e
$(2,0)$	2	e
$(2,0)$	3	r
$(2,0)$	4	e
$(2,1)$	1	e
$(2,1)$	2	e
$(2,1)$	3	a
$(2,2)$	1	r
$(2,2)$	2	a
$(2,3)$	1	e
$(3,0)$	4	s
$(3,1)$	3	r
$(3,2)$	2	e
$(3,3)$	1	s

Substitution into the fifteen square relations

$$\sigma_{v,i}\sigma_{v+e_i,j} = \sigma_{v,j}\sigma_{v+e_j,i}$$

uses only $r^{-1}r = e$ and $as = r$ besides identities with the same factor on both sides. Flatness of all square faces implies compatibility on the two 3-cube boundaries. Endpoint products are e, r, s on the three optimal boxes, so the commutator image is $(021) \neq e$.

91.7 Ancestry model and sharp local-memory bounds

A buffered ancestry refinement of capacity b augments finite control with at most b temporary handles. A one-top microstep may move the visible cell's ancestry into a handle, attach a stored handle to the rewritten visible cell, or push a new decorated cell carrying a stored handle. A closed macro is ancestry-bijective when every initial distinguished ancestry has exactly one final descendant.

Lemma 91.6 (Bottom-move bound)**.** *If an ancestry-bijective top-access macro on n distinguished cells changes the ancestry occupying the bottom distinguished position, then $b \geq n-1$. The bound is sharp.*

Proof. Before the original bottom cell can be inspected, all $n-1$ cells above it have been removed from the live stack. Their ancestry classes must remain available for the final bijection, hence all $n-1$ must be retained externally at that instant. Conversely, pop the upper $n-1$ ancestries into handles, rewrite the exposed bottom, and rebuild the prefix in the target order. □

For the free three-slot carrier, the generator $s = (02)$ moves the third distinguished cell, so two handles are necessary and sufficient. The strict-streaming no-go on the full

interval follows because, before the third ancestry is visible, every second-layer carrier update lies in a common order-two stabilizer. Flatness then forces the two endpoint generators into a common conjugate of that stabilizer, hence forces them to commute. The noncommuting pair r, s therefore requires transient path memory in addition to the two ancestry handles.

In the deferred-carrier subclass, the three histories 222, 231, and 411 reach the common preterminal vertex with accumulated products s, a, e and share the same final input. They must be distinguished before the final rewrite, giving a lower bound of three transient path states. The edge table above has local accumulated-product sets of size at most three, so three states suffice.

91.8 Global torus-descent audit

For the toral residual, a normalized flat endpoint quadruple has the form

$$(U, V, V, U), \qquad UV = VU.$$

Relative to the raw tuple

$$A = (1243), \quad B = (24), \quad C = (1234), \quad D = (23),$$

at least three entries must change: $A \neq D$, $B \neq C$, and every possible retained pair from $\{A, D\} \times \{B, C\}$ has nontrivial commutator. Three changes are attained by (A, e, e, A).

For Cayley distance d_T with all transpositions as generators,

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V) \geq 4.$$

Equality is attained by

$$U = (243), \qquad V = (234) = U^{-1},$$

through

$$A(12) = U, \qquad B(34) = V, \qquad C(12) = V, \qquad D(34) = U.$$

If exactly three raw comparisons are altered, one raw edge is retained. The four centralizer cases give:

retained	fixed generator	min F	min max supp
A	$U = A$	6	4
D	$U = D$	6	4
B	$V = B$	6	4
C	$V = C$	6	4

so the nondominated profiles are $(3, 6, 4)$ and $(4, 4, 2)$. The minimum-distance pair generates C_3 and acts on only three distinguished slots; therefore it reuses the exact height-d three-slot carrier with two ancestry handles. The full toral quotient has vertex set

$$\{(0, 0)\} \cup \{(1, s) : 0 \leq s \leq 3\} \cup \{(2, s) : 0 \leq s \leq 5\} \cup \{(3, s) : 2 \leq s \leq 5\} \cup \{(4, 5)\}$$

and f-vector $(16, 38, 28, 6)$. The representation $\rho_T(ag + bh) = U^{a-b}$ therefore gives a flat connection on all 40 shortest words.

91.9 Topology-to-LIFO synthesis

At active rank two the intrinsic topology and the physical transport layer are now cleanly separated. The group π_1 and displacement kernel determine which pure loop sectors exist; the chosen transport category determines whether those sectors appear as nonabelian holonomy, literal carrier depth, transient ancestry cost, or toral correction. No claim is made that topology alone forces a unique physical debt law.

Higher-rank continuation. The present chapter fixes the physical category and establishes the rank-two free/toral transfer in sharp form. The next chapter first raises the intrinsic completion geometry to arbitrary active rank and then returns to this transport layer: the displacement-marked group, exact product/prism/apex kernel calculus, finite physical pure-image transfer, general carrier girth, support-based ancestry law, connection-memory multiplicity, repair optimization, and an all-rank S_3/A_3 carrier family are developed there without changing the category-sensitive interpretation established here.

Chapter 92

Lattice-Convex Realization, Higher-Rank Topology, and Marked Holonomy

Research branch — Higher-Rank Optimal-Fiber Realization, Rank Raising, and Marked Transfer. *The rank-two classification established a rigid planar picture but did not characterize which finite lattice configurations can occur as optimal fibers or how topology and physical transport behave after active rank is raised. This branch supplies the exact lattice-convex realization theorem and the product/apex/prism rank-raising calculus, then upgrades displacement to a marked-group invariant and transfers its pure-loop sector through finite physical holonomy, literal carrier locality, connection memory, and repair cost. The final all-rank family separates active rank from finite holonomy complexity while retaining an exact carrier-height law.*

For a finite spanning lattice-convex configuration $P \subset L$, let $Q(P)$ denote the periodic translation quotient of the optimal-fiber nerve introduced in the preceding rank-two branch. The completion baseline already proved that every normalized optimal fiber is lattice-convex. The first theorem below proves the converse and therefore turns the intrinsic lattice model into an exact realization theorem.

92.1 Intrinsic periodic-fiber model

A finite set $P \subset L$ in a rank-s lattice is *lattice-convex* when $P = \mathrm{conv}(P) \cap L$, and it spans L when $L = \langle P - P \rangle_{\mathbb{Z}}$. After translation assume $0 \in P$. Let $\mathcal{N}(P)$ be the simplicial complex on vertex set L in which a finite $F \subset L$ is a simplex exactly when $c - F \subseteq P$ for some $c \in L$, and set $Q(P) = |\mathcal{N}(P)|/L$.

For $0 \in P$, the quotient has the difference presentation

$$\begin{aligned}\pi_1(Q(P)) \cong \big\langle X_p \ (p \in P),\ X_0 = 1 \ \big| \\ X_p^{-1} X_q = X_s^{-1} X_t \ \text{ whenever } q - p = t - s \big\rangle.\end{aligned} \tag{92.1}$$

and a canonical displacement homomorphism

$$\delta : \pi_1(Q(P)) \longrightarrow L, \qquad \delta(X_p) = p. \tag{92.2}$$

This is the presentation established in the preceding optimal-fiber topology branch and will be used below without rederiving the rank-two classification.

92.2 Every lattice-convex configuration is an optimal fiber

The necessity in the completion baseline proved in the preceding branch has a converse.

Definition 92.1 (Completion-total system)**.** An integer counting matrix A is *completion-total on its integer image* if every $b \in A\mathbb{Z}^d$ has a nonnegative preimage. A sufficient condition is the existence of a strictly positive vector $h \in \ker_{\mathbb{Z}} A$: any integer solution can then be translated by a large nonnegative multiple of h.

Theorem 92.2 (Lattice-convex realization)**.** *Let P be a finite lattice-convex configuration spanning a rank-s lattice L. Then there exist an integer matrix A, a completable residual r, and a strictly positive $h \in \ker_{\mathbb{Z}} A$ such that the normalized optimal fiber $P_A(r)$ is affinely lattice-isomorphic to P and its active lattice has rank exactly s.*

Proof. Identify L with $\mathbb{Z}^s$ and translate so that $0 \in P$. Since $\operatorname{conv}(P)$ is a lattice polytope, choose integral affine facet functions

$$\ell_j(z) = b_j - a_j^{\top} z \qquad (1 \le j \le m)$$

such that

$$\operatorname{conv}(P) = \{z \in \mathbb{R}^s : \ell_j(z) \ge 0 \ \forall j\}.$$

Choose $B \in \mathbb{Z}^s$ so large coordinatewise that $z + B \in \mathbb{N}^s$ for every $z \in P$. Define the integral affine embedding

$$E_0(z) = \big(z + B, \ell_1(z), \dots, \ell_m(z)\big) \in \mathbb{Z}^{s+m}.$$

For lattice points $z \in \mathbb{Z}^s$ we have

$$E_0(z) \ge 0 \quad \Longleftrightarrow \quad z \in P.$$

Indeed, points of P satisfy every facet inequality and the chosen coordinate shifts; a lattice point outside $P = \operatorname{conv}(P) \cap \mathbb{Z}^s$ violates at least one facet inequality.

Choose an integer

$$C \ge \max_{z \in P} \mathbf{1}^{\top} E_0(z)$$

and append one balancing coordinate:

$$E(z) = \big(E_0(z),\ C - \mathbf{1}^{\top} E_0(z)\big) \in \mathbb{Z}^n, \qquad n = s + m + 1.$$

Then

$$E(z) \ge 0 \Longleftrightarrow z \in P, \qquad \mathbf{1}^{\top} E(z) = C \quad (z \in \mathbb{Z}^s). \tag{92.3}$$

Let

$$D = \{E(z) - E(0) : z \in \mathbb{Z}^s\} \subset \mathbb{Z}^n.$$

The first s coordinates of $E(z) - E(0)$ are exactly z, so projection onto those coordinates is a left inverse. Hence $D \cong \mathbb{Z}^s$ is a saturated direct summand of $\mathbb{Z}^n$. By (92.3), every $d \in D$ has coordinate sum zero.

Append one final coordinate and set

$$h = (1, \dots, 1) \in \mathbb{Z}^{n+1}, \qquad K = (D \times \{0\}) + \mathbb{Z}h \subset \mathbb{Z}^{n+1}.$$

The last coordinate of h is one, so quotienting first by $\mathbb{Z}h$ identifies $\mathbb{Z}^{n+1}/\mathbb{Z}h$ with $\mathbb{Z}^n$; the image of K is the saturated lattice D. Thus K is saturated. Consequently there is an integer matrix A with

$$\ker_{\mathbb{Z}} A = K.$$

The vector h is strictly positive, so A is completion-total on its integer image.

Put

$$x_0 = (E(0), 0) \in \mathbb{N}^{n+1}, \qquad r = -Ax_0.$$

Any integer point in the fiber of Ax_0 has a unique expression

$$x = x_0 + (d, 0) + th, \qquad d \in D,\ t \in \mathbb{Z}.$$

Its last coordinate is t, so nonnegativity forces $t \geq 0$. Writing $d = E(z) - E(0)$ gives

$$\mathbf{1}^\top x = C + t(n+1).$$

Hence every minimum-length point has $t = 0$. At $t = 0$, nonnegativity is equivalent by (92.3) to $z \in P$. Therefore

$$\mathcal{O}_A(r) = \{(E(z), 0) : z \in P\}.$$

The active difference lattice is $D \times \{0\}$, which has rank s, and the map $z \mapsto (E(z), 0)$ is an affine lattice isomorphism from P onto the normalized optimal fiber. □

Corollary 92.3 (Exact fiber characterization)**.** *Up to affine lattice equivalence, the normalized optimal fibers of completion-total integer counting systems are exactly the finite lattice-convex configurations.*

Proof. Necessity is the completion baseline proved in the preceding branch; sufficiency is theorem 92.2. □

92.3 Minimal fibers and free groups in every rank

A spanning rank-s configuration has at least $s + 1$ points. At equality the quotient is completely rigid.

Theorem 92.4 (Minimal active fiber)**.** *Let $P \subset L$ be a normalized, spanning, lattice-convex configuration with* $\operatorname{rank} L = s$ *and $|P| = s + 1$. Then*

$$Q(P) \simeq \bigvee^{s} S^1, \qquad \pi_1(Q(P)) \cong F_s, \qquad H_j(Q(P); \mathbb{Z}) = 0 \quad (j \geq 2).$$

Moreover the abelianized displacement map

$$H_1(Q(P); \mathbb{Z}) \longrightarrow L$$

is an isomorphism, so $K_{\mathrm{lin}}(P) = 0$.

Proof. The $s + 1$ points are affinely independent. No positive-dimensional subset can be a nontrivial translate of another subset of P: such a translation would give either a repeated nonzero difference among four vertices or three collinear vertices, in either case an affine dependence. Thus the only identifications in a fundamental simplex are the singleton vertices, which form one translation orbit. Consequently

$$Q(P) \cong \Delta^s / P^{(0)}.$$

Collapsing the $s + 1$ vertices of a contractible simplex to one point gives the homotopy cofiber of the inclusion of a discrete $(s + 1)$-point set into Δ^s, hence a wedge of s circles.

After translating one point to 0, the remaining s points generate L by definition. Their affine independence makes them linearly independent, so they form a $\mathbb{Z}$-basis of L. The s circle generators map to this basis under displacement, proving the final assertion. □

Proposition 92.5 (Explicit scalar realization)**.** *For every $s \geq 1$, the scalar weight vector*

$$\omega_s = (\underbrace{1, \ldots, 1}_{s+1 \text{ entries}}, -(s+1))$$

with residual $r = -1$ has minimum completion length one and exactly the first $s+1$ unit vectors as optimal completions. Its active rank is s, and the all-ones vector is a positive kernel vector.

Proof. The equation $\omega_s x = 1$ has the $s+1$ positive unit vectors as length-one solutions; the final coordinate cannot occur in a length-one solution because its weight is negative. Thus the minimum is one and the displayed unit vectors are the complete optimum. Their difference lattice has rank s, while the sum of all $s+2$ columns is zero, so the all-ones vector lies in the kernel. □

At $s = 3$ this gives a genuine active-rank-three quotient with $\pi_1 \cong F_3$ and $K_{\text{lin}} = 0$.

92.4 Products and higher tori

The periodic nerve is compatible with Cartesian products in a form that is especially useful for raising active rank.

Theorem 92.6 (Optimal-fiber product theorem)**.** *Let $P \subset L$ and $R \subset M$ be finite lattice-convex configurations spanning their lattices. Then $P \times R$ is lattice-convex in $L \oplus M$, has active rank* $\operatorname{rank} L + \operatorname{rank} M$*, and*

$$Q(P \times R) \simeq Q(P) \times Q(R).$$

The equivalence is $(L \oplus M)$-equivariant before passage to the quotient.

Proof. Write $K = \mathcal{N}(P)$ and $H = \mathcal{N}(R)$. A finite set $F \subset L \times M$ is a simplex of $\mathcal{N}(P \times R)$ exactly when some $(c, d) \in L \times M$ satisfies

$$(c, d) - F \subset P \times R.$$

This is equivalent to

$$c - \pi_L(F) \subset P, \qquad d - \pi_M(F) \subset R.$$

Hence $\mathcal{N}(P \times R)$ is the simplicial categorical product whose simplices are precisely the finite sets whose two coordinate projections are simplices of K and H.

On the posets of nonempty faces define

$$\phi(F) = (\pi_L F, \pi_M F), \qquad \psi(A, B) = A \times B.$$

Then $\phi\psi = \mathrm{id}$ and

$$F \subseteq \psi\phi(F)$$

for every face F. Pointwise comparable poset maps induce homotopic maps on order complexes [63]. Therefore ϕ and ψ induce mutually inverse homotopy equivalences after barycentric subdivision. The order complex of the product face poset is the standard barycentric triangulation of the product of the two barycentric subdivisions, so its realization is naturally homeomorphic to $|K| \times |H|$. The construction and the comparison $F \subseteq \psi\phi(F)$ commute with translations, so the equivalence is $(L \oplus M)$-equivariant. Quotienting gives the asserted product law.

Finally,

$$\operatorname{conv}(P \times R) = \operatorname{conv}(P) \times \operatorname{conv}(R),$$

so

$$\big(\operatorname{conv}(P \times R)\big) \cap (L \times M) = (\operatorname{conv}(P) \cap L) \times (\operatorname{conv}(R) \cap M) = P \times R.$$

The difference lattice is $\langle P - P \rangle \oplus \langle R - R \rangle$, proving the rank statement. □

Corollary 92.7 (Iterated products)**.** *For finitely many spanning lattice-convex configurations $P_i \subset L_i$,*

$$Q\left(\prod_i P_i\right) \simeq \prod_i Q(P_i).$$

Every such product is realized as a genuine completion-total optimal fiber by theorem 92.2.

The most symmetric instance is the Boolean cube.

Theorem 92.8 (Higher-torus realization)**.** *For every $s \geq 1$ there is a completion-total integer counting system with active rank s and normalized optimal fiber affinely lattice-isomorphic to $\{0,1\}^s$. Its translation quotient is*

$$Q(\{0,1\}^s) \simeq T^s,$$

so

$$H_j(Q;\mathbb{Z}) \cong \mathbb{Z}^{\binom{s}{j}} \qquad (0 \leq j \leq s).$$

Proof. The topological statement follows immediately from theorem 92.6: $\{0,1\}^s$ is the product of s rank-one intervals, and $Q(\{0,1\}) \simeq S^1$ by theorem 92.4. Thus its quotient is T^s.

For a direct completion-total realization, use $2s+1$ letters grouped into pairs (a_i, b_i), $1 \leq i \leq s$, together with a padding letter c. Let A_s have s rows. For $1 \leq i < s$, row i has coefficient 1 on a_i, b_i, coefficient -1 on a_s, b_s, and 0 elsewhere. The final row has coefficient 1 on the first $2s$ letters and coefficient $-2s$ on c.

Take residual

$$r = (0, \dots, 0, -s),$$

so the completion equation $A_s x = -r$ has last coordinate s. If $t_i = x_{a_i} + x_{b_i}$ and $z = x_c$, the first $s-1$ equations force

$$t_1 = \cdots = t_s = t,$$

while the last equation gives

$$st - 2sz = s, \qquad t = 1 + 2z.$$

The total length is

$$st + z = s + (2s+1)z,$$

so it is minimized exactly at $z = 0$, $t = 1$. Hence an optimal completion chooses exactly one letter from each pair, giving 2^s optima and normalized cube fiber. The within-pair swaps generate an active lattice of rank s. The all-ones vector is a strictly positive kernel vector because every difference row has total coefficient zero and the last row has $2s - 2s = 0$. □

Corollary 92.9 (First third homology)**.** *Active rank three admits*

$$Q \simeq T^3, \qquad (\beta_0, \beta_1, \beta_2, \beta_3) = (1, 3, 3, 1).$$

This is the first possible active rank for nonzero third homology, because theorem 88.2 gives $H_j = 0$ for $j \geq 3$ in active rank two.

92.5 Unequal simplex-strand product rigidity

The periodic nerve, marked quotient and Cartesian-product results of this chapter apply to lattice-convex optimal fibers. The following family has a toric product homotopy type even when it is not a Cartesian product of lattice configurations. The entire face preimage, including horizontal simplices, is retained in the argument.

92.5.1 Lattice normalization, full faces, and marked displacement

Fix $s \geq 1$, the transverse vertices $q_0 = 0$, $q_j = e_j$ of the unimodular simplex in $\mathbb{Z}^s$, and nondegenerate integral intervals $I_j = [a_j, b_j] \cap \mathbb{Z}$. The configuration is

$$P = \bigcup_{j=0}^{s} (I_j \times \{q_j\}) \subset \mathbb{Z} \oplus \mathbb{Z}^s. \tag{92.4}$$

The widths need not coincide; in particular, P need not be lattice-affinely a product. The periodic nerve and quotient $Q(P)$ retain the definitions established above.

Throughout, $X \simeq Y$ signifies that there exist continuous maps $u : X \to Y$ and $v : Y \to X$ together with continuous one-parameter families of maps connecting $v \circ u$ to id_X and $u \circ v$ to id_Y. This convention fixes precisely what is meant by equivalence under continuous deformation; it is stronger than an equality of fundamental groups or of integral singular groups. Let L be a finitely generated free abelian group and $P \subset L$ a finite set spanning the lattice $\langle P - P \rangle_{\mathbb{Z}} = L$. We call P *lattice-convex* if $P = \mathrm{conv}(P) \cap L$. Define

$$\begin{aligned} \mathcal{N}(P) &= \{F \subset L : F \text{ finite and } c - F \subseteq P \text{ for some } c \in L\}, \\ Q(P) &= |\mathcal{N}(P)|/L. \end{aligned} \tag{92.5}$$

As usual the empty face is included. L acts on $\mathcal{N}(P)$ by translation. A nonzero lattice translation cannot preserve a finite simplex setwise, so the action is free on simplices and $Q(P)$ is a CW complex. A choice of $0 \in P$ leads to the known difference presentation

$$\pi_1 Q(P) = \left\langle X_p \ (p \in P),\ X_0 = 1 \ \middle|\ X_p^{-1} X_q = X_u^{-1} X_v \text{ if } q - p = v - u \right\rangle. \tag{92.6}$$

The displacement group map $\delta_P : \pi_1 Q(P) \to L$ takes X_p to p; the difference presentation and marking are established in [186, 188].

Lemma 92.10 (Lattice-convexity and alignment)**.** *The configuration P from* (92.4) *is lattice-convex and spans $\mathbb{Z}^{s+1}$. There is an affine unimodular change of coordinates mapping it to*

$$P' = \bigcup_{j=0}^{s} \big(\{0, 1, \ldots, m_j\} \times \{q_j\}\big), \qquad m_j = b_j - a_j \geq 1.$$

Proof. The projection of $\mathrm{conv}(P)$ to $\mathbb{R}^s$ is the unimodular simplex $\mathrm{conv}\{q_0, \ldots, q_s\}$. Its only integer points are the q_j, and the slice above each vertex is exactly $[a_j, b_j]$. Hence $\mathrm{conv}(P) \cap \mathbb{Z}^{s+1} = P$. A horizontal unit difference is present in each strand. Differences between the strand starts, together with that horizontal vector, generate $\mathbb{Z}^{s+1}$, since the $q_j - q_0$ form a $\mathbb{Z}$-basis in the transverse component. Finally the integral shear

$$(t, y_1, \ldots, y_s) \mapsto \left(t - a_0 - \sum_{j=1}^{s} (a_j - a_0) y_j, y_1, \ldots, y_s \right)$$

has determinant one and aligns all left endpoints to zero. □

From now on assume all strands are aligned, put $m_j \geq 1$ and $M = \max_j m_j$, and write $T = \{q_0, \dots, q_s\} \subset \mathbb{Z}^s$. Let $K = \mathcal{N}(T)$, $H = \mathbb{Z}$ be the horizontal translation subgroup, and $G = \mathbb{Z}^s$ the transverse subgroup. Projection on the transverse coordinate is a simplicial map

$$p : \mathcal{N}(P) \longrightarrow K,$$

with compatible H- and G-translation actions.

92.6 The corrected full-face lemma

For a nonempty simplex σ of K, let X_σ be the *full induced subcomplex* of $\mathcal{N}(P)$ on all vertices whose transverse coordinate belongs to the vertex set of σ. This is the actual geometric inverse image of $|\sigma|$, not a choice of mixed simplices.

Write B_q^m for the horizontal simplicial complex on $\mathbb{Z} \times \{q\}$ whose simplices are the finite sets of horizontal diameter at most m. It is the union of the full translated horizontal simplices of width m.

Lemma 92.11 (Horizontal interval complexes)**.** *For any integer $m \geq 1$, B_q^m is contractible. The quotient $|B_q^m|/H$ is a $K(\mathbb{Z}, 1)$, and the horizontal barycenter modulo $\mathbb{Z}$ is an equivalence under continuous deformation with S^1.*

Proof. Cover B_q^m by full simplices supported on the intervals $[h-m, h] \cap \mathbb{Z}$, $h \in \mathbb{Z}$. Every nonempty finite intersection is a simplex. Their abstract cover nerve consists of finite nonempty $D \subset \mathbb{Z}$ with $\max D - \min D \leq m$. This nerve is also the nerve of the intervals $[h - m, h] \subset \mathbb{R}$, which cover the real line and have contractible intersections. The two applications of the nerve theorem show $|B_q^m| \simeq \mathbb{R}$, hence it is contractible. Horizontal H acts freely; the quotient has universal cover B_q^m, so is a $K(\mathbb{Z}, 1)$. Barycenter changes by 1 under the deck generator and induces an isomorphism on π_1, giving the asserted equivalence under continuous deformation. □

Lemma 92.12 (Full transverse-face preimage)**.** *For every nonempty simplex $\sigma \subset K$ the full complex X_σ is contractible. If σ is a vertex $\{q\}$, then $X_\sigma = B_q^M$. If $|\sigma| \geq 2$ in vertices, there is a contractible mixed-row core C_σ and uniquely assigned indices $j(q) \in \{0, \dots, s\}$ such that*

$$X_\sigma = C_\sigma \ \cup \bigcup_{q \in \sigma} B_q^M, \qquad C_\sigma \cap B_q^M = B_q^{m_{j(q)}}. \tag{92.7}$$

The distinct B_q^M are disjoint, and all inclusions in (92.7) *are cellular cofibrations.*

Proof. For a singleton transverse vertex, that vertex can assume every role among the translates of T; its horizontal simplices consequently arise from all strand widths, whose union is the width-M complex.

For clarity, the uniqueness claim used for mixed faces follows directly from $T = \{0, e_1, \dots, e_s\}$: every nonzero difference $q_i - q_j$ uniquely determines the ordered pair (i, j). Indeed, vectors of the forms $\pm e_i$ and $e_i - e_j$ are distinct unless their ordered indices agree (the assertion also holds for $s = 1$). Thus two different transverse translates of T cannot contain the same ordered pair of distinct vertices. A mixed transverse simplex has one role assignment, whereas a singleton has several.

Suppose σ contains at least two vertices. A nonzero difference $q_i - q_j$ with $q_i, q_j \in T$ uniquely specifies the ordered pair (i, j): the simplex vertices $\{0, e_1, \dots, e_s\}$ are affinely independent. Therefore σ belongs to a unique maximal translate of the transverse simplex K and assigns a unique row role $j(q)$ to each $q \in \sigma$. Let C_σ be the subcomplex generated by the mixed-row full simplices

$$D_h(\sigma) = \bigcup_{q \in \sigma} \big(([h - m_{j(q)}, h] \cap \mathbb{Z}) \times \{q\} \big), \quad h \in \mathbb{Z}.$$

Their finite intersections are simplices and are nonempty exactly when the horizontal centers have diameter at most $M_\sigma = \max_{q\in\sigma} m_{j(q)} \geq 1$. The same interval-nerve argument as in Lemma 92.11 gives $C_\sigma \simeq *$.

Every face involving at least two distinct transverse vertices belongs to this core by the uniqueness of its transverse maximal translate. A face involving just one transverse vertex may instead be witnessed by another translate of T and need not belong to the mixed core: it lies precisely in the full-width horizontal complex B_q^M. These two cases exhaust the actual simplices of X_σ, proving the union identity. Restricting a core simplex to the strand at q gives horizontal diameter at most $m_{j(q)}$, and every such horizontal simplex is present in the core; thus the stated intersection is exact.

By Lemma 92.11, C_σ, $B_q^{m_{j(q)}}$, and B_q^M are all contractible. A subcomplex inclusion $B_q^{m_{j(q)}} \hookrightarrow B_q^M$ is therefore an equivalence under continuous deformation and a cofibration. Attaching the width-M strand along that inclusion preserves the deformation class of the core by the CW gluing theorem for equivalences along cofibrations. Repeating for the finitely many $q \in \sigma$ proves $X_\sigma \simeq *$. □

Remark 92.13 (Why the full preimage matters). The tempting identification $X_\sigma = C_\sigma$ is false for unequal intervals. For example, take $s = 2$, strand widths $(m_0, m_1, m_2) = (1, 1, 4)$ and transverse face $\{0, -e_1\}$. Its mixed core uses the first two short strands, yet the horizontal edge with endpoints $(0, 0)$ and $(4, 0)$ is a genuine simplex of X_σ witnessed by the third, longer role. Formula (92.7) accounts for that edge. This repair is essential for the proof of Theorem 92.14.

92.7 Product rigidity and marked displacement

Theorem 92.14 (Simplex-strand product rigidity). *For every $s \geq 1$ and every configuration P of the form* (92.4) *with all $|I_j| \geq 2$,*

$$Q(P) \simeq Q(T) \times S^1 \simeq \left(\bigvee_{j=1}^{s} S^1 \right) \times S^1. \tag{92.8}$$

The displacement-marked fundamental group is $(F_s \times \mathbb{Z}, \mathrm{ab})$, up to the intrinsic integral basis of the active lattice.

Proof. After Lemma 92.10, write $L = H \oplus G$, with $H = \mathbb{Z}$ horizontal and $G = \mathbb{Z}^s$ transverse, and set

$$Y = |\mathcal{N}(P)|/H.$$

The affine horizontal barycentric coordinate η on every simplex obeys $\eta(hx) = \eta(x) + h$ and is fixed under transverse translations. Consequently the map

$$f = (p, \eta \bmod \mathbb{Z}) : Y \longrightarrow |K| \times S^1$$

is continuous and strictly G-equivariant, with G translating the first factor.

For a transverse face σ, put $Y_\sigma = |X_\sigma|/H$. By Lemma 92.12, X_σ is contractible. Horizontal translation acts freely on it, so Y_σ is a $K(\mathbb{Z}, 1)$. Since a width-one horizontal edge occurs in every X_σ, $\eta \bmod \mathbb{Z}$ maps the deck generator to winding number one. Therefore $\eta : Y_\sigma \to S^1$ is an equivalence under continuous deformation, and hence so is

$$f|_{Y_\sigma} : Y_\sigma \longrightarrow |\sigma| \times S^1.$$

Importantly, these maps are restrictions of the *same* global f; they commute strictly with face inclusions, without any independently chosen facewise contractions.

For a finite transverse subcomplex $K' \subset K$, write $Y[K'] = p^{-1}(|K'|) \subset Y$. The preimage of $|\sigma|$ for any face σ is exactly Y_σ; since it is the full geometric preimage, inverse images of two faces intersect over $\sigma \cap \tau$. When a maximal simplex σ is attached to a finite subcomplex containing its entire boundary, the two squares

$$\begin{array}{ccc} Y[\partial\sigma] & \longrightarrow & Y_\sigma \\ \downarrow & & \downarrow \\ Y[K'_{\mathrm{old}}] & \longrightarrow & Y[K'_{\mathrm{old}} \cup \sigma] \end{array} \qquad \begin{array}{ccc} |\partial\sigma| \times S^1 & \longrightarrow & |\sigma| \times S^1 \\ \downarrow & & \downarrow \\ |K'_{\mathrm{old}}| \times S^1 & \longrightarrow & |K'_{\mathrm{old}} \cup \sigma| \times S^1 \end{array}$$

are cellular pushouts, with horizontal and vertical attaching maps subcomplex inclusions and hence cofibrations. The natural map f induces a map of squares. On the full simplex and, by induction, on the boundary and the previously attached part it is a weak equivalence. The CW gluing principle for weak equivalences makes it so on the new union. Induction over faces of finite K', followed by exhaustion by finite subcomplexes (compact maps from spheres have images in a finite CW subcomplex), gives a weak equivalence

$$f : Y \xrightarrow{\simeq} |K| \times S^1.$$

We still need to descend this statement to the actual periodic quotient. The transverse complex K is locally finite because T is finite. No nonzero element of G preserves a simplex of K, so the translation action of G on $|K|$ is free and properly discontinuous: each point has a neighborhood U whose distinct G-translates are disjoint. Equivariance of $p : Y \to |K|$ gives the same property for $p^{-1}(U)$ in Y. Thus both quotient projections are regular covering maps with deck group G, not merely free cellular quotients. The map f induces the identity on that deck group. Both covering spaces are connected: K is connected because the edge differences of the unimodular simplex generate G, and Y is connected by the horizontal edges and the mixed simplices. Their regular covering sequences therefore give a commuting diagram

$$\begin{array}{ccccccccc} 1 & \to & \pi_1(Y) & \to & \pi_1(Y/G) & \to & G & \to & 1 \\ & & \downarrow\cong & & \downarrow & & \| & & \\ 1 & \to & \pi_1(|K| \times S^1) & \to & \pi_1((|K| \times S^1)/G) & \to & G & \to & 1. \end{array}$$

The middle arrow is an isomorphism by a group-extension diagram chase; the higher groups π_j also agree, because covering spaces preserve them for $j \geq 2$. The quotient map is therefore a weak equivalence and hence, by the CW Whitehead theorem, an equivalence under continuous deformation:

$$Q(P) = Y/G \simeq (|K|/G) \times S^1 = Q(T) \times S^1.$$

For the primitive simplex T, every maximal simplex of K is a translate of $-T$. Two different translates cannot share an edge: equality of a nonzero ordered edge difference determines its unique ordered pair of vertices of T, and hence determines the translation. More generally, no two distinct non-vertex faces of $-T$ lie in the same translation orbit. Consequently $Q(T)$ is the quotient of one geometric s-simplex by identifying its $s+1$ vertices and making no further identifications. Let A denote this finite vertex subcomplex. Since $A \hookrightarrow \Delta^s$ is a CW cofibration and Δ^s is contractible, its cofiber Δ^s/A has the homotopy type of the unreduced suspension of A, which is a wedge of s circles. Thus $Q(T) \simeq \bigvee^s S^1$; this also recovers the primitive-simplex case of [187].

Finally, the displacement homomorphism $\delta_P : \pi_1 Q(P) \to L$ is surjective: the differences $P - P$ generate the full active lattice L, and the difference presentation (92.6) realizes these differences as displacements of loops. By the product equivalence just proved, $H_1(Q(P); \mathbb{Z}) \cong \mathbb{Z}^{s+1}$, while $L \cong \mathbb{Z}^{s+1}$. Because L is abelian, δ_P factors through

a surjection between free abelian groups of the same finite rank; that map is an isomorphism. Hence the marking is exactly abelianization, up to the intrinsic unimodular choice of coordinates on L. □

Corollary 92.15 (Integral invariants)**.** *In the setting of Theorem 92.14,*

$$\pi_1 Q(P) \cong F_s \times \mathbb{Z}, \quad H_0(Q(P);\mathbb{Z}) \cong \mathbb{Z},$$
$$H_1(Q(P);\mathbb{Z}) \cong \mathbb{Z}^{s+1}, \qquad H_2(Q(P);\mathbb{Z}) \cong \mathbb{Z}^s,$$

$H_j(Q(P);\mathbb{Z}) = 0$ *for* $j \geq 3$*, and* $\pi_j Q(P) = 0$ *for all* $j \geq 2$*. In particular* $Q(P)$ *is aspherical and has no torsion in its integral singular groups. The linear displacement kernel* $\ker(H_1 \to L)$ *vanishes; the full zero-displacement subgroup is* $[F_s, F_s] \times \{0\}$*.*

Proof. A wedge of circles is a graph and hence aspherical. Its product with S^1 is aspherical. Integral Künneth gives the stated integral singular groups, and the marking follows from Theorem 92.14. □

92.8 An exact shortest-completion example with a topological transition

Consider the fixed integer source with six genuine unit-cost actions

$$A = \begin{pmatrix} -1 & 2 & 1 & 0 & 3 & -2 \\ -2 & 3 & 0 & 1 & 2 & -1 \end{pmatrix}. \tag{92.9}$$

Its image is $\mathbb{Z}^2$, since columns $(1,0)$ and $(0,1)$ occur; the strictly positive vector $(2,1,1,1,1,2)$ lies in $\ker_{\mathbb{Z}} A$. Consequently every integer residual has a nonnegative source completion: for an integral preimage u one may add a large positive multiple of the strictly positive kernel vector to make u nonnegative. The source is a genuine rank-two counting system, not a globally recognized one-counter language.

For target $r = (x, y)$ set $S = x + y$ and $D = x - y$. The six action vectors in (D, S) coordinates are precisely $\{\pm 1\} \times \{-3, 1, 5\}$.

Proposition 92.16 (Exact length and counting profile)**.** *For any* $r = (x, y) \in \mathbb{Z}^2$ *and* $N \geq 0$*, a source Parikh vector* $z \in \mathbb{N}^6$ *with* $Az = r$ *and* $|z| = N$ *exists exactly when*

$$N \geq |D|, \qquad -3N \leq S \leq 5N, \qquad N \equiv S \pmod 4. \tag{92.10}$$

When this holds, put $P = (N + D)/2$*,* $R = (N - D)/2$*,* $K = (S + 3N)/4$ *and*

$$F_j(t) = \sum_{a_0 + a_1 + a_2 = j} t^{a_1 + 2a_2} = \begin{bmatrix} j+2 \\ 2 \end{bmatrix}_t.$$

Then the number of Parikh vectors *of length* N *at* r *equals*

$$c_A(r, N) = [t^K] F_P(t) F_R(t). \tag{92.11}$$

Thus $L_A(r)$ *is the least integer* $N \geq \max\{|D|, S/5, -S/3, 0\}$ *satisfying* $N \equiv S \pmod 4$*.*

Proof. Every action has $D = \pm 1$ and $S = -3 + 4j$, $j = 0, 1, 2$, proving necessity. Conversely, (92.10) makes P, R nonnegative integral, with $P + R = N$, and makes $0 \leq K \leq 2N$ integral. Decompose K into N digits from $\{0, 1, 2\}$ and assign P of them to the $D = +1$ columns and the remaining R to $D = -1$ columns; each sign group offers all three digit choices. This produces the required vector. At fixed P, R the choices of multiplicities of each digit in each sign group are independent, hence the generating-function product and coefficient formula. This counts unordered source-letter multiplicities, not word permutations. □

Let $r_N = (N-1, 1)$ for $N \geq 3$. Then $S = N$, $D = N-2$, and (92.10) gives $L_A(r_N) = N$. The length-N fiber has one $D = -1$ letter and $N-1$ $D = +1$ letters. Index the chosen negative letter by digit $j \in \{0, 1, 2\}$ and put $t = z_5$, the multiplicity of the original fifth action. Solving the remaining equations gives

$$z_1 = t + j - 1, \qquad z_3 = N - j - 2t, \qquad z_5 = t,$$

with exactly one of z_6, z_4, z_2 equal to one, corresponding to $j = 0, 1, 2$ respectively. Its integral affine fiber coordinates are

row j	allowed t
0	$1 \leq t \leq \lfloor N/2 \rfloor$
1	$0 \leq t \leq \lfloor (N-1)/2 \rfloor$
2	$0 \leq t \leq \lfloor (N-2)/2 \rfloor$.

(92.12)

This bijection preserves the full active difference lattice, and the total number of optimal vectors is $\lfloor 3N/2 \rfloor$. For $N = 3$ the row sizes are $(1, 2, 1)$. The *entire* shortest Parikh fiber, in the original six-action coordinates, consists of

$$(0,0,1,0,1,1), \qquad (0,0,2,1,0,0),$$
$$(1,0,0,1,1,0), \qquad (1,1,1,0,0,0).$$

Using the first point as origin, the three difference vectors have a 3×3 minor of determinant $+1$ on original coordinates $1, 2, 3$. They therefore form an active *primitive* three-simplex (the full active lattice has rank three). For every $N \geq 4$, each of the three complete rows in (92.12) contains at least two consecutive points and their transverse coordinates form a primitive triangle.

Corollary 92.17 (Target-indexed topological transition)**.** *For the* same fixed source (92.9)*, the true full shortest-completion fibers satisfy*

$$Q(P_A(r_N)) \simeq \begin{cases} \bigvee^3 S^1, & N = 3, \\ (S^1 \bigvee S^1) \times S^1, & N \geq 4. \end{cases}$$

Thus H_2 changes from 0 at $N = 3$ to $\mathbb{Z}^2$ for all $N \geq 4$, even though active rank is three throughout and the optimal Parikh multiplicity grows as $\lfloor 3N/2 \rfloor$.

Proof. The $N = 3$ case is the primitive-minimal-fiber theorem. For $N \geq 4$ apply Theorem 92.14 to the three true rows (92.12). The integral-group conclusion follows from Corollary 92.15. □

92.9 Toric-fiber interpretation and theorem hypotheses

Every signed integer counting source can be embedded into a pointed nonnegative semigroup by adjoining a degree coordinate. Given $A = (a_1, \ldots, a_n)$, choose $c \in \mathbb{Z}^d$ with $a_i + c \geq 0$ for every i, and form B_c with columns $(a_i + c, 1)$. The tautological identity

$$\{z \in \mathbb{N}^n : Az = r,\ |z| = N\} = \{z \in \mathbb{N}^n : B_c z = (r + Nc, N)\}$$

identifies fixed-length signed Parikh sets with degree-graded pointed toric fibers. The associated scalar counts belong to established vector partition theory [191]; in our example the Gaussian coefficient expression is an explicit specialization of standard fiber-counting methods. Conversely, our periodic nerve $Q(P)$ is built from lattice translations in the

full minimum-grade fiber and is not, by definition, the indispensable complex attached to a toric ideal [192].

Theorem 92.14 is stated for unimodular transverse simplices and intervals containing at least two consecutive lattice points. The target-indexed formula in Corollary 92.17 concerns the minimum-grade fibers of the fixed counting matrix (92.9); the input source, rather than its parameters, is held constant. The result is an intrinsic statement about the periodic quotient of these exact Parikh fibers, without requiring a physical pushdown realization.

92.10 Two one-rank extensions: apex versus prism

Product with the interval is one natural one-rank extension. A second extension has a different homotopy effect.

Theorem 92.18 (Translation-isolated apex)**.** *Let $P \subset L$ be finite, spanning, and lattice-convex. In $L \oplus \mathbb{Z}e$ put*

$$P^+ = (P \times \{0\}) \cup \{e\},$$

where e has last coordinate one. Then P^+ is lattice-convex, has active rank $\operatorname{rank} L + 1$*, and*

$$Q(P^+) \simeq Q(P) \vee S^1.$$

Proof. A lattice point of the pyramid $\operatorname{conv}(P^+)$ has last coordinate 0 or 1. At height zero the slice is $\operatorname{conv}(P)$, whose lattice points are exactly P; at height one the slice is only the apex. Thus P^+ is lattice-convex, and its difference lattice is $L \oplus \mathbb{Z}e$.

No positive-dimensional face containing the apex has a nontrivial translate inside P^+. Indeed, a translation with last coordinate zero must fix the unique height-one point and is therefore zero; a translation with last coordinate ± 1 sends one of the two height levels outside $\{0, 1\}$. Hence every nontrivial face identification is confined to the base simplex, although all vertices, including the apex, belong to one translation orbit.

Choose the simplex of the periodic nerve corresponding to the translate P^+ itself as a fundamental simplex; it is the cone on the base simplex $\Delta(P)$. Every simplex orbit has a representative in this cone. The preceding height argument has two consequences for its face identifications. First, all identifications between positive-dimensional base faces are exactly those already defining $Q(P)$. Second, a positive-dimensional face containing the apex cannot be identified with a distinct face containing the apex: a translating vector with last coordinate zero would have to fix the unique height-one vertex and hence be zero, while a vector with nonzero last coordinate cannot preserve the two-level pattern of a mixed face. Only the apex vertex itself is identified with the common vertex orbit, because the translation action is transitive on vertices.

Consequently the quotient is obtained from the cone $C\Delta(P)$ by applying the old base quotient

$$q : \Delta(P) \longrightarrow Q(P)$$

and identifying the cone point with the vertex orbit of $Q(P)$. Equivalently it is the homotopy coequalizer of q and the constant map $\Delta(P) \to Q(P)$ at that vertex. Since $\Delta(P)$ is nonempty and contractible, q is homotopic to the constant map. The homotopy coequalizer of two homotopic maps from a nonempty contractible space is homotopy equivalent to $Q(P) \vee S^1$ (for example by the standard mapping-cylinder model). This proves the theorem. □

Corollary 92.19 (Prism law)**.** *For every finite spanning lattice-convex P,*

$$Q(P \times \{0, 1\}) \simeq Q(P) \times S^1.$$

Proof. Apply theorem 92.6 and $Q(\{0,1\}) \simeq S^1$. □

Thus apex and prism both add one active lattice direction but act differently:

$$\pi_1(Q(P^+)) \cong \pi_1(Q(P)) * \mathbb{Z}, \qquad \pi_1(Q(P \times \{0,1\})) \cong \pi_1(Q(P)) \times \mathbb{Z}.$$

For torsion-free homology, the Künneth formula also gives

$$H_j(Q(P) \times S^1) \cong H_j(Q(P)) \oplus H_{j-1}(Q(P)),$$

while apex extension preserves H_j for $j \geq 2$.

Corollary 92.20 (Rank-three branches)**.** *Both operations produce genuine completion-total active-rank-three families.*

(a) *Apexing a rank-two F_2 fiber gives F_3; apexing a rank-two F_3 fiber gives F_4.*

(b) *Apexing a rank-two toral fiber gives fundamental group $\mathbb{Z}^2 * \mathbb{Z}$ and, for the basic torus, homotopy type $T^2 \vee S^1$.*

(c) *Prisming the rank-two minimal F_2 branch gives an aspherical $F_2 \times \mathbb{Z}$ quotient with Betti vector $(1,3,2)$. Prisming a rank-two F_3 wedge branch gives an aspherical $F_3 \times \mathbb{Z}$ quotient with Betti vector $(1,4,3)$.*

92.11 Fixed-rank higher homology and displacement

We now combine the rank-raising calculus with an explicit rank-two seed.

Proposition 92.21 (Rank-two seed family)**.** *For every $m \geq 2$ there is a completion-total scalar system with active rank two and a normalized fiber affinely equivalent to*

$$P_m = \{(t,0) : 0 \leq t \leq m\} \cup \{(0,1), (m-1,-1)\} \subset \mathbb{Z}^2. \tag{92.13}$$

Its quotient satisfies

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2, \qquad H_1(Q(P_m);\mathbb{Z}) \cong \mathbb{Z}^2, \qquad H_2(Q(P_m);\mathbb{Z}) \cong \mathbb{Z}^{m-1},$$

and has no homology above degree two. Equivalently, its Betti polynomial is

$$1 + 2t + (m-1)t^2.$$

Proof. Take $\alpha = 2m+1$ and the four scalar weights

$$\omega_m = \big(1 - m\alpha,\ 1 - (m-1)\alpha,\ 1,\ 1 + \alpha\big),$$

with residual

$$r_m = \alpha m(m-1) - 2m.$$

The strictly positive vector

$$h_m = \big(1, 1, (2m-2)\alpha - 3, 1\big)$$

lies in the kernel. All four weights are 1 modulo α, so every completion length satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m+1}.$$

At length $2m$, the complete optimum is

$$A_t = (t, m-t, m-t, t) \quad (0 \leq t \leq m),$$

$$B = (0, m+1, 0, m-1), \qquad C = (m-1, 0, m+1, 0).$$

Indeed, eliminating the common congruence term gives

$$x_2 + mx_3 + (m+1)x_4 = m(m+1),$$

and reduction modulo m forces $x_2 + x_4 \in \{0, m, 2m\}$, yielding exactly C, the A_t, and B.

The equal-length kernel has basis

$$g = (1, -1, -1, 1), \qquad h = (0, 1, -m, m-1),$$

and relative to A_0 the optimal points have coordinates

$$A_t - A_0 = tg, \qquad B - A_0 = h, \qquad C - A_0 = (m-1)g - h,$$

which is (92.13). Its convex hull is the lattice quadrilateral

$$(0,0),\ (m-1,-1),\ (m,0),\ (0,1).$$

All four boundary edges are primitive. The Euclidean area is m, so Pick's theorem gives $m-1$ interior lattice points. These are exactly $(1,0), \dots, (m-1,0)$; together with the four boundary vertices this is precisely P_m. Hence P_m is lattice-convex.

We next compute the fundamental group directly from the difference presentation (92.1), without invoking a general planar classification. Write

$$x = X_{(1,0)}, \qquad y = X_{(0,1)}, \qquad z = X_{(m-1,-1)}.$$

Repeated horizontal differences give $X_{(t,0)} = x^t$. The repeated difference $(0,1)$ between the pairs $((0,0),(0,1))$ and $((m-1,-1),(m-1,0))$ gives

$$y = z^{-1}x^{m-1}, \qquad z = x^{m-1}y^{-1}.$$

For $1 \le t \le m-1$, the repeated difference $(-t,1)$ between

$$((t,0),(0,1)) \quad \text{and} \quad ((m-1,-1),(m-1-t,0))$$

gives

$$x^{-t}y = yx^{-t}.$$

Taking $t = 1$ shows $xy = yx$. Thus the presentation is generated by the commuting pair x, y. The displacement map (92.2) sends them to the standard basis of $\mathbb{Z}^2$, so it rules out any further relation and yields

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2.$$

In particular $H_1(Q(P_m);\mathbb{Z}) \cong \mathbb{Z}^2$. The rank-two homological baseline theorem 88.2 gives no homology above degree two and makes H_2 free abelian.

It remains to compute the Euler characteristic by simplex orbits. Put $H = \{(t,0) : 0 \le t \le m\}$. A horizontal-only orbit can be normalized to have minimum horizontal coordinate zero, so its alternating contribution is

$$\sum_{k\ge 1} (-1)^{k-1} \binom{m}{k-1} = 0.$$

For either fixed off-line point, faces containing that point and a nonempty horizontal subset contribute

$$\sum_{k=1}^{m+1} (-1)^k \binom{m+1}{k} = -1,$$

while faces containing both off-line points have total alternating contribution

$$\sum_{k=0}^{m+1}(-1)^{k+1}\binom{m+1}{k}=0.$$

The only cross-identifications between the two one-off-line families are the m edge pairs

$$\{(0,1),(t,0)\}\sim\{(m-1,-1),(m-1-t,0)\},\qquad 0\le t\le m-1.$$

A nonzero vertical translation can match at most one horizontal point because each off-line level contains only one point, so there is no higher-dimensional cross-identification. Identifying each duplicate edge orbit raises the alternating count by one. Therefore

$$\chi(Q(P_m))=0-1-1+m=m-2.$$

Since $\beta_0=1$, $\beta_1=2$, and higher homology vanishes,

$$m-2=1-2+\beta_2,$$

so $\beta_2=m-1$. Torsion-freeness has already been supplied by theorem 88.2. □

Corollary 92.22 (Apex branch with unbounded second homology)**.** *Let P_m^+ be the apex extension of the seed in theorem 92.21. Then P_m^+ has active rank three and*

$$Q(P_m^+)\simeq Q(P_m)\vee S^1,\qquad \pi_1(Q(P_m^+))\cong\mathbb{Z}^2*\mathbb{Z}.$$

Moreover

$$H_2(Q(P_m^+);\mathbb{Z})\cong\mathbb{Z}^{m-1},\qquad H_j(Q(P_m^+);\mathbb{Z})=0\quad(j\ge 3).$$

Hence β_2 is unbounded at fixed active rank three and fixed fundamental group $\mathbb{Z}^2*\mathbb{Z}$.

Proof. Apply theorem 92.18 to theorem 92.21. Wedging on a circle changes only H_1 and takes the fundamental group to the free product with $\mathbb{Z}$. □

The prism operation behaves differently: it moves the seed's second homology into degree three. This leads to the main all-rank phenomenon.

Theorem 92.23 (Fixed rank and fixed fundamental group with unbounded higher homology)**.** *For every $s\ge 2$ there is a one-parameter family of completion-total integer counting systems whose normalized optimal fibers have active rank exactly s and whose quotients $Q_{m,s}$ satisfy*

$$\pi_1(Q_{m,s})\cong\mathbb{Z}^s,$$

while every Betti number $\beta_j(Q_{m,s})$, $2\le j\le s$, is unbounded as $m\to\infty$.

More precisely, take

$$P_{m,s}=P_m\times\{0,1\}^{s-2}.$$

Then

$$Q(P_{m,s})\simeq Q(P_m)\times T^{s-2}$$

and its Betti polynomial is

$$\boxed{\big(1+2t+(m-1)t^2\big)(1+t)^{s-2}}.\tag{92.14}$$

Equivalently,

$$\boxed{\beta_j=\binom{s-2}{j}+2\binom{s-2}{j-1}+(m-1)\binom{s-2}{j-2}},\tag{92.15}$$

with out-of-range binomial coefficients interpreted as zero. In particular

$$\beta_1=s,\qquad \beta_s=m-1.$$

Proof. The active ranks add under products, so $P_{m,s}$ has rank $2 + (s-2) = s$. By theorems 92.6 and 92.8,

$$Q(P_{m,s}) \simeq Q(P_m) \times T^{s-2}.$$

Its fundamental group is

$$\mathbb{Z}^2 \times \mathbb{Z}^{s-2} = \mathbb{Z}^s.$$

The two factors have torsion-free homology, so the integral Künneth formula has no Tor terms and multiplies Betti polynomials. This gives (92.14) and (92.15). For $2 \le j \le s$, the coefficient

$$\binom{s-2}{j-2}$$

of $m-1$ is positive, so every such β_j is unbounded. Finally, theorem 92.2 realizes each product configuration as a completion-total optimal fiber. □

Corollary 92.24 (Unbounded third homology in rank three)**.** *At active rank three there are completion-total systems with*

$$\pi_1 \cong \mathbb{Z}^3, \qquad (H_0, H_1, H_2, H_3) \cong (\mathbb{Z}, \mathbb{Z}^3, \mathbb{Z}^{m+1}, \mathbb{Z}^{m-1}).$$

Thus $\beta_3 = m-1$ is unbounded at fixed active rank and fixed fundamental group. The case $m = 2$ is T^3.

92.11.1 The displacement map contains information beyond π_1

The minimal rank-three fiber from theorem 92.4 has

$$\pi_1 \cong F_3, \qquad H_1 \cong \mathbb{Z}^3 \xrightarrow{\cong} L \cong \mathbb{Z}^3,$$

so $K_{\mathrm{lin}} = 0$. For a rank-two comparison, take

$$P_F = \{(0,0), (0,1), (1,1), (3,2)\} \subset \mathbb{Z}^2.$$

Its convex hull is the triangle with vertices $(0,0), (0,1), (3,2)$ and normalized area 3; Pick's theorem shows that $(1,1)$ is its unique interior lattice point, so P_F is lattice-convex. The six unordered nonzero pair differences

$$(0,1), (1,1), (3,2), (1,0), (3,1), (2,1)$$

are distinct up to sign. Hence the presentation (92.1) has no nontrivial repeated-difference relation and is free on the three nonzero vertices:

$$\pi_1(Q(P_F)) \cong F_3.$$

But the displacement map on abelianization is a surjection

$$\mathbb{Z}^3 \longrightarrow \mathbb{Z}^2,$$

so its kernel is primitive of rank one. By theorem 92.2, P_F also occurs as a genuine completion-total optimal fiber. We therefore obtain the following intrinsic conclusion.

Corollary 92.25 (Same fundamental group, different displacement visibility)**.** *The abstract group $\pi_1(Q(P))$ does not determine the active rank or the linear zero-displacement subgroup $K_{\mathrm{lin}}(P)$. In particular F_3 occurs with both $K_{\mathrm{lin}} = 0$ and $K_{\mathrm{lin}} \cong \mathbb{Z}$.*

This statement is purely about the intrinsic completion quotient. It does not assert a physical stack-carrier lower bound; such claims require a separately specified transport model. Part VIII takes up exactly that representation-independent physical question.

92.12 The displacement-marked group in arbitrary rank

The preceding sections show that the abstract group $\pi_1(Q(P))$ and the active lattice L both matter. The correct intrinsic package retains the map between them.

Definition 92.26 (Displacement-marked group)**.** For a finite spanning lattice-convex configuration $P \subset L$, put

$$G_P = \pi_1(Q(P)), \qquad \mathfrak{M}(P) = (G_P, L, \delta_P),$$

where $\delta_P : G_P \twoheadrightarrow L$ is the displacement homomorphism of (92.2). Define

$$K_0(P) = \ker \delta_P, \qquad K_{\mathrm{lin}}(P) = \ker\big(H_1(Q(P);\mathbb{Z}) \xrightarrow{\delta_P^{\mathrm{ab}}} L\big).$$

The first is the full pure zero-displacement sector; the second is its abelian shadow.

Theorem 92.27 (Affine-lattice invariance of the marked group)**.** *Let $f : P \to P'$ be an affine lattice isomorphism $f(z) = Az + b$ from a spanning lattice-convex $P \subset L$ to $P' \subset L'$. Then the induced quotient equivalence satisfies*

$$\delta_{P'} \circ f_* = A \circ \delta_P.$$

Hence $\mathfrak{M}(P)$ is an affine-lattice invariant up to marked isomorphism.

Proof. Translate both configurations so that the chosen base vertex is zero. A generator X_p of the difference presentation maps to the generator associated with Ap. Since $\delta_P(X_p) = p$ and $\delta_{P'}(X_{Ap}) = Ap$, the displayed identity holds on generators and therefore on all of G_P. □

Theorem 92.28 (Universal visibility exact sequences)**.** *For every finite spanning lattice-convex $P \subset L$ there are exact sequences*

$$\boxed{1 \longrightarrow [G_P, G_P] \longrightarrow K_0(P) \longrightarrow K_{\mathrm{lin}}(P) \longrightarrow 0}$$

and

$$\boxed{0 \longrightarrow K_{\mathrm{lin}}(P) \longrightarrow H_1(Q(P);\mathbb{Z}) \xrightarrow{\delta_P^{\mathrm{ab}}} L \longrightarrow 0.}$$

Equivalently,

$$K_0(P)/[G_P, G_P] \cong K_{\mathrm{lin}}(P).$$

Proof. Because the target lattice is abelian, $[G_P, G_P] \subseteq K_0(P)$. Abelianization identifies $H_1(Q(P);\mathbb{Z})$ with $G_P/[G_P, G_P]$, and the image of $K_0(P)$ in that quotient is exactly the kernel of δ_P^{ab}. This gives the first sequence. The second is the defining kernel sequence, and surjectivity follows from the fact that P spans L. □

Corollary 92.29 (Linear visibility excess)**.** *Since L is free abelian, the second sequence splits noncanonically. Thus*

$$H_1(Q(P);\mathbb{Z}) \cong K_{\mathrm{lin}}(P) \oplus L, \qquad \operatorname{rank} K_{\mathrm{lin}}(P) = \beta_1(Q(P)) - \operatorname{rank} L.$$

In particular, if $K_{\mathrm{lin}}(P) = 0$ then $K_0(P) = [G_P, G_P]$; and $K_0(P) = 1$ exactly when G_P is abelian and $K_{\mathrm{lin}}(P) = 0$.

Positioning. The use of a fundamental-group representation as monodromy is classical in covering-space theory [11]; voltage and gain graphs provide related group-valued cycle bookkeeping [117, 118], and periodic-net theory likewise attaches translation data to quotient cycles [119, 120]. The completion-specific point here is the exact marked package $\mathfrak{M}(P)$ together with the product, prism, apex, and physical-transfer laws below.

92.13 Marked product, prism, and apex transfer

Define

$$(G, L, \delta) \boxtimes (H, M, \varepsilon) = (G \times H, L \oplus M, \delta \oplus \varepsilon).$$

Theorem 92.30 (Marked product transfer)**.** *For finite spanning lattice-convex $P \subset L$ and $R \subset M$,*

$$\boxed{\mathfrak{M}(P \times R) \cong \mathfrak{M}(P) \boxtimes \mathfrak{M}(R).}$$

Consequently

$$K_0(P \times R) \cong K_0(P) \times K_0(R), \qquad K_{\mathrm{lin}}(P \times R) \cong K_{\mathrm{lin}}(P) \oplus K_{\mathrm{lin}}(R).$$

Proof. The topological product theorem already gives $Q(P \times R) \simeq Q(P) \times Q(R)$. At the presentation level, with $A_p = X_{(p,0)}$ and $B_r = X_{(0,r)}$, repeated-difference relations give

$$X_{(p,r)} = A_p B_r, \qquad [A_p, B_r] = 1.$$

Thus the fundamental group splits as the direct product, and the displacement of $X_{(p,r)}$ is (p, r). The kernel formulas follow immediately. □

Corollary 92.31 (Prism preserves the pure sector)**.** *Let $I = \{0, 1\}$. Then*

$$\mathfrak{M}(P \times I) \cong \mathfrak{M}(P) \boxtimes (\mathbb{Z}, \mathbb{Z}, \mathrm{id}),$$

so

$$\boxed{K_0(P \times I) \cong K_0(P), \qquad K_{\mathrm{lin}}(P \times I) \cong K_{\mathrm{lin}}(P).}$$

A prism adds one displacement-visible direction but no new pure-loop sector.

Theorem 92.32 (Apex pure-holonomy creation)**.** *For the translation-isolated apex*

$$P^+ = (P \times \{0\}) \cup \{e\} \subset L \oplus \mathbb{Z}e,$$

write $G = \pi_1(Q(P))$ and let t denote the new apex circle. Then

$$\pi_1(Q(P^+)) \cong G * \langle t \rangle,$$

with marked displacement

$$\delta_+(g) = (\delta_P(g), 0), \qquad \delta_+(t) = (0, 1),$$

and

$$\boxed{K_0(P^+) = \langle\langle K_0(P), [G, t] \rangle\rangle .}$$

Moreover

$$\boxed{K_{\mathrm{lin}}(P^+) \cong K_{\mathrm{lin}}(P).}$$

Thus apex rank-raising can create new nonabelian pure holonomy while leaving the linear pure sector unchanged.

Proof. The topological apex theorem gives the free product. Every element of $K_0(P)$ and every cross-commutator $[G, t]$ has zero displacement, so their normal closure lies in $\ker \delta_+$. Conversely, quotient first by $K_0(P)$, obtaining $L * \langle t \rangle$, and then by all cross-commutators, obtaining $L \oplus \mathbb{Z}$. The resulting quotient is exactly the displacement target, proving the kernel formula. Abelianization of $G * \mathbb{Z}$ is $H_1(Q(P)) \oplus \mathbb{Z}$, and the new summand maps isomorphically to the new lattice direction, so the linear kernel is unchanged. □

Corollary 92.33 (Apex–prism separation)**.** *Prism and apex both raise active rank by one and preserve K_{lin}, but they have different full pure-loop behavior: prism preserves K_0, while apex adjoins the normal closure of cross-commutators. In particular, the minimal rank-s fiber has*

$$\pi_1 \cong F_s, \qquad K_{\mathrm{lin}} = 0, \qquad K_0 = [F_s, F_s] \quad (s \geq 2),$$

whereas the Boolean cube I^s has

$$\pi_1 \cong \mathbb{Z}^s, \qquad K_0 = 1, \qquad K_{\mathrm{lin}} = 0.$$

Corollary 92.34 (Fixed marked group with unbounded higher homology)**.** *For the family $P_{m,s} = P_m \times I^{s-2}$ of theorem 92.23,*

$$\boxed{\mathfrak{M}(P_{m,s}) \cong (\mathbb{Z}^s, \mathbb{Z}^s, \mathrm{id})}$$

for every m, hence $K_0 = 1$ and $K_{\mathrm{lin}} = 0$, while every β_j, $2 \leq j \leq s$, is unbounded with m.

This strengthens the earlier fixed-π_1 statement: even the full displacement-marked group can be held fixed while all higher Betti numbers vary without bound. Together with theorem 92.25, it gives the exact intrinsic hierarchy

$$\pi_1 \not\Rightarrow (L, K_{\mathrm{lin}}), \qquad \mathfrak{M}(P) \Rightarrow (K_0, K_{\mathrm{lin}}), \qquad \mathfrak{M}(P) \not\Rightarrow H_{\geq 2}(Q(P)).$$

92.14 Finite physical holonomy and literal carrier costs

The marked group identifies which pure loops exist. A physical implementation must still choose how those loops act.

Definition 92.35 (Marked finite-permutation transport)**.** Choose a homomorphism

$$\rho : G_P \longrightarrow S_n$$

and put

$$G_{\mathrm{phys}} = L \times S_n, \qquad \mathrm{Hol}_\rho(g) = (\delta_P(g), \rho(g)).$$

The physically realized pure image is

$$\Pi_\rho(P) = \rho(K_0(P)).$$

Proposition 92.36 (Flat local realization of a finite representation)**.** *Let X be a connected finite CW complex and $\rho : \pi_1(X, x_0) \to H$ a homomorphism to a finite group. After a finite subdivision, the oriented edges of X admit an H-labeling with trivial ordered product around every 2-cell and based loop holonomy exactly ρ.*

Proof. Choose a maximal tree, label its edges by the identity, and label each non-tree edge by the image under ρ of the associated based generator loop. Every 2-cell attaching word is null in π_1, hence has trivial label product. Tree collapse shows that loop holonomy recovers ρ. □

Theorem 92.37 (Physical pure-image theorem)**.** *For every marked finite-permutation transport,*

$$\boxed{\mathrm{Hol}_\rho(K_0(P)) = \{0\} \times \Pi_\rho(P).}$$

Thus $K_0(P) \neq 1$ means that pure loops are intrinsically available, not that every representation must realize them nontrivially.

Theorem 92.38 (Physical prism and apex transfer)**.** *For a prism extension, any homomorphic extension $\rho^{\square} : G_P \times \langle u \rangle \to S_n$ has $h = \rho^{\square}(u)$ centralizing $\rho(G_P)$ and*

$$\boxed{\Pi_{\rho^{\square}}(P \times I) = \Pi_{\rho}(P).}$$

For an apex extension, choose $h \in S_n$ and extend ρ to $\rho_h : G_P * \langle t \rangle \to S_n$ by $\rho_h(t) = h$. Then

$$\boxed{\Pi_{\rho_h}(P^+) = \langle\!\langle \Pi_{\rho}(P), [\rho(G_P), h] \rangle\!\rangle_{\langle \rho(G_P), h \rangle}.}$$

In particular, an apex creates no new realized pure holonomy when h centralizes the old image, and can create it exactly through failure of centralization.

The same separation appears at the carrier level. Let A be the integer counting matrix of a realization and let $z \in \mathbb{N}^d$ be a nonzero zero-semantic count vector, $Az = 0$. The number of distinct literal orderings with this Parikh vector is

$$M(z) = \frac{|z|!}{\prod_i z_i!}.$$

For $k \geq 2$ define the order-carrier girth

$$\boxed{g_k^{\mathrm{ord}}(A) = \min\{|z| : z \neq 0,\ Az = 0,\ M(z) \geq k\},}$$

with value $+\infty$ if no such z exists.

Theorem 92.39 (Persistent literal carrier bound)**.** *In the buffered literal-order category, if k persistent sheets share one zero-semantic Parikh vector z, then*

$$H = |z| \geq g_k^{\mathrm{ord}}(A).$$

In particular a regular-sheet realization of a finite image $H_\rho = \operatorname{im} \rho$ satisfies

$$H \geq g_{|H_\rho|}^{\mathrm{ord}}(A).$$

Proof. Persistent sheet identity is stored only in literal order, so one Parikh vector z can support at most $M(z)$ sheets. Hence $k \leq M(z)$ and the definition of $g_k^{\mathrm{ord}}(A)$ gives the bound. □

Theorem 92.40 (Exact buffered ancestry law)**.** *Let $H \leq S_n$ be a finite permutation image and let*

$$a(H) = \left| \bigcup_{\sigma \in H} \operatorname{supp}(\sigma) \right|.$$

In the ancestry-bijective one-top buffered macro category,

$$\boxed{b_{\min}(H) = \max\{a(H) - 1, 0\}.}$$

For nontrivial H this is exactly $a(H) - 1$.

Proof. Place the active support contiguously at the top of the distinguished block. Some group element moves the deepest active cell. Before that cell can be inspected, all $a(H)-1$ cells above it have been removed from the live stack and their ancestries must be stored externally, giving the lower bound. Conversely, pop those ancestries into handles, rewrite the deepest active cell, and rebuild the block in the target order. The trivial image requires no handle. □

Relation to structured pushdown models. Visibly pushdown and height-deterministic pushdown automata constrain stack-height evolution in different ways [122, 123]. The present carrier law concerns a different frozen category: persistent sheet identity lives in literal order, while temporary ancestry handles are cleared at macro endpoints. The numerical bounds should therefore be read relative to this category, not as unrestricted PDA lower bounds.

92.15 Connection memory and repair

Carrier height and ancestry support do not determine all transient control. The local connection may merge histories that must later be distinguished.

Definition 92.41 (Terminal collision multiplicity)**.** For a deferred-carrier connection certificate $\mathcal{C}$, let $\chi(\mathcal{C})$ be the largest number of histories that reach one preterminal semantic vertex with the same exposed persistent sheet and the same next action, but carry pairwise distinct accumulated physical multipliers that require distinct terminal updates.

Theorem 92.42 (Deferred path-memory lower bound)**.** *Every deterministic deferred-carrier realization of $\mathcal{C}$ requires*

$$\boxed{q_{\mathrm{tr}} \geq \chi(\mathcal{C})}$$

transient path states. If at every preterminal vertex there are at most $\chi(\mathcal{C})$ accumulated-product classes, the bound is attained by recording that class in finite control.

Proof. Histories counted by $\chi(\mathcal{C})$ expose identical persistent data and the same next action. If two shared one transient state, determinism would force the same terminal update, contradicting the definition. The upper construction stores exactly the accumulated-product class. □

For repair, fix a finite cell model X, a finite physical group H, and a raw oriented-edge label $a : E^{\mathrm{or}}(X) \to H$ with $a(\bar{e}) = a(e)^{-1}$. Let $Z^1(X;H)$ be the set of flat labelings, equivalently nonabelian 1-cocycles. For a conjugation-invariant generating metric d_H define

$$C_0(a) = \min_{b \in Z^1(X;H)} |\{e : a_e \neq b_e\}|,$$

$$C_1(a) = \min_{b \in Z^1(X;H)} \sum_e d_H(a_e, b_e).$$

Proposition 92.43 (Repair is nearest-flat nonabelian cocycle optimization)**.** *Flatness is determined by the 2-skeleton of X: higher cells impose no additional relations on group-valued edge transport. The numerical values of C_0 and C_1 depend on the raw labeling, chosen trivialization, cell presentation, physical group, and metric, and therefore are not invariants of $\mathfrak{M}(P)$ alone.*

Proof. The fundamental group of a CW complex is presented by the 1-skeleton modulo the attaching relations of the 2-cells; higher cells do not change π_1. Thus group-valued edge flatness is exactly the satisfaction of the 2-cell relations. The optimization objectives explicitly contain the raw labels and the chosen metric, so changing those data can change the minimum without changing the intrinsic marked group. This is the same cycle-consistency distinction that appears in group synchronization, though the present objective is tied to the completion transport category [121]. □

The higher-rank transfer is therefore layered:

$$\boxed{\begin{array}{c}\text{marked topology} \to \text{chosen holonomy representation} \\ \to \text{carrier/support constraints} \to \text{connection memory} \\ \to \text{raw-policy repair cost}\end{array}}$$

92.16 An all-rank exact finite-holonomy family

Let $P_s = \{0, e_1, \dots, e_s\} \subset \mathbb{Z}^s$ be the minimal active rank-s configuration. By theorem 92.4,

$$G_s = \pi_1(Q(P_s)) \cong F_s, \qquad \delta_s : F_s \twoheadrightarrow \mathbb{Z}^s$$

is the abelianization map, so $K_0(P_s) = [F_s, F_s]$ and $K_{\mathrm{lin}}(P_s) = 0$.

Theorem 92.44 (Constant finite physical image in every rank)**.** *For every $s \geq 2$ there is a permutation representation $\rho_s : F_s \to S_3$ with*

$$\boxed{\operatorname{im}\rho_s = S_3, \qquad \Pi_{\rho_s}(P_s) = A_3.}$$

Hence genuine pure physical holonomy persists in arbitrary active rank with constant finite permutation image.

Proof. Let $r = (123)$ and $q = (13)$, and set

$$\rho_s(x_1) = r, \qquad \rho_s(x_2) = q, \qquad \rho_s(x_i) = 1 \quad (i \geq 3).$$

The image is S_3. Since $K_0(P_s) = [F_s, F_s]$, its image is the derived subgroup $[S_3, S_3] = A_3$; already $[x_1, x_2]$ maps to a nontrivial 3-cycle. □

Use the scalar realization

$$\omega_s = (\underbrace{1, \dots, 1}_{s+1 \text{ entries}}, -(s+1)).$$

A nonzero zero-weight vector $z \in \mathbb{N}^{s+2}$ satisfies

$$\sum_{i=1}^{s+1} z_i = (s+1) z_{s+2},$$

so

$$|z| = (s+2) z_{s+2} \geq s+2.$$

The all-ones vector attains equality and has order capacity $(s+2)! \geq 6$.

Theorem 92.45 (All-rank exact buffered carrier family)**.** *For every $s \geq 2$, the minimal active rank-s fiber admits a marked finite-permutation transport and buffered literal-order realization with*

$$\boxed{\operatorname{im}\rho_s = S_3, \qquad \Pi_{\rho_s}(P_s) = A_3,}$$

$$\boxed{H_{\min} = s+2, \qquad b_{\min} = 2.}$$

Thus active rank is unbounded while full image size 6*, pure image size* 3*, and ancestry cost* 2 *remain constant; the linear carrier-height growth comes from the zero-weight arithmetic of the chosen scalar realization rather than from increasing permutation complexity.*

Proof. The preceding theorem supplies the S_3 transport, and theorem 92.36 realizes it by a flat local connection. Every nonempty zero-weight literal carrier for ω_s has height at least $s + 2$, while the all-ones vector supplies at least six orderings at exactly that height. Choose three literal types as an active top block, use its six orderings as the regular S_3 sheets, and keep the remaining $s - 1$ types as a common suffix. This realizes the persistent transport at height $s + 2$. The image acts on exactly three active cells, so theorem 92.40 gives $b_{\min} = 3 - 1 = 2$. □

Remark 92.46 (Real-time scope). The preceding exact bounds hold in the buffered literal-order macro category. They do not by themselves assert the same all-rank bounds in the stricter no-ε real-time Stable-LIFO subclass. A separate scheduling construction would be required there. This is a category boundary, not an unresolved point in the marked-group calculus.

The chapter now separates five levels that were previously easy to conflate: intrinsic displacement-marked topology, realized finite holonomy, persistent literal carrier capacity, transient connection memory, and raw-policy repair. Higher homology lies outside this one-type transport chain: theorem 92.34 keeps the entire marked group fixed while higher Betti numbers diverge.

Part VIII

Representation-Independent Physical Complexity and Resource Synthesis

Part guide. **Question.** What resource lower bounds remain when the opponent is allowed to choose an arbitrary deterministic pushdown representation? **Main result.** The Part proves exact unrestricted state–stack endpoints, an exact one-work-symbol bounded-replacement staircase, filtered-collapse/source-stack exposure bounds, and protected-contract Pareto frontiers. **Why here.** It turns the semantic/physical distinction into representation-independent complexity theorems and extends the language-visible resource viewpoint introduced by the hierarchy theorem of Part I. **Dependency.** It uses the semantic invariants and collapse-capacity viewpoint developed earlier and leads directly to the decomposition question of Part IX.

Chapter 93

Protected Phase Pumping and Deferred Exposure

Fix coprime positive integers p, q and

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

The question is now unrestricted: an arbitrary deterministic pushdown recognizer may distribute the residual among control, visible stack, deep stack, long rewrites, and deterministic ε-normalization. The proof below reconstructs no canonical representation. Instead it uses quotient rigidity to label reachable physical configurations semantically and then forces synchronized phases above protected stack cuts.

93.1 Model and semantic quotient

93.1.1 Pushdown model

A deterministic pushdown automaton has finite control Q, finite stack alphabet Γ, and transitions that inspect the current state, the visible top stack symbol, and either one input symbol or ε. The stack top is written on the right. Stack words, including regular languages of stacks used later, are therefore written and read from bottom to top. Thus a transition

$$\delta(q, X, a) = (q', \beta)$$

acts as

$$(q, \eta X, av) \vdash (q', \eta\beta, v).$$

A single transition may replace X by an arbitrary fixed finite word $\beta \in \Gamma^*$. Determinism means that for each state/top pair there is at most one move on each input letter, and if an ε-move is enabled then no input-consuming move is enabled at that state/top pair.

For the lower bound we use a *regular-target* acceptance convention. Fix a regular set $T \subseteq Q \times \Gamma^*$ of accepting control–stack configurations. After the finite binary input has been consumed, the word is accepted exactly when the deterministic post-input computation reaches T without consuming any further binary input. Reaching T in zero post-input steps is allowed. Ordinary final-state, empty-stack, and final-state-plus-empty-stack acceptance are obtained by the corresponding regular choices of T. A mandatory right-endmarker convention is handled by treating the endmarker as the final symbol of the continuation. The fixed-continuation regularity consequence needed later is proved separately in Theorem 93.12; it is not part of the model definition.

The matching upper bound in Section 94.2 is stated explicitly for a mandatory right endmarker.

Remark 93.1 (Meaning of "unrestricted"). The lower-bound opponent may use an arbitrary finite stack alphabet, arbitrary finite replacement words, deterministic ε-moves, noncanonical intermediate configurations, and any physical encoding compatible with correct recognition. No raw-conjugacy, homogeneous-tail, or canonical normal-form assumption is imposed.

93.1.2 Residuals

For $w \in \{0,1\}^*$ define

$$\rho(w) = qN_1(w) - pN_0(w).$$

Then

$$w \in L_{p,q} \iff \rho(w) = 0,$$

and

$$\rho(wa) = \begin{cases} \rho(w) - p, & a = 0, \\ \rho(w) + q, & a = 1. \end{cases}$$

Lemma 93.2 (Integer residual realizability)**.** *If* $\gcd(p,q) = 1$*, then for every* $t \in \mathbb{Z}$ *there exists* $z \in \{0,1\}^*$ *with*

$$\rho(z) = t.$$

Proof. Choose integers x_0, y_0 with $qx_0 - py_0 = t$. All integer solutions are

$$x = x_0 + pk, \qquad y = y_0 + qk.$$

For sufficiently large k, both x, y are nonnegative. Any binary word with $N_1 = x$ and $N_0 = y$ then has residual t. □

Lemma 93.3 (Residual classification of right quotients)**.** *For all* $u, v \in \{0,1\}^*$*,*

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \quad \iff \quad \rho(u) = \rho(v).$$

Proof. If $\rho(u) = \rho(v)$, then for every continuation z,

$$uz \in L_{p,q} \iff \rho(z) = -\rho(u) \iff \rho(z) = -\rho(v) \iff vz \in L_{p,q}.$$

Conversely, if $\rho(u) \neq \rho(v)$, choose by Theorem 93.2 a word z with $\rho(z) = -\rho(u)$. Then $uz \in L_{p,q}$ while

$$\rho(vz) = \rho(v) - \rho(u) \neq 0,$$

so $vz \notin L_{p,q}$. □

Lemma 93.4 (Physical configurations determine residuals)**.** *If two consumed prefixes* u, v *lead to the same reachable physical configuration of a deterministic recognizer of* $L_{p,q}$*, then*

$$\rho(u) = \rho(v).$$

Consequently every reachable physical configuration C *has a well-defined semantic label* $\Theta(C) \in \mathbb{Z}$*.*

Proof. Starting from one and the same deterministic physical configuration, the set of accepting continuations is identical. Thus

$$u^{-1}L_{p,q} = v^{-1}L_{p,q},$$

and Theorem 93.3 gives $\rho(u) = \rho(v)$. □

For every reachable input-consuming transition,

$$\Theta \xrightarrow{1} \Theta + q, \qquad \Theta \xrightarrow{0} \Theta - p,$$

while ε-moves preserve Θ.

93.2 Unary nonrecurrence and height escape

Consider the computation induced by the infinite unary input 1^ω. For every $n \geq 0$, both 1^n and 1^{n+1} have accepting finite extensions: their residuals nq and $(n+1)q$ admit cancelling continuations by Theorem 93.2. Consequently, after consuming 1^n the deterministic machine cannot halt, reject permanently, or become trapped in an infinite ε-computation before consuming the next 1; otherwise no extension beginning with that next 1 could ever be accepted. Hence the unary computation consumes arbitrarily many input symbols, and every finite consumed unary prefix is live.

Lemma 93.5 (Raw nonrecurrence on the positive unary ray)**.** *No full raw configuration occurs twice on the live computation over 1^ω.*

Proof. Suppose the same configuration C occurs twice. If the two occurrences have consumed different numbers $m < n$ of input symbols, then Theorem 93.4 assigns C both residuals mq and nq, impossible.

If the two occurrences have consumed the same number of input symbols, the intervening segment consists entirely of ε-moves. Determinism then repeats the same ε-cycle forever, so the next input symbol can never be read. This contradicts liveness of the current unary prefix. □

Corollary 93.6 (Unary height escape)**.** *Along the computation over 1^ω, stack height eventually leaves every fixed finite bound.*

Proof. For each fixed height bound H, there are only finitely many state/stack configurations with height at most H. If infinitely many raw positions remained in that bounded region, some full configuration would repeat, contradicting Theorem 93.5. □

Definition 93.7 (Suffix minimum)**.** A raw position C_i on an infinite computation is a suffix minimum if

$$h(C_i) \leq h(C_t) \qquad \text{for every } t \geq i.$$

The following elementary consequence will be used repeatedly.

Lemma 93.8 (Unbounded suffix minima)**.** *If an integer-valued height sequence tends to infinity, then it has suffix-minimum positions at arbitrarily large heights.*

Proof. For every raw position u, the tail $\{h(t) : t \geq u\}$ has a minimum because $h(t) \to \infty$. Choose a position at which this tail minimum is attained. As u tends to infinity, the selected positions cannot remain in a finite set, and their heights must become arbitrarily large. □

93.3 Protected stack pumping and synchronized phases

A nonempty stack configuration $(q, \eta X)$ has *mode* (q, X).

Lemma 93.9 (Protected unary pump)**.** *There exist a state $r \in Q$, a symbol $X \in \Gamma$, an integer $c \geq 1$, a nonempty word $U \in \Gamma^+$, and a lower context η such that*

$$(r, \eta X) \xRightarrow{1^c} (r, \eta U X),$$

where the displayed macro includes all forced ε-moves and no raw configuration of the macro has stack height below $|\eta X|$.

Moreover the macro is context-independent: replacing the untouched lower context η by any other lower context leaves the raw transition sequence unchanged as long as that context is not exposed.

Proof. By Theorem 93.8, suffix-minimum positions occur at arbitrarily large heights. Since only $|Q||\Gamma|$ state/top modes exist, some mode (r, X) occurs at suffix minima of arbitrarily large height. Choose two such positions $t_1 < t_2$ with the same mode and with the later height strictly larger. Write the earlier stack as ηX.

Because t_1 is a suffix minimum, the computation from t_1 to t_2 never drops below height $|\eta X|$. Exposing any symbol of the lower prefix η would first require popping the visible X and reaching height $|\eta| < |\eta X|$. Hence η is never exposed or modified. At t_2 the state and visible top are again r, X, so the larger stack has the form

$$\eta U X$$

for some $U \in \Gamma^+$.

Let c be the number of input symbols 1 consumed between t_1 and t_2. Every transition of the segment sees only the active suffix above the untouched lower context, so replacing η by any other lower context reproduces the same deterministic raw transition sequence. This proves context independence.

To prove $c \geq 1$, suppose instead that $c = 0$. Then the segment is a context-independent, height-increasing ε-macro from mode (r, X) back to itself:

$$(r, \eta X) \overset{\varepsilon}{\Rightarrow} (r, \eta U X).$$

Context independence would then repeat the same forced macro from $(r, \eta U^m X)$ for every $m \geq 0$, producing an infinite deterministic ε-computation and preventing the next symbol of the live unary input from ever being consumed. This is impossible. Therefore $c \geq 1$. □

93.3.1 Iteration and protected cuts

By context independence, for every $k \geq 0$,

$$(r, \eta U^k X) \overset{1^c}{\Longrightarrow} (r, \eta U^{k+1} X).$$

Each copied raw segment is identical to the original segment above its lower context, and hence never goes below the height of its own starting configuration. Therefore the cuts

$$H_k = \eta U^k$$

are protected throughout the entire future computation on 1^ω starting at

$$C_k = (r, H_k X).$$

After discarding finitely many initial k and rebasing the index, we may assume every H_k is nonempty.

This explicit repetition argument is important: we do not infer protection of the synthetic endpoints merely from the fact that the original two selected positions were suffix minima.

Lemma 93.10 (Synchronized p-phase family)**.** *For each $j \in \{0, \ldots, p-1\}$ there exist a state q_j and a nonempty upper stack word W_j, independent of k, with the following property. Starting from C_k, consume exactly j additional symbols 1 and then take every forced ε-move until the computation is again ready to inspect the next unread input symbol. The resulting raw configuration is*

$$D_{k,j} = (q_j, H_k W_j).$$

There is a constant $A \in \mathbb{Z}$ such that

$$\Theta(D_{k,j}) = A + kcq + jq.$$

Proof. The future 1^ω-computation from every C_k remains above H_k. Hence the relative computation above the cut depends only on the common state/top data and not on k. Liveness excludes an infinite ε-divergence after the jth consumed symbol, so the stated next-input boundary is well defined. For fixed j its state and upper stack are therefore independent of k; call them q_j, W_j. The upper word is nonempty because exposing H_k would contradict protection of the cut along the 1^ω future. The residual formula follows because each pump block contributes cq and each of the additional j symbols contributes q. □

93.4 Exposure, deferral, and the diagonal obstruction

Fix a phase j and follow the unique computation on 0^ω from each $D_{k,j}$. Until the common buried cut H_k is exposed, the machine sees only the state and the stack portion above H_k. Thus the relative raw computation is independent of k.

We say that H_k is *exposed* when the full stack is exactly H_k at a raw position. This event cannot be skipped downward. If the current stack is H_kZ with a nonempty upper suffix, one PDA transition rewrites only the visible top symbol and therefore lowers total stack height by at most one. Consequently any computation that ever reaches below the height of H_k must pass through a raw configuration whose stack is exactly H_k first.

Lemma 93.11 (Uniform exposure dichotomy)**.** *For each phase j, exactly one of the following holds.*

(i) There exist $z_j \geq 0$ and $s_j \in Q$ such that for every k the first exposure of H_k occurs after exactly z_j consumed zeros, at configuration

$$(s_j, H_k).$$

(ii) For every k, the computation on 0^ω from $D_{k,j}$ never exposes H_k.

Proof. Before exposure, every transition acts strictly above H_k. The initial relative configuration (q_j, W_j) is independent of k, so the entire relative raw computation is the same for every k. Either its upper stack becomes empty after one fixed raw history and a fixed number z_j of consumed zeros, or it never becomes empty. In the former case the state at that moment is also fixed. □

93.4.1 Fixed-continuation regularity

Lemma 93.12 (Fixed-continuation stack regularity)**.** *Fix a control state s and a finite continuation v (including the mandatory right endmarker, when that convention is used). Let*

$$R_{s,v} = \{S \in \Gamma^* : \textit{starting from } (s, S),\ \textit{the fixed continuation } v \textit{ leads to acceptance}\}.$$

Then $R_{s,v}$ is regular.

Proof. Write $v = v_1 \cdots v_m$. Augment the finite control by an input-position coordinate $i \in \{0, \ldots, m\}$. Original ε-moves leave i fixed; at position $i < m$, only a transition consuming the required letter v_{i+1} advances the coordinate to $i + 1$. Acceptance of the fixed word v is therefore reduced to predecessor reachability of a regular target set in an ordinary pushdown system. If one uses the standard bounded-push presentation of pushdown systems, each transition that writes a longer fixed word is compiled into finitely many auxiliary control states; this normalization does not change the predecessor set on the original configurations. Regular sets of pushdown configurations are effectively closed

under predecessor reachability by the standard saturation theorem [4]. Restricting the resulting regular configuration set to the fixed control state s yields a regular language of stacks. □

Lemma 93.13 (Deferred exposure is impossible)**.** *No phase $j \in \{0, \dots, p-1\}$ can satisfy alternative (ii) of Theorem 93.11.*

Proof. Assume phase j is deferred. Fix one k. The 0^ω-computation remains strictly above H_k. Every finite consumed zero-prefix is live, because its current residual has a finite cancelling continuation by Theorem 93.2. Hence the computation consumes arbitrarily many zeros and cannot become trapped in an ε-cycle.

No full raw configuration can repeat: a repetition after different numbers of consumed zeros would assign one physical configuration two residuals differing by a nonzero multiple of p, while a repetition at the same consumed input position would create a deterministic ε-cycle. Thus the relative upper-stack height above H_k escapes every finite bound.

Apply Theorem 93.9, with 0 in place of 1, to this relative zero computation. There exist a state s, a visible symbol Y, an integer $d \geq 1$, a nonempty word $V \in \Gamma^+$, and a fixed upper word β such that

$$(s, H_k\beta Y) \overset{0^d}{\Longrightarrow} (s, H_k\beta VY).$$

Because the complete relative zero computation before exposure is identical for every k, the same second pump appears above every H_k. Iterating it gives reachable configurations

$$E_{k,\ell} = (s, \eta U^k \beta V^\ell Y), \qquad k, \ell \geq 0.$$

There is a constant $B \in \mathbb{Z}$ such that

$$\Theta(E_{k,\ell}) = B + cqk - pd\ell.$$

Choose, using Theorem 93.2, a fixed binary word v_0 with $\rho(v_0) = -B$. If the model uses a mandatory right endmarker, append it and write $v = v_0 \dashv$; otherwise put $v = v_0$. Since every $E_{k,\ell}$ is reachable and the automaton recognizes $L_{p,q}$,

$$E_{k,\ell} \overset{v}{\Rightarrow} \text{accept} \quad \Longleftrightarrow \quad cqk = pd\ell.$$

By Theorem 93.12, the stack language $R_{s,v}$ is regular. Hence so is

$$K = \{a^k \# b^\ell : \eta U^k \beta V^\ell Y \in R_{s,v}\}.$$

Indeed, if a DFA recognizes $R_{s,v}$, reading each fixed block η, U, β, V, Y induces a fixed transformation of the DFA state set. A finite automaton on the parameter alphabet $\{a, \#, b\}$ can therefore simulate the transformations corresponding to $\eta U^k \beta V^\ell Y$, proving that K is regular.

On the other hand correctness gives

$$K = \{a^k \# b^\ell : cqk = pd\ell\}.$$

With $g = \gcd(cq, pd)$, set

$$\alpha = \frac{pd}{g}, \qquad \beta = \frac{cq}{g}.$$

Then

$$K = \left\{ a^{\alpha n} \# b^{\beta n} : n \geq 0 \right\}.$$

This language is nonregular. Indeed, if N were a pumping length, the word $a^{\alpha N} \# b^{\beta N}$ would have every admissible pumping segment lying wholly in the initial a-block; pumping that segment changes the number of a's while leaving the number of b's fixed, destroying the equation $\beta k = \alpha \ell$. This contradiction eliminates the deferred alternative. □

Chapter 94

Exact Unrestricted State Complexity of $L_{p,q}$

Permanent deferral has now been ruled out, so every synchronized phase must expose the common protected cut. This immediately converts the arithmetic phases into distinct finite-control states. A symmetric argument gives the opposite side, and a three-symbol right-endmarker construction supplies the matching upper bound.

94.1 The unrestricted lower bound

Theorem 94.1 (Positive-side lower bound)**.** *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq p.$$

Proof. By Theorem 93.13, every phase $j \in \{0, \ldots, p-1\}$ exposes the common cut. Fix one k. For each phase,

$$D_{k,j} \xRightarrow{0^{z_j}} (s_j, H_k).$$

If $s_j = s_t$ for distinct j, t, the two exposure configurations are physically identical, because their stacks are both exactly H_k. By Theorem 93.4, their residuals are equal:

$$A + kcq + jq - pz_j = A + kcq + tq - pz_t.$$

Thus

$$q(j - t) = p(z_j - z_t).$$

Since $\gcd(p, q) = 1$, this implies $p \mid (j - t)$, impossible for $0 < |j - t| < p$. Therefore $s_0, \ldots, s_{p-1}$ are pairwise distinct and $|Q| \geq p$. □

Theorem 94.2 (Negative-side state lower bound)**.** *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq q.$$

Proof. Apply the proof of Theorem 94.1 after interchanging the two input letters, interchanging p and q, and replacing the residual by its negative. □

Theorem 94.3 (Unrestricted state lower bound)**.** *For coprime $p, q \geq 1$, every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated regular-target model satisfies*

$$\boxed{|Q| \geq \max(p, q)}.$$

Proof. Combine Theorems 94.1 and 94.2. □

Remark 94.4. The theorem is stronger than the product lower bound

$$|Q||\Gamma| \geq \max(p, q)$$

that one might first seek. The argument charges the ratio parameter directly to finite control, without any restriction on $|\Gamma|$.

94.2 A matching three-symbol construction

We now give a self-contained upper bound. This section is independent of the lower-bound representation and is included so that the exact state-complexity statement is proved entirely within the present Part.

Assume first that $p \geq q$. Let

$$Q = \{R_0, \ldots, R_{p-1}\}, \qquad \Gamma = \{A, B, Z_0\}.$$

We encode each residual $r \in \mathbb{Z}$ as follows. For $r \geq 0$, write uniquely

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\zeta(r) = (R_s, Z_0 A^k).$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad h \geq 1, \quad 0 \leq u < q,$$

and define

$$\zeta(r) = (R_u, Z_0 B^h).$$

For $0 \leq s < p$, put

$$s^+ = (s + q) \bmod p, \qquad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

and

$$u_s = (s - p) \bmod q, \qquad h_s = \frac{u_s + p - s}{q} \geq 1.$$

For $0 \leq u < q$, put

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \geq 1.$$

With the stack top on the right, define the following binary rules:

$$\delta(R_s, A, 0) = (R_s, \varepsilon), \qquad \delta(R_s, A, 1) = (R_{s^+}, A^{c_s+1}),$$

$$\delta(R_s, Z_0, 1) = (R_{s^+}, Z_0 A^{c_s}), \qquad \delta(R_s, Z_0, 0) = (R_{u_s}, Z_0 B^{h_s}),$$

and for $0 \leq u < q$,

$$\delta(R_u, B, 1) = (R_u, \varepsilon), \qquad \delta(R_u, B, 0) = (R_{u^-}, B^{d_u+1}).$$

No ε-transition is used.

A mandatory right endmarker $\dashv$ is accepted only from

$$\zeta(0) = (R_0, Z_0).$$

For example, declare R_0 final and define only

$$\delta(R_0, Z_0, \dashv) = (R_0, Z_0)$$

among endmarker moves.

Theorem 94.5 (Three-symbol residual realization)**.** *For every coprime $p, q \geq 1$, there is an ε-free right-endmarker DPDA recognizing $L_{p,q}$ with*

$$|Q| = \max(p, q), \qquad |\Gamma| = 3.$$

Proof. Assume $p \geq q$. We verify

$$\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)$$

for every residual r.

Let $r = pk + s \geq 0$. If $k \geq 1$, input 0 pops one A, giving

$$pk + s - p = p(k - 1) + s.$$

For input 1,

$$pk + s + q = p(k + c_s) + s^+,$$

so the rule replacing A by A^{c_s+1} gives exactly $\zeta(r + q)$. When $k = 0$, the 1-rule uses the same carry identity, while

$$s - p = u_s - qh_s,$$

so the boundary 0-rule enters exactly the negative encoding.

Now let $r = u - qh < 0$. Input 1 pops one B. If $h > 1$ this gives $u - q(h - 1)$; if $h = 1$ it exposes Z_0 and leaves $(R_u, Z_0) = \zeta(u)$. For input 0,

$$u - p = u^- - qd_u,$$

and hence

$$r - p = u^- - q(h + d_u),$$

which is implemented by replacing one visible B by B^{d_u+1}.

Therefore every binary input symbol realizes the correct residual translation in one ordinary transition. Starting from $\zeta(0)$, after reading w the machine is in $\zeta(\rho(w))$. The right endmarker is accepted exactly at $\zeta(0)$, so the recognized language is $L_{p,q}$.

If $q > p$, exchange 0 and 1, exchange p and q, exchange A and B, and negate the residual. The resulting machine has $q = \max(p, q)$ states. □

94.3 Exact state complexity

Theorem 94.6 (Exact unrestricted right-endmarker state complexity)**.** *For coprime $p, q \geq 1$, among all deterministic pushdown automata in the lower-bound model whose inputs are terminated by the mandatory symbol $\dashv$,*

$$\boxed{Q_{\min}(L_{p,q}) = \max(p, q)}$$

under deterministic pushdown recognition with a mandatory right endmarker.

Proof. Theorem 94.3 gives the lower bound and Theorem 94.5 gives the matching upper bound. □

Corollary 94.7 (Arbitrary positive parameters)**.** *For arbitrary $p, q \geq 1$, let $d = \gcd(p, q)$. Since*

$$L_{p,q} = L_{p/d,q/d},$$

the right-endmarker state complexity is

$$\boxed{Q_{\min}(L_{p,q}) = \max\left(\frac{p}{d}, \frac{q}{d}\right).}$$

Chapter 95

Exact Unrestricted Bounded-Replacement State Complexity

The preceding unrestricted-state branch removed the assumption that a recognizer must expose the residual in a prescribed physical coordinate system. The remaining bounded-replacement question is sharper: if the work-stack alphabet has only one non-bottom symbol and one transition may replace the visible work symbol by a word of length at most B, what is the exact state cost when *all* deterministic encodings and normalization histories are allowed?

Within the raw residual-conjugacy model, the bounded question has an exact staircase. The theorem developed here shows that the same law is representation-independent. Its proof retains only the language-level residual quotient and generic top-access locality; the decisive ingredients are protected one-symbol phases, finite-depth descent, outward first-hit bands, and a synchronization argument that converts bounded local stack growth into an exact global state lower bound.

Position relative to pushdown resource complexity. The resource parameter used here should not be presented as if bounded stack growth or bounded replacement were new ideas. Pushdown descriptional complexity already studies tradeoffs involving finite control and pushdown-alphabet size, including hierarchies obtained by limiting the number of pushdown symbols [85, 16]. Pushdown height and broader memory-use measures have also been studied quantitatively [87, 88], and push complexity counts the number of push operations used by accepting computations [86]. In standard PDA presentations that allow one transition to replace the top symbol by an arbitrary word, the maximum one-step replacement length is likewise part of the machine description; it can be normalized to unit pushes at a descriptional cost [87]. These neighboring parameters are not the theorem below. The claim of this chapter is the exact *representation-independent* state law, for the fixed rational-balance family $L_{p,q}$ and one work symbol, as a function of the per-transition replacement bound B, even against arbitrary deterministic ε-behavior and history-dependent encodings.

Put

$$w_{\max} = \max(p, q), \qquad w_{\min} = \min(p, q).$$

The chapter culminates in the exact law

$$\boxed{Q^{(B)}_{\mathrm{unrestricted,1work}}(p, q) = w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right)} \qquad (B \geq 2),$$

and therefore upgrades the former typed raw staircase to an unrestricted theorem.

95.1 Model and semantic rigidity

95.1.1 Bounded-replacement one-work-symbol pushdown recognizers

A deterministic right-endmarker ε-pushdown automaton has a finite control set Q, binary input alphabet $\{0,1\}$ followed by one mandatory symbol $\dashv$, and stack alphabet

$$\Gamma = \{Z_0, A\},$$

where Z_0 is a permanent bottom marker and A is the unique work symbol; the initial stack is Z_0. Permanence means that Z_0 is never removed and never occurs above the bottom: a rule reading Z_0 replaces it only by a word of the form Z_0A^t. Acceptance is by final state after the mandatory endmarker has been consumed. The stack top is written on the right. The *work height* of Z_0A^h is h. A raw transition has the form

$$(q, \eta X, av) \vdash (q', \eta\beta, v)$$

or its ε-analogue, where X is the visible top symbol. Determinism means that for each state/top pair there is at most one move on each symbol in $\{0,1,\dashv\}$ and, whenever an ε-move is enabled, no input-consuming move is enabled at that state/top pair.

Fix $B \geq 2$. The machine is *B-bounded* if every replacement word β has length at most B. Thus one raw transition can decrease work height by at most one and can increase it by at most $B-1$. The opponent may otherwise use arbitrary deterministic ε-moves, non-canonical intermediate configurations, and arbitrary history-dependent representations. A reachable configuration is called *settled* when no forced ε-move is enabled, so the machine is ready either to consume the next input symbol or the right endmarker. When comparing runs on different inputs, we call the pair consisting of the current control state and the complete stack Z_0A^h the *physical configuration*; thus "physically identical" suppresses the unread input suffix but requires equality of both state and complete stack.

The lower bound uses only the binary-prefix behavior of this model and the following liveness consequence: every finite prefix occurring below has a finite binary continuation followed by $\dashv$ that is accepted. Hence deterministic ε-divergence cannot permanently prevent the next required input symbol from being consumed.

Definition 95.1 (Unrestricted bounded one-work-symbol state complexity)**.** Let

$$Q^{(B)}_{\text{unrestricted,1work}}(p,q)$$

denote the least $|Q|$ over deterministic right-endmarker ε-pushdown recognizers of $L_{p,q}$ that are B-bounded, have a permanent bottom marker, and use exactly one non-bottom work symbol.

95.1.2 Residual quotient

For $w \in \{0,1\}^*$ define

$$\rho(w) = qN_1(w) - pN_0(w).$$

Then $w \in L_{p,q}$ iff $\rho(w) = 0$.

Lemma 95.2 (Integer residual realizability)**.** *For every $t \in \mathbb{Z}$ there is a word $z \in \{0,1\}^*$ with $\rho(z) = t$.*

Proof. By Bezout, choose integers x_0, y_0 with $qx_0 - py_0 = t$. Replacing (x_0, y_0) by $(x_0 + pn, y_0 + qn)$ for sufficiently large n makes both entries nonnegative. Then $z = 1^{x_0+pn}0^{y_0+qn}$ has residual t. □

Lemma 95.3 (Residuals classify right quotients)**.** *For binary prefixes u, v,*

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \quad \Longleftrightarrow \quad \rho(u) = \rho(v).$$

Proof. If $\rho(u) = \rho(v)$, then for every continuation z,

$$uz \in L_{p,q} \iff \rho(z) = -\rho(u) \iff \rho(z) = -\rho(v) \iff vz \in L_{p,q}.$$

Conversely, if $\rho(u) \neq \rho(v)$, choose by Theorem 95.2 a word z with $\rho(z) = -\rho(u)$. Then $uz \in L_{p,q}$ but $vz \notin L_{p,q}$. □

Lemma 95.4 (Physical residual rigidity)**.** *If two reachable raw configurations of a deterministic recognizer are physically identical, then the consumed binary prefixes have the same residual.*

Proof. From one physical configuration, determinism gives the same future behavior on every finite continuation. Hence the two consumed prefixes have equal right quotients. Apply Theorem 95.3. □

It is therefore legitimate to write $\mathrm{Res}(C)$ for the unique residual carried by a reachable raw configuration C. Input-consuming transitions satisfy

$$\mathrm{Res}(C \xrightarrow{1} C') = \mathrm{Res}(C) + q, \qquad \mathrm{Res}(C \xrightarrow{0} C') = \mathrm{Res}(C) - p,$$

while ε-moves preserve Res.

95.2 A matching bounded one-work-symbol construction

By symmetry it is enough to assume $p \geq q$. Put

$$b = b_B := \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

Use positive states $P_0, \ldots, P_{p-1}$ and negative states $N_0, \ldots, N_{b-1}$. Encode residuals as follows. For $r \geq 0$, write

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and set

$$\eta(r) = (P_s, Z_0A^k).$$

For $r < 0$, write uniquely

$$r = u - bh, \qquad 0 \leq u < b, \quad h \geq 1,$$

and set

$$\eta(r) = (N_u, Z_0A^{h-1}).$$

Proposition 95.5 (Bounded residual construction)**.** *For $p \geq q$ and $B \geq 2$, the encoding above is realized by an ε-free deterministic B-bounded one-work-symbol pushdown automaton with exactly*

$$p + b_B$$

states. Hence, for arbitrary coprime p, q,

$$Q^{(B)}_{\mathrm{unrestricted,1work}}(p, q) \leq w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right).$$

Proof. Take (P_0, Z_0) as the initial configuration. We verify that the two residual translations can be implemented by one top rewrite.

On the positive side, input 0 sends $pk + s$ to $p(k-1) + s$ when $k \geq 1$, so one A is popped. At the boundary $k = 0$, write

$$s - p = u - bh, \qquad 0 \leq u < b, \quad h = \left\lceil \frac{p-s}{b} \right\rceil.$$

Since $p \leq b(B-1)$, one has $h \leq B-1$; replacing Z_0 by $Z_0 A^{h-1}$ has length at most B.

For input 1, put

$$s' = (s+q) \bmod p, \qquad c = \frac{s+q-s'}{p} \in \{0,1\}.$$

Then $pk+s+q = p(k+c)+s'$. Thus the stack height changes by at most one, implemented either by $A \mapsto A^{1+c}$ or by the corresponding bottom-marker rule.

On the negative side, write $r = u - bh$. For input 0, let

$$u' = (u-p) \bmod b, \qquad d = \frac{u'-u+p}{b} = \left\lceil \frac{p-u}{b} \right\rceil.$$

Then

$$r - p = u' - b(h+d).$$

Again $d \leq B-1$, so an existing top A is rewritten as A^{d+1}, or at height zero the bottom marker is rewritten as $Z_0 A^d$; both replacement words have length at most B.

Finally, because $b \geq q$, input 1 changes the negative quotient height by at most one. If $u+q < b$, the height is unchanged and the remainder becomes $u+q$. If $u+q \geq b$ and $h \geq 2$, one work symbol is popped and the remainder becomes $u+q-b$. If $u+q \geq b$ and $h = 1$, then

$$0 \leq u+q-b < q \leq p,$$

so the transition crosses directly to the positive state P_{u+q-b} while leaving the stack Z_0 unchanged. All binary rules are therefore deterministic, ε-free, and B-bounded, and they realize $r \mapsto r-p$ and $r \mapsto r+q$ exactly. Finally designate P_0 as the sole final state and, among right-endmarker moves, define only

$$\delta(P_0, Z_0, \dashv) = (P_0, Z_0).$$

Since $\eta(0) = (P_0, Z_0)$ and no nonzero residual representative has both state P_0 and top Z_0, the mandatory endmarker accepts exactly residual zero without introducing another state. □

95.3 Protected unary-stack phases and finite-depth descent

The first-hit block constructed later must be compared with a representation-independent family of opposite-sign phases living over one common unary stack. We therefore isolate the protected unary interface needed for the synchronization argument. All statements in this section concern an arbitrary deterministic recognizer in the model of Section 95.1; no bounded replacement property is used until Section 95.4.

Consider the live raw computation on 1^ω. Every finite prefix 1^n has a finite accepting extension by Theorem 95.2. Hence the run consumes arbitrarily many input symbols and cannot become trapped in a deterministic ε-cycle before the next required 1 is read.

Lemma 95.6 (Unary-stack height escape)**.** *Along the raw computation on 1^ω, work height eventually leaves every finite bound.*

Proof. No full raw state/stack configuration can recur. A recurrence after different numbers of consumed 1's would assign the same physical configuration two different residuals by Theorem 95.4; a recurrence at the same input position would create a deterministic ε-cycle and prevent the next live input symbol from being read. If work height failed to leave some finite bound, finitely many control states and unary stacks would force such a recurrence. □

Because the integer-valued height tends to infinity in the preceding sense, there are suffix minima at arbitrarily large heights. At a suffix minimum, the remainder of the 1^ω run never goes below the current complete unary stack.

Lemma 95.7 (Protected unary-stack pump)**.** *There exist a state r, integers $a, c, u \geq 1$, and an infinite family of reachable configurations*

$$(r, Z_0 A^{h_k}), \qquad h_k = a + uk,$$

such that one fixed raw macro consuming exactly c symbols 1 maps

$$(r, Z_0 A^{h_k}) \xRightarrow{1^c} (r, Z_0 A^{h_k+u})$$

for every $k \geq 0$. The macro never exposes a source cell below height h_k and is therefore independent of the buried unary context.

Proof. Among suffix minima of arbitrarily large height, some state/top mode occurs twice, the later occurrence at a strictly larger height. Between the two occurrences the computation never drops below the earlier suffix minimum, so the buried stack below that cut is not inspected. Since the work alphabet is unary, the later stack differs from the earlier one by $u \geq 1$ additional copies of A. The raw transition sequence is therefore valid over every copy of the same buried context and can be iterated. It must consume at least one input symbol: otherwise a deterministic height-increasing ε-macro from one mode back to itself would iterate forever, contradicting liveness. Writing the first chosen height as a gives $h_k = a + uk$. □

The next lemma is the unary form of finite-depth exposure. It is the point at which arbitrary history-dependent representations are neutralized: a fixed finite layer above a protected cut cannot remain hidden forever. We say that a family of configurations is obtained from the protected family by one *fixed relative history above height h_k* if one finite raw transition pattern, with the same sequence of input-consuming and ε-moves and the same rewrite rules, translates over every cut $Z_0 A^{h_k}$, never visits height below h_k, and ends in the stated common state and fixed height offset. Because all symbols above a positive cut are A, such a history is independent of the buried unary context.

Lemma 95.8 (Finite-depth unary-stack descent)**.** *Fix $d, D \geq 0$ and a reachable family*

$$(q, Z_0 A^{h_k+d}), \qquad h_k = a + uk \longrightarrow \infty,$$

obtained from the protected family by one fixed relative history above height h_k. Then, after discarding finitely many k so that $h_k \geq D$, there are an integer $e_{d,D} \geq 0$ and a state $q_{d,D}$, independent of k, such that following 0^ω until the first exposure of the shifted cut at height $h_k - D$ gives

$$(q, Z_0 A^{h_k+d}) \xRightarrow{0^{e_{d,D}}} (q_{d,D}, Z_0 A^{h_k-D}).$$

The displayed macro includes all forced ε-moves. In particular, $D = 0$ gives descent to the original protected cut.

Proof. Put $t_k = h_k - D$. Before the shifted cut at height t_k is exposed, the relative configuration above it starts from the fixed pair (q, A^{d+D}) and follows a deterministic raw history independent of k. Hence either this relative stack is emptied after one fixed finite history for every sufficiently large k, or it is never emptied for any sufficiently large k. Because one raw transition decreases work height by at most one, first exposure of the cut occurs at height exactly t_k.

Assume the second alternative. The relative 0^ω computation is live. Raw recurrence is impossible by Theorem 95.4, so its relative height escapes. Applying the suffix-minimum argument of Theorem 95.7 with 0 in place of 1 to this relative computation produces a second context-independent pump, now consuming $e \geq 1$ zeros and adding $v \geq 1$ work cells. After a fixed prefix, variation of the original protected-family parameter k and independent iteration of the new 0-pump yield reachable configurations in one fixed control state with residual

$$C + kcq - \ell ep$$

and complete unary height

$$H + uk + v\ell$$

for all sufficiently large k, ℓ, where the constants C, H absorb the fixed relative history and the shift D.

Compare the parameter pairs $(k+v, \ell)$ and $(k, \ell+u)$. Their unary heights, hence their complete stacks, are identical, and their control states are identical; but their residuals differ by

$$vcq + uep > 0.$$

This contradicts physical residual rigidity. Permanent nonexposure is therefore impossible, so the shifted cut is reached after one fixed relative history. The number $e_{d,D}$ of consumed zeros and the exposure state $q_{d,D}$ are consequently independent of k. □

We now extract the p positive residual phases used later. Starting at $(r, Z_0 A^{h_k})$, consume $j \in \{0, \ldots, p-1\}$ further symbols 1 and all forced ε-moves. Since the future on 1^ω is obtained by iterating the same protected macro, the computation above the cut is a fixed relative history for each j; thus for some state q_j and integer $d_j \geq 0$, independent of k, the resulting configuration is

$$(q_j, Z_0 A^{h_k + d_j}).$$

Apply Theorem 95.8 with $(d, D) = (d_j, 0)$.

Proposition 95.9 (Protected positive phases and simultaneous descent)**.** *There are affine protected heights $h_k = a + uk$ and p reachable configurations*

$$(s_j, Z_0 A^{h_k}), \qquad j = 0, \ldots, p-1,$$

whose states $s_0, \ldots, s_{p-1}$ are pairwise distinct. Their residuals tend to $+\infty$ as $k \to \infty$.

Moreover, for every fixed $D \geq 0$, after discarding finitely many k there are states $s_{j,D}$ and integers $z_{j,D} \geq 0$, independent of k, such that

$$(s_j, Z_0 A^{h_k}) \xRightarrow{0^{z_{j,D}}} (s_{j,D}, Z_0 A^{h_k - D}),$$

and the p states $s_{j,D}$ are pairwise distinct. For each fixed D, the descended residuals also tend to $+\infty$ with k. The conclusion holds simultaneously for all D in any prescribed finite set.

Proof. The first family is obtained from the preceding construction by descending the fixed upper depth d_j for each phase. If phases $j \neq t$ exposed the common stack $Z_0 A^{h_k}$ in the same state, and if z_j, z_t are the corresponding numbers of consumed zeros, then residual rigidity would give

$$q(j-t) = p(z_j - z_t).$$

Coprimality forces $p \mid (j-t)$, impossible for $0 < |j-t| < p$. Hence the exposure states are pairwise distinct. The protected pump contributes kcq to every phase residual, so all phase residuals tend to $+\infty$.

For a further fixed depth D, each exposed phase family is still obtained from the original protected family by one fixed relative history ending at height h_k. Apply Theorem 95.8 with $(d, D) = (0, D)$ to each of the p exposed phase families. Equality of two descended states at the common stack would again make their residuals equal. The residual difference between phases j and t always has the form

$$q(j-t) - pn$$

for some integer n, namely the difference of the total numbers of consumed zeros along the two phase histories. Equality would therefore force $q(j-t) = pn$, impossible by coprimality and $0 < |j-t| < p$. Thus phase separation survives descent. Since the additional zero history for a fixed D is independent of k, the common kcq term still drives every descended residual to $+\infty$. For a finite set of depths, take the maximum of the finitely many eventuality thresholds; all required descents then hold simultaneously. □

By exchanging 0 with 1 and p with q, one obtains a symmetric protected negative family with q phases and affine protected heights. We use that symmetric family only in the baseline $p+q$ theorem; the new bounded-replacement argument will construct its own negative block by first-hit geometry.

95.4 Outward first-hit bands

Assume $p \geq q$. Such settled configurations exist at arbitrarily negative residuals: start from a sufficiently long prefix 0^n and follow its forced ε-moves to the next settled configuration, which must be reached by liveness. Choose one whose residual R is sufficiently negative that

$$R + (p-1)q < 0.$$

For $j = 0, \ldots, p-1$, consume 1^j and all forced ε-moves until the machine is again ready to inspect the next binary input symbol. Denote the resulting reachable configuration by C_j. Its residual is

$$\mathrm{Res}(C_j) = R + jq < 0.$$

From every C_j follow the unique live computation on 0^ω, including all forced ε-moves. After n consumed zeros the residual is

$$R + jq - np.$$

The settled heights on each such ray are unbounded: otherwise finitely many states and bounded unary heights would force a repeated physical configuration carrying two different residuals.

Lemma 95.10 (Outward first-hit band)**.** *Let K exceed the heights of $C_0, \ldots, C_{p-1}$. For each j, let $E_j(K)$ be the first raw configuration on the outward 0^ω run from C_j whose work height is at least K. Then*

$$\mathrm{ht}(E_j(K)) \in \{K, K+1, \ldots, K+B-2\}.$$

Consequently there is a subset $J_K \subseteq \{0, \dots, p-1\}$ *with*

$$|J_K| \geq \left\lceil \frac{p}{B-1} \right\rceil$$

such that all configurations $E_j(K)$, $j \in J_K$, *have one common complete unary stack height. Their control states are pairwise distinct.*

Proof. The first-hit configuration exists because the settled heights on the ray are unbounded. The raw configuration immediately preceding it has work height at most $K-1$. One B-bounded top rewrite can increase work height by at most $B-1$, so the first-hit height is at most $K+B-2$. This proves the band statement.

There are p phases and only $B-1$ possible first-hit heights, so pigeonhole gives the stated subset. Suppose two distinct indices $j, t \in J_K$ have the same control state. Since their stacks are unary and have the same height, the two full physical configurations are identical. Let z_j, z_t be the numbers of consumed zeros at the respective first-hit positions. Residual rigidity gives

$$R + jq - pz_j = R + tq - pz_t,$$

so

$$q(j-t) = p(z_j - z_t).$$

Because $\gcd(p,q) = 1$, one has $p \mid (j-t)$, impossible for $0 < |j-t| < p$. Hence the states are pairwise distinct. □

Remark 95.11. The lemma uses the replacement bound only through the finite overshoot band of width $B-1$. It makes no assumption about a residual section, a homogeneous tail, the timing of ε-moves, or the history by which the current unary height was created.

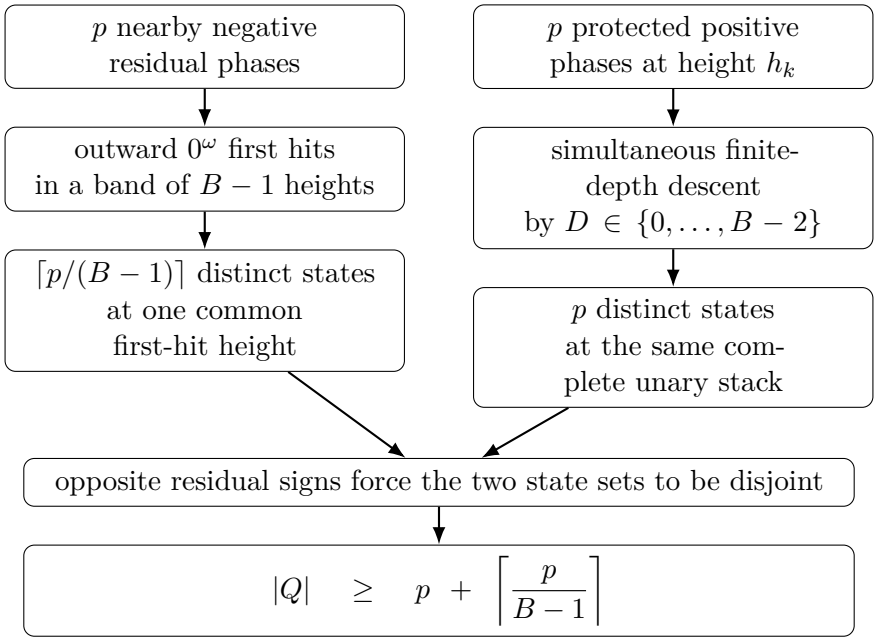


Figure 95.1: Architecture of the new unrestricted bounded-replacement lower bound for $p \geq q$. The bounded rewrite length is used only to create the finite first-hit band; all representation freedom before that crossing is absorbed by residual rigidity and protected finite-depth descent.

95.5 Synchronization and the exact theorem

We now synchronize the negative first-hit block with the positive protected family at one complete physical stack.

Theorem 95.12 (First-hit/protected-phase synchronization)**.** *Assume $p \geq q$. Every unrestricted deterministic B-bounded one-work-symbol recognizer of $L_{p,q}$ satisfies*

$$\boxed{|Q| \geq p + \left\lceil \frac{p}{B-1} \right\rceil.}$$

Proof. Take arbitrarily high positive protected heights h_k from Theorem 95.9. Put

$$K_k = h_k - (B-2).$$

For k large, K_k lies above all initial negative phase heights C_j, so Theorem 95.10 applies. Its common first-hit height has the form

$$K_k + d_k = h_k - D_k, \qquad 0 \leq d_k \leq B-2, \qquad D_k = (B-2) - d_k \in \{0, \ldots, B-2\}.$$

By the simultaneous finite-depth form of Theorem 95.9, once k is large enough the positive protected family can be descended by whichever D_k occurs. Since only the finitely many depths $0, \ldots, B-2$ are involved, enlarge k once more so that all descended phase residuals at all these depths are positive. At the common complete stack

$$Z_0 A^{h_k - D_k}$$

there are therefore

(i) p reachable positive-residual configurations in pairwise distinct states, and

(ii) at least $\lceil p/(B-1) \rceil$ reachable negative-residual first-hit configurations in pairwise distinct states.

The two state sets are disjoint. Indeed, equality of a positive state and a negative state at the same complete unary stack would identify two reachable physical configurations carrying residuals of opposite sign, contradicting Theorem 95.4. The displayed lower bound follows. □

We also use the representation-independent one-work-symbol bound.

Theorem 95.13 (Unrestricted one-work-symbol baseline)**.** *Every deterministic right-endmarker ε-pushdown recognizer of $L_{p,q}$ with a permanent bottom marker and one work symbol satisfies*

$$\boxed{|Q| \geq p + q.}$$

Proof. Let the positive and negative protected cut heights be

$$h_k^+ = a + uk, \qquad h_\ell^- = b + v\ell, \qquad u, v \geq 1.$$

Put $g = \gcd(u, v)$. Choose sufficiently large indices k_0, ℓ_0 and define

$$k_n = k_0 + \frac{v}{g}n, \qquad \ell_n = \ell_0 + \frac{u}{g}n.$$

Then

$$h_{k_n}^+ - h_{\ell_n}^- = D$$

is independent of n. If $D \geq 0$, apply the finite-depth descent theorem to all p positive phases with depth D and leave the q negative phases at depth zero. For all sufficiently large n, the resulting $p+q$ configurations have the identical complete unary stack

$$Z_0 A^{h^-_{\ell_n}}.$$

The p positive states are pairwise distinct by phase separation, and the symmetric argument separates the q negative states. Their residuals have opposite signs for sufficiently large n, so a positive and a negative configuration cannot share a state at the common stack by Theorem 95.4. Hence $|Q| \geq p+q$. If $D < 0$, descend the negative family by $-D$ instead. □

Theorem 95.14 (Exact unrestricted bounded-replacement one-work-symbol complexity)**.** *For every coprime $p, q \geq 1$ and every $B \geq 2$, with*

$$w_{\max} = \max(p,q), \qquad w_{\min} = \min(p,q),$$

one has

$$\boxed{Q^{(B)}_{\text{unrestricted,1work}}(p,q) = w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right).}$$

Proof. Assume first $p \geq q$. By Theorem 95.12,

$$|Q| \geq p + \left\lceil \frac{p}{B-1} \right\rceil,$$

while Theorem 95.13 gives $|Q| \geq p+q$. Therefore

$$|Q| \geq p + \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

The case $q > p$ is obtained by symmetry. The matching upper bound is Theorem 95.5. □

Theorem 95.15 (Complete unrestricted state–stack frontier)**.** *For coprime $p, q \geq 1$, under deterministic right-endmarker recognition with a permanent bottom marker, the Pareto-minimal pairs*

$$(|Q|, |\Gamma|)$$

are exactly

$$\boxed{\text{Pareto}_{\text{unrestricted}}(L_{p,q}) = \{(p+q, 2), (w_{\max}, 3)\}.}$$

Here $|\Gamma|$ counts the permanent bottom marker.

Proof. With only one total stack symbol the store cannot carry unbounded information, so the resulting device has only finitely many physical configurations and cannot recognize the nonregular language $L_{p,q}$. With two total stack symbols there is exactly one work symbol; Theorem 95.13 gives $|Q| \geq p+q$, while Theorem 95.14 attains $p+q$ once $B \geq B_{\text{crit}}$. With three or more total stack symbols, the representation-independent state lower bound Theorem 94.3 gives $|Q| \geq w_{\max}$, and the three-symbol construction Theorem 94.5 attains $(w_{\max}, 3)$. Any point with more than three stack symbols and at least $w_{\max}$ states is dominated by $(w_{\max}, 3)$. Finally $w_{\max} < p+q$, so the two displayed endpoints are incomparable and hence both Pareto-minimal. □

95.6 The unrestricted state–push staircase

Put

$$Q_B := w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{B-1} \right\rceil\right).$$

The exact theorem turns the bounded replacement parameter into a completely explicit staircase. The restriction $B \geq 2$ is the first nontrivial regime: with $B = 1$, the initial work height zero can never increase, so only finitely many physical configurations are reachable. By Theorem 95.3, $L_{p,q}$ has infinitely many right quotients and is therefore nonregular; thus $B = 1$ cannot recognize it in this model.

Corollary 95.16 (The first two bounded levels)**.** *For every coprime p, q,*

$$\boxed{Q_2 = 2w_{\max},}$$

and

$$\boxed{Q_3 = w_{\max} + \max\left(w_{\min}, \left\lceil \frac{w_{\max}}{2} \right\rceil\right).}$$

Corollary 95.17 (Critical replacement threshold)**.** *The unbounded one-work-symbol optimum $w_{\max} + w_{\min} = p + q$ is attained if and only if*

$$\boxed{B \geq B_{\text{crit}} := 1 + \left\lceil \frac{w_{\max}}{w_{\min}} \right\rceil.}$$

Proof. By Theorem 95.14, $Q_B = w_{\max} + w_{\min}$ exactly when

$$\left\lceil \frac{w_{\max}}{B-1} \right\rceil \leq w_{\min},$$

which is equivalent to $B - 1 \geq w_{\max}/w_{\min}$. □

Corollary 95.18 (Inverse state-to-replacement law)**.** *Fix a state budget $S \geq w_{\max} + w_{\min}$. The least replacement bound permitting an unrestricted one-work-symbol recognizer with at most S states is*

$$\boxed{B_{\min}(S) = \begin{cases} 2, & S \geq 2w_{\max}, \\ 1 + \left\lceil \dfrac{w_{\max}}{S - w_{\max}} \right\rceil, & w_{\max} + w_{\min} \leq S < 2w_{\max}. \end{cases}}$$

Proof. If $S \geq 2w_{\max}$, Theorem 95.16 shows that $B = 2$ already suffices, and no smaller nontrivial replacement bound exists. Suppose $w_{\max} + w_{\min} \leq S < 2w_{\max}$. Since $S - w_{\max} \geq w_{\min}$, Theorem 95.14 gives $Q_B \leq S$ exactly when

$$\left\lceil \frac{w_{\max}}{B-1} \right\rceil \leq S - w_{\max},$$

which is equivalent to

$$B - 1 \geq \frac{w_{\max}}{S - w_{\max}}.$$

The least integer B satisfying this is the displayed value. □

Theorem 95.19 (Exact unrestricted one-work-symbol state–push Pareto staircase)**.** *At fixed total stack alphabet size $|\Gamma| = 2$, the Pareto-minimal pairs $(|Q|, B)$ are exactly*

$$\boxed{\{(Q_B, B) : 2 \leq B \leq B_{\text{crit}},\ B = 2 \text{ or } Q_B < Q_{B-1}\}.}$$

Proof. A plateau value is dominated by the same state count with the smaller replacement bound. Every point after B_{crit} is dominated by the critical-threshold point. At every strict drop, reducing B forces a larger state count by Theorem 95.14; hence every displayed breakpoint is Pareto-minimal. □

For the motivating 2:1 case, $w_{\max} = 2$ and $w_{\min} = 1$, so

$$Q_B = 2 + \max\left(1, \left\lceil \frac{2}{B-1} \right\rceil\right).$$

Thus $Q_2 = 4$, while $Q_B = 3$ for every $B \geq 3$. The bounded-push transition from four states to the unrestricted one-work-symbol optimum of three states therefore occurs exactly at $B = 3$.

95.7 Why the lower bound is genuinely unrestricted

The proof of Theorem 95.14 never assumes that the competitor stores the residual in a chosen state/height coordinate. In particular it uses none of the following:

- raw residual conjugacy;
- a prescribed translation-invariant high-stack regime;
- realtime computation;
- a history-independent representative of a residual;
- a prescribed normalization boundary;
- a canonical relationship between residual magnitude and stack height.

The bounded replacement parameter enters only at the first crossing of a physical height threshold. That crossing has at most $B - 1$ possible overshoots. The semantic phase information is then separated by residual rigidity and coprimality, while the protected family supplies the opposite-sign block at the same complete unary stack. This is why a representation-relative state–push staircase becomes a language-intrinsic unrestricted theorem.

95.8 A separate three-symbol bounded-push optimum

The optimal three-symbol point can also be made replacement-at-most-two. Its interface is different from the one-work-symbol theorem above: a normalized macro-step consists of one input-consuming move is followed, when necessary, by a uniquely forced terminating ε-burst, after which the stable residual representative has been restored.

Assume $p \geq q$ and use the $w_{\max} = p$-state, three-symbol stable encoding from the preceding unrestricted-state construction. For $0 \leq u < q$, let

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q},$$

and for a target remainder $v \in \mathbb{Z}_q$ put

$$\mathrm{pre}(v) = (v + p) \bmod q, \qquad L_v = d_{\mathrm{pre}(v)} - 1.$$

Since addition by p permutes $\mathbb{Z}_q$,

$$\sum_{v=0}^{q-1} L_v = \sum_{u=0}^{q-1} (d_u - 1) = p - q.$$

Introduce abstract suffix roles $H_{v,j}$, $1 \le j \le L_v$, meaning "push exactly j further copies of the negative work symbol B, then enter R_v." On a visible B their forced rules are

$$\delta(H_{v,1}, \varepsilon, B) = (R_v, BB), \qquad \delta(H_{v,j}, \varepsilon, B) = (H_{v,j-1}, BB) \quad (j > 1).$$

These are roles rather than new state names. Stable negative configurations use top B only in $R_0, \ldots, R_{q-1}$, so the $p - q$ state/top cells

$$(R_i, B), \qquad q \le i < p,$$

are unused by stable semantics. Fix a bijection from the suffix roles onto those cells. A long negative input-0 write is replaced by one consuming $B \mapsto BB$ move into the appropriate suffix role; positive-to-negative boundary writes enter a suffix of the same chain. If $u_s = (s - p) \bmod q$ and $h_s = (u_s + p - s)/q$, then

$$d_{\mathrm{pre}(u_s)} = h_s + \left\lfloor \frac{s}{q} \right\rfloor,$$

so $h_s - 1 \le L_{u_s}$ and every required boundary suffix exists.

Theorem 95.20 (State-preserving push-at-most-two macro compiler)**.** *For every coprime p, q, there is a deterministic right-endmarker DPDA whose stable configurations realize the residual graph and satisfy*

$$\boxed{|Q| = w_{\max}, \qquad |\Gamma| = 3, \qquad \max |\alpha| \le 2.}$$

Every nontrivial normalization burst is a uniquely forced finite ε-suffix chain. In particular, for every replacement bound $B \ge 2$, the minimum state count among deterministic recognizers with at most three total stack symbols is exactly $w_{\max}$.

Proof. The role overlay above proves the case $p \ge q$. Every long negative write is split into one consuming push and a strictly descending suffix chain. The number of roles is exactly $p - q$, all fit into state/top cells unused by stable negative semantics, and those cells carry no competing input transition, so determinism is preserved. Boundary crossings use suffixes of the same chains, every chain terminates, and its macro-result equals the original stable residual transition. The case $q > p$ is symmetric. The baseline lower bound $|Q| \ge w_{\max}$ from theorem 94.3 proves state optimality. For $B = 1$ stack height cannot increase, leaving only finitely many physical configurations, so no nonregular $L_{p,q}$ can be recognized. □

Corollary 95.21 (The bounded three-symbol point is Pareto-minimal)**.** *In the broad deterministic permanent-bottom model, the resource triple*

$$\boxed{(w_{\max}, 3, 2)}$$

is Pareto-minimal for $(|Q|, |\Gamma|, B)$.

Proof. It is attained by theorem 95.20. A dominating triple cannot use fewer than $w_{\max}$ states by theorem 94.3; with only two total stack symbols and replacement bound $B = 2$, theorem 95.14 requires $2w_{\max} > w_{\max}$ states; and $B = 1$ cannot recognize the nonregular language. Hence no strict coordinate improvement dominates $(w_{\max}, 3, 2)$. □

95.9 Resource-theoretic synthesis

The one-work-symbol staircase above and the three-symbol push-at-most-two compiler answer complementary resource questions. The former is now representation-independent at fixed $|\Gamma| = 2$; the latter shows that one additional work symbol collapses the state requirement back to $w_{\max}$ while keeping $B = 2$. The protected-contract synthesis later in the monograph uses these endpoint theorems without reintroducing a raw residual-section hypothesis.

Chapter 96

Filtered Collapse Capacity for Unrestricted Pushdown Stores

Research branch — Retained Information, Physical Exposure, and Protected Resource Synthesis. *The General Theory now contains the abstract filtered collapse-capacity theorem. This branch supplies the representation-independent pushdown specialization, then uses universal higher-order geodesic fibers as exponentially large semantic collapse witnesses before passing to protected resource synthesis.*

Chapter 62 established the abstract collapse interface: a finite source family $\mathcal{S} \subseteq U_a^{-1}(m^*)$, collapse load $N = |\mathcal{S}|$, debt count D, canonical count $K = N - D$, nested source observations ω_h of capacity $V(h)$, and the number X_h of canonical sources that cannot remain h-confined. Its Filtered Collapse-Capacity Theorem (57.1) states

$$\boxed{D + X_h + V(h) \geq N}$$

under hidden-tail noncoalescence.

The present chapter does not re-prove that counting theorem. Its task is the model-specific step that the abstract theory deliberately leaves open: prove hidden-tail noncoalescence for an unrestricted deterministic top-access ε-pushdown realization and compute the resulting observation-growth function from source-local state, alphabet, and exposure depth.

Position relative to stack-space and store-complexity literature. The exposure depth introduced below is deliberately narrower than classical stack-space measures. Pushdown height and its constant/nonconstant regimes are established resource measures [87, 88], and related stack-automaton models have been studied through weak, accepting, and strong space complexity [90]. Pushdown store languages and the descriptional complexity of representing reachable store contents are also classical objects [17]. None of those measures records which cells of the *initial source representation* must become visible during one exact semantic macrostep. The contribution here is therefore not a new claim about stack height in general; it is a representation-independent collapse-to-exposure theorem for this provenance-sensitive source-access resource, followed by sharp universal-k-fiber witnesses.

For this specialization we write the physical projection as

$$\pi : P \to M$$

and the chosen reachable section as

$$\tau : M \to P, \qquad \pi\tau = \mathrm{id},$$

to keep the pushdown notation of the original exposure argument distinct from the book-wide symbols Θ, σ. The one-symbol physical action is the settled macrostep of the General Theory: it may contain uniquely forced finite internal moves, consumes exactly the selected semantic action, consumes no later input, and ends at the next settled boundary. All quantities $\mathcal{S}, N, D, K, X_h, V(h)$ therefore have exactly the meanings fixed in Chapter 62.

96.1 Unrestricted deterministic pushdown exposure

96.1.1 Raw model

We use the standard deterministic top-access ε-pushdown interface. A physical configuration is

$$(q, \alpha), \qquad q \in Q, \quad \alpha \in \Gamma^*,$$

with the leftmost symbol of α regarded as the top. A raw transition inspects the finite-control state, the visible top symbol when one exists, and either one input symbol or ε, and replaces the visible top symbol by an arbitrary fixed finite word. Determinism includes the usual exclusion that an enabled ε-move prevents a competing input-consuming move at the same state/top mode. No bound is imposed on replacement length. A permanent bottom marker may be used and is not counted among the work symbols; the same argument also covers the empty-stack convention. The lift may contain arbitrarily many history-dependent physical representatives of the same semantic state. The section τ merely selects one reachable reference representative for the collapse interface; uniqueness of representation is not assumed.

Only the initial source stack is charged to the exposure resource. Conceptually tag the work cells present at the source boundary, from top to bottom, by immutable provenance labels $1, 2, \ldots$. These labels are bookkeeping devices only and are not available to the machine. When a transition replaces a visible cell by a finite word, the removed source cell keeps no descendant provenance: every newly written cell is administrative, even if it carries the same stack symbol. Thus arbitrary replacement words cannot create new source cells for purposes of the resource measure.

Definition 96.1 (Source-stack exposure depth)**.** For a canonical source x, define $e(x)$ to be the largest provenance index of an initial source-stack cell that becomes physically visible as the top cell during the one-symbol macrostep, up to and including the settled canonical hit. If no initial work cell becomes visible, put $e(x) = 0$.

Thus $e(x)$ is not maximum stack height and not the number of pushes. It is a depth of *read access to the initial semantic representation.*

For the chosen canonical sources, let

$$q_{\mathrm{src}} = \big|\{\text{source control state of } \tau(x)\}\big|$$

and let g_{src} be the number of work-stack symbols that actually occur in their initial source stacks, with the convention $g_{\mathrm{src}} = 1$ if every canonical source stack is work-empty. If $K = 0$, set $q_{\mathrm{src}} = 0$ and $g_{\mathrm{src}} = 1$; then every exposure inequality below reduces immediately to $D = N$. Administrative states and symbols used only after leaving the source interface are not charged to these source-local parameters.

For $h \geq 0$ define the depth-h source signature as follows: record the source control state and, if the initial stack has fewer than h work cells, record it completely together with its end; otherwise record its top h source work symbols. The number of possible

signatures is at most

$$C_h^{\mathrm{src}} = q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^r.$$

Lemma 96.2 (Hidden suffix cannot coalesce)**.** *Two distinct canonical sources with the same depth-h source signature cannot both reach the same fixed canonical target while exposing at most h initial source cells.*

Proof. Suppose two source representatives share the same depth-h signature and both computations remain within the first h source cells. Until a deeper source cell becomes visible, every raw step sees the same control state, the same currently visible top symbol, and the same unread input status on both computations. Determinism therefore forces the same raw rule whenever the visible prefixes agree.

A raw rule replaces only the visible top symbol by a fixed finite word. Consequently it cannot alter the untouched source suffix below the exposed cut. Pushed administrative material may be created and later removed, but the hidden suffix remains literally unchanged until a deeper source cell is exposed. Hence two distinct hidden suffixes cannot be transformed into one identical complete target stack by computations that never expose those suffixes. If the hidden suffixes were already identical, equality of the complete source configurations together with section injectivity would identify the semantic sources. Both alternatives contradict distinctness. Therefore at most one shallow canonical source occupies each depth-h signature. □

Theorem 96.3 (Source-Stack Exposure Profile)**.** *For every finite semantic collapse family realized by an unrestricted deterministic ε-pushdown system and every $h \geq 0$,*

$$\boxed{D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^r \geq N.}$$

No bounded-replacement, realtime, homogeneous-tail, canonical residual, or unique-representation hypothesis is required.

Proof. Use the depth-h source signatures as the filtration in Theorem 62.1. Lemma 96.2 verifies hidden-tail noncoalescence, and the displayed signature count bounds $V(h)$. □

The global machine-resource form follows immediately from $q_{\mathrm{src}} \leq |Q|$ and $g_{\mathrm{src}} \leq |\Gamma|$.

96.2 Optimal exposure spectrum and pointwise sharpness

Let $K = N - D$ be the number of canonical sources and sort their exposure depths:

$$e_{(1)} \leq e_{(2)} \leq \cdots \leq e_{(K)}.$$

Define

$$d_r = \min \left\{ h \geq 0 : q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^j \geq r \right\}.$$

Theorem 96.4 (Exact exposure-rank lower bound)**.** *For every $1 \leq r \leq K$,*

$$\boxed{e_{(r)} \geq d_r.}$$

For $g_{\mathrm{src}} = g \geq 2$,

$$d_r = \left\lceil \log_g \left(1 + \frac{(g-1)r}{q_{\mathrm{src}}} \right) \right\rceil - 1,$$

whereas for $g_{\mathrm{src}} = 1$,

$$d_r = \left\lceil \frac{r}{q_{\mathrm{src}}} \right\rceil - 1.$$

Proof. At most C_h^{src} canonical sources can have exposure at most h by Theorem 96.3. If $e_{(r)} < d_r$, then with $h = e_{(r)}$ there are at least $r > C_h^{\mathrm{src}}$ shallow canonical sources, a contradiction. The closed forms invert the geometric sum. □

Corollary 96.5 (Aggregate exposure)**.**

$$\boxed{\sum_{x \text{ canonical}} e(x) \geq \sum_{r=1}^{K} d_r = \sum_{h \geq 0} \left(K - q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \right)_{+}.}$$

For fixed $g \geq 2$ *and* $K/q_{\mathrm{src}} \to \infty$,

$$\bar{e} \geq \log_g(K/q_{\mathrm{src}}) - O_g(1),$$

while for $g = 1$ *one has* $\bar{e} = \Omega(K/q_{\mathrm{src}})$.

Theorem 96.6 (Pointwise sharpness)**.** *The rank lower bound is pointwise sharp. For every* $q \geq 1$, $g \geq 1$, *and* $H \geq 1$, *there exists a zero-debt deterministic* ε*-pushdown collapse family with*

$$K = q \sum_{r=0}^{H} g^r, \qquad q_{\mathrm{src}} = q, \qquad g_{\mathrm{src}} = g,$$

for which

$$\boxed{e_{(r)} = d_r \qquad (1 \leq r \leq K).}$$

Initial-segment truncations give every intermediate family size and remain rank-sharp with their actual source-local alphabet. The canonicalization uses no push operation.

Proof. Take source control states $q_1, \ldots, q_q$, one cleanup state c, a permanent bottom marker Z_0, and a work alphabet Γ of size g. Use every pair

$$(q_i, wZ_0), \qquad w \in \Gamma^*, \quad |w| \leq H,$$

as a distinct semantic source. One common semantic action a sends all sources to one target.

If $w \neq \varepsilon$, the physical a-transition removes the first source work symbol and enters c; deterministic ε-rules in c remove the remaining source word. If $w = \varepsilon$, the a-transition on Z_0 goes directly to the common canonical target. No rule pushes material. A source represented by a word of length r has exposure exactly r. There are exactly qg^r source representatives at depth r, hence exactly

$$q \sum_{j=0}^{h} g^j$$

with exposure at most h for $0 \leq h \leq H$. The ordered spectrum therefore equals the generalized inverse of this count at every rank. Truncating the list in nondecreasing word length preserves the same rank saturation; if the truncation has not yet used every work symbol, its actual g_{src} is smaller and the same counting argument applies to that actual source alphabet. □

Remark 96.7 (Why exposure is a distinct resource). The witness in Theorem 96.6 has zero push complexity during canonicalization but unbounded source exposure. Conversely, a machine may push arbitrarily much administrative material above the source without exposing any deeper source cell. Source exposure is therefore neither push count nor maximum current stack height.

96.3 Universal k-fibers as the semantic witness family

The universal witness need not be reconstructed here. Chapter 19 already proved the Universal k-Fiber Theorem theorem 19.30: for every $s \geq k \geq 2$, one fixed completion-total periodic action system contains $2^{\binom{s}{k}}$ labeled states with common exact distance and common complete compatibility through order $k-1$, while the order-k faces on a distinguished structural s-simplex range over every possible pattern.

For the present physical question, retain only the following interface. Write the distinguished actions as $a_1, \ldots, a_s$, and let v_A denote the witness corresponding to the k-uniform pattern $A \subseteq \binom{[s]}{k}$. For a structural prefix J, its action map has the periodic-translation form

$$T_J(z, \phi) = \big(z + G_J(\phi), \tau_J(\phi)\big),$$

so it is injective in the lattice coordinate. Adjoin a dead semantic state $\perp$ and define the geodesic-prefix transition by sending a distinguished action to its translated state when it remains on a shortest path and to $\perp$ otherwise. This converts the already-established universal fiber into a deterministic many-to-one semantic collapse family without repeating its construction.

96.3.1 From universal freedom to a common collapse

Turn geodesic compatibility into an ordinary deterministic semantic system by adjoining a dead state $\perp$. For a distinguished action a_i define

$$U_{a_i}(v) = \begin{cases} T_i v, & c_i + d(T_i v) = d(v), \\ \perp, & \text{otherwise}, \end{cases} \qquad U_{a_i}(\perp) = \perp.$$

A structural action word remains live exactly while it is a shortest prefix.

Lemma 96.8 (Affine prefix retention)**.** *For every executable action word w from a fixed phase ϕ,*

$$T_w(z, \phi) = \big(z + G_w(\phi), \tau_w(\phi)\big).$$

Hence T_w is injective in z. In particular, every lower-order structural prefix T_J preserves all distinctions among the universal witness states v_A.

Proof. Composition of translations is translation by the sum of the phase-dependent displacement vectors encountered along the fixed phase word. Subtracting the common displacement recovers the initial lattice coordinate. □

Theorem 96.9 (Universal face-collapse retention)**.** *Let a full universal periodic k-fiber on an s-simplex be witnessed by $\{v_A : A \subseteq \binom{[s]}{k}\}$. Fix $|J| = k-1$, $j \notin J$, and $E = J \cup \{j\}$. Then the sources*

$$x_A = T_J v_A, \qquad E \notin A,$$

are pairwise distinct and all satisfy

$$U_{a_j}(x_A) = \perp.$$

Therefore every full universal periodic k-fiber has a one-action collapse family of exact size

$$\boxed{N_{s,k}^{\text{face}} = 2^{\binom{s}{k}-1}},$$

and retained information

$$\boxed{I_{\text{ret}} = \binom{s}{k} - 1 \ \textit{bits.}}$$

Proof. Exactly half of the 2^m patterns omit E. By Theorem 96.8, applying T_J preserves their pairwise distinctness. Full universality makes every $(k-1)$-prefix J geodesic and makes the completed k-face E nongeodesic exactly on the branch $E \notin A$. Since J itself is geodesic,

$$c_J + d(T_J v_A) = d(v_A).$$

Thus geodesicity of the next action a_j at $T_J v_A$ is equivalent to geodesicity of the whole face E at v_A. On the branch $E \notin A$ the geodesic-prefix semantic transition therefore goes to $\bot$. □

Corollary 96.10 (Fixed-action star amplification)**.** *In the explicit universal construction, fix $j \in [s]$ and take all $(k-1)$-sets $J \subseteq [s]\setminus\{j\}$. Their collapse-source families are disjoint, and the common final action a_j has a collapse family of size*

$$\boxed{N^{\star}_{s,k} = \binom{s-1}{k-1} 2^{\binom{s}{k}-1}.}$$

Proof. The post-prefix private-coordinate pattern records J: coordinate p_i equals 0 exactly when $i \in J$ and remains -1 otherwise. Thus source families for distinct J are disjoint. Each family has the face-collapse size from Theorem 96.9. □

Chapter 97

Universal Retained Information Forces Physical Exposure

Combining Theorem 96.3 with Theorem 96.9 gives the main representation-independent bridge.

Theorem 97.1 (Universal Information-to-Physical Exposure Theorem)**.** *For every $s \geq k \geq 2$, every full universal periodic k-fiber, every chosen face-collapse interface, and every unrestricted deterministic exact ε-pushdown realization,*

$$\boxed{D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq 2^{\binom{s}{k}-1}}$$

for every $h \geq 0$.

In particular, the global machine-resource bound

$$D + X_h + |Q| \sum_{r=0}^{h} |\Gamma|^{r} \geq 2^{\binom{s}{k}-1}$$

holds independently of the chosen stack encoding and normalization history.

Proof. Apply Theorem 96.3 to the face-collapse family supplied by Theorem 96.9, whose cardinality is $N = 2^{\binom{s}{k}-1}$. This gives the first displayed inequality. The global machine-resource form follows from $q_{\mathrm{src}} \leq |Q|$ and $g_{\mathrm{src}} \leq |\Gamma|$. □

Corollary 97.2 (Zero-debt bit budget)**.** *Assume zero debt and $g_{\mathrm{src}} = g \geq 2$. If every source has exposure at most h, then*

$$q_{\mathrm{src}} \frac{g^{h+1} - 1}{g - 1} \geq 2^{\binom{s}{k}-1}.$$

Equivalently, the exact depth bound is

$$h \geq \left\lceil \log_g \left(1 + \frac{(g-1)2^{\binom{s}{k}-1}}{q_{\mathrm{src}}} \right) \right\rceil - 1.$$

In particular,

$$\log_2 q_{\mathrm{src}} + (h+1) \log_2 g > \binom{s}{k} - 1 + \log_2(g-1).$$

Corollary 97.3 (Universal exposure spectrum)**.** *In the zero-debt case and $g \geq 2$,*

$$e_{(r)} \geq \left\lceil \log_g \left(1 + \frac{(g-1)r}{q_{\mathrm{src}}} \right) \right\rceil - 1, \qquad 1 \leq r \leq 2^{\binom{s}{k}-1}.$$

For fixed q_{src} and fixed $g \geq 2$, every fixed positive quantile has exposure $\Omega\big(\binom{s}{k}\big)$. In particular, for fixed k this is $\Omega(s^k)$. For $g = 1$, fixed positive quantiles are exponential in $\binom{s}{k}$.

Corollary 97.4 (Star-amplified pushdown bound)**.** *For the explicit universal witness and fixed final action a_j,*

$$D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq \binom{s-1}{k-1} 2^{\binom{s}{k}-1}.$$

97.0.1 The graph case

For $k = 2$, the universal fiber realizes every graph on s distinguished vertices while exact distance and all order-1 compatibility data remain fixed. A single edge query retains

$$2^{\binom{s}{2}-1}$$

sources before collapse. Hence fixed finite control and fixed nonunary source alphabet force exposure $\Omega(s^2)$ on every fixed positive quantile. In the star-amplified construction, one final vertex action simultaneously supports

$$(s-1)2^{\binom{s}{2}-1}$$

collapse sources across its incident edge contexts.

Example 97.5 (The smallest nontrivial graph fiber)**.** Take $s = 3$ and $k = 2$. Then $m = 3$, so one common order-1 signature contains all eight graphs on the three distinguished vertices. Fix the edge $E = \{1, 2\}$, use prefix $J = \{1\}$, and test the final action a_2. Exactly four graphs omit E; after the prefix their four semantic states remain distinct, and all four collapse to $\perp$ on a_2. Thus $I_{\mathrm{ret}} = 2$ bits.

If a zero-debt realization uses one source control state and a binary source alphabet, the rank theorem gives

$$(d_1, d_2, d_3, d_4) = (0, 1, 1, 2).$$

Hence at least one of the four sources must expose two initial work cells. The example is intentionally small: the general theorem replaces these four sources by $2^{\binom{s}{k}-1}$.

Chapter 98

Strong Converse and Simultaneous-Query Hardness

98.0.1 Strong converse

For a universal face-collapse family put

$$m = \binom{s}{k}, \qquad N = 2^{m-1}, \qquad C_h^{\mathrm{src}} = q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r}.$$

Then Corollary 62.3 becomes

$$\boxed{\frac{D + X_h}{2^{m-1}} \geq \left(1 - \frac{C_h^{\mathrm{src}}}{2^{m-1}}\right)_+.}$$

If

$$\log_2 C_h^{\mathrm{src}} \leq m - 1 - \lambda,$$

then

$$\boxed{\frac{D + X_h}{2^{m-1}} \geq 1 - 2^{-\lambda}.}$$

Thus a deficit of λ physical observation bits forces all but a $2^{-\lambda}$ fraction of the retained semantic sources to pay canonical debt or deeper exposure.

98.0.2 A direct-sum theorem across all k-face queries

For every $E \in \binom{[s]}{k}$ choose once and for all a decomposition

$$E = J_E \cup \{j_E\}, \qquad |J_E| = k - 1,$$

and use the face-collapse query obtained by applying J_E and then testing a_{j_E}. For a fiber pattern $A \subseteq \binom{[s]}{k}$, the query E is an *absent-face query* when $E \notin A$. Call an absent-face incidence (A, E) *h-hard* when its realization either has canonical debt or requires exposure $> h$.

Assume one common upper bound $V(h)$ on the depth-h source observation count for every face query. Put

$$\eta_h = \frac{V(h)}{2^{m-1}}.$$

Theorem 98.1 (Simultaneous-query direct sum)**.** *There exists a universal witness pattern A with at least*

$$\boxed{\frac{m}{2}(1-\eta_h)_+}$$

h-hard absent-face queries. In particular, if $V(h) \le 2^{m-1-\lambda}$, then one witness is hard on at least

$$\boxed{\frac{m}{2}(1-2^{-\lambda})}$$

absent-face queries.

Proof. For each face E, exactly 2^{m-1} fiber patterns omit E. By the filtered capacity theorem, at most $V(h)$ of those absent-branch incidences can be both canonical and h-confined. Hence the number of hard incidences for this face is at least $\big(2^{m-1} - V(h)\big)_+$. Summing over the m faces gives at least

$$m\big(2^{m-1} - V(h)\big)_+$$

hard incidences. Averaging over the 2^m witness patterns yields one A with at least

$$\frac{m}{2}(1-\eta_h)_+$$

hard queries. □

Theorem 98.2 (Typical-case direct sum)**.** *Let A be uniformly random among the 2^m universal fiber patterns. For every $0 < \rho < 1/2$ and $0 < \zeta < 1/2 - \rho$, at least a fraction*

$$\boxed{\left(1 - e^{-2\rho^2 m} - \frac{\eta_h}{2\zeta}\right)_+}$$

of all patterns have at least

$$\boxed{m\left(\frac{1}{2} - \rho - \zeta\right)}$$

h-hard absent-face queries.

Proof. The number $Z(A)$ of absent faces is distributed as $\mathrm{Bin}(m, 1/2)$. Hoeffding's inequality [89] gives

$$\Pr\left[Z(A) < m\left(\frac{1}{2} - \rho\right)\right] \le e^{-2\rho^2 m}.$$

Across all patterns and faces, the number of easy absent incidences is at most $mV(h)$. Therefore the expected number $Y(A)$ of easy absent queries of a random pattern is at most

$$\mathbb{E}Y(A) \le \frac{mV(h)}{2^m} = \frac{m\eta_h}{2}.$$

Markov's inequality yields

$$\Pr[Y(A) > \zeta m] \le \frac{\eta_h}{2\zeta}.$$

Outside the union of these exceptional events, the number of hard absent queries is at least

$$Z(A) - Y(A) \ge m\left(\frac{1}{2} - \rho - \zeta\right).$$

□

Corollary 98.3 (Asymptotically almost-everywhere hardness)**.** *If $s \to \infty$ and the depth-h physical capacity deficit $\lambda = \lambda(s)$ tends to infinity so that $\eta_h \le 2^{-\lambda}$, choose for example $\rho = m^{-1/4}$ and, for all sufficiently large s, $\zeta = 2^{-\lambda/2}$. Then $\rho, \zeta \to 0$ and*

$$e^{-2\rho^2 m} + \frac{2^{-\lambda}}{2\zeta} \to 0.$$

Consequently, asymptotically almost every universal witness pattern is h-hard on

$$\left(\frac{1}{2} - o(1)\right)\binom{s}{k}$$

of its absent-face queries.

98.1 Transition to protected-contract synthesis

The conclusion of this branch is not that every semantic collapse must be realized in one canonical stack representation. It is the opposite: arbitrary exact pushdown encodings are allowed, and the invariant is a three-way alternative between source-local capacity, deeper exposure, and debt. The following protected-contract chapters use that distinction to compare resource points without confusing harmless recodings with genuine reductions.

Chapter 99

Contract Refinement and Protected-Information Capacity

Research branch — Protected Contracts and Pareto Synthesis. *Earlier chapters established exact semantic lifts, protected quotient/recoding theory, representation-independent state complexity, and several endpoint resource theorems. This branch reads those results together. Protected contracts order the information a realization must preserve; contract refinement then induces exact monotonicity and strictness principles for the attainable state–stack region.*

The minimal-core theory becomes quantitatively useful once the protected interface is promoted to part of a representation contract. Refining the contract can only shrink the admissible realization class, and finite protected-role capacity gives explicit certificates that the Pareto frontier must move.

99.1 Protected representation contracts and refinement

Observable refinement alone cannot compare genuinely different physical re-coordinations. The representation class must also be part of the data.

Definition 99.1 (Observable refinement)**.** On one representation class, write

$$\lambda' \succeq_{\mathrm{obs}} \lambda$$

if there is a map h with

$$\lambda = h \circ \lambda'.$$

Then λ' remembers at least as much protected information as λ.

Proposition 99.2 (Quotient antitonicity)**.** *If $\lambda' \succeq_{\mathrm{obs}} \lambda$, every λ'-safe surjective quotient is λ-safe.*

Proof. If $\lambda' = \bar{\lambda}' F$, then $\lambda = h\bar{\lambda}' F$. □

Definition 99.3 (Protected representation contract)**.** A protected contract is

$$\mathfrak{K} = \big(\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}\big),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned pushdown lifts of one semantic system and each $\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda$ is compatible with BBR conjugacy. We suppress the object subscript when harmless. Write $\mathfrak{K}' \succeq \mathfrak{K}$ if $\mathfrak{C}' \subseteq \mathfrak{C}$ and the protected interfaces refine objectwise on every lift of $\mathfrak{C}'$. Section 59.2 records the typing explicitly.

Define

$$\mathcal{R}(\mathfrak{K}) = \{(|Q_{\text{use}}(\mathcal{L})|, |\Gamma_{\text{use}}(\mathcal{L})|) : \mathcal{L} \in \mathfrak{C}\}$$

and

$$\text{Pareto}(\mathfrak{K}) = \text{Min}\, \mathcal{R}(\mathfrak{K}).$$

For finite antichains $A, B \subset \mathbb{N}^2$, write $A \preceq_P B$ if every $b \in B$ is componentwise dominated by some $a \in A$.

Theorem 99.4 (Contract-refinement Pareto monotonicity)**.** *If $\mathfrak{K}' \succeq \mathfrak{K}$, then*

$$\mathcal{R}(\mathfrak{K}') \subseteq \mathcal{R}(\mathfrak{K})$$

and

$$\boxed{\text{Pareto}(\mathfrak{K}) \preceq_P \text{Pareto}(\mathfrak{K}').}$$

Thus a stronger physical contract cannot improve the Pareto frontier.

Proof. Class inclusion gives resource-set inclusion. For any finer-contract Pareto point b, the same point belongs to the coarser resource set. Descending componentwise inside $\mathbb{N}^2$ yields a coarser minimal point $a \leq b$. □

99.2 Strictness from protected-information capacity

Weak monotonicity is automatic from class inclusion. Strictness requires an actual obstruction.

Definition 99.5 (Resource down-set)**.** For $r \in \mathbb{N}^2$, let

$$\downarrow r = \{s \in \mathbb{N}^2 : s \leq r \text{ componentwise}\}.$$

Theorem 99.6 (Strictness by down-set exclusion)**.** *Let $\mathfrak{K}' \succeq \mathfrak{K}$. If there exists an attainable coarse point $r_0 \in \mathcal{R}(\mathfrak{K})$ such that*

$$\mathcal{R}(\mathfrak{K}') \cap \downarrow r_0 = \varnothing,$$

then

$$\boxed{\text{Pareto}(\mathfrak{K}) \prec_P \text{Pareto}(\mathfrak{K}').}$$

Proof. Theorem 99.4 gives weak dominance. Since r_0 is coarse-attainable, some coarse Pareto point a satisfies $a \leq r_0$. Reverse dominance would give a fine Pareto point $b \leq a \leq r_0$, contradicting the down-set exclusion. □

Corollary 99.7 (Protected-role capacity certificate)**.** *Suppose a fine contract forces N protected roles to occupy distinct local-observation cells, while every realization with resource vector at most r has at most* cap(r) *such cells, with* cap *monotone. If a coarse attainable point r_0 satisfies*

$$N > \text{cap}(r_0),$$

then r_0 is a down-set exclusion witness and Pareto refinement is strict.

This is the main conceptual bridge:

$$\boxed{\begin{array}{c}\text{protected-information separation} \\ \Longrightarrow \text{ resource down-set exclusion} \\ \Longrightarrow \text{ strict Pareto penalty.}\end{array}}$$

Chapter 100

Exact Pareto Frontiers for Rational Balance Lifts

We now apply the contract order to the rational-balance endpoint theorems already established in the monograph. The point is synthesis rather than re-proving those endpoint results: broad, deferred, raw residual-faithful, and phase-local architectures become comparable only after their protected information and recoding conventions have been typed.

100.1 Rational balance application I: an exact contract state premium

The abstract SCQ/BBR/protected-contract results above are proved entirely within this development. The rational-balance applications now combine that new comparison framework with four exact endpoint resource theorems from the preceding normalization theory. Section 100.3 states those inputs and records their proof interfaces and scope; in particular, the unrestricted state lower bound and the raw/phase-local optima are not being re-labelled as new results of the present chapter sequence.

Fix coprime $p, q \geq 1$ and a mandatory right-endmarker convention. Let $\mathfrak{K}_{\mathrm{broad}}$ be the broad exact-pushdown contract for the residual semantics of $L_{p,q}$. The previously established broad-class state theorem gives

$$Q_{\min}(\mathfrak{K}_{\mathrm{broad}}) = \max(p, q).$$

A three-symbol exact realization attains this value.

Let $\mathfrak{K}_{\mathrm{def}}$ be the canonical deferred-SEARCH literal-demand contract. Its previously established exact state minimum is

$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$

also attained with three total stack symbols.

Theorem 100.1 (Exact broad/deferred state premium)**.** *The exact state premium of the deferred literal-demand contract over the broad exact-pushdown contract is*

$$\boxed{\Delta_Q(p, q) = D_{p,q} - \max(p, q) = \min(p, q) - \mathbf{1}_{\{p+q \text{ even}\}}.}$$

It is strict except when $\min(p, q) = 1$ *and* $p + q$ *is even.*

Proof. If $p+q$ is odd, $D_{p,q} = p+q$, giving $\Delta_Q = \min(p, q)$. If $p+q$ is even, $D_{p,q} = p+q-1$, giving $\Delta_Q = \min(p, q) - 1$. The equality classification follows immediately. □

The two attaining constructions use the same total stack alphabet size 3, so the displayed witness points isolate a control-state premium rather than a stack-alphabet artifact.

100.2 Rational balance application II: exact two-resource frontiers

Let

$$w_{\max} = \max(p, q).$$

In raw residual-faithful Model R, let C be the number of work symbols; the total stack alphabet has size $C + 1$ including Z_0. The exact per-C state minimum is

$$S_{\mathrm{raw}}(C) = \begin{cases} p + q, & C = 1, \\ w_{\max}, & C \geq 2. \end{cases}$$

Theorem 100.2 (Exact raw two-resource frontier)**.** *Across all finite work-alphabet sizes,*

$$\boxed{\mathrm{Pareto}(\mathfrak{K}_{\mathrm{raw}}) = \{(p + q, 2), (w_{\max}, 3)\}.}$$

Proof. The point $C = 1$ gives $(p + q, 2)$. The point $C = 2$ gives $(w_{\max}, 3)$ and dominates every $C > 2$ point $(w_{\max}, C + 1)$. Since $w_{\max} < p + q$, the two displayed points are incomparable. □

The same two endpoint pairs are now known to be the complete *unrestricted* state–stack frontier for $L_{p,q}$ by Theorem 95.15. The raw theorem above is retained because this chapter compares protected representation contracts and therefore needs the stronger statement that the endpoint is attained inside the raw residual-faithful category, not merely somewhere in the unrestricted category.

Now fix $B \geq w_{\max}$ and write every residual as $r = s + Bz$, $0 \leq s < B$. Define

$$h_+(B) = \frac{p}{\gcd(B, p)}, \qquad h_-(B) = \frac{q}{\gcd(B, q)}, \qquad H(B) = h_+(B) + h_-(B).$$

The normalized phase-local contract requires the current signed quotient phase to be visible through control state plus one work symbol. Its exact per-C state minimum is

$$Q^{\min}_{\mathrm{phase}}(B, C) = B \left\lceil \frac{H(B)}{C} \right\rceil.$$

Theorem 100.3 (Exact phase-local Pareto staircase)**.** *Let $H = H(B)$ and*

$$\mathcal{B}_H = \{1\} \cup \left\{ 2 \leq C \leq H : \left\lceil \frac{H}{C} \right\rceil < \left\lceil \frac{H}{C - 1} \right\rceil \right\}.$$

Then

$$\boxed{\mathrm{Pareto}(\mathfrak{K}_{\mathrm{phase}}(B)) = \left\{ \left(B \left\lceil \frac{H}{C} \right\rceil, C + 1 \right) : C \in \mathcal{B}_H \right\}.}$$

Proof. The state coordinate is nonincreasing in C, while the alphabet coordinate is strictly increasing. A point inside a plateau of $\lceil H/C \rceil$ is dominated by the smallest C in that plateau. At every strict drop, no earlier point has as small a state coordinate and no later point has as small an alphabet. For $C > H$ the state coordinate is already B, so $C = H$ dominates all later points. Exact attainability holds for every C by the normalized phase-local constructions. □

At fixed C, the phase state premium is

$$\Delta_Q^{\text{phase}}(B,C) = B\left\lceil \frac{H(B)}{C} \right\rceil - S_{\text{raw}}(C) \geq 0.$$

For $C \geq 2$, equality holds exactly when

$$B = w_{\max}, \qquad C \geq H(B).$$

Since p, q are coprime, $B = w_{\max}$ gives

$$H(B) = \min(p,q) + 1.$$

Thus additional stack symbols can absorb the stronger protected phase information while retaining raw-optimal state count.

100.2.1 The audited raw-safe nesting

There is one important scope distinction. Model R forbids epsilon-normalization on encoded residual configurations. The phase-local optimum is raw and epsilon-free for $C \geq 2$, but the $C = 1$ optimum uses one forced bottom-cell normalization. Hence full $C \geq 1$ phase-local and raw classes are not literally nested.

Restrict to at least two work symbols. Let $\mathfrak{K}_{\text{raw}}^{\geq 2}$ be Model R with $C \geq 2$, and let $\mathfrak{K}_{\text{phase}}^{\text{raw}}(B)$ be the phase-local contract restricted to $C \geq 2$.

Theorem 100.4 (Exact nested raw-safe comparison)**.** *The restricted phase-local contract refines the restricted raw contract, and*

$$\text{Pareto}(\mathfrak{K}_{\text{raw}}^{\geq 2}) = \{(w_{\max}, 3)\}.$$

The two restricted Pareto frontiers are equal exactly when

$$\boxed{B = w_{\max} \quad \textit{and} \quad \min(p,q) = 1.}$$

In every other case the refinement is strict.

Proof. The $C \geq 2$ phase constructions are raw, so the contract inclusion is genuine. Every phase point has state at least $B \geq w_{\max}$ and alphabet at least 3, hence $(w_{\max}, 3)$ dominates. Reverse dominance requires a phase point at most $(w_{\max}, 3)$, so $C = 2$, $B = w_{\max}$, and $\lceil H/2 \rceil = 1$, equivalently $H = 2$. At $B = w_{\max}$, coprimality gives $H = \min(p,q) + 1$, hence $\min(p,q) = 1$. □

Theorem 100.5 (Direct full-frontier comparison)**.** *Over all $C \geq 1$,*

$$\text{Pareto}(\mathfrak{K}_{\text{raw}}) \preceq_P \text{Pareto}(\mathfrak{K}_{\text{phase}}(B)),$$

and the dominance is strict except exactly at

$$(p,q,B) = (1,1,1),$$

where both frontiers equal

$$\{(2,2),(1,3)\}.$$

Proof. For the $C = 1$ phase point,

$$BH(B) \geq 2B \geq 2M \geq p + q,$$

so the raw point $(p+q, 2)$ dominates it. For every phase point with $C \geq 2$, the raw point $(w_{\max}, 3)$ dominates because its state coordinate is at least $B \geq w_{\max}$. Reverse dominance forces a phase point below $(p+q, 2)$, necessarily the $C = 1$ point with $BH(B) \leq p + q$. Equality throughout the displayed chain occurs only at $(1,1,1)$. □

100.3 Endpoint resource theorems used in the rational-balance application

The rational-balance application combines previously established endpoint optima with the new protected-contract formalism. The logical inputs are:

1. in the broad deterministic pushdown/right-endmarker exact-lift class, the exact state minimum is $\max(p, q)$ and is attained with two work symbols plus the bottom marker;

2. in the canonical deferred-SEARCH literal-demand class, the exact state minimum is
$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$
again attainable with three total stack symbols;

3. in raw residual-faithful Model R, with C work symbols,
$$S_{\mathrm{raw}}(C) = \begin{cases} p+q, & C = 1, \\ \max(p, q), & C \geq 2; \end{cases}$$

4. in the normalized phase-local class,
$$Q^{\min}_{\mathrm{phase}}(B, C) = B\left\lceil \frac{H(B)}{C} \right\rceil, \qquad H(B) = \frac{p}{\gcd(B, p)} + \frac{q}{\gcd(B, q)}.$$

For reviewability, the proof interfaces are as follows. The deferred lower bound exposes the $2h+1$ residual debts $-hm, \ldots, 0, \ldots, hm$ at the same exhausted work stack; a common distinguishing continuation then forces distinct control states, while the centered rail attains the bound. The raw frontier uses positive- and negative-side synchronization to give $|Q| \geq \max(p, q)$ for every nonempty work alphabet, strengthens this to $p + q$ with one work symbol by cross-sign unary separation, and is attained by the one- and two-work-symbol residual encodings. The normalized phase-local formula counts, in each residue class modulo B, the $H(B)$ signed phase roles that must be exposed through C work symbols and then sums the sharp per-class bound; the matching construction is raw for $C \geq 2$. The broad unrestricted state theorem combines the positive- and negative-side phase-exposure bounds $|Q| \geq p$ and $|Q| \geq q$ with the explicit three-symbol residual realization having $\max(p, q)$ states.

These endpoint results are inputs, not re-labelled contributions of the present chapter sequence. They are included here at the level needed to audit every numerical comparison, while the new results are the SCQ/BBR reduction theory, protected-contract order, Pareto monotonicity/strictness criteria, and the synthesis of the endpoint optima into exact contract frontiers. The $C = 1$ phase optimum uses one forced bottom-cell normalization and is therefore compared directly with the full raw frontier rather than being treated as a literal subclass of raw Model R. For $C \geq 2$ the phase construction is raw, which is precisely the nested scope of Theorem 100.4.

100.4 Synthesis and scope

The resulting frontiers are exact inside their declared contracts. In particular, the raw Model-R frontier and the normalized phase-local staircase are not promoted to an unrestricted two-resource theorem for all recognizers. The $C = 1$ phase optimum uses one

forced bottom-cell normalization, so the full phase-local class is compared directly with raw Model R rather than asserted to be nested inside it; genuine raw/phase nesting is used only for the audited $C \geq 2$ subfamily. This closes the representation-Pareto synthesis. The next Part changes the question from resource minimization to existential decomposition and locates the Prime-DOCA algorithmic boundary; only after that boundary is established does the monograph return to the original 2:1 machine.

Part IX

State Decomposition and the Algorithmic Boundary

Part guide. **Question.** Beyond optimizing a fixed presentation, what is the algorithmic boundary between existential decomposition and comparison of fixed one-counter presentations? Can one decide whether a deterministic one-counter language presentation is an intersection of strictly smaller superlanguage presentations, how many factors are needed for hardness, what happens when a fixed deterministic one-counter net is compared directly with a fixed nondeterministic one-counter net, and what changes when an explicit zero test is restored on the deterministic side? **Main result.** Totzke's Prime-DOCA problem is undecidable and the hardness already appears at the smallest nontrivial fixed budget $k = 2$. In contrast, both fixed-comparison problems treated here are decidable: Problem 24.04(B) for deterministic versus nondeterministic one-counter nets, and Problem 24.04(A) for a deterministic zero-testing one-counter automaton versus a nondeterministic one-counter net; the latter remains decidable under the stated epsilon-transition extension. **Why here.** This is an algorithmic boundary, not another resource formula: fixed comparison, unrestricted existential decomposition, and fixed-budget decomposition separate sharply even within closely related one-counter models. **Dependency.** It follows the state/resource complexity program. The B branch introduces exact envelope/requirement and WQO machinery under global net monotonicity; the A branch then replaces that monotonicity by zero-cut/first-return compactness and, for silent moves, finite weak macros and stable/reset normalization.

Chapter 101

From State Complexity to State-Primality

Research branch — From exact resource cost to decomposition. *Exact state complexity asks how small one recognizer may be. State-primality asks whether a given presentation can instead be decomposed into finitely many strictly smaller superlanguage recognizers. The threshold is the number of control states of the presented automaton, so this is a presentation-relative decomposition problem.*

Totzke's Prime-DOCA formulation asks whether this property is decidable for deterministic one-counter automata [99]. The theorem below gives the unrestricted boundary result. Chapters 105–106 sharpen it by showing that unbounded factor supply is unnecessary: undecidability already holds for exactly two smaller factors.

Theorem 101.1 (Main theorem)**.** *The Prime-DOCA decision problem is undecidable. More precisely, from a deterministic two-counter Minsky machine M one can effectively construct a DOCA T_M such that*

$$\boxed{M \text{ halts} \iff T_M \text{ is prime.}}$$

The target T_M can be chosen complete, real-time, epsilon-free, with one work symbol, a permanent bottom marker, all control states final, and only unit counter updates $-1, 0, +1$.

The proof does not encode two source counters simultaneously in the target counter. Instead, it separates a proposed source computation into three predicates and lets the *intersection structure itself* enforce simultaneous correctness. This is the main conceptual shift.

The proof is by an effective reduction from deterministic two-counter halting. Its two structural ingredients are developed in the next chapter: a transcript decomposition into one regular and two one-counter predicates, and an elementary-abelian permutation core that is prime with one rejected point but admits exact half-size quotient factors as soon as an additional point fails.

101.1 Preliminaries

101.1.1 Deterministic one-counter automata

We use a counter presentation equivalent to a deterministic pushdown automaton with a permanent bottom marker Z and one work symbol A. A configuration is a pair

$$(q, h) \in Q \times \mathbb{N},$$

where h is the number of A symbols above Z. A transition can inspect whether $h = 0$ or $h > 0$, consume either one input symbol or ε, change the control state, and update the counter by an allowed finite amount. In all automata constructed in this branch, transitions are real-time and use only updates $-1, 0, +1$.

For the ambient model we allow epsilon transitions. We assume the standard deterministic pushdown convention: from a fixed control/top situation there is at most one enabled epsilon transition, and if an epsilon transition is enabled then no input-consuming transition is enabled from the same situation. Thus, once the unread input is fixed, the continuation from a physical configuration is unique.

Acceptance is by final control and zero counter after the entire input has been consumed. Formally, after the last input symbol has been consumed, the unique deterministic post-input epsilon path (finite or infinite) is followed; a word is accepted exactly when some configuration on that path has the form $(q, 0)$ with q final. This is the reachability formulation of final-state-plus-zero acceptance and makes post-prefix epsilon cuts explicit. The constructed target and factors are epsilon-free and have every control state final, so for them acceptance is exactly the condition that the real-time run ends at height zero.

Definition 101.2 (Prime and composite DOCA instances)**.** Let A be a DOCA with $n = |Q_A|$ control states. We call A *composite* if there exist an integer $k \geq 1$ and DOCAs $B_1, \ldots, B_k$ such that

$$L(A) \subseteq L(B_i), \qquad |Q_{B_i}| < n \quad (1 \leq i \leq k),$$

and

$$L(A) = \bigcap_{i=1}^{k} L(B_i).$$

Otherwise A is *prime*. The decision problem asking whether a given DOCA is prime is denoted $\mathrm{PRIME}_{\mathrm{DOCA}}$.

Definition 101.3 (Fixed-budget DOCA compositionality)**.** Let A be a DOCA with $n = |Q_A|$. For a fixed integer $k \geq 1$, call A *k-factor composite* if there are DOCAs $B_1, \ldots, B_k$ such that

$$|Q_{B_i}| < n \quad (1 \leq i \leq k), \qquad L(A) = \bigcap_{i=1}^{k} L(B_i).$$

We write $\mathsf{Factor\text{-}DOCA}_k$ for the corresponding decision problem. The equality automatically implies $L(A) \subseteq L(B_i)$ for every factor. No distinctness condition is imposed on the factors.

Thus unrestricted primality quantifies over every finite factor budget, whereas $\mathsf{Factor\text{-}DOCA}_k$ fixes the budget in advance. The case $k = 2$ is the smallest budget capable of expressing a genuine intersection decomposition.

The superlanguage condition is stated separately because it is part of the original formulation; once the intersection equality holds it is of course automatic. The size measure is the number of control states of the presented automaton, exactly as in the original problem formulation [99]. We do not assume that an arbitrary input DOCA is minimal. Thus primality here is a property of the presentation $(A, |Q_A|)$ rather than an invariant defined using the minimal DOCA size of $L(A)$. In the halting branch of our reduction the constructed presentation will in fact be state-minimal (Corollary 104.3).

101.1.2 Deterministic two-counter machines

A deterministic two-counter Minsky machine has a finite control set and two nonnegative counters c_1, c_2. We use instructions of the forms

$$\begin{aligned}&\text{increment } c_j \text{ and go to } q',\\ &\text{if } c_j = 0 \text{ go to } q_0, \text{ else decrement } c_j \text{ and go to } q_+,\\ &\text{halt.}\end{aligned}$$

The machine starts in a designated state p_0 with both counters zero. Halting of deterministic two-counter machines from zero is r.e.-complete, hence undecidable; this is a standard consequence of the universality of two-counter Minsky machines [100]. The fixed-input r.e.-hardness form follows by compiling the finite input into the control of the simulated machine.

We use a normalization that makes the accepting counter values explicit. After the original halt instruction, enter fresh deterministic drain control, decrement c_1 to zero, then decrement c_2 to zero, and only then enter a fresh state p_f with no outgoing machine transition. Therefore

$$M \text{ halts} \iff (p_f, 0, 0) \text{ is reached by the normalized machine.} \tag{101.1}$$

We write Q_M for the control states of this normalized machine.

Chapter 102

Transcript Encoding and the Elementary-Abelian Prime Core

A single one-counter target does not attempt to simulate both source counters simultaneously. Instead the proposed source computation is separated into three independently checkable predicates, while the intersection structure of the eventual factors enforces conjunction. The finite prime/composite switch is supplied by an elementary-abelian translation core.

102.1 A transcript lemma for deterministic two-counter halting

The reduction will not attempt to store both source counters in one target run. Instead, a proposed source computation is split into one regular predicate and two one-counter predicates.

102.1.1 Branch-labelled transition alphabet

Let Δ_M contain one symbol for every *branch-labelled* transition of the normalized machine. An increment instruction on counter j from p to q contributes

$$\tau = (p, \mathrm{inc}_j, q).$$

A conditional instruction

$$\text{if } c_j = 0 \text{ goto } q_0 \text{ else } c_j := c_j - 1;\ \text{goto } q_+$$

contributes two distinct symbols

$$\tau_0 = (p, \mathrm{zero}_j, q_0), \qquad \tau_+ = (p, \mathrm{dec}_j, q_+).$$

For $\tau \in \Delta_M$, let $\mathrm{src}(\tau)$ and $\mathrm{tgt}(\tau)$ denote its source and target control states. For $j \in \{1, 2\}$, let $\mathrm{op}_j(\tau) \in \{\mathrm{inc}, \mathrm{zero}, \mathrm{dec}, \mathsf{skip}\}$ be the operation induced on counter c_j: a transition acting on the other counter has operation skip on c_j.

102.1.2 The three predicates

Definition 102.1 (Regular control predicate)**.** A word $z = \tau_1 \cdots \tau_k \in \Delta_M^*$ with $k \geq 1$ belongs to R_M if

$$\mathrm{src}(\tau_1) = p_0, \qquad \mathrm{tgt}(\tau_i) = \mathrm{src}(\tau_{i+1})\ (1 \leq i < k), \qquad \mathrm{tgt}(\tau_k) = p_f.$$

We normalize so that $p_0 \neq p_f$; in particular the empty word is not in R_M. Clearly R_M is regular.

Definition 102.2 (Counter predicates)**.** For $j \in \{1, 2\}$, define a deterministic height trajectory on $z = \tau_1 \cdots \tau_k$. Start with $h_0 = 0$. At step i:

- if $\mathrm{op}_j(\tau_i) = \mathrm{inc}$, set $h_i = h_{i-1} + 1$;
- if $\mathrm{op}_j(\tau_i) = \mathrm{zero}$, require $h_{i-1} = 0$ and set $h_i = 0$;
- if $\mathrm{op}_j(\tau_i) = \mathrm{dec}$, require $h_{i-1} > 0$ and set $h_i = h_{i-1} - 1$;
- if $\mathrm{op}_j(\tau_i) = \mathsf{skip}$, set $h_i = h_{i-1}$.

We put $z \in C_j$ if every guard is satisfied and $h_k = 0$.

Each C_j is a deterministic one-counter language. In partial form it needs only one logical checker state; a complete version routes illegal behavior to a permanent-positive poison role.

Lemma 102.3 (Two-Counter Transcript Lemma)**.** *For the normalized deterministic two-counter machine M,*

$$\boxed{M \text{ halts} \iff R_M \cap C_1 \cap C_2 \neq \varnothing.}$$

Proof. If M halts, take the branch-labelled transition sequence of the unique normalized computation from $(p_0, 0, 0)$ to $(p_f, 0, 0)$. The finite-control path lies in R_M, and its projections onto the two counters satisfy exactly the definitions of C_1 and C_2.

Conversely, let $z = \tau_1 \cdots \tau_k \in R_M \cap C_1 \cap C_2$. We prove by induction on i that after $\tau_1 \cdots \tau_i$, the heights maintained by C_1 and C_2 equal the two counter values of the genuine machine computation following the control path forced by R_M. The claim is immediate for increments and for operations on the other counter. A symbol τ with $\mathrm{op}_j(\tau) = \mathrm{zero}$ is admitted by C_j exactly when the corresponding height is zero, hence exactly when the zero branch is enabled in M. A symbol with $\mathrm{op}_j(\tau) = \mathrm{dec}$ is admitted exactly at positive height and then both semantics decrement once. Thus every branch-labelled transition appearing in z is genuinely enabled. The path ends at p_f, and both projected counters end at zero. By (101.1), the original machine halts. □

102.2 The elementary-abelian prime core and failure lattice

Choose a power of two

$$N = 2^d$$

large enough that

$$N - 3 \geq |Q_M|, \qquad N/2 \geq |Q_M| + 1. \tag{102.1}$$

Let

$$V = (\mathbb{Z}_2)^d, \qquad |V| = N,$$

and let $E = \{e_1, \ldots, e_d\}$ be an alphabet indexed by the standard basis. Each core letter e_i acts on V by translation $v \mapsto v + e_i$.

Define

$$P_V = \{x \in E^* : \delta(0, x) \neq 0\}. \tag{102.2}$$

Lemma 102.4 (Elementary-Abelian Prime-Core Lemma)**.** *The translation DFA for P_V is minimal of index N, is a permutation DFA, has exactly $N - 1$ accepting states, and P_V is DFA-prime.*

Proof. Every vector of V is reachable because the basis translations generate V. Let $u \neq v$ be two states. Choose a core word whose total translation is u. Since every element of V is self-inverse, appending this word sends u to 0 and sends v to $v + u \neq 0$. Hence the states are distinguishable and the DFA has index N.

Every input letter acts as a permutation, and the accepting set is $V \setminus \{0\}$, so there are exactly $N-1$ accepting states. Theorem 7.13 of Kupferman and Mosheiff states that a permutation DFA of index n with $n-1$ accepting states is prime [98]. Applying it with $n = N$ proves the claim. □

The next observation is elementary but is the structural heart of the reduction.

Lemma 102.5 (Quotient Failure-Factor Lemma)**.** *Let $R \subseteq V$ be a rejecting set with $0 \in R$. If $R = \{0\}$, the corresponding core section is P_V and is prime. If $|R| \geq 2$, then the section accepting $V \setminus R$ is an intersection of superlanguages recognized by quotient DFAs with exactly $N/2$ states.*

Proof. For every nonzero $r \in R$, put

$$H_r = \langle r \rangle = \{0, r\}.$$

The quotient translation action on V/H_r has $N/2$ states. Let the quotient factor reject exactly the zero coset H_r and accept all other cosets. Because $H_r \subseteq R$, every state accepted by the original section is accepted by this quotient factor, so the factor is a superlanguage.

Moreover,

$$\bigcup_{r \in R \setminus \{0\}} H_r = \bigcup_{r \in R \setminus \{0\}} \{0, r\} = R.$$

Hence the intersection of these quotient factors rejects exactly R. □

Remark 102.6 (Why exponent two is ideal). The identity $H_r = \{0, r\}$ is what makes every extra rejected point generate a two-point failure coset while keeping the quotient exactly half-size. No subgroup-intersection calculation is hidden in the argument: the rejected set is recovered simply as a union of these two-element subgroups.

Chapter 103

The Target DOCA: Primality, Failure Factors, and Exact Decomposition

We now couple the transcript predicates to the finite group core. In the halting case a fixed-suffix zero cut exposes the prime core section even against epsilon-enabled smaller factors. In the nonhalting case every nonzero failure point supplies an explicit quotient factor with exactly half the target states, and these factors intersect back to the target language.

103.1 Construction of the target DOCA

103.1.1 Coloring and alphabet

Partition the nonzero core points as

$$V \setminus \{0\} = G_R \sqcup G_1 \sqcup G_2, \tag{103.1}$$

with $G_1, G_2 \neq \varnothing$ and $|G_R| \geq |Q_M|$. This is possible by (102.1). Fix an injection

$$\phi : Q_M \hookrightarrow G_R.$$

The target state set is exactly V; there are no auxiliary control states. Every target state is final.

The full alphabet is

$$\Sigma = E \sqcup \{@\} \sqcup \Delta_M \sqcup \{\$\}. \tag{103.2}$$

The markers @ and \$ package a source transcript. The counter-color modes do not need a separate finite-control bit recording whether @ has appeared. This is intentional: the reduction needs exact behavior on valid transcript suffixes and exact factor simulation on arbitrary words, not a separate global syntax recognizer in each color.

We now define T_M. Its counter height is the number of work symbols above the permanent bottom marker. A positive configuration in control state 0 is the permanent poison mode: on every input symbol it remains in state 0 and preserves positive height.

103.1.2 Core and poison transitions

For every $v \in V$ and $e_i \in E$,

$$(v, 0) \xrightarrow{e_i} (v + e_i, 0). \tag{103.3}$$

At positive height, every core letter enters permanent poison while keeping the counter positive.

At the zero anchor $(0, 0)$, every non-core symbol enters permanent-positive poison. In particular,

$$(0, 0) \xrightarrow{@} (0, 1).$$

A transition symbol or \$ at $(0, 0)$ is totalized in the same way.

103.1.3 Counter-color transitions

Fix $v \in G_j$, where $j \in \{1, 2\}$. At zero height the control state remains v except for poison or successful reset:

input	zero-height action at $v \in G_j$	purpose
@	stay $(v, 0)$	begin/ignore marker
τ: $\mathrm{op}_j(\tau) = \mathrm{inc}$	$(v, 0) \to (v, 1)$	increment checked counter
τ: $\mathrm{op}_j(\tau) = \mathrm{zero}$	stay $(v, 0)$	legal zero branch
τ: $\mathrm{op}_j(\tau) = \mathrm{dec}$	enter positive poison	illegal decrement at zero
τ: $\mathrm{op}_j(\tau) = \mathsf{skip}$	stay $(v, 0)$	projection ignores other counter
\$	reset to $(0, 0)$	successful zero terminal

At positive height while the control is v:

input	positive-height action	purpose
τ: $\mathrm{op}_j(\tau) = \mathrm{inc}$	increment	checked increment
τ: $\mathrm{op}_j(\tau) = \mathrm{dec}$	decrement	checked positive branch
τ: $\mathrm{op}_j(\tau) = \mathrm{zero}$	poison, preserve positivity	illegal zero branch
τ: $\mathrm{op}_j(\tau) = \mathsf{skip}$	preserve height	projection ignores other counter
@, \$, or $e_i \in E$	poison, preserve positivity	forbidden in positive checker mode

If a legal decrement returns to height zero, the next input symbol is governed again by the zero-height rules at control state v.

103.1.4 Regular-color transitions

Fix an anchor $v \in G_R$. At zero height, only @ starts regular mode:

$$(v, 0) \xrightarrow{@} (\phi(p_0), 1). \tag{103.4}$$

Any $\tau \in \Delta_M$ or \$ at zero height enters poison.

At positive height one, if the control is $\phi(p)$, then on a transition symbol τ with $\mathrm{src}(\tau) = p$ the target moves to $\phi(\mathrm{tgt}(\tau))$ and keeps height one. A wrong transition symbol, @ or a core letter enters poison. On \$, the machine pops the unique mode symbol and resets to $(0, 0)$ exactly when $p = p_f$; otherwise it enters poison. Thus regular mode uses counter height one only as a mode bit.

Every transition not explicitly specified above is routed to state 0 while making or keeping the counter positive.

Lemma 103.1 (Target Construction Lemma)**.** *T_M is a complete real-time deterministic one-counter automaton with exactly N control states, one work symbol above a permanent bottom marker, all control states final, and counter updates in $\{-1, 0, +1\}$.*

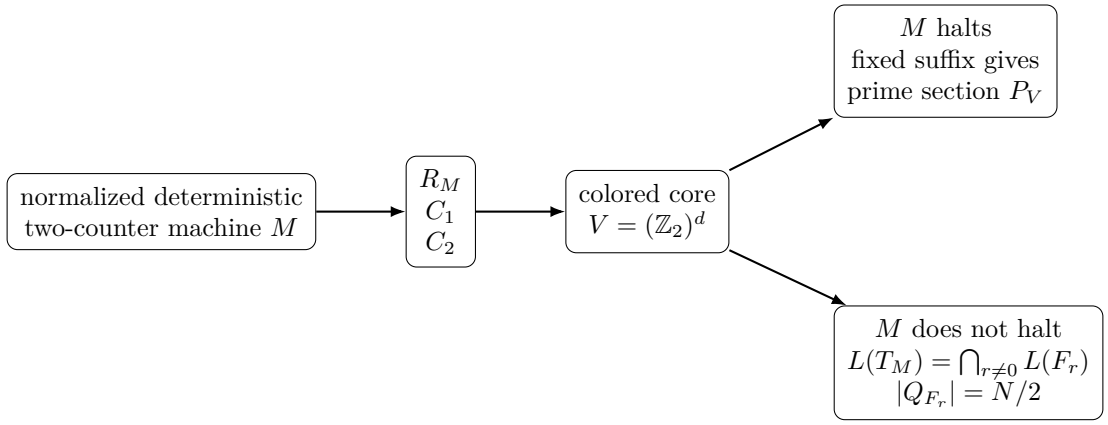


Figure 103.1: Proof architecture. The target never stores both source counters simultaneously. The halting direction extracts a prime finite section; the nonhalting direction is witnessed by explicit quotient factors.

Proof. The state set is exactly V, hence has size N. The cases above are disjoint by control state, zero/positive status, and input symbol. Every unspecified case is totalized into the existing positive poison role. No epsilon transition or extra control state is introduced, and every update is a unit increment, unit decrement, or no-op. □

103.2 Halting implies primality

Assume M halts. By Lemma 102.3, choose

$$z = \tau_1 \cdots \tau_k \in R_M \cap C_1 \cap C_2$$

and define the fixed suffix

$$u = @z\$. \tag{103.5}$$

For a core word $x \in E^*$, let $v = \delta(0, x)$ in the translation core.

If $v = 0$, the first symbol @ enters target poison, so xu is rejected. If $v \in G_1$, the target follows the valid counter-1 projection and returns to zero on \$. The case $v \in G_2$ is symmetric. If $v \in G_R$, regular mode follows the valid control path from p_0 to p_f and returns to zero on \$. Hence

$$\boxed{xu \in L(T_M) \iff \delta(0, x) \neq 0.} \tag{103.6}$$

Thus the fixed-suffix section is exactly P_V.

The remaining issue is that a hypothetical smaller factor in a Prime-DOCA decomposition may use epsilon transitions even though T_M does not. We therefore isolate the section argument carefully.

Lemma 103.2 (Epsilon-Robust Zero-Cut Section Lemma)**.** *Let B be a deterministic one-counter automaton in the ambient model, possibly with epsilon transitions, and suppose $E^* \subseteq L(B)$. Fix a suffix $u \in \Sigma^*$. Then*

$$S_u(B) = \{x \in E^* : xu \in L(B)\}$$

is recognized by a DFA with at most $|Q_B|$ states.

Proof. If $u = \varepsilon$, then $S_u(B) = E^*$ and the claim is trivial, so assume $u \neq \varepsilon$.

For each $x \in E^*$, consider the unique post-input epsilon trajectory that starts immediately after the last symbol of x has been consumed. Since $x \in L(B)$, this trajectory visits a final zero configuration. Let

$$c_x = (q_x, 0)$$

be the first such configuration.

We first isolate the point on which epsilon-robustness depends.

Forced-cut claim. For every continuation $y \in \Sigma^*$, the run of B on xy reaches the same physical configuration c_x after consuming exactly x and before consuming any symbol of y.

Indeed, every transition taken between the configuration immediately after x and the occurrence of c_x is an epsilon transition on the post-input run of x. Each such transition is forced by determinism; and whenever it is enabled, epsilon/input exclusivity forbids an input-consuming transition from the same control/top situation. Pending input therefore cannot pre-empt any step of this finite prefix of the epsilon trajectory. Thus the run on every extension xy follows exactly the same epsilon prefix and reaches c_x before reading y.

Now put

$$S = \{q_x : x \in E^*\} \subseteq Q_B.$$

We define a DFA D over E with state set S and initial state q_ε. For $q_x \in S$ and $e \in E$, set

$$\delta_D(q_x, e) = q_{xe}.$$

This is well-defined. If $q_x = q_y$, then $c_x = c_y = (q, 0)$ is literally the same physical configuration. By the forced-cut claim, the runs on xe and ye reach this same cut before reading e; from that point, determinism forces the same continuation on e and the same subsequent epsilon behavior. Hence their canonical first final zero cuts coincide, so $q_{xe} = q_{ye}$. The transition is defined for every $e \in E$ because $xe \in E^* \subseteq L(B)$.

Declare $q \in S$ accepting in D exactly when B, started from the physical configuration $(q, 0)$, accepts the fixed continuation u. An induction on $x \in E^*$ gives that the state reached by D after x is q_x. Applying the forced-cut claim once more, now with continuation u, shows that the run on xu reaches c_x before processing u. Therefore

$$x \in L(D) \iff B \text{ accepts } u \text{ from } (q_x, 0) \iff xu \in L(B).$$

Thus D recognizes $S_u(B)$ and has $|S| \leq |Q_B|$ states. □

Lemma 103.3 (Halting Implies Prime)**.** *If M halts, then T_M is prime.*

Proof. Suppose toward contradiction that

$$L(T_M) = \bigcap_{i=1}^{k} L(B_i), \qquad |Q_{B_i}| < N,$$

where every B_i is a DOCA superlanguage of $L(T_M)$. Every core-only word belongs to $L(T_M)$, so $E^* \subseteq L(B_i)$ for every i. Apply Lemma 103.2 to each B_i with the fixed suffix u from (103.5); this yields DFAs D_i with

$$|Q_{D_i}| \leq |Q_{B_i}| < N.$$

By (103.6),

$$P_V = \bigcap_{i=1}^{k} L(D_i).$$

This contradicts the DFA-primality of P_V from Lemma 102.4. Hence T_M is prime. □

103.3 Explicit half-size factors in the nonhalting case

Assume from now on that M does not halt. For every nonzero $r \in V$, let

$$H_r = \{0, r\}$$

and define a factor F_r with control-state set

$$Q_r = V/H_r.$$

Thus

$$|Q_r| = N/2 < N. \tag{103.7}$$

Every factor state is final. Write

$$S_r = H_r$$

for the special zero coset.

103.3.1 Core behavior and the special coset

At counter height zero, core letters act by quotient translation:

$$(v + H_r, 0) \xrightarrow{e_i} (v + e_i + H_r, 0). \tag{103.8}$$

Consequently, when the exact target zero anchor is v, the factor is in its special quotient state exactly when

$$v + H_r = H_r \iff v \in \{0, r\}.$$

At a nonzero anchor v, exactly one factor—F_v—is special.

103.3.2 Non-special zero cosets

At a zero-height coset $C \neq S_r$, the factor deliberately overaccepts query symbols:

- on @ or any $\tau \in \Delta_M$, remain at $(C, 0)$;
- on \$, reset to $(S_r, 0)$;
- on a core letter, use quotient translation (103.8).

No source correctness is checked in a non-special coset.

103.3.3 Special counter factors

Suppose $r \in G_j$. At $(S_r, 0)$, the factor uses the same counter-j transition rules as the target uses at a state of G_j, except that a successful \$ resets to $(S_r, 0)$. Legal positive counter behavior remains in control state S_r.

Choose one existing quotient state $\pi_r \in Q_r$ as a positive poison role. If illegal behavior occurs, enter (π_r, h) with $h > 0$ and thereafter preserve positive height forever. Height distinguishes this poison role from the ordinary zero-height quotient meaning of π_r, so no additional control state is required. Since $N/2 \geq 2$, such a distinct quotient state is available.

103.3.4 Special regular factors

Suppose $r \in G_R$. By (102.1),

$$N/2 \geq |Q_M| + 1.$$

Therefore choose an injection

$$\psi_r : Q_M \cup \{\bot\} \hookrightarrow Q_r,$$

where $\bot$ is a positive poison role. At $(S_r, 0)$, only @ starts regular mode:

$$(S_r, 0) \xrightarrow{@} (\psi_r(p_0), 1).$$

At height one, $\psi_r(p)$ follows exactly those branch-labelled transition symbols whose source is p. A successful \$ from $\psi_r(p_f)$ pops the mode symbol and resets to $(S_r, 0)$. Wrong transitions, malformed markers, core letters while positive, or a premature \$ enter $\psi_r(\bot)$ at positive height; this poison role preserves positivity forever.

Every otherwise undefined transition of F_r is routed to an existing positive poison role while making or preserving positive height.

Lemma 103.4 (Explicit Factor Construction)**.** *For every $r \neq 0$, F_r is a complete real-time DOCA with exactly $N/2$ control states, one work symbol, all states final, and unit counter updates.*

Proof. The quotient state set has size $N/2$. Counter factors use one quotient state as the legal special state and another, distinguished only at positive height, as poison. Regular factors fit an encoding of all Q_M states plus one poison role into Q_r by the size choice. Every remaining transition is totalized into an existing positive poison role. No new state is added. □

103.4 Exactness of the factorization

We first show that every F_r is a superlanguage, independently of the nonhalting assumption.

Lemma 103.5 (Zero-height quotient and owner invariant)**.** *Fix $r \neq 0$. For every consumed prefix $y \in \Sigma^*$ for which the target run is at a zero-height configuration $(v, 0)$ after reading y, the run of F_r on the same prefix is at*

$$(v + H_r, 0).$$

Consequently, at every nonzero zero anchor v the factor F_v is the unique special factor, while at anchor 0 every factor is special. If the target leaves a nonzero anchor v to execute a positive checker episode, the owner factor F_v follows the same checker semantics until the next successful zero return/reset or the corresponding permanent-positive failure; ownership remains v even when a regular checker changes its finite-control state through $\phi(Q_M)$.

Proof. We prove the zero-height statement by induction over successive target configurations of height zero. It holds initially at $(0, 0)$ because every factor starts at $(H_r, 0)$.

Assume the target is at $(v, 0)$ and F_r is at $(v + H_r, 0)$. A core letter sends the target to $(v + e_i, 0)$ and the factor to $(v + e_i + H_r, 0)$, so the invariant is preserved immediately. If $v = 0$, every non-core symbol sends the target to permanent-positive poison; hence no later target zero configuration can arise from that move, and there is no further zero-height obligation to verify.

Suppose next that $v \neq 0$. If $v = r$, then $v + H_r = H_r = S_r$, so F_r is special. For $v \in G_1 \cup G_2$, the special factor uses exactly the target's counter-checker operations. A zero-preserving step leaves both at their corresponding zero anchors; a legal positive segment is mirrored operation for operation and, if it returns to zero, returns the target to v and the factor to $S_r = v + H_r$; a successful \$ resets them to 0 and H_r, respectively. Any illegal positive behavior that sends the target to permanent-positive poison sends the owner factor to permanent-positive poison as well. For $v \in G_R$, the same statement holds for the regular checker: @ starts the two corresponding positive regular modes, the branch-labelled control path is mirrored under ϕ and ψ_r, and a successful \$ resets both sides to their zero anchors. The owner is still the original anchor $v = r$ throughout this positive episode, regardless of the target's current state in $\phi(Q_M)$.

If instead $v \notin \{0, r\}$, then $v + H_r \neq H_r$, so F_r is non-special. On @ and transition symbols it deliberately stays at the same zero coset, and on \$ it resets to H_r. Thus, while the target performs a legal counter or regular episode launched from v, the non-owner factor overaccepts at height zero; when the target next returns to zero at v or resets to 0, the factor is respectively at $v + H_r$ or H_r. If the target enters permanent-positive poison, again no later target zero configuration exists. This proves renewal of the invariant at every possible next zero configuration.

Finally,

$$v + H_r = H_r \iff v \in H_r = \{0, r\}.$$

Hence at a nonzero anchor v exactly F_v is special, whereas at anchor 0 all factors are special. The owner statement was established in the $v = r$ case above. □

Lemma 103.6 (Forward inclusion)**.** *For every nonzero $r \in V$,*

$$L(T_M) \subseteq L(F_r).$$

Proof. Fix $r \neq 0$ and let $w \in L(T_M)$. Because every target state is final, acceptance means that the real-time run of T_M ends at some zero-height configuration $(v, 0)$. Lemma 103.5, applied to the full prefix w, places F_r at $(v + H_r, 0)$ after the same input. Every state of F_r is final, so F_r accepts w. Since r was arbitrary, $L(T_M) \subseteq L(F_r)$ for every $r \neq 0$. □

The reverse inclusion uses the nonhalting assumption. It is useful to formulate it in terms of the *final positive episode*: because all target states are final, a target word is rejected exactly when its real-time run ends at positive height. Let a be the exact target control state at the last zero-height configuration before this final positive episode.

Lemma 103.7 (Nonhalting Implies Composite)**.** *If M does not halt, then*

$$\boxed{L(T_M) = \bigcap_{r \in V \setminus \{0\}} L(F_r)}.$$

Consequently T_M is composite.

Proof. The inclusion from left to right is Lemma 103.6. For the converse, take a word $w \notin L(T_M)$. The target ends at positive height. Since the run starts at height zero, there is a last zero-height configuration before the final positive episode; write its exact target control state as a. Lemma 103.5 fixes the simultaneous factor configurations at this anchor.

Case 1: $a \neq 0$. By Lemma 103.5, F_a is the unique special factor at that anchor and is the owner of the ensuing checker episode.

If $a \in G_1 \cup G_2$, the owner factor F_a performs exactly the same one-counter operations as the target during the final positive episode. Illegal branches, a forbidden marker or core letter while positive, or an unfinished positive trajectory leave both target and owner factor positive. Hence F_a rejects w.

If $a \in G_R$, the owner factor is still F_a even though the target finite control may later change through states $\phi(Q_M)$. There are two subcases. If the first symbol of the final positive episode is not @ (necessarily a transition symbol or \$), both the target and F_a enter positive poison immediately. If the episode starts with @ then both enter regular mode, and from that point they follow the same branch-labelled control path. A wrong transition, a second @ while positive, a core letter while positive, a premature \$, or an unfinished regular payload leaves the owner factor positive. In either subcase F_a rejects w.

Case 2: $a = 0$. By Lemma 103.5, every factor is simultaneously at its special zero coset S_r. Because a is the last zero anchor, the symbol that begins the final positive episode is non-core; the target enters poison on that symbol.

If that symbol is not @ then every regular-color factor immediately enters positive poison, so at least one factor rejects permanently.

Suppose the first non-core symbol is @. Every G_R-indexed factor starts the regular checker, every G_1-indexed factor starts the counter-1 checker, and every G_2-indexed factor starts the counter-2 checker. Now inspect the suffix after this @ until the first \$, if any.

If the input ends first, every regular-color factor is still positive and rejects. If an @ or a core letter occurs before the first \$, every regular-color factor enters permanent positive poison. If a transition symbol violates the control path, the regular-color factors enter poison as well. Hence the only remaining possibility is a block

$$@z\$, \qquad z \in \Delta_M^*,$$

whose transition sequence is control-consistent up to the terminal marker. At this first \$, a regular-color factor returns to zero exactly when $z \in R_M$, and a counter-j factor returns to zero exactly when $z \in C_j$. Thus all factors could return to zero simultaneously only if

$$z \in R_M \cap C_1 \cap C_2.$$

By Lemma 102.3, this would imply that M halts, contradicting the assumption. Therefore before or at the first terminal marker at least one factor becomes permanently positive. It remains positive on all later input, so it rejects w.

Thus every target rejection is witnessed by at least one F_r, proving

$$\bigcap_{r \neq 0} L(F_r) \subseteq L(T_M).$$

By (103.7), every factor has strictly fewer than N states, so the target is composite. □

Remark 103.8 (Owner factor versus current regular state). The owner of a query is determined by the zero anchor where the query began. In regular mode the target control subsequently moves through $\phi(Q_M)$, but this does not change the owner. Confusing the current regular control state with the owner index is an easy proof error; the construction and Lemma 103.7 avoid it explicitly.

Chapter 104

Undecidability of State-Primality and the Verification–Decomposition Boundary

The construction is now complete. The first section states the undecidability theorem and its exact restricted-class consequences; the second records the proof features most likely to be confused when the result is compared with the earlier exact-lift verification hierarchy.

104.1 Undecidability and consequences

Theorem 104.1 (Prime-DOCA Undecidability)**.** *The mapping $M \mapsto T_M$ is computable and satisfies*

$$\boxed{M \text{ halts} \iff T_M \text{ is prime.}}$$

Hence $\mathrm{PRIME}_{\mathrm{DOCA}}$ *is undecidable.*

Proof. Lemma 103.3 gives the forward implication. Lemma 103.7 gives

$$M \text{ does not halt} \Rightarrow T_M \text{ is composite,}$$

which is the reverse implication. The construction of N, the coloring, transition tables, and factors is effective. Undecidability follows from deterministic two-counter halting. □

Corollary 104.2 (Restricted-class undecidability)**.** *Prime-DOCA remains undecidable when the input presentation is restricted to complete real-time deterministic one-counter automata that use one work symbol and a permanent bottom marker, have all control states final, and use only updates $-1, 0, +1$. The same reduction also proves undecidability for the stricter variant in which admissible smaller factors are required to lie in this subclass as well.*

Proof. Every target T_M lies in the stated subclass. In the halting case Lemma 103.3 excludes all smaller ambient DOCAs, hence in particular all smaller factors from the subclass. In the nonhalting case the explicit factors F_r themselves lie in the subclass. Therefore both interpretations of the restriction are covered by the same mapping. □

Corollary 104.3 (State-minimality in the prime case)**.** *If M halts, then no DOCA with fewer than N control states recognizes $L(T_M)$.*

Proof. Otherwise let B be such a smaller recognizer. Since $E^* \subseteq L(T_M) = L(B)$, Lemma 103.2 applied to the valid halting suffix u would yield a DFA with fewer than N states recognizing the fixed section P_V. This contradicts the fact that P_V has minimal DFA index N. □

Corollary 104.4 (Recursion-theoretic consequences)**.** *There is a many-one reduction*

$$\mathrm{HALT}_2 \leq_m \mathrm{PRIME}_{\mathrm{DOCA}}.$$

Consequently:

(a) $\mathrm{PRIME}_{\mathrm{DOCA}}$ *is r.e.-hard and undecidable;*

(b) $\mathrm{PRIME}_{\mathrm{DOCA}}$ *is not co-r.e.;*

(c) $\mathrm{COMPOSITE}_{\mathrm{DOCA}}$ *is not r.e.*

No upper semidecidability classification is claimed.

Proof. The reduction is Theorem 104.1. If $\mathrm{PRIME}_{\mathrm{DOCA}}$ were co-r.e., then its computable preimage HALT_2 would be co-r.e., contradiction. Likewise,

$$M \notin \mathrm{HALT}_2 \iff T_M \in \mathrm{COMPOSITE}_{\mathrm{DOCA}},$$

so r.e.-ness of $\mathrm{COMPOSITE}_{\mathrm{DOCA}}$ would imply r.e.-ness of deterministic two-counter nonhalting. □

104.2 Proof anatomy, edge cases, and verification points

The reduction is compact once the elementary-abelian core is in place, but several details are structurally important. This section records them explicitly both for readability and for independent verification.

104.2.1 Why the target does not need two counters

The same-witness coupling problem is delegated to the intersection. A single target run beginning at a nonzero anchor checks only one of R_M, C_1, C_2. This keeps the target within one-counter power. The common zero anchor is where all quotient factors become special simultaneously; only there does the intersection enforce all three predicates at once. Thus conjunction exists semantically across factors, not physically inside one automaton.

104.2.2 Why the state budget is exact

The target uses exactly the N core points. Counter colors need no separate checker state: the anchor state itself is the checker control, and positive height stores the unbounded counter value. Regular colors encode Q_M inside G_R while positive height serves as the mode bit. State 0 at positive height is reused as target poison.

Each factor has exactly $N/2$ quotient states. Counter factors reuse one quotient state as positive poison; regular factors reuse the positive-height meanings of quotient states to encode Q_M plus one poison role. The inequality $N/2 \geq |Q_M| + 1$ is precisely the factor-side packing condition.

104.2.3 Malformed and unfinished inputs

The target language is not defined by a global "well-formed transcript" grammar. In counter colors, when the counter happens to be zero the machine may process symbols according to its zero-height transition table even if a human reader would regard the current substring as malformed. This causes no problem: the reduction only requires exact target/factor agreement on arbitrary inputs and exact source semantics on the fixed valid suffix used in the prime direction.

In the nonhalting reverse inclusion, malformed behavior launched from a nonzero anchor is caught by the owner factor because it mirrors the target transition system, not an abstract syntax predicate. At anchor zero, regular-color factors provide the strict syntax/control witness needed to prevent all factors from accepting a malformed payload simultaneously.

104.2.4 Underflow and zero tests

A decrement branch for counter j is legal only at positive height; a zero branch is legal only at zero. Illegal cases go to permanent-positive poison. Therefore the counter predicates implement exact Minsky guards rather than merely net counter effects.

104.2.5 Epsilon-enabled adversarial factors

The target and explicit factors are epsilon-free, but prime decompositions quantify over all smaller DOCAs in the ambient problem model. This is why Lemma 103.2 is stated for epsilon-enabled factors. The key object is a physical accepting zero configuration reached after a consumed core prefix. Determinism and epsilon/input exclusivity make that configuration a continuation-independent cut. No assumption is made about the factor's internal representation of the core translation.

104.2.6 One reject point versus many

The proof is not simply "embed a hard regular language." That alone would prove a lower bound in the halting case but would not produce explicit decompositions in the nonhalting case. The elementary-abelian core simultaneously supplies both sides of the reduction: the single-reject section is prime, while each additional reject point carries its own canonical half-size quotient factor. This dual role is the feature that the earlier residual-rigidity route was missing.

104.3 Related work and positioning

Kupferman and Mosheiff introduced intersection primality for regular languages and developed structural and algorithmic tools for DFA decomposition [98]. Their size threshold is the language index, i.e. the number of states of the minimal DFA. This distinction matters for the present problem: Totzke's Prime-DOCA formulation is presentation-relative, comparing every factor with the number of control states in the given DOCA presentation. Spenner later called the analogous presentation-relative DFA notion S-primality [103]. Our halting construction happens to remove this distinction internally, because the prime target is state-minimal by Corollary 104.3.

The regular-language theory has continued to develop after the original prime-language paper. Unary primality admits a structural LogSpace algorithm [101]; permutation automata admit specialized decomposition algorithms, including low-complexity results for commutative permutation DFAs [102]; primality of acyclic DFAs

and finite languages is characterized in [103]; and the general Prime-DFA problem was recently shown NP-hard [104]. Our finite obstruction lies inside the especially structured commutative permutation class, but it is used only as a fixed-suffix section: the undecidable part comes from asking whether an infinite-state DOCA presentation has any finite intersection decomposition by smaller DOCAs.

The algorithmic theory of deterministic one-counter automata is classical. Valiant and Paterson proved decidability of language equivalence for DOCAs [106], and Böhm, Göller, and Jančar later proved the problem NL-complete [107]. These results make an instructive contrast with state-primality: equivalence of two given DOCAs is decidable, while deciding whether one DOCA language admits an exact finite intersection decomposition into arbitrary smaller DOCA superlanguages is undecidable.

Almagor, Avni, Sinclair-Banks, and Yeshurun study primality for counter nets and prove undecidability when "smaller" means lower counter dimension [105]. Counter nets differ from DOCAs in both the resource measure and operational model: they may have several counters, have no explicit zero tests, and compare dimensions rather than finite-control state counts. Their theorem and ours therefore establish undecidability for distinct decomposition axes.

104.4 The verification–decomposition boundary

The contrast with the earlier verification hierarchy is structural. Equivalence of two given deterministic one-counter automata is decidable [106, 107]; by contrast, the theorem above shows that deciding whether one presented DOCA language admits an exact finite intersection decomposition into arbitrary strictly smaller DOCA superlanguages is undecidable. Thus a decision procedure for comparing two fixed presentations does not extend to a decision procedure for existential decomposition over an unbounded family of smaller presentations.

This distinction is the reason the present branch belongs after the General Theory and after the exact state/resource frontiers. It is not an ingredient used to build those theories; it identifies the algorithmic decomposition boundary of the representation program. With optimization and decomposition now separated, the final Part can return to the original 2023 machine and synthesize which structures were forced, chosen, or subsequently generalized.

Chapter 105

From Unrestricted Decomposition to Two Factors

Research branch — The fixed-budget sharpening. *The unrestricted Prime-DOCA theorem allows an arbitrary finite family of smaller factors. The new question is whether that factor supply is essential. It is not: alternating sentinels distribute the natural regular-control witness across two one-counter factors, so the smallest genuine intersection budget already carries the undecidability.*

The transcript, elementary-abelian prime core, and epsilon-robust zero-cut arguments have already been established in Chapters 102–103. They are reused here rather than restated. In particular, Lemma 102.3 gives M halts iff $R_M \cap C_1 \cap C_2 \neq \varnothing$, Lemma 102.4 supplies the prime section P_V, and Lemma 103.2 applies against arbitrary smaller epsilon-enabled DOCAs. The genuinely new task is to enforce the three transcript predicates with only two smaller one-counter factors.

105.1 Phase coding and alternating sentinels

The natural three-checker split assigns R_M, C_1, C_2 to three separate factors. To use only two factors we must let the factors share the syntax/control burden without ever requiring one DOCA to store both unbounded counters.

For every $\tau \in \Delta_M$, introduce six phase-labelled symbols

$$O^1_\tau,\ A^1_\tau,\ C^1_\tau,\ O^2_\tau,\ A^2_\tau,\ C^2_\tau.$$

For $z = \tau_1 \cdots \tau_m$, let

$$\widehat{z} = \prod_{\ell=1}^{m} O^1_{\tau_\ell} A^1_{\tau_\ell} C^1_{\tau_\ell} O^2_{\tau_\ell} A^2_{\tau_\ell} C^2_{\tau_\ell}. \tag{105.1}$$

Add a query-start marker @ and terminal symbols $T_1, Z_1, T_2, Z_2, \$$. The canonical complete query is

$$u(z) = @\widehat{z}\, T_1 Z_1 T_2 Z_2 \$. \tag{105.2}$$

Both checkers represent a source counter value c_j at stable points by physical height

$$h_j = c_j + 1.$$

The extra unit is a sentinel. To test a source zero/decrement guard, checker j temporarily removes its sentinel, making the physical height exactly c_j. The construction staggers these exposures: checker 2 remains positive while checker 1 is exposed, and checker 1 remains positive while checker 2 is exposed.

105.1.1 Checker 1

Checker 1 carries the source-control state and verifies counter 1. Its logical roles are

$$P_p\ (p \in Q_M), \qquad K_\tau^A, K_\tau^C\ (\tau \in \Delta_M), \qquad U_1, U_2, U_3, \qquad \bot_1,$$

together with one raw/special role S_1.

At $(S_1, 0)$, @ starts the checker in $(P_{p_0}, 1)$. From (P_p, h) with $h > 0$, the symbol O_τ^1 is legal only when $\mathrm{src}(\tau) = p$; it removes the sentinel and moves to raw role S_1. At raw S_1, A_σ^1 performs $\mathrm{op}_1(\sigma)$ on the exposed height, with illegal zero/decrement branches entering permanent-positive poison. A symbol C_ρ^1 restores the sentinel and moves to $P_{\mathrm{tgt}(\rho)}$.

During the second half, checker 1 becomes the lexical witness: O_τ^2 moves from P_p to K_τ^A, only A_τ^2 moves to K_τ^C, and only C_τ^2 returns to $P_{\mathrm{tgt}(\tau)}$. The height stays positive throughout this half.

105.1.2 Checker 2

Checker 2 verifies counter 2 and carries the complementary lexical memory. Its logical roles are

$$B, \qquad A_\tau, C_\tau\ (\tau \in \Delta_M), \qquad W_1, V_2, \qquad \bot_2,$$

together with raw/special role S_2.

At $(S_2, 0)$, @ starts $(B, 1)$. From B, O_τ^1 stores τ in A_τ; only A_τ^1 reaches C_τ. The symbol C_τ^1 preserves that lexical role, and only O_τ^2 then removes checker 2's sentinel and enters raw S_2. At raw S_2, A_σ^2 performs $\mathrm{op}_2(\sigma)$, and C_ρ^2 restores the sentinel and returns to B. Checker 1 forces equality of the second-half labels.

The apparent reuse of C_τ before and after C_τ^1 causes no ambiguity: if C_τ^1 is omitted, checker 1 is still raw and rejects a premature O_τ^2.

105.1.3 Terminal block

From P_p, checker 1 permits T_1 only for $p = p_f$, removes its sentinel, and enters raw S_1. The next Z_1 is legal only at height zero; it restores the sentinel and enters U_1. Checker 1 then reads T_2, Z_2 through U_1, U_2, U_3 and finally removes the restored sentinel on \$.

Checker 2 reads T_1, Z_1 through B, W_1, B, then T_2 removes its sentinel and enters raw S_2. The next Z_2 is legal only at height zero; it restores the sentinel in V_2, and \$ removes it. For the checker pair used as the two quotient factors, a successful final \$ returns checker i to its special zero role S_i. When the same checker routine is instantiated inside the target, the successful final \$ instead returns the target to the exact core zero state 0. This is the only terminal-destination difference between the target checker and its quotient-factor copy. In either implementation, simultaneous terminal success is equivalent to source control p_f and both source counters zero.

105.2 Synchronization lemmas

Define a synchronized stable pair

$$\mathcal{G}(p, a, b) = ((P_p, a + 1), (B, b + 1)).$$

Lemma 105.1 (Block synchronization)**.** *Suppose the two checkers start in $\mathcal{G}(p, a, b)$, consume a non-poisoning source-transition macro, and next return to a synchronized stable pair. Then the consumed macro is uniquely*

$$O_\tau^1 A_\tau^1 C_\tau^1 O_\tau^2 A_\tau^2 C_\tau^2$$

for a single $\tau \in \Delta_M$ with $\mathrm{src}(\tau) = p$. Both source-counter guards are legal at (a, b), and the new pair is

$$\mathcal{G}(\mathrm{tgt}(\tau), a', b'),$$

where (a', b') are exactly the source-counter values after τ.

Proof. Checker 1 source-checks the opening O^1_τ. Checker 2 stores that same label and forces A^1_τ, C^1_τ, so checker 1 performs exactly the counter-1 operation of τ. Checker 1 rejects O^2 while still raw, hence the closing C^1_τ cannot be omitted. Checker 2 then permits only O^2_τ and exposes counter 2. Checker 1 stores the same label and forces A^2_τ, C^2_τ, so checker 2 performs exactly the counter-2 operation. Both sentinels are restored and checker 1's control role is $P_{\mathrm{tgt}(\tau)}$. □

Lemma 105.2 (Common-legal-symbol property)**.** *Fix a non-poison paired phase after the opening @ and before the first simultaneous zero return. Apart from the choice of the next source transition τ at a stable source boundary, there is at most one next phase label that both checkers can process without poison. For a chosen τ, the common sequence is forced to be*

$$O^1_\tau \to A^1_\tau \to C^1_\tau \to O^2_\tau \to A^2_\tau \to C^2_\tau.$$

At a terminal boundary it is forced to be

$$T_1 \to Z_1 \to T_2 \to Z_2 \to \$.$$

Proof. This follows by inspecting the paired roles. After O^1_τ, checker 2's state A_τ forces A^1_τ. Next C_τ permits C^1_τ, while checker 1 rejects a premature O^2. Once C^1_τ restores checker 1's sentinel, checker 2 still remembers τ and forces O^2_τ. Checker 1 then forces A^2_τ and C^2_τ. The terminal roles force the terminal sequence analogously. A core letter during an exposed raw-zero phase may be non-poisoning for the exposed checker, but the complementary checker is positive and therefore poisons on that same core letter; hence it is never a common legal continuation. □

Lemma 105.3 (Alternating sentinel)**.** *After the opening @ and before the final common reset \$, every proper non-poison prefix leaves at least one of the two checker counters strictly positive.*

Proof. Checker 1 removes its sentinel only during the O^1A^1 interval, while checker 2 keeps its sentinel. The C^1 symbol restores checker 1's sentinel before checker 2 removes its sentinel on O^2. During O^2A^2, checker 1 therefore remains positive, and C^2 restores checker 2's sentinel. The terminal block is staggered in the same order: T_1Z_1 opens and recloses checker 1 before T_2Z_2 opens checker 2. Only the final \$ removes both restored sentinels on the same input symbol. □

Lemma 105.4 (First simultaneous zero return)**.** *Start both checkers in their special zero configurations and consume an opening @. A later prefix is the first prefix after that opening at which both checkers are again at height zero without poison if and only if that prefix is*

$$u(z) = @\widehat{z}T_1Z_1T_2Z_2\$$$

for some $z \in R_M \cap C_1 \cap C_2$.

Proof. By Lemma 105.2, every common non-poison continuation is forced block by block. By Lemma 105.1, each complete block realizes one genuinely enabled source transition and updates the two represented counters correctly. By Lemma 105.3, no proper prefix before the final common reset can have both counters zero. The terminal block succeeds for both checkers exactly at source state p_f with both source counters zero. Conversely, any normalized halting transcript follows all these transitions. □

Combining Lemmas 102.3 and 105.4 gives

$$\boxed{M \text{ halts} \iff \text{a first simultaneous zero-return query exists.}} \tag{105.3}$$

Chapter 106

Two Half-Size Factors and the Sharp Factor-Budget Boundary

The alternating-sentinel mechanism now replaces the unrestricted family of failure factors by two explicit quotient DOCAs. The halting branch still exposes the same prime finite section, while the nonhalting branch becomes an exact two-factor decomposition.

106.1 Target and two half-size factors

Let

$$q = |Q_M|, \qquad m = |\Delta_M|.$$

Choose a power of two $N = 2^d$ satisfying

$$N/2 \geq q + 2m + 5. \tag{106.1}$$

Choose distinct nonzero vectors $r_1, r_2 \in V$ and put

$$H_i = \{0, r_i\}, \qquad Q_i = V/H_i, \qquad S_i = H_i.$$

Thus $|Q_i| = N/2$.

Checker 1 needs at most $q + 2m + 5$ control names: q source-control roles, $2m$ lexical roles, three terminal roles, one poison role, and one raw/special role. Checker 2 needs at most $2m+5$ names. Hence (106.1) packs both checkers into their quotient state sets. The target can use disjoint positive-role families of total size at most $q + 4m + 10 \leq N$. The same underlying state names may carry unrelated zero-height core meanings and positive-height checker meanings; physical height disambiguates them, so no extra control states are needed.

106.1.1 The target T_M

The target state set is exactly V, every state is final, and the input alphabet is the disjoint union of the core alphabet E and the phase/query alphabet.

At height zero, every core letter acts by translation. At exact anchor 0, every non-core symbol enters permanent-positive poison. At r_i, the special zero table of checker i applies. Every other nonzero zero-height state is generic: all non-reset query symbols leave the target at the same zero anchor, while the final reset \$ sends it to 0. A successful owner-checker terminal block also sends the target to exact state 0: after checker 1 reaches $(U_3, 1)$, its final \$ transition is $(U_3, 1) \xrightarrow{\$} (0, 0)$, and after checker 2 reaches $(V_2, 1)$, its final \$ transition is $(V_2, 1) \xrightarrow{\$} (0, 0)$. At positive height, the packed logical role determines the

appropriate checker transition; every unspecified positive transition enters permanent-positive poison.

A core symbol read at height zero always performs core translation, even if the target has reached r_i as a temporary raw-zero checker configuration. A core symbol read at positive height poisons. Thus no hidden query-mode bit is assumed at height zero.

106.1.2 The factors F_1, F_2

Factor F_i has control set $Q_i = V/H_i$ and all states final. At height zero, core letters act by quotient translation:

$$(v + H_i, 0) \xrightarrow{e} (v + e + H_i, 0).$$

At the special coset S_i, checker i's special/raw transition table applies. In particular, a successful checker-i terminal block removes the restored sentinel on its final \$ and returns F_i to $(S_i, 0)$. Every non-special zero coset deliberately overaccepts query symbols: non-reset query symbols preserve that zero coset, while \$ resets to S_i. Thus every successful factor reset returns to the quotient zero state $H_i = S_i$, whereas every successful target reset returns to the exact zero state 0. Positive logical roles and a permanent-positive poison role are packed into existing quotient state names. No additional state is introduced.

Lemma 106.1 (Size and syntactic restrictions)**.** *T_M is a complete real-time ε-free DOCA with exactly N control states. Each F_i is a complete real-time ε-free DOCA with exactly $N/2$ control states. All three automata use one work symbol, have all states final, and use only updates in $\{-1, 0, +1\}$.*

Proof. The state counts follow from the construction and (106.1). Each transition case is determined by the current control role, zero/positive status, and input class. Undefined cases are totalized into an existing permanent-positive poison role, using $+1$ at zero and 0 at positive height. Hence no sink state or ε-transition is needed. □

106.2 The quotient invariant and forward inclusion

Lemma 106.2 (Quotient invariant)**.** *Fix $i \in \{1, 2\}$. Whenever, after a common consumed prefix, the target is at zero height in exact core state v, factor F_i is at*

$$(v + H_i, 0),$$

unless an earlier target rejection has already made the implication irrelevant.

Proof. Core letters preserve the relation by translation. At a generic nonzero anchor, the target and every non-special factor remain at their zero states throughout non-reset query symbols; a successful \$ sends the target to exact state 0 and every factor F_i to $H_i = 0 + H_i$.

At owner anchor r_i, factor F_i is special and mirrors the target checker throughout the body of the query. If a raw counter action returns to height zero before the query is complete, the target is again at exact owner state r_i and the owner factor is at $H_i = r_i + H_i$, so the quotient relation is preserved. On a successful terminal \$, the target resets to 0 while the owner factor resets to H_i, again giving the required relation $H_i = 0 + H_i$.

At the other active anchor r_j, $j \neq i$, factor F_i is non-special and remains at zero in coset $r_j + H_i$ while the target runs checker j; whenever checker j temporarily returns to raw zero, its exact state is again r_j, so the relation still holds. If checker j completes successfully, the target's final \$ resets to 0, while the non-owner factor's non-special reset sends it to $H_i = 0 + H_i$.

If a core letter interrupts an exposed raw-zero phase, the target translates from the exact owner anchor and every zero-height factor performs the corresponding quotient translation. Finally, a non-core symbol from exact anchor 0 makes the target permanently positive and creates no later zero-height obligation. Thus every successful query reset, whether from a generic anchor or either active anchor, renews the exact zero/quotient-zero synchronization, and these cases exhaust all ways to pass between zero-height target configurations. □

Remark 106.3 (Reset synchronization). The distinction between exact zero and quotient zero is intentional. A successful target query always ends at $(0, 0)$, whereas a successful run of factor F_i ends at $(H_i, 0)$. Since $H_i = 0+H_i$, this is precisely the destination required by the quotient invariant. In particular, after any successful query episode, subsequent core letters or later query episodes start from a freshly synchronized exact/quotient zero configuration.

Lemma 106.4 (Forward inclusion). *For $i = 1, 2$,*

$$L(T_M) \subseteq L(F_i).$$

Proof. An accepted target word ends at zero height in some exact state v. By Lemma 106.2, factor F_i ends at zero height in quotient state $v + H_i$. Since every factor state is final, F_i accepts. □

106.3 Halting exposes a prime section

Assume M halts and choose $z \in R_M \cap C_1 \cap C_2$. Fix the suffix $u = u(z)$ from (105.2). For a core word $x \in E^*$ let $v = \delta(0, x)$.

If $v = 0$, the first @ enters permanent-positive poison. If $v = r_1$, checker 1 successfully processes u and the final \$ resets the target to exact state 0; if $v = r_2$, checker 2 does the same. At every other nonzero anchor the query is deliberately overaccepted and the final \$ also resets the target to 0. Hence

$$\boxed{xu \in L(T_M) \iff \delta(0, x) \neq 0.} \tag{106.2}$$

The fixed-suffix section is exactly P_V.

Lemma 106.5 (Halting implies primality). *If M halts, then T_M is prime: there is no finite family of DOCAs $B_1, \ldots, B_t$ with $|Q_{B_j}| < N$ and*

$$L(T_M) = \bigcap_{j=1}^{t} L(B_j).$$

Proof. Suppose such a family exists. Every core word belongs to $L(T_M)$, hence to every $L(B_j)$. Apply Lemma 103.2 to each B_j with the fixed suffix u. This yields DFAs D_j with

$$|Q_{D_j}| \leq |Q_{B_j}| < N$$

and, by (106.2),

$$P_V = \bigcap_{j=1}^{t} L(D_j).$$

This contradicts the DFA-primality of P_V from Lemma 102.4. □

106.4 Nonhalting gives an exact two-factor decomposition

Assume from now on that M does not halt. By (105.3), no query launched with both factors special can have a first simultaneous zero return.

Lemma 106.6 (Reverse inclusion)**.** *If M does not halt, then*

$$L(F_1) \cap L(F_2) \subseteq L(T_M).$$

Proof. Take $w \notin L(T_M)$. Since the target is complete, real-time, and all states are final, its run on w ends at positive height. Let a be the last zero-height exact target state before the final positive episode.

If $a = r_i$, then by Lemma 106.2 factor F_i is special at that point. The target and owner factor use identical checker transitions throughout the query body; their only successful terminal difference is that the target's final \$ goes to exact zero 0 while the factor's goes to quotient zero H_i. Since a was chosen as the last zero-height target state before the final positive episode, no successful terminal reset occurs afterward. Hence target and owner factor follow the same positive behavior throughout this final episode, and the owner factor also remains positive and rejects. This includes the case where $a = r_i$ is a temporary raw-zero configuration reached inside a malformed or unfinished checker episode.

A generic nonzero a is impossible: generic non-core behavior never creates positive height.

It remains to consider $a = 0$. Both factors are then simultaneously special. A non-core symbol different from @ sends at least one special factor to permanent-positive poison; in particular, \$ is not an immediate reset from a special zero state. Suppose the final positive episode begins with @. By Lemma 105.2, a wrong phase, label mismatch, repeated marker, or premature terminal symbol poisons at least one factor. An illegal source zero/decrement branch poisons the exposed checker. A core symbol during an exposed raw-zero interval translates the exposed checker at height zero, but the complementary checker is still positive and therefore enters permanent-positive poison. If the input ends before a common reset, Lemma 105.3 leaves at least one factor positive. Finally, if both factors were to return to zero, their first simultaneous zero return would contradict nonhalting by Lemma 105.4. Thus at least one factor rejects w. □

Combining Lemmas 106.4 and 106.6 gives the exact decomposition.

Lemma 106.7 (Two factors suffice in the nonhalting case)**.** *If M does not halt, then*

$$\boxed{L(T_M) = L(F_1) \cap L(F_2),} \qquad |Q_{F_1}| = |Q_{F_2}| = N/2 < N.$$

106.5 Undecidability and consequences

Theorem 106.8 (Two-factor undecidability)**.** *The mapping $M \mapsto T_M$ is computable and satisfies*

$$M \text{ halts} \Longrightarrow T_M \text{ is prime},$$

while

$$M \text{ does not halt} \Longrightarrow L(T_M) = L(F_1) \cap L(F_2)$$

for two explicit factors with $N/2$ states each. Consequently Factor-DOCA$_2$ *is undecidable, even under the restrictions of Lemma 106.1.*

Proof. The halting implication is Lemma 106.5; the nonhalting implication is Lemma 106.7. A decider for Factor-DOCA$_2$ would therefore decide halting of deterministic two-counter machines. □

Corollary 106.9 (Unrestricted Prime-DOCA recovered from the two-factor theorem)**.** *The unrestricted Prime-DOCA problem is undecidable under the same restricted target class. This recovers Theorem 104.1 from the sharper two-factor construction.*

Proof. The reduction of Theorem 106.8 maps halting machines to prime targets and nonhalting machines to targets that already decompose into two smaller factors. Hence a decider for unrestricted primality would decide deterministic two-counter halting. □

Corollary 106.10 (Every fixed budget $k \geq 2$)**.** *For every fixed integer $k \geq 2$,* Factor-DOCA$_k$ *is undecidable under the same restrictions.*

Proof. The case $k = 2$ is Theorem 106.8. For $k > 2$, repeat either of the two factors. Repetition preserves the intersection language and Definition 101.3 does not require distinct factors. □

Corollary 106.11 (Bounded-factor input)**.** *The problem that receives a DOCA A together with an integer k and asks whether A is k-factor composite is undecidable, even when every reduction instance has $k = 2$.*

Corollary 106.12 (Non-semidecidability)**.** *The set of 2-factor-composite DOCA presentations is not recursively enumerable.*

Proof. The construction gives

$$M \notin \mathrm{HALT}_2 \iff T_M \in \text{Factor-DOCA}_2.$$

Deterministic two-counter nonhalting is not recursively enumerable, so recursive enumerability of Factor-DOCA$_2$ would give a recursive enumeration of nonhalting instances by computable preimage. □

106.6 Interpretation

The result identifies the exact nontrivial factor-budget boundary for this decomposition problem. Unrestricted primality permits an arbitrary finite number of smaller factors, but the fixed-budget theorem shows that unbounded factor supply is not the source of the hardness. Two factors, the smallest budget capable of expressing a genuine intersection decomposition, already support undecidability.

The finite prime obstruction and the infinite-state synchronization play different roles. The elementary-abelian core handles the halting branch by exposing a fixed regular section that no intersection of smaller finite-state sections can realize. The alternating-sentinel code handles the nonhalting branch by making two one-counter factors jointly certify a two-counter transcript. Neither factor stores two unbounded counters. Instead, each factor stores lexical information exactly while the other temporarily gives up its sentinel to expose an exact zero test. In this sense, the natural third regular checker has not been simulated wholesale inside either counter factor; its responsibility has been distributed across the pair.

This gives a sharp contrast with the finite-state landscape. Bounded-factor decomposition remains within classical complexity classes for DFAs, including NP-completeness for commutative permutation DFAs [102]. For deterministic one-counter automata, by contrast, the same bounded formulation is already undecidable at $k = 2$.

106.7 Construction reference tables

The tables below retain the non-poisoning phase transitions so the construction can be reconstructed entirely from the monograph. Every omitted non-core transition from a special zero state enters permanent-positive poison with update +1; every omitted transition at positive height enters permanent-positive poison with update 0. Core letters always translate at height zero and poison at positive height. At non-special factor cosets, query symbols overaccept at height zero and \$ resets to the special coset. The checker body is shared by target and factor implementations, but the successful final terminal reset is deliberately different: the target goes to exact state 0, whereas factor F_i goes to special coset $S_i = H_i$.

106.7.1 Checker 1

Mode	Input	Action
S_1, zero	@	$(P_{p_0}, +1)$
P_p, positive	O^1_τ, $\mathrm{src}(\tau) = p$	$(S_1, -1)$
S_1, raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathrm{inc}$	raw, +1
S_1, raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathsf{skip}$	raw, 0
S_1, zero raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathrm{zero}$	raw, 0
S_1, positive raw	A^1_τ, $\mathrm{op}_1(\tau) = \mathrm{dec}$	raw, −1
S_1, raw	C^1_τ	$(P_{\mathrm{tgt}(\tau)}, +1)$
P_p, positive	O^2_τ	$(K^A_\tau, 0)$
K^A_τ, positive	A^2_τ	$(K^C_\tau, 0)$
K^C_τ, positive	C^2_τ	$(P_{\mathrm{tgt}(\tau)}, 0)$
P_{p_f}, positive	T_1	$(S_1, -1)$
S_1, zero raw	Z_1	$(U_1, +1)$
U_1, positive	T_2	$(U_2, 0)$
U_2, positive	Z_2	$(U_3, 0)$
U_3, height 1	\$	target: $(0, -1)$; factor F_1: $(S_1, -1)$

106.7.2 Checker 2

Mode	Input	Action
S_2, zero	@	$(B, +1)$
B, positive	O^1_τ	$(A_\tau, 0)$
A_τ, positive	A^1_τ	$(C_\tau, 0)$
C_τ, positive	C^1_τ	$(C_\tau, 0)$
C_τ, positive	O^2_τ	$(S_2, -1)$
S_2, raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathrm{inc}$	raw, +1
S_2, raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathsf{skip}$	raw, 0
S_2, zero raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathrm{zero}$	raw, 0
S_2, positive raw	A^2_τ, $\mathrm{op}_2(\tau) = \mathrm{dec}$	raw, −1
S_2, raw	C^2_τ	$(B, +1)$
B, positive	T_1	$(W_1, 0)$
W_1, positive	Z_1	$(B, 0)$
B, positive	T_2	$(S_2, -1)$
S_2, zero raw	Z_2	$(V_2, +1)$
V_2, height 1	\$	target: $(0, -1)$; factor F_2: $(S_2, -1)$

106.8 Edge conditions for repeated queries

Several edge conditions are part of the construction rather than optional proof commentary. They are collected here to make malformed inputs, raw-zero interruptions, and repeated query episodes explicit.

1. The target never stores both source counters; conjunction is enforced by the two factors at the zero anchor.
2. The factor budget is exactly two, and both factors have exactly $N/2$ states.
3. No hidden query-mode bit is assumed at zero height: raw zero and ordinary owner zero are the same physical configuration.
4. During every raw-zero interval, the complementary factor is positive and carries the lexical/sentinel witness.
5. Core interruption at raw zero cannot create a false common success: the zero factor translates, while the positive factor poisons.
6. A premature O^2 before C^1 is rejected by raw checker 1; hence checker 2 may safely reuse C_τ before and after C_τ^1.
7. The terminal transition $B \xrightarrow{T_2,-1} S_2$ is explicit.
8. Immediate \$ is not a legal reset from a special zero state.
9. Successful reset renews the quotient invariant for repeated query episodes: the target returns to exact state 0, every owner factor returns to its special coset H_i, and every non-owner factor resets from its non-special coset to the same $H_i = 0+H_i$.
10. The halting/primality argument is self-contained for arbitrary smaller deterministic one-counter factors, including ε-enabled factors, via the physical zero-cut Lemma 103.2.

Chapter 107

Fixed One-Counter Language Equivalence: Problems 24.04(B) and 24.04(A)

Research branch — The comparison boundary after decomposition. *The preceding chapters show that existential decomposition of a deterministic one-counter presentation is algorithmically hard even at the smallest nontrivial factor budget. This chapter keeps the two fixed-presentation comparison problems distinct. It first establishes the solution of Automata Exchange Problem 24.04(B), comparing a deterministic one-counter net with a nondeterministic one-counter net. It then adds the stronger zero-testing branch, Problem 24.04(A), where the deterministic machine is a one-counter automaton with an explicit zero test. Both fixed-comparison problems are decidable, but by different compactness mechanisms; the second branch also survives the stated epsilon-transition extension.*

The distinction is essential. Prime-DOCA and fixed-budget factorization quantify over unknown smaller recognizers. Language equivalence fixes both recognizers in advance. Nondeterminism enlarges the forward semantic state from one counter value to a finite envelope, but the monotonicity of one-counter nets still leaves enough order structure for a computable small-witness theorem.

The first fixed-presentation comparison theorem answers Automata Exchange Problem 24.04(B) [109]. The model is the one-counter *net*: the counter is nonnegative, transitions may be disabled by underflow, and there is no explicit zero test. This theorem itself does not cover an explicit zero test. The zero-testing asymmetric problem is taken up later in this same chapter as Problem 24.04(A); its proof requires a different compactness architecture rather than a cosmetic modification of the net argument.

107.1 Model, acceptance, and integer-weight normalization

A one-counter net (OCN) is a tuple

$$A = (Q, \Sigma, \delta, q_0, F),$$

where

$$\delta \subseteq Q \times \Sigma \times \mathbb{Z} \times Q.$$

A transition (q, a, z, r) is enabled from counter value $c \in \mathbb{N}$ exactly when $c + z \geq 0$, and then moves to $(r, c + z)$. A word is accepted when some run from $(q_0, 0)$ ends in a state of F; the final counter value is unrestricted. Determinism means that for every control state and input letter there is at most one outgoing transition.

For the structural proof it is convenient to work in unit-step form.

Lemma 107.1 (Unit-step normalization)**.** *Every finite integer-weight OCN can be effectively transformed, over the same input alphabet, into a unit-step OCN whose updates lie in* $\{-1, 0, +1\}$ *and which recognizes the same final-state language. Determinism and nondeterminism are preserved.*

Proof. Let W be the largest absolute transition weight and choose $M > W$; when every weight is zero take $M = 1$. Write each original counter value uniquely as

$$n = Mc + r, \qquad c \in \mathbb{N}, \quad 0 \le r < M.$$

Store r in finite control and c in the physical counter. For an original update z and stored remainder r, write

$$r + z = kM + r', \qquad 0 \le r' < M.$$

Since $|z| < M$, one has $k \in \{-1, 0, +1\}$. Replace (q, a, z, q') by

$$((q, r), a, k, (q', r')).$$

If $k = -1$ and the quotient counter is zero, the unit-step transition is disabled exactly when the original update would make $n + z < 0$. Otherwise the encoding invariant is preserved. Finality depends only on the original state component, so accepted words are unchanged. Refining the source control state by r cannot introduce nondeterministic choices into a deterministic presentation. □

From now until the final assembly, D denotes a deterministic unit-step OCN and N a nondeterministic unit-step OCN over the same alphabet, both started at counter value zero.

107.2 Exact residual semantics for the nondeterministic side

For a prefix u and $q \in Q_N$, define the exact forward envelope

$$H_u(q) = \max\{h \in \mathbb{N} : (q_{0,N}, 0) \xrightarrow{u} (q, h)\},$$

with value $\perp$ when q is unreachable. For a suffix v, define the backward completion requirement

$$R_v(q) = \min\{h \in \mathbb{N} : (q, h) \xrightarrow{v} (f, c) \text{ for some } f \in F_N, c \in \mathbb{N}\},$$

with value ∞ when no such counter exists.

The pair (H, R) is the finite-dimensional replacement for the scalar residuals used earlier in the monograph.

Proposition 107.2 (Exact cut semantics)**.** *For every factorization* $w = uv$*,*

$$uv \in L(N) \iff \exists q \in Q_N : \; H_u(q) \ge R_v(q).$$

Hence

$$uv \notin L(N) \iff H_u(q) < R_v(q) \quad (\forall q \in Q_N).$$

Proof. An accepting run cut after u has some state q and counter h with $h \le H_u(q)$ and $h \ge R_v(q)$. Conversely, if $H_u(q) \ge R_v(q)$, take a prefix run attaining the envelope value and execute an accepting v-run that needs only $R_v(q)$ units of credit. One-counter monotonicity permits the same transition sequence from any larger initial credit. □

For a fixed factor x, let T_x be the exact forward-envelope transformer and S_x the exact backward-requirement transformer. If a path $\pi : q \xrightarrow{x} r$ has total effect $\mathrm{eff}(\pi)$ and minimum entry requirement $\mathrm{req}(\pi)$, then

$$(T_x H)(r) = \max_{\substack{q,\pi: q \xrightarrow{x} r \\ H(q) \geq \mathrm{req}(\pi)}} \big(H(q) + \mathrm{eff}(\pi)\big),$$

and

$$(S_x R)(q) = \min_{r,\pi: q \xrightarrow{x} r} \max\{\mathrm{req}(\pi), R(r) - \mathrm{eff}(\pi)\}.$$

Theorem 107.3 (Exact residuation)**.** *For every finite envelope H, requirement vector R, and word x,*

$$T_x(H) < R \iff H < S_x(R)$$

componentwise, with the usual conventions for $\bot$ and ∞.

Proof. Fix a source coordinate q. The inequality $H(q) < S_x(R)(q)$ says that for every x-path $\pi : q \to r$, either $H(q)$ is below its entry requirement or the resulting credit remains below $R(r)$. This is exactly the assertion that no contribution from q can make any coordinate of $T_x(H)$ reach the corresponding completion requirement. □

107.3 Shortest counterexamples and bounded-height compactness

Assume the reverse-inclusion promise

$$L(N) \subseteq L(D)$$

and choose a shortest word

$$w \in L(D) \setminus L(N).$$

Let the unique deterministic run visit (p_t, d_t) and let H_t be the exact N-envelope at the same cut.

Lemma 107.4 (Mixed-domination deletion)**.** *Suppose $i < j$ satisfy*

$$p_i = p_j, \qquad d_i \geq d_j, \qquad H_i \leq H_j.$$

Then deleting the factor between the two cuts produces a shorter word in $L(D) \setminus L(N)$.

Proof. The unchanged deterministic suffix starts from at least the original height d_j, so it remains feasible and accepting. On the nondeterministic side, rejection at cut j gives $H_j < R_j$ for the exact requirement of the unchanged suffix. Since $H_i \leq H_j$, the shortened word is still rejected. □

This immediately gives the first effective compactness theorem.

Theorem 107.5 (Effective bounded-height compression)**.** *For every fixed B there is a total computable $F(B, D, N)$ such that every shortest counterexample with*

$$d_t \leq B \qquad (0 \leq t \leq |w|)$$

has length at most $F(B, D, N)$.

Proof. Encode $\perp$ by zero and a finite envelope value h by $h+1$. At cut t record

$$X_t = (p_t, B - d_t, \widehat{H}_t).$$

For each fixed deterministic control state, order the last $1 + |Q_N|$ coordinates componentwise. Dickson's lemma makes the finite disjoint union of these orders a well-quasi-order. A comparable pair $X_i \le X_j$ would satisfy the hypotheses of Lemma 107.4, so a shortest counterexample yields a bad sequence.

For effectivity, at depth t every finite forward-envelope coordinate is at most t in the unit-step model. Hence the tree of all bad sequences satisfying the depth-dependent coordinate bounds and $d_t \in \{0, \dots, B\}$ is effectively finitely branching. Dickson excludes an infinite branch and König therefore makes the whole tree finite. Breadth-first generation computes its maximum depth. □

Thus all remaining difficulty is concentrated in high deterministic excursions.

107.4 Promise geometry: sources, sinks, and slope budgets

Long high regions expose only finitely many primitive synchronized control-cycle types: deterministic sign cycles, D/N relative cycles, $D/N/N$ strand-differential cycles, and bounded support/boundary events. Exposure-free connectors have bounded length because their normalized backward requirement profile ranges over a finite set and any repeated connector color either deletes or creates a local recurrence outcome.

The standing promise now imposes a numerical restriction on recurrent sources and sinks. Consider a compatible ordered route with an earlier synchronized source

$$X : (a, +A)$$

and a later synchronized sink

$$Z : (-b, -B),$$

where $A, B, b > 0$ and compatibility means that fixed bounded connectors lead from the prefix to X, from X to Z, and from Z to one accepting N-suffix.

Lemma 107.6 (Promise source sign)**.** *If a synchronized loop has positive N-effect and can be pumped to cross a fixed accepting-suffix requirement, then its deterministic effect is nonnegative.*

Proof. A negative deterministic effect together with positive N-effect would, after enough repetitions, supply the nondeterministic suffix while forcing deterministic underflow. That would produce a word in $L(N) \setminus L(D)$, contradicting the promise. □

Theorem 107.7 (Promise efficiency inequality)**.** *Every compatible ordered source–sink route satisfies*

$$aB \ge bA.$$

When $a > 0$ this is

$$\frac{A}{a} \le \frac{B}{b}.$$

Proof. Let δ be the fixed N-credit needed by the connector and accepting suffix. For n sink repetitions take

$$m_n = \left\lceil \frac{nB + \delta}{A} \right\rceil$$

source repetitions. The selected nondeterministic route is then feasible. If $aB < bA$, the deterministic net effect $m_n a - nb$ tends to $-\infty$, so sufficiently large n again yields a word accepted by N but blocked in D. □

Choosing a rational slope $\lambda = p/q$ between every realized source ratio and compatible sink cost gives the adjusted drift

$$\Psi = q\,\Delta N - p\,\Delta D.$$

Every recurrent source/sink block cycle has $\Psi \le 0$. Inside a strongly connected mode graph this yields a max-plus potential; noncritical edges spend at least one unit of adjusted budget. A two-resource cone-separation argument pushes any still-unbounded entry credit backward along the synchronized SCC condensation DAG. Since the DAG is finite, unresolved funding cannot retreat forever. This is the mechanism that localizes large relative credit to bounded-entry source episodes.

The quantitative form needed later is worth recording explicitly.

Corollary 107.8 (Quantitative relative-source localization)**.** *For every bounded deterministic entry range $0 \le d \le B$ there is a computable threshold*

$$R_{\mathrm{rel}}(D, N, B)$$

such that, whenever a reachable envelope coordinate satisfies

$$H(q) - d > R_{\mathrm{rel}}(D, N, B),$$

its provenance contains an earlier bounded-entry synchronized source P with

$$\Delta D(P) \ge 0, \qquad \Delta N(P) > 0, \qquad \Delta N(P) - \Delta D(P) > 0.$$

Both source-entry counters are bounded effectively from D, N, B.

Proof. For every synchronized SCC whose cycle cone misses the target cone

$$\{(d, r) : d \ge 0, > 0\}, \qquad r = \Delta N - \Delta D,$$

rational separation bounds relative gain by a computable affine function of deterministic entry credit plus a bounded simple-path/connector transient. The condensation DAG has finite depth, and there are only finitely many SCC, connector, and regime types. Taking the maximum of these effective bounds over all provenance chains with deterministic entry at most B produces R_{rel}. Above it, some SCC must realize a positive-relative cone direction, which can be converted, after bounded connectors, into the asserted closed source macro. □

107.5 Full-envelope control inside same-level high returns

Selected-run source/sink arguments are insufficient by themselves: a different nondeterministic run may become the maximizing envelope owner later. The proof therefore works in the synchronized triple graph

$$G_3 = Q_D \times Q_N \times Q_N$$

with differential weight

$$g = \Delta N(\pi) - \Delta N(\rho).$$

If a triple SCC has no negative g-cycle, a vertex potential bounds how far one strand can lose relative to another; dually for no positive cycle. If both signs occur, bounded simple positive and negative differential cycles can be combined into a same-base zero-differential package. Choosing canonical maximizing predecessors at envelope updates

then gives an effective finite switch-signature set outside mixed-sign SCCs. Repeated signatures expose zero-differential recurrence rather than unbounded owner switching.

Together with Corollary 107.8, this yields a canonical positive-relative source cut whose *entire* exact forward envelope is bounded. Between such cuts, the only unresolved recurrent lag/catch-up pair has the sign pattern

$$C_{\mathrm{lag}} : (+a, -A), \qquad C_{\mathrm{up}} : (-b, +B),$$

with $a, A, b, B > 0$ and nonpositive useful N-mean on the catch-up side. Positive-N residual modes either contradict the promise or return to the bounded source–sink program.

These finitely many surviving event modes are assembled into a raw graph M_{raw}. Each edge carries its deterministic effect $d(e)$, tracked nondeterministic effect $n(e)$, finite control/regime data, internal requirements, and a literal-length bound.

Zero-D closed-mode recurrence is handled from the *concrete exact entry envelope*: the local search state is a finite mode vertex together with the exact envelope. Dickson plus König gives an effective finite node-dependent search tree unless mixed-domination deletion occurs.

Positive-N cycles are then removed by sign.

Lemma 107.9 (Chronological compatible matching)**.** *In one same-level bounded-return excursion, after cycle erasure every recurrent positive deterministic unit can be matched in chronological order to later negative deterministic consumption, up to a computably bounded correction absorbed by the simple residual skeleton and bounded endpoint fragments. Every matched source/sink pair belongs to one of finitely many compatible ordered route classes.*

Proof. Give each positive-D recurrent occurrence one token per gained unit and each negative-D occurrence one consumption slot per lost unit. Scan from left to right and assign each slot to the earliest available earlier token. Since the excursion returns to its starting deterministic level, only the bounded residual skeleton can account for unmatched tokens or initial deficits. Erasing recurrent subwalks between one matched source and sink leaves a simple synchronized connector. Accessibility comes from the original prefix and coaccessibility from the original suffix, while support, requirement regime, critical residue, and internal requirements range over finite event data. Hence only finitely many compatible route classes arise. □

A negative-D/positive-N useful cycle contradicts the promise. A zero-D/positive-N cycle belongs to the exact-entry WQO system. A positive-D/positive-N cycle is matched by Lemma 107.9 to later negative-D consumption and is handled by the strict or exact source–sink slope alternatives. Exact slopes use finite critical relation periods together with balanced deletion: above an effective protection threshold, one source period and one later sink period can be deleted while rejoining the same deterministic physical configuration and the same critical N relation state seen by the unchanged suffix.

After replacing these discharged episodes by finitely many exact bounded representatives, and retaining bounded-entry deferral transfers with exact scalar entry/exit labels, one obtains a finite reduced graph M_{red}.

Theorem 107.10 (No positive cycle in the reduced graph)**.** *Every directed cycle* Γ *of* M_{red} *satisfies*

$$n(\Gamma) \leq 0.$$

Proof. Expand each reduced macro edge to its exact bounded representative. If the resulting closed useful walk had positive tracked N-effect, simple-cycle decomposition would expose a positive useful recurrent cycle of the raw graph. The preceding trichotomy sends

such a recurrence to promise contradiction, zero-D exact-entry compactness, bounded strict/exact source–sink reduction, or bounded-entry deferral. The first three are removed rather than recurrent in M_{red}; retained deferral edges are split by exact scalar transfer, so any closed cycle of transfers telescopes and cannot create positive net scalar gain. □

Theorem 107.11 (Arbitrary-interleaving envelope bound)**.** *Fix a same-level bounded-return excursion entered with deterministic control p and exact envelope H_0. There is a computable constant*

$$C_{\mathrm{BR}}(D, N, p, H_0)$$

such that every suffix-relevant forward-envelope coordinate throughout any retained unresolved excursion is at most

$$\max_{q:H_0(q)\neq\bot} H_0(q) + C_{\mathrm{BR}}(D, N, p, H_0).$$

Proof. Inside each SCC of M_{red}, Theorem 107.10 gives a max-plus potential bounding every path's positive tracked gain by an endpoint constant. The SCC condensation graph is acyclic, so inter-SCC gain contributes only a finite DAG correction. Every collapsed zero-D, source–sink, or retained deferral edge has a computable internal peak above its endpoints. Taking the maximum internal correction over the finite reduced graph, rather than multiplying by the number of traversals, gives the asserted bound under arbitrary interleaving. □

107.6 Context-safe recurrence and exact-entry bounded-return compactness

The full-envelope bound makes the exact active color set finite at canonical cuts. An active color records the deterministic control, finite completion support A, and the exact restricted envelope $H|_A$.

Lemma 107.12 (Exact active-color deletion)**.** *If two canonical cuts of a rejected word have the same exact active color and the intervening deterministic effect is nonpositive, deleting that factor preserves deterministic acceptance and nondeterministic rejection of the unchanged suffix.*

Proof. The deterministic control state agrees, and nonpositive effect means the earlier cut has at least as much deterministic credit as the later cut. On the N side the active-envelope coordinates are identical, while states outside the completion support cannot accept the unchanged suffix. Thus the same strict cut separator survives deletion. □

A retained exact-color cycle therefore has strictly positive integral deterministic effect. This yields a global event bound inside a same-level return without any cumulative N-loss assumption.

Theorem 107.13 (Positive-return total multiplicity bound)**.** *For every concrete exact same-level entry (p, H_0), the total number of canonical event edges in a shortest retained bounded-return excursion is effectively bounded, uniformly over arbitrary interleaving of the remaining reduced modes.*

Proof. Let G_{col} be the finite exact-color graph induced after Theorem 107.11. Apply ordinary cycle erasure to the traced vertex sequence. It decomposes into a simple residual

path σ and directed simple cycles $\Gamma_1, \dots, \Gamma_m$. Let C_{col} be the number of color vertices and D_{edge} the largest absolute deterministic effect of one reduced edge. Then

$$d(\sigma) \geq -(C_{\mathrm{col}} - 1)D_{\mathrm{edge}}.$$

The complete excursion has deterministic net effect zero, up to a fixed bounded endpoint correction C_{end}, so

$$\sum_{j=1}^{m} d(\Gamma_j) \leq (C_{\mathrm{col}} - 1)D_{\mathrm{edge}} + C_{\mathrm{end}}.$$

Lemma 107.12 makes every retained cycle effect a positive integer. Hence m is bounded, and each simple cycle has at most C_{col} edges. This bounds the whole canonical event trace. □

Neutral zero-D powers are bounded node-by-node by exact-envelope self-cover: in the computable orbit $H, T_x H, T_x^2 H, \dots$, Dickson gives $i < j$ with $T_x^i H \leq T_x^j H$, and the intervening power deletes. Negative-mean block recurrences have an effective linear kill threshold. Zero-mean recurrences are not falsely declared finite merely because their critical relation powers are periodic: zero-D recurrence returns to exact-entry WQO deletion, while positive-D zero-mean recurrence is retained as deterministic funding and transferred to later negative-D exposure or, if unmatched, to the terminal level ladder.

Combining these mechanisms gives the local theorem on which the global construction rests.

Theorem 107.14 (Exact-entry bounded-return compactness)**.** *For every concrete same-level high entry (p, H) there is a total computable integer*

$$L^{\dagger}_{\mathrm{BR}}(D, N, p, H)$$

such that every same-level bounded-return factor in a shortest hard-direction counterexample has length at most this number, unless the local analysis already produces endpoint-preserving shortening, a contradiction to $L(N) \subseteq L(D)$, or a bounded decisive continuation from the current exact node.

Proof. First compute the exact-entry zero-D search trees and the positive-cycle reduction, then the reduced-graph potential constant C_{BR}, then the full-envelope bound, then the finite exact-color graph and its maximum local literal edge length, and finally the event-edge bound of Theorem 107.13. None of these quantities uses the terminal ladder, the later LOW-state tree, or the final witness bound. Multiplying the finite event-edge bound by the finite local edge-length bound and adding the bounded endpoint fragments gives $L^{\dagger}_{\mathrm{BR}}$. □

107.7 Terminal growth and the exact LOW continuation tree

Choose a computable low threshold B_0 above every primitive deterministic effect and every bounded internal requirement used by the local event decomposition. A cut is LOW when the deterministic counter is at most B_0.

After the last LOW cut, if the run never returns to the low band, let c_ℓ be the last occurrence of deterministic level ℓ. Between c_ℓ and $c_{\ell+1}$ there is one $+1$ deterministic step followed by a same-level bounded-return excursion. Thus Theorem 107.14 makes the successor set from one exact ladder state (p_ℓ, H_ℓ) effectively finite.

Lemma 107.15 (Terminal ladder deletion)**.** *If $\ell < \ell'$ satisfy*

$$p_\ell = p_{\ell'}, \qquad H_\ell \le H_{\ell'},$$

then deleting the whole factor from c_ℓ to $c_{\ell'}$ produces a shorter word in $L(D) \setminus L(N)$.

Proof. The deleted factor raises the deterministic level by $e = \ell' - \ell$. By last-occurrence minimality, the original suffix after $c_{\ell'}$ never falls below ℓ', so after deletion it remains nonnegative when shifted down by e. Rejection follows from $H_\ell \le H_{\ell'} < R_{\ell'}$. □

Hence the terminal ladder is a bad sequence in finitely many copies of $\mathbb{N}^{|Q_N|}$. Effective finite branching plus Dickson and König produce a total computable terminal-tail bound

$$L^\dagger_{\mathrm{term}}(D, N, p, H).$$

For the nonterminal part, record at every LOW cut the exact state

$$X = (p, d, H) \in Q_D \times \{0, \dots, B_0\} \times (\mathbb{N} \cup \{\bot\})^{Q_N}.$$

Order LOW states by equality of p, d and componentwise domination of H. A comparable pair deletes by the same mixed-domination argument.

From one concrete LOW state the next LOW successor set is finite and computable. A one-letter transition that remains LOW is immediate. If the deterministic run exits from B_0 to $B_0 + 1$, compute the exact post-exit envelope and enumerate all first-return high interiors of length at most $L^\dagger_{\mathrm{BR}}$; if no LOW return occurs, use the terminal bound above. This yields an effectively finitely branching tree $T^\dagger_{\mathrm{low}}$ of LOW bad sequences.

Theorem 107.16 (Hard-direction small-witness theorem)**.** *Under the promise*

$$L(N) \subseteq L(D),$$

there is a total computable function $B^\dagger_{\mathrm{hard}}(D, N)$ such that

$$L(D) \not\subseteq L(N) \Longrightarrow \exists w \in L(D) \setminus L(N) : |w| \le B^\dagger_{\mathrm{hard}}(D, N).$$

Proof. Every branch of $T^\dagger_{\mathrm{low}}$ is a bad sequence in a finite disjoint union of copies of $\mathbb{N}^{|Q_N|}$, so Dickson excludes infinite branches. Since every node has an explicitly computable finite successor set, König makes the entire generated tree finite. Attach each edge its literal realization length. Take the maximum accumulated LOW-tree weight, the maximum terminal-tail weight obtained after a high exit, and the maximum already-traversed prefix plus any genuinely bounded decisive continuation produced by the local analysis. These are maxima over a finished finite tree. Finally take the maximum with the independent bounded-height bound of Theorem 107.5. This yields $B^\dagger_{\mathrm{hard}}$. □

Corollary 107.17 (Hard inclusion under the reverse-inclusion promise)**.** *Under $L(N) \subseteq L(D)$, the inclusion $L(D) \subseteq L(N)$ is decidable.*

Proof. Compute $B^\dagger_{\mathrm{hard}}$ and exhaustively test all words up to that length. If none belongs to $L(D) \setminus L(N)$, Theorem 107.16 excludes any counterexample. □

107.8 Equality and the resolved comparison problem

The opposite inclusion orientation is classical. Hofman and Totzke proved decidability of deterministic-right OCN trace inclusion and their compact-witness method applies to monotone final-state acceptance relations [108]. Thus for nondeterministic N and deterministic D one can decide

$$L(N) \subseteq L(D).$$

Only decidability is used here; no complexity bound for the full equivalence algorithm is claimed.

Theorem 107.18 (Deterministic–nondeterministic OCN language equivalence)**.** *Given a deterministic one-counter net D and a nondeterministic one-counter net N over the same finite alphabet, with arbitrary integer transition weights, both started at counter value zero and accepting by final control state, it is decidable whether*

$$\boxed{L(D) = L(N).}$$

Proof. Normalize both nets by Lemma 107.1. First decide the deterministic-right inclusion

$$L(N) \subseteq L(D)$$

using the Hofman–Totzke theorem. If it fails, the languages are unequal. If it holds, the standing promise required by Corollary 107.17 is satisfied, so decide the reverse inclusion

$$L(D) \subseteq L(N).$$

The two normalized languages are equal exactly when both inclusions hold, and normalization preserves the original final-state languages. □

This resolves the fixed-presentation mixed equivalence question recorded as Automata Exchange Problem 24.04(B) [109]. The result should be read together with the decomposition chapters immediately preceding it:

fixed deterministic/nondeterministic OCN comparison is decidable

existential decomposition of a deterministic one-counter presentation is undecidable

These are different quantifier structures, not competing statements.

107.9 Scope of the net theorem and the zero-test frontier

The proof uses monotonicity at every scale: forward envelopes are upward-credit monotone, mixed-domination deletion lowers or preserves the relevant resources, and the terminal/LOW WQO arguments depend on the fact that added counter credit cannot invalidate a fixed transition sequence. An explicit zero test breaks exactly this global monotonicity. Therefore the theorem in this first branch does not by itself decide the corresponding problem in which the deterministic machine is a zero-testing OCA and the other machine is a nondeterministic counter net.

Conceptually, this is the one-counter analogue of a recurring distinction throughout the monograph. A physical feature may look syntactically small while changing the semantic order structure on which a normalization or compactness theorem depends.

Here the absent zero test is not a cosmetic restriction: it is what makes the exact envelope order compatible with deletion and makes the two global bad-sequence trees possible. The next branch closes this zero-test frontier by introducing deterministic zero cuts, positive first-return compactness, and—for silent transitions—a finite weak-macro calculus.

Problem 24.04(A): deterministic zero-testing OCA versus nondeterministic OCN

Research branch — Closing the zero-test comparison frontier. *The preceding branch proves Problem 24.04(B) for deterministic versus nondeterministic one-counter nets, where global credit monotonicity drives the compactness argument. We now restore an explicit zero test on the deterministic side. This changes the proof architecture: zero cuts and positive first returns replace the global net-only deletion scheme, and the epsilon extension requires finite threshold-affine weak macros together with stable/reset deterministic normalization. The branch is logically independent of the proof of Problem 24.04(B), but it occupies the adjacent comparison frontier and completes the two Automata Exchange questions treated in this monograph.*

The earlier state-primality and two-factor chapters concern existential decomposition of one fixed deterministic presentation. Here both machines are supplied as input and the question is language equality. The left-hand device may test zero and is deterministic; the right-hand device has no zero test and may branch nondeterministically. The proof therefore requires a different invariant: exact forward envelopes and backward requirements replace the decomposition quotients, and shortest-counterexample compactness replaces factor construction.

107.10 Main theorems and proof architecture

A one-counter automaton (OCA) is a pushdown automaton with a singleton stack alphabet. Unlike a one-counter net (OCN), it may distinguish counter value 0 from positive values. We study the asymmetric language-equivalence problem in which the zero-testing machine is deterministic and the net is nondeterministic.

Problem 24.04(A) in the Automata Exchange collection asks whether language equivalence between a deterministic one-counter automaton and a nondeterministic one-counter net is decidable [109]. The letter-labelled formalization used in the surrounding one-counter-net literature is real time; we first solve that formulation and then prove that the same decidability survives a standard deterministic-epsilon extension on the OCA side together with arbitrary epsilon moves on the OCN side.

Theorem 107.19 (Real-time main theorem)**.** *Let A be a deterministic real-time unit-step one-counter automaton with zero tests and let N be a nondeterministic real-time unit-step one-counter net over the same finite alphabet. Both start with counter value* 0 *and accept finite words by final control state. It is decidable whether $L(A) = L(N)$.*

Theorem 107.20 (Real-time global small witness)**.** *There is a total computable function B such that for every pair (A, N) as in theorem 107.19,*

$$L(A) \not\subseteq L(N) \Longrightarrow \exists w \in L(A) \setminus L(N) \text{ with } |w| \leq B(A, N).$$

Theorem 107.21 (Epsilon-move extension)**.** *Given a deterministic unit-update zero-testing one-counter automaton A with standard deterministic epsilon moves and a nondeterministic unit-update one-counter net N with arbitrary epsilon moves and no zero tests, both initialized at counter* 0 *and using finite-word final-state acceptance, it is decidable whether $L(A) = L(N)$.*

Theorem 107.22 (Epsilon hard inclusion)**.** *For the machines of theorem 107.21, it is decidable whether $L(A) \subseteq L(N)$. Moreover a finite bound $B^{\varepsilon}_{\mathrm{global}}(A, N)$ is effectively computable such that failure of inclusion has a witness of visible length at most this bound.*

Theorem 107.23 (Epsilon reverse inclusion)**.** *For the machines of theorem 107.21, it is decidable whether* $L(N) \subseteq L(A)$*.*

Proof architecture. The real-time proof has four layers: exact residual semantics, positive first-return compactness, an effectively finite zero-state tree, and a terminal drain reduction. The epsilon extension does not eliminate silent moves syntactically. Instead it compiles each weak visible OCN step into a finite threshold-affine macro system, proves an internal guarded-macro robustness theorem for the compactness argument, and combines this with the standard stable/reset normalization of deterministic epsilon-OCA. The reverse inclusion in both settings is handled independently by one-zero-test VASS reachability [115, 116].

Relation to known results and novelty scope. Valiant and Paterson proved decidability of equivalence for deterministic one-counter automata [106], and Böhm, Göller, and Jančar later obtained the NL-completeness bound in the real-time setting [107]. This should be contrasted with deterministic-OCA language inclusion, which is already undecidable [110]; hence the present equality problem does not follow from a two-way inclusion test on deterministic OCAs.

On the net side, trace inclusion from a nondeterministic OCN into a deterministic OCN is decidable (indeed NL-complete in the standard trace setting), whereas general OCN trace inclusion is undecidable [108]; strong and weak simulation over OCNs remain decidable and PSPACE-complete [112]. General OCN determinizability is undecidable under most natural notions [113]. History-determinism gives a different positive frontier: history-deterministic OCNs can be converted to language-equivalent deterministic OCAs and their language inclusion/equivalence problems are decidable in PSPACE [114]. The same work notes that its HD-OCN results extend, with weak-simulation machinery, to epsilon-transitions; thus silent moves themselves are not the novelty claimed here.

The comparison class in this chapter is different from all of the preceding settings: the right-hand machine is an arbitrary nondeterministic OCN, not assumed deterministic or history-deterministic, while the left-hand machine may test zero and is deterministic. To the best of our knowledge, we are not aware of a prior published decidability theorem for finite-word final-state language equality for this exact asymmetric pair, either in the real-time formulation or under the epsilon conventions of theorem 107.21. We cite the Automata Exchange entry only as a record of Problem 24.04(A), not as evidence of priority or novelty [109]. Problem 24.04(B), proved in the preceding branch, is logically separate and is not used in any argument of the present zero-testing proof.

Organization. Sections 107.12–107.19 of this branch prove the real-time theorem. The subsequent epsilon-extension sections develop weak envelopes, guarded macros, reset-ending returns, the stable-zero tree, terminal tails, and the asynchronous one-zero-test product. The final ledger sections record both effectivity ledgers and the module-by-module transfer certificate.

107.11 Machines and acceptance

107.11.1 Deterministic one-counter automata

A *real-time unit-step one-counter automaton* is a tuple

$$A = (Q_A, \Sigma, \delta_A, q_A^0, F_A),$$

where the counter ranges over $\mathbb{N}$. Transitions are separated into zero and positive rules. A zero rule is enabled exactly when the current counter is 0; a positive rule is enabled exactly when the current counter is positive. Each rule reads one input letter and changes the counter by -1, 0, or $+1$, subject to nonnegativity. We assume that A is deterministic: for each control state, input letter, and zero/positive regime, at most one transition is enabled.

A configuration is $(p, c) \in Q_A \times \mathbb{N}$. The initial configuration is $(q_A^0, 0)$, and a word is accepted when the unique run is defined and ends in a control state in F_A, with arbitrary final counter value.

107.11.2 Nondeterministic one-counter nets

A *one-counter net* is a tuple

$$N = (Q_N, \Sigma, \delta_N, q_N^0, F_N)$$

with transitions (q, a, d, r), $d \in \{-1, 0, +1\}$. A transition is enabled at counter value c iff $c + d \geq 0$. There is no zero test. Acceptance is again by final control state with arbitrary final counter value.

For a word w, we write $L(A)$ and $L(N)$ for the corresponding final-state languages from initial counter 0.

107.11.3 Model scope and normalization

The theorem is stated for the standard *real-time* labelled one-counter model: every transition consumes exactly one input symbol. This is the convention used throughout the proof because literal factors of a shortest counterexample are repeatedly deleted and replaced. The Automata Exchange record for Problem 24.04(A) describes the left machine as "a pushdown automaton with one stack alphabet" and the right machine as a one-counter automaton with no zero tests, but does not itself print a transition syntax. The formal OCA/OCN definition used by Prakash and Thejaswini [114] matches the present convention: OCA transitions are given by

$$Q \times \{\mathsf{zero}, \mathsf{nonzero}\} \times \Sigma \times \{-1, 0, 1\} \times Q,$$

and OCN transitions by $Q \times \Sigma \times \{-1, 0, 1\} \times Q$, with determinism a partial function on the letter-labelled OCA rules. Their surrounding generic LTS preliminaries allow ε-labels, and one displayed OCA sentence inherits the notation Σ_ε; however the OCA transition relation itself is letter-labelled over Σ, and their paper explicitly distinguishes realtime LTSs. We therefore treat Theorem 107.19 as the standard real-time formulation of Problem 24.04(A). The stronger convention with silent transitions is treated separately later in the chapter: the deterministic side is first put into stable/reset normal form, while arbitrary silent behavior of the net is compiled into a weak threshold-affine macro calculus.

The unit-step assumption is inessential for finite integer weights.

Proposition 107.24 (Integer-weight normalization)**.** *Let M be a deterministic real-time one-counter automaton, or a nondeterministic real-time one-counter net, whose counter updates belong to a finite subset of $\mathbb{Z}$. One can effectively construct a language-equivalent unit-step machine of the same deterministic/nondeterministic and zero-testing/net type.*

Proof. Choose an integer K larger than every absolute transition weight and represent a counter value uniquely as $n = Kc + r$ with $0 \leq r < K$. Store r in finite control and use c as the physical counter. For a transition of weight z, write $r + z = Kk + r'$ with

$k \in \{-1, 0, 1\}$; update the physical counter by k and the finite remainder to r'. Underflow of the physical counter when $k = -1$ is exactly original underflow. For an OCA, original zero is characterized by the conjunction $c = 0$ and $r = 0$, which is expressible because r is finite control and $c = 0$ is the unique zero test. If $r > 0$, the original counter is positive irrespective of c, so the corresponding original positive rule is copied into both physical zero and physical positive regimes; if $r = 0$, it is copied only to the physical positive regime. Determinism is preserved because the remainder refines the source state. For an OCN no zero test is introduced. Acceptance controls are copied for every remainder. □

Remark 107.25 (Scope of Theorem 107.19). Consequently the real-time theorem covers arbitrary finite integer transition weights. Silent transitions are not folded into this normalization; they are handled by the separate epsilon-extension machinery developed later in the chapter.

107.12 Exact residual semantics of the net

The proof uses exact finite-dimensional summaries of the nondeterministic net, rather than a selected run.

Definition 107.26 (Forward envelope)**.** For a prefix u and $q \in Q_N$, define

$$H_u(q) = \max\{c \in \mathbb{N} : (q_N^0, 0) \xrightarrow{u} (q, c)\},$$

with $H_u(q) = \bot$ if q is unreachable. We order $\bot < h$ for every $h \in \mathbb{N}$.

Definition 107.27 (Backward requirement)**.** For a suffix v and $q \in Q_N$, define

$$R_v(q) = \min\{c \in \mathbb{N} : \exists f \in F_N,\ (q, c) \xrightarrow{v} (f, c')\},$$

with $R_v(q) = \infty$ if no such counter value exists.

Proposition 107.28 (Exact cut semantics)**.** *For every factorization $w = uv$,*

$$w \in L(N) \iff \exists q \in Q_N :\ H_u(q) \geq R_v(q).$$

Hence

$$w \notin L(N) \iff H_u(q) < R_v(q) \quad \textit{for all } q \in Q_N.$$

Proof. If w is accepted, cut an accepting run after u. If the cut configuration is (q, c) then $c \leq H_u(q)$ and $c \geq R_v(q)$. Conversely, if $H_u(q) \geq R_v(q)$, choose a prefix run realizing $H_u(q)$ and use monotonicity of the net to execute an accepting v-run that needs only $R_v(q)$ units of credit. □

For a fixed word x, let T_x denote the exact forward-envelope transformer and S_x the exact backward-requirement transformer.

Theorem 107.29 (Exact residuation)**.** *For every forward envelope H, backward requirement R, and word x,*

$$T_x(H) < R \iff H < S_x(R)$$

componentwise.

Proof. Fix a source state q. The inequality $H(q) < S_x(R)(q)$ says that every x-path starting at q either requires more initial counter than $H(q)$ or exits below the corresponding requirement in R. This is exactly the statement that no contribution from q can make $T_x(H)$ meet R. Taking all source and target states gives the equivalence. □

107.13 Shortest counterexamples and zero cuts

Assume $L(A) \not\subseteq L(N)$ and choose a shortest word

$$w \in L(A) \setminus L(N).$$

Let (p_t, d_t) be the unique A-configuration after the prefix of length t, and let H_t and R_t be the exact net residuals at that cut. Proposition 107.28 gives

$$H_t < R_t$$

at every cut.

Lemma 107.30 (Zero-cut deletion)**.** *Suppose $i < j$, $d_i = d_j = 0$, $p_i = p_j$, and $H_i \le H_j$ componentwise. Then deleting the factor between cuts i and j produces a shorter word in $L(A) \setminus L(N)$.*

Proof. The deterministic run rejoins the identical physical configuration $(p_i, 0) = (p_j, 0)$, so the unchanged suffix remains accepting. At cut j, rejection gives $H_j < R_j$. Since $H_i \le H_j$, the unchanged suffix also rejects from H_i. □

Thus the zero-cut states (p_t, H_t) of a shortest counterexample form a bad sequence under equality of p and componentwise order on H.

Between consecutive zero cuts, a nontrivial factor has the form

$$0 \xrightarrow{+1} \text{strictly positive interior} \xrightarrow{-1} 0.$$

The interior contains no zero test. We call such a factor a *positive first return*.

107.14 Positive first-return compactness

This section contains the main technical theorem. We state it in a suffix-parametric form because the proof is local to one exact zero entry and uses the future suffix only through its exact backward requirement. This stronger formulation will also be needed for reset-ending returns in the epsilon extension.

Definition 107.31 (Residual-minimal first return)**.** Fix a concrete exact zero entry (C_0, H_0), a deterministic zero exit configuration C_1, and a backward requirement vector

$$R \in (\mathbb{N} \cup \{\infty\})^{Q_N}.$$

A positive first-return word P from C_0 to C_1 is *R-minimal* if its exact exit envelope H_P satisfies $H_P < R$ and there is no strictly shorter word Q that takes A from C_0 to the same physical exit configuration C_1 and whose exact exit envelope satisfies $H_Q < R$. The competitor Q need not itself be a first return.

Theorem 107.32 (Suffix-parametric positive first-return compactness)**.** *Fix a concrete exact zero-cut entry consisting of an A zero configuration C_0 and an exact net envelope H_0. There is a total computable integer*

$$B_{\mathrm{FR}}(A, N, C_0, H_0)$$

such that, for every deterministic zero exit configuration C_1 and every backward requirement vector R, every R-minimal positive first-return word from (C_0, H_0) to C_1 has literal length at most B_{FR}. In particular, every positive first-return corridor occurring from this entry inside a globally shortest word of $L(A) \setminus L(N)$ has length at most the same bound.

Indeed, if a corridor P occurs in a globally shortest counterexample and R is the exact backward requirement of the unchanged suffix at its exit, then any shorter competitor Q from the same entry to the same physical zero exit with $H_Q < R$ would splice with that suffix and produce a shorter global counterexample. Thus P is R-minimal. Conversely, the proof below uses global shortestness only through this residual-minimality implication. Throughout this section, "shortest witness" may therefore be read as "R-minimal witness" for one fixed but arbitrary exit requirement R.

107.14.1 Replacement principles and anchored shells

Two different operations occur in the proof and must not be conflated.

Definition 107.33 (Trace-preserving deletion)**.** A trace-preserving deletion removes one or more literal contiguous blocks from the *observed* R-minimal word. Every removed block is identified at its actual location in that word. The deterministic run on the remaining letters must be checked directly.

Definition 107.34 (Endpoint replacement)**.** An endpoint replacement substitutes an observed factor P by a possibly different word Q. It is valid when Q starts at the same cut, reaches the identical deterministic A exit configuration, and its exact N exit envelope H_Q satisfies $H_Q \le H_P$.

Lemma 107.35 (Extensional splice principle)**.** *Let P and Q start at the same exact entry and reach the same deterministic exit configuration, with $H_Q \le H_P$. For every backward requirement vector R,*

$$H_P < R \quad \Longrightarrow \quad H_Q < R.$$

Hence an R-minimal witness admits no strictly shorter endpoint-dominating replacement. Applied with R equal to the exact requirement of an actual suffix, this is precisely the usual splice argument in a globally shortest counterexample.

Proof. Componentwise $H_Q \le H_P < R$ immediately gives $H_Q < R$. The deterministic endpoint is identical, so the same future deterministic computation is available when R comes from a concrete suffix. □

Choose a peak cut z of a positive first return. For each level h below the peak, let u_h be the last height-h cut before z and v_h the first height-h cut after z. Then

$$u_h < u_k < z < v_k < v_h \qquad (h < k).$$

The inner factor from u_k to v_k stays at height at least k and can therefore be translated downward by $k - h$ without activating a zero rule.

For every finite envelope coordinate put

$$F_t(q) = H_t(q) - d_t.$$

For every fixed OCN word transformer T_x and $c \ge 0$,

$$T_x(H + c\mathbf{1}) \ge T_x(H) + c\mathbf{1}. \tag{107.1}$$

Lemma 107.36 (Anchored shell deletion)**.** *Let $1 \le h < k$. Assume the corresponding ascent and descent boundary controls agree and*

$$F_{u_h} \le F_{u_k}, \qquad F_{v_k} \le F_{v_h}.$$

Then translating the inner shell to the outer level gives a strictly shorter word reaching the same deterministic exit and still rejected against the fixed requirement R, contradicting R-minimality.

Proof. The translated factor follows the same positive-regime deterministic control path and has enough vertical reserve, hence rejoins the exact deterministic outer exit. The first anchored inequality gives no larger effective input envelope after lowering the inner shell; applying the same literal inner word and (107.1), then using the second anchored inequality, yields an outer exit envelope no larger than the original one. Lemma 107.35 finishes the shortening. □

Thus a deep deletion-free tower forces noncomparability of anchored endpoint data. All subsequent machinery exists to turn that noncomparability into either lower-rank escape or a bounded context.

107.14.2 Finite owner/provenance refinement and macro systems

The later guard and wall constructions use a finite owner-resolved macro system. We make that object explicit here.

Definition 107.37 (Bounded macro type)**.** Fix a positive-mode proof state. A *bounded macro type* records a fixed finite literal transition skeleton between two refined cuts: its source and target deterministic controls, its literal input word, its deterministic counter effect and internal entry requirement, and the finite family of N-paths over that word between states of the current finite support. Low counter values at which one of those paths changes feasibility are unfolded into finite control. At every proof state only finitely many bounded macro types are used.

For a macro e from support S to support S', exact max-envelope propagation has the form

$$H'(r) = \max_{q \in S} \{H(q) + w_e(q, r) : H(q) \geq \lambda_e(q, r)\}, \qquad r \in S',$$

where $w_e(q, r)$ and the finite entry requirement $\lambda_e(q, r)$ are effectively computable from the fixed literal macro. Fix once and for all an order on Q_N and break ties by the least maximizing predecessor.

Definition 107.38 (Owner frame and provenance color)**.** At an owner-resolved macro cut, the *owner map*

$$\sigma_e : S' \to S$$

sends each target coordinate to its canonical maximizing predecessor. The *owner frame* records the current support, the owner map on live coordinates, and—after rank stabilization—the induced permutation of live provenance classes. Starting from a concrete entry support S_0, give every $q \in S_0$ a distinct provenance atom $[q]$. The composed map

$$\mathrm{Prov}_t : S_t \to S_0$$

assigns to every live coordinate at cut t the entry atom obtained by composing the intervening owner maps. A *provenance color* records the current support, the partition of current coordinates by equality of Prov_t, and the finite permutation/frame data.

All these colors are finite: Q_N is finite, the macro alphabet is finite, there are finitely many supports, maps between finite supports, set partitions, threshold regimes, and permutations. On one fixed owner-resolved edge the chosen maximizing predecessor is exact, so the live owner values satisfy an affine update

$$x' = P_e x + c_e, \tag{107.2}$$

with P_e the 0–1 matrix induced by the owner map. The validity of that owner choice is exactly a finite conjunction of affine difference inequalities comparing the selected predecessor with all competitors. Thus the refinement is an exact encoding of the envelope, not a selected-run approximation.

Definition 107.39 (Provenance rank)**.** The provenance rank at a refined cut is

$$\nu(t) := |\operatorname{im} \mathrm{Prov}_t|.$$

A *provenance-rank drop* means that one entry provenance atom represented before a macro step is no longer represented afterwards. A raw support change that merely creates or removes coordinates carrying already represented atoms is a finite-mode change, not a rank drop.

Lemma 107.40 (Finite-depth provenance descent)**.** *Along owner-resolved macro cuts, ν is monotone nonincreasing. If the induced owner action on the live provenance classes is non-surjective, then ν decreases strictly. Hence there are at most $|Q_N| - 1$ strict provenance-rank drops before the rank stabilizes; after stabilization every owner update induces a permutation of the live provenance classes.*

Proof. For consecutive macro cuts $t \to t'$, the provenance map factors as

$$S_{t'} \xrightarrow{\sigma_e} S_t \xrightarrow{\mathrm{Prov}_t} S_0,$$

so $\operatorname{im} \mathrm{Prov}_{t'} \subseteq \operatorname{im} \mathrm{Prov}_t$. Strict loss of a represented class gives strict inequality of cardinalities. If the cardinality is stable, the induced map on the finite set of live provenance classes is surjective and therefore bijective. Recording that bijection is the finite permutation-frame refinement. □

Proposition 107.41 (Fixed-route terminal module)**.** *Fix one owner-resolved bounded-expression macro schema*

$$\alpha_0 \beta_1^{n_1} \alpha_1 \cdots \beta_m^{n_m} \alpha_m$$

from a concrete exact entry to one exact deterministic target configuration. Then feasibility of the schema is an effective Presburger condition on $(n_1, \ldots, n_m)$, the endpoint owner vector is affine in the exponents, and the componentwise-minimal reachable endpoint set together with shortest representatives is effectively computable. Consequently this fixed-route family is shortest-witness compact.

Proof. After the finite owner/mode refinement, each powered block β_i begins and ends in the same refined mode and acts on its live owner vector by an affine map

$$x \longmapsto P_i x + c_i,$$

where P_i is a 0–1 functional matrix induced by the owner map. The finite monoid generated by P_i is effectively eventually periodic: compute μ_i, λ_i with $P_i^{\mu_i+\lambda_i} = P_i^{\mu_i}$. Refining the exponent by the finitely many cases

$$n_i < \mu_i \quad \text{or} \quad n_i = \mu_i + s_i + \lambda_i k_i, \qquad 0 \le s_i < \lambda_i,$$

makes the endpoint vector affine in the quotient variable k_i. Indeed

$$P_i^{n_i} x + \sum_{j<n_i} P_i^j c_i$$

has a fixed residue-dependent linear part plus k_i times one fixed period-sum vector.

Every threshold guard, owner comparison, and deterministic-positivity condition at an intermediate repetition is an affine inequality in the current owner vector. After the same residue refinement its truth at repetition j is therefore a Presburger formula in the exponent quotients and the bounded repetition index. Requiring it at every repetition is still Presburger, since Presburger arithmetic is effectively closed under bounded and unbounded first-order quantification. The exact deterministic target is handled in the same way. Taking the finite disjunction over all residue branches yields an effective Presburger description of the feasible exponent family, and on each branch the endpoint owner vector is affine in the quotient variables (equivalently Presburger-piecewise affine in the original exponents).

Presburger arithmetic therefore decides exact endpoint slices and computes the finite semilinear feasible exponent family. Recursively minimize endpoint coordinates using exact-slice queries, exactly as in Proposition 107.47 below, and enumerate shortest realizing words. Endpoint domination then invokes Lemma 107.35. □

107.14.3 Exact projective modes and finite-projective compression

Let $E = \operatorname{supp}(H)$ be the finite support of an exact envelope. If $E \neq \varnothing$, define its common scalar and normalized profile by

$$z(H) = \min_{q \in E} H(q), \qquad \theta(H) = H - z(H)\mathbf{1}_E.$$

Thus $\min \theta(H) = 0$. A *projective mode* records the finite data that determine how the exact envelope is read from (z, θ): the deterministic positive-mode control, the support E, the maximizing-owner frame used by the current finite macro system, and the normalized profile θ when that profile belongs to a fixed finite palette.

Definition 107.42 (Finite exact projective mode system)**.** A positive factor lies in a finite exact projective mode system $\mathcal{P}$ if there is a finite set V of modes and fixed vectors θ_v such that at every cut in mode v,

$$H = z\mathbf{1} + \theta_v$$

on the recorded support, and every macro transition of the factor updates the deterministic A height and the common scalar z additively while changing only the finite mode.

The phrase *exact* means that support, threshold regime, and the affine readout of every finite coordinate are part of the mode; no selected-run approximation is used.

Proposition 107.43 (Finite-projective compression)**.** *Fix a concrete exact entry, a finite exact projective system $\mathcal{P}$, and one exact deterministic target configuration. There is an effective bound on every shortest factor staying in $\mathcal{P}$ and reaching that target. Moreover one can compute an endpoint-dominating shortest representative.*

Proof. Use deterministic height and the common scalar z as two ordinary VASS counters; the projective mode is finite control. Reachability sets of two-dimensional VASS with states are effectively semilinear [111]; hence, after fixing the target mode and exact deterministic target height, the one-dimensional set of reachable target scalars z is effectively semilinear and its emptiness and least element are decidable. Let $z_{\min}$ be that least element whenever the target slice is nonempty. Exhaustive word enumeration then computes a shortest word reaching (target mode, target A configuration, $z_{\min}$). Any observed factor exiting with scalar $z \geq z_{\min}$ is endpoint-dominated by this representative. Taking the maximum over the finitely many nonempty target modes gives the bound; empty target slices contribute nothing. □

Remark 107.44 (Safe use of alternative paths). Proposition 107.43 is exactly the place where an alternative VASS path is legitimate: it is used to *construct a replacement word* with the same deterministic endpoint and a smaller exact envelope. No claim is made that the observed literal word itself belongs to a particular VASS path scheme.

107.14.4 Owner guards, recurrent guard rank, and the rank-one base

After finite support/provenance and maximizing-owner refinement, the remaining owner changes are governed by finitely many affine comparisons

$$g_j(x) = \ell_j(x) - k_j,$$

where x denotes the projective owner/resource vector. Fix one recurrent finite owner/chamber strongly connected component S. Let W_S be the rational span of projective displacements of directed cycles of S, after common translation is factored out.

Definition 107.45 (Recurrent guard rank)**.** The recurrent guard rank of S is

$$\rho(S) = \dim_{\mathbb{Q}} \operatorname{span}\{\ell_j|_{W_S} : j \text{ switches recurrently in } S\}.$$

Only restrictions to the recurrent cycle space count. Many syntactic guards may therefore have rank one.

The base of the induction is stronger than mere compactness.

Lemma 107.46 (Common scalar for rank-one guards)**.** *If $\rho(S) = 1$, there is an effectively computable integer linear functional μ on the cycle lattice and integers a_j such that*

$$\ell_j(w) = a_j \mu(w)$$

for every recurrent cycle displacement $w \in W_S$.

Proof. The restricted functionals span a one-dimensional rational subspace of W_S^*. Choose a rational generator and clear denominators simultaneously. □

For an edge $e : u \to v$ of the finite owner SCC with translation c_e, the residual

$$r_j(e) = \ell_j(c_e) - a_j \mu(c_e)$$

has zero sum on every directed cycle. Hence it is a finite-control coboundary: $r_j(e) = \phi_j(v) - \phi_j(u)$. Along a path from a fixed entry, with

$$T = \mu(x - x_0),$$

every recurrent guard therefore has the form

$$g_j(x) = a_j T + b_{j,v} \tag{107.3}$$

for a finite table of constants $b_{j,v}$.

Proposition 107.47 (Rank-one compilation)**.** *Every recurrent owner SCC of guard rank at most one can be compiled effectively into a VASS with one zero-tested counter. Consequently, from a concrete entry and exact deterministic target configuration, exact target-slice nonemptiness and finite Pareto-minimal endpoint bases are computable; in particular the SCC is shortest-witness compact.*

Proof. For rank zero all recurrent guard truths are fixed after finite refinement and their slacks are ordinary counters. For rank one, the finitely many affine thresholds in (107.3) partition $\mathbb{Z}$ into two unbounded rays and finitely many bounded cells. On an unbounded ray store the distance of T from its boundary in one distinguished counter s; bounded cells store the exact value of T in finite control. A macro edge changes T by a fixed integer. A cell crossing can occur only when s belongs to a computably bounded source band. Drain s through a bounded finite-control chain, use the unique zero test to verify the crossing value, remember the consumed bounded amount in finite control, and reconstruct the target slack. All other resources are ordinary VASS counters. Reachability for VASS with one zero test is decidable. An endpoint slice fixing some counters exactly and leaving the others free reduces to one exact-configuration query by entering a private cleanup phase that decrements only the free counters and leaves the fixed counters unchanged; the queried cleanup target fixes the prescribed counters and sets every formerly free counter to zero. Hence exact endpoint-slice nonemptiness is decidable from the same oracle.

Effective Pareto algorithm. The basis computation uses exact-slice reachability directly and does not rely on detecting when a growing product box is large enough. More generally, let R_F be the reachable endpoint set after fixing some owner coordinates F exactly. First decide whether R_F is empty. If it is nonempty, enumerate one reachable endpoint $a \in R_F$. Every componentwise-minimal $x \in R_F$ satisfies either $x = a$ or $x_i < a_i$ for at least one still-free coordinate i. For every such i and every integer $0 \le \zeta < a_i$, recursively solve the lower-dimensional exact slice $R_{F \cup \{x_i = \zeta\}}$. Each recursive call fixes one additional coordinate and each branching range is finite, so the recursion terminates. Taking the union of the returned candidates with a and removing dominated vectors gives exactly $\mathrm{Min}(R_F)$. At the root this computes the full finite Pareto basis. For each minimum, breadth-first word enumeration yields a shortest representative. Thus effectivity follows from the one-zero-test exact-slice oracle, not from Dickson finiteness alone.

Since the exact envelope is an affine readout of the owner resources in the fixed refined mode, a Pareto-minimal representative endpoint-dominates any longer competitor and Lemma 107.35 applies. □

107.14.5 Directional walls and exact finite wall signatures

Assume now $\rho \ge 2$. Choose a basis $\mu_1, \dots, \mu_\rho$ for the recurrent guard directions. A guard is μ_1-sensitive if its restriction has a nonzero μ_1 coefficient. A μ_1 *wall event* is a transition at which the truth value of at least one μ_1-sensitive guard changes.

Every such switch occurs in a finite source band: the guard changes by a fixed integer on a fixed macro edge, so its source value must lie between two consecutive bounded thresholds.

Definition 107.48 (Exact wall signature)**.** An exact μ_1-wall signature records

(a) the source and target refined finite modes and deterministic controls;

(b) the set of μ_1-sensitive guards that change truth and their orientations;

(c) the exact bounded source guard/slack values at the crossing;

(d) the current support/provenance color and owner frame;

(e) the crossing macro edge.

The set $\mathcal{W}$ of exact wall signatures is finite and effectively enumerable.

Finiteness follows because all control/frame/support components are finite and every switching guard source value belongs to an edge-dependent finite threshold band.

Lemma 107.49 (Wall-free rank descent)**.** *Every maximal μ_1-wall-free strip has recurrent guard rank at most $\rho - 1$.*

Proof. Every μ_1-sensitive guard has fixed truth throughout the strip and is carried as an ordinary nonnegative slack. Only guards whose μ_1 coefficient is zero can switch, and their restrictions lie in $\operatorname{span}(\mu_2, \dots, \mu_\rho)$. □

A bounded number of walls can also be handled without creating a second zero test.

Proposition 107.50 (Bounded-wall compilation)**.** *Fix K. A rank-ρ factor containing at most K μ_1-wall events yields either a responsible rank-$\leq \rho - 1$ escape or an effective endpoint-dominating length bound, assuming the lower-rank compactness invariant.*

Proof. The factor is a concatenation of at most $K + 1$ wall-free strips and K crossing macros. If normalized/projective/anchored/common-shift data escape in a strip, Lemma 107.49 localizes a genuine lower-rank escape. Otherwise every strip belongs to a finite lower-rank bounded-context palette and is compact by induction; the crossing macros are finite. Summing the finitely many bounds gives the factor bound. If one wants an exact compilation rather than merely an extensional replacement, an exact bounded wall value $s = c$ can be checked without a fresh zero test by subtracting c, retiring the old ordinary slack counter, and requiring it to be zero only at the final exact-reachability target. This is needed only finitely many times because K is fixed. □

Hence a genuinely rank-ρ obstruction must have unbounded wall multiplicity.

107.14.6 Dense same-signature atoms and carrier colors

Let $B = |\mathcal{W}|$. Write the wall-signature word of a factor as $w_1 \cdots w_N$.

Lemma 107.51 (Dense same-signature returns)**.** *If $N \geq 4B$, some signature $\sigma \in \mathcal{W}$ has at least*

$$\left\lfloor \frac{N}{4B} \right\rfloor - 1$$

pairwise interior-disjoint consecutive σ-to-σ return intervals, each containing at most

$$K_{\mathrm{dense}} = 4B + 1$$

wall events.

Proof. Some σ occurs at least $\lceil N/B \rceil$ times. The sum of the consecutive occurrence gaps is $< N$. If more than half of these gaps exceeded $4B$, their sum would exceed N. The stated coarse linear lower bound follows. □

Returning to the same exact signature fixes one independent μ_1-sensitive wall equation. Thus the endpoint displacement of every dense atom lies in $W_S \cap \ker \ell_\sigma$, and its endpoint recurrent guard rank is at most $\rho - 1$.

Definition 107.52 (Carrier color)**.** For a dense atom, a left or right *carrier* is the bounded neighborhood from the exact wall endpoint to the canonical shell/interface cut used for comparison. Under a fixed bounded-context cutoff, its carrier color records the exact wall signature, side (left/right), deterministic control, support/provenance color, owner frame, bounded vertical offset from the shell level, and all normalized envelope/guard data in the prescribed finite cutoff box.

Carrier colors are therefore finite whenever the context is bounded. If the boundary-to-canonical correction leaves this finite palette, the first escaping quantity occurs in one of finitely many wall-free strip positions inside a bounded-wall carrier, and Lemma 107.49 localizes the escape below rank ρ.

107.14.7 High walls and translated arches

If the heights of all recurrent walls are bounded, the exact physical wall contexts belong to a finite set and Proposition 107.50 closes the branch. Thus the residual case has arbitrarily high wall centers. Around a wall center t of height a, for each $1 \le \ell < a$ define the last/first height-ℓ cuts

$$u_\ell^t < t < v_\ell^t.$$

Their intervening factors are nested positive arches. Combining the dense atom skeleton, finite carrier colors, and ordinary finite Ramsey refinement yields an arbitrarily deep family

$$J_\ell = e_\ell^L \longrightarrow e_\ell^R$$

with fixed left and right carrier colors and fixed offsets

$$d(e_\ell^L) = \ell + r_L, \qquad d(e_\ell^R) = \ell + r_R.$$

The carriers have recurrent rank at most $\rho - 1$.

The next lemma makes the finite-color argument explicit.

Lemma 107.53 (Translated-arch localization)**.** *Let $J_{h_0}, \dots, J_{h_m}$ be a sufficiently long fixed-carrier translated-arch family with strictly increasing levels. Then either*

(i) two levels satisfy the anchored deletion inequalities of Lemma 107.36; or

(ii) after passing to a homogeneous subfamily, one fixed side $X \in \{L, R\}$ and one fixed live coordinate $q \in Q_N$ satisfy a strict chronological chain

$$F_{x_0}(q) > F_{x_1}(q) > \cdots > F_{x_n}(q),$$

where x_i are the selected canonical endpoints/carrier cuts on side X.

In case (ii), the escape is either carried inside one lower-rank carrier strip or is visible on the same-signature lower-rank endpoint system.

Proof. For each pair of levels $h < k$, first ask whether both anchored comparisons required by Lemma 107.36 hold. If yes, case (i) occurs. Otherwise color the pair by the first failed side (left or right) and the least envelope coordinate witnessing that failure. Support and side are already fixed, so this is a finite coloring by at most $2|Q_N|$ colors. Ramsey's theorem gives a long homogeneous set. If the homogeneous color is, for example, "left, q", then for every earlier/later pair the left comparison fails in the same orientation, hence integrality and total order on the selected coordinate give a strict monotone chain after reversing orientation if necessary. The right-side case is symmetric.

Now compare each canonical endpoint with the corresponding exact wall endpoint through its fixed carrier. There are finitely many strip positions. If the correction in coordinate q or any other normalized coordinate is unbounded, one fixed wall-free strip position carries unbounded variation and has rank at most $\rho - 1$. If all corrections are bounded, pass to a finite refinement fixing them; the strict drift then transfers to the exact same-signature wall endpoints. Their endpoint displacement lies in the wall kernel, so their recurrent endpoint rank is at most $\rho - 1$. □

Absolute shell height is not itself a new resource on a fixed carrier: once the vertical range about level ℓ is bounded and all zero tests are inactive, translating the whole carrier upward preserves its transition structure. Hence only the normalized/projective, anchored-relative, common-shift, and relative A-transfer quantities can escape.

107.14.8 Finite scalar/projective reserve split and the vector shell defect

Lemma 107.53 leaves a deletion-free translated-arch family in which one endpoint side carries an arbitrarily long strict anchored drift. The final proof uses a finite reserve split; no infinitary Dickson extraction on normalized profiles is used.

Assume for definiteness that the responsible side is the left side. At the selected left endpoints write

$$F_\ell^L = H_{e_\ell^L} - d(e_\ell^L)\mathbf{1}.$$

After stable-support refinement, finite pigeonhole fixes one live coordinate r attaining the minimum of F_ℓ^L on an arbitrarily long subfamily. Put

$$a_\ell := F_\ell^L(r), \qquad \theta_\ell := F_\ell^L - a_\ell\mathbf{1}. \tag{107.4}$$

Then $\theta_\ell \in \mathbb{N}^S$, $\theta_\ell(r) = 0$, and $F_\ell^L = a_\ell\mathbf{1} + \theta_\ell$. By Lemma 107.53, on the selected chain one fixed coordinate q satisfies

$$F_{h_0}^L(q) > F_{h_1}^L(q) > \cdots > F_{h_n}^L(q).$$

Lemma 107.54 (Effective projective cutoff)**.** *For the concrete first-return entry and fixed finite support/regime palette there is a computable constant C_{proj} with the following property. If a normalized endpoint profile has*

$$\max_{p\in S} \theta(p) > C_{\mathrm{proj}},$$

then a bounded nonzero N/N differential core is exposed on a fixed refined route, unless a provenance-rank drop occurs first. The branch therefore transfers to the projective/lower-rank localization machinery. In the complementary branch every normalized profile lies in the finite box $[0, C_{\mathrm{proj}}]^S$.

Proof. Track a fixed difference $H(p) - H(r)$ in the finite synchronized owner/product graph. After cycle erasure, a simple residual has effectively bounded length and hence bounded differential effect. If the difference exceeds the concrete entry spread plus twice the maximal simple-residual contribution (enlarged over the finite support/regime event palette), some primitive recurrent cycle has nonzero differential effect. Finite refinement fixes its sign and route type. If the responsible owner/provenance data cannot be stabilized before that extraction, the provenance branch has already descended. Taking the maximum of these finitely many simple-path/event bounds gives C_{proj}. □

Lemma 107.55 (Finite scalar/projective reserve split)**.** *For every n-step strict endpoint chain as above,*

$$(a_{h_0} - a_{h_n}) + (\theta_{h_0}(q) - \theta_{h_n}(q)) \geq n. \tag{107.5}$$

Consequently either the chain crosses the projective cutoff of Lemma 107.54, or finite pigeonhole extracts an arbitrarily long subchain on which one exact profile

$$\theta_\ell = \bar{\theta}$$

is fixed and the scalar sequence a_ℓ decreases strictly.

Proof. Integrality gives $F^L_{h_0}(q) - F^L_{h_n}(q) \geq n$. Substitute $F^L(q) = a + \theta(q)$ to obtain (107.5). If some profile leaves the finite projective box, Lemma 107.54 transfers the branch. Otherwise only finitely many exact profiles occur, so finite pigeonhole—not Dickson's lemma—gives an arbitrarily long constant-profile subchain. Strict decrease of $F^L(q)$ then forces strict decrease of a. □

From now on we work in this canonical no-projective-escape branch and write $\theta_\ell = \bar{\theta}$. Take $h < k$ in the fixed-profile subchain and put

$$c := k - h, \qquad \delta := a_h - a_k > 0.$$

The fixed endpoint offsets give the exact left-envelope identity

$$H^L_k = H^L_h + (c - \delta)\mathbf{1}. \tag{107.6}$$

Lemma 107.56 (Translated input domination)**.** *If $\delta \leq c$ and H' is the exact net envelope obtained by reading the literal inner word J_k from the outer left envelope H^L_h, then*

$$H' \leq H^R_k - (c - \delta)\mathbf{1}. \tag{107.7}$$

Proof. Equation (107.6) gives

$$H^L_h + (c - \delta)\mathbf{1} = H^L_k.$$

For a one-counter net, common input credit is monotone under a fixed word: for every $t \geq 0$,

$$T_x(H + t\mathbf{1}) \geq T_x(H) + t\mathbf{1}.$$

Applying this to $x = J_k$ gives

$$H' + (c - \delta)\mathbf{1} \leq T_{J_k}(H^L_k) = H^R_k.$$

□

Define the *vector shell defect*

$$Y_\ell := F^R_\ell - a_\ell\mathbf{1}, \qquad F^R_\ell := H^R_\ell - d(e^R_\ell)\mathbf{1}. \tag{107.8}$$

Because $d(e^R_k) - d(e^R_h) = c$,

$$H^R_k - (c - \delta)\mathbf{1} = H^R_h + (Y_k - Y_h). \tag{107.9}$$

Lemma 107.57 (Vector shell-defect pinning)**.** *Let $h < k$ belong to the fixed-profile subchain and assume $\delta \leq c$. Exactly one of the following sufficient outcomes occurs.*

(i) $Y_k \leq Y_h$ componentwise, in which case replacing the outer arch by the translated inner arch gives an endpoint-dominating strict shortening;

(ii) there is a live coordinate s such that

$$1 \leq R^R_h(s) - H^R_h(s) \leq Y_k(s) - Y_h(s). \tag{107.10}$$

Proof. Combine Lemma 107.56 with (107.9). If $Y_k \leq Y_h$, the translated exit envelope is at most H^R_h, so Lemma 107.35 shortens the shortest witness. Otherwise the translated word is still accepted by A and is strictly shorter, hence R-minimality forces its exit envelope to satisfy the requirement. At the outer right cut the original word has $H^R_h < R^R_h$, while the translated exit reaches the suffix requirement in some coordinate s, giving (107.10). □

A pair is *nonsuperlevel* when $\delta \leq c$ and *superlevel* when $\delta > c$. Finite Ramsey refinement yields a long homogeneous subfamily of one kind. In the nonsuperlevel case, Lemma 107.57 and a finite coloring by the least positive coordinate of $Y_k - Y_h$ give a fixed coordinate with strict Y-growth. The superlevel case is threshold-sensitive and is developed next.

107.14.9 Superlevel activation and the vector-defect/activation split

Fix a superlevel pair $h < k$ and put

$$d := \delta - c > 0. \tag{107.11}$$

Then (107.6) becomes

$$H_k^L = H_h^L - d\mathbf{1}, \tag{107.12}$$

and the corresponding right identity is

$$H_k^R = H_h^R - d\mathbf{1} + (Y_k - Y_h). \tag{107.13}$$

Suppose first that $Y_k \le Y_h$. Then the inner right envelope is strictly dominated by the outer right envelope. Let

$$Q_{h,k} := S_{J_k}(R_h^R) \tag{107.14}$$

be the exact entry requirement for reading the literal inner word J_k and then the inherited outer suffix.

Lemma 107.58 (Exact superlevel activation certificate)**.** *If $h < k$ is superlevel and $Y_k \le Y_h$, then*

$$H_k^L < Q_{h,k}$$

componentwise, while shortestness implies $H_h^L \not< Q_{h,k}$. Hence there is a live coordinate p such that

$$H_k^L(p) < Q_{h,k}(p) \le H_h^L(p). \tag{107.15}$$

Writing

$$\eta := Q_{h,k}(p) - H_k^L(p),$$

one has the exact bound

$$1 \le \eta \le d. \tag{107.16}$$

Proof. Since $Y_k \le Y_h$, equation (107.13) gives $H_k^R < H_h^R < R_h^R$. Exact residuation, Theorem 107.29, yields $H_k^L < S_{J_k}(R_h^R) = Q_{h,k}$. Replacing J_h by the downward translation of J_k is a legal strict deterministic shortening. Hence R-minimality forces the shortened replacement to meet the inherited requirement, equivalently $H_h^L \not< Q_{h,k}$. Choose p satisfying (107.15). Since $H_h^L(p) - H_k^L(p) = d$ by (107.12), the gap satisfies (107.16). □

This is an exact threshold statement: the actual inner input is short of the virtual continuation threshold by η, while the outer input crosses that threshold. It is not yet a statement about a forward maximizing strand of the original rejected word.

For every sufficiently separated superlevel pair color the pair by

- A if $Y_k \le Y_h$ componentwise;
- otherwise V_s, where s is the least coordinate with $Y_k(s) > Y_h(s)$.

There are finitely many colors.

Lemma 107.59 (Activation-or-vector-defect dichotomy)**.** *Every arbitrarily deep superlevel tower contains arbitrarily long subfamilies of exactly one of the following forms.*

(i) Activation-critical*: for every earlier/later pair Y is componentwise nonincreasing and Lemma 107.58 supplies an exact threshold gap η;*

(ii) Vector-defect*: one fixed coordinate s satisfies*

$$Y_{h_0}(s) < Y_{h_1}(s) < \cdots .$$

Proof. Apply finite Ramsey to the coloring above. Homogeneous color A gives the first alternative and homogeneous color V_s gives the second. □

The vector-defect alternative is not left abstract. Fix a live right coordinate s and, for each ℓ, choose one maximizing N-path $\pi_\ell : p_\ell \to s$ over the literal arch J_ℓ realizing $H^R_\ell(s)$. Pass to a subsequence with fixed input state p. Put

$$\kappa_A := r_R - r_L = \Delta_A(J_\ell).$$

Because the left projective profile is fixed to $\bar{\theta}$, the exact owner/path identity is

$$Y_\ell(s) = \bar{\theta}(p) + \omega_\ell, \qquad \omega_\ell := \Delta_N(\pi_\ell) - \kappa_A. \tag{107.17}$$

Lemma 107.60 (Actual maximizing-strand source of vector defect)**.** *If $Y_\ell(s)$ is unbounded above in the canonical no-projective-escape branch, then $\omega_\ell \to \infty$ along a subsequence. Cycle erasure on the actual maximizing synchronized paths exposes a fixed primitive synchronized cycle with positive relative effect $\Delta_N - \Delta_A > 0$, unless a provenance-rank drop occurs first.*

Proof. The term $\bar{\theta}(p)$ in (107.17) is fixed, so unbounded Y is exactly unbounded positive relative gain on the chosen literal maximizing strands. Cycle erasure in the finite synchronized positive-mode product leaves a bounded simple residual. Hence recurrent primitive cycles carry unbounded positive relative contribution, and finite pigeonhole fixes one positive type unless the owner/provenance refinement changes first. □

Thus only the activation-critical branch is new.

107.14.10 Literal activation annuli and fixed affine cells

Let $h < k$ be an activation-critical pair. The outer arch factors literally as

$$J_h = L_{h,k}\, J_k\, R_{k,h},$$

where $L_{h,k}$ runs from e^L_h to e^L_k and $R_{k,h}$ from e^R_k to e^R_h. The deterministic effects are $+c$ and $-c$, respectively.

Lemma 107.61 (Matched anchored drift in an activation annulus)**.** *For every activation-critical pair,*

$$F^L_k(r) - F^L_h(r) = -\delta,$$

and, in chronological order on the right flank,

$$F^R_h - F^R_k = (Y_h - Y_k) + \delta\mathbf{1} \geq \delta\mathbf{1}.$$

Equivalently, in absolute envelopes,

$$H^L_h(r) - H^L_k(r) = d, \qquad H^R_h - H^R_k \geq d\mathbf{1}. \tag{107.18}$$

Proof. The left identity follows from $F^L_\ell(r) = a_\ell$. The right identity follows from $F^R_\ell = Y_\ell + a_\ell\mathbf{1}$ and $Y_h \geq Y_k$. Restoring the deterministic height difference c converts δ into $d = \delta - c$ in the absolute envelopes. □

The two literal flanks of one annulus therefore carry a matched credit valley: the left loses anchored mass and the right restores at least the corresponding common amount. If adjacent cells have unbounded values of δ or unbounded endpoint increments, the responsible mass is already localized to one of finitely many bounded-wall flank positions and hence to lower rank. We may therefore restrict to bounded cells.

After finite refinement fix integers

$$c, \delta, d = \delta - c > 0, \qquad p, \qquad 1 \le \eta \le d$$

and define the right adjacent-cell increment

$$v_i := Y_i - Y_{i+1} \ge 0.$$

The left profile is already fixed by Lemma 107.55, so the endpoint recurrences are

$$H^L_{i+1} = H^L_i - d\mathbf{1}, \qquad H^R_i = H^R_{i+1} + d\mathbf{1} + v_i. \tag{107.19}$$

Lemma 107.62 (Finite affine-cell refinement)**.** *If some coordinate of v_i is unbounded, a bounded-height annulus carries unbounded normalized right-endpoint change and hence localizes to a lower-rank projective/relative escape or a provenance-rank drop. Otherwise, after finite refinement, $v_i = v$ is one fixed vector for every recurrent cell.*

Proof. A cell has fixed shell width c and fixed wall/carrier types, hence only finitely many wall-free strip positions. Unbounded normalized endpoint change forces one fixed position to carry unbounded change; Lemma 107.49 lowers the recurrent rank there. In the no-escape case v_i ranges over a finite box and may be fixed by finite refinement. □

Fixed endpoint affine data do not imply that the cell interiors form one lower-rank machine. We use only the endpoint recurrences and the extensional replacement principle.

107.14.11 Pure-scalar recurrence and the virtual threshold shadow

The terminal fixed-cell survivor simplifies further.

Lemma 107.63 (Scalarization of stable cells)**.** *In a recurrent fixed affine cell family with no endpoint-visible projective escape or provenance-rank drop, one has*

$$v = \lambda\mathbf{1}$$

for some fixed $\lambda \ge 0$. Hence

$$H^L_{i+1} = H^L_i - d\mathbf{1}, \qquad H^R_{i+1} = H^R_i - (d+\lambda)\mathbf{1}. \tag{107.20}$$

Proof. The left normalized profile is already fixed. If two coordinates of v differ, repeated cells produce linearly unbounded right projective spread, contradicting the no-projective-escape hypothesis. Hence v is a common scalar vector. □

Let

$$Q_i := S_{J_{i+1}}(R^R_i)$$

be the replacement-entry requirement of cell i, and retain the fixed activation coordinate p with

$$Q_i(p) - H^L_{i+1}(p) = \eta.$$

Equation (107.20) gives the exact threshold ladder

$$Q_{i+1}(p) = Q_i(p) - d, \qquad H^L_i(p) - Q_i(p) = d - \eta. \tag{107.21}$$

Put $K := d - \eta \ge 0$ and $q_i := Q_i(p)$. Then

$$H^L_{i+1}(p) = q_i - \eta, \qquad H^L_i(p) = q_i + K. \tag{107.22}$$

For each i choose canonically an N-path $\pi_i : p \to s_i$ over J_{i+1} attaining the minimum in the backward transformer at coordinate p:

$$q_i = \max\{\mathrm{req}(\pi_i), R_i^R(s_i) - \mathrm{eff}(\pi_i)\}. \tag{107.23}$$

After finite refinement the exit state $s_i = s$ is fixed.

There are two formal ways for the maximum in (107.23) to equal q_i. If $\mathrm{req}(\pi_i) = q_i$, the threshold run reaches counter 0 at an internal prefix. If instead the terminal term equals q_i and $\mathrm{req}(\pi_i) \le q_i - 1$, minimality forces exact tightness along the path.

Lemma 107.64 (Virtual tight-shadow principle)**.** *In the terminal-tight case, let $C_i(x)$ be the counter value of the threshold run on π_i at a cut x, and let $B_i(x)$ be the exact backward requirement at that path state for the remaining virtual continuation J_{i+1} followed by the inherited outer suffix. Then*

$$B_i(x) = C_i(x) \tag{107.24}$$

for every cut x of π_i.

Proof. The chosen suffix is feasible from $C_i(x)$, so $B_i(x) \le C_i(x)$. If $B_i(x) \le C_i(x) - 1$, then because the whole selected path has internal requirement at most $q_i - 1$, starting the prefix one unit lower remains feasible up to x, and the exact continuation is feasible from the resulting $C_i(x) - 1$. This contradicts the minimality of $q_i = Q_i(p)$. □

Thus the accepting translated replacement, which starts K units above threshold, stays exactly K above the virtual backward requirement along a terminal-tight path.

107.14.12 Collapse of the tight shadow to a bounded high contact

The tight-shadow alternative is not a separate terminal branch.

Lemma 107.65 (Terminal tightness forces a contact)**.** *In the terminal-tight case,*

$$q_i - \eta < \mathrm{req}(\pi_i) \le q_i - 1.$$

Consequently some prefix of π_i attains threshold counter

$$j_i := q_i - \mathrm{req}(\pi_i) \in \{1, \dots, \eta - 1\}.$$

At that cut the exact virtual backward requirement equals j_i, while the actual translated replacement path has counter $K + j_i$.

Proof. Assume instead that $\mathrm{req}(\pi_i) \le q_i - \eta = H_{i+1}^L(p)$. Then the selected path is internally feasible from the actual inner envelope coordinate. Its endpoint counter is $R_i^R(s) - \eta$. But the pure-scalar right recurrence raises the outer right envelope over the inner one by $d + \lambda$, and $\eta \le d$, giving $H_i^R(s) \ge R_i^R(s)$, contradicting rejection at the outer right cut. The rest follows from Lemma 107.64. □

In the internal-contact case the threshold run itself reaches counter 0, so the virtual requirement at that cut is 0 and the actual replacement counter is exactly K. Combining both cases yields the following uniform normal form.

Theorem 107.66 (Bounded virtual contact normal form)**.** *After finite refinement, every recurrent terminal activation cell contains a cut on a concrete accepting translated-replacement path with*

- *one fixed N contact state t;*

- *one fixed virtual backward requirement* $b \in \{0, \dots, \eta - 1\}$ *at* t*;*
- *one fixed actual replacement counter* $b + K$*.*

Moreover the deterministic A height at these contacts tends to infinity along every arbitrarily deep translated-arch tower.

Proof. The first three items follow from finite refinement of the two contact cases above. High-shell geometry gives a constant M_* such that every translated inner arch in the fixed carrier family stays at height at least its shell level minus M_*. The shell levels tend to infinity, so the contact height does as well. □

Let B_i be the full exact backward-requirement vector at the virtual contact. Since $B_i(t) = b$ is fixed, either some normalized difference $B_i(q) - B_i(t)$ escapes, in which case the backward differential argument localizes a lower-rank/projective escape, or the complete vectors B_i lie in a finite set and may be fixed to one exact vector B_*. In the latter branch the contact has fixed finite N context but arbitrarily high deterministic height.

107.14.13 From bounded contacts to an actual-trace relative valley

The threshold ladder (107.21) implies that in a tower of $n + 1$ recurrent cells the outer threshold is at least nd. Thus along a selected outer sequence

$$q_i \longrightarrow \infty. \tag{107.25}$$

Fix the bounded contact normal form of Theorem 107.66. Let P_i be the literal prefix of the translated inner arch from its translated start to the contact. On the chosen accepting replacement path, the N counter starts at $q_i + K$ and reaches $b + K$, so

$$\Delta_N(P_i) = b - q_i. \tag{107.26}$$

Let $\alpha_i := \Delta_A(P_i)$. High-shell geometry bounds α_i from below by a constant independent of i. Therefore

$$\omega(P_i) := \Delta_N(P_i) - \Delta_A(P_i) \longrightarrow -\infty. \tag{107.27}$$

Lemma 107.67 (Contact-prefix negative relative mass)**.** *Every arbitrarily deep terminal activation tower contains literal synchronized prefixes* P_i *of the translated/original inner-arch words with* $\omega(P_i) \to -\infty$*. Cycle erasure on these actual synchronized paths exposes a fixed primitive cycle type with negative relative effect, unless the responsible mass has already localized to a lower-rank wall-free strip or support/provenance drop.*

Proof. Equation (107.27) is exact. The synchronized positive-mode product is finite. Removing all cycle occurrences leaves a simple path of uniformly bounded relative effect, so unbounded negative relative mass must be carried by recurrent primitive cycles. Finite pigeonhole fixes one responsible type. If its occurrences have unbounded internal wall-free variation, Lemma 107.49 already gives the lower-rank alternative; otherwise bounded-wall packet refinement applies. □

Let T_i be the literal suffix of the same concrete accepting translated-replacement run from the bounded contact to the zero-return cut z terminating the positive first-return corridor. At the contact the deterministic counter is $D_i \to \infty$ and the selected N counter is the fixed value $b + K$. At z the deterministic counter is 0, while the selected accepting N run has some counter $c_i \geq 0$. Hence

$$\Delta_A(T_i) = -D_i, \qquad \Delta_N(T_i) = c_i - (b + K) \geq -(b + K),$$

and therefore

$$\omega(T_i) = \Delta_N(T_i) - \Delta_A(T_i) \geq D_i - (b + K) \longrightarrow +\infty. \tag{107.28}$$

This is the canonical post-contact argument: no split according to the A-transfer to the contact is needed.

Theorem 107.68 (Relative valley)**.** *After excluding strict lower-rank/projective/support escapes, every terminal activation tower contains, on one concrete accepting translated-replacement run and in chronological order,*

$$P_i \longrightarrow (t, b + K;\ A = D_i) \longrightarrow T_i$$

with a fixed bounded N contact, $D_i \to \infty$, and

$$\omega(P_i) \to -\infty, \qquad \omega(T_i) \to +\infty.$$

The suffix T_i ends at the actual zero-return cut of the positive first-return corridor. All factors are literal factors of the same accepting translated-replacement word; no alternative VASS path is used.

Proof. The negative side is Lemma 107.67. The positive side is the exact zero-return calculation above. The fixed contact state/counter and diverging deterministic height are supplied by Theorem 107.66. No path-scheme or fixed-transfer assumption enters the argument. □

Theorem 107.68 is the only input needed for the actual-trace cycle analysis below. Importantly, it does not say that the observed trace belongs to a fixed path scheme.

107.14.14 Trace fidelity: actual cycles versus alternative paths

The activation branch needs a second distinction in addition to the projective-mode and recurrent-guard-rank definitions. A generic 2-VASS flattening theorem says that if one synchronized configuration reaches another, some alternative path belonging to a finite family of path schemes reaches it. It does *not* imply that the observed literal input factor itself belongs to one of those schemes. Therefore path-scheme periods may not be deleted from the observed word unless their literal occurrences have been independently located.

We use the following two rules throughout.

(T1) **Observed-word operation.** Any literal deletion package is extracted by cycle erasure on the actual synchronized trace. Its blocks occur at identified positions of the original word.

(T2) **Replacement-word operation.** VASS reachability or flattening may construct a different word only when Lemma 107.35 is invoked at the common deterministic endpoint, as in Proposition 107.43.

Consider now an actual synchronized positive-mode trace before a bounded contact. For a simple synchronized cycle C put

$$v(C) = (\Delta_N(C), \omega(C)), \qquad \omega(C) = \Delta_N(C) - \Delta_A(C).$$

Cycle-erasing the actual trace leaves a bounded simple residual and a multiset of primitive simple cycle occurrences. Let $C_1, \ldots, C_M$ be the finite simple-cycle types and $x_{i,j}$ their multiplicities in the ith actual pre-contact path. Pass to a subsequence so that, for each j, either $x_{i,j}$ is bounded or $x_{i,j} \to \infty$. Let I_∞ be the latter types and put

$$K_\infty = \mathrm{cone}_{\mathbb{Q}_{\geq 0}}\{v(C_j) : j \in I_\infty\}.$$

Lemma 107.69 (Actual-trace unbounded-multiplicity cone)**.** *If $\Delta_N(P_i) \to -\infty$ and $\omega(P_i) \to -\infty$ on the actual pre-contact paths, then K_∞ meets the open southwest quadrant*

$$\mathrm{SW} = \{(x, y) : x < 0,\ y < 0\}.$$

Every generator used in K_∞ occurs with unbounded actual multiplicity in the observed literal traces.

Proof. The bounded-multiplicity cycle types and the simple residual contribute only $O(1)$. If K_∞ were disjoint from the open southwest quadrant, the rational polyhedral separation theorem would give a nonzero nonnegative linear functional that is nonnegative on K_∞. Applied to the cycle decomposition, that functional would bound from below the corresponding nonnegative combination of $(\Delta_N(P_i), \omega(P_i))$, contradicting that both coordinates tend to $-\infty$. The last statement is the definition of I_∞. □

Lemma 107.70 (Support-two southwest reduction)**.** *If a finitely generated rational cone in $\mathbb{R}^2$ meets* SW*, it contains a southwest vector supported on at most two generators. If the support is two and neither generator is southwest, then after relabeling the two actual cycle types S, D satisfy*

	Δ_A	Δ_N	ω
S	> 0	≥ 0	< 0
D	< 0	< 0	≥ 0.

Proof. Apply the conic form of Carathéodory's theorem in dimension two to the cone K_∞ from Lemma 107.69. The selected one or two generators therefore belong to I_∞ and occur with unbounded actual multiplicity. For a support-minimal two-generator southwest combination, neither generator can have both coordinates negative. One must supply the negative N coordinate and the other the negative relative coordinate; substituting $\Delta_A = \Delta_N - \omega$ gives the stated opposite deterministic signs. □

Write $a = \Delta_A(S) > 0$, $d = -\Delta_A(D) > 0$, $x = \Delta_N(S) \geq 0$, and $y = -\Delta_N(D) > 0$. Define

$$\rho = x/a, \qquad \kappa = y/d.$$

A positive combination is southwest exactly when $\rho < \kappa$. With $g = \gcd(a, d)$ set

$$m = d/g, \qquad n = a/g.$$

Then the primitive deterministic balance

$$\Pi = mS + nD$$

satisfies

$$\Delta_A(\Pi) = 0, \qquad \Delta_N(\Pi) = \frac{ad}{g}(\rho - \kappa) < 0. \tag{107.29}$$

After clearing rational coefficients, m, n are fixed positive integers. By Lemma 107.69 and the proof of Lemma 107.70, both selected types lie in I_∞; sufficiently deep actual traces therefore contain the required finite number of literal occurrences of each type.

Proposition 107.71 (Deep trace-faithful neutral package deletion)**.** *In a sufficiently high translated arch, delete from the* original word *the selected actual occurrences of one package Π, leaving the intervening connectors untouched. The modified deterministic run stays positive, rejoins the exact original A configuration after the last removed block, and strictly shortens the word.*

Proof. Each removed block is an actual A-control cycle in positive mode. The package contains a fixed number of fixed cycle types, so the maximum intermediate height discrepancy caused by deleting them in chronological order is bounded by a package constant. High-shell reserve eventually exceeds that constant. Equation (107.29) gives zero total deterministic effect, hence exact rejoining after the final deletion. □

No statement about preserving an N run is made. Instead shortestness converts the A-safe deletion into a full-envelope certificate.

Lemma 107.72 (Critical deletion certificate)**.** *Let H be the original exact envelope and H' the deleted-word envelope at the common splice after the last removed block; let R be the backward requirement of the unchanged suffix. Then for some live coordinate q,*

$$H(q) < R(q) \le H'(q).$$

Proof. The original factor is rejected against the fixed residual, so $H < R$. The deletion word is shorter and still accepted by A; R-minimality therefore gives $H' \not< R$. The same coordinate witnesses both inequalities. □

The singleton southwest case with $\Delta_A = 0$ is identical. Suppose instead that one actual primitive southwest cycle C occurs with unbounded multiplicity and $\Delta_A(C) \neq 0$. The literal translated endpoint arch containing these occurrences has one fixed total deterministic effect κ_A, independent of shell depth. Cycle-erasing that same actual synchronized arch leaves a simple residual of uniformly bounded deterministic effect and finitely many primitive cycle types. In the exact identity

$$\kappa_A = \mathrm{res}_A(i) + \sum_j N_j(i)\Delta_A(C_j),$$

the contribution of C tends to one infinity sign. Therefore the sum of contributions of the opposite deterministic sign tends to the other; by finiteness one fixed actual primitive cycle type D of opposite A-effect also has unbounded multiplicity. With

$$g = \gcd(|\Delta_A(C)|, |\Delta_A(D)|), \quad m = |\Delta_A(D)|/g, \quad n = |\Delta_A(C)|/g,$$

the sparse literal package $mC + nD$ has total deterministic effect zero. Its cardinality is fixed, so high-shell reserve makes its chronological deletion A-safe and exact-rejoining. No sign condition on its net N effect is needed: shortestness alone yields Lemma 107.72. Thus the nonzero-A singleton folds into the same critical-deletion branch without any trace-changing path scheme.

107.14.15 Activation gain, threshold contacts, and noncircular return

At a critical deletion splice fix the decisive coordinate and write

$$\kappa_{\mathrm{act}} = H'(q) - H(q) \ge 1.$$

There are two cases.

Bounded activation gain. If $\kappa_{\mathrm{act}} \le M$, then

$$1 \le R(q) - H(q) \le M.$$

The chronological rejection margin before the splice is therefore confined to a finite band. After support and normalized-profile refinement, every no-escape segment lies in finitely many exact diagonal/projective phases. Absorbing their finite interfaces produces one finite exact projective mode system, which closes by Proposition 107.43.

Unbounded activation gain. The deleted package contains only a fixed number of fixed literal blocks. Telescoping between successive deletion variants selects one fixed block whose removal causes unbounded later envelope sensitivity. Immediately after a fixed word C, two envelopes with stable support and bounded normalized entry data differ by at most an effective constant L_C: above the finite entry thresholds, every C-path is affine in the common scalar. Moreover, as long as two one-letter transformers enable the same transition family, max-plus evaluation is sup-norm nonexpansive. Therefore unbounded amplification must have a first step at which enabled-edge families differ, or support changes.

Lemma 107.73 (First-threshold contact)**.** *Excluding an earlier support/projective escape, unbounded activation gain forces an aligned cut of the actual original shortest word, a fixed N state r, and an effective constant B_* such that*

$$H(r) \le B_*.$$

The corresponding deterministic height tends to infinity along the high-shell family.

Proof. Choose the first suffix step where the previous uniform sup-distance bound fails. If support has not changed, some transition is enabled from the larger envelope and disabled from the smaller. Its fixed negative requirement pins the source coordinate of both envelopes into a bounded threshold band. There are only finitely many earlier deleted slots; bounded pairwise differences before the first responsible slot transfer the bounded coordinate back to the actual original witness by the triangle inequality. High-shell geometry supplies the diverging A height. □

Thus at the contact

$$F(r) = H(r) - d \longrightarrow -\infty.$$

If the normalized N profile relative to r stays bounded, the contact belongs to a finite exact projective context. If it is unbounded, a genuine N/N projective difference escapes and localizes through a bounded-wall carrier to rank at most $\rho - 1$ or to a support/provenance drop.

Finally transfer the bounded contact to the canonical left endpoint of the translated arches. If that bounded correction itself escapes, it is already a lower-rank carrier escape. Otherwise after finite refinement one fixed reference coordinate has

$$H_\ell^L(r) = b, \qquad d(e_\ell^L) = \ell + r_L.$$

Hence

$$a_\ell := F_\ell^L(r) = b - (\ell + r_L).$$

For $h < k$, with $c = k - h$,

$$a_h - a_k = c. \tag{107.30}$$

Lemma 107.74 (Contact-anchor noncircularity)**.** *The superlevel activation condition $a_h - a_k > k - h$ cannot recur after a threshold contact is returned to the canonical endpoint system.*

Proof. Equation (107.30) gives equality for every pair. □

This is the strict well-foundedness invariant missing from a naive recurring-wall argument.

107.14.16 Bounded-context domination and the rank lift

The induction does not need a global lower-rank relation or an arbitrary exact-slice oracle. It needs only the endpoint replacement guaranteed by Lemma 107.35.

Definition 107.75 (Bounded-context domination compactness, WCI(ρ))**.** Fix an actual positive-mode factor P in an R-minimal witness, one concrete exact entry, and one exact deterministic A exit. Assume its recurrent owner-guard rank is at most ρ, and assume that all normalized/projective, anchored-relative, common-shift, support/provenance, and deterministic transfer/range data relevant to the current carrier belong to fixed finite palettes. Then WCI(ρ) asserts that there is an effective uniform bound on $|P|$; equivalently, every longer P has a strictly shorter endpoint replacement Q with the same exact deterministic exit and $H_Q \le H_P$.

If one of the listed quantities leaves its finite palette, that first exit is not hidden inside WCI: it is declared a responsible escape and localized separately.

Proposition 107.47 proves WCI(ρ) for $\rho \le 1$.

Theorem 107.76 (Structural rank lift)**.** *For every finite $\rho \ge 1$,*

$$\mathrm{WCI}(\rho - 1) \Longrightarrow \mathrm{WCI}(\rho).$$

Proof. Fix a rank-ρ bounded-context factor.

Wall-free branch. Lemma 107.49 lowers the rank. If a tracked normalized quantity escapes, this is a responsible lower-rank escape; otherwise WCI($\rho - 1$) bounds the strip.

Bounded-wall branch. Proposition 107.50 gives the same dichotomy for every fixed wall budget.

Unbounded-wall branch. Lemma 107.51 supplies linearly many bounded-wall same-signature atoms, with lower-rank endpoint systems. Bounded carrier corrections belong to the finite palette of Definition 107.52; unbounded corrections localize to a lower-rank wall-free strip.

Translated-arch branch. Lemma 107.53 gives either immediate anchored deletion or one fixed-coordinate endpoint/carrier escape. The latter is lower rank unless the only remaining mass is the activation/debt configuration.

Activation/contact branch. Sections 107.14.8–107.14.13 derive the activation mechanism from exact translated-arch endpoint identities, reduce the terminal survivor to a bounded high contact, and obtain the actual contact-centered relative valley of Theorem 107.68. Section 107.14.14 then extracts literal cycle packages from the observed synchronized trace, without trace-changing flattening. Bounded activation enters a finite projective system. Unbounded activation produces an actual threshold contact, and Lemma 107.74 prevents return to the same-rank superlevel branch. The return therefore ends in finite-projective compactness, support/provenance decrease, or a genuine lower-rank projective/relative escape.

All branches either terminate in Proposition 107.43, strictly lower recurrent guard rank, strictly lower provenance rank, or advance to a later same-(ν, ρ) stage in the finite stage order made explicit in Table 107.1. A responsible escape is therefore not a same-state recursive call: at its first occurrence it is reclassified as one of these successor types. Lemma 107.74 rules out the only possible return from the contact stage to the earlier activation stage at unchanged (ν, ρ). Hence the rank-ρ bounded-context factor is structurally reduced to terminal compactness or a strictly smaller proof state. The computability of the resulting numerical bound is established separately in Proposition 107.77. □

107.14.17 Quantitative effectivity and the proof-state order

The preceding theorem is structural. We now explain why its bound is a total computable integer rather than only an abstract finiteness statement.

Every tracked integer observable changes by at most a computable constant on the finite primitive macro alphabet. Therefore, for any cutoff M, the *first* exit from $[-M, M]$ occurs in a finite band

$$M < |Z| \leq M + D$$

for a computable drift bound D. Before this first exit the factor lies in a finite normalized context box, so the relevant lower-rank WCI bound applies. Wall-density constants are explicit, exact wall and carrier colors are finite, and every Ramsey or pigeonhole refinement is invoked only after choosing a finite required chain length.

A potentially dangerous infinitary Dickson extraction is avoided. Along a strict endpoint chain, after choosing a fixed reference coordinate, one obtains

$$(a_0 - a_n) + (\theta_0(q) - \theta_n(q)) \geq n. \tag{107.31}$$

Thus either common anchored scalar debt is large or projective spread is large. Projective spread above the explicit synchronized-product simple-path bound exposes a bounded nonzero differential core; below that bound normalized profiles lie in a finite box.

For complete algorithmic bookkeeping define the lexicographic measure

$$\mathcal{M} = (\nu, \rho, s),$$

where ν is the provenance rank from Definition 107.39, ρ is recurrent guard rank, and s belongs to the following finite stage set, ordered from top to bottom. Support changes that preserve the represented provenance atoms are finite-mode refinements; a recursive "support/provenance drop" means a strict decrease of ν as in Lemma 107.40.

s	Stage	Allowed same-(ν, ρ) successor
7	bounded-context entry/refinement	wall decomposition or terminal finite projective
6	directional wall decomposition	fixed-K terminal or unbounded-wall skeleton
5	dense same-signature atoms	translated high-wall carrier stage
4	translated arches / vector shell defect	anchored deletion, lower rank, or activation cells
3	activation cells / bounded contact / actual trace	relative valley or lower-rank terminal
2	cycle package / activation gain	critical deletion, finite-projective phase, or threshold contact
1	contact return	finite-projective terminal or strict lower-rank/support escape
0	terminal compactness	no recursive same-rank call

Table 107.1: Finite same-rank stage component of the proof-state measure.

A genuine provenance-rank drop strictly decreases ν by Lemma 107.40; wall-free responsible escapes strictly decrease ρ; every other nonterminal transition in Table 107.1 decreases s. Lemma 107.74 is precisely what prevents a contact return from jumping back from stage 1 to stage 4 at the same (ν, ρ).

Proposition 107.77 (Effective proof recursion)**.** *For a concrete exact first-return entry, all thresholds and terminal bounds can be computed by well-founded recursion on $\mathcal{M}$. In particular, the bound $B_{\mathrm{FR}}(A, N, C_0, H_0)$ is total computable.*

Proof. The terminal modules are effective: rank at most one uses Proposition 107.47; finite-projective branches use Proposition 107.43; fixed owner-resolved bounded-expression route schemas use Proposition 107.41; fixed wall and carrier alphabets are explicit finite sets. At a nonterminal proof state, compute the finitely many smaller-measure successor bounds first, choose the finite first-exit/Ramsey/wall thresholds required to force one of those successors, and take the finite maximum/sum dictated by the decomposition. Lexicographic well-foundedness guarantees termination. □

Theorem 107.32 follows from Theorem 107.76 and Proposition 107.77.

107.15 Effective next-zero successors

Fix a concrete zero state

$$Z = (p, H),$$

meaning deterministic configuration $(p, 0)$ and exact net envelope H. The first-return theorem is node dependent: the bound used at Z is computed from this exact H, not from a global uniform envelope bound.

Definition 107.78 (Next-zero factor)**.** A nonempty word u is a next-zero factor from Z if the unique A-run starts at $(p, 0)$, ends at some $(p', 0)$, and has no zero cut at a strict internal word boundary.

Thus a next-zero factor is either a one-letter zero-to-zero transition or a nontrivial positive first return. Put

$$B_0(Z) = \max\{1, B_{\mathrm{FR}}(A, N, Z) + 2\},$$

where the harmless $+2$ absorbs whether the boundary exit/return letters are counted inside the local first-return convention.

For a fixed finite word $u = a_1 \cdots a_k$, the exact transformer $T_u(H)$ is computable without any reachability oracle. For each source coordinate q and each length-k N-path π labelled by u, let $\mathrm{eff}(\pi)$ be its total counter effect and let

$$\mathrm{req}(\pi) = \max\{0, -\min \text{ prefix effect of } \pi\}$$

be the least initial credit required to execute it. Then

$$(T_u H)(r) = \max_{\substack{q,\pi: q \xrightarrow{u} r \\ H(q) \geq \mathrm{req}(\pi)}} \big(H(q) + \mathrm{eff}(\pi)\big).$$

The maximum is over a finite set of paths.

Definition 107.79 (Raw zero-successor set)**.** $\mathrm{RawSucc}_0(Z)$ is the set of records

$$e = (u, |u|, p_u, H_u)$$

obtained by enumerating all words $u \in \Sigma^{1 \ldots B_0(Z)}$, retaining exactly the next-zero factors, and putting $H_u = T_u(H)$ and $(p_u, 0)$ for the exact deterministic exit.

Theorem 107.80 (Effective zero-successor enumeration)**.** $\mathrm{RawSucc}_0(Z)$ *is finite, effectively computable, and literally complete for consecutive zero cuts of every shortest counterexample passing through* Z*.*

Proof. The enumeration is finite. Exact deterministic simulation decides the next-zero property and the formula above computes H_u. If u is the actual factor to the next zero cut of a shortest counterexample, then either $|u| = 1$ or Theorem 107.32 bounds its first-return length by B_{FR}; hence u occurs in the enumeration. □

There is also a safe finite pruning order, useful for implementations though not needed in the global proof.

Definition 107.81 (Cost-envelope order)**.** For two raw edges with the same target zero control,

$$e_i = (u_i, \ell_i, p', H_i),$$

write $e_1 \preceq_0 e_2$ when $\ell_1 \le \ell_2$ and $H_1 \le H_2$ componentwise.

Lemma 107.82 (Zero-edge domination)**.** *If $e_1 \preceq_0 e_2$ and $\ell_1 < \ell_2$, then e_2 cannot be used between consecutive zero cuts of a shortest counterexample. If $\ell_1 = \ell_2$, replacing e_2 by e_1 preserves shortestness and rejection.*

Proof. Both factors start from the same zero configuration and end in the same deterministic zero control. The smaller envelope preserves rejection by Lemma 107.35. A strict cost decrease contradicts shortestness; equal cost gives another shortest witness. □

The finite set of $\preceq_0$-minimal edges may therefore be used as a canonical successor antichain when only existence of a shortest witness matters. For the global tree below we keep the raw set, so the actual shortest-witness zero prefix embeds literally rather than only up to replacement.

107.16 The finite zero-state tree

The local bound of Section 107.15 is converted into a global bound by a second well-quasi-order argument.

Definition 107.83 (Zero-state order)**.** For zero states (p, H) and (p', H') put

$$(p, H) \le_Z (p', H') \quad \Longleftrightarrow \quad p = p' \text{ and } H \le H'.$$

Encode $\bot$ by 0 and a finite value h by $h + 1$. For each fixed control p, this is isomorphic to componentwise order on $\mathbb{N}^{|Q_N|}$.

Hence the finite disjoint union of all zero-control components is a WQO by Dickson's lemma.

Construct a rooted edge-labelled tree T_0 as follows. The root is the initial exact zero state. At a node Z, compute $\mathrm{RawSucc}_0(Z)$. For a candidate child $Z' = (p', H')$, inspect only the ancestors on the current root-to-node path. If some ancestor (p', H_a) satisfies $H_a \le H'$, prune Z'; otherwise insert it and retain the actual edge word and its cost.

Lemma 107.84 (No infinite branch)**.** *The pruned zero tree has no infinite branch.*

Proof. Along a branch no later zero state is above an earlier zero state with the same deterministic control. If the branch were infinite, some control would occur infinitely often. Dickson's lemma applied to the corresponding infinite sequence of encoded envelopes would produce an increasing pair, contradicting the pruning rule. □

Theorem 107.85 (Finite zero tree)**.** *The pruned tree T_0 is finite and effectively constructible. The complete zero-cut sequence of every shortest counterexample embeds literally as one root-to-node path of T_0.*

Proof. Theorem 107.80 gives finite branching at every generated node. Lemma 107.84 rules out infinite branches. König's lemma therefore implies finite depth; finite branching at each finite depth gives a finite tree. Exhaustive BFS or DFS terminates because it is enumerating this finite object.

For literal completeness, let $Z_0, \dots, Z_m$ be the actual zero states of a shortest counterexample. Theorem 107.80 inserts the actual factor $Z_i \to Z_{i+1}$ whenever Z_i is reached. Lemma 107.30 says that no Z_j can be pruned against an earlier Z_i with the same control, because such a pair would itself shorten the witness. Induction on i embeds the whole zero prefix. □

Two computable quantities are extracted from the explicit finite tree:

$$D_0 = \max\{\text{root-to-node edge depth}\},$$

and, retaining literal edge costs,

$$L_0 = \max\{\text{root-to-node total input length}\}.$$

Thus every shortest counterexample has at most $D_0 + 1$ zero cuts, and its prefix through the last zero cut has length at most L_0.

Remark 107.86 (Why König is effective here). König's lemma is used only to prove that the explicitly generated finitely branching tree is finite. The algorithm itself is ordinary exhaustive generation; it does not require a closed-form Dickson length function. Once mathematical finiteness is established, the generation necessarily reaches an empty frontier.

107.17 The terminal positive tail

A shortest counterexample may end strictly after its last zero cut. This case has no final zero return, so it is not directly covered by Theorem 107.32. A fresh-symbol construction turns it into one.

Fix a zero node $Z = (p, H)$ of T_0.

Definition 107.87 (Positive terminal tail)**.** A word u is a positive terminal tail from Z if the A-run starts in $(p, 0)$, is positive at every noninitial cut, and ends in an accepting control at height $h \geq 1$. It is rejecting if N has no accepting continuation over u from exact input envelope H.

Introduce three fresh letters $\#, d, \$$. Form A_Z^{tail} with a start-zero copy of p and positive-phase copies q^+ of the original controls. The start copy permits exactly the original zero rules that leave zero; the positive copies permit exactly the original positive rules. From each accepting f^+ add

$$f^+ \xrightarrow{\#,0} r,$$

where r is a drain control. At r add the positive rule $r \xrightarrow{d,-1} r$ and the zero rule $r \xrightarrow{\$,0} \mathsf{acc}$. The only accepting control is acc.

Extend N to N^{tail} by adding zero-effect self-loops on $\#, d, \$$ at every state, leaving the final states unchanged.

Lemma 107.88 (Identity of the fresh suffix)**.** *For every original-alphabet word u and every $k \geq 0$,*

$$T^{N^{\text{tail}}}_{u\#d^k\$}(H) = T^N_u(H).$$

In particular the extended net accepts $u\#d^k\$$ from H iff the original net accepts u from H.

Proof. Each fresh letter has exactly one zero-effect self-loop at every net control and no other fresh-letter transition, so its exact envelope transformer is the identity. □

Lemma 107.89 (Exact tail/corridor correspondence)**.** *A rejecting positive terminal tail u from Z ending at deterministic height h exists iff*

$$u\#d^h\$ \in L(A_Z^{\mathrm{tail}}) \setminus L_H(N^{\mathrm{tail}}),$$

where the net side starts from envelope H. Every accepted word of the extended deterministic machine has this unique form.

Proof. The extended control graph forces the original positive phase, then the marker from an accepting original control, then exactly h decrements to reach zero, then the zero-only final marker. Lemma 107.88 gives the net equivalence. □

Theorem 107.90 (Terminal-tail reduction)**.** *For every zero node $Z = (p, H)$ there is a total computable integer $B_{\mathrm{tail}}(Z)$ such that, if any rejecting positive terminal tail exists from Z, then one exists with length at most $B_{\mathrm{tail}}(Z)$.*

Proof. If a rejecting tail exists, choose a shortest extended counterexample $u\#d^h\$$ given by Lemma 107.89. The factor $u\#d^h$ is a strictly positive first return from zero to zero: the original part stays positive, the marker has zero effect, and only the last drain letter reaches zero. Therefore Theorem 107.32 applied to the extended instance gives

$$|u\#d^h| \le B_{\mathrm{FR}}(A_Z^{\mathrm{tail}}, N^{\mathrm{tail}}, (s_p, 0), H).$$

Take this first-return bound as $B_{\mathrm{tail}}(Z)$; then $|u| \le B_{\mathrm{tail}}(Z)$. □

The theorem is existential: it need not bound the literal terminal suffix of every shortest original witness. For global decidability it is enough that any obstruction at the same exact last-zero state has a bounded replacement tail.

Because T_0 is finite, compute

$$B_{\mathrm{tail}}^{\max} = \max_{Z \in T_0} B_{\mathrm{tail}}(Z).$$

107.18 Global small witnesses and decidability

Define

$$B_{\mathrm{global}} = L_0 + B_{\mathrm{tail}}^{\max}.$$

Theorem 107.91 (Global small witness, detailed form)**.** *If $L(A) \not\subseteq L(N)$, then there is a word $w \in L(A) \setminus L(N)$ with*

$$|w| \le B_{\mathrm{global}}.$$

Proof. Choose a shortest counterexample. If it ends at a zero cut, its entire word is a root-to-node path in T_0, hence has length at most L_0. Otherwise write it xu, where x ends at the last zero state $Z = (p, H)$ and u is a rejecting positive terminal tail. The literal zero prefix x embeds in T_0, so $|x| \le L_0$. Theorem 107.90 supplies another rejecting positive tail v from the same exact deterministic configuration $(p, 0)$ and the same exact net envelope H, with $|v| \le B_{\mathrm{tail}}(Z)$. Consequently xv is still accepted by A and rejected by N, and

$$|xv| \le L_0 + B_{\mathrm{tail}}^{\max}.$$

□

Corollary 107.92 (Hard inclusion)**.** *It is decidable whether $L(A) \subseteq L(N)$.*

Proof. Compute the finite integer B_{global} and enumerate $\Sigma^{\leq B_{\mathrm{global}}}$. Deterministic membership in A is direct. Membership in N is a finite dynamic program over the word. If one enumerated word lies in $L(A) \setminus L(N)$, report noninclusion; if none does, Theorem 107.91 proves inclusion. □

107.19 The reverse inclusion

The hard part of the chapter is $L(A) \subseteq L(N)$. Equality also requires the opposite orientation; we spell out the reduction because the deterministic machine has zero tests.

Proposition 107.93 (Deterministic-right inclusion)**.** *It is decidable whether* $L(N) \subseteq L(A)$.

Proof. **1. Total deterministic complement.** The deterministic OCA A may be partial. Add one dead control † and, for every missing letter transition in either the zero or positive regime, add a zero-effect transition to †; add zero-effect self-loops on all letters in both regimes at †. The resulting deterministic OCA is total on every word. Complement the set of final controls to obtain $\overline{A}$, so

$$L(\overline{A}) = \Sigma^* \setminus L(A).$$

2. Synchronous product. Now

$$L(N) \not\subseteq L(A) \iff L(N) \cap L(\overline{A}) \neq \varnothing.$$

Synchronize one transition of N with the unique $\overline{A}$ transition on each letter. The product has two nonnegative counters: the N counter is never zero-tested and the A counter is the only tested counter.

3. Compile the positive regime. A positive-regime OCA rule must be disabled at A-counter value zero. Simulate it by first decrementing the A counter by one, which is possible exactly when it is positive, while performing the synchronized N update. Then use a bounded finite-control chain to add back the amount needed to realize the original update $-1, 0$, or $+1$ (equivalently use the corresponding fixed VASS vector if arbitrary finite vectors are allowed). Thus positivity is enforced without any zero test.

4. Serialize all zero-regime rules through one zero-test gate. The product may contain many zero-regime rules, whereas the standard VAS_0 presentation uses one distinguished zero-test transition. This difference is harmless and we spell out the reduction. For every product zero-rule e, introduce one fresh ordinary tag counter t_e. From the source control of e, an ordinary transition moves to a common pre-gate control and places one token in t_e, without changing the tested A counter. A single distinguished transition

$$g_0 : \; q_{?0} \longrightarrow q_{=0}$$

is enabled exactly when the A counter is zero. After g_0, the unique post-gate transition corresponding to e consumes the token in t_e, performs the synchronized N update and the legal zero-regime A update (necessarily 0 or $+1$ in unit-step form), and enters the target product control. All other post-gate branches are disabled because their tag counter is zero. If the source A counter was positive, the run can enter the pre-gate gadget but cannot pass g_0, so it produces no spurious reachable target. Hence every zero-regime product edge is simulated exactly while all tests share one single distinguished zero-test transition.

The resulting counter system is therefore effectively a VASS/VAS with one zero-test in the sense of Reinhardt and Bonnet, for which exact configuration reachability is decidable [115, 116].

5. Final counters need not be zero. The product accepts upon reaching a control pair in $F_N \times F_{\overline{A}}$ with arbitrary nonnegative counter values. From every such product control enter a drain phase with ordinary decrement loops for each counter. Then some accepting product control is reachable with arbitrary counters iff one fixed drain target with both counters equal to zero is exactly reachable. No additional zero test is required because exact reachability asks for the target vector itself.

Therefore noninclusion is decidable, and so is $L(N) \subseteq L(A)$. □

Combining Proposition 107.93 and Corollary 107.92 proves Theorem 107.19.

107.20 The epsilon-move extension: models and deterministic reset normalization

107.20.1 Deterministic epsilon one-counter automata

A deterministic epsilon one-counter automaton (epsilon-OCA) has a finite control set Q_A, an initial control q_A^0, a set F_A of accepting controls, a nonnegative counter initialized to 0, a visible alphabet Σ, and transitions labelled by $\Sigma \cup \{\varepsilon\}$. Transitions may distinguish counter zero from counter positive. We use the standard unit-update OCA model, with counter effects in $\{-1, 0, +1\}$. This unit-update convention is part of the theorem statement; no unproved expansion of larger deterministic updates is used below.

The determinism convention is the standard one: in a fixed control/sign regime, if an epsilon transition is enabled then no visible transition competes with it. A visible word is accepted when the unique complete deterministic computation is defined, includes all forced silent behavior, and ends in a control of F_A; the final counter may be any nonnegative integer. If the computation becomes undefined, the word is rejected.

We use the standard stable/reset normalization of deterministic epsilon-OCA from the deterministic one-counter equivalence literature [107]. In this presentation, stable controls consume visible letters and reset controls perform forced silent behavior. A reset control s carries a finite period per_s and a map

$$\mathrm{goto}_s : \{0, \ldots, \mathrm{per}_s - 1\} \to Q_A^{\mathrm{st}},$$

with semantics

$$(s, n) \xrightarrow{\varepsilon^*} \big(\mathrm{goto}_s(n \bmod \mathrm{per}_s), 0\big).$$

Initial and accepting configurations are arranged stable. We take this normalization as an external standard input and perform the remainder of the hard-direction proof in the stable/reset presentation.

Remark 107.94. The deterministic side of theorem 107.21 is not allowed arbitrary nondeterministic silent behavior. The theorem covers the standard deterministic-epsilon convention described above.

107.20.2 Nondeterministic epsilon one-counter nets

A nondeterministic epsilon one-counter net (epsilon-OCN) has a finite control set Q_N, an initial control q_N^0, a set F_N of accepting controls, one nonnegative counter initialized to 0, visible and epsilon transitions, and no zero test. Primitive transitions have unit effects $d \in \{-1, 0, +1\}$ and are enabled exactly when the resulting counter remains nonnegative. Weak macro effects introduced below are arbitrary integers because they sum finitely many primitive effects. A word is accepted when some weak run ends in a control of F_N, with arbitrary nonnegative final counter value.

For a visible word w, a weak run is any run whose visible projection is exactly w; arbitrary epsilon moves may occur before, between, and after visible letters. The empty visible word is included.

107.20.3 Path requirement and effect

For a finite OCN path π with counter effects $d_1, \dots, d_m$, define

$$\operatorname{eff}(\pi) = \sum_{i=1}^{m} d_i$$

and

$$\operatorname{req}(\pi) = \max\left(0, -\min_{0 \le j \le m} \sum_{i=1}^{j} d_i\right).$$

Lemma 107.95. *A path π is feasible from counter c if and only if $c \ge \operatorname{req}(\pi)$. In that case its exit counter is $c + \operatorname{eff}(\pi)$.*

Proof. Let $s_j = \sum_{i=1}^{j} d_i$, with $s_0 = 0$. Starting from counter c, the counter after the first j transitions is $c + s_j$. Thus the whole path is feasible exactly when $c + s_j \ge 0$ for every j, equivalently when $c \ge -\min_j s_j$ and $c \ge 0$. This is precisely $c \ge \operatorname{req}(\pi)$. The final counter is $c + s_m = c + \operatorname{eff}(\pi)$. □

This elementary formula is the basis of the weak-envelope construction.

107.20.4 Stable cuts and visible length

A stable cut is taken after one visible input letter has been consumed and every forced deterministic reset closure triggered by that letter has completed. A stable zero configuration is a stable deterministic configuration $(p, 0)$.

All length bounds in this chapter count visible input letters only. Silent microsteps have zero word cost.

107.21 Weak epsilon-OCN envelope calculus

For controls $p, q \in Q_N$ and $a \in \Sigma$, write

$$(p, c) \Rightarrow_a (q, d)$$

when there is an OCN path of shape

$$\varepsilon^* \, a \, \varepsilon^*$$

from (p, c) to (q, d). Define

$$U_{p,a,q}(c) = \sup\{d : (p, c) \Rightarrow_a (q, d)\},$$

with value $\perp$ when no such weak path exists and ω when reachable exit counters are unbounded.

107.21.1 Cycle deletion and pumping

Lemma 107.96 (Nonpositive epsilon-cycle deletion)**.** *If a weak-a path contains an epsilon cycle C with $\mathrm{eff}(C) \le 0$, deleting that literal occurrence of C does not increase the path's entry requirement and does not decrease its exit counter.*

Proof. Write the path as $\alpha C \beta$ and let $e = \mathrm{eff}(C) \le 0$. Prefix sums before C are unchanged after deletion. Prefix sums after the deleted block increase by $-e \ge 0$, while the internal prefixes of C disappear. Hence the minimum prefix sum cannot decrease, so the entry requirement cannot increase. The total effect increases by $-e$, and therefore the exit counter does not decrease. □

Lemma 107.97 (Positive epsilon-cycle pumping)**.** *If a feasible weak-a path contains an epsilon cycle C with $\mathrm{eff}(C) > 0$ in its epsilon prefix or suffix, then above an effectively computable entry threshold the corresponding target upper envelope is unbounded.*

Proof. Write one weak path containing the positive epsilon cycle as $\rho C \tau$, where C is silent and $g = \mathrm{eff}(C) > 0$. Put

$$r_C = \mathrm{req}(\rho C).$$

For every source credit $c \ge r_C$, the prefix ρ and one traversal of C are feasible. After that traversal the counter at the cycle base has increased by g, so the same cycle remains feasible; after k repetitions it has gained kg. The fixed continuation τ has a finite entry requirement, hence sufficiently many repetitions make τ feasible and the target counter can be made arbitrarily large. Thus r_C is the exact entry threshold for this fixed pump witness.

Conversely, if pumping is possible from a source credit c, some feasible weak run contains a positive epsilon closed walk. Decomposing that walk into simple cycles yields a simple positive epsilon cycle. Choose the first such cycle on the run and delete all preceding nonpositive silent cycles; this produces one of the finitely enumerable pump witnesses with $r_C \le c$. Taking the minimum r_C over all such witnesses therefore gives the exact onset of the pumping branch. □

Theorem 107.98 (Finite weak-envelope normal form)**.** *For every p, a, q one can effectively compute:*

- *an optional pumping threshold r_∞, and*
- *a finite set $S_{p,a,q} \subseteq \mathbb{N} \times \mathbb{Z}$ of threshold/effect pairs,*

such that

$$U_{p,a,q}(c) = \omega$$

whenever the pump threshold applies, and otherwise

$$U_{p,a,q}(c) = \max\{c + e : (r, e) \in S_{p,a,q},\ r \le c\},$$

with $\bot$ when the set of enabled clauses is empty.

Proof. Positive epsilon cycles give the pumping branch by theorem 107.97. Below all pump thresholds, take any feasible weak a-path and repeatedly delete every nonpositive epsilon cycle using theorem 107.96. If an epsilon prefix or suffix still contained a closed walk, it would contain a simple cycle; because positive cycles are excluded below the pump threshold, such a cycle would be nonpositive and could be deleted. Thus both silent parts become simple. There are only finitely many resulting simple $\varepsilon^* a \varepsilon^*$ skeletons. By theorem 107.95, each skeleton contributes one pair $(\mathrm{req}(\pi), \mathrm{eff}(\pi))$, and every deleted path is dominated by its reduced skeleton. Taking the maximum over enabled skeletons gives the displayed finite branch. Enumerating simple connectors and simple positive epsilon cycles makes both the finite clause set and the pumping thresholds effective; Section 107.32 gives the explicit construction. □

For all later proof bookkeeping we use a *skeleton-refined palette*: if several simple weak paths induce the same threshold/effect pair, they are retained as distinct macro types whenever their concrete skeleton identity is needed. Likewise a pump type retains one concrete lasso certificate (prefix, positive silent cycle, and suffix) witnessing its threshold. This finite refinement does not change the envelope transformer.

107.21.2 Forward envelopes and backward requirements

Let

$$D = \{\bot\} \cup \mathbb{N} \cup \{\omega\},$$

ordered by

$$\bot < 0 < 1 < 2 < \cdots < \omega.$$

A forward envelope is a vector $H \in D^{Q_N}$. The value $H(q) = \omega$ means that arbitrarily large finite counters are reachable at q; it does not mean that every sufficiently large integer is reachable.

For $a \in \Sigma$, define

$$T_a^{\varepsilon}(H)(q) = \max_{p \in Q_N} U_{p,a,q}(H(p)),$$

with the natural interpretation of a source value ω through the finite threshold/pump description.

For a visible suffix v and state q, define

$$R_v(q) = \min \left\{ c \in \mathbb{N} : \begin{array}{l} \text{some weak run from } (q,c) \text{ reading } v \\ \text{reaches a final control} \end{array} \right\},$$

with $R_v(q) = \infty$ when the set is empty.

Definition 107.99 (Acceptance predicate)**.** For a forward envelope H and requirement vector R, write $\mathrm{Acc}(H, R)$ if there exists q such that $R(q) < \infty$ and either $H(q) = \omega$ or $H(q) \in \mathbb{N}$ with $H(q) \geq R(q)$. Write $\mathrm{Rej}(H, R)$ for the negation.

We deliberately do not encode rejection by a global expression $H < R$ once ω and ∞ coexist.

Theorem 107.100 (Exact weak residuation)**.** *For every visible letter a, there is an effective backward transformer S_a^{ε} such that*

$$\mathrm{Acc}(T_a^{\varepsilon}(H), R) \iff \mathrm{Acc}(H, S_a^{\varepsilon}(R)).$$

Equivalently,

$$\mathrm{Rej}(T_a^{\varepsilon}(H), R) \iff \mathrm{Rej}(H, S_a^{\varepsilon}(R)).$$

Proof. For each source control p, define $S_a^{\varepsilon}(R)(p)$ semantically as the least finite counter from which some weak a-run reaches a target q with enough exit credit to satisfy the finite requirement $R(q)$; use ∞ if no such counter exists. The feasible source set is upward closed because N has no zero tests. Therefore a forward envelope H can realize an R-accepting weak a-continuation exactly when one of its coordinates meets this least source requirement; an ω coordinate meets every finite requirement because it represents unbounded reachable finite credit. This is precisely the displayed equivalence. Effectivity follows from the finite simple-skeleton and pump description of theorem 107.98; the concrete formula is recorded below and in Section 107.32. □

For a finite simple weak skeleton $\pi : p \Rightarrow_a q$ and finite target requirement $R(q)$, the corresponding source requirement is

$$\max\bigl(\mathrm{req}(\pi),\, R(q) - \mathrm{eff}(\pi),\, 0\bigr).$$

A pump lasso contributes one finite entry threshold independent of the magnitude of the finite target requirement.

107.21.3 Initial and terminal epsilon closure

Let H_ε be the exact epsilon saturation of the initial N configuration. Let

$$R_\varepsilon(q)$$

be the least counter from which an epsilon-only run reaches an accepting control, or ∞ if impossible. If $w = a_1 \cdots a_n$, then

$$H_w = T^\varepsilon_{a_n} \cdots T^\varepsilon_{a_1}(H_\varepsilon),$$

and

$$w \in L(N) \iff \mathrm{Acc}(H_w, R_\varepsilon).$$

The same formula handles the empty visible word.

Lemma 107.101 (Active-omega exclusion)**.** *If uv is rejected and $R_v(q) < \infty$, then $H_u(q) \neq \omega$.*

Proof. If $H_u(q) = \omega$, arbitrarily large finite counters are reachable at q after u. Since $R_v(q) < \infty$, choose one reachable counter $c \geq R_v(q)$. By definition of $R_v(q)$, the suffix v then has an accepting weak continuation, contradicting rejection of uv. □

107.22 From unit edges to threshold-affine weak macros

This section records the epsilon-specific transfer lemmas needed to lift the real-time positive first-return compactness theorem.

107.22.1 Epsilon separator margin

For $c \in \mathbb{N}$, define $H + c$ by leaving $\bot$ and ω unchanged and adding c to finite coordinates. At a rejected cut set

$$m_\varepsilon(H, R) = \min\{c \in \mathbb{N} : \mathrm{Acc}(H + c, R)\},$$

with value ∞ if no such c exists.

Lemma 107.102 (Weak shift inequality)**.** *For every visible factor x and $c \geq 0$,*

$$T^\varepsilon_x(H + c) \geq T^\varepsilon_x(H) + c.$$

Proof. Every concrete weak run feasible from a finite source counter h remains feasible from $h + c$ and exits exactly c higher. Taking coordinatewise suprema preserves the inequality. If a coordinate on the right is ω, then arbitrarily large finite outputs from H remain arbitrarily large after the source shift, so the left coordinate is also ω. □

Lemma 107.103 (Margin monotonicity)**.** *If*

$$H' = T_x^{\varepsilon}(H), \qquad R = S_x^{\varepsilon}(R'),$$

and the factor lies in a rejected word, then

$$m_{\varepsilon}(H, R) \le m_{\varepsilon}(H', R').$$

Proof. Let $c = m_{\varepsilon}(H', R') < \infty$. Then $\mathrm{Acc}(H' + c, R')$. By theorem 107.102, $T_x^{\varepsilon}(H + c) \ge H' + c$, and acceptance is monotone in the forward envelope. Hence $\mathrm{Acc}(T_x^{\varepsilon}(H + c), R')$. Applying exact residuation theorem 107.100 gives $\mathrm{Acc}(H + c, S_x^{\varepsilon}(R')) = \mathrm{Acc}(H + c, R)$, so $m_{\varepsilon}(H, R) \le c$. The case $c = \infty$ is immediate. □

107.22.2 Threshold contacts and protected propagation

A finite weak macro is a tuple

$$e = (p, a, q, r, \delta, \sigma),$$

where r is its entry threshold, δ its integer effect, and σ identifies one concrete simple weak skeleton $\varepsilon^* a \varepsilon^*$ realizing the clause. Pump clauses are recorded separately.

Lemma 107.104 (Pump-or-threshold contact)**.** *Suppose a state q is critical at one step of a constant finite separator margin $\eta \ge 1$. If a backward-minimizing witness is a pump clause with threshold r_{pump}, then*

$$H(q) = r_{\mathrm{pump}} - \eta < r_{\mathrm{pump}}.$$

Hence the source lies in a computably finite threshold band.

Proof. A pump witness can satisfy any finite target requirement once its source counter reaches its fixed threshold, so its backward candidate is exactly r_{pump}. If it attains the backward minimum at the critical source, then $R(q) = r_{\mathrm{pump}}$. Criticality gives $R(q) - H(q) = \eta$, hence $H(q) = r_{\mathrm{pump}} - \eta < r_{\mathrm{pump}}$. □

Lemma 107.105 (Protected finite-macro propagation)**.** *Suppose a backward-minimizing witness is a finite macro with threshold r and the critical source satisfies $H(q) \ge r$. Then the macro is forward-enabled, forward-maximizing, backward-tight, and transports the critical gap exactly to the next cut.*

Proof. Let the macro go from q to q' with effect δ, and let R' be the next-cut requirement. At the critical source $R(q) = H(q) + \eta > H(q) \ge r$. Since the macro attains the backward minimum, its candidate is $\max(r, R'(q') - \delta) = R(q)$. The maximum therefore cannot be attained by r, and so $R(q) = R'(q') - \delta$. The source threshold is satisfied, hence the macro is forward-enabled and

$$H'(q') \ge H(q) + \delta = R'(q') - \eta.$$

Constant separator margin η at the next cut gives the opposite inequality $H'(q') \le R'(q') - \eta$. Thus equality holds: the macro is forward-maximizing, the backward equality is tight, and q' is critical with the same gap η. □

Lemma 107.106 (Threshold-contact recurrence is margin-independent)**.** *Work inside one constant-margin phase. If a backward-minimizing finite macro with threshold r is not forward-enabled at a critical source, then its source credit is one of the finitely many values*

$$c = H(q) \in \{0, \dots, r - 1\}.$$

For recurrence/pigeonhole bookkeeping it is sufficient to color such a contact by the refined finite mode, the critical source state, the chosen macro type, the bounded value c, and the finite support/omega data. The numerical separator margin η need not be included in the contact color. Consequently, if deterministic height and projective spread relative to the pinned value c are bounded, only finitely many exact contact states occur; a repeated exact state admits the usual shortest-witness deletion. All contact multiplicity bounds are therefore independent of the future backward requirement.

Proof. The inequality $H(q) < r$ gives the finite source-credit set. Inside the chosen phase the separator margin is already constant, so coloring by its numerical value contributes nothing to identifying the forward exact state. After fixing the finite mode and omega mask, bounded deterministic height and bounded projective spread around the pinned finite coordinate c bound every remaining finite envelope coordinate. Hence the exact state space is finite. If the same deterministic configuration and exact envelope recur, deleting the intervening visible factor rejoins both machines at the same exact forward state and preserves rejection against the inherited suffix, contradicting shortestness. No numerical bound on η is used. □

Thus every constant-margin factor either exposes a bounded threshold contact or carries a coherent tight critical strand through finite weak macros.

107.22.3 High-zone pump obstruction

Lemma 107.107 (High-zone pump obstruction)**.** *Inside a bounded projective/relative mode, a pump macro into a backward-live target is either disabled below a computable finite deterministic-height threshold or contradicts rejection. Consequently, above a computable height, all backward-live active coordinates evolve only by finite threshold-affine macros.*

Proof. In a fixed bounded projective/relative mode, every finite present source has form $H(p) = d + \kappa_p$, where d is the deterministic height parameter and κ_p ranges over a finite computable set. Consider a pump macro from p to a target q with threshold r_p. If q is backward-live and the pump is enabled, then the next forward value at q is ω, contrary to theorem 107.101 for a rejected continuation. Hence every such pump must satisfy $d + \kappa_p < r_p$, which bounds d by $r_p - 1 - \kappa_p$. Taking the maximum over the finite pump palette gives the claimed computable high-zone cutoff. Above it, only finite affine macros can contribute to backward-live targets. □

107.22.4 First amplification and trace fidelity

Lemma 107.108 (Regime nonexpansiveness and first split)**.** *Fix a visible letter a. Let H, K have the same finite support/omega mask and enable the same finite and pump macro clauses. On every target coordinate that is finite in both outputs,*

$$\left|T_a^{\varepsilon}(H)(q) - T_a^{\varepsilon}(K)(q)\right| \le \|H - K\|_{\infty}$$

where the norm ranges over their common finite source coordinates. Moreover, if two previously identical finite semantic regimes first diverge on the next visible step, then the divergence is caused by a support/omega event or by crossing one fixed finite or pump threshold; if the pre-split finite sup-distance is at most L, the responsible source values lie in a computably bounded L-neighborhood of that threshold.

Proof. For a finite target in a fixed regime, the weak transformer is the maximum of the same finite family of affine candidates $H(p) + \delta_e$; taking a maximum of functions with slope one in one source coordinate is 1-Lipschitz in the sup norm. Enabled pump clauses

produce ω on both sides and therefore do not enter the finite-coordinate estimate. If the next regimes differ for the first time, the two inputs had the same support/omega mask and the same clause status immediately before that step. Hence the first disagreement is either a support/omega event or the enabling status of some clause with fixed threshold r. In the latter case one source value is below r and the other is at least r; a pre-split distance at most L places both values in the finite band $[r-L, r+L]$ after intersecting with $\mathbb{N}$. □

Lemma 107.109 (Exact finite-composition closure)**.** *Fix a refined route $e_1 \cdots e_k$ of weak macro choices with matching controls. If every e_j is finite, write r_j, δ_j for its threshold and effect and $E_{j-1} = \sum_{i<j} \delta_i$. Then the composition is exactly one threshold-affine route*

$$c \longmapsto c + \sum_{j=1}^{k} \delta_j \quad \textit{for} \quad c \geq r_{\mathrm{comp}} := \max\Big(0, \max_{1\leq j\leq k} (r_j - E_{j-1})\Big).$$

If the first pump on the route occurs at position j, then the whole route is a pump route with exact source threshold

$$r_{\mathrm{comp}} = \max\Big(0, \max_{i<j}(r_i - E_{i-1}), r_j^{\mathrm{pump}} - E_{j-1}\Big).$$

The construction is effective. A finite composition retains the concatenated concrete weak skeleton; a pump composition retains the finite prefix/suffix skeletons together with one concrete positive-cycle lasso certificate.

Proof. Before the first pump, the entry credit at macro e_j is $c + E_{j-1}$, so simultaneous feasibility of all finite prefixes is exactly the displayed maximum of shifted thresholds. With no pump, the endpoint formula follows by summing effects. If the first pump is enabled, it produces arbitrarily large reachable finite credit at its target. Every fixed remaining suffix has only finitely many finite thresholds (or another pump), so sufficiently large pumped credit executes that suffix and leaves unbounded final credit. Conversely, before the first pump no smaller source credit can violate any of the listed prefix/first-pump requirements. The skeleton and lasso certificates are obtained by concatenation. □

Lemma 107.110 (Residue-refined powered-schema realization)**.** *Fix an owner-resolved bounded-expression schema over the skeleton-refined finite weak-macro palette. After an effective finite refinement of every powered exponent into residue classes, its threshold feasibility, owner comparisons, support/mode constraints, and deterministic positivity are Presburger conditions on the exponent variables; on each residue branch the endpoint finite owner vector is affine in the quotient exponent variables. The refinement and all formulas are independent of the future backward requirement.*

Proof. First apply theorem 107.109 to each fixed connector and powered block. On a fixed owner-resolved block β, the finite owner vector has an affine update $x \mapsto Px + c$ for one effectively known 0–1 functional matrix P. Since the powers of a finite matrix chosen from this finite transformation monoid are eventually periodic, compute μ, λ with $P^{\mu+\lambda} = P^{\mu}$. For $n = \mu + s + \lambda k$ with $0 \leq s < \lambda$,

$$P^n x + \sum_{j<n} P^j c$$

is a fixed residue-dependent affine function of x and k: the tail of the matrix-power sequence is periodic and each full period contributes one fixed integer vector. The finitely many cases $n < \mu$ are handled separately.

Every macro threshold and every owner comparison is an affine inequality in the current finite owner vector. After splitting the repetition index by the same finite residues, its validity at an intermediate repetition is a Presburger formula in the quotient exponents and the repetition index. Requiring validity for all repetitions is therefore Presburger by effective closure under first-order quantification. Repeating this construction for the finitely many powered blocks and taking the finite product of residue branches proves the claim. No backward requirement appears in the construction. □

For trace arguments, macro types retain the concrete simple skeleton identifier σ. Concatenating selected actual macro occurrences thus gives one actual N micro-run over the visible factor. In particular, an actual synchronized macro cycle consumes a nonempty contiguous visible word. No generic alternative VASS path is used to infer literal structure of the observed word.

107.23 Guarded-macro robustness of positive first-return compactness

The real-time compactness theorem has already been proved in theorem 107.32. We now isolate the exact features of that proof that are used by the epsilon compilation. This turns the epsilon extension into a self-contained internal robustness theorem.

Definition 107.111 (Admissible guarded macro system)**.** Fix the two machines and one concrete exact zero entry. An *admissible guarded macro system* is a finite, effectively computed family of visible macro types (for all visible letters, not for one preselected corridor) such that:

(G1) every finite macro has a source threshold r, an integer endpoint effect δ, and a concrete weak skeleton, and acts as $c \mapsto c + \delta$ whenever $c \geq r$;

(G2) every pump macro has one finite enabling threshold and one concrete lasso certificate and, when enabled, produces unbounded exit credit, represented by ω;

(G3) the induced forward and backward transformers are exact and monotone and satisfy exact residuation; moreover, on a constant-margin rejected phase, every enabled finite backward-minimizing critical macro is simultaneously forward-maximizing and transports the same tight separator equality to the next cut;

(G4) common-credit shifts are monotone: for every visible factor x and $c \geq 0$, $T_x(H+c) \geq T_x(H) + c$ on finite coordinates, with $\perp$ and ω fixed under the shift;

(G5) on each fixed finite support/omega/threshold regime the transformer is nonexpansive in finite coordinates, and a first regime split is localized to a support/omega event or to a bounded neighborhood of one of finitely many thresholds; threshold-contact recurrence uses only the bounded source credit and finite mode data, not the numerical separator margin;

(G6) every tracked integer observable used by the compactness proof has a computable one-macro drift bound; moreover every fixed bounded-expression macro schema $\alpha_0\beta_1^{n_1}\alpha_1 \cdots \beta_m^{n_m}\alpha_m$ has effectively Presburger threshold/guard/owner feasibility, and after an effective finite exponent-residue refinement its endpoint data are affine in the quotient exponent variables;

(G7) support, owner/provenance, guard, threshold, and omega data range over finite effectively enumerable mode alphabets, while bounded-height exact-state boxes are well-quasi-ordered after fixing the finite omega mask;

(G8) every macro used in a literal deletion argument retains a concrete skeleton from the observed weak run, so a selected synchronized macro cycle represents a nonempty contiguous visible factor of the observed word;

(G9) the palette is effectively closed under every fixed finite composition used as a proof macro: after finitely refining clause and pump choices, each composed route is again either threshold-affine with one computable source threshold and total integer effect, or a pump route; finite routes retain the concatenated concrete skeleton, while pump routes retain a concrete lasso certificate and finite prefix/suffix skeletons.

All palettes, mode alphabets, drift bounds, and thresholds in (G1)–(G9) are computed from the machines and the concrete exact zero entry and are independent of the future suffix and its backward requirement.

Theorem 107.112 (Weak-palette realization)**.** *For the skeleton-refined weak semantics of the epsilon-OCN, the macro palette computed from the machines satisfies (G1)–(G9) of theorem 107.111. The construction is uniform over all ordinary positive first-return corridors from the fixed exact zero entry and is independent of the future suffix, exit requirement, and numerical separator margin.*

Proof. The skeleton-refined finite weak-envelope normal form theorem 107.98 gives (G1) and (G2), including concrete finite skeletons and pump lasso certificates. The exact forward transformer is a finite maximum of those clauses, and theorem 107.100 gives the exact monotone backward transformer. On a constant-margin rejected phase, theorem 107.105 shows that every forward-enabled finite backward minimizer is also forward-maximizing and transports the same tight gap; this is precisely (G3). The shift property (G4) is theorem 107.102.

For (G5), fixed enabled-clause/support/omega regimes are nonexpansive by theorem 107.108; the first loss of that regime is a support/omega event or a crossing of one of finitely many finite/pump thresholds. Theorem 107.106 shows that recurrent threshold contacts are colored by bounded source credit and finite mode data only—the numerical separator margin is not a color and therefore cannot enter the bound. Pump minimizers are threshold contacts by theorem 107.104, while theorem 107.107 removes enabled pumps from backward-live protected high zones.

The finite macro palette gives a computable drift bound for every fixed integer observable. The powered part of (G6) is exactly theorem 107.110: eventual periodicity of the finite owner-map monoid yields a finite exponent-residue refinement on which endpoint data are affine and all threshold/owner/positivity constraints are Presburger. For (G7), support masks, threshold regimes, owner/provenance frames, guard data, and omega masks range over finite effectively enumerable sets. After fixing the bottom/omega masks, the remaining finite coordinates lie in a finite power of $\mathbb{N}$ and are well-quasi-ordered by Dickson; bounded context boxes are effectively finite when the relevant integer ranges are bounded.

For (G8), every finite target value is a maximum over a finite family of finite skeleton clauses and is therefore attained. Choosing the canonical maximizing predecessor and one attaining skeleton at each observed macro occurrence makes consecutive owner edges concatenate to an actual weak run; no omega coordinate is used on a backward-live rejected strand by theorem 107.101. Finally, theorem 107.109 proves (G9), with exact shifted thresholds for finite routes and for the first pump on a pump route. Every construction in this paragraph uses only the machines, the fixed exact entry for its provenance naming, and finite syntactic refinements; no backward requirement or future suffix is an input. □

Theorem 107.113 (Guarded-macro transfer theorem)**.** *Fix a concrete exact zero entry for the deterministic machine and an admissible guarded macro system in the sense of theorem 107.111. Then the suffix-parametric real-time positive first-return compactness proof of section 107.14 remains effective after replacing the unit net edges by these guarded macros. More precisely, one can compute from the machines and the exact entry a single*

integer B_{GM} such that, for every deterministic zero exit configuration C_1 and every exact backward requirement R at that exit, every strictly positive first-return corridor that is R-minimal in the sense of theorem 107.31 has visible length at most B_{GM}. In particular, B_{GM} is independent of R and hence of the future suffix.

Proof. The real-time proof uses only the interfaces listed in (G1)–(G9); the quantitative replacements are recorded in Section 107.33. Briefly, (G1), (G6), (G7), and (G9) provide the finite owner/provenance data, additive guarded effects, composition closure, Presburger powered-route checks, and bounded drift; (G4) supplies the common-credit shift inequalities; (G3) and (G5) supply exact critical propagation together with bounded threshold/contact localization; and (G8) preserves literal observed-trace deletion. Hence the wall, carrier, translated-arch, activation, contact, bounded-context, and WQO modules of Sections 107.14.2–107.14.16 carry over after finite rescaling.

The terminal modules remain in the same decidability classes: a finite route with threshold r and effect δ is compiled by subtracting r and then adding $r + \delta$, so threshold guards add no zero test to the finite-projective 2-VASS module or to the rank-one one-zero-test module. Pumps create no new recursive branch: a critical pump is a threshold contact, a protected high-zone pump into a backward-live target contradicts rejection, and an ω coordinate in a rejected factor is backward-dead. Therefore the same lexicographic proof measure (ν, ρ, s) is well founded and the effective recursion of theorem 107.77 computes one bound from the machines and exact entry alone. Since the real-time theorem is already suffix-parametric, the backward requirement R is never a recursive bound parameter. This yields the stated uniform B_{GM}. □

Theorem 107.114 (Robust positive first-return compactness)**.** *Fix a concrete stable zero entry*

$$Z = (p, H)$$

of the deterministic stable/reset automaton together with the exact ε-saturated net envelope H. There is a total computable integer

$$B^{\varepsilon}_{\mathrm{FR}}(Z)$$

such that, for every stable zero exit configuration C_1 and every exact backward requirement vector R, every ordinary positive first-return factor P from Z to C_1 that is visible-length-minimal among all words from the same exact entry to C_1 whose exact exit envelope is rejected against R has visible length at most $B^{\varepsilon}_{\mathrm{FR}}(Z)$. In particular, the same bound holds for every ordinary positive first return occurring inside a globally shortest rejected continuation from Z.

Proof. By theorem 107.112, the skeleton-refined weak palette is an admissible guarded macro system satisfying (G1)–(G9), uniformly in the future requirement. Apply theorem 107.113 at the exact entry Z. Its bound B_{GM} is already uniform over all stable zero exits and all exact backward requirements, so we may take $B^{\varepsilon}_{\mathrm{FR}}(Z) = B_{\mathrm{GM}}$. □

107.24 Reset-ending first returns and effective zero successors

We now return to the deterministic stable/reset structure. The local compactness theorem theorem 107.114 applies directly to ordinary positive first returns. The only new deterministic phenomenon is a visible step that enters a reset control and thereby returns silently to stable counter zero. We show that this case can be reduced to the same compactness theorem by a genuine auxiliary construction that stays inside the epsilon-OCA/epsilon-OCN model.

107.24.1 Consecutive stable zero cuts

Lemma 107.115 (Stable-zero trichotomy)**.** *Let $u \neq \varepsilon$ be the visible factor between two consecutive stable zero cuts of the deterministic stable/reset automaton. Exactly one of the following occurs.*

1. Direct zero step: $|u| = 1$*, and after the unique visible step and its forced reset closure, if any, the stable counter is zero.*
2. Ordinary positive first return: *after the first visible letter the stable counter is positive, all strict internal stable cuts are positive, and the final visible step reaches stable zero without entering a reset control.*
3. Reset-ending positive first return: *after the first visible letter the stable counter is positive, all strict internal stable cuts are positive, and the final visible letter enters a reset control whose forced silent behavior ends at stable zero.*

Proof. If a reset occurs at a strict internal stable boundary, then its forced silent closure produces an earlier stable zero cut, contrary to consecutive- ness. Hence all strict internal stable cuts are positive and reset-free. The final visible step either reaches stable zero directly or enters a reset control. The remaining case is the one-letter direct zero step. □

107.24.2 Reset signatures

Fix a reset-ending factor of the form ua. Immediately before the final letter a, suppose the deterministic machine is in stable control p at height $h > 0$. Let the a-rule have deterministic counter effect j, enter reset control s, and let

$$m = \mathrm{per}_s, \qquad r \equiv h + j \pmod{m}, \qquad q = \mathrm{goto}_s(r).$$

The finite tuple

$$\sigma = (p, a, j, s, m, r, q)$$

is called the *reset signature*. There are only finitely many such signatures. While reading the reset-free prefix u, the residue $h \bmod m$ can be carried in finite control.

107.24.3 A genuine reset-drain auxiliary instance

Fix a stable zero entry $Z = (p_0, H)$ and a reset signature σ as above. Construct a deterministic auxiliary automaton $A^{\mathrm{reset}}_{Z,\sigma}$ as follows.

From the fresh stable-zero entry representing Z, the prefix phase first copies the deterministic zero-regime visible step of A whenever its forced closure reaches a stable positive configuration without entering a reset. Thereafter it copies only the positive-regime, reset-free stable dynamics of A, while carrying the current counter residue modulo m in finite control. Every visible transition entering any reset control before the designated final marker is omitted, and no zero-regime transition is provided from a positive-copy control. Consequently, if the copied run reaches counter zero before the marker, it is stuck and cannot later reach the private drain exit. From the copy of the designated pre-reset control p, and only in the positive regime and in residue classes satisfying

$$h + j \equiv r \pmod{m},$$

a fresh visible marker # with counter effect 0 enters a private drain control. The marker deliberately does not reproduce the original update j: the reset discards the post-a value $h+j$, while the only information about that value needed to select the correct reset target is its residue, which is already enforced by the guard above. Keeping effect 0 preserves the pre-reset height h so that the drain exposes it as an ordinary visible first return. At

the drain control the fresh letter d is available only in the positive regime, decrements the counter by one, and remains in the drain control. Thus every run reaching the drain-zero exit is automatically a strictly positive first return, and its first return to counter zero occurs after a word

$$u \# d^h,$$

where the corresponding original factor ua has reset signature σ. The zero drain control is declared stable and is regarded as the fixed auxiliary exit representing the stable reset target q.

The net side realizes the final visible a semantically. Construct $N_{a\mapsto\#}$ by retaining all original epsilon transitions and, for every original visible transition

$$x \xrightarrow{a,w} y,$$

adding the copied transition

$$x \xrightarrow{\#,w} y.$$

All original visible transitions on the whole alphabet Σ are retained, independently of the eventual witness u, and d is a zero-effect self-loop at every N-control.

Lemma 107.116 (Copied-marker/drain identity)**.** *For every epsilon-saturated envelope K,*

$$T^{\varepsilon}_{\#}(K) = T^{\varepsilon}_{a}(K), \qquad T^{\varepsilon}_{d}(K) = K.$$

Consequently, whenever $u\#d^h$ is an auxiliary reset-drain first return,

$$T^{\varepsilon}_{u\#d^h}(H) = T^{\varepsilon}_{ua}(H).$$

Proof. A weak # path in the auxiliary net is exactly a path $\varepsilon^* \# \varepsilon^*$ whose unique visible edge is a copied a-edge with the same source, target, and counter effect as in N; replacing it by the original a-edge gives a bijection preserving requirements and exit credit. The fresh letter d is a zero-effect self-loop at every N-control, so on an already epsilon-saturated envelope its weak transformer is the identity. □

Lemma 107.117 (Reset-auxiliary uniformity certificate)**.** *For fixed machines A, N, a concrete stable zero entry $Z = (p_0, H)$, and a reset signature σ, the auxiliary pair*

$$(A^{\mathrm{reset}}_{Z,\sigma}, N_{a\mapsto\#})$$

and its exact entry envelope are effectively determined before any reset-ending witness, pre-reset height, or future backward requirement is chosen. In particular, the height h is not compiled into the machines: it is the runtime counter value, while its residue modulo m is tracked in finite control. Thus the guarded-macro palette and the compactness bound for the auxiliary pair are uniform over all factors of signature σ and all inherited suffix requirements.

Proof. The deterministic auxiliary control graph is obtained from the finite stable/reset presentation of A by product with the residue set $\mathbb{Z}/m\mathbb{Z}$, deletion of reset-entering prefix edges, and addition of the fixed marker/drain gadget. The net auxiliary is obtained from N by copying the finitely many a-edges to # and adding the fixed d-loops. All of these operations depend only on A, N, Z, σ. The exact entry is Z with envelope H. Neither the literal prefix u, nor its runtime height h, nor the later backward vector R enters the construction. The suffix-parametric guarded-macro theorem therefore computes one bound for this fixed auxiliary instance, uniformly in R. □

For an auxiliary reset-drain first return put

$$\mathrm{rcost}(u\#d^h) = |u| + 1.$$

Thus d-letters have zero reset cost, and the reset cost is exactly the visible length of the corresponding original factor ua.

Lemma 107.118 (Residual-minimal reset comparison)**.** *Let a reset-ending factor ua of signature σ occur between consecutive stable zero cuts of a globally shortest word in $L(A) \setminus L(N)$, and let R be the exact weak backward requirement of the unchanged suffix after the reset. If the pre-reset height is h, then $P = u\#d^h$ is reset-cost-minimal among all auxiliary positive first returns Q from the same exact entry to the drain-zero exit satisfying*

$$\mathrm{Rej}(T^{\varepsilon}_{Q}(H), R).$$

Proof. By theorem 107.116, the exit envelope of P is exactly the original envelope after ua, so it rejects the requirement R. Every auxiliary positive first return to the drain-zero exit has the form $Q = u'\#d^{h'}$. The residue guard on # guarantees that the original word $u'a$ has the same reset signature σ and therefore reaches the same stable zero control q after the forced reset. Again by theorem 107.116, its net envelope is exactly the exit envelope of Q.

If such a Q rejected R and had $\mathrm{rcost}(Q) < \mathrm{rcost}(P)$, replacing ua by $u'a$ and keeping the original suffix would give a strictly shorter word accepted by A and rejected by N, contradicting global shortestness. Hence P is reset-cost-minimal. □

Theorem 107.119 (Effective reset-ending first-return bound)**.** *For every concrete stable zero entry Z and reset signature σ there is a total computable bound $B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)$ such that every reset-ending factor of signature σ occurring between consecutive stable zero cuts of a globally shortest counterexample has visible length at most this bound.*

Proof. Fix such a factor and its suffix requirement R. Among all auxiliary positive first returns from the concrete auxiliary entry to the drain-zero exit that reject R, choose one, say Q_*, of minimum ordinary visible length. It exists because $P = u\#d^h$ is one such return. The suffix-parametric form of robust compactness, theorem 107.114, applied to the fixed auxiliary pair $(A^{\mathrm{reset}}_{Z,\sigma}, N_{a\mapsto\#})$, gives a computable number $\widehat{B}^{\varepsilon}_{\mathrm{FR}}(Z, \sigma)$ such that

$$|Q_*| \le \widehat{B}^{\varepsilon}_{\mathrm{FR}}(Z, \sigma).$$

By theorem 107.117, this number is computed uniformly from A, N, Z, σ and does not depend on the chosen factor, its pre-reset height, or R.

By theorem 107.118, P is reset-cost-minimal. Therefore

$$|ua| = \mathrm{rcost}(P) \le \mathrm{rcost}(Q_*) \le |Q_*| \le \widehat{B}^{\varepsilon}_{\mathrm{FR}}(Z, \sigma).$$

Taking the last quantity as $B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)$ proves the claim. Notice that this argument does not assert that the particular drain word $u\#d^h$ is ordinary-length-minimal; this is exactly why the separate reset cost is introduced. □

107.24.4 Effective next-zero successors

For a concrete stable zero node $Z = (p, H)$, let $B^{\varepsilon}_{\mathrm{ord}}(Z)$ be the suffix-independent ordinary first-return bound from theorem 107.114. For each syntactically feasible reset signature σ of A, let $B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)$ be the bound from theorem 107.119; if there is no such signature, the corresponding maximum below is defined as 0. Define

$$B^{\varepsilon}_{0}(Z) = \max\Big(1, B^{\varepsilon}_{\mathrm{ord}}(Z), \max_{\sigma} B^{\varepsilon}_{\mathrm{reset}}(Z, \sigma)\Big),$$

up to the fixed boundary-letter convention already absorbed into the auxiliary constructions.

Theorem 107.120 (Effective next-zero successors)**.** *From every concrete stable zero node $Z = (p, H)$ one can effectively compute a finite set*

$$\mathrm{Succ}_0^{\varepsilon}(Z)$$

of exact next-stable-zero successors, complete for every shortest word in $L(A) \setminus L(N)$ passing through Z.

Proof. Enumerate every visible word u with $1 \le |u| \le B_0^{\varepsilon}(Z)$. Simulate the deterministic stable/reset machine exactly, retaining u iff it is a literal factor from the current stable zero configuration to the next stable zero configuration and contains no earlier stable zero cut. By theorems 107.115 and 107.119, every consecutive-zero factor of a shortest counterexample is covered by the direct, ordinary, or reset-ending bound and therefore appears in this finite enumeration.

For each retained word compute the exact target envelope

$$H' = T_u^{\varepsilon}(H)$$

using theorems 107.98 and 107.100, and record the target stable control, exact envelope, literal word, and visible cost. All steps are effective and the enumeration is finite. □

107.25 The finite stable-zero tree

We next organize the finitely branching zero-successor relation by an extended Dickson order.

Definition 107.121 (Stable zero state)**.** A stable zero state is a pair

$$Z = (p, H),$$

where p is a stable deterministic zero control and $H \in D^{Q_N}$. For equal deterministic control define

$$(p, H) \preceq_0 (p, K) \quad \Longleftrightarrow \quad H \le K$$

componentwise. States with different deterministic controls are incomparable.

Lemma 107.122 (Extended Dickson property)**.** *For every finite set of N-controls, D^{Q_N} is a well quasi-order under componentwise order.*

Proof. Take an infinite sequence of vectors in D^{Q_N}. There are only finitely many masks describing the coordinates equal to ω, so pass to an infinite subsequence with one fixed omega mask. On the remaining coordinates, $D \setminus \{\omega\} = \{\bot\} \cup \mathbb{N}$ is order-isomorphic to $\mathbb{N}$ via $\bot \mapsto 0$ and $n \mapsto n + 1$. Dickson's lemma gives two comparable vectors on those coordinates, while the omega coordinates agree. □

The key deletion statement uses only monotonicity of weak N semantics, not arithmetic involving ω and ∞.

Lemma 107.123 (Stable zero-cut domination)**.** *Let $i < j$ be two stable zero cuts on a rejected deterministic accepting run. Suppose the deterministic stable zero control is the same at both cuts and that the exact envelopes satisfy*

$$H_i \le H_j.$$

Then deleting the visible factor between the two cuts preserves deterministic acceptance and N-rejection.

Proof. The deterministic run rejoins at the identical physical configuration $(p, 0)$, so its entire later behavior, including every forced reset closure, is unchanged. Let v be the inherited visible suffix after cut j. If v were accepted from H_i, monotonicity and $H_i \leq H_j$ would imply acceptance from H_j, contradicting rejection of the original word. Thus the shortened word remains rejected. □

Starting from the initial stable zero context, generate the raw successor tree using theorem 107.120. Prune a candidate node (p, H') if an ancestor on the same branch has the same deterministic control p and an envelope $H \leq H'$.

Theorem 107.124 (Finite stable-zero tree)**.** *The pruned zero-successor tree $\mathcal{T}_0^{\varepsilon}$ is finite and effectively constructible. Moreover, the complete stable-zero sequence of every shortest counterexample embeds literally as one root-to-node path in $\mathcal{T}_0^{\varepsilon}$.*

Proof. Every node has finitely many children by theorem 107.120. An infinite unpruned branch would be a bad sequence for the wqo of theorem 107.122, which is impossible. Hence the finitely branching tree has no infinite branch and is finite by König's lemma. Exhaustive generation therefore terminates.

For completeness, consider the literal stable-zero sequence of a shortest counterexample. Each consecutive factor appears among the raw successors by theorem 107.120. If a later literal node were pruned by an earlier dominated node on that same path, theorem 107.123 would give a strictly shorter counterexample, contradicting shortestness. □

Let

$$D_0^{\varepsilon}$$

be the maximum root-to-node edge depth of $\mathcal{T}_0^{\varepsilon}$, and let

$$L_0^{\varepsilon}$$

be the maximum sum of visible edge costs along a root-to-node path. Both are total computable integers. Every shortest counterexample has at most $D_0^{\varepsilon} + 1$ stable zero cuts, and its visible prefix through the last stable zero cut has length at most L_0^{ε}.

107.26 Terminal tails and the global hard-inclusion bound

It remains to control the suffix after the last stable zero cut. Such a suffix cannot contain a reset: every reset ends at stable counter zero and would create a later stable zero cut.

107.26.1 A syntactic terminal auxiliary automaton

Fix a stable zero node $Z = (p, H)$ from $\mathcal{T}_0^{\varepsilon}$. Construct a deterministic auxiliary automaton A_Z^{term} syntactically. From a fresh zero start representing p, copy exactly those zero-regime visible transitions whose target is a stable positive configuration. Omit transitions entering reset controls or returning immediately to stable zero. For every stable control q, create a positive copy q^+ and copy every positive-regime visible transition whose target is stable; transitions entering reset controls are omitted. If a copied decrement reaches zero, the run is stuck because no zero-regime original transitions are available inside the positive copy.

From every copied accepting stable control f^+ add a fresh marker # *only in the positive regime*, with counter effect 0, into a fresh stable drain control. At the drain control, the fresh letter d is available only in the positive regime and decrements the counter by one; the fresh letter \$ is available only in the zero regime and enters the unique accepting control. No zero-regime # rule is present at any positive-copy state.

Hence a copied decrement that reaches zero before the marker is truly stuck, while an accepted run reaches zero for the first time during the private drain. Thus the accepted words are exactly

$$u \# d^h \$,$$

where u is a reset-free positive terminal tail ending at an original accepting stable control at height $h > 0$.

On the N side retain all original transitions and add zero-effect self-loops on $\#, d, \$$ at every control. Both auxiliary machines are constructed uniformly from (A, N, Z) alone; neither the particular terminal tail, its ending height, nor any future backward requirement is used in their control graphs.

Lemma 107.125 (Fresh identity letters)**.** *If K is epsilon-saturated and $x \in \{\#, d, \$\}$ is one of the fresh identity letters, then*

$$T_x^{\varepsilon}(K) = K.$$

Consequently,

$$T_{u\#d^h\$}^{\varepsilon}(H) = T_u^{\varepsilon}(H).$$

Proof. A weak x path has the form $\varepsilon^* x \varepsilon^*$ and the visible x-edge is a zero-effect self-loop at every control. Hence its transformer is $C_\varepsilon \circ \mathrm{Id} \circ C_\varepsilon$, which equals the identity on an already epsilon-saturated envelope. Iterating over the fresh suffix gives the second equality. □

Theorem 107.126 (Effective terminal-tail bound)**.** *For every concrete zero node Z, one can compute an integer $B_{\mathrm{tail}}^{\varepsilon}(Z)$ such that, if there exists any accepting terminal tail from Z rejected by N, then there exists one of visible length at most $B_{\mathrm{tail}}^{\varepsilon}(Z)$.*

Proof. Choose a shortest counterexample of the auxiliary terminal instance. It has shape $u\#d^h\$$. The prefix $u\#d^h$ starts at stable zero, is positive after its first visible letter, contains no internal reset or stable zero cut, and reaches zero for the first time on the final d. It is therefore an ordinary positive first-return corridor. By theorem 107.114, its visible length is bounded by a total computable integer depending only on the concrete auxiliary entry. By theorem 107.125, rejection of the auxiliary word is equivalent to rejection of the original tail u from the same exact envelope H. Taking the first-return bound as $B_{\mathrm{tail}}^{\varepsilon}(Z)$ proves the claim. □

Define

$$B_{\mathrm{tail,max}}^{\varepsilon} = \max_{Z \in \mathcal{T}_0^{\varepsilon}} B_{\mathrm{tail}}^{\varepsilon}(Z)$$

and

$$B_{\mathrm{global}}^{\varepsilon} = L_0^{\varepsilon} + B_{\mathrm{tail,max}}^{\varepsilon}.$$

Both are computable because the zero tree is finite.

Theorem 107.127 (Global small counterexample)**.** *If*

$$L(A) \not\subseteq L(N),$$

then there exists

$$w \in L(A) \setminus L(N)$$

with

$$|w| \le B_{\mathrm{global}}^{\varepsilon}.$$

Proof. Take a shortest counterexample w. Let x be the prefix ending at its last stable zero cut, with zero node $Z = (p, H)$, and write $w = xu$. By theorem 107.124, $|x| \le L_0^\varepsilon$.

If $u = \varepsilon$, the claim follows immediately. Otherwise u is a reset-free positive terminal tail from Z. By theorem 107.126, there exists an accepting tail v from the same deterministic physical configuration and the same exact N envelope H, rejected by N, with $|v| \le B_{\mathrm{tail}}^\varepsilon(Z)$. Thus xv is again in $L(A) \setminus L(N)$ and

$$|xv| \le L_0^\varepsilon + B_{\mathrm{tail,max}}^\varepsilon = B_{\mathrm{global}}^\varepsilon.$$

□

Proof of theorem 107.22. Compute $B_{\mathrm{global}}^\varepsilon$ and enumerate all visible words of lengths

$$0, 1, \ldots, B_{\mathrm{global}}^\varepsilon.$$

The length-0 case is included explicitly. For each word, simulate the unique deterministic stable/reset run of A and compute the exact weak N envelope from theorem 107.98; final N-acceptance is tested after trailing epsilon saturation. By theorem 107.127, the inclusion fails if and only if one of these finitely many words belongs to $L(A) \setminus L(N)$. □

107.27 Reverse inclusion

We now decide $L(N) \subseteq L(A)$. This direction does not use the small- witness theorem.

107.27.1 A marked noninclusion reduction

Introduce a fresh visible endmarker $\dashv \notin \Sigma$. First totalize the visible stable semantics of A by sending every missing stable visible rule to one fresh stable dead control, with dead self-loops on all original visible letters in both sign regimes.

Construct a marked deterministic rejection automaton A_{bad} by adding $\dashv$-rules from stable controls: an original accepting stable control reads $\dashv$ into a nonaccepting sink GOOD, whereas every nonaccepting stable control, including the dead control, reads $\dashv$ into an accepting sink BAD. The marked sinks have no outgoing transitions.

Construct N_{end} by retaining all original transitions and, from every original accepting N-control q, adding

$$q \xrightarrow{\dashv,0} f_{\mathrm{end}},$$

where f_{end} is the unique marked accepting control.

Lemma 107.128 (Marked noninclusion)**.**

$$L(N) \not\subseteq L(A)$$

if and only if there exists $w \in \Sigma^*$ such that both marked systems have accepting runs on $w \dashv$.

Proof. The totalized deterministic stable/reset machine has one unique stable endpoint after every visible word. Hence A_{bad} accepts $w \dashv$ exactly when the original A rejects w. On the net side, arbitrary trailing epsilon moves may occur before the endmarker, so N_{end} accepts $w \dashv$ exactly when N accepts w. The case $w = \varepsilon$ is represented by the one-letter marked word $\dashv$. □

107.27.2 Reset decompression

For a reset control s of period p, replace the semantic reset

$$(s, n) \xRightarrow{\varepsilon^*} (\mathrm{goto}_s(n \bmod p), 0)$$

by finite micro-controls

$$s[0], \dots, s[p-1].$$

At positive counter add

$$s[i] \xrightarrow{\varepsilon,-1} s[(i+1) \bmod p],$$

and at zero add

$$s[i] \xrightarrow{\varepsilon,0} \mathrm{goto}_s(i).$$

Starting from height n, exactly n silent decrements occur and the zero exit is taken in phase $n \bmod p$.

Lemma 107.129 (Commutation with N-epsilon moves)**.** *Inside the asynchronous product, N-epsilon moves commute with the decompressed A-reset microsteps without changing component endpoints or visible projection.*

Proof. An N-epsilon move changes only the N-control and N-counter, while an A-reset microstep changes only the A-control and A-counter. Neither consumes visible input. Their feasibility conditions therefore concern disjoint counters, so adjacent independent moves may be swapped. Repeating the swap moves all N-epsilon steps to either side of the finite A-reset microsequence while preserving both component endpoints. □

107.27.3 The asynchronous marked product

A genuine product control stores one control of N_{end} and one stable or reset-micro control of A_{bad}. The two main counters are denoted by x_N and x_A.

N-epsilon moves change only (q_N, x_N). Decompressed A-epsilon moves change only (q_A, x_A). An original visible letter synchronizes one chosen N-edge with the unique deterministic A-edge in the current A sign regime. The endmarker $\dashv$ is synchronized in the same way. By theorem 107.129, allowing arbitrary interleavings of the two silent components introduces no new visible behavior.

Lemma 107.130 (Product/run correspondence)**.** *A marked accepting product control is reachable if and only if there exists $w \in L(N) \setminus L(A)$.*

Proof. Projecting a product run gives weak component runs with the same visible word $w \dashv$. Conversely, interleave the silent moves of any two such component runs and synchronize their visible letters; theorem 107.129 handles reset microsteps. Apply theorem 107.128. □

107.27.4 Compilation to one zero-test VASS

The low-level construction is given in Section 107.34; we record here only the correctness interfaces used by the theorem. A positive-regime synchronized edge with effects (d_N, d_A) is compiled through a private intermediate control as

$$(d_N, -1) \quad \text{then} \quad (0, d_A + 1),$$

so ordinary nonnegativity enforces $x_A > 0$ and the private control makes the gadget atomic. Positive reset microsteps already have effect $(0, -1)$.

All zero-regime semantic edge types are serialized through one common gate. Each type e has an ordinary tag counter t_e: its source transition records e, one distinguished transition g_0 tests only $x_A = 0$, and the tagged post-gate branch consumes t_e, performs the fixed ordinary vector update of e, and returns to a genuine product control.

Lemma 107.131 (Single-gate exactness)**.** *A complete gate path for a semantic zero edge e reaches its genuine target if and only if the edge is taken at its proper source with $x_A = 0$ and its ordinary counter update is feasible. At every genuine product control all tag counters are zero. Thus g_0 is the unique zero-test transition.*

Proof. The unique tag identifies the chosen edge; all other completion branches are disabled by ordinary nonnegativity. If the vector update is infeasible the run is trapped inside the private gadget. A successful completion consumes the tag and restores the all-zero tag invariant. See Section 107.34 for the explicit transition schema. □

From the marked accepting product control, a private cleanup phase supplies ordinary decrement loops for every counter and one final zero-effect edge to a fixed target.

Lemma 107.132 (Cleanup equivalence)**.** *Some marked accepting product configuration with arbitrary nonnegative counters is reachable if and only if the fixed all-zero cleanup target is exactly reachable.*

Proof. Drain all counters before taking the final edge. Conversely cleanup is reachable only from marked acceptance, and taking the final edge early cannot reach the queried all-zero configuration. □

Exact reachability for VASS/Petri nets with one zero test (equivalently, one inhibitor arc in the standard formulation) is decidable [115, 116]. Our compiled system matches that model exactly: g_0 is the single zero-test transition, all remaining counters are ordinary VASS counters, and every other transition is an ordinary vector addition.

Proof of theorem 107.23. By theorems 107.130 to 107.132, $L(N) \not\subseteq L(A)$ if and only if the fixed all-zero cleanup target is reachable in the compiled VASS with one zero test. The latter reachability problem is decidable, so $L(N) \subseteq L(A)$ is decidable. □

Proof of theorem 107.21. By theorem 107.22, the inclusion $L(A) \subseteq L(N)$ is decidable. By theorem 107.23, the reverse inclusion $L(N) \subseteq L(A)$ is decidable. Therefore one decides $L(A) = L(N)$ by deciding both inclusions and accepting exactly when both hold. □

107.28 Conclusion

Language equality between deterministic zero-testing one-counter automata and nondeterministic one-counter nets is decidable in the standard real-time setting and remains decidable under the epsilon-transition conventions of theorem 107.21. The common mechanism is a shortest-counterexample compactness theory organized around exact residual semantics and deterministic zero returns. Silent net behavior is absorbed by a finite threshold-affine weak-macro calculus, while deterministic silent behavior is handled by stable/reset normalization.

The resulting bounds are effective but intentionally complexity-agnostic. They combine well-founded rank recursions, finite Ramsey and Dickson arguments, Presburger checks, and one-zero-test reachability subroutines.

107.29 Real-time effectivity ledger

The real-time global small-witness theorem needs a *computable* witness bound. The following list records the external or finite-search ingredients used to compute that real-time bound. The epsilon extension does *not* use the finite-path statement in item 1; its corresponding effectivity is supplied instead by the exact weak-envelope construction and guarded-macro ledgers in Appendices 107.32 and 107.33.

1. Exact net transformers T_x and S_x for a fixed word are finite dynamic programs over the finitely many paths of length $|x|$.

2. Support, owner frames, primitive synchronized simple cycles, exact wall signatures, and bounded-context carrier colors are finite and effectively enumerable.

3. Rank-one recurrent owner switching is compiled to a VASS with one zero-tested scalar (Proposition 107.47); exact reachability for that class is decidable [115, 116].

4. Finite-projective systems reduce to exact 2-VASS reachability and shortest representative search (Proposition 107.43).

5. The dense-wall constants are explicit: $K_{\mathrm{dense}} = 4|\mathcal{W}| + 1$, and a requested number of dense atoms is forced by a computable linear wall-count threshold.

6. Every finite Ramsey use is on an explicitly finite coloring (for example side $\times$ envelope coordinate in Lemma 107.53); only finite Ramsey numbers are required.

7. Every tracked integer observable has a computable one-macro drift bound; first exit from a cutoff occurs in a finite band.

8. The effectivity proof avoids an uncontrolled infinite Dickson subsequence by the reserve split (107.31); below the explicit projective cutoff the normalized profile space is finite.

9. The translated-arch activation bridge is quantitative: bounded-cell refinement fixes the finite tuple $(c, \delta, d, p, \eta, u, v)$; the pure-scalar survivor fixes (d, λ, η, K); the bounded-contact state, requirement, and counter lie in finite computable sets. Hence the contact/relative-valley transition introduces no infinitary choice.

10. The proof-state recursion uses the explicit finite stage set in Table 107.1 and the lexicographic measure (ν, ρ, s).

11. Next-zero successors are obtained by literal word enumeration up to the computed first-return bound; exact envelopes are computed by dynamic programming.

12. The pruned zero-state tree is effectively generated. It terminates because it is finitely branching and every branch is a Dickson bad sequence.

13. Terminal tails reduce to another concrete first-return instance by the fresh-marker drain construction, after which a finite maximum is taken over the finite zero tree.

107.30 Real-time canonical dependency graph and excluded proof routes

The real-time compactness proof uses only the corrected canonical dependency chain below. The epsilon theorem invokes this core only through the guarded-macro transfer

theorem of Section 107.23. Exploratory routes that were later superseded are not premises of any theorem in this chapter.

exact residuals and extensional splice
$\Downarrow$
shell deletion + finite-projective terminal module + rank-one base
$\Downarrow$
wall-free descent + bounded-wall compactness
$\Downarrow$
dense same-signature atoms + translated arches
$\Downarrow$
monotone endpoint debt + vector shell defect
$\Downarrow$
activation cells + pure-scalar recurrence + bounded high contact
$\Downarrow$
actual contact-centered relative valley + literal cycle packages
$\Downarrow$
critical deletion + activation/contact noncircularity
$\Downarrow$
WCI rank lift + effective B_{FR}
$\Downarrow$
finite next-zero successors + finite zero tree
$\Downarrow$
terminal drain + global small witness.

For referee clarity, several historically tempting routes are *not* dependencies of the theorem.

Excluded shortcut	**Why it is insufficient**	**Canonical replacement**
Uniform separator "gap spending"	the rejection margin is monotone but need not decrease as a consumable budget	endpoint domination, anchored shell deletion, and rank descent
Global compilation of recurrent wall equality into one lower-rank machine	unboundedly many exact wall equalities may require independent equality checks	fixed-K compilation plus dense same-signature returns and translated arches
Treating 2-VASS flattening as trace preserving	reachability endpoints may be represented by a different input word	actual-trace cycle erasure for literal deletions; VASS alternatives only for endpoint replacements
Strong arbitrary lower-rank exact-slice/Pareto oracle	stronger than the shortest-witness induction can justify at every rank	bounded-context domination WCI(ρ) plus responsible-escape alternative
Infinitary Dickson extraction for normalized endpoint profiles	abstract WQO alone does not yield the desired quantitative threshold without growth control	explicit scalar/projective reserve split and finite profile boxes

Excluded shortcut	Why it is insufficient	Canonical replacement
Nested-zero-test shortcut at wall boundaries	simultaneous-zero compatibility of high-rank wall tests with lower-rank tests is not established	extensional fixed-K wall handling and literal high-shell deletion

107.31 Real-time trace-preservation checklist

Every use of cycle or VASS machinery in the real-time compactness proof falls into one of the following two audited categories. The epsilon transfer preserves the same distinction by retaining concrete weak skeletons, as recorded in Section 107.33.

1. **Literal operation on the observed witness.** The factor to be removed is an actual contiguous cycle occurrence (or a finite sparse set of such occurrences) found by cycle erasure on the observed synchronized trace. Deterministic safety is verified on that literal word. No alternative path scheme is substituted silently.

2. **Constructive endpoint replacement.** A reachability theorem may return a different word. The proof records its exact deterministic endpoint and exact N exit envelope and invokes Lemma 107.35. Proposition 107.43 and the bounded-projective terminal branches are of this form.

The support-two southwest reduction and the A-neutral package in Section 107.14.14 belong to category 1. Finite-projective compression belongs to category 2. No theorem crosses from category 2 to a literal deletion claim without an independent trace-faithful extraction.

107.32 Effective weak-envelope construction

This ledger section supplies the algorithmic details behind theorems 107.98 and 107.100. The purpose is not to eliminate silent moves syntactically, but to compile their exact effect on upper envelopes and backward requirements into a finite collection of clauses.

107.32.1 Effective finite and pump enumeration

Fix p, q and a visible letter a. Enumerate every simple epsilon-prefix, one a-edge, and every simple epsilon-suffix. Each resulting skeleton π contributes the exact finite clause

$$c \mapsto c + \mathrm{eff}(\pi) \qquad (c \geq \mathrm{req}(\pi)).$$

For pumping, enumerate a simple positive epsilon cycle together with simple connectors from the source, through the visible edge, and to the target, with the cycle on either silent side. For a lasso $\rho C \tau$, the pump threshold is $\mathrm{req}(\rho C)$. The proofs of theorems 107.96 to 107.98 show that these finite families are complete: every finite maximizing run is dominated by a simple skeleton, while every pumpable run contains one of the enumerated positive-cycle witnesses. Hence both the finite clause palette and the exact onset of the pumping branch are effectively computable.

107.32.2 Exact propagation from an omega source

If $H(p) = \omega$, the set of reachable finite counters at p is unbounded. For any finite clause with threshold r, some reachable finite source counter is at least r, and applying the fixed clause to arbitrarily large reachable source counters produces arbitrarily large target counters. Thus the target value is ω. No congruence or eventual-density assumption is used.

107.32.3 Backward transformer

Let R be a target requirement vector. For a finite simple skeleton $\pi : p \Rightarrow_a q$ with $R(q) < \infty$, the least source credit that both executes π and leaves enough target credit is

$$b(\pi, q, R) = \max\{\mathrm{req}(\pi),\, R(q) - \mathrm{eff}(\pi),\, 0\}.$$

A pump witness leading to a target with finite $R(q)$ contributes its pump entry threshold itself: after the positive cycle is executable, enough repetitions can manufacture any prescribed finite target credit. Targets with $R(q) = \infty$ contribute no candidate. Therefore

$$S_a^{\varepsilon}(R)(p)$$

is the minimum over a finite effectively computable candidate set, or ∞ when that set is empty. Upward closure of OCN feasibility then gives

$$\mathrm{Acc}(T_a^{\varepsilon}(H), R) \iff \mathrm{Acc}(H, S_a^{\varepsilon}(R)),$$

including the ω case. This also proves that the initial saturation H_{ε} and terminal requirement R_{ε} are effectively computable by the same finite epsilon-path/cycle analysis with the visible edge omitted.

107.33 Macro-transfer and constant-rescaling ledger

We collect here the quantitative interface between the guarded-macro transfer theorem theorem 107.113 and the weak-macro system of section 107.22. The real-time argument is contained earlier in this chapter: owner/provenance and finite-projective setup in Sections 107.14.2–107.14.4, wall and translated-arch localization in Sections 107.14.5–107.14.9, threshold/contact and literal-trace analysis in Sections 107.14.10–107.14.15, the structural rank lift in theorem 107.76, and the quantitative recursion in theorem 107.77. The point of this ledger section is to isolate every place where those arguments used unit real-time semantics and to give its weak-macro replacement.

Let $\mathcal{M}$ be the finite palette of finite weak macro types after refining by source, target, visible letter, threshold, effect, and concrete simple skeleton identity. Let $\mathcal{P}$ be the finite palette of pump witness types. Put

$$R_* = \max\big(\{r_e : e \in \mathcal{M}\} \cup \{r_p : p \in \mathcal{P}\} \cup \{0\}\big)$$

and

$$E_* = \max\big(\{|\delta_e| : e \in \mathcal{M}\} \cup \{0\}\big).$$

Both are effectively computable.

107.33.1 Per-step observable bounds

On a fixed finite regime:

- one N/N projective difference changes by at most $2E_*$;
- one anchored A/N relative coordinate changes by at most $E_* + 1$, since the deterministic stable step has unit effect;
- a common anchored N-credit observable changes by at most E_*;
- support, provenance, owner, and omega masks range over finite sets.

Thus one conservative universal first-exit constant is

$$D_* = \max\{1, 2E_* + 2\}.$$

Whenever a tracked integer observable first exits $[-M, M]$, it lies in the finite band

$$M < |z| \le M + D_*.$$

This is the only change required in first-exit localization.

107.33.2 Threshold events

Every finite macro enabling change occurs at one of the finitely many thresholds $r_e \le R_*$. Every pump enabling change occurs at one of the finitely many pump thresholds $r_p \le R_*$. Hence a first semantic-regime split is localized either by a finite support/omega event or by a source counter in a bounded band around one of these thresholds. For a recurrent threshold contact the color records this bounded source credit and finite mode data, but not the numerical separator margin; by theorem 107.106 the resulting contact alphabet is finite and independent of the future backward requirement.

107.33.3 Projective cutoffs

Any real-time estimate that bounded the accumulated effect along a simple synchronized path by the number of vertices is replaced by the same path length multiplied by the appropriate macro differential bound. In particular, if a simple residual path contains at most $|V| - 1$ macro edges, its N/N differential contribution has absolute value at most

$$(|V| - 1)\, 2E_*.$$

Thus every explicit projective-spread cutoff remains computable after one finite rescaling.

107.33.4 Transfer certificate

The structural use of these constants is summarized in the proof of theorem 107.113. No additional semantic assumption is hidden in this ledger section: all owner/provenance, wall/contact, WQO, terminal-oracle, and powered-route requirements are exactly (G1)–(G9). The purpose of the present ledger is only to make the finite rescalings R_*, E_*, and D_* explicit.

107.33.5 Literal-trace interface

The transferred proof may delete only cycles/packages extracted from the actual observed trace. Accordingly, each macro edge used by an owner/provenance strand stores one concrete simple weak skeleton. Consecutive selected macro edges concatenate to an actual N micro-run. A nonempty synchronized macro cycle therefore consumes a nonempty contiguous visible factor. Alternative VASS paths may be used for decidability or fixed-route checks, but never as a substitute for the literal observed trace in a shortest-word deletion.

107.34 Formal one-zero-test product gadget

This ledger section records the low-level compilation used in section 107.27. It is included to make the exact match with the external one-zero-test reachability theorem transparent.

107.34.1 Semantic product edges

Before compilation, a semantic product edge records:

- one source product control and one target product control;
- the chosen N transition, when the edge is visible;
- the unique deterministic A rule in the current sign regime;
- the ordinary vector effect on (x_N, x_A);
- whether the A source guard is positive or zero.

N-epsilon and positive reset-decrement edges are already ordinary VASS transitions and need no special treatment.

107.34.2 Positive-source gadget

For a positive-source semantic edge with effect (d_N, d_A) and $d_A \in \{-1, 0, +1\}$, introduce one private intermediate control and replace the edge by

$$(d_N, -1) \quad \text{followed by} \quad (0, d_A + 1).$$

The first transition is enabled exactly when the original N update is feasible and $x_A \geq 1$. The second transition only adds a nonnegative amount to x_A. No other transition is incident from the private intermediate control, so the gadget is atomic with respect to all asynchronous behavior.

107.34.3 Unique zero-test gate

Let E_0 be the finite set of zero-source semantic product-edge types. For each $e \in E_0$ introduce an ordinary tag counter t_e, initially zero. Every genuine product configuration has all tags zero.

For an edge e from genuine source s_e to genuine target t_e^{ctl}:

1. an ordinary source transition from s_e enters the common pre-gate control and increments only the tag counter t_e;
2. the common pre-gate has one outgoing distinguished transition g_0, enabled exactly when $x_A = 0$, with zero vector effect;
3. the common post-gate has one completion branch per semantic edge type; branch e decrements the tag t_e, performs the original ordinary vector update of e, and enters t_e^{ctl}.

All unchosen tags are zero, so their completion branches are disabled by ordinary VASS nonnegativity. The pre/post-gate controls expose no product, epsilon, visible, endmarker, or cleanup transitions other than the gadget moves just listed.

If the ordinary N component of the semantic update is infeasible, a run may cross g_0 but cannot complete the chosen branch and is trapped away from all genuine accepting controls. Hence incomplete gate attempts create no spurious accepting reachability.

It follows that a genuine source-to-target zero-edge simulation exists if and only if the chosen semantic edge is at its proper source, $x_A = 0$, and its ordinary vector update is feasible. The compiled system therefore has exactly one zero-test transition, namely g_0.

107.34.4 Reset decompression and asynchronous commutation

For a reset state of period p, the phase controls

$$s[0], \dots, s[p-1]$$

perform positive silent decrements

$$s[i] \xrightarrow{\varepsilon,-1} s[i+1 \bmod p]$$

until $x_A = 0$, followed by the appropriate zero exit. N-epsilon steps modify only x_N and the N control; reset microsteps modify only x_A and the A control. Since neither consumes visible input, any interleaving of these two families commutes to an interleaving with the same component endpoints and counter values. Thus decompression does not change the synchronized visible language of the product.

107.34.5 Exact cleanup target

After the unique marked accepting genuine product control, enter a private cleanup phase. Each ordinary counter, including x_N, x_A, and all tags, has a decrement loop. A zero-effect edge leads to one fixed target control. The reachability query is for that control with the exact all-zero vector. Taking the final edge before complete drainage does not satisfy the query, and no edge returns from cleanup to the product. Hence arbitrary nonnegative accepting counter values are equivalent to exact reachability of one fixed all-zero configuration.

Part X

Unified Realization Theory and External Scalar-Field Generativity

Part guide. **Question.** Can the geometric, chronological, normalization, protected-information, and holonomy branches be placed in one typed exact-realization architecture without identifying objects that carry different mathematical meanings, and does that architecture generate useful structure outside pushdown automata? **Main result.** USNRT supplies exact path-outcome factorization, fixed-template section debt, protected-obligation capacity, product and transfer calculus, same-model saturation, typed holonomy/resource bridges, and four-axis source recovery; EF–FG then provides an independent scalar-field theory with observer-record normal form, exact common-realization semantics, a polynomial/NP-complete local–global boundary, and all-orders obstruction families; its second stage adds shallow spatial-frame NP-completeness, universality of finite common-realization complexes, fixed-parameter tractable ambiguity regimes, correlated rational-polyhedral semantics, and finite layered transfer. **Why here.** Every source axis required by the recovery theorem has now been developed, so the synthesis can be stated without forward dependence. **Dependency.** USNRT depends only on previously proved source structures; EF–FG is logically independent as a scalar-field theory and is used afterward as an external generativity test.

Chapter 108

Exact Realizations, Path Outcomes, and Fixed-Template Debt

Research branch — Unified Semantic Normalization and Realization Theory. *The preceding General Theory and post-theory branches developed exact lifts, completion geometry, chronology, protected information, topology, and physical exposure separately. This chapter begins a typed synthesis. Its purpose is not to replace those source theories, but to state which objects may be identified, which must remain distinct, and which exact factorization statements survive after the source-specific coordinates are forgotten.*

The exact-realization object below is the same semantic/physical architecture already developed in the General Theory: a reachable deterministic semantic action system, a deterministic physical realization, a surjective semantic projection, domain reflection, and exact action intertwining. We restate the data because the present Part uses them as a typed host for several earlier branches. A canonical section remains optional extra structure rather than semantic identity.

108.1 Typed architecture and theorem scope

An enriched USNRT object may carry, in addition to the exact realization, independent geometry, chronology, observation, protected-contract, category, and resource data, which we denote schematically by

$$\mathfrak{U} = (R; \mathcal{G}_\bullet, \mathcal{J}_\bullet, \mathcal{O}_\bullet, \mathcal{B}, \mathsf{Cat}, \varrho).$$

The notation is packaging only: the enrichments are not required simultaneously by every theorem. Geometry may be absent from a purely chronological specialization; chronology may be absent from a purely geometric one; a resource evaluation is meaningful only after a physical category has been declared.

Proposition 108.1 (Typing and non-collapse principle)**.** *The USNRT axioms do not identify any of the following pairs by definition: semantic state and physical representative; representation fiber and compatibility fiber; section debt and behavioral failure; exact chronology and its shadow; intrinsic marked topology and physical transport; protected obligation pressure and resource cost. Any theorem transferring information between two such layers must state the required bridge data explicitly.*

Proof. The primitive data are typed independently. The exact realization contains only the semantic/physical projection and action laws. A section is optional extra structure. Geometry, chronology, observation, protected contracts, physical category, and resource evaluation are independent enrichments. The derived objects used later—path-outcome fibers, obligation complexes, holonomy orbits, and Pareto/resource bounds—are formed only after the corresponding enrichments are supplied. Therefore none of the listed identifications follows from the base definitions. □

This proposition is the main scope discipline of the present Part. It is deliberately negative, but it is mathematically useful: it tells us exactly which hypotheses must appear when a geometric, chronological, or topological distinction is converted into protected information or physical cost.

108.2 Semantic action systems and exact realizations

Definition 108.2 (Semantic action system)**.** A rooted deterministic semantic action system is a tuple

$$S = (M, m_0, \Sigma, U),$$

where M is a set, $m_0 \in M$, Σ is an action alphabet, and each $U_a : M \rightharpoonup M$ is a partial function. We restrict all statements to the reachable part of M.

A semantic morphism $f : S \to S'$ preserves roots, definedness on reachable states, and actions: $f(U_a(m)) = U'_a(f(m))$ whenever $U_a(m)$ is defined.

Definition 108.3 (Exact realization)**.** An exact realization of S is

$$R = (S; C, c_0, \delta; \Theta),$$

where C is a physical state set with root c_0, each $\delta_a : C \rightharpoonup C$ is deterministic and partial, and

$$\Theta : C_{\mathrm{reach}} \to M_{\mathrm{reach}}$$

is surjective, root preserving, domain reflecting, and action intertwining:

$$c \in \mathrm{Dom}(\delta_a) \iff \Theta(c) \in \mathrm{Dom}(U_a), \qquad \Theta(\delta_a(c)) = U_a(\Theta(c)).$$

An exact-realization morphism $(f, g) : R \to R'$ consists of a semantic morphism f and a rooted physical map g that intertwines physical actions and satisfies $\Theta' g = f\Theta$.

Proposition 108.4. *Exact realizations and exact-realization morphisms form a category* Real_Σ.

Proof. Identity pairs satisfy the defining conditions. If $(f, g) : R \to R'$ and $(f', g') : R' \to R''$, then roots and physical intertwining are preserved by composition and

$$\Theta'' g' g = f' \Theta' g = f' f \Theta.$$

Associativity is inherited from function composition. □

Theorem 108.5 (Identity realization embedding)**.** *The assignment*

$$I(S) = (S; M, m_0, U; \mathrm{id}_M), \qquad I(f) = (f, f)$$

defines a fully faithful functor $I : \mathsf{SemAct}_\Sigma \to \mathsf{Real}_\Sigma$.

Proof. Exactness is immediate for the identity projection. A morphism between identity realizations is determined by its semantic component, so the hom-set map is bijective. □

This embedding is useful conceptually: semantics is not a quotient imposed after the fact. It sits fully faithfully inside the realization category, while nontrivial realizations add physical multiplicity.

108.2.1 Sections are optional structure

Definition 108.6 (Sectioned realization)**.** A sectioned realization is an exact realization together with a map

$$\sigma : M_{\text{reach}} \to C_{\text{reach}}, \qquad \Theta\sigma = \mathrm{id}.$$

When root-zero debt is required we also impose $\sigma(m_0) = c_0$.

The representation fiber is

$$\mathrm{Fib}_{\text{rep}}(m) = \Theta^{-1}(m).$$

A section chooses one representative from each fiber, but no theorem identifies this choice with semantic identity. This distinction will matter below.

108.3 Path outcomes, canonicalization, and fixed-template debt

Fix a rooted exact realization. For an admissible word w, let

$$e_M(w) = U_w(m_0), \qquad e_C(w) = \delta_w(c_0).$$

Define two equivalence relations on rooted admissible words by equality of semantic and physical endpoints, respectively. Exactness gives

$$e_C(u) = e_C(v) \Longrightarrow e_M(u) = e_M(v),$$

but the converse may fail.

For a reachable semantic state m, define its path-outcome fiber

$$H_{c_0}(m) = \{\delta_w(c_0) : U_w(m_0) = m\}.$$

This is distinct from the full representation fiber $\Theta^{-1}(m)$: it records only physical representatives actually produced by rooted semantic paths.

Theorem 108.7 (Rooted endpoint factorization)**.** *The following are equivalent.*

(a) There is a rooted transition-closed section σ through c_0.

(b) The physical endpoint map factors through the semantic endpoint map: $e_C = \sigma e_M$.

(c) Semantic endpoint equivalence equals physical endpoint equivalence.

(d) Every path-outcome fiber $H_{c_0}(m)$ is a singleton.

When these conditions hold, the section is uniquely determined by c_0.

Proof. Transition closure implies $\delta_w(c_0) = \sigma(U_w(m_0))$, giving the factorization. The factorization identifies physical endpoints whenever semantic endpoints agree; the reverse implication already follows from exactness. Equality of endpoint equivalences is precisely singleton path-outcome fibers. Finally, if each fiber is a singleton, define $\sigma(m)$ to be its unique rooted path outcome; exact intertwining makes the section transition closed. □

Assume the relevant fibers are finite and define $\mu_{c_0}(m) = |H_{c_0}(m)|$. For a finite semantic test set T, set

$$\Delta_T(c_0) = \sum_{m \in T} (\mu_{c_0}(m) - 1).$$

For any rooted section σ, let $\mathrm{Debt}_T(\sigma)$ count path outcomes over T that do not lie on the selected section.

Theorem 108.8 (Exact section-defect formula)**.** *For every finite test region T,*

$$\min_{\sigma} \mathrm{Debt}_T(\sigma) = \Delta_T(c_0).$$

Proof. At a state m, a single section value can cover at most one element of $H_{c_0}(m)$, so at least $\mu_{c_0}(m) - 1$ outcomes remain off-section. Summing gives the lower bound. Choose one path outcome at each $m \in T$ as the section value and extend arbitrarily on other reachable states; then exactly one path outcome at each m lies on the section. □

Corollary 108.9 (Fixed-template debt)**.** *If $\mu_{c_0}(m) \geq 2$ for some reachable m, then no transition-closed section through c_0 exists. Refining only the observation map cannot remove this defect while the physical transition template and root representative remain fixed.*

Thus canonicalization failure is a path-factorization obstruction, not an observation-capacity statement. Observation becomes relevant only after additional protected future distinctions are specified.

Chapter 109

Protected Obligations, Products, and Same-Model Saturation

Research branch — Protected obligation and transfer calculus. *The abstract collapse-capacity theorem and the later protected-contract/resource chapters already show that semantic distinctions can force physical pressure. The present chapter packages that mechanism as an obligation complex, develops its product and asymptotic calculus, and then separates abstract Cartesian product realization from simultaneous compilation inside one restricted physical category.*

109.1 Protected obligations and compatibility complexes

A protected obligation problem adds an observation level, confinement semantics, and a protected future contract to an exact realization. We suppress inessential data and write P_h for a finite protected problem with finite semantic source family S.

Definition 109.1 (Obligation complex)**.** A subset $A \subseteq S$ is *one-cell compatible at level h* if the sources in A can be realized within one common level-h observation cell, under the declared confinement condition and while satisfying the protected contract. Empty sets and singletons are compatible by convention. The downward-closed family of compatible subsets is the obligation complex

$$\mathrm{Ob}_h(P_h).$$

Its face-cover number $\kappa_h(P_h)$ is the minimum number of faces whose union covers S.

Definition 109.2 (Protected distinguishability certificate)**.** A map $d : S \to Z$ is a protected distinguishability certificate if every face of $\mathrm{Ob}_h(P_h)$ lies in one fiber of d.

Let D be the number of source vertices allowed to leave the declared canonical interface, X_h the number of remaining canonical sources whose obligation-realizing computation is not level-h confined, and

$$V_S(h) = |\omega_h(\sigma(S))|$$

the number of observation cells occupied by canonical confined sources.

Theorem 109.3 (Obligation-cover capacity)**.** *If every actual level-h observation class of canonical confined sources is a face of* $\mathrm{Ob}_h(P_h)$*, then*

$$D + X_h + V_S(h) \geq \kappa_h(P_h).$$

Proof. Remove the D sources permitted to leave the canonical interface and the X_h canonical sources not satisfying confinement. The remaining sources are partitioned by their actual observation cells, and each such cell is a face by hypothesis. Together with the exceptional singletons, these faces cover all source vertices, so their number is at least the minimum face-cover number. □

Corollary 109.4 (Distinguishability-capacity transfer)**.** *If $d : S \to Z$ is a protected distinguishability certificate, then*

$$\kappa_h(P_h) \geq |d(S)|, \qquad D + X_h + V_S(h) \geq |d(S)|.$$

Proof. A face lies inside one fiber of d, so a face cover must contain at least one face for each represented value of d. Apply the preceding theorem. □

This inequality is intentionally not a physical resource lower bound by itself. A resource conclusion requires a category-specific capacity envelope relating $D+X_h+V_S(h)$ to the resource evaluation.

109.1.1 Refinement and resource exclusion

Strengthening a protected contract can delete compatible faces but cannot create new ones. Hence obligation complexes are contravariantly monotone under contract refinement, while their minimum face-cover number is nondecreasing. If a physical category has a resource bound r forcing

$$D + X_h + V_S(h) \leq B_h(r),$$

then any problem with $\kappa_h(P_h) > B_h(r)$ is excluded below resource r. This is the generic route from protected incompatibility to a category-specific Pareto exclusion.

109.2 Products, amplification, and asymptotic obligation rate

Given finite simplicial complexes K on S and L on T, define their rectangular compatibility product $K \boxtimes L$ on $S \times T$ by declaring $A \subseteq S \times T$ a face when both coordinate projections are faces.

Theorem 109.5 (Rectangular product cover bounds)**.** *For nonempty finite complexes,*

$$\max\{\kappa(K), \kappa(L)\} \leq \kappa(K \boxtimes L) \leq \kappa(K)\kappa(L).$$

Both bounds are sharp in general.

The upper bound comes from products of faces in optimal covers. Projecting any cover of $K \boxtimes L$ yields covers of each factor, giving the lower bound. No universal exact product law exists without additional structure.

For repeated rectangular powers $K^{\boxtimes n}$, submultiplicativity implies existence of the asymptotic rate

$$h_{\mathrm{obl}}(K) = \lim_{n\to\infty} \frac{1}{n} \log \kappa(K^{\boxtimes n}).$$

Let $\kappa_f(K)$ be the fractional face-cover number. The classical linear-programming fractional cover formalism [144, 147] yields exact multiplicativity under rectangular powers. A standard probabilistic rounding argument then gives the exact asymptotic identity

$$h_{\mathrm{obl}}(K) = \log \kappa_f(K). \tag{109.1}$$

The combinatorial shell of (109.1) is classical; its role here is to quantify repeated protected-realization pressure.

The equality (109.1) is a statement about the derived compatibility complex. A physical interpretation requires an additional product-realization theorem.

Definition 109.6 (Iterated rectangular realization)**.** Let P_h be a protected problem with $K = \mathrm{Ob}_h(P_h)$. An iterated rectangular realization family is a sequence $P_h^{(n)}$ such that

$$\mathrm{Ob}_h(P_h^{(n)}) \cong K^{\boxtimes n}$$

for every $n \geq 1$, with the declared protection and confinement semantics preserved coordinatewise.

Theorem 109.7 (Asymptotic capacity-rate exclusion)**.** *Suppose P_h admits an iterated rectangular realization family and suppose resource bounds r_n imply certified capacity bounds*

$$D_n + X_n + V_n \leq B_n.$$

If

$$\limsup_{n\to\infty} \frac{1}{n} \log B_n < h_{\mathrm{obl}}(K) = \log \kappa_f(K),$$

then, for all sufficiently large n, no admissible n-fold realization can satisfy resource bound r_n.

Proof. Let $\kappa_n = \kappa(K^{\boxtimes n})$. By (109.1), $(1/n)\log \kappa_n$ converges to $h_{\mathrm{obl}}(K)$. Choose a number λ strictly between the two exponential rates. Eventually $\kappa_n > e^{\lambda n} > B_n$. This contradicts the one-shot obligation-capacity inequality $\kappa_n \leq D_n + X_n + V_n \leq B_n$ for any admissible realization. □

Corollary 109.8 (Mixed protected rate)**.** *If one protected problem carries two jointly independent finite certificates of ranks m and n, and their iterated physical product realization exists, then*

$$h_{\mathrm{obl}} \geq \log m + \log n.$$

This is a protected-realization rate statement, not an identification with Shannon entropy.

109.3 Transfer calculus and product realization

A finite distinction transfer $\tau : X \to Y$ has logarithmic loss

$$L(\tau; X) = \log |X| - \log |\tau(X)|.$$

This measures cardinality loss only; reversibility is a stronger structured property.

Theorem 109.9 (Exact transfer-loss composition)**.** *For a chain of effective-image-normalized transfers*

$$X_0 \xrightarrow{\tau_1} X_1 \xrightarrow{\tau_2} \cdots \xrightarrow{\tau_k} X_k,$$

$$\log |X_k| = \log |X_0| - \sum_{i=1}^{k} L(\tau_i; X_{i-1}).$$

Equivalently, retention ratios multiply.

Proof. The logarithmic differences telescope. □

When the surviving terminal distinctions are protected, the obligation-capacity theorem converts the surviving rank into physical pressure. This gives a precise way to account for lossy bridges rather than asserting that information is "preserved" informally.

109.3.1 Exact product realization

Two exact realizations admit an asynchronous stutter-completed product: semantic and physical states are Cartesian products, and actions update one or both coordinates with the other coordinate stuttering. The projection and, when present, the section are coordinatewise products.

Theorem 109.10 (Exact product realization and protected product)**.** *The asynchronous product is an exact realization. For coordinate-separable protected contracts,*

$$\mathrm{Ob}_h(P_1 \otimes P_2) = \mathrm{Ob}_h(P_1) \boxtimes \mathrm{Ob}_h(P_2).$$

Proof. Surjectivity, domain reflection, and action intertwining hold coordinatewise. For protected compatibility, any product witness projects to witnesses in the factors. Conversely, factor witnesses can be run asynchronously with stutters, preserving coordinatewise confinement, observation, and protected obligations. □

Suppose the first problem has protected certificate $g : S_G \to G$ with $|g(S_G)| = m$, and the second has $j : S_J \to J$ with $|j(S_J)| = n$. Then the product certificate (g, j) has full image $g(S_G) \times j(S_J)$ and hence rank mn. Therefore

$$D + X_h + V_h \geq mn.$$

This is an abstract mixed-axis theorem. It does not assert that a restrictive one-stack pushdown category is closed under such products.

109.4 Same-model transfer saturation

The product theorem separates abstract coordinate products from compilation inside one restrictive physical model. Fix finite protected label sets A, B, with $|A| = m$ and $|B| = n$, and fix a declared realization category Cat together with its confinement and repair contract.

Definition 109.11 (Same-model compiler class)**.** An object of $\mathrm{Comp}_{\mathsf{Cat}}(A, B)$ is a tuple

$$P = (R_P, S_P, \lambda_P, \Pi_A, \Pi_B)$$

where R_P is one realization object in Cat, S_P is a finite protected source family inside that same realization, and

$$\lambda_P : S_P \longrightarrow A \times B$$

is a joint protected-label map. The contracts Π_A and Π_B require the coordinate projections $\pi_A\lambda_P$ and $\pi_B\lambda_P$ to realize the prescribed marginal protected families under the declared confinement/repair semantics. No product closure of Cat is assumed. The joint image is

$$J_P := \lambda_P(S_P) \subseteq A \times B.$$

Define

$$\mathrm{Sat}_{\mathsf{Cat}}(A, B) = \max_{P \in \mathrm{Comp}_{\mathsf{Cat}}(A,B)} |J_P|,$$

when the class is nonempty and the finite maximum is attained. The normalized saturation ratio is $\mathrm{Sat}_{\mathsf{Cat}}/(mn)$, and the logarithmic loss of one compiler is $\log(mn/|J_P|)$. Thus the optimal same-model loss is

$$L^*_{\mathsf{Cat}} = \log(mn) - \log \mathrm{Sat}_{\mathsf{Cat}}(A, B).$$

Proposition 109.12 (Full-saturation criterion)**.** *The following are equivalent:*

(a) $\mathrm{Sat}_{\mathsf{Cat}}(A, B) = mn$*;*

(b) the optimal loss is zero;

(c) the category contains a marginal-preserving compiler whose joint image is all of $A \times B$*.*

Product closure is sufficient but not necessary: a category may contain a bespoke full-rank compiler for one pair without admitting arbitrary Cartesian products.

109.4.1 Pairwise defect versus assembly defect

Let

$$E_{\mathsf{Cat}}(A, B) = \bigcup_{P \in \mathrm{Comp}_{\mathsf{Cat}}(A,B)} J_P$$

be the pairwise envelope and $p_{\mathsf{Cat}} = |E_{\mathsf{Cat}}(A, B)|$. Define

$$\delta_{\mathrm{pair}} = mn - p_{\mathsf{Cat}}, \qquad \delta_{\mathrm{asm}} = p_{\mathsf{Cat}} - \mathrm{Sat}_{\mathsf{Cat}}(A, B).$$

Proposition 109.13 (Saturation-defect identity)**.**

$$mn - \mathrm{Sat}_{\mathsf{Cat}}(A, B) = \delta_{\mathrm{pair}} + \delta_{\mathrm{asm}}.$$

Both defects are nonnegative, and full saturation occurs exactly when both vanish.

Proof. By definition,

$$mn - \mathrm{Sat}_{\mathsf{Cat}} = (mn - p_{\mathsf{Cat}}) + (p_{\mathsf{Cat}} - \mathrm{Sat}_{\mathsf{Cat}}).$$

The envelope contains every joint image, so $p_{\mathsf{Cat}} \geq \mathrm{Sat}_{\mathsf{Cat}}$; also $p_{\mathsf{Cat}} \leq mn$. Hence both defects are nonnegative. Their simultaneous vanishing is equivalent to $p_{\mathsf{Cat}} = \mathrm{Sat}_{\mathsf{Cat}} = mn$. □

The interpretation is important. A positive pairwise defect means that some label pairs are never jointly realizable. A positive assembly defect means that every pair may occur somewhere, while no single physical realization assembles all available pairs simultaneously. Thus pairwise compatibility and simultaneous compilation are different questions even when they use the same marginal labels.

Proposition 109.14 (Elementary saturation bounds)**.** *Whenever the compiler class is nonempty and every compiler is marginal-surjective,*

$$\max\{m, n\} \leq \mathrm{Sat}_{\mathsf{Cat}}(A, B) \leq mn.$$

Moreover, if $\mathsf{Cat}_1 \subseteq \mathsf{Cat}_2$ *with the same protected witness interpretation, then*

$$\mathrm{Sat}_{\mathsf{Cat}_1}(A, B) \leq \mathrm{Sat}_{\mathsf{Cat}_2}(A, B).$$

Proof. For any marginal-surjective joint image $J \subseteq A \times B$, both coordinate projections are onto, so $|J| \geq m$ and $|J| \geq n$. The upper bound is $|J| \leq |A \times B| = mn$. Enlarging the admissible compiler class can only enlarge the set over which the maximum is taken. □

109.4.2 Shared-interface bounds

If every compiler factors its joint certificate through an interface of capacity at most C, then

$$\mathrm{Sat}_{\mathsf{Cat}}(A, B) \le C.$$

Hence $C < mn$ forces positive same-model information loss, with

$$L^*_{\mathsf{Cat}} \ge \log \frac{mn}{C}.$$

More generally, if a resource bound r gives a capacity envelope $B_h(r)$, then the resource-bounded saturation profile obeys

$$\mathrm{Sat}_{\mathsf{Cat}}(r; A, B) \le B_h(r).$$

Proposition 109.15 (Optimal same-model transfer loss)**.** *The minimum logarithmic loss among marginal-preserving compilers is exactly*

$$L^*_{\mathsf{Cat}} = \log \frac{mn}{\mathrm{Sat}_{\mathsf{Cat}}(A, B)}.$$

An independent capacity bound $\mathrm{Sat}_{\mathsf{Cat}} \le C$ *gives a lower bound* $L^*_{\mathsf{Cat}} \ge \log(mn/C)$*; an explicit compiler of joint rank* R *gives* $L^*_{\mathsf{Cat}} \le \log(mn/R)$*. Matching bounds certify exact loss optimality.*

Proof. Every admissible compiler has joint rank $R \le \mathrm{Sat}_{\mathsf{Cat}}$ and hence loss at least

$$\log \frac{mn}{\mathrm{Sat}_{\mathsf{Cat}}}.$$

By definition of saturation, some admissible compiler attains rank $\mathrm{Sat}_{\mathsf{Cat}}$ whenever the finite maximum is attained, and its loss is exactly the displayed value. The capacity and construction estimates follow monotonically from the same formula. □

109.4.3 A source-stack capacity specialization

The abstract theory does not prescribe a stack or memory measure. The required capacity law has already been proved inside this monograph. The filtered collapse-capacity theorem, Theorem 62.1, gives

$$D + X_h + V(h) \ge N$$

for a collapse family of size N under hidden-tail noncoalescence. The representation-independent pushdown specialization, Theorem 96.3 in Chapter 96, then gives

$$\boxed{D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g^j_{\mathrm{src}} \ge N.}$$

Its proof verifies hidden-tail noncoalescence for deterministic top-access ε-pushdown sources and counts depth-h source signatures.

Choose one protected source representative for each distinct joint label in a same-stack compiler of joint rank R; the resulting collapse family has size $N = R$. Applying Theorem 96.3 gives the necessary inequality

$$\boxed{R \le D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g^j_{\mathrm{src}}.}$$

Consequently, a zero-debt, h-confined compiler with full joint rank mn must satisfy

$$q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq mn.$$

For $g_{\mathrm{src}} \geq 2$, this yields

$$h \geq \left\lceil \log_{g_{\mathrm{src}}} \left(1 + \frac{(g_{\mathrm{src}} - 1)mn}{q_{\mathrm{src}}} \right) \right\rceil - 1.$$

Remark 109.16 (Necessary, not sufficient). The source-stack exposure theorem is a lower-bound/capacity statement. It neither constructs a one-stack compiler meeting the bound nor asserts full rectangular independence for any particular pair of geometric and chronological witnesses. Those are category-specific construction questions.

109.5 Source-native shortest fibers and protected paired-session comparison

The existence of optimal fibers and their marked comparison groups was established in the geometric chapters. A single unit-cost source can realize both a prescribed support complex and its literal shortest Parikh fiber, with a separate operational question of reading marked comparisons from genuine source words. A paired-session service has a designated root reset and fixed block length; this is not an unframed single-start machine.

109.5.1 Native source and comparison semantics

Let $A \in \mathbb{Z}^{d \times n}$ be an integer counting source on letters $a_1, \ldots, a_n$, with each letter costing one. For $b \in \mathbb{Z}^d$ with nonnegative solutions set

$$\ell_A(b) = \min\{\mathbf{1}^{\mathsf{T}} z : z \in \mathbb{N}^n,\ Az = b\},$$
$$P_A(b) = \{z \in \mathbb{N}^n : Az = b,\ \mathbf{1}^{\mathsf{T}} z = \ell_A(b)\}.$$

The source is *completion-total* when $Ah = 0$ for some strictly positive integer vector h: any integral solution may then be shifted by a positive multiple of h to become nonnegative. For an optimal fiber P, its active lattice is $L(P) = \langle P - P \rangle_{\mathbb{Z}}$. Choose a vertex $p_0 \in P$ and write $\widehat{P} = P - p_0 \subset L(P)$. Its periodic nerve and quotient are

$$\mathcal{N}(\widehat{P}) = \{F \subset L(P) : F \text{ finite},\ \exists c \in L(P),\ c - F \subset \widehat{P}\},$$
$$Q(P) = |\mathcal{N}(\widehat{P})|/L(P).$$

The quotient is independent of the chosen translation vertex, up to its natural identification. The geometric compression of shortest fibers and the marked difference presentation for $Q(P)$ are established in [194, 195, 196]. In particular, a primitive fiber with m affinely independent vertices has a comparison group F_{m-1}, with displacement onto the rank-$(m-1)$ active lattice given by abelianization; this background is not reproved as a new topological theorem here. Prior work on semigroup realizations and isolated factorizations supplies related constructions and structural restrictions [197, 198]; it does not by itself assert the exact simultaneous source-and-service contract defined below.

Three types of operational statements must be distinguished. A *whole-language recognizer* answers whether $A\,\mathrm{Parikh}(w) = 0$ for an arbitrary word. A *paired-session comparator* receives fixed-length successive rooted shortest sessions, with externally specified session resets. A *single uninterrupted source trace* has neither those resets nor a separate mode selector. A pushdown realization in one category is not silently promoted to either of the others.

109.6 An exact facet-and-fiber compiler

Let K have labeled vertices $1, \dots, s$ and facets $F_0, \dots, F_{m-1}$, with $m \geq 1$. Write $v_{ji} = \mathbf{1}_{\{i \in F_j\}}$ and $d_i = \sum_j v_{ji}$. On $(x_i^+, x_i^-)_{i=1}^s, (y_j)_{j=0}^{m-1}, t$, impose

$$x_i^+ - \sum_j v_{ji} y_j + (d_i - 1)t = 0, \tag{109.2}$$

$$x_i^- - \sum_j (1 - v_{ji}) y_j + (m - d_i - 1)t = 0, \tag{109.3}$$

$$\sum_j y_j - mt = 1. \tag{109.4}$$

These are the row equations for a single $(2s+1) \times (2s+m+1)$ integer matrix A_K and one target b.

Theorem 109.17 (Simultaneous source-native realization)**.** *The source A_K is completion-total with the all-ones kernel. Its unit-cost shortest completion length is $s+1$, and the entire optimal Parikh fiber consists of exactly m vectors $p_0, \dots, p_{m-1}$, with*

$$t = 0, \quad y_j = 1,\ y_k = 0\ (k \neq j), \quad x_i^+ = v_{ji}, \quad x_i^- = 1 - v_{ji}.$$

On the designated letters x_i^+, the optimal support complex is precisely K. The fiber is a primitive integral $(m-1)$-simplex in its active lattice. Consequently its native quotient has $\pi_1 Q(P) \cong F_{m-1}$ with natural displacement the abelianization to $\mathbb{Z}^{m-1}$.

Proof. Every row vanishes at the all-ones vector. Adding (109.2) and (109.3) gives $x_i^+ + x_i^- = 1 + 2t$, while (109.4) gives $\sum_j y_j = 1 + mt$. Thus every nonnegative solution has length

$$s(1 + 2t) + (1 + mt) + t = (s + 1) + (2s + m + 1)t.$$

The minimum forces $t = 0$, where the selector y is one-hot, giving exactly the stated p_j. Its positive-letter supports are exactly the facets, and their downward closure is K. The differences $p_j - p_0$ are an integral basis of their span because their selector coordinates $y_1, \dots, y_{m-1}$ project to the standard basis. The quotient and its marked group follow from the minimal-active-fiber theorem theorem 92.4, proved earlier, with the basis $p_j - p_0$ fixing the displacement coordinates. □

109.7 Whole-language recognition after lossless scalar folding

For a matrix source $A \in \mathbb{Z}^{d \times n}$, target b, and known shortest length L, let

$$M = \max\{1, \max_{j,i} |A_{ji}|, \max_j |b_j|\}, \qquad R > (L+1)M$$

with integer R. Define scalar letter effects and scalar target

$$\lambda_i = \sum_{j=0}^{d-1} A_{ji} R^j, \qquad \beta = \sum_{j=0}^{d-1} b_j R^j. \tag{109.5}$$

Theorem 109.18 (Literal preservation of every shortest completion)**.** *For every nonnegative z of length at most L, $\sum_i \lambda_i z_i = \beta$ if and only if $Az = b$. Hence the scalar source has the same shortest length and* exactly the same *optimal Parikh fiber, supports, active lattice, and marked periodic quotient. If $Ah = 0$ with $h > 0$, then also $\lambda h = 0$.*

Proof. Write $e_j = \sum_i A_{ji} z_i - b_j$. Then $|e_j| \le (L+1)M < R$. The scalar equality is $\sum_j e_j R^j = 0$. Taking residues modulo R forces $e_0 = 0$. Divide by R and repeat; every $e_j = 0$. All original length-L solutions persist and no shorter scalar solutions are introduced. The remaining claims are literal invariance of the fiber and linearity of (109.5). □

Proposition 109.19 (Real-time one-stack zero-return recognition)**.** *For any integer scalar weights $(\lambda_1, \dots, \lambda_n)$, the language $\{w : \sum_i \lambda_i \#_{a_i}(w) = 0\}$ has a deterministic real-time one-stack recognizer, using at most one push or pop per input symbol and no empty-input transitions.*

Proof. Set $D = \max(1, \max_i |\lambda_i|)$. Represent the running total uniquely as $Dq + r$ with $0 \le r < D$. Keep r in finite control and q as a signed-unary stack above a bottom marker. On input letter a_i, write $r + \lambda_i = D\delta + r'$ with $0 \le r' < D$. Since $0 \le r < D$ and $|\lambda_i| \le D$, necessarily $\delta \in \{-1, 0, 1\}$. Thus the quotient changes by at most one; make the corresponding push, pop, or no-op and update the remainder to r'. The sign is represented by the stack alphabet $\{\bot, +, -\}$. Only the zero residue and empty signed stack accept. □

Remark 109.20. For general completion-total vector sources of rational rank at least two, whole-language context-freeness already fails by a standard two-independent-counters reduction, consistent with the group-language literature [199]. The folding theorem changes the matrix and retains only the specified shortest fiber: it does not preserve the entire vector zero-return language. Standard base encoding and one-counter recognition themselves are not claimed as new methods.

109.8 A source-faithful protected comparison service

Take the primitive optimal fiber $P = \{p_0, \dots, p_{m-1}\}$ from Theorem 109.17 or its folded scalar source, with minimum length L. Fix the source-native base vertex p_0. One *paired block* consists of two successive genuine shortest source words, each exactly L letters, and an interface reset between them to the same root. Restrict to pairs (p_0, p_j) and (p_j, p_0), $j \ge 1$, assigning symbols X_j and X_j^{-1} respectively. These are comparison labels extracted from source words, not new source input letters. Let $B = \bigcup_{p \in P} \{z \in \mathbb{N}^n : z \le p\}$ be the finite set of possible shortest-prefix Parikh vectors.

Theorem 109.21 (Protected paired-session comparison)**.** *With this explicitly selected service, there is a deterministic real-time single-stack machine on the* original source alphabet *that checks all session words, updates the freely reduced word in F_{m-1}, and accepts precisely the full paired streams representing the identity. It uses at most $(m+1)|B|+1$ verifier states, stack alphabet of size $1 + 2(m-1)$, and at most one push or pop on the final letter of a pair, with no empty-input moves. For $m \ge 3$ the service records nonidentity comparisons with zero natural displacement.*

Proof. The first-half verifier stores a prefix Parikh vector $z \in B$ in finite control, rejecting an impossible extension. The completed first half uses $|B|$ states; the second-half states are pairs (i, z) with $i \in \{0, \dots, m-1\}$ and $z \in B$, plus one rejecting sink. Thus the advertised $(m+1)|B|+1$ bound includes both phases and the sink. After L letters, z must equal one unique p_i; remember i and restart the verifier component at zero by the service clock. Check the second half similarly, deriving p_j. If the pair is allowed, read its single $X_k^{\pm 1}$ comparison label at the final source-letter transition. Cancel against the inverse top stack letter or push it; this maintains free reduction inductively. Accept at a completed block boundary if and only if only the bottom marker remains. There is no

input alphabet extension and one stack suffices. The chosen group generators correspond to the source's primitive optimal-fiber comparisons. For $m \geq 3$, $[X_1, X_2]$ is nontrivial in the free group but has zero displacement since its exponents sum to zero on every generator. □

Proposition 109.22 (Unbounded comparison requires memory)**.** *For $m \geq 2$, the identity language of complete admissible paired blocks cannot be recognized by a finite-state-only device, although the single-stack construction above recognizes it.*

Proof. Select two fixed genuine paired blocks U, V that produce X_1 and X_1^{-1}. The concatenation $U^a V^b$ represents identity precisely when $a = b$. For distinct a, a', the prefixes U^a and $U^{a'}$ are distinguished by suffix V^a. Thus infinitely many finite-state residuals are needed by the Myhill–Nerode criterion. The proof concerns faithful unbounded pair streams and does not give a minimum number of pushdown states or stack symbols. □

Proposition 109.23 (Sharp semantic restrictions)**.** *Suppose a shortest rooted session has nonzero target displacement b. A concatenation of $k \geq 2$ such sessions has displacement kb, and cannot be one original rooted shortest execution to b. Further, for any completion-total source with $h > 0$, every finite source word is the prefix of some zero-return source word. Thus no finite prefix can be reserved exclusively for irreversible comparison-mode routing while still recognizing every zero-return continuation.*

Proof. Displacements and lengths add, so the concatenation ends at kb and costs kL, not b in length L. For the prefix claim, if a word has Parikh vector c, choose N with $Nh \geq c$ and append any suffix of Parikh vector $Nh - c$. The completed word has net effect $ANh = 0$. The routing consequence follows immediately. □

Remark 109.24 (Precisely what the construction certifies). The whole-language counter recognizer and the paired comparison machine are *host-selected alternative services* using one physical stack per run. The theorem does not prove one fixed-start device that concurrently emits both outputs for every unframed raw input, or a physical directed path literally executing formal inverses. Its push and state counts are constructive upper bounds, not general resource minima. These type distinctions are part of the result, not merely implementation commentary.

109.9 A seven-letter exact certificate

The folded unit-cost scalar source

$$\lambda = (1, 13, 169, 2041, 2029, 2197, -6450), \qquad \beta = 2211$$

has $\lambda \cdot (1, \ldots, 1) = 0$ and precisely three shortest Parikh vectors of length three:

$$p_0 = (1, 1, 0, 0, 0, 1, 0), \quad p_1 = (1, 0, 1, 1, 0, 0, 0), \quad p_2 = (0, 1, 1, 0, 1, 0).$$

The corresponding source supports are $\{1, 2, 6\}$, $\{1, 3, 4\}$, $\{2, 3, 5\}$; the first three action types induce the boundary of a triangle. The active fiber is a primitive rank-two triangle, and its comparison group is the free group $F(X, Y)$ with the natural rank-two displacement marking. The global zero-return language is recognized by Proposition 109.19 using $D = 6450$ remainder-control states and signed-unary stack. The paired comparison service has $|B| = 19$, at most 77 verifier states and five stack symbols. Four pairs built from the genuine words $a_1a_2a_6$, $a_1a_3a_4$, and $a_2a_3a_5$ yield the reduced word $XYX^{-1}Y^{-1}$, which is nontrivial but has zero displacement. Indeed, the seven-coordinate nonnegative vectors of length at most three number $\sum_{j=0}^{3} \binom{6+j}{j} = 120$. Direct substitution into

$\lambda \cdot z = \beta$ gives no solutions of length below three and exactly the three displayed solutions at length three. Each has six distinct orderings; their downsets have union of size 19, and the four allowed ordered types of pairs therefore yield $4 \cdot 6^2 = 144$ six-letter words. The general assertions do not rely on this finite example.

Chapter 110

Marked Transport, Four-Axis Recovery, and Non-Transfer

Research branch — Typed synthesis and recovery. *A geometric fiber, a chronological shadow, a path-outcome fiber, a protected compatibility class, and a physical resource coordinate are not the same object. This chapter records the exact bridges that permit information to pass between them, proves source-faithful recovery for four theorem-bearing branches already developed in the monograph, and records the corresponding non-transfer principles.*

110.1 Marked topology, transport, and holonomy

Intrinsic topology does not itself specify a physical action. Let a marked displacement system be

$$\mathfrak{M} = (G, L, \partial),$$

where G is a loop group, L is a displacement group, and $\partial : G \to L$ is a homomorphism. The pure sector is $K_0 = \ker \partial$. A chosen physical transport representation is a homomorphism $\rho : G \to H$.

Theorem 110.1 (Pure-holonomy factorization)**.** *The following are equivalent:*

(a) $\rho(K_0) = \{e\}$*;*

(b) $K_0 \subseteq \ker \rho$*;*

(c) there is a unique homomorphism $\bar{\rho} : \mathrm{im}(\partial) \to H$ *such that* $\rho = \bar{\rho}\,\partial$*.*

Now suppose loops act on a physical carrier set Ω, and fix $c \in \Omega$. The orbit $A = G \cdot c$ is a set of physical path outcomes with a common semantic endpoint.

Theorem 110.2 (Orbit-to-section defect)**.** *If the orbit A is finite, then the path-outcome multiplicity at the common semantic state is at least $|A|$, and every section of the fixed realization incurs at least $|A| - 1$ off-section outcomes there. If the action is transitive on the orbit,*

$$|A| = [G : \mathrm{Stab}_G(c)].$$

A resource conclusion requires another bridge.

Theorem 110.3 (Protected holonomy-orbit transfer)**.** *Assume the orbit alternatives carry a protected readout $\beta : A \to Y$ whose distinct values cannot share one level-h observation cell under the declared confined repair contract. Then*

$$D + X_h + V_A(h) \geq |\beta(A)|.$$

If β is injective, the right side is $|A|$.

Thus the valid chain is

$$\begin{aligned} \text{marked topology} &\longrightarrow \text{chosen transport} \longrightarrow \text{orbit/holonomy} \\ &\longrightarrow \text{protected readout} \longrightarrow \text{obligation pressure} \\ &\longrightarrow \text{resource, if a capacity law is supplied.} \end{aligned}$$

Deleting any of the middle hypotheses is not justified in general.

110.1.1 A typed transfer theorem

The previous branches can be summarized without collapsing them into one implication.

Theorem 110.4 (Typed transfer chain)**.** *Let X be a finite structural witness family. Suppose a declared bridge produces a finite protected label image Y, and suppose every level-h observation class of confined canonical sources lies in one protected-compatible face. Then*

$$D + X_h + V_h \geq |Y|.$$

If, in addition, a category-specific resource envelope gives

$$D + X_h + V_h \leq B_h(r)$$

for all realizations of resource at most r, then $|Y| > B_h(r)$ excludes resource r. The conclusion depends on the witness-to-label bridge and on the resource envelope; neither is a consequence of the bare structural witness alone.

Proof. The bridge produces a protected distinguishability certificate on Y, so the obligation-cover theorem gives the first inequality. The second statement is immediate by contradiction from the category-specific upper envelope. □

Three common instances are now transparent. A geometric compatibility witness requires a protected realization/readout. An exact chronological distinction requires a protected decoder or future obligation; a common shadow alone is insufficient. A holonomy class requires a physical transport action, then an orbit, then a protected readout. In all three cases the last step to a physical resource is category-sensitive.

110.2 Recovery of four source theories

USNRT was extracted from four previously developed theorem packages. The recovery claim in this section is therefore not that USNRT re-proves those source theorems from weaker axioms. It is a source-faithfulness statement: the source data, hypotheses, and theorem-bearing distinctions embed into typed USNRT components and are recovered by an explicit forgetful map. The source results used in this audit occur in earlier Parts of this monograph; their external provenance is recorded at the corresponding source locations and in the bibliography where appropriate.

110.2.1 Recovery datum

For a source class Src_i, let USNRT_i be the typed specialization carrying exactly the source structures listed below. A recovery datum consists of

$$I_i : \mathrm{Src}_i \longrightarrow \mathrm{USNRT}_i, \qquad R_i : \mathrm{im}(I_i) \longrightarrow \mathrm{Src}_i$$

with $R_i I_i \simeq \mathrm{id}$ up to the native source isomorphism. In addition, for each audited source theorem we record the source statement, the USNRT component that carries each hypothesis, and the reconstruction that returns the original theorem.

110.2.2 Axis I: completion geometry

Source. The source package is the periodic shortest-path compatibility theory developed in Chapters 20–23. The audited source statements are the exact universal row-complexity and zero-shell-defect theorem, the universal k-fiber theorem, and the extension-fiber completeness consequences.

USNRT data and bridge. The fixed Bellman/completion system, exact distance, lower-order compatibility signature, higher-order extension fibers, and optional marked data are copied into the geometric enrichment $\mathcal{G}_\bullet$. Purely geometric instances use the identity realization; no physical resource is inferred until a separate protected-readout/capacity bridge is supplied.

Reconstruction. The map R_I forgets unused realization/protection layers and returns the original periodic system, exact distance, compatibility hierarchy, and extension fibers. In particular a universal k-fiber is not identified with a representation fiber. The Bellman coefficient construction and common-shell hypotheses remain source-specific.

110.2.3 Axis II: exact pushdown normalization

Source. The source package is the exact-normalization and General-Theory development of the preceding Parts, together with the confluence specialization in Chapter 60. The audited theorem family comprises finite-observation/common-target collapse capacity, the bounded-local three-stage obstruction decomposition, and rooted path-outcome coherence.

USNRT data and bridge. Semantic dynamics, physical configurations, the exact projection, a chosen section when present, bounded observations, protected future obligations, and local schedule/action data are carried literally by the exact-realization and protection layers. Under the source tail-injectivity/noncoalescence and transition-closure hypotheses, the same-target predecessor family becomes a complete-conflict protected-obligation instance, so the USNRT capacity inequality returns the source finite-observation bound. The later allocation and local-alignment stages remain source-specific rather than consequences of the cover bound.

Reconstruction. The forgetful map returns the original semantic/physical policy, section, observation map, and local action data. It preserves the distinctions "representation debt versus behavioral correctness", "observation collision versus hidden-tail noncoalescence", and "capacity screen versus exact local alignment".

110.2.4 Axis III: chronological arithmetic

Source. The source is the CCR-k General Theory in Chapter 34. The audited package consists of terminal-block separation, higher-band lower-Hessenberg realization, the subdiagonal light-cone theorem, normalized forward transport, inverse source reconstruction, the stationary parameter lattice, and the universal raw Stirling shadow.

USNRT data and bridge. The deterministic chronology is placed in $\mathcal{J}_\bullet$: exact connected source sequences, Hessenberg bands, normalized transport, inverse reconstruction, stationary parameters, and raw shadow maps remain separate typed coordinates. A common shadow is not itself a protected-information conclusion; a protected decoder/readout is an additional bridge.

Reconstruction. Forgetting unrelated realization layers returns the exact source sequences and shadow maps separately. Thus exact chronological data are not identified with their mod-p shadow, and the CCR-k theorem retains the source hypotheses under which its parameter lattice and shadow statements are valid.

110.2.5 Axis IV: information and resource complexity

Source. The source is the retained-information and physical-exposure branch beginning at Chapter 96. Its filtered collapse-capacity theorem gives the inequality

$$D + X_h + V(h) \geq N$$

under hidden-tail noncoalescence, and its source-stack exposure profile gives the deterministic top-access pushdown specialization

$$D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq N.$$

USNRT data and bridge. The source family, protected contract, observation filtration, confinement predicate, and debt/escape counts are represented by the protected-obligation branch. In the same-stack saturation specialization, choose one protected source representative for each distinct joint label. Then the collapse-family size is $N = R$, where R is the compiler's joint rank, and the source-stack exposure theorem gives exactly the bound used in Chapter 109.

Reconstruction. The map returns the source family, observation filtration, exposure depths, debt counts, control/alphabet parameters, and protected recoding contract. Source exposure is not identified with maximum stack height, and a capacity inequality is not identified with a matching compiler.

Theorem 110.5 (Source-faithful four-axis recovery)**.** *For the four cited source packages above, the displayed interpretations define recovery data (I_i, R_i) satisfying*

$$R_i I_i \simeq \mathrm{id}_{\mathrm{Src}_i}.$$

For each audited theorem family, every source hypothesis is assigned to an explicit USNRT component or to an explicitly retained source-specific enrichment, and reconstruction returns the source theorem without adding a converse.

Proof. The four interpretations are componentwise injections of the source data listed above. In each case R_i is the projection onto those components, so $R_i I_i$ is the identity up to the source's native renaming/isomorphism. The theorem-level claim is checked axis by axis: Bellman/common-shell data are retained on Axis I; tail noncoalescence, transition closure, and local alignment are retained on Axis II; exact Hessenberg transport and shadow maps remain distinct on Axis III; and the observation filtration plus top-access capacity law are retained on Axis IV. Since no source-specific hypothesis is deleted or promoted to a generic USNRT axiom, reconstruction yields the original implication and no additional converse. □

110.2.6 Recovery/transfer consistency

The recovery maps also preserve the direction of the cross-axis bridges used in this synthesis. Four checks are especially important.

(R1) *Axis I to Axis IV.* A universal compatibility fiber becomes a physical lower bound only after a protected collapse/readout bridge and the Axis-IV capacity theorem are inserted.

(R2) *Axis II to obligation capacity.* Common-target finite-observation collapse is the complete-conflict specialization of protected-obligation capacity; the subsequent allocation and local-alignment stages remain source-specific.

(R3) *Axis III to protected information.* Exact chronological alternatives with one common raw shadow create protected pressure only when a future decoder/readout distinguishes those exact alternatives.

(R4) *Axis I topology to transport/resource.* Marked topology is distinct from a chosen physical action; resource consequences additionally require protected readout and a category-specific capacity envelope.

Corollary 110.6 (Non-collapse of the unification)**.** *On the cited source packages, USNRT is a typed host rather than a quotient that identifies theorem-bearing source data. The recovery statement does not assert that arbitrary objects satisfying the bare USNRT axioms must satisfy the source-specific Bellman, pushdown, CCR-k, or exposure theorems.*

110.2.7 Dependency boundary

The proved USNRT core uses only the source packages and hypotheses explicitly cited above. Later arithmetic refinements, unresolved resource-frontier calculations, or category-specific compiler constructions are specialization questions and are not premises of the exact-realization, protected-obligation, saturation, holonomy, or recovery results in this Part.

110.3 Non-transfer principles and scope

Several tempting implications are invalid without additional hypotheses. They are recorded here because the boundary of a transfer theorem is part of the theorem.

(N1) **Topology alone does not imply physical cost.** A physical transport representation must first be chosen; a protected readout and capacity law are additionally required for resource consequences.

(N2) **Chronological shadow loss alone does not imply physical cost.** Lost exact chronology becomes a resource issue only if the lost distinctions remain protected by a future decoder/readout contract.

(N3) **Geometry alone does not imply physical cost.** A geometric witness needs a protected-realization bridge and then a category-specific capacity envelope.

(N4) **Pairwise availability does not imply global assembly.** This is measured exactly by the assembly defect δ_{asm}.

(N5) **A capacity lower bound does not imply a matching construction.** Lower bounds are necessary conditions; exact optimality additionally needs an upper-bound construction.

(N6) **Abstract product realization does not imply one-stack pushdown product closure.** A Cartesian product may carry two independent physical memories.

These non-transfer principles are not limitations of the framework; they are the mechanism by which it prevents category errors between semantic, informational, topological, and physical statements.

110.4 Relation to neighboring frameworks

USNRT has substantial classical neighbors, and the distinctions matter for the scope of the contribution. Refinement mappings explicitly relate lower-level state machines to higher-level specifications by state-space maps preserving steps and behaviors [139]; refinement calculi formalize correctness-preserving program and data refinement [140]; abstract interpretation constructs composable hierarchies of semantics through abstraction maps [141]; and coalgebra provides a broad categorical theory of transition and dynamical systems [142]. Accordingly, the existence of an abstract/concrete state map, a semantics-preserving refinement relation, or a categorical transition-system shell is not claimed as new here.

Rewriting theory studies confluence, normal forms, and canonicalization [143, 61]. Fractional covers and zero-error/confusability methods provide classical combinatorial and information-theoretic lower-bound shells [144, 145, 146, 147]. Sheaf-theoretic contextuality identifies local/global incompatibility with global-section obstructions [148]; valuation algebras generalize disagreement across information sources [149]; bundle approaches relate contextuality and holonomy [150]; and resource theories formalize monotones and free operations in operational settings [151]. Thus simplicial compatibility, local/global obstruction, holonomy, fractional covering, and resource monotonicity are likewise not claimed as inventions of USNRT.

The contribution of this synthesis Part is the combined typed architecture and its theorem package: (i) exact realization with an optional rather than intrinsic section; (ii) an exact path-outcome criterion and debt formula for fixed-template canonicalization; (iii) protected-obligation compatibility as a separate information layer; (iv) explicit conditional bridges from geometry, chronology, or transported holonomy into that layer; and (v) category-relative transfer saturation that separates missing pairwise compatibility from failure of simultaneous assembly in one physical model. The novelty claim is therefore a claim about this specialized synthesis and the resulting transfer/non-transfer theorems, not about the classical ingredients individually.

To the best of our knowledge after the targeted literature comparison summarized above, no single neighboring framework contains this complete typed package with the same theorem roles. We do not treat that comparison as a historical-priority certificate:

later prior art may revise the novelty classification without changing the mathematical statements proved here.

110.5 Theorem-scope and bridge ledger

The purpose of this ledger is to make the logical status of the main theorem families explicit. A statement is *intrinsic/formal* when its conclusion follows from the typed mathematical data stated in the theorem and does not need a model-specific physical capacity theorem. A statement is *conditional* when an explicit bridge must be supplied. A statement is *category-sensitive* when its numerical physical consequence depends on the admissible realization category or resource evaluation.

110.5.1 Intrinsic/formal results

The following results belong to the formal USNRT core once their displayed inputs are given:

(I1) the exact-realization category and the fully faithful identity-realization embedding;

(I2) the rooted endpoint-factorization theorem and the fixed-template section-defect formula;

(I3) formation of the protected obligation complex and its finite face-cover invariants;

(I4) rectangular compatibility-product bounds and the fractional-cover rate identity at the derived-complex level;

(I5) exact finite transfer-loss telescoping;

(I6) abstract asynchronous product realization for coordinate products;

(I7) transfer-saturation arithmetic, pair/assembly defect decomposition, and optimal same-model loss once a compiler category has been fixed.

None of these results, by itself, asserts a stack-height, state-count, alphabet-size, exposure, or Pareto lower bound.

110.5.2 Conditional bridges

Four bridge types recur throughout this Part.

(C1) A geometric witness produces protected information only when a witness constructor and protected realization/readout are provided.

(C2) Exact chronology produces protected information only when a future decoder or equivalent protected obligation distinguishes the exact alternatives. A common shadow is insufficient.

(C3) Path-outcome multiplicity produces obligation pressure only when the alternatives receive protected repair or continuation labels.

(C4) Marked loops produce physical orbit information only after a transport action on a carrier is specified.

These hypotheses are not technical decoration. Removing them changes the type of the statement and can make the intended conclusion false.

110.5.3 Category-sensitive conclusions

A protected-information lower bound becomes a physical resource statement only after a category-specific capacity envelope is supplied. Thus observation capacity, debt/escape envelopes, source-stack exposure, state/stack/alphabet lower bounds, and Pareto exclusions are category-sensitive. The source-stack formula in Section 109.4.3 is one such specialization; it is not promoted to an invariant of bare semantics.

Likewise, a physical repeated-product interpretation of h_{obl} requires an iterated rectangular realization family. Without that hypothesis, $h_{\mathrm{obl}}(K) = \log \kappa_f(K)$ remains a theorem about the derived protected compatibility complex.

110.6 Four-axis recovery ledger

This section gives the recovery audit in the same five-field format for each axis. It is intentionally a verification ledger rather than a second proof of the source theorems.

110.6.1 Axis I: completion geometry

Source. Chapters 20–23.
Audited theorem family. Exact universal row complexity, zero shell defect, the universal k-fiber theorem, and extension-fiber completeness.
Embedding. Fixed periodic/Bellman system, exact distance, compatibility hierarchy, lower-order signature, higher-order extension fibers, and marked data are placed in the geometric enrichment $\mathcal{G}_\bullet$; pure geometric examples use the identity realization.
Bridge. No physical conclusion is drawn without an additional protected-readout/capacity bridge.
Reconstruction. Forgetting unrelated USNRT layers returns the source system and its theorem-bearing compatibility data; in particular, compatibility fibers are not identified with representation fibers.

110.6.2 Axis II: exact pushdown normalization

Source. The preceding exact-normalization/General-Theory Parts and Chapter 60.
Audited theorem family. Finite-observation/common-target collapse capacity, the bounded-local three-stage obstruction decomposition, and rooted path-outcome coherence.
Embedding. Semantic dynamics, physical configurations, exact projection, chosen section, bounded observations, protected obligations, and local action/schedule data are carried by the realization and protection layers.
Bridge. Tail-injectivity/noncoalescence and transition closure turn the same-target predecessor family into the complete-conflict protected-obligation specialization; later allocation and local-alignment stages remain source-specific.
Reconstruction. Projection returns the original policy, section, observations, and local action data while preserving the distinction between a capacity screen and exact local realizability.

110.6.3 Axis III: chronological arithmetic

Source. Chapter 34.
Audited theorem family. Terminal-block separation, higher-band lower-Hessenberg realization, subdiagonal light cone, normalized forward transport, inverse source reconstruction, stationary parameter lattice, and raw Stirling shadow.

Embedding. Exact connected sources, Hessenberg bands, transport/inverse data, stationary parameters, and shadow maps are stored as separate chronological coordinates in $\mathcal{J}_\bullet$.
Bridge. A common shadow creates protected-information pressure only when a future decoder/readout distinguishes the exact alternatives.
Reconstruction. Forgetting unrelated layers returns exact and shadow data separately, with all source hypotheses for stationarity, divisibility, and transport retained.

110.6.4 Axis IV: information and resource complexity

Source. The retained-information/exposure branch beginning at Chapter 96.
Audited theorem family. Filtered collapse-capacity and source-stack exposure theorems.
Embedding. The collapse source family, protected contract, observation filtration, confinement predicate, debt/escape counts, and resource parameters are represented in the protected-obligation/resource layers.
Bridge. The abstract collapse theorem gives $D + X_h + V(h) \geq N$ under hidden-tail noncoalescence, while the pushdown specialization gives $D + X_h + q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq N$ for deterministic top-access pushdown sources. Selecting one source representative per distinct joint protected label sets $N = R$ in the saturation specialization.
Reconstruction. Projection returns the source filtration, exposure depths, debt data, and source-local state/alphabet parameters; exposure is not identified with stack height and a capacity inequality is not identified with a matching compiler.

110.6.5 Ledger conclusion

For every axis, the reconstruction map returns the audited source package after interpretation, while source-specific hypotheses remain explicit. The ledger therefore certifies source faithfulness; it does not assert that bare USNRT axioms imply any of the four source theorem families.

110.7 Right-turn spectra and deterministic recognition limits

For tree-walking automata, determinism, unambiguity, and visible-pebble access must be distinguished as recognition contracts. The three-marker non-determinization witness of Bojanczyk and Colcombet [156] provides a fixed starting obstruction, not a new separation claim. For this family the relations $\mathsf{DTWA} \subseteq \mathsf{UTWA} \subsetneq \mathsf{NTWA} \subsetneq \mathsf{REG}$ are known, while strictness of $\mathsf{DTWA} \subseteq \mathsf{UTWA}$ is not established here. The one-pebble class is not an intermediate member of that chain.

110.7.1 Spectrum model and exact regularity boundary

Let $\mathcal{W}_3$ be the set of finite ordered full binary trees whose internal nodes are blank and whose leaves are labelled a or b, with exactly three a-labelled leaves. Denote those three leaves, in left-to-right order, by a_1, a_2, a_3. A basic deterministic tree-walking automaton recognizes $\mathcal{W}_3$: perform a standard complete depth-first tour and saturate the number of marked leaves at four.

For $t \in \mathcal{W}_3$, define

$$\begin{aligned} &r(t) = \mathrm{lca}(a_1, a_2, a_3), && q(t) = \mathrm{lca}(a_2, a_3), \\ &\kappa(t) = \#\{\text{right-child edges on } r(t) \rightsquigarrow q(t)\}. \end{aligned} \tag{110.1}$$

The downward path can be empty. In the left-pair branching case $((a_1, a_2), a_3)$, $q = r$ and $\kappa = 0$; in the right-pair case $(a_1, (a_2, a_3))$, q lies in the right subtree of r, so $\kappa \geq 1$. This remains true with arbitrary blank subtrees inserted. For $E \subseteq \mathbb{N} = \{0, 1, 2, \ldots\}$, define

$$\mathcal{L}_E = \{t \in \mathcal{W}_3 : \kappa(t) \in E\}, \quad \mathcal{Z} = \mathcal{L}_{\{0\}}, \quad \mathcal{B} = \mathcal{L}_{\{1,2,3,\ldots\}}.$$

The three-marker separation from [156] is $\mathcal{B} \notin \mathsf{DTWA}$. Because $\mathcal{W}_3 \in \mathsf{DTWA}$ and deterministic tree-walking languages are Boolean-closed [159], also $\mathcal{Z} \notin \mathsf{DTWA}$.

A set $E \subseteq \mathbb{N}$ is *ultimately periodic* if there are $N \geq 0$ and $p \geq 1$ with $\mathbf{1}_E(n+p) = \mathbf{1}_E(n)$ for all $n \geq N$.

Theorem 110.7 (Exact regularity characterization)**.** *For every $E \subseteq \mathbb{N}$,*

$$\boxed{\mathcal{L}_E \in \mathsf{REG} \iff E \text{ is ultimately periodic.}}$$

If N, p witness ultimate periodicity, a deterministic bottom-up automaton with at most $3 + 2(N + p)$ states can be constructed.

Proof. For necessity let $D = \mathrm{Node}(a, a)$, $C[z] = \mathrm{Node}(b, z)$, and $R[z] = \mathrm{Node}(a, z)$, where each displayed b is an unmarked leaf. For $n \geq 0$ let $T_n = R[C^n[D]]$. The three marked leaves are ordered as required and $\kappa(T_n) = n + 1$. In any deterministic bottom-up finite automaton, the state reached on $C^n[D]$ evolves by iterating a single self-map of a finite state set. Its state sequence is eventually periodic. Acceptance of the fixed outer context $R[-]$ is then eventually periodic, so $\mathbf{1}_E(n + 1)$, and thus $\mathbf{1}_E(n)$, is ultimately periodic.

Conversely, choose an ultimately periodic representation with threshold N and period p. Let $\rho(v)$ retain v exactly for $v < N$ and retain the residue v mod p in a disjoint periodic part for $v \geq N$. The domain $\mathcal{D}_{N,p}$ contains $N + p$ values, supports a deterministic increment, and determines whether $v \in E$.

At each subtree, store its number c of marked leaves capped at four. When $c = 2$, additionally store $\rho(d)$, where d is the number of right-child edges from the subtree root to the LCA of its two marked leaves. When $c = 3$, instead store $\rho(\kappa)$ for its three marked leaves. The other three count classes $c \in \{0, 1, \geq 4\}$ need no auxiliary value. The complete transition rules at an internal node with left and right child counts c_L, c_R are

count split for $c = 2$	$(1,1)$	$(2,0)$	$(0,2)$
d_{new}	0	d_L	$1 + d_R$

count split for $c = 3$	$(3,0)$	$(0,3)$	$(2,1)$	$(1,2)$
κ_{new}	κ_L	κ_R	0	$1 + d_R$

The additions are performed in $\mathcal{D}_{N,p}$. At the root accept precisely if the marked-leaf count is three and the represented statistic belongs to E. There are at most $3 + 2(N + p)$ states. □

Theorem 110.8 (Linear-order bottom-up state complexity)**.** *Suppose E is ultimately periodic. Let $p \geq 1$ be its* least eventual period*, and let $N \geq 0$ be the least threshold from which membership is periodic with period p. If s_E denotes the minimum number of states of a deterministic bottom-up tree automaton recognizing $\mathcal{L}_E$, then*

$$\boxed{p + \max\{N - 1, 0\} \;\leq\; s_E \;\leq\; 3 + 2(N + p).}$$

Consequently the deterministic bottom-up state complexity of this family has tight linear order *in $N + p$; this is not a lower bound on the state count of a deterministic tree-*walking *automaton.*

Proof. The upper bound is the explicit construction in Theorem 110.7. For the lower bound retain the fixed binary tree $D = \mathrm{Node}(a, a)$ and unary context $C[z] = \mathrm{Node}(b, z)$ from that proof. If a deterministic bottom-up recognizer has state set Q, the state u_n at $C^n[D]$ is obtained from u_0 by n iterations of a *single* map $f : Q \to Q$. Its acceptance by the fixed outer context $R[z] = \mathrm{Node}(a, z)$ is precisely

$$\mathbf{1}_E(n+1) \qquad (n \geq 0).$$

The state orbit of a finite map has an initial tail of length μ followed by a cycle of length λ, using exactly $\mu + \lambda$ distinct states. The displayed Boolean output sequence then has eventual period dividing λ and becomes periodic no later than index μ. Its *least* eventual period is p, while its least preperiod is $\max\{N-1, 0\}$: shifting a unary sequence by one index removes at most its initial exceptional value. Indeed, any periodicity of the shifted sequence from an earlier index, possibly with a nonminimal period, would extend the original eventually-p periodic tail back to an index earlier than N. Thus $p \mid \lambda$ and $\mu \geq \max\{N-1, 0\}$. Therefore

$$|Q| \geq \mu + \lambda \geq p + \max\{N-1, 0\}.$$

This holds for every deterministic bottom-up recognizer and hence for the minimum s_E. □

Corollary 110.9 (Growing numerical thresholds need growing bottom-up control)**.** *For the threshold spectrum $E = \{0, \ldots, B\}$ with $B \geq 1$, a deterministic bottom-up recognizer of $\mathcal{L}_E$ needs at least $B + 1$ states and the construction of Theorem 110.7 uses at most $2B + 7$ states. For the pure modulus-m spectrum $E = m\mathbb{N}$, $m \geq 2$, every such recognizer needs at least m states.*

Proof. For the threshold spectrum the least preperiod is $N = B + 1$ and eventual period is $p = 1$; for $m\mathbb{N}$ they are $N = 0$ and $p = m$. Substitute in Theorem 110.8. □

Corollary 110.10. *If E is not ultimately periodic, $\mathcal{L}_E$ is not recognized by any ordinary or finite-pebble tree-walking automaton. In particular it is not in* DTWA.

Proof. Ordinary tree-walking languages are regular [156, 158]; languages of the standard nested finite-pebble model are regular [160]. Apply Theorem 110.7. □

110.8 Right-edge dilation and a nonrecognizability criterion

For $d \geq 1$, define $R_0(z) = z$, $R_{j+1}(z) = \mathrm{Node}(b, R_j(z))$ and the tree transformation

$$H_d(a) = a, \quad H_d(b) = b, \qquad H_d(\mathrm{Node}(u, v)) = \mathrm{Node}\big(H_d(u), R_{d-1}(H_d(v))\big). \tag{110.2}$$

Every original right-child edge becomes a chain of d right-child edges with blank left sibling leaves; original left-child edges are unchanged.

Lemma 110.11 (Exact dilation)**.** *H_d preserves the number of marked leaves and, for every $t \in \mathcal{W}_3$, $\kappa(H_d(t)) = d\,\kappa(t)$.*

Proof. The marked-leaf order and the corresponding original lowest common ancestors are unchanged. The path joining the two relevant ancestor nodes is obtained from the old path by replacing each right-child edge with exactly d such edges; a left-child edge remains left-child. □

Lemma 110.12 (Finite-control inverse simulation)**.** *For fixed $d \geq 1$, $K \in \mathsf{DTWA}$ implies $H_d^{-1}(K) \in \mathsf{DTWA}$.*

Proof. A machine reading t can simulate a deterministic walker on the virtual tree $H_d(t)$ without constructing that tree. Encode a virtual head position by a physical input node plus one of finitely many tags: an original node, one of the $d-1$ wrapper nodes replacing a right edge out of the current original node, or one of the corresponding blank left leaves. For a wrapper node or its blank left leaf, the physical position is the original parent of the expanded edge. Local parent/left/right/stay transitions of the virtual machine translate into a bounded number of physical movements and tag changes, determined by local input node type and the fixed d. Labels and node types in the virtual tree can also be recovered from this representation. Consequently, the product of the original finite control with the tag set supplies a deterministic simulation. This is a direct special-case inverse transformation argument, consonant with [161]. □

Theorem 110.13 (Spectrum dilation obstruction)**.** *Let $E \subseteq \mathbb{N}$. If some $d \geq 1$ satisfies*

$$\mathbf{1}_E(dn) = 1 - \mathbf{1}_E(0) \qquad (n \geq 1), \tag{110.3}$$

then $\mathcal{L}_E \notin \mathsf{DTWA}$.

Proof. Inside $\mathcal{W}_3$, Lemma 110.11 identifies $H_d^{-1}(\mathcal{L}_E)$ with $\mathcal{Z}$ if $0 \in E$ and with $\mathcal{B}$ otherwise. Either case is outside DTWA by the classical separation and Boolean closure. Were $\mathcal{L}_E$ deterministically recognizable, Lemma 110.12 would recognize this preimage, a contradiction. □

Corollary 110.14 (Uniform threshold nonrecognizability)**.** *For every fixed $B \geq 0$,*

$$\boxed{\mathcal{L}_{\{0,\ldots,B\}}, \quad \mathcal{L}_{\{B+1,B+2,\ldots\}} \in \mathsf{REG} \setminus \mathsf{DTWA}.}$$

Proof. For each spectrum use $d = B + 1$ in Theorem 110.13, and apply Theorem 110.7 for regularity. □

Proposition 110.15 (Sharp factor within uniform dilation)**.** *For every $d \geq 1$ and $B \geq 0$,*

$$H_d^{-1}(\mathcal{L}_{\{0,\ldots,B\}}) = \mathcal{L}_{\{0,\ldots,\lfloor B/d \rfloor\}}.$$

Consequently, the least dilation factor whose inverse image is exactly $\mathcal{Z}$ is $B + 1$.

Proof. The inequality $d\,\kappa(t) \leq B$ is equivalent to $\kappa(t) \leq \lfloor B/d \rfloor$. This bound is zero precisely when $d > B$. Every nonnegative value of κ occurs in $\mathcal{W}_3$, so smaller d cannot give exactly $\mathcal{Z}$. □

Corollary 110.16 (Periodic spectra, including infinite examples)**.** *For every $m \geq 2$ and $R \subseteq \{0, \ldots, m-1\}$ not containing 0, the spectrum*

$$E = \{0\} \cup \{n \geq 1 : n \bmod m \in R\}$$

satisfies $\mathcal{L}_E \in \mathsf{REG} \setminus \mathsf{DTWA}$. The same holds for $E = \{n \geq 1 : n \bmod m \in R\}$ whenever $0 \in R$.

Proof. Set $d = m$ in Theorem 110.13 and use ultimate periodicity for regularity. In the first displayed family the choice $R = \varnothing$ gives the finite spectrum $\{0\}$; to obtain an infinite spectrum there take $R \neq \varnothing$. □

110.9 Exactly what the dilation method proves

For ultimately periodic E with threshold N and period p, put

$$b_E = \mathbf{1}_E(0), \qquad e_E = \mathbf{1}_E(np) \quad \text{for sufficiently large } n.$$

Let $\mathcal{A}_E$ be the Boolean algebra of subsets of $\mathcal{W}_3$ generated by $H_d^{-1}(\mathcal{L}_E)$ for all positive integers d, with complements taken relative to $\mathcal{W}_3$. For clarity, this construction allows *only* these inverse images and finite Boolean operations; it does not encompass arbitrary DTWA proof techniques.

Theorem 110.17 (Exact reach of the dilation–Boolean method)**.** *For every ultimately periodic E, the following are equivalent:*

(i) $b_E \neq e_E$*;*
(ii) *for some $d \geq 1$, $H_d^{-1}(\mathcal{L}_E)$ is $\mathcal{Z}$ or $\mathcal{B}$;*
(iii) $\mathcal{Z} \in \mathcal{A}_E$ *(equivalently, $\mathcal{B} \in \mathcal{A}_E$).*

Proof. If $b_E \neq e_E$, choose $d \geq N$ divisible by p. At $\kappa = 0$, the inverse image has membership b_E; at every positive κ, $d\kappa$ is beyond the threshold and congruent to 0 modulo p, so the membership is $e_E = 1 - b_E$. This proves (i)⇒(ii), and (ii)⇒(iii) follows by Boolean closure.

For the converse suppose $b_E = e_E$. Choose a positive multiple q of p with $q \geq N$. For every $d \geq 1$,

$$\mathbf{1}_E(d \cdot 0) = b_E = e_E = \mathbf{1}_E(dq).$$

Thus each generator, and every Boolean combination of them, agrees on the nonempty fibers $\kappa = 0$ and $\kappa = q$. Neither $\mathcal{Z}$ nor $\mathcal{B}$ does. Therefore (iii) fails. □

Remark 110.18 (A method frontier, not a class characterization). Theorem 110.17 says exactly when this particular construction can recover the known branching separator. If $b_E = e_E$, it does *not* follow that $\mathcal{L}_E$ belongs to DTWA. In particular the pure even spectrum survives this test. The saturated-pattern arguments of [156] alone do not supply a parity-changing equivalent pair by the reductions used here.

110.10 One visible pebble recognizes every regular spectrum

In the standard deterministic one-pebble tree-walking model, the machine has a single movable head and finite control, and can place one visible pebble at its current node, detect it when revisiting that node, and lift it there. The pebble must be lifted before acceptance. Let 1pDTWA denote the resulting language class. The nested-pebble regularity inclusion and early anchored-walk techniques were established by Engelfriet and Hoogeboom [160]; a later account of deterministic one-pebble branching tests is given in [157]. The following theorem is a classification of our *particular* numerical family, not the first known zero-/one-pebble class gap.

Lemma 110.19 (Bounded subtree probe)**.** *Given its head at an internal node v, a deterministic one-pebble walker can count, up to any fixed bound, the marked leaves in either chosen child subtree and return to v with its pebble removed.*

Proof. Place the pebble at v and enter the chosen child. Execute a complete local depth-first tour with a finite saturated count. The first upward return to the pebble identifies the boundary v; lift the pebble and resume. The local DFS requires no unbounded stack since the parent/left/right node type is visible at each step. □

Theorem 110.20 (Exact deterministic one-pebble characterization)**.** *For every $E \subseteq \mathbb{N}$,*

$$\boxed{\mathcal{L}_E \in \mathsf{1pDTWA} \iff \mathcal{L}_E \in \mathsf{REG} \iff E \text{ is ultimately periodic.}}$$

Proof. Assume E is ultimately periodic. Begin with a deterministic full-tree DFS verifying membership in $\mathcal{W}_3$. On failure reject. On success, start a fresh preorder DFS, with a finite counter of the marked leaves already visited. At each internal node v, probe both child subtrees via Lemma 110.19; the global preorder counter is untouched during the probes.

If the global counter is one and each child subtree contains exactly one marked leaf, then $v = q(t)$: the visited leaf is a_1 and the two unvisited marked leaves are separated at v. Conversely, in the right-pair case this situation occurs at q, and in the left-pair case no node satisfies it. In the latter case $r = q$, hence $\kappa = 0$, and the machine decides according to $0 \in E$.

In the right-pair case, suspend preorder at q and move upwards. On an ascent from a *left* child, do not probe that parent and do not increment the statistic. On an ascent from a *right* child, increment the finite counter and probe the left subtree of the parent v. This distinction is essential: after a left-child ascent below r, the parent's left subtree can contain a_2, a_3, whereas for a right-child ascent strictly below r the left sibling subtree has no marked leaf. At r the path from r to q begins with a right-child edge and the left subtree contains a_1; therefore the first positive probe performed after a right-child ascent identifies r. The increments are exactly the right-child edges on $q \rightsquigarrow r$, read downward, so their number is $\kappa(t)$.

Implement the counter by exact values below the threshold N and residues modulo p after that threshold, as in Theorem 110.7; increments are monotone on the ascent, and the finite-state summary determines membership in E. The pebble is always lifted after each subtree probe, so only one pebble is used. If the acceptance convention requires a final visit to the root, move there while retaining the decision in one state bit.

Conversely, every language recognized by the standard one-pebble model is regular [160], and Theorem 110.7 then forces E to be ultimately periodic. □

Corollary 110.21 (Explicit deterministic pebble gap)**.** *For every fixed $B \geq 0$,*

$$\boxed{\mathcal{L}_{\{0,\dots,B\}} \in \mathsf{1pDTWA} \setminus \mathsf{DTWA}.}$$

Proof. Combine Theorem 110.20 and Corollary 110.14. □

110.11 The pure even spectrum: resolved and unresolved aspects

The spectrum $E_{\text{even}} = 2\mathbb{N}$ is purely periodic with period two, and contains zero. Hence

$$\mathcal{L}_{\text{even}} = \{t \in \mathcal{W}_3 : \kappa(t) \equiv 0 \pmod 2\} \in \mathsf{REG} \cap \mathsf{1pDTWA} \tag{110.4}$$

by Theorems 110.7 and 110.20. Its membership is also unproblematic for a different *zero-pebble* model that accepts according to the parity of the number of accepting nondeterministic computations; this is not ordinary deterministic acceptance.

Lemma 110.22 (Right-turn witness bijection)**.** *On every $t \in \mathcal{W}_3$, the number of left-to-right DFS turns $v : v_L \to v_R$ made after exactly one marked leaf has been visited and with a_3 in the right subtree of v is precisely $\kappa(t)$.*

Proof. If $\kappa(t) = 0$, the leftmost two marked leaves lie on the same side of the top branching node and no such right-subtree turn occurs after seeing only a_1. Otherwise the a_2, a_3 pair lies in the right child subtree of r, and its common ancestor is q. A successful turn's right subtree contains a_3 and occurs after exactly a_1 was visited; it must therefore also contain a_2 (which precedes a_3) and cannot contain a_1. Thus the turning node is an ancestor of q at which the path to q follows the right child, and is at or below r. Conversely, every right-child edge on the path $r \rightsquigarrow q$ produces exactly one such turn: at that time a_1 has been visited, whereas a_2, a_3 remain in the corresponding right subtree. This gives the bijection. □

Proposition 110.23 (Exact ambiguity and an auxiliary parity-of-runs model)**.** *There is an ordinary nondeterministic tree-walking automaton A_{turn} with no pebbles such that*

$$\#\operatorname{Acc}_{A_{\mathrm{turn}}}(t) = \begin{cases} \kappa(t), & t \in \mathcal{W}_3, \\ 0, & t \notin \mathcal{W}_3. \end{cases}$$

Consequently an automaton with odd-number-of-accepting-runs semantics and no pebbles recognizes $\mathcal{L}_{\mathrm{even}}$. This conclusion is not *deterministic recognizability.*

Proof. For completeness, a zero-pebble nondeterministic walker can be arranged to have exactly $\kappa(t)$ accepting runs on $t \in \mathcal{W}_3$. After checking the three-leaf promise, it performs a deterministic left-to-right DFS. At each transition from a left subtree to the corresponding right subtree, provided exactly one marked leaf has been seen, it may guess that turn, then nondeterministically descend to a marked leaf in the right subtree. Once a marked leaf z is reached, the walker continues the ordinary left-to-right depth-first tour from z to the end of the tree. This continuation starts in the *leave-leaf* phase and accepts the guess precisely when no later marked leaf is encountered. Thus z must be the rightmost marked leaf a_3. Both the successor tour and its one-bit flag for a later marked leaf use finite control, require no pebble, and have no further branching. For a fixed candidate turn, the unique downward path to a_3 is therefore the only successful guessed path, if a_3 lies in its right subtree. By Lemma 110.22, the successful turns are in bijection with the right edges on $r \rightsquigarrow q$, giving $\kappa(t)$ accepting runs. Adding one independent accepting run yields $1 + \kappa(t)$ runs, odd precisely when $\kappa(t)$ is even. No pebble is involved, but the acceptance semantics is *parity-of-runs*, not DTWA. □

The exact remaining question within the family is therefore:

Question 110.24 (Unannotated, zero-pebble deterministic parity)**.** *Is $\mathcal{L}_{\mathrm{even}}$ recognized by an ordinary deterministic tree-walking automaton with one head, no pebbles, and standard single-run acceptance?*

$$\boxed{\mathcal{L}_{\mathrm{even}} \overset{?}{\in} \mathsf{DTWA}.}$$

Theorem 110.17 explains why uniform dilation and Boolean combinations do not decide Question 110.24: here $b_E = e_E = 1$. The one-pebble construction does not decide it either, since its reusable subtree probe relies on an explicitly marked return point. The preceding parity-of-runs construction proves membership only in a different computational model. These are precise limitations of the present arguments, *not* an assertion of general literature-wide openness.

110.12 Discussion, limitations and future work

The family $\{\mathcal{L}_E : E \subseteq \mathbb{N}\}$ illustrates how a simple numerical statistic can separate three questions: regularity, zero-pebble deterministic recognition, and one-pebble deterministic recognition. Exact regularity and one-pebble membership both reduce to ultimate periodicity. The dilation obstruction transfers the classical three-leaf deterministic lower bound to fixed thresholds and many infinite spectra; its exact Boolean closure boundary is expressible by the value of the spectrum at zero and at sufficiently large multiples of an eventual period.

The proof dependencies matter. The fundamental three-leaf nonrecognizability is due to [156]; Boolean closure and inverse-transformation principles are established machinery [159, 161]; finite-pebble regularity and the general utility of pebbles were known by [160, 157]. The explicit 2025 one-pebble example [163, Theorem 20] concerns a different language; it is not a previous proof of our full spectrum classification. The explicit right-turn family sharpens the parameterized consequences of those tools. Its deterministic lower bounds transfer an established separation rather than constituting a new baseline separation of machine classes.

The results leave the separate DTWA–UTWA equality question untouched: a language outside DTWA is not a counterexample to that equality unless it is also shown to belong to UTWA.

Future work is deliberately limited to Question 110.24: either a concrete ordinary zero-pebble DTWA for $\mathcal{L}_{\text{even}}$ or a proof that every such device fails on some parity-distinguishing tree pair. No positive or negative answer to that question is claimed here, and its resolution is not a prerequisite for the theorems proved here.

A small comparison of the established boundaries. For the fixed threshold spectrum $E = \{0, \dots, B\}$, $\mathcal{L}_E$ is regular and one-pebble recognizable but not in DTWA. For the pure even spectrum $E = 2\mathbb{N}$, regularity and one-pebble recognition are established, while zero-pebble deterministic membership is undecided by the present results. Finally, if E is not ultimately periodic, $\mathcal{L}_E$ lies outside all standard finite-pebble tree-walking language classes because it is nonregular. The distinction between these three cases prevents the general regularity theorem from being mistaken for a DTWA membership decision.

110.13 Finite-control realization details

The following two virtual-head and pebble implementation lemmas supply the finite-control details used in the preceding proofs.

110.13.1 Local virtual-head transitions for dilation

Fix $d \geq 2$. An original input node x is represented by (x, O). For every original internal node x, the $d-1$ wrapper nodes inserted on the edge from x to its right child are represented by (x, W_j) for $1 \leq j < d$ (numbered from top to bottom). The blank left leaf attached to wrapper j is represented by (x, D_j). This uses precisely $1 + 2(d-1)$ virtual-position tags. The virtual downward neighbors are

virtual position	left child	right child
(x, O), x internal	(x_L, O)	(x, W_1)
(x, W_j), $j < d-1$	(x, D_j)	(x, W_{j+1})
(x, W_{d-1})	(x, D_{d-1})	(x_R, O)

Their inverse parent operations are

virtual position	parent
(x, D_j)	(x, W_j)
(x, W_1)	(x, O)
$(x, \mathsf{W}_j),\ j > 1$	(x, W_{j-1})
$(x, \mathsf{O}),\ x \neq \text{root},\ x = x_{\text{par},L}$	$(x_{\text{par}}, \mathsf{O})$
$(x, \mathsf{O}),\ x \neq \text{root},\ x = x_{\text{par},R}$	$(x_{\text{par}}, \mathsf{W}_{d-1})$

A virtual movement between original nodes induces the corresponding physical parent/child move; a movement inside a fixed original right-edge gadget only changes the finite tag. The virtual node label, root/left/right-child type, and leaf/internal type are all functions of the physical local type and tag. All virtual transitions can therefore be implemented by a bounded microprogram. For $d = 1$ only the tag O is needed, and the simulator is the original input walker. This supplies the promised deterministic implementation of Lemma 110.12 with no pebble or stack.

110.13.2 Anchored one-pebble probes and the monotone ascent

The one-pebble recognizer has a fixed sequence of phases: (i) check $\mathcal{W}_3$ by root-to-root DFS; (ii) root-to-root preorder with a saturated count of a leaves visited, interrupting at a candidate node for two bounded subtree probes; (iii) if q is found, abandon preorder and climb from q to r, probing left siblings only after an ascent along a right-child edge; and (iv) lift the pebble, if present, return to the root if required, and halt according to the final finite counter. Every subtree probe begins by placing the sole pebble at the candidate parent v and ends on the first return to v from the chosen child, when that pebble is lifted. Thus the ordinary preorder count and the probe count belong to disjoint phases and are not conflated.

The control can be organized into finitely many traversal phases: preorder entry, anchored left-child probe, anchored right-child probe, and upward ascent. A probe saves the saturated preorder count and the ascent summary in finite control; its own temporary marked-leaf count is reset when the pebble is placed and discarded after the return to that pebble. The ascent phase inspects the type of the child just left, which the standard tree-walking model exposes: leaving a left child continues upward without either testing the parent or incrementing the right-edge count; leaving a right child increments that count and starts a fresh anchored probe of the parent's left child. No probe remains active across successive ascent edges, and acceptance is tested only after an eligible right-child ascent reveals the leftmost marker. Thus there is no unbounded recursion or second pebble.

The test for q uses the preorder count 1 and child counts $(1, 1)$. For three marks, it can first succeed only at $q = \operatorname{lca}(a_2, a_3)$ in the right-pair case. During the climb, a positive left-sibling probe on a right-child ascent occurs first exactly at r; all lower such left siblings contain no marked leaf. Each qualifying ascent increments the finite E-counter once, and no backtracking probe changes that counter. Its final value is therefore the right-edge count on $r \rightsquigarrow q$. This justifies the one-pebble implementation independently of any bottom-up automaton construction.

110.13.3 Reproducibility scope

The finite consistency range contains $C_{n-1}\binom{n}{3}$ marked ordered full binary trees with n leaves, where $C_j = \frac{1}{j+1}\binom{2j}{j}$. Hence there are exactly $\sum_{n=3}^{8} C_{n-1}\binom{n}{3} = 29{,}646$ inputs for $3 \leq n \leq 8$. Direct lowest-common-ancestor counting, the bottom-up recurrence, the one-pebble search-and-ascent invariant, and the right-turn witness bijection describe the

same statistic. For $d = 1, 2, 3, 4$ there are 118,584 corresponding finite dilation identities. These finite counts illustrate the constructions; the unrestricted statements are proved by the symbolic arguments above.

110.14 Bridge to the external scalar-field realization

The next three chapters provide a deliberately non-pushdown test of this architecture. Extrema–Frame Foreground–Background theory is developed as an independent scalar-field theory, not as an axiom of USNRT. After its native theorems are proved, the all-orders common-realization obstruction families will be read back through the protected-obligation language of this chapter sequence. No EF–FG theorem is used as a premise of the USNRT core above.

Chapter 111

Extrema–Frame Semantics, Observer Records, and Exact Realizations

Research branch — External generativity: finite scalar fields. *This chapter begins a theory that is independent of pushdown automata. Its role in the monograph is twofold: it develops a self-dual extrema-based foreground/background semantics in its own native language, and it tests whether the exact-realization discipline of the preceding USNRT chapters transfers to a genuinely different physical setting. The resulting scalar-field theorems do not depend on the original p:q automata.*

Relation to neighboring scalar-field theories. Merge trees, hierarchical region trees, self-dual morphology, extinction/dynamics, and uncertainty in scalar-field topology are established subjects [125, 126, 127, 128, 129, 130, 131, 136, 135]. The contribution here is narrower: observer-relative foreground ownership under exact same-realization semantics, the observer-record normal form, the local/global complexity separation on shallow graph classes, and explicit all-orders common-realization obstruction families.

111.1 EF-FG model

111.1.1 Filtration conventions and birth atoms

For $t \in \mathbb{R}$, let $G^{\geq t}$ be the induced subgraph on $\{v : F(v) \geq t\}$. A **birth plateau** at level B is a connected local-maximum plateau $P \subseteq F^{-1}(B)$ whose superlevel component appears when the threshold reaches B. The corresponding component identity is a **primitive birth atom** λ, with birth level $B_\lambda = B$. A **birth-carrier representative** is any vertex of P; all such representatives have the same maximin attachment to any observer because they are connected inside $G^{\geq B}$.

As the threshold decreases, birth components merge. The elder rule keeps the atom of larger birth level. If several atoms have equal birth level, an **admissible elder/tie resolution** is a total order on those equal-birth atoms, used consistently at later merges. Exactly one atom survives to the bottom of the connected filtration; it is the **essential atom**. Every other atom is **finite** (or **nonessential**), and its **elder death level** $d(\lambda)$ is the merge level at which it is killed by an older atom. The dark filtration is the same construction applied to $-F$.

111.1.2 Scalar fields, births, and maximin attachment

Fix a finite connected graph $G = (V, E)$ and a scalar field $F : V \to \mathbb{R}$. For the bright polarity, let λ range over primitive nonessential birth atoms of the superlevel filtration. Write

$$B_\lambda = \text{birth level of } \lambda, \qquad d(\lambda) = \text{elder death level.}$$

For an observer $x \in V$, define

$$a_\lambda(x) = \kappa_F(\lambda, x),$$

where κ_F is maximin connectivity. If $O(\lambda)$ denotes the elder atoms of λ, define the observer elder pressure

$$D_x(\lambda) = \max_{\mu \in O(\lambda)} a_\mu(x).$$

The dark polarity is the bright construction applied to $-F$.

111.1.3 Observer records

A nonessential birth λ is an **observer record at** x if

$$a_\lambda(x) > D_x(\lambda).$$

Thus records are precisely the births that strictly raise the running elder attachment available at the observer.

A basic maximin identity drives the theory.

Lemma 111.1 (maximin equality)**.** *If*

$$\kappa_F(p, x) > \kappa_F(q, x),$$

then

$$\boxed{\kappa_F(p, q) = \kappa_F(q, x).}$$

Proof sketch. The strong-triangle inequality gives the lower bound $\kappa_F(p, q) \geq \kappa_F(q, x)$. If the left side were larger, concatenating a maximizing p–q path with a q–x path would force q to attach to x above $\kappa_F(q, x)$, a contradiction. A complete proof is given in the observer-record proof package later in this chapter. □

111.1.4 Observer-core saliency

Fix $0 < \theta < 1$. A finite birth λ contributes to the observer core when

$$a_\lambda(x) > \theta B_\lambda + (1 - \theta) D_x(\lambda).$$

The one-polarity observer saliency is

$$\Sigma_\theta(x; F) = \max\Big(\{B_\lambda - D_x(\lambda) : a_\lambda(x) > \theta B_\lambda + (1 - \theta) D_x(\lambda)\} \cup \{0\}\Big).$$

Set

$$\Sigma_\theta^+(x; F) = \Sigma_\theta(x; F), \qquad \Sigma_\theta^-(x; F) = \Sigma_\theta(x; -F).$$

For a significance threshold $\Delta \geq 0$, the polarity-core ownership label is

$$\mathrm{Own}_{\Delta,\theta}(x;F) \in \{\text{bright}, \text{dark}, \text{background}, \text{ambiguous}\},$$

with background defined by

$$\max\{\Sigma_\theta^+(x;F), \Sigma_\theta^-(x;F)\} \le \Delta.$$

The remaining labels are determined by which polarity exceeds Δ, with an explicit ambiguous state when both do.

111.1.5 Spatial-frame projection

Fix a spatial seed threshold $\Delta_{\text{seed}} \ge 0$. For one polarity, the selected seed family consists of the finite birth atoms with persistence $B_\lambda - d(\lambda) > \Delta_{\text{seed}}$; the essential atom is excluded. Choose one birth-carrier representative for each selected atom p and put

$$\kappa_p(v) = \kappa_F(p, v), \qquad \beta(v) = \max_r \kappa_r(v), \qquad W(v) = \{p : \kappa_p(v) = \beta(v)\}.$$

The **rooted co-winner basin** $\mathcal{B}_p$ is the connected component containing p in $\{v : p \in W(v)\}$. Two selected seeds p, q are **carrier-adjacent** when $\mathcal{B}_p \cap \mathcal{B}_q \neq \varnothing$. Their gate scalar is

$$s(p, q) = \max_{v \in \mathcal{B}_p \cap \mathcal{B}_q} \beta(v),$$

and the connected components attaining this maximum are the top gate components. Route incidence through an observer, midpoint margins, designated boundary terminals, and correlated residual terminal regions then form the spatial-frame state; their exact definitions are given in the spatial-frame construction later in this chapter.

Scope principle. Polarity-core ownership is an observer/persistence projection, whereas spatial frame is a carrier/sector projection. The two are evaluated from the same scalar realization, but neither determines the other in general. The full fixed-realization EF-FG state stores both rather than identifying them.

111.1.6 Interval uncertainty and exact semantics

An interval-valued scalar field is

$$\mathcal{P} = \prod_{v \in V} [\ell_v, u_v].$$

A **realization** consists of one $F \in \mathcal{P}$ together with an admissible resolution of ties/elder order. A property is **possible** if one realization satisfies it and **robust** if every realization satisfies it.

This quantifier discipline is essential. If $A(\omega)$ and $B(\omega)$ are observables on the realization space, then generally

$$\{(A(\omega), B(\omega)) : \omega \in \Omega\} \subsetneq \{A(\omega) : \omega \in \Omega\} \times \{B(\omega) : \omega \in \Omega\}.$$

Thus one must **join before projecting**: a possible ownership label cannot in general be inferred by combining independently attainable extrema of births, attachments, or saliencies.

111.2 Hierarchy-free observer normal form

The full merge hierarchy can be eliminated from pointwise observer-core evaluation.

111.2.1 Exact observer-record death

Theorem 111.2 (exact observer-record death)**.** *Let λ be a nonessential primitive birth atom. If*

$$a_\lambda(x) > D_x(\lambda),$$

then λ is an observer record and

$$\boxed{d(\lambda) = D_x(\lambda).}$$

If instead $a_\lambda(x) \le D_x(\lambda)$, then λ cannot make a positive observer-core contribution.

Proof sketch. For a record, choose the elder atom attaining $D_x(\lambda)$. Lemma 111.1 identifies the merge level between the younger record and that elder atom with the elder's attachment at x, so the ordinary elder death is exactly $D_x(\lambda)$. If λ is not a record, its attachment does not exceed elder pressure and therefore cannot satisfy the strict core inequality. □

A complete proof is given in the observer-record proof package.

An immediate consequence is the formula already used in the observer-record definition above: the saliency maximum can be taken over all births because nonrecords eliminate themselves automatically.

111.2.2 Record chains

Order the finite observer records by elder-to-younger birth order,

$$r_1, r_2, \dots, r_m,$$

and write $a_i = a_{r_i}(x)$. Include the essential elder supply as a_0. Then

$$\boxed{d(r_i) = a_{i-1}.}$$

Thus the complete finite record history is a chain of adjacent transitions. If B_i is the birth of r_i, then the i-th potential contribution is determined by the triple

$$(B_i, a_{i-1}, a_i).$$

For an H-safety query, a record is dangerous precisely when

$$B_i - a_{i-1} > H$$

and

$$a_i > \theta B_i + (1-\theta)a_{i-1}.$$

Equivalently, define the required predecessor supply

$$L_i(H) = \min\left\{B_i - H, \frac{a_i - \theta B_i}{1-\theta}\right\}$$

with the obvious convention that only the active branch matters. Then r_i is H-safe iff a_{i-1} exceeds the corresponding threshold. This is the local origin of the later scheduling formulation.

111.2.3 A single elder-supply profile

The record chain may be represented by one rooted supply profile $E_x(b)$: the greatest observer attachment supplied by births older than threshold b. Its jumps are exactly the observer records. At a finite jump (B, D, A),

$$D = \text{exact elder pressure}, \qquad A = \text{record attachment},$$
$$B - D = \text{exact persistence}.$$

Hence

$$\boxed{\Sigma_\theta(x; F) \le H \iff \text{every finite jump } (B, D, A) \text{ satisfies } B - D \le H \quad \text{or} \quad A \le \theta B + (1-\theta)D.}$$

On rooted trees the profile obeys a max/min recurrence over child subtrees, giving a direct deterministic evaluation algorithm. A proof of the record-chain reduction and the rooted supply recurrence is given in the observer-record proof package.

111.2.4 Self-duality

All statements above are polarity-normalized. Applying them to $-F$ gives the dark theory with no change of form. In particular, grayscale inversion transports the entire common-realization relation bijectively: bright-possible becomes dark-possible, bright-robust becomes dark-robust, and the full ownership relation is conjugated realization by realization.

111.3 Exact common-realization perturbation theory

111.3.1 Why marginal envelopes are insufficient

Suppose two admissible realizations ω_0, ω_1 give

$$(A(\omega_0), B(\omega_0)) = (0, 1), \qquad (A(\omega_1), B(\omega_1)) = (1, 0).$$

Then each coordinate individually attains both 0 and 1, but $(0, 0)$ and $(1, 1)$ are unrealizable. This elementary observation becomes structural in EF-FG because births, attachments, elder pressure, record identity, and ownership are coupled by the same field values.

We therefore define the **exact EF-FG realization relation** by evaluating all requested coordinates on one realization first and projecting only afterward. Possible and robust states are existential and universal statements over that relation.

111.3.2 Ownership relation

For a fixed observer x, a realization determines

$$\big(\Sigma_\theta^+(x), \Sigma_\theta^-(x), \mathrm{Own}_{\Delta,\theta}(x), \mathsf{SF}_x, \text{record data}\big).$$

The exact perturbation object is the image of the realization space under this joint map. In particular, a claim that x is possibly background means

$$\exists F, \rho: \ \Sigma_\theta^+(x; F, \rho) \le \Delta \quad \text{and} \quad \Sigma_\theta^-(x; F, \rho) \le \Delta$$

for the **same** F and resolution ρ.

111.3.3 Realization-wise self-dual conjugacy

Theorem 111.3 (realization-wise self-duality)**.** *Grayscale inversion $F \mapsto -F$ induces a bijection between the bright exact relation over $\mathcal{P}$ and the dark exact relation over $-\mathcal{P}$. Consequently possibility, robustness, background status, and deterministic bright/dark ambiguity are preserved under polarity exchange in the expected self-dual sense.*

The theorem follows because inversion is a bijection on realizations and every dark construction is defined as the bright construction on $-F$.

Proof details are included in the observer-record proof package.

111.3.4 Local witness versus global synchronization

The exact relation already suggests two algorithmic regimes. A local exceedance needs one feature witness and a bounded set of separator constraints. A global safety condition quantifies over the entire realized record history. Later sections make this distinction exact: the former is polynomial on interval-valued cacti, while the latter is NP-complete already on radius-two trees.

111.4 Proof package: observer records and exact common-realization conjugacy

This proof package supplies the details used by the observer-record and exact-common-realization results above.

111.4.1 Maximin equality

For vertices p, q, x, if $\kappa_F(p, x) > \kappa_F(q, x)$, then

$$\boxed{\kappa_F(p, q) = \kappa_F(q, x).}$$

Indeed, the maximin strong-triangle inequality gives

$$\kappa_F(p, q) \geq \min\{\kappa_F(p, x), \kappa_F(q, x)\} = \kappa_F(q, x).$$

If the inequality were strict, applying the same inequality to $q \to p \to x$ would give

$$\kappa_F(q, x) \geq \min\{\kappa_F(q, p), \kappa_F(p, x)\} > \kappa_F(q, x),$$

a contradiction.

111.4.2 Proof of Theorem 111.2

Assume first that λ is an observer record, so $a_\lambda(x) > a_\mu(x)$ for every elder atom μ. Choose birth-carrier representatives e_λ, e_μ. By A.1,

$$\kappa_F(e_\lambda, e_\mu) = a_\mu(x).$$

The ordinary elder death of λ is the largest merge level at which it meets an elder atom, hence

$$d(\lambda) = \max_{\mu \in O(\lambda)} \kappa_F(e_\lambda, e_\mu) = \max_{\mu \in O(\lambda)} a_\mu(x) = D_x(\lambda).$$

If λ is not a record, choose an elder μ with $a_\mu(x) = D_x(\lambda) \geq a_\lambda(x)$. Then

$$\kappa_F(e_\lambda, e_\mu) \geq \min\{a_\lambda(x), a_\mu(x)\} = a_\lambda(x),$$

so $d(\lambda) \geq a_\lambda(x)$. Its observer-relative depth is therefore nonpositive and it cannot satisfy the strict core inequality for any $\theta > 0$. This proves both record death and nonrecord elimination.

111.4.3 Adjacent-record chain and supply profile

Order the records $r_1, \ldots, r_m$ from elder to younger. Because r_i strictly raises the running attachment, its elder pressure is the previous record value: $D_x(r_i) = a_{i-1}$. By A.2,

$$\boxed{d(r_i) = a_{i-1}.}$$

Thus each finite contribution is determined by (B_i, a_{i-1}, a_i). The danger conditions are exactly

$$B_i - a_{i-1} > H, \qquad a_i > \theta B_i + (1-\theta)a_{i-1}.$$

Define $E_x(b)$ to be the greatest observer attachment supplied by births older than level b. The function is monotone stepwise; a jump occurs exactly when a younger birth strictly raises the running maximum, i.e. exactly at an observer record. Its left value is D, its right value is A, and A.2 gives persistence $B-D$. Hence the one-profile criterion in the supply-profile section above follows by checking the displayed two danger inequalities at every finite jump. On a rooted tree, maximin connectivity to a parent is the minimum of the edge/vertex bottleneck and the child-subtree value, while independent child branches combine by maximum; induction on the rooted tree gives the stated max/min recurrence.

111.4.4 Self-dual conjugacy

The map $F \mapsto -F$ is a bijection between realizations of $\mathcal{P}$ and $-\mathcal{P}$. Superlevel births of $-F$ are precisely sublevel births of F; maximin connectivity, elder order, record status, and all strict/equality cases are transported under the polarity relabeling. Therefore the joint tuple

$$(\Sigma_\theta^+, \Sigma_\theta^-, \mathrm{Own}_{\Delta,\theta}, \mathsf{SF}, \text{record data})$$

is conjugated realization by realization. Existential and universal projections of this exact relation are consequently exchanged in the claimed bright/dark fashion. This proves Theorem 111.3.

111.5 Exact spatial-frame construction

The ownership projection does not encode route geometry. This proof package fixes one polarity-normalized realization and gives the second EF-FG projection used in the spatial-frame synthesis chapter.

Fix a spatial seed threshold $\Delta_{\mathrm{seed}} \geq 0$. Let S be the finite set of nonessential birth atoms satisfying

$$B_p - d(p) > \Delta_{\mathrm{seed}}.$$

Choose one representative vertex for each $p \in S$. For $v \in V$ define

$$\kappa_p(v) = \kappa_F(p, v), \qquad \beta(v) = \max_{r \in S} \kappa_r(v), \qquad W(v) = \{p \in S : \kappa_p(v) = \beta(v)\}.$$

The *rooted co-winner basin* $\mathcal{B}_p$ is the connected component containing p in the induced set $\{v : p \in W(v)\}$. The rooted unique-winner basin is defined analogously from $W(v) = \{p\}$. Selected seeds p, q are *carrier-adjacent* iff

$$\mathcal{B}_p \cap \mathcal{B}_q \neq \varnothing.$$

For an adjacent pair put

$$s(p,q) = \max_{v \in \mathcal{B}_p \cap \mathcal{B}_q} \beta(v).$$

The *top gate carrier* is the induced set

$$\{v \in \mathcal{B}_p \cap \mathcal{B}_q : \beta(v) = s(p,q)\},$$

retained componentwise rather than collapsed to one arbitrary point.

Fix an ordered pair (p,q) and $x \in \mathcal{B}_p$. For a top gate component C, a simple path in $\mathcal{B}_p$ from p to a vertex of C is an admissible top route. The component is *weakly incident through* x when at least one such route contains x, and *strongly incident through* x when every such route contains x. Thus strong incidence implies weak incidence; on a cactus the two conditions reduce respectively to route existence and a no-bypass (vertex-dominator) test in the rooted carrier subgraph.

For a bright sector define the midpoint margin

$$\mu^{+}_{p,q,x} = 2F(x) - F(p) - s(p,q).$$

The three signs > 0, $= 0$, and < 0 are retained, and are interpreted together with weak/strong incidence. The dark sector state is the same construction applied to $-F$.

To make boundary information explicit, let $T \subseteq V$ be a designated terminal set supplied with the spatial query. An exposed terminal for seed p is a vertex $t \in T \cap \mathcal{B}_p$. Its gate is

$$s_\partial(p,t) = \kappa_{F|\mathcal{B}_p}(p,t).$$

Put $s = s_\partial(p,t)$ and, for $v \in \mathcal{B}_p$,

$$\alpha(v) = \kappa_{F|\mathcal{B}_p}(p,v), \qquad \gamma(v) = \kappa_{F|\mathcal{B}_p}(t,v).$$

Vertices with $\alpha > s$ are forced to the peak side; vertices with $\gamma > s$ are forced to the terminal side; only the tie zone $\alpha = \gamma = s$ can be assigned by residual resolution. A *residual terminal resolution* is a connected terminal-side region R containing t, containing every forced terminal-side vertex in its connected terminal component, excluding forced peak-side vertices, and using only tie-zone vertices for the remaining choice. The intrinsic family of all such regions is denoted $\mathfrak{R}^{\mathrm{res}}_{p,t}$.

For $R \in \mathfrak{R}^{\mathrm{res}}_{p,t}$ define its floor and cap

$$L(R) = \min_{z \in R} F(z), \qquad U(R) = \max_{z \in R} F(z).$$

For the canonical terminal side $L(R) = s_\partial(p,t)$. At threshold Δ, define

$$T^{+}_{\Delta}(L,U) = \max\left\{L + \Delta, \frac{L+U}{2}\right\}, \qquad T^{-}_{\Delta}(L,U) = \min\left\{U - \Delta, \frac{L+U}{2}\right\}.$$

A query value y is terminal-dark when $y < T^{-}_{\Delta}$, neutral when $T^{-}_{\Delta} \le y \le T^{+}_{\Delta}$, and terminal-bright when $y > T^{+}_{\Delta}$. Grayscale inversion exchanges the bright and dark inequalities and fixes the neutral band.

Under interval uncertainty, these quantities must remain correlated. The exact residual object is therefore

$$\mathrm{ResRel}_{p,t,x} = \left\{ (s, U, y, \chi) : \begin{array}{l} \text{one admissible realization and one } R \in \mathfrak{R}^{\mathrm{res}}_{p,t} \\ \text{realize the tuple} \end{array} \right\},$$

where $y = F(x)$ and $\chi = \mathbf{1}[x \in R]$. Independent extrema of s, U, y, χ may not be recombined. On cactus graphs the block-cut decomposition gives constant-width route interfaces, so pairwise gates, weak/strong incidence, terminal exposure, residual connectedness, and the correlated floor/cap/query relation admit polynomial exact dynamic programs. This is a spatial-frame statement and is distinct from the polarity-core complexity theorems below.

Chapter 112

Threshold-Band Scheduling and Ownership Complexity

Research branch — Local witness versus global synchronization. *Exact common-realization semantics creates an algorithmic boundary. A local exceedance witness can be certified through bounded separator data on cactus graphs, whereas one globally synchronized realization that suppresses every dangerous record is NP-complete already on radius-two trees. The proof passes through a two-coordinate scheduling kernel and an exact physical lift.*

Book-wide notation. We use τ for the local critical birth floor and call the corresponding scheduling kernel the *τ-band*, or *threshold-band*. This distinguishes the local floor from the global balance-ratio parameter q.

112.1 The τ-band owner-scheduling kernel

We now isolate the arithmetic core of one-polarity common-realization safety.

112.1.1 Hard-band state

Write the running state as (M, D), where M is the current birth cap and D is the current elder supply. For one event i, let its birth interval be $[\ell_i, u_i]$ and its attachment interval be $[\alpha_i, \beta_i]$. In the hard band, safety is

$$a \le \theta B + (1-\theta)D.$$

Fix a critical birth floor τ. Events with $\ell_i = \tau$ must be resolved before the cap drops below τ. When $\tau \le D + H$, the band is easy: an H-safe suffix is immediately reachable. The nontrivial regime is therefore

$$\tau > D + H.$$

For a hard-band event define the maximal safe choice

$$U_i = \min\{M, u_i\}, \qquad a_i^* = \min\{\beta_i, \theta U_i + (1-\theta)D\}.$$

Every other τ-preserving safe realization of the event is dominated by this choice.

112.1.2 τ-band coordinates

Set

$$h = M - \tau, \qquad \xi = \tau - D,$$

and for event i define

$$h_i = u_i - \tau, \qquad A_i = \tau - \alpha_i, \qquad B_i = \tau - \beta_i.$$

For general θ let $c = 1 - \theta$ and $s_i = \min\{h, h_i\}$. Then an owner use is legal exactly when

$$\boxed{B_i < \xi}$$

and

$$\boxed{c\xi - \theta s_i \le A_i},$$

with deterministic transition

$$\boxed{(h, \xi) \longmapsto (s_i, \max\{B_i, c\xi - \theta s_i\})\,.}$$

For the hardness specialization $\theta = 1/2$ this becomes

$$(h, \xi) \longmapsto \left(s_i, \max\left\{B_i, \frac{\xi - s_i}{2}\right\}\right),$$

legal when

$$B_i < \xi, \qquad \frac{\xi - s_i}{2} \le A_i.$$

112.1.3 Fixed-order reachability

The scalar transition is monotone for a fixed effective headroom, but a used event can lower the future headroom. Consequently one scalar minimum does not suffice for an arbitrary fixed order. The correct state retains the current headroom as well.

Theorem 112.1 (fixed-order absorbing-tail reachability)**.** *For any fixed event order, whether some admissible subsequence reaches the absorbing safe tail $\xi \le H$ is decidable by a headroom-indexed dynamic program using $O(n^2)$ event-state transitions. If additionally*

$$h_0 \ge h_1 \ge \cdots \ge h_n,$$

then every used event has effective headroom h_i, independent of previous skips, and the one-pass prefix-minimum greedy is exact in $O(n)$ transitions.

A proposed owner word is still verified in $O(n)$ exact transitions, so NP membership does not rely on the stronger greedy special case.

Remark 112.2 (why the headroom hypothesis matters). At $\theta = 1/2$, $H = 0$, start from $(h, \xi) = (10, 10)$ and fix the order

$$e_1 = (0, 10, 0), \qquad e_2 = (10, 0, 0).$$

Using e_1 first gives $(0, 5)$, after which e_2 is infeasible because $(5 - 0)/2 > 0$. Skipping e_1 and using e_2 gives $(10, 0)$ and reaches the safe tail. Hence a single prefix minimum in ξ is not valid without retaining headroom information.

The dynamic program and the nonincreasing-headroom greedy proof are given in the threshold-band derivation later in this chapter.

112.2 NP-completeness of τ-band owner scheduling

112.2.1 Decision problem

The τ-band owner-scheduling completion problem asks whether a finite event family admits an order and legal owner-use pattern that reaches the absorbing safe tail. We use the fixed specialization

$$\theta = \frac{1}{2}, \qquad H = 0.$$

Membership in NP follows from a rational schedule certificate: one checks the event order and the deterministic τ-band transitions in polynomial time.

112.2.2 Reduction spine

NP-hardness is by reduction from Monotone NAE-3-Sat-E4 [134]. The main logical spine is as follows; the complete reduction, including the binding scale and phase-leak repairs, is given in the complete threshold-band reduction later in this chapter.

Each variable module has exactly two successful traversal modes. The choice designates one of two payload banks for deletion and leaves the other available for clause auditing. A two-gate **headroom certificate** prevents a payload token from being replaced by a token belonging to another variable or another truth bank. Clause modules use a three-token not-all-high audit, together with directed self-policing, so that a clause succeeds exactly when its three incoming truth states are not all equal. A homogeneous scale hierarchy isolates variables from one another and global audit floors prevent cross-module substitutions. A final finite control wrapper closes premature phase entry and early-pivot leakage.

Completeness maps a satisfying NAE assignment to the corresponding branch of each variable module and then executes all clause/audit modules. Soundness reconstructs a Boolean choice from every successful schedule and uses the headroom certificates plus the self-policing clause gadgets to show that every clause is NAE-satisfied.

Theorem 112.3 (τ-band NP-completeness)**.** *τ-band owner-scheduling completion is NP-complete even for*

$$\boxed{\theta = \frac{1}{2}, \qquad H = 0,}$$

with rational dyadic event parameters and a linear number of events.

The result isolates the source of hardness: **global order selection under one-use owner resources**, not evaluation of a fixed order.

The complete iff reduction and NP-membership proof are in the complete threshold-band reduction later in this chapter.

112.3 Exact lift to interval-valued scalar trees

The scheduling kernel is not merely an abstract surrogate. It is exactly realized by radius-two scalar graphs.

112.3.1 Affine rectangle conjugacy

Choose τ larger than all τ-band coordinates and apply

$$M = \tau + h, \qquad D = \tau - \xi,$$

and, for every event,

$$\ell_i = \tau, \qquad u_i = \tau + h_i, \qquad \alpha_i = \tau - A_i, \qquad \beta_i = \tau - B_i.$$

Under this change of variables, the maximal-safe rectangle transition is conjugate to the τ-band transition. Before the physical tree is built we apply the positive-floor normalization proved in the physical-lift proof package later in this chapter: every convenience floor $B_i = 0$ is replaced by one common rational $\varepsilon > 0$ chosen below every intended positive transition scale. the physical-lift proof package later in this chapter proves by a dominance coupling that this normalization preserves YES/NO equivalence; in particular it neither destroys the intended completeness schedules nor creates a false YES instance. After normalization,

$$0 < \varepsilon \le B_i \le A_i$$

for every event.

112.3.2 Radius-two physical realization

For each event create a branch

$$x - s_i - p_i,$$

where p_i is an interval-valued peak and s_i an interval-valued saddle. After positive-floor normalization,

$$F(s_i) \le \tau - \varepsilon = F(x) < \tau \le F(p_i).$$

Since the branch path is unique and the observer is no lower than the saddle,

$$\kappa_F(p_i, x) = F(s_i).$$

An initial essential branch sets the starting elder pressure, and a fixed observer guard prevents unintended critical changes. Peak heights encode the schedule order; saddle intervals encode the admissible attachments.

Theorem 112.4 (exact physical lift)**.** *For the constructed radius-two interval tree,*

$$\boxed{\begin{array}{c}\textit{threshold-band instance is YES}\\ \Longleftrightarrow\\ \textit{interval-tree one-polarity observer-safety instance is YES.}\end{array}}$$

The construction is polynomial and uses rational endpoints.

The exact conjugacy and tree-realization proof are in the physical-lift proof package later in this chapter.

112.3.3 Physical complexity theorem

A YES instance has a polynomial-bit rational realization and polynomial-size elder/tie certificate. On a tree, all maximin attachments are unique-path minima, so verification is polynomial.

Theorem 112.5 (physical observer-safety complexity)**.** *The problem of deciding whether an interval-valued rooted tree and observer x admit one realization and one admissible elder/tie resolution for which the one-polarity EF-FG saliency is H-safe at x is NP-complete even when*

$$\boxed{\theta = \frac{1}{2}, \qquad H = 0,}$$

the tree has radius two about the observer, and all interval endpoints are rational.

Thus the hardness does not require cycles, deep tree topology, or variable model parameters.

Polynomial witness size and verification are proved in the physical-lift proof package later in this chapter.

112.4 Self-dual ownership complexity

112.4.1 Dark-anchor neutralization

To lift one-polarity safety to the full self-dual ownership predicate, add one constant-size low-valued leaf d, the **dark anchor**. It is chosen so that it is not a bright primitive maximum and hence leaves every bright observer quantity unchanged.

In the inverted field $-F$, the anchor becomes the essential dark atom. Every other primitive dark birth reaches the observer through a path whose bottleneck equals the observer value. Consequently every finite dark birth has the same observer attachment as the anchor's running elder pressure.

Theorem 112.6 (dark-anchor neutralization)**.** *For every realization and every admissible resolution,*

$$\boxed{\Sigma_\theta^-(x;F) = 0.}$$

The statement is realization-wise; no independent choice of a dark realization is made.

A realization-wise proof is given in the dark-anchor proof package later in this chapter.

112.4.2 Possible background ownership

At $\Delta = 0$, after adding the dark anchor,

$$\mathrm{Own}_{0,\theta}(x;F) = \text{background} \iff \Sigma_\theta^+(x;F) = 0.$$

Thus the one-polarity safe-realization problem is exactly the self-dual possible-background decision problem.

Membership in NP follows from a polynomial-bit scalar realization and polynomial-size resolved elder/tie data for both polarities.

Theorem 112.7 (possible-background ownership complexity)**.** *Possible Background Polarity-Core Ownership is NP-complete even on interval-valued radius-two trees with fixed*

$$\boxed{\theta = \frac{1}{2}, \qquad \Delta = 0.}$$

The only additional construction beyond Theorem 112.5 is the constant-size dark anchor.

The reduction equivalence and NP-membership argument are in the dark-anchor proof package later in this chapter.

112.5 The local/global algorithmic boundary

The complexity theorem is best understood by contrasting it with local witness problems.

112.5.1 Local possible exceedance

For fixed θ and H, a positive saliency witness can be represented by one candidate birth together with a bounded set of separator/connectivity conditions. On interval-valued cactus graphs those constraints are decidable by polynomial exact routines.

Theorem 112.8 (possible one-polarity exceedance)**.** *On interval-valued cactus graphs,*

$$\exists F, \rho : \Sigma_\theta^\sigma(x; F, \rho) > H$$

is polynomially decidable for either polarity $\sigma \in \{+, -\}$.

A proof by a finite separator witness and cactus block-cut dynamic program is given in the cactus-witness proof package later in this chapter.

112.5.2 Complexity separation

Combining the local witness theorem, the fixed-order collapse, and the hardness results gives:

Theorem 112.9 (EF-FG local/global separation)**.** *For polarity-core saliency:*

1. *possible one-polarity exceedance is polynomially decidable on interval-valued cacti;*
2. *arbitrary fixed-order absorbing-tail reachability is polynomial (quadratic by Theorem 112.1), and the nonincreasing-headroom subclass is linear;*
3. *global possible one-polarity safety is NP-complete on radius-two trees;*
4. *self-dual possible-background ownership is NP-complete on radius-two trees.*

The transition is therefore neither "tree versus cactus" nor merely "deterministic versus uncertain." It occurs when one realization must coordinate all potential observer records under a global no-danger requirement.

The conclusion follows from Theorem 112.8, Theorem 112.1, and the hardness proofs completed in Appendices D–F.

112.6 Proof package: threshold-band derivation and fixed-order theorem

Let $c = 1 - \theta$. In a critical band with floor τ, no schedule may lower the running birth cap below τ while an event with minimum birth τ remains unresolved. If $\tau \le D + H$, every remaining birth $B \le \tau$ is persistence-safe, so only the hard regime $\tau > D + H$ matters.

For one hard-band event, put $U_i = \min\{M, u_i\}$. It has a τ-preserving safe realization iff

$$\alpha_i \le \theta U_i + cD.$$

The coordinatewise maximal safe attachment is

$$a_i^* = \min\{\beta_i, \theta U_i + cD\},$$

and every other τ-preserving realization is dominated by $(U_i, \max\{D, a_i^*\})$.

Translate by

$$h = M - \tau, \qquad \xi = \tau - D,$$

and $h_i = u_i - \tau$, $A_i = \tau - \alpha_i$, $B_i = \tau - \beta_i$. With $s_i = \min\{h, h_i\}$, owner legality becomes

$$B_i < \xi, \qquad c\xi - \theta s_i \le A_i,$$

and the transition is

$$(h, \xi) \mapsto (s_i, \max\{B_i, c\xi - \theta s_i\}).$$

The absorbing safe tail is $\xi \le H$. For a fixed legal owner word $i_1, \dots, i_k$ with realized headrooms s_t, repeated substitution gives

$$\xi_k = \max\left\{ B_{i_k}, cB_{i_{k-1}} - \theta s_k, \dots, c^k \xi_0 - \theta \sum_{t=1}^{k} c^{k-t} s_t \right\}.$$

Thus evaluation of a fixed word is linear-size max-affine arithmetic.

For an arbitrary fixed order, let

$$\mathcal{H} = \{h_0\} \cup \{\min(h_0, h_i) : 1 \le i \le n\}.$$

After every prefix, the current headroom belongs to $\mathcal{H}$. For $\eta \in \mathcal{H}$, let $g_j(\eta)$ be the minimum hard deficit $\xi > H$ reachable after processing the first j events with current headroom exactly η; use $+\infty$ if no such state exists, and retain one absorbing flag when some subsequence has reached $\xi \le H$. Initially $g_0(h_0) = \xi_0$.

Process event i from every finite state (η, g) in two ways. The skip transition retains (η, g). For the use transition set

$$s = \min\{\eta, h_i\}.$$

It is legal exactly when

$$B_i < g, \qquad cg - \theta s \le A_i,$$

and then outputs

$$y = \max\{B_i, cg - \theta s\}.$$

If $y \le H$, set the absorbing flag; otherwise update $g_i(s) \leftarrow \min\{g_i(s), y\}$.

For a fixed headroom η, a smaller hard deficit dominates every larger one for future absorbing-tail reachability. If an event is feasible at the smaller value, monotonicity gives no larger output there. If the smaller value is at or below an event floor B_i, any use from a larger live predecessor outputs at least B_i and cannot beat the smaller state. If the smaller value violates the upper feasibility inequality, every larger value violates it as well. Hence keeping only $g_j(\eta)$ is exact. There are at most $n + 1$ headroom values and n prefixes, so the dynamic program uses $O(n^2)$ event-state transitions. This proves the first part of Theorem 112.1.

Now assume

$$h_0 \ge h_1 \ge \cdots \ge h_n.$$

Whenever event i is used, its effective headroom is exactly h_i, independent of earlier skips. Put

$$U_i = \frac{A_i + \theta h_i}{c}, \qquad T_i(\xi) = \max\{B_i, c\xi - \theta h_i\}.$$

At a hard state event i is usable exactly when $B_i < \xi \le U_i$, and then $T_i(\xi) < \xi$. Scan left to right and use the event whenever it is usable. If the current greedy value is g, every alternative predecessor is at least g. If i is usable at g, monotonicity of T_i makes

$T_i(g)$ the smallest successor. If $g \le B_i$, every live use from a larger state outputs at least $B_i \ge g$. If $g > U_i$, the event is infeasible from every larger state. Induction proves the prefix-minimum invariant and the $O(n)$ greedy specialization.

112.7 Complete threshold-band NP-completeness reduction

We now give the reduction underlying Theorem 112.3. Fix $\theta = c = 1/2$ and $H = 0$, so

$$(h, \xi) \mapsto \left(s_i, \max\left\{ B_i, \frac{\xi - s_i}{2} \right\} \right), \qquad s_i = \min\{h, h_i\},$$

with legality $B_i < \xi$ and $(\xi - s_i)/2 \le A_i$. We reduce from Monotone NAE-3-SAT-E4.

112.7.1 The normalized Boolean interface

We first give the exact-one control used for a Boolean variable. Work at three consecutive caps

$$h_0, \qquad h_1 = h_0 - 1, \qquad h_2 = h_0 - 2,$$

and let both marker banks have size $m = 8$. Fix $Z = 1$, take

$$h_2 = 2^{80}, \qquad h_0 = h_2 + 2,$$

and set

$$y_0 = 2^{28}(Z + h_0 + 10), \qquad X = y_0 - h_0.$$

The module enters at (h_0, X). Each A-marker has headroom h_0 and each B-marker headroom h_1; all markers have the common large attachment threshold

$$A_{\text{big}} = X.$$

Choose their floors in two disjoint ordered intervals far below every assignment-time core value, with every B-marker floor larger than every A-marker floor. Distinct dyadic values in, say, $(-2, -3/2)$ for the A bank and $(-1, -1/2)$ for the B bank suffice for the local construction. Since every intended marker step is then core-limited, the numerical values of these floors do not enter the displayed recurrences. In the final composed reduction they may be translated to a small positive ordered interval, still below all assignment-time core outputs; this preserves every comparison below and permits the exact cleanup used in Section D.4.

At cap h_0, put $y = \xi + h_0$. An A-marker acts by $y \mapsto y/2$, so after t such markers

$$\xi_A(t) = \frac{y_0}{2^t} - h_0.$$

The **early A pivot** is

$$E_A : \qquad h = h_1, \quad B = X - 1, \quad A = A_{\text{big}}.$$

At prefix zero it sends (h_0, X) to $(h_1, X - 1)$. After even one A-marker we have $\xi_A(1) < X - 1$, so E_A is permanently owner-dead. Thus E_A is exactly the retain-A exit.

For the **late A pivot**, define

$$C_A(t) = \frac{\xi_A(t) - h_1}{2}$$

and choose

$$B_{L_A} = 0, \qquad A_{L_A} = \frac{C_A(7) + C_A(8)}{2}.$$

Hence L_A is infeasible before all eight A-markers have been consumed and becomes feasible at $t = 8$. The two first-fork outputs are therefore

$$\xi_H = X - 1$$

on the retain-A branch and

$$\boxed{\xi_L = C_A(8) = \frac{y_0}{2^9} - h_0 + \frac{1}{2}}$$

on the consume-A branch.

Now work at cap h_1. After t B-markers write

$$\xi_H(t) = \frac{\xi_H + h_1}{2^t} - h_1, \qquad \xi_L(t) = \frac{\xi_L + h_1}{2^t} - h_1.$$

The chosen scale gives $\xi_H(8) > \xi_L$. We use the sharpened second fork needed later in the global reduction. Set

$$B_{E_B} = \xi_L - \frac{1}{4}, \qquad U_{E_B} = \xi_H(8) - \frac{1}{4}, \qquad A_{E_B} = \frac{U_{E_B} - h_2}{2}.$$

At the low input ξ_L, the early pivot E_B is live and feasible and is floor-limited, so it sends the branch exactly to

$$\boxed{\xi_- = \xi_L - \frac{1}{4}}.$$

After one low-branch B-marker it is owner-dead. On every high-branch prefix $0 \le t \le 8$, the scalar is at least $\xi_H(8) > U_{E_B}$, so the corresponding pivot candidate exceeds A_{E_B} and E_B is infeasible there. Thus the consume-A branch can complete only by retaining the whole B bank through immediate E_B.

For the **late B pivot**, put

$$B_{L_B} = \xi_L + 1, \qquad C_H(t) = \frac{\xi_H(t) - h_2}{2}, \qquad A_{L_B} = \frac{C_H(7) + C_H(8)}{2}.$$

The floor B_{L_B} is above every scalar on the low branch, so L_B is owner-dead there. On the high branch it is infeasible through the first seven B-markers and becomes feasible after the eighth. At that point

$$C_H(8) = \xi_L + 1 - 2^{-8} < B_{L_B},$$

so it is floor-limited and sends the branch exactly to

$$\boxed{\xi_+ = \xi_L + 1}.$$

Consequently the only two local completions to cap h_2 are

$$E_A,\ B_1, \ldots, B_8,\ L_B$$

and

$$A_1, \ldots, A_8,\ L_A,\ E_B.$$

The first retains the entire A bank and consumes the entire B bank; the second consumes the entire A bank and retains the entire B bank. There is no completion retaining both banks, consuming both banks, or retaining a proper nonempty subbank. Thus the module is an exact-one Boolean selector.

The two exits share cap h_2 and satisfy

$$\xi_+ - \xi_- = \frac{5}{4}.$$

Consume eighteen copies of the explicit plateau normalizer

$$N: \qquad h_N = h_2, \qquad B_N = 0, \qquad A_N = \frac{\xi_+ - h_2}{2}.$$

At every intended normalizer input $\xi \le \xi_+$, its core candidate is $(\xi - h_2)/2 \le A_N$; it is positive throughout these eighteen uses, so the zero floor is inactive. Hence, in shifted coordinate $y = \xi + h_2$, each copy acts exactly by $y \mapsto y/2$. The normalized pair is

$$S_- = (h_2, \ell), \qquad S_+ = (h_2, \ell + \delta),$$

where

$$\delta = \frac{5}{2^{20}}, \qquad \ell = h_2 + 26 - \frac{7}{2^{20}}.$$

Finally choose the low-branch adapter cap $C = h_2 - 1$. A low adapter with threshold strictly between the two branch candidates is feasible only on S_-, while a high adapter with floor $\ell + \delta/2$ is owner-dead on S_- and live on S_+. The resulting transfer-mode entry scalars are

$$a = \frac{27}{2} - \frac{7}{2^{21}}, \qquad b = h_2 + 26 - \frac{9}{2^{21}},$$

with $b/a > 2^{75}$. Hence a completed variable selects exactly one Boolean transfer mode and designates exactly one of two eight-token payload banks for deletion, leaving the other for the final clause audits.

112.7.2 Exact headroom certification

At a transfer state (C, X) let the intended payload have effective headroom $0 < h < C$, every wrong candidate have headroom $t \ne h$, and assume $X > 2h$. A candidate gives $y(t) = (X - t)/2$. After a high-headroom first gate the core candidate is $z(t) = (X - 3t)/4$. Choose

$$\max\{\max_{t>h} y(t), \max_{t \le h} z(t)\} < Q < y(h).$$

A first gate with floor Q kills every $t > h$ substitute and floor-normalizes all $t \le h$ survivors to common scalar Q. The target then has next core candidate $c_h = (Q - h)/2$, whereas every $t < h$ gives $c_t = (Q - t)/2 > c_h$. Choose

$$c_h < R < A_2 < \min_{t<h} c_t.$$

A second gate with floor R and threshold A_2 accepts and floor-normalizes only the target. Thus payload identity is certified entirely by headroom; its own floor B remains free for clause logic.

For the eight-step deletion word use, at stage j,

$$X_j = \frac{S}{16^{j-1}}, \qquad h_j = \frac{X_j}{4},$$

$$Q_j = \frac{17}{64}X_j, \qquad R_j = \frac{1}{16}X_j, \qquad A_j^{(2)} = \frac{3}{32}X_j.$$

For the first gate choose the explicit attachment threshold

$$A_j^{(1)} = X_j.$$

Every intended or competing Gate-1 core candidate considered here is at most $X_j/2 < A_j^{(1)}$, so this threshold never removes an intended Gate-1 use.

The intended branch satisfies

$$\frac{X_j}{128} < \frac{X_j}{16} < \frac{3X_j}{32} < \frac{X_j}{8},$$

so the second gate accepts it and outputs $X_{j+1} = X_j/16$, while higher substitutes are rejected by Gate 1 and lower substitutes by Gate 2. This deletes exactly the designated eight-token bank.

112.7.3 Clause audit

At cap $h = 0$, an audit token acts by $\xi \mapsto \max\{B, \xi/2\}$. For a clause scale H let high tokens have floor H and low tokens floor $L \le H/2$, and enter at $\xi_0 = 4H$. Three high tokens fail because

$$4H \to 2H \to H$$

and the third high token is owner-dead. If at least one token is low, all three designated uses are possible and the state reaches $H/2$; a progress event at floor/threshold $H/4$ certifies completion. Hence one orientation checks *not all True*. Reversing the high/low truth orientation checks *not all False*. The conjunction is exactly NAE.

Audit stages are chained by

$$H_{s+1} = H_s/16, \qquad L_s = H_s/2.$$

After the two audit copies define the final positive audit output

$$\boxed{\varepsilon := H_{2m}/4.}$$

The explicit scale inequalities of the control construction place every intended input and every intended core output of a convenience zero-floor event at or above this final scale, with strict input liveness and attachment allowance $A_i \ge \varepsilon$. The successful word terminates with τ-deficit exactly ε.

Earlier-stage tokens are already owner-dead at later stages. Therefore any foreign substitution points strictly forward. Following the missing-token deficit reaches a maximal stage with no later substitute, so no globally successful schedule can have net foreign-token displacement.

112.7.4 Global scale separation and phase-leak repair

The τ-band kernel is positively homogeneous. Put

$$\lambda_i = \sigma^{i-1}, \qquad \sigma = 2^{-512},$$

and let $X_0 = X$ denote the normalized variable-input scalar from Section D.1.

Lemma 112.10 (scale separation and exact inter-variable composition)**.** *For every variable $i < n$, the two truth branches can be reconverged at one explicit positive raw port and transferred to the input port of variable $i+1$ by a constant number of τ-band events. Moreover, at every certified deletion stage of variable i, payloads belonging to earlier or later variables cannot substitute for the intended payload. All event coordinates are dyadic rationals of polynomial bit length.*

Proof. After the eight certified deletions, the high and low transfer modes have scalars

$$S_H = \frac{b}{16^8}, \qquad S_L = \frac{a}{16^8}.$$

Define the normalized raw port

$$C_{\mathrm{raw}} = \frac{S_L}{2}, \qquad P_{\mathrm{raw}} = \frac{3S_L}{4}.$$

Use eighty copies of the reconvergence normalizer

$$N_{\mathrm{rec}}: \qquad h = C_{\mathrm{raw}}, \qquad B = P_{\mathrm{raw}}, \qquad A = S_H.$$

On the low branch the first copy is floor-limited and reaches $(C_{\mathrm{raw}}, P_{\mathrm{raw}})$. On the high branch, with $y = \xi + C_{\mathrm{raw}}$, the recurrence is

$$y \longmapsto \max\left\{P_{\mathrm{raw}} + C_{\mathrm{raw}}, \frac{y}{2}\right\}.$$

Since $b < 2^{81}$, $a > 2^3$, and hence $S_H/S_L = b/a < 2^{78}$, eighty copies are more than enough to reach the same floor. Once the floor is reached, the unused copies are owner-dead. Thus both truth branches reach exactly the same raw port.

Scale variable i by λ_i. From the normalized constants,

$$2^{-31} < C_{\mathrm{raw}} < P_{\mathrm{raw}} < 2^{-28}, \qquad h_0 < 2^{81},$$
$$2^{107} < X_0 < 2^{108}, \qquad X_0 + h_0 = y_0 > 2^{108}.$$

Hence

$$\sigma h_0 < 2^{-431} < C_{\mathrm{raw}}, \qquad \sigma X_0 < 2^{-404} < P_{\mathrm{raw}},$$

so the first tail event is both cap-lowering and owner-live. Append 384 copies of

$$T_i: \qquad h_T = \lambda_{i+1} h_0, \quad B_T = \lambda_{i+1} X_0, \quad A_T = \lambda_i P_{\mathrm{raw}}.$$

The first and every later intended use is attachment-feasible because its core candidate is below $\lambda_i P_{\mathrm{raw}}/2 < A_T$. After the first copy the cap is $\lambda_{i+1} h_0$; in the shifted coordinate $y = \xi + \lambda_{i+1} h_0$ the recurrence is exactly

$$y \longmapsto \max\left\{\lambda_{i+1}(X_0 + h_0), \frac{y}{2}\right\}.$$

At the start of the tail,

$$y < \lambda_i 2^{-27},$$

whereas the target floor satisfies

$$\lambda_{i+1}(X_0 + h_0) > \lambda_i 2^{-404}.$$

Consequently

$$\frac{y}{2^{384}} < \lambda_i 2^{-411} < \lambda_{i+1}(X_0 + h_0),$$

so the target floor is reached by or before the 384th copy. At that moment the state is exactly

$$\lambda_{i+1}(h_0, X_0),$$

and every unused tail copy is owner-dead.

For separation, every transfer headroom of variable i lies in

$$\lambda_i(2^{-30}, 2^{80}).$$

Hence every future-variable transfer headroom is at most

$$2^{80}\lambda_{i+1} = 2^{-432}\lambda_i,$$

whereas every current intended transfer headroom exceeds $2^{-30}\lambda_i$. The ratio is therefore greater than 2^{402}. Future payloads are lower/over-drop candidates and fail Gate 2. A retained payload from the immediately preceding variable already has headroom greater than $2^{-30}\lambda_{i-1} = 2^{482}\lambda_i$, while the current cap is below $h_0\lambda_i < 2^{81}\lambda_i$. Hence every earlier retained payload appears at the current cap as a higher/no-drop candidate and fails Gate 1. Thus no foreign payload can replace an intended deletion token. Finally, the exponent of λ_i is $512(i-1)$, and all other constants have fixed dyadic encoding, so the complete construction has polynomial bit length. □

The same scale bounds keep future transfer and audit gates attachment-infeasible throughout the preceding handoff: before the next canonical input one has $\xi > \lambda_{i+1}X_0$, with $X_0 > 2^{107}$, while the normalized transfer/audit headrooms and attachment thresholds are below 2^{81}. Thus their core candidate is greater than $2^{105}\lambda_{i+1}$, far above the corresponding attachment threshold.

It remains to exclude premature use of the high-threshold assignment controls. Choose the residual RN136 marker floors positive and in their required order, but below

$$\varepsilon_{\mathrm{ctl}} = \frac{P_{\mathrm{raw}}}{2^{13}}.$$

At the common raw port use one drop event

$$D: \qquad h_D = 0, \quad B_D = 0, \quad A_D = P_{\mathrm{raw}}/2,$$

which sends the canonical scalar to $P_{\mathrm{raw}}/2$. Eight residual controls are then core-limited and yield

$$\xi_8 = \frac{P_{\mathrm{raw}}}{2^9}.$$

The cleanup gate

$$G_{\mathrm{ctl}}: \qquad h = 0, \quad B = 0, \quad A = \frac{P_{\mathrm{raw}}}{2^{10}}$$

is feasible after eight controls and infeasible after only seven. Therefore premature consumption of an assignment marker destroys both future truth branches. Likewise, the branch adapter requires exactly eighteen common post-choice normalizers; using one during the preceding handoff leaves only seventeen after the scale floor erases its scalar effect, so the certified adapter input cannot be reached.

The only remaining high-threshold leak is a premature early pivot E_A. Write its output as

$$\xi_E = X - 1 + \xi, \qquad 0 \le \xi \le 1.$$

After it fires, all remaining assignment markers act as $y \mapsto y/2$ in shifted coordinate. No truth exit is possible before eight contractions, and more than eight destroys the cleanup count. After exactly eight contractions, either the low route $L_A \to E_B$ reaches exactly ξ_-, or the direct mixed E_B value lies in

$$\xi_L + 1 - \frac{1}{256} \le \xi_{\text{mid}} \le \xi_L + 1 - \frac{1}{512}.$$

Since the truth midpoint is $\xi_L + 3/8$, every mixed state lies strictly on the high side and the branch adapter routes it to the canonical high mode. Thus premature phase entry cannot create a third truth value or retain both payload banks.

112.7.5 Correctness and NP membership

Given a NAE assignment, choose the corresponding truth branch in each variable, delete the opposite bank by the certified eight-step words, traverse the scale ladder, and run both audit copies. Every clause contains both truth values, so both not-all audits pass.

Conversely, every successful completion induces exactly one retained truth bank per variable. Headroom certification prevents foreign substitution during deletion; phase-leak repair preserves the two-valued interpretation; directed audit self-policing removes net foreign displacement. The first audit copy implies no clause is all True and the second no clause is all False. Hence the induced assignment is NAE-satisfying.

Each variable and clause contributes only constantly many events, so the construction has $O(n + m)$ events. The dyadic scale exponents and all rational coordinates have polynomial encoding length. A YES certificate is the owner-event order. Verification simulates at most N exact rational transitions; at $\theta = 1/2$ denominator exponent grows by at most one per non-floor step. Therefore the problem is in NP. This proves Theorem 112.3.

112.8 Proof package: exact physical lift and physical NP membership

The radius-two realization requires event saddles not to exceed the observer. The hardness construction initially uses a few zero floors only for notational convenience, so we first remove them.

Lemma 112.11 (positive-floor normalization)**.** *For the τ-band hardness family, there is a polynomial-bit rational $\varepsilon > 0$ such that replacing every convenience floor $B_i = 0$ by $B_i = \varepsilon$ preserves YES/NO equivalence. After normalization,*

$$0 < \varepsilon \le B_i \le A_i$$

for every event.

Proof. Use the same final audit scale fixed in the complete threshold-band reduction later in this chapter,

$$\varepsilon = H_{2m}/4.$$

It is a positive dyadic rational of polynomial encoding length. The explicit scale inequalities in the complete threshold-band reduction later in this chapter give, for every intended use of a convenience zero-floor event, input $\xi > \varepsilon$, attachment allowance $A_i \ge \varepsilon$, and

intended core output at least ε. Thus the intended completeness schedules are literally unchanged after replacing $B_i = 0$ by $B_i = \varepsilon$; in particular the successful word still ends at τ-deficit exactly ε.

For soundness, denote the original and normalized floors by $B_i^{(0)}$ and $B_i^{(\varepsilon)}$, respectively. Thus

$$B_i^{(0)} \le B_i^{(\varepsilon)},$$

with equality except for convenience zero-floor events, where $B_i^{(0)} = 0$ and $B_i^{(\varepsilon)} = \varepsilon$. Compare any normalized trajectory with the original instance and couple states in the product order

$$h^{(0)} \ge h^{(\varepsilon)}, \qquad \xi^{(0)} \le \xi^{(\varepsilon)}.$$

This holds initially. Suppose the normalized schedule next uses event i. If $B_i^{(0)} < \xi^{(0)}$, then the event is owner-live in the original state; moreover

$$\frac{\xi^{(0)} - \min\{h^{(0)}, h_i\}}{2} \le \frac{\xi^{(\varepsilon)} - \min\{h^{(\varepsilon)}, h_i\}}{2} \le A_i,$$

so the original use is feasible. Its new deficit is no larger because both its floor and its core candidate are no larger, while its new headroom is no smaller. If instead $\xi^{(0)} \le B_i^{(0)}$, then $B_i^{(0)} \le A_i$ implies $\xi^{(0)} \le A_i$, so event i is already passively resolved in the original instance and may be skipped. The product order is preserved in either case. Keeping the larger old headroom only makes every later core-feasibility inequality easier, while the smaller old deficit never worsens terminal safety. Induction therefore turns every normalized successful completion into a successful completion of the original instance. Hence normalization cannot create a false YES. Together with the unchanged intended completeness schedules, YES/NO equivalence is preserved. □

Choose

$$\tau > \max\{\xi_0, \max_i A_i\} + 1$$

and truncate h_i to $\bar{h}_i = \min\{h_i, h_0\}$, which changes no reachable effective headroom. Map

$$M = \tau + h, \qquad D = \tau - \xi,$$

and

$$\ell_i = \tau, \qquad u_i = \tau + \bar{h}_i, \qquad \alpha_i = \tau - A_i, \qquad \beta_i = \tau - B_i.$$

By Lemma 112.11,

$$0 \le \alpha_i \le \beta_i \le \tau - \varepsilon < \tau = \ell_i \le u_i.$$

Thus the physical birth/attachment constraint is automatic and every event lies in the same critical τ-band.

At $(M, D) = (\tau + h, \tau - \xi)$, with $s_i = \min\{h, \bar{h}_i\}$,

$$U_i = \min\{M, u_i\} = \tau + s_i.$$

The physical core-feasibility inequality

$$\alpha_i \le \frac{1}{2}U_i + \frac{1}{2}D$$

is exactly $(\xi - s_i)/2 \le A_i$, while the event raises the owner exactly when

$$\beta_i > D \iff B_i < \xi.$$

The owner-changing update translates to

$$\xi' = \max\left\{B_i, \frac{\xi - s_i}{2}\right\}, \qquad h' = s_i.$$

Hence rectangle-event and τ-band owner dynamics are exactly conjugate; owner-dead events correspond to $\beta_i \le D$ and may be processed passively.

For the tree realization create an observer x with

$$F(x) = \tau - \varepsilon$$

and, for every event, a branch $x - s_i - p_i$ with

$$F(p_i) \in [\tau, \tau + \bar{h}_i], \qquad F(s_i) \in [\tau - A_i, \tau - B_i].$$

Then

$$F(s_i) \le \tau - \varepsilon = F(x) < \tau \le F(p_i).$$

Thus p_i is a primitive local maximum, s_i is not, and the unique p_i–x path has minimum exactly $F(s_i)$:

$$\boxed{\kappa_F(p_i, x) = F(s_i).}$$

Different event rectangles are independent because their branches are vertex-disjoint away from x.

Put $M_0 = \tau + h_0$ and $D_0 = \tau - \xi_0$. Add an essential branch $x - s_0 - e$ with $F(e) = M_0$ and $F(s_0) = D_0$, resolving e before equal-height event peaks. Add one guard leaf g adjacent to x with $F(g) = \tau$ and resolve it after event peaks born at τ. Event peak heights realize the schedule order and saddle values realize the attachments. After a prefix $i_1, \dots, i_t$ the observer elder pressure is

$$D_t = \max\{D_0, F(s_{i_1}), \dots, F(s_{i_t})\},$$

exactly the rectangle running supply. By the observer-record proof package, a peak that raises this maximum is an observer record whose death is the previous elder pressure, so its safety inequality is exactly the rectangle/τ-band inequality.

In the successful normalized construction the final τ-deficit is ε, hence final elder pressure is $\tau - \varepsilon = F(x)$. The guard has observer attachment $F(x)$ and is therefore a nonrecord. It creates no positive saliency. Conversely, the guard only adds a constraint, so any observer-safe tree realization already induces a feasible rectangle schedule. Therefore

$$\boxed{\tau\text{-band YES} \iff \text{radius-two interval-tree observer-safety YES.}}$$

The construction uses $2N + O(1)$ vertices and polynomial rational endpoints.

For NP membership, use as certificate a rational scalar realization, total preorder/tie resolution, bottleneck identities, and record/nonrecord pattern. Once these discrete choices are fixed on a tree, all remaining conditions are rational linear equalities and inequalities: interval bounds, order/tie relations, path-minimum identities, local-maximum conditions, record inequalities, and either $B - D \le H$ or $a \le \theta B + (1-\theta)D$. Replace strict inequalities by one common positive rational slack δ. A nonempty rational polyhedron has a basic feasible rational point of polynomial bit length. The verifier checks all path minima and inequalities exactly in polynomial time. This proves Theorems 7.1–7.2.

112.9 Proof package: dark-anchor lift and ownership NP-completeness

Start from the bright hardness tree and attach one fixed sufficiently low leaf d at the observer-side construction so that it is never a bright primitive maximum. Therefore all bright births, attachments, records, and saliencies are unchanged.

In the inverted field $-F$, the leaf d is the essential dark atom. Every finite dark birth reaches the observer through a path whose bottleneck is the observer value, so all finite dark observer attachments equal the running essential dark pressure. Consequently no finite dark atom strictly raises the running elder supply. By the nonrecord elimination theorem in the observer-record proof package,

$$\boxed{\Sigma_\theta^-(x;F)=0}$$

for every realization and every admissible tie/elder resolution.

At $\Delta = 0$ we therefore have, realization by realization,

$$\mathrm{Own}_{0,\theta}(x;F)=\text{background} \iff \Sigma_\theta^+(x;F)=0.$$

The construction adds only constantly many vertices and leaves the radius-two bound unchanged. Thus the bright one-polarity safe-realization problem reduces exactly to Possible Background Polarity-Core Ownership. NP membership uses the same polynomial-bit realization certificate as the physical-lift proof package later in this chapter, evaluated for both polarities. This proves Theorems 8.1–8.2.

112.10 Proof package: local possible-exceedance witness on cactus graphs

Fix one polarity and one finite birth atom λ. Write

$$B=B_\lambda, \qquad a=a_\lambda(x), \qquad D=D_x(\lambda).$$

By the observer-record proof package, λ contributes more than H iff

$$D<B-H, \qquad D<\frac{a-\theta B}{1-\theta}.$$

Equivalently, there exists a separator scalar ℓ such that

$$D<\ell, \qquad B-\ell>H, \qquad a>\theta B+(1-\theta)\ell. \tag{G.1}$$

The last inequality means that a marked representative p and the observer x lie in the same component of

$$\{v : F(v)>t_\theta(B,\ell)\}, \qquad t_\theta(B,\ell)=\theta B+(1-\theta)\ell.$$

The inequality $D<\ell$ means that the x-component of $\{F\ge\ell\}$ contains no elder capable of suppressing the marked birth. Ties require only two modes: either a strict-higher elder exists, or one marked disconnected equal-B plateau is resolved elder. Thus a witness uses a constant number of marked vertices and the two retained parameters (B,ℓ).

Root the cactus block-cut tree at the block containing x. For each threshold graph in the certificate, a block interface stores only:

(i) whether each articulation port is active;

(ii) the connectivity partition of active ports;

(iii) whether the active component is connected to the marked root;

(iv) constant-many flags for the required presence/absence of an elder/high vertex and for the equal-height tie mode.

The number of thresholds and marks is constant, so every interface has constant state width. A bridge has a constant transition table. A simple cycle is scanned as a two-port finite automaton; child-block messages combine without increasing the state width.

For a vertex interval $[l_v, u_v]$, local modes are determined by comparisons with the affine forms

$$B, \qquad \ell, \qquad \theta B + (1-\theta)\ell,$$

together with the fixed offset $B - H$. Their truth pattern changes only when one of these forms meets l_v or u_v, or on the relevant equality faces. Hence $O(n)$ rational lines in the two-dimensional (B, ℓ) plane induce an arrangement with $O(n^2)$ cells and faces. On each cell every vertex has a fixed finite local-mode set, so one bottom-up block-cut DP decides feasibility in $O(n)$ state transitions.

Enumerate the marked birth representative p and, in equal-elder mode, one elder mark q. There are $O(n^2)$ marked choices. For each, inspect the $O(n^2)$ arrangement cells/faces and run the $O(n)$ cactus DP. A straightforward implementation therefore uses

$$\boxed{O(n^5)}$$

rational arithmetic/state-transition operations; no optimality of this bound is claimed. Coefficients have polynomial bit length, so rational cell representatives and comparisons have polynomial bit complexity.

Soundness. An accepted state supplies an actual birth plateau, an admissible strict/equal elder mode, and a separator ℓ satisfying (G.1), so the marked atom contributes saliency $> H$.

Completeness. If one realization has $\Sigma_\theta(x) > H$, choose an atom attaining the contribution, a representative of its birth plateau, and a separator ℓ strictly between D and the two upper bounds in (G.1). That realization lies in one enumerated arrangement cell/face and induces an accepted block-cut state.

This proves Theorem 112.8. Applying the same construction to $-F$ gives the opposite polarity. The local/global separation theorem then follows from Theorem 112.1 and Appendices D–F.

Chapter 113

All-Orders Common-Realization Obstructions and EF–FG Synthesis

Research branch — Unbounded obstruction order and external generativity. *The complexity theorem is not the only global obstruction. This chapter constructs protected strict-record families whose feasible subfamilies form simplex boundaries of arbitrary dimension, first under the canonical threshold-band section and then in a section-free radius-two scalar realization. The chapter closes by translating those native EF–FG families into the USNRT protected-obligation language.*

113.1 Unbounded common-realization obstruction order

NP-completeness is an algorithmic statement. EF-FG also exhibits an all-orders **structural** local-to-global obstruction.

113.1.1 Protected record complexes

Let P be a finite family of named strict observer-record obligations. Define

$$K_{\mathrm{raw}}(P) = \left\{ A \subseteq P : \begin{array}{l} \text{one admissible EF–FG realization} \\ \text{satisfies every obligation in } A \end{array} \right\}.$$

When the native maximal-safe hard-band rule is imposed, define

$$K_{\sigma}(P) = \left\{ A \subseteq P : \begin{array}{l} \text{one admissible canonical realization} \\ \text{satisfies every obligation in } A \end{array} \right\}.$$

Both are downward-closed simplicial complexes. Their minimal nonfaces measure the order at which local joint feasibility ceases to assemble into one realization.

113.1.2 Canonical boundary theorem

Fix $r \geq 1$. Start from

$$h_0 = 1, \qquad x_0 = 2^r,$$

and take $r+1$ parameter-identical events

$$(h_i, A_i, B_i) = (0, 2^{r-1}, 1).$$

For a canonical owner use,

$$x' = \max\{1, x/2\}.$$

After k protected uses,

$$x_k = 2^{r-k}, \qquad 0 \le k \le r.$$

Thus any proper protected subfamily, having size at most r, may be scheduled first and every selected event is a strict canonical record. After r uses, however, $x = 1 = B_i$, so one further strict owner record would require $1 < 1$.

Theorem 113.1 (canonical record boundary)**.** *For every $r \ge 1$ there exists an interval-valued radius-two EF-FG tree with $|P_r| = r + 1$ such that*

$$\boxed{K_\sigma(P_r) = \partial\Delta^r.}$$

This theorem is genuinely section-dependent for the symmetric family: with arbitrary safe attachments, the analogous raw claim can fail already at $r = 2$.

The displayed construction is the full proof: every proper face uses at most r protected events, while a $(r + 1)$st strict owner use is blocked by $B_i = x = 1$.

113.1.3 Raw boundary theorem

The section-free construction uses shared maximin geometry. Fix $r \ge 1$ and form a radius-two graph with observer x, gates $g_1, \dots, g_r$, individual peaks $p_1, \dots, p_r$, and one shared peak p_*. Each p_i is connected to x through g_i, while p_* is connected through every gate. Add one fixed essential branch to initialize observer pressure.

Choose

$$F(x) = 5, \qquad F(s_0) = 1, \qquad F(e) = 8, \qquad F(g_i) = u_i \in [2, 3],$$

and allow event peak values in $[6, 7]$. Then

$$\kappa_F(p_i, x) = u_i, \qquad \kappa_F(p_*, x) = \max_i u_i.$$

Let

$$P_r = \{p_1^{\mathrm{ob}}, \dots, p_r^{\mathrm{ob}}, p_*^{\mathrm{ob}}\}.$$

Every proper subfamily is realizable. If the shared obligation is absent, choose distinct selected gates and process the selected individual peaks in increasing attachment order. If the shared obligation is present, at least one individual obligation is absent; let an omitted gate supply the maximum value 3, choose all selected individual gate values distinct below 3, process them first, and then process p_*.

The full family is impossible. In every realization, some index j attains the maximum gate value, so

$$\kappa_F(p_*, x) = u_j = \kappa_F(p_j, x).$$

All $r+1$ events cannot then be strict records in one elder order, because strict records require strictly increasing and hence pairwise distinct attachments.

Theorem 113.2 (raw protected-record boundary)**.** *For every $r \ge 1$,*

$$\boxed{K_{\mathrm{raw}}(P_r) = \partial\Delta^r.}$$

Hence every proper protected observer-record family admits a common scalar realization while the full family does not.

The preceding proper-face construction and duplicate-maximum argument form the full proof.

113.1.4 Consequence

For every fixed locality order k, choose $r > k$. Every protected subfamily of size at most k is realizable, yet the whole family is not. Therefore

Local-to-global consequence. No bounded order of protected local-feasibility tests certifies global common realizability in general.

The statement is specific to the EF-FG observer-record semantics; it does not claim novelty for simplex boundaries, Helly-type language, or local/global inconsistency in the abstract.

113.2 Spatial-frame layer and complete EF-FG output

The complexity results above concern the polarity-core ownership projection. EF-FG retains a second, spatial projection because scalar foreground/background structure contains information that observer saliency alone does not encode.

For a fixed realization, competitor extrema determine maximin gates toward the observer. Weak/strong sector incidence, terminal directions, and residual terminal regions produce a spatial label recording which side of a local extremal frame the observer occupies. On trees, much of this structure is rigid; on cactus graphs, multiple route channels require a correlated residual state. The spatial layer is self-dual under $F \mapsto -F$.

The **complete deterministic EF-FG state** is therefore a pair

$$\boxed{\big(\mathsf{PO}_x(F), \mathsf{SF}_x(F)\big),}$$

accompanied when needed by witness/certificate data. Under interval uncertainty the exact state is not the Cartesian product of marginal possibilities; both projections must be computed from one common realization before projection.

The exact spatial-frame construction in Chapter 111 supplies pairwise gates, route incidence, terminal thresholds, correlated residual states, and the cactus closure statement. None of those details is used to strengthen the ownership hardness claim; they are retained because they are part of the EF-FG theory rather than an implementation artifact.

113.3 Finite evaluation and structural consistency checks

The fixed-realization evaluation and the interval-valued decision questions are mathematically distinct. For a specified grayscale field, the deterministic evaluation consists of locating bright and dark extrema, their persistence and record chains, observer attachments, saliency values, polarity-core ownership, route incidence, and the resulting fixed spatial frame. Every comparison is made for the same specified realization, so the associated construction is polynomial; this does not settle the existential ownership questions for interval inputs.

113.3.1 Fixed-realization evaluation

The evaluation data are:

- bright and dark extrema and persistence values;
- observer attachments, record chains, and saliency;
- polarity-core ownership and fixed-realization spatial-frame output;
- finite witnesses for the stated route and terminal conditions.

These objects are determined by the definitions and the indicated finite comparisons. Their existence is separate from the global NP-completeness statement about choosing a common realization.

113.3.2 Interval-valued instances

For interval-valued inputs, local witness conditions and globally coupled common-realization conditions must be treated separately. A local possible-exceedance predicate admits the stated finite witness criterion; correlated safety and ownership conditions demand one common choice of all uncertain quantities, not independent coordinatewise extrema. The interval-cactus theorem is proved by its combinatorial decomposition; no implementation hypothesis is part of its statement.

113.3.3 Finite consistency checks

The finite checks address observer-record death identities, grayscale-inversion duality, ownership partitions, record examples, deterministic ambiguity, weak/strong route incidence, terminal sectors, interval path and cactus cases, common-realization/order instances, and the smallest raw boundary seed at $r = 2$. These ranges provide consistency evidence for the definitions and boundary conventions. They do not replace the symbolic proofs or extend any theorem beyond its stated hypothesis.

113.4 Discussion

113.4.1 What the theory adds

EF-FG is not a replacement for merge trees, persistence, mathematical morphology, or uncertain scalar-field topology. It uses those neighboring structures to define a specific foreground/background semantics. Three aspects are central.

First, observer-relative saliency admits a hierarchy-free record normal form. Second, uncertainty is treated by exact common realization rather than by independent coordinate ranges. Third, the same semantics supports both computational and structural local/global separation theorems.

The NP-completeness result and the simplex-boundary theorem are complementary. The former says that choosing one globally safe realization can encode hard combinatorial synchronization. The latter says that even arbitrarily high-order local feasibility can fail to certify global feasibility. Neither theorem is implied by the other.

113.4.2 Why radius two matters

Hardness on radius-two trees rules out an explanation based on graph-topological complexity. The reduction uses shallow geometry; the combinatorial burden lies in coordinating the scalar realization and event order. Conversely, the raw boundary theorem uses radius-two fan graphs because shared gates are the mechanism that forces the duplicate maximum. The canonical boundary theorem already works on radius-two trees.

113.4.3 Scope and limitations

Several boundaries should remain explicit.

- The NP-complete ownership predicate is the **polarity-core ownership projection**, not the full spatial-frame mask.
- Deterministic evaluation of one fixed scalar field is not NP-complete.
- The tractable local-exceedance result is proved for the stated structured graph class; it is not a universal polynomial theorem for arbitrary graphs and arbitrary EF-FG predicates.
- The raw boundary theorem is category-specific; it does not assert that every simplicial complex is realizable as an EF-FG obligation complex.
- The present image-analysis experiments are validation and demonstration, not a benchmark claim against learned segmentation systems.

113.4.4 Second-stage program

The foundational theorem package leaves five explicit directions: characterize which finite complexes occur as EF–FG common-realization complexes; determine parameterized complexity in the number of incomparable records or uncertain gates; extend exact perturbation semantics beyond boxes; develop multichannel and spatiotemporal versions; and determine the complexity of the full spatial-frame projection under uncertainty.

These questions are retained here as the historical handoff from EF–FG I, but they are not left open in the monograph. The three chapters immediately following this one form EF–FG II and close all five within explicit finite scopes: shallow global spatial-frame NP-completeness, raw and canonical finite-complex universality, two fixed-parameter tractable ambiguity regimes, correlated rational-polyhedral common-realization semantics, and finite multichannel/spatiotemporal transfer.

A broader exact-realization viewpoint also motivates comparing EF-FG obligation complexes with protected-realization invariants developed in other domains in earlier chapters. The EF–FG theorems proved here, however, are self-contained and require no external realization-theory axiom.

113.5 Conclusion

EF-FG provides a self-dual mathematical theory in which foreground and background structure is organized by extrema, persistence, maximin observer geometry, and exact common-realization uncertainty. Its two projections are deliberately separate: polarity-core ownership captures persistent observer-relative explanation, while the spatial-frame layer captures carrier and terminal geometry.

For the ownership layer, observer records eliminate the need to carry the full merge hierarchy. This normal form leads to an exact threshold-band scheduling kernel. Fixed schedules are easy to evaluate, but choosing a globally safe order is NP-complete. The kernel lifts exactly to interval-valued radius-two scalar trees, and a constant-size dark anchor yields NP-completeness of Possible Background Polarity-Core Ownership at fixed $\theta = 1/2$ and $\Delta = 0$.

The same common-realization semantics also supports an all-orders structural obstruction: for every r, explicit EF-FG families realize the simplex boundary $\partial\Delta^r$ as a

protected record-feasibility complex, both canonically and, with shared maximin geometry, in a section-free raw form. Hence no bounded local consistency order can certify global common realizability in general.

Together these results identify a precise boundary: **individual danger may be witnessed locally, but global absence of danger and global assembly of protected records can require genuinely global coordination inside one scalar realization.**

113.6 External generativity: interval-valued scalar fields

The EF–FG scalar-field theory proved in the preceding chapters supplies a downstream test of the USNRT architecture. The application is intentionally asymmetric: EF-FG provides native scalar-field theorems, while USNRT supplies a typed interpretation of their common-realization obstruction; the USNRT layer does not claim those scalar-field boundary constructions as a second primary home.

For a finite protected family P, let $K_\sigma(P)$ denote EF-FG feasibility under its native maximal-safe canonical section and $K_{\mathrm{raw}}(P)$ raw common-realization feasibility without imposing that section. The preceding EF–FG development proves two all-orders boundary results:

$$K_\sigma(P_r) = \partial\Delta^r, \qquad K_{\mathrm{raw}}(P_r) = \partial\Delta^r \quad (r \geq 1),$$

using respectively the canonical threshold-band family and the raw radius-two fan construction.

Proposition 113.3 (USNRT interpretation of the EF-FG boundary families)**.** *Under the protected-obligation interpretation, both EF-FG families are instances of unbounded common-realization obstruction order: every proper protected subfamily is jointly realizable, while the full family is not. The canonical family additionally depends on a chosen section, whereas the raw family does not. No new USNRT axiom is required to state this distinction.*

Proof. The preceding EF–FG development supplies the realizability statements and their proofs. In USNRT, place the protected observer-record obligations as vertices of $\mathrm{Ob}_h(P_r)$. A face is present exactly when one common EF-FG realization satisfies the corresponding protected obligations. Therefore the two cited EF-FG identities translate directly to $\mathrm{Ob}_h(P_r) \cong \partial\Delta^r$. The only difference between the two cases is whether the EF-FG section is part of the realization contract. □

This application is evidence of expressiveness, not a logical premise of the core theory. It also avoids theorem duplication: the scalar constructions, physical lift, and NP-completeness results remain in the preceding EF–FG chapters, while the present USNRT interpretation records only the typed consequence relevant to its obligation-complex language.

Chapter 114

EF–FG II: Spatial-Frame Complexity and Finite-Complex Universality

The preceding EF–FG chapters established the foundational semantics, the observer-record normal form, exact common-realization uncertainty, the threshold-band ownership kernel, shallow-tree NP-completeness, and all-orders obstruction families. They also isolated five theorem-sized directions. The present second-stage development closes those directions within explicit finite scopes.

The first two closures concern the spatial projection and the topology of common realizability. Complete finite categorical spatial-frame realization is NP-complete even on shallow bipartite graphs, while protected-obligation complexes are universal: every finite simplicial complex occurs under both raw and canonical EF–FG common-realization semantics. These results sharpen the earlier local/global boundary without attributing hardness to deep graph topology.

Relation to prior work. Merge trees, component trees, and trees of shapes provide established scalar-field and morphological summaries [125, 126, 130, 131]. Uncertainty in scalar-field topology has been studied through mandatory critical points and statistical/ensemble summaries [128, 129, 136, 135]. Multivariate topology uses genuinely multivariate objects such as Reeb spaces and related simplifications [152], while temporal scalar-field analysis includes merge-tree alignment, tracking, and time-varying contour topology [153, 154]. The claims here concern the specific EF–FG common-realization decision problems, finite-complex universality inside EF–FG semantics, and the exact parameterized/polyhedral/layered transfer theorems developed in the following chapters.

114.1 Preliminaries from EF–FG I

We use the scalar-field and observer semantics of the preceding foundational EF–FG chapters. Only the ingredients required for the new theorems are recalled here.

Let $G = (V, E)$ be a finite graph and $F : V \to \mathbb{R}$ a scalar field. For vertices u, v, the maximin connectivity is

$$\kappa_F(u, v) = \max_{\pi : u \rightsquigarrow v} \min_{w \in \pi} F(w).$$

A resolved bright birth atom has a birth value, an elder order, and an observer attachment. Dark quantities are defined by applying the same construction to $-F$. The observer-record theorem from EF–FG I reduces pointwise polarity-core saliency to a record chain:

a finite birth is relevant exactly when its observer attachment strictly raises the running elder pressure.

For spatial-frame structure, fix the selected finite birth atoms. For a seed p define

$$\kappa_p(v) = \kappa_F(p, v), \qquad \beta(v) = \max_r \kappa_r(v), \qquad W(v) = \{p : \kappa_p(v) = \beta(v)\}.$$

The rooted co-winner basin B_p is the component containing p in $\{v : p \in W(v)\}$. If p, q are carrier-adjacent, their gate floor is

$$s(p, q) = \max_{v \in B_p \cap B_q} \beta(v).$$

For a boundary terminal $t \in B_p$, the terminal gate is

$$s_\partial(p, t) = \kappa_{F|B_p}(p, t).$$

For an observer x, the bright midpoint margin against a gate floor s is

$$\mu^+_{p,q,x} = 2F(x) - F(p) - s,$$

with the dark analogue obtained from $-F$. Weak incidence means that at least one admissible top route passes through x, while strong incidence means that every admissible top route does.

Under uncertainty, a claimed state must arise from one common realization. For a realization family $\mathcal{P}$ and a finite query Q,

$$\text{Possible}_{\mathcal{P}}(Q) \iff \exists F \in \mathcal{P}, \exists \rho : \ Q(F, \rho),$$

where ρ denotes the finite elder/tie resolution. Robust semantics use universal quantification over the same common-realization space.

For polarity-core complexity we also use the threshold-band kernel. At $\theta = 1/2$, a hard-band state is (h, x) and event i has parameters (h_i, A_i, B_i). With

$$s_i = \min\{h, h_i\},$$

owner use is legal exactly when

$$B_i < x, \qquad \frac{x - s_i}{2} \le A_i,$$

and the transition is

$$(h, x) \mapsto \left(s_i, \max\left\{ B_i, \frac{x - s_i}{2} \right\} \right).$$

For a fixed admissible event order, the prefix-minimum theorem of EF–FG I evaluates existential completion in linear time. The NP-completeness therefore lies in choosing a successful order among incomparable one-use owner events, not in evaluating a proposed order.

114.2 Global spatial-frame realization complexity

The ownership hardness of EF–FG I does not determine the complexity of the spatial-frame projection. We now classify a natural finite categorical realization problem for that projection.

114.2.1 Terminal mixedness

For one polarity and observer x, let

$$M_T(x) = (W^+, W^-, S^+, S^-)$$

be the terminal peak-side mask. The bit W^+ is one iff some weakly incident exposed terminal sector has positive midpoint margin; W^- is one iff some such sector has nonpositive margin; S^+ and S^- use strong incidence.

Definition 114.1 (Possible Terminal-Mixed Mask (PTMM))**.** The input consists of a finite graph, rational interval scalar values, a boundary-terminal set T, an observer set X, and a rational seed threshold. The question is whether one common scalar realization and one elder/tie resolution satisfy

$$M_T(x) = (1, 1, 1, 1) \qquad \text{for every } x \in X.$$

Theorem 114.2 (Terminal-mask hardness)**.** *PTMM is NP-complete. NP-hardness holds simultaneously when the graph is bipartite, has radius two about a specified hub, the seed threshold is fixed at* 2*, uncertainty occurs only on boundary terminals, every uncertain terminal has interval* $[5, 7]$*, and every other scalar value belongs to* $\{0, 8, 10, 20\}$*.*

Proof. Reduce from Monotone NAE-3-SAT-E4 [134]. Let h be a hub with $F(h) = 0$ and let e be an essential peak with $F(e) = 20$. For each clause c, create an observer x_c with value 8 and a clause peak p_c with value 10, with edges

$$h - x_c - p_c.$$

For each Boolean variable i, create one shared boundary terminal z_i with

$$F(z_i) \in [5, 7],$$

and join x_c to z_i iff variable i occurs in clause c.

The graph is bipartite with hub radius two. The selected finite seeds are exactly the clause peaks. At x_c the unique rooted seed is p_c: its score is 8, while any other clause peak reaches x_c through a bottleneck at most 7. A terminal z_i belongs to B_{p_c} exactly when $i \in c$. In that case the path $p_c - x_c - z_i$ lies in the basin, the terminal gate is

$$s_\partial(p_c, z_i) = F(z_i),$$

and x_c is strongly incident because no basin route can bypass it.

Therefore the terminal midpoint margin is

$$\mu_{c,i} = 2F(x_c) - F(p_c) - s_\partial(p_c, z_i) = 16 - 10 - F(z_i) = 6 - F(z_i).$$

Hence a clause observer has mask 1111 exactly when its three terminals contain at least one value below 6 and at least one value at least 6. A NAE assignment gives such a realization by mapping the two truth values to 5 and 7. Conversely, any successful realization induces a Boolean assignment by the threshold $F(z_i) < 6$ versus $F(z_i) \geq 6$, and every clause is NAE-satisfied.

NP membership follows from a polynomial-bit rational realization certificate together with a polynomial-size deterministic comparison/maximin trace. Once the trace is fixed, all scalar requirements are rational linear equalities and strict/non-strict inequalities; a nonempty trace cell has a polynomial-bit rational point. Verification of the deterministic spatial mask is polynomial. □

114.2.2 Complete categorical spatial-frame realization

PTMM constrains a native coordinate block but is still a cylinder query. We now lift it to a fully prescribed finite categorical state.

For one polarity define

$$\mathsf{CSF}^{\sigma}(x) = (\rho^{\sigma}(x), M^{\sigma}_{\mathrm{aug}}(x), N^{\sigma}(x), R^{\sigma}(x)),$$

where ρ records unique-rooted/co-winner/no-finite-root status, M_{aug} is the complete competitor+terminal four-bit mask, N is the exact-neutral flag, and R is the categorical residual-terminal state. The complete self-dual state is

$$\mathsf{CSF}(x) = (\mathsf{CSF}^{+}(x), \mathsf{CSF}^{-}(x)).$$

Exact real-valued gate/cap coordinates are deliberately excluded from this finite quotient.

Definition 114.3 (Complete Categorical Spatial-Frame Realization (CCSFR))**.** Given an interval-valued graph, boundary terminals, observer set X, fixed seed threshold, and one prescribed finite categorical target τ_x for each observer, decide whether one common realization and one elder/tie resolution satisfy

$$\mathsf{CSF}(x) = \tau_x \qquad (x \in X).$$

Theorem 114.4 (Complete spatial-state hardness)**.** *CCSFR is NP-complete. Hardness holds on bipartite graphs of radius at most three with fixed seed threshold* 2*, uncertainty only on identical terminal intervals* $[5, 7]$*, and all fixed scalar values in the constant rational set*

$$\{-8, 0, 5/2, 4, 8, 19/2, 12, 20\}.$$

Proof. Use the same Boolean coordinates $z_i \in [5, 7]$ in two mirror clause gadgets. In the bright gadget choose

$$F(x_c^{+}) = 8, \qquad F(p_c^{+}) = 19/2.$$

The active terminal margin is

$$\mu^{+}_{c,i} = \frac{13}{2} - z_i.$$

Thus the required bright nonpositive bit exists exactly when some clause variable has $z_i > 6.5$ once exact neutrality is separately excluded.

For the dark mirror use inversion constant 12 with

$$F(x_c^{-}) = 4, \qquad F(p_c^{-}) = 5/2.$$

In dark-normalized coordinates the corresponding margin is

$$\mu^{-}_{c,i} = z_i - \frac{11}{2},$$

so the required dark nonpositive bit exists exactly when some clause variable has $z_i < 5.5$.

The remaining categorical coordinates are frozen throughout $z_i \in [5, 7]$: the selected seed family is unchanged; same-polarity query roots remain unique; opposite-polarity query roots remain co-winners/nonunique; the observer stays on the forced peak side of each active terminal gate, so the residual coordinate is empty; and no unintended neutral sector appears under the prescribed target. Hence each bright observer can be prescribed the full target

$$((\text{unique}, 1111, 0, \text{empty}), (\text{co-winner}, 0000, 0, \text{empty}))$$

and each dark observer the mirror target.

If the NAE instance is satisfiable, map the two truth values to 7 and 5. Every clause then has one value above 6.5 and one below 5.5, so all full targets are realized. Conversely, a realization of every full target gives in each clause one $z_i > 6.5$ and one $z_j < 5.5$. Assign these two ranges opposite Boolean values and assign middle values arbitrarily. Every clause already has both truth values and is NAE-satisfied.

Membership in NP uses the same polynomial trace-cell argument as Theorem 114.2. □

Corollary 114.5. *Exact arbitrary querying of the richer numerical full spatial-frame realization relation is NP-hard.*

The mechanism of Theorems 114.2–114.4 is independent of polarity-core owner scheduling. It uses terminal exposure, gate floors, route incidence, midpoint signs, and synchronization through one common scalar realization.

114.3 Finite-complex realization universality

Let P be a finite family of named protected strict observer-record obligations. As in EF–FG I, define

$$K_{\mathrm{raw}}(P) = \left\{ A \subseteq P : \begin{array}{c}\text{one admissible raw realization}\\ \text{satisfies every obligation in } A\end{array} \right\}$$

and let $K_\sigma(P)$ be the corresponding complex under the canonical maximal-safe rule. Both are finite simplicial complexes.

Earlier constructions produced all simplex boundaries. We now classify the realizability classes completely.

114.3.1 Facet-selector construction

Let K be a finite simplicial complex on vertices $1, \dots, n$ with facets

$$F_1, \dots, F_m.$$

Create one observer x with $F(x) = 4$. For every facet create a selector gate g_j with uncertain value

$$y_j := F(g_j) \in [2, 3].$$

Choose rational caps

$$2 < c_1 < \cdots < c_n < 3.$$

For every complex vertex i, create a protected peak p_i and a private elder blocker q_i. Their maximin attachments to x are arranged so that, for every selector realization y,

$$a_i(y) = \kappa(p_i, x) = \min\left(c_i, \max_{j : i \in F_j} y_j\right),$$

and

$$b_i(y) = \kappa(q_i, x) = \min\left(c_i, \max\left(2, \max_{j : i \notin F_j} y_j\right)\right).$$

Fix an essential elder with observer attachment 1 and choose finite births in $(6, 7)$ with elder order

$$q_1 > p_1 > q_2 > p_2 > \cdots > q_n > p_n.$$

The graph is bipartite and has radius at most three about the observer.

Lemma 114.6 (Maximizer-facet soundness)**.** *For every selector realization, every protected peak that is a strict record belongs to every facet attaining the global maximum selector value. In particular the protected record set is a face of K.*

Proof. If p_i is a strict record against its private blocker, then $a_i > b_i$. Since the same cap c_i is applied to both sides, this forces

$$\max_{j:i\in F_j} y_j > \max\left(2, \max_{j:i\notin F_j} y_j\right).$$

Let F_{j^*} be any facet with globally maximal y_{j^*}. If $i \notin F_{j^*}$, the right-hand maximum is at least y_{j^*}, while the left-hand maximum cannot exceed it, contradicting the strict inequality. Thus every strict protected record lies in every maximizing facet. □

Lemma 114.7 (Facet completeness)**.** *For every facet F_t there is a realization whose protected strict-record set is exactly F_t.*

Proof. Set

$$y_t = 3, \qquad y_j = 2 \quad (j \neq t).$$

Then for $i \in F_t$ we obtain $a_i = c_i$ and $b_i = 2$, while for $i \notin F_t$ the blocker is not below the protected attachment. With increasing caps and the fixed elder birth order, exactly the p_i with $i \in F_t$ are strict observer records. □

Theorem 114.8 (Raw universality)**.** *For every finite simplicial complex K there exists a finite interval-valued EF–FG graph and protected obligation family P such that*

$$\boxed{K_{\mathrm{raw}}(P) \cong K.}$$

The construction may be chosen with one observer, a bipartite graph of radius at most three, uncertainty only on facet selectors, and identical selector interval $[2, 3]$.

Proof. Lemma 114.6 gives $K_{\mathrm{raw}}(P) \subseteq K$. Lemma 114.7 realizes every facet, hence every subface, so $K \subseteq K_{\mathrm{raw}}(P)$. □

114.3.2 Canonical freezing

The raw construction becomes canonical because its geometry already forces every relevant attachment once the selector vector is fixed.

Lemma 114.9 (Forced-safe canonical fiber)**.** *Fix one selector realization y. For every protected peak and private blocker, the feasible attachment fiber is the singleton $\{a_i(y)\}$ or $\{b_i(y)\}$. Moreover every such attachment lies strictly below the canonical hard-band safety ceiling at $\theta = 1/2$ and $H = 0$.*

Proof. The maximin construction itself fixes the attachment values. Every attachment is below 3, whereas all finite births exceed 6 and the essential elder pressure is 1. Hence the canonical safety ceiling satisfies

$$\frac{B + D}{2} > \frac{6 + 1}{2} = 3.5 > 3.$$

Thus every geometrically forced attachment is strictly safe. The maximal-safe canonical section has no alternative value to choose and therefore equals the forced raw attachment. □

Theorem 114.10 (Canonical universality)**.** *For every finite simplicial complex K there exists a finite interval-valued EF–FG instance with*

$$\boxed{K_\sigma(P) \cong K.}$$

Consequently

$$\boxed{\{K_{\mathrm{raw}}(P)\} = \{K_\sigma(P)\} = \{\textit{all finite simplicial complexes}\}.}$$

Proof. By Lemma 114.9, realization by realization the canonical and raw elder scans coincide on the protected family. The proof of Theorem 114.8 therefore transfers verbatim. □

Remark 114.11. The equality in Theorem 114.10 is an equality of realizability classes. It does not assert $K_{\mathrm{raw}}(P) = K_\sigma(P)$ for every fixed EF–FG instance. The canonical/raw distinction remains genuine when an instance has nontrivial intermediate attachment freedom.

Chapter 115

EF–FG II: Parameterized Ownership and Correlated Uncertainty

The shallow-tree hardness results show that global common-realization synchronization can remain difficult even when the ambient graph is simple. This chapter separates that hardness from two finite ambiguity parameters and then enlarges the uncertainty model from independent interval boxes to correlated rational-polyhedral realization spaces.

115.1 Parameterized complexity of the ownership kernel

The threshold-band kernel is NP-complete, but fixed-order evaluation is linear. We now identify two natural parameters that measure the remaining global choice.

115.1.1 Elder-order ambiguity

For event birth intervals

$$I_i = [\ell_i, u_i],$$

define forced elder precedence

$$i \prec_B j \iff \ell_i > u_j.$$

Let C_B be the set of events incomparable with at least one other event under this relation, and put

$$k = |C_B|.$$

Lemma 115.1 (Linear extensions are exactly realizable elder orders)**.** *A total event order is realizable by scalar choices from the birth intervals iff it is a linear extension of $\prec_B$.*

Proof. Necessity is immediate. For sufficiency, write the proposed elder-to-younger order as $i_1, \dots, i_n$ and construct birth values backwards. Set the last value to any point in its interval and recursively choose

$$b_{i_t} = \max\{\ell_{i_t}, b_{i_{t+1}}\}.$$

If this exceeded u_{i_t}, then $b_{i_{t+1}} > u_{i_t}$, and some later event would be forced elder to i_t, contradicting the linear-extension property. Ties are resolved in the proposed order. □

Lemma 115.2 (Ambiguity-core enumeration)**.** *The events outside C_B form one total chain and are comparable with every event. Fixing a linear order of the k ambiguity-core events determines at most one full linear extension. Hence the number of admissible elder orders is at most $k!$.*

Proof. If two events outside C_B were incomparable, each would belong to C_B. Thus the outside set is a chain. For each core event, its comparisons with the outside chain determine a unique insertion slot. A chosen core permutation is either inconsistent with the internal precedence constraints or yields one full extension. □

Theorem 115.3 (Order-ambiguity FPT)**.** *Threshold-band completion is fixed-parameter tractable in k:*

$$\boxed{T(n,k) = O(k!\,\mathrm{poly}(n)).}$$

For one hard band the bound is $O(k!\,n)$ after the admissible orders have been generated.

Proof. Enumerate the at most $k!$ core permutations, reconstruct the corresponding full linear extension when one exists, and evaluate that fixed order by the EF–FG I prefix-minimum scan. □

115.1.2 Uncertain attachment gates

For event $e_i = (h_i, A_i, B_i)$, the corresponding physical attachment interval is

$$[q - A_i, q - B_i].$$

Call it uncertain iff $A_i > B_i$, and let

$$g = \#\{i : A_i > B_i\}.$$

Lemma 115.4 (Deterministic-gate reset)**.** *If $A_i = B_i = b_i$ and event i is owner-legal, then the post-event deficit is exactly*

$$x' = b_i.$$

Proof. Owner legality gives $(x - s_i)/2 \le b_i$, so

$$x' = \max\{b_i, (x - s_i)/2\} = b_i.$$

□

Lemma 115.5 (Deterministic-gate chain)**.** *If two deterministic-gate events satisfy $B_i > B_j$, then any legal owner word using both must place i before j. Equal-B deterministic events cannot both be owner uses.*

Proof. If j were used first, Lemma 115.4 would leave $x = B_j$, after which event i would require $B_i < x = B_j$, impossible. For equal B, the first reset makes the second fail the strict owner-live inequality. □

Define state dominance by

$$(h_1, x_1) \succeq (h_2, x_2) \iff h_1 \ge h_2, \quad x_1 \le x_2.$$

A dominating state can simulate every successful future continuation of the dominated state, possibly skipping an event that has already become owner-dead.

Theorem 115.6 (Uncertain-gate FPT)**.** *Threshold-band completion is fixed-parameter tractable in g:*

$$\boxed{T(n, g) = 2^g \operatorname{poly}(n).}$$

A straightforward exact dynamic program runs in $O(2^g g n^3)$ arithmetic operations.

Proof. Sort the deterministic-gate events into the forced chain $d_1, \ldots, d_m$ from Lemma 115.5, and let U be the g uncertain-gate events. Use DP cells

$$DP[j, S], \qquad 0 \le j \le m, \quad S \subseteq U,$$

storing the Pareto frontier of reachable hard states after deciding the first j deterministic events and using exactly the uncertain owner events in S. From each state one may skip d_{j+1}, use it if legal, or use any unused uncertain event if legal. Every legal owner word is represented because its deterministic subsequence must respect the fixed chain.

Reachable headroom values belong to the finite set $\{h_0, h_1, \ldots, h_n\}$. For a fixed headroom, only the smallest deficit survives dominance pruning, so each frontier has at most $n + 1$ states. There are $(m + 1)2^g$ cells and at most $g + 1$ owner choices per state, yielding the displayed FPT bound. □

Corollary 115.7 (Parameterized boundary)**.** *On the threshold-band/radius-two family carrying the ownership hardness theorem:*

$$\boxed{\textit{small } k \textit{ or small } g \Longrightarrow \textit{FPT.}}$$

By contrast, band count, graph treewidth, and graph radius alone are para-NP-hard parameters: NP-hardness already holds at one hard band, treewidth one, and radius two.

Proof. The positive statements are Theorems 115.3–115.6 and transfer through the exact radius-two physical lift and dark-anchor lift of EF–FG I. The negative statements follow immediately from the restrictions of the existing NP-completeness construction. □

115.2 Correlated rational-polyhedral uncertainty

Independent interval boxes are useful but cannot express shared calibration, conservation, monotonicity, or other cross-vertex correlations. We now replace the box by an exact correlated class.

Definition 115.8 (Rational-polyhedral realization model)**.** Let $F \in \mathbb{R}^V$ be the scalar field and $z \in \mathbb{R}^r$ auxiliary latent variables. A realization model is given in extended form by

$$\widehat{\mathcal{P}} = \{(F, z) : AF + Bz \le c,\ CF + Dz = e\},$$

with rational data. The admissible scalar fields are

$$\mathcal{P} = \operatorname{proj}_F \widehat{\mathcal{P}}.$$

The class contains interval boxes, shared offsets, linear calibration models, Lipschitz constraints, monotonicity constraints, affine conservation laws, and finite intersections of rational halfspaces and hyperplanes.

Theorem 115.9 (Polyhedral trace theorem)**.** *Fix a finite categorical or rational-threshold EF–FG query Q. There exists a finite family of discrete deterministic traces $\mathfrak{T}_Q$ such*

that the realization set of Q is a finite union of projections of relatively open rational polyhedra:

$$\{F \in \mathcal{P} : \exists \rho\ Q(F, \rho)\} = \bigcup_{\tau \in \mathfrak{T}_Q} \operatorname{proj}_F \widehat{\mathcal{P}}_\tau.$$

For a fixed trace τ, feasibility of $\widehat{\mathcal{P}}_\tau$ is decidable in polynomial time by rational linear programming.

Proof. The deterministic EF–FG evaluator is a polynomial-time finite comparison program built from scalar coordinates, rational affine combinations, min/max operations, comparisons, graph searches whose active subgraphs are comparison-determined, and finite elder/tie choices. A trace records every comparison branch, chosen min/max attainer, elder/tie decision, and resulting query branch. Once the trace is fixed, every numerical intermediate value is a rational affine function of F. Trace compatibility is therefore a rational system

$$A_\tau F \le b_\tau, \qquad C_\tau F = d_\tau, \qquad S_\tau F < t_\tau.$$

Intersecting with the extended uncertainty model gives a relatively open rational polyhedron.

For strict inequalities introduce one common slack $0 \le \delta \le 1$ and replace $r < s$ by $r + \delta \le s$. Maximizing δ yields strict feasibility iff the optimum is positive. This is a rational LP of polynomial size for a fixed trace. □

Corollary 115.10 (Polynomial rational witnesses)**.** *Every feasible existential finite EF–FG query under rational-polyhedral uncertainty has a polynomial-bit rational common realization and polynomial-size finite trace/tie certificate. Hence such possible queries lie in NP whenever deterministic verification is polynomial. Their universal robust complements lie in coNP.*

Proof. A feasible strict rational trace LP with positive slack has a polynomial-bit basic feasible solution by standard determinant bounds for rational linear systems. Verification replays the deterministic trace and checks the joint polyhedral constraints exactly. □

Proposition 115.11 (Self-duality)**.** *If $\mathcal{P}$ is rational-polyhedral, then*

$$-\mathcal{P} = \{-F : F \in \mathcal{P}\}$$

is rational-polyhedral, and the realization-wise EF–FG duality

$$\mathcal{E}(-F) = \mathcal{D}(\mathcal{E}(F))$$

extends exactly to possible and robust polyhedral semantics.

Proof. Substitute $F = -F'$ in the rational extended formulation and use the deterministic inversion theorem layerwise. □

Corollary 115.12 (Complexity transfer)**.** *All interval-box NP-hardness and NP-completeness results proved in EF–FG I and Section 114.2 remain valid for rational-polyhedral uncertainty.*

Proof. An interval box is a rational polytope. NP membership follows from Corollary 115.10. □

Remark 115.13 (Scope). The old interval-cactus dynamic programs exploit coordinatewise independence and local separator factorization. Arbitrary dense linear correlations can couple distant blocks, so we do not claim that those specific DPs remain polynomial. The generic exact replacement is finite trace enumeration with LP feasibility.

Chapter 116

EF–FG II: Layered Semantics and Closure of the Open Program

The final second-stage chapter transfers exact common-realization semantics to finitely many coupled scalar layers and then records the closure of the five directions posed by the foundational EF–FG stage. The layered construction is deliberately finite and scalar-by-layer: it does not identify EF–FG with a multivariate Reeb-space theory, but couples ordinary scalar EF–FG layers through one joint realization and explicit cross-layer predicates.

116.1 Finite multichannel and spatiotemporal transfer

We now extend the scalar theory to a finite family of coupled scalar layers without imposing a vector-valued total order.

Definition 116.1 (Finite layered EF–FG)**.** Let $\Lambda = \{1, \dots, L\}$ be a finite layer set. Layer λ has graph $G_\lambda = (V_\lambda, E_\lambda)$, scalar field F_λ, and ordinary deterministic EF–FG output $\mathcal{E}_\lambda(F_\lambda, \rho_\lambda)$. The uncoupled product output is

$$\mathcal{E}_\Lambda(F, \rho) = (\mathcal{E}_1, \dots, \mathcal{E}_L).$$

All layer scalar coordinates and optional latent variables are jointly constrained by one rational extended polyhedron.

A finite cross-layer specification Γ may contain categorical relation atoms

$$(s_{\lambda,a}, s_{\mu,b}) \in R$$

for an explicitly supplied finite relation R, and rational affine atoms in named numerical observables,

$$\sum_j q_j y_j \ \bowtie \ b, \qquad \bowtie \in \{<, \leq, =, \geq, >\}.$$

For time layers, finite correspondence relations between named observers/features may be supplied as part of the input. No automatic feature-identity theorem is assumed.

Theorem 116.2 (Deterministic layered transfer)**.** *For fixed realizations and elder/tie resolutions, finite layered EF–FG is evaluable in*

$$\boxed{O\left(\sum_{\lambda=1}^{L} T_\lambda + |\Gamma|\right),}$$

where T_λ is the deterministic evaluation time of layer λ.

Proof. Evaluate every scalar layer independently and then evaluate the finite cross-layer predicates. □

Theorem 116.3 (Joint layered trace theorem)**.** *Fix one deterministic trace for each layer and one Boolean branch of every predicate in* Γ*. The joint realization cell is a relatively open rational polyhedron in the concatenated layer and latent variables. Fixed joint-trace feasibility is rational LP feasibility, and every feasible possible query has a polynomial-bit rational joint witness.*

Proof. Apply Theorem 115.9 to each layer. On a fixed layer trace every named numerical observable is rational affine in that layer's scalar coordinates. The cross-layer affine atoms remain affine in the concatenated vector; categorical relation atoms become fixed truth values. One common slack handles all strict inequalities. Corollary 115.10 then gives a polynomial-bit rational joint solution. □

Theorem 116.4 (Layered self-duality)**.** *Under simultaneous inversion*

$$\mathfrak{I}(F_1, \dots, F_L) = (-F_1, \dots, -F_L),$$

finite layered EF–FG satisfies

$$\boxed{\mathcal{E}_\Lambda(\mathfrak{I}F) = \mathcal{D}_\Lambda(\mathcal{E}_\Lambda(F)),}$$

with $\mathcal{D}_\Lambda = \mathcal{D}_1 \times \cdots \times \mathcal{D}_L$*. If* Γ *is transformed by the corresponding duality action, possible and robust statements are preserved; if* Γ *is duality-invariant, the entire model is self-dual.*

Proof. Apply the scalar inversion theorem to every layer and the linear substitution $F \mapsto -F$ to the joint polyhedral model. □

Theorem 116.5 (Independent-product factorization)**.** *Suppose*

$$\mathcal{P} = \prod_{\lambda=1}^{L} \mathcal{P}_\lambda$$

and the query is a conjunction $Q = \bigwedge_\lambda Q_\lambda$ *of layer-local predicates with no cross-layer atom. Then*

$$\mathrm{Possible}_{\mathcal{P}}(Q) \iff \bigwedge_\lambda \mathrm{Possible}_{\mathcal{P}_\lambda}(Q_\lambda),$$

and

$$\mathrm{Robust}_{\mathcal{P}}(Q) \iff \bigwedge_\lambda \mathrm{Robust}_{\mathcal{P}_\lambda}(Q_\lambda).$$

Proof. For possible semantics, combine independent layer witnesses into their product realization; the converse is projection. The robust statement follows from universal quantification over a Cartesian product. □

Theorem 116.6 (Two-layer replication transfer)**.** *Let* Q *be any single-layer EF–FG decision query. Construct two layer copies and impose exact equality*

$$F_2(v) = F_1(v)$$

for corresponding scalar coordinates. Require the same target query in both layers. Then the single-layer and two-layer instances are equivalent. Hence every NP-hard single-layer EF–FG query remains NP-hard already with exactly two channels or exactly two time slices; when the single-layer query is in NP, the replicated version is NP-complete.

Proof. A single-layer witness can be copied to both layers with the same resolution. Conversely, a successful two-layer realization projects to a successful realization of either layer. The equality coupling is rational-polyhedral and the transformation is polynomial. □

A temporal state-persistence requirement is a special categorical relation. For example, if S is a finite queried state space, the relation

$$R_{\mathrm{id}} = \{(s, s) : s \in S\}$$

enforces exact state persistence across adjacent frames. Theorem 116.6 remains valid with this explicit temporal consistency predicate.

116.2 Synthesis: the closed theorem package

The new results complete the theorem-sized directions left by EF–FG I.

Theorem 116.7 (EF–FG II completion theorem)**.** *Within the finite scopes defined in this second-stage development, the following statements hold.*

(i) Full finite categorical spatial-frame realization is NP-complete on severely restricted shallow bipartite graphs (Theorem 114.4).

(ii) Every finite simplicial complex occurs as a raw EF–FG common-realization complex and as a canonical EF–FG common-realization complex (Theorems 114.8–114.10).

(iii) Threshold-band ownership completion is FPT both in elder-order ambiguity k and in uncertain-gate count g, while one hard band, treewidth one, and radius two alone do not remove NP-hardness (Corollary 115.7).

(iv) Exact common-realization semantics extends from independent interval boxes to correlated rational-polyhedral uncertainty with polyhedral trace cells, LP fixed-trace feasibility, polynomial-bit rational witnesses, and preserved self-duality (Theorem 115.9 and Proposition 115.11).

(v) Finite multichannel and spatiotemporal EF–FG admits exact joint-realization semantics, deterministic product evaluation, polyhedral trace certification, self-duality, factorization in the uncoupled case, and two-layer complexity transfer (Theorems 116.2–116.6).

These statements sharpen the original local/global picture. Global difficulty does not come from deep graph topology: ownership hardness survives on radius-two trees, while spatial-frame hardness survives on shallow bipartite graphs. At the same time, difficulty is not uniform. Order ambiguity and gate uncertainty expose finite-parameter routes back to exact tractability. The realization-complex theorem shows that structural compatibility is maximally expressive at finite scale, while the polyhedral and layered theorems show that exact common-realization semantics is stable under a substantial enlargement of the uncertainty and data model.

The scope boundaries are part of the theorem package. The finite layered theory is not a multivariate Reeb-space theory, and the polyhedral theorem does not assert preservation of interval-cactus factorization under dense correlations. Likewise, the equality of raw and canonical realizability classes does not identify the two complexes for each individual instance. Stating these distinctions explicitly keeps the sequel within the semantics actually proved.

116.3 Conclusion

This second-stage development turns the open theorem program of EF–FG I into a closed second-stage theory. The spatial-frame projection now has a direct NP-completeness classification at the complete finite categorical level. Protected common-realization complexes are universal among finite simplicial complexes under both raw and canonical semantics. The ownership hardness kernel admits two exact FPT escape parameters, distinguishing genuine order/gate ambiguity from shallow graph structure. Common-realization semantics extends exactly to correlated rational-polyhedral uncertainty, and finite scalar layers can be coupled into multichannel or spatiotemporal models without abandoning self-duality or exact certification.

The combined picture is therefore not simply that global EF–FG questions are hard. Rather, the theory identifies three complementary regimes: local or fixed-order calculations with direct polynomial algorithms; globally synchronized realization problems that are NP-complete even on shallow graph families; and intermediate instances whose global uncertainty is concentrated in a small ambiguity core and is fixed-parameter tractable. At the structural level, finite compatibility patterns are fully universal. At the modeling level, exact semantics survives correlated and layered coupling by remaining faithful to one common realization before projection.

Part XI

Synthesis and Return to the Original Machine

Part guide. **Question.** What does the original 2023 machine look like after the full semantic, geometric, structural, enumerative, resource, topological, and algorithmic theory has been developed? **Main result.** The final Part reconstructs the research chain, distinguishes forced structure from coordinate choice, and identifies which theorem families genuinely descend from the original completion invariant. **Why here.** The monograph ends by explaining the starting machine rather than merely stopping after the last branch. **Dependency.** This Part synthesizes all preceding Parts and introduces no new foundational assumptions.

Chapter 117

The 2023 DPDA Revisited

The monograph began with two diagrams.

They represented the same $C/L/R$ recognition core but different empty-word wrappers:

$$\boxed{L_{2,1} \setminus \{\varepsilon\}}$$

and

$$\boxed{L_{2,1}.}$$

At the beginning, the figures were concrete machines to be understood.

We now return to them after developing:

- shortest-completion geometry and exact residual-faithful resource frontiers;
- representation-independent hierarchy methods;
- higher-dimensional completion geometry;
- finite geodesic compilation, higher-order optimal-compatibility structure theory, and full simplicial universality with exact Boolean-lattice-width row complexity;
- cancellation/defect geometry and chronological arithmetic refinement;
- the general real-time p:q machine and WSC Structure Theory;
- exact semantic lifts and physical lift freedom;
- arbitrary-coprime causal-schedule compilation and exact fatal-support classification;
- strict coordinate quotients, protected interfaces, bounded bi-recoding, and minimal cores;
- General Theory;
- operational CANCEL/WRITE enumeration and exact one-counter compression;
- explicit rational-Dyck run–return and ratio-run–return enumeration, followed by contact-preserving barrier compression;
- Farey-recursive shortest-completion arithmetic and universal spectra;
- quantitative LIFO chronology;
- rank-two topology, physical holonomy, and sharp carrier costs;
- exact unrestricted state–stack and bounded-replacement complexity;
- filtered collapse capacity, retained-information bounds, and source-stack exposure;
- protected-contract Pareto synthesis;
- Unified Semantic Normalization and Realization Theory: rooted path outcomes, protected-obligation capacity, same-model transfer saturation, typed holonomy/resource bridges, and four-axis source recovery;

- Extrema–Frame Foreground–Background theory: observer-record normal form, exact common-realization scalar uncertainty, the cactus/radius-two local–global complexity boundary, shallow spatial-frame NP-completeness, finite-complex universality, parameterized ambiguity bounds, correlated polyhedral uncertainty, finite layered semantics, and all-orders obstruction families;
- DOCA state-primality and exact two-factor decomposition, followed by decidability of deterministic zero-testing OCA–nondeterministic OCN language equivalence, including the epsilon extension, and the resulting separation between fixed comparison and existential decomposition.

The purpose of this chapter is not to redraw the machine as an abstract object.

It is to show what every part of the original design means after the full theory has been developed.

117.1 The two origin figures remain distinct

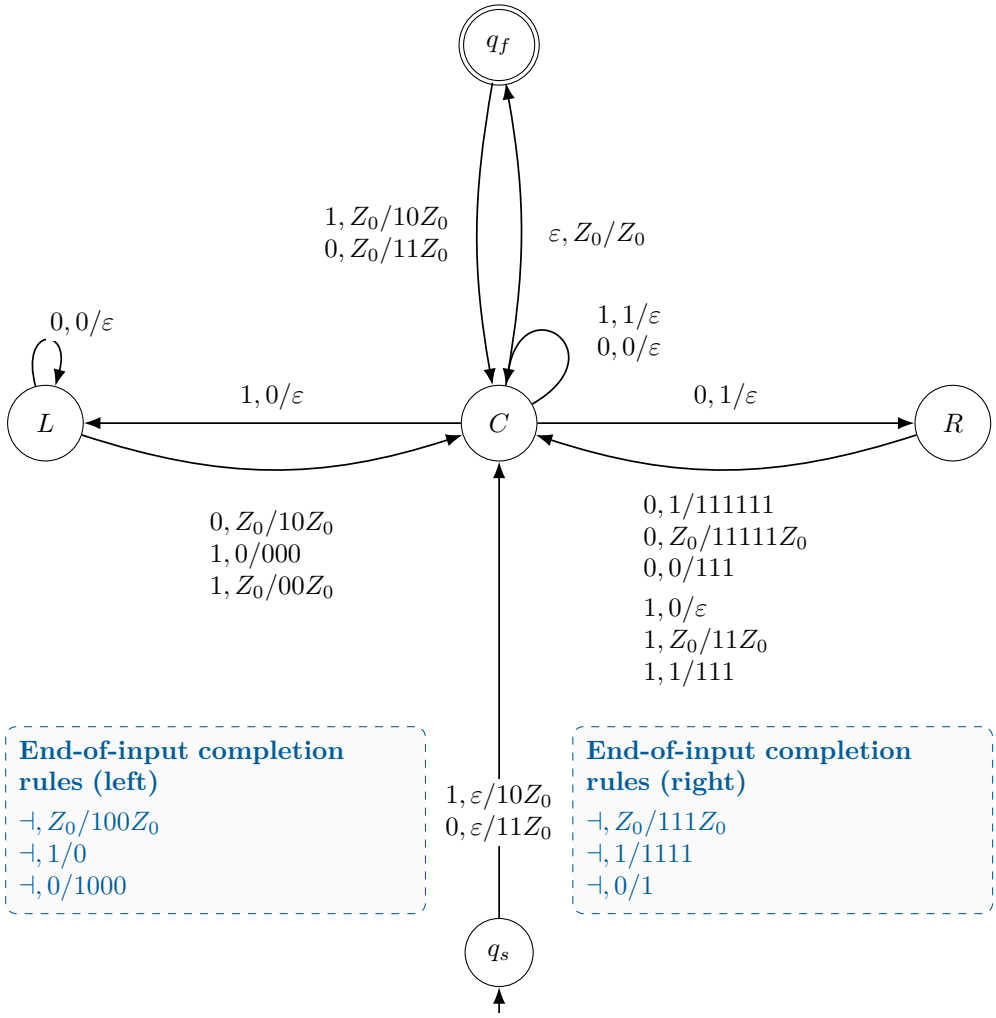


Figure 117.1: The empty-input-excluded original DPDA revisited after the general theory. This is the same formalized machine as Figure 1.1, deliberately repeated so that the final interpretation can be read against the original object.

This variant recognizes

$$L_{2,1} \setminus \{\varepsilon\}.$$

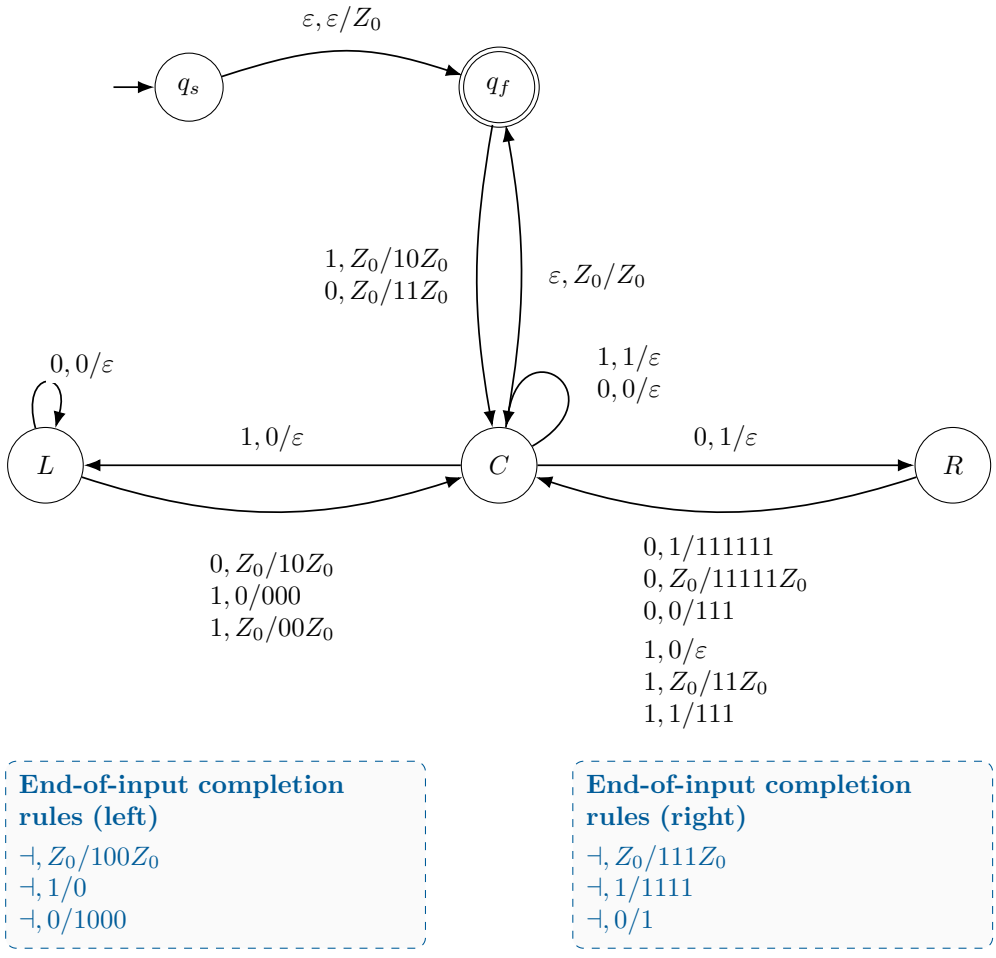


Figure 117.2: The empty-input-included original DPDA revisited. This is the same recognition/completion core as Figure 1.2; the wrapper accepts ε.

This variant recognizes

$$L_{2,1}.$$

The common $C/L/R$ operational core is the same.
The difference is the acceptance/initialization wrapper.
This is the first lesson of the completed theory:

$$\boxed{\text{acceptance convention} \neq \text{semantic core.}}$$

117.2 What the stack was storing

The original stack looked like an operational memory structure.
Chapter 1 proved that its reachable form is highly restricted.
The state-potential invariant then gave

$$\boxed{\rho(w) = D(S) + \Phi(q).}$$

For 2:1,

$$D(S) = 2N_0(S) - N_1(S)$$

under the original machine's stack-weight convention, while the state potentials are

$$\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.$$

Thus the machine stores the arithmetic imbalance **distributed between control and stack**.
That observation became the prototype for every later semantic/physical decomposition.

117.3 What the completion rules were doing

The right-endmarker completion rules do not create new semantic information.
They materialize information already present in the final state/stack configuration.
After normalization, the physical word decodes a shortest completion demand.
The machine therefore contains an implicit map

$$\boxed{\begin{array}{c}\text{physical configuration} \to \text{residual} \to \text{minimum demand} \\ \to \text{canonical shortest continuation.}\end{array}}$$

This was the discovery that opened the entire completion-geometry branch.

117.4 Why depth two appeared

The original normalized stores lie in only a few structural families.
Therefore the qualitative question

> Which demand symbols occur anywhere in the remaining stack?

is determined by the first two cells.

The general centered $p{:}q$ machine later recovered this two-cell qualitative visibility through a completely different canonical ordering.

Thus the original depth-two property was not merely a graphical accident.

It was an early manifestation of a broader principle:

unbounded quantitative demand can have bounded qualitative visibility.

117.5 The cancellation transition

The transition

$$\boxed{(L, 0S) \xrightarrow{0} (L, S)}$$

acquired three meanings.

Machine meaning

Delete one matching stack symbol.

Geodesic meaning

Consume one required zero along a shortest completion.

Combinatorial meaning

In the nonnegative 2-Dyck sector, mark a down-step landing at an exact odd-height family.

That third interpretation generated:

$$\begin{aligned}\text{correction signatures} &\to \text{defect} \to \text{Catalan boundary} \\ &\to \text{binary skeleton} \to \text{enumeration.}\end{aligned}$$

Thus one local transition produced an entire independent research branch.

117.6 Why the improved raw DPDA looked so different

The three-symbol raw residual-conjugate DPDA of Chapters 7–8 is not a cleaned-up drawing of the original machine.

It solves a different physical representation problem.

It asks:

> If the residual itself is the semantic state, what is the most efficient stable raw encoding within the stated model?

That representation moves finite phase information into control and leaves unary magnitude on the stack.

It achieves the exact raw residual-faithful frontier.

The original machine instead stores a richer literal completion-demand process with normalization chronology.

Therefore the two machines are not rivals for one undefined notion of optimality.

They are optimal or explanatory in different representation categories.

117.7 Why the general p:q machine returned to the original architecture

The centered real-time p:q construction deliberately preserved the operational idea of the original machine:

- literal demand on the stack;
- one-cell LIFO blocker displacement;
- signed debt in finite control;
- later WRITE materialization.

When

$$(p, q) = (2, 1),$$

the general debt rail becomes

$$D_{-1}, D_0, D_1$$

with

$$\boxed{D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L.}$$

The general potentials become

$$-3, 0, +3.$$

The historical WRITE arithmetic is recovered from the kernel-carry compiler.
Thus the original architecture was not merely generalized by analogy.
It was reconstructed from the general arithmetic.

117.8 Why the historical table was not literally the centered specialization

Two reachable right-state matching families remained different.
The centered machine keeps debt lazily in R.
The historical machine immediately rematerializes the same semantic demand in C.
Therefore:

$$\boxed{\text{historical} = \text{eager normalization},}$$

$$\boxed{\text{centered specialization} = \text{lazy normalization}.}$$

This small transition-table discrepancy turned out to be one of the most important facts in the research program.
It proved concretely that:

$$\boxed{\text{same semantics} \not\Rightarrow \text{unique physical chronology}.}$$

That observation led directly to Structure Theory and Uniqueness Theory.

117.9 C/L/R after Structure Theory

For 2:1,

$$r = p + q = 3.$$

The WSC exchange-sheet coordinate gives

$$\boxed{s(C) = 0, \qquad s(L) = 1, \qquad s(R) = -1.}$$

Thus the original states become SEARCH sheets.

The characteristic stack-length changes

$$-1, +2, +5$$

become materialization levels

$$M = 0, 1, 2.$$

And the historical identity

$$b + 2c = n$$

becomes a specialization of the global materialization budget.

The original table has therefore been reinterpreted from local syntax into semantic/physical accounting.

117.10 C/L/R after Uniqueness Theory

The original eager machine is one reduced lift.

The centered lazy machine is another.

They are:

- semantically equivalent;
- section-preserving incomparable;
- coordinate-reduced;
- non-isomorphic.

Within the selected three-site family, there are six admissible reduced normalization schedules.

The historical machine is therefore not “the” unique physical realization of the semantics.

It is a distinguished point in a nontrivial lift space.

117.11 The original machine after General Theory

The General Theory gives a final vocabulary.

The original configuration space is a physical lift.

The residual is semantic state.

The canonical completion word is a section choice.

The side-state configurations occupy noncanonical semantic fibers.

The one-top LIFO rule creates locality constraints.

Leaving the canonical section creates representation debt.

Some schedule choices preserve the behavioral hull.

Some expose fatal frontiers.

Alternative coordinate systems can remove obstructions that are essential in the literal policy.

Thus the full research program can be read backward into the original figure.

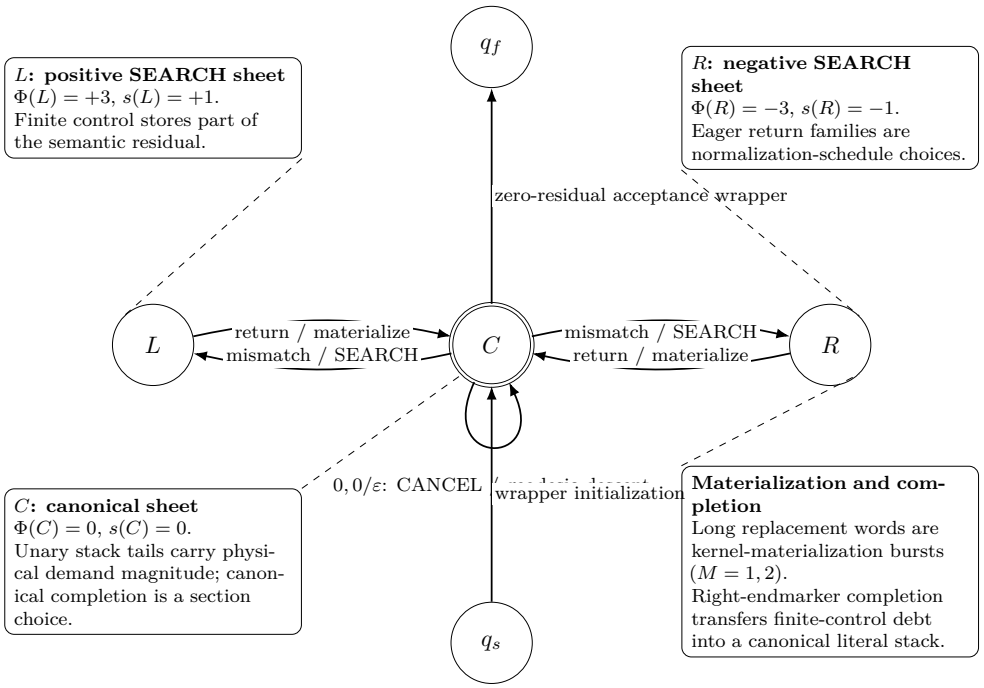


Figure 117.3: The original $C/L/R$ architecture annotated by the complete theory. The figure is interpretive rather than a replacement transition table: it overlays the residual/state-potential invariant, WSC SEARCH-sheet coordinates, literal-demand magnitude, kernel materialization, end-of-input completion, schedule freedom, and the semantic-lift viewpoint on the original machine architecture.

Suggested annotations:

- C: canonical sheet / zero SEARCH charge;
- L: positive SEARCH sheet;
- R: negative SEARCH sheet;
- unary stack tails: physical demand magnitude;
- $0, 0/\varepsilon$: CANCEL / geodesic descent / correction event;
- long replacement words: kernel-materialization bursts;
- right-endmarker completion: final canonical materialization;
- eager right-state families: schedule choice;
- wrapper: acceptance convention, not semantic core.

The final visual object of the monograph should be this reinterpreted original machine.

Chapter 118

What Was Forced, What Was Chosen, What Became General

The entire monograph can now be organized by one question:

$$\boxed{\text{Which features were mathematically forced, and at what level?}}$$

The answer requires several categories.

118.1 Language-semantic necessities

These are determined by $L_{p,q}$ itself.

Residual quotient

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).}$$

The right-residual system is canonical.

Minimum completion demand

Each residual determines one unique minimum nonnegative completion Parikh vector.

Primitive kernel

$$\boxed{K = (q, p).}$$

Every equivalent completion vector differs by a kernel packet.

Semantic carry chronology

For a fixed input word,

$$(\rho_t), \quad (\kappa_t), \quad (\mu_t), \quad (\chi_t)$$

are unique.

These facts do not depend on a pushdown representation.

118.2 Necessities of a chosen locality category

These are not language-intrinsic, but become forced once the physical category is fixed.

Examples include:

One-top suffix obstruction

A hidden suffix cannot be arbitrarily rearranged by one top rewrite.

Literal-policy fiber branching

For nontrivial $p{:}q$, the fixed canonical literal section cannot remain transition-closed.

Deferred-SEARCH control width

$$2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1$$

states are exact in the canonical deferred-SEARCH model.

Bounded-local collapse capacity

Too many semantic predecessors cannot be distinguished by too few local observations.

These are **representation-category necessities**.

They must always be stated with their category.

118.3 Coordinate artifacts

Some difficulties disappear when the physical coordinates change.

The clearest example is the four-symbol normalized lift.

Its

$$p+q-1$$

raw boundary defects were exact in those coordinates.

But a boundary-compatible coordinate shift removed them and produced a three-symbol raw DPDA.

Therefore:

$$\boxed{\text{locality defect} \not\Rightarrow \text{semantic necessity.}}$$

A defect may belong to the coordinate system.

118.4 Normalization-schedule choices

The semantic chronology can be unique while physical materialization chronology varies.

Examples:

- historical eager 2:1;
- centered lazy 2:1;
- the six admissible reduced 2:1 schedules.

The total kernel budget remains rigid even when its timing changes.

Thus:

$$\boxed{\text{semantic WRITE arrivals are forced;}}$$

$$\boxed{\text{physical WRITE timing may be chosen.}}$$

118.5 Special 2:1 features

Several concrete facts are specializations rather than universal principles.

For

$$p = 2, \qquad q = 1 :$$

$$K = (1, 2),$$

$$m = 3,$$

$$h = 1,$$

so the centered debt rail has exactly three states.

These become:

$$R, C, L.$$

The state potentials are

$$-3, 0, +3.$$

The historical materialization changes are

$$-1, +2, +5.$$

The correction transition in the nonnegative path sector lands at

$$3, 5, 7, \ldots$$

These numerical values are 2:1-specific.

The principles they instantiate are broader.

118.6 What generalized to all coprime p:q

The following extend uniformly:

- residual semantics;
- shortest completion demand;
- primitive kernel;
- geodesic rectangles;
- bounded canonical action visibility;
- raw residual-conjugate DPDA;
- exact raw residual-faithful frontier;
- centered real-time literal-demand DPDA;
- SEARCH debt;
- minimal kernel-carry WRITE;
- canonical-segment invariant;
- WSC semantic carry factorization;
- materialization/backlog accounting.

The original 2:1 machine was therefore a genuine first member of a general structural family.

118.7 What generalized beyond rational binary counting

Several branches moved beyond $L_{p,q}$.

Matrix completion geometry

The unique minimum pair became an optimal Parikh fiber.

The rectangle became a cubical quotient by the equal-length kernel.

Topological hierarchy

Kernel rank controlled local/global shortest-schedule topology.

Finite geodesic compilation

The shortest-completion viewpoint generalized further to positive-cost finite-phase periodic systems. Finite arithmetic data compile exact distance, optimal branching, arbitrary shortest-prefix membership, and the complete local optimal-compatibility complex. Across the class, that compatibility hierarchy is strictly non-collapsing at every prescribed order and admits Boolean amplification under products. The universal k-fiber theorem sharpens this: even after exact distance and the complete hierarchy through order $k-1$ are fixed, all $2^{\binom{s}{k}}$ next-order patterns on a distinguished structural s-simplex can occur, including arbitrarily overlapping k-faces. The all-rank theorem removes the distinguished rank altogether: one fixed coefficient bank of exact size $\binom{s}{\lfloor s/2 \rfloor}$ realizes every labeled simplicial complex, the same optimum survives common-shell synchronization, and one periodic Bellman system realizes the entire simplicial universe at one canonical distance.

General exact-lift theory

Semantic fibers, canonical sections, bounded-local obstruction, behavioral hulls, containment, causality, synthesis, and verification no longer depend specifically on weighted binary balance.

Thus the project eventually generalized not just the machine, but the distinction that the machine exposed:

> what must be remembered semantically
>
> *versus*
>
> how a LIFO machine can physically remember it.

118.8 The dependency structure in one chain

The completed monograph has one main mathematical spine together with consequence branches that reuse the same semantic objects without being prerequisites of the earlier construction:



The displayed branches are consequences and syntheses rather than a single linear causation claim. The large consequence array appears before USNRT because the recovery theorem uses theorem-bearing geometry, chronology, and exposure branches developed there. USNRT then acts as a typed synthesis of those already established branches; EF–FG is a logically independent scalar-field theory placed after that synthesis because it supplies a genuine external generativity test rather than another reformulation of the pushdown model. The geodesic-compiler branch and its full-simplicial completion are placed immediately after the higher-dimensional completion geometry to which they mathematically belong; the other consequence branches occur after the General Theory. State-primality and its two-factor sharpening follow the exact state/resource results because they mark an algorithmic decomposition boundary rather than a prerequisite of the General Theory; the one-counter comparison chapter then preserves the fixed OCN/OCN

solution of Problem 24.04(B) and adds the zero-testing OCA/OCN solution of Problem 24.04(A), completing the contrast with two fixed-presentation comparison problems that remain decidable. The A branch moreover survives the stated epsilon-transition extension. This order preserves mathematical prerequisites and narrative continuity: operational statistics precede intrinsic path statistics, topology precedes physical holonomy, and endpoint resource theorems precede Pareto synthesis.

Six points on this map are explicit resolutions of questions posed in the literature: the two Dai–Fu–Qiu joint-enumeration questions (run with return and ratio-run with return) [23], the fixed-state stack-alphabet hierarchy case left open after Masopust's realtime theorem [16], Totzke's Prime-DOCA decidability question [99], Automata Exchange Problem 24.04(B) on deterministic–nondeterministic one-counter-net language equivalence, and Automata Exchange Problem 24.04(A) on deterministic zero-testing one-counter automata versus nondeterministic one-counter nets [109]. The remaining branches are stated as new consequences, abstractions, or synthesis results rather than retroactively being labeled as external open-problem resolutions.

118.9 Final perspective

The research program began with a concrete deterministic pushdown automaton.

Its most productive feature was not simply that it recognized

$$L_{2,1}.$$

It distributed one arithmetic quantity across:

- finite control;
- a LIFO store;
- completion operations.

Trying to understand that distribution revealed:

$$\boxed{\text{semantic information} \neq \text{physical representation.}}$$

Everything that followed can be traced to that separation. The same semantic extraction later reached beyond the first pushdown representation itself: shortest-completion geometry admitted a finite cubical geodesic compiler whose local optimal-compatibility hierarchy is strict at every prescribed order; normalization history became an operational enumerative object; the residual became rational-path rank and boundary contact; the shortest-completion map became a Farey-recursive arithmetic object with universal scalar spectra and then an exact Cohn–Markoff completion geometry; optimal-fiber topology acquired a rank-raising realization calculus, a displacement-marked pure-loop calculus, and a distinct finite physical holonomy/carrier/connection/repair layer, while unequal simplex-strand optimal fibers obey marked product rigidity beyond literal Cartesian decompositions; chronological carry progressed from its third-order refinement through an exact fourth-order deformation parameter to an all-order CCR-k source–band–transport–shadow theory with an explicit integral Stirling obstruction lattice, exceptional local folds, Galois specializations and adelic resonance laws; and the state lower-bound branch closed to an unrestricted state–stack frontier and then an exact unrestricted bounded-replacement staircase, while universal higher-order semantic freedom acquired a representation-independent source-stack exposure cost. The geodesic branch reaches full

simplicial universality with exact Boolean-lattice-width row complexity and zero common-shell defect. A source-native facet construction realizes the entire shortest Parikh fiber, preserves it under bounded scalar folding, and admits both a real-time zero-return recognizer and an explicitly clocked, separately selected paired-session comparison service; neither contract implies a universal unframed dual-service machine. The final protected-contract synthesis then makes explicit which resource reductions are genuine and which are only bounded changes of coordinates; the subsequent DOCA state-primality theorem and its two-factor sharpening show that existential decomposition is undecidable already at the smallest genuine intersection budget, while the final one-counter comparison chapter first proves deterministic–nondeterministic OCN language equivalence decidable for two fixed net presentations and then closes the adjacent zero-testing frontier by proving deterministic OCA–nondeterministic OCN language equivalence decidable in real time and under the stated epsilon-transition extension. The contrast isolates the quantifier boundary rather than attributing hardness to one-counter storage alone.

A separate tree-walking boundary establishes an exact regularity criterion for right-turn spectra, family-specific deterministic nonrecognizability and a one-visible-pebble characterization, while leaving the UTWA–DTWA question unresolved. After those domain-specific and algorithmic branches are in place, the late USNRT synthesis makes the bookkeeping principle itself mathematical: representation fibers, path-outcome fibers, protected-obligation complexes, transported holonomy, and resource coordinates remain typed separately, with transfer theorems stating exactly which bridges are needed. EF–FG then supplies the deliberately external test: the same common-realization discipline organizes observer records in finite scalar fields, separates polynomial local witnessing from NP-complete global synchronization on shallow graphs, classifies complete categorical spatial-frame realization as NP-complete on shallow bipartite graphs, realizes every finite simplicial complex as a raw and canonical common-realization complex, exposes two fixed-parameter tractable ambiguity regimes, extends exact semantics to correlated rational-polyhedral uncertainty and finite coupled layers, and supports obstruction families of unbounded order. This last step is not a new encoding of the original pushdown language; it is evidence that the realization viewpoint extracted from that language has become reusable mathematics.

The final lesson is therefore not one specific DPDA construction or one specific theorem.

It is a method:

extract the semantic object from the machine,

identify which physical constraints are truly forced,

change coordinates when the obstruction is representational,

retain debt when locality forces it,

and distinguish semantic correctness from normalization history.

One additional lesson of the geodesic-compiler synthesis is that

shortest-path geometry can be finite in description

while

its optimal compatibility is unbounded in informational order across the class.

The original 2023 DPDA remains the first concrete object in that story—and, after the full theory, the final object through which the story can be read.

Appendix A

Complete Transition Audit of the Original 2023 DPDA

A.1 Purpose and scope

This appendix closes the finite verification step that is deliberately compressed in Chapter 1.

The object audited here is the **original 2023 completion-aware** 2:1 **DPDA**, in the formalized presentation used throughout this monograph. The language convention is

$$L_{2,1} = \{\, w \in \{0,1\}^* : N_1(w) = 2N_0(w) \,\},$$

with signed prefix residual

$$\boxed{\rho(w) = N_1(w) - 2N_0(w).}$$

For a logical stack word $S \in \{0,1\}^*$, define

$$\boxed{D(S) = 2N_0(S) - N_1(S)}$$

and assign the core-state potentials

$$\boxed{\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.}$$

The Chapter 1 invariant is

$$\boxed{\rho(w) = D(S) + \Phi(q)} \tag{A.1}$$

for every processed prefix w, where (q, S) is the corresponding core-projected configuration.

Since reading a binary symbol changes the residual by

$$\Delta\rho = \begin{cases} -2, & a = 0, \\ +1, & a = 1, \end{cases} \tag{A.2}$$

it is enough to verify, for every reachable recognition transition,

$$\boxed{\Delta D + \Delta\Phi = \Delta\rho.} \tag{A.3}$$

The audit also checks that every transition preserves the state-dependent reachable-store families

$$\boxed{\mathcal{S}_C = 0^* \cup 1^* \cup \{10\}, \qquad \mathcal{S}_L = 0^*, \qquad \mathcal{S}_R = 1^* \cup \{0\}.} \tag{A.4}$$

The bottom marker Z_0 is omitted from logical-stack words below. The leftmost logical symbol is the top.

A.2 Wrapper projection

The two formal variants differ only in how initialization and empty-input acceptance are wrapped around the semantic core.

The projection used throughout the monograph is

$$\pi(C, SZ_0) = (C, S), \qquad \pi(L, SZ_0) = (L, S), \qquad \pi(R, SZ_0) = (R, S),$$

together with

$$\boxed{\pi(q_s, \varepsilon) = (C, \varepsilon), \qquad \pi(q_f, Z_0) = (C, \varepsilon).} \tag{A.5}$$

Thus the wrappers carry no additional residual semantics.

Empty-input-excluded wrapper

The first binary symbol is consumed directly at the source wrapper.

input	projected source	projected target	ΔD	$\Delta\Phi$	$\Delta(D+\Phi)$	$\Delta\rho$
1	(C, ε)	$(C, 10)$	$+1$	0	$+1$	$+1$
0	(C, ε)	$(C, 11)$	-2	0	-2	-2

Indeed,

$$D(10) = 2 - 1 = 1, \qquad D(11) = -2.$$

Empty-input-included wrapper

The initialization move

$$\varepsilon, \varepsilon / Z_0$$

takes the physical source state to q_f. Under (A.5), both endpoints represent (C, ε), so this move has semantic change 0.

When the first binary symbol is later consumed from q_f,

$$1, Z_0 / 10 Z_0 \quad \text{or} \quad 0, Z_0 / 11 Z_0,$$

the same two projected checks above apply.

Finally, the core-to-final move

$$\varepsilon, Z_0 / Z_0$$

is available only at the balanced empty logical stack and projects

$$(C, \varepsilon) \longrightarrow (C, \varepsilon).$$

It therefore also preserves (A.1).

A.3 Center-state recognition transitions

A.3.1 Matching 1-cancellation

$$(C, 1T) \xrightarrow{1} (C, T).$$

One 1 is popped, hence

$$\Delta D = +1, \qquad \Delta\Phi = 0.$$

Therefore

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

The target remains in $0^* \cup 1^* \cup \{10\}$: from a homogeneous 1-stack it remains homogeneous, while the exceptional source 10 becomes 0.

A.3.2 Matching 0-cancellation

$$(C, 0T) \xrightarrow{0} (C, T).$$

One 0 is popped, so

$$\Delta D = -2, \qquad \Delta\Phi = 0,$$

and therefore

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

Reachability forces the source to lie in 0^+, so the target lies in 0^*.

A.3.3 Center-to-left mismatch

$$(C, 0T) \xrightarrow{1} (L, T).$$

Popping the top 0 gives

$$\Delta D = -2.$$

The state potential changes by

$$\Delta\Phi = \Phi(L) - \Phi(C) = 3.$$

Thus

$$\boxed{\Delta(D + \Phi) = -2 + 3 = +1 = \Delta\rho.}$$

Because the source is a reachable center configuration with top 0, it belongs to 0^+; hence $T \in 0^*$, exactly the required left-state family.

A.3.4 Center-to-right mismatch

$$(C, 1T) \xrightarrow{0} (R, T).$$

Popping one 1 yields

$$\Delta D = +1,$$

while

$$\Delta\Phi = \Phi(R) - \Phi(C) = -3.$$

Therefore

$$\boxed{\Delta(D + \Phi) = 1 - 3 = -2 = \Delta\rho.}$$

There are two reachable source shapes:

$$1^{m+1} \longmapsto 1^m,$$

which enters the 1^* part of $\mathcal{S}_R$, and

$$10 \longmapsto 0,$$

which produces the exceptional right-state store 0.
These are exactly the two right-state families in (A.4).

A.4 Left-state recognition transitions

Every reachable left-state logical store lies in 0^*.

A.4.1 Left-state 0-cancellation

$$(L, 0T) \xrightarrow{0} (L, T).$$

The stack change is one popped 0:

$$\Delta D = -2, \qquad \Delta\Phi = 0.$$

Hence

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

The target remains in 0^*.

A.4.2 Empty-left state on input 0

The bottom-marker rule is

$$0, Z_0/10Z_0,$$

so logically

$$(L, \varepsilon) \xrightarrow{0} (C, 10).$$

The source value is

$$D(\varepsilon) + \Phi(L) = 3.$$

The target value is

$$D(10) + \Phi(C) = 1.$$

Thus

$$\Delta(D + \Phi) = 1 - 3 = -2 = \Delta\rho.$$

The target 10 is the exceptional center word permitted by $\mathcal{S}_C$.

A.4.3 Nonempty-left state on input 1

For $T \in 0^*$,

$$(L, 0T) \xrightarrow{1} (C, 000T).$$

Replacing one 0 by three 0's adds two zeros, so

$$\Delta D = +4.$$

The state potential drops by 3:

$$\Delta\Phi = 0 - 3 = -3.$$

Hence

$$\boxed{\Delta(D + \Phi) = 4 - 3 = +1 = \Delta\rho.}$$

The target is again homogeneous 0^*.

A.4.4 Empty-left state on input 1

The bottom-marker rule

$$1, Z_0/00Z_0$$

gives

$$(L, \varepsilon) \xrightarrow{1} (C, 00).$$

Now

$$\Delta D = +4, \qquad \Delta\Phi = -3,$$

so again

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

The target lies in 0^*.

A.5 Right-state recognition transitions

Every reachable right-state logical store lies in

$$1^* \cup \{0\}.$$

A.5.1 Nonempty homogeneous-right state on input 0

For $T \in 1^*$,

$$(R, 1T) \xrightarrow{0} (C, 111111T).$$

Replacing one 1 by six 1's adds five ones, therefore

$$\Delta D = -5.$$

The state potential rises from -3 to 0:

$$\Delta\Phi = +3.$$

Thus

$$\boxed{\Delta(D + \Phi) = -5 + 3 = -2 = \Delta\rho.}$$

The target is in 1^*.

A.5.2 Empty-right state on input 0

The bottom-marker rule

$$0, Z_0/11111Z_0$$

gives

$$(R, \varepsilon) \xrightarrow{0} (C, 11111).$$

The stack contribution changes by

$$\Delta D = -5,$$

and the potential changes by $+3$. Hence

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

A.5.3 Exceptional right store 0 on input 0

The rule

$$0, 0/111$$

gives

$$(R, 0) \xrightarrow{0} (C, 111).$$

The stack weight changes from

$$D(0) = 2$$

to

$$D(111) = -3,$$

so

$$\Delta D = -5.$$

Again $\Delta\Phi = +3$, and

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

A.5.4 Exceptional right store 0 on input 1

The rule

$$1, 0/\varepsilon$$

gives

$$(R, 0) \xrightarrow{1} (C, \varepsilon).$$

The stack contribution changes by

$$\Delta D = -2,$$

while the state-potential change is $+3$. Therefore

$$\boxed{\Delta(D + \Phi) = -2 + 3 = +1 = \Delta\rho.}$$

A.5.5 Empty-right state on input 1

The bottom-marker rule

$$1, Z_0/11Z_0$$

gives

$$(R, \varepsilon) \xrightarrow{1} (C, 11).$$

Here

$$\Delta D = -2, \qquad \Delta\Phi = +3,$$

and therefore

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

A.5.6 Nonempty homogeneous-right state on input 1

For $T \in 1^*$,

$$(R, 1T) \xrightarrow{1} (C, 111T).$$

Replacing one 1 by three 1's adds two ones:

$$\Delta D = -2.$$

Together with $\Delta\Phi = +3$,

$$\boxed{\Delta(D + \Phi) = +1 = \Delta\rho.}$$

The target remains homogeneous 1^*.

A.6 Complete recognition-phase audit table

The preceding cases may be compressed into one table.

source	input / top	logical action	target	ΔD	$\Delta\Phi$	total	required
C	$1/1$	pop 1	C	$+1$	0	$+1$	$+1$
C	$0/0$	pop 0	C	-2	0	-2	-2
C	$1/0$	pop 0	L	-2	$+3$	$+1$	$+1$
C	$0/1$	pop 1	R	$+1$	-3	-2	-2
L	$0/0$	pop 0	L	-2	0	-2	-2
L	$0/Z_0$	write 10	C	$+1$	-3	-2	-2
L	$1/0$	$0 \mapsto 000$	C	$+4$	-3	$+1$	$+1$
L	$1/Z_0$	write 00	C	$+4$	-3	$+1$	$+1$
R	$0/1$	$1 \mapsto 111111$	C	-5	$+3$	-2	-2
R	$0/Z_0$	write 11111	C	-5	$+3$	-2	-2
R	$0/0$	$0 \mapsto 111$	C	-5	$+3$	-2	-2
R	$1/0$	pop 0	C	-2	$+3$	$+1$	$+1$
R	$1/Z_0$	write 11	C	-2	$+3$	$+1$	$+1$
R	$1/1$	$1 \mapsto 111$	C	-2	$+3$	$+1$	$+1$

Thus every reachable recognition transition satisfies (A.3).

This proves the finite transition step omitted from the short proof in Chapter 1.

A.7 Reachable-store closure

The same table proves the state-dependent store normal form.

Center

The center can contain only

$$0^*, \qquad 1^*, \qquad 10.$$

Its two matching loops shorten an existing block. A mismatch can leave the center only by popping the visible blocking symbol. Every side-to-center transition returns to a homogeneous block except the left-empty/input-0 case, which creates exactly 10.

Left

The only transition entering L is

$$(C, 0T) \xrightarrow{1} (L, T).$$

Reachability forces the source center stack to belong to 0^+, so

$$T \in 0^*.$$

Therefore no reachable left configuration has top 1.
This proves, in particular, that the listed completion rule

$$\dashv, 1/0$$

has an unreachable source condition.

Right

The only transition entering R is

$$(C, 1T) \xrightarrow{0} (R, T).$$

If the source is 1^{m+1}, then $T = 1^m$.
If the source is the exceptional center word 10, then $T = 0$.
Hence

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

No other right-state store can be produced.

A.8 Audit of the end-of-input completion rules

The endmarker $\dashv$ is not a binary input symbol, so completion normalization must preserve the residual already accumulated by the prefix.

The side-state potential is transferred into literal stack content.

A.8.1 Left-state completion

Every reachable left store is 0^m.

Empty left store.

$$(L, \varepsilon) \xrightarrow{\dashv} 100.$$

Before completion,

$$D(\varepsilon) + \Phi(L) = 0 + 3 = 3.$$

After completion,

$$D(100) = 4 - 1 = 3.$$

Nonempty left store. For $m \geq 1$,

$$(L, 0^m) \xrightarrow{\dashv} 1\,0^{m+2}.$$

Indeed the physical rule replaces the top 0 by 1000, leaving 0^{m-1} untouched. Before completion,

$$D(0^m) + \Phi(L) = 2m + 3.$$

After completion,

$$D(1\,0^{m+2}) = 2(m+2) - 1 = 2m + 3.$$

Thus left completion preserves the represented residual exactly.
The listed left rule with top 1 is unreachable and plays no role in any run.

A.8.2 Right-state completion

There are three reachable right forms.

Empty right store.

$$(R, \varepsilon) \xrightarrow{\dashv} 111.$$

Before completion,

$$D(\varepsilon) + \Phi(R) = -3,$$

and after completion,

$$D(111) = -3.$$

Nonempty homogeneous right store. For $m \geq 1$,

$$(R, 1^m) \xrightarrow{\dashv} 1^{m+3}.$$

Before completion,

$$D(1^m) + \Phi(R) = -m - 3.$$

After completion,

$$D(1^{m+3}) = -m - 3.$$

Exceptional right store.

$$(R, 0) \xrightarrow{\dashv} 1.$$

Before completion,

$$D(0) + \Phi(R) = 2 - 3 = -1.$$

After completion,

$$D(1) = -1.$$

Therefore every reachable endmarker normalization rule preserves the prefix residual.

A.9 From residual preservation to the canonical shortest completion

After the side-state correction, every normalized logical stack belongs to

$$0^* \cup 1^* \cup 10^+.$$

The three residual cases are therefore:

$$\widehat{S}(w) = \begin{cases} 0^m, & \rho(w) = 2m, \quad m \geq 0, \\ 1\,0^{m+1}, & \rho(w) = 2m+1, \quad m \geq 0, \\ 1^m, & \rho(w) = -m, \quad m \geq 1. \end{cases} \tag{A.6}$$

For any continuation containing x zeros and y ones,

$$wz \in L_{2,1} \quad \Longleftrightarrow \quad 2x - y = \rho(w). \tag{A.7}$$

The words in (A.6) realize the unique minimum nonnegative Parikh solution of (A.7):

- $2m$: $(x, y) = (m, 0)$;
- $2m+1$: $(x, y) = (m+1, 1)$;
- $-m$: $(x, y) = (0, m)$.

Thus the transition audit not only proves recognition-phase arithmetic consistency; together with the endmarker audit it recovers the shortest-completion semantics of the original design.

A.10 Two representative runs

A.10.1 Accepted word 110

Using the core projection and omitting Z_0,

$$(C, \varepsilon) \xrightarrow{1} (C, 10) \xrightarrow{1} (C, 0) \xrightarrow{0} (C, \varepsilon).$$

The residual sequence is

$$0,\ 1,\ 2,\ 0,$$

while $D(S) + \Phi(q)$ gives the same sequence:

$$0,\ 1,\ 2,\ 0.$$

The final empty center stack therefore represents zero residual and the word is accepted.

A.10.2 Prefix 111

The run is

$$(C, \varepsilon) \xrightarrow{1} (C, 10) \xrightarrow{1} (C, 0) \xrightarrow{1} (L, \varepsilon).$$

The residual is

$$\rho(111) = 3.$$

The final logical stack is empty, but

$$D(\varepsilon) + \Phi(L) = 3.$$

The endmarker rule transfers this finite-control contribution into the literal completion word:

$$(L, \varepsilon) \xrightarrow{\dashv} 100.$$

Since

$$2N_0(100) - N_1(100) = 4 - 1 = 3,$$

the normalized stack represents exactly the outstanding residual. Appending 100 gives

$$111100 \in L_{2,1}.$$

This is the canonical shortest completion selected by the machine.

A.11 Determinism of the completion extension

The completion callouts should not be interpreted as ordinary ε-moves.

For a fixed state/top pair:

- binary recognition transitions are selected by next input 0 or 1;
- completion is selected only by the distinct right endmarker $\dashv$.

Direct all completion rules to a dedicated completion sink with no outgoing transitions.

Then no configuration has two enabled successors.

Hence the completion-extended formalization is deterministic in the standard DPDA sense.

A.12 Finite transition consistency checks

The symbolic audit above is complete: it covers every reachable transition family.

Independent finite-range checks complement the symbolic transition audit. Their scope comprises the following cases and invariants:

- 524,287 original-machine binary inputs;
- 8,912,898 prefix transition checks;
- the residual invariant;
- the reachable-store normal form;
- completion normalization;
- independent shortest-completion length;
- depth-two support visibility.

All listed comparisons agree with the symbolic formulas.

The computational audit is supporting evidence only; the proof of (A.1) is the finite symbolic transition audit in Sections A.2–A.8.

A.13 Conclusion

The original 2023 design is not merely a correct recognizer.

Its state and stack form an exact split representation of the prefix residual:

$$\boxed{\rho = D + \Phi.}$$

The center state stores the residual entirely in the stack.

The side states store part of the same residual in finite control:

$$\boxed{\Phi(L) = +3, \qquad \Phi(R) = -3.}$$

Every binary transition preserves that semantic quantity with the correct residual increment, and every reachable endmarker rule transfers the finite-control offset back into literal stack form.

Thus the three operational ideas visible in the original machine—

$$\boxed{\text{cancellation,} \qquad \text{side-state mismatch,} \qquad \text{completion,}}$$

—are already an exact semantic normalization mechanism.

This is the local machine-level fact from which the later completion geometry, residual quotients, WRITE–SEARCH–CANCEL theory, lift theory, and General Theory are developed.

Appendix B

General Construction Transition Schemas

B.1 Purpose and model separation

The main text deliberately develops the general constructions conceptually before presenting every transition rule. This appendix collects the exact machine syntax in one place so that the book remains reproducible without forcing long transition tables into the narrative chapters.

Fix coprime integers $p, q \geq 1$ and write

$$\rho(w) = qN_1(w) - pN_0(w), \qquad T_0(r) = r - p, \qquad T_1(r) = r + q.$$

The primitive zero-weight kernel is

$$K = (q, p).$$

Three constructions must be kept separate.

1. The *paired-sign four-symbol lift* realizes the residual graph exactly after a deterministic normalization macro-step. It is a normalized residual–configuration conjugacy, not a raw one-step realization.

2. The *boundary-compatible three-symbol lift* changes coordinates so that every residual edge is one ordinary binary-input transition. It is ε-free and gives raw residual–configuration conjugacy with $\max(p, q)$ states.

3. The *centered literal-demand construction* stores the unique shortest completion demand itself, uses a finite-control debt rail to repair the LIFO ordering obstruction, consumes exactly one binary symbol per transition, and is real-time during binary input.

The exact resource-frontier theorems later compare these constructions only inside their stated representation models. In particular, no table below should be read as an unrestricted minimum-state theorem for every DPDA recognizing $L_{p,q}$.

B.2 Four-symbol normalized paired-sign lift

This is the first compact general physical lift used for the residual-geometry and configuration-rectangle results.

Put

$$m_{\mathrm{st}} = \max(p, q), \qquad Q_4 = \{R_0, \ldots, R_{m_{\mathrm{st}}-1}\}, \qquad \Gamma_4 = \{0, 1, X, Z_0\}.$$

For $0 \le s < p$ and $0 \le t < q$, define the finite residual bases

$$a_s = \begin{cases} 0, & s = 0, \\ s - p, & 1 \le s < p, \end{cases} \qquad b_t = \begin{cases} 0, & t = 0, \\ q - t, & 1 \le t < q. \end{cases}$$

The stable encoding $\eta_4 : \mathbb{Z} \to Q_4 \times \Gamma_4^*$ is

$$\eta_4(0) = (R_0, Z_0).$$

If $r > 0$, let $s = r \bmod p$ and write uniquely

$$r = pk + a_s, \qquad k \ge 1;$$

then

$$\eta_4(r) = (R_s, 0^k Z_0).$$

If $r < 0$, let $t = (-r) \bmod q$ and write uniquely

$$r = -qk + b_t, \qquad k \ge 1;$$

then

$$\eta_4(r) = (R_t, 1^k X).$$

The two sign roles may share a control state; their work symbols and sentinels keep the stable encodings disjoint.

Positive stable rules

For $0 \le s < p$, put

$$s' = (s + q) \bmod p, \qquad c_s = \frac{a_s + q - a_{s'}}{p}.$$

Then

$$\boxed{\delta(R_s, 0, 0) = (R_s, \varepsilon)}$$

and

$$\boxed{\delta(R_s, 1, 0) = (R_{s'}, 0^{c_s+1})}.$$

When the last positive work symbol is removed, a unique forced boundary normalization is used:

$$\boxed{\delta(R_s, \varepsilon, Z_0) = \eta_4(a_s)} \qquad (1 \le s < p).$$

For $s = 0$, popping the final positive 0 already exposes the zero encoding (R_0, Z_0), so no boundary normalization is required.

Negative stable rules

For $0 \le t < q$, put

$$t' = (t + p) \bmod q, \qquad d_t = \frac{p - b_t + b_{t'}}{q}.$$

Then

$$\boxed{\delta(R_t, 1, 1) = (R_t, \varepsilon)}$$

and

$$\boxed{\delta(R_t, 0, 1) = (R_{t'}, 1^{d_t+1})}.$$

When the negative sentinel is exposed,

$$\boxed{\delta(R_t, \varepsilon, X) = \eta_4(b_t)}.$$

Zero rules and acceptance

At residual zero,

$$\boxed{\delta(R_0, 1, Z_0) = \eta_4(q)}, \qquad \boxed{\delta(R_0, 0, Z_0) = \eta_4(-p)}.$$

A mandatory right endmarker is accepted only at the stable zero configuration $\eta_4(0)$.

An expression of the form $\delta(\cdot) = \eta_4(r')$ is semantic shorthand: the exposed boundary symbol is replaced by the stack word of the stable encoding while control enters the corresponding state. Long fixed writes may be compiled into deterministic forced ε-chains if a bounded-write PDA syntax is required. Such a compilation preserves the stable target but is no longer a raw one-binary-transition realization.

Define a normalized input macro-step $c \overset{a}{\Rightarrow} c'$ to mean one consumed binary symbol followed by the unique forced normalization chain. The transition schema satisfies

$$\boxed{\eta_4(r) \overset{0}{\Rightarrow} \eta_4(r-p), \qquad \eta_4(r) \overset{1}{\Rightarrow} \eta_4(r+q).}$$

This is the exact normalized residual–configuration conjugacy used in the configuration-rectangle chapters.

B.3 One-work-symbol raw conjugacy

The one-work-symbol construction is included because it is the matching upper bound for the exact $p+q$-state result in the permanent-bottom raw-conjugacy model.

Let

$$\Gamma_2 = \{A, Z_0\},$$

with disjoint state families

$$P_0, \ldots, P_{p-1}, \qquad N_0, \ldots, N_{q-1}.$$

For $r \geq 0$, write

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\xi(r) = (P_s, A^k Z_0).$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad 0 \leq u < q, \quad h \geq 1,$$

and define

$$\xi(r) = (N_u, A^{h-1} Z_0).$$

For $0 \leq s < p$, set

$$s^+ = (s+q) \bmod p, \qquad c_s = \frac{s+q-s^+}{p} \geq 0,$$

and write

$$s - p = u_s - qh_s, \qquad 0 \leq u_s < q, \quad h_s \geq 1.$$

The positive and boundary rules are

$$\boxed{\delta(P_s, 0, A) = (P_s, \varepsilon)}, \qquad \boxed{\delta(P_s, 1, A) = (P_{s^+}, A^{c_s+1})},$$

$$\boxed{\delta(P_s,1,Z_0)=(P_{s^+},A^{c_s}Z_0)},\qquad \boxed{\delta(P_s,0,Z_0)=(N_{u_s},A^{h_s-1}Z_0)}.$$

For $0\le u<q$, put

$$u^-=(u-p)\bmod q,\qquad d_u=\frac{u^--u+p}{q}\ge 0.$$

The negative rules are

$$\boxed{\delta(N_u,1,A)=(N_u,\varepsilon)},\qquad \boxed{\delta(N_u,0,A)=(N_{u^-},A^{d_u+1})},$$

$$\boxed{\delta(N_u,1,Z_0)=\xi(u)},\qquad \boxed{\delta(N_u,0,Z_0)=(N_{u^-},A^{d_u}Z_0)}.$$

The mandatory right endmarker is accepted only at

$$\xi(0)=(P_0,Z_0).$$

There are no ε-transitions. Direct substitution gives

$$\boxed{\delta(\xi(r),0)=\xi(r-p),\qquad \delta(\xi(r),1)=\xi(r+q).}$$

The construction uses exactly $p+q$ states and two total stack symbols, one of which is the permanent bottom marker.

B.4 Boundary-compatible three-symbol raw DPDA

The four-symbol normalized lift shows that the residual graph has a compact physical realization, but it has a finite set of boundary edges that require normalization. The following coordinate change removes those defects completely.

Assume first that $p\ge q$. Let

$$Q_3=\{R_0,\dots,R_{p-1}\},\qquad \Gamma_3=\{0,1,Z_0\}.$$

For $r\ge 0$, write uniquely

$$r=pk+s,\qquad k\ge 0,\quad 0\le s<p,$$

and set

$$\boxed{\zeta(r)=(R_s,0^kZ_0)}.$$

For $r<0$, write uniquely

$$r=u-qh,\qquad h\ge 1,\quad 0\le u<q,$$

and set

$$\boxed{\zeta(r)=(R_u,1^hZ_0)}.$$

The state sets overlap deliberately. The sign is visible from the top of stack: negative encodings have top 1, whereas nonnegative encodings have top 0 or Z_0.

For $0\le s<p$, define

$$s^+=(s+q)\bmod p,\qquad c_s=\frac{s+q-s^+}{p}\in\{0,1\},$$

$$u_s=(s-p)\bmod q,\qquad h_s=\frac{u_s+p-s}{q}\ge 1.$$

For $0\le u<q$, define

$$u^-=(u-p)\bmod q,\qquad d_u=\frac{u^--u+p}{q}\ge 1.$$

Complete transition table for $p \geq q$

region	input/top	target	replacement
nonnegative interior	$0/0$	R_s	ε
nonnegative interior	$1/0$	R_{s^+}	0^{c_s+1}
nonnegative boundary	$1/Z_0$	R_{s^+}	$0^{c_s}Z_0$
nonnegative boundary	$0/Z_0$	R_{u_s}	$1^{h_s}Z_0$
negative	$1/1$	R_u	ε
negative	$0/1$	R_{u^-}	1^{d_u+1}

Equivalently,

$$\boxed{\begin{aligned}\delta(R_s,0,0) &= (R_s,\varepsilon),\\ \delta(R_s,1,0) &= (R_{s^+},0^{c_s+1}),\\ \delta(R_s,1,Z_0) &= (R_{s^+},0^{c_s}Z_0),\\ \delta(R_s,0,Z_0) &= (R_{u_s},1^{h_s}Z_0),\\ \delta(R_u,1,1) &= (R_u,\varepsilon),\\ \delta(R_u,0,1) &= (R_{u^-},1^{d_u+1}).\end{aligned}}$$

The right endmarker is accepted only at

$$\zeta(0) = (R_0, Z_0).$$

No ε-transition is used. Every binary transition consumes exactly one input symbol, and

$$\boxed{\delta(\zeta(r),0) = \zeta(r-p), \qquad \delta(\zeta(r),1) = \zeta(r+q).}$$

Hence the raw binary-input configuration graph is label-isomorphic to the residual graph.

If $q > p$, use the symmetric construction obtained by simultaneously exchanging

$$0 \leftrightarrow 1, \qquad p \leftrightarrow q, \qquad r \leftrightarrow -r.$$

The resulting state count is always

$$\boxed{|Q_3| = \max(p,q)},$$

with three total stack symbols including the permanent bottom marker.

B.5 Centered real-time literal-demand construction

The raw residual encodings above store a signed residual coordinate. The centered construction solves a different problem: it keeps a physical representation whose semantic projection is the unique shortest completion demand and repairs one-cell LIFO ordering conflicts by moving a blocker into finite control.

Let

$$m = p + q, \qquad h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

For a residual r, let

$$\mu(r) = (x_r, y_r)$$

be the unique minimum nonnegative solution of

$$px - qy = r.$$

For a demand vector (x, y) use the canonical two-visible ordering

$$\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}$$

For debt $d \in \mathbb{Z}$, define the virtual demand

$$V_d = \mu(dm), \qquad \Phi(D_d) = dm.$$

The centered state set is

$$\boxed{Q_{\mathrm{ctr}} = \{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}\},}$$

with $D_0 = C$, and

$$\Gamma_{\mathrm{ctr}} = \{0, 1, Z_0\}.$$

For a logical stack S, the semantic projection is

$$\Theta(D_d, S) = \mathrm{red}_{p,q}\big(P(S) + V_d\big).$$

The residual invariant is

$$\boxed{ho = W(P(S)) + dm.}$$

MATCH

For $z \in \{0, 1\}$ and every $|d| \leq h_{\mathrm{ctr}}$,

$$\boxed{(D_d, zT) \xrightarrow{z} (D_d, T).}$$

Interior SEARCH

For $d < h_{\mathrm{ctr}}$,

$$\boxed{(D_d, 0T) \xrightarrow{1} (D_{d+1}, T),}$$

and for $d > -h_{\mathrm{ctr}}$,

$$\boxed{(D_d, 1T) \xrightarrow{0} (D_{d-1}, T).}$$

The displaced blocker is not restored immediately. Its signed effect is represented by the debt coordinate d.

Outer-boundary WRITE

Put

$$U_+ = \eta(\mu((h_{\mathrm{ctr}} + 1)m))\,, \qquad U_- = \eta(\mu(-(h_{\mathrm{ctr}} + 1)m))\,.$$

Then

$$\boxed{(D_{h_{\mathrm{ctr}}}, 0T) \xrightarrow{1} (C, U_+ T),}$$

and

$$\boxed{(D_{-h_{\mathrm{ctr}}}, 1T) \xrightarrow{0} (C, U_- T).}$$

Bottom WRITE

For every $|d| \leq h_{\mathrm{ctr}}$,

$$\boxed{(D_d, Z_0) \xrightarrow{1} (C, \eta(\mu(dm + q))Z_0)\,,}$$

$$\boxed{(D_d, Z_0) \xrightarrow{0} (C, \eta(\mu(dm - p))Z_0)\,.}$$

Acceptance

A mandatory right endmarker is permitted only at the balanced empty configuration:

$$\boxed{(C, Z_0) \xrightarrow{\dashv} (C, \varepsilon).}$$

There is no right-endmarker rule from any other reachable configuration. Every binary transition consumes exactly one input symbol; the centered construction has no ε-move during binary processing.

B.6 Minimal kernel-carry compiler identity

The WRITE strings in the centered construction are not ad hoc. Suppose a transition consumes input symbol a, removes visible top symbol z, changes state from D_s to D_t, and writes a replacement word u. Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Ignoring the untouched suffix, invariant preservation forces

$$P(u) = e_z - e_a + V_s - V_t + \kappa K.$$

At the bottom marker, omit the term e_z. If

$$B = e_z - e_a + V_s - V_t,$$

then the smallest kernel carry making the replacement nonnegative is

$$\boxed{\kappa_* = \max\left(\left\lceil -\frac{B_0}{q} \right\rceil, \left\lceil -\frac{B_1}{p} \right\rceil\right).}$$

The unique shortest compatible replacement vector is

$$\boxed{R_* = B + \kappa_* K.}$$

Every other nonnegative invariant-preserving replacement is $R_* + \ell K$ for some $\ell \geq 1$ and is longer by exactly $\ell(p + q)$ symbols.

This identity is the common arithmetic compiler behind the large literal WRITE words of the original 2:1 machine and the general centered family.

B.7 The 2:1 specialization and the eager–lazy distinction

For $(p, q) = (2, 1)$,

$$m = 3, \qquad h_{\mathrm{ctr}} = 1,$$

so

$$D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L,$$

with potentials

$$-3, \quad 0, \quad +3.$$

The centered schema reproduces the historical center/left/right control skeleton, the center mismatch pops, the left cancellation family, and the arithmetic of the characteristic WRITE words.

Two reachable right-state matching families differ in schedule, not semantics. The centered machine keeps the resulting debt in finite control (lazy materialization), whereas the historical machine immediately materializes the same canonical demand on the stack (eager materialization). If $\mathcal{N}$ denotes canonical materialization, then on exactly those families

$$\delta_{\mathrm{hist}}(c, a) = \mathcal{N}(\delta_{\mathrm{cent}}(c, a)) .$$

Accordingly, direct specialization is not claimed to be literal transition-table identity. The remaining difference is precisely the eager–lazy normalization-schedule choice.

B.8 Determinism and implementation conventions

The constructions in this appendix use three different transition disciplines and should not be conflated.

- In the four-symbol paired-sign lift, forced ε-normalization is enabled only on transient boundary pairs where binary input is disabled. The normalization macro-step is therefore deterministic.
- In the one-work-symbol and boundary-compatible three-symbol raw lifts, there are no ε-transitions at all; the right endmarker is the only nonbinary input symbol.
- In the centered literal-demand construction, every binary step is real-time and the right endmarker is enabled only at the balanced empty configuration.

All replacement words are fixed finite constants once (p, q) is fixed. If a separate PDA convention imposes a bound on replacement length, long fixed writes can be compiled using auxiliary deterministic microstates. Such compilation changes the physical transition granularity and must not be silently substituted into a theorem whose statement concerns raw one-input transitions or exact state counts in a fixed model.

B.9 Summary table

construction	states	stack alphabet	exact role
paired-sign normalized lift	$\max(p, q)$	$\{0, 1, X, Z_0\}$	exact residual conjugacy after a unique normalization macro-step
one-work-symbol raw lift	$p + q$	$\{A, Z_0\}$	exact raw residual conjugacy; matching upper bound for the one-work-symbol state theorem
boundary-compatible raw lift	$\max(p, q)$	$\{0, 1, Z_0\}$	ε-free raw residual conjugacy by ordinary binary transitions
centered literal-demand lift	$2\lfloor(p + q - 1)/2\rfloor + 1$	$\{0, 1, Z_0\}$	real-time shortest-demand normalization with finite-control SEARCH debt; exact width only in the canonical deferred-SEARCH model

The purpose of the table is organizational. The constructions optimize different interfaces, and their state counts are not interchangeable unrestricted complexity claims.

Appendix C

Scoped Lower Bounds and Finite Certificates

C.1 Why this appendix is necessary

Several chapters use the word *optimal*, but they do so in different representation categories. The purpose of this appendix is to make the scope of every principal lower bound explicit and to place the one genuinely computer-assisted finite elimination in an explicitly auditable finite form.

The central distinction is:

$$\boxed{\text{representation-relative lower bound} \neq \text{unrestricted recognition lower bound.}}$$

A machine may recognize the same language while abandoning the residual coordinates, stack-tail form, literal-demand interface, or normalization schedule used by another construction. A lower bound obtained by counting coordinates in one architecture therefore cannot automatically be promoted to all DPDAs.

C.2 Lower-bound scope table

result	scope, exact bound, and excluded overclaim
one-work-symbol raw residual conjugacy	**Opponent:** permanent bottom, one work symbol, ε-free raw residual–configuration conjugacy. **Bound:** $\lvert Q\rvert \geq p + q$. **Not:** an unrestricted recognition bound for $L_{p,q}$.
homogeneous-tail raw frontier	**Opponent:** permanent bottom, sign-separated homogeneous tails, at least two work symbols. **Bound:** $\lvert Q\rvert \geq \max(p, q)$. **Not:** a theorem about arbitrary stack encodings.
arbitrary three-symbol raw residual encoding	**Opponent:** permanent bottom, three total stack symbols, arbitrary injective raw residual encoding. **Bound:** $\lvert Q\rvert \geq \max(p, q)$ for every coprime (p, q). **Not:** an unrestricted recognition bound; Theorem 8.7 is representation-shape-free within Model R.
canonical deferred-SEARCH width	**Opponent:** literal shortest-demand representation with deferred SEARCH interface. **Bound:** $2\lfloor (p + q - 1)/2 \rfloor + 1$ states. **Not:** an unrestricted state minimum for every DPDA recognizing $L_{p,q}$.

result	scope, exact bound, and excluded overclaim
phase-local capacity	**Opponent:** specified local observation depth and remainder architecture. **Bound:** $\|Q\|g^{\lambda} \geq BH(B)$. **Not:** a universal Pareto frontier.
binary stack-alphabet hierarchy witness	**Opponent:** arbitrary general DPDA in the stated single-initial-symbol model; deterministic ε-moves allowed. **Bound:** $\|Q\|\|\Gamma\| \geq mk + 1$ for $B_{m,k}$. **Not:** automatically a bound for $L_{p,q}$; it applies to the engineered witness language.
unrestricted rational-balance state complexity	**Opponent:** arbitrary deterministic right-endmarker ε-DPDA in the stated regular-target model. **Bound:** $\|Q\| \geq w_{\max}$, attained with three total stack symbols. **Scope:** representation-independent for $L_{p,q}$; no residual coordinate is assumed.
unrestricted rational-balance state–stack frontier	**Opponent:** the same broad class, with the permanent-bottom convention. **Frontier:** $\{(p+q, 2), (w_{\max}, 3)\}$. **Scope:** the explicit global endpoint, superseding any raw-only reading of the same pair.
unrestricted bounded replacement, one work symbol	**Opponent:** arbitrary history-dependent deterministic encodings, one work symbol, replacement bound $B \geq 2$. **Bound:** $w_{\max} + \max(w_{\min}, \lceil w_{\max}/(B-1) \rceil)$. **Scope:** exact only for this one-work-symbol bounded-replacement resource question.
filtered collapse / source exposure	**Opponent:** exact deterministic lifts with a finite collapse family and hidden-tail noncoalescence; the pushdown specialization is arbitrary. **Bounds:** $D + X_h + V(h) \geq N$ and $V(h) \leq q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r}$. **Not:** a state minimum for every language; this is an information/exposure tradeoff.

C.3 Raw residual-conjugacy frontier

For the permanent-bottom raw residual-conjugacy models developed in the resource chapters, the exact structured frontier is:

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2, \end{cases}}$$

where g denotes the number of work symbols above the permanent bottom marker in the homogeneous-tail model.

The upper bounds are realized by the constructions collected in Appendix B. The lower bounds rely on the raw-conjugacy interface: every residual must have one injective physical configuration, and each binary residual edge must be realized by one ordinary input transition.

The $g = 1$ lower bound is exact because a one-work-symbol store can encode unbounded magnitude only through height, while the p positive-end and q negative-end arithmetic phases require disjoint recurrent state resources. The resulting state requirement is

$$\boxed{|Q| \geq p + q.}$$

With two distinct work symbols, sign can be placed in the visible top symbol. The

boundary-compatible construction then attains

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3}$$

including the permanent bottom marker.

The homogeneous-tail hypothesis is removed completely in Chapter 8. The representation-shape-free Model-R argument proves the same three-symbol state optimum $|Q| = \max(p, q)$ for every coprime pair. The earlier small-ratio eliminations remain useful finite certificates, but they are no longer the endpoint of the theory.

C.4 The 3:2 finite reduction

The first genuinely two-sided arbitrary-encoding case is $(p, q) = (3, 2)$. The theorem concerns the full permanent-bottom three-symbol model

$$\Gamma_3 = \{A, B, Z_0\}$$

with no ε-moves and an injective raw residual–configuration conjugacy. No homogeneous-tail or sign-separated assumption is imposed.

Assume, for contradiction, that two control states suffice. Choose an encoded configuration of minimum work-stack height and write the two parity chains relative to its residual r_* as

$$A_k = \eta(r_* + 2k), \qquad B_k = \eta(r_* + 1 + 2k).$$

Let G denote the input-1 action and F the input-0 action. Since input 1 adds 2 to the residual and input 0 subtracts 3,

$$G(A_k) = A_{k+1}, \qquad G(B_k) = B_{k+1},$$

$$F(A_{k+2}) = B_k, \qquad F(B_{k+1}) = A_k.$$

Therefore, on the encoded residual graph,

$$\boxed{F^2G^3 = G^3F^2 = \mathrm{id}\,.} \tag{C.1}$$

Equation (C.1) is the key finiteness mechanism. One consuming PDA transition deletes at most one work symbol. If a work-top rule writes m work symbols in place of one, its first step increases work height by $m - 1$. The remaining four moves of either five-step inverse identity can delete at most four symbols before the original configuration must be recovered. Hence

$$\boxed{m \leq 5}$$

for every work-top replacement encountered in the search. From the bare bottom, the first step creates rather than replaces its work word, so every bottom write has length at most four.

Thus an a priori infinite transition search reduces to a finite deterministic rule-synthesis problem.

C.5 Normalization of the minimum configuration

There are exactly two minimum-height cases.

Positive minimum height. The suffix strictly below the visible top of a minimum encoded configuration can never be exposed by another encoded configuration; it is inert and may be stripped. Rename states and work symbols so that

$$\boxed{A_0 = (0, A\#),}$$

where # denotes the fixed inert suffix.

Zero minimum height. Normalize instead to

$$\boxed{A_0 = (0, \#),}$$

where # is the bare bottom marker.

These two cases exhaust the possible minimum work heights.

C.6 Legal branching and pruning rules

The finite certificate assigns a transition rule only when a state/top pair is encountered for the first time. At that first occurrence it branches over every legal target state and every replacement word satisfying the proven length bounds. A repeated state/top pair must reuse the rule already fixed by determinism.

Branches are rejected only by consequences already forced by the raw conjugacy and the inverse identities:

1. a prescribed F-relation or G-relation has the wrong target;
2. suffix preservation makes the required replacement impossible;
3. two distinct residuals obtain the same encoded configuration, violating injectivity;
4. a configuration is too tall to return to a prescribed target in the remaining number of one-pop transitions.

Representative deletion-capacity bounds are

$$F^2(A_3) = A_0, \qquad F(B_1) = A_0,$$

which imply

$$|A_3| \le |A_0| + 2, \qquad |B_1| \le |A_0| + 1.$$

More generally,

$$F(B_j) = A_{j-1}, \qquad F(A_{j+2}) = B_j,$$

and the corresponding two-deletion relations

$$F^2(B_j) = B_{j-3}, \qquad F^2(A_{j+2}) = A_{j-1}$$

provide the remaining legal height-pruning bounds.

No heuristic pruning is permitted: every discarded branch violates one of these proved necessary conditions.

C.7 Exact survivor vectors

For positive minimum height, the exact numbers of surviving partial rule systems at successive synthesis stages are

$$\boxed{1, 123, 11643, 1691, 2120, 1591, 348, 43, 13, 0.} \tag{C.2}$$

The search examines exactly

$$\boxed{1{,}130{,}430}$$

branch attempts in this case.

For minimum height zero, the survivor vector is

$$\boxed{1, 61, 7556, 1134, 2146, 2966, 6846, 7032, 2376, 650, 190, 26, 0.} \tag{C.3}$$

This case examines exactly

$$\boxed{1{,}696{,}120}$$

branch attempts.

Both exhaustive searches terminate with zero survivors. Since the symbolic reduction proves that every possible two-state raw encoding must occur in one of the two finite searches, two states are impossible. The explicit three-state boundary-compatible construction supplies the matching upper bound. Hence

$$\boxed{|Q|_{\min} = 3}$$

for the arbitrary permanent-bottom three-symbol raw-conjugacy model at $(3, 2)$, and symmetrically at $(2, 3)$.

C.8 Computational certificate validation

An independent finite evaluation of the structure and resource conditions yields

$$\boxed{\text{80,717,789 checks – PASS.}}$$

Among the independently checked components are

component	count
brute minimum-demand cases	11,011
broad arithmetic cases	1,548,547
compact normalized macro cases	667,110
normalized rectangle vertices	1,859,269
normalized rectangle diamonds	1,286,746
raw four-symbol immediate cases	1,111,110
one-work-symbol raw cases	1,111,110
three-symbol raw cases	1,111,110
raw rectangle classifications	61,455
three-symbol raw rectangle vertices	731,091
three-symbol raw rectangle diamonds	474,684

The same validation also confirms the exact branch counts and survivor vectors in (C.2)–(C.3).

The later resource-frontier computational validation was likewise rerun and terminates with

$$\boxed{79{,}606{,}679 \text{ checks} - \text{PASS.}}$$

These computational audits support the symbolic results and certify the bounded search. Except for the explicitly computer-assisted $(3, 2)$ elimination, they are regression evidence rather than substitutes for proof.

C.9 Architecture-relative deferred-SEARCH width

For the centered literal-demand machine, put

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor.$$

In the canonical deferred-SEARCH model, after a witness demand block has been consumed, the remaining work stack is the same bare bottom marker for every debt value

$$-h_{\mathrm{ctr}}, \ldots, h_{\mathrm{ctr}}.$$

Distinct residuals are continuation-distinguishable. Determinism therefore forces distinct control states for all these debt values. The exact width is

$$\boxed{|Q|_{\min}^{\mathrm{deferred}} = 2h_{\mathrm{ctr}} + 1 = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1.}$$

The centered construction attains the bound.

The deferred condition is essential: the final consumed witness cell is a pure pop, and materialization is postponed to a later transition. Eager encodings, alternate stack coordinates, or machines that do not store literal minimum demand are outside this lower-bound category.

C.10 Representation-independent hierarchy product bound

A different chapter deliberately leaves the counting-language representation class. For every $m \geq 2$ and $k \geq 1$, the binary witness language $B_{m,k}$ is recognized by a realtime DPDA with exactly m states and $k+1$ pushdown symbols. Every competing DPDA in the stated *general* deterministic model—including opponents with arbitrary deterministic ε-moves and unrelated internal stack encodings—satisfies

$$\boxed{|Q|\,|\Gamma| \geq mk + 1.} \tag{C.4}$$

Consequently the fixed-state stack-alphabet classes obey the strict hierarchy

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1} \qquad (m \geq 2,\ k \geq 1).}$$

The proof does not count residual phases in a prescribed code. It isolates $mk+1$ externally distinguishable affine completion slopes. Repetition, height-escape, and step-pump arguments show that one recurrent state/top-symbol mode cannot realize two distinct slopes. Since a DPDA has at most $|Q||\Gamma|$ nonempty modes, (C.4) follows.

This is representation-independent for the witness language $B_{m,k}$, but it must not be retroactively read as an unrestricted product lower bound for the rational counting languages $L_{p,q}$.

C.11 Final scope rule

The lower-bound hierarchy in this monograph is intentionally layered, and the final endpoint contains two genuinely unrestricted branches:

coordinate-relative exact frontiers and protected contracts

$\Downarrow$

architecture-relative literal-demand and local-observation bounds

$\Downarrow$

representation-independent witness hierarchy

$\Downarrow$

unrestricted $L_{p,q}$ state/state–stack/bounded-replacement theorems

$\Downarrow$

representation-independent collapse/exposure tradeoffs.

The arrows record the research route rather than logical implication. In particular, raw residual-faithful frontiers remain useful typed results, but they are no longer the strongest statements available for the natural rational-balance family. The unrestricted endpoint is stated explicitly in Chapter 94 and Theorem 95.15; the abstract information-capacity endpoint is Theorem 62.1 and its pushdown specialization. Every lower bound remains attached to the opponent model printed in its theorem.

Appendix D

Computational Validation

D.1 Role of computation

The symbolic proofs are the mathematical basis of the monograph. Computational audits serve three narrower purposes:

1. detect transcription errors in long transition tables and closed forms;
2. test boundary cases across large finite parameter ranges;
3. certify a finite bounded elimination when a theorem explicitly has a computer-assisted final step.

The discussion below records the scope and numerical outcome of each audit. Only the mathematical method, finite ranges, and resulting certificates are reported; no implementation-specific identifiers are needed.

D.2 Validation-status terminology

Three statuses are used throughout this appendix.

Independently rerun.
: The reported validation was repeated during the monograph audit and the stated outcome was reproduced.

Independently recomputed.
: The stated finite mathematical conditions were evaluated afresh from the explicit definitions, and the recorded counts were reproduced.

Reported finite count.
: The stated finite count is recorded as supporting evidence but has not been independently reproduced in the present review.

This distinction concerns the provenance of the computational evidence only; it does not alter the symbolic theorem statements.

D.3 Independently rerun audits

Completion/frontier/original-machine audit

The independent audit passes and includes:

check family	count
original-machine binary inputs	524,287
prefix transition checks	8,912,898
independent quotient-distance checks	25,281,628
compact-DPDA runs	1,282,610
parameter formula audits	3,931

The audit checks the original-machine residual invariant, reachable-store structure, completion normalization, shortest-completion formulas, compact realization, and related finite-frontier claims.

Structure/resource audit

The independent audit terminates with

80,717,789 checks – PASS.

Principal reported subcounts are:

check family	count
brute minimum-demand cases	11,011
broad arithmetic cases	1,548,547
compact macro cases	667,110
configuration-rectangle vertices	1,859,269
configuration-rectangle diamonds	1,286,746
phase samples	44,047
raw four-symbol cases	1,111,110
raw four-symbol defects encountered	16,247
one-work-symbol cases	1,111,110
raw three-symbol cases	1,111,110
raw rectangle cases	61,455
raw three-symbol rectangle vertices	731,091
raw three-symbol rectangle diamonds	474,684
ladder paths	11,778

The same audit contains the bounded $(3, 2)$ rule-synthesis certificate recorded in Appendix C.

Exact resource-frontier audit

The independent audit terminates with

79,606,679 checks – PASS.

It independently rechecks the minimum-demand arithmetic, normalized and raw configuration rectangles, raw-conjugacy boundaries, and the state–alphabet resource formulas used in the exact frontier analysis.

Centered real-time family audit

The independent audit covers every ordered coprime pair

$$1 \le p, q \le 20$$

and reports

$$\boxed{255 \text{ ordered coprime ratios}},$$

$$\boxed{1{,}909{,}191 \text{ reached configurations}},$$

and

$$\boxed{3{,}738{,}864 \text{ binary edges}}.$$

The audit checks transition totality on reached binary configurations, the residual invariant, residual-collision consistency, two-cell visibility, and zero-residual correctness. The rerun passes.

D.4 Independent finite-range comparisons

WRITE–SEARCH–CANCEL audit

The WRITE–SEARCH–CANCEL validation study reports

$$\boxed{477{,}033 \text{ assertion-level checks.}}$$

These checks evaluate the stated finite mathematical conditions directly over the ranges below:

- all coprime $1 \le p, q \le 8$ and all binary prefixes of length at most nine for the closed-form minimum-demand formula, semantic carry criterion, and demand clock;
- the bounded canonical one-top direct-realizability classification;
- WSC defect-lattice coordinate formulas;
- count-shadow fiber inequalities;
- physical-factorization reconstruction formulas;
- bounded carry-order debt instances;
- supplementary kernel-fiber identities.

The combined number of checked conditions is:

$$\boxed{477{,}033 \text{ checks} - \text{PASS}.}$$

These are finite consistency comparisons; the symbolic arguments establish the general claims.

Semantic-uniqueness/lift audit

The semantic-uniqueness/lift validation study reports

458,776 assertion-level checks.

The same finite method applies to the schedule and causality conditions, including:

- the six admissible and two inadmissible 2:1 schedules;
- pairwise distinct reduced schedule fingerprints;
- the two-maxima/non-lattice schedule-poset structure;
- the directed $A \rightsquigarrow B$ causal certificate;
- the one-kernel excess at the eager B successor;
- the first forbidden recognition witness.

The number of checked conditions is:

458,776 checks – PASS.

The finite checks support the stated symbolic conclusions but do not replace their proofs.

D.5 General-Theory computational support

The General Theory is formulated and proved symbolically. Its verification chapter imports classical decision mechanisms only after reducing correctness to an exact structural target. No external computation is required to establish the main General-Theory proofs.

Finite catalogues used as stress tests, including the 3:2 and odd-p:2 schedule families, are recorded separately in the General-Theory catalogue appendix rather than being treated as substitutes for the abstract theorems.

D.6 Enumerative and arithmetic computational audits

The enumerative and arithmetic results have symbolic proofs as their primary evidence. Their numerical audits are recorded below as source-reported unless an independent rerun is explicitly stated.

Explicit run–return determinants

The source audit reports direct coarea enumeration against the weighted determinant formula and an independent coefficient-free check. The final source record lists 33 sampled-run determinant cases and 86 coefficient-free cases with no mismatches; an earlier full audit record additionally reports an extended 220-case determinant/direct-enumeration pass. These computations audit indexing and signs and are not proof substitutes.

Contact-preserving barrier compression

The source audit reports 88,032 determinant/enumeration comparisons for the general exact-contact theorem. Its rational-path specializations report 755 ordinary fixed-composition instances with 9,983 statistic queries, 628 ratio-run instances with 54,664 exact-return queries, and 501 simultaneous instances with 246,392 queries, all without discrepancy. The fixed-signature audit covers 2,357 classes over 12,081 paths. A larger $(23, 11)$ stress test enumerates all 63,920 fixed-composition paths; 616 paths of signature 11111110 split into four classes of 154 each, matching the determinant.

Farey recursion and spectra

The reviewer-oriented source audit exhaustively checks all 24,064 ordered coprime pairs with $2 \le p, q \le 200$ and adds 30,000 randomized coprime pairs with $p, q \le 5000$ for the unit-lift laws, closed profiles, all four ordered-recursion identities, and the complementary index partitions, with no counterexample. The scalar audit covers all 13,715 coprime ratios with $1 \le p, q \le 150$ through residual shells $k = 0, \ldots, 7$, again with no counterexample. The novelty stress test separately confirms that the local walk is a unimodular image of classical mechanical/Christoffel geometry and therefore keeps the novelty claim on the cross-parameter minimizer recursion.

D.7 Interpretation rule

Throughout the book, the phrase "validated computationally" must be read according to the hierarchy

symbolic theorem first, computational evidence second.

The sole exception is an explicitly labeled computer-assisted theorem whose symbolic argument reduces the remaining obligation to a finite exhaustive search. In that case, the finite certificate is part of the proof and its pruning assumptions, survivor vectors, and terminal zero-survivor result must be preserved together.

D.8 Post-theory validation

D.8.1 Fixed-Parikh quantitative LIFO chronology

The fixed-semantics chronology results are symbolic. Computation was used only as an independent transcription and boundary-case audit of the finite physical-lift kernels and small-parameter constants. Direct probability propagation and kernel calculations were compared on overlapping horizons and parameter pairs; the numerical checks do not enter any proof premise.

D.8.2 Optimal-fiber rank-two topology

The rank-two optimal-fiber classification is proved symbolically. Finite-grid computations were used as consistency checks for active-fiber saturation, nerve compression, cellular homology, and the explicit scalar realization families. These calculations serve only as audits of examples and boundary cases.

D.8.3 Unrestricted state complexity

The unrestricted state lower bound is entirely symbolic. During preparation, the matching three-symbol construction was independently sanity-checked on finite ranges of coprime parameters and residuals by verifying the two translation identities for every tested configuration. No computational assertion is used in the proof of the lower bound or exactness theorem.

D.9 Geodesic-compiler validation

Prescribed-order separation and Boolean amplification

The geodesic-compiler and non-factorization theorems are proved symbolically. Independent finite checks use the explicit translation actions in the theorem statements. Exact

shortest distances are obtained by reverse breadth-first search and compared with, rather than inferred from, the stated closed distance formulas.

For every $1 \le k \le 10$, the audit verifies the Binary-Deficit Distance Lemma on every pair

$$0 \le D < 2^k, \qquad k \le \ell \le 2k,$$

for a total of

$$\boxed{\text{20,480 binary-deficit instances.}}$$

For the same ten blocks it reconstructs the complete structural/optimal face data and confirms

$$\mathcal{H}(r) = 2^{A_-}, \qquad \mathcal{H}(r') = 2^{A_-} \setminus \{J_\star, A_-\},$$

including exact agreement through order $k-1$ and the unique order-k discrepancy.

A separate product-graph reverse-BFS audit checks

$$(k, m) = (1, 3), (2, 2), (3, 2), (4, 2).$$

In every case all 2^m selected product states have the same exact distance and the same order-$(k-1)$ signature, while the order-k signatures are pairwise distinct and each coordinate toggles exactly its distinguished face $J_\star^{(t)}$. The finite comparison gives

$$\boxed{\text{OVERALL PASS.}}$$

These computations are independent regression evidence only and are not premises of the compiler, non-factorization, or amplification theorems.

D.10 Universal k-fiber validation

The Universal k-Fiber Theorem and exact capacity theorem in Chapter 19 are proved symbolically. As an independent regression check, the explicit two-phase construction was exhaustively evaluated for several small parameter pairs. For every witness state in each tested family, the audit checked the common closed-form distance, reconstructed the complete structural/optimal truncation through order $k-1$, and tested every distinguished k-face against the Bellman equality.

s	k	$\binom{s}{k}$	witnesses	common distance	patterns
2	2	1	2	6	2
3	2	3	8	11	8
4	2	6	64	18	64
5	2	10	1024	27	1024
3	3	1	2	8	2
4	3	4	16	18	16
5	3	10	1024	37	1024
4	4	1	2	10	2
5	4	5	32	27	32

In every tested case, all witness states had one common exact distance and one common complete lower-order compatibility truncation, while the distinguished order-k layer ranged over all $2^{\binom{s}{k}}$ prescribed patterns. The $s = 3, k = 2$ row independently reproduces all eight simple graphs in the worked universal fiber of Chapter 19. These computations are regression evidence only and are not premises of any theorem.

Appendix E

Lagrange–Bürmann Derivation of the Exact Correction Triangle

E.1 The marked first-return grammar

This appendix supplies the coefficient derivation deferred from Chapter 27. Fix an integer $d \geq 1$. Let $F(z,t)$ denote the correction-marked excursion generating function at baseline zero, and let $A(z,t)$ and $B(z,t)$ be the two periodic tail classes introduced by the first-return decomposition. They satisfy

$$\boxed{F = 1 + zA^d F,} \tag{E.1}$$

$$\boxed{A = 1 + zA^d B,} \tag{E.2}$$

$$\boxed{B = 1 + ztA^{d-1}B^2.} \tag{E.3}$$

The variable z marks down-steps and t marks the distinguished correction events.

Define

$$\boxed{Q_d(n,k) = [z^n t^k]F(z,t).}$$

The goal is to derive the closed form for $Q_d(n,k)$ directly from (E.1)–(E.3).

E.2 The auxiliary kernel variable

Set

$$Y = zA^{d-1}B.$$

Equation (E.2) gives

$$A = 1 + AY,$$

so

$$\boxed{A = \frac{1}{1-Y}.} \tag{E.4}$$

Similarly, divide (E.3) by B and use $Y = zA^{d-1}B$:

$$1 = B^{-1} + tY,$$

whence

$$\boxed{B = \frac{1}{1-tY}.} \tag{E.5}$$

Substituting (E.4)–(E.5) into the definition of Y yields

$$Y = \frac{z}{(1-Y)^{d-1}(1-tY)}.$$

Equivalently,

$$\boxed{Y(1-Y)^{d-1}(1-tY) = z.} \tag{E.6}$$

Thus Y is the unique formal power series with zero constant term satisfying

$$\boxed{Y = z\,\Phi(Y), \qquad \Phi(y) = (1-y)^{-(d-1)}(1-ty)^{-1}.} \tag{E.7}$$

This is exactly the form required for Lagrange inversion.

E.3 Coefficient formula for the auxiliary kernel

Define

$$K_d(z,t) = \frac{Y}{z}.$$

Since

$$[z^n t^k]K_d = [z^{n+1}t^k]Y,$$

the basic Lagrange formula applied to (E.7) gives

$$[z^{n+1}t^k]Y = \frac{1}{n+1}[y^n t^k]\Phi(y)^{n+1}.$$

Now

$$\Phi(y)^{n+1} = (1-y)^{-(d-1)(n+1)}(1-ty)^{-(n+1)}.$$

Extracting the t^k coefficient gives

$$[t^k](1-ty)^{-(n+1)} = \binom{n+k}{k} y^k.$$

Therefore

$$[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}[y^{n-k}](1-y)^{-(d-1)(n+1)}.$$

Using

$$[y^r](1-y)^{-a} = \binom{a+r-1}{r},$$

we obtain

$$\boxed{[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}\binom{dn+d-k-2}{n-k}.} \tag{E.8}$$

For $d = 2$ this reduces to

$$\boxed{[z^n t^k]K_2 = \frac{1}{n+1}\binom{n+k}{k}\binom{2n-k}{n-k}.} \tag{E.9}$$

This is the auxiliary kernel triangle identified in the main text. It is not the actual correction distribution.

E.4 Recovering the correction generating function

From (E.1),

$$F = \frac{1}{1 - zA^d}.$$

Using (E.4),

$$\boxed{F = \sum_{m \geq 0} z^m A^{dm} = \sum_{m \geq 0} z^m (1 - Y)^{-dm}.} \tag{E.10}$$

This expansion is the convenient starting point for the exact correction triangle.

For fixed $n \geq 1$,

$$Q_d(n, k) = \sum_{m=0}^{n} [z^{n-m} t^k](1 - Y)^{-dm}. \tag{E.11}$$

The term $m = n$ contains no positive power of Y, so it contributes

$$\boxed{\mathbf{1}_{\{k=0\}}.} \tag{E.12}$$

The term $m = 0$ contributes nothing when $n \geq 1$. It remains to treat $1 \leq m \leq n - 1$.

E.5 Lagrange–Bürmann extraction for a fixed summand

Fix $1 \leq m \leq n - 1$ and set

$$N = n - m > 0.$$

Apply Lagrange–Bürmann to

$$f(y) = (1 - y)^{-dm}.$$

Since

$$f'(y) = dm(1 - y)^{-dm-1},$$

we obtain

$$[z^N](1 - Y)^{-dm} = \frac{dm}{N} [y^{N-1}](1 - y)^{-dm-1} \Phi(y)^N.$$

Substitute (E.7):

$$[z^N](1 - Y)^{-dm} = \frac{dm}{N} [y^{N-1}](1 - y)^{-dm-1-(d-1)N} (1 - ty)^{-N}. \tag{E.13}$$

Because $N = n - m$,

$$dm + 1 + (d - 1)N = (d - 1)n + m + 1.$$

Thus

$$[z^N t^k](1 - Y)^{-dm} = \frac{dm}{N} [y^{N-1} t^k](1 - y)^{-((d-1)n+m+1)} (1 - ty)^{-N}.$$

Now

$$[t^k](1 - ty)^{-N} = \binom{N + k - 1}{k} y^k,$$

so

$$[z^N t^k](1 - Y)^{-dm} = \frac{dm}{N} \binom{N + k - 1}{k} [y^{N-k-1}](1 - y)^{-((d-1)n+m+1)}.$$

A final binomial extraction gives

$$[y^{N-k-1}](1 - y)^{-((d-1)n+m+1)} = \binom{dn - k - 1}{N - k - 1}.$$

Hence

$$\boxed{[z^{n-m}t^k](1-Y)^{-dm} = \frac{dm}{n-m}\binom{n-m+k-1}{k}\binom{dn-k-1}{n-m-k-1}.} \tag{E.14}$$

The coefficient is automatically zero unless

$$n - m \geq k + 1.$$

E.6 Exact correction triangle

Insert (E.12) and (E.14) into (E.11). Make the change of variable

$$j = n - m.$$

Then

$$m = n - j,$$

and the nonzero range is

$$k + 1 \leq j \leq n - 1.$$

Therefore

$$\boxed{Q_d(n,k) = \mathbf{1}_{\{k=0\}} + \sum_{j=k+1}^{n-1} \frac{d(n-j)}{j}\binom{j+k-1}{k}\binom{dn-k-1}{j-k-1}.} \tag{E.15}$$

This is the formula stated in Chapter 27.

The upper support bound follows immediately. If $k > n - 2$, then there is no integer j with

$$k + 1 \leq j \leq n - 1,$$

and the indicator term vanishes because $k > 0$. Thus

$$\boxed{Q_d(n,k) = 0 \qquad (k > n - 2).} \tag{E.16}$$

E.7 Equivalent elimination through $U = A - 1$

For completeness, the scalar equation quoted in the main chapter follows directly from the same grammar. Put

$$U = A - 1.$$

From (E.2),

$$U = zA^dB = AY.$$

Since $A = 1 + U$, this gives

$$Y = \frac{U}{1+U}.$$

Equation (E.3) can be written as

$$B = 1 + tYB,$$

so

$$B = \frac{1}{1-tY} = \frac{1+U}{1+(1-t)U}.$$

Substituting into $U = zA^dB$ with $A = 1 + U$ gives the convenient form

$$\boxed{U\big(1 + (1-t)U\big) = z(1+U)^{d+1}.} \tag{E.17}$$

Equation (E.17) and the kernel equation (E.6) are related by

$$\boxed{Y = \frac{U}{1+U}.}$$

Thus the two inversion routes are exactly the same formal system in different coordinates.

E.8 Row sum

Setting $t = 1$ forgets the correction marking. Equation (E.17) becomes

$$U = z(1+U)^{d+1}.$$

At $t = 1$, the correction generating function simplifies to

$$F(z, 1) = 1 + U.$$

Lagrange inversion gives, for $n \geq 1$,

$$[z^n]U = \frac{1}{n}[u^{n-1}](1+u)^{(d+1)n} = \frac{1}{n}\binom{(d+1)n}{n-1}.$$

Using

$$\frac{1}{n}\binom{(d+1)n}{n-1} = \frac{1}{dn+1}\binom{(d+1)n}{n},$$

we recover the Fuss–Catalan row sum

$$\boxed{\sum_{k\geq 0} Q_d(n,k) = \frac{1}{dn+1}\binom{(d+1)n}{n}.} \tag{E.18}$$

E.9 Why the kernel and correction arrays differ

The kernel coefficients (E.8) arise from the single implicit series Y and record the algebraic kernel introduced to solve the periodic grammar. The correction coefficients (E.15) arise only after the full zero-baseline excursion series

$$F = \sum_{m\geq 0} z^m (1-Y)^{-dm}$$

is reconstructed.

Therefore

$$\boxed{K_d \text{ is an algebraic waypoint, whereas } Q_d \text{ is the path statistic.}}$$

The distinction is not terminological: the two coefficient arrays have different combinatorial meanings and diverge as soon as the correction marking becomes nontrivial.

Appendix F

General-Theory Stress-Test Catalogues

F.1 Purpose and scope

The abstract General Theory developed in the General Theory macro-Part is not defined by the weighted-balance family. Its semantic system may be non-additive and need not have a residual space isomorphic to $\mathbb{Z}$. The centered $p{:}q$ constructions are instead a deliberately rigid stress-test family in which safe-reset domains, source-fiber expansion, directed exposure, and robust fatal supports can all be calculated exactly.

This appendix records the full finite 3:2 certificate and the infinite odd-p:2 arithmetic/source catalogues used in the stress-test theorems. The purpose is reproducibility: the main chapters can state the structural consequences without interrupting the narrative with long case catalogues.

Throughout this appendix, for a canonical literal-demand pair (x, y) we use

$$\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}$$

For the weighted-balance residual, the primitive nonnegative kernel vector is $K = (q, p)$.

F.2 The exact centered 3:2 certificate

Set

$$p = 3, \qquad q = 2, \qquad m = p + q = 5,$$

and use the centered states

$$D_{-2}, D_{-1}, \mathsf{C}, D_1, D_2.$$

The six natural eager sites are indexed by the current centered state, the next input symbol, and the visible top symbol. Their eager replacement words are

site	(d, a, X)	γ
A	$(-2, 0, 0)$	11111
B	$(-2, 1, 0)$	10111
C	$(-2, 1, 1)$	11111
D	$(1, 0, 0)$	10100
E	$(1, 1, 0)$	10000
F	$(2, 0, 0)$	10000

(F.1)

F.2.1 Finite reset repertoires

Define

$$\boxed{\begin{aligned}\mathcal{R}_0^{(3,2)} = {} & \{(2,2)\} \cup \{(0,y) : y \geq 4\} \cup \{(1,y) : y \geq 3\} \\ & \cup \{(x,1) : x \geq 3\} \cup \{(x,0) : x \geq 4\}\end{aligned}} \tag{F.2}$$

and

$$\boxed{\mathcal{R}_D^{(3,2)} = \mathcal{R}_0^{(3,2)} \cup \{(x,2) : x \geq 3\}.} \tag{F.3}$$

Before the first reset event of an episode begun from one of these canonical pairs, the relevant eager-source shapes are

local context	$\mathcal{R}_0^{(3,2)}$	$\mathcal{R}_D^{(3,2)}$
D_{-2}, top 0	0	0^k
D_{-2}, top 1	1^k	1^k
D_1, top 0	0^k	0^k
D_2, top 0	0^k	0^k

$(k \geq 1)$. (F.4)

The bottom-WRITE targets are

d	0	1
-2	$(1,8)$	$(0,4)$
-1	$(0,4)$	$(1,3)$
0	$(1,3)$	$(2,2)$
1	$(2,2)$	$(3,1)$
2	$(3,1)$	$(4,0)$

(F.5)

and every one lies in $\mathcal{R}_0^{(3,2)}$.

Writing the unary source tail as $t \geq 0$, the remaining reset targets are

reset action	canonical target pair
positive boundary WRITE	$(t+5, 0)$
negative boundary WRITE	$(1, t+9)$
A	$(0,5)$
B	$(1,4)$
C	$(0, t+5)$
D	$(t+3, 2)$
E	$(t+4, 1)$
F	$(t+4, 1)$

(F.6)

These tables give two closed reset regimes:

(a) with D disabled, every baseline reset and every enabled eager action among A, B, C, E, F returns to $\mathcal{R}_0^{(3,2)}$;

(b) with D enabled and A, B disabled, every baseline reset and every enabled eager action among C, D, E, F returns to $\mathcal{R}_D^{(3,2)}$.

The only new reset ray in the second regime is

$$\{(x,2) : x \geq 3\},$$

and this is exactly the ray that enlarges the D_{-2}-top-0 source from the singleton stack 0 to the unary family 0^k.

F.2.2 Direct verification of reset closure

The canonical words in $\mathcal{R}_0^{(3,2)}$ are respectively

$$1010, \qquad 1^y, \qquad 10\,1^{y-1}, \qquad 1\,0^x, \qquad 0^x,$$

while the added D-ray has

$$\eta(x,2) = 101\,0^{x-1}.$$

Before the first reset, MATCH and interior SEARCH only remove the visible top cell, so the current stack is always a suffix of the starting canonical word. The source catalogue in (F.4) follows immediately. In particular, without the D-ray only the word 1010 can reach D_{-2} while a zero remains visible, and exactly its final zero remains.

For the non-bottom reset events, the centered boundary blocks and the six eager replacement words yield the canonical words

$$0^{t+5}, \quad 10\,1^{t+8}, \quad 1^5, \quad 10\,1^3, \quad 1^{t+5}, \quad 101\,0^{t+2}, \quad 1\,0^{t+4}.$$

These are precisely the encodings listed in (F.6). Every row except D lies in $\mathcal{R}_0^{(3,2)}$; the D row is precisely the additional ray of $\mathcal{R}_D^{(3,2)}$.

F.2.3 Exact schedule classification and fatal witnesses

The admissible schedules are exactly

$$\boxed{\mathrm{Adm}_{3,2} = \{H : \{A,D\} \not\subseteq H,\ \{B,D\} \not\subseteq H\}\,.} \tag{F.7}$$

Therefore

$$\boxed{\text{40 schedules are admissible and 24 are inadmissible.}} \tag{F.8}$$

The minimal directed fatal fork is

$$\boxed{D \to A, \qquad D \to B.} \tag{F.9}$$

The maximal admissible faces are

$$\{A,B,C,E,F\}, \qquad \{C,D,E,F\},$$

so the admissible complex is non-pure and the schedule poset is not a lattice. Its independence polynomial is

$$\boxed{(1+x)^3(1+3x+x^2).} \tag{F.10}$$

Sufficiency follows from the two reset-closed regimes above: the $D = 0$ regime contributes $2^5 = 32$ schedules, while $D = 1$ with $A = B = 0$ contributes $2^3 = 8$ more.

Necessity is certified by two complete balanced words,

$$\boxed{w_B = 1111000011}, \qquad \boxed{w_A = 011111110000011}. \tag{F.11}$$

Their critical portions are

branch	reaching prefix	victim step	diagnostic
$D \to B$	$11110000 : (D_{-2}, 00)$	$1 : (\mathsf{C}, 101110)$	$1 : (\mathsf{C}, 01110)$
$D \to A$	$011111110000 : (D_{-2}, 000)$	$0 : (\mathsf{C}, 1111100)$	$11 : (\mathsf{C}, 11100)$

(F.12)

In the first row, the earlier eager D step occurs on prefix 11110; in the second it occurs on prefix 011111110. Before the victim step the only eager optional site encountered is therefore D. After the victim step the listed diagnostic uses only baseline center transitions. Both terminal configurations have semantic residual zero but a nonempty center stack, so they are nonaccepting under the fixed centered acceptance convention. The witnesses are therefore extension-stable under arbitrary choices at C, E, F and at the other victim site.

F.3 The infinite odd-p:2 family

Fix

$$\boxed{p = 2h - 1, \qquad q = 2, \qquad h \geq 2,} \tag{F.13}$$

so that

$$m = p + q = 2h + 1.$$

For $t \geq 1$ set

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + p\, k_t,$$

and define

$$\boxed{\Gamma_t = \eta(\epsilon_t, \beta_t).} \tag{F.14}$$

The six eager sites are

$$\begin{aligned}
&A : (D_{-h}, 0T) \xrightarrow{0} (\mathsf{C}, \Gamma_h T),\\
&B : (D_{-h}, 0T) \xrightarrow{1} (\mathsf{C}, \Gamma_{h-1} T),\\
&C : (D_{-h}, 1T) \xrightarrow{1} (\mathsf{C}, \Gamma_h T),\\
&D : (D_{h-1}, 0T) \xrightarrow{0} (\mathsf{C}, (10)^h 0T),\\
&E : (D_{h-1}, 0T) \xrightarrow{1} (\mathsf{C}, (10)^{h-1} 000T),\\
&F : (D_h, 0T) \xrightarrow{0} (\mathsf{C}, (10)^{h-1} 000T).
\end{aligned} \tag{F.15}$$

F.3.1 Arithmetic catalogue

For the minimum-demand map μ the exact identities are:

(i) for every $t \geq 1$,

$$\boxed{\mu(-tm) = (\epsilon_t, \beta_t);} \tag{F.16}$$

(ii) for $0 \leq j \leq h$,

$$\boxed{\mu(jm + 2) = (j + 2, p - j - 1);} \tag{F.17}$$

(iii) for $1 \leq t \leq h + 1$,

$$\boxed{\mu(-tm + 2) = (\epsilon_t, \beta_t - 1);} \tag{F.18}$$

(iv) at the positive outer boundary,

$$\boxed{\mu((h + 1)m) = (h + 3, h - 2).} \tag{F.19}$$

For completeness, if $t = 2r$ then $(0, \beta_t)$ has weight

$$-2(2r + pr) = -2r(p + 2) = -tm,$$

while if $t = 2r + 1$ then $(1, \beta_t)$ has weight

$$p - 2(2r + 1 + p(r + 1)) = -(2r + 1)(p + 2) = -tm.$$

In both cases the first coordinate is below the kernel step $q = 2$, so the pair is kernel-reduced. Equation (F.17) follows from

$$p(j + 2) - 2(p - j - 1) = j(p + 2) + 2 = jm + 2,$$

with second coordinate below p. Equation (F.18) is obtained by lowering the second coordinate in (F.16) by one, and (F.19) follows from

$$p(h+3) - 2(h-2) = (h+1)(2h+1).$$

The complete family of canonical baseline reset pairs is therefore

$$\boxed{\begin{aligned}\mathcal{R}_h^{\mathrm{base}} = {} & \{\mu(jm+2) : -h-1 \le j \le h\} \\ & \cup \{(x, h-2) : x \ge h+3\} \\ & \cup \{(\epsilon_{h+1}, y) : y \ge \beta_{h+1}\}.\end{aligned}} \tag{F.20}$$

The two schedule-regime repertoires are

$$\boxed{\begin{aligned}\mathcal{R}_{\neg D,h} = {} & \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ & \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \ge h+2\},\end{aligned}} \tag{F.21}$$

and

$$\boxed{\begin{aligned}\mathcal{R}_{D,h} = {} & \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ & \cup \{(x, h-1) : x \ge h+2\} \cup \{(x, h) : x \ge h+1\}.\end{aligned}} \tag{F.22}$$

Every pair in either repertoire has minority size at most h.

F.3.2 Source-geometry catalogue

Consider an episode beginning at a canonical center pair in one of the two repertoires and stopping at its first reset event. The exact source geometry is:

(i) in $\mathcal{R}_{\neg D,h}$,

$$D_{-h} \text{ with top } 0 \Longrightarrow \boxed{\text{complete remaining stack } 0}; \tag{F.23}$$

(ii) in $\mathcal{R}_{D,h}$,

$$D_{-h} \text{ with top } 0 \Longrightarrow \boxed{\text{complete remaining stack } 0^n,\ n \ge 1}, \tag{F.24}$$

and lengths $n \ge 2$ arise only from the additional D-ray $\{(x, h) : x \ge h+1\}$;

(iii) in either repertoire,

$$\begin{aligned} D_{-h} \text{ with top } 1 &\Longrightarrow 1^n, \\ D_{h-1} \text{ with top } 0 &\Longrightarrow 0^n, \\ D_h \text{ with top } 0 &\Longrightarrow 0^n \end{aligned} \tag{F.25}$$

for some $n \ge 1$.

The key observation is that, before the first reset event, MATCH and interior SEARCH only delete the visible top cell, so the current stack is a suffix of the canonical reset word from which the episode began. Reaching D_{-h} requires at least h more negative than positive SEARCHes. In $\mathcal{R}_{\neg D,h}$ the only canonical pair with enough of both symbols to leave a zero visible at that boundary is (h, h), whose word is $(10)^h$; all h one-cells must have been negative SEARCHes, leaving exactly its final zero. The only new family in $\mathcal{R}_{D,h}$ that can leave a longer zero suffix is (x, h), whose word is $(10)^h 0^{x-h}$.

The remaining unary-source conclusions follow from the same minority-size bound. A suffix containing the opposite symbol beyond the visible boundary cell would force minority size at least $h+1$, contradicting the defining families in (F.21)–(F.22).

F.3.3 Reset-target catalogue and closure

Writing the unary source stack length as $n \geq 1$, all eager reset targets are

site	source stack	canonical target pair
A	0	(ϵ_h, β_h)
B	0	$(\epsilon_{h-1}, \beta_{h-1})$
C	1^n	$(\epsilon_h, \beta_h + n - 1)$
D	0^n	$(n + h, h)$
E	0^n	$(n + h + 1, h - 1)$
F	0^n	$(n + h + 1, h - 1)$

(F.26)

The identities are literal: for instance

$$(10)^h 0\, 0^{n-1} = \eta(n + h, h)$$

and

$$(10)^{h-1} 000\, 0^{n-1} = \eta(n + h + 1, h - 1).$$

Consequently:

(a) if D is disabled, baseline resets and enabled actions among A, B, C, E, F preserve $\mathcal{R}_{\neg D,h}$;

(b) if D is enabled and A, B are disabled, baseline resets and enabled actions among C, D, E, F preserve $\mathcal{R}_{D,h}$.

F.3.4 Uniform D-induced source-fiber expansion

For every $x \geq h + 1$,

$$\boxed{(\mathsf{C}, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}).} \tag{F.27}$$

Indeed,

$$\eta(x, h) = (10)^h 0^{x-h}.$$

The first $2h - 2$ zeros remove $h - 1$ complete 10 pairs, and the final zero SEARCHes through the last 1, leaving the last alternating zero together with the zero tail. Thus enabling D expands the common A/B source fiber from

$$\boxed{\{0\}}$$

to

$$\boxed{\{0, 00, 000, \ldots\}}. \tag{F.28}$$

F.3.5 Uniform reachability of the length-three frontier

There is, for every $h \geq 2$, a word u_h that reaches

$$\boxed{(D_{-h}, 000)} \tag{F.29}$$

from the initial centered configuration and encounters no eager optional site other than D.

Define

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0\, 1^{\,2h+(h-2)m/2}, & h \text{ even}. \end{cases} \tag{F.30}$$

Then

$$\boxed{u_h = v_h\, 1^{\,2h-1}\, 0^{\,2h}.} \tag{F.31}$$

The useful increment block is

$$(D_d, \varepsilon) \xrightarrow{1^m} (D_{d+2}, \varepsilon), \qquad 0 \le d \le h-2. \tag{F.32}$$

The first symbol bottom-WRITEs

$$\mu(dm+2) = (d+2, p-d-1),$$

whose canonical word has length $p+1 = 2h$; the remaining $2h$ ones erase the word, and its $d+2$ zeros contribute the required positive SEARCHes. This builds v_h so that the machine reaches (D_{h-1}, ε). The next 1^{2h-1} segment leaves $(D_{h-1}, 00)$, the following zero invokes D and produces $(\mathsf{C}, (10)^h 00)$, and the remaining 0^{2h-1} reaches $(D_{-h}, 000)$. No site among A, B, C, E, F is encountered before that endpoint.

F.3.6 Uniform kernel-suffix fatality

For every $t \ge 1$, Γ_t has suffix 1^p. Hence there is a unique word z_t such that

$$\boxed{\Gamma_t = z_t 1^p.} \tag{F.33}$$

If $t = 2r$ then

$$\Gamma_t = 1^{\beta_t}, \qquad \beta_t = rm \ge m > p,$$

while if $t = 2r+1$ then

$$\Gamma_t = 10\, 1^{\beta_t - 1}, \qquad \beta_t - 1 = p + rm \ge p.$$

Thus the suffix claim is uniform in t.

At the common source $(D_{-h}, 000)$, eager A and B yield

$$A : (D_{-h}, 000) \xrightarrow{0} (\mathsf{C}, \Gamma_h 00) = (\mathsf{C}, z_h 1^p 00),$$

$$B : (D_{-h}, 000) \xrightarrow{1} (\mathsf{C}, \Gamma_{h-1} 00) = (\mathsf{C}, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} from the center uses only MATCHes and reaches

$$\boxed{(\mathsf{C}, 1^p 00).} \tag{F.34}$$

The stack has two zeros and p ones, hence weight

$$p \cdot 2 - 2 \cdot p = 0,$$

but is nonempty. Under centered acceptance it is therefore a site-free fatal diagnostic.

F.3.7 Exact odd-p:2 classification

For every odd $p \ge 3$,

$$\boxed{\mathrm{Adm}_{p,2} = \mathrm{Ind}\big\{\{A, D\}, \{B, D\}\big\}.} \tag{F.35}$$

Equivalently,

$$\boxed{H \in \mathrm{Adm}_{p,2} \iff \neg(A \wedge D) \wedge \neg(B \wedge D).} \tag{F.36}$$

Every member of the family therefore has exactly 40 admissible and 24 inadmissible schedules, with facets

$$\{A, B, C, E, F\}, \qquad \{C, D, E, F\},$$

and the same fatal fork

$$\boxed{D \to A, \qquad D \to B.} \tag{F.37}$$

Sufficiency is the reset-closure induction: $D = 0$ gives $2^5 = 32$ schedules, and $D = 1$ with $A = B = 0$ gives another $2^3 = 8$. Necessity uses the common reaching prefix u_h from (F.31) together with the complete counterexample words

$$\boxed{w_A^{(h)} = u_h\, 0\, z_h,} \qquad \boxed{w_B^{(h)} = u_h\, 1\, z_{h-1}.} \tag{F.38}$$

The prefix u_h encounters no optional eager site except D, and the post-victim suffixes are site-free. The two forbidden supports are therefore robust under arbitrary additions of C, E, F and of the other victim site.

F.4 Interpretive boundary

The finite 3:2 calculation and the infinite odd-p:2 family have a single role in the monograph: they certify that the abstract mechanisms of the General Theory macro-Part are not artifacts of one numerical instance. They exhibit, in exact families,

$$\begin{array}{c}\text{safe reset} \longrightarrow \text{fiber expansion} \\ \longrightarrow \text{directed exposure} \longrightarrow \text{robust fatal support.}\end{array}$$

They do *not* define the General Theory. The General Theory applies to a rooted exact physical lift over a deterministic partial semantic system. Its settled macro-updates reflect defined domains exactly and satisfy

$$\Theta\delta_a = U_a\Theta$$

on that common domain, with no requirement that the semantic system be a weighted-balance residual space or even an additive group.

Bibliography

Bibliography

[1] A. Adelberg and T. Lengyel, New results on the p-adic valuation of Stirling numbers, arXiv:2111.08766 [math.NT], 2021.

[2] R. Alur, V. Kumar, P. Madhusudan, and M. Viswanathan, Congruences for visibly pushdown languages, in *ICALP 2005*, Lecture Notes in Computer Science, vol. 3580, Springer, 2005, pp. 1102–1114. doi:10.1007/11523468_89.

[3] J. Autry, P. Graves, J. Loucks, C. O'Neill, V. Ponomarenko, and S. Yih. Squarefree divisor complexes of certain numerical semigroup elements. *Involve*, 14(1):1–9, 2021.

[4] A. Bouajjani, J. Esparza, and O. Maler, Reachability analysis of pushdown automata: Application to model-checking, in *CONCUR 1997: Concurrency Theory*, Lecture Notes in Computer Science, vol. 1243, Springer, 1997, pp. 135–150. doi:10.1007/3-540-63141-0_10.

[5] A. Carayol and O. Serre. Marking shortest paths on pushdown graphs does not preserve MSO decidability. *Information Processing Letters*, 116(10):638–643, 2016. doi:10.1016/j.ipl.2016.04.015.

[6] D. Chistikov, W. Czerwiński, P. Hofman, M. Pilipczuk, and M. Wehar. Shortest paths in one-counter systems. *Logical Methods in Computer Science*, 15(1):19:1–19:28, 2019. doi:10.23638/LMCS-15(1:19)2019.

[7] P. Diaconis and J. Fulman, Carries, shuffling, and symmetric functions, arXiv:0902.0179 [math.PR], 2009.

[8] O. Gauwin, A. Muscholl, and M. Raskin. Minimization of visibly pushdown automata is NP-complete. *Logical Methods in Computer Science*, 16(1):14, 2020. doi:10.23638/LMCS-16(1:14)2020.

[9] J. Goldstine, M. Kappes, C. M. Kintala, H. Leung, A. Malcher, and D. Wotschke. Descriptional complexity of machines with limited resources. *Journal of Universal Computer Science*, 8(2):193–234, 2002. DOI: 10.3217/jucs-008-02-0193.

[10] J. Goldstine, J. K. Price, and D. Wotschke. On reducing the number of stack symbols in a PDA. *Mathematical Systems Theory*, 26:313–326, 1993. DOI: 10.1007/BF01189852.

[11] A. Hatcher. *Algebraic Topology*. Cambridge University Press, 2002.

[12] S. Hong and M. Qiu, On the p-adic properties of Stirling numbers of the first kind, *Acta Mathematica Hungarica* 161 (2020), 366–395.

[13] P. Jančar. A short decidability proof for DPDA language equivalence via first-order grammars. CoRR abs/1010.4760, 2011.

[14] D. Kozlov. *Combinatorial Algebraic Topology*. Algorithms and Computation in Mathematics, vol. 21, Springer, 2008.

[15] O. Martynova. Exact descriptional complexity of determinization of input-driven pushdown automata. In *Implementation and Application of Automata (CIAA 2024)*, pp. 249–260, 2024. arXiv:2404.10516.

[16] T. Masopust. A note on limited pushdown alphabets in stateless deterministic pushdown automata. *International Journal of Foundations of Computer Science*, 24(3):319–328, 2013. DOI: 10.1142/S0129054113500068; arXiv:1208.5002.

[17] A. Malcher, K. Meckel, C. Mereghetti, and B. Palano. Descriptional complexity of pushdown store languages. *Journal of Automata, Languages and Combinatorics*, 17(2–4):225–244, 2012. DOI: 10.25596/jalc-2012-225.

[18] D. A. Moj, Biorthogonal eigenvectors of the Holte carry matrix and cascade-free enumeration, arXiv:2604.04591 [math.CO], 2026.

[19] The OEIS Foundation, The On-Line Encyclopedia of Integer Sequences, A371395, triangle $T(n,k) = \binom{n+k}{k}\binom{2n-k}{n-k}/(n+1)$, accessed August 2026.

[20] G. Pighizzini. Deterministic pushdown automata and unary languages. *International Journal of Foundations of Computer Science*, 20(4):629–645, 2009. doi:10.1142/S0129054109006784.

[21] G. Sénizergues. $L(A) = L(B)$? Decidability results from complete formal systems. *Theoretical Computer Science*, 251(1–2):1–166, 2001. doi:10.1016/S0304-3975(00)00285-1.

[22] D. Armstrong, N. A. Loehr, and G. S. Warrington, Rational parking functions and Catalan numbers, *Annals of Combinatorics* 20 (2016), 21–58. doi:10.1007/s00026-015-0293-6.

[23] L. Dai, S. Fu, and D. Qiu, Symmetric statistics on rational Dyck paths, *Advances in Applied Mathematics* 177 (2026), 103066. doi:10.1016/j.aam.2026.103066; arXiv:2503.20155.

[24] I. Gessel and G. Viennot, Binomial determinants, paths, and hook length formulae, *Advances in Mathematics* 58 (1985), no. 3, 300–321. doi:10.1016/0001-8708(85)90121-5.

[25] G. Kreweras, Sur une classe de problèmes de dénombrement liés au treillis des partitions des entiers, *Cahiers du Bureau universitaire de recherche opérationnelle, Série Recherche* 6 (1965), 9–107.

[26] R. Pemantle and H. S. Wilf, Counting nondecreasing integer sequences that lie below a barrier, *Electronic Journal of Combinatorics* 16 (2009), no. 1, Research Paper R60. doi:10.37236/149.

[27] J. Berstel, A. Lauve, C. Reutenauer, and F. V. Saliola, *Combinatorics on Words: Christoffel Words and Repetitions in Words*, CRM Monograph Series, vol. 27, American Mathematical Society, 2009.

[28] V. Berthé, A. de Luca, and C. Reutenauer, On an involution of Christoffel words and Sturmian morphisms, *European Journal of Combinatorics* 29 (2008), no. 2, 535–553. doi:10.1016/j.ejc.2007.03.001.

[29] F. de Vieilleville and J.-O. Lachaud, Revisiting digital straight segment recognition, in *Discrete Geometry for Computer Imagery (DGCI 2006)*, LNCS 4245, Springer, 2006, pp. 355–366. doi:10.1007/11907350_30.

[30] R. Graham and K. O'Bryant, A discrete Fourier kernel and Fraenkel's tiling conjecture, *Acta Arithmetica* 118 (2005), no. 3, 283–304. doi:10.4064/aa118-3-4.

[31] H. Rademacher and E. Grosswald, *Dedekind Sums*, Carus Mathematical Monographs, vol. 16, Mathematical Association of America, 1972.

[32] R. Klette and A. Rosenfeld, Digital straightness—a review, *Discrete Applied Mathematics* 139 (2004), no. 1–3, 197–230. doi:10.1016/j.dam.2002.12.001.

[33] J.-O. Lachaud and M. Said, Two efficient algorithms for computing the characteristics of a subsegment of a digital straight line, *Discrete Applied Mathematics* 161 (2013), no. 15, 2293–2315. doi:10.1016/j.dam.2012.08.038.

[34] L. Takács, On generalized Dedekind sums, *Journal of Number Theory* 11 (1979), no. 2, 264–272. doi:10.1016/0022-314X(79)90044-1.

[35] L. Takács. On the fluctuations of election returns. *Journal of Applied Probability*, 7(1):114–123, 1970. doi:10.2307/3212153.

[36] F. B. Knight. The uniform law for exchangeable and Lévy process bridges. In *Hommage à P. A. Meyer et J. Neveu*, *Astérisque* 236, pages 171–188, 1996. doi:10.24033/ast.337.

[37] P. J. Fitzsimmons and R. K. Getoor. Occupation time distributions for Lévy bridges and excursions. *Stochastic Processes and their Applications*, 58(1):73–89, 1995. doi:10.1016/0304-4149(95)00013-W.

[38] C. Banderier and M. Wallner. Local time for lattice paths and the associated limit laws. In *Proceedings of GASCom 2018*, pages 69–78, 2018. arXiv:1805.09065.

[39] G. Latouche and V. Ramaswami. *Introduction to Matrix Analytic Methods in Stochastic Modeling*. ASA-SIAM Series on Statistics and Applied Probability, SIAM, Philadelphia, 1999.

[40] P. Flajolet and R. Sedgewick. *Analytic Combinatorics*. Cambridge University Press, Cambridge, 2009.

[41] N. Alon, G. Kalai, J. Matoušek, and R. Meshulam, Transversal numbers for hypergraphs arising in geometry, *Advances in Applied Mathematics* **29** (2002), no. 1, 79–101. DOI: 10.1016/S0196-8858(02)00003-9.

[42] I. Peeva and B. Sturmfels, Syzygies of codimension 2 lattice ideals, *Mathematische Zeitschrift* **229** (1998), no. 1, 163–194. DOI: 10.1007/PL00004645.

[43] R. G. Swan, Projective modules over Laurent polynomial rings, *Transactions of the American Mathematical Society* **237** (1978), 111–120. DOI: 10.1090/S0002-9947-1978-0469906-4.

[44] D. Bayer and B. Sturmfels, Cellular resolutions of monomial modules, *Journal für die reine und angewandte Mathematik* **502** (1998), 123–140.

[45] T. Fernós, D. Futer, and M. Hagen, Homotopy equivalent boundaries of cube complexes, *Geometriae Dedicata* **218** (2024), article 33. DOI: 10.1007/s10711-023-00877-w.

[46] J. R. Norris. *Markov Chains*. Cambridge University Press, Cambridge, 1997.

[47] T. Tao and V. Vu, *Additive Combinatorics*, Cambridge Studies in Advanced Mathematics, vol. 105, Cambridge University Press, Cambridge, 2006.

[48] R. P. Stanley, *Enumerative Combinatorics, Vol. 1*, 2nd ed., Cambridge University Press, 2012.

[49] L. E. Dickson, "Finiteness of the odd perfect and primitive abundant numbers with n distinct prime factors," *American Journal of Mathematics* 35 (1913), 413–422.

[50] A. Asinowski, A. Bacher, C. Banderier, and B. Gittenberger, *Analytic Combinatorics of Lattice Paths with Forbidden Patterns, the Vectorial Kernel Method, and Generating Functions for Pushdown Automata*, Algorithmica 82 (2020), 386–428. DOI: 10.1007/s00453-019-00623-3.

[51] D. Armstrong, B. Rhoades, and N. Williams, *Rational Associahedra and Noncrossing Partitions*, Electronic Journal of Combinatorics 20(3) (2013), P54. DOI: 10.37236/3432.

[52] C. Banderier and P. Flajolet, *Basic analytic combinatorics of directed lattice paths*, Theoretical Computer Science 281 (2002), 37–80. DOI: 10.1016/S0304-3975(02)00007-5.

[53] D. Birmajer, J. B. Gil, and M. D. Weiner, *On rational Dyck paths and the enumeration of factor-free Dyck words*, Discrete Applied Mathematics 244 (2018), 36–43. DOI: 10.1016/j.dam.2018.02.020.

[54] M. T. L. Bizley, *Derivation of a new formula for the number of minimal lattice paths from* $(0,0)$ *to* (km,kn) *having just* t *contacts with the line* $my = nx$ *and having no points above this line; and a proof of Grossman's formula for the number of paths which may touch but do not rise above this line*, Journal of the Institute of Actuaries 80(1) (1954), 55–62. DOI: 10.1017/S002026810005424X.

[55] J. E. Bonin, A. de Mier, and M. Noy, *Lattice path matroids: enumerative aspects and Tutte polynomials*, Journal of Combinatorial Theory, Series A 104(1) (2003), 63–94. DOI: 10.1016/S0097-3165(03)00122-5.

[56] P. Brändén and J. Huh, *Lorentzian polynomials*, Annals of Mathematics 192(3) (2020), 821–891. DOI: 10.4007/annals.2020.192.3.4.

[57] P. Duchon, *On the enumeration and generation of generalized Dyck words*, Discrete Mathematics 225 (2000), 121–135. DOI: 10.1016/S0012-365X(00)00150-3.

[58] M. Latteux, *Langages à un compteur*, Journal of Computer and System Sciences 26(1) (1983), 14–33. DOI: 10.1016/0022-0000(83)90018-1.

[59] G. N. Raney, *Functional composition patterns and power series reversion*, Transactions of the American Mathematical Society 94 (1960), 441–451.

[60] B. Lindström, *On the Vector Representations of Induced Matroids*, Bulletin of the London Mathematical Society 5(1) (1973), 85–90. DOI: 10.1112/blms/5.1.85.

[61] F. Baader and T. Nipkow, *Term Rewriting and All That*, Cambridge University Press, 1998.

[62] M. H. A. Newman, "On theories with a combinatorial definition of equivalence," *Annals of Mathematics* 43 (1942), 223–243.

[63] D. Quillen, “Homotopy properties of the poset of nontrivial p-subgroups of a group,” *Advances in Mathematics* 28 (1978), 101–128.

[64] G.-C. Rota, “On the foundations of combinatorial theory I: Theory of Möbius functions,” *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 2 (1964), 340–368.

[65] T. P. Speed, “Cumulants and partition lattices,” *Australian Journal of Statistics* 25 (1983), 378–388.

[66] J. Coussement and W. Van Assche, “Gaussian quadrature for multiple orthogonal polynomials,” *Journal of Computational and Applied Mathematics* 178 (2005), 131–145.

[67] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics: A Foundation for Computer Science*, 2nd ed., Addison–Wesley, 1994.

[68] R. J. Parikh, “On context-free languages,” *Journal of the ACM* 13(4) (1966), 570–581. DOI: 10.1145/321356.321364.

[69] S. Ginsburg and E. H. Spanier, “Semigroups, Presburger formulas, and languages,” *Pacific Journal of Mathematics* 16(2) (1966), 285–296. DOI: 10.2140/pjm.1966.16.285.

[70] F. Höfting and E. Wanke, “Minimum cost paths in periodic graphs,” *SIAM Journal on Computing* 24(5) (1995), 1051–1067. DOI: 10.1137/S0097539792234378.

[71] W. Bruns and J. Herzog, “Semigroup rings and simplicial complexes,” *Journal of Pure and Applied Algebra* 122(3) (1997), 185–208. DOI: 10.1016/S0022-4049(97)00051-0.

[72] R. Ghrist and V. Peterson, “The geometry and topology of reconfiguration,” *Advances in Applied Mathematics* 38(3) (2007), 302–323. DOI: 10.1016/j.aam.2005.08.009.

[73] A. Botea, “Fast, optimal pathfinding with compressed path databases,” in *Proceedings of the Fifth Annual Symposium on Combinatorial Search*, vol. 3, no. 1, 2012, pp. 204–205. DOI: 10.1609/socs.v3i1.18251.

[74] S. Bhattacharya, “Towards optimal path computation in a simplicial complex,” *The International Journal of Robotics Research* 38(8) (2019), 981–1009. DOI: 10.1177/0278364919855422.

[75] T. Bogart, J. Goodrick, and K. Woods, “Periodic behavior in families of numerical and affine semigroups via parametric Presburger arithmetic,” *Semigroup Forum* 102(2) (2021), 340–356. DOI: 10.1007/s00233-021-10164-3.

[76] B. Shen, M. A. Cheema, D. D. Harabor, and P. J. Stuckey, “Contracting and compressing shortest path databases,” in *Proceedings of the Thirty-First International Conference on Automated Planning and Scheduling*, 2021, pp. 322–330.

[77] A. Stjerna and P. Rümmer, “A constraint solving approach to Parikh images of regular languages,” *Proceedings of the ACM on Programming Languages* 8(OOPSLA1) (2024), 1235–1263. DOI: 10.1145/3649855.

[78] C. Haase, S. N. Krishna, K. Madnani, O. S. Mishra, and G. Zetzsche, "An efficient quantifier elimination procedure for Presburger arithmetic," in *51st International Colloquium on Automata, Languages, and Programming (ICALP 2024)*, LIPIcs 297, Article 142, 2024, pp. 142:1–142:17. DOI: 10.4230/LIPIcs.ICALP.2024.142.

[79] M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture: A Mathematical Journey from Irrational Numbers to Perfect Matchings*, Springer, Cham, 2013. doi:10.1007/978-3-319-00888-2.

[80] H. Cohn, "Approach to Markoff's minimal forms through modular functions," *Annals of Mathematics* 61 (1955), 1–12. doi:10.2307/1969618.

[81] H. Cohn, "Markoff forms and primitive words," *Mathematische Annalen* 196 (1972), 8–22.

[82] C. Lagisquet, E. Pelantová, S. Tavenas and L. Vuillon, "On the Markov numbers: fixed numerator, denominator, and sum conjectures," *Advances in Applied Mathematics* 130 (2021), 102227. doi:10.1016/j.aam.2021.102227.

[83] C. Reutenauer, *From Christoffel Words to Markoff Numbers*, 2nd ed., Oxford University Press, Oxford, 2026. doi:10.1093/9780197907535.001.0001.

[84] A. P. Veselov, "Markov fractions and Cohn matrices," arXiv:2604.17401, 2026. doi:10.48550/arXiv.2604.17401.

[85] A. Meduna, L. Vrábel, and P. Zemek, "An Infinite Hierarchy of Language Families Resulting from Stateless Pushdown Automata with Limited Pushdown Alphabets," in *Descriptional Complexity of Formal Systems (DCFS 2012)*, Lecture Notes in Computer Science 7386, Springer, 2012, pp. 236–243. doi:10.1007/978-3-642-31623-4_18.

[86] G. Pighizzini, *Push complexity: Optimal bounds and decidability*, Theoretical Computer Science 1071 (2026), 115839. `https://doi.org/10.1016/j.tcs.2026.115839`

[87] G. Pighizzini and L. Prigioniero, "Pushdown Automata and Constant Height: Decidability and Bounds," *Acta Informatica* 60 (2023), 123–144. doi:10.1007/s00236-022-00434-0.

[88] G. Pighizzini and L. Prigioniero, *Pushdown and one-counter automata: Constant and non-constant memory usage*, Information and Computation 306 (2025), 105329. `https://doi.org/10.1016/j.ic.2025.105329`

[89] W. Hoeffding, *Probability inequalities for sums of bounded random variables*, Journal of the American Statistical Association 58 (1963), 13–30.

[90] O. H. Ibarra, J. Jirásek, I. McQuillan, and L. Prigioniero, *Space complexity of stack automata models*, International Journal of Foundations of Computer Science 32(6) (2021), 801–823. `https://doi.org/10.1142/S0129054121420090`

[91] E. Sperner, Ein Satz über Untermengen einer endlichen Menge, *Mathematische Zeitschrift* 27 (1928), 544–548.

[92] N. G. de Bruijn, C. van Ebbenhorst Tengbergen, and D. Kruyswijk, On the set of divisors of a number, *Nieuw Archief voor Wiskunde*, Ser. 2, 23 (1951), 191–193.

[93] D. Lubell, A short proof of Sperner's lemma, *Journal of Combinatorial Theory* 1(2) (1966), 299.

[94] D. J. Kleitman, On Dedekind's problem: The number of monotone Boolean functions, *Proceedings of the American Mathematical Society* 21(3) (1969), 677–682.

[95] R. Geoghegan, *Topological Methods in Group Theory*, Graduate Texts in Mathematics 243, Springer, New York, 2008.

[96] M. O. Rabin, Recursive unsolvability of group theoretic problems, *Annals of Mathematics* 67(1) (1958), 172–194.

[97] M. Filakovský, P. Franek, U. Wagner, and S. Zhechev, Computing simplicial representatives of homotopy group elements, *Journal of Applied and Computational Topology* 2 (2018), 177–231.

[98] O. Kupferman and J. Mosheiff, Prime languages, *Information and Computation* 240 (2015), 90–107. doi:10.1016/j.ic.2014.09.010.

[99] P. Totzke, Prime DOCAs, Automata Exchange, Problem 20.03, July 1, 2020.

[100] M. L. Minsky, *Computation: Finite and Infinite Machines*, Prentice-Hall, Englewood Cliffs, NJ, 1967.

[101] I. Jecker, O. Kupferman, and N. Mazzocchi, Unary prime languages, in *MFCS 2020*, LIPIcs 170, Article 51, 2020. doi:10.4230/LIPIcs.MFCS.2020.51.

[102] I. Jecker, N. Mazzocchi, and P. Wolf, Decomposing permutation automata, *Journal of Computer and System Sciences* 156 (2026), Article 103721. doi:10.1016/j.jcss.2025.103721.

[103] D. A. Spenner, Decomposing finite languages, in *MFCS 2023*, LIPIcs 272, Article 83, 2023. doi:10.4230/LIPIcs.MFCS.2023.83.

[104] D. A. Spenner, Deciding DFA-primality is NP-hard, in *ICALP 2026*, LIPIcs 374, Article 192, 2026. doi:10.4230/LIPIcs.ICALP.2026.192.

[105] S. Almagor, G. Avni, H. Sinclair-Banks, and A. Yeshurun, Dimension-minimality and primality of counter nets, in *FoSSaCS 2024*, LNCS 14575, Springer, 2024, pp. 229–249. doi:10.1007/978-3-031-57231-9_11.

[106] L. G. Valiant and M. S. Paterson, Deterministic one-counter automata, *Journal of Computer and System Sciences* 10(3) (1975), 340–350. doi:10.1016/S0022-0000(75)80005-5.

[107] S. Böhm, S. Göller, and P. Jančar, Equivalence of deterministic one-counter automata is NL-complete, in *Proceedings of STOC 2013*, ACM, 2013, pp. 131–140. doi:10.1145/2488608.2488626.

[108] P. Hofman and P. Totzke, Trace inclusion for one-counter nets revisited, *Theoretical Computer Science* 735 (2018), 50–63. doi:10.1016/j.tcs.2017.05.009.

[109] A. Prakash, *Problem 24.04: Language Equivalence between Nondeterministic and Deterministic One-Counter Machines*, Automata Exchange, 2024. Problem 24.04 webpage.

[110] L. G. Valiant, *Decision Procedures for Families of Deterministic Pushdown Automata*, Ph.D. thesis, University of Warwick, 1973.

[111] R. R. Howell, L. E. Rosier, D. T. Huynh, and H.-C. Yen, Some complexity bounds for problems concerning finite and 2-dimensional vector addition systems with states, *Theoretical Computer Science* 46 (1986), 107–140. doi:10.1016/0304-3975(86)90026-5.

[112] P. Hofman, S. Lasota, R. Mayr, and P. Totzke, Simulation problems over one-counter nets, *Logical Methods in Computer Science* 12(1) (2016), Article 6. doi:10.2168/LMCS-12(1:6)2016.

[113] S. Almagor and A. Yeshurun, Determinization of one-counter nets, in *CONCUR 2022*, LIPIcs 243, Article 18, 2022. doi:10.4230/LIPIcs.CONCUR.2022.18.

[114] A. Prakash and K. S. Thejaswini, On history-deterministic one-counter nets, in *FoSSaCS 2023*, LNCS 13992, Springer, 2023, pp. 218–239. doi:10.1007/978-3-031-30829-1_11.

[115] K. Reinhardt, Reachability in Petri nets with inhibitor arcs, *Electronic Notes in Theoretical Computer Science* 223 (2008), 239–264. doi:10.1016/j.entcs.2008.12.042.

[116] R. Bonnet, The reachability problem for vector addition system with one zero-test, in *MFCS 2011*, LNCS 6907, Springer, 2011, pp. 145–157. doi:10.1007/978-3-642-22993-0_16.

[117] J. L. Gross and T. W. Tucker, Generating all graph coverings by permutation voltage assignments, *Discrete Mathematics* 18(3) (1977), 273–283. doi:10.1016/0012-365X(77)90131-5.

[118] T. Zaslavsky, Biased graphs. I. Bias, balance, and gains, *Journal of Combinatorial Theory, Series B* 47(1) (1989), 32–52. doi:10.1016/0095-8956(89)90063-4.

[119] J.-G. Eon, Graph-theoretical characterization of periodicity in crystallographic nets and other infinite graphs, *Acta Crystallographica Section A* 61(5) (2005), 501–511. doi:10.1107/S0108767305019963.

[120] O. Delgado-Friedrichs and M. O'Keeffe, Crystal nets as graphs: Terminology and definitions, *Journal of Solid State Chemistry* 178(8) (2005), 2480–2485. doi:10.1016/j.jssc.2005.06.011.

[121] G. Lerman and Y. Shi, Robust Group Synchronization via Cycle-Edge Message Passing, *Foundations of Computational Mathematics* 22 (2022), 1665–1741. doi:10.1007/s10208-021-09532-w.

[122] R. Alur and P. Madhusudan, Visibly pushdown languages, in *Proceedings of STOC 2004*, ACM, 2004, pp. 202–211. doi:10.1145/1007352.1007390.

[123] D. Nowotka and J. Srba, Height-Deterministic Pushdown Automata, in *MFCS 2007*, LNCS 4708, Springer, 2007, pp. 125–134. doi:10.1007/978-3-540-74456-6_13.

[124] A. Björner, Topological methods, in R. L. Graham, M. Grötschel, and L. Lovász (eds.), *Handbook of Combinatorics*, Vols. 1–2, Elsevier, Amsterdam, 1995, pp. 1819–1872.

[125] P. Monasse and F. Guichard, Fast Computation of a Contrast-Invariant Image Representation, *IEEE Transactions on Image Processing* 9(5) (2000), 860–872. doi:10.1109/83.841532.

[126] T. Liu, M. Seyedhosseini, and T. Tasdizen, Image Segmentation Using Hierarchical Merge Tree, *IEEE Transactions on Image Processing* 25(10) (2016), 4596–4607. doi:10.1109/TIP.2016.2592704.

[127] A. G. Silva and R. de Alencar Lotufo, Efficient Computation of New Extinction Values from Extended Component Tree, *Pattern Recognition Letters* 32(1) (2011), 79–90. doi:10.1016/j.patrec.2010.07.019.

[128] D. Günther, J. Salmon, and J. Tierny, Mandatory Critical Points of 2D Uncertain Scalar Fields, *Computer Graphics Forum* 33(3) (2014), 31–40. doi:10.1111/cgf.12359.

[129] L. Yan, Y. Wang, E. Munch, E. Gasparovic, and B. Wang, A Structural Average of Labeled Merge Trees for Uncertainty Visualization, *IEEE Transactions on Visualization and Computer Graphics* 26(1) (2020), 832–842. doi:10.1109/TVCG.2019.2934242.

[130] T. Géraud, N. Boutry, S. Crozet, E. Carlinet, and L. Najman, A Proof of the Tree of Shapes in n-D, arXiv:2206.05109, 2022. doi:10.48550/arXiv.2206.05109.

[131] J. Mendes Forte, N. Passat, A. Shimizu, and Y. Kenmochi, Consistent Connected Operators Based on Trees of Shapes, *SIAM Journal on Imaging Sciences* 18(4) (2025), 2547–2579. doi:10.1137/25M174704X.

[132] E. Carlinet, B. Esteban, N. Boutry, and G. Romero-García, Leveraging the Geodesic Distance Transform on Interval-Valued Maps to Compute the Tree of Shapes, Research Square preprint, 2026. doi:10.21203/rs.3.rs-8888553/v1.

[133] D. Attali, U. Bauer, O. Devillers, M. Glisse, and A. Lieutier, Homological Reconstruction and Simplification in $\mathbb{R}^3$, *Computational Geometry* 48(8) (2015), 606–621. doi:10.1016/j.comgeo.2014.08.010.

[134] A. Darmann and J. Döcker, On a simple hard variant of Not-All-Equal 3-Sat, *Theoretical Computer Science* 815 (2020), 147–152. doi:10.1016/j.tcs.2020.02.010.

[135] T. M. Athawale, Z. Wang, D. Pugmire, K. Moreland, Q. Gong, S. Klasky, C. R. Johnson, and P. Rosen, Uncertainty Visualization of Critical Points of 2D Scalar Fields for Parametric and Nonparametric Probabilistic Models, *IEEE Transactions on Visualization and Computer Graphics* 31(1) (2025), 108–118. doi:10.1109/TVCG.2024.3456393.

[136] T. M. Athawale, D. Maljovec, L. Yan, C. R. Johnson, V. Pascucci, and B. Wang, Uncertainty Visualization of 2D Morse Complex Ensembles Using Statistical Summary Maps, *IEEE Transactions on Visualization and Computer Graphics* 28(4) (2022), 1955–1966. doi:10.1109/TVCG.2020.3022359.

[137] A. Nigmetov and D. Morozov, Local–Global Merge Tree Computation with Local Exchanges, in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '19)*, Art. 60, 13 pp., 2019. doi:10.1145/3295500.3356188.

[138] J. Brüggemann, On Merge Trees and Discrete Morse Functions on Paths and Trees, *Journal of Applied and Computational Topology* 7 (2023), 103–138. doi:10.1007/s41468-022-00101-w.

[139] M. Abadi and L. Lamport, The existence of refinement mappings, *Theoretical Computer Science* 82 (1991), 253–284. doi:10.1016/0304-3975(91)90224-P.

[140] R.-J. Back and J. von Wright, *Refinement Calculus: A Systematic Introduction*, Springer, New York, 1998. doi:10.1007/978-1-4612-1674-2.

[141] P. Cousot, Constructive design of a hierarchy of semantics of a transition system by abstract interpretation, *Electronic Notes in Theoretical Computer Science* 6 (1997), 77–102. doi:10.1016/S1571-0661(05)80168-9.

[142] J. J. M. M. Rutten, Universal coalgebra: a theory of systems, *Theoretical Computer Science* 249 (2000), 3–80. doi:10.1016/S0304-3975(00)00056-6.

[143] G. Huet, Confluent reductions: abstract properties and applications to term rewriting systems, *Journal of the ACM* 27 (1980), 797–821. doi:10.1145/322217.322230.

[144] L. Lovász, On the ratio of optimal integral and fractional covers, *Discrete Mathematics* 13 (1975), 383–390. doi:10.1016/0012-365X(75)90058-8.

[145] M. Karchmer, E. Kushilevitz, and N. Nisan, Fractional covers and communication complexity, *SIAM Journal on Discrete Mathematics* 8 (1995), 76–92. doi:10.1137/S0895480192238482.

[146] H. S. Witsenhausen, The zero-error side information problem and chromatic numbers, *IEEE Transactions on Information Theory* 22 (1976), 592–593. doi:10.1109/TIT.1976.1055607.

[147] E. R. Scheinerman and D. H. Ullman, *Fractional Graph Theory*, Wiley, 1997.

[148] S. Abramsky and A. Brandenburger, The sheaf-theoretic structure of non-locality and contextuality, *New Journal of Physics* 13 (2011), 113036. doi:10.1088/1367-2630/13/11/113036.

[149] S. Abramsky and G. Carù, Non-locality, contextuality and valuation algebras: a general theory of disagreement, *Philosophical Transactions of the Royal Society A* 377 (2019), 20190036. doi:10.1098/rsta.2019.0036.

[150] S. B. Montanhano, Contextuality in the bundle approach, n-contextuality, and the role of holonomy, arXiv:2105.14132, 2021.

[151] B. Amaral, Resource theory of contextuality, *Philosophical Transactions of the Royal Society A* 377 (2019), 20190010. doi:10.1098/rsta.2019.0010.

[152] A. Chattopadhyay, H. Carr, D. Duke, Z. Geng, and O. Saeki, Multivariate Topology Simplification, *Computational Geometry* 58 (2016), 1–24. doi:10.1016/j.comgeo.2016.05.006.

[153] W. Köpp and T. Weinkauf, Temporal Merge Tree Maps: A Topology-Based Static Visualization for Temporal Scalar Data, *IEEE Transactions on Visualization and Computer Graphics* 29(1) (2023), 1157–1167. doi:10.1109/TVCG.2022.3209387.

[154] M. Li, X. Yan, L. Yan, T. Needham, and B. Wang, Flexible and Probabilistic Topology Tracking With Partial Optimal Transport, *IEEE Transactions on Visualization and Computer Graphics* 31(10) (2025), 7951–7969. doi:10.1109/TVCG.2025.3561300.

[155] A. E. Bütün, *Semantic Normalization and Physical Realization: A Unified Theory—Geometry, Chronology, Information, and Resources*, research monograph, 2026, Chapters 108–109. doi:10.5281/zenodo.23221519.

[156] M. Bojańczyk and T. Colcombet, "Tree-walking automata cannot be determinized," *Theoretical Computer Science* **350** (2006), 164–173. doi:10.1016/j.tcs.2005.10.031.

[157] M. Bojańczyk, "Tree-walking automata," survey in *Language and Automata Theory and Applications (LATA 2008)*. author-hosted preprint.

[158] M. Bojańczyk and T. Colcombet, "Tree-walking automata do not recognize all regular languages," *SIAM Journal on Computing* **38** (2008), 658–701. doi:10.1137/050645427.

[159] A. Muscholl, M. Samuelides, and L. Segoufin, "Complementing deterministic tree-walking automata," *Information Processing Letters* **99** (2006), 33–39. doi:10.1016/j.ipl.2005.09.017.

[160] J. Engelfriet and H. J. Hoogeboom, "Tree-walking pebble automata," in *Jewels Are Forever*, Springer (1999), 72–83. doi:10.1007/978-3-642-60207-8_7.

[161] O. Martynova and A. Okhotin, "Homomorphisms and inverse homomorphisms on graph-walking automata," *Theoretical Computer Science* **979** (2023), 114197. doi:10.1016/j.tcs.2023.114197.

[162] M. Bojańczyk, "It is undecidable if two regular tree languages can be separated by a deterministic tree-walking automaton," *Fundamenta Informaticae* **154** (2017), 37–46. doi:10.3233/FI-2017-1551.

[163] O. Martynova and A. Okhotin, "Nondeterministic tree-walking automata are not closed under complementation," *ICALP 2025*, LIPIcs **334**, article 168. doi:10.4230/LIPIcs.ICALP.2025.168.

[164] A. E. Bütün, *Chronological Carry Refinement and p-Adic Root Geometry: A Third-Order Arithmetic Lift of DPDA-Derived Correction Geometry*, Zenodo preprint (2026), doi:10.5281/zenodo.22709258.

[165] A. E. Bütün, *Fourth-Order Chronological Carry: Connected Quadruples, Extension Parameters, and the Stirling Shadow*, Zenodo preprint (2026), doi:10.5281/zenodo.22709298.

[166] A. E. Bütün, *CCR-k General Theory: Higher-Order Chronological Carry, Hessenberg Hierarchies, and Universal Stirling Shadows*, Zenodo preprint (2026), doi:10.5281/zenodo.22734188.

[167] F. Bencherif and T. Garici, *On a property of Stirling polynomials*, Publ. Inst. Math. (Beograd) 102(116) (2017), 149–153. doi:10.2298/PIM1716149B.

[168] A. T. Lundell, *A divisibility property for Stirling numbers*, J. Number Theory 10 (1978), 35–54. doi:10.1016/0022-314X(78)90005-7.

[169] A. Adelberg, *A finite difference approach to degenerate Bernoulli and Stirling polynomials*, Discrete Math. 140 (1995), 1–21. doi:10.1016/0012-365X(93)E0188-A.

[170] A. Adelberg, *Arithmetic properties of the Nörlund polynomial* $B_n^{(x)}$, Discrete Math. 204 (1999), 5–13. doi:10.1016/S0012-365X(98)00363-X.

[171] NIST Digital Library of Mathematical Functions, §26.8, Eq. 26.8.36 (signed surjection/Stirling-number identity), `https://dlmf.nist.gov/26.8.E36`.

[172] NIST Digital Library of Mathematical Functions, §24.16, *Nörlund polynomials*, `https://dlmf.nist.gov/24.16`.

[173] A. D. Sokal, *Multiple orthogonal polynomials, d-orthogonal polynomials, production matrices, and branched continued fractions*, Trans. Amer. Math. Soc. Ser. B 11 (2024), 762–797. doi:10.1090/btran/133.

[174] A. Krelifa and E. Zerouki, *Riordan arrays and d-orthogonality*, Linear Algebra Appl. 515 (2017), 331–353. doi:10.1016/j.laa.2016.11.039.

[175] R. Cluckers and A. Herremans, *The fundamental theorem of prehomogeneous vector spaces modulo p^m (with an appendix by F. Sato)*, Bull. Soc. Math. France 135 (2007), 475–495, §3 (a p-adic analogue of Morse's lemma). `https://www.numdam.org/article/BSMF_2007__135_4_475_0.pdf`.

[176] A. Dwivedi and N. Saxena, *Computing Igusaś local zeta function of univariates in deterministic polynomial-time*, preprint (2020), arXiv:2006.08926.

[177] Mertens' theorem on reciprocal primes and the corresponding Euler product; see e.g. Z. Teitler, *Analytic Number Theory Notes*, Section 6.3, `https://zteitler.github.io/analytic-number-theory-notes/sec-mertens.html`.

[178] *Chebotarev density theorem*, classical theorem; see, e.g., H. W. Lenstra Jr., "Chebotarev and his density theorem".

[179] V. P. Kostov, *A property of discriminants*, Vietnam J. Math. (2019), doi:10.1007/s10013-018-0296-9.

[180] S. D. Cohen, *The Galois group of a polynomial with two indeterminate coefficients*, Pacific J. Math. 90 (1980), 63–76, doi:10.2140/pjm.1980.90.63.

[181] S. D. Cohen, *Corrections to: The Galois group of a polynomial with two indeterminate coefficients*, Pacific J. Math. 97 (1981), 483–486, doi:10.2140/pjm.1981.97.483.

[182] F. Naccarato, *The arithmetic of critical values I: equicritical quartic polynomials*, International Mathematics Research Notices 2026 (2026), rnag141, doi:10.1093/imrn/rnag141.

[183] J.-P. Serre, *Topics in Galois Theory*, 2nd ed., A K Peters, 2008, §4.4.

[184] J. C. Rosales and P. A. García-Sánchez, *Numerical Semigroups*, Springer, 2009.

[185] A. E. Bütün, "Higher-Dimensional Completion Geometry for Linear Counting Languages: Shortest-Prefix Quotients, Rank-One Topology, and Eventual Geodesic Structure," 2026. `https://doi.org/10.5281/zenodo.22709372`.

[186] A. E. Bütün, "Beyond Kernel Rank: Optimal-Fiber Geometry and Rank-Two Topology of Minimum-Completion Quotients," 2026. `https://doi.org/10.5281/zenodo.22709322`.

[187] A. E. Bütün, "Lattice-Convex Realization and Rank-Raising Topology of Optimal Completion Fibers," 2026. `https://doi.org/10.5281/zenodo.22709306`.

[188] A. E. Bütün, "Higher-Rank Completion Holonomy: Displacement-Marked Optimal Fibers and Physical LIFO Transport," 2026. https://doi.org/10.5281/zenodo.22863525.

[189] U. Bauer, M. Kerber, F. Roll, and A. Rolle, "A unified view on the functorial nerve theorem and its variations," *Expositiones Mathematicae* **41** (2023), art. 125503, https://doi.org/10.1016/j.exmath.2023.04.005.

[190] E. J. González and J. González, "Equivariant Nerve Lemma, simplicial difference, and models for configuration spaces on simplicial complexes," *Topology and its Applications* **341** (2024), art. 108749, https://doi.org/10.1016/j.topol.2023.108749.

[191] B. Sturmfels, "On vector partition functions," *Journal of Combinatorial Theory, Series A* **72** (1995), 302–309, https://doi.org/10.1016/0097-3165(95)90067-5.

[192] H. Charalambous, A. Katsabekis, and A. Thoma, "Minimal systems of binomial generators and the indispensable complex of a toric ideal," *Proceedings of the American Mathematical Society* **135** (2007), 3443–3452, https://doi.org/10.1090/S0002-9939-07-09037-5.

[193] A. E. Bütün, *Higher-Dimensional Completion Geometry for Linear Counting Languages: Shortest-Prefix Quotients, Rank-One Topology, and Eventual Geodesic Structure* (2026), https://doi.org/10.5281/zenodo.22709372.

[194] A. E. Bütün, *Beyond Kernel Rank: Optimal-Fiber Geometry and Rank-Two Topology of Minimum-Completion Quotients* (2026), https://doi.org/10.5281/zenodo.22709322.

[195] A. E. Bütün, *Lattice-Convex Realization and Rank-Raising Topology of Optimal Completion Fibers* (2026), https://doi.org/10.5281/zenodo.22709306.

[196] A. E. Bütün, *Higher-Rank Completion Holonomy: Displacement-Marked Optimal Fibers and Physical LIFO Transport* (2026), https://doi.org/10.5281/zenodo.22863525.

[197] W. Bruns and J. Herzog, "Semigroup rings and simplicial complexes," *Journal of Pure and Applied Algebra* **122** (1997), 185–208, https://doi.org/10.1016/S0022-4049(97)00051-0.

[198] P. A. García-Sánchez and A. Herrera-Poyatos, "Isolated factorizations and their applications in simplicial affine semigroups," *Journal of Algebra and Its Applications* **19** (2020), https://doi.org/10.1142/S0219498820500826.

[199] S. A. M. Jones and R. M. Thomas, "Word problems of groups: Formal languages, characterizations and decidability," *Theoretical Computer Science* (2018), https://doi.org/10.1016/j.tcs.2018.05.007.

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